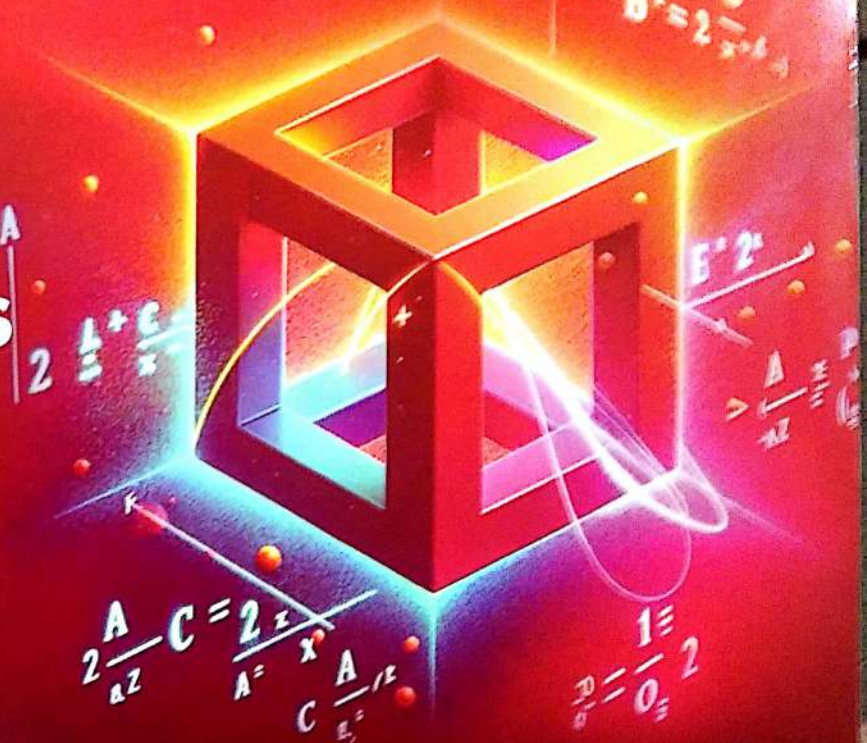
PHYSICS WALLAH

ARJUNA JEE



Class-XI

SOLUTIONS

By Aditya Mohod
username - muje_kabhi_dekha_nahi

MATHEMATICS

ARJUNA JEE

MATHEMATICS

CLASS-XI

FULL COURSE STUDY MATERIAL

SOLUTIONS

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CHAPTER 1

Basic Mathematics and Logarithm

Prarambh (Topicwise)

1. (b) It is a property.

2. (b) x and y are rational and
$(x+y)+(x-2y)\sqrt{2}=2x-y+(x-y-1)\sqrt{6}$
Since real part of both sides are equal and coefficient of irrational parts should be zero in both sides
$\therefore \quad x-2y=0 \Rightarrow x=2y$
and $x-y-1=0 \Rightarrow y=1 \therefore x=2$
$x=2, y=1$

3. (d) It is a property

4. (a) Here $(x-1)^2+(x-2)^2+(x-3)^2=0$
Here sum of three + ve number can not be equal to zero then all three number independently be zero.
$\therefore$ There are no any real root.

5. (c) Let $P(x)=x^3-a^2x+x+2$ be the given polynomial
Then by factor theorem, $(x-a)$ is a factor of $P(x)$ iff $P(A)=0$
$\Rightarrow \quad a^3-a^2.a+a+2=0$
$\Rightarrow \quad a+2=0 \Rightarrow a=-2$

6. (a) Every irrational number can be expressed on the number line. This statement is always true.

7. (d) (rational) $\times$ (irrational) = irrational except when $x=0$

8. (b) $2x^2-4xy+xy-2y^2=7$
$2x(x-2y)+y(x-2y)=7$
$(x-2y)(2x+y)=7$
x, y are integers $\Rightarrow x-2y, 2x+y$ are also integers
Four cases are possible
Case I: $x-2y=1, 2x+y=7 \Rightarrow x=3, y=1$
$|x+y|=4$
Case II: $x-2y=7, 2x+y=1 \Rightarrow x=\frac{9}{5}$ rejected
Case III: $x-2y=-1, 2x+y=-7$
$\Rightarrow \quad x=-3, y=-1$
$|x+y|=4$
Case IV: $x-2y=-7, 2x+y=-1 \Rightarrow x=-\frac{9}{5}$ rejected
Hence $|x+y|=4$

9. (b) $a(a-b)+b(b-c)+c(c-a)=0$
$\Rightarrow a^2+b^2+c^2-ab-bc-ca=0$
Multiplying & dividing by 2,
$\frac{1}{2}[(a^2+b^2-2ab)+(b^2+c^2-2bc)+(a^2+c^2-2ac)]=0$
$\Rightarrow \quad \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]=0$
$\Rightarrow \quad (a-b)^2+(b-c)^2+(c-a)^2=0 \Rightarrow a=b=c$

10. (a) $2x^3-5x^2+x+2$
$=ax^3+(-b-2a)x^2+(-1+2b)x+2$
On comparing coefficient of x^3 & x, we get
$a=2$ & $b=1$

11. (b) $[e]-[-\pi]$
here $e \approx 2.7$
$\pi \approx 3.14$
$\therefore [2.7]-[-3.14]$
$=2-(-4)=2+4=6$

12. (d)
$$L=\frac{1}{\sqrt{7}-\sqrt{8}}+\frac{1}{\sqrt{7}-\sqrt{6}}+\frac{1}{3-\sqrt{8}}+\frac{1}{\sqrt{5}+2}+\frac{1}{\sqrt{5}-\sqrt{6}}$$
$$=\frac{\sqrt{7}+\sqrt{8}}{-1}+\frac{\sqrt{7}+\sqrt{6}}{1}+\frac{3+\sqrt{8}}{1}+\frac{\sqrt{5}-2}{1}+\frac{\sqrt{5}+\sqrt{6}}{-1}$$
$$=-\sqrt{7}-\sqrt{8}+\sqrt{7}+\sqrt{6}+3+\sqrt{8}+\sqrt{5}-2-\sqrt{5}-\sqrt{6}$$
$$=1$$
$\Rightarrow a=0, b=0$

13. (d) $\frac{(a-b)^3+(b-c)^3+(c-a)^3}{(a-b)(b-c)(c-a)}=?$
Let $a-b=A, (b-c)=B, (c-a)=C$
[$\because A+B+C=0 \Rightarrow A^3+B^3+C^3=3ABC$]
$\therefore \quad \frac{(a-b)^3+(b-c)^3+(c-a)^3}{(a-b)(b-c)(c-a)}=\frac{3ABC}{ABC}=3$

14. (c) Putting $x=1$, remainder = 7

15. (d)
$$\frac{1}{\log_b^b+\log_b^a+\log_b^c}+\frac{1}{\log_c^c+\log_c^a+\log_c^b}+\frac{1}{\log_a^a+\log_a^b+\log_a^c}$$
$$=\frac{1}{\log_b abc}+\frac{1}{\log_c abc}+\frac{1}{\log_a abc}$$
$$=\log_{abc} b+\log_{abc} c+\log_{abc} a=\log_{abc} abc$$
$$=1$$

16. (c) $\log_7 \log_7 \sqrt{7\sqrt{7\sqrt{7}}}$
$=\log_7 \log_7 7^{7/8}=\log_7(7/8)$
$=\log_7 7-\log_7 8=1-\log_7 2^3=1-3\log_7 2.$

17. (b) Applying base change theorem,
$=\log_{abc}\sqrt{bc}+\log_{abc}\sqrt{ca}+\log_{abc}\sqrt{ab}$
$=\log_{abc}\sqrt{bc}\cdot\sqrt{ca}\cdot\sqrt{ab}=\log_{abc} abc=1$

18. (d) Given, $\log_x \log_{18}(\sqrt{2}+\sqrt{8})=\frac{1}{3}$
$\Rightarrow \log_{18}(\sqrt{2}+2\sqrt{2})=x^{1/3}$
$\Rightarrow \log_{18} 3\sqrt{2}=x^{1/3}$
$\Rightarrow \log_{18}(\sqrt{2}+2\sqrt{2})=x^{1/3}$
$\Rightarrow \log_{18} 3\sqrt{2}=x^{1/3}$
$2x^{1/3}=1 \Rightarrow x^{1/3}=\frac{1}{2} \Rightarrow x=\frac{1}{8}$
$\Rightarrow 1000x=125$

19. (c) $\sqrt{\log_{10}(-x)}=\log_{10}\sqrt{x^2}$
$\Rightarrow x<0$
$\sqrt{\log_{10}(-x)}=\log_{10}(-x)$
$\Rightarrow \log^2{}_{10}(-x)-\log_{10}(-x)=0$
$\Rightarrow \log_{10}(-x)=0, 1 \Rightarrow x=-1, -10$
two solution

20. (c) $\log_2 15 . \log_{1/6} 2 . \log_3 1/6$
$=\frac{\log_e 15}{\log_e 2}\times\frac{\log_e 2}{\log_e 1/6}\times\frac{\log_e 1/6}{\log_e 3}$
$=\frac{\log_e(3\times5)}{\log_e 3}=1+\log_3 5$
$\therefore [1+\log_3 5]=2$

21. (d) $=\frac{2^{\log_2(a^4)}-3^{\log_3(a^2+1)}-2a}{7^{\log_7(a^2)}-a-1}$
$=\frac{a^4-(a^2+1)-2a}{a^2-a-1}$
$=\frac{(a^2)^2-(a+1)^2}{(a^2-a-1)}=a^2+a+1$

22. (*b*) $3^{\log_3(x^2)} - 2x - 3 = 0$

$\Rightarrow x^2 - 2x - 3 = 0$

$\Rightarrow (x-3)(x+1) = 0$

$\Rightarrow x = -1, 3$, ($x = -1$ reject $\because x > 0$)

Number of values of x is one

23. (*c*) $\log_2 7 \Rightarrow \log_2 4 < \log_2 7 < \log_2 8$

$\Rightarrow 2 < \log_2 7 < 3$ i.e. not integer

Let $\log_2 7 = \frac{p}{q}$ (where p and q are coprime)

$\Rightarrow 2^{p/q} = 7 \Rightarrow 2^p = 7^q$

which is not possible so $\log_2 7$ is an irratoinal number

24. (*c*) $\Rightarrow \text{antilog}_{16} 0.75 = (16)^{0.75}$

$= (16)^{3/4} = (2^4)^{3/4} = 2^3 = 8$

25. (*d*) $2\log x - \log(2x - 75) = 2$

$\frac{x^2}{2x-75} = 100 \Rightarrow x^2 - 200x + 7500 = 0$

26. (*b*) $\log_{\sqrt{0.9}} \log_5 (\sqrt{x^2+5+x}) > 0$

$\log_5 (\sqrt{x^2+5+x}) < 1$

$(x^2 + 5 + x)^{1/2} < 5$ and $x^2 + x + 5 > 0$

$\Rightarrow \quad x^2 + 5 + x < 25$

$\Rightarrow \quad x^2 + x - 20 < 0$

$\Rightarrow \quad (x+5)(x-4) < 0$

$\Rightarrow \quad x \in (-5, 4)$

$\therefore \quad n = 8$

27. (*a*) $x^2 - 4 > 0 \Rightarrow x \in (-\infty, -2) \cup (2, \infty)$

$\log_5(x^2-4) > 0 \Rightarrow x^2 - 4 > 1 \Rightarrow x^2 - 5 > 0$

$\Rightarrow x \in (-\infty, -\sqrt{5}) \cup (\sqrt{5}, \infty)$

Now

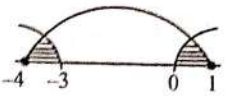
$\log_{0.5} \log_5 (x^2-4) > \log_{0.5} 1 \Rightarrow \log_5 (x^2-4) < 1$

$x^2 - 4 < 5 \Rightarrow x^2 - 9 < 0, x \in (-3, 3)$

$\therefore \quad (-3, -\sqrt{5}) \cup (\sqrt{5}, 3)$

28. (*b*) $2 - \log_2 (x^2 + 3x) \geq 0$

$\log_2 (x^2 + 3x) \leq 2$



$\Rightarrow x^2 + 3x \leq 4$

$\Rightarrow x^2 + 3x - 4 \leq 0$

$\Rightarrow (x+4)(x-1) \leq 0$

$\Rightarrow x \in [-4, 1]$

and $x^2 + 3x > 0 \Rightarrow x \in (-\infty, -3) \cup (0, \infty)$

Ans. : $[-4, -3) \cup (0, 1]$

29. (*c*) When $x \leq -\frac{3}{4}$; $-4x - 3 - 3x + 4 = 12$

$\Rightarrow x = -\frac{11}{7}$

When $-\frac{3}{4} < x \leq \frac{4}{3}$; $4x + 3 - 3x + 4 = 12$

$x = 5$ will not satisfy

When $x > \frac{4}{3}$ $4x + 3 + 3x - 4 = 12$

$7x = 13 \Rightarrow x = \frac{13}{7}$

30. (*a*) Since $(x^2 + x - 2) - (x^2 - 2x - 8) = 3x + 6 = 3(x+2)$

$\therefore \quad (x^2 - 2x - 8)(x^2 + x - 2) \leq 0$

i.e. $(x-4)(x+2)(x+2)(x-1) \leq 0$

$\therefore$ Solution set is $[1,4] \cup \{-2\}$

31. (*b*) $||x-1|-1| \leq 1$

$\Rightarrow -1 \leq |x-1| - 1 \leq 1$.

$\Rightarrow 0 \leq |x-1| \leq 2$

$\Rightarrow -2 \leq x - 1 \leq 2$

$\Rightarrow -1 \leq x \leq 3$

$\therefore x \in [-1, 3]$

32. (*d*) $|x|^2 - 3|x| + 2 = 0$

$\Rightarrow (|x|-2)(|x|-1) = 0$

$\Rightarrow |x| = 1, 2$

$\Rightarrow x = \pm 1, \pm 2$

$\therefore$ number of real roots is 4.

33. (*b*) Taking log on both sides,

$(3x^2 - 10x + 3) \log |x-3| = 0$

$\log |x-3| = 0$ or $3x^2 - 10x + 3 = 0$

$x - 3 \neq 0$; $|x-3| = 1$ or $(x-3)(3x-1) = 0$

$x \neq 3$; $(x-3) = \pm 1$ or $x = 3$; $x = \frac{1}{3}$

$\therefore x = 2, 4, 1/3 \quad [\because x \neq 3]$

Hence, three real solutions

34. (*a*) Use base change property.

35. (*d*) $x^2 - (y-z)^2 + (x+y-z)$

36. (*c*) $\frac{3x^4}{x^2-2x-3} = \frac{5x^4}{2x^2-7x+3}$

37. (*b*) $\frac{x^a + x^b + x^c}{x^a + x^b + x^c}$, Multiply x^a in first term, x^b in second and x^c in third term.

38. (*d*) Take log both side and simplify.

39. (*b*) $\log a = k(b-c)$, $\log b = k(c-a)$, $\log c = k(a-b)$

40. (*d*) $\frac{x^2 - 6x - 16}{2x+5} < 0$

Prabal (JEE Main Level)

1. (*c*) A & B are two rational number then $\frac{A}{B}$ is

Also rational number if $B \neq 0$.

2. (*a*) $x^{x \cdot x^{1/3}} = \left(x.x^{1/3}\right)^x \Rightarrow x^{x^{\frac{4}{3}}} = \left(x^{1+\frac{1}{3}}\right)^x$

$\Rightarrow x^{x^{\frac{4}{3}}} = \left(x^{4/3}\right)^x = x^{x^{\frac{4}{3}}} = x^{\frac{4}{3}x} \Rightarrow x^{\frac{4}{3}} = \frac{4}{3}x$,

Also 1 is an obvious solution.

3. (*a*) $4^{(x^2+2)} - 9.2^{(x^2+2)} + 8 = 0$

$\Rightarrow \left(2^{(x^2+2)}\right)^2 - 9.2^{(x^2+2)} + 8 = 0$

Put $2^{(x^2+2)} = y$. Then $y^2 - 9y + 8 = 0$, which gives $y = 8, y = 1$.

when $y = 8 \Rightarrow 2^{x^2+2} = 8 \Rightarrow 2^{x^2+2} = 2^3$

$\Rightarrow x^2 + 2 = 3$

$\Rightarrow x^2 = 1 \Rightarrow x = 1, -1$.

when $y = 1 \Rightarrow 2^{x^2+2} = 1 \Rightarrow 2^{x^2+2} = 2^0$

$\Rightarrow x^2 + 2 = 0 \Rightarrow x^2 = -2$, which is not possible

4. (*a*) Let x be the required logarithm, then by definition

$\log_{2\sqrt{2}} 32\sqrt[5]{4} = x$

$\left(2\sqrt{2}\right)^x = 32\sqrt[5]{4} \Rightarrow \left(2.2^{1/2}\right)^x = 2^5.2^{2/5}$;

$\therefore \quad 2^{\frac{3x}{2}} = 2^{5+\frac{2}{5}}$

Here, by equating the indices, $\frac{3}{2}x = \frac{27}{5}$

$\therefore x = \frac{18}{5} = 3.6$.

5. (*b*) $x = \log_a bc \Rightarrow 1 + x = \log_a a + \log_a bc = \log_a abc$

$\therefore (1+x)^{-1} = \log_{abc} a$

$\therefore (1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1}$

$= \log_{abc} a + \log_{abc} b + \log_{abc} c = \log_{abc} abc = 1$.

6. (*c*) $\log_7 \log_5 \left(\sqrt{x^2+5+x}\right) = 0 = \log_7 1$

$\Rightarrow \log_5 \left(x^2+5+x\right)^{1/2} = 1 = \log_5 5$

$\Rightarrow (x^2 + 5 + x)^{1/2} = 5$

$\Rightarrow (x^2 + x + 5) = 25 \Rightarrow x^2 + x - 20 = 0$

$\Rightarrow (x-4)(x+5) = 0 \Rightarrow x = 4, -5 \Rightarrow x = 4$

7. (*a*) $(0.05)^{\log_{\sqrt{20}}(0.1+0.01+\ldots\ldots)} = \left(\frac{1}{20}\right)^{2\log_{20}\left(\frac{0.1}{1-0.1}\right)}$

$= 20^{-2\log_{20}(1/9)} = 20^{2\log_{20} 9} = 20^{\log_{20} 9^2} = 9^2 = 81$

8. (*b*) $\log_2 . \log_3 \ldots . \log_{99} \log_{100} 100^{99^{98^{\cdot^{\cdot^{\cdot^{2^1}}}}}}$

$= \log_2 \log_3 \ldots . \log_{99}^{99^{98^{\cdot^{\cdot^{\cdot^{2^1}}}}}}$

$[\log_{100} 100 = 1]$

$= \log_2 . \log_3 \ldots . \log_{98}^{98^{97^{\cdot^{\cdot^{\cdot^{2^1}}}}}}$

$= \log_2 . \log_3 \ldots . \log_{97}^{97^{96^{\cdot^{\cdot^{\cdot^{2^1}}}}}} = \log_2 \log_3 3^{2^1}$

$= \log_2 2 \cdot \log_3 3 = \log_2 2 = 1$.

9. (*d*) $\log_2(x+5) = 6 - x$

$\Rightarrow x + 5 = 2^{6-x}$

$\Rightarrow x + 5 = 64.2^{-x}$

Let $y = x + 5$, $y = 64.2^{-x}$ will intersect at one point.

Number of solutions = 1.

10. (*c*) $y = 3^{12} \times 2^8 \Rightarrow \log_{10} y = 12 \log_{10} 3 + 8 \log_{10} 2$

$= 12 \times 0.47712 + 8 \times 0.30103$

$= 5.72544 + 2.40824 = 8.13368$

$\therefore$ Number of digits in y = 8 + 1 = 9.

11. (*d*) $\log_{|x|}(x^2 + x + 1) \geq 0$

$D : |x| \neq 0, 1$

Case-I: if $|x| < 1$

$x^2 + x + 1 \leq 1 \Rightarrow x(x+1) \leq 0$

$x \in (-1, 0)$

Case-II: if $|x| > 1$

$x^2 + x + 1 > 1 \Rightarrow x(x+1) \geq 0$

$\Rightarrow \quad x \in (-\infty, -1) \cup (0, \infty)$

$\therefore \quad x \in (-\infty, -1) \cup (1, \infty)$

12. (*b*) $\log_{1/2}\left(x^2 - 6x + 12\right) \geq -2 \quad \ldots (i)$

For log to be defined, $x^2 - 6x + 12 > 0$

$\Rightarrow (x-3)^2 + 3 > 0$, which is true $\forall\, x \in R$.

From (i), $x^2 - 6x + 12 \leq \left(\frac{1}{2}\right)^{-2}$

$\Rightarrow x^2 - 6x + 12 \leq 4 \Rightarrow x^2 - 6x + 8 \leq 0$

$\Rightarrow (x-2)(x-4) \leq 0 \Rightarrow 2 \leq x \leq 4$

$\therefore x \in [2, 4]$.

13. (*c*) $\log_{0.04}(x-1) \geq \log_{0.2}(x-1) \quad \ldots (i)$

For log to be defined $x - 1 > 0 \Rightarrow x > 1$

From (i), $\log_{(0.2)^2}(x-1) \geq \log_{0.2}(x-1)$

$\Rightarrow \frac{1}{2}\log_{0.2}(x-1) \geq \log_{0.2}(x-1)$

$\Rightarrow \sqrt{x-1} \leq (x-1)$

$\Rightarrow \sqrt{x-1}(1 - \sqrt{x-1}) \leq 0 \Rightarrow 1 - \sqrt{x-1} \leq 0$

$\Rightarrow \sqrt{x-1} \geq 1 \Rightarrow x \geq 2 \therefore x \in [2, \infty)$

14. (*a*) $\log_{0.3}(x-1) < \log_{(0.3)^2}(x-1)$

$\log_{0.3}(x-1) < \frac{1}{2}\log_{(0.3)}(x-1)$

$\log_{0.3}(x-1) < \log_{(0.3)}(x-1)^{1/2}$

Here base is less than 1, therefore the inequality is reversed

$(x-1) > (x-1)^{1/2}$

$(x-1)^2 > (x-1)$

$x^2 1 - 2x - x + 1 > 0$

$x^2 - 3x + 2 > 0$

$x^2 - 2x - x + 2 > 0$

$x(x-2) - 1(x-2) > 0$

$(x-1)(x-2) > 0$

$x \in (-\infty, 1) \cup (2, \infty)$

$\because x - 1 > 0 \Rightarrow x > 1$

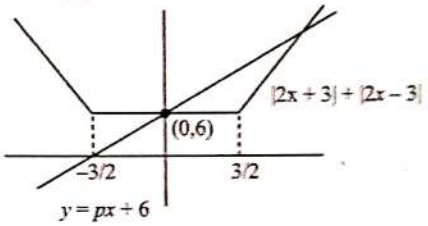
than $x \in (2, \infty)$

15. (*b*) $f(x) = |x-1| + |x-2| + |x-3|$

$$= \begin{cases} -x+1-x+2-x+3 = 6-3x, & x \leq 1 \\ x-1-x+2-x+3 = 4-x & 1 < x \leq 2 \\ x-1+x-2-x+3 = x & 2 < x \leq 3 \\ x-1+x-2+x-3 = 3x-6 & x > 3 \end{cases}$$

$\min f(x) = 2$.

16. (*d*)



$y = px + 6$

For more than 2 solution $p = 0$

17. (*b*) $3^a = 4$; $a = \log_3 4$

Similarly $b = \log_4 5$ etc.

Hence, $abcdef = \log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$

$= \log_3 9 = 2$

18. (*a*) Changing on base 2, we get

$$\frac{1}{1+\log_2 x} + \frac{\log_2 2x}{3} = -\frac{3}{2}$$

$$\Rightarrow \frac{1}{1+\log_2 x} + \frac{1+\log_2 x}{2} = -\frac{3}{2}$$

let $1 + \log_2 x = t$

$$\frac{1}{t} + \frac{t}{2} = -\frac{3}{2} \Rightarrow 2 + t^2 + 3t = 0$$

$\Rightarrow t^2 + 3t + 2 = 0$

$t_1 + t_2 = -3$

$(1 + \log_2 x_1) + (1 + \log_2 x_2) = -3$

$\log_2 x_1 x_2 = -5 \Rightarrow x_1 x_2 = \frac{1}{32}$

19. (*b*)

$$\frac{\sqrt{x^2+1}}{x^2+2} = \frac{1}{\frac{x^2+1+1}{\sqrt{x^2+1}}} = \frac{1}{\sqrt{x^2+1} + \frac{1}{\sqrt{x^2+1}}} \leq \frac{1}{2}$$

(as $t + \frac{1}{t} \geq 2$ for $t \to 1$)

$$\log_2(2x^2 + \sqrt{2}) \leq \frac{1}{2} \Rightarrow 2x^2 + \sqrt{2} \leq (2)^{1/2}$$

$\Rightarrow 2x^2 \leq 0$

which is possible only for $x = 0$

20. (*d*) $\therefore \quad x = \pm 1$ or $x^2 + 12 = 16$

$\therefore \quad x \pm 2$ also $x = 0$

$\therefore$ 5 solutions.

21. (*b*) $\log_{10}(0.15)^{20} = 20(\log_{10} 15 - 2)$

$= -16.478$

22. (*b*) $k + 2k + 3k \ldots 24k = n^3$

$\Rightarrow \quad k \cdot (1 + 2 + 3 \ldots\ldots 24) = n^3$

$\Rightarrow \quad k \cdot \frac{24 \cdot 25}{2} = n^3 \Rightarrow 2^2 \cdot 3 \cdot 5^2 \cdot k$

$= n^3 \Rightarrow$ smallest value of $k = 2 \cdot 3^2 \cdot 5 = 90$

23. (*c*) $\frac{a^3}{b^3} + \frac{b^3}{c^3} + \frac{c^3}{a^3} - 3 =$

$$\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a}\right)\left(\frac{a^2}{b^2} + \frac{b^2}{c^2} + \frac{c^2}{a^2} - \frac{a}{c} - \frac{b}{a} - \frac{c}{b}\right)$$

$$\Rightarrow \quad 6 \cdot \left[\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a}\right)^2 - 3\left(\frac{a}{c} + \frac{b}{a} + \frac{c}{b}\right)\right]$$

$\Rightarrow 6[6^2 - 3.6] = 6.12 = 72$

24. (*d*) $x, y \in I \Rightarrow x = y = 0$ $x, y \in I$ $3^x = 5^y$

$\Rightarrow \quad 5^{x \log_5 3} = 5^y \quad \Rightarrow \quad x = 1, y = \log_5 3^x$

$= 2$

or, $y = \log_5 9$

$\therefore$ Infinite ordered pairs are there

25. (*c*)

$\log_2 x + \frac{1}{2}\log_2 y + \frac{1}{2}\log_2 z = 2 \Rightarrow x\sqrt{y}\sqrt{z} = 4$

$\log_3 y + \frac{1}{2}\log_3 x + \frac{1}{2}\log_3 z = 2 \Rightarrow \sqrt{x} \cdot y \cdot \sqrt{z} = 9$

$\log_4 z + \frac{1}{2}\log_4 x + \frac{1}{2}\log_4 y = 2 \Rightarrow \sqrt{x} \cdot \sqrt{y} \cdot z = 16$

$\Rightarrow \quad xyz = 24$

26. (*c*) If $\alpha > 1$

$\log_{10}\alpha < \log_e \alpha < \log_2 \alpha$

Also $\log_2 \alpha > \log_5 \alpha$

27. (*c*) $\sqrt{x+2} - \sqrt{x-2} < 0.1$

$\Rightarrow \quad \sqrt{x+2} < 0.1 + \sqrt{x-2}$

Squaring both sides

$\Rightarrow \quad x + 2 < 0.01 + (x-2) + 0.2\sqrt{x-2}$

$\Rightarrow \quad 0.2\sqrt{x-2} > 3.99$

$\Rightarrow \quad x - 2 > 398.0025$

$\Rightarrow \quad x > 400.0025$

28. (*a*) Simplify using log properties.

29. (*a*) Clearly, $\left(\frac{1}{\log_{10}\pi}\right)^3 = (\log_\pi 10)^3 > 8$

$$\frac{1}{\log_{10}\sqrt{\pi}} = 2\log_\pi 10 > 4$$

$0 < \log_{10}\pi < 1$

$\sqrt{\log_{10}\pi^2} = \sqrt{2\log_{10}\pi} > \log_{10}\pi$

30. (*a*) $x^2 - 2x > 2$. Compare inequality on both side.

31. (*a*) Simplify using log base change properties.

32. (*a*) $xy = 18$, $x + y = 9$. Form two equations in x and y.

33. (*b*) Compare log with difference base and argument.

34. (*d*) Use basic identities.

35. (*a*) When $a+b+c=0$, then $a^3+b^3+c^3=3abc$

36. (*b*) $(\ln x)^2=(\ln x)-(\ln x)^2+1$

37. (*d*) $x-2>0,\ 1-x>0$

38. (*b*) $\frac{x^2-1}{x+3}\geq 0$ and $x^2-5x+2\leq 0$

$\frac{(x-1)(x+1)}{(x+3)}\geq 0$ and $x^2-5x+2\leq 0$

$x=5\pm\sqrt{17}$ and

$\left(x-\frac{5+\sqrt{1}}{2}\right)\left(x-\frac{5-\sqrt{17}}{2}\right)\leq 0$

$x\in(-3,-1)\cup[1,\infty)$

so, $x\in\left[1,\frac{5+\sqrt{17}}{2}\right]$

39. (*d*) $x^2+9<(x+3)^2<8x+25$

$x^2+9<x^2+9+6x<8x+25$

$\Rightarrow x^2+9<x^2+9+6x$

$\Rightarrow\ 6x>0, x>0$

and $x^2+9<6x+8x+25$

$x^2-2x-16<0$

$(x-1)^2-17<0$

$(x-1+\sqrt{17})(x-1-\sqrt{17})<0$

Number of integeral solutions = 5

40. (*d*) $\frac{2}{x^2-x+1}-\frac{1}{x+1}-\frac{2x-1}{x^3+1}\geq 0$

$\Rightarrow\frac{2(x+1)-\left(x^2-x+1\right)-(2x-1)}{(x+1)\left(x^2-x+1\right)}\geq 0$

$\Rightarrow\frac{2x+2-x^2+x-1-2x+1}{(x+1)\left(x^2-x+1\right)}$

$-\frac{(x-2)(x+1)}{(x+1)\left(x^2-x+1\right)}\geq 0$

$\Rightarrow\frac{(x-2)}{\left(x^2-x+1\right)}\leq 0\Rightarrow x\leq 2$

41. (*a*) $\log_{(3x^2+1)}2<\frac{1}{2}\log_{(3x^2+1)}\left(3x^2+1\right)$

$2<\left(3x^2+1\right)^{1/2}$

$\Rightarrow 4<3x^2+1$

$\Rightarrow\ 3<3x^2$

$\Rightarrow\ x^2>1$

$x\in(-\infty,-1)\cup(1,\infty)$

42 (*b*) $\log 4+\left(1+\frac{1}{2x}\right)\log 3=\log\left(3^{1/x}+27\right)$

$3^{1/2x}=t$

$\log 4+\log(3t)=\log(t^2+27)$

$\Rightarrow 12t=t^2+27$

$t=9,3\Rightarrow x=\frac{1}{4},\frac{1}{2}$

43. (*a*) $\log_7\left(\frac{2x-6}{2x-1}\right)>0$

$\frac{2x-6}{2x-1}>0$ and $\frac{2x-6}{2x-1}>1$

$x\in\left(-\infty,\frac{1}{2}\right)\cup(3,\infty)$ and $\frac{2x-6-2x+1}{2x-1}>0$

$\frac{1}{2x-1}<0$

$x<\frac{1}{2}$

Common solution $x<\frac{1}{2}$

44. (*c*) $\log_{(x-3)}(2x-3)$

$2x-3>0\ \ x>3/2$ and

$x-3\neq 1$ and $x-3>0$

$x\neq 4$ and $x>3$

$x\in(3,\infty)-\{4\}$

45. (*a*) Use property

If $a+b+c=0$, then $(a^3+b^3+c^3)=3abc$

46. [3] $\log_e x+\log_e y+\log_e z=0$

$\Rightarrow\log_e(xyz)=0$

$\Rightarrow xyz=1$

And

$\log_e k=\left(\frac{1}{\log_1 y}+\frac{1}{\log_e z}\right)\log_e x+\left(\frac{1}{\log_e z}+\frac{1}{\log_e x}\right)\log_e y$

$+\left(\frac{1}{\log_2 x}+\frac{1}{\log_e y}\right)\log_e z$

47. [89] $\log_2\left(\log_3\left(\log_4 x\right)\right)=0$

$\Rightarrow\log_3\left(\log_4 x\right)=1$

$\log_4 x=3$

$\Rightarrow x=4^3=64$

$\log_3\left(\log_4\left(\log_2(y)\right)\right)=0$

$\Rightarrow\log_4\left(\log_2(y)\right)=1$

$\Rightarrow\ \log_2 y=4$

$\boxed{y=16}$

$\log_4\left(\log_2\left(\log_3(2)\right)\right)=0$

$\log_2\left(\log_3(z)\right)=1$

$\log_3(z)=2$

$\boxed{z=9}$

48. [75] $\log_2 x+\log_4 x+\log_8 x=\log_k x$

$\log_2 x+\frac{1}{2}\log_2 x+\frac{1}{3}\log_2 x=\log_k x$

$\log_2 x\left(1+\frac{1}{2}+\frac{1}{3}\right)=\log_k x$

$(6+3+2)\log_2 x=\log_k x$

$\log_{\frac{1}{2^{11}}}(x)=\log_k x$

$\Rightarrow 1/\log^{1/1}$

$\Rightarrow K=2^{1/11}$

$a+b=2+11=13$

49. [1] $\frac{2}{\log_4(2000)^6}+\frac{3}{\log_5(2000)^6}$

$=\frac{2}{6}\log_{2000}(4)+\frac{3}{6}\log_{2000}(5)$

$=\frac{1}{3}\log(2000)^4+\frac{1}{2}^{\log_{(2000)}5}$

$=\log_{2000}\left(4^{1/3}\cdot 5^{1/2}\right)=\frac{1}{6}$

50. [2] $N=7^{\log_{49}(q-0)}$

$A=2^{\log_2 4}+3^{\log_2 4}+4^{\log_2 2}-4^{\log_2 3}$

$D=(\log_5 49)(\log_7 125)$

$N=7^{\log_7(30)^2}=7^{\log_7(30)}=30$

$A=4+9+4-9=8$

$D=\frac{\log 49}{\log 5}\times\frac{\log 125}{\log 7}=\frac{2\times 3}{1}=6$

Now, $A-\frac{N}{10}=8-3=5$

and $|N|+A+D+6|=|30+8+6+6|=50$

$P=\log_5 50-\log_5 2=\log_5 25+\log_5 2$

$-\log_5 2=2$

51. [1] $a+b+c=1$

$a^2+b^2+c^2=9$

$a^3+b^3+c^3=1$

$(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ac)$

$1=9+2(ab+bc+ca)$

$\Rightarrow ab+bc+ca=-4$

Now,

$a^3+b^3+c^3-3abc$

$=(a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right)$

$=1\ (9+4)$

$1-3abc=13$

$\Rightarrow\ 3abc=-12$

$\boxed{abc=-4}$

$\Rightarrow\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=\frac{ab+bc+ca}{abc}=\frac{-4}{-4}=1$

52. [400] $a+b+c=10, \quad ab+bc+ca=20$

$a^3+b^3+c^3-3abc$

$=[a+b+c]\left[a^2+b^2+c^2-ab-bc-ca\right]$

$=10\left[a^2+b^2+c^2-20\right]$

now,

$(a+b+c)^2-2(ab+bc+ca)=a^2+b^2+c^2$

$\Rightarrow \quad 100-40=60=a^2+b^2+c^2$

so,

$a^3+b^3+c^3-3abc=10\times[60-20]=400$

53. [5] Use property

when $a+b+c=0$

$a^3+b^3+c^3=3abc$

54. [5] $\sqrt{5\sqrt{5\sqrt{5\sqrt{5}}}}$

$x=\sqrt{5\cdot x}$

$x^2=5x$

$\Rightarrow x=0$ or $x=5$

55. [0] $x=\sqrt[3]{7+5\sqrt{2}}-\dfrac{1}{\sqrt[3]{7+5\sqrt{2}}}$

$(5\sqrt{2}+7)(5\sqrt{2}-7)=1$

$x=(7+5\sqrt{2})^{1/3}-(5\sqrt{2}-7)^{1/3}$

$x^3=(7+5\sqrt{2})-(5\sqrt{2}-7)-3(5\sqrt{2}+7)(5\sqrt{2}-7)(x)$

$x^3=14-3x$

$\Rightarrow x^3+3x-14=0$

56. [-1/3] $\left[\left\{\left(\dfrac{1}{7^2}\right)^{-2}\right\}^{-1/3}\right]^{1/4}=7^x$

$\Rightarrow \quad 7^{4x-\frac{1}{3}\times\frac{1}{4}}=7^x$

$\Rightarrow \quad 7^{-1/3}=7^x$

$\Rightarrow \quad x=-\dfrac{1}{3}$

57. [9] $y=\left(\dfrac{5}{4}\right)^{-100}$

$\log_{10}y=(\log_{10}5-\log_{10}4)(-100)$

$\Rightarrow \log_{10}y=-9.691$

no. of cyphers is 9

58. [1] $\log_5\left(2+\log_3(x+3)\right)=0$

$\Rightarrow 2+\log_3(x+3)=1$

$\log_3(x+3)=-1$

$x+3=3^{-1}=\dfrac{1}{3}$

$x=-3+\dfrac{1}{3}=-\dfrac{8}{3}$

59. [10] $4^A+9^B=10^C$

$A=\log_{16}4, \quad B=\log_3 9; \quad C=\log_x 83$

$4^A=4^{\log_{16}4}=4^{\frac{1}{2}}=2$

$9^B=9^{\log_3 9}=81$

$10^C=10^{\log_x 83}=81+2=83$

$\Rightarrow x=10$

Parikshit (JEE Advanced Level)

1. (a) $\log_p\log_p(p)^{\frac{1}{p^n}}$

$=\log_p\left(\dfrac{1}{p^n}\right)=-n$

2. (c) $\log_{1/2}\left(x^2-3x+2\right)\geq 2$

$x^2-3x+2\leq\dfrac{1}{4}$

$4x^2-12x+7\leq 0$

and $x^2-3x+2>0$

3. (a) $\log_4\log_3\log_2\left(x^2-2x+a\right)\geqslant 0$

$\log_3\left(\log_2\left(x^2-2x+a\right)\right)\geqslant 1$

$\log_2\left(x^2-2x+a\right)\geqslant 8$

$x^2-2x+a\geqslant 8$

$x^2-2x+(a-8)\geqslant 0$

$D\leqslant 0$

4. (a) $\dfrac{(16)^{\frac{1}{x}}}{2^{x+3}}>1$

$\dfrac{2^{\frac{4}{x}}}{2^{x+3}}>1$

$2^{\frac{4}{x}-(x+3)}>1$

$\dfrac{4}{x}-x+3>0$

$\dfrac{4-x^2+3x}{x}>0$

$\dfrac{x^2-3x-4}{x}<0$

5. (d) $\dfrac{(3^x-4^x)\ln(x+2)}{x^2-3x-4}\leq 0$

$\dfrac{\left(\left(\dfrac{3}{4}\right)^x-1\right)\ln(x+2)}{x^2-3x-4}\leq 0$

$\dfrac{\left(\left(\dfrac{3}{4}\right)^x-1\right)\ln(x+2)}{(x-4)(x+1)}\leq 0$

$x\in(-2,-1)\cup(-1,0)\cup(4,\infty)$

6. (d) $\dfrac{\ln\sqrt{x^2+1}-x^2}{\ln(x)}=x$

$\Rightarrow 0=x$

$\Rightarrow x=0$

But $x>0$

7. (d) $A=2\log_5 5^{\frac{1}{2^{10}}}$

$A=\dfrac{2}{2^{10}}$

$1024A=2$

$1024A+1=3$

8. (b, c)

$\left(\dfrac{1}{2}+\log_{\frac{1}{3}}27+\log_{9\sqrt{3}}81\right).\log_7 49=\log_7|x|$

$\Rightarrow \left(\dfrac{1}{2}-3\log_7 3+\dfrac{4}{3}\log_7 3\right).2=\log_7|x|$

$\Rightarrow \left(\dfrac{1}{2}-\dfrac{5}{3}\log_7 3\right).2=\log_7|x|$

$\Rightarrow 1-\dfrac{10}{3}\log_7 3=\log_7|x|$

$\Rightarrow 1=\log_7\left(3^{\frac{10}{3}}.|x|\right) \Rightarrow 3^{\frac{10}{3}}.|x|=7$

$\Rightarrow |x|=\dfrac{7}{3^{10/3}} \Rightarrow x=\pm\dfrac{7}{3^{10/3}}$

9. (a, d)

(a) L.H.S > 1 and R.H.S < 1 (true)

(b) L.H.S < 2 and R.H.S > 2 (false)

(c) $\log_4\log_3 4>\dfrac{1}{2}\Rightarrow\log_3 4>(4)^{1/2}$

$\Rightarrow 4>3^2$

which is false

(d) Let $\log_{10}3=a$ and $\log_{10}2=b$

$\therefore 2a^2-3b^2-ab=2a^2-3ab+2ab-3b^2$

$=(2a-3b)(a+b)$

$=(2\log_{10}3-3\log_{10}2)(\log_{10}3+\log_{10}2)$

>0

$\Rightarrow$ L.H.S > R.H.S

10. (a, b, c)

$x^{\frac{3}{4}(\log_2 x)^2+\log_2 x-\frac{5}{4}}=\sqrt{2}$

$\left[\dfrac{3}{4}(\log_2 x)^2+\log_2 x-\dfrac{5}{4}\right]\log_2 x=\log_2\sqrt{2}=\dfrac{1}{2}$

Let $\log_2 x=t$

$(3t^2+4t-5)t=2\Rightarrow 3t^3+4t^2-5t-2=0$

$t=1,-2,-\dfrac{1}{3}=\log_2 x$

$x=2,\dfrac{1}{4},2^{-\frac{1}{3}}$

11. (a, b, c, d)

Taking $\log_3$ on both sides

$$[(\log_3 x)^2 - \frac{9}{2}\log_3 x + 5](\log_3 x) = \frac{3}{2}$$

Let $\log_3 x = \alpha$

$$\Rightarrow \frac{(2\alpha^3 - 9\alpha^2 + 10\alpha)}{2} = \frac{3}{2}$$

$$\Rightarrow 2\alpha^3 - 9\alpha^2 + 10\alpha - 3 = 0$$

$$\Rightarrow (\alpha - 1)(2\alpha^2 - \alpha + 3) = 0$$

$$\Rightarrow (\alpha - 1)(\alpha - 3)(2\alpha - 1) = 0 \Rightarrow \alpha = 1, 3, \frac{1}{2}$$

$$\therefore \log_3 x = 1;\ \log_3 x = 3;\ \log_3 x = \frac{1}{2}$$

$$\Rightarrow x = 3;\ x = 3^3 = 27;\ x = \sqrt{3}$$

Exactly three solution, one is irrational solution and every real number is also complex.

12. (b, c)

$$(\log_2 x)^4 - \left(\log_2\left(\frac{x}{2}\right)^3\right)^2 + 9\left[\log_2 32 - \log_2 x^2\right] < 4(\log_2 x)^2$$

(note $x > 0$)

$$(\log_2 x)^4 - (3\log_2 x - 3)^2 + 45 - 18\log_2 x < 4(\log_2 x)^2$$

Let $\log_2 x = t$

$$t^4 - (3t-3)^2 + 45 - 18t < 4t^2$$

$$\Rightarrow t^4 - (9t^2 + 9 - 18t) - 18t + 45 < 4t^2$$

$$\Rightarrow t^4 - 13t^2 + 36 < 0$$

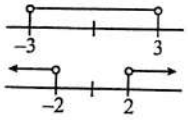
$$\Rightarrow (t^2 - 4)(t^2 - 9) < 0$$

$$\Rightarrow 4 < t^2 < 9$$

$t^2 < 9 \Rightarrow -3 < t < 3$

and $t^2 > 4 \Rightarrow t > 2$ or $t < -2$

hence, $t \in (-3, -2) \cup (2, 3)$



$$x \in \left(\frac{1}{8}, \frac{1}{4}\right) \cup (4, 8) \Rightarrow B, C$$

13. $(a,b,c,)$

(i) $$\frac{(81)^{\log_4 5} + 3^{3\log_3\sqrt{6}}}{409}\left(25 - 6^{\frac{3}{2}}\right)$$

$$= \frac{25 + (6)^{3/2}}{409} \times \left(25 - 6^{3/2}\right)$$

$$= \frac{625 - 6^3}{409} = 1$$

(ii) $5^{\log_{1/5}\left(\frac{1}{2}\right)} = 5^{\log_5 2} = 2$

and $\log_{\sqrt{2}}\left(\sqrt{7} - \sqrt{3}\right) - \log_2 \frac{1}{(\sqrt{7} + \sqrt{3})^2}$

$$= \log_{12}\left(\sqrt{7} - \sqrt{3}\right) - 2\log_2\left(\sqrt{7} - \sqrt{3}\right) + 4\log_2 2 = 4$$

So, Ans is $4 + 2 = 6$

(iii) $\sqrt{10^{2+\frac{1}{2}\log 16}} = \sqrt{10^{12} + \frac{1}{2}4\log 2} = \sqrt{10^{\log 10}} = 4$

14. (a,b,c,d)

(i) $\log_2\left(\log_{1/2} x\right) < 2$

for existence $\log_{1/2} x > 0$

$\Rightarrow x \in (0, 1)$ ...(i)

now,

$\log_{1/2} x < 4$

$$x > \left(\frac{1}{2}\right)^4$$

$$x > \frac{1}{16} \quad \text{...(ii)}$$

from (i) and (ii)

$$x \in \left(\frac{1}{16}, 1\right)$$

(ii) $\log_{1/2}\left((\log_3 x)\right) > 3$

$$\log_3 x < \left(\frac{1}{2}\right)^3$$

$$x < 3^{1/8}$$

$$(\log_2 x - 1)(\log_3 x - 2) \le 0$$

$$\Rightarrow (\log_e x - \log_e 2)(\log_e x - \log_e 9) \le 0$$

(iii) $\log_e 2 \le \log_e x \le \log_e 9$

$\Rightarrow 2 \le x \le 9$

(iv) $(\log_2 x - 1)(\log_2 x + 2) \ge 0$

$\log_2 x \le -2$ OR $\log_2 x \ge 1$

Sol. [15 to 17]

$$\left(\sqrt{x}\right)^{\log_5 x - 1} = 5 \Rightarrow (\log_5 x - 1)\frac{1}{2}\log_5 x = 1$$

$$\Rightarrow \log_5 x (\log_5 x - 1) = 2$$

Put $\log_5 x = y$

$\Rightarrow y^2 - y - 2 = 0 \Rightarrow (y-2)(y+1) = 0$

$\Rightarrow y = 2$ or $-1 \Rightarrow \log_5 x = 2$ or -1

$$x = 25 \text{ or } x = \frac{1}{5}$$

$$\alpha = 25 \text{ and } \beta = \frac{1}{5}$$

15. (b) Now, let $N = (25)^{10}$

$\log_{10} N = 10 \cdot 2\log_{10} 5 = 20\,[1 - 0.3010] = 20\,[0.6990] = 13.980 = 13.98$

$\Rightarrow N$ is a 14 digit number.

16. (d) Let $x = \left(\frac{1}{5}\right)^{10}$

$\log x = 10\,(-\log 5) = -10\,(1 - \log 2) = -10\,[0.699]$

$= -6.99$

Hence, there are 6 zeroes after decimal before a significant figure.

17. (a) $$\left(\frac{1}{5}\right)^{\log_{25} 9} = \left(\frac{1}{5}\right)^{\log_5 3} = \frac{1}{5^{\log_5 3}} = \frac{1}{3}.$$

18. (c) A → q, r, s, t; B → p; C → q, r, s, t; D → p, r, s

A. Let $\log_2(x-1) = A$ & $\log_2 x = B$ then

$\Rightarrow (A + B)^2 - 4AB = 1$

$\Rightarrow (A - B)^2 = 1$

$\Rightarrow (A - B) = \pm 1$

$$\therefore \log_2\left(\frac{x-1}{x}\right) = 1 \text{ or } -1 \quad \therefore x = -1 \text{ or } 2$$

Only 2 is solution.

B. $(8x - 3) = 13$ $\quad [a^{\log_a b} = b]$

$(8x = 16)$

$\therefore x = 2$

C. $N = \left(\frac{1}{\log_2 \pi} + \frac{1}{\log_6 \pi}\right) \Rightarrow (\log_\pi 2 + \log_\pi 6)$

$\Rightarrow \log_\pi \pi^2 < \log_\pi 12 < \log_\pi \pi^3$

$\Rightarrow 2 < \log_\pi 12 < 3$

D. $l = (\log_3 4 + \log_2 9)^2 - (\log_3 4 - \log_2 9)^2$

$m = 0.8^{\,(1 + 9^{\log_3 8})^{\log_{65} 5}}$

$\therefore \quad l = (A+B)^2 - (A-B)^2 = 4AB = 4\log_3 4 \cdot \log_2 9$

$$\Rightarrow 4 \cdot \frac{\log 4}{\log 3} \times \frac{\log 9}{\log 2} \Rightarrow 16$$

$l = 16$

$m = 0.8^{\,(1 + 3^{\log_3 8^2})^{\log_{65} 5}}$

$\Rightarrow (0.8)^{\,(1+64)^{\log_{65} 5}} \Rightarrow (0.8)^{\,(65)^{\log_{65}(5)}}$

$\Rightarrow (0.8)(5) = 4$

$\therefore m = 4$

so, $l + m = 16 + 4 = 20$

19. (b) A → s, B → r, C → q, D → p, E → x, F → w, G → v, H → u, I → t

Use Basic log property.

20. (c) (A) $A \to r, C \to P, D \to P, B \to S$

$$\sqrt{8 + 2\sqrt{7}} = \sqrt{(\sqrt{7}+1)^2} = \sqrt{7} + 1$$

$$\sqrt{8 - 2\sqrt{7}} = \sqrt{(\sqrt{7}-1)^2} = \sqrt{7} - 1$$

$a = 3(2) = 6$

$b = \sqrt{42 \times 30 + 36}$

$= \sqrt{6 \times 6(7 \times 5 + 1)}$

$= \sqrt{36 \times 36} = 36$

$\log_6(36) = 2$

(B) $= \sqrt{4 + 2\sqrt{3}} = \sqrt{(\sqrt{3}+1)^2} = \sqrt{3} + 1$

$a = \sqrt{3} + 1 - \sqrt{3} + 1 = 2$

$b = 36$

$\log_2 36 = \log_2 4 + \log_2 9 = 2 + 2\log_2 3$

(C) $a=\sqrt{3+2\sqrt{2}}=\sqrt{2}+1$

$b=\sqrt{2}-1=\frac{1}{\sqrt{2}+1}=\frac{1}{a}$

$\log_a\left(\frac{1}{a}\right)=-1$

(D) $a=\sqrt{7+\sqrt{48}}=\sqrt{7+4\sqrt{3}}=2+\sqrt{3}$

$b=2-\sqrt{3}=\frac{1}{2+\sqrt{3}}=\frac{1}{a}$

$\log_a\left(\frac{1}{a}\right)=-1$

21. [1] Let $x+|x-2|=y$

$\therefore$ Equation becomes

$\log_y y^2=\log_y(5y-6)$

$\Rightarrow y^2=5y-6 \Rightarrow y^2-5y+6=0$

$\Rightarrow y=2$ or 3

If $y=2$

then $x+|x-2|=2$

$\Rightarrow 0<x<1 \cup 1<x\le 2$

If $y=3$

then $x+|x-2|=3 \Rightarrow x=\frac{5}{2}$ only

Hence number of integral solutions is 1.

22. [7] As, $\frac{1}{\log_a\left(2-\sqrt{3}\right)}+\frac{1}{\log_b\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)}$

$=\log_{2-\sqrt{3}} a+\log_{\frac{\sqrt{3}-1}{\sqrt{3}+1}} b$

$=\log_{2-\sqrt{3}} a+\log_{2-\sqrt{3}} b=\log_{2-\sqrt{3}}(ab)$

Now, $\left(2+\sqrt{3}\right)^{\log_{2-\sqrt{3}}(ab)}=\frac{1}{12}$

$\Rightarrow \left(2-\sqrt{3}\right)^{\log_{2-\sqrt{3}}\left(\frac{1}{ab}\right)}=\frac{1}{12}$

$\Rightarrow \frac{1}{ab}=\frac{1}{12} \Rightarrow ab=12$

As a, b are co-prime numbers, so either $a=4, b=3$ or $a=3, b=4$.

Hence, $(a+b)=7$.

23. [2] $2\log_8(2x)+\log_8(x^2-2x+1)=\frac{4}{3}$

$\Rightarrow 2\log_8(2x)+\log_8(x-1)^2=\frac{4}{3}$

$\Rightarrow 2\log_8(2x)+2\log_8|x-1|=\frac{4}{3}$

$\Rightarrow \log_8(2x|x-1|)=\frac{2}{3} \Rightarrow 2x.|x-1|=8^{2/3}=4$

$\Rightarrow x|x-1|=2$

Now, when $x\ge 1$

$x(x-1)=2 \Rightarrow x^2-x-2=0$

$\Rightarrow x=2,-1$

$x\ge 1$

$\therefore x=2$ will be a solution but $x=-1$ will be rejected.

when $x<1$

$\Rightarrow x-x^2=2 \Rightarrow x^2-x+2=0$

Clearly the equation has no real solution

Hence, only solution of equation is $x=2$ and therefore sum of solution is 2.

24. [10] $(\log_{10} x)^2\ge \log_{10} x+2$

$(\log_{10} x)^2-\log_{10} x-2\ge 0$

$(\log_{10} x-2)(\log_x+1)\ge 0$

$\log_{10} x\in(-\infty,-1]\cup[2,\infty)$

$x\in\left(0,\frac{1}{10}\right]\cup[100,\infty)$

25. [17] $\log_{1/2}(x+5)^2>\log_{1/2}(3x-1)^2$

$(x+5)^2<(3x-1)^2$

$(x+5+3x-1)(x+5-3x+1)<0$

$(4x+4)(x-3)>0$

and $x+5>0$ and $3x-1>0$

26. [2] $(x^2-x)=2 \Rightarrow x^2-2x+x-2$

$(x-2)(x+2)=0 \Rightarrow x=2$

27. [2] $(0.3)^{\log_{1/3}\log_2\left(\frac{3x+6}{x^2+2}\right)}>1$

$\log_{1/3}\log_2\left(\frac{3x+6}{x^2+2}\right)<0$

$\log_2\left(\frac{3x+6}{x^2+2}\right)>1$

$\frac{3x+6}{x^2+2}>2 \Rightarrow 3x+6>2x^2+4$

$2x^2-3x-2<0$

$2x^2-4x+x-2$

$(x-2)(2x+1)<0$

$x\left(-\frac{1}{2},2\right)$

28. [1] $\log_{0.5}\log_6\left(\frac{x^2+x}{x+4}\right)<0$

$\Rightarrow \log_6\left(\frac{x^2+x}{x+4}\right)>1 \Rightarrow \frac{x^2+x}{x+4}>6$

$\Rightarrow \frac{x^2+x-6x-24}{x+4}>0$

$\Rightarrow \frac{x^2-5x-24}{x+4}>0$

29. [6] $\log\frac{|x^2-4x|+3}{x^2+4}\ge 0$

$\Rightarrow \log\frac{|x^2-4x|+3}{x^2+|x-5|}\ge 1$

$\Rightarrow |x^2-4x|+3x^2+|x-5|$

when $x<0$

$+(x^2-4x)+3\ge x^2-x+5$

$-3x+3\ge 0$

$3x+2\le 0$

when $0\le x<4$

$-x^2+4x+3\le x^2-x+5$

$\Rightarrow 2x^2-5x+2\le 0$

when $4\le x<5$

$-x^2+4x+3\ge x^2-x+5 \Rightarrow 2x^2-5x+2\le 0$

when $x\ge 5$

$x^2-4x+3\ge x^2+x-5$

$5x-8\le 0$

30. [2] $(0.4)^{\log_{\frac{1}{5}}^2 x+1}=(6.25)^{2-p\log_{10} x}$

$\left(\frac{2}{5}\right)^{\log_{\frac{2}{5}}^2 x+1}=\left(\left(\frac{2}{5}\right)^{-2}\right)^{2-2\log_{10} x}$

$(\log_{10} x)^2-4\log_{10} x+5=0$

no solution $m=0$

If $P=3$

$\left(\frac{2}{5}\right)^{(\log_{10} x)^2+1}=\left(\left(\frac{5}{2}\right)^2\right)^{2-3\log_{10} x}$

$(\log_{10} x)^2+1=6\log_{10} x-4$

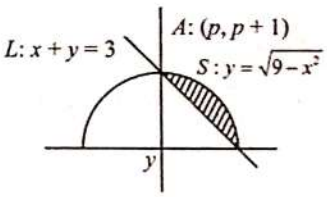
$(\log_{10} x-1)(\log_{10} x-5)=5$

$x=10, 10^5$

$x=2$

PYQ's (Past Year Questions)

1. [3] $3-x\le y\le\sqrt{9-x^2}; 0\le x\le 3$



$L(A)>0 \Rightarrow p+p+1-3>0 \Rightarrow p>1$ ...(i)

$S(A)<0 \Rightarrow p+1-\sqrt{9-p^2}<0$

$\Rightarrow p+1<\sqrt{9-p^2}$

$\Rightarrow p^2+2p+1<9-p^2$

$\Rightarrow 2p^2+2p-8<0$

$\Rightarrow p^2+p-4<0$

$\Rightarrow p\in\left(\frac{-1-\sqrt{17}}{2},\frac{-1+\sqrt{17}}{2}\right)$ ...(ii)

From eqn (i) $\cap$ eqn (ii), we get

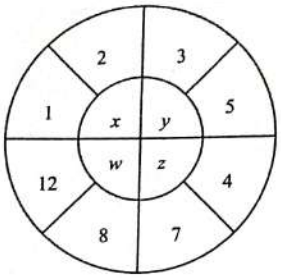
$p\in\left(1,\frac{\sqrt{17}-1}{2}\right)\equiv(a,b)$

2. [4] $x=(2-1)^{1!}=1$

$w=(12-8)^{4!}=4^{24}$

$z=(7-4)^{3!}=3^6$

Hence $y=(5-3)^{2!}=2^2=4$



3. (*d*) $2^x = t > 0$

$5 + |t-1| = t(t-2)$

when $t \geq 1$

$\Rightarrow 5 + t - 1 = t^2 - 2t$

$\Rightarrow t^2 - 3t - 4 = 0$

$\Rightarrow (t-4)(t+1) = 0$

$\Rightarrow t = 4$

Now $t < 1$

$5 - t + 1 = t^2 - 2t$

$\Rightarrow t^2 - t - 6 = 0 \Rightarrow (t-3)(t+2) = 0$, No solution

Hence, $2^x = 4 \Rightarrow x = 2$

4. (*a*) $\log_{\left(x+\frac{7}{2}\right)}\left(\frac{x-7}{2x-3}\right)^2 \geq 0$

Feasible region : $x + \frac{7}{2} > 0 \Rightarrow x > -\frac{7}{2}$

And $x + \frac{7}{2} \neq 1 \Rightarrow x \neq -\frac{5}{2}$

And $\frac{x-7}{2x-3} \neq 0$ and $2x - 3 \neq 0$

$\Downarrow$ $\Downarrow$

$x \neq 7$ $x \neq \frac{3}{2}$

Taking intersection : $x \in \left(\frac{-7}{2}, \infty\right) - \left\{-\frac{5}{2}, \frac{3}{2}, 7\right\}$

Now $\log_a b \geq 0$ if $a > 1$ this implies $b \geq 1$

Or

$a \in (0, 1)$ this implies $b \in (0, 1)$

Case-I: $x + \frac{7}{2} > 1$ and $\left(\frac{x-7}{2x-3}\right)^2 \geq 1$

$x > -\frac{5}{2}$ and $(2x-3)^2 - (x-7)^2 \leq 0$

$\Rightarrow (2x - 3 + x - 7)(2x - 3 - x + 7) \leq 0$

$\Rightarrow (3x - 10)(x + 4) \leq 0$

$\Rightarrow x \in \left[-4, \frac{10}{3}\right]$

Intersection : $x \in \left(\frac{-5}{2}, \frac{10}{3}\right]$

Case-II: $x + \frac{7}{2} \in (0,1)$ and $\left(\frac{x-7}{2x-3}\right)^2 \leq 1$

$0 < x + \frac{7}{2} < 1$ and $\left(\frac{x-7}{2x-3}\right)^2 \leq 1$

$-\frac{7}{2} < x < \frac{-5}{2}$ and $(x-7)^2 \leq (2x-3)^2$

$\Rightarrow x \in (-\infty, -4) \cup \left(\frac{10}{3}, \infty\right)$

No common values of x.

Hence intersection with feasible region

We get $x \in \left(\frac{-5}{2}, \frac{10}{3}\right] - \left\{\frac{3}{2}\right\}$

Integral value of x are $\{-2,-1,0,1,2,3\}$

Number of integral values = 6

5. [45] $e^{2x} - 11e^x - 45e^{-x} + \frac{81}{2} = 0$

$\Rightarrow (e^x)^3 - 11(e^x)^2 - 45 + \frac{81e^x}{2} = 0$

Put $e^x = t$

$\Rightarrow 2t^3 - 22t^2 + 81t - 90 = 0 \Rightarrow t_1 t_2 t_3 = 45$

$\Rightarrow e^{x_1} \cdot e^{x_2} \cdot e^{x_3} = 45$

$\Rightarrow e^{x_1 + x_2 + x_3} = 45$

$\Rightarrow \log_e e^{x_1 + x_2 + x_3} = \log_e 45$

$\Rightarrow x_1 + x_2 + x_3 = \log_e 45$

$\Rightarrow \log_e p = \log_e 45 \quad \Rightarrow p = 45$

6. [1] $\log_4(x-1) = \log_2(x-3)$

$x > 1$ & $x > 3 \Rightarrow x > 3$ (according to definition of log)

$\Rightarrow \log_4(x-1) = \log_2(x-3)$

$\frac{1}{2}\log_2(x-1) = \log_2(x-3)$

$\log_2(x-1) = \log_2(x-3)^2$

$\Rightarrow (x-1) = (x-3)^2$

$\Rightarrow x^2 + 9 - 6x = x - 1$

$\Rightarrow x^2 - 7x + 10 = 0$

$\Rightarrow (x-2)(x-5) = 0 \Rightarrow x = 2$ or 5

But $x > 3$

So, $x = 5$

Only one solution.

7. (*b*) $\log_{10}\sin x + \log_{10}\cos x = -1$

$\log_{10}(\sin x \cos x) = -1$

$\sin x \cos x = 10^{-1} = \frac{1}{10}$ ...(*i*)

Also, $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$

$\log_{10}(\sin x + \cos x)^2 = \log_{10}\left(\frac{n}{10}\right)$

$\sin^2 x + \cos^2 x + 2\sin x \cos x = \frac{n}{10}$

$1 + 2 \cdot \frac{1}{10} = \frac{n}{10} \Rightarrow n = 12$

8. (*a, c*)

Given, $y = 5^{\log x}$

$\Rightarrow \log_5 y = \log x$

$x = 10^{\log_5 y} = y^{\log_5 10} = y^{\frac{1}{\log 5}}$

$\Rightarrow y = f^{-1}(x), y = x^{\frac{1}{\log 5}} \Rightarrow x = y^{\log 5}$

9. (*c*)

$x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$

$\Rightarrow x + 1 + \log_2\left[\frac{10.2^x - 1}{(3+2^x)^2}\right] - x = 0$

$\Rightarrow 1 + \log_2\left[\frac{10.2^x - 1}{(3+2^x)^2}\right] = 0$

$\Rightarrow \frac{10.2^x - 1}{9 + (2^x)^2 + 6.2^x} = \frac{1}{2}$

$\Rightarrow (2^x)^2 - 14.2^x + 11 = 0$

Let $2^x = y$

$y^2 - 14y + 11 = 0$

$y = 7 \pm \frac{\sqrt{152}}{2}$

$2^{x_1} = y_1 = 7 + \frac{\sqrt{152}}{2}, y_2 = 7 - \frac{\sqrt{152}}{2} = 2^{x_2}$

$x_1 = \log_2\left(7 + \frac{\sqrt{152}}{2}\right)$

$x_2 = \log_2\left(7 - \frac{\sqrt{152}}{2}\right)$

$\therefore x_1 + x_2 = \log_2\left(49 - \frac{152}{4}\right)$

Hence, $x_1 + x_2 = \log_2 11$

10. [1]

$\frac{\log((2x+5)(x+1))}{\log(x+1)} + \frac{\log(x+1)}{\log(2x+5)} - 4 = 0$

$\Rightarrow \frac{\log(2x+5)}{\log(x+1)} + \frac{\log(x+1)}{\log(2x+5)} - 3 = 0$

$\Rightarrow x \in (-1, 0) \cup (0, \infty)$

$t = \frac{\log(2x+5)}{\log(x+1)} \Rightarrow t + \frac{2}{t} - 3 = 0$

$\Rightarrow t^2 - 3t + 2 = 0 \Rightarrow t = 1$ or 2

$\frac{\log(2x+5)}{\log(x+1)} = 1 \Rightarrow x = -4$ (not possible)

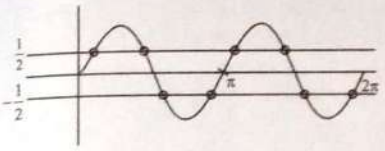
and $\frac{\log(2x+5)}{\log(x+1)} = 2 \Rightarrow x = 2$

Hence only one solution is possible.

11. [8] $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$

$\log_{1/2}|\sin x \cos x| = 2$

$|\sin x \cos x| = \frac{1}{4}$, $\sin 2x = \pm\frac{1}{2}$



Number of solution = 8

12. [8] $\frac{3^{x_1}+3^{x_2}+3^{x_3}}{3} \geq \left[3^{(x_1+x_2+x_3)}\right]^{\frac{1}{3}}$

$\Rightarrow 3^{x_1}+3^{x_2}+3^{x_3} \geq 3^4$

$\Rightarrow \log_3\left(3^{x_1}+3^{x_2}+3^{x_3}\right) \geq 4$

$\Rightarrow m = 4$

Also, $\frac{x_1+x_2+x_3}{3} \geq \sqrt[3]{x_1x_2x_3}$

$\Rightarrow x_1x_2x_3 \leq 27$

$\Rightarrow \log_3 x_1 + \log_3 x_2 + \log_3 x_3 \leq 3 \Rightarrow M = 3$

Thus, $\log_2(m^3) + \log_3(M^2) = 6 + 2 = 8$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER

2 Sets

Prarambh (Topicwise)

1. **(d)** Since, intelligency is not defined for students in a class *i.e.*, Not a well defined collection.

2. **(b)** Since $x^2 + 1 = 0$, gives $x^2 = -1$

$\Rightarrow x = \pm i$

$\therefore x$ is not real but x is real (given)

No value of x is possible.

3. **(a)** $x^2 = 16 \Rightarrow x = \pm 4$

$2x = 6 \Rightarrow x = 3$

There is no value of x which satisfies both the above equations. Thus, $A = \phi$.

4. **(c)** Number of subsets of $A = {}^nC_0 + {}^nC_1 + \ldots \ldots + {}^nC_n = 2^n$.

5. **(b)** Number of proper subsets of the set $\{1, 2, 3) = 2^3 - 1 = 7$.

6. **(a)** Since $= 8^n - 7n - 1 = (7 + 1)^n - 7n - 1 = 7^n + {}^nC_1\, 7^{n-1} + {}^nC_2\, 7^{n-2} + \ldots\ldots + {}^nC_{n-1}\, 7 + {}^nC_n - 7n - 1$

${}^nC_2 + {}^nC_2 7^3 + \ldots\ldots + {}^nC_n 7^n ({}^nC_0 = {}^nC_n, {}^nC_1 = {}^nC_{n-1}$ etc.)

$= 49[{}^nC_2 + {}^nC_3\,(7) + \ldots\ldots + {}^nC_n\, 7^{n-2}]$

$\therefore 8^n - 7n - 1$ is a multiple of 49 for $n \in N$.

For $n = 1$, $8^n - 7n - 1 = 8 - 7 - 1 = 0$;

For $n = 2$, $8^n - 7n - 1 = 64 - 14 - 1 = 49$

$\therefore 8^n - 7n - 1$ is a multiple of 49 for all $n \in N$.

$\therefore X$ contains elements which are multiples of 49 and clearly Y contains all multiplies of 49.

$\therefore X \subseteq Y$.

7. **(b)** $B \cap C = \{4\}$, $\therefore A \cup (B \cap C) = \{1, 2, 3, 4\}$.

8. **(c)** Since $\Rightarrow A \subseteq B \Rightarrow A \cup B = B$.

9. **(a)** $A \cap B \subseteq A$. Hence $A \cup (A \cap B) = A$.

10. **(d)** $A \cap (A \cap B)^c = A \cap (A^c \cup B^c) = (A \cap A^c) \cup (A \cap B^c)$

$= \phi \cup (A \cap B^c) = A \cap B^c$.

11. **(b)** $N_3 \cap N_4 = \{3,6,9,12,15.....\} \cap \{4,8,12,16,20,...\}$

$= \{12, 24, 36.....\} = N_{12}$

Trick: $N_3 \cap N_4 = N_{12}$ [$\because 3$, 4 are relatively prime numbers]

12. **(a)** bN = the set of positive integral multiples of b, cN = the set of positive integral multiplies of c.

$\therefore bN \cap cN$ = the set of positive integral multiples of $bc = b \subset N$ [$\because b, c$ are prime]

$\therefore d = bc$.

13. **(c)** Since $y = \frac{1}{x}$, $y = -x$ meet when $-x = \frac{1}{x} \Rightarrow x^2 = -1$, which does not give any real value of x

Hence $A \cap B = \phi$.

14. **(b)** $A = [x : x \in R, -1 < x < 1]$

$B = [x : x \in R : x - 1 \le -1$

or $x - 1 \ge 1] = [x : x \in R : x \le 0$ or $x \ge 2]$

$\therefore A \cup B = R - D$

Where $D = [x : x \in R, 1 \le x < 2]$

15. **(c)** Since, $y = e^x$ and $y = x$ do not meet for any $x \in R$.

16. **(b)** Since, $4^n - 3n - 1 = (3+1)^n - 3n - 1$

$= 3^n + {}^nC_1 3^{n-1} + {}^nC_2 3^{n-2} + \ldots + {}^nC_{n-1} 3 + {}^nC_n - 3n - 1$

$= {}^nC_2 3^2 + {}^nC_3 3^3 + \ldots\ldots + {}^nC_n 3^n$

${}^nC_0 = {}^nC_n$, ${}^nC_1 = {}^nC_{n-1}$ etc.)

$= 9[{}^nC_2 + {}^nC_3(3) + \ldots\ldots + {}^nC_n 3^{n-1}]$

$\therefore 4^n - 3n - 1$ is a multiple of 9 for $n \in N$.

For $n = 1$, $4^n - 3n - 1 = 4 - 3 = 0$, For $n = 2$, $4^n - 3n$

$= 16 - 6 - 1 = 9$

$\therefore 4^n - 3n - 1$ is a multiple of 9 for all $n \in N$

$\therefore X$ contains elements which are multiples of 9 and clearly Y contains all multiples of 9.

$\therefore X \subseteq Y$, $\therefore X \cup Y = Y$.

17. **(b)** $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 3 + 6 - n(A \cap B)$

Since maximum number of elements in $A \cap B = 3$

Minimum number of elements in $A \cup B = 9 - 3 = 6$.

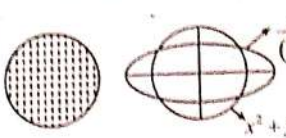
18. **(d)** $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow 110 = 70 + 60 - n(A \cap B)$

$\therefore n(A \cap B) = 130 - 110 = 20$.

19. **(c)** $n(A^c \cap B^c) = n[(A \cup B)^c] = n(U) - n(A \cup B) = n(U) - [n(A) + n(B) - n(A \cap B)] = 700 - [200 + 300 - 100] = 300$.

20. **(d)** A = Set of all values $(x, y) : x^2 + y^2 = 25 = 5^2$



$B = \frac{x^2}{144} + \frac{y^2}{16} = 1$ *i.e.*, $\frac{x^2}{(12)^2} + \frac{y^2}{(4)^2} = 1$.

Clearly, $A \cap B$ consists of four points.

21. **(c)** $n(C) = 20$, $n(B) = 50$, $n(C \cap B) = 10$

Now, $n(C \cup B) = n(C) + n(B) - n(C \cap B)$

$= 20 + 50 - 10 = 60$.

Hence, required number of persons = 60%.

22. **(d)** $n(M) = 23$, $n(P) = 24$, $n(C) = 19$

$n(M \cap P) = 12$, $n(M \cap C) = 9$, $n(P \cap C) = 7$

$n(M \cap P \cap C) = 4$

We have to find $n(M \cap P' \cap C')$, $n(P \cap M' \cap C')$, $n(C \cap M' \cap P')$

Now $n(M \cap P' \cap C') = n[M \cap (P \cup C)']$

$= n(M) - n(M \cap (P \cup C))$

$= n(M) - n((M \cap P) \cup (M \cap C)]$

$= n(M) - n(M \cap P) - n(M \cap C) + n(M \cap P \cap C)$

$= 23 - 12 - 9 + 4 = 27 - 21 = 6$

$n(P \cap M' \cap C') = n[P \cap (M \cup C)']$

$= n(P) - n[P \cap (M \cup C)]$

$= n(P) - n[(P \cap M) \cup (P \cap C)]$

$= n(P) - n(P \cap M) - n(P \cap C) + n(P \cap M \cap C)$

$= 24 - 12 - 7 + 4 = 9$

$n(C \cap M' \cap P') = n(C) - n(C \cap P) - n(C \cap M) + n(C \cap P \cap M) = 19 - 7 - 9 + 4 = 23 - 16 = 7$

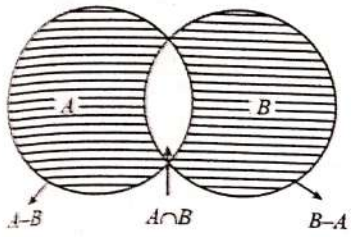
23. **(b)** $A \times (B \cap C) = (A \times B) \cap (A \times C)$. It is distributive law.

24. **(a)** It is distributive law.

25. **(b)** It is De' Morgan law.

26. **(b)** $A - B = A \cap B^c = A \cap \overline{B}$

27. (c) $(A-B)\cup(B-A)=(A\cup B)-(A\cap B)$.



28. (d) By the definition of cartesian product of sets
Clearly, $A\times B=\{(0,1),(0,0),(1,1),(1,0)\}$.

29. (b) $A\times B=\{(2,7),(2,8),(2,9),(4,7),(4,8),(4,9),(5,7),(5,8),(5,9)\}$
$n(A\times B)=n(A).n(B)=3\times3=9$.

30. (c) $n(A\times B)=pq$.

31. (c) $B\cup C=\{c,d\}\cup(d,e\}=\{c,d,e\}$
$A\times(B\cup C)=\{a,b\}\times\{c,d,e\}=\{(a,c),(a,d),(a,e),(b,c),(b,d),(b,e)\}$.

32. (a) Clearly, $A=\{2,3\}, B=\{2,4\}, C=\{4,5\}$
$B\cap C=\{4\}$
$A\times(B\cap C)=\{(2,4),(3,4)\}$.

33. (b) $R\times(P^c\cup Q^c)^c=R\times[(P^c)^c\cap(Q^c)^c]$
$=R\times(P\cap Q)=(R\times P)\cap(R\times Q)$
$=(R\times Q)\cap(R\times P)$

34. (a) There is number 'x' that satisfy this equation.
$\therefore S=$ null set

35. (b) $=4x+2^{x+1}-8=0$
$2^{2x}+2.2^x-8=0$
Let, $y=22$
$y^2+2y-8=0$
$(y-2)(y+4)=0$
$y=2,-4$
$2x=2\Rightarrow x=1$
$2x=-4\Rightarrow$ not possible

36. (b) $(A\cap B')'\cup(B\cap C)$
$(A'\cup B)\cup(B\cap C)$
$\Rightarrow A'\cup B$

37. (d) $A\cap(A\cap B)^c$
$\Rightarrow A\cap(A^c\cup B)^c$
$\Rightarrow A\cup B^c$

38. (b) $A\cup B=p=1$
$A\cap C=q=2$
$q-p=1$

39. (b) Number of students failed in Physics = at least 70%
Number of students failed in Chemistry = at least 72%
Number of students failed in Math = at least 80%
Number of students failed in English = at least 85%
That means,
Number of students passed in Physics = at least 30%
Number of students passed in Chemistry = at least 28%
Number of students passed in Math = at least 20%
Number of students passed in English = at least 15%
$\therefore$ Total number of students passed in all subjects
= at most $(30+28+20+15)\%=93\%$
Total number of students failed in all the subjects
at least $(100-93)\%=7\%$

40. (a) The correct option (a) is 1 only
$(N\cup B)\cap Z=(N\cap Z)\cup(B\cap Z)$
$=N\cup(B\cap Z)$
$\therefore$ (1) is correct.
$A=\{3,6,9,12,15,18,21,24\}$
and there exist subset B of N such that the number of elements in A is equal to the number of elements in B.
Clearly, we can find $B\subset N$ such that,
$n(A)=n(B)$
$\therefore$ (2) is not correct.

41. (b) $198=(90+130-x)+212-2x$
$3x=432-198$
$x=78$

Prabal (JEE Main Level)

1. (b) $A\cap B$ is $\{x:x\in R, 2\le x<4\}$

2. (c) $A=\{\phi,\{\phi\}\}$
$P(A)=$ set containing all subsets
$=\{\phi,\{\phi\},\{\{\phi\}\},\{\phi,\{\phi\}\}$
$=\{\phi,\{\phi\},\{\{\phi\}\},A\}$

3. (a) $A=\{2,3\}; B=\{1,2\}$
$A\times B=\{(2,1),(2,2),(3,1),(3,2)\}$

4. (c) $n(A\cup B)=n(A)+n(B)-n(A\cap B)$
$=200+300-100$
$n(A\cap B)=400$
Now $n(A'\cap B')=U-n(A\cup B)$
(De marganistans)
$=700-400=300$

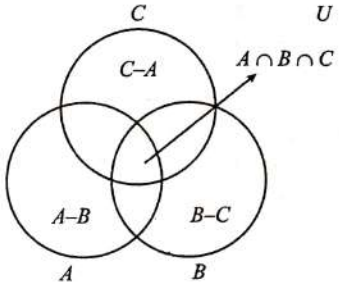
5. (c) for $x=2, y=3\in N$
$x=4, y=2\in N$
$x=6,\ y=1\in N$

6. (d) Given, $A=\{x\in N:30<x^2<70\}=\{6,7,8\}$
$B=\{x:x$ is a prime number less than $10\}$
$=\{2,3,5,7\}$
$\therefore A\cup B=\{2,3,5,6,7,8\}$ and $A\cup B=\{7\}$
$A-B=\{6,8\}$ and $B-A=\{2,3,5\}$
$\therefore A\,\Delta\,B=(A-B)\cup(B-A)$
$=\{6,8\}\cup\{2,3,5\}=\{2,3,5,6,8\}$

7. (b) $\because n(A\cup B)=n(A)+n(B)-n(A\cap B)$
$\therefore n(A\cup B)=200+300-100=400$
$\therefore n(A'\cap B')=n(A\cup B)'=n(X)-n(A\cup B)$
$\Rightarrow 300=n(X)-400$
$\therefore n(X)=700$

8. (b) $n\{(A\times B)\cap(B\times A)\}$
$=n\{(A\cup B)\times(B\cap A)\}$
$=n(A\cap B)\times n(B\cap A)$
$=n(A\cap B)\times n(A\cap B)$
$=80\times80=80^2$

9. (c) From Venn Euler's diagram,



It is clear that, $\{(A-B)\cup(B-C)\cup(C-A)\}'$
$=A\cap B\cap C$

10. (a) $(A\cup B\cup C)\cap(A\cap B'\cup C')\cap C'$
$=(A\cup B\cup C)\cap(A'\cup B\cup C)\cap C'=(\phi\cup B\cup C)\cap C'$
$=(B\cup C)\cap C'=(B\cap C')\cup\phi=B\cap C'$

11. (b) Given, $n(B)=21, n(H)=26, n(F)=29$, $n(H\cap B)=14, n(H\cap F)=15, n(F\cap B)=12$ and $n(B\cap H\cap F)=8$
$\therefore n(B\cup H\cup F)=n(B)+n(H)+n(F)-n(B\cap H)$
$-n(H\cap F)-n(B\cap F)+n(B\cap H\cap F)$
$21+26+29-14-15-12+8=43$

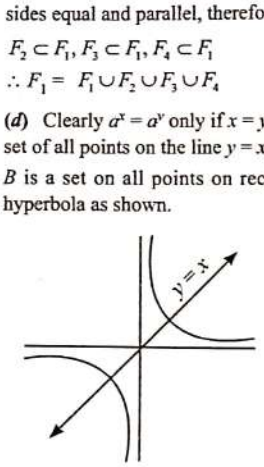
12. (d) The number of elements in power set of A is 1.
$\therefore P\{P(A)\}\ 2^1=2$
$\Rightarrow P\{P\{P(A)\}\}=2^2=4$
$\Rightarrow P\{P\{P\{P(A)\}\}\}=2^4=16$

13. (c) For $n=10$
Between 10 and 20, prims numbers are 11, 13, 17, 19

14. (b) $2^{333x-2}+2^{10x+1}=2^{222x+2}+1$
Let, $y=210x$
$\frac{y^3}{4}+2y=4y^2+1$
$y^3-16y^2+8y-4=0$
Let x_1, x_2, x_3 are roots of equation,
$2^{111(x_1+x_2+x_3)}=4$
$x_1+x_2+x_3=\frac{2}{111}$

15. **(d)** Given, F_1 = the set of parallelograms
F_2 = the set of rectangles
F_3 = the set of rhombuses
F_4 = the set of squares
and F_5 = the set of trapeziums
By definition of parallelogram, opposite sides are equal and parallel. In rectangles, rhombuses and squares, all have opposite sides equal and parallel, therefore,
$F_2 \subset F_1, F_3 \subset F_1, F_4 \subset F_1$
$\therefore F_1 = F_1 \cup F_2 \cup F_3 \cup F_4$

16. **(d)** Clearly $a^x = a^y$ only if $x = y$. So A is set of all points on the line $y = x$.
B is a set on all points on rectangular hyperbola as shown.



(*i*) None of the sets is a subset of other.
(*ii*) Both are infinite sets.
(*iii*) $A \neq B, A \subseteq B, B \subseteq A$ hence A and B are non-comparable.

17. **(d)** By observing Veinn diagram

18. **[3]**
$U = \{x : x^5 + 6x^4 + 11x^3 - 6x^2 = 0\} = \{0,1,2,3\}$
$A = \{x : x^2 - 5x + 6 = 0\} = \{2,3\}$
and $B = \{x : x^2 - 3x + 2 = 0\} = \{2,1\}$
$\therefore (A \cap B)' = U - (A \cap B) = \{0,1,2,3\} - \{2\} = \{0,1,3\}$

19. **[3]** Minimum possible value of $n(B \cap C)$ is $n(A \cap B \cap C) = 3$.

20. **[8]** Each subset will contain 3 and any number of elements from the remaining 3 elements 1, 2 and 4. So, required number of elements $= 2^2 = 8$.

21. **[4]** We have,
$$\frac{n^3 + 5n^2 + 2}{n} = n^2 + 5n + \frac{2}{n}$$
$\therefore \frac{n^3 + 5n^2 + 2}{n}$ is an integer, if $\frac{2}{n}$ is an integer
$\Rightarrow n = \pm 1, \pm 2$
$\Rightarrow$ A consists of four elements viz. $-1, 1, -2, 2$

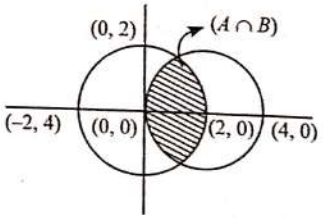
22. **[14]** Let M, P and C be the sets of students taking examinations in Mathematics, Physics and Chemistry respectively.
We have,
$n(M \cup P \cup C) = 50$, $n(M) = 37$, $n(P) = 24$, $n(C) = 43$
$n(M \cap P) < 19$, $n(M \cap C) \leq 29$, $n(P \cap C) \leq 20$
Now,
$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P)$
$-n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$
$\Rightarrow 50 = 37 + 24 + 43 - \{n(M \cap P) + n(M \cap C) + n(P \cap C)\} + n(M \cap P \cap C)$
$\Rightarrow n(M \cap P \cap C)$
$= n(M \cap P) + n(M \cap C) + n(P \cap C) - 54$
$\Rightarrow n(M \cap P) + n(M \cap C) + n(P \cap C)$
$= n(M \cap P \cap C) + 54$ ...(i)
Now,
$n(M \cap P) \leq 19$, $n(M \cap C) \leq 29$, $n(P \cap C) \leq 20$
$\Rightarrow n(M \cap P) + n(M \cap C) + n(P \cap C) \leq 19 + 29 + 20$ [Using (i)]
$\Rightarrow n(M \cap P \cap C) + 54 \leq 68$
$\Rightarrow n(M \cap P \cap C) \leq 14$

Parikshit (JEE Advanced Level)

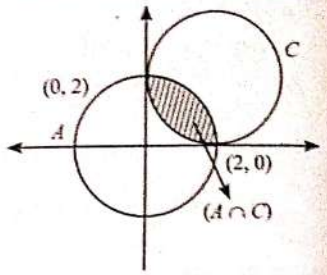
1. **(c)** Sum of elements in $A \cap B$
$$\underbrace{(2+4+6+\ldots+200)}_{\text{Multiple of 2}} - \underbrace{(6+12+\ldots+198)}_{\text{Multiple of 2 \& 3 i.e. 6}}$$
$$- \underbrace{(10+20+\ldots+200)}_{\text{Multiple of 5 \& 2 i.e. 10}} + \underbrace{(30+60+\ldots+180)}_{\text{Multiple of 2, 5 \& 3 i.e. 30}}$$
$= 5264$.

2. **(b)** $n^2 - n - 10000 \leq 0$
$$n \in \left(\frac{1-\sqrt{40001}}{2}, \frac{1+\sqrt{40001}}{2}\right)$$
But $n \in N$
$\therefore A = \{1, 2, 3 \ldots\ldots 100\}$
and $B - C = \{3k + 1 \mid k \in \text{even}\}$
$\Rightarrow B - C = \{7, 13, 19, \ldots\ldots, 97, \ldots.\}$
$\therefore A \cap (B - C) = \{7, 13, 19, \ldots., 97\}$
Sum of all elements $= \frac{16}{2}[7 + 97] = 832$.

3. **(c)** The set A and B are represented as:



$\therefore A \cap B = \{(0, 0), (1, 0), (2, 0), (1, 1), (1, -1)\}$
The set A and set C are represented as:



$\therefore A \cap C = \{(1, 1), (2, 0), (2, 1), (2, 2), (3, 2)\}$
Total number relations from $A \cap B$ to $A \cap C = 2^{5 \times 5}$.
$\therefore p = 25$

4. **(a, b, c)**
Required progression is 13, 19, ... a_n
$a_n = 13 + (n - 1)\,6 < 100$
$n < 15.5$
$n = 15$ $n \in N$.

5. **(a, b, c)**
$\alpha = 107$,
Sum of digits = 8
divisible by 2, 4, 8.

6. **(a)** Total students = 100
Only English = 18
English but not Hindi = 23
English and Sanskrit = 8
English = 26
Sanskrit = 48
Sanskrit and Hindi = 8
No language = 24
From above diagram we can see that number of students studying Hindi = 10 + 5 +3 = 18

7. **(a)** Number of students studying English and Hindi = 3
E-English, H-Hindi, S-Sanskrit
We have, Only English = $a = 18$
English but not Hindi = $a + d = 23$
English and Sanskrit = $d + e = 8$
English = $a + b + d + e = 26$
$d + e + f + g = 48$
$a + b + c + d + e + f + g = 100 - 24 = 76$
$\therefore a = 18, b = 0, c = 10, d = 5, e = 3, f = 5$ and $g = 35$
$n(H \cap E) = b + e = 3$

8. **(b)** Let P, C and AI be the sets of students who have taken physics, chemistry and mathematics respectively.
Let a, b, c, d, e, f and g denotes the number of students in the respectively regions, as shown in the adjoining Venn diagram.

As per data given, we have

$$\begin{aligned} a+b+c+d&=12\\ b+c+e+f&=11\\ c+d+f+f&=15\\ b+c&=4\\ c+d&=9\\ c+f&=5\\ c&=3 \end{aligned}$$

From these equations, we get

$c=3, f=2, d=6, b=1$

Now, $c+d+f+g=15$

$\Rightarrow 3+6+g=15 \Rightarrow g=14$

$b+c+e+f=11$

$\Rightarrow 1+3+e+2=11 \Rightarrow e=5$

$a+b+c+d=12$

$\Rightarrow a+1+3+6=12 \Rightarrow a=2$

$\therefore a=2, b=1, c=3, d=6, e=5, f=2$ and $g=4$

Number of students who offered mathematics only $g=4$

9. (a) Let P, C and AI be the sets of students who have taken physics, chemistry and mathematics respectively.

Let a, b, c, d, e, f and g denotes the number of students in the respectively regions, as shown in the adjoining Venn, diagram.

As per data given, we have

$$\begin{aligned} a+b+c+d&=12\\ b+c+e+f&=11\\ c+d+f+f&=15\\ b+c&=4\\ c+d&=9\\ c+f&=5\\ c&=3 \end{aligned}$$

From these equations, we get

$c=3, f=2, d=6, b=1$

Now, $c+d+f+g=15$

$\Rightarrow 3+6+g=15 \Rightarrow g=14$

$b+c+e+f=11$

$\Rightarrow 1+3+e+2=11 \Rightarrow e=5$

$a+b+c+d=12$

$\Rightarrow a+1+3+6=12 \Rightarrow a=2$

$\therefore a=2, b=1, c=3, d=6, e=5, f=2$ and $g=4$

Number of students who offered physics and mathematics but not chemistry = $d=6$

10. (b) A → q, B → r, C → p, D → s

A. $((A'\cup B')-A)=((A\cap B')-A)$

$=(\phi)'=U$

B. $(B\cup(B'-A'))=B\cup(B'-A')=B$

C. $(A-B)-(B-C)$

As, $(A-B)\cap(B-C)=\phi$

$\therefore (A-B)(B-C)=A-B$

D. $(A-B)(C-CB)=(A\cap C)-B$

11. (c) A → q, B → r, C → s, D → p

A. $3^{2n}-8n-1; n\in N$

For $n=2$ it is 64

B. $2^{3n}-1; n\in N$

For $n=1$ it is 7

C. $3^{2n}-1; n\in N$

For $n=1$ it is 8

D. $2^{3n}-7n-1; n\in N$

For $n=2$ it is 49

12. [4] We have

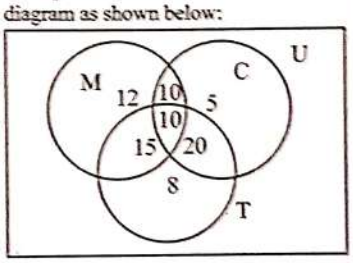
$n(U)=100$, where U stands for universal set

$n(M\cap C\cap T)=10; n(M\cap C)=20;$

$n(C\cap T)=30; n(M\cap T)=25;$

$n(\text{only } M)=12; n(\text{only } C)=5; n(\text{only } T)=8$

Filling all the entries we obtain the Venn diagram as shown below:



$\therefore n(M\cup C\cup T)=12+10+5+15+10+20+8=80$

$\therefore n(M\cup C\cup T)'=100-80=20=n.$

So, $\frac{n}{5}=\frac{20}{5}=4$

13. [7] Numbers which are divisible by 5 are 5, 10, 15, 20, 25, 30, 40, 45, 50, 55, 60, 65, 70, 75, 80 they are 16 in numbers.

Now, Numbers which are divisible by 7 are 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77 they are 11 in numbers.

Also, total odd numbers = 40

Let C represents the students who opt. for cricket, F for football and H for hockey.

$\therefore$ we have $n(C)=40, n(F)=16, n(H)=11$

Now, $C\cap F$ = Odd numbers which are divisible by 5.

$C\cap H$ = Odd numbers which are divisible by 7.

$F\cap H$ = Numbers which are divisible by both 5 and 7.

$n(C\cap F)$, 8, $n(C\cap H)=6$, $n(F\cap H)=2, n(C\cap F\cap H)=1$

We know

$n(C\cap F\cup H)=n(C)+n(F)+n(H)-n(C\cap F)-n(C\cap H)-n(C\cap H\cap F)$

$n(C\cap F\cup H)=67-16+1=52$

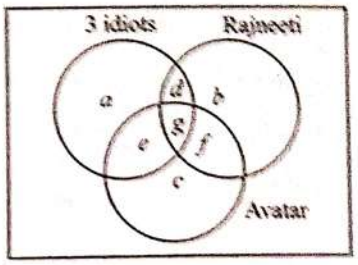
$\therefore n(C'\cap F'\cap H')$

= Total students $-n(C\cap F\cup H)$

$n(C'\cap F'\cap H')=80-52=28$, So,

$\frac{n}{4}=\frac{28}{4}=7$

14. [7] The given is as follows-



We know that $\{(a+d+e+g)(b+d+f+g)+(c+e+f+g)\}-(d+e+f)-2g$

$=a+b+c+d+e+f+g$

or $61x+46x+29x-25x-2g=97x$

or $2g=14x$ or $g=7x$. So 7% of people watched all the three movies)

15. [4] $-n(A\cap B)$ is minimum when $A\subseteq B$. In this case, $(A\cup B)=B$ and hence minimum value of $n(A\cup B)=n(B)=7$.

$n(A\cup B)$ is maximum when A and B are disjoint.

$\therefore$ Maximum value of $n(A\cup B)=4+7=11$.

So $11-7=4$

16. [5] $s=\{x\in R\}: |x-2|^{10x^2-x}=|x-2|^{3x}$

$s=\left\{-\frac{1}{5},\frac{1}{2},1,3,2\right\}$

$\Rightarrow n(s)=5$

17. [1] $5^x\cdot 8^{\frac{x-1}{x}}=500$

$\Rightarrow 5^x\cdot 8^{\frac{x-1}{x}}=53\ 22$

$\Rightarrow s=\{3\}$

$\Rightarrow n(s)=1$

18. [52] Let 'A' be the set of people who read newspaper H.

Let 'B' be the set of people who read newspaper T.

Let 'C' be the set of people who read newspaper I.

Given: $n(A)=25, n(B)=26$

and $n(C)=26$

$n(A\cap C)=9, n(A\cap B)=11$

and $(B\cap C)=8$

$n(A\cap B\cap C)=3$

Let 'U' be the set of people who took part in the survey.

$n(A\cup B\cup C)$

$= n(A)+n(B)+n(C)-n(A\cap B)$
$-n(B\cap C)-n(C\cap A)+n(A\cap B\cap C)$
$= 25+26+26-11-8-9+3$
$= 52$

19. **[3]** Time → T
Mirror → M
Sun → S
Set 'T' is subset of set 'S'
Everone either reads sun or does not read mirror for a person.
⇒ either we reads sun as well as mirror
⇒ or reads sun but does not reads mirror
⇒ or reads neither
If $M\cap T=0$
Then number of people which read, $S=6.5$
x : Number of people who read only S.
b : Number of people who read T and S but not M.
c : Number of people who read all three.
d : Number of people who read M and S but not T.
$a+b=11$ ...(1)
$b+c=8$ ...(2)
$a+c+d=10$ ...(3)
$a+b+c=14$ ...(4)
Equations (4) and (1), $a+b+c=a+b$
$c=14-11=3$
$c=33$
Therefore, $a=5$, $b=6$
$c=3$, $d=2$
From the, $a+x=9$
$x=9-5=4$
Number of people who read only, $S=5$
Number of people who read T and M both $=3$

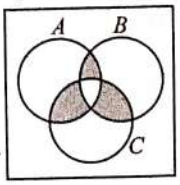
20. **[3]** $5^x+\frac{125}{5^x}=30$
$5^x+5^{3x}=30$
$x=1,2$

21. **[1]** $\frac{x+5}{x-7}-5=\frac{4x-40}{13-x}$
$x=10$
$\therefore T=\{10\}$

22. **[50]**
(i) $y=x-13$ (ii) $\max(x)=25$ $\max(y)=125$
$\min(x)=12$ $\max(y)=0$
(i) $n(A'\cap B')=n((A\cup B)'$
$=n(u)-n(A\cup B)$
$=n(u)-(n(A)+n(B)-n(A\cap B))$
$=40-(28+25-x)$
$y=x-13$
(ii) $\max(n(A\cap B)=\min(n(A)\,n(B))$
$\max(x)=25$
$\Rightarrow \max(y)=25-13=12$
$\min(n(A\cap B)=28+25-40$
$\min(x)=13$
$\min(y)=13-13=0$

PYQ's (Past Year Questions)

1. **(c)** $n(A)=48$
$n(B)=25$
$n(C)=18$
$n(A\cup B\cup C)=60$ [Total]
$n(A\cap B\cap C)=5$



$n(A\cup B\cup C)=\sum n(A)-\sum n(A\cap B)+n(A\cap B\cap C)$
$\sum n(A\cap B)=48+25+18+5-60=36$
Number of men who received exactly 2 medals
$\sum n(A\cap B)-3n(A\cap B\cap C)$
$36-15=21$

2. **[15]** $\{n\in[10,100]$
3^n-3 is multiple of 7
$3^n=7\lambda+3$
$n=6k+1$ form
$n=1,7,13,19.....97$
Number of possible values of $n=15$

3. **[6]** $-6<n^2-10n+19<6$
$\Rightarrow n^2-10n+25>0$ and $n^2-10n+13<0$
$\Rightarrow (n-5)^2>0$ and $n\in\left[5-2\sqrt{3},5+2\sqrt{3}\right]$
$n\in R-\{5\}$...(i) and $n=\{2,3,4,5,6,7,8\}$ ...(ii)
Taking intersection of (i) & (ii), we get
$\Rightarrow n=2,3,4,6,7,8$
$\therefore$ Number of values of $n=6$

4. **(d)** **Case-I:**
$|x|-2>0$ and $|x+3|-1\geq 0$
$\Rightarrow |x|>2\Rightarrow x>2$ or $x<-2$
and $|x+3|\geq 1\Rightarrow (x+3)\geq 1$ or $x+3\leq -1$
$\Rightarrow x\geq -2$ or $x\leq -4$
$\Rightarrow x\in[-6,-4]\cup(2,3]$
Case-II:
$|x|-2<0$ and $|x+3|-1\leq 0$
$\Rightarrow |x|<2\Rightarrow -2<x<2$ and $|x+3|\leq 1$
$\Rightarrow -1\leq x+3\leq 1$
$\Rightarrow -4\leq x\leq -2$
$\therefore$ No common solution exists
$\Rightarrow S=\{x\in[-6,-4]\ (2,3]\}$
$T=\{x\in z: x^2-7|x|+9\leq 0\}$
$x^2-7x+9\leq 0$ for $x\geq 0\Rightarrow x\in\{2,3,4,5\}$
$x^2+7x+9\leq 0$ for $x\leq 0\Rightarrow x\in\{-5,-4,-3,-2\}$
$T=\{\pm2,\pm3,\pm4,\pm5\}$
Thus $S\cap T=\{-5,-4,3\}$
Hence, number of elements in $S\cap T$ is 3.

5. **[107]** $\because (B\cup C)'=B'\cap C'$
B' is a set containing sub sets of A containing element 1 and not containing 2.
And C is a set containing subsets of A whose sum of elements is not prime.
So, we need to calculate number of subsets of
$\{3,4,5,6,7\}$ whose sum of elements plus 1 is composite.
Number of such 5 elements subset = 1
Number of such 4 elements subset = 3(except selecting 3 or 7)
Number of such 3 elements subset = 6(except selecting $\{3,4,5\}$, $\{3,6,7\}$, $\{4,5,7\}$ or $\{5,6,7\}$)
Number of such 2 elements subset = 7(except selecting $\{3,7\}$, $\{4,6\}$, $\{5,7\}$)
Number of such 1 elements subset = 3(except selecting $\{4\}$ or $\{6\}$)
Number of such 0 elements subset = 1
$n(B'\cap C')=21\Rightarrow n(B\cup C)=2^7-21=107$

6. **[112]** $C\subseteq A$ and $C\cap B=\phi$
If C is formed only by $\{1,2,3,4,5,6,7\}$ total number of subsets of $A=2^7$.
when C is formed by $\{1,2,4,5\}=2^4$
$\therefore$ Number of subsets where $C\cap B\neq\phi$
$=2^7-2^4=112$

7. **[1633]**
$\{a\in(1,2,3,\ldots,100):\text{HCF}(a,24)=1\}$
HCF of $(a, 24)=1$ $\therefore$ $a=1,5,7,11,13,17,19,23$
sum of these numbers = 96.
$\therefore$ There are four such blocks and a number 97 is there upto 100.
$\therefore$ complete sum $=96+(24\times 8+96)+(48\times 8+96)+(72\times 8+96)+97=1633$

8. **[5264]**
Sum of elements in $A\cap B$
$=\underbrace{(2+4+6+\ldots+200)}_{\text{Multiple of 2}}-\underbrace{(6+12+\ldots+198)}_{\text{Multiple of 2 \& 3 i.e. 6}}$
$\underbrace{-(10+20+\ldots+200)}_{\text{Multiple of 5 \& 2 i.e. 10}}+\underbrace{(30+60+\ldots+180)}_{\text{Multiple of 2, 5 \& 3 i.e. 30}}$
$=5264$

9. **[11]** $T=\{9,10,11,12,.....,1000\}$
$S=\{4,6,9\}$
$A=\{a_1+a_2+.....+a_k: k\in N, a_i\in S\}$

Let 4 appear x no. of times

6 appear y no. of times

9 appear z no. of times

Then the set A has n element $4x + 6y + 9z$

Concept: Let a and b are co-prime numbers, then members from $(a-1)(b-1)$ and more can be expressed in the form $ax + by$ where $x, y \in \{0, 1, 2, ...\}$.

So, all the numbers of the form

$2y + 3z$ are $(2-1)\cdot(3-1)$, ...

i.e. 6,7,8,9,10,11, ...

form : $2 + t$, $t = 0, 1,2,3$, ...

So, $6y + 9z = 3(2y + 3z) = 3(2 + 1) = 6 + 3t$, $t = 0,1,2,3$

$\Rightarrow 4x + 6y + 9z$

$= 4x + 6 + 3t$

$= 4x + 3t + 6$

Now, all the numbers of the form $4x + 3t$ start from $(4-1)(3-1) = 6$ and 6, 7, 8, 9, ... $\rightarrow$ form: $6 + k$

$k = 0, 1, 2, 3, ...$

So, $4x + 6y + 9z = 4x + 3t + 6 = 6 + k + 6$

$= 12 + k$ $(k = 0, 1, 2, ...)$

$\Rightarrow$ Numbers of the form $4x + 6y + 9z$ are 12, 13, 14, 15 ...

But $9 \le 4x + 6y + 9z \le 1000$ and $9 = 4(0) + 6(0) + 9(1)$ & $10 = 4(1) + 6(1) + 9(0)$

But 11 can't be written in form $4x + 6y + 9z$.

$\Rightarrow$ A = {9, 10, 12, 13, 14, 15, 1000}

T = {9, 10, 11, 12, 13, 14, 15,....,1000}

T – A = {11} $\Rightarrow$ sum of element of T – A is 11

10. **[27]** $\frac{(x-3)^2}{16} + \frac{(y-4)^2}{9} \le 1, x, y \in N$, $(x-7)^2 + (y-4)^2 \ge 36, x, y \in R$

Total number of common point

$= 1 + 5 + 7 + 5 + 5 + 3 + 1 = 27$

11. **[256]**

$|x-2| > 1$ gives $x - 2 < -1$ or $x - 2 > 1$

i.e., $x < 1$ or $x > 3$ .....(*i*) represent set A

$\sqrt{x^2-3} > 1$ gives $x^2 - 3 > 1$ or $x^2 - 4 > 0$

$\Rightarrow x < -2$ or $x > 2$.....(*ii*) represent set B

$|x-4| \ge 2$ gives $x - 4 \le -2$ or $x - 4 \ge 2$

$x \le 2$ or $x \ge 6$...(*iii*) represent set C

Now, $A \cap B \cap C = \{x : x < -2 \text{ or } \ge 6\}$

so, $(A \cap B \cap C)^c = \{x: -2 \le x < 6\}$

so, Now, $(A \cap B \cap C)^c \cap z = \{-2, -1, 0, 1, 2, 3, 4, 5\}$

Hence, number of subset $= 2^8 = 256$

12. (*b*) $(|x| - 3)(|x + 4|) = 6$

Case-I: When $x \ge 0$

$(x-3)(x+4) = 6$

$\Rightarrow x^2 + x - 18 = 0$

$x = \frac{-1 \pm \sqrt{73}}{2}$

But $x \ge 0$

$\therefore x = \frac{-1+\sqrt{73}}{2}$

Only 1 solution

Case-II: When $-4 \le x < 0$

$(-(x) - 3)(x + 4) = 6$

$(x + 3)(x + 4) = -6$

$x^2 + 7x + 18 = 0$

$x = \frac{-7 \pm \sqrt{49-72}}{2}$ No real root.

Case-III: When $x < -4$

$(-1)^2(x + 3)(x + 4) = 6$

$x^2 + 7x + 6 = 0$

$(x + 1)(x + 6) = 0$

$x = -1$ or -6

But $x < -4$, So $x = -6$

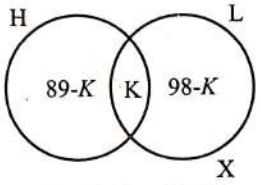
In total, there are 2 real solution.

13. (*d*) $89 - K + K + 98 - K + x = 100$

$\Rightarrow 187 - K + x = 100$

$\Rightarrow K = 87 + x$

$\Rightarrow K \ge 87$



Also, $89 - K \ge 0 \Rightarrow K \le 89$

14. (*b*) None of above Venn diagrams represent the required.

15. **[80]** {1,2,3,4,5,6,9}

Elements of type $3n \rightarrow 3, 6, 9$

Type $3n + 1 \rightarrow 1,4$

$3n + 2 \rightarrow 2, 5$

Number of subset of S containing one element which are not divisible by 3 $= {}^2C_1 + {}^2C_1 = 4$

Number of subset of S containing two numbers whose sum is not divisible by 3 $= {}^3C_1 \times {}^2C_1 + {}^3C_1 \times {}^2C_1 + {}^2C_2 + {}^2C_2 = 14$

Number of subset of S containing 3 elements whose sum is not divisible by $3 = {}^3C_2 \times {}^2C_1 + {}^3C_2 \times {}^2C_1 + {}^2C_2 \times {}^3C_1 + {}^2C_2 \times {}^3C_1 + ({}^2C_2 \times {}^2C_1) \times 2 = 22$

Number of subset containing 4 elements whose sum is not divisible by 3

$= {}^3C_2 \times {}^4C_1 + {}^3C_2\left({}^2C_2 + {}^2C_2\right) + \left({}^3C_1 \times {}^2C_1 \times {}^2C_2\right) \times 2 = 14$

Number of subset of S containing 6 elements = 4

Hence, total subset = 80

16. **[96]** Given, $11^n - 9^n > 10^n$

$\Rightarrow (10+1)^n - (10-1)^n > 10^n$

$\Rightarrow 2\left[{}^nc_1 10^{n-1} + {}^nc_3 10^{n-3} + ...\right] > 10^n$

$\Rightarrow$ For $n^3 5$ the inequality satisfies.

For $n = 4$, we have $2\left[4 \times 10^3 + 4 \times 10\right] > 10^4$

$\Rightarrow 8 \times 1010 > 10^4$ which is a contradiction

$\therefore n \ge 5$ all values satisfies.

Hence possible values of n is 96

17. **[5]** Given, sum of elements

$A \cap (B \cup C) = 274 \times 400$

In set B numbers of the form $9k + 2$ are {101, 109, ... 992}

$\therefore$ sum $= \frac{100}{2}(101 + 992) = \frac{100 \times 1093}{2}$

Another possible number is $9k + 5$ forms are {104, ...995}

$\therefore$ sum $= \frac{100}{2}(104 + 995) = \frac{100}{2} \times 1099$ ...

$\therefore$ Total $= \frac{100}{2} \times [1093 + 1099] = 100 \times 1096$

$= 274 \times 4 \times 100 = 274 \times 400$

$\therefore$ Required value of $\ell = 5$

18. **[832]** A = {1, 2, 3, 4, 5 ... 100}

B = {4, 7, 10, 13, 16, 19....}

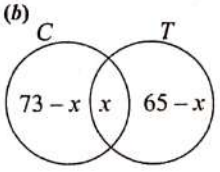
C = {2, 4, 6, 8, 10...}

$B - C$ = {7, 13, 19, ... 97}

$A \cap (B\text{-}C)$ = {7, 13, 19, ... 97}

$\therefore$ Sum of elements $= \frac{16}{2}(7 + 97) = 832$

19. (*b*)



$73 + 65 - x \le 100$

$\Rightarrow x \ge 38$ and $x \le 65$

So, $38 \le x \le 65$.

20. (*a*) $B - A = R - (-2, 5)$

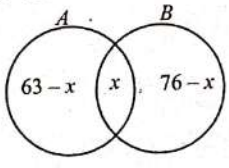
21. (*c*) $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$

$\Rightarrow \frac{10 \times 50}{20}$

$= \frac{5n}{6}$ $\left\{\begin{array}{l} \because n(x_i) = 10, n(Y_i) = 5 \\ So, \bigcup_{i=1}^{50} X_i = 500, \bigcup_{i=1}^{n} Y_i = 5n \end{array}\right\}$

$\Rightarrow n = 30$

22. (*b*)



Here $63 - x + x + 76 - x + y = 100$

$39 + y = x$

$\therefore\ 39 \le x \le 63$

$\therefore$ Possible value of x is 55.

23. [28]

Number of subsets of $A = 2^m$

Number of subsets of $B = 2^n$

Given $= 2^m - 2^n = 112$

$\Rightarrow (m, n) = (7, 4)$

$\Rightarrow mn = 28$

24. [29] $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 25 + 7 - 3 = 29$

25. (d) $n(P) = \left[\frac{140}{3}\right] = 46$ (Here [.] is GIF)

$$n(C) = \left[\frac{140}{5}\right] = 28$$

$$n(M) = \left[\frac{140}{2}\right] = 70$$

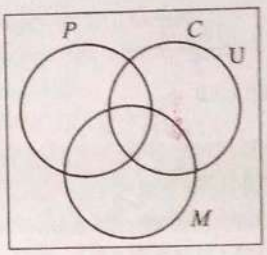
$$n(P \cup C \cup M) = n(P) + n(C) + n(M)$$

$$- n(P \cap C) - n(C \cap M) - n(M \cap P) + n(P \cap M \cap C)$$

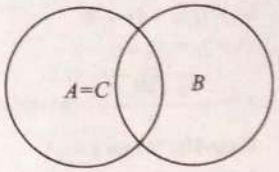
$$= 46 + 28 + 70 - \left[\frac{140}{15}\right] - \left[\frac{140}{10}\right] - \left[\frac{140}{6}\right] + \left[\frac{140}{30}\right]$$

$= 144 - 9 - 14 - 23 + 4 = 102$

So required number of student $= 140 - 102 = 38$



26. (a)



for $A = C$, $A - C = \phi$

$\Rightarrow \phi \subseteq B$

But $A \not\subseteq B$

$\Rightarrow$ option (a) is NOT true

Let $x \in (C \cup A) \cap (C \cup B)$

$\Rightarrow x \in (C \cup A)$ and $x \in (C \cup B)$

$\Rightarrow (x \in C$ or $x \in A)$ and $(x \in C$ or $x \in B)$

$\Rightarrow x \in C$ or $x \in (A \cap B)$

$\Rightarrow x \in C$ or $x \in C$ (as $A \cap B \subseteq C$)

$\Rightarrow x \in C$

$\Rightarrow (C \cup A) \cap (C \cup B) \subseteq C$ .....(i)

Now $x \in C \Rightarrow x \in (C \cup A)$ and $x \in (C \cup B)$

$\Rightarrow x \in (C \cup A) \cap (C \cup B)$

$\Rightarrow C \subseteq (C \cup A) \cap (C \cup B)$ ...(ii)

$\Rightarrow$ from (i) and (ii)

$C = (C \cup A) \cap (C \cup B)$

option (b) is ture

Let $x \in A$ and $x \notin B$

$\Rightarrow x \in (A - B)$

$\Rightarrow x \in C$ (as $A - B \subseteq C$)

Let $x \in A$ and $x \in B$

$\Rightarrow x \in (A \cap B)$

$\Rightarrow x \in C$ (as $A \cap B \subseteq C$)

Hence $x \in A \Rightarrow x \in C$

$\Rightarrow A \subseteq C$

$\Rightarrow$ option (c) is true

as $C \supseteq (A \cap B)$

$\Rightarrow B \cap C \supseteq (A \cap B)$

as $A \cap B \neq \phi$

$\Rightarrow B \cap C \neq \phi$

$\Rightarrow$ Option (d) is true.

27. (a) $A = \{x \in z : 2^{(x+2)(x^2-5x+6)} = 1\}$

$2^{(x+2)(x^2-5x+6)} = 2^0 \Rightarrow x = -2, 2, 3$

$A = \{-2, 2, 3\}$

$B = \{x \in Z : -3 < 2x - 1 < 9\}$

$B = \{0, 1, 2, 3, 4\}$

Hence, $A \times B$ has is 15 elements.

So number of subsets of $A \times B$ is 2^{15}

28. (b) Product is even when atleast one element of subset is even.

Hence required number of subsets = total subsets − number of subsets all whose elements are odd

$= 2^{100} - 2^{50}$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

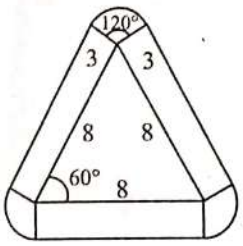
CHAPTER

3

Trigonometric Ratios and Identities

Prarambh (Topicwise)

1. (*a*) Area $= 3\cdot(8\cdot 3)+3\cdot\frac{1}{2}r^2\theta$



$= 72+\frac{3}{2}\cdot 9\cdot\frac{2\pi}{3} = 72+9\pi = 9(8+\pi)$

2. (*a*) Let radius of circle be r.
length of sector be l.
Then, $l = r\theta$
Also, according to question,

$2r + l = \frac{1}{2}.2\pi r$

$2r + r\theta = \pi r$

$\theta = \frac{(\pi-2)}{r}$

$\theta = (\pi - 2)$ radian

3. (*d*) $\cos\theta = 0.707 \quad \theta \in (0, 2\pi)$

$\cot\theta = \frac{1}{\sqrt{2}} \quad \theta \in (0, 2\pi)$

$\theta = \frac{\pi}{4}, \frac{7\pi}{4}$

Number of value of $\theta = 2$

4. (*b*) $\frac{2\sin\alpha}{1+\sin\alpha+\cos\alpha} = \lambda$

$\frac{2\sin\alpha}{1+\sin\alpha+\cos\alpha}\cdot\frac{1+\sin\alpha-\cos\alpha}{1+\sin\alpha-\cos\alpha} = \lambda$

$\frac{2\sin\alpha+2\sin^2\alpha-2\sin\alpha\cos\alpha}{1+\sin^2\alpha+2\sin\alpha-\cos^2\alpha} = \lambda$

$\frac{2\sin\alpha+2\sin^2\alpha-2\sin\alpha\cos\alpha}{2\sin^2\alpha+2\sin\alpha} = \lambda$

$\frac{2\sin\alpha(1+\sin\alpha-\cos\alpha)}{2\sin(\sin\alpha+1)} = \lambda$

$\frac{1+\sin\alpha-\cos\alpha}{1+\sin\alpha} = \lambda$

5. (*a*) $\sin A, \cos A, \tan A$ are in G.P.
$\cos^2 A = \sin A\cdot\tan A$.

$\cos^2 A = \frac{\sin^2 A}{\cos A} \Rightarrow \cos^3 A = 1-\cos^2 A$

$\Rightarrow \cos^3 A + \cos^2 A = 1$

6. (*c*) $\operatorname{cosec} A + \cot A = \frac{11}{2}$ ... (i)

$\operatorname{cosec} A - \cot A = \frac{2}{11}$ (as $\operatorname{cosec}^2 A - \cot^2 A = 1$) ... (ii)

Subtracting (ii) from (i),

$2\cot A = \frac{11}{2}-\frac{2}{11} = \frac{121-4}{22} = \frac{117}{22}$

$\cot A = \frac{117}{44}$

$\tan A = \frac{44}{117}$

7. (*d*) $\cot\left(\frac{\pi}{2}\right) = 0$

8. (*b*) $\frac{\sin^3\theta-\cos^3\theta}{\sin\theta-\cos\theta}-\frac{\cos\theta}{\sqrt{1+\cot^2\theta}}-$
$2\tan\theta\cot\theta = -1$

$\Rightarrow 1+\sin\theta\cos\theta-\cos\theta\,|\sin\theta|-1=0$

$\Rightarrow \sin\theta \geq \theta \Rightarrow \theta\in\left(\frac{\pi}{2},\pi\right)$

9. (*c*) $\frac{\tan^3 A}{1+\tan^2 A}+\frac{\cot^3 A}{1+\cot^2 A}$

$= \frac{\sin^3 A}{\cos A}+\frac{\cos^3 A}{\sin A}$

$= \frac{\sin^4 A+\cos^4 A}{\sin A\cos A} = \frac{1-2\sin^2 A\cos^2 A}{\sin A\cos A}$.

$\sec A\operatorname{cosec} A - 2\sin A\cos A$

10. (*b*) Given expression
$= 3[\cos^4\alpha+\sin^4\alpha]-2[\cos^6\alpha+\sin^6\alpha]$
$= 3[1+2\cos^2\alpha\sin^2\alpha]-2[1+3\cos^2\alpha\sin^2\alpha]$
$= 3+6\cos^2\alpha\sin^2\alpha-2-6\cos^2\alpha\sin^2\alpha = 1$

11. (*a*) $\cos(540°-\theta)-\sin(630°-\theta)$
$= -\cos\theta-(-\cos\theta) = 0$

12. (*a*) $\sin(\pi+\theta).\sin(\pi-\theta).\operatorname{cosec}^2\theta$
$= (-\sin\theta)(\sin\theta)\operatorname{cosec}^2\theta$
$= -\sin^2\theta\operatorname{cosec}^2\theta = -1$

13. (*d*) $\cos 0+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$

$= 1+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\left(\pi-\frac{3\pi}{7}\right)+\cos\left(\pi-\frac{2\pi}{7}\right)+\cos\left(\pi-\frac{\pi}{7}\right)$

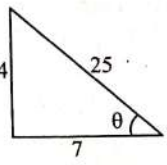
$= 1+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}-\cos\frac{3\pi}{7}-\cos\frac{2\pi}{7}-\cos\frac{\pi}{7} = 1$

14. (*a*) $\cos 10° - \sin 10°$
$\because \cos 10° > 0$ & $\sin 10° > 0$
$\therefore \cos 10° + \sin 10° > 0$
$\Rightarrow (\cos 10° - \sin 10°)(\cos 10° + \sin 10°)$
$\Rightarrow \cos^2 10° - \sin^2 10°$
$\Rightarrow \cos 20° > 0$
$\Rightarrow \cos 10° - \sin 10° > 0$

15. (*b*) $\sin\theta = \cos\theta, \theta$ is in 3rd quadrant

$\theta = \frac{5\pi}{4}$

16. (*c*)



$\sec\theta = \frac{25}{7}$

$\tan\theta = \frac{24}{7}$

$\operatorname{Sec}\theta + \tan\theta = \frac{25}{7}+\frac{24}{7} = \frac{25+24}{7} = \frac{49}{7}$

Lies in second Quadrent than $\sec\theta + \tan\theta$
$= -7$

17. (*d*) $3\tan A + 4 = 0 \Rightarrow \tan A = -\frac{4}{3}$

$\Rightarrow \sin A = \pm\frac{\tan A}{\sqrt{1+\tan^2 A}} = \pm\frac{-4/3}{\sqrt{1+16/9}} = \frac{4}{5}$

(Q A is in 2nd quadrant)

and $\cos A = -\frac{3}{5}$,

Thus, $2\cot A - 5\cos A + \sin A$

$= 2\left(-\frac{3}{4}\right) - 5\left(-\frac{3}{5}\right) + \frac{4}{5} = \frac{23}{10}$

18. (*d*) $e^{\log_{10}\tan 1^\circ + \log_{10}\tan 2^\circ + \log_{10}\tan 3^\circ + \ldots + \log_{10}\tan 89^\circ}$

$= e^{\log_{10}(\tan 1^\circ \tan 2^\circ \tan 3^\circ \ldots \tan 89^\circ)} = e^{\log_{10} 1} = e^0 = 1$

19. (*b*) $\cos A + \sin(270^\circ + A) - \sin(270^\circ - A) + \cos(180^\circ + A)$

$\cos A - \cos A + \cos A - \cos A = 0$

20. (*d*) $\sin^2\frac{\pi}{8} + \sin^2\frac{3\pi}{8} + \sin^2\frac{5\pi}{8} + \sin^2\frac{7\pi}{8}$

$= \sin^2\frac{\pi}{8} + \sin^2\frac{3\pi}{8} + \sin^2\frac{3\pi}{8} + \sin^2\frac{\pi}{8}$

$= 2\left(\sin^2\frac{\pi}{8} + \sin^2\frac{3\pi}{8}\right) = 2 \times 1 = 2$

21. (*a*) We know that $A + C = 180^\circ$

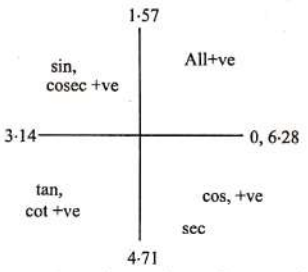
$\Rightarrow A = 180^\circ - C$

$\Rightarrow \cos A + \cos C = 0$ ... (i)

$\Rightarrow \cos B + \cos D = 0$ ... (ii)

From (i) and (ii) $\cos A - \cos B + \cos C - \cos D = 0$

22. (*b*)



$\sin 1 \cdot \cos 2 \cdot \tan 3 \cdot \cot 4 \cdot \sec 5 \cdot \operatorname{cosec} 6$

sign of above expression

= (+) (–) (–) (+) (+) (–)

= (–) ve

Negetive

23. (*a*) $S = \frac{1}{1+\tan^3 0^\circ} + \frac{1}{1+\tan^3 10^\circ} + \ldots + \frac{1}{1+\tan^3 80^\circ}$

Now, $\frac{1}{1+\tan^3\theta} + \frac{1}{1+\tan^3(90-\theta)}$

$= \frac{1}{1+\tan^3\theta} + \frac{1}{1+\cot^3\theta}$

$= \frac{1}{1+\tan^3\theta} + \frac{\tan^3\theta}{1+\tan^3\theta}$

$= \frac{1+\tan^3\theta}{1+\tan^3\theta} = 1$

Hence, $S = 1 + (1 + 1 + 1 + 1) = 5$

24. (*b*) $a = \cos(2012\,\pi) = 1$

$b = \sec(2013\,\pi) = -1$

$c = \tan(2014\,\pi) = 0$

$\therefore b < c < a$

25. (*a*) $\frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \cos 4^\circ}{\sin 21^\circ \cos 39^\circ - \sin 39^\circ \cos 21^\circ}$

$= \frac{\sin(24^\circ - 6^\circ)}{\sin(21^\circ - 39^\circ)} = \frac{\sin 18^\circ}{\sin(-18^\circ)} = -1$

26. (*d*) $\because 3\sin\alpha = 5\sin\beta$

$\Rightarrow \frac{\sin\alpha}{\sin\beta} = \frac{5}{3} \Rightarrow \frac{\sin\alpha + \sin\beta}{\sin\alpha - \sin\beta} = \frac{8}{2}$

$\Rightarrow \frac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)} = 4$

27. (*a*) $\tan A + \tan B = a$

$\tan A \tan B = b$

$\Rightarrow \tan(A + B) = \frac{a}{1-b}$

$\therefore \sin^2(A+B) = \left[\frac{|a|}{\sqrt{a^2 + (1-b)^2}}\right]^2$

$= \frac{a^2}{a^2 + (1-b)^2}$

28. (*b*) $\because \tan A < 0$ and $A + B + C = 180^\circ$

$\Rightarrow A > 90^\circ$

$\Rightarrow B + C < 90^\circ$

$\Rightarrow \tan(B + C) > 0 \Rightarrow \frac{\tan B + \tan C}{1 - \tan B \tan C} > 0$

$\Rightarrow 1 - \tan B \tan C > 0 \Rightarrow \tan B \tan C < 1$

29. (*c*) $\tan A - \tan B = x$

$\cot B - \cot A = y$

$\frac{\tan A - \tan B}{\tan A \tan B} = y$

$\Rightarrow \tan A \tan B = \frac{x}{y}$

Now $\cot(A - B) = \frac{1}{\tan(A-B)} = \frac{1 + \tan A \tan B}{\tan A - \tan B}$

$= \frac{1 + \frac{x}{y}}{x} = \frac{1}{x} + \frac{1}{y}$

30. (*a*) $\frac{\tan(180^\circ - 25^\circ) - \tan(90^\circ + 25^\circ)}{1 + (\tan(180^\circ - 25^\circ)\tan(90^\circ + 25^\circ)}$

$= \frac{-\tan 25^\circ + \frac{1}{\tan 25^\circ}}{2} = \frac{1 - x^2}{2x}$

31. (*d*) $\sin\alpha \sin\beta - \cos\alpha \cos\beta + 1 = 0$

$\Rightarrow -\cos(\alpha + \beta) + 1 = 0$

$\Rightarrow \cos(\alpha + \beta) = 1$

$\Rightarrow \sin(\alpha + \beta) = 0$

then $1 + \cot\alpha \tan\beta = 1 + \frac{\cos\alpha \sin\beta}{\sin\alpha \cos\beta}$

$= \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\sin\alpha\cos\beta} = \frac{\sin(\alpha+\beta)}{\sin\alpha\cos\beta} = 0$

32. (*d*) $\tan^2\theta = 2\tan^2\phi + 1$ ... (i)

$\cos 2\theta + \sin^2\phi = \frac{1 - \tan^2\theta}{1 + \tan^2\theta} + \sin^2\phi$

$= \frac{1 - 2\tan^2\phi - 1}{1 + 2\tan^2\phi + 1} + \sin^2\phi$

$= \frac{-2\tan^2\phi}{2(1 + \tan^2\phi)} + \sin^2\phi$

$= -\sin^2\phi + \sin^2\phi = 0$

which is independent of ϕ

33. (*b*) $\left(1 + \cos\frac{\pi}{10}\right)\left(1 + \cos\frac{3\pi}{10}\right)\left(1 - \cos\frac{3\pi}{10}\right)\left(1 - \cos\frac{\pi}{10}\right) = \left(1 - \cos^2\frac{\pi}{10}\right)\left(1 - \cos^2\frac{3\pi}{10}\right)$

$\sin^2\frac{\pi}{10} \cdot \sin^2\frac{3\pi}{10}$

$= \left(\frac{\sqrt{5}-1}{4} \cdot \frac{\sqrt{5}+1}{4}\right)^2 = \left(\frac{4}{16}\right)^2 = \frac{1}{16}$

34. (*b*) $= \frac{(\cos 6x + \cos 4x) + 5(\cos 4x + \cos 2x) + 10(\cos 2x + 1)}{\cos 5x + 5\cos 3x + 10\cos x}$

$= \frac{2\cos 5x \cos x + 10\cos 3x \cos x + 20\cos^2 x}{\cos 5x + 5\cos 3x + 10\cos x}$

$= \frac{2\cos x[\cos 5x + 5\cos 3x + 10\cos x]}{[\cos 5x + 5\cos 3x + 10\cos x]}$

$= 2\cos x$

35. (*c*) Given, $\cos(A - B) = \frac{3}{5}$

$\cos A \cos B + \sin A \sin B = \frac{3}{5}$ ... (i)

Also, $\tan A \tan B = 2$

$\Rightarrow 2\cos A \cos B - \sin A \sin B = 0$ ... (ii)

From (i) and (ii), we get

$\cos A \cos B = \frac{1}{5}$ & $\sin A \sin B = \frac{2}{5}$

$\therefore \cos A \cos B - \sin A \sin B = \frac{1}{5} - \frac{2}{5}$

$\Rightarrow \cos(A + B) = -\frac{1}{5}$

36. (*b*) $\frac{\sin^2 A - \sin^2 B}{\sin A \cos A - \sin B \cos B}$

$= \frac{2\sin(A+B)\sin(A-B)}{\sin 2A - \sin 2B}$

$= \frac{2\sin(A+B)\sin(A-B)}{2\cos(A+B)\sin(A-B)} = \tan(A+B)$

37. (*a*) Since $\tan 3A = \frac{\tan A + \tan 2A}{1 - \tan A \tan 2A}$

$\Rightarrow \tan 3A - \tan 2A - \tan A$

$= \tan 3A \tan 2A \tan A$

38. (*d*) $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$

As max value of $\sin\theta = 2$

$\sin\theta_1 = \sin\theta_2 = \sin\theta_3 = 1$

$\cos\theta_1 = \cos\theta_2 = \cos\theta_3 = 0$

$\cos\theta_1 + \cos\theta_2 + \cos\theta_3 = 0$

39. (*c*)

$$\sin\left(\frac{\pi}{10}\right)\cdot\sin\left(\frac{13\pi}{10}\right) = \sin\left(\frac{\pi}{10}\right)\sin\left(\pi+\frac{3\pi}{10}\right)$$

$$= \sin\left(\frac{\pi}{10}\right)\left(-\sin\left(\frac{3\pi}{10}\right)\right) = -\sin 18^\circ \sin 54^\circ$$

$$= -\left(\frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4}\right) = \frac{-4}{16} = \frac{-1}{4}$$

40. (*a*) $8x^2 - 26x + 15 = 0$

$\rightarrow$ roots $= \tan\frac{\alpha}{2}, \tan\frac{\beta}{2}$

$$\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} = \frac{26}{8} = \frac{13}{4}; \quad \tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2} = \frac{15}{8}$$

$$\tan\left(\frac{\alpha}{2}+\frac{\beta}{2}\right) = \frac{\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}}{1+\tan\frac{\alpha}{2}\tan\frac{\beta}{2}} = \frac{\frac{26}{8}}{1-\frac{15}{8}} = \frac{26}{8}\left(\frac{-8}{7}\right)$$

$$\tan\left(\frac{\alpha+\beta}{2}\right) = \frac{-26}{7}$$

$$\cos(\alpha+\beta) = \frac{1-\tan^2\left(\frac{\alpha+\beta}{2}\right)}{1+\tan^2\left(\frac{\alpha+\beta}{2}\right)} = \frac{1-\frac{676}{49}}{1+\frac{676}{49}}$$

$$\cos(\alpha+\beta) = \frac{-627}{725}$$

41. (*a*) $N^r = 2$

$$\left[\left(\sin 1^\circ + \sin 89^\circ\right) + \left(\sin 2^\circ + \sin 88^\circ\right) + \cdots + \left(\sin 44^\circ + \sin 46^\circ\right) + \sin 45^\circ\right]$$

$$\Rightarrow \frac{N^r}{D^r} = 2\{\sin 45^\circ[2(\cos 44^\circ + \cos 43^\circ + \cdots + \cos 1^\circ)] + 1\}$$

$$= 2\sin 45^\circ$$

$$= \sqrt{2}$$

42. (*c*) $\tan y = \dfrac{1+\sqrt{1-x}}{1+\sqrt{1+x}}$

If $x = \cos\theta$ then

$\sqrt{1-x} = \sqrt{2}\sin(\theta/2), \sqrt{1+x} = \sqrt{2}\cos(\theta/2)$

$$\Rightarrow \tan y = \frac{\sqrt{2}\left[\frac{1}{\sqrt{2}}+\sin\frac{\theta}{2}\right]}{\sqrt{2}\left[\frac{1}{\sqrt{2}}+\cos\frac{\theta}{2}\right]} = \frac{\sin\frac{\pi}{4}+\sin\frac{\theta}{2}}{\cos\frac{\pi}{4}+\cos\frac{\theta}{2}}$$

$$= \frac{2\sin\left(\frac{\pi}{8}+\frac{\theta}{4}\right)\cos\left(\frac{\pi}{8}-\frac{\theta}{4}\right)}{2\cos\left(\frac{\pi}{8}+\frac{\theta}{4}\right)\cos\left(\frac{\pi}{8}-\frac{\theta}{4}\right)} = \tan\left(\frac{\pi}{8}+\frac{\theta}{4}\right)$$

$$\Rightarrow 4y = \frac{\pi}{2}+\theta$$

$$\Rightarrow \sin 4y = \cos\theta = x$$

43. (*c*) $\cos A = \dfrac{3}{4}$

$$16\cos^2\frac{A}{2} - 32\sin\frac{A}{2}\sin\frac{5A}{2}$$

$$= \frac{16(1+\cos A)}{2} - 16(\cos 2A - \cos 3A)$$

$$= \frac{16(1+\cos A)}{2} - 16\{(2\cos^2 A - 1)\} - (4\cos^3 A - 3\cos A)\}$$

$$= 8\left(1+\frac{3}{4}\right) - 16\left\{2\times\frac{9}{16} - 1 - 4\times\frac{27}{64} + 3\times\frac{3}{4}\right\} = 3$$

44. (*c*) $A = \tan 6^\circ \tan 42^\circ$, $B = \cot 66^\circ \cot 78^\circ$

$$\frac{A}{B} = \tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ$$

$$\Rightarrow \frac{A}{B} = \frac{\tan 6^\circ \tan(60^\circ-6^\circ)\tan(60^\circ+6^\circ)}{\tan 54^\circ}\cdot\tan 78^\circ$$

$$\Rightarrow \frac{A}{B} = \frac{\tan 18^\circ.\tan(60^\circ-18^\circ)\tan(60^\circ+18^\circ)}{\tan 54^\circ}\cdot\tan 42^\circ$$

$$= \frac{\tan 54^\circ}{\tan 54^\circ} \Rightarrow \frac{A}{B} = 1 \Rightarrow A = B$$

45. (*d*) $\cos\frac{\pi}{10}\cdot\cos\frac{2\pi}{10}\cdot\cos\frac{4\pi}{10}\cdot\cos\frac{8\pi}{10}\cdot\cos\frac{16\pi}{10}$

$$= \frac{\sin 2^5\frac{\pi}{10}}{2^5\sin\frac{\pi}{10}} = \frac{1}{32}\frac{\sin\frac{32\pi}{10}}{\sin\frac{\pi}{10}} = \frac{1}{32}\frac{\sin\left(3\pi+\frac{2\pi}{10}\right)}{\sin\left(\frac{\pi}{10}\right)}$$

$$= -\frac{1}{32}\cdot\frac{2\sin\frac{\pi}{10}\cos\frac{\pi}{10}}{\sin\frac{\pi}{10}} = -\frac{1}{16}\cos\frac{\pi}{10}$$

$$= -\frac{1}{64}\sqrt{10+2\sqrt{5}}$$

46. (*a*) We have, $\sin\alpha + \sin\beta = \dfrac{-21}{65}$

& $\cos\alpha + \cos\beta = \dfrac{-27}{65}$

Squaring and adding both sides, we get

$$\Rightarrow 2 + 2\cos(\alpha-\beta) = \frac{21^2+27^2}{65^2}$$

$$\Rightarrow 2(1+\cos(\alpha-\beta)) = \frac{21^2+27^2}{65^2}$$

$$\Rightarrow 2\,.\,2\cos^2\left(\frac{\alpha-\beta}{2}\right) = \frac{3^2(7^2+9^2)}{65^2}$$

$$\Rightarrow \cos\left(\frac{\alpha-\beta}{2}\right) = \pm\frac{3\sqrt{130}}{130} = \pm\frac{3}{\sqrt{130}}$$

$$\because \frac{\pi}{2} < \left(\frac{\alpha-\beta}{2}\right) < \frac{3\pi}{2} \therefore \cos\left(\frac{\alpha-\beta}{2}\right) < 0$$

Hence $\cos\left(\dfrac{\alpha-\beta}{2}\right) = \dfrac{-3}{\sqrt{130}}$

47. (*c*) $\tan x + \tan\left(\frac{\pi}{3}+x\right) + \tan\left(\frac{2\pi}{3}+x\right)$

$$= \tan x + \frac{\tan x+\sqrt{3}}{1-\sqrt{3}\tan x} + \frac{\tan x-\sqrt{3}}{1+\sqrt{3}\tan x}$$

$$= \tan x + \frac{8\tan x}{1-3\tan^2 x} = \frac{3(3\tan x - \tan^3 x)}{1-3\tan^2 x}$$

$$= 3\tan 3x$$

Therefore, the given equation is $3\tan 3x = 3$

$\Rightarrow \tan 3x = 1$.

48. (*c*) $\sqrt{2+\sqrt{2+2\cos 4\theta}}$

$$= \sqrt{2+\sqrt{2.2\cos^2 2\theta}}$$

$$= \sqrt{2+2\cos 2\theta} = \sqrt{4\cos^2\theta} = 2\cos\theta$$

49. (*b*) We have $x + \dfrac{1}{x} = 2\cos\theta$,

Now $x^3 + \dfrac{1}{x^3} = \left(x+\dfrac{1}{x}\right)^3 - 3x\dfrac{1}{x}\left(x+\dfrac{1}{x}\right)$

$$= (2\cos\theta)^3 - 3(2\cos\theta) = 8\cos^3\theta - 6\cos\theta$$

$$= 2(4\cos^3\theta - 3\cos\theta) = 2\cos 3\theta$$

Trick: Put $x = 1 \Rightarrow \theta = 0^\circ$

Then $x^3 + \dfrac{1}{x^3} = 2 = 2\cos 3\theta$

50. (*a*) $\tan\theta = \sqrt{n}$

$$\Rightarrow \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{1-n}{1+n} = \text{rational}$$

51. (*a*) $\sin 2\theta = \cos 3\theta$

$\Rightarrow 2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta$

$\Rightarrow 2\sin\theta = 4(1-\sin^2\theta) - 3$

$\Rightarrow 4\sin^2\theta + 2\sin\theta - 1 = 0$

$$\Rightarrow \sin\theta = \frac{\sqrt{5}-1}{4}$$

52. (*c*) $\because k = \dfrac{2\tan x}{1+\tan^2 x} = \sin 2x$

$\Rightarrow \sin 2C = \sin 2B$

$\angle B < \angle C$

$\Rightarrow 2C = \pi - 2B \Rightarrow B + C = \dfrac{\pi}{2}$

$\therefore \angle A = \dfrac{\pi}{2}$

53. (*a*) $\because \alpha+\beta+\gamma = 2\pi \Rightarrow \frac{\alpha}{2}+\frac{\beta}{2}+\frac{\gamma}{2} = \pi$

$\Rightarrow \tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2}$ ta

54. (*d*) If $A+B+C = \dfrac{3\pi}{2}$ then

$2\cos(A+B)\cos(A-B) + \cos 2C$

$= -2\sin C\cos(A-B) + \cos 2C$

$= -2\sin C\cos(A-B) + 1 - 2\sin^2 C$

$= 1 - 2\sin C[\cos(A-B) + \sin C]$

$= -2\sin C[\cos(A-B) + \sin[3\pi/2 - (A+B)]]$

$= 1 - 2\sin C[\cos(A-B) - \cos(A+B)]$

$= 1 - 2\sin C \cdot 2\sin A \sin B$

$= 1 - 4\sin A \sin B \sin C$

55. (*c*) $A+B+C=\pi \Rightarrow A=\pi-(B+C)$

$\cos A = \cos(\pi-(B+C)) = -\cos(B+C)$ ... (i)

Also, $\cos A = \cos B \cdot \cos C$ ... (ii)

$-\cos(B+C) = \cos B \cdot \cos C$ (from (i) & (ii))

$\sin B \sin C - \cos B \cos C = \cos B \cos C$

$\sin B \sin C = 2\cos B \cos C$

$\tan B \cdot \tan C = 2.$

56. (*c*) $f(0) = \sin^4 0 + \cos^2 0 = \sin^2 0(1-\cos^2 0) + \cos^2 0$

$= \sin^2 0 + \cos^2 0 - \sin 0 \cos^2 0$

$f(0) = 1 - \frac{1}{4}\sin^2 2\theta$

$\because 0 \le \sin^2 2\theta \le 1$

$f(0)_{max} = 1$

$f(0)_{min} = 1 - \frac{1}{4} = 3/4$

$\therefore$ Range is $\left[\frac{3}{4}, 1\right]$

57. (*d*) Maximum value of

$f(x) = \sqrt{1^2+1^2} = \sqrt{2}$

58. (*b*) Since,

$-\sqrt{a^2+b^2} \le a\sin x + b\cos x \le \sqrt{a^2+b^2}$

$\therefore -\sqrt{74} \le 7\cos x + 5\sin x \le \sqrt{74}$

So. $-\sqrt{74} < 2k+1 < \sqrt{74}$

Therefore, $2k+1 = \pm 8, \pm 7, \pm 6, \pm 1, 0$

So, $k = -4, \pm 3, \pm 2, \pm 1, 0$, so, 8 values of k.

59. (*a*) $f(x) = 3\cos x + 5\sin(x - \pi/6)$

$= \frac{1}{2}\cos x + 5 \times \frac{\sqrt{3}}{2}\sin x$

Then,

$$-\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \le f(x) \le \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2}$$

$$-\sqrt{19} \le f(x) \le \sqrt{19}$$

60. (*c*) If $A+B+C = n\pi$

then $\tan A + \tan B + \tan c = \tan A \tan B \tan C$

$\tan(\alpha+\beta+\gamma) = \tan(\beta+\gamma-\alpha) + (\gamma+\alpha-\beta) + (\alpha+\beta-\gamma)$

$\tan(\beta+\gamma-\alpha) + \tan(\gamma+\alpha-\beta) + \tan(\alpha+\beta-\gamma)$

$= \tan(\beta+\gamma-\alpha)\ \tan(\gamma+\alpha-\beta)\tan(\alpha+\beta-\gamma)$

61. (*c*) $A+B=\alpha$

$A-B=x$

$\frac{\tan A}{\tan B} = \frac{k}{1}$

$\frac{\sin A\cos B}{\cos A \sin B} = \frac{k}{1}$

$\frac{\sin A\cos B + \cos A \sin B}{\sin A\cos B - \cos A \sin B} = \frac{k+1}{k-1}$

$\frac{\sin(A+B)}{\sin(A-B)} = \frac{k+1}{k-1}$

$\frac{\sin\alpha}{\sin x} = \frac{k+1}{k-1}$

$\sin x = \left(\frac{k-1}{k+1}\right)\sin\alpha$

62. (*b*) $\tan x + \tan y = 42$

$\cot x + \cot y = 49 \quad \frac{\tan x + \tan y}{\tan x \tan y} = 49$

$\tan x \tan y = \frac{42}{49} = \frac{6}{7}$

$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$

$= \frac{42}{1-\frac{6}{7}} = 42.7 = 294$

63. (*b*) $-\sin x - \cos x + \sin x + 18\cos x + \cos x + 9\sin x$

$= 18\cos x + 9\sin x$

$a = 9 \quad b = 18$

$a + b = 27$

64. (*b*) $\cos A(\sin(B-C)) + \cos B\sin(C-A) + \cos C\sin(A-B)$

$= \cos A(\sin B\cos C - \sin C\cos B)$

$+ \cos B(\sin C\cos A - \sin A\cos C)$

$= \sin 0\cos B\cos C - \sin C\cos A\cos B$

$+ \sin C\cos A\cos B - \sin A\cos B\cos C$

$= \sin B\cos B\cos C - \sin C\cos A\cos B\cos$

$+ \sin A\cos B\cos C - \sin B\cos A\cos C$

$= 0$

65. (*a*) $\frac{\sin(\alpha+\beta)}{\sin(\alpha-\beta)} = \frac{p}{q}$ then $\tan\alpha\cot\beta$

Using compenent = divendo

$\frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{\sin(\alpha+\beta)-\sin(\alpha-\beta)} = \frac{p+q}{p-q}$

$\frac{2\sin\alpha\cos\beta}{2\sin\beta\cos\alpha} = \frac{p+q}{p-q}$

$\tan\alpha \cdot \cot\beta = \frac{p+q}{p-q}$

Prabal (JEE Main Level)

1. (*a*) $\tan\left(7\frac{1}{2}^\circ\right) = \tan 75^\circ = \frac{\sin(7.5^\circ)}{\cos(7.5^\circ)}$

$= \frac{2\sin^2(7.5^\circ)}{2\sin(7.5^\circ)\cos(7.5^\circ)} = \frac{1-\cos(2\times 7.5^\circ)}{\sin(2\times 7.5^\circ)}$

$\left(\begin{matrix}2\sin^2\theta = 1-\cos 2\theta \\ 2\sin\theta\cos\theta = \sin 2\theta\end{matrix}\right)$

$= \frac{1-\cos 15^\circ}{\sin 15^\circ}$ ... (i)

$\sin 15^\circ = \sin(45^\circ - 30^\circ)$

$= \sin 45^\circ \cos 30^\circ - \sin 30^\circ \cos 45^\circ$

$= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} - \frac{1}{2}\cdot\frac{1}{\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}}$ ... (ii)

$\cos 15^\circ = \cos(45^\circ - 30^\circ)$

$= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$

$= \frac{1}{\sqrt{2}}\cdot\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}\cdot\frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$ ... (iii)

From (*i*), (*ii*) and (*iii*)

$$\tan\left(7\frac{1^\circ}{2}\right) = \frac{1-\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\frac{\sqrt{3}-1}{2\sqrt{2}}} = \frac{2\sqrt{2}-\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{2\sqrt{2}-\sqrt{3}-1}{\sqrt{3}-1}, \frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$= \frac{2\sqrt{6}+2\sqrt{2}-3-\sqrt{3}-\sqrt{3}-1}{3-1}$$

$$= \frac{2\sqrt{6}-2\sqrt{3}+2\sqrt{2}-4}{2} = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{2}+1}$$

2. (*d*) $a\sin x + b\cos(c+x) + b\cos(c-x) = \alpha$

$a\sin x + b(\cos(c+x) + \cos(c-x) = \alpha$

$a\sin x + b.\ 2\cos\frac{(c+x+c-x)}{2}\cos\left(\frac{c+x-c+x}{2}\right) = \alpha$

$a\sin x + 2b\cos c\cos x = \alpha$

$\cos c = \frac{\alpha - a\sin x}{2b\cos x} = \frac{1}{2b}(\alpha\sec x - a\tan x)$

Let $f(x) = |\cos c|$

for minimum value of $f(x) = \frac{1}{2b}(\alpha\sec x - a\tan x)$

Differentiating.

$f'(x) = \frac{1}{2b}(\alpha\sec x\tan x - a\sec^2 x)$

$f'(x) = 0 \Rightarrow \alpha\sec x\tan x - a\sec^2 x = 0$

$\sec x(\alpha\tan x - a\sec x) = 0 \Rightarrow \sec x = 0$

or $\alpha\tan x - a\sec x = 0$

$\alpha\cdot\frac{\sin x}{\cos x} = \frac{a}{\cos x}$

$\Rightarrow \sin x = \frac{a}{\alpha}$

$\sin x = \frac{a}{\alpha}, \quad \cos x = \sqrt{1-\frac{a^2}{\alpha^2}} = \frac{\sqrt{\alpha^2-a^2}}{\alpha}$

$$\cos c = \frac{\alpha - a\sin x}{2b\cos x} = \frac{\alpha - a.\frac{a}{\alpha}}{2b\frac{\sqrt{\alpha^2-a^2}}{\alpha}}$$

$$= \frac{\alpha^2-a^2}{2b\sqrt{\alpha^2-a^2}} = \frac{\sqrt{\alpha^2-a^2}}{2b} = \sqrt{\frac{\alpha^2-a^2}{4b^2}}$$

3. (c) $\frac{\sin A}{\sin B}=\frac{\sqrt{3}}{2}$ ... (i)

$\frac{\cos A}{\cos B}=\frac{\sqrt{5}}{2}$ ... (ii)

$0<A.B<\frac{\pi}{2}$

(i) ÷ (ii)

$\frac{\tan A}{\tan B}=\frac{\sqrt{3}}{\sqrt{5}} \Rightarrow \frac{\tan A}{\sqrt{3}}=\frac{\tan B}{\sqrt{5}}=k$(Let); $k>0$

$\tan A=\sqrt{3}k; \quad \tan B=\sqrt{5}k$

Also, $\frac{\sin A}{\sin B}=\frac{\sqrt{3}}{2} \Rightarrow 2\sin A=\sqrt{3}\sin B$

$\Rightarrow \frac{2\sin A}{\cos A}.\cos A=\frac{\sqrt{3}\sin B}{\cos B}.\cos B$

$\frac{2\tan A}{\sec A}=\frac{\sqrt{3}\tan B}{\sec B}$

$\Rightarrow \frac{2\tan A}{\sqrt{1+\tan^2 A}}=\frac{\sqrt{3}\tan B}{\sqrt{1+\tan^2 B}}$

$\Rightarrow \frac{2\sqrt{3}k}{\sqrt{1+3k^2}}=\frac{\sqrt{3}.\sqrt{5}k}{\sqrt{1+5k^2}}$

$\Rightarrow \frac{2k}{\sqrt{1+3k^2}}=\frac{\sqrt{5}k}{\sqrt{1+5k^2}}$

$\Rightarrow \frac{4k^2}{1+3k^2}=\frac{5k^2}{1+5k^2}$

$4k^2+20k^4=5k^2+15k^4$

$5k^4-k^2=0 \Rightarrow k^2(5k^2-1)=0$

$\Rightarrow k^2=0$ or $k^2=\frac{1}{5}$

But $k>0$, so $k=\frac{+1}{\sqrt{5}}$

$\tan A+\tan B=\sqrt{3}k+\sqrt{5}k=\frac{\sqrt{3}+\sqrt{5}}{\sqrt{5}}$

4. (c)

$\frac{\sin\alpha-\sin\gamma}{\cos\gamma-\sin\alpha}=\frac{2\sin\left(\frac{\alpha-\gamma}{2}\right)\cos\left(\frac{\alpha+\gamma}{2}\right)}{2\sin\left(\frac{\alpha+\gamma}{2}\right)\sin\left(\frac{\alpha-\gamma}{2}\right)}$

$=\frac{\cos\left(\frac{\alpha+\gamma}{2}\right)}{\sin\left(\frac{\alpha+\gamma}{2}\right)}=\cot\left(\frac{\alpha+\gamma}{2}\right)$

$=\cot(\beta)$

5. (c) $\log_3(\sin x)-\log_3(\cos x)-\log_3(1-\tan x)-\log_3(1+\tan x)=-1$

$\log_3\left(\frac{\sin x}{\cos x(1-\tan x)(1+\tan x)}\right)=-1$

$\log_3\left(\frac{\tan x}{1-\tan^2 x}\right)=-1$

$\frac{\tan x}{1-\tan^2 x}=3^{-1} \Rightarrow \frac{2\tan x}{1-\tan^2 x}=\frac{2}{3}$

$\Rightarrow \tan(2x)=\frac{2}{3}$

6. (c) $\sum_{r=1}^{n-1}\cos^2\frac{r\pi}{n}=\cos^2\frac{\pi}{n}+\cos^2\frac{2\pi}{n}+\cos^2\frac{3\pi}{n}+...+\cos^2\frac{(n-1)\pi}{n}$

$\frac{1}{2}\left(1+\cos\frac{2\pi}{n}\right)+\frac{1}{2}\left(1+\cos\frac{4\pi}{n}\right)+\frac{1}{2}\left(1+\cos\frac{6\pi}{n}\right)+...+\frac{1}{2}\left(1+\cos 2\frac{(n-1)\pi}{n}\right)$

$=\frac{1}{2}\left\{(n-1)+\left(\cos\frac{2\pi}{n}+\cos\frac{4\pi}{n}+\cos\frac{6\pi}{n}+...(n-1)\text{term}\right)\right\}$

$=\frac{n-1}{2}+\frac{1}{2}\frac{\sin\left(\frac{n-1\pi}{n}\right)}{\sin\left(\frac{\pi}{n}\right)}.\cos\left(\frac{2\pi}{n}+(n-2)\frac{\pi}{n}\right)$

$=\frac{n-1}{2}+\frac{1}{2}\frac{\sin\left(\frac{\pi}{n}\right)}{\sin\left(\frac{\pi}{n}\right)}.\cos\pi=\frac{n-1}{2}-\frac{1}{2}$

$=\frac{n-2}{2}=\frac{n}{2}-1$

7. (b) $2\sin 2°+4\sin 4°+6\sin 6°+...+178\sin 178°+180\sin 180°$

$=2\sin 2°+4\sin 4°+6\sin 6°+...+178\sin 2°+180.0$

$=(2+178)\sin 2°+(4+176)\sin 4°+(6+174)\sin 6°+...+90\sin 90°$

$=180(\sin 2°+\sin 4°+\sin 6°+...\sin 88°)+90.1$

$=180\times\sin\frac{\left(44.\frac{2}{2}\right)}{\sin\left(\frac{2}{2}\right)}\times\sin\left(2+(44-1)\frac{2}{2}\right)+90$

$=180\times\frac{\sin 44°}{\sin 1°}\times\sin 45°+90$

$=180\times\frac{\sin(45°-1)}{\sin 1°}\times\frac{1}{\sqrt{2}}+90$

$=\frac{180}{\sqrt{2}}\frac{(\sin 45°\cos 1°-\sin 1°\cos 45°)}{\sin 1°}+90$

$=\frac{180}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}.\cot 1°-\frac{1}{\sqrt{2}}\right)+90$

$=\frac{180}{2}\cot 1°-\frac{180}{2}+90=90\cot 1°$

8. (a) $\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)(1+\sec 2^2\theta)(1+\sec 2^3\theta).....(1+\sec 2^n\theta)$

Let us solve it for $n=1$

$\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)$

$=\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\left(1+\frac{1}{\cos\theta}\right)\left(1+\frac{1}{\cos 2\theta}\right)$

$=\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\left(\frac{1+\cos\theta}{\cos\theta}\right)\left(\frac{1+\cos 2\theta}{\cos 2\theta}\right)$

$=\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}.\frac{2\cos^2\frac{\theta}{2}}{\cos\theta}.\frac{2\cos^2\theta}{\cos 2\theta}$

$=\frac{\sin\frac{\theta}{2}.2\cos\frac{\theta}{2}.2\cos\theta}{\cos 2\theta}=\frac{\sin\theta.2\cos\theta}{\cos 2\theta}$

$=\frac{\sin 2\theta}{\cos 2\theta}=\tan 2\theta$

for $n=n$,

$\tan 2^n\theta$.

9. (c) $f(x)=27^{\cos 3x}.81^{\sin 3x}$

$-\sqrt{3^2+4^2} \le 3\cos 3x+4\sin 3x \le +\sqrt{3^2+4^2}$

$-5 \le 3\cos 3x+4\sin 3x \le +5$

$3^{-5} \le 3^{3\cos 3x+4\sin 3x} \le +3^5$

$\frac{1}{243} \le f(x) \le 243$

Hence $f(x)|_{\text{min.}}=\frac{1}{243}$

10. (d) $\sin x(\sin x+\cos x)=a$

$\sin^2 x+\sin x\cos x=a$

$2\sin^2 x+2\sin x\cos x=2a$

$1-\cos 2x+\sin 2x=2a$

$\sin 2x-\cos 2x=2a-1$

$-\sqrt{1^2+(-1)^2} \le \sin 2x-\cos 2x \le \sqrt{1^2+(-1)^2}$

$-\sqrt{2} \le \sin 2x-\cos 2x \le \sqrt{2}$

$-\sqrt{2} \le 2a-1 \le \sqrt{2}$

$1-\sqrt{2} \le 2a \le 1+\sqrt{2}$

$\frac{1-\sqrt{2}}{2} \le a \le \frac{1+\sqrt{2}}{2}$

11. (a) $\frac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=\frac{\sin\theta+2\sin\theta\cos\theta}{(1+\cos 2\theta)+\cos\theta}$

$=\frac{\sin\theta(1+2\cos\theta)}{2\cos^2\theta+\cos\theta}$

$=\frac{\sin\theta(1+2\cos\theta)}{\cos\theta(1+2\cos\theta)}=\frac{\sin\theta}{\cos\theta}=\tan\theta$

As $\theta\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right) \Rightarrow \tan\theta\in(-\infty,\infty)$

12. (a) $f(\alpha,\beta)=\cos^2\alpha+\sin^2\alpha\cdot\cos 2\beta$

$=1-\sin^2\alpha+\sin^2\alpha\cdot\cos 2\beta$

$=1-\sin^2\alpha(1-\cos 2\beta)$

$=1-\sin^2\alpha\cdot 2\sin^2\beta$

$=1-2\sin^2\alpha\cdot\sin^2\beta$

$f(\beta,\alpha)=1-2\sin^2\beta\cdot\sin^2\alpha$

$f(\alpha,\beta)=f(\beta,\alpha)$

13. (*a*) $f(x) = \cos 10x + \cos 8x + 3\cos 4x + 3\cos 2x$

$g(x) = 8\cos x \cos^3 3x$

$f(x) = \cos 10x + \cos 2x + \cos 8x + \cos 4x + 2\cos 4x + 2\cos 2x$

$= 2\cos\left(\frac{10x+2x}{2}\right)\cos\left(\frac{10x-2x}{2}\right)$

$+ 2\cos\left(\frac{8x+4x}{2}\right)\cos\left(\frac{8x-4x}{2}\right) + 2\cos 4x + 2\cos 2x$

$= 2\cos 6x \cos 4x + 2\cos 6x \cos 2x + 2\cos 4x + 2\cos 2x$

$= 2\cos 6x(\cos 4x + \cos 2x) + 2(\cos 4x + \cos 2x)$

$= (\cos 4x + \cos 2x)(2\cos 6x + 2)$

$= 2\cos\left(\frac{4x+2x}{2}\right)\cos\left(\frac{4x-2x}{2}\right) 2\cdot 2\cos^2 3x$

$= 2\cos 3x \cos x \cdot 2 \cdot 2\cos^2 3x$

$= 8\cos x \cos^3 3x = g(x)$

Hence, $f(x) = g(x)$

14. (*b*) $\cot 70° + 4\cos 70°$

$= \frac{\cos 70° + 4\sin 70°\cos 70°}{\sin 70°}$

$= \frac{\cos 70° + 2\sin 140°}{\sin 70°}$

$= \frac{\cos 70° + 2\sin(180° - 40°)}{\sin 70°}$

$= \frac{\sin 20° + \sin 40° + \sin 40°}{\sin 70°}$

$= \frac{2\sin 30°\cos 10° + \sin 40°}{\sin 70°}$

$= \frac{\sin 80° + \sin 40°}{\sin 70°}$

$= \frac{2\sin 60°\cos 20°}{\sin 70°} = \sqrt{3}$

15. (*c*) We have $\cos^2 A + \cos^2 B - (1 - \cos^2 C) = 0$

$\Rightarrow \cos^2 A + \cos^2 B - \sin^2 C - 0$

$\Rightarrow \cos^2 A + \cos(B + C)\cos(B - C) = 0$

$\Rightarrow 2\cos A \cos B \cos C = 0$

Hence, either A or B or C is 90°.

16. (*a*) $f_k(\theta) = \sin^k\theta + \cos^k(\theta)$

$f_6(\theta) = \sin^6\theta + \cos^6\theta$

$= (\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)$

$f_6(\theta) = 1 - 3\sin^2\theta\cos^2\theta$

similarly $f_4(\theta) = 1 - 2\sin^2\theta\cos^2\theta$

$\frac{1}{6}f_6(\theta) - \frac{1}{4}f_4(\theta) = -\frac{1}{12}$

17. (*b*) When $270° < \theta < 360°$, we have

$\sqrt{2(1+\cos\theta)} = \sqrt{\left(2\cos^2\frac{\theta}{2}\right)}$

which is non-negative. Now, the above equation can be written as

$\sqrt{2(1+\cos\theta)} = 2\left|\cos\frac{\theta}{2}\right|$

$= -2\cos\frac{\theta}{2}$

$\left(\because \cos\frac{\theta}{2} < 0 \text{ when } 135° < \frac{\theta}{2} < 180°\right)$

Now, let us consider that

$\sqrt{2+\sqrt{2(1+\cos\theta)}}$

which is not-negative. That is,

$\sqrt{2+\sqrt{2(1+\cos\theta)}} = \sqrt{2-2\cos\frac{\theta}{2}}$

$= \sqrt{2}\sqrt{1-\cos\frac{\theta}{2}} = \sqrt{2}\sqrt{2\sin^2\frac{\theta}{4}} = 2\left|\sin\frac{\theta}{4}\right|$

$= 2\sin\frac{\theta}{4}$

$\left(\because \sin\frac{\theta}{4} > 0 \text{ when } \frac{135°}{2} < \frac{\theta}{4} < 90°\right)$

18. (*a*) $\tan\alpha + \tan\beta = -p$, $\tan\alpha\tan\beta = q$

$\tan(\alpha+\beta) = \frac{-p}{1-q} = \frac{p}{q-1}$

$\frac{1}{1+\tan^2(\alpha+\beta)}$ $[\tan^2(\alpha+\beta) + p\tan(\alpha+\beta) + q]$

$= \frac{1}{1+\frac{p^2}{(q-1)^2}}$ $\left[\frac{p^2}{(q-1)^2} + \frac{p^2}{(q-1)} + q\right]$

$= \frac{1}{(q-1)^2 + p^2}[p^2 + p^2(p-1) + q(q-1)^2]$

$= \frac{1}{p^2 + (q-1)^2}[p^2 q + q(q-1)^2]$

$= q\left[\frac{p^2 + (q-1)^2}{p^2 + (q-1)^2}\right] = q$

19. (*b*) $(\cos^4 1° + \cos^4 2° + \cos^4 3° + \ldots + \cos^4 179°)$

$-(\sin^4 1° + \sin^4 2° + \sin^4 3° + \ldots + \sin^4 179°)$

$= \cos 2° + \cos 4° + \cos 6° + \ldots + \cos(358°)$

$= \cos\frac{\left(\frac{2°+358°}{2}\right)\cdot\sin(179\times 1°)}{\sin 1°}$

$= \cos(180°) = -1.$

20. (*b*) $\because \cos 3A + \cos 3B + \cos 3C = 1$

$\Rightarrow \cos 3A + \cos 3B + \cos 3C - 1 = 0$

$\Rightarrow \cos 3A + \cos 3B + \cos 3C + \cos 3\pi = 0$

$\Rightarrow 2\cos\left(\frac{3A+3B}{2}\right)\cos\left(\frac{3A-3B}{2}\right) + 2\cos\left(\frac{3\pi+3C}{2}\right)\cos\left(\frac{3\pi-3C}{2}\right) = 0$

$\Rightarrow 2\cos\left(\frac{3\pi-3C}{2}\right)\left\{\cos\left(\frac{3A-3B}{2}\right) + \cos\left(\frac{3\pi+3C}{2}\right) = 0\right\}$

$2\cos\left(\frac{3\pi}{2} - \frac{3C}{2}\right)\cdot 2\cos\left(\frac{3\pi+3C+3A-3B}{4}\right)\cdot\cos\left(\frac{3\pi+3C-3A+3B}{4}\right) = 0$

$\Rightarrow 2\cos\left(\frac{3\pi}{2} - \frac{3C}{2}\right)\cdot 2\cos\left(\frac{3\pi}{2} - \frac{3B}{2}\right)\cdot\cos\left(\frac{3\pi}{2} - \frac{3A}{2}\right) = 0$

$\Rightarrow -4\sin\left(\frac{3A}{2}\right)\sin\left(\frac{3B}{2}\right)\sin\left(\frac{3C}{2}\right) = 0$

$\Rightarrow \sin\left(\frac{3A}{2}\right)\sin\left(\frac{3B}{2}\right)\sin\left(\frac{3C}{2}\right) = 0$

$\therefore \frac{3A}{2} = \pi \text{ or } \frac{3B}{2} = \pi \text{ or } \frac{3C}{2} = \pi$

$\therefore A = \frac{2\pi}{3} \text{ or } B = \frac{2\pi}{3} \text{ or } C = \frac{2\pi}{3}$

21. (*b*) Given expression is $2\sin^2\theta + 4\cos(\theta + \phi)\sin\theta\sin\phi + 2\cos 2(\theta+\phi) - 1$

$= (1 - \cos 2\phi) + 4\cos(\theta+\phi)\sin\theta\sin\phi + 2\cos 2(\theta+\phi) - 1$

$= -\cos 2\phi + 4\cos(\theta+\phi)\sin\theta\sin\phi + 2\cos^2(\theta+\phi)$

$= -\cos 2\phi + 2\cos(\theta+\phi)[\cos(\theta+\phi) + 2\sin\theta\sin\phi]$

$= -\cos 2\phi + 2\cos(\theta+\phi)[\cos\theta\cos\phi + \sin\theta\sin\phi]$

$= -\cos 2\phi + 2\cos(\theta+\phi)\cos(\theta-\phi)$

$= -\cos 2\phi + \cos 2\theta + \cos 2\phi = \cos 2\theta$

22. (*b*) As ΔABC is right angle traingle so orthocentre will be one of its vertex.

$\sin\theta < 0$ In IVth quadrant

$\cos 1 - \sin 1 < 0$

$\sin 4 + \sin 6 < 0$

learning plus trigo ratio

23. (*c*) $\frac{1}{\sin 10°} + \frac{1}{\sin 50°} - \frac{1}{\sin 70°}$

$= \frac{\sin 50°\sin 70° + \sin 10°\sin 70° - \sin 10°\sin 50°}{\sin 10°\sin 50°\sin 70°}$

$= \frac{\frac{1}{2}(\cos 20° - \cos 120° + \cos 60° - \cos 80° - \cos 40° + \cos 60°)}{\frac{1}{4}\sin 30°}$

$= \frac{\frac{1}{2}\left(\frac{3}{2} + \cos 20° - 2\cos 60°\cos 20°\right)}{\frac{1}{4}\sin 30°} = 6$

24. (*b*) $0 < \sin\alpha < \cos\alpha < 1$

$(\sin\alpha)^{\cos\alpha} < (\sin\alpha)^{\sin\alpha}$

$(\cos\alpha)^{\cos\alpha} < (\cos\alpha)^{\sin\alpha}$

25. (*b*) $a\cos x + b\sin x = c$

$\frac{a\left(1-\tan^2\frac{x}{2}\right)}{1+\tan^2\frac{x}{2}} + \frac{2b\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} = c$

$$(c+a)\tan^2\frac{x}{2}-2b\tan\frac{x}{2}+c-a=0$$

$$\tan\frac{x_1}{2}+\tan\frac{x_2}{2}=\frac{2b}{c+a}\text{ and }\tan\frac{x_1}{2}\tan\frac{x_2}{2}=\frac{c-a}{c+a}$$

$$\tan\left(\frac{x_1+x_2}{2}\right)=\frac{\frac{2b}{c+a}}{1-\frac{c-a}{c+a}}=\frac{2b}{2a}=\frac{b}{a}$$

26. (*d*) We have $4x^2-16x+15<0$

$\Rightarrow \frac{3}{2}<x<\frac{5}{2}$

Therefore, the integral solution of $4x^2-16x+15<0$ is $x=2$

Thus, $\tan\alpha=2$. It is given that $\cos\beta=\tan 45^\circ=1$.

$\therefore \sin(\alpha+\beta)\sin(\alpha-\beta)=\sin^2\alpha-\sin^2\beta$

$$=\frac{1}{1+\cot^2\alpha}-\left(1-\cos^2\beta\right)=\frac{1}{1+\frac{1}{4}}-0=\frac{4}{5}$$

27. (*a*) $\sum_{r=1}^{n}\sin\left(3^{r+1}\theta\right)\sec\left(3^r\theta\right)$

$$=\sum_{\gamma=1}^{n}\sin\left(3^{r-1}\theta\right)\sec\left(3^r\theta\right)\times\frac{2\cos\left(3^{r-1}\theta\right)}{2\cos\left(3^{r-1}\theta\right)}$$

$$=\frac{1}{2}\sum_{r=1}^{n}\frac{\sin\left(3^r\theta-3^{r-1}\theta\right)}{\cos\left(3^{r-1}\theta\right)\cos\left(3^r\theta\right)}$$

$$=\frac{1}{2}\sum_{r=1}^{n}\frac{\sin\left(3^r\theta\right)\cos\left(3^{r-1}\theta\right)-\cos\left(3^r\theta\right)\sin\left(3^{r-1}\theta\right)}{\cos\left(3^{r-1}\theta\right)\cos\left(3^r\theta\right)}$$

$$=\frac{1}{2}\sum_{r=1}^{n}\left(\tan\left(3^r\theta\right)-\tan\left(3^{r-1}\theta\right)\right)$$

$$=\frac{1}{2}\left(\tan 3^n\theta-\tan\theta\right)$$

28. (*a*) $\frac{\sin x}{\cos 3x}+\frac{\sin 3x}{\cos 9x}+\frac{\sin 9x}{\cos 27x}$

$$=\frac{1}{2}\left[\frac{2\sin x}{\cos 3x}+\frac{2\sin 3x}{\cos 9x}+\frac{2\sin 9x}{\cos 27x}\right]$$

$$=\frac{1}{2}\left[\frac{2\sin x\cos x}{\cos 3x\cos x}+\frac{2\sin 3x\cos 3x}{\cos 9x\cos 3x}+\frac{2\sin 9x\cos 9x}{\cos 27x\cos 9x}\right]$$

$$=\frac{1}{2}\left[\frac{\sin 2x}{\cos 3x\cos x}+\frac{\sin 6x}{\cos 9x\cos 3x}+\frac{\cos 18x}{\cos 27x\cos 9x}\right]$$

$$=\frac{1}{2}\left[\frac{\sin(3x-x)}{\cos 3x\cos x}+\frac{\sin(9x-3x)}{\cos 9x\cos 3x}+\frac{\sin(27x-9x)}{\cos 27x\cos 9x}\right]$$

$$=\frac{5}{2}[\tan 27x-\tan 9x+\tan 9x-\tan 3x+\tan 3x-\tan x]$$

$$=\frac{1}{2}[\tan 27x-\tan x]$$

29. (*a*) $\tan\frac{\theta}{2}=\sqrt{\frac{a-b}{a+b}}\tan\frac{\varphi}{2}$ ...(i)

N o w ,

$$\cos\theta=\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=\frac{1-\frac{a-b}{a+b}\tan^2\frac{\varphi}{2}}{1+\frac{a-b}{a+b}\tan^2\frac{\varphi}{2}}$$

$$=\frac{1-\frac{a-b}{a+b}\cdot\frac{\sin^2\frac{\varphi}{2}}{\cos^2\frac{\varphi}{2}}}{1+\frac{a-b}{a+b}\cdot\frac{\sin^2\frac{\varphi}{2}}{\cos^2\frac{\varphi}{2}}}$$

$$=\frac{(a+b)\cos^2\frac{\varphi}{2}-(a-b)\sin^2\frac{\varphi}{2}}{(a+b)\cos^2\frac{\varphi}{2}+(a-b)\sin^2\frac{\varphi}{2}}$$

$$a\left(\cos^2\frac{\varphi}{2}-\sin^2\frac{\varphi}{2}\right)$$

$$=\frac{+b\left(\cos^2\frac{\varphi}{2}+\sin^2\frac{\varphi}{2}\right)}{a\left(\cos^2\frac{\varphi}{2}+\sin^2\frac{\varphi}{2}\right)}$$

$$+b\left(\cos^2\frac{\varphi}{2}-\sin^2\frac{\varphi}{2}\right)=\frac{a\cos\varphi+b}{a+b\cos\varphi}$$

30. (*c*) $a\sin^2\theta+b\sin\theta\cos\theta+\cos^2\theta$

$$=\frac{a}{2}\sin^2\theta+\frac{c}{2}\cos^2\theta+\frac{a}{2}\sin^2\theta+\frac{c}{2}\cos^2\theta+b\sin\theta\cos\theta$$

$$=\left(\frac{a}{2}-\frac{c}{2}\right)\sin^2\theta+\frac{c}{2}\sin^2+\frac{c}{2}\cos^2\theta+\left(\frac{c}{2}-\frac{a}{2}\right)\cos^2\theta+\frac{a}{2}\sin^2\theta+\frac{b}{2}(2\sin\theta\cos\theta)$$

$$=\left(\frac{a}{2}-\frac{c}{2}\right)\sin^2\theta+\frac{c}{2}+\left(\frac{c}{2}-\frac{a}{2}\right)\cos^2\theta+\frac{a}{2}+\frac{b}{2}\sin 2\theta$$

$$=\left(\frac{a+c}{2}\right)+\left(\frac{a-c}{2}\right)\left(\sin^2\theta-\cos^2\theta\right)+\frac{b}{2}\sin 2\theta$$

$$=\frac{a+c}{2}+(\cos 2\theta)\frac{(c-a)}{2}+\frac{b}{2}\sin 2\theta$$

$$\sqrt{A^2+B^2}\le A\cos\alpha+B\sin\alpha\le+\sqrt{A^2+B^2}$$

$$\therefore -\sqrt{\left(\frac{c-a}{2}\right)^2+\left(\frac{b}{c}\right)^2}\le(\cos 2\theta)\frac{(c-a)}{2}+\frac{b}{2}\sin 2\theta\le\sqrt{\left(\frac{c-a}{2}\right)^2+\left(\frac{b}{2}\right)^2}$$

$$\therefore\frac{-1}{2}\sqrt{b^2(a-c)^2}\le\left(\frac{c-a}{2}\right)\cos 2\theta+\frac{b}{2}\sin 2\theta\le\frac{b}{2}\sin 2\theta$$

$$=\left(\frac{a+c}{2}\right)+\left(\frac{a-c}{2}\right)\left(\sin^2\theta-\cos^2\theta\right)+\frac{b}{2}\sin 2\theta$$

$$=\frac{a+c}{2}+(\cos 2\theta)\frac{(c-a)}{2}+\frac{b}{2}\sin 2\theta$$

$$\sqrt{A^2+B^2}\le A\cos\alpha+B\sin\alpha\le+\sqrt{A^2+B^2}$$

$$\therefore -\sqrt{\left(\frac{c-a}{2}\right)^2+\left(\frac{b}{2}\right)^2}\le(\cos 2\theta)\frac{(c-a)}{2}+\frac{b}{2}\sin 2\theta\le\sqrt{\left(\frac{c-a}{2}\right)^2+\left(\frac{b}{2}\right)^2}$$

$$\therefore\frac{-1}{2}\sqrt{b^2(a-c)^2}\le\left(\frac{c-a}{2}\right)\cos 2\theta+\frac{b}{2}\sin 2\theta\le\frac{1}{2}\sqrt{b^2+(a-c)^2}$$

$$\therefore\frac{a+c}{2}-\frac{\sqrt{b^2+(a-c)^2}}{2}\le\underbrace{\frac{a+c}{2}(2\cos\theta)\frac{(c-a)}{2}+\frac{b}{2}\sin 2\theta}_{a\sin^2\theta+b\sin\theta\cos\theta+c\cos^2\theta}\le\frac{a+c}{2}+\frac{1}{2}\sqrt{b^2+(a-c)^2}$$

31. (*b*) $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$

$$=\frac{2\sin\frac{\pi}{7}}{2\sin\frac{\pi}{7}}\left(\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}\right)$$

$$=\frac{\sin\frac{3\pi}{7}-\sin\frac{\pi}{7}+\sin\frac{5\pi}{7}-\sin\frac{3\pi}{7}+\sin\pi-\sin\frac{5\pi}{7}}{2\sin\frac{\pi}{7}}$$

$$=-\frac{1}{2}$$

32. (*c*) $y=\sqrt{\frac{1-\sin\alpha}{1+\sin\alpha}}=\sqrt{\frac{(1-\sin\alpha)^2}{(1+\sin\alpha)(1-\sin\alpha)}}$

$$=\left|\frac{1-\sin\alpha}{\cos\alpha}\right|=|\sec\alpha-\tan\alpha|$$

Given $y=\sec\alpha-\tan\alpha>0$

That means, $\sec\alpha-\tan\alpha>0$

$\Rightarrow(1-\sin\alpha)/\cos\alpha>0\Rightarrow\cos\alpha>0$

therefore solution of α is

$$a\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$$

33. (*a*) We have

$$\sin^6x+\cos^6x=\left(\sin^2x\right)^3+\left(\cos^2x\right)^3$$

$$=\left(\sin^2x+\cos^2x\right)^3-3\sin^2x\cos^2x\left(\sin^2x+\cos^2x\right)$$

$$=1-3\sin^2x\cos^2x=1-\frac{3}{4}\cdot 4\sin^2x\cos^2x$$

$$=1-\frac{3}{4}(\sin 2x)^2$$

$\Rightarrow$ maximum value of $\sin^6x+\cos^6x$ is $1-\frac{3}{4}\times$

$0=1$ and minimum values is $1-\frac{3}{4}\times 1=\frac{1}{4}$

34. (*a*) $(\sin\alpha+\sin 5\alpha)+(\sin 2\alpha+\sin 4\alpha)$

using

$$\sin C+\sin D=2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$2\sin\left(\frac{\alpha+5\alpha}{2}\right)\cos\left(\frac{\alpha-5\alpha}{2}\right)+$$

$$2\sin\left(\frac{2\alpha+4\alpha}{2}\right)\cos\left(\frac{2\alpha-4\alpha}{2}\right)$$

$$2\sin 3\alpha\cdot\cos 2\alpha+2\sin 3\alpha\cdot\cos\alpha$$

$$2\sin 3\alpha[\cos 2\alpha+\cos\alpha]$$

using

$$\cos C+\cos D=2\cos\left(\frac{c+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$2\sin 3\alpha\left[2\cos\left(\frac{2\alpha+\alpha}{2}\right)\cdot\cos\left(\frac{2\alpha-\alpha}{2}\right)\right]$$

$$4\sin 3\alpha\cdot\cos\left(\frac{3\alpha}{2}\right)\cdot\cos\left(\frac{\alpha}{2}\right)$$

35. (*a*) As we know that

$$\cos(a+b)\cos(a-b)=\cos^2 a-\sin^2 b$$

$$\cos(36°-\mathrm{A})\cos(36°+\mathrm{A})+\cos(54°+\mathrm{A})$$

$$\cos(54°-A)=\cos^2 36°-\sin^2 A+\cos^2 54°-\sin^2 A$$

$$=\cos^2 36°+\cos^2(90°-36°)-2\sin^2 A$$

$$\cos^2 36°+\sin^2 36°-2\sin^2\mathrm{A}$$

$$=1-2\sin^2 A=\cos 2A$$

36. (*a*) $\operatorname{cosec}\theta=\dfrac{1}{\sin\theta}\times\dfrac{\sin\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$

$$=\frac{\sin\left(\theta-\frac{\theta}{2}\right)}{\sin(\theta)\sin\left(\frac{\theta}{2}\right)}$$

$$=\frac{\sin\theta\cos\left(\frac{\theta}{2}\right)-\cos\theta\sin\left(\frac{\theta}{2}\right)}{\sin\theta\cdot\sin\left(\frac{\theta}{2}\right)}$$

$$\operatorname{cosec}\theta=\cot\left(\frac{\theta}{2}\right)-\cot\theta$$

similarly $\operatorname{cosec}2\theta=\cot(\theta)-\cot(2\theta)$

$$\operatorname{cosec}(2^{n-1}\theta)=\cot(2^{n-2}\theta)-\cot(2^{n-1}\theta)$$

adding above

$$=\cot\left(\frac{\theta}{2}\right)-\cot\left(2^{n-1}\theta\right)$$

37. (*d*) $A+B+C=180°$

$$C=180°-(A+B)\Rightarrow\frac{C}{2}=\frac{\pi}{2}-\frac{A+B}{2}$$

$$\tan(C/2)=\tan(\pi/2-(A+B)/2)$$

$$=\frac{1}{\tan((A+B)/2)}$$

$$=\frac{1-\tan(A/2)\tan(B/2)}{\tan(A/2)+\tan(B/2)}$$

Let

$$a=\tan\left(\frac{A}{2}\right),b=\tan\left(\frac{B}{2}\right),c=\tan\left(\frac{C}{2}\right)$$

all positive the constraint becomes

$$c=\frac{(1-ab)}{(a+b)}$$

Which is equivalent to $ab+bc+ca=1$

and we know that,

$$a^2+b^2+c^2\geq ab+bc+ca\Rightarrow a^2+b^2+c^2\geq 1$$

$$\Rightarrow\tan^2(\mathrm{A}/2)+\tan^2(\mathrm{B}/2)+\tan^2(\mathrm{C}/2)\geq 1$$

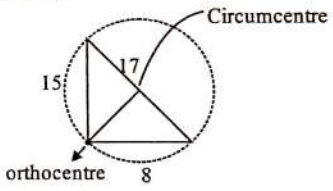
38. (*b*) $\dfrac{2\cos(B+C)\cos A+2\cos A\cos(C-B)}{2\sin(B+c)\cos A+2\cos A\sin(\theta-c)}$

$$=\frac{\cos(\theta+C)+\cos(C-B)}{\sin(\theta+C)+\sin(\theta-C)}$$

$$=\frac{2\cos(C)\cos(B)}{2\sin B\cos(C)}$$

$$=\cot B$$

39. (*b*) Distance between orthocentre & circumcentre in a triangle with sides 17, 15 & 8.



Distance between orthocentre and circumcentre

$$=\frac{17}{2}=8.5$$

40. (*c*)

$$\text{L.H.S.}=\frac{1}{2}\left[(2\sin°\sin 66°)(2\sin 42°\sin 78°)\right]$$

$$=\frac{1}{4}\left[(\cos 60°-\cos 72°)(\cos 36°-\cos 120°)\right]$$

$$=\frac{1}{4}\left[\left(\frac{1}{2}-\frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4}\frac{1}{2}\right)\right]$$

$$=\frac{1}{4}\frac{3-\sqrt{5}}{4}\cdot\frac{3+\sqrt{5}}{4}=\frac{1}{64}\cdot(9-5)=\frac{1}{16}.$$

41. (*a*) $\sin x+\sin^2 x=1\Rightarrow\sin x=1-\sin^2 x$

$$\Rightarrow\sin x=\cos^2 x \quad \ldots(i)$$

$$\cos^{12}x+3\cos^{10}x+3\cos^8 x+\cos^6 x-1$$

$$=\sin^6 x+3\sin^5 x+3\sin^4 x+\sin^3 x-1 \quad \text{(from ...(i))}$$

$$=\sin^3 x(\sin^3 x+3\sin^2 x+3\sin x+1)-1$$

$$=\sin^3 x(\sin x+1)^3)-1=0$$

42. (*c*) $\tan\alpha=\dfrac{\sqrt{3}+1}{\sqrt{3}-1}=\dfrac{\sqrt{3}+1}{\sqrt{3}-1}\cdot\dfrac{\sqrt{3}+1}{\sqrt{3}+1}$

$$=\frac{3+1+2\sqrt{3}}{3-1}$$

$$=\frac{4+2\sqrt{3}}{2}=2+\sqrt{3}=\tan 75°$$

$\alpha=75°$

Now, $\cos 2\alpha+(2+\sqrt{3})\sin 2\alpha=$

$\cos(2\times 75°)+(2+\sqrt{3})\sin(2\times 75°)$

$=\cos 150°+(2+\sqrt{3})\sin(150°)$

$$=-\frac{\sqrt{3}}{2}+(2+\sqrt{3})\cdot\frac{1}{2}=\frac{1}{2}(-\sqrt{3}+2+\sqrt{3})$$

$$=\frac{1}{2}\cdot 2=1$$

43. (*c*) $f(\theta)=1+\sin\left(\frac{\pi}{4}+\theta\right)+2\cos\left(\frac{\pi}{4}-\theta\right)$

$$1+\sin\frac{\pi}{4}\cos\theta+\cos\frac{\pi}{4}\sin\theta+2$$

$$\left(\cos\frac{\pi}{4}\cos\theta+\sin\frac{\pi}{4}\sin\theta\right)$$

$$=1+\frac{\cos\theta}{\sqrt{2}}+\frac{\sin\theta}{\sqrt{2}}+\frac{2\cos\theta}{\sqrt{2}}+\frac{2\sin\theta}{\sqrt{2}}$$

$$=1+\frac{3}{\sqrt{2}}(\sin\theta+\cos\theta)$$

$\sin\theta+\cos\theta\in\left[-\sqrt{2},+\sqrt{2}\right]$

$$\frac{3}{\sqrt{2}}(\sin\theta+\cos\theta)\in\left[-\sqrt{2}.\frac{3}{2},+\sqrt{2}.\frac{3}{\sqrt{2}}\right]$$

$$1+\frac{3}{\sqrt{2}}(\sin\theta+\cos\theta)\in[1-3,\ 1+3]$$

$f(\theta)\in[-2,+4]\ \Rightarrow\ f(\theta)\max=4$

44. (*b*) $\cos(\alpha+\beta)+\sin(\alpha-\beta)=0$

$\Rightarrow\ \cos\alpha\cos\beta-\sin\alpha\sin\beta+\sin\alpha\cos\beta-\sin\beta\cos\alpha=0$

$\cos\alpha\cos\beta-\sin\beta\cos\alpha=\sin\alpha\sin\beta-\sin\alpha\cos\beta$

$\cos\alpha(\cos\beta-\sin\beta)=\sin\alpha(\sin\beta-\cos\beta)$

$\Rightarrow\ \cos\alpha=-\sin\alpha$

$\tan\alpha=-1$

$(\cos\beta\neq\sin\beta)$ as $\tan\beta=\dfrac{1}{2009}$

$$\tan 3\alpha=\frac{3\tan\alpha-\tan^3\alpha}{1-3\tan^2\alpha}=\frac{3.(-1)-(-1)^3}{1-3(-1)^2}$$

$$=\frac{-3+1}{1-3}$$

$$=\frac{-2}{-2}=1$$

45. (*c*) We have $\tan 9°-\tan 27°-\tan 63°+\tan 81°$

$=(\tan 9°+\tan 81°)-(\tan 27°+\tan 63°)$

$$=\frac{1}{\sin 9°\cos 9°}-\frac{1}{\sin 27°\cos 27°}$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ}$$

$$= 2\left[\frac{\sin 54^\circ - \sin 18^\circ}{\sin 54^\circ \sin 18^\circ}\right]$$

$$= 2\left[\frac{2\cos 36^\circ \sin 18^\circ}{\sin 18^\circ \cos 36^\circ}\right] = 4$$

46. (a) $\frac{1}{2}\left[\frac{2\sin 0\cos 0}{\cos 0\cos 30} + \frac{2\sin 30\cos 30}{\cos 90\cos 30} + \frac{2\sin 90\cos 90}{\cos 90\cos 270} + \frac{2\sin 270\cos 270}{\cos 270\cos 810}\right]$

$$= \frac{1}{2}\left[\frac{\sin(30-0)}{\cos 0\cos 30} + \frac{\sin(90-30)}{\cos 30\cos 90} + \frac{\sin(270-90)}{\cos 90\cos 270} + \frac{\sin(810-270)}{\cos 270\cos 810}\right]$$

$$= \frac{1}{2}[\tan 810 - \tan 0] = \frac{1}{2}\left[\frac{\sin 800}{\cos 0\cos 810}\right]$$

47. (b) $\cot(A+B) = \frac{\cot A \cdot \cot B - 1}{\cot A + \cot B}$

$\cot(60) = \cot(16+44)$

$\cot(16^\circ)\cot(44^\circ) = \frac{\cot(16^\circ)+\cot(44^\circ)}{\sqrt{3}} + 1$ ...(i)

$\cot(44^\circ)\cot(76^\circ) = \frac{\cot(44^\circ)+\cot(76^\circ)}{-\sqrt{3}}$ ...(ii)

$\cot(76^\circ)\cot(16^\circ) = \frac{\cot(16^\circ)-\cot(76^\circ)}{\sqrt{3}} - 1$ ...(iii)

(i) + (ii) − (iii)

$\cot(16^\circ)\cot(44^\circ) + \cot(76^\circ) - \cot(76^\circ)$

$\cot(16^\circ) = 3$

48. (d) $a_1 + a_2\sin x + a_3\cos x + a_4\sin 2x + a_5\cos 2x = 0$

Put $x = 0$

$a_1 + a_3 + a_5 = 0$

put $x = \pi$

$a_1 - a_3 + a_5 = 0$

$\Rightarrow a_3 = 0$ & $a_1 + a_5 = 0$

put $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$

$a_1 + a_2 - a_5 = 0$

$a_1 - a_2 + a_5 = 0$

$\Rightarrow a_2 = 0$ & $a_1 - a_5 = 0$

$\boxed{a_1 = a_2 = a_3 = a_5 = 0}$

$\Rightarrow a_4 \sin 2x = 0$ for all x

$\Rightarrow a_4 = 0$

$(a_1, a_2, a_3, a_4, a_5) = (0, 0, 0, 0, 0)$

49. (c) Multiplying and dividing by $2\sin\theta$

$$\Rightarrow 1 + \frac{1}{2\sin\theta}(2\sin\theta\cos 2\theta + 2\cos 4\theta\sin\theta + 2\cos 6\theta\sin\theta + 2\cos 8\theta\sin\theta + 2\cos 10\theta\sin\theta)$$

$$\Rightarrow 1 + \frac{1}{2\sin\theta}(\sin 3\theta - \sin\theta + \sin 5\theta - \sin 3\theta + \sin 7\theta - \sin 5\theta + \sin 9\theta - \sin 7\theta + \sin 11\theta - \sin 9\theta)$$

$(\sin 2\theta = 2\sin\theta\cos\theta)$

$$\Rightarrow 1 + \frac{1}{2\sin\theta}(\sin 11\theta - \sin\theta)$$

$$\Rightarrow \frac{2\sin\theta + \sin 11\theta - \sin\theta}{2\sin\theta}$$

$$\Rightarrow \frac{\sin 11\theta + \sin\theta}{2\sin\theta} \Rightarrow \frac{(2\cos 5\theta\sin 6\theta)}{2\sin\theta}$$

50. (c) $\log[\sec(2022\pi) + \sec(2023\pi) + \cos(12345678\pi)]$

$= \log[1 - 1 + 1] = \log 1 = 0$

$a + b + c + d + e + f + g$

$= 0 + 0 + 0 - 1 + 1 + 1 - 1 + 1 - 1 = 0$

51. (c) $a + b + c + d + e + f + g$

$= 0 + 0 + 0 - 1 + 1 + 1 - 1 = 0$

Parikshit (JEE Advanced Level)

1. (a) By the given conditions

$\tan\frac{\pi}{9} + \tan\frac{5\pi}{18} = 2x$

$\tan\frac{\pi}{9} + \tan\frac{7\pi}{18} = 2y$

$\Rightarrow 2x = \tan 20^\circ + \tan 50^\circ$

$$= \frac{\sin 20^\circ}{\cos 20^\circ} + \frac{\sin 50^\circ}{\cos 50^\circ}$$

$$= \frac{\sin 20^\circ\cos 50^\circ + \cos 20^\circ\sin 50^\circ}{\cos 20^\circ\cos 50^\circ}$$

$$= \frac{\sin 70^\circ}{\cos 20^\circ\cos 50^\circ}$$

$$= \frac{\cos 20^\circ}{\cos 20^\circ\cos 50^\circ} = \frac{1}{\cos 50^\circ} = \frac{1}{\sin 40^\circ}$$

$= \text{cosec } 40^\circ$

$2y = \tan 20^\circ + \tan 70^\circ$

$$= \frac{\sin 20^\circ}{\cos 20^\circ} + \frac{\sin 70^\circ}{\cos 70^\circ} = \frac{\sin 90^\circ}{\cos 20^\circ\cos 70^\circ}$$

$$= \frac{1}{\cos 20^\circ\cos 70^\circ} = \frac{1}{\cos 20^\circ\sin 20^\circ}$$

$$= \frac{2}{2\sin 20^\circ\cos 20^\circ} = \frac{2}{\sin 40^\circ} = 2\text{cosec}40^\circ$$

$\therefore 2y = 2(2x) \Rightarrow y = 2x$

2. (d) Let $y = \frac{\sqrt{3}\sin(\alpha+\beta) - \frac{2}{\cos\frac{\pi}{6}}\cos(\alpha+\beta)}{\sin\alpha}$

$$= \frac{\sqrt{3}\sin(\alpha+\beta) - \frac{4}{\sqrt{3}}\cos(\alpha+\beta)}{\sin\alpha}$$

$$= \frac{3\sin(\alpha+\beta) - 4\cos(\alpha+\beta)}{\sqrt{3}\sin\alpha}$$

Case I :

If β lies in I quadrant i.e. $\tan\beta > 0$, then

$$y = \frac{3\sin\alpha\left(\frac{3}{5}\right) + 3\cos\alpha\left(\frac{4}{5}\right) - 4\cos\alpha\left(\frac{3}{5}\right) + 4\left(\frac{4}{5}\right)}{\sqrt{3}\sin\alpha}$$

$$= \frac{5\sin\alpha}{\sqrt{3}\sin\alpha} = \frac{5}{\sqrt{3}}$$

Case II :

If β lies in II quadrant, i.e. $\tan\beta < 0$ then

$$y = \left(\frac{\frac{7}{5}\sin\alpha + \frac{24}{5}\cos\alpha}{\sqrt{3}\sin\alpha}\right) = \frac{\sqrt{3}(7 + 24\cot\alpha)}{15}.$$

3. (c) $S = \sum_{x=2}^{44}(2\sin x\sin 1)[1 + \sec(x-1)\sec(x+1)]$

$$= \sum_{x=2}^{44}[\cos(x-1) - \cos(x+1)](1 + \sec(x-1)\sec(x+1))$$

$$= \sum_{X=2}^{44}[\cos(x-1) - \cos(x+1)] + \sum_{r=2}^{44}[\sec(x+1) - \sec(x-1)]$$

$= \cos 1 + \cos 2 - \cos 44 - \cos 45^\circ - [\sec 1 + \sec 2 - \sec 44 - \sec 45^\circ]$

$$= \left[\cos 1 - \frac{1}{\cos 1}\right] + \left(\left(\cos 2 - \frac{1}{\cos 2}\right) + \left(\frac{1}{\cos 44} - \cos 44\right) + \left(\frac{1}{\cos 45^\circ} - \cos 45^\circ\right)\right)$$

$$= \frac{-\sin^2 1}{\cos 1} + \frac{\sin^2 44}{\cos 44} - \frac{\sin^2 2}{\cos 2} + \frac{\sin^2 45^\circ}{\cos 45^\circ}$$

$$s = \sum_{n=1}^{4}(-1)^n\frac{\sin^2\theta_n}{\cos\theta_n}$$

$\theta_1 = 1 \quad \theta_2 = 44 \quad \theta_3 = 2 \quad \theta_4 = 45^\circ$

$|1 + 2 - 44 - 45| = |-86^\circ| = 86^\circ$

4. (c) We know that

$\sqrt{3} + \tan(30^\circ - x)$

$$= \sqrt{3} + \frac{\tan 30^\circ - \tan x}{1 + \tan 30^\circ\cdot\tan x}$$

$$= \sqrt{3} + \frac{1 - \sqrt{3}\tan x}{\sqrt{3} + \tan x} = \frac{4}{\sqrt{3} + \tan x}$$

Now,

$$\left(\sqrt{3} + \tan 1^\circ\right)\left(\sqrt{3} + \tan 29^\circ\right)$$

$$= \frac{4}{\sqrt{3} + \tan 29^\circ} \times \left(\sqrt{3} + \tan 29^\circ\right) = 4$$

Similarly

$\left(\sqrt{3} + \tan 2^\circ\right)\left(\sqrt{3} + \tan 28^\circ\right) = 4$

So,

$\left(\sqrt{3} + \tan 1^\circ\right)\left(\sqrt{3} + \tan 2^\circ\right)\left(\sqrt{3} + \tan 3^\circ\right)\cdots(\sqrt{3} + \tan 2$

$= 4^{14}\cdot\left(\sqrt{3} + \tan 15^\circ\right)$

$$= 4^{14}\cdot\left(\sqrt{3}+\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$$

$$= 4^{14}\cdot\left(\frac{2+2\sqrt{3}}{\sqrt{3}+1}\right) = 2^{29}$$

5. (*d*) $x^4 - x^3\sin 2\beta + x^2\cos 2\beta - x\cos\beta - \sin\beta = 0$

$$\tan(\theta_1+\theta_2+\theta_3+\theta_4) = \frac{S_1 - S_3}{1 - S_2 + S_4}$$

$$= \frac{\sin 2\beta - \cos\beta}{1 - \cos 2\beta - \sin\beta}$$

$$= \frac{(2\sin\beta - 1)\cos\beta}{2\sin^2\beta - \sin\beta} = \frac{\cos\beta}{\sin\beta} = \cot\beta$$

6. (*d*) Take $\alpha = \beta = \gamma = \frac{-\pi}{4}$

Then $12\sin^2(\alpha-\beta) + 15\sin^2(\beta-\gamma) - 7\sin^2(\gamma-\alpha)$

$= 12\sin^2(0) + 15\sin^2(0) - 7\sin^2(0) = 0$

7. (*a,b*) Let $\cos 4\theta = \frac{1}{3}$

$$\Rightarrow 2\cos^2(2\theta) - 1 = \frac{1}{3}$$

$$\Rightarrow 2\cos^2(2\theta) = 1 + \frac{1}{3} = \frac{4}{3}$$

$$\Rightarrow \cos^2(2\theta) = \frac{2}{3} \Rightarrow \cos(2\theta) = \pm\sqrt{\frac{2}{3}}$$

$$\Rightarrow 2\cos^2\theta - 1 = \pm\sqrt{\frac{2}{3}}$$

$$\Rightarrow 2\cos^2\theta = 1 \pm \sqrt{\frac{2}{3}}$$

Now,

$$f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta} = \frac{2\cos^2\theta}{2\cos^2\theta - 1}$$

$$\Rightarrow f\left(\frac{1}{3}\right) = \frac{2\left(1 \pm \sqrt{\frac{2}{3}}\right)}{2\left(1 \pm \sqrt{\frac{2}{3}}\right) - 1} \Rightarrow = 1 \pm \sqrt{\frac{3}{2}}$$

8. (*b,d*)

(A) $\tan 1 > 1$ and $\sin 1 < 1$, then $\log_{\sin 1}\tan 1 < 0$

(B) $1 + \tan 3 < 1$ and $\cos 1 < 1$, then $\log_{\cos 1}(1 + \tan 3) > 0$

(C) $\cos\theta + \sec\theta > 2$ and $\log_{10} 5 < 1$, then $\log_{\log_{10}5}(\cos\theta + \sec\theta) < 0$

(D) $2\sin 18° < 1$ and $\tan 15° < 1$, then $\log_{\tan 15°} 2\sin 18° > 0$

9. (*a,b,c,d*)

$$P(k) = \left(1+\cos\frac{\pi}{4k}\right)\left(1+\cos\left(\frac{\pi}{2}-\frac{\pi}{4k}\right)\right)\left(1+\cos\left(\frac{\pi}{2}+\frac{\pi}{4k}\right)\right)\left(1+\cos\left(\pi-\frac{\pi}{4k}\right)\right)$$

$$\left(1+\cos\frac{\pi}{4k}\right)\left(1+\sin\frac{\pi}{4k}\right)\left(1-\sin\frac{\pi}{4k}\right)\left(1-\cos\frac{\pi}{4k}\right)$$

$$= \left(1-\cos^2\frac{\pi}{4k}\right)\left(1-\sin^2\frac{\pi}{4k}\right) = \frac{4\sin^2\frac{\pi}{4k}\cdot\cos^2\frac{\pi}{4k}}{4}$$

$$P(k) = \frac{1}{4}\sin^2\left(\frac{\pi}{2k}\right) \Rightarrow P(3) = \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}$$

$$\Rightarrow P(4) = \frac{\pi}{4}\sin^2\frac{\pi}{2k} = \frac{1}{4}\sin^2\frac{\pi}{8}$$

$$= \frac{1}{8}\left(1-\cos\frac{\pi}{4}\right) = \frac{2-\sqrt{2}}{16}$$

$$\Rightarrow P(5) = \frac{1}{4}\sin^2\frac{\pi}{10} = \frac{1}{8}\left(2\sin^2\frac{\pi}{10}\right)$$

$$= \frac{1}{8}(1-\cos 36°)$$

$$= \frac{1}{8}\left(1-\frac{\sqrt{5}+1}{4}\right) = \frac{3-\sqrt{5}}{32}$$

$$\Rightarrow P(6) = \frac{1}{4}\sin^2\frac{\pi}{12} = \frac{1}{8}\left(2\sin^2\frac{\pi}{12}\right)$$

$$= \frac{1}{8}\left(1-\cos\frac{\pi}{6}\right)$$

$$= \frac{1}{8}\left(1-\frac{\sqrt{3}}{2}\right) = \frac{2-\sqrt{3}}{16}$$

10. (*b, c*)

$y = a\cos^2 x + 2b\sin x\cos x + c\sin^2 x$ & $\tan x = \frac{2b}{a-c}$

$z = a\sin^2 x - 2b\sin x\cos x + c\cos^2 x$

$\Rightarrow y + z = a + c$

and $y - z = (a-c)(\cos^2 x - \sin^2 x) + 4b\sin x\cos x$

$= (a-c)\cos 2x + 2b\sin 2x$ ($\because 2b = (a-c)\tan x$)

$= (a-c)[\cos 2x + \tan x.\sin 2x] = (a-c)\left[2\cos 2x + \frac{\sin x}{\cos x}\sin 2x\right] = \frac{(a-c)\cos(2x-x)}{\cos x}$

$= (a-c)$.

11. (*b, c*)

$$\left\{\frac{2\cos\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right).\sin\left(\frac{A-B}{2}\right)}\right\}^n + \left\{\frac{2\sin\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right)}{-2\sin\left(\frac{A+B}{2}\right).\sin\left(\frac{A-B}{2}\right)}\right\}^n$$

$$= \cot^n\left(\frac{A-B}{2}\right) + (-1)^n\cot^n\left(\frac{A-B}{2}\right)$$

$$= \begin{cases} 0; & n \in \text{odd} \\ 2\cot^n\left(\frac{A-B}{2}\right); & n \in \text{even} \end{cases}$$

12. (*a, b*)

$\tan A + \tan B + \tan C = 6$, $\tan A\tan B = 2$

In any ΔABC,

$\tan A + \tan B + \tan C = \tan A\tan B\tan C$

$\Rightarrow 6 = 2\tan C \Rightarrow \tan C = 3$

$\therefore \tan A + \tan B + 3 = 6$

$\Rightarrow \tan A + \tan B = 3$ & $\tan A\tan B = 2$

Now $(\tan A - \tan B)^2 = (\tan A + \tan B)^2 - 4\tan A\tan B = 9 - 8 = 1$

$\Rightarrow \tan A - \tan B = \pm 1$

$\therefore \tan A - \tan B = 1$ or $\tan A - \tan B = -1$

$\tan A + \tan B = 3$ $\tan A + \tan B = 3$

on solving on solving

$\tan A = 2$ $\tan A = 1$

$\tan B = 1$ $\tan B = 2$

13. (*c*) Given $\cos\alpha + \cos\beta = a$

$$\Rightarrow 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = a \quad \ldots(i)$$

and $\sin\alpha + \sin\beta = b$

$$\Rightarrow 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = b \quad \ldots(ii)$$

by (i) & (ii)

$$\tan\left(\frac{\alpha+\beta}{2}\right) = \frac{b}{a} \Rightarrow \tan\theta = \frac{b}{a}$$

$\therefore \sin 2\theta + \cos 2\theta$

$$= \frac{2\left(\frac{b}{a}\right)}{1+\frac{b^2}{a^2}} + \frac{1-\frac{b^2}{a^2}}{1+\frac{b^2}{a^2}} = \frac{2ab}{a^2+b^2} + \frac{a^2-b^2}{a^2+b^2}$$

$$= \frac{a^2+b^2-2b^2+2ab}{a^2+b^2} = 1 + \frac{2b(a-b)}{a^2+b^2}$$

$\therefore n = 2$

14. (*a*) $\sin^n A = x \Rightarrow \sin^2 A = x$

$\therefore \sin A\sin 2A\sin 3A\sin 4A$

$= \sin A(2\sin A\cos A)(3\sin A - 4\sin^3 A)(4\sin A\cos A(1 - 2\sin^2 A))$

$= 8\sin^4 A(1 - \sin^2 A)(1 - 2\sin^2 A(3 - 4\sin^2 A)$

If we put $\sin^2 A = x$, then given expression is a polynomial of degree 5 in x.

15. (*b*) If $p = 5$

$\therefore \sin x + (p-5), \cos x, \tan x$

$\Rightarrow \sin x\cos x, \tan x$ are in G.P.

$\therefore \cos^2 x = \sin x\tan x$

$\cos^3 x = \sin^2 x$

$\therefore \cos^3 x = 1 - \cos^2 x$

$\therefore \cos^3 x + \cos^2 x = 1$

taking cube both sides

$\therefore \cos^9 x + \cos^6 x + 3\cos^5 x = 1$

$\Rightarrow \cos^9 x + \cos^6 x + 3\cos^5 x - 1 = 0$

16. (*b*) A → q; B → r; C → p; D → q

(A) $\Rightarrow \dfrac{\sin 22^\circ}{\sin 56^\circ \cos 56^\circ \dfrac{\sin 22^\circ}{\cos 22^\circ}}$

$\Rightarrow \dfrac{\cos 22^\circ}{\frac{1}{2}\times 2\sin 56^\circ \cos 56^\circ}$

$\Rightarrow \dfrac{2\cos 22^\circ}{\sin(2\times 56^\circ)} \Rightarrow \dfrac{2\cos 22^\circ}{\sin(90^\circ+22^\circ)}$

$\Rightarrow \dfrac{2\cos 22^\circ}{\cos 22^\circ}$

(B) $\left(\cos 65^\circ + \sqrt{3}\sin 5^\circ + \cos 5^\circ\right)^2$

$= \left(\sqrt{3}\sin 5^\circ + 2\cos 35^\circ \cos 30^\circ\right)^2$

$= \left(\sqrt{3}\sin 5^\circ + \sqrt{3}\cos 35^\circ\right)^2$

$= 3\left(\sin 5^\circ + \sin 55^\circ\right)^2$

$= 3\left(2\sin 30^\circ \cos 25^\circ\right)^2$

$= 3\cos^2 25^\circ$

(C) $f(\theta) = \dfrac{1}{4}[4\sin 2\theta - 1 + \cos 2\theta + 3(1+\cos 2\theta)]$

$= \dfrac{1}{4}[2 + 4\sin 2\theta + 4\cos 2\theta]$

$= 1 + 2(\sin 2\theta + \cos 2\theta)$

$y_{max} = 1 + 2\sqrt{2}, y_{min} = 1 - 2\sqrt{2}$

(D) $f(\theta) = \sqrt{2}\sin\left(\dfrac{\pi}{4} + \sin\theta\right)$

17. (*a*) $a\cos\theta + b\sin\theta = c$

$a\cos\theta = c - b\sin\theta$

on squaring

$a^2\cos^2\theta = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$a^2(1-\sin^2\theta) = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$a^2 - a^2\sin^2\theta = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$(a^2+b^2)\sin^2\theta - 2bc\sin\theta + c^2 - a^2 = 0$

$\sin\alpha + \sin\beta = \dfrac{2bc}{a^2+b^2}$,

$\sin\alpha\sin\beta = \dfrac{c^2-a^2}{a^2+b^2}$

$a\cos\theta + b\sin\theta = c$

$a\left(\dfrac{1-\tan^2(\theta/2)}{1+\tan^2\left(\frac{\theta}{2}\right)}\right) + b\left(\dfrac{2\tan\left(\frac{\theta}{2}\right)}{1+\tan^2\left(\frac{\theta}{2}\right)}\right) = c$

$a - a\tan^2\left(\dfrac{\theta}{2}\right) + 2b\tan\left(\dfrac{\theta}{2}\right) = C + C\tan^2\left(\dfrac{\theta}{2}\right)$

$(a+c)\tan^2\left(\dfrac{\theta}{2}\right) - 2b\tan\left(\dfrac{\theta}{2}\right) + (c-a) = 0$

$\tan\left(\dfrac{\alpha}{2}\right) + \tan\left(\dfrac{B}{2}\right) = \dfrac{2b}{a+c}$

$\cdot \tan\left(\dfrac{\alpha}{2}\right)\tan\left(\dfrac{\beta}{2}\right) = \dfrac{c-a}{a+c}$

18. (*a*) (A) $\cot 35^\circ + \cot 145^\circ + 3$

$= \cot 35^\circ - \cot 35^\circ + 3 = 3$

(B) $\dfrac{1}{\sec^2 120^\circ} + \dfrac{1}{\operatorname{cosec}^2 120^\circ}$

$= \sin^2 120^\circ + \cos^2 120^\circ = 1$

(C) $(\sin^2)^x + (\cos 2)^x = 1 \Rightarrow x = 2$

(D) $\tan\left(\dfrac{\pi}{\pi}\right) + \tan\left(\dfrac{2\pi}{11}\right) + \tan\left(\dfrac{4\pi}{\pi}\right) + \tan\left(\dfrac{9\pi}{11}\right)\tan\left(\dfrac{7\pi}{11}\right) + \tan\left(\dfrac{10\pi}{11}\right)$

$= \tan\left(\dfrac{\pi}{\pi}\right) + \tan\left(\dfrac{2\pi}{\pi}\right) + \tan\left(\dfrac{4\pi}{\pi}\right) + \tan\left(\pi - \dfrac{2\pi}{\pi}\right) + \tan\left(\pi - \dfrac{4\pi}{\pi}\right) + \tan\left(\pi - \dfrac{\pi}{\pi}\right)$

$= \tan\left(\dfrac{\pi}{11}\right) + \tan\left(\dfrac{2\pi}{11}\right) + \tan\left(\dfrac{4\pi}{11}\right) - \tan\left(\dfrac{2\pi}{11}\right) - \tan\left(\dfrac{4\pi}{11}\right) - \tan\left(\dfrac{\pi}{11}\right) = 0$

(E) $\cos 10^\circ + \cos 20^\circ + \cos 30^\circ + \ldots + \cos 80^\circ + \cos 100^\circ + \ldots + \cos 150^\circ + \cos 160^\circ + \cos 170^\circ + 4$

$= \cos 10^\circ + \cos 20^\circ + \ldots + \cos 80^\circ + \cos(180^\circ - 100^\circ) + \ldots + \cos(180^\circ - 20^\circ) + \cos(180^\circ - 10^\circ) + 4$

$\cos 10^\circ + \cos 20^\circ + \cdots + \cos 80^\circ - \cos 80^\circ - \cos 70^\circ - \ldots - \cos 10^\circ + 4 = 4$

19. [1] We have $\operatorname{cosec}\theta - \sin\theta = a^3$

$\Rightarrow \cos^2\theta = a^3\sin\theta$ & $\sec\theta - \cos\theta = b^3$

$\Rightarrow \sin^2\theta = b^3\cos\theta$ on simplifying

$\Rightarrow a^2b^2(a^2+b^2) = 1$

20. [7] We have, $\tan(55^\circ)\tan(65^\circ)\tan(75^\circ)$

$= \dfrac{1}{\tan(5^\circ)} \times [\tan(55^\circ)\tan(5^\circ)\tan(65^\circ)] \times \tan(75^\circ)$

$= \dfrac{1}{\tan(5^\circ)} \times \tan(15^\circ) \times \tan(75^\circ)$

$= \dfrac{1}{\tan(5^\circ)} \times \tan(15^\circ) \times \cot(15^\circ) = \dfrac{1}{\tan(5^\circ)} = \cot(5^\circ)$

Thus, $x = 5$

Also, $\tan(27^\circ) + \tan(18^\circ) + \tan(27^\circ)\tan(18^\circ)$

$= \tan(27^\circ + 18^\circ) = \tan(45^\circ) = 1$

So, $m = 1$

Hence, the value of $(m + x + 1)$ is 7.

21. [2] $a^2\sec^2 190^\circ = c^2\tan^2 190^\circ + d^2 + 2cd\tan 190^\circ$

$b^2\sec^2 190^\circ = c^2 + d^2\tan^2 190^\circ - 2cd\tan 190^\circ$

$\Rightarrow a^2 + b^2 = c^2 + d^2$

$(a\sec 190^\circ - c\tan 190^\circ)^2 = d^2$

$(b\sec 190^\circ + d\tan 190^\circ)^2 = c^2$

$\Rightarrow (c^2+d^2)(\sec^2 190^\circ + \tan^2 190^\circ) + (2bd - 2ac)\sec 190^\circ \tan 190^\circ = c^2 + d^2$

$\Rightarrow (c^2+d^2)(2\tan^2 190^\circ) = (2ac - 2bd)\sec 190^\circ \tan 190^\circ$

$\Rightarrow \dfrac{2(c^2+d^2)}{ac-bd} = \dfrac{2\sec 190^\circ}{\tan 190^\circ} = \dfrac{2}{\sin 190^\circ} = \dfrac{-2}{\sin 10^\circ}$

Therefore, $\left(\dfrac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 10^\circ = 2$

22. [27] $P = \sec^6\theta - \tan^6\theta - 3\sec^2\theta\tan^2\theta$

$= \left(\sec^2\theta - \tan^2\theta\right)^3 + 3\sec^2\theta\tan^2\theta\left(\sec^2\theta - \tan^2\theta\right) - 3\sec^2\theta\tan^2\theta$

$= 1 + 3 - 3 = 1$

similarly $Q = R = 1$

$(P+Q+R)^{(P+Q+R)} = (3)^3 = 27$

23. [1] $\dfrac{\sin\beta}{\sin(2\alpha+\beta)} = \dfrac{n}{m}$

$\dfrac{\sin(2\alpha+\beta)}{\sin\beta} = \dfrac{m}{n}$

$\dfrac{\sin\beta}{\sin(2\alpha+\beta)+\sin\beta} = \dfrac{m+n}{m-n}$

$\dfrac{2\sin(\alpha+\beta)\cos(\alpha)}{2\sin(\alpha)\cos(\alpha+\beta)} = \dfrac{m+n}{m-n}$

$\dfrac{\tan(\alpha+\beta)}{\tan\alpha} = \dfrac{m+n}{m-n}$

$\dfrac{\tan\alpha + \tan\beta}{1-\tan\alpha\cdot\tan\beta}\cdot\dfrac{1}{\tan\alpha} = \dfrac{m+n}{m-n}$

$1 + \dfrac{\tan\beta}{\tan\alpha} = \dfrac{1-\tan\alpha\tan\beta}{m-n}$

$\dfrac{1-\tan\alpha\tan\beta}{m+n} = \dfrac{1-\tan\alpha\tan\beta}{m-n}$

24. [1] $\displaystyle\sum_{n=0}^{88} \frac{1}{\cos k\cdot\cos(n+1)k}$

$\left[\dfrac{1}{\cos k\cdot\cos k} + \dfrac{1}{\cos k\cdot\cos 2k} + \ldots + \dfrac{1}{\cos 88k\cdot\cos 8}\right]$

Now $\dfrac{1}{\cos nk\cdot\cos(n+1)k}$

$= \dfrac{1}{\sin k}\left[\dfrac{\sin k}{\cos nk\cdot\cos(n+1)k}\right]$

$= \dfrac{1}{\sin k}\left[\dfrac{\sin[(n+1)k - nk]}{\cos nk\cos(n+1)k}\right]$

$= \dfrac{1}{\sin k}\left[\dfrac{\sin(n+1)k\cos nk - \cos(n+1)k\sin nk}{\cos nk\cos(n+1)k}\right]$

$= \dfrac{1}{\sin x}[\tan(n+1)k - \tan nk]$.

$$\therefore \frac{1}{\cos ok \cdot \cos k} = (\tan k - \tan ok) \times \frac{1}{\sin k}$$

$$= \frac{1}{\cos k \cos 2k} = (\tan 2k - \tan k) \times \frac{1}{\sin k}$$

$$\frac{1}{\cos 88k \cos 89k} = (\tan 89k - \tan 88k) \times \frac{1}{\sin k}$$

$$\sum_{n=0}^{88} \frac{1}{\cos k \cdot \cos(n+1)k} = \tan 89k \times \frac{1}{\sin k}$$

$\cos k = 1^\circ$

$$\therefore \text{sum} = \tan 89^\circ \times \frac{1}{\sin 1^\circ} = \cot 1^\circ \times \frac{1}{\sin 1^\circ}$$

$$= \frac{\cos 1^\circ}{(\sin 1^\circ)^2}$$

$$= \frac{\cos k}{(\sin k)^2} = \text{RHS}$$

25. **[1]** $\alpha+\beta+\gamma = \pi$

$\beta+\gamma-\alpha = \pi-2\alpha$

$\gamma+\alpha-\beta = \pi-2\beta$

$\alpha+\beta-\gamma = \pi-2\gamma$

$$\tan\left(\frac{\pi}{4}-\frac{\alpha}{2}\right)\tan\left(\frac{\pi}{4}-\frac{\beta}{2}\right)\tan\left(\frac{\pi}{4}-\frac{\gamma}{2}\right) = 1$$

$$\frac{1-\tan\frac{\alpha}{2}}{1+\tan\frac{\alpha}{2}} \cdot \frac{1-\tan\frac{\beta}{2}}{1+\tan\frac{\beta}{2}} \frac{1-\tan\frac{\gamma}{2}}{1+\tan\frac{\gamma}{2}}$$

$$= \left(1-\tan\frac{\alpha}{2}\right)\left(1-\tan\frac{\beta}{2}\right)\left(1-\tan\frac{\gamma}{2}\right)$$

$$= \left(1+\tan\frac{\alpha}{2}\right)\left(1+\tan\frac{\beta}{2}\right)\left(1+\tan\frac{\gamma}{2}\right)$$

$$1-\sum\tan\frac{\alpha}{2}+\sum\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$$

$$-\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$$

$$= 1+\sum\tan\frac{\alpha}{2}+\sum\tan\frac{\alpha}{2}\tan\frac{\beta}{2} 2+\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$$

$$2\left(\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2}+\tan\frac{\gamma}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}\right)$$

$$\frac{\sin\frac{\alpha}{2}}{\cos\frac{\alpha}{2}}+\frac{\sin\frac{\beta}{2}}{\cos\frac{\beta}{2}}+\frac{\sin\left(\frac{\gamma}{2}\right)}{\cos\left(\frac{\gamma}{2}\right)}+\frac{\sin\frac{\alpha}{2}\text{San}\frac{\beta}{2}\sin\frac{\gamma}{2}}{\cos\frac{\alpha}{2}\cos\frac{\beta}{2}\cos\frac{\gamma}{2}}=0$$

$$\left(\sin\frac{\alpha}{2}\cos\frac{\beta}{2}\cos\frac{\gamma}{2}+\sin\frac{\beta}{2}\cos\frac{\alpha}{2}\cos\frac{\gamma}{2}\right)$$

$$+\left(\sin\frac{\gamma}{2}\cos\frac{\alpha}{2}\text{con}\frac{\beta}{2}+\sin\frac{\alpha}{2}\sin\frac{\beta}{2}\sin\frac{\gamma}{2}\right)=0$$

$$\cos\frac{\gamma}{2}\left[\sin\frac{\alpha}{2}\cos\frac{\beta}{2}+\cos\frac{\alpha}{2}\sin\frac{\beta}{2}\right]$$

$$+\sin\frac{\gamma}{2}\left[\cos\frac{\alpha}{2}\cos\frac{\beta}{2}+\sin\frac{\alpha}{2}\sin\frac{\beta}{2}\right]=0$$

$$\cos\frac{\gamma}{2}\sin\left(\frac{\alpha+\beta}{2}\right)+\sin\left(\frac{\gamma}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)=0$$

$$\cos^2\left(\frac{\gamma}{2}\right)+\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)=0$$

$$\frac{1+\cos\gamma}{2}+\frac{1}{2}[\cos\alpha+\cos\beta]=0$$

$$\boxed{\cos\alpha+\cos\beta+\cos\gamma+1=0}$$

26. **[4]** Given expression

$\sin(S-A)+\sin(S-B)+\sin(S-C)-\sin S$

using $\sin C + \sin D$ formula using $\sin C - \sin D$ formula

$$2\sin\frac{(S-A+S-B)}{2}\cdot\cos\frac{(S-A-S+B)}{2}+$$

$$2\cos\frac{(S-C+S)}{2}\sin\frac{(S-C-S)}{2}$$

$$=2\sin\frac{(2S-A-B)}{2}\cdot\cos\frac{(-A+B)}{2}+$$

$$2\cos\frac{(2S-C)}{2}\cdot\sin\frac{(-C)}{2}$$

Substituting $2S = A+B+C$

$$=2\sin\left(\frac{(A+B+C-A-B)}{2}\right)\cdot\cos\left(\frac{(-A+B)}{2}\right)+$$

$$2\cos\left(\frac{(A+B+C-C)}{2}\right)\left\{-\sin\frac{C}{2}\right\}$$

$$=2\sin\frac{C}{2}\cdot\cos\left(\frac{B-A}{2}\right)-2\cos\left(\frac{A+B}{2}\right)\cdot\sin\frac{C}{2}$$

$$=2\sin\frac{C}{2}\left[\cos\left(\frac{B-A}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right]$$

Using $\cos C - \cos D$ formula

$$=2\sin\frac{C}{2}\left[2\sin\frac{\left(\frac{B-A}{2}+\frac{A+B}{2}\right)}{2}\cdot\right.$$

$$\left.\sin\frac{\left(\left(\frac{A+B}{2}-\left(\frac{B-A}{2}\right)\right)\right)}{2}\right]$$

$$=2\sin\frac{C}{2}\left[2\sin\frac{B}{2}\cdot\sin\frac{A}{2}\right]$$

$$=4\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2}$$

27. **[4]** $1+4\cos 5\cos(S-A)\cos(S-B)$

$$\cos(S-C)$$

$$=1+2\cos 5\cos(S-A)\quad 2\cos(5-B)\cos(5-1)$$

$$=1+(\cos(25-A)+\text{con}\,A)(\cos(2s-(B+C))$$

$$+\cos(B-C))$$

$$=1+(\cos(B+C)+\cos A)(\cos A+\cos(B-C))$$

$$=1+\cos A(\cos(B+C)+\cos(\theta-C))+\cos^2 B-$$

$$\sin^2 C+\cos^2 A$$

$$=2\cos A\cos B\cos C+\cos^2 B+(1-\sin^2 C)+$$

$$\cos^2 A$$

$$=\cos^2 A+\cos^2 B+\cos^2 C+2\cos A\cos B\cos C$$

28. **[4]** $\cos(A+B+C)+\cos(-A+B+C)+$

$\cos(A-B+C)+\cos(A+B-C)$

$$2\cos\left(\frac{-A+B+C+A-B+C}{2}\right)$$

$$\cos\left(\frac{-A+B+C-A+B-C}{2}\right)$$

$$+2\cos\left(\frac{A+B+C+A+B-C}{2}\right)$$

$$\cos\left(\frac{A+B+C+A-B+C}{2}\right)$$

$$\Rightarrow 2\cos C\cos(B-A)+2\cos(A+B)\cos C.$$

$$\Rightarrow 2\cos C[\cos(B-A)+\cos(A+B)]$$

$$\Rightarrow 2\cos C[2\cos B\cos A]$$

$$=4\cos A\cos B\cos C.$$

29. **[3]** $\tan(3A+3B+3C)=0$

$$\therefore \frac{S_1-S_3}{1-S_2}=0 \text{ so that } S_1=S_3$$

or $\tan 3A+\tan 3B+\tan 3C=\tan 3A\cdot\tan 3B\cdot 3C$

Now put

$$\tan 3A=\frac{3\tan A-\tan^3 A}{1-3\tan^2 A}=\frac{3x-x^3}{1-3x^2}$$

Similarly we can write for $\tan 3B$ and $\tan 3C$

$$\therefore \frac{3x-x^3}{1-3x^2}+\frac{3y-y^3}{1-3y^2}+\frac{3z-z^3}{1-3z^2}=\frac{3x-x^3}{1-3x^2}\cdot$$

$$\frac{3y-y^3}{1-3y^2}\cdot\frac{3z-z^3}{1-3z^2}$$

30. **[4]** $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}$

$$=2\sin\frac{A+B}{4}\cos\frac{A-B}{4}+\sin\frac{c}{2}$$

$$=2\sin\frac{A+B}{4}\cos\frac{A-B}{4}+1-2\sin^2\frac{A+B}{4}$$

$[\because\ A+B+C=\pi]$

$$\left[\frac{C}{2}=\frac{\pi}{2}-\left(\frac{A}{2}+\frac{B}{2}\right)\right]$$

$$\therefore\ 1+2\sin\left(\frac{\pi}{4}-\frac{C}{4}\right)\left[\cos\left(\frac{A-B}{4}\right)-\right.$$

$$\left.\cos\left(\frac{\pi}{2}-\left(\frac{A+B}{4}\right)\right.\right.$$

$$\Rightarrow 4+2\sin\left(\frac{\pi}{4}-\frac{C}{4}\right)2\sin\left(\frac{\pi}{4}-\frac{A}{4}\right)\sin\left(\frac{\pi}{4}-\frac{B}{4}\right)$$

$$\Rightarrow 1+4\sin\frac{\pi-C}{4}\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}$$

PYQ's (Past Year Questions)

1. *(a)* Given

$$\tan 15^\circ+\frac{1}{\tan 75}+\frac{1}{\tan 105}+\tan 195=2$$

$\Rightarrow \tan 15 + \frac{1}{\tan(90-15)} + \frac{1}{\tan(90+15)} + \tan(180+15) = 2a$

$\Rightarrow \tan 15 + \frac{1}{\cot 15} - \frac{1}{\cot 15} + \tan 15 = 2a$

$\Rightarrow 2\tan 15 = 2a \Rightarrow a = \tan 15 = 2 - \sqrt{3}$

$\therefore a + \frac{1}{a} = 2 - \sqrt{3} + \frac{1}{2-\sqrt{3}} = 4$

2. (*a*) $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{2^2\pi}{33}\cos\frac{2^3\pi}{33}\cos\frac{2^4\pi}{33}$

$\because \cos A \cos 2A \cos 2^2 A \cos 2^3 A \ldots \cos 2^{n-1} A = \frac{\sin(2^n A)}{2^n \sin A}$

Here, $A = \frac{\pi}{33}, n = 5$

$\frac{96\sin\left(2^5\frac{\pi}{33}\right)}{2^5\sin\left(\frac{\pi}{33}\right)} = \frac{96\sin\left(\frac{32\pi}{33}\right)}{32\sin\left(\frac{\pi}{33}\right)}$

$= \frac{3\sin\left(\pi - \frac{\pi}{33}\right)}{\sin\left(\frac{\pi}{33}\right)} = 3$

3. (*b*) Given, $16\sin 20^\circ \sin 40^\circ \sin 80^\circ$

$= 16\sin 40^\circ \sin 20^\circ \sin 80^\circ$

$4(4\sin(60-20)\sin(20)\sin(60+20))$

$= 4\sin(3\times 20^\circ)$

$[\because \sin 3\theta = 4\sin(60-\theta)\sin\theta\sin(60+\theta)]$

$= 4 \times \sin 60^\circ$

$= 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}$

4. (*b*) Given trigonometric equation is

$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}$

$= \left(\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)\right) + \cos\left(\frac{4\pi}{7}\right)$

$= \cos\left(\frac{4\pi}{7}\right)\left[2\cos\left(\frac{2\pi}{7}\right) + 1\right]$

$= \cos\left(\frac{4\pi}{7}\right)\left[2\left(1 - 2\sin^2\left(\frac{\pi}{7}\right)\right) + 1\right]$

$= \cos\left(\frac{4\pi}{7}\right)\left[3 - 4\sin^2\left(\frac{\pi}{7}\right)\right]$

$= \frac{\sin\left(3\times\frac{\pi}{7}\right)}{\sin\frac{\pi}{7}} \times \cos\left(\frac{4\pi}{7}\right)$

$= \frac{2\sin\left(\frac{3\pi}{7}\right)}{2\sin\frac{\pi}{7}} \times \cos\left(\frac{4\pi}{7}\right)$

Apply $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

$= \frac{\sin\left(\frac{7\pi}{7}\right) + \sin\left(\frac{-\pi}{7}\right)}{2\sin\frac{\pi}{7}}$, Here, $\sin(\pi) = 0$.

$= \frac{-\sin\frac{\pi}{7}}{2\sin\frac{\pi}{7}} = -\frac{1}{2}$

5. (*b*) Since,

$\sin\left(\frac{\pi}{22}\right) = \cos\left(\frac{\pi}{2} - \frac{\pi}{22}\right) = \cos\frac{5\pi}{11} = -\cos\frac{16\pi}{11}$

$\sin\frac{3\pi}{22} = \cos\frac{4\pi}{11}, \sin\frac{5\pi}{22} = \cos\frac{3\pi}{11} = -\cos\frac{8\pi}{11}$

$\sin\frac{7\pi}{22} = \cos\frac{2\pi}{11}, \sin\frac{9\pi}{11} = \cos\frac{\pi}{11}$

Now, $2\sin\frac{\pi}{22}.\sin\frac{3\pi}{22}.\sin\frac{5\pi}{22}.\sin\frac{7\pi}{22}.\sin\frac{9\pi}{22}$

$2\cos\frac{\pi}{11}.\cos\frac{2\pi}{11}.\cos\frac{4\pi}{11}.\cos\frac{8\pi}{11}\cos\frac{16\pi}{11}$

$\frac{2\times\sin 2^5\frac{\pi}{11}}{2^5\sin\frac{\pi}{11}} = \frac{1}{16}\left(\because \sin 2^5\frac{\pi}{11} = \sin\frac{\pi}{11}\right)$

6. (*a*) We are given that,

$\cot\alpha = 1, \sec\beta = \frac{-5}{3}$,

$\Rightarrow \cos\beta = -\frac{3}{5}, \Rightarrow \tan\beta = -\frac{4}{3}$, and $\tan\alpha = 1$

$\therefore \tan(\alpha+\beta) = \frac{1-\frac{4}{3}}{1+\frac{4}{3}\times 1} = \frac{-1}{7}$

$\tan(\alpha+\beta) = \frac{-1}{7}$ lies in IVth quadrant

$\because \left[\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\cdot\tan\beta}\right]$

7. (*d*) Given trigonometric equation is

$\sin 12^\circ + \sin 12^\circ - \sin 72^\circ$

$= \sin 12^\circ - 2\cos 42^\circ \sin 30^\circ$

$\left[\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)\right]$

$= \sin 12^\circ - \sin(90^\circ - 42^\circ) = \sin 12^\circ - \sin 48^\circ$

$= -2\cos 30^\circ \sin 18^\circ$

$= -2\times\frac{\sqrt{3}}{2}\times\frac{\sqrt{5}-1}{4} = \frac{\sqrt{3}}{4}\left(1-\sqrt{5}\right)$

8. **[80]** $(\sin 10^\circ \cdot \sin 50^\circ \cdot \sin 70^\circ)(\sin 10^\circ \cdot \sin 20^\circ \cdot \sin 40^\circ)$

$= \left(\frac{1}{4}\sin 30^\circ\right)\cdot\left[\frac{1}{2}\sin 10^\circ\left(\cos 20^\circ - \cos 60^\circ\right)\right]$

$= \frac{1}{16}\left[\sin 10^\circ\left(\cos 20^\circ - \frac{1}{2}\right)\right]$

$= \frac{1}{32}\left[2\sin 10^\circ\cdot\cos 20^\circ - \sin 10^\circ\right]$

$= \frac{1}{32}\left[\sin 30^\circ - \sin 10^\circ\right]$

$= \frac{1}{64} - \frac{1}{16}\sin 10^\circ$

Clearly $\alpha = \frac{1}{64}$

Hence $16 + \alpha^{-1} = 16 + 64 = 80$

9. **[1]** Rearrange the given expression

$\left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\alpha}{\sin\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\sin\beta}{\cos\alpha}\right)^2$

$= \left(\frac{\cos(\alpha-\beta)}{\sin\beta\cos\beta} + \frac{\cos(\alpha-\beta)}{\sin\alpha\cdot\cos\alpha}\right)^2$

$= \left(\frac{4}{3}\left\{\frac{1}{\sin 2\beta} + \frac{1}{\sin 2\alpha}\right\}\right)^2$

$= \frac{16}{9}\left[\frac{\sin 2\alpha + \sin 2\beta}{\sin 2\alpha\cdot\sin 2\beta}\right]$

$= \frac{64}{9}\left(\frac{2\sin(\alpha+\beta)\cdot\cos(\alpha-\beta)}{2\sin 2\alpha\cdot\sin 2\beta}\right)^2$

$= \frac{64}{9}\left(\frac{2\cdot\frac{1}{3}\cdot\frac{2}{3}}{\cos(2\alpha-2\beta) - \cos(2\alpha+2\beta)}\right)^2$

$= \frac{64}{9}\left(\frac{\frac{4}{9}}{2\cos^2(\alpha-\beta) - 1 - 1 + 2\sin^2(\alpha+\beta)}\right)^2$

$= \frac{64}{9}\left(\frac{\frac{4}{9}}{\frac{8}{9} - 2 + \frac{2}{9}}\right)^2 = \frac{64}{9}\left(\frac{1}{-2}\right)^2 = \left[\frac{16}{9}\right]$

$= [1.7] = 1$

10. (*a*) Given that $\cos x + \cos y - \cos(x+y) = \frac{3}{2}$

$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) - \left(2\cos^2\left(\frac{x+y}{2}\right) - 1\right) = \frac{3}{2}$

$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\left\{\cos\left(\frac{x-y}{2}\right) - \cos\left(\frac{x+y}{2}\right)\right\} = \frac{3}{2} - 1$

$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\cdot 2\sin\left(\frac{x}{2}\right)\sin\left(\frac{y}{2}\right) = \frac{1}{2}$

$\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x}{2}\right)\sin\left(\frac{y}{2}\right) = \frac{1}{8}$

As $0 < x, y < \pi$, possibility is $\frac{x}{2} = 30°$;

$\frac{y}{2} = 30°$

$\Rightarrow x = 60°; y = 60°$

Then, $\sin x + \cos y = \sin 60° + \cos 60° = \frac{1+\sqrt{3}}{2}$

11. (*b*) $\cot\theta = \frac{1+\cos 2\theta}{\sin 2\theta}$

Converting $\frac{\pi}{24}$ into degree i.e., $\frac{\pi}{24} \times \frac{180}{\pi} = \frac{15°}{2}$

$\Rightarrow \cot\left(\frac{15}{2}\right)^\circ = \frac{1+\cos 15°}{\sin 15°}$

Now, $\cos 15° = \cos(45° - 30°) = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Similarly, $\sin 15° = \frac{\sqrt{3}-1}{2\sqrt{2}}$

$\therefore \cot\left(\frac{15}{2}\right)^\circ = \frac{1+\frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{\sqrt{3}-1}{2\sqrt{2}}} = \frac{2\sqrt{2}+\sqrt{3}+1}{\sqrt{3}-1}$

$= \sqrt{6}+\sqrt{2}+\sqrt{3}+2$

12. (*b*)

13. (*d*) We have,

$2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$

$2\sin\frac{\pi}{8}\sin\frac{2\pi}{8}\sin\frac{3\pi}{8}\sin\left(\pi-\frac{5\pi}{8}\right)\sin\left(\pi-\frac{6\pi}{8}\right)\sin\left(\pi-\frac{7\pi}{8}\right)$

$= 2\sin\frac{\pi}{8}\sin\frac{2\pi}{8}\sin\frac{3\pi}{8}\sin\frac{3\pi}{8}\sin\frac{2\pi}{8}\sin\frac{\pi}{8}$

$= 2\sin^2\frac{\pi}{8}\sin^2\frac{2\pi}{8}\sin^2\frac{3\pi}{8}$

$= \left(\frac{1}{\sqrt{2}}\right)^2 \cdot 2\sin^2\frac{\pi}{8}\sin^2\frac{3\pi}{8}$

$= \frac{1}{4}\cdot\left(2\sin\frac{\pi}{8}\sin\frac{3\pi}{8}\right)^2$

$= \frac{1}{4}\left[\cos\frac{\pi}{4} - \cos\frac{\pi}{2}\right]^2 = \frac{1}{4}\times\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{8}$

14. (*b*) $L + M = 1 - 2\sin^2\frac{\pi}{8} = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ ...(*i*)

and $L - M = -\cos\frac{\pi}{8}$ ...(*ii*)

By (*i*) and (*ii*)

$L = \frac{1}{2}\left(\frac{1}{\sqrt{2}} - \cos\frac{\pi}{8}\right) = \frac{1}{2\sqrt{2}} - \frac{1}{2}\cos\frac{\pi}{8}$

and $M = \frac{1}{2}\left(\frac{1}{\sqrt{2}} + \cos\frac{\pi}{8}\right) = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\frac{\pi}{8}$

15. [1] $\frac{\sqrt{2}\sin\alpha}{\sqrt{2}\cos\alpha} = \frac{1}{7}$

and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$

$\Rightarrow \frac{\sqrt{2}\sin\beta}{\sqrt{2}} = \frac{1}{\sqrt{10}}$

$\tan\alpha = \frac{1}{7}$ and $\sin\beta = \frac{1}{\sqrt{10}}$; $\tan\beta = \frac{1}{3}$

Now, $\tan 2\beta = \frac{2\tan\beta}{1-\tan^2\beta} = \frac{2\cdot\frac{1}{3}}{1-\frac{1}{9}} = \frac{\frac{2}{3}}{\frac{8}{9}} = \frac{3}{4}$

So, $\tan(\alpha+2\beta) = \frac{\tan\alpha+\tan 2\beta}{1-\tan\alpha\tan 2\beta}$

$= \frac{\frac{1}{7}+\frac{3}{4}}{1-\frac{1}{7}\cdot\frac{3}{4}} = \frac{\frac{4+21}{28}}{\frac{25}{28}} = 1$

16. (*a*) $\cos^3\frac{\pi}{8}\left[4\cos^3\frac{\pi}{8} - 3\cos\frac{\pi}{8}\right] + \sin^3\frac{\pi}{8}\left[3\sin\frac{\pi}{8} - 4\sin^3\frac{\pi}{8}\right]$

$= 4\cos^6\frac{\pi}{8} - 4\sin^6\frac{\pi}{8} - 3\cos^4\frac{\pi}{8} + 3\sin^4\frac{\pi}{8}$

$= 4\left[\left(\cos^2\frac{\pi}{8} - \sin^2\frac{\pi}{8}\right)\right]\left[\left(\sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8} + \sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right)\right]$

$-3\left[\left(\cos^2\frac{\pi}{8} - \sin^2\frac{\pi}{8}\right)\right]$

$= \cos\frac{\pi}{4}\left[4\left(1-\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right) - 3\right]$

$= \frac{1}{\sqrt{2}}\left[1-\frac{1}{2}\right] = \frac{1}{2\sqrt{2}}$

17. (*d*) $0 < \alpha+\beta = \frac{\pi}{2}$ and $\frac{-\pi}{4} < \alpha-\beta < \frac{\pi}{4}$

if $\cos(\alpha+\beta) = \frac{3}{5}$ then $\tan(\alpha+\beta) = \frac{4}{3}$

and if

$\sin(\alpha-\beta) = \frac{5}{13}$ then $\tan(\alpha-\beta) = \frac{5}{12}$

(since $\alpha-\beta$ here lies in the first quadrant)

Now $\tan(2\alpha) = \tan\{(\alpha+\beta)+(\alpha-\beta)\}$

$= \frac{\tan(\alpha+\beta)+\tan(\alpha-\beta)}{1-\tan(\alpha+\beta).\tan(\alpha-\beta)} = \frac{\frac{4}{3}+\frac{5}{12}}{1-\frac{4}{3}\cdot\frac{5}{12}} = \frac{63}{16}$

18. (*d*) $(\sin 10°\sin 50°\sin 70°)\sin 30°$

$= \frac{1}{4}(\sin 30°)^2 = \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}$

$\left[\because \frac{1}{4}\sin 3\theta = \sin(60-\theta)\sin\theta.\sin(60+\theta)\right]$

19. (*b*) $\cos^2 10° - \cos 10°\cos 50° + \cos^2 50°$

$= \frac{1}{2}(2\cos^2 10° - 2\cos 10°\cos 50° + 2\cos^2 50°)$

$= \frac{1}{2}(1 + \cos 20° - (\cos 60° + \cos 40°) + 1 + \cos 100°)$

$= \frac{1}{2}\left(\frac{3}{2} + \cos 20° + 2\sin 70°\sin(-30°)\right)$

$= \frac{1}{2}\left(\frac{3}{2} + \cos 20° - \sin 70°\right) = \frac{3}{4}$

20. (*b*) $2y = 2\sin x\sin(x+2) - 2\sin^2(x+1)$

[multiply by 2 on both sides]

$2y = \cos 2 - \cos(2x+2) - (1-\cos(2x+2))$

[$\because$ 2sinAsinB = cos(A–B) – cos(A+B)]

$= \cos 2 - 1$

$2y = -2\sin^2 1$

$y = -\sin^2 1 < 0$

21. (*b*) $3(1-\sin 2\theta)^2 + 6(1+\sin 2\theta) + 4\sin^6\theta$

$= 3(1 - 2\sin 2\theta + \sin^2 2\theta) + 6 + 6\sin 2\theta + 4\sin^6\theta$

$= 9 + 3\sin^2 2\theta + 4\sin^6\theta$

$= 9 + 12\sin^2\theta\cos^2\theta + 4(1-\cos^2\theta)^3$

$= 9 + 12(1-\cos^2)\cos^2\theta + 4(1 - 3\cos^2\theta + 3\cos^4\theta - \cos^6\theta)$

$= 13 + 12\cos^2\theta - 12\cos^4\theta - 12\cos^2\theta + 12\cos^4\theta - 4\cos^6\theta$

$= 13 - 4\cos^6\theta$

22. (*a*) $\cos\frac{\pi}{2^2}.\cos\left(\frac{\pi}{2^3}\right).\cos\left(\frac{\pi}{2^4}\right)\ldots\cos\left(\frac{\pi}{2^{10}}\right).\sin\left(\frac{\pi}{2^{10}}\right)$,

we know

$\cos A\cos 2A\cos 4A\ldots\cos 2^{n-1}.A = \frac{\sin 2^n A}{2^n\sin A}$

Let $\frac{\pi}{2^{10}} = A$

So, $\sin A\cos A\cos 2A\cos 4A\ldots\cos 2^8 A$

$= \sin A.\frac{\sin 2^9 A}{2^9\sin A}$

$= \frac{\sin 2^9 A}{2^9} = \frac{\sin\left(\frac{\pi}{2}\right)}{2^9} = \frac{1}{512}$

23. (*a*, *b*, *d*)

$$f(n)=\frac{\sum_{k=0}^{n}\left(\cos\left(\frac{\pi}{n+2}\right)-\cos\left(\frac{2k+3}{n+3}\right)\pi\right)}{\sum_{k=0}^{n}\left(1-\cos\left(\frac{2k+2}{n+2}\right)\pi\right)}$$

$$f(n)=\frac{(n+1)\cos\left(\frac{\pi}{n+2}\right)-\left(\sum_{k=0}^{n}\cos\left(\frac{2k+3}{n+2}\right)\pi\right)}{(n+1)-\left(\sum_{k=0}^{n}\cos\left(\frac{2k+2}{n+2}\right)\pi\right)}$$

$$f(n)=\frac{(n+1)\cos\frac{\pi}{n+2}-\left(\frac{\sin\left(\frac{(n+1)\pi}{n+2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}.\cos\left(\frac{n+3}{n+3}\right)\pi\right)}{(n+1)-\left(\frac{\sin\left(\frac{(n+1)\pi}{n+2}\right)}{\sin\left(\frac{\pi}{n+2}\right)}.\cos\left(\frac{2(n+2)\pi}{2(n+2)}\right)\right)}$$

$$f(n)=\frac{(n+1)\cos\left(\frac{\pi}{n+2}\right)+\cos\left(\frac{\pi}{n+2}\right)}{(n+1)+1}$$

$$\Rightarrow g(x)=\cos\left(\frac{\pi}{n+2}\right)$$

(*a*) $\sin\left(7\cos^{-1}\cos\frac{\pi}{7}\right)=\sin\pi=0$

(*b*) $f(4)=\cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$

(*c*) $\lim_{x\to\infty}\cos\left(\frac{\pi}{n+2}\right)=1$

(*d*) $\alpha=\tan\left(\cos^{-1}\cos\frac{\pi}{8}\right)=\sqrt{2}-1$

$\Rightarrow \alpha+1=\sqrt{2}$

$\alpha^2+2\alpha-1=0$

24. (*d*) $\lambda=\cos^2 2x-2\sin^4 x-2\cos^2 x$

convert all in to cos*x*.

$\lambda=(2\cos^2 x-1)^2-2(1-\cos^2 x)^2-2\cos^2 x$

$=4\cos^4 x-4\cos^2 x+1-2(1-2\cos^2 x+\cos^4 x)-2\cos^2 x$

$=2\cos^4 x-2\cos^2 x-1$

$=2\left[\cos^4 x-\cos^2 x-\frac{1}{2}\right]$

$=2\left[\left(\cos^2 x-\frac{1}{2}\right)^2-\frac{3}{4}\right]$

$\lambda_{max}=2\left[\frac{1}{4}-\frac{3}{4}\right]=2\times\left(-\frac{2}{4}\right)=-1$

(Maximum Value)

$\lambda_{max}=2\left[0-\frac{3}{4}\right]=-\frac{3}{2}$ (Minimum Value)

So, Range $=\left[-\frac{3}{2},-1\right]$

25. (*a*) Given expression is

$$2\cos\left(\frac{x^2+x}{6}\right)=4^x+4^{-x}$$

L.H.S. ≤ 2 and R.H.S. ≥ 2

when $x=0$ then $2\cos\left(\frac{x^2+x}{6}\right)=2$ and

$4^x+4^{-x}=2$ Therefore, $x=0$ is the only solution.

26. (*a*) Let $5\sin\left(\theta-\frac{\pi}{6}\right)+3\cos\theta=f(\theta)$

$\Rightarrow f(\theta)=5\left(\sin\theta\cos\frac{\pi}{6}-\cos\theta\sin\frac{\pi}{6}\right)+3\cos\theta$

$=\frac{5\sqrt{3}}{2}\sin\theta+\frac{1}{2}\cos\theta$

$-\sqrt{a^2+b^2}\le a\cos\theta+b\sin\theta\le\sqrt{a^2+b^2}$

Maximum value is

$\sqrt{\left(\frac{5\sqrt{3}}{2}\right)^2+\left(\frac{1}{2}\right)^2}=\sqrt{\frac{76}{4}}=\sqrt{19}$

By Aditya Mohod

username - muje_kabhi_dekha_nahi

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 4

Trigonometric Equations

Prarambh (Topicwise)

1. (*d*) $\left(\cos^2 x - \sin^2 x\right)\left(2\sin\frac{x}{2} - 1\right) = 0$

$\cos 2x = 0$ or $\sin\frac{x}{2} = \frac{1}{2}$

2. (*c*) $2\tan^2\theta = \sec^2\theta \Rightarrow 2\tan^2\theta = \tan^2\theta + 1$

$\Rightarrow \tan^2\theta = 1 = \tan^2\left(\frac{\pi}{4}\right) \Rightarrow \theta = n\pi \pm \frac{\pi}{4}$

3. (*b*) $\sqrt{3}\tan 2\theta + \sqrt{3}\tan 3\theta + \tan 2\theta \tan 3\theta = 1$

$\Rightarrow \frac{\tan 2\theta + \tan 3\theta}{1 - \tan 2\theta \tan 3\theta} = \frac{1}{\sqrt{3}}$

$\Rightarrow \tan 5\theta = \tan\frac{\pi}{6}$

$\Rightarrow 5\theta = n\pi + \frac{\pi}{6} \Rightarrow \theta = \left(n + \frac{1}{6}\right)\frac{\pi}{5}$

4. (*d*) $\tan 2\theta = \cot \Rightarrow \tan 2\theta = \tan\left(\frac{\pi}{2} - \theta\right)$

$\Rightarrow 2\theta = n\pi + \frac{\pi}{2} - \theta \Rightarrow \theta = \frac{n\pi}{3} + \frac{\pi}{6}$

5. (*b*) $4 + 2\sin 2x = 5$

$\Rightarrow \sin^2 x = \frac{1}{2} = \sin^2\frac{\pi}{4} \Rightarrow x = n\pi \pm \frac{\pi}{4}$

6. (*a*) We have

$\frac{\pi}{4}\cot\theta = \frac{\pi}{2} - \frac{\pi}{4}\tan\theta \Rightarrow \tan\theta + \cot\theta = 2$

$\Rightarrow \sin 2\theta = 1 = \sin\frac{\pi}{2} \Rightarrow \theta = n\pi + \frac{\pi}{4}$

7. (*c*) $2\sqrt{3}\cos^2\theta = \sin\theta \Rightarrow 2\sqrt{3}\sin^2\theta + \sin\theta - 2\sqrt{3} = 0$

$\Rightarrow \sin\theta = \frac{-1 \pm 7}{4\sqrt{3}} \Rightarrow \sin\theta = \frac{-8}{4\sqrt{3}}$, (Impossible)

and $\sin\theta = \frac{6}{4\sqrt{3}} = \frac{\sqrt{3}}{2}$

$\Rightarrow \theta = n\pi + (-1)^n\frac{\pi}{3}$.

8. (*b*) $(2\cos x - 1)(3 + 2\cos x) = 0$

Then, $\cos x = \frac{1}{2}$ as $\cos x \neq \frac{-3}{2}$

$\Rightarrow x = 2n\pi \pm \frac{\pi}{3}; \begin{cases} \text{for } n = 0,\ x = \frac{\pi}{3}, \frac{5\pi}{3} \\ \text{for } n = 1,\ x = \frac{5\pi}{3} \end{cases}$

9. (*b*) $\tan(\cot x) = \cot(\tan x)$

$\Rightarrow \tan(\cot x) = \tan\left(\frac{\pi}{2} - \tan x\right)$

$\Rightarrow \cot x = n\pi + \frac{\pi}{2} - \tan x$

$\Rightarrow \cot x + \tan x = n\pi + \frac{\pi}{2}$

$\Rightarrow \frac{2}{\sin 2x} = n\pi + \frac{\pi}{2} \Rightarrow \sin 2x = \frac{2}{n\pi + \frac{\pi}{2}}$

$= \frac{4}{(2n+1)\pi}$

10. (*d*) $4\sin\theta.\cos\theta - 2\cos\theta - 2\sqrt{3}\sin\theta + \sqrt{3} = 0$

$\Rightarrow 2\cos\theta(2\sin\theta - 1) - \sqrt{3}(2\sin\theta - 1) = 0$

$\Rightarrow (2\sin\theta - 1)(2\cos\theta - \sqrt{3}) = 0$

$\Rightarrow \sin\theta = \frac{1}{2}, \cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{11\pi}{6}$

11. (*b*) $2\sin\theta + \tan\theta = 0 \Rightarrow \sin\theta = 0$

or $2 + \frac{1}{\cos\theta} = 0$

$\Rightarrow \theta = n\pi$ or $2 = -\frac{1}{\cos\theta} \Rightarrow \cos\theta = -\frac{1}{2}$

$\Rightarrow \theta = 2m\pi \pm \frac{2\pi}{3}$.

12. (*a*) $2\cos 2(\pi + x) + 3\sin(\pi + x) = 0$

$\Rightarrow 2\cos 2x - 3\sin x = 0 \Rightarrow 2 - 2\sin 2x - 3\sin x = 0$

$\Rightarrow 2\sin 2x + 3\sin x - 2 = 0 \Rightarrow \sin x = -2, \frac{1}{2}$

$\Rightarrow \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$

13. (*a*) $\cos 2\theta + 3\cos\theta = 0 \Rightarrow 2\cos 2\theta + 3\cos\theta - 1 = 0$

$\Rightarrow \cos\theta = \frac{3 \pm \sqrt{9+8}}{4} = \frac{-3 \pm \sqrt{17}}{4}$

As $-1 \le \cos\theta \le 1$ *a* $\therefore$ $\cos\theta = \frac{-3+\sqrt{17}}{4}$ only

$\Rightarrow \theta = 2n\pi \pm \alpha$ where $\cos\alpha = \frac{\sqrt{17}-3}{4}$

14. (*b*) Given, $2\cos 2x = 3.2\cos^2 x - 4$

$\Rightarrow 2\cos 2x = 3(\cos 2x + 1) - 4$

$\Rightarrow \cos 2x = 1 \Rightarrow 2x = 2n\pi;\ n \in I$

$\Rightarrow x = n\pi;\ n \in I$

15. (*c*)

$\sin 2\pi x = \sqrt{2}\cos\pi x$

$2\sin\pi x\cos\pi x = \sqrt{2}\cos\pi x$

$\cos\pi x(2\sin\pi x - \sqrt{2}) = 0$

$\cos\pi x = 0$

or $\sin\pi x = \frac{1}{\sqrt{2}}$

[As $x \in (0, 1)$, $\pi x \in (0, \pi)$]

$\pi x = \frac{\pi}{2}$ or $\pi x = \frac{\pi}{4}, \frac{3\pi}{4}$

$x = \frac{1}{2}$ $\quad$ $x = \frac{1}{4}, \frac{3}{4}$

$x = \frac{1}{2}, \frac{1}{4}, \frac{3}{9}$

Greatest root $= \frac{3}{4}$

16. (*b*) $\sec\theta + \tan\theta = \sqrt{3}$ ...(*i*)

$\sec\theta - \tan\theta = \frac{1}{\sqrt{3}}$ ...(*ii*)

(*i*) + (*ii*),

$2\sec\theta = \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{3+1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$

$\sec\theta = \frac{2}{\sqrt{3}} \Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$

$\cos\theta = \cos\frac{\pi}{6} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{6}$ $n \in z$

17. (*b*) $\sin^2\theta\sec\theta + \sqrt{3}\tan\theta = 0$

$\frac{\sin^2\theta}{\cos\theta} + \frac{\sqrt{3}\sin\theta}{\cos\theta} = 0$

$\Rightarrow \frac{\sin\theta}{\cos\theta}(\sin\theta + \sqrt{3}) = 0$

$\Rightarrow \sin\theta = 0$ or $\sin\theta = -\sqrt{3}$ (Not possible)

$\sin\theta = 0$

$\theta = n\pi,\ n \in z$

18. (*b*) $\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{3\pi}{4}, \frac{5\pi}{4}; \tan\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

$\therefore$ The general value is $2n\pi + \frac{5\pi}{4}$ or $(2n+1)\pi + \frac{\pi}{4}$

19. (*b*) $\tan\theta = \frac{-1}{\sqrt{3}} = \tan\left(\pi - \frac{\pi}{6}\right)$, $\sin\theta$

$= \frac{1}{2} = \sin\left(\pi - \frac{\pi}{6}\right)$

and $\cos\theta = \frac{-\sqrt{3}}{2} = \cos\left(\pi - \frac{\pi}{6}\right)$

Hence principal value is $\theta = \frac{5\pi}{6}$

20. (*c*) Given, $\tan\theta = -1$ & $\cos\theta = \frac{1}{\sqrt{2}}$

principal solution

$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$ & $\theta = \frac{\pi}{4}, \frac{7\pi}{4}$

common principal solutoin is $\frac{7\pi}{4}$

then general solution is $\theta = 2n\pi + \frac{7\pi}{4}$, $n \in I$

21. (*d*) $\sin\theta = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right) = \sin\left(\pi + \frac{\pi}{6}\right)$

$\tan\theta = \frac{1}{\sqrt{3}} = \tan\left(\frac{\pi}{6}\right) = \tan\left(\pi + \frac{\pi}{6}\right)$

$\Rightarrow \theta = \left(\pi + \frac{\pi}{6}\right)$

Hence general value of θ is $2n\pi + \frac{7\pi}{6}$

22. (*c*) $\sin A = \sin B$

$A = n\pi + (-1)^n B, n \in Z$

$A = B, \pi - B, 2\pi + B, 3\pi - B, 4\pi + B \ldots$ ...(*i*)

Also, $\cos A = \cos B$

$A = 2n\pi \pm B, n \in Z$

$A = B, 2\pi + B, 4\pi + B, 6\pi + B \ldots$ ...(*ii*)

From (i) and (ii), common solution is

$A = 2n\pi + B, n \in Z$

23. (*c*) $\sec^2\theta + \tan^2\theta = \frac{5}{3}$, also sec2 θ – tan2 $\theta = 1$

$\Rightarrow \tan^2\theta = \frac{1}{3} = \tan^2\left(\frac{\pi}{6}\right) \Rightarrow \theta = n\pi \pm \frac{\pi}{6}$.

24. (*c*) $\cot\theta + \tan\theta = 2\,\text{cosec}\,\theta$

$\Rightarrow \frac{2}{\sin\theta} = \frac{1}{\sin\theta\cos\theta}$

$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}$

25. (*a*) $\sin 2x + \cos 4x = 2$

$\sin 2x + 1 - 2\sin^2 2x = 2$ ($\cos 2\theta = 1 - 2\sin^2\theta$)

$2\sin^2 2x - \sin 2x + 1 = 0$

$D = (1)^2 - 4.2.1 = 1 - 8 = -7$

As discriminant of the quadratic equation is negative. Hences No solution.

Number of solution = 0

26. (*a*) $\tan^2\theta - \tan\theta - \sqrt{3}\tan\theta + \sqrt{3} = 0$

$\Rightarrow \tan\theta(\tan\theta - 1) - \sqrt{3}(\tan\theta - 1) = 0$

$\Rightarrow (\tan\theta - \sqrt{3})(\tan\theta - 1) = 0$

$\Rightarrow \theta = n\pi + \frac{\pi}{3}, n\pi + \frac{\pi}{4}$

27. (*a*) $4 - 4\cos^2\theta + 2(\sqrt{3}+1)\cos\theta = 4 + \sqrt{3}$

$\Rightarrow 4\cos^2\theta - 2(\sqrt{3}+1)\cos\theta + \sqrt{3} = 0$

$\Rightarrow \cos\theta = \frac{2(\sqrt{3}+1) \pm \sqrt{4(\sqrt{3}+1)^2 - 16\sqrt{3}}}{8}$

$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$ or $\cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{6}$ or $2n\pi \pm \pi/3$

28. (*a*) $2 - 2\cos^2\theta + \sqrt{3}\cos\theta + 1 = 0$

$\Rightarrow 2\cos^2\theta - \sqrt{3}\cos\theta - 3 = 0$

$\Rightarrow \cos\theta = \frac{\sqrt{3} \pm \sqrt{3+24}}{4} = \frac{\sqrt{3}(1 \pm 3)}{4} = \sqrt{3}\left(-\frac{1}{2}\right)$

$\Rightarrow \theta = \frac{5\pi}{6}$

29. (*c*) $12\cot^2\theta - 31\,\text{cosec}\,\theta + 32 = 0$

$12(\text{cosec}^2\theta - 1) - 31\,\text{cosec}\,\theta + 32 = 0$

$12\,\text{cosec}^2\theta - 31\,\text{cosec}\,\theta + 20 = 0$

$12\,\text{cosec}^2\theta - 16\,\text{cosec}\,\theta - 15\,\text{cosec}\,\theta + 20 = 0$

$(4\,\text{cosec}\,\theta - 5)(3\,\text{cosec}\,\theta - 4) = 0$

$\text{cosec}\,\theta = \frac{5}{4}, \frac{4}{3}; \therefore \sin\theta = \frac{4}{5}, \frac{3}{4}$

30. (*c*) $\sin 4\theta = \cos\theta - \cos 7\theta \Rightarrow \sin 4\theta = 2\sin(4\theta)\sin(3\theta)$

$\Rightarrow \sin 4\theta = 0 \Rightarrow 4\theta = n\pi$ or $\sin 3\theta = \frac{1}{2} = \sin\left(\frac{\pi}{6}\right)$

$\Rightarrow 3\theta = n\pi + (-1)^n\frac{\pi}{6} \Rightarrow \theta = \frac{n\pi}{4}, \frac{n\pi}{3} + (-1)^n\frac{\pi}{18}$

31. (*d*) $\cos\theta + \cos 2\theta + \cos 3\theta = 0$

$\Rightarrow (\cos\theta + \cos 3\theta) + \cos 2\theta = 0$

$\Rightarrow 2\cos 2\theta\cos\theta + \cos 2\theta = 0$

$\Rightarrow \cos 2\theta(2\cos\theta + 1) = 0$

$\Rightarrow \cos 2\theta = 0 = \cos\frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4} \Rightarrow \theta = 2m\pi \pm \frac{\pi}{4}$

or $\cos\theta = \frac{-1}{2} = \cos\frac{2\pi}{3} \Rightarrow \theta = 2m\pi \pm \frac{2\pi}{3}$.

32. (*c*) $\sin x + \sin 5x = \sin 2x + \sin 4x$

$\Rightarrow 2\sin 3x\cos 2x = 2\sin 3x\cos x$

$\Rightarrow \sin 3x = 0$ or $\cos 2x = \cos x$

$\Rightarrow 3x = n\pi$ or $2x = 2n\pi \pm x$

$\Rightarrow x = \frac{n\pi}{3}, 2n\pi, \frac{2n\pi}{3}$

$\Rightarrow x = \frac{n\pi}{3}$ (It includes all three possible)

33. (*d*) $\frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta = \frac{\sqrt{2}}{2}$

{dividing by $\sqrt{(\sqrt{3})^2 + 1^2} = 2$}

$\Rightarrow \sin\left(\theta + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}} = \sin\left(\frac{\pi}{4}\right)$

$\Rightarrow \theta = n\pi + (-1)^n\frac{\pi}{4} - \frac{\pi}{3}$

34. (*d*) Given equation is $\sqrt{3}\sin x + \cos x = 4$ which is of the form $a\sin x + b\cos x = c$ with

$a = \sqrt{3}, b = 1, c = 4$.

Here $a^2 + b^2 = 3 + 1 = 4 < c^2$, therefore the given equation has no solution.

35. (*d*) $3\cos x + 4\sin x = 6$

$\Rightarrow \frac{3}{5}\cos x + \frac{4}{5}\sin x = \frac{6}{5} \Rightarrow \cos(x - \theta) = \frac{6}{5}$

[where $\theta = \cos - 1(3/5)$]

So, that equation has no solution.

36. (*d*) Given

$\sin\theta + 2\sin 2\theta + 3\sin\theta + 4\sin 4\theta = 10$ in $(0, \pi)$

Using boundary of SM

It's only possible if $\Rightarrow 1 + 2 + 3 + 4 = 10$

(Using boundary values)

$\Rightarrow \sin\theta = 1$ & $\sin 2\theta = 1$ & $\sin 3\theta = 1$ & $\sin 4\theta = 1$

$\theta = 2m\pi + \frac{\pi}{2}, 2\theta = 2n\pi + \frac{\pi}{2}, 3\theta = 2k\pi + \frac{\pi}{2}$,

$4\theta = 2p\pi + \frac{\pi}{2}$,

$m \in I\, n \in I\, k \in I\, p \in I$

$\theta = \frac{\pi}{2}, \theta = \frac{\pi}{4}, \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \theta = \frac{\pi}{8}, \frac{5\pi}{8}$.

$\therefore x \in \left\{\frac{\pi}{2}\right\} \cap \left\{\frac{\pi}{4}\right\} \cap \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\} \cap \left\{\frac{\pi}{8}, \frac{5\pi}{8}\right\}$

$x \in \{\}$ number of solutoins is zero.

37. (*a*) $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$

$-\sqrt{1^2+1^2} \le \sin x + \cos x \le \sqrt{1^2+1^2}$

$-\sqrt{2} \le \text{LHS} \le \sqrt{2}$ ...(*i*)

Also, $y + \frac{1}{y} \ge 2$

$\sqrt{y + \frac{1}{y}} \ge \sqrt{2}$

RHS $\ge \sqrt{2}$ ...(*ii*)

From (*i*) and (*ii*)

$\sin x + \cos x = \sqrt{2}$

$\Rightarrow \left(\frac{1}{\sqrt{2}}\sin x + \cos x \frac{1}{\sqrt{2}} = 1 \sin\left(\frac{\pi}{4} + x\right) = 1\right.$

$\frac{\pi}{4} + x = n\pi + (-1)^n \frac{\pi}{2}$; As $x \in (0, \pi)$

$\Rightarrow x = \frac{\pi}{4}$ Also, $y + \frac{1}{y} = 2 \Rightarrow y = 1$

38. (*a*) $f(x) = \cos x - x + \frac{1}{2}, f(0) = \frac{3}{2} > 0$

$f\left(\frac{\pi}{2}\right) = 0 - \frac{\pi}{2} + \frac{1}{2} = \frac{1-\pi}{2} < 0 \left(\because \pi = \frac{22}{7} \text{ nearly}\right)$

$\therefore$ One root lies in the interval $\left[0, \frac{\pi}{2}\right]$.

39. (*c*) $\sec x \cos 5x = -1 \Rightarrow \cos 5x = -\cos x$

$\Rightarrow 5x = 2n\pi \pm (\pi - x)$

$\Rightarrow x = \frac{(2n+1)\pi}{6}$ or $\frac{(2n-1)\pi}{4}$

Hence,

$x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{6}, \frac{5\pi}{4}, \frac{7\pi}{6}, \frac{7\pi}{4}, \frac{9\pi}{6}, \frac{11\pi}{6}$

40. (*b*) $1 + \sin^2 x\theta = \cos\theta$ ...(*i*)

$0 \le \sin^2 x\theta \le 1$

$1 \le 1 + \sin^2 x\theta \le 2$

$1 \le \cos\theta \le 2$ (From (*i*))

...(*ii*)

But $\cos\theta \ge = 1; -1 \le \cos\theta \le 1$

...(*iii*)

From (*ii*) and (*iii*)

$\cos\theta = 1 \Rightarrow \theta = 2m\pi, m \in z$

$1 + \sin^2 x\theta = 1$

$\sin^2 x\theta = 0$

$x\theta = n\pi, n \in z$

$x = \frac{n\pi}{\theta}$

$x = \frac{n\pi}{2m\pi} = \frac{n}{2m}$; x is rational number.

41. (*a*) $3\sin x + 4\cos x = x^2 + 16$ ...(*i*)

$-\sqrt{3^2+4^2} \le 3\sin x + 4\cos x \le \sqrt{3^2+4^2}$

$-5 \le 3\sin x + 4\cos x \le 5$

Also, $x^2 \ge 0$

$x^2 + 16 \ge 16$

In equation (*i*),

LHS $\in [-5, 5]$

RHS $\in [16, \infty]$

So, no solution.

No. of solution = 0.

42. (*d*) $\sin 4x = 1 + \tan^8 x$ ...(*i*)

$0 \le \sin^4 x \le 1$

Also, $\tan^8 x \ge 0$

$1 + \tan^8 x \ge 1$

LHS $\in [0, 1]$; RHS = $[1, \infty)$

Only possibility

$\sin^4 x = 1 = 1 + \tan^8 x$

$\sin^4 x = 1$ and $1 = 1 + \tan^8 x$

$\sin^2 x = 1$ and $\tan^8 x = 0$

$\sin^2 x = \sin^2 \frac{\pi}{2}$

$\tan x = 0; x = m\pi, n \in I$

$x = n\pi \pm \frac{\pi}{2}, n \in I$

There will be no common solution.

43. (*a*) We have $|4\sin x - 1| < \sqrt{5}$

$\Rightarrow -\sqrt{5} < 4\sin x - 1 < \sqrt{5}$

$\Rightarrow -\left(\frac{\sqrt{5}-1}{4}\right) < \sin x < \left(\frac{\sqrt{5}+1}{4}\right)$

$\Rightarrow -\sin\frac{\pi}{10} < \sin x < \cos\frac{\pi}{5}$

$\Rightarrow \sin\left(-\frac{\pi}{10}\right) < \sin x < \sin\left(\frac{\pi}{2} - \frac{\pi}{5}\right)$

$\Rightarrow \sin\left(-\frac{\pi}{10}\right) < \sin x < \sin\frac{3\pi}{10}$

$\Rightarrow x \in \left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$

44. (*d*) $4\sin^2 x - 12\sin x + 5 \le 0; 0 \le x \le 2\pi$

$\Rightarrow 4\sin^2 x - 10\sin x - 2\sin x + 5 \le 0$

$\Rightarrow 2\sin x(2\sin x - 5) - 1(2\sin x - 5) \le 0$

$\Rightarrow (2\sin x - 5)(2\sin x - 1) \le 0$

$\Rightarrow \frac{1}{2} \le \sin x \le \frac{1}{5} \Rightarrow \frac{1}{2} \le \sin x \le 1$

$\therefore$ The solution set is $x \in \left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

45. (*b*) We have,

$\sin x \cdot \cos^3 x - \cos x \cdot \sin^3 x; 0 \le x \le 2\pi$

$\Rightarrow \sin x \cos^3 x - \cos x \cdot \sin^3 > 0$

$\Rightarrow \sin x \cos x (\cos^2 x - \sin^2 x) > 0$

$\Rightarrow 1/2 \sin 2x \cdot \cos 2x > 0 \Rightarrow 1/4 (2\sin 2x \cos 2x) > 0$

$\Rightarrow 1/4 \sin 4x > 0 \Rightarrow \sin 4x > 0$

$\Rightarrow 4x \in (2n\pi, (2n+1)\pi) \forall n \in Z$

$\Rightarrow x \in \left(\frac{n\pi}{2}, (2n+1)\frac{\pi}{4}\right) \forall n \in Z$

But $x \in [0, 2\pi]$

For $n = 0; x \in (0, \pi/4)$

For $n = 1, x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

For $n = 2, x \in \left(\pi, \frac{5\pi}{4}\right)$

For $n = 3, x \in \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

46. (*a*) $[\sin x] + \left[\sqrt{2}\cos x\right] = -3 \quad x \in (0, 2\pi)$

$-1 \le \sin x \le 1$

Only possible solution $[\sin x] = -1$

$\left[\sqrt{2}\cos x\right] = -2$

$[\sin x] = -1$

when $\sin x < 0$

$\rightarrow x \in (\pi, 2\pi)$

$\left[\sqrt{2}\cos x\right] = -2$

$x \in \left(\pi, \frac{5\pi}{4}\right)$

Prabal (JEE Main Level)

1. (*c*) $|\sin x| = |\cos 3x| \quad x \in [-2\pi, 2\pi]$

Case (I): $\sin x = \cos 3x$

$\cos\left(\frac{\pi}{2} - x\right) = \cos 3x$

$3x = 2n\pi \pm \left(\frac{\pi}{2} - x\right) \quad n \in I$

$3x = 2n\pi + \frac{\pi}{2} - x$ and $3x = 2n\pi - \frac{\pi}{2} + x$

$4x = (4n+1)\frac{\pi}{2} \quad 2x = (4n-1)\frac{\pi}{2}$

$x = (4n+1)\frac{\pi}{8} \quad x = (4n-1)\frac{\pi}{4}$

$-2 \le \frac{4n+1}{8} \le 2 \quad -2 \le \frac{4n-1}{4} \le 2$

$-\frac{17}{4} \le n \le \frac{15}{4} \quad -\frac{7}{4} \le n \le \frac{9}{4}$

$n = -4, -3, -2, -1, 0, 1, 2, 3, n = -1, 0, 1, 2$

no. of sol. 8 no. of sol. 4

Case (II): $-\sin x = \cos 3x$

$\cos\left(\frac{\pi}{2} + x\right) = \cos 3x$

$3x = 2n\pi \pm \left(\frac{\pi}{2} + x\right)$ $\quad \because n \in I$

$3x = 2n\pi + \frac{\pi}{2} + x$ $\qquad$ $3x = 2n\pi - \frac{\pi}{2}$

$x = \frac{(4n+1)\pi}{4}$ $\qquad$ $x = \frac{(4n-1)\pi}{8}$

$-2 \le \frac{4x+1}{4} \le 2$ $\qquad$ $-2 \le \frac{4x-1}{8} \le 2$

$-\frac{9}{4} \le n \le \frac{7}{4}$ $\qquad$ $-\frac{15}{4} \le x \le \frac{17}{4}$

$-\frac{9}{4} \le n \le \frac{7}{4}$ $\qquad$ $-\frac{15}{4} \le x \le \frac{17}{4}$

$n = -2, -1, 0, 1$ $\qquad$ $n = -3, -2, -1, 0, 1, 2, 3, 4$

no. of sol. 4 $\qquad$ no. of sol. 8

Total no. of sol. is $= 8 + 4 + 4 + 8 = 24$

2. (*a*) We have, $\tan(\cot x) = \cot(\tan x)$

$\Rightarrow \tan(\cot x) = \tan\left(\frac{\pi}{2} - \tan x\right)$

$\Rightarrow \cot x = n\pi + \frac{\pi}{2} - \tan x$

$\Rightarrow \cot x + \tan x = n\pi + \frac{\pi}{2}$

$\Rightarrow \frac{2}{\sin 2x} = n\pi + \frac{\pi}{2}$

$\Rightarrow \sin 2x = \frac{2}{n\pi + \frac{\pi}{2}} = \frac{4}{(2n+1)\pi}$

$\Rightarrow \operatorname{cosec} 2x = \frac{(2n+1)\pi}{4}$

$\Rightarrow \tan(\cot x) = \tan\left(\frac{\pi}{2} - \tan x\right)$

$\Rightarrow \cot x = n\pi + \frac{\pi}{2} - \tan x$

$\Rightarrow \cot x + \tan x = n\pi + \frac{\pi}{2}$

$\Rightarrow \frac{2}{\sin 2x} = n\pi + \frac{\pi}{2}$

$\Rightarrow \sin 2x = \frac{2}{n\pi + \frac{\pi}{2}} = \frac{4}{(2n+1)\pi}$

$\Rightarrow \operatorname{cosec} 2x = \frac{(2n+1)\pi}{4}$

3. (*a*) $\because A.M. \ge G.M.$

$\therefore \frac{2^{2\sin x} + 2^{2\cos x}}{2} \ge \sqrt{2^{2\sin x} 2^{2\cos x}}$

$\therefore 2^{2\sin x} + 2^{2\cos x} \ge 2\sqrt{2^{2(\sin x + \cos x)}}$. But minimum

Value of $\cos x + \sin x$ is $-\sqrt{2}$

$\therefore 2^{2\sin x} + 2^{2\cos x} \ge 2\sqrt{2^{-2\sqrt{2}}} = 2^{1-\sqrt{2}}$

But the given equation is

$2^{2\sin x} + 2^{2\cos x} = 2^{1-\sqrt{2}}$

Which can hold only if $2^{2\sin x} = 2^{2\cos x}$

$\Rightarrow \sin x = \cos x \Rightarrow \tan x = 1 \Rightarrow x = m\pi + \frac{\pi}{4}$

4. (*a*) $4\sin^2 x + \tan^2 x + \operatorname{cosec}^2 x + \cot^2 x - 6 = 0$

$(2\sin x - \operatorname{cosec} x)^2 + (\tan x - \cot x)^2 = 0$

$\Rightarrow \sin^2 x = \frac{1}{2} \cap \tan^2 x = 1 \Rightarrow x = m\pi \pm \frac{\pi}{4}$

5. (*d*) $\cot(\theta) + \cot\left(\theta + \frac{\pi}{3}\right) + \cot\left(\theta - \frac{\pi}{3}\right) + \cot(3\theta) = 0$

Put $\theta = \frac{\pi}{2} - \alpha$

$\tan\alpha + \tan\left(\alpha - \frac{\pi}{3}\right) + \tan\left(\alpha + \frac{\pi}{3}\right) + \tan 3\alpha = 0$

$\tan\alpha + \frac{\tan\alpha - \tan\frac{\pi}{3}}{1 + \tan\alpha\tan\frac{\pi}{3}} + \frac{\tan\alpha + \tan\frac{\pi}{3}}{1 - \tan\alpha\tan\frac{\pi}{3}} + \tan 3\alpha = 0$

$\Rightarrow 3\left(\frac{3\tan\alpha - \tan^3\alpha}{1 - 3\tan^2\alpha}\right) + \tan 3\alpha = 0$

$\Rightarrow 4\tan 3\alpha = 0$

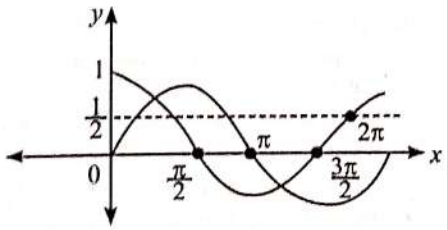
$\Rightarrow \tan 3\alpha = 0$

$\Rightarrow 3\alpha = n\pi$

$\Rightarrow \alpha = \frac{n\pi}{3}$

$\Rightarrow \alpha = \frac{\pi}{3}, \frac{2\pi}{3}$

6. (*d*)



If $[0, 2\pi]$ max. $(\sin x, \cos x) = \frac{1}{2}$ has two solutions.

7. (*b*)

$\sin^4 x - \cos^2 x \sin x + 2\sin^2 x + \sin x = 0$

$\Rightarrow \sin x [\sin^3 x - \cos^2 x + 2\sin x + 1] = 0$

$\Rightarrow \sin x [\sin^3 x - 1 + \sin^2 x + 2\sin x + 1] = 0$

$\Rightarrow \sin x [\sin^3 x + \sin^2 x + 2\sin x] = 0$

$\Rightarrow \sin^2 x = 0$ or $\sin^2 x + \sin x + 2 = 0$ (not possible for real x)

$\Rightarrow \sin x = 0$

Hence, the solutions are $x = 0, \pi, 2\pi, 3\pi$.

8. (*d*) We have $(\sin x + \sin 3x) + \sin 2x = (\cos x + \cos 3x) + \cos 2x$

$\Rightarrow 2\sin 2x \cos x + \sin 2x = 2\cos 2x \cos x + \cos 2x$

$\Rightarrow \sin 2x (2\cos x + 1) = \cos 2x (2\cos x + 1)$

$\Rightarrow (2\cos x + 1)(\sin 2x - \cos 2x) = 0$

$\Rightarrow \cos x = -1/2$ or $\sin 2x - \cos 2x = 0$

$\Rightarrow x = 2n\pi \pm (2\pi/3)$ or $\tan 2x = 1 = \tan(\pi/4)$

$= 2n\pi \pm (2\pi/3)$ or $x = (4n+1)\pi/8$, $n \in Z$

But here $0 \le x \le 2\pi$

Hence, $x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{9\pi}{8}, \frac{4\pi}{3}, \frac{13\pi}{8}$

9. (*a*) $\cos x = \sqrt{1 - \sin 2x} = |\sin x - \cos x|$

(a) $\sin x < \cos x \Rightarrow x \in \left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$ ...(i)

Then the given equation is $\cos x = \cos x - \sin x \Rightarrow \sin x = 0 \Rightarrow x = \pi, 2\pi$

$\Rightarrow x = 2\pi$ [from Eq. (i)]

(b) $\sin x \ge \cos x$ $\Rightarrow x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$

$\Rightarrow \cos x = \sin x - \cos x$

$\Rightarrow \tan x = 2$

$\Rightarrow x = \tan^{-1} 2$ ...(ii)

Hence, there are two solutions.

10. (*a*) $\tan^4\theta + 2\tan^2\theta = (\tan^2\theta + 1)^2 - 1 = (\sec^2\theta)^2 - 1 = \sec^4\theta - 1$

Put $\sec^2\theta = t$

$\Rightarrow \left(\sqrt{3}\right)^t = t^2 - 1$

$\Rightarrow t = 2$ is only solution as $t > 1$

$\therefore$ There will be 2 values of θ in given interval.

11. (*c*) Consider, $\tan^2\frac{x}{2} = 1 + \sec x$

$\Rightarrow 8\left(\frac{1 - \cos x}{1 + \cos x}\right) = 1 + \frac{1}{\cos x}$

$\Rightarrow 8\cos x - \cos^2 x = (1 + \cos x)^2$

$\Rightarrow 8\cos x - 8\cos^2 x = 1 + \cos^2 x + 2\cos x$

$\Rightarrow 9\cos^2 x - 6\cos x + 1 = 0$

$\Rightarrow (3\cos x - 1)^2 = 0$

$\Rightarrow \cos x = \frac{1}{3} = \cos\alpha$ [By using $\cos\theta = \cos\alpha$

$\Rightarrow 2n\pi \pm \alpha$]

$\Rightarrow x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

12. (*c*) We have $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta . \tan 4\theta . \tan 7\theta$

$\Rightarrow \tan 0 + \tan 4\theta = -\tan 7\theta + \tan\theta.\tan 4\theta.\tan 7\theta$

$\Rightarrow \tan\theta + \tan 4\theta = -\tan 7\theta\,(1 - .\tan\theta.\tan 4\theta)$

$\Rightarrow \dfrac{\tan\theta + \tan 4\theta}{1 - \tan\theta \cdot \tan 4\theta} = -\tan 7\theta$

$\Rightarrow \tan(\theta + 4\theta) = -\tan 7\theta$

$\Rightarrow \tan 5\theta = \tan(-7\theta)$

$\Rightarrow 5\theta = n\pi + (-7\theta)$

$\Rightarrow 12\theta = n\pi \Rightarrow \theta = \dfrac{n\pi}{12}$ (where $n \in I$)

13. (*d*) Consider: 1st equation i.e., $2\sin^2 x + 3\sin x - 2 > 0$

$\Rightarrow (2\sin x - 1)(\sin x + 2) > 0$

$\Rightarrow (2\sin x - 1) > 0$

[As $\sin x + 2 > 0\ \forall\ x \in R$]

$\Rightarrow \sin x > \dfrac{1}{2} \Rightarrow x \in \left(\dfrac{\pi}{6}, \dfrac{5\pi}{6}\right)$ ...(i)

...(i)

consider $x^2 - x - 2 < 0$

$\Rightarrow (x-2)(x+1) < 0 \Rightarrow -1 < x < 2$ ...(ii)

From Eqs. (i) and (ii),

Now, as, $2 < \dfrac{5\pi}{6}$, we obtain that 'x' must lie in $\left(\dfrac{\pi}{6}, 2\right)$

14. (*b*) $\sin^{100} x = 1 + \cos^{100} x \Rightarrow \sin^{100} x = 1$ and $\cos^{100} x = 0$

15. (*c*) $\cos x + \cos 2x + \cos 3x + \cos 4x + \cos 5x = 5$

$\Rightarrow \cos x = 1 \cap \cos 2x = 1 \cap \cos 3x = 1 \cap \cos 4x = 1 \cap \cos 5x = 1$

$x = 2n\pi \cup x = n\pi \cap x = \dfrac{2n\pi}{3} \cap x = \dfrac{2n\pi}{4} \cap x = \dfrac{2n\pi}{5}$

$\Rightarrow x = 2n\pi$

16. (*a*) $[\tan(x+y) - \cot(x+y)]^2 + (x+1)^2 = 0$

$\Rightarrow x = -1$ and $\tan^2(x+y) = 1$

$x + y = n\pi \pm \dfrac{\pi}{4}$

17. (*b*)

$3\cos\theta + 4\sin\theta = 5\left[\dfrac{3}{5}\cos\theta + \dfrac{4}{5}\sin\theta\right]$

$= 5\cos(\theta - \alpha)$

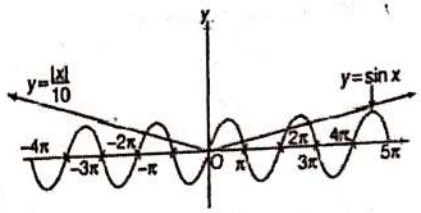
Where $\cos\alpha = 3/5$, $\sin\alpha = 4/5$

Now, $3\cos\theta + 4\sin\theta = k$

$\therefore 5\cos(\theta - \alpha) = \pm 5 \Rightarrow \cos(\theta - \alpha) = \pm 1$

$\Rightarrow \theta - \alpha = 0^\circ, 180^\circ \Rightarrow \theta = \alpha, 180^\circ + \alpha$

18. (*b*) Graphs of $y = \sin x$ and $y = \dfrac{|x|}{10}$ meet exactly six times. Hence, there are six solutions.



19. (*a*) We have,

$\cos 2x - 3\cos x + 1 = \dfrac{1}{(\cot 2x - \cot x)\sin(x - \pi)}$

$2\cos^2 x - 1 - 3\cos x + 1 = \dfrac{1}{\dfrac{\sin(x-2x)}{\sin 2x.\sin x} \cdot (-\sin x)}$

$\left[\text{using formula, } \cot A - \cot B = \dfrac{\sin(B-A)}{\sin A \cdot \sin B}\right]$

$2\cos^2 x - 3\cos x - 2\cos x = 0$

$2\cos^2 x - 5\cos x = 0$

$\cos x\,(2\cos x - 5) = 0$

Either $2\cos x - 5 = 0$

$\Rightarrow \cos x = \dfrac{5}{2}$ (which is not possible)

Then, $\cos x = 0$

20. (*a*) $[\sin x + \cos x] = 3 + [-\sin x] + [-\cos x]$

Maximum value of left-hand side is 1 and minimum of right hand side is also 1.

$\Rightarrow [\sin x + \cos x] = 3 + [-\sin x] + [-\cos x] = 1$

$\Rightarrow x \in \pi \pm \pi/4$

$\Rightarrow [\sin x + \cos x] = 1, [-\sin x] = -1, [-\cos x] = -1.$

Which is not possible.

21. (*d*) Given

$\sin 3\theta = 4\sin\theta.\sin 2\theta \sin 4\theta$, where $0 \le \theta \le \pi$

$\Rightarrow 3\sin\theta - 4\sin^3\theta - 4\sin\theta \sin 2\theta \sin 4\theta = 0$

$\Rightarrow \sin\theta\,[3 - 4\sin^2\theta - 4\sin 2\theta \sin 4\theta]$

$\Rightarrow \sin\theta\,[3 - 2 + 2\cos 2\theta - 2\cos 2\theta + 2\cos 6\theta]$

$\Rightarrow \sin\theta\,[1 + 2\cos 6\theta] = 0$

$\Rightarrow \sin\theta = 0$ or $\cos 6\theta = -\dfrac{1}{2}$

$\theta = 0, \pi$ or $0 \le \theta \le \pi$

$0, \pi \in [0, \pi]$ or $0 \le 6\theta \le 6\pi$ (3 rounds)

Number of solutions in 1 round $= 2$

total number of solutions in $\theta \in [0, \pi]$

$= 2 + 6 = 8$

22. (*b*) Given $4\cos^3 x - 4\cos^2 x - \cos(\pi + x) - 1 = 0$

$\Rightarrow 4\cos^3 x - 4\cos^2 x + \cos x - 1 = 0$

A.M. of roots $x \in [0, 315]$

$\Rightarrow 4\cos^2 x(\cos x - 1) + (\cos x - 1) = 0$

$\Rightarrow (\cos x - 1)(4\cos^2 x + 1) = 0$

$\Rightarrow \cos x = 1$ or $\cos^2 x = -\dfrac{1}{4}$

(not possible)

$x = 2n\pi,\ n \in I$

$0 \le 2n \le 100,\ x \in [0, 315]$

$0 \le n \le 50$ or $[0, 100\pi]$

$x = 0, 2\pi, 4\pi, 6\pi, \ldots, 100\,\pi$

$\text{A.M.} = \dfrac{2\pi[0+1+2+3+\ldots+50]}{51}$

$= \left(\dfrac{2\pi}{51}\right)\dfrac{50\times 51}{2} = 50\pi$

23. (*b*) $\sin\theta + 7\cos\theta = 5$

$\Rightarrow \dfrac{2t}{1+t^2} + \dfrac{7(1-t^2)}{1+t^2} = 5$

where $t = \tan\left(\dfrac{\theta}{2}\right)$

$\Rightarrow 2t + 7 - 7t^2 = 5 + 5t^2$

$\Rightarrow \tan\dfrac{\theta}{2}$ is root of $12t^2 - 2t - 2$

$= 0$ or $6t^2 - t - 1 = 0$.

24. (*c*) Given, $\dfrac{\sin 3\theta}{2\cos 2\theta + 1} = \dfrac{1}{2}$

$\Rightarrow \dfrac{3\sin\theta - 4\sin^3\theta}{2 - 4\sin^2\theta + 1} = \dfrac{1}{2}$

$\Rightarrow 2\sin\theta\,[3 - 4\sin^2\theta] = (3 - 4\sin^2\theta)$

$\Rightarrow (2\sin\theta - 1)(3 - 4\sin^2\theta) = 0$

$\therefore \sin\theta = \dfrac{1}{2} \Rightarrow \theta = n\pi + (-1)^n \dfrac{\pi}{6}, n \in I$

or $\sin^2\theta = \dfrac{3}{4} \Rightarrow \theta = n\pi \pm \dfrac{\pi}{3}, m \in I$

which does not satisfy the given equation

25. (*c*) $\tan x + \tan\left(x + \dfrac{\pi}{3}\right) + \tan\left(x + \dfrac{2\pi}{3}\right) = 3$

$\Rightarrow 3\tan 3x = 3$

$\Rightarrow \tan 3x = 1$

$\Rightarrow x = \dfrac{n\pi}{3} + \dfrac{\pi}{12}, n \in I$

26. (*b*) Given, $\sin(2A + B) = \dfrac{1}{2}$

$\Rightarrow \sin(2A + B) = \sin\dfrac{\pi}{6}$ or $\sin\dfrac{5\pi}{6}$

But $2A + B = \dfrac{\pi}{6}$ is not possible

$\therefore 2A + B = \dfrac{5\pi}{6}$ ...(i)

Given that
$2B = A + C$ ...(ii)
We know
$A + B + C = \pi$ .... (iii)
From (ii) & (iii) $3B = \pi \Rightarrow B = \frac{\pi}{3}$

From (i) $2A = \frac{5\pi}{6} - \frac{\pi}{3} \Rightarrow A = \frac{\pi}{4}$

From (iii) $C = \pi - (A + B)$
$= \pi - \left(\frac{\pi}{3} + \frac{\pi}{4}\right) \Rightarrow C = \frac{5\pi}{12}$

27. (*d*) $a_1 + a_2 \cos 2x + a_3 \sin^2 x = 0$ for all x

$a_1 + a_2 \cos 2x + a_3 \left(\frac{1-\cos 2x}{2}\right) = 0$

$\left(a_1 + \frac{a_3}{2}\right) + \cos 2x \left(a_2 - \frac{a_3}{2}\right) = 0$

As it is valid for all x,

$a_1 + \frac{a_3}{2} = 0, a_2 - \frac{a_3}{2} = 0$, *Let* $a_3 = k$

$a_1 = -\frac{a_3}{2} = \frac{-k}{2}; a_2 = \frac{a_3}{2} = \frac{k}{2}$

$(a_1, a_2, a_3) \equiv \left(\frac{-k}{2}, \frac{k}{2}, k\right)$ where $k \in R$

$\Rightarrow$ Infinite number of triplets.

28. (*b*) Given $4 \operatorname{cosec}^2(\pi(a + x)) + a^2 - 4a = 0$

$\Rightarrow 4 \operatorname{cosec}^2(\pi(a + x)) = 4a - a^2$

$\Rightarrow \sin^2(\pi(a + x)) = \frac{4}{4a - a^2}$

$0 \le \sin^2 x \le 1$

$0 \le \frac{4}{4a - a^2} \le 1$

$0 \le 4 \le 4a - a^2$

$\therefore \quad 4a - a^2 \ge 4$

$\therefore \quad a = 2$

29. (*d*) Given $|\cos x| = \cos x - 2\sin x$

Case-I : if $\cos x \ge 0$, then $\cos x = \cos x - 2 \sin x$

$\Rightarrow \sin x = 0$

$\Rightarrow x = n\pi$ but $\cos x \ge 0$

so only even integral multiple of π is acceptable

$\therefore x = 2n\pi \; ; \; n \in I$

Case-II :

if $\cos x < 0$, then $-\cos x = \cos x - 2 \sin x - 2 \cos x = -2\sin x$

$\Rightarrow \tan x = 1$

$\Rightarrow x = n\pi + \frac{\pi}{4}$ (x can be in I & III quadrant)

But $\cos x < 0$ So x will be in III quadrant s.t. n should be odd integer

$x = (2m + 1)\pi + \frac{\pi}{4}, m \in I$

30. (*d*) $n|\sin x| = m|\sin x|$

$(n - m)|\sin x| = 0$

As $n > m$

$\Rightarrow n - m \ne 0$

$\Rightarrow |\sin x| = 0$

$\sin x = 0$

$x = n\pi, n \in Z$

In $[0, 2\pi]$, $x = 0, \pi, 2\pi$

No. of solution = 3.

31. (*b*) $\cos(\pi\sqrt{x-4}) \cos(\pi\sqrt{x}) = 1$

Only possibility is

$\cos(\pi\sqrt{x-4}) = 1$ and $\cos(\pi\sqrt{x}) = 1$

Also, $x - 4 \ge 0 \Rightarrow x \ge 4$

and from $\sqrt{x}, x \ge 0 \Rightarrow x \ge 4$

$\cos(\pi\sqrt{x-4}) = 1$ and $\cos(\pi\sqrt{x}) = 1$

$\pi\sqrt{x-4} = 0 \Rightarrow x = 4$

Also, $\cos(\pi\sqrt{x}) = \cos(2\pi) = 1$

Thus, $x = 4$ is only solution.

32. (*a*) $\theta \in [0, 2\pi]$

$r \sin\theta = \sqrt{3} \quad \Rightarrow \quad \sin\theta = \frac{\sqrt{3}}{r}$

$r + 4\sin\theta = 2(\sqrt{3} + 1) \Rightarrow r + \frac{4\sqrt{3}}{r} = 2(\sqrt{3} + 1)$

$r^2 - 2r - 2\sqrt{3}r + 4\sqrt{3} = 0$

$\Rightarrow \quad r = 2 \quad \text{or} \quad 2\sqrt{3}$

$\sin\theta = \frac{\sqrt{3}}{r} \Rightarrow \sin\theta = \frac{\sqrt{3}}{2} \; or \; \frac{\sqrt{3}}{2\sqrt{3}}$

$= \frac{\sqrt{3}}{2}$ or $\frac{1}{2}$

As $\theta \in [0, 2\pi]$

No. of solution = 4.

33. (*b*) $4\sin^4 x + \cos^4 x = 1$

$4\sin^4 x + (1 - \sin^2 x)^2 = 1$

$4\sin^4 x + 1 + \sin^4 x - 2\sin^2 x = 1$

$5\sin^4 x - 2\sin^2 x = 0$

$\sin^2 x (5\sin^2 x - 2) = 0$

$\sin^2 x = 0 \quad \text{or} \quad \sin^2 x = \frac{2}{5}$

$x = n\pi \quad \text{or} \quad x = n\pi \pm \theta,$

where $\sin^2\theta = \frac{2}{5}$ and $\cos^2\theta = \frac{3}{5}$

34. (*a*) $\sin x \cdot \cos x (\cos x - \sin x)^2 (\sin x + \cos x) = \lambda$

$(2\sin x \cos x)(\cos x - \sin x)(\cos^2 x - \sin^2 x) = 2\lambda$

$\sin 2x \cdot (\cos x - \sin x) \cdot \cos 2x = 2\lambda$

$(2\sin 2x \cos 2x)(\cos x - \sin x) = 4\lambda$

$(\sin 4x)\left(\frac{1}{\sqrt{2}}\cos - \frac{1}{\sqrt{2}}\sin x\right) = \frac{4\lambda}{\sqrt{2}}$

$(\sin 4x)\left(\sin\left(\frac{\pi}{4} - x\right)\right) = 2\sqrt{2}\lambda$

As $\lambda > \frac{1}{2\sqrt{2}} \Rightarrow 2\sqrt{2}\lambda > 1 \Rightarrow$ RHS > 1

But LHS ≤ 1

Hences no solution.

Number of solution = 0

35. (*d*) As $x \in \left(\frac{-\pi}{2}, \pi\right)$

$\cos 2x > |\sin x|$

Case 1: $\sin x \ge 0$ i.e. $x \in [0, \pi)$

$\cos 2x > \sin x$

$1 - 2\sin^2 x > \sin x$

$2\sin^2 x + \sin x - 1 < 0$

$(\sin x + 1)(2\sin x - 1) < 0$

$-1 < \sin x < \frac{1}{2}$ ...(i)

Case 2: $\sin x < 0$

$\cos 2x > -\sin x$

$1 - 2\sin^2 x > -\sin x$

$2\sin^2 x - \sin x - 1 < 0$

$(\sin x - 1)(2\sin x + 1) < 0$

$-\frac{1}{2} < \sin x < 1$ ...(ii)

From (i) and (ii), $-\frac{1}{2} < \sin x < \frac{1}{2}$

$\Rightarrow x \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \pi\right)$

36. (*c*) $x \in [0, 2\pi]$

$\sin x \sqrt{8\cos^2 x} = 1$

$\sin^2 x \cdot 8\cos^2 x = 1$

$2(2\sin x \cos x)^2 = 1$

$2(\sin 2x)^2 = 1$

$2\sin^2 2x = 1$

$-\cos 4x = 1$

$\cos 4x = 0$

$4x = (4n + 1) \cdot \frac{\pi}{2}$

$x = (4n + 1)\frac{\pi}{8}, n \in Z$

$x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8} \ldots$

Common diff. $= d = \frac{5\pi}{8} - \frac{\pi}{8} = \frac{4\pi}{8} = \frac{\pi}{2}$

37. (*c*) $a\cos\theta + b\sin\theta = c$

$a\cos\theta = c - b\sin\theta$

$a^2\cos^2\theta = (c - b\sin\theta)^2$

$a^2(1 - \sin^2\theta) = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$(a^2 + b^2)\sin^2\theta - 2bc\sin\theta + c^2 - a^2 = 0$

$$\sin\alpha\sin\beta = \frac{c^2-a^2}{a^2+b^2} \quad \ldots(i)$$

Also, $b\sin\theta = c - a\cos\theta$

$b^2\sin^2\theta = c^2 + a^2\cos^2\theta - 2ac\cos\theta$

$b^2(1-\cos^2\theta) = c^2 + a^2\cos^2\theta - 2ac\cos\theta$

$\cos^2\theta(a^2+b^2) - 2ac\cos\theta + c^2 - b^2 = 0$

$$\cos\alpha\cos\beta = \frac{c^2-b^2}{a^2+b^2}$$

$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$$= \frac{c^2-b^2-c^2+a^2}{a^2+b^2} = \frac{a^2-b^2}{a^2+b^2}$$

38. **(a)** $\log_{\cos x^2}(3-2x) < \log_{\cos x^2}(2x-1)$

$\cos x^2 \in [-1, 1]$, for this equation $\cos x^2 \in (0, 1)$

$\Rightarrow 3-2x > 2x-1$

Also, $3-2x > 0 \Rightarrow x < \frac{3}{2}$

$2x-1 > 0 \Rightarrow x > \frac{1}{2}$

$\therefore \quad x \in \left(\frac{1}{2}, 1\right)$

39. **(b)** Given $\sin\alpha = \sin\beta = \sin\gamma - \sin\delta = k > 0$

and $\alpha > \beta > \gamma$

α is acute

$\beta = \pi - \alpha$

$\gamma = 2\pi + \alpha$

$A \cdot B > 0$

Hence, $\frac{(x+1)^1}{x} > 0$

$$= 4\sin\left(\frac{\alpha}{2}\right) + 3\sin\left(\frac{\pi-\alpha}{2}\right) + 2\sin\left(\frac{2\pi+\alpha}{2}\right) + \sin\left(\frac{3\pi-\alpha}{2}\right)$$

$$= 4\sin\left(\frac{\alpha}{2}\right) + 3\sin\left(\frac{\pi}{2}-\frac{\alpha}{2}\right) + 2\sin\left(\pi+\frac{\alpha}{2}\right) + \sin\left(\frac{3\pi}{2}-\frac{\alpha}{2}\right)$$

$$= 4\sin\left(\frac{\alpha}{2}\right) + 3\cos\left(\frac{\alpha}{2}\right) - 2\sin\left(\frac{\alpha}{2}\right) - \cos\left(\frac{\alpha}{2}\right)$$

$$= 2\left(\sin\left(\frac{\alpha}{2}\right) + \cos\left(\frac{\alpha}{2}\right)\right)$$

$$= 2\left(\sqrt{\left(\sin\left(\frac{\alpha}{2}\right) + \cos\left(\frac{\alpha}{2}\right)\right)^2}\right)$$

$= 2(\sqrt{1+\sin\alpha}) = 2\sqrt{1+k}$

40. **(b)** $\Rightarrow \log_{\cos x}\left[\frac{\sqrt{3}}{2}\sin x\right] = 2 + \log_{\cos x}(\tan x)$

$$\log_{\cos x}\left[\frac{\sqrt{3}\sin x}{\tan x \cdot 2}\right] = 2$$

$$\Rightarrow \log_{\cos x}\left[\frac{\sqrt{3}}{2}\cos x\right] = 2$$

$$\log_{\cos x}\left(\frac{\sqrt{3}}{2}\right) + 1 = 2$$

$$\therefore \cos x = \frac{\sqrt{3}}{2}$$

$$\Rightarrow x = \frac{\pi}{6}$$

41. **(b)** $\log_9(a) = \log_{12} b = \log_{16}(a+b) = R$

$\log_q(a) = k$

$\Rightarrow a = 9^k = 3^{2k} \Rightarrow \sqrt{a} = 3^k$

$\log_{12} b = k$

$\Rightarrow b = 12^k = 3^k 4^k \Rightarrow b = \sqrt{a}\,2^{2k}$

$\log_{1b}(a+b) = k$

$$a+b = 16^k = 2^{4k} = \left(\frac{b}{\sqrt{a}}\right)^2$$

$$\Rightarrow (a+b)^2 = \frac{b^2}{a}$$

$\Rightarrow a^2 + ab = b^2$

$$\Rightarrow \left(\frac{b}{a}\right)^2 - \left(\frac{b}{a}\right) - 1 = 0$$

$$= \frac{b}{a} = \frac{1\pm\sqrt{5}}{2}$$

As, $a, b > 0$

$$\therefore \frac{b}{a} = \frac{1+\sqrt{5}}{2}$$

$$\therefore 2\sin\theta = \frac{1+\sqrt{5}}{2}$$

$$\sin\theta = \frac{1+\sqrt{5}}{4}$$

$$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

42. **(a)** $8\sin\theta\sin 3\theta\left(\frac{\sin 8\theta}{4\sin 2\theta}\right) = \cos 6\theta$

$\sin 3\theta\sin 8\theta = \cos 6\theta\cos\theta$

$\cos 5\theta - \cos 11\theta = \cos 7\theta + \cos 5\theta$

$\cos 7\theta + \cos 11\theta = 0$

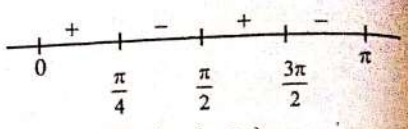
$2\cos 9\theta \cdot \cos 2\theta = 0$

43. **(a)** $\sin x \cdot \cos^3 x > \cos x \cdot \sin^3 x$

$\Rightarrow \sin x \cdot \cos x(\cos^2 x - \sin^2 x) > 0$

$\Rightarrow \sin x \cdot \cos x \cdot \cos 2x > 0$

$\Rightarrow \cos x \cdot \cos 2x > 0$



$\therefore x \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

44. **(d)** $|\sin|x|| = x + |x|$

when $x > 0$

$|\sin|x|| = 2x$

$\Rightarrow |\sin x| = 2x$

$\Rightarrow \sin x = +2x \,|\, -2x \Rightarrow x = 0$

and when $x < 0$

$|\sin(-x) = 0$

$|\sin x| = 0$

$\Rightarrow x = x\pi$

$x = -\pi, -2\pi$

45. **(b)** $8\sin x\cos^5 x - 8\sin^5 x\cos x = 1$

$\Rightarrow 8\sin x\cos x\left(\cos^4 x - \sin^4 x\right) = 1 \Rightarrow 4$

$\cdot 2\sin x\cos x\left[(\cos^2 x + \sin^2 x)\left(\cos^2 x - \sin^2\right.\right.$

$\Rightarrow 4\sin 2x[1(\cos 2x)] = 1$

$\Rightarrow 2 \cdot 2\sin 2x\cos 2x = 1$

$\Rightarrow \sin 4x = \frac{1}{2}$

$\Rightarrow \sin 4x = \sin\left(\frac{\pi}{6}\right)$ and $\sin 4x$

$= \sin\left(\frac{5\pi}{6}\right)$

$\Rightarrow 4x = \frac{\pi}{6}$ and $4x = \frac{5\pi}{6}$

$\Rightarrow x = \frac{\pi}{24}$ and $x = \frac{5\pi}{24}$

$\Rightarrow x = 7\frac{1}{2^o}$ and $x = 37\frac{1}{2^o}$

46. **(a)** $\tan^2 2x = 2\tan 2x \cdot \tan 3x + 1$

$\Rightarrow \tan^2 2x - 2\tan 2x\tan 3x = 1$

$\Rightarrow \tan 2x[\tan 2x - \tan 3x] = 1 + \tan 2x\tan 3x$

$$\Rightarrow \tan 2x\left[\frac{\tan 2x - \tan 3x}{1+\tan 2x\tan 3x}\right] = 1$$

$\Rightarrow \tan 2x(\tan(-x)) = 1$

$\Rightarrow \tan x\tan 2x = -1$

$$\Rightarrow \tan x\frac{(2\tan x)}{1-\tan^2 x} = -1$$

$\Rightarrow 2\tan^2 x = \tan^2 x - 1$

$\Rightarrow \tan^2 x = -1$

$\therefore$ no solution

47. **[5]** $p\sin\theta+\cos 2\theta=2p-7$

$\Rightarrow \quad p\sin\theta+1-2\sin^2\theta=2p-7$

$2\sin^2\theta-p\sin\theta+(2p-8)=0$

$\sin x=\dfrac{p\pm(p-8)}{2}=\dfrac{p-4}{2},$

4 (not possible)

$\Rightarrow 1\le\dfrac{p-4}{2}\le 1$

$\Rightarrow p\in[2,6]$

48. **[2]** $4\sin^2 x-8\sin x+3\le 0$

$\Rightarrow(2\sin x-3)(2\sin x-1)\ge 0$

But we know $-1\le\sin x\le 1\Rightarrow 2\sin x-3<0$

$\Rightarrow 2\sin x-1\ge 0\Rightarrow\dfrac{1}{2}\le\sin x\le 1$

Hence solution in given interval is

$x\in\left[\dfrac{\pi}{6},\dfrac{5\pi}{6}\right]$

49. **[1]**

$3^{\sin 2x+2\cos^2 x}+3^{1-\sin 2x+2\left(1-\cos^2 x\right)}=28$

$3^{\sin 2x+2\cos^2 x}+\left(\dfrac{3^3}{3^{\sin 2x}-2\cos^2 x}\right)=28$

Let $3^{\sin 2x+2\cos^2 x}=t$

$\therefore t+\left(\dfrac{27}{t}\right)=28$

$t^2+27=28t$

$t^2-28t+27=0$

$t^2-27t-t+27=0$

$t(t-27)-1(t-27)=0$

$(t-27)(t-1)=0$

$\therefore t=27$ or $t=1$

if $t=27$, then $\sin^2x+2\cos^2x=3$

Which is not possible

If $t=1$, then $\sin^2x+2\cos^2x=1$

at $X=\left(\dfrac{\pi}{2}\right)$

50. **[5]** Given, $3\sin x+4\cos x=5$

$\Rightarrow\left(\dfrac{3}{5}\right)\sin x+\left(\dfrac{4}{5}\right)\cos x=1$ From general triangle $\cos 53°=\dfrac{3}{5}$ and

$\cos 53°\sin x+\sin 53°\cos x=1$

Using formula,

$\sin A\cos B+\cos A\sin B=\sin(A+B)$

We have $\sin(x+53°)=1$

$\sin(x+53°)=\sin 90°\left(\because x\in\left(0,\dfrac{\pi}{2}\right)\right)$

$x+53°=90°\ x=37°$ to find,

$2\sin x+\cos x+4\tan x$

$=2\sin x+\cos x+4\cdot\left(\dfrac{\sin x}{\cos x}\right)$

$=2\sin 37°+\cos 37°+4\cdot\left(\dfrac{\sin 37°}{\cos 37°}\right)$

$=2\cdot\dfrac{3}{5}+\dfrac{4}{5}+4\cdot\dfrac{3}{4}=\dfrac{6}{5}+\dfrac{4}{5}+3$

$=\dfrac{10}{5}+3=2+3=5$

Parikshit (JEE Advanced Level)

1. **(a)** $2^{\sin x}+2^{\cos x}\le 2^{1-\frac{1}{\sqrt{2}}}\ x\in R$

By AM ≥ GM in equality

$\dfrac{2^{\sin x}+2^{\cos x}}{2}\ge 2^{\frac{1}{2}(\sin x+\cos x)}$

$2^{\sin x}+2^{\cos x}\ge 2.2^{\frac{1}{2}(\sin x+\cos x)}$

$=\sqrt{2}\sin\left(x+\dfrac{\pi}{4}\right)$

From equation

R.H.S

$\Rightarrow 2.2^{\frac{1}{2}}(\sin x+\cos x)=2^{1+\frac{1}{2}\left(\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\right)}$

$\ge 2^{1-\frac{1}{\sqrt{2}}}$

$-1\le\sin\left(x+\dfrac{\pi}{4}\right)\le 1$

$2^{\sin x}+2^{\cos x}\ge 2^{1+\frac{1}{\sqrt{2}}\sin\left(x+\frac{\pi}{4}\right)}\ge 2^{1-\frac{1}{\sqrt{2}}}$

2. **(a)** $\sqrt{2}\left(\dfrac{\sin x}{\sqrt{2}}+\dfrac{\cos x}{\sqrt{2}}\right)$

$\dfrac{10\sin x}{\sqrt{5}\sin x}=\dfrac{10(4\sin^2 x+1)}{10\sin x}$

$\Rightarrow 4\sin^2 x-2\sqrt{5}\sin x+1=0$

$\Rightarrow\sin x=\dfrac{\sqrt{5}\pm 1}{4}$

$\Rightarrow x=\dfrac{\pi}{10},\dfrac{3\pi}{10}$

$|\alpha-\beta|=\dfrac{\pi}{5}$

3. **(d)** Given that

$[1+\sin x]+[1-\cos x]=\sin x$

Using $[x\pm n]=[x]\pm n\forall n\in Z.$

$\Rightarrow 1+[\sin x]+1+[-\cos x]=\sin x$

$\Rightarrow 2+[\sin x]+[-\cos x]=\sin x$

$\Rightarrow 2+[-\cos x]=\sin x-[\sin x]$

Now, $\sin x-[\sin x]\in[0,1)$

And, $2-[-\cos x]\in\{1,2,3\}$

Hence, the equation will have no solution.

4. **(c)** Given equation $8^{2\sin^2\theta}+8^{2-2\sin^2\theta}=16$

$y+\dfrac{64}{y}=16; y^2-16y+64=0$

Then, $y=8\ \therefore\ \sin^2\theta=\dfrac{1}{2}$

Taken

$n(S)+\Sigma_{\theta\in S}\left(\sec\left(\dfrac{\pi}{4}+2\theta\right)\operatorname{cosec}\left(\dfrac{\pi}{4}+2\theta\right)\right)$

$=4+(-2)4=-4$

5. **(c)** $\left(2\cos\beta-\sqrt{3}\right)(2\sin\alpha+1)=0$

$\therefore\cos\beta=\dfrac{\sqrt{3}}{2}$ or $\sin\alpha=-\dfrac{1}{2}$

$\beta=\dfrac{\pi}{6},\dfrac{11\pi}{6}$ or $\alpha=\dfrac{7\pi}{6},\dfrac{11\pi}{6}$

$\therefore L=\alpha-\beta=\begin{cases}\text{maximum where}\quad \alpha=\dfrac{11\pi}{6}\quad \beta=\dfrac{\pi}{6}\\ \text{maximum where}\quad \alpha=\dfrac{7\pi}{6}\quad \beta=\dfrac{11\pi}{6}\end{cases}$

$L_{\text{maximum}}=\dfrac{5\pi}{3}; L_{\text{minimum}}=\dfrac{-2\pi}{3}$

6. **(b)** $\dfrac{\dfrac{1}{1-\sin x}}{\dfrac{1}{1+\sin x}}=\dfrac{1+\sin x}{1-\sin x}=4\cos^2 x$

$=\dfrac{1-\sin^2 x}{(1-\sin x)}=4\cos^2 x$

$\cos^2 x=0$ or $1-\sin x=\pm\dfrac{1}{2}\rightarrow\sin x=\dfrac{1}{2}$

$x=(2n+1)\dfrac{\pi}{2}$ or $x=n\pi+(-1)^n\dfrac{\pi}{6}$

7. **(a, c)**

If we consider $\tan\alpha/2 = x$ and $\tan\beta/2 = y$, then

$2(\cos\beta-\cos\alpha)+\cos\alpha\cos\beta=1$

$\Rightarrow 2\left[\dfrac{1-y^2}{1+y^2}-\dfrac{1-x^2}{1+x^2}\right]=1-\dfrac{(1-x^2)(1-y^2)}{(1+x^2)(1+y^2)}$

$\Rightarrow 2\left[(1+x^2)(1-y^2)-(1-x^2)(1+y^2)\right]$

$=(1+x^2)(1+y^2)-(1-x^2)(1-y^2)$

$\Rightarrow 4(x^2-y^2)=2(x^2+y^2)$

$x^2=3y^2\Rightarrow x=\pm\sqrt{3}y\Rightarrow\tan\dfrac{\alpha}{2}\pm\sqrt{3}\tan\dfrac{\beta}{2}=0$

8. **(a, b, c)**

$|\cos x|^{\sin^2 x-\frac{3}{2}\sin x+\frac{1}{2}}=1$

$\sin^2 x-\frac{3}{2}\sin x+\frac{1}{2}=0$ & $\cos x\neq 0$

$\Rightarrow 2\sin^2 x-3\sin x+1=0$

$(2\sin x-1)(\sin x-1)$

$\Rightarrow \sin x=\frac{1}{2}$ or 1

$\sin x=\frac{1}{2}$

$x=n\pi+(-1)^n\frac{\pi}{6}$

or $\cos^2 x=1$

$x=n\pi$

9. **(a, b, c, d)** Given that

$\cos x\cos y=\frac{\sqrt{3}}{4}$ ...(i)

$\sin x\sin y=\frac{\sqrt{3}}{4}$ ... (ii)

adding equation (i) and (ii)

$\cos x\cos y+\sin x\sin y=\frac{\sqrt{3}}{2}$

$\Rightarrow \cos(x-y)=\frac{\sqrt{3}}{2}$

$\Rightarrow \cos(x-y)=\cos\frac{\pi}{6}$

$\Rightarrow x-y=2n\pi\pm\frac{\pi}{6}$ ... (iii)

subtracting (ii) from (i)

$\cos x\cos y-\sin x\sin y=0$

$\Rightarrow \cos(x+y)=0$

$\Rightarrow x+y=k\pi+\frac{\pi}{2}$ ...(iv)

adding (iii) & (iv)

$2x=(2n+k)\pi+\frac{\pi}{2}\pm\frac{\pi}{6}$

$\Rightarrow x=(2n+k)\frac{\pi}{2}+\frac{\pi}{4}\pm\frac{\pi}{12}$

$\Rightarrow x=(2n+k)\frac{\pi}{2}+\frac{\pi}{3}$ or

$(2n+k)\frac{\pi}{2}+\frac{\pi}{6}$

Now,

$y=k\pi+\frac{\pi}{2}-x$

$y=k\pi+\frac{\pi}{2}-(2n+k)\frac{\pi}{2}-\frac{\pi}{3}$

$=(k-2n)\frac{\pi}{2}+\frac{\pi}{6}$

or $y=k\pi+\frac{\pi}{2}-(2n+k)\frac{\pi}{2}-\frac{\pi}{6}$

$=(k-2n)\frac{\pi}{2}+\frac{\pi}{3}$

10. **(c,d)** for $0<\theta<\frac{\pi}{2}$

$\sum_{m=1}^{6}\operatorname{cosec}\left(\theta+\frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta+\frac{m\pi}{4}\right)=4\sqrt{2}$

$\Rightarrow \sum_{m=1}^{6}\frac{1}{\sin\left(\theta+\frac{(m-1)\pi}{4}\right)\sin\left(\theta+\frac{m\pi}{4}\right)}=4\sqrt{2}$

$\Rightarrow \sum_{m=1}^{6}\frac{\sin\left[\theta+\frac{m\pi}{4}-\left(\theta+\frac{(m-1)\pi}{4}\right)\right]}{\sin\frac{\pi}{4}\left\{\sin\left(\theta+\frac{(m-1)\pi}{4}\right)\sin\left(\theta+\frac{m\pi}{4}\right)\right\}}=4\sqrt{2}$

$\Rightarrow \sum_{m=1}^{6}\frac{\cot\left(\theta+\frac{(m-1)\pi}{4}\right)-\cot\left(\theta+\frac{m\pi}{4}\right)}{1/\sqrt{2}}=4\sqrt{2}$

$\Rightarrow \sum_{m=1}^{6}\left[\cot\left(\theta+\frac{(m-1)\pi}{4}\right)-\cot\left(\theta+\frac{m\pi}{4}\right)\right]=4$

$\Rightarrow \cot(\theta)-\cot\left(\theta+\frac{\pi}{4}\right)+\cot\left(\theta+\frac{\pi}{4}\right)-\cot\left(\theta+\frac{2\pi}{4}\right)+$

$\ldots+\cot\left(\theta+\frac{5\pi}{4}\right)-\cot\left(\theta+\frac{6\pi}{4}\right)=4$

$\Rightarrow \cot\theta-\cot\left(\frac{3\pi}{2}+\theta\right)=4$

$\Rightarrow \cot\theta+\tan\theta=4\Rightarrow\tan^2\theta-4\tan\theta+1=0$

$\Rightarrow (\tan\theta-2)^2-3=0$

$\Rightarrow (\tan\theta-2+\sqrt{3})(\tan\theta-2-\sqrt{3})=0$

$\Rightarrow \tan\theta=2-\sqrt{3}$ or $\tan\theta=2+\sqrt{3}$

$\Rightarrow \theta=\frac{\pi}{12};\ \theta=\frac{5\pi}{12}$

$\because \theta\in\left(0,\frac{\pi}{2}\right)$

11. **(a,d)** $2\sin x+3\cos y=3$ ...(i)

$3\sin y+2\cos x=4$ ...(ii)

squaring and adding

$\Rightarrow \sin(x+y)=1$

$\therefore x+y=(4n+1)\frac{\pi}{2}$, $n\in I$

Now, Let $x+y=\frac{\pi}{2}$

$y=\frac{\pi}{2}-x$

Put in equation (i) and (ii)

We get $\sin x=\frac{3}{5}$ and $\sin y=\frac{4}{5}$

$\therefore y>x$

12. **(a,b,c,d)**

$2^{(2\sin^2 x-3\sin x+1)}+2^{3-(2\sin^2 x-3\sin x+1)}=9$

Let $2^{(2\sin^2 x-3\sin x+1)}=t \Rightarrow t+\frac{8}{t}=9$

$\Rightarrow t^2-9t+8=0\Rightarrow t=1,8$

$\Rightarrow 2\sin^2 x-3\sin x+1=0$

or $2\sin^2 x-3\sin x+1=0$

$\Rightarrow \sin x=-\frac{1}{2},\ \sin x=\frac{1}{2},\sin x=1$

13. **(a,b)**

$\frac{\sqrt{5}-1}{4}\times\frac{1}{\sin\theta}+\frac{\sqrt{10+2\sqrt{5}}}{4}\times\frac{1}{\cos\theta}=2$

$\frac{\sqrt{5}-1}{4}\cos\theta+\frac{\sqrt{10+2\sqrt{5}}}{4}\times\sin\theta=2\sin\theta\cdot\cos\theta$

$\sin\left[\theta+\frac{\pi}{10}\right]=\sin 2\theta$

$\theta+\frac{\pi}{10}=2\theta,\theta=\frac{\pi}{10}=18°$

or $\sin\left(\theta+\frac{\pi}{10}\right)=\sin(\pi-2\theta)$

$\Rightarrow \theta+\frac{\pi}{10}=\pi-2\theta$

$\Rightarrow 3\theta=\pi-\frac{\pi}{10}=\frac{9\theta}{10}\quad\Rightarrow\theta=\frac{3\pi}{10}=54°$

14. **(b,c)** $(t-[|\sin x|])!=3!5!7!$

if $x=n\pi+\frac{\pi}{2}(n\in I)$

then $(t-1)!=3!5!7!$

$\Rightarrow (t-1)!=10!\ \Rightarrow\ t-1=10\ \Rightarrow\ t=11$

If $x\neq n\pi+\frac{\pi}{2}(n\in I)$, then

$(t-0)!=10!\ \Rightarrow t=10$

15. **(a,b,c,d)**

(a) For $x\in\left(0,\frac{\pi}{4}\right)$, $\tan x<\cot x$

Also $\ln(\sin x)<0$

$\Rightarrow (\tan x)^{\ln(\sin x)}>(\cot x)^{\ln(\sin x)}$

(b) For $x\in\left(0,\frac{\pi}{2}\right)$, $\operatorname{cosec} x\geq 1$

$\Rightarrow \ln(\operatorname{cosec} x)\geq 0\Rightarrow 4^{\ln\operatorname{cosec} x}<5^{\ln(\operatorname{cosec} x)}$

16. **(a, b, c, d)**

$2\sin\frac{x}{2}\cdot\cos^2 x+\sin^2 x=2\sin\frac{x}{2}\cdot\sin^2 x+\cos^2 x$

$\Rightarrow 2\sin\frac{x}{2}(\cos^2 x-\sin^2 x)=\cos^2 x-\sin^2 x$

$\Rightarrow \sin 2x=1$ or -1 and $\cos x=1-2\sin^2\frac{x}{2}$

$=1-2\left(\frac{1}{4}\right)=\frac{1}{2}$

$\therefore \cos 2x=2\cos^2 x-1=2\times\frac{1}{4}-1=-\frac{1}{2}$

17. **(a, c)**

$5\sin^2$

Case-I

$\therefore x$

Case -

$\therefore$ 5ta

$\Rightarrow$ ta

18. **(b,c)**

using

$4\cos^3\theta-3$

$+4\sin^2\theta\cos$

$\Rightarrow$ cos

$\Rightarrow (\cos\theta-\sin$

[Using

$=(a-b)\left(a^2\right.$

$\Rightarrow$ (co

$\Rightarrow\theta\in\left(2n\pi-\right.$

19. **(a)**

$t=1$

$(t-1)$

$t=1$

$\sin x$

so $x=$

20. **(b)**

$\sin x(2$

$\sin x($

-1 (c

$\Rightarrow$

$\therefore$

$\Rightarrow$

21. **(c)**

$\Rightarrow$

$\Rightarrow$

Let

u^2-

$u=2$

$\therefore$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

17. (a, c)

$5\sin^2 x + \sqrt{3}\sin x\cos x + 6\cos^2 x = 5$

Case-I $\cos x = 0 \Rightarrow 5 + 0 + 0 = 5$

$\therefore \quad x = n\pi + \frac{\pi}{2}$

Case -II $\cos x \neq 0$

$\therefore \quad 5\tan^2 x + \sqrt{3}\tan x + 6 = 5(1 + \tan^2 x)$

$\Rightarrow \quad \tan x = -\frac{1}{\sqrt{3}}$

18. (b,c) $\cos 3\theta + \sin 3\theta + (2\sin 2\theta - 3)(\sin\theta - \cos\theta) > 0$

using the formula of $\cos 3\theta, \sin 3\theta \sin 2\theta$

$4\cos^3\theta - 3\cos\theta + 3\sin\theta - 4\sin^3\theta$

$+4\sin^2\theta\cos\theta - 3\sin\theta - 4\sin\theta\cos^2\theta + 3\cos\theta$

$\Rightarrow \cos^3\theta - \sin^3\theta + \cos\theta - \sin\theta > 0$

$\Rightarrow (\cos\theta - \sin\theta)(1 + \sin\theta\cos\theta) + \cos\theta - \sin\theta >$ [Using formula $(a^3 - b^3) = (a-b)\left(a^2 + ab + b^2\right)$ and $\sin^2\theta + \cos^2\theta = 1$]

$\Rightarrow (\cos\theta - \sin\theta)(2 + \sin\theta\cos\theta) > 0$

$\Rightarrow \theta \in \left(2n\pi - \frac{\pi}{4}, 2n\pi + \frac{\pi}{6}\right), \left(2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3}\right)$

19. (a) Given $\sin x + \cos x = 1 + \sin x \cos x$

$t = 1 + \frac{t^2 - 1}{2}$

$(t-1)^2 = 0$

$t = 1$

$\sin x + \cos x = 1$

so $x = 2n\pi$ and $2n\pi + \frac{\pi}{2}$

20. (b) $(\cos x - \sin x)(2\sin x + 1) + 2\cos x = 0$

$\sin x(2\cos x - 1) + 3\cos x - 2(1 - \cos^2 x) = 0$

$\sin x(2\cos x - 1) + 2\cos x(\cos x + 2) - 1(\cos x + 2) = 0$

$\Rightarrow \quad (\sin x + \cos x + 2)(2\cos x - 1) = 0$

$\therefore \quad \cos x = \frac{1}{2}$

$\Rightarrow \quad x = 2n\pi \pm \frac{\pi}{3}$

21. (c) Given $\sin^4 x + \cos^4 x = \sin x \cos x$

$\Rightarrow \quad 1 - \frac{(t^2-1)^2}{2} = t^2 - 1$

$\Rightarrow \quad t^4 - t^2 - 2 = 0$

Let $t^2 = u$

$u^2 - u - 2 = 0$

$u = 2 \quad$ or $\quad u \neq -1$

$\therefore \quad t = \pm\sqrt{2}$

$\Rightarrow \quad \sin x + \cos x = \pm\sqrt{2}$

$\Rightarrow \quad \sin(x - \pi/4) = 1$

$\Rightarrow \quad x = n\pi + \frac{\pi}{4}$

22. (a) A-(q,s); B-(p); C-(q); D-(r)

(A) $\cos 3x.\cos^3 x + \sin 3x.\sin^3 x = 0$

$\Rightarrow (4\cos^3 x - 3\cos x)\cos^3 x + (3\sin x - 4\sin^3 x)\sin^3 x = 0$

$\Rightarrow 4(\cos^6 x - \sin^6 x) - 3(\cos^4 x - \sin^4 x) = 0$

$\Rightarrow 4(\cos^6 x - \sin^6 x) - 3(\cos^2 x - \sin^2 x) = 0$

$\Rightarrow (\cos^2 x - \sin^2 x)(1 - 4\cos^2 x \sin^2 x) = 0$

$\Rightarrow \cos 2x = 0$ or $\sin^2 2x = 1 = \sin\frac{\pi}{2}$

$\Rightarrow \quad 2x = 2n\pi \pm \frac{\pi}{2}$ or $2x = n\pi \pm \frac{\pi}{2}$

$\Rightarrow \quad x = n\pi \pm \frac{\pi}{4}$ or $x = \frac{n\pi}{2} \pm \frac{\pi}{4} \Rightarrow$ (q),(s)

(B) $\Rightarrow \sin 3\alpha = 2\sin\alpha\left[\frac{\cos 2\alpha - \cos 2x}{1}\right]$

$\Rightarrow 3\sin\alpha - 4\sin^3\alpha = 2\sin\alpha\left[\frac{\cos 2\alpha - \cos 2x}{1}\right]$

($\because \sin\alpha \neq 0$ if $\alpha \neq n\pi$)

$\therefore 3 - 4\sin^2\alpha = 2\cos 2\alpha - 2\cos 2x$

$\Rightarrow 3 - 2(1 - \cos 2\alpha) = 2\cos 2\alpha - 2\cos 2x$

$\Rightarrow 3 - 2 + 2\cos 2\alpha = 2\cos 2\alpha - 2\cos 2x$

$\Rightarrow \quad \cos 2x = -\frac{1}{2}$

$x = n\pi \pm \frac{\pi}{3} \Rightarrow$ (p)

(C) $|2\tan x - 1| + |2\cot x - 1| = 2$

$\Rightarrow$ sum of two non-negative quantities = 2

$\Rightarrow 2\tan x - 1 = 1$

$\Rightarrow \tan x = 1$

$\Rightarrow x = n\pi + \pi/4$

$\Rightarrow$ (q)

(D) $\sin^{10} x + \cos^{10} x = \frac{29}{16}\cos^4 2x$

$\Rightarrow \left(\frac{1 - \cos 2x}{2}\right)^3 + \left(\frac{1 + \cos 2x}{2}\right)^3 = \frac{29}{16}\cos^4 2x$

$\Rightarrow \frac{1}{16}(1 + 10\cos^2 2x + 5\cos^4 2x) = \frac{29}{16}\cos^4 2x$

$\Rightarrow 24\cos^4 2x - 10\cos^2 2x - 1 = 0$

$\Rightarrow (12\cos^2 2x + 1)(2\cos^2 2x - 1) = 0$

$\Rightarrow \cos^2 2x = \frac{1}{2}$ or $\cos^2 2x = \cos^2\left(\frac{\pi}{4}\right)$

$\Rightarrow 2x = n\pi \pm \frac{\pi}{4}$ or $x = \frac{n\pi}{2} \pm \frac{\pi}{8} \Rightarrow$ (r)

23. **[5]** $(\sin x - 1)^3 + (\cos x - 1)^3 + (\sin x)^3 = (2\sin x + \cos x - 2)^3$

Let $\sin x - 1 = a$, $\cos x - 1 = b$, $\sin x = c$

$\Rightarrow a^3 + b^3 + c^3 = (a + b + c)^3$

$\Rightarrow a + b = 0$ or $b + c = 0$ or $c + a = 0$

$\Rightarrow \sin x + \cos x = 2$ or $\sin x + \cos x = 1$ or $\sin x = \frac{1}{2}$

(i) $\sin x = \frac{1}{2}$

$\therefore \quad x = \frac{\pi}{6}, \frac{5\pi}{6}$ $\quad$ (as $x \in [0, 2\pi]$)

(ii) $\sin x + \cos x = 1$

$\therefore \quad x = 0, \frac{\pi}{2}, 2\pi$ $\quad$ (as $x \in [0, 2\pi]$)

(iii) $\sin x + \cos x = 2$

No solution.

24. **[4]** Here,

$1 \leq |\sin(2x)| + |\cos(2x)| \leq \sqrt{2}$ and $|\sin(y)| \leq 1$

It is possible only when $|\sin(y)| = 1$

$\Rightarrow \sin(y) = \pm 1 \quad \Rightarrow \quad y = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}.$

Thus, the number of values of y is 4.

25. **[8]** $2\sin^2 x + 4\sin^2 x\cos^2 x = 2$

$2\sin^4 x - 3\sin^2 x + 1 = 0$

$\Rightarrow (\sin^2 x - 1)(2\sin^2 x - 1) = 0$

$\sin x = \pm\frac{1}{\sqrt{2}}, \pm 1$

$\sin 2x + \cos 2x = \tan x$

$\frac{2\tan x}{1 + \tan^2 x} + \frac{1 - \tan^2 x}{1 + \tan^2 x} = \tan x$

$\Rightarrow \tan^3 x + \tan^2 x - \tan x - 1 = 0$

$(\tan x + 1)^2(\tan x - 1) = 0$

$2\cos^2 x + \sin x \leq 2$

$\sin x(2\sin x - 1) \geq 0$

26. **[4]**

$1 - \sin^2 x + \frac{\sqrt{3}+1}{2}\sin x - \frac{-\sqrt{3}}{4} - 1 = 0$

$\sin^2 x - \frac{-\sqrt{3}}{4}\sin x + \frac{-\sqrt{3}}{4} = 0;$

$4\sin^2 x - 2\sqrt{3}\sin x - 2\sin x + \sqrt{3} = 0$

On solving we get

$\sin x = 1/2;\ \frac{\sqrt{3}}{2} = (\pi/6, 5\pi/6; \pi/3, 2\pi/3]$

27. **[15]**

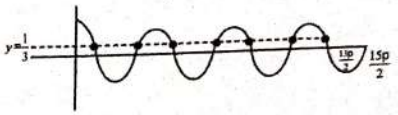
$2\tan^2 x - 5\sec x - 1 = 0$

$\Rightarrow 2(\sec^2 x - 1) - 5\sec x - 1 = 0$

$\Rightarrow 2\sec^2 x - 5\sec x - 3 = 0$

$\Rightarrow \sec x = \frac{6}{2}, \frac{-1}{2} = 3, \frac{-1}{2}$

$\Rightarrow \sec x = 3\left(\sec x \neq \frac{-1}{2}\right)$



$\Rightarrow \cos x = \frac{1}{3}$

$\Rightarrow$ 7 solutions in $\left[0, \frac{15\pi}{2}\right]$

$\therefore n = 15.$

28. **[0]** $\sin^2\left(x-\frac{\pi}{4}\right)=\frac{(\sin x-\cos x)^2}{2}$.

$\cos 2x = (\cos x+\sin x)(\cos x-\sin x)$

$\Rightarrow \frac{\sin^2(x-\pi/4)}{\cos 2x}=\frac{1}{2}\frac{\cos x-\sin x}{\cos x+\sin x}$

$=\frac{1}{2}\frac{1-\tan x}{1+\tan x} =-\frac{1}{2}\tan\left(x-\frac{\pi}{4}\right)$

Let, $\tan\left(x-\frac{\pi}{4}\right)=t$

$2^t-2\cdot\left(\frac{1}{4}\right)^{-t/2}+1=0$

$\Rightarrow 2^t-2\cdot 2^t+1=0\Rightarrow 2^t=1$

$t=0\Rightarrow x=\frac{\pi}{4}$

no solution

29. **[6]** $\sin x+\sin y=\sin(x+y)$

We can write

$2\sin\frac{1}{2}(x+y)\cos\frac{1}{2}(x-y)$

$=2\sin\frac{1}{2}(x+y)\cos\frac{1}{2}(x+y)$

or $2\sin\frac{1}{2}(x+y)\left\{\cos\frac{1}{2}(x-y)-\cos\frac{1}{2}(x+y)\right\}=0$

or $2\sin\frac{1}{2}(x+y)\cdot 2\sin\frac{1}{2}x\sin\frac{1}{2}y=0$

Either

$2\sin\frac{x+y}{2}=0$ or $\sin\frac{x}{2}=0$ or $\sin\frac{y}{2}=0$

$\Rightarrow x+y=0, x=0, y=0$ and $|x|+|y|=1$

$\Rightarrow x+y=1, x-y=1$

$x+y=-1, x-y=-1$

When $x+y=0$, We have to reject $x+y=1$ or -1 and solve it with $x-y=1$ or $x-y=1$ which gives $\left(\frac{1}{2},-\frac{1}{2}\right)$ or $\left(-\frac{1}{2},\frac{1}{2}\right)$ as the possible solution.

Again solving with $x=0$, we get $(0,\pm1)$ and by solving with $y=0$, we get $(\pm1, 0)$ as the other solution. Thus we have 6 pairs of solutions for x and y

30. **[765]** $\sin\sqrt{x}=-1 \Rightarrow \sin\sqrt{x}=-\sin\frac{\pi}{2}$

$\sqrt{x}=2n\pi-\frac{\pi}{2} \quad n\in I^+$

$\sqrt{x}=(4n-1)\frac{\pi}{2} \quad n+I^+$

$x=(4n-1)^2\frac{\pi^2}{4} \quad n\in I^+$

$\therefore$ roots less than $100\,\pi^2$ are

$9\frac{\pi^2}{4}, 49\frac{\pi^2}{4}\ldots(19)^2\frac{\pi^2}{4}$

sum $=\sum_{n=1}^{5}(4n-1)^2\frac{\pi^2}{4}$

$=\frac{\pi^2}{4}\sum_{n=1}^{5}\left(16n^2-8n+1\right)$

$=\frac{\pi^2}{4}\left[16\times\frac{5\times5+1)(10+1)}{6}-\frac{8\times5\times6}{2}+5\right]$

$=\frac{\pi^2}{4}[880-120+5] =\frac{765\pi^2}{4}$

sum of square roots of these roots

$=\sum_{n=1}^{5}(4n-1)\frac{\pi}{2} =\frac{\pi}{2}\sum_{n=1}^{5}(4n-1)$

$=\frac{\pi}{2}\left[4\times\frac{5\times6}{2}-5\right] =\frac{\pi}{2}\cdot[55]=\frac{55\pi}{2}$

$\cos\sqrt{x}=0$

$\sqrt{x}=(2n+1)\frac{\pi}{2} \quad n\in I^+\cup\{0\}$

$x=(2n+1)^2\frac{\pi^2}{4} \quad n\in I^+U\{0\}$

$x=\frac{\pi^2}{4},\frac{9\pi^2}{4},\frac{25\pi^2}{4},\frac{49\pi^2}{4}\ldots$

$\therefore$ all the roots of $\cos\sqrt{x}=0$ are not the same as those of $\sin\sqrt{x}=-1$

PYQ's (Past Year Questions)

1. ***(b)*** $S=\left\{\theta\in[0,\pi]: f'(\theta)=-\frac{\sqrt{3}}{2}\right\}$

$\Rightarrow f(\theta)=3\left(\cos^4\theta+\sin^4\theta\right)-2\cos^2 2\theta$

$\Rightarrow f(\theta)=3\left(1-\frac{1}{2}\sin^2 2\theta\right)-2\cos^2 2\theta$

$\Rightarrow f(\theta)=3-\frac{3}{2}\sin^2 2\theta-2\cos^2 2\theta$

$=\frac{3}{2}-\frac{1}{2}\cos^2 2\theta=\frac{3}{2}-\frac{1}{2}\left(\frac{1+\cos 4\theta}{2}\right)$

$f(\theta)=\frac{5}{4}-\frac{\cos 4\theta}{4}$

$f'(\theta)=\sin 4\theta$

Also given, $f'(\theta)=\sin 4\theta=-\frac{\sqrt{3}}{2}$

$4\theta=\pi+\frac{\pi}{3},2\pi-\frac{\pi}{3},3\pi+\frac{\pi}{3},4\pi-\frac{\pi}{3}$

$\sum_{\theta\in s}4\theta=10\pi\Rightarrow\sum_{\theta\in s}\theta=\frac{10\pi}{4}=\frac{5\pi}{2}=4\beta$

$\Rightarrow\beta=\frac{5\pi}{8}$

$\Rightarrow\beta=\frac{5\pi}{8}\Rightarrow f(\beta)=\frac{5}{4}-\frac{\cos\frac{5\pi}{2}}{4}=\frac{5}{4}$

2. ***(d)*** $\log_{\cos x}\cot x+4\log_{\sin x}\tan x=1$

$\Rightarrow\frac{\ln(\cos x)-\ln(\sin x)}{\ln(\cos x)}+4\frac{\ln(\sin x)-\ln(\cos x)}{\ln(\sin x)}=1$

$\Rightarrow(\ln\sin x)^2-4(\ln\sin x)(\ln\cos x)+4(\ln\cos x)^2=1$

$\Rightarrow\ln\sin x=2\ln\cos x$

$\Rightarrow\sin^2 x+\sin x-1=0\Rightarrow\sin x=\frac{-1+\sqrt{5}}{2}$

$\Rightarrow x=\sin^{-1}\left(\frac{-1+\sqrt{5}}{2}\right)$

$\alpha=-1$ & $\beta=5$

$\therefore\alpha+\beta=4$

3. ***(c)*** From the question Let

$\alpha=\theta+(m-1)\frac{\pi}{6}$ and $\beta=\theta+\frac{m\pi}{6}$

$\therefore\ \beta-\alpha=\frac{\pi}{6}$

$\sum_{m=1}^{9}\sec\alpha\cdot\sec\beta=\sum_{m=1}^{9}\frac{1}{\cos\alpha\cdot\cos\beta}$

$=\frac{1}{\sin(\beta-\alpha)}\sum\frac{\sin(\beta-\alpha)}{\cos\alpha\cos\beta}$

$=2\sum_{m=1}^{9}(\tan\beta-\tan\alpha)$

$=2\sum_{m=1}^{9}\left[\tan\left(\theta+m\frac{\pi}{6}\right)-\tan\left(\theta+(m-1)\frac{\pi}{6}\right)\right]$

telescopic form

Sum $=2\left(\tan\left(\theta+9\frac{\pi}{6}\right)-\tan\theta\right)$

$\frac{-8}{\sqrt{3}}=2(-\cot\theta-\tan\theta)$

$\Rightarrow\tan\theta+\cot\theta=\frac{4}{\sqrt{3}}$

$\theta=\frac{\pi}{6},\frac{\pi}{3}$

So, $S=\left\{\frac{\pi}{6},\frac{\pi}{3}\right\}$

$\therefore\ \sum_{\theta\in S}\theta=\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$

4. [3] $2\sin^2\theta = \cos 2\theta = 1 - 2\sin^2\theta$

$[\because 1 - \cos 2\theta = 2\sin^2\theta]$

$\Rightarrow 4\sin^2\theta = 1 \Rightarrow \sin\theta = \pm\frac{1}{2}$

$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

[In each Quadrant we will have one value]

Now, $2\cos^2 + 3\sin\theta = 0$

$2\sin^2\theta - 3\sin\theta - 2 = 0$

$2\sin 2\theta - 4\sin\theta + \sin\theta - 2 = 0$

$(2\sin\theta + 1)(\sin\theta - 2) = 0$

$\Rightarrow \sin\theta = -\frac{1}{2}, \theta = \frac{7\pi}{6}, \frac{11\pi}{6}$

Sum $= \frac{7\pi}{6} + \frac{11\pi}{6} = 3\pi \Rightarrow k = 3$

5. (*b*) We have

$\left\{x \in \left[\frac{-2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$

$\cos x + \sin x = 1$

$\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x = \frac{1}{\sqrt{2}}$

$\sin\left(\frac{\pi}{4} + x\right) = \frac{1}{\sqrt{2}}$

$\frac{\pi}{4} + x = n\pi + (-1)^n\frac{\pi}{4}$

$x = n\pi + (-1)^n\frac{\pi}{4} - \frac{\pi}{4}$

$\therefore x$ has 2 elements $\rightarrow$ (P)

We have II

$\left\{x \in \left[\frac{-5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}; \sqrt{3}\tan 3x = 1$

$\Rightarrow \tan 3x = \frac{1}{\sqrt{3}}$

$\Rightarrow x = (6n+1)\frac{\pi}{18}; n \in Z \Rightarrow 3x = n\pi + \frac{\pi}{6}$

or $x = \frac{n\pi}{3} + \frac{\pi}{18}$

$\therefore x$ has 2 elements (*P*)

We have III

$\left\{x \in \left[\frac{-6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos 2x = \sqrt{3}\right\}$

$2\cos 2x = \sqrt{3}$

$\Rightarrow \cos 2x = \frac{\sqrt{3}}{2} \Rightarrow 2x = 2n\pi \pm \frac{\pi}{6}; n \in Z$

or $x = n\pi \pm \frac{\pi}{12}; n \in Z$

$\Rightarrow x \in \left\{\pm\frac{\pi}{12}, -\pi \pm \frac{\pi}{12}, \pi \pm \frac{\pi}{12}\right\}$

$\therefore x$ has 6 elements $\rightarrow$ (*t*)

We have (IV)

$\left\{x \in \left[\frac{-7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$

$\sin x - \cos x = 1$

$\frac{1}{\sqrt{2}}\sin(x) - \frac{1}{\sqrt{2}}\cos(x) = \frac{1}{\sqrt{2}}$

$\sin\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\pi - \frac{\pi}{4} = n\pi + (-1)^n\frac{\pi}{4}$

$x = n\pi + (-1)^n\frac{\pi}{4} + \frac{\pi}{4}$

$x \in \left\{\frac{\pi}{2}, \frac{-3\pi}{2}, -\pi, \pi\right\}$

$\therefore x$ has 4 elements $\rightarrow R$

6. (*b*) Given,

$\sin x + \sin 2x + \sin 3x + \sin 4x = 0 x \in [0, 2\pi]$

$\Rightarrow (\sin x + \sin 3x) + (\sin 2x + \sin 4x) = 0$

$\Rightarrow 2\sin 2x\cos x + 2\sin 3x\cos x = 0$

$\Rightarrow 2\cos x(\sin 2x + \sin 3x) = 0$

$\Rightarrow 4\cos x\sin\frac{5x}{2}\cos\frac{x}{2} = 0$

$\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$

$\sin\frac{5x}{2} = 0 \Rightarrow \frac{5x}{2} = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$

$\Rightarrow x = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}, 2\pi$

$\cos\frac{x}{2} = 0 \Rightarrow \frac{x}{2} = \frac{\pi}{2}, \Rightarrow x = \pi$

$\therefore$ Sum of all values of $x = 2\pi + 6\pi + \pi = 9\pi$

7. (*a*) $\sin^4\theta + \cos^4\theta = -\lambda$

$\Rightarrow 1 - 2\sin^2\theta\cos^2\theta = -\lambda$

$\Rightarrow \lambda = \frac{(\sin 2\theta)^2}{2} - 1$

as $\sin^2 2\theta \in [0,1]$

$\Rightarrow \lambda \in \left[-1, -\frac{1}{2}\right]$

8. (*a*) $\cos 2x + \alpha\sin x = 2\alpha - 7$

$\Rightarrow 2\sin^2 x - \alpha\sin x + 2\alpha - 8 = 0$

$\sin^2 x - \frac{\alpha}{2}\sin x + \alpha - 4 = 0$

$\Rightarrow \sin x = 2$ (rejected) or

$\sin x = \frac{\alpha - 4}{2} \Rightarrow \left|\frac{\alpha - 4}{2}\right| \le 1$

$\Rightarrow \alpha \in [2,6]$

9. (*a*) $\sin x - \sin 2x + \sin 3x = 0$

$\Rightarrow (\sin x + \sin 3x) - \sin 2x = 0$

$\Rightarrow 2\sin 2x.\cos x - \sin 2x = 0$

$[\because \text{Sin}C + \text{Sin}D = 2\text{Sin}\left(\frac{C+D}{2}\right).\text{Cos}\left(\frac{C-D}{2}\right)]$

$\Rightarrow \sin 2x(2\cos x - 1) = 0$

$\Rightarrow \sin 2x = 0$ or $\cos x = \frac{1}{2} \Rightarrow x = 0, \frac{\pi}{3}$

10. (*c*)

11. (*b*)

$f(x) = \sin(\pi\cos x)$

$X: \{x : f(x) = 0\}$

$f(x) = 0 \Rightarrow \sin(\pi\cos x)\, x = n \Rightarrow \cos = 1, -1.0$

$\Rightarrow x = \frac{n\pi}{2}$

$X = \left\{\frac{n\pi}{2} : n \in N\right\} = \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi.....\right\}$

$g(x) = \cos(2\pi\sin x)$

$Z = \{x : g(x) = 0\}$

$\cos(2\pi\sin x) = 0 \Rightarrow 2\pi\sin x = (2n+1)\frac{\pi}{2}$

$\Rightarrow \sin x\ \frac{(2n+1)}{4}$

$\sin x = -\frac{1}{4}, \frac{1}{4}, \frac{-3}{4}, \frac{3}{4}$

$Z = \left\{n\pi \pm \sin^{-1}\left(\frac{1}{4}\right), n\pi \pm \sin^{-1}\left(\frac{3}{4}\right), n \in I\right\}$

$Y = \{X : f'(x) = 0\}$

$f(x) = \sin(\pi\cos x) = 0$

$\Rightarrow \pi\cos x = (2n+1)\frac{\pi}{2} \Rightarrow \cos x = \frac{(2n+1)}{2}$

$\Rightarrow \cos x = -\frac{1}{2}, \frac{1}{2}$

$Y = \left\{n\pi, n\pi \pm \frac{\pi}{3}\right\} = \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi.....\right\}$

$W = \{x : g'(x) = 0\}$

$g(x) = \cos(2\pi\sin x)$

$\Rightarrow g'(x) = -\sin(2\pi\sin x).(2\pi\cos x) = 0$

$\cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}$

$\sin(2\pi\sin x) = 0 \Rightarrow 2p\sin x = np$

$\Rightarrow \sin x = \frac{n}{2}$

$= -1, \frac{1}{2}, 0, \frac{1}{2}, 1$

$W = \left\{\frac{n\pi}{2}, n\pi \pm n \in 1\right\}$

$= \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{3\pi}{2}, ...\right\}$

Now check the options

12. (*c*) $2(1 - \sin^2\theta) + 3\sin\theta = 0$

$\Rightarrow 2\sin^2\theta - 3\sin\theta - 2 = 0$

$\Rightarrow (2\sin\theta + 1)(\sin\theta - 2) = 0$

$\Rightarrow \sin\theta = -\frac{1}{2}$; $\sin\theta = 2$(reject)

root : $\pi+\frac{\pi}{6}, 2\pi-\frac{\pi}{6}, -\frac{\pi}{6}, -\pi+\frac{\pi}{6}$

$\Rightarrow$ sum of values $= 2\pi$

13. *(a)* $2^{\sqrt{\sin^2 x-2\sin x+5}} \cdot 4^{-\sin^2 y} \le 1$

$\Rightarrow 2^{\sqrt{(\sin x-1)^2+4}} \le 2^{2\sin^2 y}$

$\Rightarrow \sqrt{(\sin x-1)^2+4} \le 2\sin^2 y$

$\Rightarrow \sin x = 1$ and $|\sin y| = 1$

14. *(a)* Let $9^{\tan^2 x} = t$

$\Rightarrow t^2 - 10t + 9 = 0 \Rightarrow (t-9)(t-1) = 0$

$\Rightarrow t = 1, 9 \Rightarrow 9^{\tan^2 x} = 1, 9^{\tan^2 x} = 9$

So, $\tan^2 x = 0$, $\tan^2 x = 1$

$x = 0, \pm\frac{\pi}{4} \therefore x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\beta = \tan^2(0) + \tan^2\left(+\frac{\pi}{12}\right) + \tan^2\left(-\frac{\pi}{12}\right)$

$= 2\tan^2\left(\frac{\pi}{12}\right)$

$\Rightarrow \beta = 2(2-\sqrt{3})^2$

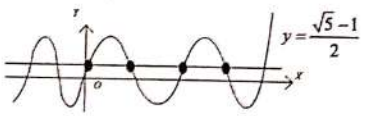
$= 2\left(7-4\sqrt{3}\right)$

$\Rightarrow \frac{(\beta-14)^2}{6} = \frac{1}{6}(14-8\sqrt{3}-14)^2 = 32$

15. **[4]** $\sin x = 1 - \sin^2 x$

$\Rightarrow \sin x = \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}$ (rejected)

draw $y = \sin x$ and $y = \frac{\sqrt{5}-1}{2}$, find their pt. of intersection.



16. **[4]** $\frac{\pi}{4} < x < \frac{7\pi}{4}$

$\frac{14}{\sin^2 x} - 2\sin^2 x = 21 - 4 + 4\sin^2 x$

Let $\sin^2 x = u$

$\frac{14}{u} - 2u = 17 + 4u$

$14 - 2u^2 = 17t + 4u^2$

$6u^2 + 17u - 14 = 0$

$u = \frac{-17 \pm \sqrt{289+336}}{12}$

$= \frac{-17+25}{12} = \frac{8}{12}, \frac{-42}{12}$

$\sin^2 x = \frac{2}{3}$, No. of solutions $= 4$

17. **[5]** $\tan\theta + \sqrt{5}\tan 2\theta \tan\theta = \sqrt{5} - \tan 2\theta$

$\Rightarrow \tan 3\theta = \sqrt{5}$

$\Rightarrow \theta = \frac{n\pi}{3} + \frac{\alpha}{3}$; $\tan\alpha = \sqrt{5}$

$\therefore$ Five solution.

18. *(d)* We have,

$\frac{\cos x}{1+\sin x} = |\tan 2x|$

$\Rightarrow \frac{\cos^2(x/2) - \sin^2(x/2)}{(\cos x/2 + \sin x/2)^2} = |\tan 2x|$

$\Rightarrow \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) = \tan 2x$

$\Rightarrow 2x = n\pi \pm \left(\frac{\pi}{4} - \frac{x}{2}\right)$

$\Rightarrow x = \frac{-3\pi}{10}, \frac{-\pi}{6}, \frac{\pi}{10}$

$\therefore$ sum of Solution

$= \frac{-3\pi}{10} - \frac{\pi}{6} + \frac{\pi}{10} = \frac{-11\pi}{30}$

19. *(b)*

20. *(c)* We have, $32^{\tan^2 x} + 32^{\sec^2 x} = 81$

$32^{\tan^2 x} + 32^{1+\tan^2 x} = 81$

$\left[\because \sec^2 x = 1 + \tan^2 x\right]$

$33.32^{\tan^2 x} = 81$

$32^{\tan^2 x} = \frac{81}{33}$

for $x \in [0, \pi/4]$, $\tan^2 x \in [0,1]$

Hence, one solution.

21. *(b)* $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$

$\Rightarrow (81)^{\sin^2 x} + \frac{(81)^1}{(81)^{\sin^2 x}} = 30$

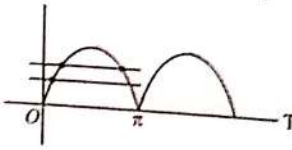
Let $(81)^{\sin^2 x} = t \Rightarrow t + \frac{81}{t} = 30$

$(t-3)(t-27) = 0$

$(81)^{\sin^2 x} = 3^1$ or $(81)^{\sin^2 x} = 3^3$

$3^{4\sin^2 x} = 3^1$ or $3^{4\sin^2 x} = 3^3$

$\sin^2 x = \frac{1}{4}$ or $\sin^2 x = \frac{3}{4}$



Total sol. $= 4$

22. **[11]** $3\sin x + 4\cos x = k + 1$

$-\sqrt{3^2+4^2} \le 3\sin x + 4\cos x \le \sqrt{3^2+4^2}$

$-5 \le k+1 \le 5$

$-6 \le k \le 4$

Integer value of $k = -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4$

No. of integer $= 11$

23. **[1]** $|\cot x| = \cot x + \frac{1}{\sin x}$

Case 1: $\cot x \ge 0$

i.e. $x \in \left(0, \frac{\pi}{2}\right] \cup \left(\pi, \frac{3\pi}{2}\right]$

$\cot x = \cot x + \frac{1}{\sin x}$

$\Rightarrow \frac{1}{\sin x} = 0$ (Not possible)

Case 2: $\cot x < 0$ i.e. $x \in \left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$

$-\cot x = \cot x + \frac{1}{\sin x}$

$\Rightarrow -2\cot x = \frac{1}{\sin x}$

$\Rightarrow -\frac{2\cos x}{\sin x} = \frac{1}{\sin x}$ $(\sin x \ne 0)$

$\Rightarrow \cos x = \frac{-1}{2} \Rightarrow x = \left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$

But x can't be $\frac{4\pi}{3}$ $[\because \cot x < 0]$

$\therefore x = \frac{2\pi}{3}$

Hence, only 1 solution exists in given interval.

24. *(d)* A.M. $\ge$ G.M.

$\frac{\sin^4\alpha + 4\cos^4\beta + 1 + 1}{4} \ge \left(\sin^4\alpha . 4\cos^4\beta . 1 . 1\right)^{\frac{1}{4}}$

$\sin^4\alpha + 4\cos^2\beta + 2 \ge 4\sqrt{2}\sin\alpha\cos\beta$

given that

$\sin^4\alpha + 4\cos^4\beta + 2 = 4\sqrt{2}\sin\alpha\cos\beta$

$\Rightarrow$ A.M. $=$ G.M. $\Rightarrow \sin^4\alpha = 1 = 4\cos^4\beta$

$\sin\alpha = \pm 1, \cos\beta = \pm\frac{1}{\sqrt{2}}$, As $\alpha\beta \in [0, \pi]$

$\Rightarrow \sin\alpha = 1, \cos\beta = \pm\frac{1}{\sqrt{2}}$

[Here $\cos\beta = \frac{1}{\sqrt{2}}$ which satisty given eq^n]

$\Rightarrow \sin\beta = \frac{1}{\sqrt{2}}$ as $\beta \in [0, \pi]$

Now, $\cos(\alpha+\beta) - \cos(\alpha-\beta) =$

$-2\sin\alpha\sin\beta = -\sqrt{2}$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER
5
Quadratic Equations

Prarambh (Topicwise)

1. (*c*) We have $4ax^2 + 3bx + 2c = 0$ Let roots are α and β

Let $D = B^2 - 4AC = 9b^2 - 4(4a)(2c)$
$= 9b^2 - 32\,ac$

Given that, $(a + b + c) = 0$

$\Rightarrow b = -(a + c)$

Putting this value, we get

$= 9(a + c)^2 - 32\,ac = 9(a - c)^2 + 4ac$

Hence roots are real.

2. (*d*) Given equation

$2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$

Let $A = 2(a^2 + b^2)$, $B = 2(a + b)$ and $C = 1$

$B^2 - 4AC = 4(a^2 + b^2 + 2ab) - 4.2\,(a^2 + b^2)1$

$\Rightarrow B^2 - 4AC = -4(a - b)^2 < 0$

Thus given equation has imaginary roots.

3. (*a*) Here $(b + c - 2a) + (c + a - 2b) + (a + b - 2c) = 0$

Therefore the roots are rational.

4. (*c*) The quadratic is $(k + 11)x^2 - (k + 3)x + 1 = 0$

Accordingly, $(k + 3)^2 - 4(k + 11)(1) = 0$

$\Rightarrow k = -7, 5$

5. (*d*) $x^2 - 3x + 2$ be factor of $x^4 - px^2 + q = 0$

Hence $(x^2 - 3x + 2) = 0 \Rightarrow (x - 2)(x - 1) = 0$

$\Rightarrow x = 2, 1$ putting these values in given equation

so $4p - q - 16 = 0$ ...(*i*)

and $p - q - 1 = 0$ ...(*ii*)

Solving (*i*) and (*ii*), we get $(p, q) = (5, 4)$

6. (*b*) check by options

$x = 1$ is root

Let other root $= a$

$\therefore$ Product of the roots $= (1)(1) = \dfrac{a-b}{b-c}$

$\Rightarrow$ roots are $1, \dfrac{a-b}{b-c}$

7. (*a*) $D = b^2 - 4ac = 20d^2$

$\sqrt{D} = 2\sqrt{5}d$ here $\sqrt{5}$ is irrational

So roots are irrational.

8. (*c*) For integral roots, D of equation should be perfect sq.

$\because D = 4(1 + n)$

By observation, for $n \in N$, D should be perfect sq. of even integer.

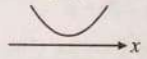
So $D = 4(1 + n) = 6^2, 8^2, 10^2, 12^2, 14^2, 16^2, 18^2, 20^2$

No. of values of $n = 8$.

9. (*a*) Let $f(x) = ax^2 - bx + 1$

Given $D < 0$ & $f(0) = 1 > 0$

$\therefore$ Possible graph is as shown



i.e. $f(x) > 0\ \forall\ x \in R$

or $f(-1) > 0$

$f(-1) = a + b + 1 > 0$

10. (*c*) $x^2 + ax + b = 0$ $\begin{matrix} a \\ b \end{matrix}$

$a + b = -a$

$\Rightarrow 2a + b = 0$

and $ab = b$

$ab - b = 0$

$b(a - 1) = 0$

$\Rightarrow$ Either $b = 0$ or $a = 1$

But $b \neq 0$(given)

$\therefore a = 1$

$\therefore b = -2$

$\therefore f(x) = x^2 + x - 2$

Least value occurs at $x = -\dfrac{1}{2}$

Least value $= \dfrac{1}{4} - \dfrac{1}{2} - 2 = -\dfrac{9}{4}$

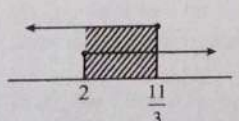
11. (*b*) $(2 - x)(x + 1) = p$

$(x - 2)(x + 1) + p = 0$

$\Rightarrow x^2 - x - 2 + p = 0$

$\dfrac{c}{a} > 0 \quad \Rightarrow p - 2 > 0$

& $D > 0 \Rightarrow 1 - 4(p - 2) > 0 \quad \Rightarrow p < \dfrac{9}{4}$



$\dfrac{-b}{2a} > 0,\ \dfrac{-1}{2(2-p)} > 0, P \in (2, \infty)$

Taking intersection of all $p \in \left(2, \dfrac{9}{4}\right)$

12. (*c*) $x^2 + px + (1 - p) = 0$

$(1 - p)^2 + p(1 - p) + (1 - p) = 0$

$(1 - p)[1 - p + p + 1] = 0 \quad \Rightarrow p = 1$

Q.E. will be $\Rightarrow x^2 + x = 0$

$\Rightarrow x(x + 1) = 0$

$\Rightarrow x = 0, -1$

Aliter

$\alpha + 1 - p = -p \quad \Rightarrow \alpha = -1$

Satisfies

$1 - p + 1 - p = 0 \Rightarrow p = 1$

$\beta = 1 - p = 0 \Rightarrow \beta = 0$

13. (*c*) Given $a + b + c = 0 \Rightarrow b = -(a + c)$

Now, $3ax^2 + 2bx + c = 0$

$D = 4b^2 - 12ac$

$= 4\left[(a + c)^2 - 3ac\right]$

$= 4\left(a^2 + c^2 - ac\right)$

[By ΔM – G.M inequaliy]

$\geq 4\left(a^2 + c^2 - ac\right)$

$\Rightarrow D \geq 0$

14. (*a*) a, b, c, are in A.P.

$\Rightarrow 2b = a + c$ ...(i)

Given $ax^2 + 2bx + c = 0$

$D = 4b^2 - 4ac$

$= (2b)^2 - 4ac$

Using (1)

$= (a + c)^2 - 40c$

$= a^2 + c^2 + 2ac - 4ac = (a - c)^2$

$D \geq 0 \Rightarrow$ roots are real and distinct

15. (*c*) As $bx^2 + cx + a = 0$ has imaginery roots so, $c^2 < 4ab$

Now, $3b^2x^2 + 6bcx + 2c^2$

$= 3(bx + c)^2 - c^2 \geq -c^2 \geq -4ab$

16. (*b*) Let first root = α and second root = $\frac{1}{\alpha}$

Then $\alpha, \frac{1}{\alpha} = \frac{k}{5} \Rightarrow k = 5$

17. (*a*) Given equation $4x^2 + 3x + 7 = 0$, therefore

$\alpha + \beta = -\frac{3}{4}$ and $\alpha\beta = \frac{7}{4}$

Now

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-3/7}{7/4} = \frac{-3}{4} \times \frac{4}{7} = -\frac{3}{7}$

18. (*b*) α, β be the roots of $x^2 - 2x + 3 = 0$, then $\alpha + \beta = 2$ and $\alpha\beta = 3$.

Now required equation whose roots are $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$ is

$x^2 - \left(\frac{1}{\alpha^2} + \frac{1}{\beta^2}\right)x + \frac{1}{\alpha^2\beta^2} = 0$

$\Rightarrow x^2 - \left(-\frac{2}{9}\right)x + \frac{1}{9} = 0$

$\Rightarrow 9x^2 + 2x + 1 = 0$

19. (*c*) According to condition

$\frac{2m-1}{m} = -1 \quad \Rightarrow 3m = 1 \Rightarrow m = \frac{1}{3}$

20. (*d*) We know that the roots of the equation

$ax^3 + bx^2 + cx + d = 0$ follows $\alpha\beta\gamma = -d/a$

Comparing above equation with given equation

We get $d = 1$, $a = 1$

So, $\alpha\beta\gamma = -1$ or $\alpha^3\beta^3\gamma^3 = -1$.

21. (*c*) $a + b = -p$

$ab = q \Rightarrow g + d = -p$

$gd = -r$

$(a-g)(a-d) = a^2 - a(g+d) + gd$

$= a^2 + pa - r = a(a+p) - r = -ab - r$

$= -q - r = -(q + r)$

22. (*c*) $ax^2 + bx + c = 0, \alpha + \beta = \frac{-b}{a}, \alpha\beta = \frac{c}{a}$

$\alpha^3 + \beta^3 = (\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]$

$\alpha^3 + \beta^3 = \left(\frac{-b}{a}\right)\left[\left(\frac{-b}{a}\right)^2 - 3\frac{c}{a}\right]$

$= \frac{-b}{a}\left[\frac{b^2}{a^2} - \frac{3c}{a}\right] = \frac{-b}{a}\frac{(b^2 - 3ac)}{a^2} = \frac{3abc - b^3}{a^3}$

23. (*d*) $\Rightarrow \alpha + \beta = -2$ .... (1)

$\Rightarrow \alpha^3 + \beta^3 = -56$ ... (2)

$\Rightarrow (\alpha+\beta)^3 = \alpha^3 + \beta^3 + 3\alpha^2\beta + 3\alpha\beta^2$

$\Rightarrow (\alpha+\beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha+\beta)$

$\Rightarrow (-2)^3 = (-56) + 3\alpha\beta(-2)$

[Using (1) and (2)]

$\Rightarrow -8 + 56 = -6\alpha\beta \Rightarrow 48 = -6\alpha\beta$

$\therefore \alpha\beta = -8$ ...(3)

The required equation ,

$x^2 - (\alpha+\beta)x + (\alpha\beta) = 0$

Now, From (1) and (3) we get

$\Rightarrow x^2 + 2x - 8 = 0$

24. (*b*) $\tan 22^\circ$ and $\tan 23^\circ$ are roots of $x^2 + ax + b = 0$

$\therefore \tan 22^\circ + \tan 23^\circ = -a$

$\tan 22^\circ . \tan 23^\circ = b$

$\tan(22 + 23) = \frac{\tan 22^\circ + \tan 23^\circ}{1 - \tan 22^\circ \tan 23^\circ}$

$1 - \tan 22^\circ \tan 23^\circ = \tan 22^\circ + \tan 23^\circ$

$1 - b = -a \Rightarrow a - b + 1 = 0$

25. (*c*) Given α, β are the roots of $(x-a)(x-b) = 0$

$\Rightarrow (x-a)(x-b) - c = (x-\alpha)(x-\beta)$

$\Rightarrow (x-a)(x-b) - c = (x-\alpha)(x-\beta) + c$

$\Rightarrow a, b$ are the roots of equation $(x-\alpha)(x-\beta) + c$

26. (*c*) α and β are the roots of $x^2 - 15x + 1 = 0$

$\therefore \alpha + \beta = 15, \alpha\beta = 1$

Now

$\left(\frac{1}{\alpha} - 15\right)^{-2} + \left(\frac{1}{\beta} - 15\right)^{-2} = (-\alpha)^{-2} + (-\beta)^{-2}$

$\left(\because \frac{1}{\alpha} = \beta, \frac{1}{\beta} = \alpha, \alpha + \beta = 15\right)$

$= \left(\frac{1}{-\alpha}\right)^2 + \left(\frac{1}{-\beta}\right)^2 = \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \alpha^2 + \beta^2 \quad (\because \alpha\beta = 1)$

$= (\alpha+\beta)^2 - 2\alpha\beta$

$= 225 - 2 = 223$

27. (*d*) It is given that, 4 is the root of the equation

$x^2 + px + 12 = 0$

$\therefore 16 + 4p + 12 = 0$

$\Rightarrow p = -7$

It is also given that, the equation $x^2 + px + q = 0$ has equal roots, So, the dicriminant of $x^2 + px + q = 0$ will be zero $\therefore p^2 - 4q = 0$

$\Rightarrow 4q = (-7)^2 = 49 \Rightarrow q = \frac{49}{4}$

28. (*b*) Let one root of $ax^2 + bx + c = 0$ be α and other be α^2 then, $\alpha + \alpha^2 = \frac{-b}{a}$

$a^3 = \frac{s}{a}$

or $\alpha = \left(\frac{s}{2}\right)^{\left(\frac{1}{3}\right)}$

Now put the value of α in $\alpha + \alpha^2 = \frac{-b}{a}$

$s^{\frac{1}{a}} + c^2 = \frac{-b}{a}$

Cubing both sides

$\left(\frac{c}{a}\right)^{\left(\frac{1}{2}\right)} + \left(\frac{\varepsilon}{a}\right)^{\left(\frac{6}{2}\right)} + 3\left(\frac{\varepsilon}{2}\right)^{\left(\frac{1}{2}\right)} \times \left(\frac{c}{a}\right)^{\left(\frac{2}{2}\right)}$

$\left(\left(\frac{c}{a}\right)^{\left(\frac{1}{2}\right)} + \left(\frac{\varepsilon}{a}\right)^{\left(\frac{2}{2}\right)}\right) = \left(\frac{-b}{a}\right)^3$

$\frac{c}{a} + \frac{c^2}{a^2} + 3\frac{c}{a}\left(\frac{-b}{a}\right) = \frac{-b^3}{a^3}$

$a^2c + ac^2 - 3abc = -b^3$

$b^3 + a^2c + a^2 = 3abc$

29. (*b*) RHS $= -3 + x - x^2$

$= -\left(x^2 - x + \frac{1}{4} - \frac{1}{4} + 3\right)$

$= -\left[\left(x - \frac{1}{2}\right)^2 + \frac{11}{4}\right]$ ...always negative

LHS $= \left(\frac{9}{10}\right)^x = (0.9)^x$ Always positive

$\therefore$ No solution

30. (*b*) $x^2 - bx + c = 0$ $\begin{cases} \alpha \\ \alpha + 1 \end{cases}$

$\alpha + \alpha + 1 = b \quad \alpha(\alpha+1) = c$

$2\alpha + 1 = b \quad \alpha^2 + \alpha = c$

$\Rightarrow \alpha^2 + \alpha = c \Rightarrow \left(\frac{b-1}{2}\right)^2 + \left(\frac{b-1}{2}\right) = c$

$\Rightarrow \frac{b^2 - 1}{4} - c = 0 \Rightarrow b^2 - 4c = 1$

31. (*c*) We know that if α, β are the roots of the quardatic equation $ax^2 + bx + c = 0$ then

$\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$

Therefore for the given equation $x^2 + px + q = 0$

$\tan\alpha + \tan\beta = -p$ ...(1)

$\tan\alpha \tan\beta = q$ ...(2)

$\tan(\alpha+\beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\cdot\tan(\beta)}$

$= \frac{-p}{1-q}$

$= \frac{p}{q-1}$ ...(3)

Therefore option (*c*) is correct.

32. (*a*) Let α is the common root,

So $\alpha^2 + p\alpha + q = 0$ ...(*i*)

and $\alpha^2 + q\alpha + p = 0$ ...(*ii*)

from (*i*) – (*ii*),

$\Rightarrow (p-q)\alpha + (q-p) = 0 \Rightarrow \alpha = 1$

Put the value of α in (*i*), $p + q + 1 = 0$.

33. (*a*) $D_1 = 4a^2b^2 - 8a^2b^2 = -4a^2b^2 < 0$ img. root

$D_2 = 4p^2q^2 - 4p^2q^2 = 0$ equal, real roots

So no common roots.

34. (*c*) Consider, $x^2 + 3x + 5 = 0$

$a = 1, b = 3, c = 5$

Let p, q be the two roots, Now, by vieta;'s formulas

$p + q = 3 = -ba$ and $pq = 5 = ca$

Therefore , b = 3a, c = 5a

Hence $a + b + c + = a + 3a + 5a = 9a$ But $a, b, c \in N$.

$\therefore$ The minimum value of $a + b + c = 9$

35. (*b*) **Case I:** When $x + 2 \geq 0$ i.e. $x \geq -2$

Then given inequality becomes

$x^2 - (x+2) + x > 0 \Rightarrow x^2 - 2 > 0$

$\Rightarrow |x| > \sqrt{2}$

$\Rightarrow x < -\sqrt{2}$ or $x > \sqrt{2}$

As $x \geq -2$, therefore, in this case the part of the solution set is $[-2, -\sqrt{2}) \cup (\sqrt{2}, \infty)$.

Case II: When $x + 2 \leq 0$ *i.e.* $x \leq -2$,

Then given inequality becomes $x^2 + (x+2) + x > 0$

$\Rightarrow x^2 + 2x + 2 > 0 \Rightarrow (x+1)^2 + 1 > 0$, which is true for all real x

Hence, the part of the solution set in this case is $(-\infty, -2]$. Combining the two cases, the solution set is

$(-\infty, -2) \cup ([-2, -\sqrt{2}] \cup (\sqrt{2}, \infty) = (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$.

36. (*b*) For $(p^2 - 3p + 2)x^2 - (p^2 - 5p + 4)x + p - p^2 = 0$ to be an identify

$p^2 - 3p + 2 = 0 \Rightarrow p = 1, 2$ ...(1)

$p^2 - 5p + 4 = 0 \Rightarrow p = 1, 4$ ...(2)

$p - p^2 = 0 \Rightarrow p = 0, 1$ ...(3)

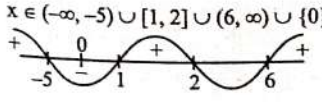
For (1), (2) & (3) to hold simultaneously $p = 1$.

37. (*b*) $\dfrac{x^2(x^2 - 3x + 2)}{x^2 - x - 30} \geq 0$

$\Rightarrow \dfrac{x^2(x-1)(x-2)}{(x+5)(x-6)} \geq 0$

$x \neq -5, 6$

$x \in (-\infty, -5) \cup [1, 2] \cup (6, \infty) \cup \{0\}$



38. (*b*) $\because (m-2)x^2 + 8x + m + 4 > 0 \; \forall x \in R$

$\Rightarrow m > 2$ & $D < 0$

$64 - 4(m-2)(m+4) < 0$

$16 - [m^2 + 2m - 8] < 0$

$\Rightarrow m^2 + 2m - 24 > 0$

$\Rightarrow (m+6)(m-4) > 0$

$m \in (-\infty, -6) \cup (4, \infty)$

But $m > 2 \Rightarrow m \in (4, \infty)$

Then least integral m is $m = 5$.

39. (*d*) $\left(\dfrac{1}{2}\right)^{x^2 - 2x} < \left(\dfrac{1}{2}\right)^2$ here base is less than zero so inequality change

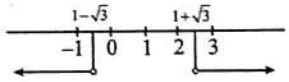
$\Rightarrow x^2 - 2x > 2 \Rightarrow x^2 - 2x - 2 > 0$

$\alpha, \beta = \dfrac{2 \pm \sqrt{4+8}}{2} = \dfrac{2 \pm 2\sqrt{3}}{2}$

$a = 1 - \sqrt{3}, b = 1 + \sqrt{3}$

$(x-a)(x-b) > 0$

$x \in \left(-\infty, 1-\sqrt{3}\right) \cup \left(1+\sqrt{3}, \infty\right)$, x can be in $(3, \infty)$



40. (*c*) Let the roots be $a, b, -b$

then $\alpha + \beta - \beta = p$

$\Rightarrow \alpha = p$ ...(1)

and $\alpha\beta - \alpha\beta - \beta^2 = q$

$\Rightarrow \beta^2 = -q$ ...(2)

also $-\alpha\beta^2 = r$

$\Rightarrow pq = r$ [using (1)].

41. (*c*) $x^3 - x - 1 = 0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$

then $\alpha^3 - \alpha - 1 = 0$. ...(1)

Let $\dfrac{1+\alpha}{1-\alpha} = y \Rightarrow \alpha = \dfrac{y-1}{y+1}$

from equation (1)

$\left(\dfrac{y-1}{y+1}\right)^3 - \left(\dfrac{y-1}{y+1}\right) - 1 = 0$

$\Rightarrow y^3 + 7y^2 - y + 1 = 0 \begin{cases} \dfrac{1+\alpha}{1-\beta} \\ \dfrac{1+\beta}{1-\beta} \\ \dfrac{1+\gamma}{1-\gamma} \end{cases}$

then $\dfrac{1+\alpha}{1-\alpha} + \dfrac{1+\beta}{1-\beta} + \dfrac{1+\gamma}{1-\gamma} = -7$

42. (*b*) $2x^2 - (a^3 + 8a - 1)x + (a^2 - 4a) = 0$

Since the roots are of opposite sign, $f(0) < 0$

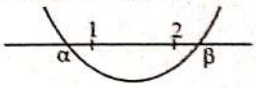
$\Rightarrow a^2 - 4a < 0 \Rightarrow a(a-4) < 0$

$\Rightarrow a \in (0, 4)$

43. (*b*) $x^2 - 2px + (8p - 15) = 0$

$f(1) < 0$ and $f(2) < 0$

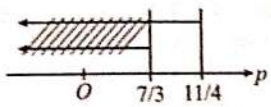
$\Rightarrow f(1) = 1 - 2p + 8p - 15 < 0$



$\Rightarrow p < 7/3$

and $f(2) = 4 - 4p + 8p - 15 < 0$

$\Rightarrow 4p - 11 < 0 \Rightarrow p < \dfrac{11}{4}$



Hence $p \in (-\infty, 7/3)$ Ans.

44. (*d*) $x^2 + 2(k-1)x + k + 5 = 0$

Case - I: (*i*) $D \ldots 0$

$\Rightarrow 4(k-1)^2 - 4(k+5) \ldots 0$

$\Rightarrow k^2 - 3k - 4 \ldots 0 \Rightarrow (k+1)(k-4) \ldots 0$

$\Rightarrow k \in (-\infty, -1] \cup [4, \infty)$

& (*ii*) $f(0) > 0 \Rightarrow k + 5 > 0$

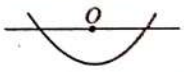
$\Rightarrow k \in (-5, \infty)$

& (*iii*) $\dfrac{-b}{2a} > 0 \Rightarrow \dfrac{-2(k-1)}{2} > 0$

$\Rightarrow k \in (-\infty, 1) \quad \therefore k \in [-5, -1]$

Case - II: $f(0) \leq 0 \Rightarrow k + 5 \leq 0$

$\Rightarrow k \in (-\infty, -5]$



Finally $k \in$ (Case - I) $\cup$ (Case - II)

$k \in (-\infty, -1]$

45. (*b*) Given $x^2 + 2(p-3)x + 9 = 0$

Roots area real

$\Rightarrow$ Discriminant $D \geq 0$

$\Rightarrow (2(p-3))^2 - 4(9) \geq 0$

$\Rightarrow (p-3)^2 - 9 \geq 0$

$(p-3-3)(p-3+3) \geq 0$

$p(p-6) \geq 0$

$p \in (-\infty, 0) \cup (6, \infty)$

For roots to exists 6 to lie between roots

$f(6) < 0$

$\Rightarrow 36 + 2(n-3)6 + 9 < 0$

$12p + 9 < 0$

$\Rightarrow p < \dfrac{-3}{4}$

$\therefore p \in \left(-\infty, \dfrac{-3}{4}\right)$

for roots to exists and to enclose 6 in between them.

46. (*a*) $\frac{(x-1)^3}{(x+2)^3}=\frac{x-4}{x+5}$

$\left(x^3-1-3x^2+3x\right)(x+5)=$

$\left(x^3+8+6x^2+12x\right)(x-4)$

$\Rightarrow x^4-x-3x^3+3x^2+5x^3-5-15x^2+15x$

$=x^4+8x+6x^3+12x^2-4x^3-32$

$-60x^2 24x^2-48x$

47. (*d*) As α, β are roots of $x^3+4x+1=0$

Gives

$s_1=\alpha+\beta+\gamma=0$

$s_2=\alpha\beta+\beta\gamma+\gamma\alpha=4$

$s_3=\alpha\beta\gamma=-1$

Now

$\alpha+\beta+\gamma=0\Rightarrow\alpha+\beta=-\gamma\Rightarrow\beta+\gamma=-\alpha$

$\Rightarrow\alpha+\gamma=-\beta$

Therefore

$(\alpha+\beta)^{-1}+(\beta+\gamma)^{-1}+(\alpha+\gamma)^{-1}$

$=(-\gamma)^{-1}+(-\alpha)^{-1}+(-\beta)^{-1}$

$=\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}$

$=\frac{-(\alpha\beta+\beta\gamma+\gamma\alpha)}{\alpha\beta\gamma}=4$

48. (*a*) $x^3-13x^2+15x+189=0$

Let roots be α, $\alpha+2$ and β

Then

$\alpha+\alpha+2+\beta=-\frac{\text{Coeff. of } x^2}{\text{Coeff. of } x^3}=-(-1)=13$

$\Rightarrow 2\alpha+\beta+2=13$

$\Rightarrow\beta=11-2a$ ...(1)

$\alpha(\alpha+2)+(\alpha+2)\beta+\alpha\beta=\frac{\text{Coeff. of } x}{\text{Coeff. of } x^3}=15$

$\Rightarrow\alpha^2+2\alpha+\alpha\beta+2\beta+\alpha\beta=15$

$\Rightarrow\alpha^2+2\alpha+2(\alpha+1)\beta=15$

$\Rightarrow\alpha^2+2\alpha+2(\alpha+1)(11-2\alpha)=15$

using 1

$\Rightarrow\alpha^2+2\alpha+2\left(11\alpha-2\alpha^2+11-2\alpha\right)=15$

$\Rightarrow\alpha^2+2\alpha+22\alpha-4\alpha^2+22-4\alpha=15$

$\Rightarrow-3\alpha^2+20\alpha+7=0$

$-3\alpha^2-20\alpha-7=0$

$\Rightarrow 3a^2+\alpha-21\alpha-7=0$

$\Rightarrow\alpha(3\alpha+1)-7(3\alpha+1)=0$

$\Rightarrow(\alpha-7)(3\alpha+1)=0$

$\Rightarrow\alpha-7=0$ or $3\alpha+1=0$

$\Rightarrow\alpha=7$ or $\alpha=-\frac{1}{3}$

So, when $\alpha=7$ then (1) becomes

$\beta=11-2(7)=11-14=-3$

and $\alpha+2=7+2=9$

and when $\alpha=-\frac{1}{3}$ then (1) becomes

$\beta=11-2\left(-\frac{1}{3}\right)=11+\frac{2}{3}=\frac{35}{3}$

and $\alpha+2=-\frac{1}{3}+2=\frac{-1+6}{3}=\frac{5}{3}$

So, roots are 7,9, –3 or $-\frac{1}{3},\frac{5}{3},\frac{35}{3}$

49. (*d*) $\frac{mx^2+3x+4}{x^2+2x+2}<5$

$\therefore mx^2+3x+4<5\left(x^2+2x+2\right)$

$\therefore mx^2+3x+4<5x^2+10x+10$

$\therefore x^2(m-5)+3x+0x+4-10<0$

$\therefore x^2(m-5)-7x-6<0$

Since above quadratic equation is always less than i) $m-5<0$

(ii) $0<0$

$\therefore m<5; b^2-4ac<0$ ...(i)

$\therefore 49-4(-6)(m-5)<0$

$\therefore 49<-24(m-5)$

$\therefore 49<-24m+120$

$\therefore 24m<120-43$

$\therefore m<\frac{71}{24}$ ...(ii)

Hence from (i) and (ii) we obtain

$m<\frac{71}{24}$

50. (*b*) Given equation can be written as $|x-1|^2+|x-1|-6=0$

$t^2+t-6=0$

$\Rightarrow t=2$ or -3 $|x-1|=2\Rightarrow x-1=\pm2$ or $x=3,-1$

Prabal (JEE Main Level)

1. (*c*) From options put $k=3$

$\Rightarrow x^2+8x+7=0$

$\Rightarrow(x+1)(x+7)=0\Rightarrow x=-1,-7$

means for $k=3$ roots are negative.

2. (*a*) Here $\alpha+\beta=-\frac{b}{a}$ and $\alpha\beta=\frac{c}{a}$

If roots are $\alpha+\frac{1}{\beta},\beta+\frac{1}{\alpha}$ then sum of roots are

$=\left(\alpha+\frac{1}{\beta}\right)+\left(\beta+\frac{1}{\alpha}\right)=(\alpha+\beta)+\frac{(\alpha+\beta)}{\alpha\beta}$

$=\frac{b}{ac}(a+c)$

and product $=\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right)$

$=\alpha\beta+1+1+\frac{1}{\alpha\beta}=2+\frac{c}{a}+\frac{a}{c}$

$=\frac{2ac+c^2+a^2}{ac}=\frac{(a+c)^2}{ac}$

Hence required equation is given by

$x^2+\frac{b}{ac}(a+c)x+\frac{(a+c)^2}{ac}=0$

$\Rightarrow acx^2+(a+c)bx+(a+c)^2=0$

Trick : Let $a=1, b=-3, c=2$, then $\alpha=1, \beta=2b$

$=-3, c=2$, then $\alpha=1, \beta=2$

$\therefore\alpha+\frac{1}{\beta}=\frac{3}{2}$ and $\beta+\frac{1}{\alpha}=3$

Therefore, required equation must be

$(x-3)(2x-3)=0$ i.e. $2x^2-9x+9=$

Here (1) gives this equation on putting

$a=1, b=-3, c=2$

3. (*a*) Let roots are α, β so, $\frac{\alpha}{\beta}=\frac{2}{3}\Rightarrow\alpha=\frac{2\beta}{3}$

$\therefore\alpha+\beta=\frac{m}{12}$

$\Rightarrow\frac{2\beta}{3}+\beta=\frac{m}{12}\Rightarrow\frac{5\beta}{3}=\frac{m}{12}$ ...(i)

and $\alpha\beta=\frac{5}{12}\Rightarrow\frac{2\beta}{3}.\beta=\frac{5}{12}\Rightarrow\beta^2=\frac{5}{8}$

$\Rightarrow\beta=\sqrt{5/8}$

Put the value of β in (*i*)

$\frac{5}{3}\sqrt{\frac{5}{8}}=\frac{m}{12}\Rightarrow m=5\sqrt{10}$.

4. (*b*) Expressions are $x^2-11x+a$ and $x^2-14x+2a$ will have a common factor then

$\Rightarrow\frac{x^2}{-22a+14a}=\frac{x}{a-2a}=\frac{1}{-14+11}$

$\Rightarrow\frac{x^2}{-8a}=\frac{x}{-a}=\frac{1}{-3}\Rightarrow x^2=\frac{8a}{3}$

and $x=\frac{a}{3}$

$\Rightarrow\left(\frac{a}{3}\right)^2=\frac{8a}{3}\Rightarrow\frac{a^2}{9}=\frac{8a}{3}\Rightarrow a=0,24$

Trick : We can check by putting the values of a from the options.

5. (*d*) Let

$f(x)=4x^2-20px+(25p^2+15p-66)=$ ...(i)

The roots of (*i*) are real if $b^2-4ac=400$

$-16(25p^2+15p-66)=16(66-15p)\geq$

$\Rightarrow p\leq 22/5$ ...(ii)

Both roots of (*i*) are less than 2. Therefore $f(2) > 0$ and sum of roots < 4.

$\Rightarrow 4.2^2 - 20p.2 + (24p^2 + 15p - 66) > 0$

and $\frac{20p}{4} < 4$

$\Rightarrow p^2 - p - 2 > 0$ and $p < \frac{4}{5}$

$\Rightarrow (p+1)(p-2) > 0$ and $p < \frac{4}{5}$

$\Rightarrow p < -1$ or $p > 2$ and $p < \frac{4}{5} \Rightarrow p < -1$ ...(*iii*)

From (*ii*) and (*iii*), we get $p < -1$ i.e. $p \in (-\infty, -1)$.

6. (*c*) If roots are α and β.

Then, $\alpha + \beta = \alpha^2 + \beta^2$

$\alpha\beta = \alpha^2\beta^2 \Rightarrow \alpha^2\beta^2 - \alpha\beta = 1$

$\alpha\beta(\alpha\beta - 1) = 0$

$\alpha\beta = 0$ or $\alpha\beta = 1 \Rightarrow \alpha = 0$ or $\beta = 0$ or $\alpha\beta = 1$

C_1: if $\alpha = 0$

$0 + \beta = 0 + \beta^2 \Rightarrow \beta^2 - \beta = 0$

$\beta(\beta - 1) = 0 \Rightarrow \beta = 0$ or 1.

Roots are 0, 0 and 0, 1

C_2: Similarly, for $\beta = 0 \Rightarrow \alpha = 0$ or 1

Roots will be 0, 0 and 1, 0

C_3: $\alpha\beta = 1$

$\alpha^2 + \beta^2 = \alpha + \beta$

$(\alpha + \beta)^2 - 2\alpha\beta = \alpha + \beta \Rightarrow (\alpha + \beta)^2 - (\alpha + \beta) - 2 = 0$

Let $\alpha + \beta = t$

$t^2 - t - 2 = 0 \Rightarrow (t-2)(t+1) = 0$

$t = 2$ or $t = -1$

$\alpha + \beta = 2$ or $\alpha + \beta = -1$

for $\alpha + \beta = 2$, $\alpha\beta = 1$

Roots are 1, 1

for $\alpha + \beta = -1$, $\alpha\beta = 1$

Roots are w, w^2.

Hences total roots are possible.

0, 0

0, 1

1, 1

ω, ω^2

7. (*a*) $ax^3 + bx^2 + cx + d = 0$

Let roots be α, β, γ.

$\alpha + \beta + \gamma = -\frac{b}{a}$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

$\alpha\beta\gamma = -\frac{d}{a}$

Now, $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$= \left(\frac{-b}{a}\right)^2 - \frac{2c}{a} = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$

As $b^2 < 2ac$

$b^2 - 2ac < 0 \Rightarrow \alpha^2 + \beta^2 + \gamma^2 < 0$

↓

Not possible if all α, β, γ are real.

And complex root occurs in pair, so two out of α, β, γ are complex & one is real.

8. (*a*) $ax^2 + bx + c = 0$

$\sec\alpha$ and $\tan\alpha$ are roots.

$\sec\alpha + \tan\alpha = -\frac{b}{a}$; $\sec\alpha \cdot \tan\alpha = \frac{c}{a}$

$(\sec\alpha - \tan\alpha)^2 = (\sec\alpha + \tan\alpha)^2 - 4\sec\alpha\tan\alpha$

$= \frac{b^2}{a^2} - \frac{4c}{a} = \frac{b^2 - 4ac}{a^2}$

$\sec\alpha - \tan\alpha = \sqrt{\frac{b^2 - 4ac}{a}}$

Now, we know that,

$\sec^2\alpha - \tan^2\alpha = 1$

$\Rightarrow (\sec\alpha - \tan\alpha)(\sec\alpha + \tan\alpha) = 1$

$\Rightarrow \left(\sqrt{\frac{b^2 - 4ac}{a}}\right)\left(\frac{-b}{a}\right) = 1$

$\Rightarrow \frac{(b^2 - 4ac)b^2}{a^4} = 1$ (Squaring both side)

$a^4 - b^4 + 4ab^2c = 0$

9. (*a*) $(l-m)x^2 + lx + 1 = 0$

Roots are α and 2α.

$\alpha + 2\alpha = \frac{-l}{l-m} \Rightarrow \alpha = \frac{-l}{3(l-3)}$ ...(*i*)

$\alpha.2\alpha = \frac{l}{l-m} \Rightarrow \alpha^2 = \frac{-l}{2(l-3)}$ ...(*ii*)

from (*i*) and (*ii*),

$\frac{l^2}{9(l-m)^2} = \frac{1}{2(l-3)}$

$\Rightarrow 2(l-m)l^2 = 9(l-m)^2 \Rightarrow 2l^2 - 9l + 9m = 0$

for real l, $D \geq 0$

$81 - 4.2.9m \geq 0 \Rightarrow 9 - 8m \geq 0$

$m \leq \frac{9}{8}$

10. (*c*) $3px^2 + 5qx + 7r = 0$

$D = (5q)^2 - 4.3p.7r = 25q^2 - 84pr$

$= 25(p+r)2 - 84pr(p + q + r = 0)$

$= 25(p^2 + r^2 + 2pr) - 84pr$

$= 25p^2 + 25r^2 + 50pr - 84pr$

$= 25p^2 - 34pr + 25r^2$

$= (5p)^2 - 2.(5p)\frac{(17r)}{5} + \frac{(17r)^2}{5} - \frac{(17r)^2}{5} + 25r^2$

As $D > 0$

Roots are real & distinct.

11. (*a*) $ax^2 + 2bx + c = 0$

$D = (2b)^2 - 4ac = 4(b^2 - ac)$

As a, b, c are in G.P. So, $b^2 = ac$

$D = b^2 - ac = 0$

Hence, above eq^n has equal roots

$x = \frac{-2b}{2a} = \frac{-b}{a}$

This root will satisfy $dx^2 + 2ex + f = 0$

So, $d\left(\frac{-b}{a}\right)^2 + 2e\left(\frac{-b}{a}\right) + f = 0$

$db^2 - 2aeb + a^2f = 0$

$dac - 2aeb + a^2f = 0$

$dac + a^2f = 2aeb \Rightarrow dc + af = 2eb$

Dividing by ac, both side

$\frac{d}{a} + \frac{f}{c} = \frac{2e}{b} \Rightarrow \frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

12. (*a*) $x^8 - 24x^7 - 18x^5 + 39x^2 + 1155 = 0$

$x^2(x^6 - 24x^5 - 18x^3 + 39) = -1155$

$= -3 \times 5 \times 7 \times 11$

As $x \in Z$.

LHS contains x^2

But in RHS, there is no perfect square.

So, there is no integral value of x.

13. (*d*) $\alpha^2 = 5\alpha - 3$; $\beta^2 = 5\beta - 3$

$\left.\begin{matrix}\alpha^2 - 5\alpha + 3 = 0 \\ \beta^2 - 5\beta + 3 = 0\end{matrix}\right\}$ α & β are roots of $x^2 - 5x + 3 = 0$

$\alpha + \beta = 5$

$\alpha\beta = 3$

Required Eq^n:

$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \left(\frac{\alpha}{\beta}.\frac{\beta}{\alpha}\right) = 0$

$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + 1 = 0$

$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

$= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{25 - 2.3}{3} = \frac{19}{3}$

$x^2 - \frac{19}{3}x + 1 = 0$

$3x^2 - 19x + 1 = 0$

14. (*c*) $x^2 - (1+\gamma)x + \gamma - 2 = 0$

α, β are roots.

$\alpha + \beta = (1 + \gamma) \Rightarrow \alpha\beta = \gamma - 2$

$\alpha + \beta - \alpha\beta = 3 \Rightarrow \alpha\beta - \gamma - \beta + 3 = 0$

$\alpha\beta - \alpha - \beta + 1 + 2 = 0$

$\alpha(\beta - 1) - (\beta - 1) = -2$

$(\alpha - 1)(\beta - 1) = -2$

Since, product is (–ve)

At least one root is +ve.

15. (c) $\frac{(\alpha+1)^2+(\beta+1)^2+(\gamma+1)^2(\delta+1)^2}{\alpha+\beta+\gamma+\delta}=4$

$(\alpha+1)^2+(\beta+1)^2+(\gamma+1)^2+(\delta+1)^2$

$=4\alpha+4\beta+4\gamma+4\delta$

$(\alpha+1)^2-4\alpha+(\beta+1)^2-4\beta+(\gamma+1)^2$

$-4\gamma+(\delta+1)^2-4\delta=0$

$(\alpha-1)^2+(\beta-1)^2+(\gamma-1)^2+(\delta-1)^2=0$

$\alpha=\beta=\gamma=\delta=1$

$a_0x^4+a_1x^3+a_2x^2+a_3x+a_4=0$

Roots are 1, 1, 1, 1

$\frac{a_2}{a_0}=S_2=$ Sum of product of roots taken two at a time.

$\frac{a_2}{a_0}=6$

16. (b) Given:

$(x-b)(r-c)+(x-a)(x-c)+(x-a)(x-b)=0$

$=3x^2-2x(a+b+c)+(ab+bc+ca)=0$

$0=\left[3(a+b+c)^2-12(ab+bc+ca)\right]$

$=(a-b)^2+(b-c)^2+(c+a)^2$

$D\geq 0$

Root are real.

17. (c) Let,

$ax^3+bx+c=\left(x^2px+1\right)(ax+\lambda)$

Then equating coefficients

$0=op+\lambda;\ b=p\lambda+a;\ c=\lambda$

$\Rightarrow P=\frac{-\lambda}{a}=\frac{-c}{a}$

18. (a) Given,

$|x|^2-3|x|+2=0$

or, $(|x|-2)(|x|-1)=0$

or, $|x|=2$ and $|x|=1$

or, $x=\pm 2$ and $x=\pm 1$

So we have 4 real solutions.

19. (c) Let the school be at a distance x km from the town A.

Then the distance of school from town B $=(60-x)$ km

Distance travelled by 150 students of town $A=150x$

Distance travelled by 50 students of town $B=50(60-x)$

So, total distance travelled by 200 students D

$=150x+3000-50x$

$D=f(x)=100\times 3000$

Clearly D is least when $x=0$

Hence, the school should be built at town A.

20. (a) Given, p, q, r are in A.P. then

$q=\frac{p+r}{2}$ ...(i)

Now $px^2+qx+r=0$ will have real root then

$q^2-4pr\geq 0.$

or, $\frac{(p+r)^2}{4}-4pr\geq 0$

or, $p^2+t^2-14pr\geq 0$

or, $r^2-14rp+49p^2\geq 48p^2$

or, $(r-7p)^2\geq\left(4\sqrt{3}p\right)^2$

or, $\left(\frac{r}{p}-7\right)^2\geq\left(4\sqrt{3}\right)^2$

[Since $p\neq 0$ for the given equation to be quadratic]

or, $\left|\frac{r}{p}-7\right|\geq 4\sqrt{3}$.

21. (c) Given equation is $px^2+qx+1=0$

For real roots, discriminant ≥ 0.

Thus $q^2-4p\geq 0\Rightarrow q^2\geq 4p$

For $p=1$, $q^2\geq 4\Rightarrow q=2,3,4$

For $p=2$, $q^2\geq 8\Rightarrow q=3,4$

For $p=3$, $q^2\geq 12\Rightarrow q=4$

For $p=4$, $q^2\geq 16\Rightarrow q=4$

Thus, total seven solutions are possible.

Hence, 7 equations can be form $px^2+qx+1=0$ has real roots.

22. (b) $\alpha+\beta=-b$

And $b>0$ Therefore,

$\alpha+\beta<0$

And

$\alpha\beta=c$ and $c<0$

Therefore,

$\alpha\beta<0$

Now,

$\alpha<\beta$

Therefore, β is positive root while α is negative root and $|\alpha|>|\beta|$ (since the sum of the root is negative).

Hence,

$\alpha<0$ and $\beta>0$

Therefore, $\alpha<0<\beta<|\alpha|$

23. (a) For the numbers $(a+b)$ & $(c+d)$, AM $\geq$ GM

$\Rightarrow\frac{(a+b)+(c+d)}{2}\geq\sqrt{(a+b)(c+d)}$

$\Rightarrow\sqrt{M}\leq 1\Rightarrow M\leq 1$

$\because (a+b)$ and $(c+d)$ are positive, $M>0$

24. (c) Let α, α^2 be the roots of $3x^2+px+3=0$

Now, $S=\alpha+\alpha^2=-\frac{p}{3}$

$P=\alpha^3=1$

$\Rightarrow\alpha=1,\omega,\omega^2$

Now, $\alpha+\alpha^2=-\frac{p}{3}$

$\Rightarrow\omega+\omega^2=-\frac{p}{3}$

$\Rightarrow -1=-\frac{p}{3}$

$p=3$

Hence $p=3$.

25. (a) Given that $a_1,a_2,\ldots a_n$ be positive real numbers such that $a_1,a_2,\ldots a_n=c$ ...(i)

We know that, A.M. $\geq$ G.M.

$\Rightarrow\frac{a_1+a_2+\ldots+a_{n-1}+2a_n}{n}\geq(a_1,a_2,\ldots a_{n-1}(2a_n))^{\frac{1}{n}}$

$\Rightarrow a_1+a_2+\ldots+a_{n-1}+2a_n\geq n(2c)^{\frac{1}{n}}$

...(from (i))

So, the minimum value of

$a_1+a_2+\ldots+a_{n-1}+2a_n$ is $n(2c)^{\frac{1}{n}}$.

26. (a) Let $k^2=n^2+96$, where n, k is a positive integer

$\Rightarrow k^2-n^2=96$

$\Rightarrow (k-n)(k+n)=96$

Both k, n must have the same parity. Hence we will try to find two factors of 96 such that both are even.

96 can be factorized as 2×48, 4×24, 6×16, 8×12

The smaller factor $=k-n$ and the bigger factor $=k+n$

Hence, the values of (n, k) are (23, 25), (10, 14), (5, 11), (2, 10)

Hence, there are 4 possible values of n

27. (a) $x^2+5=2x-4\cos(a+bx)$

$\Rightarrow x^2-2x+1+4=-4\cos(a+bx)$

$\Rightarrow (x-1)^2+4(1+\cos(a+bx))=0$

$\Rightarrow x=1$ and $1+\cos(a+bx)=0$

$\Rightarrow\cos(a+b)=-1\Rightarrow a+b=\pi\ 3\pi\ldots$

28. (b) $x^3+ax^2+bx+c=0$

$r+s+t=-a$

$\Sigma rs=b$

$rst=c$

$(r^2+s^2+t^2)^2=(r^4+s^4+t^4)+2\Sigma(rs)^2$

29. (c) Here, a and b are roots of the equation $x^2-5ax-6d=0$

$a+b=\frac{-(-5)c}{1}=5c$ ...(i)

and

$\Rightarrow a\cdot b=\frac{-6d}{1}=-6d$ ...(ii)

Now c and d are roots of the equation $x^2-5ax-6d=0$

$$\therefore c+d=\frac{-(-5a)}{1}=5a \quad ...(iii)$$

$$\Rightarrow cd=\frac{-6b}{1}=-6b \quad ...(iv)$$

Adding (i) and (ii)

$\Rightarrow a+b+c+d=5c+5a \Rightarrow b+d=4c+4a$

$\Rightarrow b+d=4(c+a)$ ...(v)

Now dividing (i) by (ii) we get

$$\Rightarrow \frac{a+b}{c+d}=\frac{5c}{5a}$$

$\Rightarrow 5a^2+5ab=5c^2+5cd$

$\Rightarrow 5a^2+5(-6d)=5c^2+5(-6b)$ [from (ii) and (iv)]

$\Rightarrow 5a^2-30d=5c^2-30b \Rightarrow 5(a^2-c^2)=-30b+30d$

$\Rightarrow (a+c)(a-c)=-6b+6d$

$\Rightarrow (a+c)(a-c)=-6(5c-a)+6(5a-c)$ [from (i) and (ii)]

$\Rightarrow (a+c)(a-c)=-30c+6a+30a-6c$

$\Rightarrow (a+c)(a-c)=-36c+36a \Rightarrow a+c=36$

Substituting value of $a+c$ in equation (v) we get,

$\Rightarrow b+d=4(36)$

$\therefore b+d=144$

30. **(a)** Let S = { 1, 2 , 3, ..., 100}

For equal roots of equation $4x^2+bx+c=0$

$D=0$

$b^2-16c=0$

$b^2=16c$

	b	c	
Possible values	4	1	} 10 Posibility
	8	4	
	|	|	
	40	100	

And Total No. of Events are 100×100

[For b and c random values form set S]

So, the probability is

$$\frac{100}{100\times100}=\frac{1}{100}=0.001$$

31. **(b)** $a_1=1/2$, $a_2=1/2$, $a_3=-1/2$, $a_4=-1/2$

$(a_1-a_2)^2+(a_2-a_3)^2+(a_3-a_4)^2+(a_4-a_1)^2$

$(0)+1+0+1+2$

$\therefore$ is (1.5, 2.5)

32. **(c)** $x^{135}+x^{125}-x^{115}+x^5+1=q(x)(x^3-x)+Ax^2+Bx+C$

Where $q(x)$ is quotient and $r(x)=Ax^2+Bx+C$ is remainder

Now, putting $x=0, 1, -1$

33. **(d)** $x^5-6x^4+11x^3-5x^2-3x+2=0$

$(x-1)(x-2)(x^3-3x+1)=0$ (factorizing)

Now, x^3-3x+1 has non-integral roots whose sum is 3.

34. **(a)** $x^2+4y^2+32^2-2x-12y-52+14$

$\rightarrow=(x^2-2x+1)+4\left(y^2-22\times\frac{3}{2}\times4+\frac{9}{4}\right)+3(2^2-22+1)+14-1-9-3$

$=(x+1)^2+4(y-3/2)^2+3(2-1)^2+(2-1)^2+1$

since the first three term of this can not be negative. ($\because(-a)^2=a^2$)

the smallest value of question is

35. **(d)** $px^2+qx+r=0$

$rx^2+qx+p=0$ (on subtract)

$\therefore x=-1$

So common root is -1

$\therefore p-q+r=0$

36. **(b)** $(k+1)\tan^2 x-\sqrt{2\lambda}\tan x+(k-1)=0$

$$\tan\alpha+\tan\beta=\frac{\sqrt{2\lambda}}{k+1}$$

$$\tan\alpha\cdot\tan\beta=\frac{k-1}{k+1}$$

$$\tan(\alpha+\beta)=\frac{\frac{\sqrt{2\lambda}}{k+1}}{1-\frac{k-1}{k+1}}=\frac{\lambda}{\sqrt{2}}$$

also given $\therefore\frac{\lambda^2}{2}=50$

$\lambda=\pm10$

37. **(d)** The given quadratic equation: $5x^2+6x-2=0$

And α and β be the roots of this equation

Then substituting $x-\alpha$ in this equation

$\Rightarrow 5\alpha2\,4-6\alpha-2-0$

$\Rightarrow 6\alpha-2=-5\alpha^2$ ... (i)

Similarly, by substituting $x=\beta$ we have,

$\Rightarrow 6\beta-2=-5\beta^2$ ...(ii)

Since given $S_n=\alpha^n+\beta^n$

Therefore.

$S_6-\alpha^6+\beta^2$ ...(iii)

$S_5-\alpha^5+\beta^4$ ...(iv)

$S_4=\alpha^4+\beta^4$ ...(v)

Now by.

$6x(iv)-2x(v)$ we have

$\Rightarrow 6S_5-2S_4=6\alpha^5-2\alpha^4-2\beta$

$=\alpha^4(6\alpha-2)+\beta^4(6\beta-2)$

$=\alpha^4(-5\alpha^2)+\beta^4(-5\beta^2)$

$=-5(\alpha^6+\beta^6)=-5S_6$

Thus. $6S_5\cdot4-5S_6=2S_4$

38. **(d)** Consider $(p^2+q^2)^2-2p^2q^2=272$

$((p+q)^2-2pq)^2-2p^2q^2=272$

$16-16pq+2p^2q^2=272$

$(pq)^2-8pq-128=0$

$(pq)^2-8pq-128=0$

$$pq=\frac{8=24}{2}=16,-8 \therefore pq=16$$

Required equation: $x^2-(2)x+16=0$

39. **(c)**

$abc+ab+a+ca+bc+b+c+1=30$

$ab(c+1)+a(c+1)+b(c+1)+c+1=30$

$(c+1)(ab+a+b+1)=30$

$(c+1)(b+1)(a+1)=30$

Now, 30 can be expressed as product of three nos (x, y, z) in following ways: [such that $x-1, y-1, z-1$ belong to set $\{0,1,2,3,4,5,6,7,8,9\}$]

case 1: $x,y,z\in\{2,3,5\}$

case 2: $x,y,z\in\{1,5,6\}$

case 3: $x,y,z\in\{1,3,10\}$

For Case 1: The digits can be assigned values in 3 ! ways.

For Case 2: Since, a in abc cannot be o [as the no will become a 2-digit no] there we have only two places to assign 1 ie. ones and tens digit. and then 2 ´ 1 for the remaining two digits. Thus, 2 ´ 2 ´ 1 = 4 ways for case #2 .

Similarly, for Case 3 : we have $2\times2\times1=4$ (Same case as Case #2).

So, the total number of all 3 digit numbers is 14

40. **[167]** $\sqrt[3]{6(5x+6)}-\sqrt[3]{5(6x-1)}=1$

Cube both sides

$\Rightarrow 30x+36-30x+55-3(1)\{6(5x+6)\}^{1/3}\{5(6x-11\}^{1/3}=1$

$\Rightarrow (30)^3=6(5x+6)\,5(6x-11)$

$\Rightarrow 900=(5x+6)(6x-11)$

$\Rightarrow x=6, -\frac{161}{30}$

41. **[99]** $mx^2+x(2-m)+3=0$

From the equation $p+q=\frac{m-2}{m}, pq=\frac{3}{m}$

Now, $\frac{p}{q}+\frac{q}{p}=\frac{2}{3}$

$\Rightarrow 3(p+q)^2=8pq$

On putting $p+q$ and pq, we get

$$3\times\left(\frac{m-2}{m}\right)^2=\frac{24}{m}$$

$m^2-12m+4=0$

So,

$m_1+m_2=12$ and $m_1m_2=4$

Now, $\frac{m_1}{m_2^2}+\frac{m_2}{m_1^2}$

$=\frac{12^2\times11}{16}=99$

42. **[2]** $\Delta_1=9-4a\Rightarrow\Delta_1+\Delta_2=a_2-4a+5$

$\Delta_2=a^2-4=(a-2)^2+1>0$

43. **[8]** Given $x^2+6x+y^2-4=0$

$=\frac{-6\pm\sqrt{52-4y^2}}{2}$

For $x\varepsilon z,\sqrt{52-4y^2}$ must be an even integer

$\Rightarrow\sqrt{52-4y^2}=2,4,6$

If $x=\pm3$

$x=\frac{-f+4}{2}\Rightarrow x=-1,-5$

and $y=\pm3\Rightarrow x,y=(-1,\pm3),(-5,13)$

If $y=\pm2$

$x=\frac{-6+6}{2}=(0,-6)\Rightarrow(x,y)=(0,\pm2),(-6,\pm2)$

$\therefore$ there are 8 solutions

44. **[9]** $x^3-3x^2+2x+\frac{2}{3}=0$

$\alpha+\beta+\gamma=3$

$\alpha\beta+\beta\gamma+\gamma\alpha=2$

$\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)=5$

$\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2=(\alpha\beta+\beta\gamma+\gamma\alpha)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)$

$=(2)^2+2\cdot\frac{2}{3}\cdot3$

$\alpha^4+\beta^4+\gamma^4=(\alpha^2+\beta^2+\gamma^2)^2-2(\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2)$

$=9$

45. **[23]**

$P(x)=Q(x)(x-1)(x-2)+2x+3$

$P(10)=72n+23$

46. **[324]**

$\alpha^2-\alpha-1=0\Rightarrow\alpha^{n+1}=\alpha^n+\alpha^{n-1}$ ...(ii)

Similarly $\beta^{n+1}=\beta^n+\beta^{n-1}$ ...(iii)

Adding equations (ii) and (iii)

$\Rightarrow\alpha^{n+1}+\beta^{n+1}=(\alpha)^n+(\beta)^n+\left[(\alpha)^{n+1}+(\beta)^{n+1}\right]$

$P_{n+1}=P_n+P_{n-1}$

$P_n=29-11$

$P_n=18$

Hence, the value of $P_n^2=324$.

47. **[191]** $x^2+3x-k=0$

$a+b=\frac{-3}{1}=-3$

$ab=\frac{(-k)}{1}=-k$

$\sqrt{(a-b)^2}=(a+b)^2-4ab$

$=\sqrt{9+4k}$

$|a-b|=\sqrt{9+4k}$

$x^2+3x-10=0$

$c+d=-3\quad cd=-10$

$(c-d)^2=(c+d)^2-4cd$

$(c-d)^2=9+40=49$

$\boxed{|c-d|=7}$

$|a-b|=2|c-d|$

$(a-b)^2=4(c-d)^2$

$9+4k=4(49)-9$

$4k=4(49)-9$

$4k=187$

$\Rightarrow k=\frac{187}{4}$.

48. **[0]** $R_1=R_1$

$2+m+1=2-(m-1)$

$m+1=-m+1$

$2m=0$

$m=0$

49. **[2]** $P(x)=x^4(x^2-1)q$

$f(x)$

Put $x=1$

$p(1)=0+f(1)$

$2=f(1)$

50. **[3]** (i) $D>0$

$4a^2-4(a^2-1)>0$

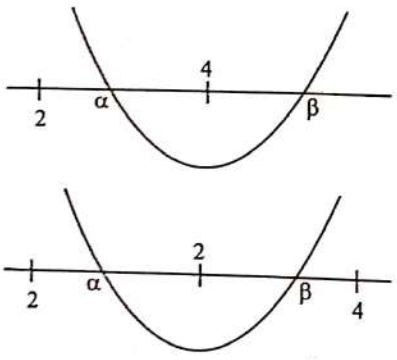
$4>0\ \forall\ x\in R$

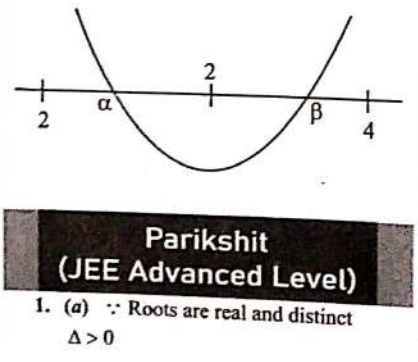
(ii) $f(2)f(4)<0$

$(4-4a+a^2-1)(16-8a+a^2-1)<0$

$(a-3)^2(a-1)(a-5)<0$

$A\in(1,5)-\{3\}$





Parikshit (JEE Advanced Level)

1. **(a)** $\because$ Roots are real and distinct

$\Delta>0$

$\Rightarrow4(a+b+c)^2-12\lambda(ab+bc+ca)>0$

$\Rightarrow4\left[(a+b+c)^2-3\lambda(ab+bc+ca)\right]>0$

$\Rightarrow a^2+b^2+c^2+2ab+2bc+2ca-3\lambda(ab+bc+ca)>0$

$\Rightarrow\sum a^2-(3\lambda-2)\sum ab>0$

$\Rightarrow\sum a^2>(3\lambda-2)\sum ab$

$\Rightarrow\frac{\sum a^2}{\sum ab}=3\lambda-2$ ...(1)

Now, in a triangle

Difference of two sides of a triangle < third side.

$|a-b|<c;|b-c|<a;|c-a|<b$

$\Rightarrow(a-b)^2+(b-c)^2+(c-a)^2<a^2+b^2+c^2$

$\Rightarrow a^2+b^2+c^2<2(ab+bc+ca)$

or $\frac{\sum a^2}{\sum ab}<2$ ...(2)

From equation (1) and (2) we get

$3\lambda-2<\frac{\sum a^2}{\sum ab}<2$

$\Rightarrow3\lambda-2<2$

$\Rightarrow3\lambda<4$ or $\lambda<\frac{4}{3}$

2. **(d)** Given equation is $x^8-x^5+x^2-x+2=0$. Clearly given equation will have no negative root. Now given equation can be written as $x^5(x^3-1)+x(x-1)+2=0$. Clearly no value of x will satisfy the given equation.

3. **(c)** Let the roots of the equation $a^2+bx+c=0$ be

α,α^n

Product of roots $\alpha^{n+1}=\frac{c}{a}$

$\Rightarrow a=\left(\frac{c}{a}\right)^{\frac{1}{n+1}}$

Sum of roots $\alpha+\alpha^n=\frac{-b}{a}$

$\Rightarrow\left(\frac{c}{a}\right)^{\frac{1}{n+1}}+\left(\frac{c}{a}\right)^{\frac{n}{n+1}}=\frac{-b}{a}$

$\Rightarrow a\left(\frac{c}{a}\right)\frac{1}{n+1}+a\left(\frac{c}{a}\right)\frac{n}{n+1}+b=0$

$\therefore\left(a^nc\right)^{\frac{1}{n+1}}+\left(ac^n\right)^{\frac{1}{n+1}}+b=0$

4. **(c)** $f(x)=x^2+ax+2$

$g(x)=x^2+2x+a$

$f(x)-g(x)=(a-2)x+(2-a)=0$

$\Rightarrow x=1$

$f(1)=0 \Rightarrow 1+a+2=0 \Rightarrow a=-3$

$f(x)+g(x)=2x^2-x-1$

5. (*d*) $(a+x)^2+64(a-x)^2+4$
$3(a^2-x^2)^{2/3}\,[5(a^2-x^2)^{1/3}]$
$=125(a^2-x)^2$
$\Rightarrow (a+x)^2+64(a-x)^2-65(a^2-x^2)=0$
$P^2+64Q^2-65PQ=0$
$\Rightarrow \quad P^2-PQ-64\,PQ+64Q^2=0$
$\Rightarrow \quad x_1=\dfrac{63Q}{65}, x_2=0$

6. (*a*) Let $f(x)=ax^2+(b-1)x+c$

$$\left.\begin{aligned} f(x_1)&=x_2-x_1\\ f(x_2)&=x_3-x_2\\ f(x_3)&=x_1-x_3 \end{aligned}\right\} \Rightarrow f(x_1)+f(x_2)+f(x_3)=0$$

$\therefore \quad af(x_1)+Af(x_2)+af(x_3)=0$ (not possible)

As $(b-1)^2-4ac<0$

$\therefore \quad af(x_1), af(x_2), af(x_3)>0$

Hence given system of equations has no real root.

7. (*a*) Given $x^4+12x^3+46x^2+ax+b=(x^2+cx+d)^2$

Can paring coefficient of x both the sides, we get $12=2c, 46=2d+c^2, a=2cd$ and $b=d^2$

$\Rightarrow c=6$ and $d=5$

$\Rightarrow a=60$ and $b=25$

8. (*c*) $k\sin^2 x-k\sin x+1\geq 0$

$\sin x=t\in[-1,1]$

$kt^2-kt+1\geq 0$

When $k>0 \quad t^2-t+\dfrac{1}{k}\geq 0$

$\Rightarrow \dfrac{1}{2}\geq\dfrac{1}{4}$

$\Rightarrow K\leq 4$

When k < 0

$t^2-t+\dfrac{1}{k}\leq 0$

$\dfrac{k}{k}\leq -2 \quad \Rightarrow k\geq -\dfrac{1}{2}$

9. (*a*) $S_1: x^2-bx+c=0$ (roots α, β)

$\because |\alpha-\beta|=1$
$\Rightarrow (\alpha-\beta)^2=1 \Rightarrow \quad b^2-4c=1.$

$S_2 \because \alpha+\beta=1$ and $\alpha\beta=3$
$\because \alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2$
$=[(\alpha+\beta)^2-2\alpha\beta]^2-2(\alpha\beta)^2$
$=(1-6)^2-2(9)=25-18=7$

$S_3 \because \Sigma\alpha=7 \Rightarrow \Sigma\alpha\beta=16 \Rightarrow \alpha\beta\gamma=12$
$\therefore \Sigma\alpha^2=(\Sigma\alpha)^2-2\,(\Sigma\alpha\beta)=49-32$
$\therefore \alpha^2+\beta^2+\gamma^2=17.$

10. (*a*)

$2\log_{1/25}(bx+28)=-\log_5\left(12-4x-x^2\right)$

$\Rightarrow \quad x^2+(b+4)x+16=0$

$D=0 \Rightarrow b=4,-12$

and only our root in domain of log

$f(-6)f(2)<0$

$b\in(-\infty,14)\cup\left(\dfrac{14}{3},\infty\right)$

$$\begin{bmatrix} 12-4x-x^2>0 \\ -6<x<2 \end{bmatrix}$$

$\Rightarrow b\in(-\infty,-14)\cup\left[\dfrac{14}{3},\infty\right)\cup\{4\}$

11. (*a,d*) Clearly $a<0$

$\dfrac{-b}{2a}>0 \Rightarrow b>0$

$\therefore$ (*a*), (*d*)

12. (*a, d*)

$20x^2+210x+400=4500 \Rightarrow 2x^2+21x-410=0$

$\Rightarrow (2x+41)(x-10)=0$

$\Rightarrow x=\dfrac{-41}{2}, x=10 \quad \Rightarrow x=-20.5, x=10$

13. (*a,b,c,d*)

$x^3+bx^2+cx-1=0$ (roots $\alpha=\frac{a}{r}$, $\beta=a$, $a=ar$)

$\dfrac{a}{r}+a+ar=-b \quad \Rightarrow a\left(\dfrac{1}{r}+1+r\right)=-b$

and $\dfrac{a}{r}\times a\times ar=1$

$a^3=1 \Rightarrow a=1$

and $\dfrac{a}{r}a+a\cdot ar+\dfrac{a}{r}\cdot ar=c$

$a^2\left(\dfrac{1}{r}+r+1\right)=c$

$\dfrac{1}{r}+r+1=-b \;\&\; \dfrac{1}{r}+r+1=c \Rightarrow b+c=0$

we know $\dfrac{1}{r}+r>2 \Rightarrow \left(\dfrac{1}{r}+r+1\right)>3$

$-b>3 \Rightarrow b<-3 \quad \Rightarrow b\in(-\infty,-3)$

and other two roots are $\dfrac{1}{r}$ and r

if $\dfrac{1}{r}>1 \Rightarrow r<1$ if $r>1 \Rightarrow r<1$

14. (*a,b,d*) $\sec^2\theta$ & $\csc^2\theta$ are the roots of the equation $ax^2+bx+c=0$

since the roots are real

$\therefore b^2-4ac\geq 0$

Sum of the roots of the equation

$=\sec^2\theta+\csc^2\theta=\dfrac{-b}{a}$

$\Rightarrow \dfrac{\sin^2\theta+\cos^2\theta}{\sin^2\theta\cos^2\theta}=\dfrac{-b}{a}$

$\dfrac{1}{\sin^2\theta\cos^2\theta}=\dfrac{-b}{a}$ .... (1)

Product of the roots of the equation

$=\sec^2\theta\csc^2\theta=\dfrac{c}{a}$

$\Rightarrow \dfrac{1}{\sin^2\theta\cos^2\theta}=\dfrac{c}{a}$ ... (2)

$\Rightarrow \sin^2 2\theta=\dfrac{4a}{c}\leq 1 \Rightarrow c\geq 4a$

Combining (1) and (2) we get,

$b+c=0$

15. (*b,c,d*) $x^4+4x^3-8x^2+k=0$,

$K\in[0,2014]$

Let $f(x)=x^4+4x^3-8x^2$

$g(x)=-k$

$f(x)=x^2\left(x^2+4x-8\right)$

OR $f'(x)=4x^3+12x^2-16x$

$=4x\left(x^2+3x-4\right)$

$=4x(x+4)(x-1)$

16. (*d*) $R(x)=x^2+ax+1$

$R(3)=2, R(4)=1$

$f(x)=(x-3)(x-4)Q(x)+R(x)$

$f(x)=(x-3)(x-3)Q(x)+Ax+B$

$f(3)=3A+B=2$

$f(4)=4A+B=1$

$A=-1, B=5$

$R(x)=-x+5$

$-x+5=x^2+ax+1$

$x^2+(a+1)x-4=0$

$D>0$

$(a+1)^2+16>0$

$a\in R$

17. (*c*) $-x+5=px^2+(q-1)x+6$

$px^2+qx+1=0$

$D\leq 0$

$3(3p+q)+1\geq 0$

$3p+q\geq -\dfrac{1}{3}$

18. (*c*) Given $R(x)$ is the remainder when unknown polynomial is divided by $(x-3)(x-4)$

Clearly, degree of $R(x)$ will be less than 2 (degree of divisor)

So, let $R(x) = ax + b$

given $R(3) = 2$

$\Rightarrow 3a + b = 2$ ...(1)

Also $R(4) = 1 \Rightarrow 4a + b = 1$

Solving (1) and (2), we get

$a = -1, b = 5$

So $R(x) = -x + 5$

Now, $y = \dfrac{-x+5}{x^2-3x+2}$

$\Rightarrow yx^2 + x(-3y+1) + (2y-5) = 0$

$\Rightarrow y^2 + 14y + 1 \geq 0$ (since x is real, $D \geq 0$)

$\Rightarrow (y + 7 + 4\sqrt{3})(y + 7 - 4\sqrt{3}) \geq 0$

$\Rightarrow y \in (-\infty, -7 - 4\sqrt{3}] \cup [-7 + 4\sqrt{3}, \infty)$

19. (c) $x^4 + 2ax^3 + x^2 + 2ax + 1 = 1$

$x^2\left(x^2 + \dfrac{1}{x^2} + 1 + 2\left(x + \dfrac{1}{x^2}\right)\right) : 0$

$\left(x + \dfrac{1}{x}\right)^2 - 2 + 1 + 2x\left(x + 1\dfrac{1}{x}\right) = 0$

put $x + \dfrac{1}{x} = t$

$t^2 + 2at - 1 = 0$

the only possible case for this equation having two distinct positive roots

$f(2) < 0$

$4 + 4a - 1 < 0$

$4a + 3 < 0$

$a < -\dfrac{3}{4}$

$a \varepsilon \left(-\infty, -\dfrac{3}{4}\right)$

20. (a) Equation having two distinct negative roots gives

$f(-2) < 0$

$4 - 4a - 1 < 0$

$a \varepsilon \left(\dfrac{3}{4}, \infty\right)$

21. (c) For real roots are not possible

22. (a) $(A) \to (R), (B) \to (P), (C) \to (S), (D) \to (P, Q)$

(a) $x^2 - 8x + k = 0$ $\begin{cases} \alpha + h \\ \alpha + 4 = \beta \end{cases}$

$\because (\beta - \alpha)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$\Rightarrow 16 = 64 - 4k \Rightarrow 4k = 48 \Rightarrow k = 12$

(b) $\because (|x| - 2)(|x| - 3) = 0$

$\Rightarrow x = \pm 2; x = \pm 3$

$\therefore n = 4 \quad \therefore \dfrac{n}{2} = 2$

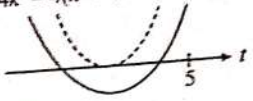
(c) $\because b = (3 - i)(3 + i)$

$b = 10$

(d) $x^2 - 2kx + (k^2 + k - 5) = 0$

(i) $D \geq 0$

$\Rightarrow 4k^2 - 4(k^2 + k - 5) \geq 0 \Rightarrow k - 5 \leq 0$

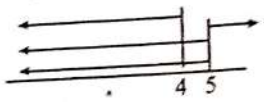


(ii) $f(5) > 0$

$\Rightarrow 25 - 10k + k^2 + k - 5 > 0$

$\Rightarrow k^2 - 9k + 20 > 0 \quad \Rightarrow (k - 5)(k - 4) > 0$

(iii) $-\dfrac{b}{2a} < 5 \Rightarrow k < 5$



$\Rightarrow k \in (-\infty, 4)$

So k may be 2, 3.

23. [2] $(x^2 + 3x + 2)(x^2 + 3x) = 120$

Let $x^2 + 3x = y$

$\Rightarrow y^2 + 2y - 120 = 0 \Rightarrow (y + 12)(y - 10) = 0$

$\Rightarrow y = -12 \quad \Rightarrow x^2 + 3x + 12 = 0$

$\Rightarrow x \in \phi$

$y = 10 \quad \Rightarrow x^2 + 3x - 10 = 0$

$\Rightarrow (x + 5)(x - 2) = 0 \quad \Rightarrow x = \{-5, 2\}$

$x = 2, -5$ are only two integer roots.

24. [8] $\left(5 + 2\sqrt{6}\right)^{x^2-3} + \dfrac{1}{\left(5 + 2\sqrt{6}\right)^{x^2-3}} = 10$

$\Rightarrow t + \dfrac{1}{t} = 10$

$\Rightarrow t^2 - 10t + 1 = 0 \quad t = \dfrac{10 \pm \sqrt{96}}{2} = 5 \pm 2\sqrt{6}$

$\Rightarrow (5 + 2\sqrt{6})^{x^2-3} = (5 + 2\sqrt{6})$ or $\dfrac{1}{5 + 2\sqrt{6}}$

$\Rightarrow x^2 - 3 = 1$ or $x^2 - 3 = -1$

$\Rightarrow x = 2$ or -2 or $-\sqrt{2}$ or $\sqrt{2}$

Product 8

25. [11] $2x^2 + 6x + a = 0$

Its roots are α, β

$\Rightarrow \alpha + \beta = -3 \quad \& \alpha\beta = \dfrac{a}{2}$

$\Rightarrow \dfrac{\alpha}{\beta} + \dfrac{\beta}{\alpha} < 2$

$\Rightarrow \dfrac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} < 2 \Rightarrow \dfrac{9 - a}{a} < 1$

$\Rightarrow \dfrac{2a - 9}{a} > 0$

$\Rightarrow a \in (-\infty, 0) \cup \left(\dfrac{9}{2}, \infty\right)$

$\Rightarrow 2a = 11$ is least prime.

26. [1]

$x^2 + px + 1 = 0$ $\begin{cases} a \\ b \end{cases}$ $a + b = -p, ab = 1;$

$x^2 + qx + 1 = 0$ $\begin{cases} c \\ d \end{cases}$ $c + d = -q, cd = 1$

$a + b = -p, ab = 1 \Rightarrow c + d = -q, cd = 1$

$RHS = (a - c)(b - c)(a + d)(b + d)$

$= (ab - ac - bc + c^2)(ab + ad + bd + d^2)$

$= (1 - ac - bc + c^2)(1 + ad + bd + d^2)$

$= 1 + ad + bd + d^2 - ac - a^2cd - abcd - acd^2 - bc - abcd - b^2cd - bcd^2 + c^2 + adc^2 + bdc^2 + c^2d^2$

$= 1 + ad + bd + d^2 - ac - a^2 - 1 - ad - bc - 1 - b^2 - bd + c^2 + ac + bc + 1$

$[\because ab = cd = 1]$

$= c^2 + d^2 - a^2 - b^2 = (c + d)^2 - 2cd - (a + b)^2 + 2ab$

$= q^2 - 2 - p^2 + 2 = q^2 - p^2 = LHS.$ **Proved.**

2nd Method :

$RHS = (ab - c(a + b) + c^2)(ab + d(ab + d(a + b) + d^2)$

$= (c^2 + pc + 1)(1 - pd + d^2)$ ...(1)

Since c & d are the roots of the equation $x^2 + qx + 1 = 0$

$\therefore c^2 + qc + 1 = 0 \Rightarrow c^2 + 1 = -qc$ & $d^2 + qd + 1 = 0$

$\Rightarrow d^2 + 1 = -qd.$

$\therefore$ (i) Becomes $= (pc - qc)(-pd - qd) = c(p - q)(-d)(p + q) = -cd(p^2 - q^2)$

$= cd(q^2 - p^2) = q^2 - p^2 =$ LHS. **Proved.**

27. [73] $\because \alpha, \beta$ are roots of $\lambda x^2 - (\lambda - 1)x + 5 = 0$

$\therefore \alpha + \beta = \dfrac{\lambda - 1}{\lambda}$ and $\alpha\beta = \dfrac{5}{\lambda}$

$\because \dfrac{\alpha}{\beta} + \dfrac{\beta}{\alpha} = 4 \quad \Rightarrow \dfrac{\alpha^2 + \beta^2}{\alpha\beta} 4$

$\Rightarrow (\alpha + \beta)^2 = 6\alpha\beta \Rightarrow \dfrac{(\lambda - 1)^2}{\lambda^2} = \dfrac{30}{\lambda}$

$\Rightarrow \lambda^2 - 32\lambda + 1 = 0$ ...(1)

$\because \lambda_1, \lambda_2$ are roots of (1)

$\therefore \lambda_1 + \lambda_2 = 32$ and $\lambda_1\lambda_2 = 1$

$\therefore \dfrac{\lambda_1}{\lambda_2} + \dfrac{\lambda_2}{\lambda_1} = \dfrac{(\lambda_1 + \lambda_2)^2 - 2\lambda_1\lambda_2}{\lambda_1\lambda_2} = \dfrac{(32)^2 - 2}{1}$

$= 1022 \Rightarrow \left(\dfrac{\dfrac{\lambda_1}{\lambda_2} + \dfrac{\lambda_2}{\lambda_1}}{14}\right) = 73$

28. [10] $\alpha.\beta = b; \gamma\delta = b - 2$

$\Rightarrow \alpha\beta\gamma\delta = b(b - 2) = 24$

$\therefore bx^2 + ax + 1 = 0$ has roots $\dfrac{1}{\alpha}, \dfrac{1}{\beta}$

$\Rightarrow \dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{-a}{b}$

$(b - 2)x^2 - ax + 1 = 0$ has root $\dfrac{1}{\gamma}, \dfrac{1}{\delta}$

$\Rightarrow \frac{1}{\gamma}+\frac{1}{\delta}=\frac{a}{b-2}$

$\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}=\frac{-a}{b}+\frac{a}{b-2}=\frac{5}{6}; \frac{+2a}{b(b-2)}=\frac{5}{6};$

$\frac{+2a}{24}=\frac{5}{6}; \ a=10.$

29. **[3]** $x^2+px+1=0$

Roots $= a, b$

$a+b=-p; ab=1$ ...(*i*)

$x^2+qx+1=0 \Rightarrow c+d=-q; cd=1$ ...(*ii*)

Roots $= c, d$

Also, $c^2+qc+1=0; d^2+qd+1=0$ ...(*iii*)

$(a-c)(b-c)(a+b)(b+d)=(ab-(a+b)c+c^2)(ab+(a+b)d+d^2)$

$=(1+p.c+c^2)(1-pd+d^2)$ from ...(*i*)

$=(c^2+1+pc)(d^2+1-pd)$

$=(-qc+pc)(-qd-pd)$ from ...(*iii*)

$=-c(p-q)(p+q)d$

$=-cd(p^2-q^2)$

$=-1(p^2-q^2)=q^2-p^2$

Required value $=$

$$\frac{(a-c)(b-c)(a+d)(b+d)}{q^2-p^2}$$

$$=\frac{q^2-p^2}{q^2-p^2}=1$$

30. **[13]** $a^3+b^3+(-9)^3=3\cdot a\cdot b\,(-9)$

$\Rightarrow a+b-9=0$ or

$a=b=-9$. Which is rejected.

As $a>b>-9$

$\Rightarrow a+b-9=0 \Rightarrow x=1$ is a root

Other root $=\frac{-9}{a}$. $\therefore \alpha=\frac{-9}{a}, \beta=1$

$\Rightarrow 4\beta-a\alpha=4-a\left(\frac{-9}{a}\right)=4+9=13.$

PYQ's (Past Year Questions)

1. **(c)** $\alpha,\beta=\frac{-\sqrt{6}\pm\sqrt{6-12}}{2}=\frac{-\sqrt{6}\pm\sqrt{6}i}{2}$

$=\frac{\sqrt{6}}{2}(-1\pm i)$

$=\sqrt{3}\left(\frac{-1}{\sqrt{2}}\pm\frac{1}{\sqrt{2}}i\right)$

$=\sqrt{3}e^{\pm\frac{3\pi i}{4}}$

Required expression

$$=\frac{(\sqrt{3})^{23}\left(2\cos\frac{69\pi}{4}\right)+(\sqrt{3})^{14}\left(2\cos\frac{42\pi}{4}\right)}{(\sqrt{3})^{15}\left(2\cos\frac{45\pi}{4}\right)+(\sqrt{3})^{10}\left(2\cos\frac{30\pi}{4}\right)}$$

$=(\sqrt{3})^8=81$

2. **(c)** Given, $14x^2-31x+3\lambda=0$

Roots are α and β.

$\therefore \alpha+\beta=\frac{31}{14}$ ...(*i*)

and $\alpha\beta=\frac{3\lambda}{14}$ ...(*ii*)

Also, we have $35x^2-53x+4\lambda=0$

Roots are α and γ.

$\alpha+\gamma=\frac{53}{35}$ ...(*iii*)

and $\alpha\gamma=\frac{4\lambda}{35}$ ...(*iv*)

On dividing equ. (*ii*) from equ. (*iv*), we get

$\frac{\beta}{\gamma}=\frac{3\times35}{4\times14}=\frac{15}{8}\Rightarrow\beta=\frac{15}{8}\gamma$

On subtracting equ.(*iii*) from equ.(*i*), we get

$\beta-\gamma=\frac{31}{14}-\frac{53}{35}=\frac{155-106}{70}=\frac{7}{10}$

$\Rightarrow\frac{15}{8}\gamma-\gamma=\frac{7}{10}\Rightarrow\gamma=\frac{4}{5}$

$\Rightarrow\beta=\frac{15}{8}\times\frac{4}{5}=\frac{3}{2}$

$\Rightarrow\alpha=\frac{31}{14}-\beta=\frac{31}{14}-\frac{3}{2}=\frac{5}{7}$ [from equ (*i*)]

$\Rightarrow\lambda=\frac{14}{3}\alpha\beta=\frac{14}{3}\times\frac{5}{7}\times\frac{3}{2}=5$

So, sum of roots $=$

$$\frac{3\alpha}{\beta}+\frac{4\alpha}{\gamma}=\left(\frac{3\alpha\gamma+4\alpha\beta}{\beta\gamma}\right)$$

$$=\frac{\left(3\times\frac{4\lambda}{35}+4\times\frac{3\lambda}{14}\right)}{\beta\gamma}=\frac{12\lambda(14+35)}{14\times35\beta\gamma}$$

$$=\frac{49\times12\times5}{490\times\frac{3}{2}\times\frac{4}{5}}=5$$

Product of roots

$$=\frac{3\alpha}{\beta}\times\frac{4\alpha}{\gamma}=\frac{12\alpha^2}{\beta\gamma}=\frac{12\times\frac{25}{49}}{\frac{3}{2}\times\frac{4}{5}}=\frac{250}{49}$$

So, required equation is $x^2-5x+\frac{250}{49}=0$

$\Rightarrow 49x^2-245x+250=0$

3. **[51]** $x^2-7x-1=0<\frac{a}{b}$

By newton's theorem

$S_{n+2}-7S_{n+1}-S_n=0$ ...(*i*)

Put $n=19$

$\Rightarrow S_{21}-7S_{20}-S_{19}=0$

Now put $n=18$ in equation (*i*)

$\Rightarrow S_{20}-7S_{19}-S_{18}=0$

And for $n=17$

$\Rightarrow S_{19}-7S_{18}-S_{17}=0$

Given expression $=$

$$\frac{S_{21}+S_{17}}{S_{19}}=\frac{S_{21}+(S_{19}-7S_{18})}{S_{19}}$$

[Using above result]

$$=\frac{S_{21}+S_{19}-7(S_{20}-7S_{19})}{S_{19}}$$

$$=\frac{50S_{19}+(S_{21}-7S_{20})}{S_{19}}=51\cdot\frac{S_{19}}{S_{19}}=51$$

4. **(c)** Let α and β be the roots of the equation $x^2+(3-a)\,x+1=2a$

Therefore $\alpha+\beta=a-3, \alpha\beta=1-2a$

$\Rightarrow\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=(a-3)^2-2(1-2a)$

$=a^2+9-6a-2+4a$

$=a^2-2a+7$

$=(a-1)^2+6\geq6$

Hence, min-value is 6.

5. **[4]** Given, $p+q=3$ and $p^4+q^4=369$

$\therefore (p+q)^2=9$

$p^2+q^2=9-2pq$

$$\frac{1}{\left(\frac{1}{p}+\frac{1}{q}\right)^2}=\frac{(qp)^2}{(q+p)^2}=\frac{(qp)^2}{9}$$ ...(*i*)

$\Rightarrow p^4+q^4=(p^2+q^2)^2-2p^2q^2$

$\Rightarrow 369=(9-2qp)^2-2(qp)^2$

$\Rightarrow 369=81+4p^2q^2-36pq-2p^2q^2$

$\Rightarrow 288=2p^2q^2-36pq$

$\Rightarrow 144=p^2q^2-18pq$

$\Rightarrow (pq)^2-2\times9\times pq+9^2=144+9^2$

$\Rightarrow (qp-9)^2=225 \Rightarrow pq-9=\pm15$

$\Rightarrow pq=\pm15+9$

$pq=24, -6$

(24 is rejected because $p^2+q^2=9-2qp$ is negative)

From eq^n (*i*) we have $\frac{(pq)^2}{9}=4$

6. **(a)** $\alpha^2-6\alpha-2=0 \Rightarrow a_{10}-6a_9-2a_8=0$

$\Rightarrow\frac{a_{10}-2a_8}{3a_9}=2$

7. **(c)** $e^{6x}-e^{4x}-2e^{3x}-12e^{2x}+e^{x}+1=0$

$\Rightarrow e^{6x}-2e^{3x}+1-e^{x}\left[e^{3x}+12e^{x}-1\right]=0$

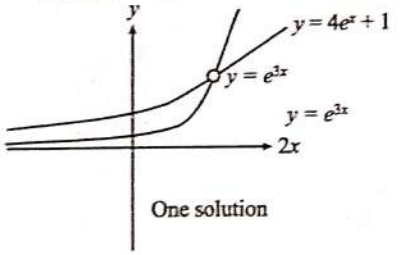
$\Rightarrow \left(e^{3x}-1\right)^2-e^{x}\left(e^{3x}-1\right)-12e^{2x}=0$

$\left(e^{3x}-1\right)^2-4e^{x}\left(e^{3x}-1\right)+3e^{x}\left(e^{3x}-1\right)-12e^{2x}=0$

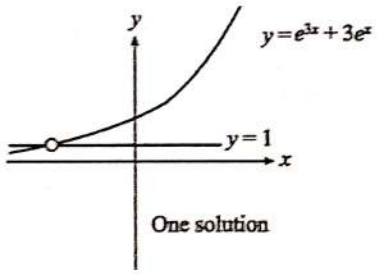
$\left(e^{3x}-1\right)\left\{e^{3x}-1-4e^{x}\right\}+3e^{x}\left\{e^{3x}-1-4e^{x}\right\}=0$

$\left(e^{3x}-4e^{x}-1\right)\left(e^{3x}-1+3e^{x}\right)=0$

Either $e^{3x}=4e^{x}+1$



OR $(e^{3x}=1-3e^{x})$



$\therefore$ The equation has total 2 solutions.

8. **[66]** After simplification : $2x^2-(6+k)x+4+3k=0$

For no solution:

$D<0 \Rightarrow 6-4\sqrt{2}<k<6+4\sqrt{2}$

Hence, s of integral values of K is $1+2+3+\ldots+11=66$

9. **(d)** $(k+1)\tan^2x-\sqrt{2}\cdot\lambda\tan x+(k-1)=0$

Sum of roots $=\tan\alpha+\tan\beta=\dfrac{\sqrt{2}\lambda}{k+1}$

Product of roots $=\tan\alpha\tan\beta=\dfrac{k-1}{k+1}$

Now, $\tan(\alpha+\beta)=\dfrac{\frac{\sqrt{2}\lambda}{k+1}}{1-\frac{k-1}{k+1}}=\dfrac{\sqrt{2}\lambda}{2}=\dfrac{\lambda}{\sqrt{2}}$

Given, $\tan^2(\alpha+\beta)=\dfrac{\lambda^2}{2}=50 \Rightarrow \lambda=10.$

10. **[8]** $D\geq 0$

$(a-10)^2-4(2)\left(\dfrac{33}{2}-2a\right)\geq 0$

$(a-10)^2-4(33-4a)\geq 0$

$a^2-4a-32\geq 0 \Rightarrow a\in(-\infty,-4]\cup[8,\infty).$

11. **(d)** $x^2+20x-2020=0$ has two roots a, $b\in R$

$x^2-20x+2020=0$ has two roots c, $d\in$ complex

$ac(a-c)+ad(a-d)+bc(b-c)+bd(b-d)$

$=a^2c-ac^2+a^2d-ad^2+b^2c-bc^2+b^2d-bd^2$

$=a^2(c+d)+b^2(c+d)-c^2(a+b)-d^2(a+b)$

$=(c+d)(a^2+b^2)-(a+b)(c^2+d^2)$

$=(c+d)((a+b)^2-2ab)-(a+b)((c+d)^2-2cd)$

$=20[(20)^2+4040]+20[(20)^2-4040]$

$=20[(20)^2+4040+(20)^2-4040]$

$=20\times 800=16000$

12. **(a)** $D<0$

$4(1+3m)^2-4(1+m^2)(1+8m)<0$

$\Rightarrow m(2m-1)^2>0 \Rightarrow m>0$

13. **(d)** Let one root is α, other is α^3

$\text{sum}=\alpha+\alpha^3=-\dfrac{K}{81}$ ...(i)

$\text{Product}=\alpha^4=\dfrac{256}{81}$

$\Rightarrow \alpha=\pm\dfrac{4}{3}$ ...(ii)

From (i) and (ii)

$\dfrac{4}{3}+\dfrac{64}{27}=\dfrac{-K}{81}$ for $\alpha=+\dfrac{4}{3}$

$K=-300$

14. **(a,b,c)**

α,β are roots of x^2-x-1

$a_{r+2}-a_r=\dfrac{(\alpha^{r+2}-\beta^{r+2})-(\alpha^r-\beta^r)}{\alpha-\beta}$

$=\dfrac{(\alpha^{r+2}-\alpha^r)-(\beta^{r+2}-\beta^r)}{\alpha-\beta}$

$=\dfrac{\alpha^r(\alpha^2-1)-\beta^r(\beta^2-1)}{\alpha-\beta}=\dfrac{\alpha^r\alpha-\beta^r\beta}{\alpha-\beta}$

$=\dfrac{\alpha^{r+1}-\beta^{r+1}}{\alpha-\beta}=a_{r+1} \Rightarrow a_{r+2}-a_{r+1}=a_r$

$\Rightarrow \sum_{r=1}^{n}a_r=a_{n+2}-a_2=a_{n+2}-\dfrac{\alpha^2-\beta^2}{\alpha-\beta}$

$a_{n+2}-(\alpha+\beta)=a_{n+2}-1$

Now $\sum_{r=1}^{\infty}\dfrac{a_n}{10^n}=\dfrac{\sum_{r=1}^{\infty}\left(\frac{\alpha}{10}\right)^n-\sum_{r=1}^{\infty}\left(\frac{\beta}{10}\right)^n}{\alpha-\beta}$

$\dfrac{\frac{\alpha}{10}-\frac{\beta}{10}}{\alpha-\beta}\cdot\dfrac{1}{\left(1-\frac{\alpha}{10}\right)\left(1-\frac{\beta}{10}\right)}=\dfrac{\alpha-\beta}{(10-\alpha)-(10-\beta)}\cdot\dfrac{1}{(\alpha-\beta)}$

$=\dfrac{10}{(10-\alpha)(10-\beta)}=\dfrac{10}{89}$

$\sum_{n=1}^{\infty}\dfrac{b_n}{10^n}=\sum_{n=1}^{\infty}\dfrac{a_{n-1}+a_{n+1}}{10^n}=\dfrac{\frac{\alpha}{10}}{1-\frac{\alpha}{10}}+\dfrac{\frac{\beta}{10}}{1-\frac{\beta}{10}}=\dfrac{12}{89}$

Further, $b_n=a_{n-1}+a_{n+1}$

$=\dfrac{(\alpha^{n-1}-\beta^{n-1})+(\alpha^{n+1}-\beta^{n+1})}{\alpha-\beta}$

(as $\alpha\beta=-1 \Rightarrow \alpha^{n-1}=-\alpha^n\beta$ & $\beta^{n-1}=-\alpha\beta^n$)

$=\dfrac{\alpha^n(\alpha-\beta)+(\alpha-\beta)\beta^n}{\alpha-\beta}=\alpha^n+\beta^n$

15. **[13]** Let common roots is k then

$k^2-5\alpha k+1=0$ ...(i)

$k^2-\alpha k-5=0$ ...(ii)

On subtracting, we get $k=\dfrac{6}{4\alpha}=\dfrac{3}{2\alpha}$

Put $k=\dfrac{3}{2\alpha}$ in equ (i), we get

$\dfrac{36}{16\alpha^2}=\dfrac{26}{4} \Rightarrow \alpha=\dfrac{3}{\sqrt{2\times 13}}=\dfrac{3}{\sqrt{2\beta}}$

Hence, $\beta=13$

16. **[272]** $(p^2+q^2)x^2-2q(p+r)x+q^2+r^2=0$

$\Rightarrow p^2x^2-2qpx+q^2+q^2x^2-2qr+r^2=0$

$\Rightarrow (px-q)^2+(qx-r)^2=0$

$\Rightarrow x=\dfrac{q}{p}=\dfrac{r}{q}$

And roots of $x^2+2x-8=0$

$(x+4)(x-2)=0$

$x=-4,2$

Now, p,q,r not have same sign.

So possible common root is -4

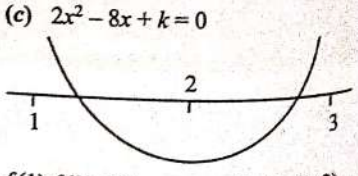
$x=\dfrac{q}{p}=\dfrac{r}{q}=-4$

Then, $\left(\dfrac{q}{p}\right)^2+\left(\dfrac{r}{p}\right)^2=16+\left(\dfrac{-4q}{p}\right)^2$

$=16+\left(\dfrac{4\times 4p}{p}\right)^2$

$=16+16^2=272$

17. **(c)** $2x^2-8x+k=0$



$f(1)f(2)<0$, one root lies in $(1,2)$

& $f(2)f(3)<0$, other root lies in $(2,3)$

$\Rightarrow (k-8)(k-6)<0$

& $(k-8)(k-6)<0$

$\Rightarrow k\in(6,8)$

Integral value of $k=7$

18. (*d*) $x^2-(m+1)x+(m+4)=0$

$D\geq 0$

$(m+1)^2-4(m+4)\geq 0$

$\Rightarrow m^2+1+2m-4m-16\geq 0$

$\Rightarrow m^2-2m-15\geq 0$

$\Rightarrow (m-1)^2-(4)^2\geq 0$

$\Rightarrow (m+3)(m-5)\geq 0$

$+\quad -\quad +$

$-3 \qquad 5$

$A=m\in(-\infty,-3)\cup[5,\infty)$

$B=[-3,5)$

$A-B=(-\infty,-3)\cup[5,\infty)$

$A\cap B=\{-3\}$

$A\cup B=R$

$B-A=(-3,5)$

19. (*d*) Equation is : $(\lambda^2+1)x^2-4\lambda x+2=0$

One root in interval (0, 1)

So, $f(0).f(1)<0$

$2.(\lambda^2+1-4\lambda+2)<0$

$(\lambda-3)(\lambda-1)<0$

$\therefore \lambda\in(1,3)$

If $\lambda=3$, then roots are 1 and $\frac{1}{5}$

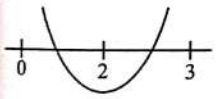
$\therefore \lambda\in(1,3]$

20. (*d*) One root lie in (0, 2)

So, $f(0)f(2)<0$

$(c-4)(c-24)<0$

$4(c-5)-4c+c-4<0\ c\in(4,24)\ldots(i)$



One root lie in (2, 3)

So, $f(2)f(3)<0$

$(c-24)(4c-49)<0$

$c\in\left(\frac{49}{4},24\right)$ ...(*ii*)

Here $(i)\cap(ii)$

$c\in\left(\frac{49}{4},24\right)$

21. (*c*) Expression is always positive if $2m+1>0$

$\Rightarrow m>-\frac{1}{2}$

and $D<0\Rightarrow 4(1+3m)^2-4(4)(1+m)$

$3-\sqrt{12}<m<3+\sqrt{12}$

$\Rightarrow m^2-6m-3<0$

$\therefore$ Common interval is $3-\sqrt{12}<m<3+\sqrt{12}$

$\therefore$ Integral value of $m=\{0,1,2,3,4,5,6\}$.

22. (*b*) $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$

$3\left[\left(x+\frac{1}{x}\right)^2-2\right]-2\left(x+\frac{1}{x}\right)+5=0$

Let $x+\frac{1}{x}=k$

$\Rightarrow 3k^2-2k-1=0$

$\Rightarrow 3k^2-3k+k-1=0$

$\Rightarrow 3k(k-1)+1(k-1)=0$

$\Rightarrow (k-1)(3k+1)=0$

$\Rightarrow k=1,-\frac{1}{3}$

$x+\frac{1}{x}=1,-\frac{1}{3}\Rightarrow$ No solution.

$\left[\because\ x+\frac{1}{x}\geq 2,\quad x+\frac{1}{x}\leq -2\right]$

23. **[5]** $|x|^2-2|x|+|\lambda-3|=0$

$|x|^2-2|x|+|\lambda-3|+1=1$

$(|x|-1)^2+|\lambda-3|=1$

At $\lambda=3$, $x=0$ and 2,

at $\lambda=4$ or 2, then

$x=1$ or -1

So maximum value of $x+\lambda=5$

24. [9] Given equation can be rearranged as

$x\left(x^6+3x^4-13x^2-15\right)=0$

Clearly $x=0$ is one of the root and other part can be observed by replacing $x^2=t$ from which we have $t^3+3t^2-13t-15=0$

$\Rightarrow (t-3)\left(t^2+6t+5\right)=0$

So, $t=3,t=-1,t=-5$

Hence, $x^2=3,x^2=-1,x^2=-5$

$\Rightarrow x=\pm\sqrt{3},x=\pm i,x=\pm\sqrt{5}i$

From the given condition $|\alpha_1|\geq|\alpha_2|\geq\ldots\geq|\alpha_7|$

Clearly $|\alpha_7|=0$ and $|\alpha_6|=\sqrt{5}=|\alpha_5|$

and $|\alpha_4|=\sqrt{3}=|\alpha_3|$ and $|\alpha_2|=1=|\alpha_1|$

So, $\alpha_1=\sqrt{5}i,\alpha_2=-\sqrt{5}i,\alpha_3=\sqrt{3}$,

$\alpha_4=-\sqrt{3},\alpha_5=i,\alpha_6=-i$

Hence, $\alpha_1\alpha_2-\alpha_3\alpha_4+\alpha_5\alpha_6=5-(-3)+1=9$

25. (*c*)

$\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+\frac{1}{(60-a)(80-a)}+\ldots+\frac{1}{(180-a)(200-a)}=\frac{1}{256}$

LHS

$=\frac{1}{20}\left(\frac{1}{20-a}-\frac{1}{40-a}\right)+\frac{1}{20}\left(\frac{1}{40-a}-\frac{1}{60-a}\right)+\ldots+\frac{1}{20}\left(\frac{1}{180-a}-\frac{1}{200-a}\right)$

$=\frac{1}{20}\left(\frac{1}{20-a}-\frac{1}{200-a}\right)=\frac{1}{20}\cdot\frac{180}{(20-a)(200-a)}$

$=\frac{9}{(20-a)(200-a)}=\frac{1}{256}$(given)

$\Rightarrow a^2-220a+4000-2304=0$

$\Rightarrow a^2-220a+1696=0$

$D=220^2-4\times 1696=16(552-424)=16\times 2601$

$\sqrt{D}=4\times 51=204$

$a=\frac{220\pm 204}{2}=212$ or 8

Hence, maximum value of $a=212$

26. **[36]** $x^4-3x^3-2x^2+3x+1=0$

$=(x^2-1)(x^2-3x-1)=0$

$\Rightarrow (x^2-1)=0$ & $(x^2-3x-1)=0$

$\Rightarrow x=\pm 1$

and let α,β are roots of $x^2-3x-1=0$

$\therefore$ Required sum $=1^3+(-1)^3+\alpha^3+\beta^3$

$=1-1+(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$=27+3(3)=36[\because\ \alpha+\beta=3,\alpha\beta=-1]$

27. **[16]** Given quadratic equation

$x^2-x-4=0$ and $P_n=\alpha^n-\beta^n$

Now, $\frac{P_{15}P_{16}-P_{14}P_{16}-P_{15}^2+P_{14}P_{15}}{P_{13}P_{14}}$

$=\frac{P_{16}(P_{15}-P_{14})-P_{15}(P_{15}-P_{14})}{P_{13}P_{14}}$

$=\frac{(P_{15}-P_{14})(P_{16}-P_{15})}{P_{13}P_{14}}$

$=\frac{4P_{13}\cdot 4P_{14}}{P_{13}\cdot P_{14}}=16$

$\therefore P_n=\alpha^n-\beta^n$

$=\alpha^{n-1}\cdot\alpha-\beta^{n-1}\cdot\beta$

$=\alpha^{n-1}(\alpha^2-4)-\beta^{n-1}(\beta^2-4)$

$P_n=\alpha^{n+1}-\beta^{n+1}-4(\alpha^{n-1}-\beta^{n-1})$

$P_n=P_{n+1}-4P_{n-1}\Rightarrow P_{n+1}-P_n=4P_{n-1}$

28. **[3]** $x^8-x^7-x^6+x^5+3x^4-4x^3-2x^2+4x-1=0$

$\Rightarrow x^7(x-1)-x^5(x-1)+3x^3(x-1)-x(x^2-1)+2x(1-x)+(x-1)=0$

$\Rightarrow (x-1)(x^7-x^5+3x^3-x(x+1)-2x+1)=0$

$\Rightarrow (x-1)(x^7-x^5+3x^3-x^2-3x+1)=0$

$\Rightarrow (x-1)(x^5(x^2-1)+3x(x^2-1)-1(x^2-1))=0$

$\Rightarrow (x-1)(x^2-1)(x^5+3x-1)=0$

$\therefore x=\pm 1$ are roots of above equation and x^5+3x-1 is a monotonic term hence vanishes at exactly one value of x other then 1 or -1.

$\therefore$ 3 real roots.

29. **[2]** Given $f(x)=2x^5+5x^4+10x^3+10x^2+10x+10$

$f(-1)=3>0$ & $f(-2)=-34<0$

So at least one root will lie in $(-2,-1)$

now $f'(x)=10x^4+20x^3+30x^2+20x+10$

$=10[x^4+2x^3+3x^2+2x+1]$

$=10\left[x^2+2x+3+\frac{2}{x}+\frac{1}{x^2}\right]x^2$

$=10x^2\left[\left(x^2+\frac{1}{x^2}\right)+2\left(x+\frac{1}{x}\right)+3\right]$

$=10x^2\left[\left(x+\frac{1}{x}\right)^2+2\left(x+\frac{1}{x}\right)+1\right]$

$=10x^2\left(x+\frac{1}{x}+1\right)^2>0\,\forall x\in R$

So, $f(x)$ be purely increasing function so exactly one root of $f(x)$ that will lie in $(-2,1)$. Hence $|a|=2$

30. **[26]** $xf(x)+2=\lambda(x-2)(x-3)(x-4)(x-5)$ ...(i)

Put $x=0\Rightarrow\lambda=\frac{1}{60}$

Put λ in equ (i),

$$xf(x)+2=\frac{1}{60}(x-2)(x-3)(x-4)(x-5)$$

Put $x=10\Rightarrow 10f(10)+2=\frac{1}{60}\times 8\times 7\times 6\times 5$

$=28$

By Aditya Mohod

username - muje_kabhi_dekha_nahi

By Aditya Mahod
username - muje_kabhi_dekha_nahi

CHAPTER 6

Complex Number-I

Prarambh (Topicwise)

1. (*b*) We know that $i^2 = -1$

$(i^2)^2 = (-1)^2 = 1 \Rightarrow i^{4n} = 1^n$

and therefore $i^{4n-1} = -i$, $i^{4n+1} = i$

2. (*b*) $\dfrac{1}{1-\cos\theta + i\sin\theta} =$

$$\frac{1}{(1-\cos\theta)+i\sin\theta} \times \frac{(1-\cos\theta)-i\sin\theta}{(1-\cos\theta)-i\sin\theta}$$

$$= \frac{(1-\cos\theta)-i\sin\theta}{(1-\cos\theta)^2+\sin^2\theta} = \frac{(1-\cos\theta)-i\sin\theta}{2(1-\cos\theta)}$$

$$= \frac{(1-\cos\theta)}{2(1-\cos\theta)} - i\frac{\sin\theta}{2(1-\cos\theta)}.$$

Therefore its real part $= \dfrac{1-\cos\theta}{2(1-\cos\theta)} = \dfrac{1}{2}$

3. (*b*) $(-1)^{\frac{1}{3}} = \dfrac{i\sqrt{3}+1}{2} = \dfrac{-\sqrt{3}+\sqrt{-1}}{\sqrt{-4}}$

4. (*b*) $\left(\dfrac{1+i}{1-i}\right)^n = \left(\dfrac{e^{i\pi/4}}{e^{-i\pi/4}}\right)^n = e^{i\frac{n\pi}{2}}$

$\Rightarrow n = 2$

5. (*b*) $\sqrt{-7-24i} = x - iy$

Squaring both sides,

$-7 - 24i = x^2 - y^2 - i(2xy)$

Equating real and imaginary parts, we get

$x^2 - y^2 = -7$ and $2xy = 24$

$\therefore x^2 + y^2 = \sqrt{49+576} = \sqrt{625} = 25$

6. (*b*) Let

$\sqrt{3-4i} = x + iy \Rightarrow 3 - 4i = x^2 - y^2 + 2ixy$

$\Rightarrow x^2 - y^2 = 3, 2xy = -4$ ... (*i*)

$\Rightarrow (x^2+y^2)^2 = (x^2-y^2)^2 + 4x^2y^2$

$= (3)^2 + (-4)^2$

$\Rightarrow x^2 + y^2 = 5$ ... (*ii*)

From equation (*i*) and (*ii*) $x^2 = 4 \Rightarrow x = \pm 2$

$y^2 = 1 \Rightarrow y = \mp 1$. Hence the square root of $(3-4i)$ is $\pm(2-i)$

7. (*a*) $(b+ia)^5 = i(-ib+a)^5 = i(\alpha - i\beta) = \beta + i\alpha$

8. (*a*) $|(2+i)(2+2i)(2+3i)\ldots(2+ni)| = |x+iy|$

$\Rightarrow 5.8.13\ldots(4+n^2) = (x^2+y^2)$

9. (*d*) $z^{1/3} = a - ib \Rightarrow z = (a-ib)^3$

$\Rightarrow x + iy = (a^3 - 3ab^2) + i(b^3 - 3a^2b)$

$\Rightarrow \dfrac{x}{a} = a^2 - 3b^2 \dfrac{y}{b} = b^2 - 3a^2$

$\dfrac{x}{a} - \dfrac{y}{b} = 4(a^2 - b^2) \Rightarrow K = 4$

10. (*c*) $z = \dfrac{(2+i)^2}{3+i} = \dfrac{3+4i}{3+i} \times \dfrac{3-i}{3-i} = \dfrac{13}{10} + i\dfrac{9}{10}$

Hence conjugate of $\dfrac{13}{10} + i\dfrac{9}{10} = \left(\dfrac{13}{10} - i\dfrac{9}{10}\right)$

11. (*c*) Given

$\left|\dfrac{z_1+z_2}{z_1-z_2}\right| = 1 \Rightarrow \dfrac{z_1+z_2}{z_1-z_2} = \cos\theta + i\sin\theta$

(say)

$\Rightarrow \dfrac{z_1}{z_2} = \dfrac{1+\cos\theta+i\sin\theta}{-1+\cos\theta+i\sin\theta} = -i\cot\dfrac{\theta}{2}$

which is zero, if $\theta = n\pi (n \in I)$, and is otherwise purely imaginary.

12. (*b*) We have $|z_1| = 1$ and be z_2 any complex number.

$\Rightarrow \left|\dfrac{z_1-z_2}{1-z_1\overline{z}_2}\right| = \dfrac{|z_1-z_2|}{\left|1-\dfrac{\overline{z}_2}{\overline{z}_1}\right|}$; $\because z_1\overline{z}_1 = |z_1|^2$

$= \dfrac{|z_1-z_2|}{|\overline{z}_1-\overline{z}_2|}|\overline{z}_1|$ Given that $\because |\overline{z}_1| = 1$

$= \dfrac{|z_1-z_2|}{|z_1-z_2|} = \dfrac{|z_1-z_2|}{|z_1-z_2|} = 1.$

13. (*b*) $\arg\left(\dfrac{13-5i}{4-9i}\right) = \arg(13-5i) - \arg(4-9i)$

$= -\tan^{-1}\left(\dfrac{5}{13}\right) + \tan^{-1}\dfrac{9}{4} = \dfrac{\pi}{4}$

14. (*d*) $|z||\omega| = 1$ ... (*i*)

and $\arg\left(\dfrac{z}{\omega}\right) = \dfrac{\pi}{2} \Rightarrow \dfrac{z}{\omega} = i\left|\dfrac{z}{\omega}\right|$ ... (*ii*)

From equation (*i*) and (*ii*)

$|z| = |\omega| = 1$ and $\dfrac{z}{\omega} + \dfrac{\overline{z}}{\overline{\omega}} = 0$; $z\overline{\omega} + \overline{z}\omega = 0$

$\overline{z}\omega = -z\overline{\omega} = \dfrac{-z}{\omega}\overline{\omega}\omega$; $\overline{z}\omega = -i|\omega|^2 = -i$.

15. (*c*) Given that $a^2 + b^2 = 1$, therefore

$\dfrac{1+b+ia}{1+b-ia} = \dfrac{(1+b+ia)(1+b+ia)}{(1+b-ia)(1+b+ia)}$

$= \dfrac{(1+b)^2 - a^2 + 2ia(1+b)}{1+b^2+2b+a^2}$

$= \dfrac{(1-a^2)+2b+b^2+2ia(1+b)}{2(1+b)}$

$= \dfrac{2b^2+2b+2ia(1+b)}{2(1+b)} = b + ia$

Trick: Put

$a = 0, b = 1, \dfrac{1+b+ia}{1+b-ia} = \dfrac{1+1+0}{1+1-0} = 1$

But opions (1) and (3) give 1.

So again put

$a = 1, b = 0, \dfrac{1+b+ia}{1+b-ia} = \dfrac{1+i}{1-i} = i$

Which gives (3) only.

16. (*b*) $z = x + iy$

$\Rightarrow z^2 \Rightarrow x^2 - y^2 + 2ixy$

$\Rightarrow \text{Re}(z^2) = x^2 - y^2, |z| = \sqrt{x^2+y^2}$

$\Rightarrow x^2 - y^2 = 0, x^2 + y^2 = 3$

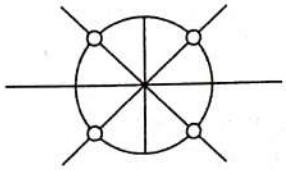
$\Rightarrow x^2 = y^2 = \dfrac{3}{2}$

$\Rightarrow x = \pm\sqrt{\dfrac{3}{2}}, y = \pm\sqrt{\dfrac{3}{2}}$

17. (*a*) $\text{Re}(z^2) = 0$

$x^2 - y^2 = 0$

$x = \pm y$



Four solution

18. (*c*) We have, $\arg\left(\dfrac{z_1}{z_2}\right) = \pi$

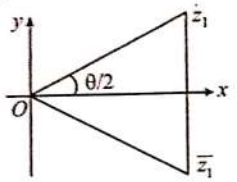
$\Rightarrow \arg(z_1) - \arg(z_2) = \pi$

$\Rightarrow \arg(z_1) = \arg(z_2) + \pi$

Let $\arg(z_2) = \theta$. Then $\arg(z_1) = \pi + \theta$.

$\therefore z_1 = |z_1|[\cos(\pi+\theta) + i\sin(\pi+\theta)]$
$= |z_1|(-\cos\theta - i\sin\theta)$
and $z_2 = |z_2|(\cos\theta + i\sin\theta) = |z_1|(\cos\theta + i\sin\theta)$
$= -z_1$ $(\because |z_1| = |z_2|)$
$\Rightarrow z_1 + z_2 = 0$

19. (*a*) $\frac{y}{x} = \tan\frac{\theta}{2} = \sqrt{2} - 1 = \tan\frac{\pi}{8}$
$= \frac{\theta}{2} = \frac{\pi}{8} \Rightarrow \theta = 45° \Rightarrow n = \frac{360°}{45°} = 8$



20. (*d*) $|iz + 3 - 4i| \le |iz| + |3 - 4i| = |z| + 5 \le 9$

21. (*c*) The expression is the sum of the distance of z from the two points $1 - 2i$ and $-3 + 4i$. The minimum value is the distance between these two points = $\sqrt{4^2 + 6^2}$
$= 2\sqrt{13}$ **Ans.**

22. (*a*) $z = \frac{1-i}{1+i}\bar{z} \Rightarrow z = -i\bar{z}$
$\Rightarrow x + iy$
$= -y - ix \Rightarrow x = -y$
so $z = x - ix, x \in R \Rightarrow x(1 - i)$
or $t(1 - i), t \in R$.

23. (*d*) $\frac{\bar{z}_1}{z_2} = \frac{r_1 e^{-i\theta_1}}{r_2 e^{i\theta_2}} = 2e^{-(\theta_1+\theta_2)i} = 2e^{-i\frac{3\pi}{2}} = 2i$

24. (*d*) $|z - 4| < |z - 2|$
$\Rightarrow z\bar{z} + 16 - 4(z + \bar{z}) < z\bar{z} + 4 - 2(z + \bar{z})$
$\Rightarrow 12 < 2(z + \bar{z}) \Rightarrow 12 < 4\text{Re}(z)$
$\Rightarrow 3 < \text{Re}(z)$

25. (*d*) $|z - i\,\text{Re}(z)| = |z - \text{In}(z)|$
$z = x + iy$
$|(x + iy + ix)| + |(x + iy + y)$
$|x + (y - x)i| = |(x - y)^2 + y|$
$x^2 + (y - x)^2 = (x - y)^2$
$\Rightarrow x^2 = y^2$
$\Rightarrow (x + >')(x _ v) = 0$
$\text{Re}(z) + \text{Re}(y) = 0$
OR $\text{Re}(z) = \text{Re}(y)$

26. (*b*) $|z - z| \le 2$
$z_0 = 5 + 3i$
$\Rightarrow z z_0 = -3 + 5i = -3 + 5i$
$|iz + z_0| = |-z + i z_0| = |z - i z_0|$
$|z - z_1|_{\max} = |z_1 - i| + 2$
$= |3 + 4i| + 2 = 7$

27. (*c*) $\text{Amp}(z_1 z_2) = 0 \Rightarrow \text{amp}\, z_1 + \text{amp}\, z_2 = 0$
$\therefore$ amp $z_1 = -$amp $z_2 =$ amp
Since $|z_1| = |z_2|$, we get $|z_1| = |\bar{z}_2|$.
So, $z_1 = \bar{z}_2$. Also, $z_1 z_2 = z_2 = |z_2|^2$
$= 1$ because $|z_2| = 1$.

28. (*a*) $\alpha + \beta = 3 + 4i \quad \gamma\delta = 13 + i\gamma$
$= \bar{\alpha}$ and $\delta = \bar{\beta}$
$\gamma + \delta = 3 - 4i$
$b = \Sigma\alpha\beta$
$= \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$
$= (\alpha + \beta)(\delta + \gamma) + \alpha\beta + \gamma\delta$
$= (3 + 4i)(3 - 4i) + 13 - i + 13 + i$
$= 9 + 16 + 26 = 51$

29. (*c*) $|Z - 2| + |Z + 2| = 8$
is an ellipse with focus (2, 0) and (02, 0) with major axis 8
$|Z - 1| + |Z + 1| = 2$
is a straight line joining (1, 0) and (–1, 0)
$\therefore$ Their are zero points of intersection.

30. (*b*) We have

$$\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|} = k(\text{let})$$

$$\Rightarrow \frac{9}{|z_2 - z_3|^2} = \frac{16}{|z_3 - z_1|^2} = \frac{25}{|z_1 - z_2|^2} = k^2$$

Now $\frac{9}{|z_2 - z_3|^2} = k^2$

$$\Rightarrow \frac{9}{z_2 - z_3} = k^2(\bar{z}_2 - \bar{z}_3) \quad ...(i)$$

[As $|z|^2 = z\bar{z}$]

Similarly, $\frac{16}{|z_3 - z_1|^2} = k^2$

$$\Rightarrow \frac{16}{z_3 - z_1} = k^2(\bar{z}_3 - \bar{z}_1) \quad ...(ii)$$

Similarly, $\frac{25}{|z_1 - z_2|^2} = k^2$

$$\Rightarrow \frac{25}{z_1 - z_2} = k^2(\bar{z}_1 - \bar{z}_2) \quad ...(iii)$$

$\therefore$ On adding (*i*), (*ii*) and (*iii*), we get

$$\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2} = k^2$$

$$(\bar{z}_2 - \bar{z}_3 + \bar{z}_3 - \bar{z}_1 + \bar{z}_1 - \bar{z}_2) = 0$$

Prabal (JEE Main Level)

1. (*d*) $z = (3p - 7q) + (7p + 3q)i$
$\because z$ is purely imaginary $3p = 7q$
$p = 7, q = 3$ for minimum value of $|z|^2$
$|z|^2 = 58 \times 58 = 3364$

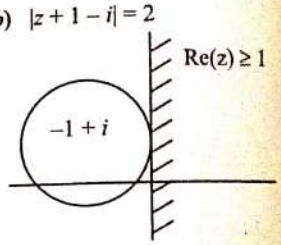
2. (*c*) Let α be the real root
$\alpha^2 - (3 + i)\alpha + m + 2i = 0$
$(\alpha^2 - 3\alpha + m) + i(2 - \alpha) = 0$
$\therefore \alpha = 2$ (real root)
$\therefore 4 - 6 + m = 0 \Rightarrow m = 2$
Product of the roots $= 2(1 + i)$ with one root as 2
non real root $= 1 + i$, addivitve inverse is $-1 - i$.

3. (*a*) $T_n = \left(\frac{\sqrt{3}+i}{2}\right)^n \Rightarrow |T_n| = 1$

4. (*b*) $x^2 + i(a - 1)x + 5 = 0$ roots of the equation are $p + iq, p - iq$ where $p, q \in$
$p + iq + p - iq = -i(a - 1)$
$\Rightarrow 2p + i(a - 1) = 0$
hence $a = 1$

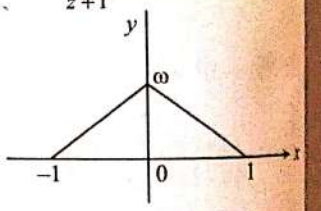
5. (*d*) $\text{Amp}\left(\frac{z-1}{z+3}\right) = 0 \Rightarrow \text{Im}\left(\frac{z-1}{z+3}\right) =$
$\Rightarrow y = 0,$
Hence $(x - 1) : y = \infty$ (does not exist)

6. (*b*) $|z + 1 - i| = 2$



Only one solution

7. (*a*) $\omega = \frac{z-1}{z+1} \Rightarrow \omega z + \omega = z - 1$



$\Rightarrow (\omega - 1)z = -1 - \omega \Rightarrow z = \frac{1+\omega}{1-\omega}$

Now $|z| = 1 \Rightarrow \left|\frac{1+\omega}{1-\omega}\right| = 1$

$\Rightarrow |\omega - (-1)| = |\omega - 1|$
$\Rightarrow \omega$ lies on the perpendicular bisector the segment joining – 1 and 1.
Thus, ω lies on the imaginary axis.

8. (*d*) $a|z_1| = b|z_2|$

$$\frac{az_1}{bz_2} = \frac{a|z_1|e^{i\theta}}{b|z_2|e^{i(\theta+\alpha)}} = e^{-i\alpha}$$

So, $\frac{e^{-i\alpha} - 1}{e^{-i\alpha} + 1} = \frac{e^{-i\alpha/2}\left(e^{-i\alpha} - 1\right)}{e^{i\alpha/2}\left(e^{-i\alpha} + 1\right)}$

$= \frac{e^{-i\alpha/2} - e^{i\alpha/2}}{e^{-i\alpha/2} + e^{ik/2}} = \frac{\bar{z} - z}{\bar{z} + z} =$ imaginary

9. (*b*) $x^6 = (4 - 3i)^5 = 5^5\left(\frac{4}{5} - \frac{3i}{5}\right)^5$
$x^6 = 5^5(\cos\theta + i\sin\theta)5$

$\Rightarrow x = 5^{5/6}(\cos 5\theta + 13\sin 5\theta)^{1/6}$

$= 5^{5/6}\left[\cos\left(\frac{2k\pi+5\theta}{6}\right)+i\sin\left(\frac{2k\pi+5\theta}{6}\right)\right]^{\frac{1}{6}}$

$k = 0, 1, 2, 3, 4, 5$

Product it $= 5^5(\cos 5\theta + i5\sin 5\theta)$

10. *(c)* Given : $\left(\frac{1+i}{1-i}\right)^n = \frac{2}{\pi}\sin^{-1}\left(\frac{1+x^2}{2x}\right)$

where $x > 0$

$\Rightarrow (1+i)^n = (1-i)^n$

$\Rightarrow \left(\frac{1+i}{1-i}\right)^n = 1$

$\Rightarrow \frac{1+i}{1-i} = \left(\frac{1+i}{1-i}\right)\left(\frac{1+i}{1-i}\right)$

$\Rightarrow \frac{1-1+2i}{1-(-1)} = \frac{2i}{2}$

$= i$

Also, $i^n = 1$

Since, $i^4 = 1$

$\Rightarrow n = 4$

11. *(a)* $|z| = 1 \quad (z-1) - ik(z+1)$

$\Rightarrow z - 1 = (iz)k + ik$

$z(1 - ik) = 1 + ik$

$\Rightarrow z = \frac{1+ik}{1-ik}$

$|z| = \frac{\sqrt{1+k^2}}{\sqrt{1+k^2}} = 1$

$k \in R$

12. *(b)* Let, $z = x + iy$.

Now, $\arg(z) = \frac{\pi}{3} \Rightarrow \tan^{-1}\frac{y}{x} = \frac{\pi}{3}$

or, $y = \sqrt{3}x$

Also, $\arg(z) = \frac{2\pi}{3} \Rightarrow y = -\sqrt{3}x$.

Also,

$\arg(z-2-2i\sqrt{3}) = \pi \Rightarrow \tan^{-1}\frac{y-2\sqrt{3}}{x-2} = \pi$

$\Rightarrow y = 2\sqrt{3}$ ...*(iii)*.

Now, the area bounded by the curves (1), (2) and (3)

$= 2 \times \Delta OAB$

$= 2 \times \frac{1}{2} \times 2 \times 2\sqrt{3} = 4\sqrt{3}$

13. *(a)* $(1 + i\sqrt{3}) = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{3}\right)$

$= 2e^{i\pi/3}$

Also, $f(z) = Re(z)$

$f(2^n e^{in\pi/2}) = 2^n \cos\frac{n\pi}{3}$

$\sum_{n=1}^{6a}\log_2\left[2^n\cos\frac{n\pi}{3}\right] = \sum_{n=1}^{6a}\left(n + \log_2\cos\frac{n\pi}{3}\right)$

$= \frac{6a(6a+1)}{2} + \sum_{n=1}^{6a}\log_2\cos\frac{n\pi}{3}$

$= 18a^2 + 3a - 4a$

$= 18a^2 - a$

14. *(d)* Let $\alpha(|\alpha| = 1)$ is a root of unity that is also a root of the equation $z^2 + az + b = 0$

$\therefore$ Alpha $+ \bar{\alpha} = -a$

$\therefore |a| = |\alpha + \bar{\alpha}| \le |\alpha| + |\bar{\alpha}| = 2$

and $b = \alpha\bar{\alpha} = 1$

Hence, we must check those equations for which $-2 \le a \le 2$ and $b = 1$.

i.e., $z^2 + 2z + 1 = 0, z^2 + z + 1 = 0$

$z^2 + 1 = 0$

$z^2 - 2z + 1 = 0, z^2 - z + 1 = 0$

Hence roots are $\pm 1, \pm i, \frac{-1 \pm \sqrt{-3}}{2}$,

$\frac{1 \pm \sqrt{-3}}{2}$

15. *(a)* $|z+1| + |z-3| = 6$ and $|z| = |z-2|$

$|2 + iy| + |-2 + iy| = 6 \Rightarrow z = 1 + iy$

$\Rightarrow \sqrt{4+y^2} + \sqrt{4+y^2} = 6$

$\Rightarrow \sqrt{4+y^2} = 3$

$y^2 = 9 - 4 = 5$

$y = \pm\sqrt{5}$

$\boxed{z = 1 \pm \sqrt{5}i}$

16. *(a)* $\alpha = 1^{1/5}$

consider $x^5 - 1 = 0$

so $2^{\left|1+\alpha+\alpha^2+\frac{\alpha^3}{\alpha^2}-\frac{\alpha^4}{\alpha^4}\right|} = 2^{|1+\alpha+\alpha^2+\alpha^3-\alpha^4|}$

$= 2^{|-\alpha^4-\alpha^4|} = 4^{|\alpha^4|} = 4$

17. *(c)* $X = 9^{\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}\cdots\infty}$

$\Rightarrow x = 9^{\frac{\frac{1}{3}}{1-\frac{1}{3}}}$

$\Rightarrow x = 9^{\frac{1}{2}}$

$\Rightarrow X = 3$

$y = 4^{\frac{1}{3}\frac{1}{3^2}\frac{1}{3^3}\cdots\infty}$

$\Rightarrow y = 4^{\frac{\frac{1}{3}}{1+\frac{1}{3}}}$

$\Rightarrow y = 4^{\frac{1}{4}}$

$\Rightarrow y = 2^{\frac{1}{2}}$

$\Rightarrow y = \sqrt{2}$

$z = \frac{1}{(1+i)^1} + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^3}\cdots\infty$

Since it is a G.P with a common ratio of $\frac{1}{(1+i)}$ we get the sum as

$= \frac{\frac{1}{1+i}}{1-\frac{1}{1+i}} = \frac{1}{i} = -i$

Hence $x + yz = 3 - \sqrt{2}i$

$\tan\theta = \frac{-\sqrt{2}}{3}$

$\theta = \tan^{-1}\left(\frac{-\sqrt{2}}{3}\right) = -\tan^{-1}\left(\frac{\sqrt{2}}{3}\right)$

18. *(b)* $\arg(z_1) = \theta_1 > \arg(z_2) = \theta_2, \arg(z_3) = \theta_3$

$A = \begin{vmatrix}\theta_1 & \theta_2 & \theta_3\\ \theta_2 & \theta_3 & \theta_1\\ \theta_3 & \theta_1 & \theta_2\end{vmatrix}$

$= \begin{vmatrix}\theta_1+\theta_2+\theta_3 & \theta_2+\theta_3+\theta_1 & \theta_1+\theta_2+\theta_3\\ \theta_2 & \theta_3 & \theta_1\\ \theta_3 & \theta_1 & \theta_2\end{vmatrix}$

$= (\theta_1+\theta_2+\theta_3)\begin{vmatrix}1 & 1 & 1\\ \theta_2 & \theta_3 & \theta_1\\ \theta_3 & \theta_1 & \theta_2\end{vmatrix}$

19. *(d)* $f(x) = x^6 + 4x^5 + 3x^4 + 2x^3 + x + 1$

Since w is the cube root of unity.

$\therefore w^3 = 1$ & $1 + w + w^2 = 0$

$f(w) = (w)^6 + 4(w)^5 + 3(w)^4 + 2(w)^3 + (w) + 1$

$f(w) = 1 + 4w^2 + 3w + 2 + w + 1 = 4 + 4w + 4w^2 = 0$

$f(w^2) = (w^2)^6 + 4(w^2)^5 + 3(w^2)^4 + 2(w^2)^3 + (w^2) + 1$

$f(w^2) = 1 + 4w + 3w^2 + 2 + w^2 + 1 = 4 + 4w + 4w^2 = 0$

$\because f(w) = 0$ & $f(w^2) = 0$

Therefore

$f(x) = x^6 + 4x^5 + 3x^4 + 2x^3 + x + 1$

is divisible by $(x - w)(x - w^2)$.

20. *(c)* $f(z) = q_1(z)(z - i) + (1 + i)$ ...*(i)*

$f(z) = q_2(z)(z + i) + (1 + i)$ ...*(ii)*

Let $f(z) = q_3(z)(z^2 + i) + az + b$ ...*(iii)*

Put $z = i$ in (i) and (iii)

we get $f(i) = 1 - i = ai + b$

Similarly put $z=-i$ in (ii) and (iii)

we get $f(-1)=1+i=-ai+b$

$\therefore a=-1$ and $b=1$

$\therefore az+b=-z+1$

21. **[6]** Let $z=e^{i\theta}$

$\therefore z^3=\cos 3\theta+i\sin 3\theta$

$\therefore \cos 3\theta=0$ ($\because z$ is purely imaginary)

$\therefore \theta=\frac{2n\pi}{3}\pm\frac{\pi}{6}$

$\therefore \theta=30°, 90°, 150°, 270°, 210°, 330°$

22. **[99]** $N=(a+ib)^3-107i$

$=(a^3-3ab^2)+i[3a^2b-b^3]-107i=$ Positive integer

$\therefore 3a^2b-b^3-107=0$

$b(3a^2-b^2)=107$

$b=1 \quad 3a^2-b^2=107 \qquad 107$ is prime

$\Rightarrow a=6$ or $b=107 \qquad 3a^2-(107)^2=1$

a is not integer not possible

$\therefore a=6 \quad b=1$

$N=216-3\times 6=216-18=198.$

23. **[32]** $z=\alpha \quad \alpha\in R$

$\alpha^3-(3+i)\alpha+m+2i=0$

$\alpha^3-3\alpha+m=0 \;\&\; -\alpha+2=0$

$\alpha=2$

$8-6+m=0 \Rightarrow m=-2$

$\Rightarrow \alpha^4+m^4=32$

24. **[275]** $f(z)=\frac{z+i}{z-i}$

$F(F(z))=\frac{F(z)+i}{F(z)-i}=\frac{z+1}{z-1}i$

$F(F(F(z)))$

$z_{k+3}=z$

$z_{2020}=z_{3\times 667+1}=z_1+1+274i$

$a+b=275$

25. **[250]** $(\sin t+i\cos t)^n=(\sin nt+i\cos nt)$

$\Rightarrow i^n(\cos t-i\sin t)^n=i^n(\cos nt-i\sin nt)$

$=\sin nt+i\cos nt$, for

$n=4k+1$

$1000\times\frac{1}{4}=250$

Parikshit (JEE Advanced Level)

1. **(a)** $|z_1|=|z_2|=|z_3|=1$

$|z_1+z_2+z_3|^2\geq 0$

$|z_1|^2+|z_2|^2+|z_1|^2+2\text{Re}(z_1\overline{z}_2+z_2\overline{z}_3+z_3\overline{z}_1)\geq 0$

$\frac{3}{2}+\text{Re}(z_1\overline{z}_2+z_2\overline{z}_3+z_3\overline{z}_1)\geq 0$

Now

$z_1\overline{z}_2(\cos(\theta_1)+i\sin\theta_1)(\cos\theta_2-i\sin\theta_2)$

$=\cos(\theta_1-\theta_2)+i\sin(\theta_1-\theta_2)$

so, $\cos(\theta_1-\theta_2)+\cos(\theta_2-\theta_3)$

$+\cos(\theta_3-\theta_1)>-\frac{3}{2}$

2. **(b)** $\sin^{-1}\left(\frac{z-1}{i}\right)$

$=\sin^{-1}\left(\frac{(x-1)+iy}{i}\right)$

$=\sin^{-1}\left(\frac{i(x-1)-y}{-1}\right)$

$\Rightarrow x-1=0$, So $\sin^{-1}(y)$

$|y|\leq 1$

3. **(a)** $|z|^n=|z|^{n-2}(z^2+z)+1$

$\Rightarrow z^2+z$ real number

$\Rightarrow \overline{z}^2+\overline{z}=$ Real number

$z^2+z=\overline{z}^2+z^-$

$z^2-z^2=\overline{z}-z$

$(z-\overline{z})(z+\overline{z})=-(z-\overline{z})$

$\Rightarrow z-\overline{z}=0$ OR $z+\overline{z}=-1$

$z=$ purely real OR $2\text{Re}(z)=-1$

$\Rightarrow y=0$ OR $\text{Re}(2)=\frac{-1}{2}$ (not possible)

4. **(a)** $u^2-2u+2=0$

$\Rightarrow u=1\pm i$

So, $\alpha=1+i$ and $\beta=1-i$

Now given that, $x=\cot\theta-1$

$\text{so,}\frac{(x+\alpha)^n-(x+\beta)^n}{\alpha-\beta}=$

$\frac{(\cot\theta-1+1+i)^n-(\cot\theta-1+1-i)^n}{2i}$

$=\frac{(\cot\theta+i)^n-(\cot\theta-i)^n}{2i}=$

$(\cos\theta+i\sin\theta)^n-(\cos\theta-\sin\theta)^n$

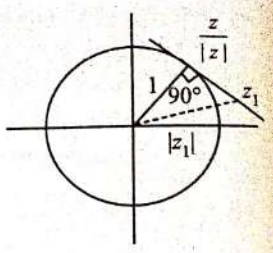
$(\sin\theta)^n(2i)$

$=\frac{e^{(in\theta)}-e^{-(in\theta)}}{(\sin\theta)^n 2i}$

$=\frac{(\cos(n\theta)+i\sin(n\theta))-(\cos(n\theta)-i\sin(n\theta))}{(\sin\theta)^n 2i}$

$=\frac{(\sin n\theta)}{(\sin\theta)^n}$

5. **(b)**



$|z_1|=\sqrt{3^2+1^2}=\sqrt{10}$

6. **(d)** It is given that, p and q are prime numbers. Hence the only common pth and qth root of unity will be the number 1. Thus there will be no common imaginary pth and qth root of unity. Hence answer is zero.

7. **(d)** Since, Z_1 and Z_2 are conjugated pairs, let $Z_1=a+bi$

and $Z_2=a-ib$

so, $Z_1\overline{Z}_2=(a+ib)(a+ib)=a_2-b_2+2abi$

so, $Z_2\overline{Z}_1=(a-ib)(a-ib)=a_2-b_2-2abi$

$Z_1-Z_2=2abi$

$(Z_1-Z_2)i=(2abi)i=-2ab=$ Purely real.

8. **(a)** $\alpha+i\beta$

$x^3+qx+k=0$

$\alpha+i\beta,\alpha-i\beta,\gamma$

sum of roots $=2\alpha+\gamma=0$

$\gamma=-2\alpha$

so, $-x^3-qx+k=0$

$x^3+qx-r=0$

9. **(a)** $\frac{a}{1-b}=\frac{b}{1-c}=\frac{c}{1-a}=k$

$a=k(1-b)$

$b=k(1-c)$

$c=k(1-a)$

$c=k(1-k(1-b))$

$c=k(1-k(1-k(1-c)))$

$c=k-k^2(1-k+kc)$

$c=k-k^2+k^3-k^3c$

$(1+k^3)c=k(k^2-k+1)$

$\Rightarrow k^2-k+1=0$

$k=\frac{1\pm\sqrt{3}i}{2}=-w_1-w^2$

10. **(b,c,d)**

$|z_1+z_2|^2=|z_1|^2+|z_2|^2$

$z_1\overline{z}_2+\overline{z}_1z_2=0$

$\Rightarrow \frac{z_1}{z_2}=-\frac{\overline{z}_1}{\overline{z}_2}$

$\Rightarrow \frac{z_1}{z_2}+\overline{\left(\frac{z_1}{z_2}\right)}=0 \Rightarrow \frac{z_1}{z_2}$ is purely imaginary

so amp $\left(\frac{z_1}{z_2}\right)$ is may be $\frac{\pi}{2}$ or $-\frac{\pi}{2}$

11. *(b,c,d)*

$z_2=i; z_3=-1+i; z_4=-i; z_5=-1+i$

$|z_{2050}|=1, |z_{2017}|=\sqrt{2}, |z_{2016}|=1, |z_{2111}|=\sqrt{2}$

12. *(a,b,c)*

If one root is i then other is $-i$

Let fourth root is α.

$2\alpha=\frac{3}{2} \Rightarrow \alpha=\frac{3}{4}$

$\frac{-a}{2}=2+i+(-i)+\frac{3}{4}=\frac{11}{4}$

$a=\frac{-11}{2}$

13. *(a,b,c,d)*

(a) $x=e^{i\theta}, y=e^{i\phi}$

$x^n+\frac{1}{x^n}=2\cos n\theta$

(b) $\frac{x}{y}+\frac{y}{x}=e^{i(\theta-\phi)}+e^{-i(\theta-\phi)}=2\cos(\theta-\phi)$

(c) $xy+\frac{1}{xy}=2\cos(\theta+\phi)$

14. *(a,c,d)* $a^3+b^3-(2c)^3=3(a)(bc)(-2c)$

When $a+b-2c=0$

$a, c, b \rightarrow AP$

OR $(a+b-2c)(a+bw-2w^2)(a+bw^2-2cw)=0$

15. *(a,d)* $|z_1+z_2|=|z_1+z_2|$

$|z_1|^2+|z_2|^2+z_1\bar{z}_2+z_2\bar{z}_1=|z_1|^2+1|z_2|^2-z_1\bar{z}_2-z_2\overline{z_1}$

$\Rightarrow z_1\bar{z}_2+z_2\bar{z}_1=0$

$\Rightarrow\left(\frac{z_1}{z_2}\right)+\left(\overline{z_1}\right)=0$

$\frac{z_1}{z_2}$ is purely imaginary

$\Rightarrow|\text{amp}(z_1)-\text{amp}(z_2)|=\frac{\pi}{2}$

16. *(b,c)* $\left(\frac{z}{|z|}\right)^2+\frac{z}{|z|}+1=0$

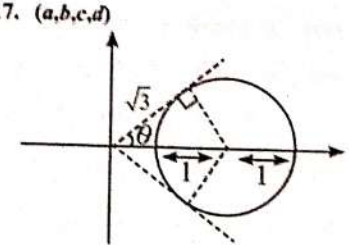
$z_1=\frac{z}{|z|}$

$z_1^2+z_1+1$

$z_1=\omega, -\omega^2$

$Z=k\omega, k\omega$

17. *(a,b,c,d)*



(b) $\max|z|=2+1=3$

(c) $\arg(z)=\pi-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(d) $\min|z|=2-1=1$

(a) $\min\arg(2)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

18. *(b,d)* $i^n=\left(e^{i\frac{\pi}{2}}\right)^{\frac{2n}{2}}=\left(e^{i\frac{\pi}{4}}\right)^{2n}=\frac{(1+i)^{2n}}{2^n}$

$i^{-n}=\frac{(1-i)^{+2n}}{2^{+n}} \therefore \text{G.E.}=\frac{(1+i)^{2n}}{2^n}+\frac{(1-i)^{2}}{2^n}$

(b)

Given that z and ω are two complex numbers. To prove

$|z|^2w-|w|^2z=z-w \Leftrightarrow z=w$ or $z\bar{w}=1$

First let us consider

$|z|^2w-|w|^2z=z-w$ ... *(i)*

$\Rightarrow z(1+|w|^2)=w(1+|z|^2)$

$\Rightarrow \frac{z}{w}=\frac{1+|z|^2}{1+|w|^2}$

$\Rightarrow \overline{\left(\frac{z}{w}\right)}=\frac{z}{w} \Rightarrow \frac{\bar{z}}{\bar{w}}=\frac{z}{w}$

$\Rightarrow \bar{z}w=z\bar{w}$ ... *(ii)*

Again from Eq. (i),

$z\bar{z}w-w\bar{w}z=z-w$

$z(\bar{z}w-1)-w(\bar{w}z-1)=0$

$z(z\bar{w}-1)-w(z\bar{w}-1)=0$ [Using Eq. (ii)]

$\Rightarrow (z\bar{w}-1)(z-w)=0$

$\Rightarrow z\bar{w}=1$ or $z=w$

20. *(c,d)* Let $z=x+iy$

from 2nd equation $x=6$ put in (1)

$3|(x-12)+yi|=5|x+(y-8)i|$

$9[36+y^2]=25[36+(y-8)^2]$

(substituting $x=6$)

$9\cdot 36+9y^2=25\cdot 36+25[y^2+64-16y]$

$16y^2-25\cdot 16y+36\cdot 16+25\cdot 64=0$

$y^2-25y+36+100=0$

$y^2-25y+136=0$

$(y-17)(y-8)=0$

then

$y=17$ or $y=8 \Rightarrow$ *(c), (d)*

21. *(a,b)* If in an equation the coefficients are real, then if roots of the equation are complex, they will occur as conjugate pairs.

Hence

z_1 is conjugate of atleast one of the roots.

Similarly z_2 is conjugate of atleast one of the roots.

In total we get 2 pair of conjugate roots $z_a, \bar{z}_a$ and $z_b, \bar{z}_b$.

Hence if z_1, z_2, z_3, z_4 are the roots of the given equation, then their conjugates are also the roots of the given equation.

22. *(a,b,c)*

$z_1=a+ib=\text{cis}(A)=\cos A+i\sin A\{|z_1|=1\}$

$z_2=c+id=\text{cis}(B)=\cos B+i\sin B \ldots\{\because|z_2|=1\}$

$\rightarrow a=\cos A$ & $b=\sin A$

$\text{Re}(\text{cis}(A)\text{cis}(-B))=0$

$\rightarrow \cos(A-B)=0$

$\rightarrow A-B=\frac{\pi}{2}$

$\rightarrow z_2=\cos\left(A-\frac{\pi}{2}\right)+i\sin\left(A-\frac{\pi}{2}\right)$

$=\sin A-i\cos A$

$\rightarrow c=\sin A$ & $d=-\cos A$

$w_1=a+ic=\cos A+i\sin A=\text{cis}A$

$\rightarrow A-B=\frac{\pi}{2}$

$\rightarrow Z_2=\cos\left(A-\frac{\pi}{2}\right)+i\sin\left(A-\frac{\pi}{2}\right)=$

$\sin A-i\cos A$

$\Rightarrow c=\sin A$ & $d=-\cos A$

$W_1=a+ic=\cos A+i\sin A=\text{cis}A$

$W_2=b+id=\sin A-i\cos A=$

$-i(i\sin A+\cos A)=-i\text{cis}(A)$

$\rightarrow|W_1|=1$ & $|W_2|=1$

$W_1\bar{W}_2=i\text{cis}A\text{cis}(-A)=i$

$\rightarrow \text{Re}(W_1W_2)=0$

Since, Z_1 and Z_2 are conjugated pairs,

let $Z_1=a+ib$

and $Z_2=a-ib$

So,

$Z_1\bar{Z}_2=(a+ib)(a-ib)=a^2-b^2+2aib$

$\bar{Z}_1Z_2=(a-ib)(a-ib)=a^2-b^2-2aib$

$Z_1-Z_2=2aib$

$(Z_1-Z_2)i=(2aib)i=-2ab=$ Purelyreal

23. **(a,b,c,d)**

$z_1 = a_1 + ib_1, \quad z_2 = a_2 + ib_2$

$a_1^2 + b_1^2 = 1; \quad a_2^2 + b_2^2 = 4$

$\text{Re}(z_1 z_2) = (a_1 + ib_1)(a_2 + ib_2)$

$= a_1a_2 + a_1b_2 i + a_2b_1 i = b_1b_2$

$= Rr\left[(a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)\right]$

$\Rightarrow \quad a_1a_2 = b_1b_2$

$\boxed{\frac{a_1}{b_1} = \frac{b_2}{a_2}} \Rightarrow \frac{a_1^2 + b_1^2}{a_1^2} = \frac{b_2^2 + a_2^2}{a_2^2}$

$\Rightarrow a_2^2 = 4b_1^2 \Rightarrow a_2 = 2b_1 \text{ or } a_2 = -2b_1$

$\frac{b_1}{a_1} = \frac{a_2}{b_2}$

$\frac{b_1^2 + a_1^2}{a_1^2} = \frac{a_2^2 + b_2^2}{b_2^2}$

$b_2^2 = 4a_1^2$

$b_2 = 2a_1$

$b_2 = 2a_1$

(c) $\omega_1\omega_2 = \left(a_1 + i\frac{a_2}{2}\right)(2b_1 + ib_2)$

$= 2a_1b_1 + ia_1b_2 + ia_2b_1 - \frac{b_2a_2}{2}$

$\text{Re}(\omega_1\omega_2) = \left[2a_1b_1 - \frac{b_2a_2}{2}\right]$

$4a_1b_1 = b_2a_2$

$4a_1b_1 - (2a_1)(2b_1) = 0$

(a) $|\omega_1|^2 = a_1^2 + \frac{a_2^2}{4} = a_1^2 + b_1^2 = 1$

(b) $|\omega_2|^2 = 4b_1^2 + b_2^2 = a_2^2 + b_2^2 = 4 \quad |\omega_2| = 2$

(d) $\text{Im}(\omega_1\omega_2) = a_1b_2 + a_2b_1$

$= 2a_1^2 + 2b_1^2 = 2(a_1^2 + b^2) = 2$

24. **(a,b,c,d)**

$|z_1 + z_2| = \left|-\frac{b}{a}\right| = 1$

$|z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + z_2\bar{z}_1 = 1 \quad \ldots (1)$

and $z_1z_2 = \frac{c}{a}$

$|z_1||z_2| = \left|\frac{c}{a}\right| = 1$

$\Rightarrow |z_2| = 1 = |z_1| \quad [\because \text{ Given } |z_1| = 1]$

so $2 + \frac{z_1}{z_2} + \frac{z_2}{z_1} = 1$

$(z_1 + z_2)^2 = z_1z_2$

$\frac{b^2}{a^2} = \frac{c}{a}$

$\Rightarrow \boxed{b^2 = ac}$

Similarly $\boxed{c^2 = ab}$

Now $|a-b| = |b-c|$

$|a|^2 + |b|^2 - a\bar{b} - b\bar{a} = |b^2 + |c|^2 - b\bar{c} - c\bar{b}$

$\Rightarrow (c-a)\bar{b} = b(\bar{a} - \bar{c})$ and $|a| = k$

$\Rightarrow a\bar{a} = k^2$

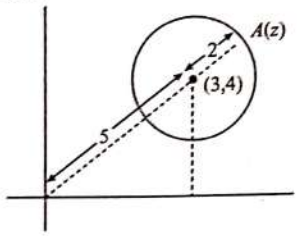
$(c-a)\bar{b} = b\left(\frac{k^2}{a} - \frac{k^2}{c}\right) = \frac{k^2b}{ac}(c-a)$

$= \frac{k^2b}{b^2}(c-a)$

$= \frac{k^2}{b} \cdot (c-a) = \bar{b}(c-a)$

25. **(b)**

(a) $|z - (3 + 4i)| = 2$ represents a circle whose centre is (3, 4) and radius is equal to 2.



From the figure it is clear that maximum value of $|z| = 7$.

(b) $|z - 12 - 6i| = |(z - i) + (12 - 5i)| \le |z - i| + |12 - 5i| < 1 + 13 = 14$

(c) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2) = 2(1 + 4) = 10$

(d) If $z = 1 + i$, then

$(z-1)^4 = i^4$

$z^4 - 4z^3 + 6z^2 - 4z + 1 = 1$

$(z^4 - 4z^3 + 7z^2 - 6z + 3) - z^2 + 2z - 2 = 1$

$z^4 - 4z^3 + 7z^2 - 6z + 3 = z^2 - 2z + 3 = (z-1)^2 + 2 = i^2 + 2 = 1$

$4(z^4 - 4z^3 + 7z^2 - 6z + 3) = 4$. **Ans.**

26. **[4]** $z = z + iy$

$\Rightarrow z^2 = x^2 - y^2 + 2ixy$

$\Rightarrow \text{Re}(z^2) = x^2 - y^2, \quad |z| = \sqrt{x^2 + y^2}$

$\Rightarrow x^2 - y^2 = 0, x^2 + y^2 = 3$

$\Rightarrow \quad x^2 = y^2 = \frac{3}{2}$

$\Rightarrow x = \pm\sqrt{\frac{3}{2}}, y = \pm\sqrt{\frac{3}{2}}$

27. **[25]** $z_1(z_1^2 - 3z_2^2) = 2 \quad \ldots (1)$

$z_2(3z_1^2 - z_2^2) = 11 \quad \ldots (2)$

Now (1) + i(2)

$(z_1 + iz_2)^3 = 2 + 11i \quad \ldots (3)$

(i) - i(2)

$(z_1 - iz_2)^3 = 2 - 11i \quad \ldots (4)$

Multiply (3) & (4) we get

$(z_1^2 + z_2^2)^3 = 125$

$z_1^2 + z_2^2 = 5 = \lambda$

28. **[2]** Given

$2|z-1| = z + \bar{z} \quad \Rightarrow \quad (x-1)^2 + y^2 = x^2$

$y^2 = 2x - 1 \quad \ldots (i)$

Also $\text{Arg}(z_1 - z_2) = \frac{\pi}{4}$ both point lies on a straight line

$y = x + c$ (say)

So curve $\quad \ldots (i)$

$\Rightarrow \quad y^2 = 2(y - c) - 1$

$\Rightarrow \quad y^2 - 2y + 2c + 1 = 0$

$\Rightarrow \quad y_1 + y_2 = 2$

$\Rightarrow \quad y^2 = 2(y - c) - 1$

$\Rightarrow \quad y^2 - 2y + 2c + 1 = 0$

$\Rightarrow \quad y_1 + y_2 = 2$

PYQ's (Past Year Questions)

1. **(b)** Consider, $z_1 = x_1 + iy_1$ and

$z_2 = x_2 + iy_2$

$\because Re(z_1z_2) = x_1x_2 - y_1y_2 = 0$

$Re(z_1 + z_2) = x_1 + x_2 = 0$

$\therefore x_1$ & x_2 are of opposite sign

y_1 & y_2 are of opposite sign.

Hence, $Im(z_1) < 0$ and $Im(z_2) > 0$ or $Im(z_1) > 0$ and $Im(z_2) < 0$

2. **[281]** $A = \frac{1967 + 1686i\sin\theta}{7 - 3i\cos\theta}$

$= \frac{281(7 + 6i\sin\theta)}{7 - 3i\cos\theta} \times \frac{7 + 3i\cos\theta}{7 + 3i\cos\theta}$

$= \frac{281(49 - 18\sin\theta\cos\theta + i(21\cos\theta + 42\sin\theta}{49 + 9\cos^2\theta}$

for positive integer

$\text{Im}(A) = 0$

$\frac{21\cos\theta + 42\sin\theta}{49 + 9\cos^2\theta} = 0$

$\sin 2\theta = \frac{2\tan\theta}{1 + \tan^2\theta}$

$21\cos\theta + 42\sin\theta = 0$

$\tan\theta = \frac{-1}{2}; \sin 2\theta = \frac{-4}{5}, \cos^2\theta = \frac{4}{5}$

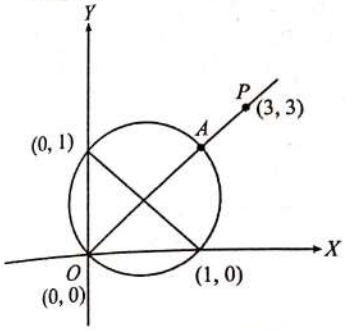
$\text{Re}(A) = \frac{281(49 - 9\sin 2\theta)}{49 + 9\cos^2\theta}$

$= \frac{281\left(49 - 9 \times \frac{-4}{5}\right)}{49 + 9 \times \frac{4}{5}} = 281$ (+ve integer)

3. (*b*) Minimum distance $AP = OP - OA$

$= 3\sqrt{2} - \sqrt{2}$

$= 2\sqrt{2}$



4. (*d*) $z = \dfrac{3+2i\sin\theta}{1-2i\sin\theta} \times \dfrac{1+2i\sin\theta}{1+2i\sin\theta}$

$z = \dfrac{(3-4\sin^2\theta)+8i\sin\theta}{1+4\sin^2\theta}$

For purely imaginary real part should be zero.

i.e., $3 - 4\sin^2\theta = 0$.

i.e., $\sin\theta = \pm\dfrac{\sqrt{3}}{2}$

$\theta = -\dfrac{\pi}{3}, \dfrac{\pi}{3}, \dfrac{2\pi}{3}$, Sum of all values is

$-\dfrac{\pi}{3} + \dfrac{\pi}{3} + \dfrac{2\pi}{3} = \dfrac{2\pi}{3}$

5. (*d*) $z = x + iy$

Given, $\dfrac{2z-3i}{2z+i}$ is purely imaginary

$\Rightarrow \text{Re}\left(\dfrac{2z-3i}{2z+i}\right) = 0$

Now, $\dfrac{2z-3i}{2z+i} = \dfrac{2(x+iy)-3i}{2(x+iy)+i}$

$= \dfrac{2x+i(2y-3)}{2x+i(2y+1)} \times \dfrac{2x-i(2y+1)}{2x-i(2y+1)}$

$\therefore \text{Re}\left[\dfrac{2z-3i}{2z+i}\right] = \dfrac{4x^2+(2y-3)(2y+1)}{4x^2+(2y+1)^2} =$

$\Rightarrow 4x^2 + (2y-3)(2y+1) = 0$

$\Rightarrow 4x^2 + 4y^2 - 4y - 3 = 0$

Given, $x + y^2 = 0 \Rightarrow x = -y^2$

$\Rightarrow 4(-y^2)^2 + 4y^2 - 4y - 3 = 0$

$\Rightarrow 4y^4 + 4y^2 - 4y - 3 = 0$

$\Rightarrow y^4 + y^2 - y = \dfrac{3}{4}$

6. [9] $z = 1 + i$

$z_1 = \dfrac{1+i\bar{z}}{\bar{z}(1-z)+\dfrac{1}{z}}$

$z_1 = \dfrac{1+i(1-i)}{(1-i)(1-1-i)+\dfrac{1}{1+i}}$

$= \dfrac{1+i-i^2}{(1-i)(-i)+\dfrac{1-i}{2}}$

$= \dfrac{\dfrac{2+i}{-3i-1}}{2} = \dfrac{4+2i}{-3i-1}$

$= \dfrac{-(4+2i)(3i-1)}{(3i)^2-(1)^2} = \dfrac{10i-10}{10} = i-1$

$\arg(z_1) = \dfrac{3\pi}{4}$

$\therefore \dfrac{12}{\pi}\arg(z_1) = \dfrac{12}{\pi} \times \dfrac{3\pi}{4} = 9$

7. (*b*) $\because |z|^3 + 2z^2 + 4\bar{z} - 8 = 0$ ...(*i*)

Take conjugate both sides

$\Rightarrow |z|^3 + 2\bar{z}^2 + 4z - 8 = 0$ ...(*ii*)

By (*i*) – (*ii*)

$\Rightarrow 2\left(z^2 - \bar{z}^2\right) + 4\left(\bar{z} - z\right) = 0$

$-2\left(\bar{z} - z\right)\left(\bar{z} + z\right) + 4\left(\bar{z} - z\right) = 0$

$-2\left(\bar{z} - z\right)\left(\bar{z} + z - 2\right) = 0$

$\Rightarrow z + \bar{z} = 2$ ...(*iii*)

$\Rightarrow \left|z + \bar{z}\right| = 2$ ...(*iv*)

Let $z = x + iy$, $\bar{z} = x - iy$

$\therefore x = 1$ $\therefore z = 1 + iy$

Put in (*i*) $|z| = \left(1 + y^2\right)^{\frac{1}{2}}$

$\Rightarrow (1+y^2)^{3/2} + 2(1 - y^2 + 2iy) + 4(1 - iy) - 8 = 0$

$\Rightarrow (1+y^2)^{3/2} + 2(1 + y^2)$

$\Rightarrow \sqrt{1+y^2} = 2 = |z|$

Also $\boxed{y = \pm\sqrt{3}}$

$\therefore z = 1 \pm i\sqrt{3}$

$\Rightarrow z - \bar{z} = \pm 2i\sqrt{3}$

$\Rightarrow \left|z - \bar{z}\right| = 2\sqrt{3}$

$\Rightarrow \left|z - \bar{z}\right| = 12$

Now $z + 1 = 2 + i\sqrt{3}$

$|z + 1|^2 = 4 + 3 = 7$

$\therefore$ P $\rightarrow$ 2 ; Q $\rightarrow$ 1 ; R $\rightarrow$ 3 ; S $\rightarrow$ 5

8. (*d*) Given, $\left|z - \dfrac{1}{z}\right| = 2$

$\because \left|\,|z| - \dfrac{1}{|z|}\,\right| \le \left|z - \dfrac{1}{z}\right| \le |z| + \dfrac{1}{|z|}$

Let $|z| = r$

$\left|r - \dfrac{1}{r}\right| \le 2 \le r + \dfrac{1}{r}$

$\left|r - \dfrac{1}{r}\right| \le 2$ & $r + \dfrac{1}{r} \ge 2$ always true

$r - \dfrac{1}{r} \ge -2$ & $r - \dfrac{1}{r} \le 2$

$\Rightarrow r^2 - 1 \le 2r$

$\Rightarrow r^2 - 2r \le 1$

$\Rightarrow (r-1)^2 \le 2$

$\Rightarrow r - 1 \le \sqrt{2}$

$\therefore |z|_{max} = 1 + \sqrt{2}$

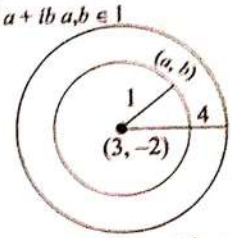
9. [40]

$1 < |z - 3 + 2i| < 4$

$1 < |z - (3 - 2i)| < 4$

It represents a circle having centre at (3, –2)

$z = a + ib$ $a, b \in I$



$\Rightarrow 1 < (a-3)^2 + (b+2)^2 < 16$

in equivalent to $1 < \alpha^2 + \beta^2 < 16$

$\alpha = a - 3 \in I$

$(\pm 2, 0)$ $(\pm 3, 0)$, $(0, \pm 2)$ $(0, \pm 3)$

$\beta = b + 2 \in I$

$(\pm 1, 1)$ $(\pm 2, 1)$, $(\pm 1, 2)$, $(\pm 2, \pm 2)$

$(\pm 3, \pm 1)$ $(\pm 1, 3)$, $(\pm 3, \pm 2)$, $(\pm 2, \pm 3)$

Total 40 such points are possible.

10. (*a*) Let $z = x + iy$

$v = |z|^2 + |z - 3|^2 + |z - 6i|^2, z \in C, z = z_0$

$\Rightarrow v = x^2 + y^2 + (x-3)^2 + y^2 + x^2 + (y-6)^2$

$\Rightarrow v = 3(x-1)^2 + (y-2)^2 + 30$

Minimum value at $z_0 = 1 + 2i$ and value is $v_0 = 30$

$z_0^2 = 1 - 4 + 4i = -3 + 4i \Rightarrow 2z_0^2 = -6 + 8i$

$z_0^3 = (-3 + 4i)(1 + 2i) = -3 - 6i + 4i - 8 = -11 - 2i$

$|2z_0 - z_0^3 + 3|^2 + v_0^2$

$|2z_0 - \bar{z}_0^3 + 3|^2 + v_0^2$

$= |2(1+2i)^2 - (1-2i)^3 + 3|^2 + 900$

$= |8 + 6i|^2 + 900$

$= 1000$

11. [0.5] $\dfrac{2+3z+4z^2}{2-3z+4z^2}$ is a real number

$\Rightarrow 1 + \dfrac{6}{\dfrac{2}{z} - 3 + 4z} \in R$

$\Rightarrow 2z+\frac{1}{z}\in R \Rightarrow 2z+\frac{1}{z}=2\bar{z}+\frac{1}{\bar{z}}$

$\Rightarrow (2\bar{z}-1)(z-\bar{z})=0$

$\Rightarrow |z|^2=\frac{1}{2}=0.50$

12. **[4]** Let $z=x+iy$

$x-iy-x^2+y^2-2ixy=$
$i\left(x-iy+x^2-y^2+2ixy\right)$

$\left(x-x^2+y^2\right)-i(y+2xy)=$
$(y-2xy)+i\left(x+x^2-y^2\right)$

$\Rightarrow x-x^2+y^2=y-2xy$ ...(*i*)

$x+x^2-y^2=-y-2xy$ ...(*ii*)

(*i*) + (*ii*) $2x=-4xy$

$\Rightarrow x=-2xy \Rightarrow x(1+2y)=0$

$\Rightarrow x=0$ or $y=-\frac{1}{2}$

Put $x=0$ in (*i*) or (*ii*) we get

$y^2=y \Rightarrow y=0,1$

$\therefore$ 2 complex numbers are possible 0 + 0i and 0 + i

put $y=-\frac{1}{2}$ in (*i*) or (*ii*)

$x-x^2+\frac{1}{4}=-\frac{1}{2}+x$

$\Rightarrow x^2=\frac{3}{4} \Rightarrow x=\pm\frac{\sqrt{3}}{2}$

$\therefore \frac{\sqrt{3}}{2}-\frac{i}{2}$ and $-\frac{\sqrt{3}}{2}-\frac{i}{2}$ are possible

$\therefore$ 4 solutions are possible.

13. (*a*) Let $\omega=I_1+iI_2$

$\Rightarrow (\bar{z})^2+\frac{1}{z^2}, I_1, I_2 \in$ Integers

$\Rightarrow |\omega|=\sqrt{I_1^2+I_2^2}$

$\Rightarrow |z|^2+\frac{1}{|z|^2}$

$\Rightarrow |z|^4+\frac{1}{|z|^4}=I_1^2+I_2^2-2$

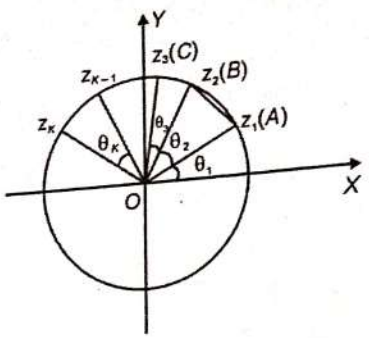
$\Rightarrow |z|^8-(I_1^2+I_2^2-2)|z|^4+1=0$

$\Rightarrow |z|^4=\frac{\left(I_1^2+I_2^2-2\right)\pm\sqrt{\left(I_1^2+I_2^2-2\right)^2-4}}{2}$

$\Rightarrow |z|=\left(\frac{43\pm3\sqrt{205}}{2}\right)^{1/4}$

where $I_1=6, I_2=3$

14. (*c*)



$|z_1|=|z_2|=\ldots|z_{10}|=1$

angle $=\frac{\text{arc}}{\text{rad}}$

$\theta_2=\text{arc}(z_1z_2)>(z_2>z_1)$

$P:|z_2-z_1|+\ldots+|z_1-z_{10}|\le\theta_1+\theta_2+\ldots+\theta_{10}$

$\Rightarrow |z_2-z_1|+\ldots+|z_1-z_{10}|\le 2\pi P$ is true

$z_1^2=e^{i2\theta_1}, z_k^2=z_{k-1}^2\cdot e^{i2\theta_k}$

Let $2\theta_k=\alpha k$

$z_1^2=e^{i\alpha_1}, z_k^2=z_{k-1}^2\cdot e^{i\alpha_k}$

$\alpha_1+\alpha_2+\ldots+\alpha_k=4\pi$

one similar sense

$\left|z_1^2-z_2^2\right|+\ldots\left|z_1^2-z_{10}^2\right|\le 4\pi$

Q is also true

15. (*c*) $z=x+iy$

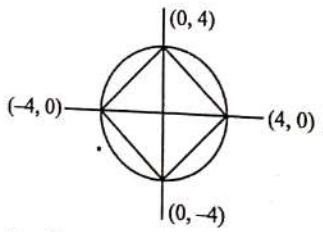
$|x|+|y|=4$

Minimum value of $|z|=2\sqrt{2}$

Maximim value of $|z|=4$

$|z|\in\left[\sqrt{8},\sqrt{16}\right]$

So $|z|$ can't be $\sqrt{7}$



16. (*c*) $x^2+(y-1)^2=x^2+(y+2)^2$

$-2y+1=4y+4$

$\Rightarrow 6y=-3 \Rightarrow y=-\frac{1}{2}$

and $|z|=\frac{5}{2}$

$\Rightarrow x^2+y^2=\frac{25}{4} \Rightarrow x^2=\frac{24}{4}=6$

$\Rightarrow z=\pm\sqrt{6}-\frac{1}{2}i$

$|z+3i|=\left|\pm\sqrt{6}+\frac{5}{2}i\right|=\sqrt{6+\frac{25}{4}}=\sqrt{\frac{49}{4}}$

$\Rightarrow |z+3i|=\frac{7}{2}$

17. **[4]** $\left(\frac{1+i}{1-i}\right)^{n/2}=\left(\frac{1+i}{1-i}\right)^{n/3}=1$

$\left(\frac{(1+i)^2}{2}\right)^{n/2}=\left(\frac{(1+i)^2}{2}\right)^{n/3}=1$

$\Rightarrow i^{n/2}=(i)^{n/3}=1$

$n_{\text{least}}=8, n_{\text{least}}=12$

GCD (8, 12) = 4

18. (*b,c*) $\left|z^2+z+1\right|=1$

$\Rightarrow \left|\left(z+\frac{1}{2}\right)^2+\frac{3}{4}\right|=1$

$\Rightarrow \left|\left(z+\frac{1}{2}\right)^2+\frac{3}{4}\right|\le\left|z+\frac{1}{2}\right|^2+\frac{3}{4}$

$\Rightarrow 1\le\left|z+\frac{1}{2}\right|^2+\frac{3}{4} \Rightarrow \left|\left(z+\frac{1}{2}\right)\right|^2\ge\frac{1}{4}$

$\Rightarrow \left|z+\frac{1}{2}\right|\ge\frac{1}{2}$

Also $|(z^2+z)+1=1\ge||z^2+z|-1|$

$\Rightarrow |z^2+z|-1\le 1$

$\Rightarrow |z^2+z|\le 2$

$\Rightarrow ||z^2|-|z||\le|z^2+z\le 2$

$\Rightarrow |r^2-r|\le 2$

$\Rightarrow r=|z|\le 2;\ \forall z\in S$

Also we can always find root of the equation $z^2+z+1=e^{i\theta};\ \forall\theta\in R$

Hence set 'S' infinite

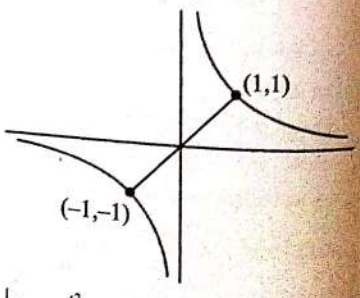
19. **[8]** Let $z=x+iy$

$z^4-|z|^4=4iz^2$

$\Rightarrow z^4-\left(z\bar{z}\right)^2=4iz^2$

$\Rightarrow z=0$ or $z^2-(\bar{z})^2=4i$

$\Rightarrow 4ixy=4i \Rightarrow xy=1$



$|z_1-z_2|^2_{\min}=8$

20. (bonus) $3|z_1|=4|z_2|$

$\Rightarrow \left|\frac{3z_1}{2z_2}\right|=2$, Let $\frac{3z_1}{2z_2}=2\cos\alpha+2(\sin\alpha)$

$\Rightarrow \frac{2z_2}{3z_1} = \frac{1}{2}\cos\alpha - \frac{1}{2}(\sin\alpha)i$

Given $z = \frac{3z_1}{2z_2} + \frac{2z_2}{3z_1} = \frac{5}{2}\cos\alpha + \frac{3}{2}(\sin\alpha)i$

Which is neither purely real non purely imaginary and $|z|$ depends on θ.

21. *(c)* Let $z = x + iy$ $|z| = 3$ represents a circle having radius 3 and centre at (0, 0).

And, $\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$

$\arg(x+iy-1) - \arg(x+iy+1) = \frac{\pi}{4}$

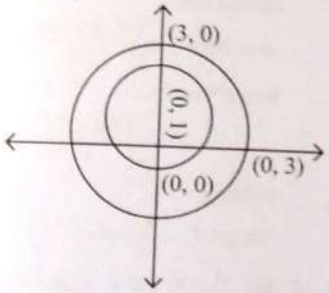
$\tan^{-1}\left(\frac{y}{x-1}\right) - \tan^{-1}\left(\frac{y}{x+1}\right) = \frac{\pi}{4}$

$\tan^{-1}\frac{\left(\frac{y}{x-1} - \frac{y}{x+1}\right)}{\left(1 + \frac{y}{x-1} \times \frac{y}{x+1}\right)} = \frac{\pi}{4}$

$\frac{yx + y - yx + y}{x^2 - 1 + y^2} = 1$

$2y = x^2 + y^2 - 1$

It represents a circle having centre at (0, 1) and radius = $\sqrt{2}$



Hence, the given curves intersect no where.

22. **[6]** Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$ & $z = x + iy$ then

and $\arg(z_1 - z_2) = \frac{\pi}{4}$ gives $\frac{y_1 - y_2}{x_1 - x_2} = 1$

or $y_1 - y_2 = x_1 - x_2$ ...(*i*)

Also, $|z-3|$

$= \text{Re}(z)$ gives $\sqrt{(x-3)^2 + y^2} = x$

$\Rightarrow y^2 - 6x + 9 = 0$ ...(*ii*)

As, z_1 & z_2 lies on equ.(*ii*) so $y_1^2 - 6x_1 + 9 = 0$ ...(*iii*)

& $y_2^2 - 6x_2 + 9 = 0$ ...(*iv*)

From equ.(*iii*) – equ.(*iv*) we get

$(y_1 + y_2)(y_1 - y_2) - 6(x_1 - x_2) = 0$

$\Rightarrow (x_1 - x_2)(y_1 + y_2 - 6) = 0$ [From equ.(*i*)]

$\Rightarrow y_1 + y_2 - 6 = 0$ i.e., $y_1 + y_2 = 6$

23. *(c)* $z = \frac{(3 + i\sin\theta)}{(4 - i\cos\theta)} \times \frac{(4 + i\cos\theta)}{(4 + i\cos\theta)}$

as z is purely real

$\Rightarrow 3\cos\theta + 4\sin\theta = 0 \Rightarrow \tan\theta = -\frac{3}{4}$

$\arg(\sin\theta + i\cos\theta) = \pi + \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right)$

$= \pi + \tan^{-1}\left(-\frac{4}{3}\right) = \pi - \tan^{-1}\left(\frac{4}{3}\right)$

24. *(b)* $|z|.|w| = 1$ $z = re^{i(\theta + \pi/2)}$ and $w = \frac{1}{r}e^{i\theta}$

$\bar{z}w = e^{-i(\theta + \pi/2)} \cdot e^{i\theta} = e^{-i(\pi/2)} = -i$

$z\bar{w} = e^{i(\theta + \pi/2)} \cdot e^{-i\theta} = e^{i(\pi/2)} = i$

25. *(b)* $(1 - \sqrt{3}i)^{200} = 2^{199}(p + iq)$

$\Rightarrow 2^{200}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)^{200} = 2^{199}(p + iq)$

$\Rightarrow 2\left(\cos\frac{200\pi}{3} - \sin\frac{200\pi}{3}\right) = (p + iq)$

$\Rightarrow 2\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = p + iq$

On comparing part, we get

$p = -1, q = -\sqrt{3}$

Now, $\alpha = p + q + q^2 = 2 - \sqrt{3}$

$\beta = p - q + q^2 = 2 + \sqrt{3}$

$\alpha + \beta = 4$

$\alpha.\beta = 1$

Hence, equation is $x^2 - 4x + 1 = 0$

[$\because x^2 - (\text{sum})x + \text{product} = 0$]

26. **[512]** Since, $z^8 - 2^8 = (z-2)(z-\alpha)(z-\alpha^2)\ldots(z-\alpha^{n-1})$

$z^8 - 2^8 = (z-2)(z-a)(z-a^2)\ldots(z-a^7)$

Put $z = 2e^{i\theta}$

$2^8(e^{i8\theta} - 1) = (2e^{i\theta} - 2)(2e^{i\theta} - \alpha)\ldots(2e^{i\theta} - \alpha^7)$ ...(*i*)

Take modus of eqn (*i*)

$2^8|e^{i8\theta} - 1| = PA_1 \, PA_2 \ldots PA_8$

$2^8|2\sin 4\theta| = PA_1 \, PA_2 \ldots PA_8$

$(PA_1 \, . \, PA_2 \ldots PA_8)_{max} = 512$

27. *(c)* $\because a_r = e^{i\frac{2r\pi}{9}}$

$\therefore a_1, a_2, a_3, \ldots\ldots$ are in G.P.

$$\begin{vmatrix} a_1 & a_1 r & a_1 r^2 \\ a_1 r^3 & a_1 r^4 & a_1 r^5 \\ a_1 r^6 & a_1 r^7 & a_1 r^8 \end{vmatrix} = a_1^3 r^9 \begin{vmatrix} 1 & r & r^2 \\ 1 & r & r^2 \\ 1 & r & r^2 \end{vmatrix} = 0$$

(*a*) $a_2 a_6 - a_4 a_8 = a_1 r \cdot a_1 r^5 - a_1 r^3 a_1 r^7 = a_1^2 r^3 - a_1^2 r^{10} \neq 0$

(*b*) $a_6 = a_1 r^5 \neq 0$

(*c*) $a_1 \cdot a_1 r^5 - a_1 r^2 \cdot a_1 r^4 = 0$

(*d*) $a_5 \neq 0$

Hence, option (*c*) is the correct answer.

28. **[3]** $(x-1)(x^2 - x + 1) = 0$

$\Rightarrow x = 1, x = \frac{1 \pm \sqrt{3}i}{2} = \frac{1}{2} + \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i$

$= e^{i\frac{\pi}{3}}, e^{-i\frac{\pi}{3}}$

Sum of 162th power of roots $= 1 + e^{i54\pi} + e^{-i54\pi}$

$= 1 + 1 + 1 = 3$

29. **[13]** $S = 21 + \sum_{k=1}^{21}\left(z^k + \frac{1}{z^k}\right)^2 = \sum\left(z^k + z^{-k}\right)^2$

$= 8\sum\left(\text{Re}(z^k)\right)^2$

(As $|z| = 1 \Rightarrow \frac{1}{z} = \bar{z}$)

$S = 2\sum_{k=1}^{21}\cos k\pi + 6\sum_{k=1}^{21}\cos\frac{k\pi}{3} + 21$

$= 2(-1) + 6\left(\frac{1}{2} - \frac{1}{2} - 1\right) + 21$

$= -8 + 21 = 13$

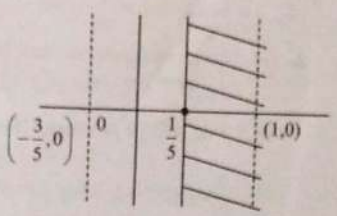
30. *(c)* $|z| < 1$

$5\omega(1 - z) = 5 + 3z$

$5\omega - 5\omega z = 5 + 3z$

$z = \frac{5\omega - 5}{3 + 5\omega}$

$|z| = 5\left|\frac{\omega - 1}{3 + 5\omega}\right| < 1$



$5(\omega - 1) < |3 + 5\omega|$

$5|\omega - 1| < 5\left|\omega + \frac{3}{5}\right|$

$|\omega - 1| < 5\left|\omega - \left(-\frac{3}{5}\right)\right|$

$\Rightarrow 5\,\text{Re}(\omega) > 1$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

By Aditya Mohad
username - muje_kabhi_dekha_nahi

CHAPTER 7

Sequences and Series

Prarambh (Topicwise)

1. (*b*) We have $\sqrt{2}+\sqrt{8}+\sqrt{18}+\sqrt{32}+...$

$=1\sqrt{2}+2\sqrt{2}+3\sqrt{2}+4\sqrt{2}+.....$

$=\sqrt{2}[1+2+3+4+.....\text{upto 24 terms}]$

$=\sqrt{2}\times\frac{24\times25}{2}=300\sqrt{2}$

2. (*c*) Suppose that first term and common difference of A.P.'s are A and D respectively.

Now, p^{th} term $=A+(p-1)D=a$ ... (*i*)

q^{th} term $=A+(q-1)D=b$ ... (*ii*)

and r^{th} term $a+(r-1)D=c$ ... (*iii*)

So, $a(q-r)+b(r-p)+c(p-q)$

$=a\left\{\frac{b-c}{D}\right\}+b\left\{\frac{c-a}{D}\right\}+c\left\{\frac{a-b}{D}\right\}$

$=\frac{1}{D}(ab-ac+bc-ab+ca-bc)=0$

3. (*c*) $T_m=a+(m-1)d=\frac{1}{n}$

and $T_n=a+(n-1)d=\frac{1}{m}$

On solving $a=\frac{1}{mn}$ and $d=\frac{1}{mn}$

$\therefore\ T_{mn}=a+(mn-1)d=\frac{1}{mn}+(mn-1)\frac{1}{mn}=1$

4. (*b*) The given number are in A.P.

$\therefore\ 2\log_9(3^{1-x}+2)=\log_3(4.3^x-1)+1$

$\Rightarrow\ 2\log_{3^2}(3^{1-x}+2)=\log_3(4.3^x-1)+\log_3 3$

$\Rightarrow\ \frac{2}{2}\log_3(3^{1-x}+2)=\log_3[3(4.3^x-1)]$

$\Rightarrow\ 3^{1-x}+2=3(4.3^x-1)$

$\Rightarrow\ \frac{3}{y}+2=12y-3$, where $y=3^x$

$\Rightarrow 12y^2-5y-3=0$

$y=\frac{-1}{3}$ or $\frac{3}{4}\Rightarrow 3^x=\frac{-1}{3}$ or $3^x=\frac{3}{4}$

$x=\log_3(3/4)\ \Rightarrow\ x=1-\log_3 4$

5. (*c*) Let $S_{Even}=2+4+6+8+.....\infty$ ... (*i*)

and $S_{Odd}=1+3+5+7+9+.....\infty$ ... (*ii*)

Sum $S_E=\frac{n}{2}[4+(n-1)2]$

$=\frac{n}{2}[2n+2]=\frac{n}{2}2(n+1)$

and $S_O=\frac{n}{2}[2+(n-1)2]=\frac{n}{2}(2n)$

Now $\frac{S_E}{S_O}=\frac{(n+1)}{n}$ or $S_E:S_O=(n+1):n$

6. (*c*) $S_{2n}-S_n=\frac{2n}{2}\{2a+(2n-1)d\}-\frac{n}{2}\{2a+(n-1)d\}$

$=\frac{n}{2}\{4a+4nd-2d-2a-nd+d\}$

$=\frac{n}{2}\{2a+(3n-1)d\}$

$=\frac{1}{3}.\frac{3n}{2}\{2a+(3n-1)d\}=\frac{1}{3}S_{3n}$

7. (*d*) Let $S=1+2+3+.....+100$

$=\frac{100}{2}(1+100)=50(101)=5050$

Let $S_1=3+6+9+12+.....+99$

$=3(1+2+3+4+.....+33)$

$=3.\frac{33}{2}(1+33)=99\times17=1683$

Let $S_2=5+10+15+.....+100$

$=5(1+2+3+.....+20)$

$=5.\frac{20}{2}(1+20)=50\times21=1050$

Let $S_2=15+30+45+.....+90$

$=15(1+2+3+.....+6)$

$=15.\frac{6}{2}(1+6)=45\times7=315$

$\therefore$ Required sum $=S-S_1-S_2+S_3$

$=5050-1683-1050+315=2632.$

8. (*b*) $S_{2n}=3S_n$

$\Rightarrow\frac{2n}{2}\{2a+(2n-1)d\}=3.\frac{n}{2}\{2a+(n-1)d\}$

$2a=(n+1)d$

Put $2a=(n+1)d$ in $\frac{S_{3n}}{S_n}$,

we get its value 6.

9. (*b*) Sum of n A-M's inserted between a, b is n' times mean between a, b.

10. (*b*) Since, a, b, c, d, e, f are six A.M.'s between 2 and 12

Therefore,

$=\frac{6}{2}(a+f)\ =\frac{6}{2}(2+12)=42$

11. (*d*) Given that x, $2x+2$, $3x+3$ are in G.P.

Therefore, $(2x+2)^2=x(2x+2)^2$

$=x(3x+3)$

$\Rightarrow x^2+5x+4=0$

$\Rightarrow(x+4)(x+1)=0\Rightarrow x=-1,-4$

Now first term a $=x$

Second term

$ar=2(x+1)\Rightarrow r=\frac{2(x+1)}{x}$

then 4^{th} term $=$

$ar^3=x\left[\frac{2(x+1)}{x}\right]^3=\frac{8}{x^2}(x+1)^3$

Putting $x=-4$

We get $T_4=\frac{8}{16}(-3)^3=-\frac{27}{2}=-13.5$

12. (*b*) $T_6=32$ and $T_8=128\Rightarrow ar^5=32$ ... (*i*)

and $ar^7=128$ ... (*ii*)

Dividing (*ii*) by (*i*), $r^2=4\ \Rightarrow r=2$

13. (*a*) Series is a G.P. with $a=0.9=\frac{9}{10}$ and $r=\frac{1}{10}=0.1$

$\therefore S_{100}=a\left(\frac{1-r^{100}}{1-r}\right)=\frac{9}{10}\left(\frac{1-\frac{1}{10^{100}}}{1-\frac{1}{10}}\right)=1-\frac{1}{10^{100}}$

14. (*b*) Given series $6+66+666+...$ upto n terms

$=\frac{6}{9}(9+99+999+.....\text{ upto terms})$

$=\frac{2}{3}(10+10^2+10^3+...+\text{ upto terms})$

$=\frac{2}{3}\left(\frac{10(10^n-1)}{10-1}-n\right)=\frac{1}{27}[20(10^n-1)-18n]$

$=\frac{2(10^{n+1}-9n-10)}{27}$

15. (*d*) Here $\frac{a}{1-r}=4$ and $ar=\frac{3}{4}$.

Dividing these,

$r(1-r)=\frac{3}{16}$ or $16r^2-16r+3=0$

or $(4r-3)(4r-1)=0$

$r=\frac{1}{4},\frac{3}{4}$ and $a=3,1$

so $(a,r)=\left(3,\frac{1}{4}\right),\left(1,\frac{3}{4}\right)$.

16. (*a*) $\because a, b, c$ are in G.P.

$\therefore \frac{b}{a}=\frac{c}{b}=r \Rightarrow \frac{b^2}{a^2}=\frac{c^2}{b^2}=r^2$

a^2, b^2, c^2 are in G.P.

17. (*b*) Let the GP be $a, ar^2, ar^3, \ldots$

Now, $ar^2=4$

$\therefore a.ar.ar^2.ar^3.ar^4=a^5r^{10}=(ar^2)^5=4^5$

18. (*a*) $S=\frac{p}{1-\frac{1}{p}}=\frac{9}{2} \Rightarrow \frac{p^2}{p-1}=\frac{9}{2}$

$\Rightarrow 2p^2-9p+9=0$

$\Rightarrow 2p^2-6p-3p+9=0$

$\Rightarrow (2p-3)(p-3)=0$

$p=3/2, 3$

19. (*b*) As given $G=\sqrt{xy}$

$\therefore \frac{1}{G^2-x^2}+\frac{1}{G^2-y^2}=\frac{1}{xy-x^2}+\frac{1}{xy-y^2}$

$=\frac{1}{x-y}\left\{-\frac{1}{x}+\frac{1}{y}\right\}=\frac{1}{xy}=\frac{1}{G^2}$.

20. (*b*) Let G_1 and G_2 are two G.M.'s between the number $a=1$ and $b=64$

$G_1=\left(a^2b\right)^{\frac{1}{3}}=(1.64)^{\frac{1}{3}}=4, G_2=\left(ab^2\right)^{\frac{1}{3}}$

$=\left(1.64^2\right)^{\frac{1}{3}}=16$

21. (*a*) Given, a, b, c are in A.P.

$\Rightarrow 2b=a+c$

$b-a, c-b, a$ are in G.P. So $(c-b)2=a(b-a)$

$\Rightarrow (b-a)^2=(b-a)a \begin{bmatrix} \because 2b=a+c \\ \Rightarrow b+b=a+c \\ \Rightarrow b-a=c-b \end{bmatrix}$

$\Rightarrow b=2a \quad [\because b \neq a]$

Put in $2b=a+c$, we get $c=3a$.

Therefore $a:b:c=1:2:3$

22. (*b*) $S=1+\frac{3}{2}+\frac{5}{2^2}+\frac{7}{2^3}+\ldots \quad \ldots(i)$

$\frac{S}{2}=\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^3}+\frac{7}{2^4}+\ldots \quad \ldots(ii)$

(i) – (ii) gives,

$\frac{S}{2}=1+\left(1+\frac{1}{2}+\frac{1}{2^2}+\ldots\right)$

$=1+\left[\frac{1}{1-\frac{1}{2}}\right]=1+2=3 \Rightarrow S=6$

23. (*c*) The sum of the first n terms is

$S_n=\left(1-\frac{1}{2}\right)+\left(1-\frac{1}{2^2}\right)+\left(1-\frac{1}{2^3}\right)+\left(1-\frac{1}{2^4}\right)$

$+\ldots\ldots+\left(1-\frac{1}{2^n}\right)=n-\left\{\frac{1}{2}+\frac{1}{2^2}+\ldots\ldots+\frac{1}{2^n}\right\}$

$=n-\frac{1}{2}\left(\frac{1-\frac{1}{2^n}}{1-\frac{1}{2}}\right)=n-\left(1-\frac{1}{2^n}\right)=n-1+2^{-n}$

Trick : Check $n=1, 2$ i.e.,

$S_1=\frac{1}{2}, S_2=\frac{5}{4}$ for and

(3) $\Rightarrow S_1=\frac{1}{2}$ and $S_2=2+2^{-2}-1=\frac{5}{4}$.

24. (*b*) S = 3 + 33 + 333 + ... to n terms

$S=3+33+\ldots\ldots$

$0=3+30+300+\ldots\ldots$ to n terms $-T_n$

(on subtracting)

$\therefore T_n=3(1+100+1000+\ldots$ to n terms

$=3\times1\cdot\frac{10^n-1}{10-1}=\frac{1}{3}\left(10^n-1\right)$

$S_n=\sum_{n=1}^{n}\frac{1}{3}\left(10^n-1\right)=\frac{1}{3}\sum_{n=1}^{n}10^n-\frac{1}{3}\sum_{n=1}^{n}1$

$=\frac{1}{3}\left(10\cdot\frac{10^n-1}{10-1}\right)-\frac{1}{3}n$

$S=\frac{1}{27}\left(10^{n+1}-9n-10\right)$

25. (*b*) $S=1+3+7+15+31+\ldots\ldots$

$S=1+3+7+15+\ldots$

$0=(1+2+4+8+16+\ldots$ to n terms$)=T_n$

(on subtracing)

$\therefore T_n=1+2+4+8+\ldots$ to n terms $=$

$1\cdot\frac{2^n-1}{2-1}=2^n-1$

$S_n=\sum_{n=1}^{n}T_n=\sum_{n=1}^{n}\left(2^n-1\right)=\sum_{n=1}^{n}2^n-\sum_{n=1}^{n}1$

$=2\cdot\left(\frac{2^n-1}{2-1}\right)-n=2^{n+1}-n-2$

26. (*d*) Suppose that x to be added then numbers 13, 15, 19 so that new numbers $x+13, 15+x, 19+x$ will be in H.P.

$\Rightarrow (15+x)=\frac{2(x+13)(19+x)}{x+13+x+19}$

$\Rightarrow x^2+31x+240=x^2+32x+247$

$\Rightarrow x=-7$

Trick: Such type of questions should be checked with the options.

27. (*a*) Here 5th term of the corresponding A.P. $=a+4d=45 \quad \ldots(i)$

and 11th term of the corresponding A.P. $=a+10d=69 \quad \ldots(ii)$

From (*i*) and (*ii*), we get $a=29, d=4$

Therefore 16th term of the correspon-ding A.P.

$=a+15d=29+15\times4=89$.

Hence 16th term of the H.P. is $\frac{1}{89}$.

28. (*d*) Considering corresponding A.P.

$a+6d=10$ and $a+11d=25$

$\Rightarrow d=3, a=-8$

Hence term of the corresponding H.P. is $\frac{1}{49}$

29. (*a*) As given $H=\frac{2pq}{p+q}$

$\therefore \frac{H}{p}+\frac{H}{q}=\frac{2q}{p+q}+\frac{2p}{p+q}=\frac{2(p+q)}{p+q}=2$

30. (*c*) A.M. $=\frac{a+b}{2}=A$ and G.M $=\sqrt{ab}=G$

on solving a and b are given by the values $A\pm\sqrt{(A+G)(A-G)}$

Trick : Let the numbers be 1, 9. Then $A = 5$ and $G = 3$. Now put these value in options.

Here (3) $\Rightarrow 5\pm\sqrt{8\times2}$ i.e. 9 and 1.

31. (*d*) Let $A=\frac{a+b}{2}, G=\sqrt{ab}$ and $H=\frac{2ab}{a+b}$.

Then $G^2=ab \quad \ldots(i)$

and $AH=\left(\frac{a+b}{2}\right).\frac{2ab}{a+b}=ab \quad \ldots(ii)$

From (*i*) and (*ii*), we have $G^2=AH \Rightarrow \beta^2=\alpha\gamma$

32. (*a*) Since a^2, b^2, c^2 be in A.P. Then

$b^2-a^2=c^2-b^2$

$\Rightarrow (b-a)(b+a)=(c-b)(c+b)$

$\Rightarrow \frac{b-a}{c+b}=\frac{c-b}{b+a}$

$\Rightarrow \frac{(b-a)(a+b+c)}{(c+a)(b+c)}=\frac{(c-b)(a+b+c)}{(a+b)(c+a)}$

$\Rightarrow \frac{b^2+bc-ac-a^2}{(c+a)(b+c)}=\frac{c^2+ac-ab-b^2}{(a+b)(c+a)}$

$\Rightarrow \frac{b}{c+a}-\frac{a}{b+c}=\frac{c}{a+b}-\frac{b}{c+a}$

Hence $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ be in A.P.

33. (*a*) ∵ A.M. = G.M.

$\Rightarrow \frac{a+b}{2} = \sqrt{ab} \Rightarrow \frac{(\sqrt{a})^2 - 2\sqrt{a}\sqrt{b} + (\sqrt{b})^2}{2} = 0$

$\Rightarrow \frac{(\sqrt{a}-\sqrt{b})^2}{2} = 0 \Rightarrow a = b$

∵ G.M = H.M

$\Rightarrow \sqrt{ab} = \frac{2ab}{a+b} \Rightarrow a + b - 2\sqrt{ab} = 0$

$\Rightarrow (\sqrt{a}-\sqrt{b})^2 = 0$

$\Rightarrow \sqrt{a} = \sqrt{b} \therefore a = b$

Thus A.M. =(G.M.) (H.M.) So $a = b$

34. (*a*) $\frac{a}{b}, \frac{b}{c}, \frac{c}{a}$ are in H.P. $\Rightarrow \frac{b}{a}, \frac{c}{b}, \frac{a}{c}$ are in A.P.

$\Rightarrow abc \times \frac{b}{a}, abc \times \frac{c}{b}, abc \times \frac{a}{c}$ are in A.P.

$\Rightarrow b^2c, ac^2, a^2b$ are in A.P.

$\therefore a^2b, c^2a, b^2c$ are in A.P.

35. (*c*) a, b, c are in G.P.

$\Rightarrow \log_x a, \log_x b, \log_x c$ are in A.P.

$\Rightarrow \frac{1}{\log_a x}, \frac{1}{\log_b x}, \frac{1}{\log_c x}$ are in A.P.

$\therefore \log_a x, \log_b x, \log_c x$ are in H.P.

36. (*a*) $\frac{1^3+2^3+3^3+4^3+.....+12^3}{1^2+2^2+3^2+4^2+.....+12^2}$

$= \frac{\left(\sum_{n=1}^{12} n^3\right)}{\left(\sum_{n=1}^{12} n^2\right)} = \left[\frac{n(n+1)}{2}\right]^2 \times \frac{6}{n(n+1)(2n+1)}$

$= \frac{3}{2} \cdot \frac{n(n+1)}{(2n+1)} = \frac{3}{2} \cdot \frac{12 \cdot 13}{25} = \frac{234}{25}$

[Putting $n = 12$].

37. (*a*) $T_n = \frac{1+2+3+.....+n}{n} = \frac{n(n+1)}{2n}$

$= \frac{1}{2}(n+1)$

Hence,

$S = \frac{1}{2}(\Sigma n + n) = \frac{1}{2}\left\{\frac{n(n+1)}{2} + n\right\} = \frac{n(n+3)}{4}$.

38. (*d*) $S = \left(\frac{1}{3\times7} + \frac{1}{7\times11} + \frac{1}{11\times15} + ...\right)$

$= \frac{1}{4}\left[\left(\frac{1}{3}-\frac{1}{7}\right) + \left(\frac{1}{7}-\frac{1}{11}\right) + \left(\frac{1}{11}-\frac{1}{15}\right) + ...\frac{1}{\infty}\right]$

$= \frac{1}{4}\left[\frac{1}{3} - \frac{1}{\infty}\right] = \frac{1}{4}\left[\frac{1}{3} - 0\right] = \frac{1}{12}$

39. (*d*) $S = \frac{2p+1}{2}[2(p^2+1)+2p]$

$= (2p+1)(p^2+1+p)$

$= 2p^3 + 3p^2 + 3p + 1 = p^3 + (p+1)^3$

40. (*d*) $S = \frac{1}{\log_2 4} + \frac{1}{\log_4 4} + \frac{1}{\log_8 4} + ... + \frac{1}{\log_n 4}$

$= \frac{1}{2} + \frac{1}{1} + \frac{1}{2/3} + ... + \frac{1}{2/n}$

$= \frac{1}{2} + 1 + \frac{3}{2} + \frac{4}{2} + ... + \frac{n}{2}$

$= \frac{n(n+1)}{4}$ **Ans**

41. (*a*) Given that

$S = 1^2 - 2^2 + 3^2 - 4^2 + ... 2002^2 + 2003^2$

$= 1 + (3^2 - 2^2) + (5^2 - 4^2) + ... + (2003^2 - 2002^2)$

$= 1 + 2 + 3 + 4 + 5 + ... + 2002 + 2003$

$= \frac{2003}{2}[1 + 2003] = 2003\,(1002)$

$= (2000 + 3)\,(1000 + 2) = 2007006$

42. (*b*) Let AP be $A, A + d, A + 2d, ...$

$T_p = a, T_q = b$

$\therefore a = A + (p-1)d$ ... (*i*)

$b = A + (q-1)d$ ... (*ii*)

$\Rightarrow$ subtract (*i*) & (*ii*) $\frac{a-b}{p-q} = d$

add (*i*) & (*ii*) $a + b = 2A + (p+q-1)d - d$

$\Rightarrow 2A + (p+q-1)d = (a+b) + d$

$S_{p+q} = [2A + (p+q-1)d]$

$= \frac{(p+q)}{2}\left[a + b + \frac{a-b}{p-q}\right]$

43. (*b*) Let the GP be $a, ar^2, ar^3, ...$

We know that sum of G.P. is possible $\Rightarrow |r| < 1$

$S = \frac{a}{1-r} \Rightarrow r = \left(1 - \frac{a}{S}\right)$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{a\left(1-\left(1-\frac{a}{S}\right)^n\right)}{\frac{a}{S}} = S\left[1-\left(1-\frac{a}{S}\right)^n\right]$

44. (*b*) a, b, c are in A.P

$\Rightarrow a - b = b - c$

$= (xz)^{b-c}\, y^{c-a} = y^{2(b-c)}\, y^{c-a}$ ($\because xz = y^2$)

x, y, z are in G.P

$y^2 = xz$

$= y^{\{2(b-c)+(c-a)\}}$

Now, the given expression becomes

$= y^{2b-a-c} = y^0 = 1$ ($\because 2b = a + c$)

$x^{b-c}, z^{b-c}, y^{c-a}$ ($\because b - c = a - b$)

45. (*a*) Let $a = x^{\log y - \log z}, b = y^{\log z - \log x}, c = z^{\log x - \log y}$

Now, $\log(abc) = \log a + \log b + \log c$

$\Rightarrow \log(abc) = (\log y - \log z)\log x + (\log z - \log x)\log y + (\log x - \log y)\log z = 0$

$\Rightarrow abc = 1$

As a, b, c are positive $AM \geq GM$

$\therefore \frac{a+b+c}{3} \geq (abc)^{\frac{1}{3}} = 1$

$\therefore a + b + c \geq 3$

Prabal (JEE Main Level)

1. (*d*) $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$

$\Rightarrow 3(a_1 + a_{24}) = 225$ (sum of terms equidistant from beginning and end are equal)

$a_1 + a_{24} = 75$

Now $a_1 + a_2 + ... + a_{23} + a_{24}$

$= \frac{24}{2}[a_1 + a_{24}] = 12 \times 75 = 900$

2. (*a*) $a + ar = 12$

$ar^2 + ar^3 = 48$

$\therefore r^2 = 4 \Rightarrow r = -2$, ∵ $r \neq -1, 2$

3. (*c*) $y = 2.\overline{357}$

$y = 2.357357357$ ... (*i*)

$1000y = 2357.357357$ ... (*ii*)

so $999y = 2355$

$y = \frac{2355}{999}$

4. (*a*) $\frac{S_{kx}}{S_x} = \frac{\frac{kx}{2}[2a + (kx-1)d]}{\frac{x}{2}[2a + (x-1)d]}$

$= k\left[\frac{2a + (kx-1)d}{2a + (x-1)d}\right]$

If $2a - d = 0$, then $\frac{S_{kx}}{S_x}$ is independent of x

So $d = 2a$

5. (*d*) $x \in R$

$5^{1+x} + 5^{1-x}, a/2, 5^{2x} + 5^{-2x}$ are in A.P

$a = (5^{2x} + 5^{-2x}) + (5^{1+x} + 5^{1-x})$

$a = (5^{2x} + 5^{-2x}) + 5(5^x + 5^{-x})$

$= (5^x - 5^{-x})^2 + 2 + 5(5^{x/2} - 5^{-x/2})^2 + 10$

$a = 12 + (5^x - 5^{-x})^2 + 5(5^{x/2} - 5^{-x/2})^2$

$\Rightarrow a \geq 12$

6. (*b*) $\frac{(54-3)}{n+1} = d$

$d = \frac{51}{n+1} \Rightarrow \frac{A_8}{A_{n-2}} = \frac{3}{5}$

$\Rightarrow \frac{3 + 8\frac{51}{n+1}}{3 + (n-2)\frac{51}{n+1}} = \frac{3}{5}$

$\Rightarrow \frac{3n + 3 + 408}{3n + 3 + 51n - 102} = \frac{3}{5}$

$\Rightarrow 15n + 2055 = 162n - 297$

$\Rightarrow 147n = 2352$

$n = 16$

7. (*a*) Let $S = \sqrt{2}+\sqrt{6}+\sqrt{18}+\sqrt{54}+$ .... + (10 terms)

$= \sqrt{2}(1+\sqrt{3}+\sqrt{9}+\sqrt{27}+$ ... + (10 terms))

$= \sqrt{2}\left(1+3^{1/2}+3^{1}+3^{3/2}+\ldots+\right)$

$= \sqrt{2}\,1.\dfrac{(1-(\sqrt{3})^{10})}{(1-\sqrt{3})} = \dfrac{\sqrt{2}((\sqrt{3})^{10}-1)}{(\sqrt{3}-1)}$

$= \dfrac{\sqrt{2}(3^5-1)(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \dfrac{\sqrt{2}}{2}242(\sqrt{3}+1)$

$= \sqrt{2}(121)(\sqrt{3}+1) = 121(\sqrt{6}+\sqrt{2})$

8. (*b*) Given,

$a_1 = 2, \& = \dfrac{a_{n+1}}{a_n} = \dfrac{1}{3}$

$= r, \sum_{r=1}^{20} a_r = \dfrac{a_1(1-r^{20})}{1-r} = \dfrac{2\left(1-\left(\frac{1}{3}\right)^{20}\right)}{\frac{2}{3}}$

$= 3\left(1-\dfrac{1}{3^{20}}\right)$

9. (*c*) Given that,

$x^2-3x+a=0\langle^{\alpha}_{\beta}\; x^2-12x+b=0\langle^{\gamma}_{\delta}$

Also, $\alpha, \beta, \gamma, \delta$ in increasing G.P.

Let $\alpha, \beta, \gamma, \delta$ be $\alpha, \alpha r, \alpha r^2, \alpha r^3$

$\alpha + \alpha r = 3$ & $\alpha r^2 + \alpha r^3 = 12$

$\alpha(1+r) = 3$ & $\alpha r^2(1+r) = 12$

& $r^2.3 = 12$

$\therefore \alpha = \dfrac{3}{1+r}$ & $r^2 = 4 \Rightarrow \alpha = \dfrac{3}{3} \Rightarrow \alpha =$ 1 $r = 2$,

$r = -2$ is rejected since

$\because$ G.P. increasing

$\alpha(\alpha r) = a$ & $(\alpha r^2)(\alpha r^3) = b$

$a = \alpha(\alpha r) = 1\,(1.2) = 2$

$b = (\alpha r^2)(\alpha r^3) = (1.4)(1.8) = 32$

10. (*d*) Let GP be $a_1, a_2, \ldots, a_k, \ldots$ with first term a & common ratio r,

$a_k = a_{k+1} + a_{k+2}\ \forall\ a_k > 0$

$\Rightarrow ar^{k-1} = ar^k + ar^{k+1} \Rightarrow r > 0$

$\Rightarrow 1 = r + r^2 \Rightarrow r^2 + r - 1 = 0$

$\Rightarrow r = \dfrac{-1 \pm \sqrt{1+4}}{2}$ {r = –ve rejected}

$\Rightarrow r = \dfrac{\sqrt{5}-1}{2} = 2\left(\dfrac{\sqrt{5}-1}{4}\right) 2\sin 18°$

11. (*a*) Let $S = \dfrac{1}{(1+p)} - \dfrac{1-p}{(1+p)^2} + \dfrac{(1-p)^2}{(1+p)^3} \ldots \infty$

$-1 < r = -\left(\dfrac{1-p}{1+p}\right) < 1 \qquad \because p > 0$

It is a GP with first term $= \dfrac{1}{1+p}$ & common ratio r

$\therefore S = \dfrac{a}{1-r} = \dfrac{\left(\frac{1}{1+p}\right)}{1+\frac{1-p}{1+p}} = \dfrac{1}{1+p+1-p} = \dfrac{1}{2}$

12. (*c*) If $x > 0$

$\log_2 x + \log_2(\sqrt{x}) + \log_2 \sqrt[4]{x} + \log_2 \sqrt[8]{x} + \log_2 \sqrt[16]{x} + \ldots = 4$

$\Rightarrow \log_2 x + \log_2 x^{1/2} + \log_2 x^{1/4} + \ldots = 4$

$\Rightarrow (\log_2 x)\,[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots] = 4$

$(\log_2 x)\,\dfrac{1}{1-\frac{1}{2}} = 4 \Rightarrow \log_2 x = 2$

$\Rightarrow x = 4$

13. (*a*) $3 + \dfrac{1}{4}(3+d) + \dfrac{1}{4^2}(3+2d) + \ldots + \infty$

$= 8$

$a = 3, r = \dfrac{1}{4}$

Sum of AGP upto ∞

$S\infty = \dfrac{a}{1-r} + \dfrac{dr}{(1-r)^2}$

$\Rightarrow 8 = \dfrac{3}{(3/4)} + \dfrac{d\left(\frac{1}{4}\right)}{3^2/4^2} \Rightarrow 8 = 4 + \dfrac{4d}{3^2}$

$\Rightarrow 4 = \dfrac{4d}{3^2} \Rightarrow d = 3^2 \Rightarrow d = 9$

14. (*c*) $\alpha+\beta = \dfrac{1}{\alpha^2} + \dfrac{1}{\beta^2} \Rightarrow \alpha+\beta = \dfrac{\alpha^2+\beta^2}{\alpha^2\beta^2}$

$\Rightarrow -\dfrac{b}{a} = \dfrac{b^2-2ac}{c^2}$

$\Rightarrow -bc^2 = ab^2 - 2a^2c$

$\Rightarrow ab^2 + bc^2 = 2a^2c \Rightarrow \dfrac{b}{c} + \dfrac{c}{a} = \dfrac{2a}{b}$

So $\dfrac{c}{a}, \dfrac{a}{b}, \dfrac{b}{c}$ are in A.P. $\Rightarrow \dfrac{c}{a}, \dfrac{a}{b}, \dfrac{b}{c}$ are in H.P.

15. (*a*) Given that,

$x = \sum_{n=0}^{\infty} a^n,\ y = \sum_{n=0}^{\infty} b^n,\ z = \sum_{n=0}^{\infty} c^n \begin{cases} |a|<1 \\ |b|<1 \\ |c|<1 \end{cases}$

$x = \dfrac{1}{1-a}, y = \dfrac{1}{1-b}, z = \dfrac{1}{1-c}$

$\because$ a, b, c are in A.P.

$\Rightarrow 1-a, 1-b, 1-c$ are also in A.P.

$\dfrac{1}{1-a}, \dfrac{1}{1-b}, \dfrac{1}{1-c}$ are in H.P.

$\Rightarrow x, y, z$ in H.P

16. (*a*) a, b, c in A.P. $\Rightarrow 2b = a + c$ ... (*i*)

p, q, r, in H.P. $\Rightarrow q = \dfrac{2pr}{p+r}$ ... (*ii*)

ap, bq, cr in G.P. $\Rightarrow b^2q^2 = acpr$ ... (*iii*)

From (ii) & (iii), we get

$\Rightarrow \dfrac{b^2.4(pr)^2}{(p+r)^2} = ac\,pr$

$\Rightarrow \dfrac{(a+c)^2\,pr}{(p+r)^2} = ac$ (from (i))

$\Rightarrow \dfrac{(p+r)^2}{pr} = \dfrac{(a+c)^2}{ac} \Rightarrow \dfrac{p^2+r^2}{pr} + 2$

$= \dfrac{a^2+c^2}{ac} + 2$

$\Rightarrow \dfrac{p}{r} + \dfrac{r}{p} = \dfrac{a}{c} + \dfrac{c}{a}$.

17. (*c*) $a^x = b^y = c^z = d^t = k$ and a, b, c, d are in G.P.

a, b, c are in G.P. $\Rightarrow$ So $b^2 = ac$

$\Rightarrow k^{2/y} = k^{1/x + 1/z} \Rightarrow \dfrac{2}{y} = \dfrac{1}{x} + \dfrac{1}{z}$

$\Rightarrow x, y, z$ are in H.P.

$\because$ b, c, d are in GP

then $= \dfrac{2}{z} = \dfrac{1}{y} + \dfrac{1}{t} \Rightarrow y, z, t$ are in HP

So x, y, z, t are in H.P.

18. (*b*) AM $= A = \dfrac{a+b+c}{3}$

GM $= G = (abc)^{1/3}$

HM $= H = \dfrac{3abc}{ab+bc+ca} = \dfrac{3G^3}{ab+bc+ca}$

Equation whose roots are a, b, c

$\Rightarrow x^3 - (a+b+c)x^2 + (\Sigma ab)x - abc = 0$

$\Rightarrow x^3 - 3Ax^2 + \geq\left(\dfrac{a+b+c}{3}\right)[(abc)^{1/3}]^3\, x$

$- G^3 = 0$ **Ans**

19. (*b*) Let $S = \sum_{r=2}^{\infty} \dfrac{1}{r^2-1}$

$= \sum_{r=2}^{\infty} \dfrac{1}{(r-1)(r+1)} = \dfrac{1}{2}\sum_{r=2}^{\infty}\left(\dfrac{1}{r-1} - \dfrac{1}{r+1}\right)$

$= \dfrac{1}{2}\left[1 - \dfrac{1}{3} + \dfrac{1}{2} - \dfrac{1}{4} + \dfrac{1}{3} - \dfrac{1}{5} + \ldots + \dfrac{1}{n-1} - \dfrac{1}{n+1} + \ldots\right]$

when $n \to \infty \Rightarrow \dfrac{1}{n+1} \to 0$

$\therefore S = \dfrac{1}{2}\left[1 + \dfrac{1}{2}\right] = \dfrac{3}{4}$.

20. (*c*) Let $\dfrac{1}{1^2} + \dfrac{1}{2^2} + \dfrac{1}{3^2} + \ldots \infty = \dfrac{\pi^2}{6}$

Now $S_{even} = \dfrac{1}{2^2} + \dfrac{1}{4^2} + \dfrac{1}{6^2} + \ldots \infty$

$= \dfrac{1}{2^2}\left[1 + \dfrac{1}{2^2} + \dfrac{1}{3^2} + \ldots \infty\right] = \dfrac{1}{2^2}\dfrac{\pi^2}{6} = \dfrac{\pi^2}{24}$

$$S_{odd} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots \infty = S - S_{even}$$

$$= \frac{\pi^2}{6} - \frac{\pi^2}{24} = \frac{3\pi^2}{24} = \frac{\pi^2}{8}$$

21. **(c)** $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \ldots$ n terms

$$= \frac{n(n+1)^2}{2} \text{ when n is even}$$

$$1^2 + 2.2^2 + 3^2 + \ldots 2n^2 = n\frac{(n+1)^2}{2}$$

when n is odd $n+1$ is even

$$1^2 + 2 \cdot 2^2 + 3^2 + \ldots n^2 + 2(n+1)^2$$

$$= (n+1)\frac{(n+2)^2}{2}$$

$$1^2 + 2 \cdot 2^2 + 3^2 + \ldots n^2 = (n+1)\left[\frac{(n+2)^2}{2} - 2(n+1)\right]$$

$$= \frac{(n+1)n^2}{2}$$

22. **(b)** Given that,

$$1^2 + 2^2 + \ldots n^2 = 1015$$

$$\frac{n(n+1)(2n+1)}{6} = 1015$$

Put $n = 15 \Rightarrow \frac{15\times16\times31}{6} = 1240$

$$\Rightarrow n \neq 15$$

Put n = 14 $\Rightarrow \frac{14\times15\times29}{6} = 1015$

$$\Rightarrow n = 14$$

23. **(b)** Let be the first term and D be the common difference of the A.P.

$$\Rightarrow a = \frac{p}{2}\left[2A + (p-1)D\right]$$

We can write

$$\frac{a}{p}(q-r) + \frac{b}{q}(r-p) + \frac{c}{r}(p-q)$$

$$= \Sigma\frac{a}{p}(q-r)$$

L.H.S. $= \Sigma\frac{a}{p}(q-r)$

$$= \Sigma\frac{1}{2}(q-r)\left[2A + (p-1)D\right]$$

$$= \frac{1}{2}\Sigma 2A(q-r) + \frac{1}{2}\Sigma(q-r)D(p-1)$$

$$= A\Sigma(q-r) + \frac{D}{2}\Sigma\left[p(q-r)\right] - \frac{D}{2}\Sigma(q-r)$$

$$= 0 + 0 - 0 = 0$$

24. **(c)** Let a be the first term and the common difference. Then $a_r = a + (r-1)d$

$$\Delta = \begin{vmatrix} a+(m-1)d & a+(n-1)d & a+(p-1)d \\ m & n & p \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a & a & a \\ m & n & p \\ 1 & 1 & 1 \end{vmatrix} + d\begin{vmatrix} m-1 & n-1 & p-1 \\ m & n & p \\ 1 & 1 & 1 \end{vmatrix}$$

$$= a\begin{vmatrix} 1 & 1 & 1 \\ m & n & p \\ 1 & 1 & 1 \end{vmatrix} + d\begin{vmatrix} m & n & p \\ m & n & p \\ 1 & 1 & 1 \end{vmatrix}$$

$$= a.0 + d.0 = 0$$

25. **(a)** We have a, the first term and d, the common difference,

$$x = \left\{2a + (p-1)d\right\}\frac{p}{2} \Rightarrow \frac{x}{p} = a + (p-1)\frac{d}{2}$$

Similarly, $\frac{y}{q} = a + (q-1)\frac{d}{2}$ and

$$\frac{z}{r} = a + (r-1)\frac{d}{2}$$

$$\therefore \frac{x}{p}(q-r) + \frac{y}{q}(r-p) + \frac{z}{r}(p-q)$$

$$= \left\{a + (p-1)\frac{d}{2}\right\}(q-r) + \left\{a + (q-1)\frac{d}{2}\right\}(r-p) + \left\{a + (r-1)\frac{d}{2}\right\}(p-q)$$

$$= a\{(q-r) + (r-p) + (p-q)\} + \frac{d}{2}\{(p-1)(q-r) + (q-1)(r-p) + (r-1)(p-q)\}$$

$$= a.0 + \frac{d}{2}[\{pq - pr - rq - pq + pr - qr - \{(q-r) + (r-p) + (p-q)\}$$

$$= 0 + \frac{d}{2}\{0-0\} = 0$$

26. **(a)** Let a and b be the two numbers

$$\therefore A_1 = a - \left(\frac{b-a}{3}\right) = \frac{2a+b}{3},$$

$$A_2 = a + 2\left(\frac{b-a}{3}\right) = \frac{a+b}{3}$$

$$G_1 = a\left(\frac{b}{a}\right)^{1/3} = a^{2/3}b^{1/3},$$

$$G_2 = a\left(\left(\frac{b}{a}\right)^{1/3}\right)^2 = a^{1/3}b^{2/3}$$

$$H_1 = \frac{1}{\frac{1}{a} + \left(\frac{1}{b} - \frac{1}{a}\right)} = \frac{3}{\frac{2}{a} + \frac{1}{b}} = \frac{3ab}{a+2b},$$

$$H_2 = \frac{3ab}{2a+b}$$

$$\therefore \frac{G_1G_2}{H_1H_2} = \frac{(a^{2/3}b^{1/3})(a^{1/3}b^{2/3})}{\frac{3ab}{a+2b}\cdot\frac{3ab}{2a+b}}$$

$$= \frac{(a+2b)(2a+b)}{9ab}$$

$$A_1 + A_2 = \frac{2a+b}{3} + \frac{a+2b}{3} = a+b$$

$$H_1 + H_2 = \frac{3ab}{a+2b} + \frac{3ab}{2a+b}$$

$$= 3ab\left(\frac{3a+b+a+2b}{(a+2b)(2a+b)}\right) = \frac{9ab(a+b)}{(a+2b)(2a+b)}$$

$$\therefore \frac{A_1 + A_2}{H_1 + H_2} = \frac{(a+2b)(2a+b)}{9ab} = \frac{G_1G_2}{H_1H_2}$$

27. **(c)** We have,

$$\frac{m}{n} = \frac{(a+b)}{2ab/(a+b)} \Rightarrow \frac{m}{n} = \frac{(a+b)^2}{4ab} = \frac{\left(\frac{a}{b}+1\right)^2}{4\frac{a}{b}}$$

$$\Rightarrow 4\frac{m}{n}\left(\frac{a}{b}\right) = \left(\frac{a}{b}+1\right)^2$$

$$\Rightarrow 2\frac{\sqrt{m}}{\sqrt{n}}\sqrt{\frac{a}{b}} = \left(1 + \frac{a}{b}\right)$$

Let $\frac{a}{b} = r^2$,

$$\frac{2\sqrt{m}}{\sqrt{n}}r = (1+r^2) \Rightarrow 2\sqrt{mr} = \sqrt{n} + \sqrt{n}r^2$$

$$\sqrt{n}r^2 - 2\sqrt{mr} + \sqrt{n} = 0$$

$$\therefore r = \frac{2\sqrt{m} \pm \sqrt{4m-4n}}{2\sqrt{n}} = \frac{\sqrt{m} \pm \sqrt{m-n}}{\sqrt{n}}$$

Considering +ve sign,

$$r = \frac{\sqrt{m} + \sqrt{m-n}}{\sqrt{n}}$$

$$= \frac{(\sqrt{m} + \sqrt{m-n})(\sqrt{m} - \sqrt{m-n})}{\sqrt{n}(\sqrt{m} - \sqrt{m-n})}$$

$$= \frac{m - (m-n)}{\sqrt{n}(\sqrt{m} - \sqrt{m-n})} = \frac{m}{\sqrt{m} - \sqrt{n}}$$

$$\therefore r^2 = \frac{\sqrt{m} + \sqrt{m-n}}{\sqrt{n}} \cdot \frac{\sqrt{n}}{\sqrt{m} - \sqrt{m-n}}.$$

Hence, $\frac{a}{b} = \frac{\sqrt{m} + \sqrt{m-n}}{\sqrt{m} - \sqrt{m-n}}$.

28. **(d)** Let us consider the diagonal and an adjacent parallel line

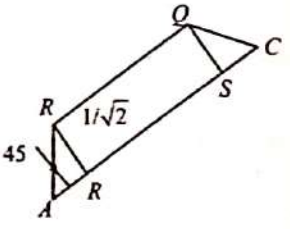
Length of the line $PQ = RS = AC - (AR + SC)$

$$= AC - 2AR$$

$$= AC - 2PR \qquad (\because AR = PR)$$

$$= n\sqrt{2} - 2\cdot\frac{1}{\sqrt{2}} = n\sqrt{2} - \sqrt{2} = (n-1)\sqrt{2}\,cm$$

Length of line adjacent to PQ, other than AC, will be $\left((n-1)-1\right)\sqrt{2} = (n-2$



$\therefore$ Sum of the lengths of all possible line segments and the diagonal

$= 2\times\left[n\sqrt{2}+(n-1)\sqrt{2}+(n-2)\sqrt{2}+.....\right]$

$-n\sqrt{2}, n\in N$

$= 2\times\sqrt{2}\left[n+(n-1)+(n-2)+.....+1\right]-n\sqrt{2}$

$= 2\times\sqrt{2}\left[n+(n-1)+(n-2)+.....+1\right]-n\sqrt{2}$

$= n^2\sqrt{2}\,cm$

29. **(a,c)** $r\times r! = (r+1-1)\times r!$

$= (r+1)! - r!$

$= V(r) - V(r-1)$

$\Rightarrow \sum_{r=1}^{30} r(r!) = V(31) - V(0) = (31)! - 1$

$\Rightarrow 1 + \sum_{r=1}^{30} r(r!) = 31!$

30. **(b)** $I(2n) = 1^4 + 2^4 + 3^4 + ... + (2n-1)^4 + (2n)^4$

$= [(1^4 + 3^4 + 5^4 + 4 - (2n-1)^4] + 2^4(1^4 + 2^4 + 3^4 + ... 4^4 + n^4)$

$= \sum_{r=1}^{n}(2r-1)^4 + 16\times I(n)$

$\sum_{r=1}^{n}(2r-1)^4 = I(2n) - 16I(n)$

31. **(a)** Clearly, $y, \frac{1}{x_1}, \frac{1}{x_2}, ..., \frac{1}{x_{20}}$ Will be in A. P. Hence,

$\frac{1}{x_2} - \frac{1}{x_1} = \frac{1}{x_3} - \frac{1}{x_2}$

$= \frac{1}{x_{r+1}} - \frac{1}{x_r} = ... = \lambda$ (say)

$\Rightarrow \frac{x_r - x_{r+1}}{x_r x_{r+1}} = \lambda \Rightarrow x_{r+1} = -\frac{1}{\lambda}(x_{r+1} - x_r)$

$\Rightarrow \sum_{r=1}^{19} x_r x_{r+1} = -\frac{1}{\lambda}\sum_{r=1}^{19}(x_{r+1} - x_r)$

$= -\frac{1}{\lambda}(x_{20} - x_1)$

Now,

$\Rightarrow \frac{x_1 - x_{20}}{x_1 x_{20}} = 19\lambda$

$\Rightarrow \frac{x_1 - x_{20}}{x_1 x_{20}} = 19\lambda$

$\sum_{r=1}^{19} x_r x_{r+1} = 19 x_1 x_{20} = 19\times 4 = 76$

32. **(a)** The general term of the given series is

$t_n = \frac{x^{2^{n-1}}}{1-x^{2^n}} = \frac{1+x^{2^{n-1}}-1}{\left(1+x^{2^{n-1}}\right)\left(1-x^{2^{n-1}}\right)}$

$\Rightarrow t_n = \frac{1}{1-x^{2^{n-1}}} - \frac{1}{1-x^{2^n}}$

Now, $S_n = \sum_{n=1}^{n} t_n$

$= \left[\left\{\frac{1}{1-x} - \frac{1}{1-x^2}\right\} + \left\{\frac{1}{1-x^2} - \frac{1}{1-x^4}\right\} + ... + \left\{\frac{1}{1-x^{2^{n-1}}} - \frac{1}{1-x^{2^n}}\right\}\right]$

$= \frac{1}{1-x} - \frac{1}{1-x^{2^n}}$

Therefore, the sum to infinite terms is

$\lim_{n\to\infty} S_n = \frac{1}{1-x} - 1$

$= \frac{x}{1-x} \qquad \left[\because \lim_{n\to\infty} x^{2^n} = 0, \text{as} |x| < 1\right]$

33. **(a)** S = (1) (2003) +(2) (2002) +(3) (2001)+...+(2003) (1)

$= \sum_{r=1}^{2003} r\left(2003-(r-1)\right) = \sum_{r=1}^{2003} r(2004-r)$

$= \sum_{r=1}^{2003} 2004r - \sum_{r=1}^{2003} r^2 = 2003\times 4007\times 334$

$= 2003\times 334\times (6012 - 4007)$

$= 2003\times 334\times 2005$

Hence, $x = 2005$.

34. **(a)** Since $x, |x+1|, |x-1|$ are in A.P.

so $2|x+1| = x + |x-1|$ ... (*i*)

Case-I If $x < -1$, then (*i*) becomes $-2(x+1) = x - (x-1)$

$\Rightarrow x = -\frac{3}{2}$

then series $-\frac{3}{2}, \frac{1}{2}, \frac{5}{2}, ...$

$\therefore S_{20} = \frac{20}{2}[-3 + (20-1)\,2] = 350$

Case-II If $-1 \le x \le 1$, then (*i*) becomes $2(x+1) = x - (x-1) \Rightarrow x = -1/2$

then series $-\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$

So $S_{20} = \frac{20}{2}$

$[-1 + (20-1)\,1]\;10\times 18 = 180$

Case-III If $x \ge 1$, then (i) becomes

$2(x+1) = x + x - 1$

$2 = -1$ impossible.

35. **(b)** $t_r = 2^{r/2} + 2^{-r/2} = \frac{1+2^r}{2^{r/2}}$

$t_r^2 = \frac{1+2^{2r}+2\cdot 2^r}{2^r}$

$\sum_{r=1}^{10} t_r^2 = \sum_{r=1}^{10} 2^{-r} + \sum_{r=1}^{10} 2^r + \sum_{r=1}^{10} 2$

$= \frac{1}{2}\left(\frac{1-\left(\frac{1}{2}\right)^{10}}{1-\frac{1}{2}}\right) + 2\frac{\left[2^{10}-1\right]}{2-1} + 2\times 10$

$= 1 - \left(\frac{1}{2}\right)^{10} + 2^{11} - 2 + 20 = \frac{2^{21}-1}{2^{10}} + 19$

36. **(a)** $a_1, a_2, ... a_n$ are in HP

$\frac{1}{a_1} + \frac{1}{a_2} - \frac{1}{a_n}$ are in AP

$a_1a_2 + a_2a_3 ... a_{n-1}a_n$

$\frac{(a_2-a_1)}{\frac{1}{a_1}-\frac{1}{a_2}} + \frac{(a_3-a_2)}{\frac{1}{a_2}-\frac{1}{a_3}} + ... \frac{a_n - a_{n-1}}{\frac{1}{a_{n-1}}-\frac{1}{a_n}}$

$= \frac{a_2-a_1}{d} + \frac{a_3-a_2}{d} ... \frac{a_n-a_{n-1}}{d}$

$= \frac{a_n-a_1}{d} = \frac{a_n-a_1}{\left(\frac{1}{a_n}-\frac{1}{a_1}\right)} \times n = na_1a_n$

37. **(c)** Let $a = A$, $b = Ar$, $C = Ar^2$

as $a + b > c$

$\therefore 1 + r > r^2$

similarly $1 + r^2 > r$ and $r + r^2 > 1$

Finding All solutions of 3 equations

we get $r \in \left[\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2}\right]$

38. **(a)** As *ab, ,c d,* are in *HP*, *b* is the HM between a and c. Also the GM between *a* and $c = \sqrt{ac}$, Now, GM > HM so that

$\sqrt{ac} > b \Rightarrow ac > b^2$ ...(*i*)

Again *a b, ,c d,* are in HP, so c is the HM between b and d. Therefore, $b\,d > c_2$ ...(*ii*)

Now multiplying Eqs. (*i*) and (*ii*), we get

$abcd > b^2c^2$ or $ad > bc$

Hence, answer (*B*) is true. Now AM between *a* and *c* = 1/2(*a* + *c*) Now as AM> HM

$a + c > 2b$ ... (*iii*)

And *c* is HM between *b* and *d* so

$b + d > 2c$ ... (*iv*)

Adding Eqs. (*iii*) and (*iv*), we get

$(a + c) + (b + d) > 2(b + c)$

$\Rightarrow a + d > b + c$

39. **(d)** $\frac{r_1+r_2+r_3+r_4}{4} = \frac{9}{4}$

$r_1r_2r_3r_4 = 16$

$\sqrt[4]{r_1r_2r_3r_4} = 2$

$\frac{9}{4} - 2 = \frac{1}{4}$

40. **(a)** $a_1, a_2 ... a_n ... a_{2n+1}$ are in AP.

$a_1 + a_{2n+1} = a_2 + a_{2n} \cdots = a_{n+2} + a_n$

$= 2a_1 + 2nd$

$a_{2n+1} - a_1 = 2nd$, $a_{2n} - a_2 = (2n-2)d$...

$a_{n+2} - a_n = 2d$

$\therefore$ our equation becomes

$2d\sum\frac{\sum n}{2a_1 + 2nd} = \frac{dn(n+1)}{2a_{n+1}}$

$= \frac{n(n+1)(a_2-a_1)}{2a_{n+1}}$

41. **(a)** Let $x_n = a_x + (n-1)d_x$

$y_n = a_y + (n-1)d_y$

$z_n = a_z + (n-1)d_z$

$w_n = a_w + (n-1)d_w$

Given $x_4 + y_4 + z_4 + w_4 = 8$

$\Rightarrow a_x + a_y + a_z + a_w + 3(d_x + d_y + d_z + d_w) = 8$ ...(i)

Also $x_{10} + y_{10} + z_{10} + w_{10} = 20$

$\Rightarrow a_x + a_y + a_z + a_w + 9(d_x + d_y + d_z + d_w) = 20$ ...(ii)

Solving equation (i) and (ii) we get

$d_x + d_y + d_z + d_w = 2$

$a_x + a_y + a_z + a_w = 2$

As the terms of the A.P. in positive using $AM \geq GM$, we get

$\frac{x_{20} + y_{20} + z_{20} + w_{20}}{4} \geq (x_{20} \cdot y_{20} \cdot z_{20} \cdot w_{20})^{1/4}$

$\Rightarrow \frac{a_x + a_y + a_z + a_w + 19(d_x + d_y + d_z + d_w)}{4} \geq (x_{20} \cdot y_{20} \cdot z_{20} \cdot w_{20})^{1/4}$

$\Rightarrow 10 \geq (x_{20} \cdot y_{20} \cdot z_{20} \cdot w_{20})^{1/4}$

$\Rightarrow x_{20} \cdot y_{20} \cdot z_{20} \cdot w_{20} \leq 10^4$

Hence, the maximum possible value of the given expression is 10^4

42. **(a)** $\frac{\left(\frac{2n(2n+1)}{2}\right)^2}{\frac{n(n+1)(2n+1)}{6}} = \frac{6n^2(2n+1)^2}{n(n+1)(2n+1)}$

$= \frac{6n(2n+1)}{(n+1)} = \frac{6n(n+1+n)}{n+1}$

$= 6n + \frac{6n^2}{n+1}$

$= 6n + 6(n-1) + \frac{6}{n+1} =$ Integer

The values of 'n' which are satisfying are $n = 1, 2, 5$ only for being integer

$\therefore$ sum $= 1 + 2 + 5 = 8$

43. **(d)** Let $x = 10a + b$ and $y = 10b + a$

$x^2 - y^2 = m^2$

$\therefore (10a+b)^2 - (10b+a)^2 = m^2$

$\therefore (10a+b+10b+a)(10a+b-10b-a) = m^2$

$\therefore 11(a+b)9(a-b) = m^2$

$\therefore 99(a+b)(a-b) = m^2$

Since a and b are digits from 1 to 9, $a + b = 11$ and $a - b = 1$ is the only possibility from m to be an integer

$\therefore a = 6$ and $b = 5$

$\Rightarrow x = 65, y = 56, m = 33$

$\therefore x + y + m = 65 + 56 + 33 = 154$

44. **(a)** $\frac{1}{1+\sqrt{2^{1947}}} + \frac{1}{2^{1947} + 2^{\frac{1947}{2}}} = \frac{1}{2^{\frac{1947}{2}}}$

Similarly & $\therefore$

$\sum_{n=0}^{1947} \frac{1}{2^4 + \sqrt{2^{1947}}} = \frac{947}{\sqrt{2^{1947}}} = \frac{487}{\sqrt{2^{1945}}}$

45. **(c)** $n + 2n + 3n + \ldots\ldots + 99n$

$n(1 + 2 + 3 + \ldots\ldots + 99)$

$n\frac{(99)(100)}{2} = 9 \times 25 \times 22 \times n$ is a perfect square when $n = 22$.

Number of digits in $n = 3$

46. **(a)** $Mn = \frac{1}{n}\sum_{r=1}^{n} a_i = \frac{1}{2}(2a + (n-1)d)$

$= a + \frac{n-1}{2}d$

$\sum M_n = 10a + \frac{d}{2}[-10] + \frac{d}{2}\left(\frac{10 \cdot 11}{2}\right)$

$= 20 - 20 + 110$

47. **(b)** $\overline{abc}$ is three digits numbers

$\Rightarrow \frac{b+c}{2} = bc \Rightarrow b + c = 2bc$

put $c = 0 \quad b + 0 = 0 \Rightarrow b = 0$

$c = 1 \quad b + 1 = 2b \Rightarrow b = 1$

$c = 2 \quad b + 2 = 4b \Rightarrow b = \frac{2}{3}$

only $c = 0, b = 0$

and $c = 1, b = 1$

and $a = 1,2,3,4,5,6,7,8,9$

$\Rightarrow 9 \times 2 \times 1$ If $\begin{bmatrix} b = 0 \to c = 0 \\ b = 1 \to c = 1 \end{bmatrix}$

$\Rightarrow$ Total $= 9 \times 2 \times 1 = 18$

48. **(d)** $\frac{ab^2 + ac^2 + bc^2 + ba^2 + ca^2 + cb^2}{6} \geq \left(a^6b^6c^6\right)^{1/6}$

So, $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) \geq 6abc$

49. **(a)** $T_r = \frac{r}{r^4 + 4} = \frac{r}{(r^2+2)^2 - 4r^2}$

$= \frac{r}{(r^2+2r+2)(r^2-2r+2)}$

$= \frac{1}{4}\left[\frac{1}{r^2-2r+2} - \frac{1}{r^2+2r+2}\right]$

$= \frac{1}{4}\left[\frac{1}{(r-1)^2+1} - \frac{1}{(r+1)^2+1}\right]$

$\Rightarrow S_n = \frac{1}{4}\left[1 + \frac{1}{2} - \frac{1}{n^2+1} - \frac{1}{(n+1)^2+1}\right]$

$\Rightarrow S_5 = \frac{1}{4}\left[1 + \frac{1}{2} - \frac{1}{26} - \frac{1}{37}\right]$

$= \frac{1}{4}\left[\frac{38 \times 37 - 26 \times 1}{26 \times 37}\right]$

$= \frac{1}{4} \times \frac{1380}{26 \times 37} = \frac{345}{26 \times 37}$

$\Rightarrow 37S_5 - \frac{7}{26} = \frac{338}{26} = 13$

50. **(c)** $S - S_n < \frac{1}{300}$

$\Rightarrow \frac{1}{1-\frac{1}{3}} - \frac{1-\frac{1}{3^n}}{1-\frac{1}{3}} < \frac{1}{300}$

$\Rightarrow \frac{3}{2}\left[1 - 1 + \frac{1}{3^n}\right] < \frac{1}{300}$

$\Rightarrow \frac{1}{3^n} < \frac{1}{300} \times \frac{2}{3} \Rightarrow \frac{1}{3^n} < \frac{1}{450} \Rightarrow 3^n > 450$

$\therefore$ Least value of n = 6

51. **(c)** We have

$b_k = \frac{n}{2}(a_k + a_{n+k-1}) = \frac{n}{2}$

$[a_1 + (k-1)d + a_1 + (n+k-2)d]$

$= \frac{n}{2}[a_n + a_1 + 2(k-1)d]$ (d is the common difference)

$\Rightarrow \sum_{k=1}^{n} b_k = \frac{n}{2}\left[na_n + na_1 + 2d\sum_{k=1}^{n}(k-1)\right]$

$= \frac{n}{2}[na_n + na_1 + d(n-1)n]$

$= \frac{n^2}{2}[a_n + a_1 + (n-1)d]$

$= \frac{n^2}{2}[a_n + a_n] = n^2 a_n$

Parikshit (JEE Advanced Level)

1. **(b)** $T_r = (-1)^r \frac{r^2 + r + 1}{r!}$

$= (-1)^r\left[\frac{r}{(r-1)!} + \frac{r}{(r-1)!} + \frac{1}{r!}\right]$

$= (-1)^r\left[\frac{1}{(r-2)!} + \frac{1}{(r-1)!} + \frac{1}{(r-1)!} + \frac{1}{r!}\right]$

$= \left[\frac{(-1)^r}{r!} + \frac{(-1)^r}{(r-1)!}\right] + \left[\frac{(-1)^r}{(r-1)!} + \frac{(-1)^r}{(r-2)!}\right]$

$= \left[\frac{(-1)^r}{r!} - \frac{(-1)^{r-1}}{(r-1)!}\right] + \left[\frac{(-1)^{r-1}}{(r-1)!} - \frac{(-1)^{r-2}}{(r-2)!}\right]$

$V(r) - V(r-1)$

$$\therefore \sum_{r=1}^{n} T_r = V(n) - V(0)$$

$$= \left[\frac{(-1)^n}{n!} - \frac{(-1)}{(n-1)!}\right] - 1$$

Therefore the sum of 20 terms is

$$\left[\frac{1}{20!} - \frac{-1}{19!}\right] - 1 = \frac{21}{20!} - 1$$

2. **(c)** $S_n = \dfrac{a(r^n - 1)}{r - 1}$

$$\Rightarrow U_n = \sum_{n=1}^{n} \frac{a(r^n - 1)}{r - 1} = \frac{a}{r-1}\sum_{n=1}^{n}(r^n - 1)$$

$$\Rightarrow U_n = \frac{a}{r-1} \qquad \{r + r^2 + \cdots + r^n - n\}$$

$$\Rightarrow = \frac{a}{r-1}\left\{\frac{r(r^n - 1)}{r - 1} - n\right\}$$

$$\Rightarrow (r-1)U_n = \frac{ar(r^n - 1)}{r-1} - an$$

$$\Rightarrow (r-1)\, U_n = rS_n - an$$

$$\Rightarrow rS_n + (1-r)\, U_n = an$$

3. **(b)** We have,

$$(OM_{n-1})^2 = (OP_n)^2 + (P_nM_{n-1})^2$$

$$= 2(OP_n)^2 = 2\alpha_n^2 \ (say)$$

Also, $(OP_{n-1})^2 = (OM_{n-1})^2 + (P_{n-1}M_{n-1})^2$

$$\Rightarrow \alpha_{n-1}^2 = 2\alpha_n^2 + \frac{1}{2}\alpha_{n-1}^2$$

$$\Rightarrow \alpha_n = \frac{1}{2}\alpha_{n-1}$$

$$\Rightarrow OP_n = \alpha_n = \frac{1}{2}\alpha_{n-1} = \frac{1}{2^2}\alpha_{n-1} = \cdots$$

$$= \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$$

4. **(a)** x, y, z are in G.P.

$\Leftrightarrow y^2 = xz$

$\Leftrightarrow x$ is a factor of y (not possible)

Taking $x = 3, y = 5, z = 7$, we have x, y, z are in A.P. Thus x, y, z may be in A.P. but not in G.P.

5. **(b)** For option (c) if $x = z = 1$ and $y = 2$ then option (c) is right. So by option (c) we can't say $x = y = z$. Remaining options implies $x = y = z$.

6. **(a)** For every, $a_i \in R$

$a_i.2^{a_i} > a_i$ and $a_i \cdot 2^{-a_i} < a_i$

$$\therefore \sum_{i=1}^{100} a_i 2^{a_i} > \sum_{i=1}^{100} a_i \ \& \sum_{i=1}^{100} a_i \cdot 2^{-a_i} < \sum_{i=1}^{100} a_i$$

$$\Rightarrow \sum_{i=1}^{100} a_i 2^{a_i} > 0 \ \& \sum_{i=1}^{100} a_i \cdot 2^{-a_i} < 0 \left(\sum_{i=1}^{100} a_i = 0, \text{ given}\right)$$

7. **(b)** $$\sum_{1 \le i < j \le m} a_i a_j = \frac{1}{2}\left\{\left(\sum_{i=1}^{m} a_i\right)^2 - \sum_{i=1}^{m} a_i^2\right\}$$

$$= \frac{1}{2}\left\{\left(\frac{a_1(1-r^m)}{1-r}\right)^2 - \frac{a_1^2(1-r^{2m})}{1-r^2}\right\}$$

$$= \frac{a_1^2}{2(1-r)}\left\{\frac{(1-r^m)^2}{1-r} - \frac{1-r^{2m}}{1+r}\right\}$$

$$= \frac{r}{1+r}\left\{\frac{a_1(1-r^{m-1})}{1-r}\right\}\left\{\frac{a_1(1-r^m)}{1-r}\right\}$$

$$= \frac{r}{1+r} S_{m-1} S_m$$

8. **(c)** In ΔABC

$b + c - a > 0, c + a - b > 0, a + b - c > 0,$

$$\frac{(b+c-a)+(c+a-b)+(a+b-c)}{3}$$

$$\ge \{(b+c-a)(c+a-b)(a+b-c)\}^{1/3}$$

$$\Rightarrow P^3 \ge 27(b+c-a)(c+a-b)(a+b-c)$$

$$\frac{(a+b+c)+(b+c-a)+(c+a-b)+(a+b-c)}{4} >$$

$$\{(a+b+c)(b+c-a)(c+a-b)(a+b-c)\}^{1/4}$$

$$\Rightarrow \frac{2P}{4} > (16A)^{1/4} \Rightarrow P > 4A^{1/4}$$

or $P^4 > 256A$

Next, we know for a given parameter, equilateral triangle has the largest area, so the area of triangle

$$A \le \frac{\sqrt{3}}{4}\left(\frac{p}{3}\right)^2 \text{ (for equilateral triangle, } a = b = c = p/3)$$

Now $p^2 = (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$$= 3(a^2 + b^2 + c^2) - 2(a^2 + b^2 + c^2 - ab - bc - ca)$$

$\le 3(a^2 + b^2 + c^2)$ (equality holds iff $a = b = c$)

Thus

$$A \le \frac{\sqrt{3}}{4}\frac{a^2 + b^2 + c^2}{3} \Rightarrow a^2 + b^2 + c^2 \ge 4\sqrt{3}A$$

9. **(c)** Let, the side of square base be 'a' and height of box be 'b', where $a, b \in I^+$

Given, Sum of Edges = Sum of areas of Six faces

i.e. $4a + 4a + 4b = 2(a^2 + ab + ab)$

$a^2 - 4a + 2b(a-1) = 0$

$2b = (a^2 - 4a)/(1-a) < 0$

10. **(a,b)** $2b = a + c$

$4b^2 = a^2 + c^2 + 2ac$

$$\Rightarrow a^2 + c^2 = 4b^2 - 2ab \ \& \ b^2 = \frac{2a^2c^2}{a^2 + c^2}$$

$\Rightarrow b^2(4b^2 - 2ac) = 2a^2c^2$

$\Rightarrow b^2(2b^2 - ac) = a^2c^2$

$\Rightarrow 2b^4 - b^2(ac) - (ac)^2 = 0$

$\Rightarrow (b^2 - ac)(2b^2 + ac) = 0$

$\Rightarrow b^2 = ac \Rightarrow a, b, c$ in G.P.

& a, b, c in A.P. $\Rightarrow a = b = c$

or $2b^2 + ac = 0 \Rightarrow b^2 = -\dfrac{ac}{2}$

$$\Rightarrow a, b, \frac{-c}{2} \text{ in G.P.}$$

11. **(a,b,c)** Given, $x = y + 2, a = 5z$

a, x, b in A.P $\Rightarrow 2x = a + b$

a, y, b in G.P $\Rightarrow y^2 = ab$

$$a, z, b \text{ in H.P} \Rightarrow z = \frac{2ab}{a+b}$$

$$\Rightarrow z = \frac{2y^2}{2x} \Rightarrow y^2 = zx$$

A.M. > G.M. > H.M.

$x > y > z$

$$\frac{a}{5} = \frac{2ab}{a+b}$$

$a^2 + ab = 10ab \Rightarrow a(a - 9b) = 0$

$\Rightarrow a \ne 0$ or $a = 9b$

$y = x - 2$

$$\therefore y = \frac{2x-4}{2} \Rightarrow y^2 = \frac{(a+b-4)^2}{4}$$

$\Rightarrow a^2 + b^2 - 2ab - 8a - 8b + 16 = 0$

$\Rightarrow 4b^2 - 5b + 1 = 0$

$\Rightarrow (b-1)(4b-1) = 0$

$\Rightarrow b = 1, a = 9$ or $b = 1/4, a = 9/4$

12. **(a,b,c,d)**

$$\sum_{r=1}^{n} r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$$

$$\sum_{r=1}^{n}(r^2 + r)(2r+3) = \sum_{r=1}^{n}(2r^3 + 5r^2 + 3r)$$

$$= 2 \cdot \frac{n^2(n+1)^2}{4} + 5\frac{n(n+1)(2n+1)}{6} + 3\frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2}\left[n(n+1) + \frac{5}{3}(2n+1) + 3\right]$$

$$= \frac{n(n+1)}{2}\left[\frac{6(n^2+n) + 10(2n+1) + 18}{6}\right]$$

$$= \frac{n(n+1)}{12}[6n^2 + 26n + 28]$$

$$= \frac{1}{12}[6n^4 + 26n^3 + 28n^2 + 6n^3 + 26n^2 + 28n]$$

$$= \frac{1}{12}[6n^4 + 32n^3 + 54n^2 + 28n]$$

$$a = \frac{6}{12}, b = \frac{32}{12}, c = \frac{54}{12}, d = \frac{28}{12} = \frac{7}{3}, e = 0$$

so $a + c = b + d$

$$b - \frac{2}{3} = \frac{32}{12} - \frac{2}{3} = \frac{24}{12}$$

$$c - 1 = \frac{42}{12}$$

so $a, b - 2/3, c - 1$ are in A.P

$$\& \frac{c}{a} = \frac{54}{6} = 9 \text{ is an integer}$$

13. **(a,c)** $\sum_{r=1}^{n} \frac{1}{\sqrt{a+rx} + \sqrt{a+(r-1)x}}$

$$= \sum_{1}^{n} \frac{\sqrt{a+rx} - \sqrt{a+(r-1)x}}{x}$$

$$\frac{1}{x} = \sum_{1}^{n} (\sqrt{a+rx} - \sqrt{a+(r-1)x})$$

$$= \frac{1}{x}[\sqrt{a+nx} - \sqrt{a}]$$

Upon rationalizing

$$= \frac{1}{x} \frac{a+nx-a}{\sqrt{a+nx} + \sqrt{a}} = \frac{n}{\sqrt{a+nx} + \sqrt{a}}$$

Upon rationalizing

$$= \frac{1}{x} \frac{a+nx-a}{\sqrt{a+nx} + \sqrt{a}} = \frac{n}{\sqrt{a+nx} + \sqrt{a}}$$

14. **(a,d)**

$$S = 1 + \frac{1}{(1+3)}(1+2)^2 + \frac{1}{(1+3+5)}(1+2+3)^2 + \frac{1}{(1+3+5+7)}(1+2+3+4)^2 + \cdots$$

$$T_r = \frac{1}{r^2}(1+2+\cdots+r)^2$$

$$= \frac{1}{r^2}\left\{\frac{r(r+1)}{2}\right\}^2$$

$$= \frac{r^2+2r+1}{4}$$

$$\therefore \quad T_7 = 16 \text{ and } S_{10} = \sum_{r=1}^{10} T_r$$

$$= \frac{1}{4}\left\{\frac{(10)(10+1)(20+1)}{6} + (10)(10+1) + 10\right\}$$

$$= \frac{505}{4}$$

15. **(c)** $G_1 G_2 \ldots G_3 = (\sqrt{1 \times 1024})^n = 2^n$

Given, $2^{5n} = 2^{45} \Rightarrow n = 9$

Hence,

$$r = (1024)^{\frac{1}{9+1}} = 2$$

$= 3\,G(=2, r=2)$

$$\Rightarrow G_1 + G_2 + \ldots + G_9 = \frac{2 \times (2^9 - 1)}{2 - 1}$$

$$= 1024 - 2 = 1022$$

16. **(b)** $A_1 + A_2 + A_3 + \ldots + A_{m-1} + 4 = 1025 \times 771$

$$m\left(\frac{-2 + 1027}{2}\right) = 1025 \times 171$$

$$\Rightarrow m = 342$$

17. **(a)** We have, $A_{171} + A_{172}$

$$= -2 + 1027 = 1025$$

$$A_{171} + A_{172} = -2 + 1027 = 1025$$

$$\Rightarrow \frac{2A_{171} + 2A_{172}}{2} = 1025 \times 171$$

Also,

$$G_5 = 1 \times 2^5 = 32$$

$$\Rightarrow G_5^2 = 1024 \Rightarrow G_5^2 = 1024$$

$$\Rightarrow G_5^2 + 1 = 1025 \text{ are in A.P..}$$

18. **(d)**

19. **(c)**

20. **(b)**

Sol. **(18 to 20)**

We have, $a + b + c = 25$ ...(i)

$2a = b + 2$ ..(ii)

$c^2 = 18b$ ...(iii)

Eliminating a from (i) and (ii), we have

$$b = 16 - \frac{2c}{3}$$

Then from (iii),

$$c^2 = 18\left(16 - \frac{2c}{3}\right)$$

$$\Rightarrow c^2 + 12c - 18 \times 16 = 0$$

$$\Rightarrow (c - 12)(c + 24) = 0$$

Now, $c = -24$ is not possible since it does not lie between 2 and 18. Hence, $c = 12$. Then from (iii), $b = 8$ and finally from (ii), $a = 5$.

Thus, $a = 5$, $b = 8$ and $c = 12$. Hence, $abc = 5 \times 8 \times 12 = 480$.

Also, equation $ax^2 + bx + c = 0$ is $5x^2 + 8x + 12 = 0$, which has imaginary roots.

If a, b, c are roots of the equation $x^3 + qx^2 + rx + s = 0$, then sum of product of roots taken two at a time is $r = 5 \times 8 + 5 \times 12 + 8 \times 12 = 196$.

21. **(b)** $\because b > \sqrt{ac}$ and $c > \sqrt{bd}$

(∵ A.M. > G.M.)

$$\Rightarrow bc > \sqrt{abcd} \Rightarrow bc > ad$$

$$\because b > \frac{2ac}{a+c} \text{ and } c > \frac{2bd}{b+d}$$

(∵ A.M. > HM.)

$$\Rightarrow bc > \frac{4abcd}{(a+c)(b+d)}$$

$$\Rightarrow (a+c)(b+d) > 4ad$$

$$\Rightarrow ab + bc + cd > 3ad$$

22. **(a)** $\because \frac{a+c}{2} > b$ and $\frac{b+d}{2} > c$

(∵ A.M. > G.M.)

$$\Rightarrow a + c > 2b \text{ and } b + d > 2c$$

23. **(c)** $\because \frac{ab+cd}{2} > bc$ (∵ A.M. > G.M.)

$$\Rightarrow (ab + cd) > 2bc$$

$$\because bc > \frac{2(ab)(cd)}{(ab+cd)} \quad (\because GM. > HM.)$$

$$\Rightarrow (ab + cd) > 2ad$$

24. **(b)** A → (p); B → (p); C → (q); D → (q)

(A) $2\log_5(2^x - 5) = \log_5 2 + \log_5 (2^x - 7/2)$

$$(2^x - 5)^2 = 2\left(2^x - \frac{7}{2}\right)$$

$t^2 - 10t + 25 = 2t - 7$ {put $2^x = t$}

$t^2 - 12t + 32 = 0$

$\therefore t = 8, 4$

$\therefore 2^x = 4$ or $2^x = 8$

$\therefore x = 2, 3$ $2^x - 5 > 0$

$\therefore 2x = 6 \quad \therefore 2^x > 5$

so only solution $x = 3$

(B) $$\frac{S_{2n}}{S_n} = \frac{\frac{2n}{2}[2a + (2n-1)d]}{\frac{n}{2}[2a + (n-1)d]} = 3$$

$$= \frac{2a + (2n-1)d}{2a + (n-1)d} = \frac{3}{2}$$

$$\therefore d = \frac{2a}{n+1}$$

$$\text{Now } \frac{S_{3n}}{S_n} = \frac{\frac{3n}{2}[2a + (3n-1)d]}{\frac{n}{2}[2a + (n-1)d]}$$

$$= \frac{3\left[2a + (3n-1)\frac{2a}{n+1}\right]}{\left[2a + (n-1)\frac{2a}{n+1}\right]}$$

$$= \frac{3[(n+1) + (3n-1)]}{(n+1) + (n-1)}$$

$$= \frac{3.4n}{2n} = 6$$

(C) $$S = 4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^3} + \ldots \frac{4n}{3^{n-1}} \quad \ldots(i)$$

$$\frac{S}{3} = \frac{4}{3} + \frac{8}{3^2} + \frac{12}{3^3} + \ldots + \frac{4m}{3^n} \quad \ldots(ii)$$

(i) – (ii) we get

$$\frac{2}{3}S = 4 + \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \ldots + \frac{4}{3^{n-1}} + \frac{4n}{3^n}$$

$$= 4\left(\frac{1 - \left(\frac{1}{3}\right)^n}{1 - \frac{1}{3}}\right) - \frac{4n}{3^n}$$

$$S=\frac{4\cdot3}{2}\cdot\frac{3}{2}\left(1-\left(\frac{1}{3}\right)^n\right)-\frac{4n}{3^n}\cdot\frac{3}{2}$$

$\therefore S\infty = 9$

(D) $\ell, \ell r, \ell r^2$

$\therefore \ell^3 r^3 = 27$ (volume)

$\ell r = 3$

surface area

$2(\ell \cdot \ell r + \ell r \cdot \ell r^2 + \ell r^2 \cdot \ell) = 78$

$\ell^2(r + r^2 + r^3) = 39$

$$\Rightarrow \ell^2\left(\frac{3}{\ell}+\frac{3^2}{\ell^2}+\frac{3^3}{\ell^3}\right)=39$$

$$3\ell + 3^2 + \frac{3^3}{\ell} = 39 \Rightarrow \ell + 3 + \frac{9}{\ell} = 13$$

$\therefore \ell^2 - 10\ell + 9 = 0 \Rightarrow \ell = 1, 9$

$\Rightarrow \ell r = 3$ and $\ell > \ell r$

$\therefore r = \frac{1}{3} \quad \therefore \ell = 9$

25. (*b*) A → (q), B. → r; C → p; D → (r).

(A) *a, b, c* are in G.P. Hence, $b^2 = ac$

$\Rightarrow 2\log_{10} b = \log_{10} a + \log_{10} c$

$$\Rightarrow \frac{2}{\log_b 10} = \frac{1}{\log_a 10} + \frac{1}{\log_c 10}$$

$$\Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

Hence, *x, y, z* are in H.P.

(B) $\frac{a-be^x}{a} = \frac{b-ce^x}{b} = \frac{c-de^x}{c}$

$$\Rightarrow \frac{2a}{a-be^x} - 1 = \frac{2b}{b-ce^x} - 1 = \frac{2c}{c-de^x} - 1$$

$$\Rightarrow 1 - \frac{b}{a}e^x = 1 - \frac{c}{b}e^x = 1 - \frac{d}{e}e^x$$

$$\Rightarrow \frac{b}{a} = \frac{c}{b} = \frac{d}{c}$$

Hence, *a, b, c, d* are in G.P.

(C) Given, $2b = a + c, x^2 = ab, y^2 = bc$.

Now, $x^2 + y^2 = b(a + c) = b\,2b = 2b^2$

$\Rightarrow x^2 + y^2 = 2b^2$

Hence, x^2, b^2, y^2 are in A.P.

(D) $x \log a - y \log b = z \log c = k$ (say).

Also, $y^2 = xz$

$$\frac{k^2}{(\log b)^2} = \frac{k^2}{\log a \log c}$$

Hence, log *a*, log *b*, log *c* are in G.P.

26. (*b*) A → p, q, r, s; B → r, s; C → p, q; D → r, s.

(A) $\Sigma n = 210$

$\Rightarrow n(n + 1) = 420$

$\Rightarrow (n - 20)(n + 21) = 0$

$\Rightarrow n = 20$

Hence, $\Sigma n^2 = \frac{n}{6}(n+1)(2n+1)$

$$= \frac{20}{6}(21)(41)$$

$= (10)(7)(41)$

Hence, the greatest prime number by which Σn^2 is divisible is 41.

(B) $4, G_1 - G_2 \ldots\ldots G_{n+1} \ldots\ldots G_{2n} G_{2n+1}$ 2916

G_{n+1} will be the middle mean of $(2n + 1)$ odd means and it will be equidistant from the first and last terms. Hence, $4, G_{n+1}$, 2916 will also be in G.P. So,

$\Rightarrow G_{n+1} = 4 \times 2916$

$= 4 \times 9 \times 324$

$= 4 \times 9 \times 4 \times 81$

$\Rightarrow G_{n+1} = 2 \times 3 \times 2 \times 9 = 108$

Hence, the greatest odd number by which G_{n+1} is divisible is 27.

(C) Terms are 40, 30, 24, 20. Now,

$$\frac{1}{30} - \frac{1}{40} = \frac{1}{120}$$

$$\Rightarrow \frac{1}{24} - \frac{1}{30} = \frac{6}{24\times30} = \frac{1}{120}$$

and $\frac{1}{20} - \frac{1}{24} = \frac{4}{20\times24} = \frac{1}{120}$

Hence, 1/30. 1/24. 1/20 are in A.P. with common difference $d = 1/120$. Hence, the next term is 1/20 + 1/120 = 7/120. Therefore, the next term of given series is $\frac{120}{7} = 17\frac{1}{7}$ is . Hence, the integral part of $17\frac{1}{7}$ is 17.

(D) $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \cdots\infty$

$$\Rightarrow \frac{1}{5}S = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + \cdots$$

$$\Rightarrow S\left(1-\frac{1}{5}\right) = 1 + 3\left[\frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \cdots\infty\right]$$

$$\Rightarrow S\left(1-\frac{1}{5}\right) = 1 + 3\left[\frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \cdots\infty\right]$$

$$\Rightarrow \frac{4}{5}S = 1 + 3\left[\frac{1/5}{1-1/5}\right] = 1 + \frac{3}{4} = \frac{7}{4}$$

$$\Rightarrow S = \frac{35}{16}$$

$\Rightarrow a = 35$ and $b = 16$

$\Rightarrow a - b = 19$

27. [1] $T_2 = 3 + d, T_{10} = 3 + 9d, T_{34} = 3 + 33d$

since T_2, T_{10}, T_{34} are in G.P

$T_{10}^2 = T_2 T_{34}$

$\Rightarrow (3 + 9d)^2 = (3 + d)(3 + 33d)$

$\Rightarrow d = 0, 1$

hence $d = 1$

28. [0] $10x^3 - nx^2 - 54x - 27 = 0$ has roots in H.P.

put $x = 1/t$

$27t^3 + 54t^2 + nt - 10 = 0$

This equation has roots in A.P., let the roots are $a - b, a$ and $a + d$

$\therefore 3a = -\frac{54}{27} \Rightarrow a = -\frac{2}{3}$

Also $(a-d)a(a+d) = \frac{10}{27}$

$$\therefore \frac{2}{3}\left(\frac{4}{9} - d^2\right) = -\frac{10}{27} \Rightarrow \left(\frac{4}{9} - d^2\right) = -\frac{5}{9}$$

$\therefore d^2 = 1 \Rightarrow d = \pm 1$

For $d = 1$, roots are $-\frac{2}{3} + 1, -\frac{2}{3}, -\frac{2}{3} - 1$

$$\Rightarrow \frac{1}{3}, -\frac{2}{3}, -\frac{5}{3}$$

for

$$\therefore \frac{n}{27} = \frac{10}{9} - \frac{5}{9} - \frac{2}{9}\,\frac{n}{27} = \frac{3}{9} \Rightarrow n = 9$$

29. [6]

$$S = 3 + \sum_{n=1}^{\infty}\frac{2n+3}{3^n} = 3 + \underbrace{\sum_{n=1}^{\infty}\frac{1}{3^{n-1}}}_{S_1} + \underbrace{\sum_{n=1}^{\infty}\frac{2n}{3^n}}_{S_2}$$

$$S_1 = \sum_{n=1}^{\infty}\frac{1}{3^{n-1}} = 1 + \frac{1}{3} + \frac{1}{3^2} + \cdots\infty$$

$$\therefore S_1 = \frac{1}{1-(1/3)} = \frac{3}{2}$$

$$S_2 = \sum_{n=1}^{\infty}\frac{2n}{3^n} = \frac{2}{3} + \frac{4}{3^2} + \frac{6}{3^3} + \frac{8}{3^4} + \cdots\infty$$

$$S_2 = \frac{2}{3} + \frac{4}{3^2} + \frac{6}{3^3} + \frac{8}{3^4} + \cdots\infty$$

Now, $\frac{S_2}{3} = + \frac{2}{3^2} + \frac{4}{3^3} + \frac{6}{3^4} + \cdots\infty$

$$\frac{2S_2}{3} = \frac{2}{3}\left[1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots\infty\right]$$

[on subtracting]

$$\therefore S_2 = \frac{1}{1-(1/3)} = \frac{3}{2}$$

Hence $S = 3 + \frac{3}{2} + \frac{3}{2} = 6$

30. [7] $ax^2 + (a + d)x + (a + 2d) = 0$

$a, a + d, a + 2d$ are in increasing A.P. $(d > 0)$

for real roots $D \geq 10$

$\Rightarrow (a + d)^2 - 4a(a + 2d) > 0$

$\Rightarrow (d - 3a)^2 - 6ad \geq 0$

$\Rightarrow (d - 3d)^2 - 12a^2 \geq 0$

$\Rightarrow \left(d - 3a - \sqrt{12}a\right)\left(d - 3a + \sqrt{12}a\right) \geq 0$

$$\Rightarrow \left[\frac{d}{a} - \left(3 + 2\sqrt{3}\right)\right]\left[\frac{d}{a} - \left(3 - 2\sqrt{3}\right)\right] \geq 0$$

$$\Rightarrow \therefore \left.\frac{d}{a}\right|_{\text{Min}} = 3 + 2\sqrt{3}$$

$\Rightarrow$ least integral value = 7

PYQ's (Past Year Questions)

1. **(a)** Let $x^{p^2} = y^{qr} = z^{p^2r} = K$

$pq^2 = \log_x K$

$qr = \log_y K$

$p^2r = \log_z K$

$$\log_y x = \frac{qr}{pq^2} = \frac{r}{pq} \quad ...(i)$$

$$\log_x z = \frac{pq^2}{p^2r} = \frac{q^2}{pr} \quad ...(ii)$$

$$\log_z y = \frac{p^2r}{qr} = \frac{p^2}{q} \quad ...(iii)$$

Now, $3, \frac{3r}{pq}, \frac{3p^2}{q}, \frac{7q^2}{pr}$ in A.P

Common difference $= \frac{3r}{pq} - 3 = \frac{1}{2}$

$$\Rightarrow r = \frac{7}{6}pq \quad ...(iv)$$

$r = pq + 1$ [Given]

$\Rightarrow pq = 6$ ...(v)

So, $r = 7$ $[pq + 1 = r]$ ...(vi)

$$\frac{3p^2}{q} = 3 + 2 \times \frac{1}{2} = 4 \quad ...(vii)$$

After solving eqn (v) and eqn (vii) $p = 2$ and $q = 3$

Hence, $r - p - q = 2$

2. **(c)** $x = \sum_{n=0}^{\infty} a^n = \frac{1}{1-a}; y = \sum_{n=0}^{\infty} b^n$

$$= \frac{1}{1-b}; \sum_{n=0}^{\infty} c^n = \frac{1}{1-c}$$

Now,

$a, b, c \rightarrow AP$

$1-a, 1-b, 1-c \rightarrow$ AP

$$\frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c} \rightarrow HP$$

$1/x$, $1/y$, $1/z$ are in AP.

3. **[1000]**

Since, required number has G..D with 18 as 3. It must be odd multiple of '3' but not a multiple of '9'.

(*i*) Now, 4-digit number which are odd multiple of '3' are,

1005, 1011, 1017,..... 9999 $\rightarrow$ 1499

(*ii*) 4-digit number which are odd multiple of 9 are,

1017, 1035, ... , 9999 $\rightarrow$ 499

$\therefore$ Required numbers = 1499 – 499 = 1000

4. **(c)** $T_{10} = \frac{1}{20} = a + 9d$ ... (*i*)

$T_{20} = \frac{1}{10} = a + 19d$ ... (*ii*)

$$\Rightarrow a = \frac{1}{200}, d = \frac{1}{200}$$

$$\Rightarrow S_{200} = \frac{200}{2}\left[\frac{2}{200} + \frac{199}{200}\right] = \frac{201}{2} = 100\frac{1}{2}$$

5. **[1]** Given $2(a_1 + a_2 + ... a_n)$

$= b_1 + b_2 + ... b_n$

$$\Rightarrow 2 \times \frac{n}{2}\left(2c + (n-2)x_2\right) = c\left(\frac{2^n - 1}{2-1}\right)$$

$\Rightarrow 2n^2 - 2n = c(2^n - 1 - 2n)$

$$\Rightarrow c = \frac{2n^2 - 2n}{2^n - 1 - 2n} \in N$$

So, $2n^2 - 2n \geq 2^n - 1 - 2n$

$\Rightarrow 2n^2 + 1 \geq 2^n \Rightarrow n < 7$

$\Rightarrow n$ can be 1, 2, 3 ....,

Checking c against these values of n we get $c = 12$ (when $n = 3$)

Hence number of such $c = 1$

6. **(d)** Two digit numbers of the from $7\lambda + 2$ are 16, 23 ..., 93

Two digit numbers of the from $7\lambda + 5$ are 12, 19 ..., 96

All numbers are in A.P.

So, sum of all the above number equals to

$$= \frac{12}{2}(16 + 93) + \frac{13}{2}(12 + 96)$$

$= 654 + 702 = 1356$

$$\left[\because S_n = \frac{n}{2}\left[2a + (n-1)d\right] = \frac{n}{2}(a + l)\right]$$

7. **[157]** We equate the general terms of three respective

A.P.'s as $1 + 3a = 2 + 5b = 3 + 7c$

$\Rightarrow 3$ divides $1 + 2b$ and 5 divides $1 + 2c$

$\Rightarrow 1 + 2c = 5, 15, 25$ etc.

So, first such terms are possible when $1 + 2c = 15$ i.e. $c = 7$

Hence, first term $= a = 52$

$d = \text{LCM}(3, 5, 7) = 105$

$\Rightarrow a + d = 157$

8. **(a)** Let a is first term and common difference $= d$

$a_7 = a + 6d = 3$, ...(*i*)

and product $P = a_1a_4$

$P = a(a + 3d)$

$= (3 - 6d)(3 - 3d)$

$P = 18d^2 - 27d + 9$

Differentiating with respect to d

$\Rightarrow 36d - 27 = 0$

$\Rightarrow d = \frac{3}{4}$, from(*i*) $a = \frac{-3}{2}$, (*P*is minimum at $d = \frac{3}{4}$)

Now, $S_n = \frac{n}{2}\left(-3 + (n-1)\frac{3}{4}\right) = 0$

$+ 3 = (n+)\frac{3}{4} \Rightarrow n - 1 = 4$

$\Rightarrow n = 5$

Now,

$n! - 4a_{n(n+2)} = 120 - 4(a_{35})$

$= 120 - 4(a + (35 - 1)d)$

$$= 120 - 4\left(\frac{-3}{2} + 34 \cdot \left(\frac{3}{4}\right)\right)$$

$$= 120 - 4\left(\frac{-6 + 102}{4}\right)$$

$= 120 - 96 = 24$

9. **[6993]**

Number of terms in 11th set 1, 3, 5, 7 ...

$a_{11} = 1 + (10)(2) = 21$

first term of 11th set will be last term of 10th set +3

So Sum of term upto 10th set

$$S_{10} = \frac{10}{2}\left[2 \times 1 + 9 \times 2\right] = 100$$

So, last term of 10th set is 300.

Therefore, 11th set first term = 303

Then sum of all terms of 11th set

$$S = \frac{21}{2}\left[2 \times 303 + 20 \times 3\right] = 6993$$

10. **(d)** Given,

$\log_{\frac{1}{9^1}} x + \log_{\frac{1}{9^2}} x + \log_{\frac{1}{9^3}} x + ...$

$+ \log_{\frac{1}{9^{22}}} x = 504$

$\Rightarrow 2\log_9 x + 3\log_9 x + 4\log_9 x + ...$

$+ 22\log_9 x = 504$

$\Rightarrow \log_9 x + 2\log_9 x + 3\log_9 x + 4\log_9 x +$

$... + 22\log_9 x - \log_9 x = 504$

$\Rightarrow (1 + 2 + 3 ... + ... + 22)\log_9 x + \log_9 x =$

$504 \Rightarrow x = 81$

11. **(a)** $S = \sum_{i=1}^{30} a_i, T = \sum_{i=1}^{15} a_{2i-1}$,

$a_5 = 27, S - 2T = 75$

Let $a_i = a + (i - 1)D$

$S = a_1 + a_2 + a_3 + ... + a_{30}$

$T = a_1 + a_3 + a_5 + ... + a_{29}$

$\therefore 2T = 2a_1 + 2a_3 + 2a_5 + ... + 2a_{29}$

$S - 2T = (a_2 - a_1) + (a_4 - a_3) + (a_6 - a_5) +$

$... + (a_{30} - a_{29})$

$= 15\,D$

But $S - 2T = 75 \Rightarrow 15D = 75$

$\Rightarrow D = 5$

Now $a_5 = 27 \Rightarrow a + 4D = 27$

$\therefore a = 27 - 20 \Rightarrow a = 7$

Now $a_{10} = a + 9D$

$= 7 + 45 = 52$

12. (*d*) Given, $ar^5 + ar^7 = 2$

$(ar^2)(ar^4) = \frac{1}{9}$

$a^2r^6 = \frac{1}{9}$

Now, $r > 0$

$ar^5(1 + r^2) = 2$

Now, $ar^3 = \frac{1}{3}$ or $-\frac{1}{3}$ (rejected)

$r^2 = 2 \Rightarrow r = \sqrt{2}$

$\therefore a = \frac{1}{6\sqrt{2}}$

Now, $6(a_2 + a_4)(a_4 + a_6) = 6(ar + ar^3)(ar^3 + ar^5)$

$= 6a^2r^4(1 + r^2)$

$= 6\left(\frac{1}{36.2}\right)(4)(9) = 3$

13. [4] $S = 1 + 3 + 3^2 + 3^3 + ... + 3^{2021}$

$= \frac{3^{2022} - 1}{2} = \frac{1}{2}\left[a^{1011} - 1\right]$

$= \frac{1}{2}\left[9^{1011} - 1\right] = \frac{1}{2}[100k + 10110 - 1 - 1]$

$= 50k_1 + 4$

14. (*b*) $T_n = \frac{1}{4n(n+1)} = \frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right)$

$S = \sum_{n=1}^{100} T_n = \frac{1}{4}\left(1 - \frac{1}{101}\right) = \frac{1}{4}\left(\frac{100}{101}\right) = \frac{25}{101}$

15. (*a*) Use $1 + r + r^2 + ... \infty = \frac{1}{1-r}, |r| < 1$

$x = 1 - \tan^2\theta + \tan^4\theta + ... \infty$

$y = 1 + \cos^2\theta + \cos^4\theta + ... \infty$

$x = \frac{1}{1 + \tan^2\theta} = \cos^2\theta$

and $y = \frac{1}{1 - \cos^2\theta} = \frac{1}{\sin^2\theta}$

So, $x + \frac{1}{y} = 1$ $[\because \sin^2\theta + \cos^2\theta = 1]$

$\Rightarrow (1 - x)y = 1$

16. (*d*) Let the numbers be $\frac{a}{r}, a, ar$

Given $a^3 = 512 \Rightarrow a = 8$

Now given $\frac{8}{r} + 4, 12, 8r$ are in A.P.

$\Rightarrow 24 = 8\left(r + \frac{1}{r} + \frac{1}{2}\right)$

$\Rightarrow \frac{5}{2} = \frac{r^2 + 1}{r}$

$\Rightarrow 2r^2 - 5r + 2 = 0$

$\Rightarrow r = \frac{1}{2}$ or 2

Numbers are 4, 8, 16 or 16, 8, 4

Sum of numbers = 4 + 8 + 16 = 28

17. [2] $10 = 1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + ...$ upto ∞

$\Rightarrow 9 = \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + ...$ upto ∞ ...(i)

Divide both sides by k

$\frac{9}{k} = \frac{4}{k^2} + \frac{8}{k^3} + \frac{13}{k^4} + ...$ upto ∞ ...(ii)

On subtracting eqn (ii) from eqn (i), we get

$S = 9\left(1 - \frac{1}{k}\right) = \frac{4}{k} + \frac{4}{k^2} + \frac{5}{k^3} + \frac{6}{k^4} + ...$

upto ∞ ...(iii)

Again divide by k

$\frac{S}{k} = \frac{4}{k^2} + \frac{4}{k^3} + \frac{5}{k^4} + ...$ upto ∞ ...(iv)

On subtracting eqn (iv) from eqn (iii), we get

$\left(1 - \frac{1}{k}\right)S = \frac{4}{k} + \frac{1}{k^3} + \frac{1}{k^4} + \frac{1}{k^5} + ... \infty$

$9\left(1 - \frac{1}{k}\right)^2 = \frac{4}{k} + \frac{\frac{1}{k^3}}{\left(1 - \frac{1}{k}\right)}$

$\left[\because S = 9\left(1 - \frac{1}{k}\right)\right]$

$\Rightarrow 9(k-1)^3 = 4k(k-1) + 1$

$\Rightarrow k = 2$

18. (*c*) Given, $A = \sum_{n=1}^{\infty} \frac{1}{\left(3 + (-1)^n\right)^n}$ and

$B = \sum_{n=1}^{\infty} \frac{(-1)^n}{\left(3 + (-1)^n\right)^n}$

$A = \frac{1}{2} + \frac{1}{4^2} + \frac{1}{2^3} + \frac{1}{4^4} + ...$

$B = \frac{-1}{2} + \frac{1}{4^2} - \frac{1}{2^3} + \frac{1}{4^4} + ...$

$A = \frac{\frac{1}{2}}{1 - \frac{1}{4}} + \frac{\frac{1}{16}}{1 - \frac{1}{16}}$, and $B = \frac{-\frac{1}{2}}{1 - \frac{1}{4}} + \frac{\frac{1}{16}}{1 - \frac{1}{16}}$

$A = \frac{11}{15}, B = \frac{-9}{15}$

$\therefore \frac{A}{B} = \frac{-11}{9}$

19. (*d*) $a, ar, ar^2, ...$

$T_2 + T_6 = \frac{25}{2} \Rightarrow ar\left(1 + r^4\right) = \frac{25}{2}$

$a^2r^2\left(1 + r^4\right)^2 = \frac{625}{4}$ ... (1)

$T_3 \cdot T_5 = 25 \Rightarrow (ar^2)(ar^4) = 25$

$a^2r^6 = 25.$ ..(2)

On dividing by (2)

$\frac{\left(1 + r^4\right)^2}{r^4} = \frac{25}{4} \Rightarrow 4r^8 - 17r^4 + 4 = 0$

$(4r^4 - 1)(r^4 - 4) = 0$

$r^4 = \frac{1}{4}, 4 \Rightarrow r^4 = 4$

(an increasing geometric series)

$a^2r^6 = 25 \Rightarrow \left(ar^3\right)^2 = 25$

$T_4 + T_6 + T_8 = ar^3 + ar^5 + ar^7$

$a = ar^3\left(1 + r^2 + r^4\right) = 5(1 + 2 + 4) = 35$

20. (*b*) $2^{\frac{1}{4} + \frac{2}{16} + \frac{3}{48} + ...\infty} = 2^{\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + ...\infty} = \sqrt{2}$

21. (*b*) $\sum_{n=1}^{\infty} \frac{2n^2 + 3n + 4}{(2n)!}$

$= \frac{1}{2}\left[\sum_{n=1}^{\infty} \frac{4n^2 - 2n + 8n + 8}{(2n)!}\right]$

$= \frac{1}{2}\sum_{n=1}^{\infty} \frac{2n(2n-1) + 8n + 8}{(2n)!}$

$= \frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{(2n-2)!} + 2\sum_{n=1}^{\infty} \frac{1}{(2n-1)!} + 4\sum_{n=1}^{\infty} \frac{1}{(2n)!}$

We know,

$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + ...$

$e^{-1} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + ...$

So, $\left(e + \frac{1}{e}\right) = 2\left(1 + \frac{1}{2!} + \frac{1}{4!} + ...\right)$

and $e - \frac{1}{e} = 2\left(1 + \frac{1}{3!} + \frac{1}{5!} + ...\right)$

Now

$\frac{1}{2}\left(\sum_{n=1}^{\infty} \frac{1}{(2n-2)!}\right) + 2\sum_{n=1}^{\infty} \frac{1}{(2n-1)!} + 4\sum_{n=1}^{\infty} \frac{1}{(2n)!}$

$= \frac{1}{2}\left[\frac{e + \frac{1}{e}}{2}\right] + 2\left[\frac{e - \frac{1}{e}}{2}\right] + 4\left(\frac{e + \frac{1}{e} - 2}{2}\right)$

$$= \frac{\left(e+\frac{1}{e}\right)}{4} + e - \frac{1}{e} + 2e + \frac{2}{e} - 4$$

$$= \frac{13}{4}e + \frac{5}{4e} - 4$$

22. **[1219]** $S = 77 + 757 + 7557 + \ldots + 75\underset{98}{\ldots}57$

$10S = 770 + 7570 + \ldots + 75\ldots570 + 755\ldots570$

$9S = -77 + \underbrace{13 + 13 + \ldots + 13}_{98 \text{ times}} + 75\underset{98}{\ldots}570$

$= -77 + 13 \times 98 + 75\underset{99}{\ldots}57 + 13$

$$S = \frac{75\underset{99}{\ldots}57 + 1210}{9}$$

On comparing with $S = \dfrac{75\underset{99}{\ldots}57 + m}{n}$

We get

$m = 1210$

$n = 9$

$m + n = 1219$

23. **(d)** $x_1 y > 0$ and $x^3y^2 = 2^{15}$

AM ≥ GM

$$\frac{3x+2y}{5} \geq \left(x^3y^2\right)^{\frac{1}{5}}$$

$\Rightarrow 3x + 2y \geq 40$

24. **(b)** $$S - \frac{1}{x-1} = \frac{-2^{101}}{\left(x^{2^{101}}\right)^2 - 1}$$

For $x = 2, S = 1 - \dfrac{2^{101}}{4^{101}-1}$

25. **(d)** *LHS* is *G.P.* of common ratio $\frac{3}{2}$

$$\therefore \frac{2^{10}\left(1-\left(\frac{3}{2}\right)^{11}\right)}{\left(1-\frac{3}{2}\right)} = S - 2^{11}$$

$$\Rightarrow 2^{10}\frac{\left(\frac{3^{11}-2^{11}}{2^{11}}\right)}{\frac{1}{2}} = S - 2^{11} \Rightarrow S = 3^{11}$$

26. **(d)** $1 + (1 - 2^2.1) + (1 - 4^2.3) + (1 - 6^2.5) + \ldots(1 - 20^2.19)$

$$S = 1 + \sum_{r=1}^{10}\left[1 - (2r)^2(2r-1)\right]$$

$$= 1 + \sum_{r=1}^{10}\left(1 - 8r^3 + 4r^2\right) = 1 + 10 - \sum_{r=1}^{10}\left(8r^3 - 4r^2\right)$$

$$= 11 - 8\left(\frac{10\times11}{2}\right)^2 + 4\times\left(\frac{10\times11\times21}{6}\right)$$

$= 11 - 2 \times (110)^2 + 4 \times 55 \times 7$

$= 11 - 220\,(110 - 7)$

$= 11 - 220 \times 103 = \alpha - 220\,\beta$

$\Rightarrow \alpha = 11, \beta = 103$

$\Rightarrow (11, 103)$

27. **(c)** $$\frac{2^{\sin x} + 2^{\cos x}}{2} \geq \left(2^{\sin x + \cos x}\right)^{\frac{1}{2}}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2.2^{\frac{\sin x + \cos x}{2}}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2^{1-\frac{1}{\sqrt{2}}}$$

(as $\sin x + \cos x \geq -\sqrt{2}$)

28. **(c)** $a_1 = 1$ and $a_n = 300$ and $d \in Z$

$300 = 1 + (n-1)d$

$$\Rightarrow d = \frac{299}{(n-1)} = \frac{23\times13}{(n-1)}$$

$\therefore n - 1 = 13$ or 23 (as d is integer)

$\Rightarrow n = 14$ or 24

$\Rightarrow n = 24$ and $d = 13$ [$\because 15 \leq n \leq 50$]

$a_{20} = 1 + 19 \cdot 13 = 248$

$$S_{20} = 20\,\frac{(248+1)}{2} = 2490$$

29. **[504]** $$\frac{1}{4}\left[\sum_{n=1}^{7}\left(2n^3 + 3n^2 + n\right)\right]$$

$$= \frac{1}{4}\left[2\left(\frac{7.8}{2}\right)^2 + 3\left(\frac{7.8.15}{6}\right) + \frac{7.8}{2}\right]$$

$$= \frac{1}{4}[2\times49\times16 + 28\times15 + 28]$$

$$= \frac{1}{4}[1568 + 420 + 28] = 504$$

30. **(a)** $$T_n = \frac{(3+(n-1)\times3)\left(1^2+2^2+\ldots+n^2\right)}{(2n+1)}$$

$$T_n = \frac{3.\frac{n^2(n+1)(2n+1)}{6}}{(2n+1)} = \frac{n^2(n+1)}{2}$$

$$S_{15} = \frac{1}{2}\sum_{n=1}^{15}(n^3 + n^2)$$

$$= \frac{1}{2}\left[\left(\frac{15(15+1)}{2}\right)^2 + \frac{15\times16\times31}{6}\right] = 782$$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 8

Binomial Theorem

Prarambh (Topicwise)

1. (*b*) Applying $T_{r+1} = {}^nC_r x^{n-r} a^r$ for $(x+a)^n$

Hence $T_6 = {}^{10}C_5 (2x^2)^5 \left(-\frac{1}{3x^2}\right)^5$

$= -\frac{10!}{5!5!} 32 \times \frac{1}{243} = -\frac{896}{27}$

2. (*c*) ${}^{20}C_{r-1} = {}^{20}C_{r+3}$

$\Rightarrow 20 - r + 1 = r + 3 \Rightarrow r = 9.$

3. (*c*) $T_{16} = {}^{17}C_{15} \left(\sqrt{x}\right)^2 \left(-\sqrt{y}\right)^{15}$

$= -\frac{17 \times 16}{2 \times 1} \times xy^{15/2} = -136xy^{15/2}$

4. (*c*) $T_r = {}^{15}C_{r-1} (x^4)^{16-r} \left(\frac{1}{x^3}\right)^{r-1}$

$= {}^{15}C_{r-1} x^{67-7r}$

$\Rightarrow 67 - 7r = 4 \Rightarrow r = 9.$

5. (*b*) We have

$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \ldots$

By hypothesis, $\frac{m(m-1)}{2}x^2 = -\frac{1}{8}x^2$

$\Rightarrow 4m^2 - 4m = -1 \Rightarrow (2m-1)^2 = 0$

$\Rightarrow m = \frac{1}{2}.$

6. (*b*) In the expansion of $\left(x^2 + \frac{a}{x}\right)^5$ the general term is

$T_{r+1} = {}^5C_r (x^2)^{5-r} \left(\frac{a}{x}\right)^r = {}^5C_r a^r x^{10-3r}$

Here, exponent of x is $10 - 3r = 1$

$\Rightarrow r = 3$

$\therefore T_{3+1} = {}^5C_3 a^3 x = 10a^3 x$

Hence coefficient of x is $10a^3$.

7. (*a*) In the expansion of $\left(x - \frac{1}{x}\right)^6$, the general term is

${}^6C_r x^{6-r} \left(-\frac{1}{x}\right)^r = {}^6C_r (-1)^r x^{6-2r}$

For term independent of x, $6 - 2r = 0 \Rightarrow r = 3$

Thus the required coefficient

$(-1)^3 \cdot {}^6C_3 = -20.$

8. (*c*) The coefficient of x^{16} in the expansion of $(x^2 - 2x)^{10}$

= The coefficient of x^{16} in $x^{10}(x-2)^{10}$

= The coefficient of x^6 in $(x-2)^{10}$

$= {}^{10}C_4 2^4, \quad (\because T_{r+1} = {}^nC_r x^{n-r} a^r)$

$= 210 \times 15 = 3360.$

9. (*c*) Let T_{r+1} term containing x^{32}.

Therefore ${}^{15}C_r x^{4(15-r)} \left(\frac{-1}{x^3}\right)^r$

Hence coefficient of x^{32} is ${}^{15}C_4$

$\Rightarrow x^{60-4r} x^{-3r} = x^{32}$

$\Rightarrow 7r = 28 \Rightarrow r = 4$

10. (*b*) x^7, x^8 will occur in T_8 and T_9.

Coefficients of T_8 and T_9 are equal.

$\therefore {}^nC_7 2^{n-7} \left(\frac{1}{3}\right)^7 = {}^nC_8 2^{n-8} \left(\frac{1}{3}\right)^8 \Rightarrow n = 55.$

11. (*d*) $(1+3x+2x^2)^6 = [1 + x(3+2x)]^6$

$= 1 + {}^6C_1 x(3+2x) + {}^6C_2 x^2 (3+2x)^2$

$+ {}^6C_3 x^3 (3+2x)^3 + {}^6C_4 x^4 (3+2x)^4$

$+ {}^6C_5 x^5 (3+2x)^5 + {}^6C_6 x^6 (3+2x)^6$

Only x^{11} gets from ${}^6C_6 x^6 (3+2x)^6$

$\because {}^6C_6 x^6 (3+2x)^6 = x^6 (3+2x)^6$

$\therefore$ Coefficient of $x^{11} = {}^6C_5 3^1 2^5 = 576.$

12. (*b*) $\left(4^{\frac{1}{3}} + \frac{1}{6^{\frac{1}{4}}}\right)^{20}$

$T_{r+1} = {}^{20}C_r (4^{1/3})^{20-r} (6^{-1/4})^r$

For rational terms

$20 - r = 3k$ & $r = 4p$, where $k, p \in I \Rightarrow r = 20$ & $r = 8$

$\therefore$ no. of rational terms = 2

$\therefore$ no. of irrational terms = 19

13. (*a*) $\left(x^{2/3} - \frac{1}{\sqrt{x}}\right)^{30}$ term x^{13}

$T_{r+1} = {}^{30}C_r x^{\frac{2}{3}(30-r)} (+x)^{-\frac{1}{2}r} (-1)^r$

$= {}^{30}C_r (-1)^r x^{\frac{120-7r}{6}}$

$\frac{120-7r}{6} = 13$

$\Rightarrow \frac{120-78}{7} = r \Rightarrow r = 6$

$= {}^{30}C_6 = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}$

which is divisibe by 29, 63, 65, term x^{14} does not exist.

14. (*a*) $\left(x^3 + 3.2^{-\log_{\sqrt{2}} x}\right)^{11} = \left(x^3 + \frac{3}{x^3}\right)^{11}$

$T_{r+1} = {}^{11}C_r (x^3)^{11-r} \; 3^r (x^3)^{-r}$

$= {}^{11}C_r 3^r (x^3)^{11-2r}$

$= {}^{11}C_r 3^r (x^3)^{11-2r} = {}^{11}C_r 3^r x^{33-6r}$

(*a*) $33 - 6r = 2 \Rightarrow \frac{31}{6} = r \Rightarrow$ Not possible

(*b*) x^2 doesn't appear

(*c*) $33 - 6r = -3 \Rightarrow 36 = 6r \Rightarrow r = 6$

(x^{-3}) term exist/appear in exp.

(*d*) for x^3, $r = 5$, & x^{-3}, $r = 6$

$\frac{T_6}{T_7} = \frac{{}^{11}C_5 \; 3^5}{{}^{11}C_6 \; 3^6} = \frac{1}{3}$

15. (*a*) 6^{th} term in the expansion of $\left[\frac{3}{2} + \frac{x}{3}\right]^n$

for $x = 3$

is numerically greatest

$\frac{n+1}{\left|\frac{x}{y}\right| + 1} - 1 \le r \le \frac{n+1}{\left|\frac{x}{y}\right| + 1}$

$\Rightarrow \frac{n+1}{\frac{3}{2}+1} - 1 \le 5 \le \frac{n+1}{\frac{3}{2}+1}$

$\Rightarrow \frac{2(n+1)}{5} - 1 \le 5 \le \frac{2(n+1)}{5}$

$\Rightarrow \frac{2(n+1)}{5} \le 6$ and $\frac{2(n+1)}{5} \ge 5$

$\Rightarrow n + 1 \le 15$ and $n + 1 \ge \frac{25}{2}$

$\Rightarrow n \le 14$ and $n + 1 \ge \frac{23}{2}$

$\Rightarrow 11.4 \le n \le 14$

$n \in N \Rightarrow n = 12, 13, 14$

for these value of n 6^{th} term is greatest term

16. (a) In the expansion of $\left(\frac{3x^2}{2}-\frac{1}{3x}\right)^9$, the general term is

$T_{r+1} = {}^9C_r \cdot \left(\frac{3x^2}{2}\right)^{9-r}\left(-\frac{1}{3x}\right)^r$

$= {}^9C_r \left(\frac{3}{2}\right)^{9-r}\left(-\frac{1}{3}\right)^r x^{18-3r}$

For the term independent of x, $18-3r=0$

$\Rightarrow r=6$

This gives the independent term

$T_{6+1} = {}^9C_6\left(\frac{3}{2}\right)^{9-6}\left(-\frac{1}{3}\right)^5 = {}^9C_3 \cdot \frac{1}{6^3}$

17. (a) $T_{r+1} = {}^{10}C_r (x^2)^{10-r}\left(\frac{-3\sqrt{3}}{x^3}\right)^r$

For term independent of x, $2-2r-3r=0$

$\Rightarrow r=4$

$\therefore T_{4+1} = {}^{10}C_4(-3)^4(\sqrt{3})^4 = 153090.$

18. (c) $(1 + {}^nC_1x + {}^nC_2x^2 \ldots) \times \left(1+\frac{{}^nC_1}{x}+\frac{{}^nC_2}{x^2}+\ldots\right)$ As in expansion, obviously the term independent of x will be

${}^nC_0 \cdot {}^nC_0 + {}^nC_1 \cdot {}^nC_1 + \ldots {}^nC_n \cdot {}^nC_n$

$= C_0^2 + C_1^2 + \ldots + C_n^2.$

19. (b) Since n is even therefore $\left(\frac{n}{2}+1\right)^{\text{th}}$ term is middle term, hence

${}^nC_{n/2}(x^2)^{n/2}\left(\frac{1}{x}\right)^{n/2} = 924x^6$

$\Rightarrow x^{n/2} = x^6 \Rightarrow n=12.$

20. (c) Middle term =

$\frac{T_{2n+2}}{2} = T_{n+1} = {}^{2n}C_n x^n = \frac{2n!}{(n!)^2}\cdot x^n.$

21. (b) Obviously the middle term

$= {}^{2n}C_n (x)^n \cdot \left(\frac{1}{2x}\right)^n = \frac{2n!}{n!\cdot n!\cdot 2^n}$

$= \frac{1.3.5\ldots(2n-1)}{n!}.$

22. (a) We know that:

$(1+x)^n = {}^nC_0 + {}^nC_1x + \ldots {}^nC_nx^n$

Also, $\left(1+\frac{1}{x}\right)^n = {}^nC_0 + {}^nC_1\left(\frac{1}{x}\right) + \ldots {}^nC_n\left(\frac{1}{x}\right)^n$

The summation aksed above

= coefficient of x^r in $(1+x)^n\left(1+\frac{1}{x}\right)^n$

= coefficient of x^r in $(1+x)^n\left(\frac{x+1}{x}\right)^n$

= coefficient of x^r in $\frac{(1+x)^{2n}}{x^n}$

= coefficient of x^{n+r} in $(1+x)^2$

$= {}^{2n}C_{n+r}$

$= \frac{2n!}{(n+r)!(2n-n-r)!}$

$= \frac{2n!}{(n+r)!(n-r)!}$

23. (c) $\int_0^1 (1+x)^n\,dx = \frac{(1+x)^{n+1}}{n+1}\Big|_0^1$

24. (c) Putting $x=1$ in $(x^2+x-3)^{319}$

We get the sum of coefficient

$= (1+1-3)^{319} = -1$

25. (c) $(1+x+x^2+x^3)^5 = (1+x)^5(1+x^2)^5$

$= (1+5x+10x^2+10x^3+5x^4+x^5)$

$\times(1+5x^2+10x^4+10x^6+5x^8+x^{10})$

Therefore the required sum of coefficients

$= (1+10+5).2^5 = 16\times 32 = 512$

Note: $2^n = 2^5 =$ Sum of all the binomial coefficients in the 2nd bracket in which all the powers of x are even.

26. (c) As we know that

${}^nC_0^2 - {}^nC_1^2 + {}^nC_2^2 - {}^nC_3^2 + \ldots + (-1)^n \cdot {}^nC_n^2 = 0$

(if n is odd) and in the question $n=15$ (odd).

27. (c) From binomial expansion, we have $(1+x)^n =$

$\left[1+nx+\frac{n(n-1)}{2}\cdot x^2 \ldots x^n\right]$

Substitute $x=n$, we get

$(1+n)^n = \left[1+nn+\frac{n(n-1)}{2}\cdot n^2 \ldots n^n\right]$

or $(1+n)^n - 1 = \left[nn+\frac{n(n-1)}{2}\cdot n^2 \ldots n^n\right]$

or $(1+n)^n - 1 = n^2\left[1+\frac{n(n-1)}{2}\ldots n^{n-2}\right]$

put n = 100

$(1+100)^{100} - 1 = 100^2$

$\left[1+\frac{100(100-1)}{2}\ldots 100^{98}\right]$

$(101)^{100} - 1 =$

$100^2\left[1+ \text{displaystyle } \frac{100(100-1)}{2}\ldots 100^{98}\right]$

clearly $(101)^{100}-1$ is divisible by 10000.

28. (c) We have $7^2 = 49 = 50-1$

Now, $7^{300} = (7^2)^{150} = (50-1)^{150}$

$= {}^{150}C_0(50)^{150}(-1)^0 + {}^{150}C_1(50)^{149}(-1)^1 + \ldots$

$+ {}^{150}C_{150}(50)^0(-1)^{150}$

Thus the last digits of 7^{300} are ${}^{150}C_{150}\cdot 1.1$ *i.e.*, 1.

29. (a) 111.....1 (91 times)

$= 1 + 10 + 10^2 + \ldots + 10^{90}$

$= \frac{10^{91}-1}{10-1} = \frac{(10^7)^{13}-1}{10-1} = \frac{t^{13}-1}{9},$

where $t = 10^7$

$= \left(\frac{t-1}{9}\right)(t^{12}+t^{11}+\ldots+t+1)$

$= \left(\frac{10^7-1}{10-1}\right)(1+t+t^2+\ldots+t^{12})$

$= (1+10+10^2+\ldots+10^6)(1+t+t^2+\ldots+t^{12})$

$\therefore$ 111 ... 1(91 times) is a composite number.

30. (d) $101^{100} - 1 = (1+100)^{100} - 1$

$= {}^{100}C_0 + {}^{100}C_1(100)^1 + {}^{100}C_2(100)^2 + \ldots + {}^{100}C_{100}(100)^{100} - 1$

$= 100\times 100 + {}^{100}C_2(100)^2 + \ldots\ldots + {}^{100}C_{100}\,100^{100}$

$= 10000\,[1 + {}^{100}C_2 + {}^{100}C_3(100) + \ldots + {}^{100}C_{100}\,100^{98}]$

$= 10000$ [Integer]

which is divisible by 100, 1000, 10000

31. (a) $7^9 + 9^7 = (8-1)^9 + (8+1)^7 = {}^9C_0(8)^9 - {}^9C_1(8)^8 + {}^9C_2(8)^7 \ldots + {}^9C_8(8) - {}^9C_9 + {}^7C_0(8)^7 + \ldots + {}^7C_6(8) + {}^7C_7$

This is divisible by 64 & 16

32. (a) Given term can be written as

$(1+x)^2(1-x)^{-2}$

$(1+2x+x^2)\left[1+2x+3x^2+\ldots+(n-1)x^{n-2}+nx^{n-1}+(n+1)x^n+\ldots\right]$

$= x^n(n+1+2n+n-1)+\ldots$

Therefore coefficient of x^n is $4n$.

33. (b) Let consecutive terms are ${}^nC_r$ and ${}^nC_{r+1}$

$\Rightarrow \frac{n!}{(n-r)!r!} = \frac{n!}{(n-r-1)!(r+1)!}$

$\Rightarrow \frac{1}{(n-r)(n-r-1)!r!} = \frac{1}{(n-r-1)!(r+1)r}$

$\Rightarrow r+1 = n-r \Rightarrow n = 2r+1.$ Hence n is odd.

34. (d) Let $f = (5\sqrt{2}-7)^{19}$

$x - f =$ an integer $\Rightarrow [x] + \{x\} - f =$ integer

$\Rightarrow \{x\} - f =$ an integer, but $-1 < \{x\} - f < 1 \Rightarrow \{x\} = f$

So. $x\{x\} = x\,f = 1^{19} = 1$

35. (b) $I + f = (4+\sqrt{15})^n$

Let $g = (4-\sqrt{15})^n$ then $0 < g < 1$

$I + f = {}^nC_0 4^n + {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}$

$+ {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots$

$g = {}^nC_0 4^n - {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15$

$- {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots$

$I+f+g=2\ ({}^nC_0 4^n + {}^nC_2 4^{n-2}15 + \ldots) =$ even integer

$\because 0<f+g<2$

$\Rightarrow f+g=1$

$\Rightarrow 1-f=g$ thus I is an odd Integer

$1-f=g=(4-\sqrt{15})^n$

$(I+f)(1-f)=(I+f)g=1$

36. *(c)* $(x+a)^{100}+(x-a)^{100}$

$=2\left({}^{100}C_0 x^{100} + {}^{100}C_2 x^{98}a^2 + \ldots + {}^{100}C_{100}\, a^{100}\right)$

Number of terms = 51 terms

37. *(a)* $T_2 = {}^nC_1 (a^{1/13})^{n-1}(a^{3/2}) = 14a^{5/2}$

$\Rightarrow n=14$

$\therefore \dfrac{{}^nC_3}{{}^nC_2}=4$

38. *(a)* $T_{r+1} = {}^{6561}C_r (7)^{\frac{6561-r}{3}} (11^{1/9})^r$

Here r should be multiple of 9

$r=0, 9, 18 \ldots 6561$

Number of terms = 730

39. *(b)* $\left.\begin{matrix} T_{2m+1} \Rightarrow {}^{10}C_{2m} \\ T_{4m+5} \Rightarrow {}^{10}C_{4m+4} \end{matrix}\right]$ equal

$2m+4m+4=10;\ 6m+4=10$

$m=1$

40. *(c)* $(2x+5y)^{13}$ greatest form for $x=10$, $y=2$

$$\frac{n+1}{\left|\frac{x}{y}\right|+1}-1 \le r \le \frac{n+1}{\left|\frac{x}{y}\right|+1}$$

$$\Rightarrow \frac{14}{\left|\frac{2x}{5y}\right|+1}-1 \le r \le \frac{14}{\left|\frac{2x}{5y}\right|+1}$$

$$\Rightarrow \frac{14}{3}-1 \le r \le \frac{14}{3} \Rightarrow \frac{11}{3} \le r \le \frac{14}{3}$$

$\Rightarrow 3.66 \ldots r \le 4.666 \Rightarrow r=4$

$\Rightarrow T_5 = {}^{13}C_4 (20)^9 (10)^4$

41. *(a)* For numerically greatest term

$$r=\left[\frac{n+1}{1+\left|\frac{x}{a}\right|}\right]$$

$$=\left[\frac{9+1}{1+\left|\frac{4}{9}\right|}\right] \Rightarrow r=6$$

Numerically greatest term $T_{r+1} = {}^9C_6 (2)^3 \left(\frac{9}{2}\right)^6$

42. *(c)* $2^{2003} = 8.(16)^{500} = 8\,(17-1)^{500}$

$\therefore$ Remainder = 8

43. *(b)* $(\sqrt{2}+1)^6 = I+f$

$(\sqrt{2}-1)^6 = f'$

$2[{}^6C_0 + {}^6C_2 . 2 + {}^6C_4 (2)^2 + \ldots] = I+f+f'$

$f+f'=1$ or $f'=1-f$

$I=2\,[{}^6C_0 + {}^6C_2 .2 + {}^6C_4 .4 + {}^6C_6 .8] - 1$

$I=2\,[1+30+60+8]-1=197$

Prabal (JEE Main Level)

1. *(c)* ${}^{2m+1}C_m \left(\frac{x}{y}\right)^{m+1}\left(\frac{y}{x}\right)^m = {}^{2m+1}C_m \left(\frac{x}{y}\right)$

Dependent upon the ratio $\frac{x}{y}$ and m.

2. *(a)* $T_6 = {}^8C_5 \left(\frac{1}{x^{8/3}}\right)^3 (x^2 \log_{10} x)^5 = 5600$

$\Rightarrow \frac{1}{x^8} x^{10} (\log_{10} x)^5 = 100$

$\Rightarrow x=10$

3. *(c)* $(1-2x^3+3x^5)\left(1+\frac{1}{x}\right)^8$

Co-efficient of $x=-2.\ {}^8C_2 + 3.\ {}^8C_4 = 154$

4. *(b)* $\left(x-\frac{1}{x}\right)\left(x^2-\frac{1}{x^2}\right)^3 = \left(x-\frac{1}{x}\right)$

$({}^3C_0 x^6 - {}^3C_1 x^2 + {}^3C_2 x^{-2} - {}^3C_3 x^{-6})$

There is no term independent of x

5. *(c)* $T_{r+1} = {}^nC_r (2)^{n-r}\left(\frac{x}{3}\right)^r = {}^nC_r \frac{2^{n-r}}{3^r} x^r$

Coeff $T_8 = T_9 \Rightarrow {}^nC_7 \frac{2^{n-7}}{3^7} = {}^nC_8 \frac{2^{n-8}}{3^8}$

$\Rightarrow \frac{{}^nC_8}{{}^nC_7} = \frac{3.2}{1} \Rightarrow \frac{n-8+1}{8}=6$

$\Rightarrow n-7=48 \Rightarrow n=55$

6. *(b)* Generaltermis $T_{r+1} = {}^{100}C_r (2)^{\frac{100-r}{2}} (3)^{\frac{r}{4}}$

The above term will be rational if exponent of 2 & 3 are integers.

i.e. $\frac{100-r}{2}$ and $\frac{r}{4}$ must be integers

the possible set of r is = {0, 4, 8, 16, ..., 100}

no. of rational terms is 26

7. *(b)* middle term = T_5

$T_5 = T_{4+1} = {}^8C_4 . k^4 = 1120$

$\Rightarrow k=2$

8. *(a)* $C_0 + C_1 + C_2 + \ldots + C_n = 2^n = 256$

$\Rightarrow 2^n = 2^8 \Rightarrow n=8$

$T_{r+1} = {}^8C_r (2x)^{8-r}\left(\frac{1}{x}\right)^r = {}^8C_r 2^{8-r} x^{8-2r}$

For Constant term $\Rightarrow 8-2r=0 \Rightarrow r=4$

$= {}^8C_4 2^4 = \frac{8\times7\times6\times5}{4\times3\times2} 2^4 = 70\times16 = 1120$

9. *(a)* sum of coeff of $(1-2x+5x^2)^n = a$

sum of coeff of $(1+x)^{2n} = b$

put $x=y=1$

$a=(1-2+5)^n = 4^n$ & $b=(1+1)^{2n} = 2^{2n} = 4^n$

$a=b$

10. *(d)* $\left(x^k + \frac{1}{x^{2k}}\right)^{3n}$, $n \in N$ Independent of x

$T_{r+1} = {}^{3n}C_r \left(x^k\right)^{3n-r}\left(\frac{1}{x^{2k}}\right)^r$

$= {}^{3n}C_r x^{3nk-rk-2kr} = {}^{3n}C_r x^{3k(n-r)}$

For Constant term $\Rightarrow 3k(n-r)=0$

$\Rightarrow n=r$

$\therefore T_{r+1} = {}^{3n}C_n$ true for any real k or $k \in R$

11. *(a)* $(3x+2)^{-1/2}$ has infinite expansion when $\left|\frac{3x}{2}\right|<1$

$\Rightarrow x \in \left(-\frac{2}{3}, \frac{2}{3}\right)$

12. *(b)* Coeff of α^t in

$(\alpha+p)^{m-1} + (\alpha+p)^{m-2}(\alpha+q) + (\alpha+p)^{m-3}(\alpha+q)^2$

$\ldots + (\alpha+q)^{m-1}$

$\because a \ne -q, p \ne q$

Let $\alpha+p=x$ & $\alpha+q=y$

$= x^{m-1} + x^{m-2}y + x^{m-3}y^2 + \ldots y^{m-1}$

$$= x^{m-1}\left[1-\left(\frac{y}{x}\right)+\left(\frac{y}{x}\right)^2 + \ldots\ldots + \left(\frac{y}{x}\right)^{m-1}\right]$$

$$= x^{m-1}\frac{\left[1-\left(\frac{y}{x}\right)^m\right]}{\left(1-\frac{y}{x}\right)}$$

$$= \frac{x^{m-1}}{x^m}\frac{x^m-y^m}{x-y}\cdot x = \frac{(\alpha+p)^m-(\alpha+q)^m}{\alpha+p-\alpha-q}$$

$$= \frac{1}{(p-q)}\left[(\alpha+p)^m-(\alpha+q)^m\right]$$

$$= \text{coeff of } \alpha^t = \left(\frac{{}^mC_t\, p^{m-t} - {}^mC_t\, q^{m-t}}{p-q}\right)$$

13. *(b)* For numerically greatest term

$$r=\left[\frac{n+1}{1+\left|\frac{x}{a}\right|}\right]=\left[\frac{34+1}{1+\left|\frac{6}{10}\right|}\right]$$

$\Rightarrow r=21.$

14. *(a)* $\sum_{r=0}^{n-1}\frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} = \sum_{r=0}^{n-1}\frac{r+1}{n+1}$

$$= \frac{1}{n+1}[1+2+\ldots+n] = \frac{1}{n+1}\times\frac{n(n+1)}{2}$$

$$= \frac{n}{2}$$

15. *(c)* ${}^{18}C_{r-2} + 2\cdot{}^{18}C_{r-1} + {}^{18}C_r \ge {}^{20}C_{13}$

$\Rightarrow {}^{18}C_{r-2} + {}^{18}C_{r-1} + {}^{18}C_{r-1} + {}^{18}C_r \ge {}^{20}C_{13}$

$\Rightarrow {}^{19}C_{r-1} + {}^{19}C_r \ge {}^{20}C_{13} \Rightarrow {}^{20}C_r \ge {}^{20}C_{13}$

$\therefore {}^{20}C_{10} > {}^{20}C_{11} > {}^{20}C_{12} > {}^{20}C_{13}$

& ${}^{20}C_{10} > {}^{20}C_9 > {}^{20}C_8 > {}^{20}C_7$

$r=7, 8, 9, 10, 11, 12, 13 \Rightarrow$ Total 7 elements

16. (b) $\frac{C_0}{1}+\frac{C_1}{2}+\frac{C_2}{3}+\ldots\ldots+\frac{C_{10}}{11}$

$\sum_{r=0}^{10} T_{r+1}=\sum_{r=0}^{10}\frac{{}^{10}C_r}{r+1}$

$=\sum_{r=0}^{10}\frac{1}{(10+1)}\cdot\frac{10+1}{r+1}\cdot{}^{10}C_r=\frac{1}{11}\sum_{r=0}^{10}{}^{11}C_{r+1}$

$=\frac{1}{11}\left[{}^{11}C_1+{}^{11}C_2+\ldots\ldots+{}^{11}C_{11}\right]$

$=\frac{1}{11}\left[{}^{11}C_0+{}^{11}C_1+{}^{11}C_2+\ldots+{}^{11}C_{11}-{}^{11}C_0\right]$

$=\frac{1}{11}\left[2^{11}-1\right]=\frac{2^{11}-1}{11}$

17. (c) ${}^{47}C_4+{}^{51}C_3+{}^{50}C_3+{}^{49}C_3+{}^{48}C_3+{}^{47}C_3={}^{52}C_4$

18. (b) ${}^{50}C_0\times{}^{50}C_1+{}^{50}C_1\times{}^{50}C_2+\ldots+{}^{50}C_{49}\times{}^{50}C_{50}$

$={}^{50}C_0\times{}^{50}C_{49}+{}^{50}C_1\times{}^{50}C_{48}+\ldots+{}^{50}C_{49}\times{}^{50}C_0$

= co-eff. of x^{49} in $(1+x)^{100}={}^{100}C_{49}={}^{100}C_{51}$

19. (d) $3^{400}=(10-1)^{200}$

${}^{200}C_0(10)^{200}+\ldots+{}^{200}C_{199}(10)(-1)+{}^{200}C_{200}$

Last two digits = 01

20. (d) coef of x^4 in $(1-x+2x^2)^{12}$

$={}^{12}C_0(1-x)^{12}(2x^2)^0+{}^{12}C_1(1-x)^{11}(2x^2)+{}^{12}C_2$

$(1-x)^{10}(2x^2)^2$ + above x^4 powers terms of x^4

$={}^{12}C_0\cdot{}^{12}C_4(-x)^4+{}^{12}C_1\,{}^{11}C_2(-x)^2\,2x^2+{}^{12}C_2\,{}^{10}C_0\,4x^4$

$={}^{12}C_4+12\cdot{}^{11}C_2\cdot2+{}^{12}C_2\cdot4$

$={}^{12}C_4+2.3\cdot\frac{12}{3}\,{}^{11}C_2+{}^{12}C_2\cdot4$

$={}^{12}C_3+{}^{12}C_2+3\left({}^{12}C_2+{}^{12}C_3\right)+{}^{12}C_3+{}^{12}C_3+{}^{12}C_4$

$={}^{12}C_3+3({}^{12}C_2+{}^{12}C_3)+{}^{12}C_2+{}^{12}C_3+{}^{12}C_3+{}^{12}C_4$

$={}^{12}C_3+3{}^{13}C_3+{}^{13}C_3+{}^{13}C_4$

$={}^{12}C_3+3{}^{13}C_3+{}^{14}C_4$

21. (b) Required sum = coefficient of x^n in $(1+x)^n(1+x)^n$

= coefficient of x^n in $(1+x)^{2n}={}^{2n}C_n$

22. (d) $\because {}^{30}C_r\,{}^{20}C_0+{}^{30}C_{r-1}\cdot{}^{20}C_1+\ldots+{}^{30}C_0\cdot{}^{20}C_r$

= coefficient of $x^r(1+x)^{30}(1+x)^{20}$

= coefficient of x^r in $(1+x)^{50}={}^{50}C_r$

$\because {}^{50}C_r$ is maximum

$\therefore\ r=\frac{50}{2}=25$

23. (b) Co-efficient of x^{50}

$S=(1+x)^{1000}+2x(1+x)^{999}+3x^2(1+x)^{998}+\ldots.+1001x^{1000}$ ... (i)

$\frac{xS}{1+x}=x(1+x)^{999}+2x^2(1+x)^{998}\ldots+1000x^{1000}+\frac{1001x^{1001}}{(1+x)}$ ... (ii)

$\frac{S}{1+x}=(1+x)^{1000}+x(1+x)^{999}+\ldots+x^{1000}-\frac{1001\,x^{1001}}{1+x}$

$\Rightarrow\ \frac{S}{1+x}=(1+x)^{1000}\left[\frac{1-\left(\frac{x}{1+x}\right)^{1001}}{1-\frac{x}{1+x}}\right]-\frac{1001\,x^{1001}}{(1+x)}$

$\Rightarrow S=(1+x)^{1002}-x^{1001}(1+x)-1001\,x^{1001}$

Co-efficient of $x^{50}={}^{1002}C_{50}$

24. (b) ${}^{39}C_{3r-1}-{}^{39}C_{r^2}={}^{39}C_{r^2-1}-{}^{39}C_{3r}$

$r^2=3r$ or $r=0,3$

25. (b) $2^{60}=(1+7)^{20}$

$={}^{20}C_0+{}^{20}C_1.7+{}^{20}C_2.7^2+\ldots.+{}^{20}C_{20}.7^{20}$

$\therefore$ The remainder $={}^{20}C_0=1$.

26. (d) $\sum_{r=1}^{101}{}^{101}C_r\,S_{r-1}=\sum_{r=1}^{101}{}^{101}C_r\,\frac{q^r-1}{q-1}$

$=\frac{1}{q-1}\left(\sum_{r=1}^{101}{}^{101}C_r\,q^r-\sum_{r=1}^{101}{}^{101}C_r\right)$

$=\frac{1}{q-1}((1+q)^{101}-1-2^{101}+1)$

$=\frac{\alpha}{2^{100}}\left(\frac{(1+q)^{101}-2^{101}}{q-1}\right)$

$\Rightarrow\alpha=2^{100}$

27. (b) $S=\sum_{i=0}^{m}{}^{10}C_i\,{}^{20}C_{m-i}$

$(1+x)^{10}={}^{10}C_0+{}^{10}C_1x+\ldots+{}^{10}C_{10}x^{10}$ ... (i)

$(1+x)^{20}={}^{20}C_0+{}^{20}C_1x+\ldots\,{}^{20}C_{20}x^{20}$ ... (ii)

$\therefore$ S represents coefficient of x^m in (i) × (ii)

Coefficient x^m in $(1+x)^{30}={}^{30}C_m$

$\therefore$ For this to be maximum $m=15$

28. (b) General term: ${}^{55}C_r(y^{1/3})^{55-r}\cdot\left(\frac{1}{x^{10}}\right)^r$: ${}^{55}C_r\cdot y^{\frac{55-r}{3}}\cdot x^{\frac{r}{10}}$.

Terms free from radical sign, when

$r=0,\ \frac{55-0}{3}=\frac{55}{3}$ (Not possible)

$r=10,\ \frac{55-10}{3}=15$

$r=20,\ \frac{55-20}{3}=\frac{35}{3}$ Not possible)

$r=40,\ \frac{55-40}{3}=5$

$r=50,\ \frac{55-50}{3}=\frac{5}{3}$ (Not possible)

Terms free from radical sign = 2.

29. (c) $f(x)=(1+x)^{20}\left[1+\frac{x}{1+x}+\left(\frac{x}{1+x}\right)^2+\ldots+\left(\frac{x}{1+x}\right)^{19}\right]$

$(1+x)^{21}\left[1-\left(\frac{x}{1+x}\right)^{19}\right]=(1+x)^{21}-(1+x)^2x^{19}$

Coefficient of x^{18} in $f(x)$ = Coefficient of x^{18} in $(1+x)^{21}={}^{21}C_{18}=1330$

30. (b) $\sum_{r=1}^{10}r(r-1)\,{}^{10}C_r$

${}^{10}C_r=\frac{10^2}{r}C_{2-1}=\frac{10}{r}\times\frac{9}{r-1}\,{}^{8}C_{r-2}$

$=\sum r(r-1)\times\frac{10}{r}\times\frac{9}{r-1}\,{}^{8}C_{r-2}$

$=90\sum_{r=2}^{10}C_{r-2}$

$=45\,[2^9]=K.2^9$

$K=45$

31. (c) $\left(1+x+x^2+x^3\right)^5=a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+\ldots..+a_{15}x^{15}$

Put $x=1$ and $x=-1$ and add

$4^5=a_0+a_1+a_2+a_3+a_4+\ldots..+a^{15}$

$0=a_0-a_1+a_2-a_3+a_4-\ldots.a^{15}$

$4^5=2\left(a_0+a_2+a_4+\ldots\ldots\ldots\ldots+a_{14}\right)$

$a_0+a_2+a_4+\ldots\ldots\ldots\ldots+a_{14}=512$

32. (b) $(1+x)^n=3+\frac{8}{3}+\frac{80}{3^3}+\frac{240}{3^4}+\ldots$

$=1+nx+\frac{n(n-1)x^2}{2}+\ldots$

On comparison, $n=-3$ and $x=-\frac{2}{3}$

33. (b) $13^{99}-19^{93}$ = odd number − odd number = even number

$\Rightarrow$ it is divisible by 2.

Also,

$=(1+12)^{99}=1+{}^{99}C_1\cdot12+{}^{99}C_2\cdot12^2+{}^{99}C_3\cdot12^3+{}^{99}C_4\cdot12^4$

$=1+99\times12+\frac{99\times98}{2}\cdot12^2+\frac{3398\times4998\times97}{3\times y}\times12^3+81I_1$

$=1+99\times12+99\times49\,.\,12^3+3\times11\times49\times97\times12^3+81I_1$

$=1+99\times12+81\{11\times49\times16+11\times49\times97\times64+I_1\}$

$13^{99} = 1 + 99\times12 + 81I_2$

$19^{93} = (1+18)^{93} = 1 + {}^{93}C_1 \cdot 18 + {}^{93}C_2 \cdot 18^2 + \ldots$

$= 1 + 93 \times 18 + 81I_3$

$\Rightarrow 13^{99} - 19^{93} = 99\times12 - 93\times18 + 81(I_2 - I_3)$

$= 27\{11\times4 - 31\times2\} + 81I_4$

$= 81\{-6 + I_4\} = 81I_5$

Hence, it is also divisble by 81.

So, it is divisible by 162

$\Rightarrow$ remaainder = 0

34. **(c)** $6^{83} + 8^{83} = (7+1)^{83} + (7-1)^{83}$

$= 2({}^{83}C_0\,(7(7)^{83} + \ldots + {}^{83}C_{82}\,(7)^1) = 7\lambda + 83 \times 7 \times 2$

Remainder is 5

35. **(b)** We know that $(t+x)^n = \sum_{k=0}^{n} {}^nC_k \cdot x^k \cdot t^{n-k}$

Differentiating above equation with respect to x, we get

$\Rightarrow n.(t+x)^{n-1} = \sum_{k=0}^{n} k\,{}^nC_k \cdot x^{k-1} \cdot t^{n-k}$ ...(i)

Now multiplying equation (i) with and then differentiating w.r.t. x,

$\Rightarrow n[x(n-1)(t+x)^{n-2} + (t+x)^{n-1}\cdot1] = \sum_{k=0}^{n} k \cdot {}^nC_k \cdot k \cdot x^{k-1} \cdot t^{n-k}$

$\Rightarrow nx[x(n-1)(t+x)^{n-2} + (t+x)^{n-1}\cdot1] = \sum_{k=0}^{n} k^2 \cdot {}^nC_k \cdot x^k \cdot t^{n-k}$

Now, replacing t by y, we get

$\Rightarrow \sum_{k=0}^{n} k^2 \cdot {}^nC_k \cdot x^k \cdot y^{n-k}$

$= nx\,[x(n-1)(y+x)^{n-2} + (y+x)^{n-1}]$

$\Rightarrow \sum_{k=0}^{n} k^2 \cdot {}^nC_k \cdot x^k \cdot y^{n-k} = nx[x(n-1)+1]$

$= nx\,[x(n-1)(y+n)^{n-2} + (y+x)^{n-1}]$ (putting $x + y = 1$)

$= nx[nx + y] = nxy + nx^2y^2$

$\Rightarrow \sum_{k=0}^{n} k^2 \cdot {}^nC_k \cdot x^k \cdot y^{n-k} = nxy + n^2x^2$

36. **(b)** Consider the expansion

$(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$ ...(i)

Putting $x = -i$ in equation (i) we have

$(1-i)^n = C_0 - C_1i - C_2 + C_3i + C_4 + \ldots + (-1)^nC_ni^n$

$\text{or}(2)^{n/2}\left(\cos\left(-\frac{n\pi}{4}\right) + i\sin\left(-\frac{n\pi}{4}\right)\right)$

$= (C_0 - C_2 + C_4 - \ldots) - i.(C_1 - C_3 + C_5 - \ldots)$ ...(ii)

equating the imaginary part in (i), we get

$C_1 - C_3 + C_5 - \ldots = 2^{n/2}\sin\frac{n\pi}{4}$

37. **(b)** $\left(5\frac{1}{5}\log\sqrt{4^x+44} + \frac{1}{5\log\sqrt{2^{x-1}+7}}\right)$

$= \left(\left(\sqrt{4^x+444}\right)^{2/5} + \left(\frac{1}{\sqrt[3]{2^{x-1}+7}}\right)\right)^8$

$= \left(\left(4^x+444\right)^{2/5} + \left(\frac{1}{\left(2^{x-1}+7\right)^{1/3}}\right)\right)^8$

Now

$T_4 = T_{3+1} = {}^8C_3\left(\left(4^x+44\right)^{1/5}\right)^{8-3}\frac{1}{\left(\left(2^{x-1}+7\right)^{1/3}\right)^3}$

Given $336 = {}^8C_3\left(\frac{4^x+44}{2^{x-1}+7}\right)$

Let $2^x = y$

$\Rightarrow 336 = {}^8C_3\left(\frac{y^2+44}{(y/2)+7}\right)$

$\Rightarrow 366 = \frac{8\times7\times6}{3\times2\times1}\left(\frac{2\left(y^2+44\right)}{y+14}\right)$

$\Rightarrow y^2 - 3y + 2 = 0 \Rightarrow y = 0, 2$

38. **(c)** In the expansion of $(1 + 3^{1/3} + 7^{1/7})^{10}$, the general term $\frac{10!}{r_1!r_2!r_3!}3^{\frac{r_2}{3}}7^{\frac{r_3}{7}}$

Now, for terms free from radical sign, r_2 must be a mulitple of 3 and r^3 must be a multiple of 7.

Now, $r_1 + r_2 + r_3 = 10$

So possible

0	3	7
10	0	0
7	3	0
3	0	7
4	6	0
1	9	0

Hence, for the above 6 combinations, there can be 6 such terms.

39. **(b)** $(19)^{100} = (20-1)^{100}$

$= 20k + {}^{100}C_{98}\cdot20^2 - {}^{100}C_{99}20 + 1$

$= 20k + 1980000 - 2000 + 1$

$= 20k + 1978001$

Since the last tree distance of $20k$ will be zeros so the last three digits of $(19)^{100}$ will be same as 1978001

So, he last three digits of $(19)^{100}$ are 001

40. **(a)** Let $(\sqrt{3}+1)^{2m} = I + f$ and $(\sqrt{3}-1)^{2m} = f'$ where $0 < f' < 1$ and $0 < f' < 1$ and I is an integer.

Thus $I + f + f' = (\sqrt{3}+1)^{2m} + (\sqrt{3}-1)^{2m}$

$= (4 + 2\sqrt{3})^m + (4 - 2\sqrt{3})^m$

$= 2^m[(2+\sqrt{3})^m + (2-\sqrt{3})^m]$

$= 2^{m+1}[2^m + {}^mC_2 \cdot 2^{m-2} \cdot (\sqrt{3})^2 + \ldots..]$

Now as in part (a) $f + f' = 1$ and so $I + f + f'$ is an integer next above $(\sqrt{3}+1)^{2m}$ which by (1) contains 2^{m+1} as a factor.

Parikshit (JEE Advanced Level)

1. **(a)** $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right) = \sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)\frac{4!}{4!}$

$= \sum_{k=0}^{4}\frac{{}^4C_k \cdot 5^{4-k}\cdot x^k}{4!} = \frac{(5+x)^4}{4!}11$

Now, $\frac{(5+x)^4}{4!} = \frac{8}{3} \Rightarrow (5+x)^4 = 64$

$x + 5 = 2^{3/2} = 2\sqrt{2} \Rightarrow x = 2\sqrt{2} - 5$

2. **(c)** $\frac{{}^nC_r}{{}^{n+4}C_4} = \frac{n!}{r!(n-r)!}\frac{4!r!}{(r+4)!}$

$= \frac{4!}{(n+1)(n+2)(n+3)(n+4)}\left({}^{n+4}C_{r+4}\right)$

$= \frac{{}^{n+4}C_{r+4}}{{}^{n+4}C_4}$

Thus,

$S = \frac{1}{{}^{n+4}C_4}\left[\sum_{r=0}^{n}3^{r+4}\left({}^{n+4}C_{r+4}\right) + \sum_{r=0}^{3}{}^{n+4}C_r3^r\right]$

$= \frac{1}{{}^{n+4}C_4}\sum_{k=0}^{n+4}{}^{n+4}C_k3^k = \frac{4^{n+4}}{{}^{n+4}C_4}$

3. **(b)** We know $x - 1 < [x] \le x\,\forall x \in R$, therefore

${}^nC_kx^2 - 1 < \left[{}^nC_kx^2\right] \le {}^nC_kx^2$

$\Rightarrow \sum_{k=0}^{n}\left({}^nC_kx^2 - 1\right) < \sum_{k=0}^{n}\left[{}^nC_kx^2\right] \le \sum_{k=0}^{n}{}^nC_kx^2$

$\Rightarrow \frac{x^2\left(2^n\right) - (n+1)}{2^n - 2} < \frac{1}{2^n-2}\sum_{k=0}^{n}\left[{}^nC_kx^2\right] \le \frac{x^2\left(2^n\right)}{2^n-2}$

Taking limit as $n \to \infty$, and using sandwich theorem, we get

$$\lim_{n\to\infty}\frac{1}{2^n-2}\sum_{k=0}^{n}\left[{}^nC_k x^2\right]=x^2$$

4. (*c*) $1234 = 3n + 1$, so 1234^{567}

$= 3m + 1, 89 = 3k - 1$, so $89^{1011} = 3p - 1$

Hence $1234^{567} + 89^{1011} = 3n$ 1234 is even,

So $1234^{567} = 4q$. $89 = 4r + 1$, so $89^{1011} = 4t + 1$.

Hence $1234^{567} + 89^{1011} = 4l + 1$. So it is $9x + 1$.

5. (*a*) We have $\sum_{r=0}^{s}\sum_{s=1}^{n} n_{c_s}\,{}^sC_r\,{}^nC_s\,{}^sC_r$

$$=\sum_{s=1}^{n}{}^nC_{c_s}\left(s_{c_0}+s_{c_1}+s_{c_2}+\ldots+{}^{s_{c_s}}\right)$$

$[\because r \le s]$

$$=\sum_{s=1}^{n} n_{c_s}(2)^s=\sum_{s=1}^{n} n_{c_s}2^n-n_{c_0}2^0$$

$$=(1+2)^n-1=3^n-1$$

6. (*b*) Let $x=2^{\frac{153}{2}}$

$N = x^{16} - 1$

$\alpha = x^2 + \sqrt{2}x + 1 \quad N = \left(x^4-1\right)\left(x^4+1\right)\left(x^8+1\right)$

$N = \left(x^4-1\right)\left(x^2+\sqrt{2}x+1\right)\left(x^2-\sqrt{2}x+1\right)\left(x^8+1\right)$

Let $y = 2^{204}$

$\beta = y^2 - y + 1$

$N = y^6 - 1 = \left(y^3-1\right)\left(y^3+1\right)$

$= \left(y^3-1\right)(y+1)\left(y^2-y+1\right)$

7. (*a*) $2^{30} \times 3^{30} = 6^{30} = (7-1)^{30}$

using Binomial expansion.All terms are divided by 7 except last term where power of 7 in 0

$\Rightarrow$ coeff remainder = 1

$$\frac{(7-1)^{30}}{7}=\frac{{}^{30}C_0}{7}7^{30}\ldots\text{----}\frac{{}^{30}C_4}{7}7^{14}\ldots\ldots\frac{{}^{30}C_{30}}{7}$$

8. (*b*) $\left(1+x+x^2+x^3+x^4\right)^n(x-1)^{n+3}$

$$=\left(x^5-1\right)^n(x-1)^3$$

$$=-\left(x^5-1\right)^n\left(x^3-1-3x(x-1)\right)$$

$$=(-1)^n\left(1-x^5\right)^n\left(x^3-1-3x^2+3x\right)$$

$$=\left((-1)^n\sum{}^nC_r(-x)^{5r}\right)\left(x^3-1-3x^2+3x\right)$$

Only possible case

$5r + 3 = 83$

$r = 16$

So coefficient of x^{83} is

$= (-1)^n\,{}^nC_{16}$

9. (*b,c*) $\dfrac{{}^nC_r}{{}^{r+2}C_r}=\dfrac{n!}{r!(n-r)!}\dfrac{r!2!}{(r+2)!}$

$$=\frac{2(n!)}{(r+2)!(n-r)!}$$

$$=\frac{2}{(n+1)(n+2)}\frac{(n+2)!}{(r+2)!\left[(n+2)-(r+2)\right]!}$$

$$=\frac{2}{(n+1)(n+2)}{}^{n+2}C_{r+2}$$

Thus, $S=\sum_{r=0}^{n}(-2)^r\left(\dfrac{{}^nC_r}{{}^{n+2}C_r}\right)$

$$=\frac{2}{(n+1)(n+2)}\sum_{r=0}^{n}(-2)^r\cdot{}^{n+2}C_{r+2}$$

$$=\frac{2}{(n+1)(n+2)}\sum_{s=2}^{n+2}(-2)^{s-2}\cdot{}^{n+2}C_s$$

[put $r + 2 = s$]

$$=\frac{2}{4(n+1)(n+2)}\sum_{s=2}^{n+2}{}^{n+2}C_s(-2)^s$$

$$=\frac{1}{2(n+1)(n+2)}\times$$

$$\left[\sum_{s=0}^{n+2}{}^{n+2}C_s(-2)^s-{}^{n+2}C_0(-2)^0-{}^{n+2}C_1(-2)^1\right]$$

$$=\frac{1}{2(n+1)(n+2)}\left[(1-2)^{n+2}-1+2(n+2)\right]$$

$$=\frac{1}{2(n+1)(n+2)}\left[2n+3+(-1)^n\right]$$

But

$$2n+3+(-1)^n=\begin{cases}2(n+2) & \text{if n is even,}\\ 2(n+1) & \text{if n is odd}\end{cases}$$

Thus, $S=\begin{cases}\dfrac{1}{n+1} & \text{if n is even,}\\ \dfrac{1}{n+2} & \text{if n is odd}\end{cases}$

10. (*a,b*) $(1 + x^2)^2 (1 + x)^n = A_0 + A_1x + A_2x^2 \ldots$

If A_0, A_1, A_2 are in A.P.

$\Rightarrow (1 + x^4 + 2x^2)(1 + x)^n = A_0 + A_1x + A_2x^2 \ldots$

If A_0, A_1, A_2 are in A.P.

$$\Rightarrow \left(1+x^4+2x^2\right)\left[1+nx+\frac{n(n-1)x^2}{2!}+\frac{n(n-1)(n-2)}{3!}x^3+\ldots\ldots\right]=A_0+A_1x+A_2x^2+\ldots$$

by comparism

$A_0 = 1$

$$A_1 = n \;\&\; A_2=\frac{n(n-1)}{2!}+2=\frac{n^2-n+4}{2}$$

$2A_1 = A_0 + A_2 \Rightarrow 2n = 1 + \dfrac{n^2-n+4}{2}$

$\Rightarrow 4n = 2 + n^2 - n + 4$

$\Rightarrow n^2 - 5n + 6 = 0$

$\Rightarrow (n - 2)(n - 3) = 0$

$\Rightarrow n = 2$ or $n = 3$

11. (*a,c,d*)

$(9+\sqrt{80})^n = I + f$

$(9-\sqrt{80})^n = f'$

$2[{}^nC_0(9)^n + {}^nC_2(9)^{n-2}(\sqrt{80})^2 + \ldots] = I + f + f'$

$\therefore \quad I = 2(\text{integer}) - 1 \quad (\because f + f' = 1)$

$\therefore \quad (I + f)(1 - f) = 1$

12. (*a,d*) Constant term in $P_1(x)$ is 4

If the constant term in $P_k(x)$ is also 4, then

$P_k(x) = 4 + a_1x + a_2x^2 + \ldots$

and $P_{k+1}(x) = (P_k(x) - 2)^2 = (a_1x + a_2x^2 + \ldots + 2)^2$

13. (*a,b,c*)

General term $= \dfrac{10!}{r_1!r_2!r_3!}(1)^{r_1}(2x)^{r_2}(3x^2)^{r_3}$

a_1 = Coeff. of x

$r_2 + 2r_3 = 1 \Rightarrow r_2 = 1, r_1 = 9, r_3 = 0$ or

$\therefore \quad a_1 = \dfrac{10!}{1!9!}(2)^1 = 20$

a_2 = Coeff. of x^2

$r_2 + 2r_3 = 2 \Rightarrow r_2 = 2, r_1 = 8, r_3 = 0$ or

$r_2 = 0, r_1 = 9, r_3 = 1$ or

$$a_2=\frac{10!}{2!8!}(2)^2+\frac{10!}{9!1!}(3)=210$$

a_4 = coeff. of x^4

$r_2 + 2r_3 = 4 \Rightarrow r_2 = 4, r_1 = 6, r_3 = 0$

$r_2 = 2, r_1 = 7, r_3 = 1$

$r_2 = 0, r_1 = 8, r_3 = 2$

$$a_4=\frac{10!}{4!6!}(2)^4+\frac{10!}{2!7!1!}(2)^2(3)+\frac{10!}{8!2!}(3)^2=8085$$

$a_{20} = 3^{10}$

14. (*a*) $\because {}^{20}C_r\,{}^{10}C_0 + {}^{20}C_{r-1}\cdot{}^{10}C_1 + \ldots + {}^{20}C_0\cdot{}^{10}C_r$

= coefficient of x^r in $(1 + x)^{20}(1 + x)^{10}$

= coefficient of $x^r(1 + x)^{30} = {}^{30}C_r$

$\because {}^{30}C_r$ is minimum

$\therefore r = 0$

15. [12] $n = 12$

For T_9 to be the numerically greatest term

$$r=\left[\frac{n+1}{1+\left|\frac{x}{a}\right|}\right]$$

$$= \left[\frac{n+1}{1+\left|\frac{1}{2}\right|}\right] = 8 \Rightarrow 8 < \frac{2(n+1)}{3} < 9$$

$\Rightarrow 11 < n < 12.5 \Rightarrow n = 12$

16. [3]

Let $(6\sqrt{6}+14)^{2n+1} = P = I + F$

where is a positive integer and $0 < F < 1$

clearly $(6\sqrt{6}-14)^{2n+1} = F'$ where $0 < F' < 1$ Subrtacting, we get

$$2[{}^{2n+1}C_1(6\sqrt{6})^{2n} \times 14 + {}^{2n+1}C_3(6\sqrt{6})^{2n-2} \times 14^3 + \ldots..]$$

$= I + F' - F'$

$\therefore \quad I + F' - F' =$ an even integer.

But since $0 < F$ 1 and $0 < F' < 1$, the only possible is that $F - f' = 0$ or $F = F'$.

Also $PF = PF'$

$= (6\sqrt{6}+14)^{2n+1}(6\sqrt{6}-14)^{2n+1}$

$= 20^{2n+1}$.

17. [7]

$$\left({}^{19}c_0(x^3)^{19} + {}^{19}c_1(px^2+2x-5)(x^3)^{18} + \ldots.\right)$$

$$\left({}^{8}c_0x^{16} + {}^{8}c_1(qx-41)x^{14} + \ldots.\right)$$

$$\left({}^{6}c_0x^{24} + {}^{6}c_1(-x^3+x-1)x^{20} + \ldots..\right)$$

$= x^{97} + 391x^{96} + a_{95}x^{95} + \ldots.$

Comparing the coefficient of x^{96} we get

$19p + 8q - 6$

$= 391 \Rightarrow 19p + 8q = 397$

Let $q = 19\lambda + k$, $0 \le k < 19$

$$p = \frac{397 - 8(19\lambda + k)}{19}$$

$$= 21 - 8\lambda - 2\frac{(4k+1)}{19}$$

$\frac{4k+1}{19}$ must be integer $\Rightarrow k = 14$,

$p = 15 - 8\lambda$

For minimum positive value of p, $\lambda = 1$

$\Rightarrow p = 7$

18. [4] $\displaystyle\sum_{r=0}^{n} \frac{{}^{n}c_r \cdot 3^{r+4}}{(r+1)(r+2)(r+3)(r+4)}$

$$= \sum_{r=0}^{n} \frac{{}^{n}c_r \cdot 3^{r+4}}{{}^{r+4}C_4 \cdot 4!}$$

$$= \sum_{r=0}^{n} \frac{n!}{r!(n-r)!} \frac{3^{r+4}}{(r+4)!}.4!$$

$$= \sum_{r=0}^{n} \frac{n!\,3^{r+4}}{(n-r)!(n+4)!}$$

$$= \sum_{r=0}^{n} \frac{n!3^{r+4}}{(n-r)!(n+4)!} \cdot \frac{(n+1)(n+2)(n+3)(n+4)}{(n+1)(n+2)(n+3)(n+4)}$$

$$= \frac{1}{(n+1)(n+2)(n+3)(n+4)}\left[\sum_{r=0}^{n} \frac{(n+4)!3^{r+4}}{(n-r)!(r+4)!}\right]$$

$$\frac{1}{(n+1)(n+2)(n+3)(n+4)}\left\{\sum_{r=0}^{n+4} C_{r+4}.3^{r+4}\right\}$$

pur $r+4 = t$

$$\frac{1}{(n+1)(n+2)(n+3)(n+4)} \left\{\sum_{r=0}^{n} n+4C_r.3^r - \sum_{t=0}^{3} 3^t.{}^{n+4}C_t\right\}$$

$$\frac{1}{(n+1)(n+2)(n+3)(n+4)}\left\{4^{n+4} - \sum_{t=0}^{3} {}^{n+4}C_t.3^t\right\}$$

19. [2] $S = \displaystyle\sum_{i=1}^{n}\sum_{j=1}^{n}(i+j){}^{n}C_i{}^{n}C_j$

S is symmetric function $S(i, j) = S(j, i)$

$$S = \sum_{n=1}^{n}\sum_{j=1}^{n}(n-x+n-1)^{n}c_1{}^{n}C_j$$

$$S = 2n\sum_{\lambda=1}^{n}\sum_{S=1}^{n} n_{C_\lambda}\,{}^{n}C_1 - S$$

$$2S = 2n\sum_{\lambda=1}^{n}\sum_{j=1}^{n} {}^{n}C_i\,{}^{n}C_J$$

$$s = n\sum_{\lambda=1}^{n} n_{c_n}\sum_{j=1}^{n} n_{c_j}$$

$$S = n\cdot\left(2^n\right)^2 = n\cdot 2^{2n}$$

20. [0] When 2222 is divided by 7 it leaves a remainder 3 . So adding & subtracting 3^{5555}, we get

$$E = \underbrace{2222^{5555} - 3^{5555}}_{E_1} + \underbrace{3^{5555} + 555^{2222}}_{E_2}$$

For E_2 : Now since $2222 - 3 = 2019$ is divisible by 7 , therefore E_1 is divisible by 7

($\because x^n - a^n$ is divisible by $x - a$)

For E_1: 5555 when devided by 7 leaves remainder 4. So adding and subtracting 4^{2222}, we get :

$E_2 = 3^{5555} + 4^{2222} + 5555^{2222} - 4^{2222}$

$= (243)^{1111} + (16)^{1111} + (5555)^{2222} - 4^{2222}$

Again $(243)^{1111} + 16^{1111}$ and $(5555)^{2222} + -4^{2222}$ are divisible by 7

($x^n + a^n$ is divisible by $x + a$ when n is odd)

Hence $2222^{5555} + 5555^{2222}$ is divisible by 7.

$2222^{5555} + 5555^{2222} = 7k + r$, then $r = 0$

21. [2] $\displaystyle\sum_{m=0}^{n} {}^{n}C_m.\sin(mix).\cos(n-m)x$

$= {}^{n}C_0 \sin(\alpha x)\cos nx + {}^{n}C_1 \sin x.\cos(n-1)x + {}^{n}C_2.\sin 2x \cos(n-2)x + {}^{n}C_3.\sin 3x.\cos(n-3)x + \ldots + {}^{n}C_n \sin nx.\cos(\alpha x)$ ...(i)

$\Rightarrow 2\displaystyle\sum_{m=0}^{n} {}^{n}C_m.\sin(mx).\cos(n-m)x =$

$[{}^{n}C_0 \sin(\alpha x).\cos nx + {}^{n}C_n.\sin(nx).\cos\alpha x]$

$+ [{}^{n}C_1 \sin x.\cos(n-1)x + nC^{n-1}\sin(n-1)x.\cos x] + \ldots + [C_n.\sin nx.\cos(ox) + {}^{n}C_0.\sin(ox).\cos(nx)]$

Using ${}^{n}C_r = {}^{n}C_{n-r}$; $0 \le r \le n$, for above equation, we get

$$\Rightarrow 2\sum_{m=0}^{n} {}^{n}C_m \cdot \sin(mx)\cdot\cos(n-m)x = {}^{n}C_0$$

$[\sin(ox)\cdot\cos(nx) + \sin(nx)\cdot\cos(ox)] + {}^{n}C_1[\sin x\cdot\cos(n-1)x + \sin(n-1)x\cos x] + \ldots + {}^{n}C_n[\sin(nx)\cos(ox) + \sin(ox)\cdot\cos(nx)]$

Uisng formula $\sin(A+B) = \sin A\cos B + \cos A\sin B$, the R.H.S. becomes

$({}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \ldots + {}^{n}C_n)\sin nx = 2^n \sin nx$

$$\Rightarrow \sum_{m=0}^{n} {}^{n}C_m.\sin(mx).\cos(n-m)x = 2^{n-1}\sin nx$$

22. [4] $32^{(32)^{32}} = 2^{5(3k+1)} = 4(1+7)^{\lambda} = 7\mu + 4$

23. [1] $\displaystyle\sum_{k=0}^{4}\left(\frac{3^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)$

$$= \sum_{k=0}^{4}\left(\frac{4!}{(4-k)!k!}3^{4-k}x^k.\frac{1}{4!}\right)$$

$$= \sum_{k=0}^{4}\frac{{}^{4}C_k.3^{4-k}.x^k}{04!} = \frac{(3+x)^4}{4!}$$

According to the question

$$\frac{(3+x)^4}{4!} = \frac{32}{3}$$

or $(3+x)^4 = 256$

or $x + 3 = 256$

or $x = 1$

24. [0]

$\Rightarrow S = {}^{n}C_0\,{}^{2n}C_n - {}^{n}C_1\,{}^{2n-2}C_n + {}^{n}C_2\,{}^{2n-4}C_n - \ldots$

$\Rightarrow S = {}^{n}C_0\left[\text{coff. of } x^n \text{ in } (1+x)^{2n}\right] - {}^{n}C_1[\text{ coff. of } x^n \text{ in}(1+x)^{2n-2}] + \ldots +$

$${}^{n}C_n\left[\text{coff. of } {}^{n} \text{ in}(1+x)^{2n-2n}\right]$$

$\Rightarrow$ s = coefficient of x^n in

$$\left[{}^{n}C_0(1+x)^{2n} - {}^{n}C_1(1+x)^{2n-2} + {}^{n}C_2(1+x)^{2n+4} + \ldots\right]$$

$$\Rightarrow S = \text{coff. of } x^n \text{ in } \left[(1+x)^2 - 1\right]^n$$

$$\Rightarrow S = \text{ coff. of } x^n \text{ in } \left[2x + x^2\right]^n$$

$$\Rightarrow S = \text{ Coff. of } x^n \text{ in } x^n[x+2]^n$$

So, Coff. of x^n in $x^n[x+2]^n = 2^n$

25. [8016]

$$N = \sum_{r=0}^{2000} r\cdot{}^{2000}c_r$$

$$N = \sum_{r=1}^{2000} r\frac{2000}{r}{}^{1999}C_{r-1}$$

$$N=\sum_{r=1}^{2000}(2000)^{1999}C_{r-1}$$

$$=2000.\,2^{1999}$$

$$=2^4\times5^3\times2^{1999}$$

$$N=2^{2003}\times5^3$$

Number of divisor = (2004)(4) = 8016

26. **[2]** In equaition(1) let us put $x-3=y$ or $x-3=y$ or $x-2=y+1$ and we get

$$\sum_{r=0}^{2n}a_r(1+y)^r=\sum_{r=0}^{2n}b_r(y)^r$$

$$\Rightarrow a_0+a_1(1+y)+....+a_{n-1}(1+y)^{n-1}+(1+y)^n+$$

$$(1+y)^{n+1}+...+(1+y)^{2n}=\sum_{r=0}^{2n}b_r(y)^r$$

$[\text{Using} a_k=1,\forall k\geq n]$

Equating the coefficients of y^n on both the sides, we get

$${}^nC_n+{}^{n+1}C_n+{}^{n+2}C_n+...+{}^{2n}C_n=b_n$$

$$\Rightarrow\left({}^{n+1}C_{n+1}+{}^{n+1}C_n\right)+{}^{n+2}C_n+....+{}^{2n}C_n=b_n$$

$[\text{Using } {}^nC_n={}^{n+1}C_{n+1}=1]$

$$\Rightarrow b_n={}^{n+2}C_{n+1}+{}^{n+2}C_n+...+{}^{2n}C_n$$

$[\text{Using } {}^mC_r+{}^mC_{r-1}={}^{m+1}C_r]$

Combining the terms in similar way, we get

$$b_n={}^{2n}C_{n+1}+{}^{2n}C_n$$

$$\Rightarrow b_n={}^{2n+1}C_{n+1}$$

27. **[4]** We have

$$\sum_{r=0}^{n}\frac{r}{{}^nC_r}=\sum_{r=0}^{n}\frac{n-r}{{}^nC_{n-r}}=\sum_{r=0}^{\infty}\frac{n-r}{{}^nC_r}$$

$$\Rightarrow\sum_{r=0}^{n}\frac{2r}{{}^nC_r}=\sum_{r=0}^{n}\frac{n}{{}^nC_r}\Rightarrow n=n^2-3n+3$$

$n=1,3$

28. **(c)** $(1+i)^n=C_0+C_1i+C_2i^2+C_3i^3+...$

$$\Rightarrow (1+i)^n=(C_0-C_2+C_4-C_6+...)$$
$$+i(C_1-C_3+C_5-....)\quad ...(i)$$

$\Rightarrow C_0-C_2+C_4-C_6+...=$ Real part of $(1+i)^n$

$$=\text{Real part of}\left(\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\right)^n$$

$$=\text{Real part of } 2^{n/2}\left(\cos\frac{n\pi}{4}+i\sin\frac{n\pi}{4}\right)=2^{n/2}\cos\frac{n\pi}{4}.$$

29. **(b)** From the previous question on taking modulus and squaring of equation ... (i)

$$(C_0-C_2+C_4-...)^2+(C_1-C_3+C_5-...)^2=((\sqrt{2})^n)^2=2^n$$

30. **(c)** Add the expansion of $(1+1)^n,(1+\omega)^n$ and $(1+\omega^2)^n$

$$(1+\omega)^n=\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^n=\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)^n$$

$$(1+\omega^2)^n=\left(\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)^n=\left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)^n$$

$$C_0+C_3+C_6+...=\frac{1}{3}\left(2^4+2\cos\frac{n\pi}{3}\right)$$

31. **(a)** $\sum\sum_{0\leq i\leq j\leq n}\left(C_i+C_j\right)^2 i=0,1,2,....,(n-1)$

Where j = 1, 2, 3, 4, ... n and $i<j$

$$n\left(C_0^2+C_1^2+...+C_n^2\right)+2\sum\sum_{0\leq i\leq j\leq n}C_iC_j$$

$$=n\,{}^{2n}C_n+$$

$$\left[\left(C_0+C_1+...+C_n\right)^2-\left(C_0^2+C_1^2+...+C_n^2\right)\right]$$

$$=n.{}^{2n}C_n+(2n)2^{2n}C_n=(n-1).{}^{2n}C_n+2^{2n}.$$

(b) $\sum\sum_{0\leq i<j\leq n}{}^nC_i.{}^nC_j=\frac{1}{2}\left[2^{2n}-{}^{2n}C_n\right]$

(c) $\frac{1}{n}\sum\Sigma(i+j)({}^nC_i.{}^nC_j)$

$$=\sum\sum_{0\leq i<j\leq n}{}^nC_i.{}^nC_j$$

(d) $\sum\sum_{0\leq i<j\leq n}i\,{}^nC_i.j\,{}^nC_j=n^2\,2^{n-1}$

PYQ's (Past Year Questions)

1. **(b)** $\frac{T_5}{T_5'}=\frac{{}^nC_4\,2^{\frac{n-4}{4}}\cdot\left(3^{\frac{-1}{4}}\right)^4}{{}^nC_4\,3^{-\left(\frac{n-4}{4}\right)}\cdot\left(2^{\frac{1}{4}}\right)^4}=\frac{\sqrt{6}}{1}$

$$\Rightarrow 2^{\frac{n-8}{4}}.(3^{1/4})^{n-8}=\sqrt{6}$$

$$\Rightarrow\frac{n-8}{4}=\frac{1}{2}$$

$$\Rightarrow n=10$$

So, $T_3={}^{10}C_2\,2^{\frac{1}{4}8}\cdot3^{\frac{1}{4}2}=\frac{45.4}{\sqrt{3}}=60\sqrt{3}$

2. **(a)** $25^{190}-19^{190}-8^{190}+2^{190}$

$$=25^{190}-8^{190}+2^{190}-19^{190}$$

$a^{2m}-b^{2m}$ is divisible by $(a-b)$

So, whole expression is divisible by 17

And $25^{190}-19^{190}=$ Even

$$2^{190}-8^{190}=\text{Even}$$

Hence, expression is also divisible by 2

Hence, expression is divisible by $17\times2=34$

3. **[29]** $(21+2)^{200}+(21-2)^{200}$

$$\Rightarrow 2\left[{}^{100}C_0\,21^{200}+200C_2\,21^{198}.2^2+.....+{}^{200}C_{198}\right.$$

$$\left.21^2.2^{198}+2^{200}\right]$$

$$\Rightarrow 2\left[49I_1+2^{200}\right]=49I_1+2^{201}$$

Now,

$$2^{201}=(8)^{67}=(1+7)^{67}=49I_2+{}^{67}C_0+{}^{67}C_1.7$$

$$=49I_2+470=49I_2+49\times9+29$$

$\therefore$ Remainder is 29

4. **[98]** In $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^l}\right)^9$ general term T_{r+1} is

$$T_{r+1}={}^9C_r\frac{\left(x^{5/2}\right)^{9-r}}{2^{9-r}}\left(\frac{-4}{x^l}\right)^r$$

$$=(-1)^r\frac{{}^9C_r}{2^{9-r}}4^r\,x^{\frac{45}{2}-\frac{5r}{2}-lr}$$

$$=45-5r-2lr=0$$

$$r=\frac{45}{5+2l}\quad ...(i)$$

Now, according to the question,

$$(-1)^r\frac{{}^9C_r}{2^{9-r}}4^r=-84$$

$$=(-1)^r\,{}^9C_r\,2^{3r-9}=-21\times4$$

Only natural value of r possible if $3r-9=0$

$\Rightarrow r=3$ and ${}^9C_3=84$

$\therefore l=5$ from equation (i)

Now, coefficient of $x^{-3l}=x^{\frac{45}{2}-\frac{5r}{2}-lr}$ at $l=5$, gives $r=5$

$$\therefore{}^9C_5\left(-1\right)\frac{4^5}{2^4}=2^\alpha\times\beta$$

$$=-63\times2^7\text{ compare with }2^\alpha\times\beta$$

$\Rightarrow\alpha=7,\beta=-63$

$\therefore$ value of $|\alpha l-\beta|=98$

5. **[221]** $\sum_{K=1}^{10}K^2({}^{10}C_K)^2=\sum_{k=1}^{10}(K^{10}C_k)^2$

$$=\sum_{k=1}^{10}(10^9C_{k-1})^2$$

$$100({}^{18}C_9)=100\left(\frac{18!}{9!9!}\right)=4862000$$

$$=22000\times221$$

$\therefore\quad L=221$

6. **[23]** Coefficient of x in $(1+x)^p(1-x)^q$

$$={}^pC_0\,{}^qC_1+{}^pC_1{}^qC_0=-3\Rightarrow p-q=1-3$$

$\Rightarrow p=q-3$

Coefficient of x^2 in $(1+x)^p(1-x)^q={}^pC_0$

$${}^qC_2-{}^pC_1{}^qC_1-{}^pC_2\,{}^qC_0=-5$$

$$\Rightarrow\frac{q(q-1)}{2}-pq+\frac{p(p-1)}{2}=-5$$

$$\Rightarrow\frac{q(q-1)}{2}-(q-3)q+\frac{(q-3)(q-4)}{2}=-5$$

$\Rightarrow -2q + 12 = -20 \Rightarrow q = 11, p = 8$

Coefficient of x^3 in $(1+x)^8 (1-x)^{11} = {}^{11}C_3 + {}^8C_1 {}^{11}C_2 - {}^8C_2 {}^{11}C_1 + {}^8C_3 = 23$

7. *(c)* $7^{2022} + 3^{2022}$

$= (49)^{1011} + (9)^{1011}$

$= (50-1)^{1011} + (10-1)^{1011}$

$= k_1 - 1 + 5k_2 - 1 = 5m - 2$

Remainder $= 5 - 2 = 3$

8. *(b)* $(1+x+2x^2)^{20} = a_0 + a_1 x + \ldots + a_{40} x^{40}$

put $x = 1$

$a_0 + a_1 + \ldots a_{40} = 4^{20}$

put $x = -1$

$a_0 - a_1 + a_2 + \ldots a_{40} = 2^{20}$

$\Rightarrow a_1 + a_3 + \ldots a_{37} = 2^{39} - 2^{19} - a_{39}$

$a_{39} = 20 \times 2^{19}$

$\Rightarrow a_1 + a_3 + \ldots + a_{37} = 2^{19}(2^{20} - 21)$

9. [45] Given

$\because 7^n = (10-3)^n = 10K + (-3)^n$

$\therefore 7^n + 3^n = 10K + (-3)^n + 3^n$

$\rightarrow 10K$ if n = odd

$\rightarrow 10K + 2.3^n$ if n = even

Let $n = 2t, t \in N$

$\therefore 3^n = 3^{2t} = (10-1)^t$

$= 10p + (-1)^t$

$= 10p \pm 1$

$\therefore$ If n = even then $7^n + 3^n$ will be multiply of 10

So if n is odd then only $7^n + 3^n$ will be multiply of 10

$\therefore n = 11, 13, 15, \ldots, 99$

$\therefore 45$

10. [210] $\left[\dfrac{x+1}{x^{2/3} - x^{1/3} + 1} - \dfrac{x-1}{\sqrt{x}(\sqrt{x}-1)}\right]^{10}$

$= \left(\left(x^{1/3}+1\right) - \left(\dfrac{\sqrt{x}+1}{\sqrt{x}}\right)\right)^{10}$

$= (x^{1/3} - x^{-1/2})^{10}$

$\therefore T_{r+1} = {}^{10}C_r (x^{1/3})^{10-r} (-x^{-1/2})^r$

For independent term,

$\dfrac{10-r}{3} - \dfrac{r}{2} = 0 \Rightarrow 20 - 2r - 3r = 0$

$\Rightarrow r = 4$

$T_5 = {}^{10}C_4 = \dfrac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$

11. [98] ${}^nC_0 + 3 \cdot {}^nC_1 + 5 \cdot {}^nC_2 + 7 \cdot {}^nC_3 + \ldots + (2n+1)\,{}^nC_n$

$\therefore T_r = (2r+1)\,{}^nC_r$

$S = \sum_{r=0}^{n} (2r+1)\,{}^nC_r$

$= \sum_{r=0}^{n} 2 \cdot r \cdot {}^nC_r + \sum_{r=0}^{n} {}^nC_r$

$= 2\sum_{r=0}^{n} r \cdot \dfrac{n}{r} \cdot {}^{n-1}C_{r-1} + \sum_{r=0}^{n} {}^nC_r$

$= 2n.(2^{n-1}) + 2^n$

$= 2^n (n+1)$

Now,

$2n(n+1) = 2^{100}.101$

$\Rightarrow n = 100$

$\therefore 2\left[\dfrac{n-1}{2}\right] = 98$

12. [315] $a^{10}b^{10}\left(\dfrac{1}{b} + \dfrac{2}{a} + 4\right)^{10}$

$\dfrac{10!\left(\dfrac{1}{b}\right)^{r_1}\left(\dfrac{2}{a}\right)^{r_2} \cdot 4^{10-r_1-r_2}}{r_1! r_2! (10-r_1-r_2)!}$

$r_1 = 2, r_2 = 3$

$a^7 b^8$ is $\dfrac{10! 2^3 \cdot 4^{10-2-3}}{2!3!5!} = \dfrac{2^{13} \cdot 10!}{2!3!5!} = 2^{16} \cdot 315$

$K = 315$

13. *(c)* $\left(3^{1/2} + 5^{1/8}\right)^n$

Let $T_{r+1} = {}^nC_r (3)^{\frac{n-r}{2}} (5)^{\frac{r}{8}}$

So, r must be 0, 8, 16, 24 ...

Now $n = t_{33} = 0 + 32 \times 8 = 256$

$\Rightarrow n = 256$

14. *(b)* $\dfrac{(1+x)^{10}\left[1 - \left(\dfrac{x}{1+x}\right)^{11}\right]}{\left(1 - \dfrac{x}{1+x}\right)}$

= Sum of G.P.

$= \dfrac{(1+x)^{10}\left[(1+x)^{11} - x^{11}\right]}{(1+x)^{11} \times \dfrac{1}{(1+x)}} = (1+x)^{11} - x^{11}$

Coefficient of x^7 is ${}^{11}C_7 = {}^{11}C_4 = 330$

15. *(d)* General term

$T_{r+1} = {}^{60}C_r, 7^{\frac{60-r}{5}} 3^{\frac{r}{10}} (-1)^r$

$\therefore$ for rational term, r = 0, 10, 20, 30, 40, 50, 60

$\Rightarrow$ number of rational terms = 7

$\therefore$ number of irrational terms = 54 [total term = 61]

16. *(c)*

$\dfrac{5^{\text{th}} \text{ term from begining}}{5^{\text{th}} \text{ term from end}} = \dfrac{{}^{10}C_4 \left(\dfrac{1}{2(3^{1/3})}\right)^4 2^{6/3}}{{}^{10}C_4 (2)^{4/3} \left(\dfrac{1}{2(3^{1/3})}\right)^6}$

$= \dfrac{2^2 3^{-4/3}}{2^4 2^{(4/3)-6} 3^{-2}} = 3^{2/3}.2^{8/3} = 4.(36)^{1/3}$

17. *(c)* $(10+x)^{50} + (10-x)^{50}$

$a_0 = (10^{50})(2)$

$a_2 = {}^{50}C_2 (10)^{48} (2)$

So, $\dfrac{a_2}{a_0} = \dfrac{{}^{50}C_2 (10)^{48} (2)}{10^{50}(2)} = 12.25$

18. *(d)* $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots + x^{500}$ are in G.P

$= (1+x)^{500} \cdot \left\{\dfrac{1 - \left(\dfrac{x}{1+x}\right)^{501}}{1 - \dfrac{x}{1+x}}\right\}$

$= (1+x)^{500} \dfrac{\left((1+x)^{501} - x^{501}\right)}{(1+x)^{501}} \cdot (1+x)$

Coefficient of x^{301} in $(1+x)^{501} - x^{501}$ is given by ${}^{501}C_{301} = {}^{501}C_{200}$

19. *(a)* $\sum_{k=1}^{31} ({}^{31}C_k)({}^{31}C_{k-1}) = \sum_{k=1}^{31} ({}^{31}C_{31-k})$

$({}^{31}C_{k-1}) = {}^{62}C_{30}$

and $\sum_{k=1}^{30} ({}^{30}C_k)({}^{30}C_{k-1}) = \sum_{k=1}^{30} ({}^{30}C_{30-k})$

$({}^{30}C_{k-1}) = {}^{60}C_{29}$

Now, ${}^{62}C_{30} - {}^{60}C_{29} = \dfrac{62!}{30!32!} - \dfrac{60!}{29!31!}$

$= \dfrac{60!}{29!31!}\left\{\dfrac{62.61}{30.32} - 1\right\}$

$= \dfrac{60!}{30!31}\left\{\dfrac{2822}{32}\right\}$

$\therefore \quad 16\alpha = 16 \times \dfrac{2822}{32} = 1411$

20. *(a)* $E = (5+x)^{500} + x(5+x)^{499} + x^2(5+x)^{498} + \ldots + x^{500}$

$= \dfrac{(5+x)^{501} - x^{501}}{(5+x) - x} = \dfrac{(5+x)^{501} - x^{501}}{5}$

$\therefore$ Coefficient of x^{101} in the expression

$= \dfrac{{}^{501}C_{101} 5^{400}}{5} = {}^{501}C_{101} 5^{399}$

21. [83] ${}^{10}C_r (2x^3)^{10-r} \left(\dfrac{3}{x}\right)^r$

$= {}^{10}C_r \cdot 2^{10-r} \cdot 3^r \cdot x^{30-3r-r}$

For $x^{30-4r} \rightarrow x^{(\text{even})}$

$r = 0,1,2,3,4,5,6,7$

But $r \in \{0,1,2,\ldots,10\}$

Sum of all such coefficient

$= 5^{10} - ({}^{10}C_8 \cdot 2^2 \cdot 3^8 + {}^{10}C_9 \cdot 2^1 \cdot 3^9 + {}^{10}C_{10} 2^0 \cdot 3^{10})$

$= 5^{10} - (45 \times 4 \times 3^8 + 20 \times 3^9 + 3^{10})$

$= 5^{10} - 3^9 \cdot (60 + 20 + 3) = 5^{10} - 3^9 (83)$

So, $\beta = 83$.

22. (*d*) $T_{r+1} = {}^{10}C_r (tx^{1/5})^{(10-r)} \left[\frac{(1-x)^{1/10}}{t}\right]^r$

$= {}^{10}C_r t^{(10-2r)} \times x^{\frac{10-r}{5}} \times (1-x)^{\frac{r}{10}}$

$\Rightarrow 10 - 2r = 0 \Rightarrow r = 5$

$T_6 = {}^{10}C_5 \sqrt{1-x} \cdot x$

$\frac{dT_6}{dx} = {}^{10}C_5 \left[\sqrt{1-x} - \frac{x}{2\sqrt{1-x}}\right] = 0$

$= 1 - x = x/2 \Rightarrow 3x = 2$

$\Rightarrow x = 2/3$

$(T_6)_{\max} = \frac{10!}{5!5!} \times \frac{2}{3\sqrt{3}}$

23. **[12]** $(1+x)^{10} = {}^{10}C_0 + {}^{10}C_1 x + {}^{10}C_2 x^2 + \ldots + {}^{10}C_{10} x^{10}$

$(1+x)^{15} = {}^{15}C_0 + {}^{15}C_1 x + \ldots {}^{15}C_{k-1} x^{k-1} + {}^{15}C_k x^k + {}^{15}C_{k+1} x^{k+1} + \ldots . {}^{15}C_{15} x^{15}$

$\sum_{i=0}^{k} (10C_i)(15C_{k-i}) = {}^{10}C_0 \cdot {}^{15}C_k + {}^{10}C_1 \cdot {}^{15}C_{k-1} + \ldots + {}^{10}Ck \cdot {}^{15}C_0$

Coefficient of x_k in $(1+x)^{25} = {}^{25}C_k$

$\sum_{i=0}^{k+1} ({}^{12}C_i)({}^{13}C_{k+1-i})$

$= {}^{12}C_0 \cdot {}^{13}C_{k+1} + {}^{12}C_1 \cdot {}^{13}C_k + \ldots {}^{12}C_{k+1} \cdot {}^{13}C_0$

Coefficient of x^{k+1} in $(1+x)^{25} = {}^{25}C_{k+1}$

${}^{25}C_k + {}^{25}C_{k+1} = {}^{26}C_{k+1}$

For maximum value

$k + 1 = 13$

$k = 12$

24. **[6]** $A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k\right]$

$A = \left(1 - \frac{1}{2}\right)^n + \left(1 - \frac{3}{4}\right)^n + \left(1 - \frac{7}{8}\right)^n + \left(1 - \frac{15}{16}\right)^n + \left(1 - \frac{31}{32}\right)^n$

$= \frac{1}{2^n}\left(\frac{1 - \left(\frac{1}{2^n}\right)^5}{1 - \frac{1}{2^n}}\right) \Rightarrow A = \frac{\left(1 - \frac{1}{2^{5n}}\right)}{\left(2^n - 1\right)}$

$\Rightarrow (2^n - 1) A = 1 - \frac{1}{2^{5n}}$, but Given $63A = 1 - \frac{1}{2^{30}}$

Clearly $5n = 30$

$\Rightarrow n = 6$

25. **[30]**

Let $(1 - x + x^2 \ldots)(1 + x + x^2 + \ldots)$

$= a_0 + a_1 x + a_2 x^2 + \ldots$

Put $x = 1$

$1(2n+1) = a_0 + a_1 + a_2 + \ldots + a_{2n} \quad \ldots (i)$

Put $x = -1$

$(2n+1)\,1 = a_0 - a_1 + a_2 + \ldots + a_{2n} \ldots (ii)$

From $(i) + (ii)$,

$4n + 2 = 2(a_0 + a_2 + \ldots) = 2 \times 61$

$\Rightarrow 2n + 1 = 61 \Rightarrow n = 30.$

26. **(*a,b,d*)**

Solving

$f(m, n, p) = \sum_{i=0}^{p} {}^mC_i \cdot {}^{n+i}C_p \cdot {}^{p+n}C_{p-i}$

${}^mC_i \cdot {}^{n+i}C_p \cdot {}^{p+n}C_{p-i}$

${}^mC_i \cdot \frac{(n+i)!}{p!(n-p+i)!} \times \frac{(n+p)!}{(p-i)!(n+i)!}$

${}^mC_i \times \frac{(n+p)!}{p!} \times \frac{1}{(n-p+i)!(p-i)!}$

${}^mC_i \times \frac{(n+p)!}{p!n!} \times \frac{n!}{(n-p+i)!(p-i)!}$

${}^mC_i \cdot {}^{n+p}C_p \cdot {}^nC_{p-i} \quad \{{}^mC_i \cdot {}^nC_{p-i} = {}^{m+n}C_p\}$

$f(m, n, p) = {}^{n+p}C_p \cdot {}^{m+n}C_p$

$\frac{f(m,n,p)}{{}^{n+p}C_p} = {}^{m+n}C_p$

Now

$g(m, n) = \sum_{p=0}^{m+n} \frac{f(m,n,p)}{{}^{n+p}C_p}$

$g(m, n) = \sum_{p=0}^{m+n} {}^{m+n}C_p$

$g(m, n) = 2^{m+n}$

(*a*) $g(m, n) = q(n, m)$

(*b*) $g(m, n+1) = 2^{m+n+1}$

$g(m+n, n) = 2^{m+1+n}$

(*d*) $g(2m, 2n) = 2^{2m+2n}$

$= (2^{m+n})^2 = (g(m, n))^2$

27. (*d*) $\sum_{i=1}^{20} \left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}}\right)^3$

Now $\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} = \frac{{}^{20}C_{i-1}}{{}^{21}C_i} = \frac{i}{21}$

$\left[\because {}^{n+1}C_{r+1} = \frac{n+1}{r+1} {}^nC_r\right]$

Let given sum be S, so

$S = \sum_{i=1}^{20} \frac{(i)^3}{21^3} = \frac{1}{(21)^3}\left(\frac{20.21}{2}\right)^2 = \frac{100}{21}$

$\left[\because \sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2\right]$

Given $S = \frac{k}{21} \Rightarrow k = 100$

28. (*d*) $\sum_{r=0}^{25} \frac{|50}{|r|50-r} \times \frac{|50-r}{|25-r|25}$

$\left[\because {}^nC_r = \frac{|n}{|r\,|n-r}\right]$

$= \sum_{r=0}^{25} \frac{|50}{|r|25-r\,|25}$

$= \frac{|50}{|25} \sum_{r=0}^{25} \frac{1}{|r|25-r}$

$= \frac{50}{|25|25} \sum_{r=0}^{25} {}^{25}C_r = {}^{50}C_{25}(2^{25}).$

29. (*d*) $2.{}^{20}C_0 + 5.{}^{20}C_1 + 8.{}^{20}C_2 + 11.{}^{20}C_3 + \ldots + 62.{}^{20}C_{20}$

$= \sum_{r=0}^{20} (3r+2)\,{}^{20}C_r$

$= 3\sum_{r=0}^{20} r\,{}^{20}C_r + 2\sum_{r=0}^{20} {}^{20}C_r$

$= 3\sum_{r=1}^{20} r\left(\frac{20}{r}\right){}^{19}C_{r-1} + 2 \cdot 2^{20}$

$= 60.2^{19} + 2.2^{20} = 2^{25}$

30. **[6.20]**

Suppose

$\begin{vmatrix} \frac{n(n+1)}{2} & n(n-1).2^{n-2} + n.2^{n-1} \\ n.2^{n-1} & 4^n \end{vmatrix} = 0$

$\frac{n(n+1)}{2} \cdot 4^n - n^2(n-1).2^{2n-3} - n^2 2^{2n-1} = 0$

$\frac{n(n+1)}{2} - \frac{n^2(n-1)}{8} - \frac{n^2}{4} = 0$

$n^2 - 3n - 4 = 0$

$n = 4$

Now $\sum_{k=0}^{4} \frac{{}^4C_k}{k+1} = \sum_{k=0}^{4} \frac{k+1}{5} \cdot {}^5C_{k+1} \frac{1}{k+1}$

$= \frac{1}{5} \cdot [{}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5]$

$= \frac{1}{5}[2^5 - 1]$

$= \frac{31}{5} = 6.20$

By Aditya Mahod
username - muje_kabhi_dekha_nahi

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 9

Permutations and Combinations

Prarambh (Topicwise)

1. (*b*) As per the given condition, digit 1 should occur at alternate places of the number and at the remaining 5 places either 2, 3, 5 or 7 should appear. Now when the number starts with 1, number of numbers = 4^5 and when the number starts with either 2, 3, 5 or 7, number of numbers = 4^5

So, total number = $2 \times 4^5 = 2048$

2. (*d*) 1 . (2) . 3 . (4) . 5 . (6) . 7

T					T

$(3 \times 2) \times (5 \times 4 \times 3 \times 2) = 6 \times 120 = 720$

3. (*a*) $1 + 2 + 3 + ... + 9 = 45 = 0 + 1 + 2 + 3 + ... + 9$

All 9 digit such numbers = 9 !

All 10 digit such numbers when '0' included = 10! – 9!

So, total = 9 ! + (10 ! – 9 !) = (10) !

4. (*c*) Total number of signals can be made from 3 flags each of different colour by hoisting 1 or 2 or 3 above.

i.e. $3 + 3 \times 2 + 3 \times 2 \times 1 = 3 + 6 + 6 = 15$

5. (*c*) Word QUEUE

E → 2, Q , U - 2

E					= 12

$\frac{4!}{2!}$

Q	E				– 3

$\frac{3!}{2!}$

Q	U	E	E	U	= 1

Q	U	E	U	E	= 1

17th rank

6. (*d*) Total number of possible arrangements is

${}^4P_2 \times {}^6P_3$.

7. (*d*) First we have to find all the arrangements of the word 'GENIUS' is

6 ! = 720

number of arrangement which in either started with G ends with S is

(5! + 5! – 4!) = (120 + 120 – 24) = 216

Hence total number of arrangement which is neither started with G nor ends with S is.

(720 – 216) = 504

8. (*a*) Total no. of arrangement if all the girls do not sit side by side is = [all arrangement – girls seat side by side]

= 8! – (6! × 3!) = 6! (56 –6) = 6! × 50 = 720 × 50

= 36000

9. (*a*) Number of words which have at least one letter repeated = total words – number of words which have no letter repeated = $10^5 - 10 \times 9 \times 8 \times 7 \times 6 = 69760$

10. (*d*) First we select 3 speaker out of 10 speaker and put in any way and rest are no restriction i.e. total number of ways = ${}^{10}C_3 \cdot 7! = \frac{10!}{3!}$

11. (*b*) upperdeck - 13 seats → 8 in upper deck.

lowerdeck - 7 seats → 5 in lower deck

Remains passengers = 7

Now Remains 5 seats in upper deck and 2 seats in lower deck

for upper deck number of ways = ${}^7C_5$

for lower deck number of ways = ${}^2C_2$

So total number of ways = ${}^7C_5 \times {}^2C_2 = \frac{7.6}{2} 21$

12. (*d*) Even place

E E E E

There are four even places and four odd digit number so total number of filling is $\frac{4!}{2!.2!}$ rest are also occupy is $\frac{5!}{3!.2!}$ ways

Hence total number of ways

$= \frac{4!}{2!.2!} \times \frac{5!}{3!.2!} = 60$

13. (*b*)

T										T

Total arrangement is $\frac{9!}{2!.2!} = 90720$

14. (*c*) NINETEEN

⇒ $N \rightarrow 3$: I, T

$E \rightarrow 3$

First we arrange the word of N, N, N, I and T

then the number of way = $\frac{5!}{3!}$.

Now total 6 number of place which are arrange E is ${}^6C_3$

Hence total number of ways = $\frac{5!}{3!} \cdot {}^6C_3$

15. (*c*) Total number of ways is

$\frac{6! \times 3!}{2!} = 720 \times 3 = 2160$

16. (*a*) First we select 5 beads from 8 different beads to ${}^8C_5$

Now total number of arrangement is

${}^8C_5 \times \frac{4!}{2!} = 672$

17. (*a*) n men; n woman

$(n-1)!\, n!$ = Alternate

Couples = $(n-1)! \times {}^2C_1$

Total ways = $(n-1)!\, n! - 2.(n-1)!$

18. (*a*) 6 men and 5 women

Arrangement of remaining 5 men = 5!

Out of 6 place, 5 place select and arrange

$= {}^6C_5 \times 5! = {}^6C_5$

So, no. of ways = $5! \times {}^6C_5 = 5! \times 6!$

19. (*b*) The no. of ways of selecting 3 out of 7 consonants = ${}^7C_3 = 35$

The no. of ways of selecting 2 out of 4 vowels = ${}^6C_5$

Now these five letters after selection can be arranged among themselves in ${}^5C_5$ i.e., 120 ways.

Hence the required no. of words = $35 \times 6 \times 120 = 25200$.

20. (*a*) The number of points of intersection of 37 straight lines is ${}^{37}C_2$. But 13 of

them pass through the point A. Therefore instead of getting ${}^{13}C_2$ points we get merely one point.

Similary 11 straight lines out of the given 37 straight lines intersect at B. Therefore instead of getting ${}^{11}C_2$ points, we get only one point. Hence, the number of intersection points of the lines is

${}^{37}C_2 - {}^{13}C_2 - {}^{11}C_2 + 2 = 535$.

21. (*b*) We can select 5 members for the committee to include at least one lady in the following four ways:

(1) 1 lady and 4 gentlemen

(2) 2 ladies and 3 gentlemen

(3) 3 ladies and 2 gentlemen

(4) 4 ladies and 1 gentlemen

Hence required number of committees

$= {}^4C_1 \times {}^6C_4 + {}^4C_2 \times {}^6C_3 + {}^4C_3 \times {}^6C_2 + {}^4C_4 \times {}^6C_1$

$= 60 + 120 + 60 + 6 = 246.$

22. (*c*) (A) ${}^nC_2 - 8 = 28 - 8 = 20$

(B) ${}^nC_2 - n = 54$

$\frac{n(n-1)}{2} - n = 54$

$n(n-3) = 108$

$n = 12$

(C) ${}^nC_4 = 70 \; \frac{n(n-1)(n-2)(n-3)}{24} = 70$

$n(n-1)(n-2)(n-3) = 8 \times 7 \times 6 \times 5$

$n = 8$

Diagonals $= {}^8C_2 - 8 = 28 - 8 = 20$

23. (*d*) Total ways $= (10 + 1)(9 + 1)(7 + 1) - 1$

$= (11)(10)(8) - 1$

$= 880 - 1 = 879$

24. (*c*) Total ways $= {}^9C_5 - {}^7C_3$

$= 126 - 35$

$= 91$

25. (*a*) Word is PROPOSITION

Here P's = 2, O's = 3, I's = 2 and T, R, N, S = 1

We have to made 5 letters words

Case I : When all 5 are different

$= {}^7P_5 = \frac{7!}{2!} = 2520$

Case II : When 2 alike, 3 different

$= {}^3C_1 \times {}^6C_3 \times \frac{5!}{2!} = 360$

Case III : When 2 alike, 2 alike, 1 different

$= {}^3C_1 \times {}^5C_1 \times \frac{5!}{2!2!}$

Case IV : When 3 alike, 2 different

$= {}^1C_1 \times {}^6C_1 \times \frac{5!}{3!} = 300$

Case V : When 3 alike, 2 alike

$= {}^1C_1 \times {}^2C_1 \times \frac{5!}{3!2!} = 20$

So total required number = 6890

26. (*d*) Total number of proper divisors is

$(p + 1)(q + 1)(r + 1)(s + 1) - 2$

(Number and 1 are not proper divisor)

27. (*a*) $N = 2^\alpha . 3^\beta 5^\gamma = 2^3 . 3^2 . 5$

$(\alpha + 1)(\beta + 1)(\gamma + 1) = 4.3.2$

$N = 360 = 2^3 . 3^2 . 5$

$\frac{4.3.2}{2} = 12$

28. (*a*) Here $21600 = 2^5 . 3^3 . 5^2$

$\Rightarrow (2 \times 5) \times 2^4 \times 3^3 \times 5^1$

Now numbers which are divisible by

$10 = (4 + 1)(3 + 1)(1 + 1) = 40$

$(2 \times 3 \times 5) \times (2^4 \times 3^2 \times 5^1)$ now numbers which are divisible by both 10 and 15

$= (4 + 1)(2 + 1)(1 + 1) = 30$

So the numbers which are divisible by only

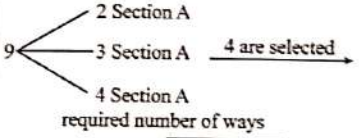
$40 - 30 = 10$

29. (*c*) They can sit in groups of either 5 and 3 or 4 and 4.

required number

$= \frac{8!}{5! \times 3!} \times 1 + \frac{8! \times 2!}{4! \times 4! \times 2!} = 126$

30. (*c*)



required number of ways

A	B	C
2	1	2
1	1	2
1	2	1

Hence ${}^2C_1 \cdot {}^3C_1 \cdot {}^4C_2 + {}^2C_1 \cdot {}^3C_2 \cdot {}^4C_1 + {}^2C_2 \cdot {}^3C_1 \cdot {}^4C_1 = 36 + 24 + 12 = 72$

31. (*d*) Peaches 5 p_1, p_2, p_3, p_4, p_5

Apples 3 a_1, a_2, a_3

Hence number of ways $= {}^3C_1 \times {}^5C_3 = 30$

32. (*a*) Coefficient x^{10} in $(x + x^2 \ldots x^5)^6 =$ coefficient of x^4 in $(x^0 + x^1 \ldots x^4)^6 = {}^{6+4-1}C_4 = {}^9C_4 = 126$

Alternatively: Give one apple to each child and then for rest 4 apples $= {}^{4+6-1}C_{6-1} = 126$

33. (*c*) $(x + y + z)^n \rightarrow$ use beggar] ${}^{n+2}C_2$

34. (*b*) $3A + 2\,O.A. = 3 . 2 = 6$; $3A + 2$ diff $= 3$;

$2A + 2\,O.A. + 1D = 3 \Rightarrow 12$

35. (*b*) $x_1 + x_2 + x_3 + x_4 = 16$

$x_1 \geq 3, x_2 \geq 3, x_3 \geq 3, x_4 \geq 4$

$a + 3 + b + 3 + c + 3 + d + 3 = 16$

$a + b + c + d = 4$

$a \geq 0, b \geq 0, c \geq 0, d \geq 0$

Total ways $= {}^{4+4-1}C_4$

$= {}^7C_4$

$= 35$

36. (*c*) If 1 be unit digit then total no. of number is $3! = 6$

Similarly so on if 3, 5, or 7 be unit digit number

then total no. of no. is $3! = 6$

Hence sum of all unit digit no. is

$= 6 \times (1 + 3 + 5 + 7) = 6 \times 16 = 96$

Hence total sum is $= 96 \times 10^3 + 96 \times 10^2 + 96 \times 10^1 + 96 \times 10^0$

$= 96000 + 9600 + 960 + 96 = 106656 = 16 \times 1111 \times 3!$

37. (*b*) Digits = 2,3,0,3,4,2,5

numbers greater than 1 million

$= \frac{6 \times \lfloor 6}{\lfloor 2 \times \lfloor 2} = 1080$

Numbers that are divisible by 5 & greater than 1 million

.....0 $\frac{\lfloor 6}{\lfloor 2 \lfloor 2} = 180$

.....5 $\frac{5 \times \lfloor 5}{\lfloor 2 \lfloor 2} = 150$

Total = 330

38. (*b*) ${}^6C_2 \times {}^4C_2 \times {}^2C_2 = 15 \times 6 = 90$

39. (*a*) $\frac{{}^{12}C_2 \times {}^{10}C_5 \times {}^5C_5}{12}$

$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6}{L^2 \times L^5 \times L^2}$

$= \frac{12!}{5!5!2!2!}$

40. (*c*) $6p_2 = 30$

41. (*b*) $(3)^6 = 729 < 900$ and $(3)^7$

$= 2187 > 900$

42. (*a*) Number of terms = Number of non-negative integral solution of the equation

$p + q + r + s = 50$

$= {}^{50+4-1}C_{50} = {}^{53}C_{50} = {}^{53}C_3$

43. (*a*) $x + y + 3 = 102$

$x, y, z > 0$

$(x-1) + (y-1) + (z-1) = 99$

$x - 1 = a;\quad y - 1 = b;\quad 2 - 1 = c,$

$a, b, c \geq 0$

$= a + b + c = 99$

Total solutions ${}^{101}C_2 = 5050$

44. (a) $\Rightarrow n(A\cup B\cup C)=n(A)+n(B)+n(C)$
$-n(A\cap B)-n(B\cap C)$
$-n(A\cap C)+n(A\cap B\cap C)$
$=3|9|2-3|8|2|2+|7|2|2|2$
Required number
$=|10-(3|9|2-3|8|2|2+|7|2|2|2)$
$=|8[90-(54-12+1)]$
$=|8\times 47$

45. (a) ${}^{2n+1}C_0+{}^{2n+1}C_1+-+{}^{2n+1}C_n=4096$
$2^{2n}=4096$
$2^{2n}=2^{12}$
$n=6$

46. (d) A garland can be made from 10 flowers in $\frac{1}{2}(9!)$ ways.
{$\because$ n flowers' garland can be made in $\frac{1}{2}(n-1)!$ ways}

47. (a) Since, $38808=8\times 4851$
$=8\times 9\times 539=8\times 9\times 7\times 7\times 11$
$=2^3\times 3^2\times 7^2\times 11$
So, number of divisors
$=(3+1)(2+1)(2+1)(1+1)=72$.
This includes two divisors 1 and 38808. Hence, the required number of divisors = $72-2=70$.

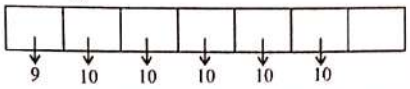
48. (b) We have, $30=2\times 3\times 5$. So, 2 can be assigned to either a or b or c i.e. 2 can be assigned in 3 ways. Similarly, each of 3 and 5 can be assigned in 3 ways. Thus, the no. of solutions are $3\times 3\times 3=27$.

49. (b) Three letters can be posted in 4 letter boxes in $4^3=64$ ways but it consists the 4 ways that all letters may be posted in same box. Hence required ways = 60.

50. (c) Let denote the exponent of 3 in n. The greatest integer less than 100 divisible by 3 is 99.
We have
$E(100!)=E(1.2.3.4....99.100)$
$=E(3.6.9....99)=E[(3.1)(3.2)(3.3)...(3.33)]$
$=33+E(1.2.3.....33)$
Now
$E(1.2.3.....33)=E(3.6.9....33)$
$=E[(3.1)(3.2)(3.3).....(3.11)]$
$=11+E(1.2.3.....11)$
and
$E(1.2.3....11)=E(3.6.9)$
$=E[(3.1)(3.2)(3.3)]$
$3+E(1.2.3)=3+1=4$
Thus $E(100!)=33+11+4=48$.

Prabal (JEE Main Level)

1. (c) $7\times 7\times 6\times 5-6\times 6\times 5\times 4=750$
2. (b) $(2^4-1)(2^4-1)(2^4-1)=(15)^3$
3. (b)



Sum of first six digits will he either even or odd so that s option are possible for last place.
Total Numbers $=9\times 10\times 10\times 10\times 10\times 10\times 5$
$=4500000$

4. (b) $(1)\,1!+2.2!+3.3!+-+n.n!$
$(2-1)1!+(3-1)2!+(4-1).3!+-+((n+17-1)n!$
$=(2!-1!)+(3!-2!)+-+((n+1!-n!)$
$=(x-1)!-1!-1!={}^{n+1}P_{n+1}-1$

5. (c) $m=3!\times 7!\ \boxed{B_1B_2B_3}\ G_1G_2...G_6$
$m={}^3C_2\,2!\,7!=3!\,7!$. Hence $k=1$.

6. (b) ${}^nC_2=153\Rightarrow\frac{n(n-1)}{2}=153\Rightarrow n=18$

7. (b) Since 5 are always to be excluded and 6 always to be included, therefore 5 players to be chosen from 14. Hence required number of ways are ${}^{14}C_5=2002$.

8. (a) Required number of ways
$={}^{12}C_1+{}^{12}C_2+{}^{12}C_3+.....+{}^{12}C_{12}=2^{12}-1$
$=4096-1=4095$

9. (b) Required number of ways
$={}^{30}C_3-{}^{15}C_3=3605$

10. (a) Total-when no one Digits'
$(10^{10}-1-1)-(9^{10}-1)$
Due to 0 Due to 1 Due to zero $=10^{10}-9^{10}-1$

11. (c) 7 letters
$P\to 1$
$E\to 2$
$R\to 1$
$F\to 1$
$C\to 1$
$T\to 1$
2 alike + 2 diff. $=1\times{}^5C_2=10$
All 4 different $={}^6C_4=15$. Total ways = 25

12. (b) Clearly, ${}^nC_3=T_n$.
So, ${}^{n+1}C_3-{}^nC_3=21$
$\Rightarrow({}^nC_3+{}^nC_2)-{}^nC_3=21$
$\therefore\ {}^nC_2=21$ or $n(n-1)=42=7.6\ \therefore\ n=7$

13. (b) ${}^9C_4$

14. (b) $S={}^{(n+1)}nC_0+{}^{(n)}nC_1+{}^{(n+1)}nC_2+..+(1)\,{}^nC_n$
$S={}^{(1)}nC_n+{}^{(2)}nC_{n-1}+{}^{(3)}nC_{n-2}+..+{}^{(n+1)}nC_0$
Add.
$2S={}^{(n+2)}({}^nC_0+{}^nC_1+{}^nC_2+-+{}^nC_n)$
$S=(n+2)2^{n-1}$

15. (a) The selection can be made in
${}^{20}C_7\times{}^5C_5+{}^{20}C_8\times{}^5C_4+{}^5C_3\times{}^{20}C_9$

16. (b) 85th SMCDW ; 86th SMCWD

17. (a) Fix up a male and the remaining 4 male can be seated in 4! ways. Now no two female are to sit together and as such the 2 female are to be arranged in five empty seats between two consecutive male and number of arrangement will be ${}^5P_2$. Hence by fundamental theorem the total number of ways is $=4!\times{}^5P_2=24\times 20=480$ ways.

18. (d) Word 'MATHEMATICS' has 2M, 2T, 2A, H, E, I, C, S. Therefore 4 letters can be hosen in the following ways.
Case I : 2 alike of one kind and 2 alike of second kind i.e., ${}^3C_2\Rightarrow$ No. of words
$={}^3C_2\frac{4!}{2!2!}=18$

Case II : 2 alike of one kind and 2 different
i.e., ${}^3C_1\times{}^7C_2\Rightarrow$ No.of words $={}^3C_1\times{}^7C_2\times\frac{4!}{2!}=756$

Case III : All are different
i.e.${}^8C_4\Rightarrow$ No. of words ${}^8C_4\times 4!=1680$
Hence total number of words are 2454.

19. (a) Since the total number of selections of r things from n things where each thing can be repeated as many times as one can, is ${}^{n+r-1}C_r$
Therefore the required number
$={}^{3+6-1}C_6=28$

20. (b) $u+v+w+x+y=7$; where $u, v, w, x, y, \geq 0$
Using beggar's method, ${}^{11}C_4=330$

21. (b) 9 people can be divided into groups of 2, 3 and 4 in $\frac{9!}{2!3!4!}$ ways. Now each group can be distributed in ${}^{10}C_3\times 3!$ Ways
$\Rightarrow\frac{9!}{2!3!4!}10\cdot 9\cdot 8=\frac{10!}{4}$
Note that in an eleven storeyed building there will be 10 floors.

22. (a) Required number of ways = (number of ways in which 16 players can be

divided in 8 couples) – (number of ways when S_1 and S_2 are in the same group)

$$= \frac{16!}{2^8 \cdot 8!} - \frac{(14!)}{2^7 \cdot 7!} = \frac{16 \cdot 15 \cdot 14!}{2 \cdot 2^7 \cdot 8 \cdot 7!} - \frac{(14!)}{2^7 \cdot 7!}$$

$$= \frac{(14!)}{2^7 \cdot 7!}\left[\frac{16 \cdot 15}{16} - 1\right] = \frac{(14) \cdot (14)!}{14 \cdot (6!) \cdot 2^6} = \frac{(14!)}{2^6 \cdot 6!}$$

23. **(b)** Number of ways is equal to number of ways of selection of 4 points out of given 18 points such that no two them are consecutive. Number of ways = ${}^{13}C_4 - {}^{13}C_2 = 1287$

24. **(b)** $N = 3^p\, 2^m\, 3^m\, 3^n\, 7^n$

$N = 2^m\, 3^{p+m+n}\, 7^n$

Odd proper Divisors = $(p + m + n + 1)(n + 1) - 1$

25. **(c)** We need a minimum of two circles to get a radical axis and 3 circles to get a radical center. If there are n circles in a plane then we get

${}^nC_2 = {}^nC_3 \Rightarrow n = 3 + 2 = 5$

26. **(a)** If 2 batsmen don't want to play then the rest of 10 players can be selected from 17 other players number of selection = ${}^{17}C_{10}$

If the particular bowler doesn't paly then number of selection = ${}^{19}C_{11}$

$\therefore$ Total number of selection = ${}^{17}C_{10} + {}^{19}C_{11}$

27. **(a)** $(HHH), (RR), \underbrace{(1, I), (P, P), AYU}_{x}$

$$= {}^{12}C_7 \cdot \frac{7!}{2!2!} \cdot 1 = (198)7!$$

28. **(d)** $B \subseteq A, \quad C \subseteq A$

Let $a \in A$, then we have 4 choice

$a \in B, \quad a \in C$

$a \notin B, \quad a \in C$

$a \in B, a \in C$

$a \notin B, \quad a \notin C$

$B \cap C = \phi$ Which has 3 options

$\therefore$ Total solutions = $3^4 = 81$

29. **(b)** Let number of men be 'm'.

Number of games played by men among themselves is $2 \times {}^mC_2 = m(m-1)$

Number of games which the men played with each of two women is $2m$.

$\therefore m(m-1) - 2(2m) = 66$

$m^2 - 5m - 66 = 0$

$m = 11, -6$

$\therefore$ Total number of participants = 13 players

Number of games is $2 \times {}^{13}C_2 = 156$

30. **(c)** Passanger has 7 choices

Their are 5 passangers

$\therefore$ total choice = ${}^7P_5$

31. **(d)** Total ways = $8 \times 9 = 72$

32. **(c)** One digit number = 6

2 digit number = $6 \times 7 = 42$

3 digit number = $6 \times 7 \times 7 = 294$

4 digit number = $6 \times 7 \times 7 \times 7 = 2058$

5 digit number starting with 1____ = $7^4 = 2401$

5 digit number starting with 2 ____ = $7^4 = 2401$

5 digit number starting with 30 ____ = $7^3 = 343$

5 digit number starting with 31 ____ = $7^3 = 343$

5 digit number starting with 320 ____ = $7^2 = 49$

5 digit number starting with 321____ = $7^2 = 49$

5 digit number starting with 3220 ____ = $7 = 7$

5 digit number starting with 32216 ___ = $7 = 7$

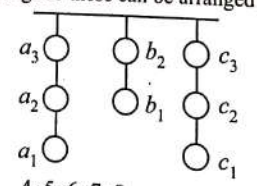
8000

Hence, the correct answer is (c).

33. **(b)** [X X X .... X (11 croses). Select 3 places for $A\,G\,J$ who can be seated in two ways and remaining 8 in 8! ways. Hence total = ${}^{11}C_3 \,.\, 2 \,.\, 8!$.]

34. **(c)** [Number of different tickets of one class = ${}^{10}C_2 = 45 \Rightarrow$ Number of different tickets of k classes = $45k \Rightarrow {}^{45k}C_6$].

35. **[6]** $(a_1, a_2, a_3), (b_1, b_2)$ and (c_1, c_2, c_3) are alike things so these can be arranged is



$$\frac{8!}{2!3!3!} = \frac{4 \cdot 5 \cdot 6 \cdot 7 \cdot 8}{2 \times 6}$$

36. **[420]**

Let $a = x_1 + x_2 + x_3$ and $b = y_1 + y_2$

$\therefore$ We have, $a, b > 0$ and $ab = 77$

Now, consider $77 = 7 \times 11 = 1 \times 77$

$\therefore$ Number of possible positive integral solution of $ab = 77$ is $2! + 2! = 4$.

But $a \neq 1$ and $b \neq 1$ as $x_1, x_2, x_3, y_1, y_2 \in N$.

$\therefore a = 7, b = 11$ or $a = 11; b = 7$

Case I: When $a = 7, b = 11$

We have, $x_1 + x_2 + x_3 = 7$ and $y_1 + y_2 = 11$

Clearly, the number of positive integral solution of

$x_1 + x_2 + x_3 = 7$ is ${}^{7-1}C_{3-1} = {}^6C_2$

$= \frac{6 \times 5}{2 \times 1} = 15$

and the number of positive integral solution of $y_1 + y_2 = 11$ is

${}^{11-1}C_{2-1} = {}^{10}C_1 = 10$

Case II: When $a = 11, b = 7$

We have $x_1 + x_2 + x_3 = 11$ and $y_1 + y_2 = 7$

the number of positive integral solution

$= {}^{11-1}C_{3-1}\, {}^{7-1}C_{2-1} = {}^{10}C_2\, {}^6C_1 = 270$

$\therefore$ Total number of solutions = $150 + 270$

$= 420$

37. **[15]**

Number divisible by 3 if sum of digits divisible

case-I If $1 + 2 + 3 + 4 + 8 = 18$ Number of ways = 120

case-II If $1 + 2 + 3 + 7 + 8 = 21$ Number of ways = 120

case-III If $2 + 3 + 4 + 7 + 8 = 24$ Number of ways = 120

case-IV If $1 + 2 + 0 + 4 + 8 = 15$ Number of ways = 96

case-V If $1 + 2 + 0 + 7 + 8 = 18$ Number of ways = 96

case-VI If $2 + 0 + 4 + 7 + 8 = 21$ Number of ways = 96

case-VII If $0 + 1 + 3 + 4 + 7 = 15$ Number of ways = 96

Total number 744

38. **[1750]**

We have, $x_1 x_2 x_3 x_4 x_5 = 240$ where x_1, x_2, x_3, x_4, x_5 are natural numbers.

Now, by using prime factorisation of 240 we can write the given equation as

$x_1 x_2 x_3 x_4 x_5 = 2^4 \times 3 \times 5$

Now, we can assign 3 and 5 to any of the unknown in 5×5 ways.

We can assign 2^4 to x_1, x_2, x_3, x_4, x_5 in $a + b + c + d + e = 4$

${}^{4+5-1}C_{5-1} = {}^8C_4$, ways

Required number of ways = ${}^8C_4 \times 25$

$= 1750$

39. **[12]**

$$\boxed{2} = \frac{5!}{2!3!} = 10$$

$$\boxed{3} = \frac{5!}{3!} = 20$$

$$\boxed{4} = \frac{5!}{2!2!} = 30$$

Hence sum of unit places is

$2 \times 10 + 3 \times 20 + 4 \times 30 = 200$

Hence required sum is

$= 200 \times (10^5 + 10^4 + 10^3 + 10^2 + 10^1 + 10^0)$

$= 200 \times (111111) = 22222200$

40. [1] Let $P(x) = x^3 + ax^2 + bx + c$

$x^2 + 1 = (x + i)(x - i)$

$P(i) = 0, P(-i) = 0$ we get

$b = 1, a = c$

No. of ways of choosing $a, b, c = 10$

$= 10 \times 1$

$\therefore K = 1$

Hence, the correct answer is (1).

41. [159]

Case I = when 1,2,3,4 are chose = 24

Case II = when two 1 and other 2 are from 2,3,4 $= \frac{{}^3C_2 \times \lfloor 4}{\lfloor 2} = 36$

Case III When '0' is chosen and other 3 from 1,2,3,4

$= {}^4C_3 \times 3 \times \lfloor 3 = 72$

Case IV When '0' is chosen and two 1's

$= {}^3C_1 \times \frac{3 \times \lfloor 3}{\lfloor 2}$

Total = 24 + 36 + 72 + 27 = 159

42. [2220]

$\frac{\lfloor 8}{\lfloor 2 \lfloor 2 \lfloor 2}\left(\frac{3\lfloor 7}{\lfloor 2 \lfloor 2} - \frac{3\lfloor 6}{\lfloor 2} + \lfloor 8\right)$

$= \lfloor 5\left[42 - \frac{63}{2} + 9 - 1\right]$

$= 120\left[50 - \frac{63}{2}\right]$

$= 60 \times 37$

$= 2220$

43. [5292]

The largest digit in the middle of the five digit number can be 4 (when numbers are formed with digits 0, 1, 2, 3, 4). 5 (when 0, 1, 2, 3, 4, 5 are used). ... , 9 (when 0, 1, 2, ... , 9 are used).

$\therefore$ Number of five digit numbers with largest digit 4 in the middle

$= \left({}^4C_4 \times 4! - {}^3C_3 \times 3!\right)$

Number of five digit numbers with largest digit 5 in the middle

$= \left({}^5C_4 \times 4! - {}^4C_3 \times 3!\right)$ and soon.

Hence required number of

$= \left({}^4C_4 \times 4! - {}^3C_3 \times 3!\right) + \left({}^5C_4 \times 4! - {}^4C_3 \times 3!\right) + \left({}^6C_4 \times 4! - {}^5C_3 \times 3!\right) + \ldots + \left({}^9C_4 \times 4! - {}^8C_3 \times 3!\right)$

44. [60] Total intersections $= 4\,{}^nC_2$

$= 4\,{}^6C_2$

$= 4 \times 15 = 60$

45. [6] $24300 = 2^2\,3^5\,5^2$ Number of odd divisors = the number of terms in $(1 + 3 + 3^2 + 3^3 + 3^4 + 3^5)(1 + 5 + 5^2)$

$= 6 + 6 + 6 = 18$

This also includes the factor 1. So the number of proper odd divisors = 17

46. [5] [${}^{10}P_6$] Let the six persons P_1, P_2, P_3, P_4, P_5 and P_6 be seated.

$_\,P_1\,_\,P_2\,_\,P_3\,_\,P_4\,_\,P_5\,_\,P_6\,_$

x_1 is the number of seats to left of P_1, x_2 is the number of seats in between P_1 and P_2 ..... x_7

is the number of seats right of P_6 such that $x_1 + x_2 + x_3 + \ldots\ldots x_7 = 9$.

i.e. coefficeient of x^9 in

$(1 + x + x^2 + \ldots)^2 (x + x^2 + x^3 + \ldots)^5$

$= {}^{15-6+1}C_6 = {}^{10}C_6$

$\therefore$ 6 persons can be arranged in 6! ways

$\therefore$ ${}^{10}P_6$ is the answer

47. [315]

AA CCDI MMn OOT

Coeff of x^7 in

$\left(1 + x + x^2\right)^3 \left(1 + x + x^2 + x^3\right)(1 + x)^4$

$= \frac{\left(1 - x^3\right)^3}{(1 - x)^3} \cdot \frac{\left(1 - x^4\right)}{(1 - x)}(1 + x)$

$= \left(1 - x^3\right)^3 \left(1 - x^4\right)(1 + x)(1 - x)^{-4} = 315$

48. [21599]

$\left[\frac{6!}{4!} \times 6! - 1 = 21599\right]$

Parikshit (JEE Advanced Level)

1. (*a*) There are 7 lists, say 1, 2, ..., 7. Suppose that list *i* has *xi* names; then,

$x_1 + \ldots + x_7 = 200$ where $x_i \geq 0$ is an integer

We need to first find the no. of solutions of this equation.

(Note that this does not complete the solution to the question as, list 1 may contain 7 names which no. would remain the same in 7! arrangements of the names)

The number of solutions is ${}^{200+7-1}C_{7-1} = {}^{206}C_6$

But corresponding to any one so lution, $(x_1 \ldots\ldots x_7)$ (i.e. list *j* contains x_j names) we can have 200! arrangements consistent with distribution of x_i names to j^{th} list

$\therefore$ No. of different sequences of 7 lists

$= {}^{200}C_6 \times 200! = \frac{206!}{6!}$.

2. (*b*) Since the train does not stop at $(p - 3)$ stations, the problem reduces to the following: In how many ways car three objects be placed among $(p - 3)$ objects in a row such that no two of them are next to each other [at most 1 object is to be placed between any two of these $(p - 3)$ objects]. Since there are $(p - 2)$ positions to place the three objects, the required number of ways $= {}^{p-2}C_3$.

3. (*a*) Number of ways of arranging 5 boys and 3 girls, i.e. 8 people on a round table would be7!

We subtract the number of ways of arranging those people when B_1 and G_1 are always together to get the required answer.

When B_1 and G_1 are together, we get 4 boys + 2 girls + 1 $(B_1 + G_1)$ i.e. 7 people and since $B_1 + G_1$ can be permuted in 2 ways, these can be arranged in 6! × 2 ways.

Subtracting, we have $7! - 6! \times 2$

$= 6!(7 - 2) = 5 \times 6!$ ways in total.

4. (*a*) $\sum_{i=1}^{k} \frac{1}{x_i} = \frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_k} = \frac{\sum x_1}{n} = \frac{75}{n}$

As L.C.M of $x_1, x_2, \ldots\ldots x_n$ is *n*)

5. (*b*) Given problem can be written as

$x_1 + x_2 + x_3 + \ldots\ldots\ldots + x_n = n^2$ such than,

$x_i \geq i, \quad i \in 1 \text{ to } n$

Let $x_i - i = y_i$

Since $x_i \geq i \Rightarrow y_i \geq 0$

Therefore original equation becomes:

$\Rightarrow (y_1 + 1) + (y_2 + 2) + \ldots + (y_n + n) = n^2, y_i \geq 0$

$\Rightarrow y_1 + y_2 + \ldots\ldots\ldots + y_n = n^2 - (n + (n-1) + \ldots\ldots + 2 + 1)$

$\Rightarrow y_1 + y_2 + \ldots\ldots + y_n = n^2 - \frac{(n + (n-1))}{2}, y_i \geq 0$

Total no. of possibilities is given by ${}^{n+r-1}C_{r-1}$ where n is the sum of r is the number of variables

There $\left(n^2 - \frac{n^2 + n}{2}\right) + {}^{n-1}C_{n-1}$

$= \left(\frac{n^2 + n - 2}{2}\right)_{C_{n-1}}$

Hence the answer $\left(\frac{n^2+n-2}{2}\right)_{C_{n-1}}$

6. **(b)** A match can have 3 results they play 5 matches $(3 \times 3 \times 3 \times 3 \times 3) = 35$ different result for the series.

since each forecast is different we need to have 35 people for one of them to get it correctly

$\therefore$ smallest group= 243.

Hence, the answer is 243.

7. **(b)** Note : $2^0 = 2^1 - 1$

$2^0 + 2^1 = 2^2 - 1$

$2^0 + 2^1 + 2^2 = 2^3 - 1$

or the sum of powers of 2 equals 1 less than the next power,

$\because 2048 = 2^{11}$

And 2047 is one less that 2^{11} $\therefore$ it is the sum of powers of 2 till n = 11.

Hence, the answer is 11.

8. **(b)** If n is odd

$3^n = 4\lambda_1 - 1, 5^n = 4\lambda_2 + 1$

$\Rightarrow 2^n + 3^n + 5^n$ divisible by r if $n \geq 2$

Thus, n = 3,5,7, 9 ....., 99 i.e. n can take 49 different values.

If n is even $3^n = 4\lambda_1 + 1, 5^n = 4\lambda_2 + 1$

$\Rightarrow 2^n + 3^n + 5^n$ is not divisible by 4

as $2^n + 3^n + 5^n$ will be in the form of $4\lambda + 2$

Thus, the total number of ways of selecting 'n' is equal to 49.

9. **(b)** 6 places can be selected in ${}^9C_6$ ways and 6 can be placed only at 5 places, except the right most of other 6 selected. Remaining numbers i.e. 7, 8, 9 in 3! ways

Hence number of ways = ${}^9C_6 \cdot 5 \cdot 3!$

10. **(a)** Perfect square = $[\sqrt{100}] - 1 = 9$ (excluding one)

Perfect cubes = $[100^{1/3}] - 1 = 3$

Perfect 4th powers = $[100^{1/4}] - 1 = 2$

Perfect 5th powers = $[100^{1/5}] - 1 = 1$

Perfect 6th powers = $[100^{1/6}] - 1 = 1$

Now, perfect 4th powers have already been counted in perfect square and perfect 6th powers have been counted with perfect squares as well as with perfect cubes.

Hence the total ways = $9 + 3 + 1 - 1 = 12$

11. **(b,d)** Required number of triangles can be determined by case work **Case I.** When the two triangles selected have none of vertices common

Number of ways

$= \frac{{}^{12}C_3 \times {}^9C_3}{2} = \frac{220 \times 84}{2} = 9240$

Case II. When exactly one vertex is common

Number of ways = ${}^{12}C_5 \cdot {}^5C_1 \cdot = \frac{{}^4C_2}{2} = 11880$

$\therefore$ Total ways = 21,120

For part C

No two consecutive vertex can be selected. First vertex can be selected in ${}^{12}C_1$ ways.

Rest 5 vertices must be selected from rest 9 (as the vertices in the neighbourhood of the selected vertices cannot be selected) so that no two vertices are consecutive which can be done in ${}^5C_5$ ways.

$\therefore$ Total ways = $\frac{{}^{12}C_1 \times {}^5C_5}{6} = 2$ ways.

For part (d)

Centre of dodecagon will not create any new line, All the lines will be created by the 12 vertices of dodecagon in ${}^{12}C_2$ ways = 66 ways.

12. **(a,b,c,d)**

(a) $11 \begin{cases} 4v \\ 7c \end{cases} \Rightarrow {}^7C_2 \cdot 6! = 3 \cdot 7 \cdot 6! = 3 \cdot 7!$

(b) No. of ways = $\frac{15!}{r!(15-r)!}$

$= 15c_r \underbrace{WW.....W}_{r}$

$\underbrace{BBB.....B}_{15-r}$

This is maximum if r = 7 or 8

(c) 12 — 4A / 5A / 3 all different

$\Rightarrow$ Total no. of combinations

$= 5 \cdot 6 \cdot 2^3 = 240$

(d) 2 alike +2 other alike + 2 other a like = 1

2 alike + 2 other alike + 2 different = ${}^3C_2 \cdot {}^4C_2 = 18$

2 alike + 4 different = ${}^3C_1 \cdot {}^5C_1 = 15$

All 6 different = 1 = 35.

13. **(a,c,d)**

1 digit plaindromes $\rightarrow 9$

2 digit palindromes $\rightarrow 9 \times 1$

3 digit palindromes $\rightarrow 9 \times 10$

4 digit palindromes $\rightarrow 9 \times 10$

5 digit palindromes $\rightarrow 9 \times 10^2$

6 digit palindromes $\rightarrow 9 \times 10^2$

1 0 0 _____ $\rightarrow 10$

1 0 1 _____ $\rightarrow 10$

$\therefore$ Number at $2(9 + 90 + 900 + 10) = 2018$, rank is 1019101

1 $\rightarrow 10^3$

20 $\rightarrow 10^2$

210 $\rightarrow 10$

211 $\rightarrow 10$

212 $\rightarrow 10$

213 $\rightarrow 10$

2140412 $\rightarrow 1$

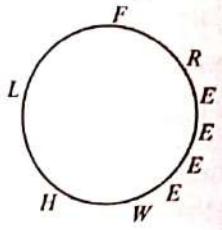
Rank of 2140412 = $2(9 + 90 + 900) + 1000 + 100 + 40 + 1$

$= 1998 + 1000 + 100 + 40 + 1$

$= 3139$

14. **(b,c,d)**

(a) False as is should be ${}^9P_5 - 1$



(b) $x \cdot 4! = 8!$

$\therefore x = \frac{8!}{4!} = {}^8P_4$

(c) Vowels $E\ E\ E\ E$ select 4 places in ${}^9C_4$ ways arrange consonants alphabatically only us one ways.

$\therefore {}^9C_4 = 126 = \frac{1}{2} \cdot 256 = \frac{1}{2} \cdot {}^{10}C_5$

(d) True

$\therefore$ correct answer are (b), (c) and (d)

15. **(a,d)** We have arrange all the letter except 'CCC' is $\frac{12!}{5!3!2!}$ now there are 13 place where 'C' can be placed ${}^{13}C_3$

Hence required number of ways is = $\frac{12!}{5!3!2!} {}^{13}C_3$

$= 11 \cdot \frac{13!}{6!}$

16. **(b,c,d)**

$|T|M|A|T|$

Number of ways = ${}^5C_2 \times \frac{4!}{2!} = 10 \times 12 = 120$.

(a) RAITHATHA

$A = 3, H = 2, T = 2, R = 1, I = 1$

$\boxed{A\ A\ A}$ $\boxed{H\ H}$ $\boxed{T\ T}$ R I

$\therefore$ Number of ways = $5! = 120$

(b) 5 — 2, 3, 3 / 3, 2, 2 / 2, 3, non-zero digit / 2, 3, 0

$= \frac{3!}{2!} + \frac{3!}{2!} + ({}^7C_1 \times 3!) + (2 \times 2) = 3 + 3 + 42 + 4 = 52.$

(c) $M_1, M_2, ..., M_7 \mid N_1, N_2, N_3$

Total ways $= 3({}^7C_1) + 3({}^7C_2) + 1\,({}^7C_3) = 21 + 63 + 35 = 119.$

Alternatively:

Number of ways $= {}^{10}C_3 - 1 = 120 - 1 = 119$

(d) Number of ways $= {}^4C_2 \times {}^6C_2 = 6 \times 15 = 90.$

17. ***(a,b,d)***

$p = {}^5C_1 \cdot 2 = 10,\ q = {}^5C_3 \cdot 2^3 \cdot 1 = 80$ and $r = {}^5C_5 \cdot 2^5 = 32$

$\therefore\ 2q = 5r = 160$. Also $8p = q = 80$

and $2(p + r) = 84 > 80 = r$

18. ***(a,d)***

Consider $(x + y + z)^9 = {}^9C_r x^{9-r} (y + z)^r$

$= {}^9C_r x^{9-r} \cdot {}^rC_p y^{r-p} z^p\,[x + (y + z)]^9$

put $r = 7;\ p = 4$

or ${}^9C_2 \cdot {}^7C_3 \cdot {}^4C_4 = \frac{9!}{2!\,3!\,4!}$

19. ***(b,c,d)***

ALTERNATIVELY triplets with

$x = y < z,\ x < y < z,\ y < z < x$

can be chosen in ${}^{n+1}C_2, {}^{n+1}C_3, {}^{n+1}C_3$ ways.

There are ${}^{n+1}C_2 + 2({}^{n+1}C_3) = {}^{n+2}C_2 + {}^{n+1}C_3 = 2({}^{n+2}C_3) - {}^{n+1}C_2$.

20. ***(a, c)*** Let person P_i get x_i number of things such that

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 30.$

If x_i is odd or $x_i = 2\lambda_i + 1$, where $\lambda_i \geq 0$, then

$2(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6) + 6 = 30$

or $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 = 12$

Then number of solutions is ${}^{12+6-1}C_{6-1} = {}^{17}C_5$. If x_i is even or $x_i = 2\lambda_i$, where $\lambda_i \geq 1$, then

$2(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6) = 30$

or $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 = 15$

Therefore, required number of ways is ${}^{15-1}C_{6-1} = {}^{14}C_5$.

21. ***(b,c)*** If a, b, c are in A.P., then a and c both are odd or both are even.

Case-I: n is even.

The number of ways of selection of two even numbers a and c is ${}^{n/2}C_2$. Number of ways of selection of two odd numbers is ${}^{n/2}C_2$. Hence the number of A.P.'s is

$$2\,{}^{\frac{n}{2}}C_2 = 2\frac{\frac{n}{2}\left(\frac{n}{2}-1\right)}{2} = \frac{n(n-2)}{4}$$

Case-II : n is odd.

The number of ways of selection of two odd numbers a and c is ${}^{\frac{(n+1)}{2}}C_2$. The number of ways of selection of two even numbers a and c is ${}^{\frac{(n-1)}{2}}C_2$. Hence, the number of A.P.'s is ${}^{\frac{(n+1)}{2}}C_2 + {}^{\frac{(n-1)}{2}}C_2$

$$= \frac{\left(\frac{n+1}{2}\right)\left(\frac{n+1}{2}-1\right)}{2} + \frac{\left(\frac{n-1}{2}\right)\left(\frac{n-1}{2}-1\right)}{2}$$

$$= \frac{1}{8}(n-1)((n+1)+(n-3)) = \frac{(n-1)^2}{4}$$

22. ***(a)***

23. ***(b)*** $3 \times {}^nP_4 = {}^{n-1}P_5$

$\Rightarrow 3 \times \frac{n!}{(n-4)!} = \frac{(n-1)!}{(n-6)!}$

$3n = (n-4)(n-5) \quad \Rightarrow 3n = n^2 - 9n + 20$

$\Rightarrow n^2 - 12n + 20 = 0 \quad \Rightarrow (n-10)(n-2) = 0$

$\Rightarrow n = 10$ as $n \neq 2$

(i) Any 3 consecutive vertices can be selected in 10 ways (any two adjacent sides) (1, 2, 3) or (2, 3, 4) or ..... or (10, 1, 2) say e.g. (1, 2, 3) now 4 and 10 cannot be selected from the remaining five vertices from 5 to 9 any one vertex can be selected in ${}^5C_1$ ways.

$\therefore$ number of such quadrilaterals $= 10 \times {}^5C_1 = 50$

(ii) Any 2 consecutive vertices can be selected in 10 ways (1, 2) or (2, 3) or ..... or (10, 1) say e.g., (1, 2) now 3 and 10 cannot be selected and from the remaining 6 vertices any two non-consecutive vertices can be selected in ${}^5C_2$ ways.

$\therefore$ number of such quadrilaterals $10 \times {}^5C_2 = 100$

24. ***(d)*** AAELLLPR

A _____ $\frac{7!}{3!}$

E _____ $\frac{7!}{3!2!}$

L _____ $\frac{7!}{2!2!}$

PAA_____ $\frac{5!}{3!}$

PAE_____ $\frac{5!}{3!}$

PAL_____ $\frac{5!}{2!}$

PARAE _____ $\frac{3!}{3!}$

PARALE _____ $\frac{2!}{2!}$

PARALLEL_____ 1, rank is 2623

Hence, the correct answer is *(d)*.

25. ***(b)*** EEIIJT

E _____ $\frac{5!}{2!} = 60$

I _____ $\frac{5!}{2!} = 60$

JE _____ $\frac{4!}{2!} = 12$

JIE _____ $3! = 6$

JIIE _____ $2! = 2$

JIITEE _____ 11. It is 141^{th} word.

Hence, the correct answer is *(b)*.

26. ***(a)***

(a) Required number of ways $= (2+1)(3+1)(4+1) - 1 = 59.$

(b) ${}^{10}C_3 = 120$

(c) Required number of ways

= Coefficient of x^{10} in $(1 + x + x^2 + ...)^4$

= Coefficient of x^{10} is $(1 - x)^{-4} = {}^{10+4-1}C_{4-1} = {}^{13}C_3 = 286$

(d) The word 'MATHEMATICS' consists of 11 letters of which 7 are consonants namely M, M, T, T, H, C, S and 4 vowels and a group of consonants can be arranged in $\frac{{}^5P_5}{2!} = \frac{5!}{2}$ ways. ($\because$ A is repeated twice)

In any such arrangement, seven consonants can be reshuffled among themselves in

$\frac{{}^7P_7}{2!\,2!} = \frac{7!}{2!\,2!}$ ways.

($\because$ Either of M and T is repeated twice)

Hence, the required number of ways $=$

$$\frac{5!}{2!} \times \frac{7!}{2!2!} = \frac{120 \times 5040}{8} = 75600.$$

27. **[7]** $x_1 + x_2 + x_3 + ... + x_7 + x_8 = 93$

$x_1 \geq 0, x_2 \geq 6, x_3 \geq 6, ... x_7 \geq 6, x_8 \geq 0$

$x_1 + x_2' + x_3' + ... + x_7' + x_8 = 57$

No. of ways $= {}^{64}C_7$.

28. **[4]** Case -I If all are different then no. of way is $= {}^6C_3 = 20$

Case-II If three each of two colours, then combination is

3 0 $\rightarrow$ 2!

2 1 $\rightarrow$ $2! = 2! + 2! = 4$ ways

Case-III If two each of three colours, then combination is

2 1 0 $\rightarrow$ 3!

1 1 1 $\rightarrow$ $1! = 3! + 1! = 7$ways

Hence required no. is $= 20 + 7 + 4 = 31$

29. **[20]** First we select one married couple out of 6 married couple i.3. ${}^6C_1$ ways total

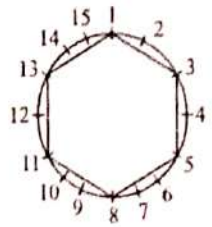
number of required case ${}^6C_1 \times {}^5C_1 \times {}^4C_1 \times 2 = 6 \times 5 \times 4 \times 2 = 240$

$N = 240$

30. [140]

Step 1. Select the initial vertex say[] '1' (in ${}^{15}C_1$ ways)

2. Now 2 and 15 cannot be selected. From the remaining vettices 3 to 14 (twelve) we have to select 5 more for our hexagon.



3. Symbolise the vertices to be taken by Ⓢ Ⓢ Ⓢ Ⓢ Ⓢ and the vertices not be taken (7 in this case) by X X X X X X X.

4. Identify the gaps between these crosses (8 in this case) and select

4	6	7	9	10	12	14
X	X	X	X	X	X	X
Ⓢ	Ⓢ		Ⓢ		Ⓢ	Ⓢ
3	5		8		11	13

any five out of these gaps in ${}^8C_5$ ways.

5. Serial number are to be alloted either to Ⓢ or to a 'X' whichever comes earlier. In the present problem the vertices corresponding to one selection are 3, 5, 8, 11, 13 and the hexagon as shown.

6. For each selection we therefore have a hexagon with two non-consecutive vertices.

7. Number of hexagons $= {}^{15}C_1 \times {}^8C_5$. However this pruticular hexagon 1, 3, 5, 8, 11, 13 will occur 6 times when we select the initial vertex as 3 or 5 or 8 or 11 or 13. Hence our answer is 6 times more.

8. Required number of hexagons $= \frac{15 \times {}^8C_5}{6} = \frac{15 \times 56}{6} = 140$

M#2: [Alternative Solution of V(b): (Similar to that in linear):

As in linear, let us open this chain to have 1, 2, 3, ..., 13, 14, 15

O O O O O O to be selected; | X | X | X | X | X | X | X | X | not to be selected

Number of ways $= {}^{10}C_6$

Required number of ways $= {}^{10}C_6$ – number of ways when 1 and 15 are included, since in circular these become consecutive

Now if 1 and 15 are already selected, 2 and 14 cannot be taken. Remaining vertices are 3, 4, 5, ..., 11, 12, 13 (11); O O O O (4); | X | X | X | X | X | X | X | (7)

Cases to be rejected $= {}^5C_4$

Required number of ways $= {}^{10}C_6 - {}^5C_4$

$\Rightarrow {}^{10}C_4 - {}^5C_4 = 210 - 70 = 140$

(vi) AA, BB, CC, D, E, G, F can be sealed on a circle if people of the same nationality are to be separated.

(vii) 16 knights are sitting on a round table.

31. [250] Total number of 7 digit number with sum 59 are

(i) 9 9 9 9 9 9 5 (7 combination)

(ii) 9 9 9 9 9 8 6 (42 combinations)

(iii) 9 9 9 9 9 7 7 (21 combinations)

(iv) 9 9 9 9 8 8 7 (105 combinations)

(v) 9 9 9 8 8 8 8 (35 combinations)

Total number of digits numbers with sum 59 is 7 + 42 + 21 + 105 + 35 = 210 numbers

Numbers in the above list that are divisible by 11

The odd numbered digits (above) must odd upto 35. They must be 9998 (${}^4C_1$)

The sum numbered digits (above) must add upto 24. they can be

9 9 6 (${}^3C_1$ combination)

9 8 7 (${}^3C_1$ combination)

8 8 8 (1 combiantion)

total number of multiples of 11 is = 4

(3 + 6 + 1) = 40

Thus the answer is 40.

PYQ's (Past Year Questions)

1. *(a)*

5	x	x	x	0
7	x	x	x	0
7	x	x	x	5
9	x	x	x	0
9	x	x	x	5

So, required numbers $= 5 \times {}^4C_3 \times 3! = 120$

2. [432] UNIVERSE

Vowels:E, E, I, U

Consonants: N, V, R, S

Case: 2 vowels different, 2 consonant different

${}^3C_2 \times {}^4C_2 \times 4! = 3 \times 6 \times 24 = 432$

3. *(d)* Let the number of chocolates given to $C_1, C_2, C_3, \& C_4$, be a, b, c, d respectively.

Given $4 \le b \le 7$

$2 \le c \le 6$

Now using these the maximum number of chocolates that can be given to C_1 or C_4 is 24 (where b & c are given 2 & 4 chocolates).

$\therefore 0 \le a \le 24$

$0 \le d \le 24$

& $a + b + c + d = 30$

So, total possible solution to the above equation

Coefficient of x^{30} in

$(x^0 + x + x^2 + ... + x^{24})(x^4 + x^5 + ... + x^{24})$
$(x^2 + x^3 + ... + x^{24})(x^0 + x + x^2 + ... + x^{24})$
$= (1 + ... + x^{24})^2 (x^4)(1 + x + ... + x^{24}) \times x^2$
$(1 + x + ... + x^4)$

$$= \left(\frac{x^{25}-1}{x-1}\right)^2 \times \left(x^{4+2}\right)\frac{\left(x^4-1\right)}{(x-1)} \times \frac{\left(x^5-1\right)}{(x-1)}$$

$= (x^{50} - 2x^{25} + 1) \times x^6 \times (x^4 - 1)(x^5 - 1) \times (x - 1)^{-4}$

$= (x^{56} - 2x^{31} + x^6) \times (x^4 - x^4 - x^5 + 1) \times (x - 1)^{-4}$

x^{56} & x^{31} can never give x^{30} so we discard them

$x^6 \times x^9 \times (x - 1)^{-4} \rightarrow {}^{15+4-1}C_{4-1} = {}^{18}C_3$

$-x^6 \times x^5 \times (x - 1)^{-4} \rightarrow {}^{19+4-1}C_{4-1} = {}^{22}C_3$

$-x^6 \times x^5 \times (x - 1)^{-4} \rightarrow {}^{19+4-1}C_{4-1} = {}^{22}C_3$

Coefficient $x^{30} \rightarrow {}^{18}C_3 - {}^{23}C_3 - {}^{22}C_3 + {}^{27}C_3$

$$= \frac{18 \times 17 \times 16}{6} - \frac{23 \times 22 \times 21}{6} - \frac{22 \times 21 \times 20}{6} + \frac{27 \times 26 \times 25}{6}$$

$= 430$

4. [569] Counting integers starting from 2

Case-I: At unit's place we can fill 2/3/4/6/7 i.e., 2 0 2 5 → 5 ways

At unit's place and ten's place we can fill digits as 3/4/6/7 and 0/2/3/4/6/7

or 2 0 4 6 → 24 ways

(Numbers except 0 or 2 in 3 place)

Case-II: If non-zero number on 2nd place

i.e., 2 5 6 6 = 180 ways

Counting integers starting from 3

3 6 6 6 = 216 ways

Counting integers starting from 4

Case-I: If 0, 2 or 3 on 2nd place

i.e., 4 3 6 6

= 108 ways

Case-II: If 4 on 2nd place

i.e., 4 4 6 6 = 36 ways

$\therefore$ Total 5 + 24 + 180 + 216 + 108 + 36 = 569 numbers

5. (*b*)

	Indian	Foreigner
Case-I:	2	4
Case-II:	3	6
Case-III:	4	8

Total number of ways

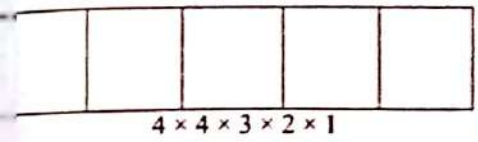
$= {}^6C_2 \times {}^8C_4 + {}^6C_3 \times {}^8C_6 + {}^6C_4 \times {}^8C_8$

$= 15 \times 70 + 20 \times 28 + 15 \times 1 = 1625$

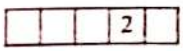
6. [96] 1st position can be filled in 4 ways as zero cannot 0, 2, 4, 6, 8 appear in 1st position.

2nd position can be filled in 4 ways and so on.

Total cases $= 4 \times 4 \times 3 \times 2 \times 1 = 96$



$4 \times 4 \times 3 \times 2 \times 1$

7. (*d*) 

Number of numbers $= 8 \times 8 \times 7 \times 6$

$= 2688 = 336\,k$

$\Rightarrow k = 8$

8. [120]

For vowels not together

1st arrange *L*, *T*, *T*, *R* in $\frac{4!}{2!}$ ways

Then put both *E* in 5 gaps formed in ${}^5C_2$ ways

$\therefore$ No. of ways $= \frac{4!}{2!}\,{}^5C_2 = 120$

9. [495]

Selection of 4 days out of 15 days such that no two of them are consecutive $= {}^{15-4+1}C_4 = {}^{12}C_4$

$\frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2} = 11 \times 5 \times 9 = 495$

10. [1080]

Required ways

$$= \frac{6!}{2!2!1!1!2!2!} \times 4! = 1080$$

11. (*c*) Number of ways = Total number of ways without restriction – When two specific boys are in team without any restriction, total number of ways of forming team is ${}^7C_3 \times {}^5C_2 = 350$. If two specific boys A, B. are in same team then total number of ways of forming team equals to ${}^5C_1 \times {}^5C_2 = 50$ ways. Total ways $= 350 - 50 = 300$ ways

12. (*c*) Starting with $5 = 6^3 = 216$

Starting with $44 = 6^2 = 36$

Starting with $45 = 6^2 = 36$

Starting with $43 = 18$ (Not using 0, 1, 2)

Starting with $432 = 4$

Total = 310

13. [171] ${}^{19}C_2 = 171$

14. [924] $N = 2^{10} \cdot 5^{10} \cdot 11^{11} \cdot 13^{13}$

There should not any power of 2

All divisors 5^{10} and 13^{13} of form $4n + 1$, so divisors of $5^{10} \times 13^{13} = 11 \times 14$

Total number of divisors of 11^{11} is of form $(4n + 1)$ can be obtained if and only if we take even power of 11.

so, number of divisors = 6{0, 2, 4, 6, 8, 10}

Total divisors of the form of $4n + 1 = 11 \times 6 \times 14 = 924$

15. [31650]

16. (*b*) Required number of ways

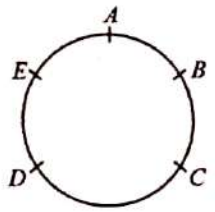
$= 6! \times {}^7C_5 \cdot 5!$

$= 720 \times \frac{7 \times 6}{2} \times 5!$

$6! \times \frac{7 \times 6}{2} \times 5! = (5!)^2 \times \frac{7 \times 6 \times 6}{2}$

$= 126 \times (5!)^2$

17. [30]



When 1*R*, 2*B*, 2*G*

${}^5C_1 \times 2 = 10$

Other possibilities

1*B*, 2*R*, 2*G*

or 1*G*, 2*R*, 2*B*

So total no. of ways $= 3 \times 10 = 30$

18. (*d*) In MATHEMATICS word, total letter = 11

M comes 2 times, A comes 2 times, T comes 2 times *C* and *S* do not come together

$= \frac{11!}{2!2!2!} - \frac{10!}{2!2!2!} \times 2!$

$= \frac{9 \times 10!}{2!2!2!} = \frac{9 \times 10 \times 9 \times 8 \times 7 \times \underline{|6}}{8}$

$= 5670\underline{|6} = k\underline{|6}$

Hence, *k* is 5670

19. [16] Let number of couples = *n*

$\therefore {}^nC_2 \times {}^{n-2}C_2 \times 2 = 840$

$\Rightarrow n(n-1)(n-2)(n-3) = 8 \times 7 \times 6 \times 5$

$\Rightarrow n = 16$

Therefore, total number of persons = 16

20. [17] ${}^bC_3 \times {}^8C_2 = 168$

$\therefore b(b-1)(b-2)(g)(g-1) = 8 \times 7 \times 6 \times 3 \times 2$

$\Rightarrow b = 8$ and $g = 3$

$\therefore b + 3g = 17$

21. [30] Let the four-digit number be $\overline{xyzt}$.

x can be 1 or 2 only.

If *x* is 1 then *zt* can be 24, 32, 36 52, 56, 64 and then *y* will have 3 choices.

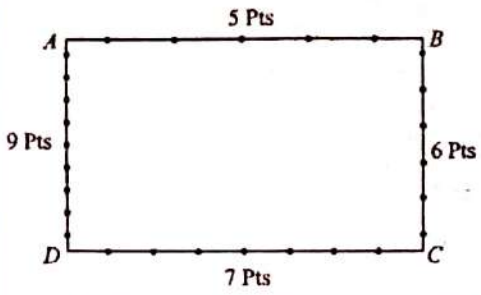
So, number of numbers $= 3 \times 6 = 18$

If *x* is 2 then *zt* can be 16, 36, 56 64 and then *y* will have 3 choices.

So, number of numbers $= 3 \times 4 = 12$

Therefore, total number of numbers $= 18 + 12 = 30$

22. (*d*)



Given, *ABCD* a rectangle having 5, 6, 7, 9 points on the sides *AB*, *BC*, *CD* and *DA* respectively

α = Number of triangles

$\alpha = 5 \cdot 6 \cdot 7 + 5 \cdot 7 \cdot 9 + 5 \cdot 6 \cdot 9 + 6 \cdot 7 \cdot 9$

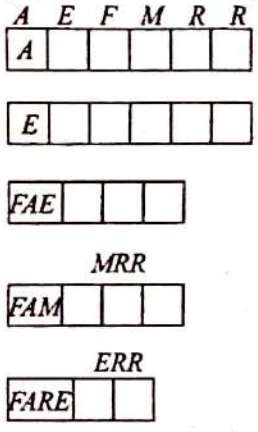
$= 210 + 315 + 270 + 378 = 1173$

β = Number of Quadrilateral

$\beta = 5 \cdot 6 \cdot 7 \cdot 9 = 1890$

$\beta - \alpha = 1890 - 1173 = 717$

23. [77]



$\frac{5!}{2!} - 4! = 36$

$\frac{5!}{2!} - 4! = 36$

$\frac{3!}{2!} - 2! = 1$

$\frac{3!}{2!} - 2! = 1$

$2! = 2$

1

Hence serial number of FARMER

$= 36 + 36 + 1 + 1 + 2 + 1 = 77$

24. **(a,b,d)**

25. **(c)** Each section has 5 questions.

$\because$ Total number of selection of 5 questions

$= 3 \times {}^5C_1 \times {}^5C_1 \times {}^5C_3 + 3 \times {}^5C_1 \times {}^5C_2 \times {}^5C_2$

$= 3 \times 5 \times 5 \times 10 + 3 \times 5 \times 10 \times 10 = 750 + 1500$

$= 2250$

26. **[135]** Select any 4 questions in ${}^6C_4$ ways which are correct

Number of ways of answering wrong question = 3

$\therefore$ Required number of ways $= {}^6C_4 \times 3^2 = 135$

27. **(c)** $k = \{4,8,12,16,20\}$

$f(k)$ can takes the values $\{3,6,9,12,15,18\}$

Number of ways $= {}^6C_5 \cdot 5!$

$\therefore$ Total number of onto functions

$= {}^6C_5 \cdot 5!(15!)$

$= (6!)\,(15!)$

28. **(b)** $x + y + z = 15$

Total no. of solution $= {}^{15+3-1}C_{3-1} = 136$ ..(i)

Let $x = y \neq z$

$2x + z = 15 \Rightarrow z = 15 - 2r \quad [\because x = y = r]$

$\Rightarrow r \in \{0, 1, 2, \ldots 7\} - \{5\}$

$\therefore$ 7 solutions

$\therefore$ there are 21 solutions in which exactly $\left[\because \frac{3!}{2!} \times 7 = 21\right]$

Two of $x_1\, y_1\, z$ are equal ...(ii)

There is one solution in which $x = y = z$ ....(iii)

Required answer = 136 –21 – 1 = 114

29. **(a)** Since ${}^{21}C_0 + \ldots + {}^{21}C_{10} + {}^{21}C_{11} + \ldots + {}^{21}C_{21} = 21^{21}$

$\Rightarrow$ but we have to select 10 objects and

${}^{21}C_0 + \ldots + {}^{21}C_{10} = {}^{21}C_{11} + \ldots {}^{21}C_{21}$

$({}^{21}C_0 + \ldots + {}^{21}C_{10}) = 2^{20}$

30. **[44]** Derangement of 5 students

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots\right)$$

$$D_5 = 5!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

$$= 120\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}\right)$$

$= 60 - 20 + 5 - 1 = 40 + 4 = 44$

By Aditya Mohod
username - maje_kabhi_dekha_nahi

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 10

Straight Lines

Prarambh (Topicwise)

1. (*a*) The vertices of triangle are the intersection points of these given lines. The vertices of Δ are $A(0, 4)$, $B(1, 2)$, $C(4, 0)$

Now, $AB = \sqrt{(0-1)^2 + (4-1)^2} = \sqrt{10}$

$BC = \sqrt{(1-4)^2 + (0-1)^2} = \sqrt{10}$

$AC = \sqrt{(0-4)^2 + (0-4)} = 4\sqrt{2}$

$\because AB = BC$;

$\therefore \Delta$ is isosceles.

2. (*b*) Mid point $= \left(\frac{1+1}{2}, \frac{3-7}{2}\right) = (1, -2)$

Therefore required line is

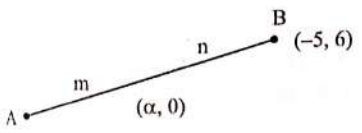
$2x - 3y = k \Rightarrow 2x - 3y = 8.$

3. (*a*) Point of intersection $y = -\frac{21}{5}$ and $x = \frac{23}{5}$

$\therefore \quad 3x + 4y = \frac{3(23) + 4(-21)}{5} = \frac{69 - 84}{5} = -3$

Hence, required line is $3x + 4y + 3 = 0$.

4. (*a*)



(3, – 4)

Suppose x-axis cuts AB at $(\alpha, 0)$ in the ratio m : n By section formula,

$0 = \frac{-4n + 6m}{m+n}$

$\Rightarrow 4n = 6m$

$\Rightarrow \frac{m}{n} = \frac{2}{3}$

5. (*c*) Given, equation of line $y - x + 2 = 0$ and coordinates of points $(x_1, y_2) = (3, -1)$ and $(x_2, y_2) = (8, 9)$. We know that if the ratio in which a line $ax + by + c = 0$ is divided by point (x_1, y_1) and (x_2, y_2) is $\lambda : 1$, then intersecting point $\left(\frac{\lambda x_2 + x_1}{\lambda+1}, \frac{\lambda y_2 + y_1}{\lambda+1}\right)$ lies on $ax + by + c = 0$. Thus any point on the line joining $(3, -1)$ and $(8, 9)$ dividing it in the ratio $\lambda : 1$ is $\left(\frac{8\lambda+3}{\lambda+1}, \frac{9\lambda-1}{\lambda+1}\right)$ and if it lies on $y - x + 2 = 0$ then $\frac{9\lambda-1}{\lambda+1} - \frac{8\lambda+3}{\lambda+1} + 2 = 0$

or

$9\lambda - 1 - (8\lambda + 3) + 2(1 + 1) = 0$

or $3\lambda - 2 = 0, \lambda = \frac{2}{3}$ i.e. ratio is 2: 3.

6. (*a*) $(h-3)^2 + (k+2)^2 = \left|\frac{5h - 12k - 13}{\sqrt{25+144}}\right|.$

Replace (h, k) by (x, y), we get

$13x^2 + 13y^2 - 83x + 64y + 182 = 0$, which is the required equation of the locus of the point.

7. (*b*) Let point be (x_1, y_1), then according to the condition

$\frac{3x_1 + 4y_1 - 11}{5} = -\left(\frac{12x_1 + 5y_1 + 2}{13}\right)$

Since the given lines are on opposite sides with respect to origin, hence the required locus is $99x + 77y - 133 = 0$

8. (*a*) Let the point be (x, y). Area of triangle with points (x, y), $(1, 5)$ and $(3, -7)$ is 21 *sq. units*

$\therefore \quad \frac{1}{2}\begin{vmatrix} x & y & 1 \\ 1 & 5 & 1 \\ 3 & -7 & 1 \end{vmatrix} = 21$

Solving; locus of point (x, y) is

$6x + y - 32 = 0.$

9. (*d*) The equation of a line passing through $(2, 2)$ and perpendicular to $3x + y = 3$ is

$y - 2 = \frac{1}{3}(x - 2)$ or $x - 3y + 4 = 0$.

Putting $x = 0$ in this equation, we obtain $y = 4/3$

So, y-intercept $= 4/3$.

10. (*a*) Take two perpendicular lines as the coordinate axes. If a, b be the intercepts made by the moving line on the coordinate axes, then the equation of the line is

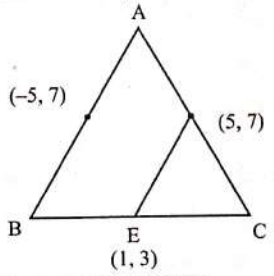
$\frac{x}{a} + \frac{y}{b} = 1$ ... (*i*)

According to the question $\frac{1}{a} + \frac{1}{b} = \frac{1}{k}$, (say)

i.e., $\frac{k}{a} + \frac{k}{b} = 1$ ... (ii)

The result (*ii*) shows that the straight line (*i*) passes through a fixed point (k, k).

11. (*b*) Slope of DE $= \frac{7-3}{5-1} = 1 \Rightarrow$ Slope of AB = 1



Hence equation of AB is

$y - 7 = (x + 5) \Rightarrow x - y + 12 = 0.$

12. (*c*) Required equation of medium is

$y + 8 = \frac{-\frac{3}{2} + 8}{-2 - 5}(x - 5)$

$\Rightarrow 13x + 14y + 47 = 0$

13. (*a*) Here intercept on x-axis is 3 and intercept on y-axis is –2. So using intercept form of the equation of line, the required line is $\frac{x}{3} - \frac{y}{2} = 1$.

14. (*d*) Given form is $3x + 3y + 7 = 0$

$\Rightarrow \frac{3}{\sqrt{3^2+3^2}}x + \frac{3}{\sqrt{3^2+3^2}}y + \frac{7}{\sqrt{3^2+3^2}} = 0$

$\Rightarrow \frac{3}{3\sqrt{2}}x + \frac{3}{3\sqrt{2}}y = \frac{-7}{3\sqrt{2}},$

$\therefore p = \left|\frac{-7}{3\sqrt{2}}\right| = \frac{7}{3\sqrt{2}}$

15. (*c*) Slope $= 2 \Rightarrow$ Equation is $2x - y = 2$.

16. (*d*) Here equation of AB is

$x + 4y - 4 = 0$ ...(*i*)

and equation of BC is $2x + y - 22 = 0$ ...(*ii*)

Thus angle between (*i*) and (*ii*) is given by

$$\tan^{-1}\left(\frac{-\frac{1}{4}+2}{1+\left(-\frac{1}{4}\right)(-2)}\right)=\tan^{-1}\left(\frac{7}{6}\right)$$

17. (*c*) $a_1a_2+b_1b_2=\frac{1}{ab'}+\frac{1}{a'b}=0$

Therefore, the lines are perpendicular

18. (*b*) $m_1=\frac{6+4}{-2-3}=\frac{10}{-5}=-2$ and

$$m_2=\frac{-18-6}{9-(-3)}=-2$$

Hence the lines are parallel.

19. (*d*) Here,

Slope of I[st] diagonal

$$=m_1=\frac{2-0}{2-0}=1\Rightarrow\theta_1=45^\circ$$

Slope of II[nd] diagonal

$$=m_2=\frac{2-0}{1-1}=\infty\Rightarrow\theta_2=90^\circ$$

$$\Rightarrow\ \theta_2-\theta_1=45^\circ=\frac{\pi}{4}$$

20. (*a*) Let the point (h, k) then $h+k=4$ ...(*i*)

and $1=\pm\frac{4h+3k-10}{\sqrt{4^2+3^2}}\Rightarrow 4h+3k=15$

...(*ii*)

and $4h+3k=5$ ...(*iii*)

On solving (*i*) and (*ii*); and (*i*) and (*iii*), we get the required points (3, 1) and (–7, 11).

Trick: Check with options. Obviously, points (3, 1) and (–7, 11) lie on $x+y=4$ and perpendicular distance of these points from $4x+3y=10$ is 1

21. (*a*) Required distance

$$=\frac{7}{\sqrt{(12)^2+5^2}}=\frac{7}{13}$$

22. (*c*) Let p be the length of the perpendicular from the vertex (2, –1) to the base $x+y=2$

Then $p=\left|\frac{2-1-2}{\sqrt{1^2+1^2}}\right|=\frac{1}{\sqrt{2}}$

If '*a*' be the length of the side of triangle, then

$$p=a\sin 60^\circ\Rightarrow\frac{1}{\sqrt{2}}=\frac{a\sqrt{3}}{2}\Rightarrow a=\sqrt{\frac{2}{3}}$$

23. (*a*) $L\equiv 2x+3y-4=0$,

$L_{(-6,2)}=-12+6-4<0$

$L'=6x+9y+8=0$,

$L'_{(-6,2)}=-36+18+8<0$

Hence the point is below both the lines.

24. (*a*) Equation of the line passing through (3, 8) and perpendicular to $x+3y-7=0$ is $3x-y-1=0$. The intersection point of both the lines is (1, 2).

Now let the image of A(3, 8) be $A'(x_1, y_1)$, then point (1, 2) will be the mid point of AA'.

$$\Rightarrow\frac{x_1+3}{2}=1\Rightarrow x_1=-1\ \text{and}\ \frac{y_1+8}{2}=2$$

$\Rightarrow y_1=-4$.

Hence the image is (–1, – 4).

25. (*b*) Required length

$$=\frac{4(3)+3(1)+20}{5}=7$$

26. (*b*) Here the lines are, $3x+4y-9=0$ ... (*i*)

and $6x+8y-15=0$ ... (*ii*)

Now distance from origin of both the lines are

$$\frac{-9}{\sqrt{3^2+4^2}}=-\frac{9}{5}\ \text{and}\ \frac{-15}{\sqrt{6^2+8^2}}=-\frac{15}{10}$$

Hence distance between both the lines are

$$\left|-\frac{9}{5}-\left(-\frac{15}{10}\right)\right|=\frac{3}{10}$$

Ailter: Put $y=0$ in the first equation, we get $x=3$ therefore, the point (3, 0) lies on it. So the required distance between these two lines is the perpendicular length of the line $6x+8y=15$ from the point (3, 0). *i.e.*, $\frac{6\times 3-15}{\sqrt{6^2+8^2}}=\frac{3}{10}$.

27. (*c*) Here the given lines are

$ax+by+c=0$ ...(*i*)

$bx+cy+a=0$ ...(*ii*)

$cx+ay+b=0$ ...(*iii*)

The lines will be concurrent, if

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$$

$$\Rightarrow\ a^3+b^3+c^3-3abc=0$$

28. (*b*) The set of lines is $4ax+3by+c=0$, where $a+b+c=0$.

Eliminating c, we get

$4ax+3by-(a+b)=0$

$\Rightarrow\ a(4x-1)+b(3y-1)=0$

This passes through the intersection of the lines $4x-1=0$ and $3y-1=0$ *i.e.* $x=\frac{1}{4}, y=\frac{1}{3}$ i.e., $\left(\frac{1}{4},\frac{1}{3}\right)$.

29. (*c*) Required line should be,

$(3x-y+2)+\lambda(5x-2y+7)=0$ ...(*i*)

$\Rightarrow\ (3+5\lambda)x-(2\lambda+1)y+(2+7\lambda)=0$

$$\Rightarrow\ y=\frac{3+5\lambda}{2\lambda+1}x+\frac{2+7\lambda}{2\lambda+1}$$ ...(*ii*)

As the equation (*ii*), has infinite slope, $2\lambda+1=0$

$\Rightarrow\lambda=-1/2$ putting $\lambda=-1/2$ in equation (*i*) we have $(3x-y+2)+(-1/2)(5x-2y+7)=0\Rightarrow x=3$.

30. (*a*) $(a-2b)x+(a+3b)y+3a+4b=0$ or

$a(x+y+3)+b(-2x+3y+4)=0$,

Which represents a family of straight lines through point of intersection of $x+y+3=0$ and $-2x+3y+4=0$ i.e., (–1, –2). **Trick:** Point (–1, –2) satisfies the given equation of straight line.

31. (*d*) $(p+2q)x+(p-3q)y=p-q$

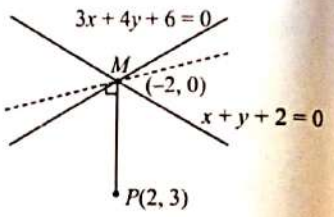
$px+py-p+2qx-3qy+q=0$

$p(x+y-1)+q(2x-3y+1)=0$

passing through intersection of $x+y-1=0$

$2x-3y+1=0$ is $\left(\frac{2}{5},\frac{3}{5}\right)$

32. (*a*)



The distance is greatest if required line is perpendicular to $PM m_{PM}=\frac{3}{4}$

$\Rightarrow$ Slope of required line $=\frac{-4}{3}$

$\Rightarrow$ Equation: $y-0=\frac{-4}{3}(x+2)$

$\Rightarrow\ 3x+3y+8=0$

33. (*a*) The equations of the bisectors of the angles between the lines are

$$\frac{x-2y+4}{\sqrt{1+4}}=\pm\frac{4x-3y+2}{\sqrt{16+9}}$$

Taking positive sign, then

$(4-\sqrt{5})x-(3-2\sqrt{5})$

$y-(4\sqrt{5}-2)=0$ ...(*i*)

and negative sign gives

$(4+\sqrt{5})x-(2\sqrt{5}+3)y+(4\sqrt{5}+2)=0$

Let 0 be the angle between the line (*i*) and one of the given line, then

$$\tan\theta=\left|\frac{\frac{1}{2}-\frac{4-\sqrt{5}}{3-2\sqrt{5}}}{1+\frac{1}{2}\cdot\frac{4-\sqrt{5}}{3-2\sqrt{5}}}\right|=\sqrt{5}+2>1$$

Hence the line (*i*) bisects the obtuse angle between the given lines.

34. (*b*) Bisectors of angles is given by

$$\frac{3x-4y+7}{5}=\pm\frac{12x+5y-2}{13}$$

$\Rightarrow 11x-3y+9=0$

and $21x+77y-101=0$

Let the angle between the line $3x-4y+7=0$ and (i) is α, then

$$\tan\alpha=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=\left|\frac{\frac{3}{4}-\frac{11}{3}}{1+\frac{3}{4}\times\frac{11}{3}}\right|=\frac{35}{45}<1$$

$\Rightarrow \alpha<45°$

Hence $11x-3y+9=0$ is the bisector of the acute angle between the given lines.

35. (*a*) Equation of angle bisectors between x and y-axis are $x+y=0$ and $x-y=0$, ($\because \theta=45°$ or 135) or $y=\pm x$.

36. (*a*) $\sqrt{3}x^2-4xy+\sqrt{3}y^2=0$

part of angle besection is $\frac{x^2-y^2}{\sqrt{3}-\sqrt{3}}=\frac{xy}{(-2)}$

$\Rightarrow x^2-y^2=0 \Rightarrow y^2-x^2=0$

37. (*a*) Let the coordinates of A be $(a, 0)$. Then the slope of the reflected ray is $\frac{3-0}{5-a}=\tan\theta$, (say)·

The slope of the incident ray

$=\frac{2-0}{1-a}=\tan(\pi-\theta)$

Since

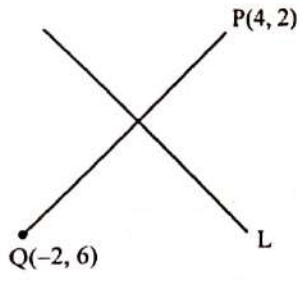
$$\tan\theta+\tan(\pi-\theta)=0\Rightarrow\frac{3}{5-a}+\frac{2}{1-a}=0$$

$$\Rightarrow 13-5a=0\Rightarrow a=\frac{13}{5}$$

Thus the coordinates of A are $\left(\frac{13}{5},0\right)$.

38. (*b*) After first transformation, the point will be (1, 4) and therefore, final point is (1 + 2, 4) = (3, 4).

39. (*a*) The mid point of P(−2, 6) and Q(4, 2) is $\left(\frac{-2+4}{2},\frac{6+2}{2}\right)$ i.e., (1, 4) and the gradient of line $PQ=\frac{2-6}{4+2}=\frac{-2}{3}$

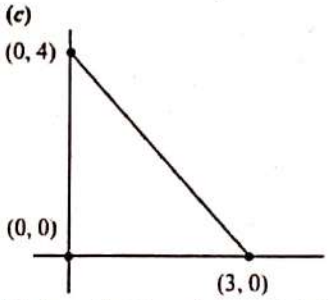


$\therefore$ The slope of $L=\frac{3}{2}$

Hence the equation of line which passes through point (1, 4) is

$$y-4=\frac{3}{2}(x-1)\Rightarrow 3x-2y+5=0$$

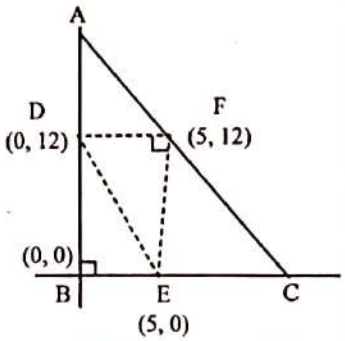
40. (*c*)



We know that for a right angled triangle, circumcenter is the mid-point of the hypotenuse

$\Rightarrow$ circumcentre is $\left(\frac{3}{2},2\right)$.

41. (*a*)



Clearly, given triangle is ABC which means orthocenter is (0, 0)

42. (*d*) $\Delta=\left|\frac{1}{2}\begin{vmatrix} a\cos\theta & b\sin\theta & 1\\ -a\sin\theta & b\cos\theta & 1\\ -a\cos\theta & -b\sin\theta & 1\end{vmatrix}\right|$

$$=\left|\frac{1}{2}\begin{vmatrix} a\cos\theta & b\sin\theta & 1\\ -a\sin\theta & b\cos\theta & 1\\ 0 & 0 & 2\end{vmatrix}\right|$$

$$=\left|\frac{1}{2}\times 2(ab\cos^2\theta+ab\sin^2\theta)\right|=ab$$

43. (*c*) By section, formula, A is

$$\left(\frac{3k-5}{k+1},\frac{5k+1}{k+1}\right)$$

$$\Delta=\left|\frac{1}{2}\begin{vmatrix}\frac{3k-5}{k+1} & \frac{5k+1}{k+1} & 1\\ 1 & 5 & 1\\ 7 & -2 & 1\end{vmatrix}\right|=2$$

$$\Rightarrow\left|\frac{7(3k-5)}{k+1}+6\frac{(5k+1)}{k+1}-37\right|=4$$

$\Rightarrow|21k-35+30k+6-37k-37|=4k+4$

$\Rightarrow|14k-66|=4k+4$

$\Rightarrow 7k-33=2k+2$ or $7k-33=-2k-2$

$\Rightarrow 5k=35$ or $9k=31$

$\Rightarrow k=7$ or $\frac{31}{9}$

44. (*d*) $\Delta=\frac{1}{2}\begin{vmatrix}2a & 3a & 1\\ 3b & 2b & 1\\ c & c & 1\end{vmatrix}=0$

$\Rightarrow(2a-c)(2b-c)-(3a-c)(3b-c)=0$

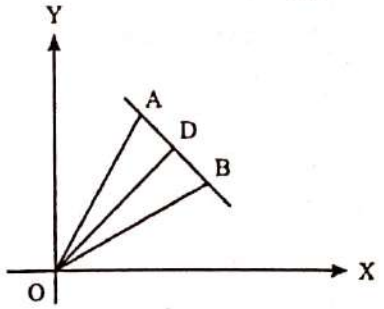
$\Rightarrow 4ab-2ac-2bc+c^2-(9ab-3ac-3bc+c^2)=0$

$\Rightarrow ac+bc-5ab=0$

$$\frac{1}{a}+\frac{1}{b}=\frac{5}{c}\Rightarrow\frac{1}{a}+\frac{1}{b}=2\left(\frac{5}{2c}\right)$$

$\therefore$ $a,\frac{2c}{5}$, b are in H.P.

45. (*b*) OA = OB = 9, $OD=\frac{15}{\sqrt{25}}=3$



Therefore,

$AB=2AD=2\sqrt{81-9}=2\sqrt{72}=12\sqrt{2}$

Hence $\Delta=\frac{1}{2}(3\times12\sqrt{2})=18\sqrt{2}$ sq. units

46. (*a*) A second degree homogeneous equation in x and y always represents a pair of straight lines passing through origin. But general equation of pair of straight lines, circle and conics are not homogenous.

47. (*a*) $$-6.10k+\frac{11.1.31}{4}-6\left(\frac{31}{2}\right)^2+10\left(\frac{10}{2}\right)^2-k\left(\frac{11}{2}\right)^2=0$$

$$\Rightarrow k\frac{361}{4}=\frac{5415}{4}\Rightarrow k=-15.$$

48. (*d*) Here $m_1+m_2=\frac{-2h}{b}$ and $m_1m_2=\frac{a}{b}$

Also, given that $4ab=3h^2$. Now we have to find $\frac{m_1}{m_2}$, therefore with the help of (i) and (ii), we get

$$(m_1-m_2)^2=\frac{4h^2-4ab}{b^2}=\frac{4h^2-3h^2}{b^2}=\frac{h^2}{b^2}$$

$$\Rightarrow m_1-m_2=\frac{h}{b}$$

Now on solving (i) and (iii), we get

$m_1=\frac{-h}{2b}$ and $m_2=\frac{-3h}{2b}$; $\therefore m_1:m_2=1:3$.

49. (a) It is a fundamental concept.

50. (a) $4x^2 + 12xy + 9y^2 = 0$

Here $h^2 - ab = 36 - 36 = 0$,

$\left(\text{from } \tan\theta = \frac{\pm 2\sqrt{h^2 - ab}}{a+b}\right)$

Hence, lines are real and coincident.

Prabal (JEE Main Level)

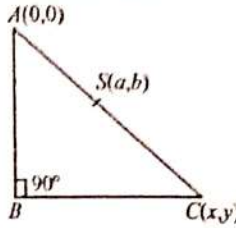
1. (a) By given information

Since in ΔABC, B is other centre. Hence $\angle B = 90°$

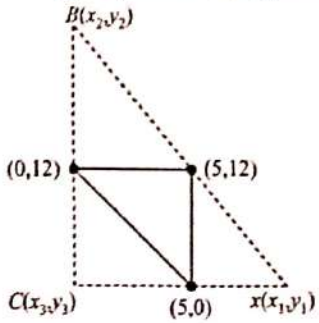
Circum centre is $S(a, b)$ $\frac{x+0}{2} = a$

$\Rightarrow x = 2a \frac{y+0}{2} = b \quad \Rightarrow \quad y = 2b$

Hence, $c(x, y) \equiv (2a, 2b)$



2. (a) $\begin{cases} x_1 + x_3 = 10 \ , & y_1 + y_3 = 0 \\ x_2 + x_3 = 0 \ , & y_2 + y_3 = 24 \\ x_1 + x_2 = 10 \ , & y_2 + y_2 = -24 \end{cases}$



$x_1 = x_2 = 10, y_1 - y_2 = -24$

$x_1 = 10, y_1 = 0$

$x_2 = 0, y_2 = 24$

$x_3 = 0, y_3 = 0$

$\left.\begin{matrix} x_1 = 10 \ , & y_1 = 0 \\ x_2 = 0 \ , & y_2 = 24 \\ x_3 = 0 \ , & y_3 = 0 \end{matrix}\right\} \Rightarrow \begin{matrix} A(10, 0) & on & x-axis \\ B(a, 24) & on & y-axis \\ C(0, 0) & is & origin \end{matrix}$

ΔABC is right angled $\Rightarrow$ orthocentre is $(0, 0)$

3. (a) $AP = \sqrt{x^2 + (y-4)^2}$

$BP = \sqrt{x^2 + (y+4)^2}$

$\because \ |AP - BP| = 6$

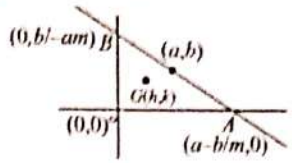
$AP - BP = \pm 6$

$\sqrt{x^2 + (y-4)^2} - \sqrt{x^2 + (y+4)^2} = \pm 6$

On squaring we get the locus of P, $9x^2 - 7y^2 + 63 = 0$

4. (a) Equation of line AB

$y - b = m(x - a)$



$\therefore \ G\left(\frac{a - \frac{b}{m}}{3}, \frac{b - am}{3}\right)$

$\Rightarrow \ h = \frac{a - \frac{b}{m}}{3}, k = \frac{b - am}{3}$

on eliminating 'm' we get required locus

$bh + ak - 3hk = 0 \Rightarrow bx + ay - 3xy = 0$

5. (d) $P(1,0), Q(-1,0), R(2,0)$, Locus of $s(h,k)$ if $SQ^2 + SR^2 2SP^2$

$\Rightarrow (h+1)^2 + k^2 + (h-2)^2 + k^2$

$= 2(h-1)^2 + 2k^2$

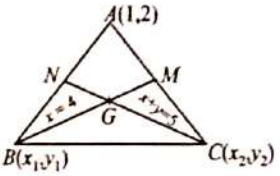
$\Rightarrow h^2 + 2h + 1 + h^2 - 4h - 4 = 2h^2 - 4h + 2$

$\Rightarrow 2h + 3 = 0$ Locus of $s(h, k)$

$\Rightarrow 2x + 3 = 0$

Parallel to y-axis.

6. (b)



$x_1 + y_1 = 5$ ...(i)

$x_2 = 4$ ...(ii)

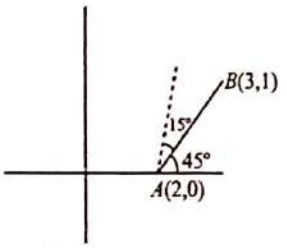
co-ordinates of G are $\equiv (4, 1)$

$\Rightarrow \frac{1 + x_1 + x_2}{3} = 4$ ...(iii)

and $\frac{y_1 + y_2 + 2}{3} = 1$ ...(iv)

solving above equations, we get B & C.

7. (c) Slope of AB is $\tan\theta = \frac{1-0}{3-2} = 1$



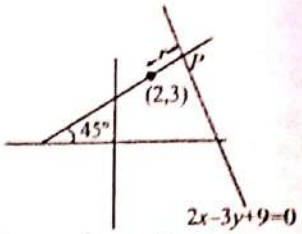
$\theta = 45°$

Hence equation of new line is

$y - 0 = \tan 60°(x - 2)$

$y = \sqrt{3}x - 2\sqrt{3} \Rightarrow \sqrt{3}x - y - 2\sqrt{3} = 0$

8. (b)



Let coordinates of point P by parametric

$P(2 + r\cos 45°, 3 + r\sin 45°)$

It satisfies the line $2x - 3y + 9 = 0$

$2\left(2 + \frac{r}{\sqrt{2}}\right) - 3\left(3 + \frac{r}{\sqrt{2}}\right) + 9 = 0$

$\Rightarrow r = 4\sqrt{2}$

9. (d) $ax + by + c = 0$

$\frac{3a}{4} + \frac{b}{2} + c = 0$

compare both $(x, y) \equiv \left(\frac{3}{4}, \frac{1}{2}\right)$

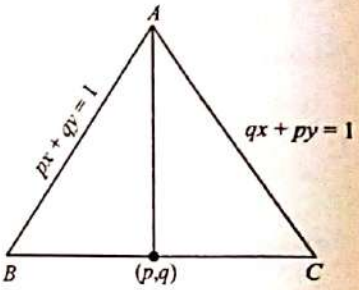
Hence given family passes through $\left(\frac{3}{4}, \frac{1}{2}\right)$

10. (c) $L_1 : Px + qy = 1$

$L_2 : qx + py = 1$

$L_1 + \lambda L_2 = 0$

$(px + qy - 1) + \lambda(qx + py - 1) = 0$



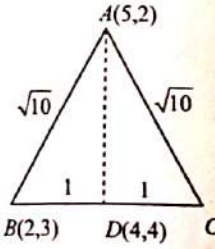
$\Rightarrow \lambda = \frac{(p^2 + q^2 - 1)}{(2pq - 1)}$

$\Rightarrow (2pq-1)(px + qy - 1)$

$= (p^2 + q^2 - 1)(qx + py - 1)$

11. (c) Equation of AD : $y - 4 = \frac{2}{-1}(x - 4)$

$\Rightarrow y - 4 = -2x + 8$

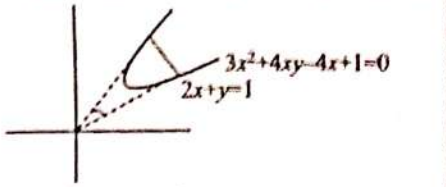


$\Rightarrow 2x + y = 12$

12. (a) Homogenize given curve with given line

$3x^2 + 4xy - 4x(2x + y) + 1(2x + y)^2 = 0$

$3x^2 + 4xy - 8x^2 - 4xy + 4x^2 + y^2 + 4xy = 0$



$-x^2 + 4xy + y^2 =$

coeff. x^2 + coeff. $y^2 = 0$

Hence angle is $90°$

13. (*c*) Point of intersection is (2, 3). Therefore, the equation of line passing through (2, 3) is

$y - 3 = m(x - 2)$

or $mx - y - (2m - 3) = 0$.

According to the condition,

$$\frac{3m-2-(2m-3)}{\sqrt{1+m^2}} = \frac{7}{5} \Rightarrow m = \frac{3}{4}, \frac{4}{3}$$

Hence the equation are $3x - 4y + 6 = 0$ and $3x - 3y + 1 = 0$.

14. (*a*) The given lines intersect at

$$\left(\frac{ab}{a+b}, \frac{ab}{a+b}\right)$$ and join of this with

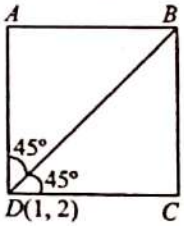
(a, b) will have slope $\frac{b^2}{a^2}$. Now proceed.

15. (*c*) Slope of BD is $\frac{8}{15}$ and angle made by BD with AD and DC is 45°. So let slope of DC be m, then

$$\tan 45° = \pm\frac{m - \frac{8}{15}}{1 + \frac{8}{15}m}$$

$$\Rightarrow (15 + 8m) = \pm(15m - 8)$$

$$\Rightarrow m = \frac{23}{7} \text{ and } -\frac{7}{23}$$



Hence the equation of DC and AD are

$$y - 2 = \frac{23}{7}(x - 1) \Rightarrow 23x - 7y - 9 = 0 \text{ and}$$

$$y - 2 = -\frac{7}{23}(x - 1) \Rightarrow 7x + 23y - 53 = 0.$$

16. (*d*) $p = \frac{ab}{\sqrt{a^2 + b^2}}$

or $\frac{a^2 + b^2}{a^2 b^2} = \frac{1}{p^2} \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$.

17. (*c*) Here $p = \left|\frac{-k}{\sqrt{\sec^2\alpha + \operatorname{cosec}^2\alpha}}\right|$, $p' = \left|\frac{-k\cos 2\alpha}{\sqrt{\cos^2\alpha + \sin^2\alpha}}\right|$

Hence

$$4p^2 + p'^2 = \frac{4k^2}{\sec^2\alpha + \operatorname{cosec}^2\alpha} + \frac{k^2(\cos^2\alpha - \sin^2\alpha)^2}{1}$$

$= 4k^2\sin^2\alpha\cos^2\alpha + k^2(\cos^4\alpha + \sin^4\alpha) - 2k^2\cos^2\alpha\sin^2\alpha$

$= k^2(\sin^2\alpha + \cos^2\alpha)^2 = k^2$.

18. (*b*) $\frac{a - at_2^2}{at_2^2 - at_1^2} = \frac{-2at_2}{2at_2 - 2at_1}$

$\Rightarrow (1 - t_2^2) = -t_2(t_2 + t_1)$ or $t_1 t_2 = -1$.

19. (*c*) The family of lines $(x - 2y + 3) + \lambda(2x - 3y + 4)$ is concurrent at point $P(1, 2)$. If the image of point $A(2, 3)$ in the above variable line is $B(h, k)$, then $AP = BP$ or

$(h-1)^2 + (k-2)^2 = (2-1)^2 + (3-2)^2$

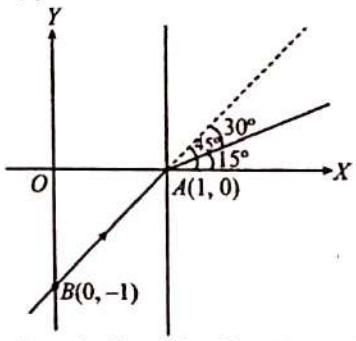
Hence, the locus of point P is

$x^2 + y^2 - 2x - 4y + 4 = 0$.

20. (*c*) $ax + by = 1$ will be one of the bisectors of the given lines. Equation of bisectors of the given lines are $\frac{3x+4y-5}{5} = \pm\left(\frac{5x-12y-10}{13}\right)$ or $64x - 8y = 115$ or $14x + 112y = 15$ or

$$a = \frac{64}{115}, b = -\frac{8}{115} \text{ or } a = \frac{14}{15}, b = \frac{12}{115}$$

21. (*d*)



From the figure, the refracted ray makes an angle of 15° with the positive direction of the x-axis and passes through the point $(1, 0)$. Its equation is $(y - 0) = \tan(45° - 30°)(x - 1)$ or $y = (2 - \sqrt{3})(x - 1)$

22. (*c*) Here, $m_1 + m_2 = -4$

and $m_1 m_2 = a$

Given that $m_1 = 3m_2$.

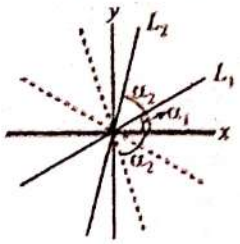
By (*i*), $3m_2 + m_2 = -4$

$\Rightarrow m_2 = -1$

Hence, $m_1 = -3$. Now, by (*ii*) $a = 3$.

23. (*a*) $ax^2 + 2hxy + by^2 = 0$

$m_1 + m_2 = \frac{-2h}{b}, m_1 m_2 = \frac{a}{b}$



Relation of slopes of image lines

$(m_1' + m_2') = -(m_1 + m_2)$

$= -\left(\frac{-2h}{b}\right) = \frac{2h}{b}\{m_1' = -\tan(\alpha_1)\}$

24. (*b*) Here the equation is $ax^2 - hxy - y^2 = 0$ and given that $m_1 = \tan\alpha$ and $m_2 = \tan\beta$ and we know that

$$m_1 + m_2 = \frac{h}{-1} = \tan\alpha + \tan\beta \text{ and}$$

$$m_1 m_2 = \frac{a}{-1} = \tan\alpha \cdot \tan\beta$$

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

$$= \frac{-h}{1-(-a)} = \frac{-h}{(1+a)}$$

25. (*b*) The distance between the parallel straight lines given by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $2\sqrt{\frac{g^2 - ac}{a(a+b)}}$.

Here, $a = 8, b = 2, c = 15, g = 13$. So, required distance

$$= 2\sqrt{\frac{169-120}{80}} = 2 \times \frac{7}{4} \cdot \sqrt{5} = \frac{7}{2}\sqrt{5}$$

26. (*c*) Sum of area of all triangles

$$= \frac{\sqrt{3}}{4}\left(1^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2 + \ldots\infty\right)$$

$$= \frac{\sqrt{3}}{4}\left(\frac{1}{1 - \frac{1}{4}}\right) = \frac{1}{\sqrt{3}}$$

27. (*d*) $A(0,3), B\left(\frac{-15}{7}, \frac{-24}{7}\right), C\left(\frac{60}{17}, \frac{6}{17}\right)$.

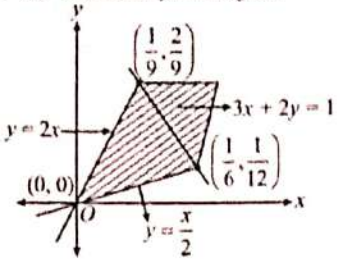
If $P(\alpha, 0)$ always lie inside the Δ then (*i*)

Point A and P are on the same side of $2x - 3y = 6$ now $2(0) - 9 - 6 =$ negative.

$\Rightarrow 2a - 6$ is also negative i.e., $2a - 6 \leq 0$ or $a \leq 3$;

(*ii*) Point B and P are on the same side of $3x + 4y - 12 = 0$

28. (*d*) ∴ Area of parallelogram



$$=2\left(\frac{1}{2}\begin{vmatrix}0 & 0 & 1\\ \frac{1}{9} & \frac{2}{9} & 1\\ \frac{2}{12} & \frac{1}{12} & 1\end{vmatrix}\right)=\frac{1}{36}$$

29. (*c*) Line $ax + by + 2 = 0$ always passes through the point $\left(2,\frac{8}{3}\right)$

$\therefore 6a + 8b + 6 = 0$ or $3a + 4b + 3 = 0$

Now $bx - ay + 4 = 0$ and $3x + 4y + 5 = 0$ meets axis in concyclic points. So, $m_1m_2 = 1$

$$\left(\frac{b}{a}\right)\cdot\left(\frac{-3}{4}\right)=1$$

$\Rightarrow 4a + 3b = 0$

Solving (1) and (2), we get $a=\frac{9}{7}, b=\frac{-12}{7}$

$\Rightarrow$ Line $ax + by + 3 = 0$ always passes through the point $(-1, 1)$

Algebraic sum of distance = 0

$$\Rightarrow \frac{a+2b+2}{\sqrt{a^2+b^2}}+\frac{2a+b+2}{\sqrt{a^2+b^2}}+\frac{3a+5b+2}{\sqrt{a^2+b^2}}=0$$

$\Rightarrow 3a + 4b + 3 = 0$

Also, the other two lines are

$$\frac{y}{4/a}-\frac{x}{4/b}=1 \text{ and } \frac{x}{-5/3}+\frac{y}{-5/4}=1$$

As the points are concyclic

$OA \cdot OC = OB \cdot OD$

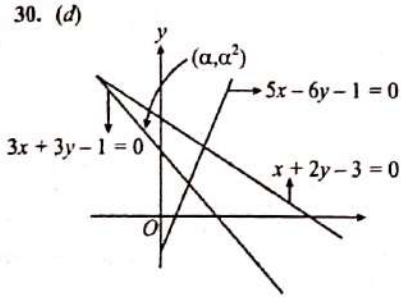
$\Rightarrow 3b = -4a$

From (1) and (2)

$a=\frac{9}{7}, b=\frac{-12}{7}$

$\Rightarrow$ Line $ax + by + 3 = 0$ always passes through the point $(-1, 1)$

30. (*d*)



O and the point (α, α^2) lie on the opposite sides with respect to $2x + 3y - 1 = 0$. Hence, $2\alpha + 3\alpha^2 - 1 > 0$

O and the point (α, α^2) lie on the same side with respect to $x + 2y - 3 = 0$. Hence, $\alpha + 2\alpha^2 - 3 < 0$

Again O and the point (α, α^2) lie on the same side with respect to $5x - 6y - 1 = 0$.

Hence, $5\alpha - 6\alpha^2 - 1\left\langle 0 \text{ or } 6\alpha^2 - 5\alpha + 1\right\rangle 0$

31. (*d*) Let point are

$P\left(t_1, 2at_1 + at_1^3\right), Q\left(t_1, 2at_2 + at_1^3\right)$

$R\left(t_3, 2at_3 + at_3^3\right)$

P, Q and R are collinear

$\Rightarrow$ area$(\Delta PQR) = 0$

$$\Rightarrow \begin{vmatrix}1 & t_1 & 2at_1+at_1^3\\ 1 & t_2 & 2at_2+at_2^3\\ 1 & t_2 & 2at_3+at_3^3\end{vmatrix}=0$$

$$\Rightarrow \begin{vmatrix}1 & t_1 & 2at_1\\ 1 & t_2 & 2a_2\\ 1 & t_3 & 2at_3\end{vmatrix}+\begin{vmatrix}1 & t_1 & at_1^3\\ 1 & t_2 & at_2^3\\ 1 & t_3 & at_3^3\end{vmatrix}=0$$

$$\Rightarrow 2a\begin{vmatrix}1 & t_1 & t_1\\ 1 & t_2 & t_2\\ 1 & t_3 & t_3\end{vmatrix}+a\begin{vmatrix}1 & t_1 & t_1^3\\ 1 & t_2 & t_2^3\\ 1 & t_3 & t_3^3\end{vmatrix}=0$$

Determinant is zero if two roows of a determinant are same

$$\Rightarrow 0+a\begin{vmatrix}1 & t_1 & t_1^3\\ 1 & t_2 & t_2^3\\ 1 & t_3 & t_3^3\end{vmatrix}=0$$

$R_2 \to R_2 - R_1 \quad R_3 \to R_3 - R_1$

$$\Rightarrow \begin{vmatrix}1 & t_1 & t_1^3\\ 1 & t_2-t_1 & t_1^3-t_1^3\\ 0 & t_2-t_1 & t_1^3-t_1^3\end{vmatrix}=0$$

$$\Rightarrow (t_2-t_1)(t_3-t_1)\begin{vmatrix}1 & t_1 & t_1^3\\ 0 & 1 & t_1^2+t_1t_2+t_2^2\\ 0 & 1 & t_1^2+t_1t_3+t_3^2\end{vmatrix}=0$$

$$\Rightarrow (t_2-t_1)(t_2-t_1)\left[t_1^2+t_1t_2+t_3^2-t_1^2-t_1t_2-t_2^2\right]=0$$

$$\Rightarrow (t_2-t_1)(t_3-t_1)\left[t_2^3+t_1t_3-t_1t_2-t_2^2\right]=0$$

$$\Rightarrow (t_2-t_1)(t_3-t_1)(t_3-t_2)(t_1+t_2+t_3)=0$$

as $t_1 \equiv t_2 \equiv t_3$

$\Rightarrow t_2 - t_1$ or $t_3 - t_1$ or $t_3 - t_2$ can't be zero

$\Rightarrow t_1 + t_2 + t_3 = 0$

32. (*a*) MDNC forms a square.

$CN = MD = 5$

$MC = DN = 4$

LEt ND be x,

For ΔCDN

By pythagorus theorem,

$C^2 = 4^2 + 5^2$

$CD = \sqrt{41}$

For Delta DNB

By pythagorus theorem,

$DB = \sqrt{4^2 + x^2}$

For Delta CDB

by pythagorus theorem

$CB^2 = CD^2 + DB^2$

$(CN + NB)^2 = 41 + \left(16 + x^2\right)$

$(5 + x)^2 = 41 + 16 + x^2$

$25 + 10x + x^2 = 53 + x^2$

$x=\frac{16}{5}$

$CB = 5 + x = 5 + \frac{16}{5}$

$CB = \frac{41}{5}$

Similarly,

Take AM y.

In Delta AMD

$AD = 25 + y^2$

In Delat ACD

$(4 + y)^2 = \left(25 + y^2\right) + 41$

$y=\frac{25}{4}$

$AC = \frac{25}{4} + 4$

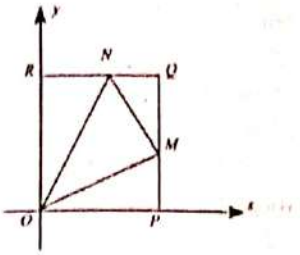
$AC = \frac{41}{4}$

33. (*b*) We can assume that OP and OR x-axis and y-axis respectively. Let $OP = a$, then ar(sq. $OPQR$) = a^2 coordinates of M and N are $\left(a,\frac{a}{2}\right)$ and $\left(\frac{a}{2},a\right)$ respectively

$$\therefore ar(\Delta OMN)=\frac{1}{2}\begin{vmatrix}a & \frac{a}{2}\\ \frac{a}{2} & a\end{vmatrix}=\frac{3a^2}{8}$$

$\therefore \frac{8}{3}=\frac{k}{6}$

$\therefore k = 16$



34. (*c*) Shift the origin to the point (0, 3) so that any point (x_1, y_1) on the reflected line is given by

$$\begin{bmatrix} x_1 \\ y_1 - 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1-q(h+3)}{p} \\ h \end{bmatrix}$$

(since $m = 0$)

$$\Rightarrow x_1 = \frac{1-q(h+3)}{p}, y_1 - 3 = -h$$

$$\Rightarrow px_1 = qy_1 + 1 - 6q$$

and hence the reflected line is $px - qy = 1 - 6q$.

35. (*d*) The triangle cannot be equilateral if coordinates of vertices of the traingle is always rational

36. (*a*) Reflection of $y = \log_{10} x$

About $y = x$ is $x = \log_{10} y$

$\Rightarrow y = 10^x$

37. (*b*) Let $y = mx$ be a line common to the given pairs of lines. The $am^2 + 2m + 1 = 0$ and $m^2 + 2m + a = 0$

$$\Rightarrow \frac{m^2}{2(1-a)} = \frac{m}{a^2-1} = \frac{1}{2(1-a)}$$

$$\Rightarrow m^2 = 1 \quad \text{and} \quad m = -\frac{a+1}{2}$$

$\Rightarrow (a + 1)^2 = 4 \Rightarrow a = 1$ or -3

But for $a = 1$, the two pairs have both the line common. So $a = -3$ and the slope m of of the line common to both pairs is 1. Now,

$$x^2 + 2xy + ay^2 = x^2 + 2xy - 3y^2$$

$$= (x - y)(y + 3y)$$

and $ax^2 + 2xy + y^2 = -3x^2 + 2xy + y^2 = -(x - y)(3x + y)$

So the equation of the required lines is $(x + 3y)(3x + y) = 0 \Rightarrow 3x^2 + 10xy + 3y^2 = 0$

38. (*a*) The above given equation $9x^2 - 24xy + 16y^2 - 12x + 16y - 12 = 0$ can be factorized as

$(3x - 4y + 2)(3x - 4y - 6) = 0$

Since both the line are parallel distance between them will be

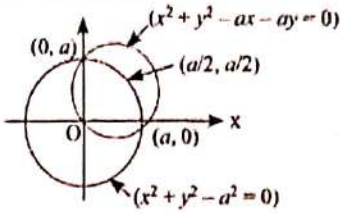
$$d = \left|\frac{C_2 - C_1}{\sqrt{3^2 + 4^2}}\right| = \left|\frac{2-(-6)}{5}\right| = \frac{8}{5}$$

39. (*a*) Applying the condition, $4\lambda h^2 = ab(1 + \lambda)^2$ Here $\lambda = 2$, therefore

$$4 \times 2 \times \left(\frac{h}{2}\right)^2 = 1 \times 2(1+2)^2$$

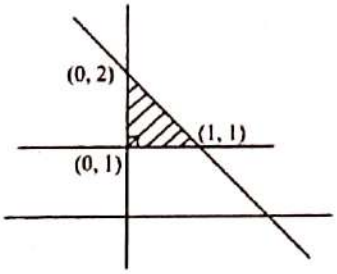
$\Rightarrow h = 9 \Rightarrow h = \pm 3$

40. (*b*) Clearly from figure $xy = 0$ represents the pair of straight lines.



i.e., combined equation of both axes.

41. (*d*)



$b = 0$

$c = 1$

$$a = \frac{1}{2} \times 1 \times 1$$

$4a + b + c = 3$

42. (*b*) The line $x + y = a$ cuts the co-ordinate axes at A $(a, 0)$, B $(0, a)$.

Circumcentre is the mid-point of AB, I.e $(a/z, a/2)$ centroid is $(a/3, a/3)$, or the centre is the origin.

All lie on $y = x$. In centre lies on the angle bisector of $\angle AOB$, which is also $y = x$.

43. (*c*) We have

$2x\sin^2\theta + y\cos^2\theta = 2(\cos^2\theta - \sin^2\theta)$

$\Rightarrow 2x\sin^2\theta + 2\sin^2\theta + y\cos^2\theta - 2\cos^2\theta = 0$

$\Rightarrow (2x + 2)\sin^2\theta + (y - 2)\cos^2\theta = 0$

$\Rightarrow (2x + 2) + (y - 2)\cot^2\theta = 0 \equiv L_1 + \lambda L_2 = 0$

$\therefore$ The line passes through fixed point $(-1, 2)$

$\Rightarrow a = -1$ and $b = 2 \quad \Rightarrow a + b = 1$

44. (*d*) Let $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ be coordinates of triangle where $x_i, y_i (i = 1, 2, 3)$ are rational numbers.

now the centroid of triangle is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$ which is

rational point.

In centre, circumcentre and orthocenter depend on sides of the triangle which may not be rational even if the co-ordinates of vertices are so.

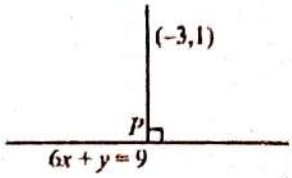
45. [100]

$6x + y = 9$

Equation of perpendicular line from $(-3, 1)$

$$y - 1 = \frac{1}{6}(x + 3)$$

$6y - 6 = x + 3$



$\Rightarrow x - 6y - 9 = 0 \Rightarrow 6x - 36y + 54 = 0$

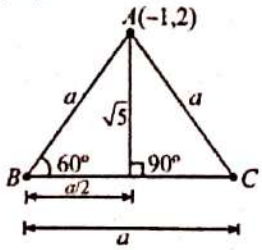
$\Rightarrow 6x + y - 9 = 0$

$-37y + 63 = 0$

$$y = \frac{63}{37} = \frac{a}{b}$$

$a + b = 100$

46. [23]



Line of BC is $2x - y = 1$

$$AD = \left|\frac{2(-1) - 2 - 1}{\sqrt{2^2 + (-1)^2}}\right|$$

$= \sqrt{5}$

$$\because \tan 60° = \frac{\sqrt{5}}{a/2} = \sqrt{3}$$

$$\Rightarrow a = \frac{2\sqrt{5}}{\sqrt{3}} = \sqrt{\frac{20}{3}}$$

47. [9] Let P be $(a, 4 - a)$

$|PA - PB|$ is minimum when $PA = PB$

$\Rightarrow (a - 4)^2 + (1 - a)^2 = (a - 6)^2 + (4 - a)^2$

$\Rightarrow 1 - a = +(a - 6)$

$$\Rightarrow a = \frac{7}{2}$$

$$\therefore P \text{ is } \left(\frac{7}{2}, \frac{1}{2}\right)$$

$$\Delta = \left|\frac{1}{2}\right| \begin{vmatrix} 4 & 3 & 1 \\ 6 & 0 & 1 \\ 7/2 & 1/2 & 1 \end{vmatrix} = \frac{13}{4} \Rightarrow m = 13, n = 4$$

Parikshit (JEE Advanced Level)

1. (*c*) Point of trisection is (5, 4). After rotation the slope of the line

$$\tan\theta_1 = \frac{\tan 45^\circ + \tan\theta_2}{1 - \tan 45^\circ \tan\theta_2} = \frac{1+\frac{1}{2}}{1-\frac{1}{2}} = 3$$

2. **(a)** Area $= \frac{1}{2}\begin{vmatrix} a & x & 1 \\ a+d & xr & 1 \\ a+2d & xr^2 & 1 \end{vmatrix}$

$$= \frac{1}{2}\begin{vmatrix} a & x & 1 \\ a & xr & 1 \\ a & xr^2 & 1 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} 0 & x & 1 \\ d & xr & 1 \\ 2d & xr^2 & 1 \end{vmatrix}$$

$$= 0 + \frac{1}{2}\begin{vmatrix} 0 & x & 1 \\ d & xr & 1 \\ 2d & xr^2 & 1 \end{vmatrix},$$ which clearly shows that area is independent of a.

3. **(b)** $9a + 4b = 5$

Using $AM \geq GM$

$$\frac{9a+4b}{2} \geq \sqrt{(9a)(4b)}$$

$$\Rightarrow \frac{5}{2} \geq 6\sqrt{ab}$$

$$\Rightarrow \sqrt{ab} \leq \frac{5}{12}$$

Now $(3\sqrt{a}+2\sqrt{b})^2 = 9a+4b+12\sqrt{ab}$

$= 5 + 12\sqrt{ab}$

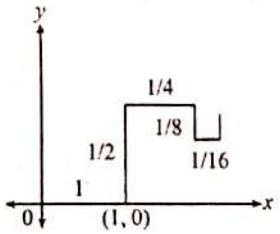
$\leq 5 + 5$

$(3\sqrt{a}+2\sqrt{b})^2 \leq 10$

$\Rightarrow 3\sqrt{a}+2\sqrt{b} \leq \sqrt{10}$

4. **(b)** x-coordinate:

$$1+\frac{1}{4}+\frac{1}{16}+\ldots\infty = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$$ y-coordinate: $\frac{1}{2}-\frac{1}{8}+\frac{1}{32}+\ldots\infty = \frac{\frac{1}{2}}{1+\frac{1}{4}} = \frac{2}{5}$



Hence $\left(\frac{4}{3}, \frac{2}{5}\right)$

5. **(a)** The equation of line passing through two given points (x_1, y_1) and (x_2, y_2) is

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Now, the given expression is

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

or $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2+x_3 & y_2+y_3 & 1 \end{vmatrix} = 0$

or $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ \frac{x_2+x_3}{2} & \frac{y_2+y_3}{2} & 1 \end{vmatrix} = 0$

This is the equation of the line passing through the points (x_1, y_1) and $((x_2 + x_3)/2, (y_2 + y_3)/2)$. This is the equation of median through vertex A.

6. **(d)** $x^3 - x^2 - x - 2 = 0 \Rightarrow (x-2)(x^2 + x + 1) = 0 \Rightarrow x = 2$ ...(1)

$xy^2 + 2xy + 4x - 2y^2 - 4y - 8 = 0$

$\Rightarrow (x-2)(y^2 + 2y + 4) = 0 \Rightarrow x = 2$ ...(2)

Both the equation represent same line. So number of triangle formed are zero.

7. **(a)** Let the common line be $y = mx$. Then it must satisfy both the equations.

Therefore, we have

$hm^2 + 2hm + a = 0$ ...(1)

$h'm^2 + 2h'm + a' = 0 = 0$ ...(2)

Solving equations (1) and (2), we get

$$\frac{m^2}{2(ha'-h'a)} = \frac{m}{ab'-a'b} = \frac{a}{2(bh'-b'h)}$$

Eliminating 'm', we get

$$\left[\frac{ab'-a'b}{2(bh'-b'h)}\right]^2 = \frac{ha'-h'a}{bh'-b'h}$$

$\Rightarrow (ab' - a'b)^2 = 4(ha' - h'a)(bh' - b'h)$

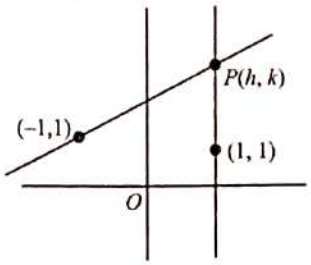
8. **(c,d)**

$|m_1 - m_2| = 2$

$m_1 = \frac{k-1}{h-1}$, $m_2 = \frac{k-1}{h+1}$

$$\Rightarrow \left(\frac{k-1}{h-1} - \frac{k-1}{h+1}\right)^2 = 4$$

$$\Rightarrow (k-1)^2\left(\frac{2}{h^2-1}\right)^2 = 4$$



$\Rightarrow (k-1)^2 = (h^2-1)^2$

$\Rightarrow (y-1) = \pm(x^2-1)$

$\Rightarrow y = x^2$ or $y = 2 - x^2$

9. **(a,b)**

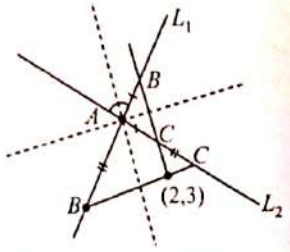
$L_1 : 2x + y = 5$ $L_2 : x - 2y = 3$

Line BC passing throug $(2, 3)$

$(y-3) = m(x-2)$

m is equal to slope of

$$\frac{2x+y-5}{\sqrt{2^2+1}} = \pm\frac{x-2y-3}{\sqrt{1+2^2}}$$



$\Rightarrow 2x \mp x + y \pm 2y = 5 \mp 3$

$x + 3y = 2 \Rightarrow m = -\frac{1}{3}$

& $3x - y = 8 \Rightarrow m = 3$

BC line

$y - 3 = (x-2) \Rightarrow 3x - y = 3$

& $y - 3 = -\frac{1}{3}(x-2)$

$\Rightarrow x + 3y = 11$

10. **(a,b)** Let 'w' be the length of the shorter segment and 'l' be the length of the longer segment. We are given that, $\frac{w}{l} = \frac{l}{w+l}$.

Cross-multiplying, we find that, $w^2 + wl = l^2$

$\Rightarrow w^2 + wl - l^2 = 0$

Now, we divide both sides by l^2 to get,

$$\left(\frac{w}{l}\right)^2 + \left(\frac{w}{l}\right) - 1 = 0$$

$\therefore R^2 + R - 1 = 0$

For this, we have, $R^2 = -R + 1$

Dividing both sides by R, we get,

$R = -1 + \frac{1}{R}$

$\Rightarrow R^{-1} = R + 1$

Therefore, $R^2 + R^{-1} = -R + 1 + R + 1 = 2$

Finally we have

11. **(a)**

12. **(b)**

13. **(c)**

Sol. **(11, 12, 13)**

Let the co-ordinates of $D(\alpha, \beta)$

then $\frac{\alpha+1+3}{3} = 3$

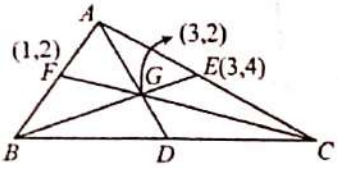
$\Rightarrow \alpha = 5$

and $\frac{\beta+2+4}{3} = 2$

$\Rightarrow \beta = 0$

$\therefore D(5, 0)$

Taking $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$



then by $\frac{x_1+x_2}{2}=1$, $\frac{x_2+x_3}{2}=5$, $\frac{x_3+x_1}{2}=3$

and $\frac{y_1+y_2}{2}=2$, $\frac{y_2+y_3}{2}=0$, $\frac{y_1+y_3}{2}=4$

we get $A(-1, 6)$, $B(3, -2)$, $C(7, 2)$

equation of AB is $2x + y = 4$

Height of altitude from

A is $= \frac{2\times \text{area}(\Delta ABC)}{BC} = 6\sqrt{2}$

14. **(d)** After solving, we get

$c + f = 4$

15. **(a)** Equation of a straight line through (2,3) and inclined at an angle of $(\pi/3)$ with y-axis ($(\pi/6)$ with x-axis) is

$$\frac{x-2}{\cos(\pi/6)}=\frac{y-3}{\sin(\pi/6)}$$

$$\Rightarrow x-\sqrt{3}y=2-3\sqrt{3}$$

Points at a distance $c + f = 4$ units from point P are

$(2 + 4\cos(\pi/6), 3 + 4\sin(\pi/6)) \equiv (2 + 2\sqrt{3}, 5)$

and $(2 - 4\cos(\pi/6), 3 - 4\sin(\pi/6)) \equiv (2 - 2\sqrt{3}, 1)$

only (A) is true out of given options

16. **(c)** Slopes of the lines

$3x + 4y = 5$ is $m_1 = -\frac{3}{4}$

and $4x - 3y = 15$ is $m_2 = \frac{4}{3}$

$\because$ $m_1 m_2 = -1$

$\therefore$ given lines are perpendicular and

$\angle A = \frac{\pi}{2}$

Now required equation of BC is

$$(y-2)=\frac{m\pm\tan(\pi/4)}{1\mp m\tan(\pi/4)}(x-1) \quad \ldots(1)$$

where m = slope of $AB = -\frac{3}{4}$

$\therefore$ equation of BC is (on solving (1))

$x - 7y + 13 = 0$ and $7x + y - 9 = 0$

$L_1 \equiv x - 7y + 13 = 0$

$L_2 \equiv 7x + y - 9 = 0$

Let required line be $x + y = a$

which is at $|b - 2a - 1| = |5 - 4 - 4\sqrt{3} - 1|$

$= 4\sqrt{3}$ units from origin

$\therefore$ required line is $x + y - 4\sqrt{6} = 0$ (since intercepts are on positive axes only)

17. **(b)** The equation of any line parallel to $2x + 6y + 7 = 0$ is $2x + 6y + k = 0$.

This meets the axes at $A(-k/2, 0)$ and $B(0, -k/6)$. By hypothesis

$$AB = 10 \Rightarrow \sqrt{\frac{k^2}{4}+\frac{k^2}{36}} = 10 \Rightarrow \sqrt{\frac{10k^2}{36}} = 10$$

$$\Rightarrow 10k^2 = 3600 \Rightarrow k = \pm 6\sqrt{10}$$

Hence, there are two lines given by

$2x + 6y \pm \sqrt{10} = 0$

18. **(d)** The equation of a line passing through (2, 2) and $\perp$ to $3x + y = 3$ is $y - 2 = 1/3(x - 2)$ or $x - 3y + 4 = 0$.

Putting $x = 0$ in this equation, we obtain $y = 4/3$.

So, y-intercept = 4/3.

19. **(a)** The equation of a line $\perp$ to $5x - y = 0$ is $x + 5y = c$. This meets with the coordinate axes at $A(c, 0)$ and $B(0, c/5)$.

So, area of $\Delta OAB = (1/2)c.c/5 = c^2/10$ It is given that area of ΔOAB = 5. sq.units.

$$\therefore \frac{c^2}{10} = 5 \Rightarrow c = \pm 5\sqrt{2}$$

Hence, the lines are, $x + 5y = \pm 5\sqrt{2}$

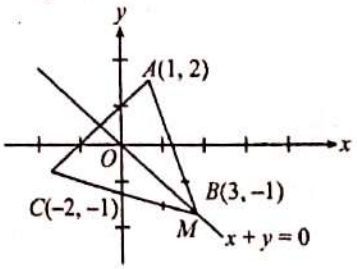
20. **(b)** Let C be the reflection of A in $L \equiv x + y = 0$

$\Rightarrow C \equiv (-2, -1)$

Now $AM + BM = CM + BM$, which is minimum if B, M and C are collinear.

Now, equation of BC is $(y + 1) = 0(x, 2)$

$\Rightarrow y + 1 = 0$



$\therefore$ On solving equation (1) with $x+y=0$, we get $M = (1, -1)$

Hence the reflection of M is the line $x = y$ is $(-1, 1)$

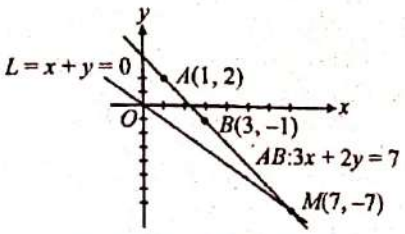
21. **(d)** We have $|AM - BM| \le AB$

$\therefore$ For $AM - BM|$ to be maximum, point A, B and

M must be collinear. Now equation of AB is

$$(y-2)=\frac{-3}{2}(x-1)$$

$$\Rightarrow 3x + 2y = 7$$



$\therefore$ On solving equation (1) with $x + y = 0$, we get

$M = (7, -7)$

Hence distance of M(7, −7) from

$$N(1,1) = MN = \sqrt{(7-1)^2+(-7-1)^2}$$

$$= \sqrt{36+64} = 10 \text{ units}$$

22. **(a)** $|AM - BM|_{\min} \Rightarrow AM = BM$.

Let M be $(h - h)$ $AM = BM$

$$\Rightarrow (h-1)^2+(-h-2)^2=(h-3)^2+(-h-1)^2$$

$$\Rightarrow -2h+1+4h+4=-6h+9+2h+1$$

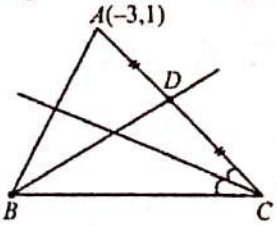
$$\Rightarrow h = 5/6 \Rightarrow M\left(\frac{5}{6},\frac{-5}{6}\right)$$ Now, find area

ΔAMB using the formula $\frac{1}{2}$ × base × height where, height = distance of M from AB

23. **(d)** $[(A) \to (R), (B) \to (S), (C) \to (Q)]$

B median $2x + y - 3 = 0$

angle bisector of C $7x - 4y - 1 = 0$



Let C on the line $7x - 4y - 100$

$$C\left(\lambda, \frac{7\lambda+1}{4}\right)$$

D is mid point of AC lie median

$$D\left(\frac{-3+\lambda}{2}, \frac{1+\frac{7\lambda-1}{4}}{2}\right)$$

$$2\left(\frac{-3+\lambda}{2}\right)+\frac{3+7\lambda}{8}-3=0$$

$-48 + 8\lambda + 3 + 7\lambda = 0 \Rightarrow \lambda = 3$

$C(3, 5)$ & $D(0, 3)$

(c) line AC is $y - 3 = \frac{2}{3}(x - 0)$

$\Rightarrow 3y - 2x - 9 = 0$

(P) will not a side Q (It's given median)

(a) Line $ABA(-3,1)$ satisfy (R) $4x+7y+5=0$ &

(b) Line BC is only (S) $18x - y - 49 = 0$

24. **(b)** [(A) → S; (B) → P, Q; (C) → R; (D) → P, Q, S]

(A) L_1 and L_3 intersect at (2, 1), if $k = 5$.

(B) As L_1 and L_3 intersect they are not parallel.

So L_2 is parallel to L_1 if $k = -9$

and L_2 is parallel to L_3 if $k = -\frac{6}{5}$

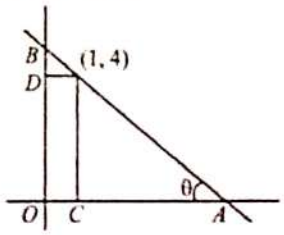
(C) The lines L_1, L_2, L_3 will form a triangle, if no two of them are parallel and they are not concurrent. Thus $k = \frac{5}{6}$.

(D) Lines L_1, L_2, L_3 will not form a triangle, if two of them are parallel or all of them are concurrent.

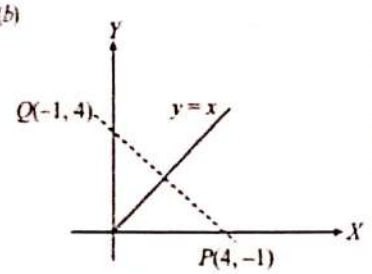
25. **(a)** (a) $OA = 1 + 4\cot\theta$, $OB = 4 + \tan\theta$

$OA + OB = 5 + 4\cot\theta + \tan\theta > 5 + 2\sqrt{4\cot\theta\tan\theta}$

$= 5 + (2 \times 2) = 9$



(b)



The reflection of $P(4,-1)$ on $y=x$ is $Q(-1,4)$.

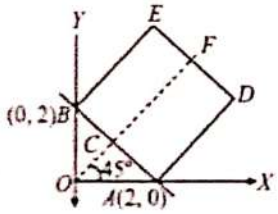
Hence,

$PQ = \sqrt{(4+1)^2 + (-1-4)^2} = \sqrt{50} = 5\sqrt{2}$

(c) $AB = 2\sqrt{2}$, $OC = \sqrt{2}$

The maximum value of d is

$OF = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$



(d) The given line is $x = 4 + \frac{1}{\sqrt{2}}\left(\frac{y+1}{\sqrt{2}}\right)$

or

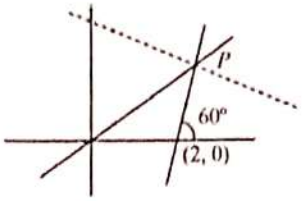
$y = 2x - 9$

Hence, the intercept made by the x-axis is 9/2.

26. **[2]**

$y - 0 = \sqrt{3}(x - 2)$ it intersect $y = x$

$\Rightarrow x = \frac{2\sqrt{3}}{\sqrt{3}-1}$



So P is $\left(\frac{2\sqrt{3}}{\sqrt{3}-1}, \frac{2\sqrt{3}}{\sqrt{3}-1}\right)$

So required line is

$y - \frac{2\sqrt{3}}{\sqrt{3}-1} = -\frac{1}{\sqrt{3}}\left(x - \frac{2\sqrt{3}}{\sqrt{3}-1}\right)$

it intersect y-axis at $x = 0$

$y = \frac{2\sqrt{3}}{\sqrt{3}-1}\left(1 + \frac{1}{\sqrt{3}}\right) = 4 + 2\sqrt{3}$

27. **[19]**

Equation of family of curves passing through intersection of C_1 & C_2 is

$-\lambda x^2 + 4y^2 - 2xy - 9x + 3 + \mu(2x^2 + 3y^2 - 4xy + 3x - 1) = 0$ ...(i)

It will give the joint equation of pair of lines passing through origin, if coefficient of $x = 0$ & Constant = 0

$\Rightarrow \mu = 3$

put $\mu = 3$ in equation (i), we get

$-\lambda x^2 + 4y^2 - 2xy + 6x^2 + 9y^2 - 12xy = 0$

It will subtend 90° at origin if coeff. of x^2 + coeff. of $y^2 = 0$

$\Rightarrow \lambda = +19$

28. **[2]** Given lines are $ax + y + 1 = 0$ ...(i)

$x + by = 0$ ...(ii)

$ax + by = 1$ ...(iii)

Joint equation of (i) and (ii) is $(ax + y + 1)(x + by) = 0$

$\Rightarrow ax^2 + by^2 + (ab + 1)xy + x + by = 0$

Making (iv) homogeneous with the help of equation (i) we have $ax^2 + by^2 + (ab + 1)xy + x(ax + by) + by(ax + by) = 0$

since angle between these two lines is 90°

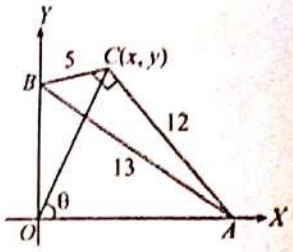
$\therefore$ Coefficient of x^2 + Coefficient of $y^2 = 0$

$2a + b + b^2 = 0$ is the required condition.

29. **[2]** Lines $(2a + b)x + (a + 3b)y + (b - 3a) = 0$ or $a(2x + y - 3) + b(x + 3y + 1) = 0$ are concurrent at the point of intersection of lines $2x + y - 3 = 0$ and $x + 3y + 1 = 0$, which is $(2, -1)$. Now, line $mx + 2y + 6 = 0$ must pass through this point.

Therefore, $2m - 2 + 6 = 0$ or $m = -2$.

30. **[5]**



Since $\angle BCA = 90°$, points A, O, B, C are concyclic.

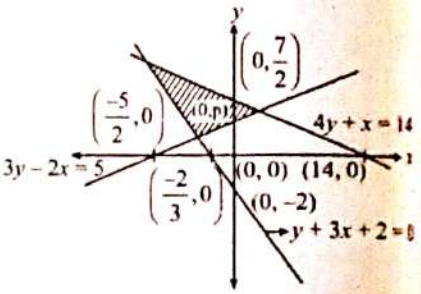
Let $\angle AOC = \theta$.

$\angle BOC = \angle BAC$

$\tan\left(\frac{\pi}{2} - \theta\right) = \frac{5}{12}$

$\frac{x}{y} = \frac{5}{12}$ or $12x - 5y = 0$

31. **[2]**

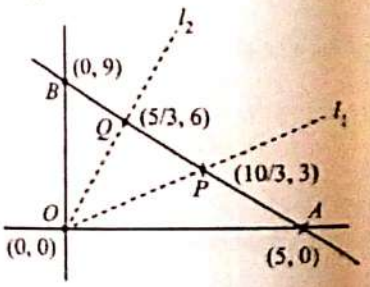


$\therefore \frac{5}{3} < p < \frac{7}{2}$

So, p integer = 2, 3.

PYQ's (Past Year Questions)

1. **(c)**

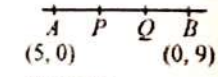


$L: 9x + 5y = 45$

$P_x = \frac{2 \times 5 + 1 \times 0}{1 + 2} = \frac{10}{3}$

$P_y = \frac{0 \times 2 + 9 \times 1}{1 + 2} = 3$

$P: (10/3, 3)$



Similarly

$Q_x = \frac{1 \times 5 + 2 \times 0}{1 + 2} = \frac{5}{3}$

$Q_x = \frac{1\times 0 + 2\times 9}{1+2} = 6$

$Q : (5\sqrt{3}, 6)$

Now, $m_1 = \frac{3-0}{10/\sqrt{3}-0} = \frac{9}{10}$

$m_2 = \frac{6-0}{5\sqrt{3}-0} = \frac{18}{5}$

Now $L : y = (m_1 + m_2)x$

$\Rightarrow y = \left(\frac{9}{2}\right)x$

$\Rightarrow 9x = 2y$ ...(iii)

from eq. (i) and eq. (iii) we get

$x = \frac{10}{7}, y = \frac{45}{7}$, which satisfy $y - x = 5$

2. (a) $L_1 : 3x - 4y + 12 = 0$

$L_2 : 8x - 6y + 11 = 0$

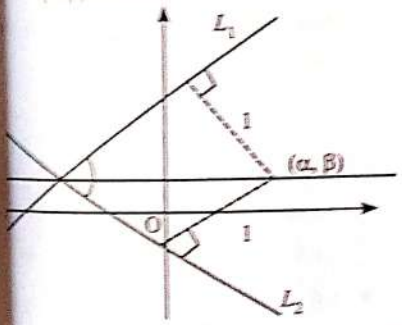
(α, β) lies on that angle which contain origin

∴ Equation of angle bisector of that angle which contain origin is

$\frac{3x-4y+12}{5} = \frac{8x+6y+11}{10}$

$\Rightarrow 2x - 14y - 13 = 0$

(α, β) lies on it



$\Rightarrow 2\alpha + 14\beta - 13 = 0$ ...(i)

Again $\frac{3\alpha - 4\beta + 12}{5} = 1$

$\Rightarrow 3\alpha - 4\beta + 7 = 0$ ...(ii)

Solving (i) and (ii)

$\alpha = -\frac{23}{25}$ and $\beta = \frac{53}{50}$

$\therefore \alpha + \beta = \frac{7}{50}$

$\Rightarrow 100(\alpha + \beta) = 14$

3. (b) Length of perpendicular from origin to

$x\cos\alpha - y\sin\alpha = k\cos 2\alpha$

$\Rightarrow x\cos\alpha - y\sin\alpha = k\cos 2\alpha$

$p = \frac{|-k\cos 2\alpha|}{\sqrt{\cos^2\alpha + \sin^2\alpha}} = \frac{k}{2}\cos 2\alpha$ ... (i)

Length of perpendicular is from origin to $x\sin\alpha + y\cos\alpha = k\sin 2\alpha$

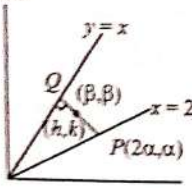
$q = \frac{|-k\sin 2\alpha|}{\sqrt{\sin^2\alpha + \cos^2\alpha}} = k\sin 2\alpha$ ... (ii)

From (i) and (ii), we get

$\frac{4p^2}{k^2} + \frac{q^2}{k^2} = 1$

$k^2 = 4p^2 + q^2$

4. (c)



Slope of $PQ = \frac{k-\alpha}{h-2\alpha} = -1 \Rightarrow k - \alpha = -h + 2\alpha$

$\Rightarrow \alpha = \frac{h+k}{3}$ ...(i)

Also $2h = 2\alpha + \beta$

$2k = \alpha + \beta$

$\Rightarrow 2h = \alpha + 2k$

$\Rightarrow \alpha = 2h - 2k$ ...(ii)

from (i) and (ii)

$\frac{h+k}{3} = 2(h-k)$

So locus is $6x - 6y = x + y \Rightarrow 5x = 7y$

5. (d) Centroid with vertices (3, –1), (1, 3) and (2, 4) is (2, 2)

Point of intersection of lines $x + 3y = 1$ and $3x - y + 1 = 0$ is $P\left(-\frac{1}{5}, \frac{2}{5}\right)$.

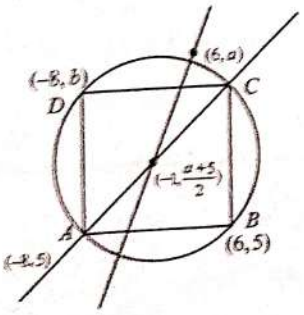
Equation of line CP

$(y-2) = \frac{\frac{2}{5}-2}{-\frac{1}{5}-2}(x-2)$

$\Rightarrow 8x - 11y + 6 = 0$

∴ $(-9, -6)$ is the only point which satisfies the equation

6. (b) In Rectangle ABCD AB is verticle line as y co-ordinate is same in A and B. So, x-co-ordinate of c is 6 as $AB \perp BC$ let $c(6, a)$



Mid-Point of AC will lie on line $3y = x + 7$

$\Rightarrow \frac{3(a+5)}{2} = -1 + 7$

$\Rightarrow a + 5 = \frac{2(6)}{3}$

$\Rightarrow a = -1$

∴ Area of rectangle $ABCD$ = AB × CB

= 14 × 6 = 84 sq. unit

7. (a) Required line is $4x - 3y + \lambda = 0$

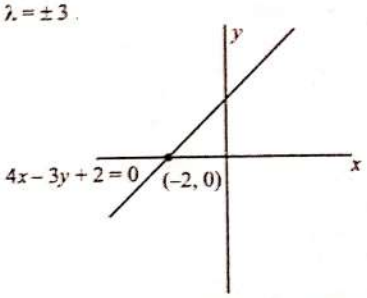
distance from origin is $\frac{3}{5}$

So, $\left|\frac{0-0+\lambda}{\sqrt{4^2+3^2}}\right| = \frac{3}{5}$

$\Rightarrow \frac{|\lambda|}{5} = \frac{3}{5}$

$|\lambda| = 3$

$\lambda = \pm 3$



So. required equation of line is $4x - 3y + 3 = 0$ and $4x - 3y - 3 = 0$

option (a) $4\left(-\frac{1}{4}\right) - 3\left(\frac{2}{3}\right) + 3 = 0$

8. (c) Let $A(\alpha, 2)$

Given $B(2, 3)$

$m_{OA} = \frac{2}{\alpha}$ and $m_{OB} = \frac{3}{2}$

$\Rightarrow \tan\frac{\pi}{4} = \left|\frac{\frac{2}{\alpha} - \frac{3}{2}}{1 + \frac{2}{\alpha}\cdot\frac{3}{2}}\right| \Rightarrow \frac{4-3\alpha}{2\alpha+6} = \pm 1$

$\Rightarrow 4 - 3\alpha = 2\alpha + 6$ & $4 - 3\alpha = -2\alpha - 6$

$\Rightarrow \alpha = \frac{-2}{5}$ & $\alpha = 10$

$\Rightarrow A\left(-\frac{2}{5}, 2\right)$ $A'(10, 2)$ and $B(2, 3)$

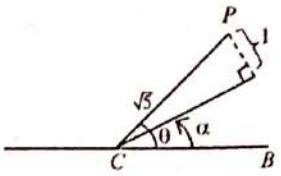
$\Rightarrow AA' = 10 + \frac{2}{5} = \frac{52}{5}$

9. (d) Given, $\angle PCB = \tan^{-1}(2)$

$\because \tan^{-1}(2) = \theta$

$\Rightarrow \tan\theta = 2$

$\sin(\theta - \alpha) = \frac{1}{\sqrt{5}}$



$$\therefore \tan(\theta-\alpha)=\frac{\frac{1}{\sqrt{5}}}{\frac{2}{\sqrt{5}}}=\frac{1}{2}$$

$$\Rightarrow \frac{\tan\theta-\tan\alpha}{1+\tan\theta\tan\alpha}=\frac{1}{2}$$

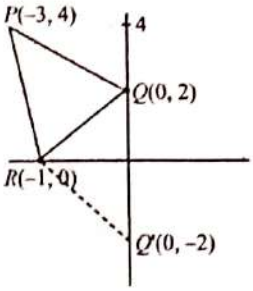
$$\Rightarrow \frac{2-\tan\alpha}{1+2\tan\alpha}=\frac{1}{2}$$

$$\Rightarrow 4-2\tan\alpha=1+2\tan\alpha \Rightarrow \tan\alpha=\frac{3}{4}$$

Hence, $\alpha=\tan^{-1}\left(\frac{3}{4}\right)$

10. (*b*)

11. [1250]

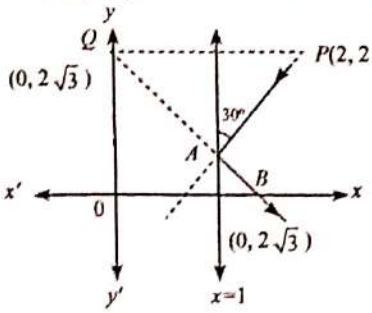


$$PR=\sqrt{(-3+1)^2+(4-0)^2}=\sqrt{20}$$

$$RQ=\sqrt{(0+1)^2+(2-0)^2}=\sqrt{5}$$

$$\therefore 50\,(PR^2+RQ^2)=50\,(20+5)=1250$$

12. (*b*) Equation of incident line AP is



$$y-2\sqrt{3}=\sqrt{3}(x-2)$$

$$\sqrt{3}x-y=0 \qquad \ldots(i)$$

Image of P w.r.t. line $x=1$ is point $Q=\left(0,2\sqrt{3}\right)$.

Equation of reflected Ray AB:

$$y-\sqrt{3}=\frac{2\sqrt{3}-\sqrt{3}}{0-1}(x-1)$$

$$\sqrt{3}x+y=2\sqrt{3}$$

$\therefore$ Point $(3,-\sqrt{3})$ lies on line AB.

13. (*d*) Line perpendicular to $2x-3y+5=0$ is $3x+2y+c=0$ Which is satisfied by point. $(7, 17)$

$$\Rightarrow 3(7)+2(17)+c=0$$

$$\Rightarrow c=-55$$

$\Rightarrow$ Equation of line is $3x+2y-55=0$ which is also passes via $(15, \beta)$

$$\Rightarrow 3(15)+2(\beta)-55=0$$

$$\Rightarrow 2\beta=55-45 \Rightarrow \beta=5$$

14. (*d*) If two lines are perpendicular then $m_1m_2=-1$.

$$\left(\frac{-1}{a-1}\right)\left(\frac{-2}{a^2}\right)=-1$$

$$\Rightarrow 2=-(a^2)(a-1)$$

$$\Rightarrow a^3-a^2+2=0$$

$$\Rightarrow (a+1)(a^2-2a+2)=0$$

$$\therefore a=-1$$

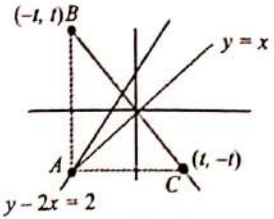
$$\left.\begin{aligned}L_1: x-2y-1=0\\ L_2: 2x+y-1=0\end{aligned}\right\}$$

Point f intersection of L_1 & L_2 is

$$P\left(\frac{3}{5},\frac{-1}{5}\right)$$

$$OP=\sqrt{\frac{1}{25}+\frac{9}{25}}=\sqrt{\frac{10}{25}}=\sqrt{\frac{2}{5}}$$

15. (*c*)



$$y-2x=2$$

Given $x=y$ and $y-2x=2$

So, $A=(-2,-2)$

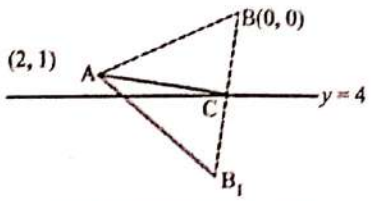
Height h of $\Delta ABC=\sqrt{(-2)^2+(-2)^2}$ $=2\sqrt{2}$

Given, ΔABC is an equilateral triangle

$$\therefore \sin 60^\circ=\frac{h}{AB} \Rightarrow AB=\frac{h}{\sin 60}$$

$$\text{Area of } \Delta=\frac{\sqrt{3}}{4}\frac{h^2}{\sin^2 60}=\frac{8}{\sqrt{3}} \text{ sq. units}$$

16. [48] A (2,1), B (0, 0), C $(t, 4)$: $t\in[0,4]$



$B_1(0, 8)$ image of B w.r.t. $y=4$

for $AC+BC+AB$ to be minimum.

$$m_{AB_1}=\frac{-7}{2}$$

line $AB_1\equiv 7x+2y=16$

$$C\left(\frac{8}{7},4\right)$$

$$\beta=\frac{8}{7}$$

For max. perimeter

$$C(4,4), A(2,1)\ \alpha=4$$

$$AB=\sqrt{5}: BC=4\sqrt{2}, AC=\sqrt{13}$$

$$6\alpha+21\beta=24+24=48$$

17. (*c*) $\because A(1,\alpha)$, $B(\alpha, 0)$ and $C(0,\alpha)$ are the vertices of ΔABC and area of ΔABC $=4$

$$\therefore \left|\frac{1}{2}\begin{vmatrix}1 & \alpha & 1\\ \alpha & 0 & 1\\ 0 & \alpha & 1\end{vmatrix}\right|=4$$

$$\Rightarrow \left|1(0-\alpha)-\alpha(\alpha)+\alpha^2\right|=8$$

$$\Rightarrow \alpha=\pm 8$$

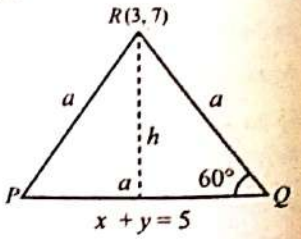
Now, $(\alpha,-\alpha),(-\alpha,\alpha)$ and (α^2,β) are collinear

$$\therefore \begin{vmatrix}8 & -8 & 1\\ -8 & 8 & 1\\ 64 & \beta & 1\end{vmatrix}=0=\begin{vmatrix}-8 & 8 & 1\\ 8 & -8 & 1\\ 64 & \beta & 1\end{vmatrix}$$

$$\Rightarrow 8(8-\beta)+8(-8-64)+1(-8\beta-8\times 64)=0$$

$$\Rightarrow 8-\beta-72-\beta-64=0 \Rightarrow \beta=-64$$

18. (*d*)



Let side of triangle $=a$

$$h=\frac{|3+7-5|}{\sqrt{1^2+1^2}}$$

$$=\frac{5}{\sqrt{2}}$$

From figure, $h=a\sin 60^\circ$

$$\Rightarrow a=\frac{2h}{\sqrt{3}}=\frac{2}{\sqrt{3}}\times\frac{5}{\sqrt{2}}=\frac{10}{\sqrt{6}}$$

$$\therefore \text{Area}=\frac{3}{4}\left(\frac{10}{\sqrt{6}}\right)^2=\frac{25}{2\sqrt{3}}$$

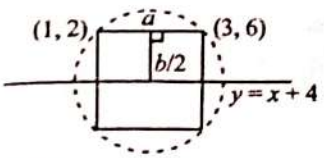
19. [16] As slope of line joining (1,2) and (3,6) is 2 given diameter is parallel to side.

$\therefore a = \sqrt{(3-1)^2 + (6-2)^2} = \sqrt{20}$

Midpoint of (1, 2), (3, 6) = (2, 4)

$\therefore \frac{b}{2} = \frac{2\times 2 - 4 + 4}{\sqrt{2^2+1^2}}$ and

$\frac{b}{2} = \frac{4}{\sqrt{5}} \Rightarrow b = \frac{8}{\sqrt{5}}$



Area of the rectangle

$= ab = 2\sqrt{5}\cdot\frac{8}{\sqrt{5}} = 16$

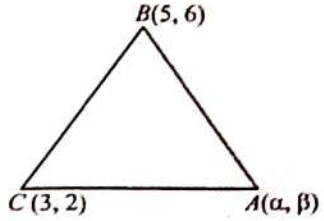
20. (*d*) Given, area = 12

$$\begin{vmatrix} \alpha & \beta & 1 \\ 5 & 6 & 1 \\ 3 & 2 & 1 \end{vmatrix} = 24$$

$4\alpha - 2\beta - 8 = \pm 24$

$4\alpha - 2\beta = 24 + 8, \quad 4\alpha - 2\beta = -24 + 8$

$2(2\alpha - \beta) = 32, \quad 2\alpha - \beta = -8$



Distance from origin

$\Rightarrow D = \sqrt{\alpha^2 + (2\alpha + 8)^2}$

$\Rightarrow D^2 = 5\alpha^2 + 32\alpha + 64$

$\Rightarrow \frac{d(D^2)}{d\alpha} = 10\alpha + 32$

$\Rightarrow 10\alpha + 32 = 0 \Rightarrow \alpha = -\frac{16}{5}$

$D = \sqrt{\left(\frac{-16}{5}\right)^2 + \left(\frac{8}{5}\right)^2} \Rightarrow D = \frac{8}{\sqrt{5}}$

If $2\alpha - \beta = 16$

Hence, $D = \frac{16}{\sqrt{5}}$

Hence, least possible length is $\frac{8}{\sqrt{5}}$

21. [80] $OA \perp OB$

$\therefore m_{OA} = \frac{0 - \frac{1}{t_1}}{0 - t_1} = \frac{1}{t_1^2}$

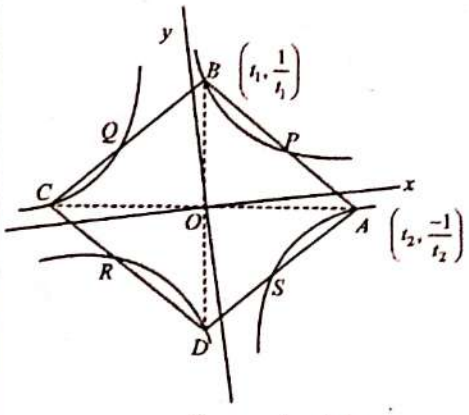
$m_{OB} = \frac{0 + \frac{1}{t_2}}{0 - t_2} = \frac{-1}{t_2^2}$

$m_{OA} \times m_{OB} = -1$

$\therefore \frac{1}{t_1^2} \times \frac{-1}{t_2^2} = -1$

$(t_1 t_2)^2 = 1$

$t_2^2 = \frac{1}{t_1^2}$



Mid point $\left(\frac{t_1 + t_2}{2}, \frac{\frac{1}{t_1} - \frac{1}{t_2}}{2}\right)$ lie on the $x^2 y^2 = 1$

$\Rightarrow (t_1 + t_2)^2 \left(\frac{1}{t_1} - \frac{1}{t_2}\right)^2 = 16$

$\Rightarrow (t_1 + t_2)^2 (t_2 - t_1)^2 = 16$

$\Rightarrow (t_1^2 - t_2^2)^2 = 16 \quad \Rightarrow t_1^2 - t_2^2 = \pm 4$

From equ. (i)

$t_1^2 - \frac{1}{t_2^2} = \pm 4$

$\Rightarrow t_1^4 \pm 4t_1^2 - 1 = 0$

$\Rightarrow t_1^2 = 2 + \sqrt{5}$ or $-2 + \sqrt{5}$

$OB^2 = t_1^2 + \frac{1}{t_1^2}$

$= 2 + \sqrt{5} + \frac{1}{2 + \sqrt{5}}$ or $-2 + \sqrt{5} + \frac{1}{-2 + \sqrt{5}} = 2\sqrt{5}$

Area $= 4\left(\frac{1}{2}\right) \times (OA) \times (OB) = 2(OB)^2$

$= 4\sqrt{5}$

Hence, square of area of

$ABCD = (4\sqrt{5})^2 = 80$

22. (*a*) $D = \frac{1}{2}\begin{vmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ x' & y' & 1 \end{vmatrix}$

$-2(1 - x') + (y' + x') = \pm 10$

$-2 + 2x' + y' + x' = \pm 10$

$3x' + y' = 12$ or $3x' + y' = -8$

$\lambda = 3, -2.$

23. [5] P will be centroid of ΔABC

$P\left(\frac{17}{6}, \frac{8}{3}\right) \Rightarrow PQ = \sqrt{\left(\frac{24}{6}\right)^2 + \left(\frac{9}{3}\right)^2} = 5$

24. (*b*) Solving the given equations we get coordinates of the triangle are (6, 8), (1, 2) and (5, −7)

Given centroid: (α, β)

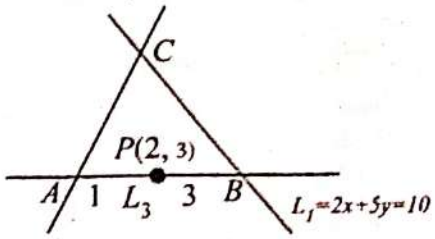
$\therefore \alpha = \frac{6 + 1 + 5}{3} = 4$ and

$\beta = \frac{8 + 2 - 7}{3} = 1$

$\therefore \alpha + 2\beta = 6$ and $2\alpha - \beta = 7$

The required eq^n. is $x^2 - 13x + 42 = 0$

25. (*b*) $L_1 : 2x + 5y = 10; L_2 : -4x + 3y = 12$



$L_2 = -4x + 3y = 12$

Solving L_1 and L_2 we get

$C \equiv \left(\frac{-15}{13}, \frac{32}{13}\right)$

Now, Let $A\left(x_1, \frac{1}{3}(12 + 4x_1)\right)$ and

$B\left(x_2, \frac{1}{5}(10 - 2x_2)\right)$

Since $P(2, 3)$ divides the line.

$\therefore \frac{3x_1 + x_2}{4} = 2$ and

$\frac{(12 + 4x_1) + \frac{10 - 2x_2}{5}}{4} = 3$

So, $3x_1 + x_2 = 8$ and $10x_1 - x_2 = -5$

So, $(x_1, x_2) = \left(\frac{3}{13}, \frac{95}{13}\right)$

$A = \left(\frac{3}{13}, \frac{56}{13}\right)$ and

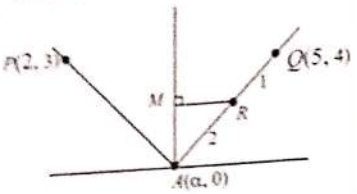
$B = \left(\frac{95}{13}, \frac{-12}{13}\right)$

$\therefore$ required area

$= \left|\frac{1}{2}\left(\frac{3}{13}\left(\frac{-44}{13}\right) - \frac{56}{13}\left(\frac{110}{13}\right) + 1\left(\frac{2860}{169}\right)\right)\right|$

$= \frac{132}{13}$ sq. units

26. [31]



During reflection of light angle of incidence is equal to the angle of reflection.

$$\frac{4}{5-\alpha}=\frac{3}{\alpha-2} \Rightarrow 4\alpha-8=15-3\alpha \Rightarrow$$

$$\alpha=\frac{23}{7}$$

Hence, $A=\left(\frac{23}{7}, 0\right)$ and $Q=(5,4)$

Since, R is dividing the line segment AQ in the ratio 2 : 1

$$R=\left(\frac{10+\frac{23}{7}}{3}, \frac{8}{3}\right)$$

$$R=\left(\frac{31}{7}, \frac{8}{3}\right)$$

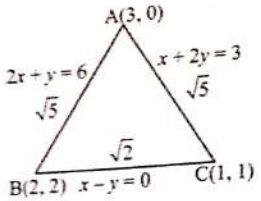
Now, bisector of angle PAQ is $X=\frac{23}{7}$

$$\Rightarrow M=\left(\frac{23}{7}, \frac{8}{3}\right)$$

So, $7\alpha+3\beta=7\left(\frac{23}{7}\right)+3\left(\frac{8}{2}\right)=31$

27. (*b*)

28. (*c*)



$x-y=0$

$x+2y=3$

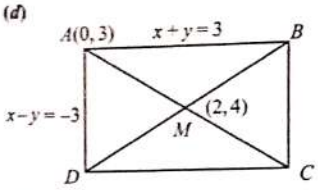
$2x+y=6$

find cordinates of point A, B and C

$AB=\sqrt{5}=BC$

$CA=\sqrt{2}$

29. (*d*)



$x+y=3$ ...(*i*)

$x-y=-3$ ...(*ii*)

Add

$2x=0 \Rightarrow x=0$

So, $y=3$

A (0, 3)

M is the mid-point of AC

So, co-ordinate of c

$$c\left(\frac{4-0}{1}, \frac{8-3}{1}\right)$$

$= c\,(4, 5)$

Equation of line BC passing via C

$x-y=\lambda$

$4-5=\lambda \Rightarrow x-y=-1$

Equation of line CD passing via C

$x+y=\lambda$

$4+5=\lambda$

So, $x+y=9$

Now for point D

$x+y=9$

$x-y=-3$

$2x=6$

$x=3$

and $y=6$

$D\,(3, 6)$

30. (*d*) Equation of the pair of angle bisector for the homogenous equation $ax^2+2hxy+by^2=0$ is given as

$$\frac{x^2-y^2}{a-b}=\frac{xy}{h}$$

Here $a=2, h=1/2$ and $b=-3$

Equation will become

$$\frac{x^2-y^2}{2-(-3)}=\frac{xy}{1/2}$$

$$x^2-y^2=10xy$$

$$x^2-y^2-10xy=0$$

By Aditya Mohad

username - maje_kabhi_dekha_nahi

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 11 Circles

Prarambh (Topicwise)

1. (a) (3, 4) and (2, 5) are ends of diameter of circle

So, Equation $(x-3)(x-2)+(y-4)(y-5)=0$

$x^2+y^2-5x-9y+26=0$

2. (c) Length of intercept on x-axis = $2\sqrt{g^2-c}$

$=2\sqrt{\frac{25}{4}+14}=2\sqrt{\frac{81}{4}}=9$

on y-axis = $2\sqrt{f^2-c}=2\sqrt{\left(\frac{13}{2}\right)^2+14}$

$=2\sqrt{\frac{169+56}{4}}=2\sqrt{\frac{225}{4}}=15$

3. (d) Given circle $x^2+y^2-4x-6y=0$

it cuts x-axis put $y=0$, $x=0, 4$

it cuts y-axis put $x=0$, $y=0, 6$

Hence mid points on x-axis (2, 0)

on y-axis (0, 3)

Equations of line $\frac{x}{2}+\frac{y}{3}=1$

$\Rightarrow \quad x+2y-6=0$

4. (c) Intersection of given lines is centre

$2x-3y-5=0$

$3x-4y-7=0$

$\frac{x}{21-20}=\frac{y}{-15+14}=\frac{1}{-8+9}$

$\Rightarrow \quad x=1, y=-1$

$(1,-1), \pi r^2=154$

$\Rightarrow \quad r^2=\frac{154}{22}\times 7 \quad \Rightarrow \quad r=7$

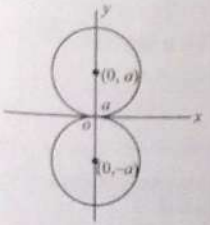
$g=-1, f=1, c=g^2+f^2-r^2$

$=1+1-49=-47$

$x^2+y^2-2x+2y-47=0$

5. (b) $x^2+(y\pm a)^2=a^2$

$x^2+y^2\pm 2ay=0$



6. (a) Centre (2, –1), radius

$=\sqrt{(3-2)^2+(6+1)^2}=\sqrt{1+49}=\sqrt{50}$

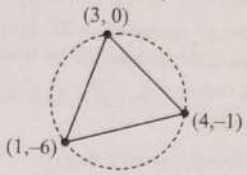
$(x-2)^2+(y+1)^2=50$

$x^2+y^2-4x+2y-45=0$

7. (d) Let the centre (a, b)

$(a-3)^2+(b)^2=(a-1)^2+(b+6)^2$

$=(a-4)^2+(b+1)$



$-6a+9=-2a+1+12b+36$

$\Rightarrow \quad 4a+12b+28=0 \quad \Rightarrow \quad a+3b+7=0$

$-6a+9=-8a+16+2b+1$

$\Rightarrow \quad 2a-2b=8 \quad \Rightarrow \quad a-b=4$

$a=\frac{5}{4}, b=-\frac{11}{4} \quad r=\sqrt{\frac{49}{16}+\frac{121}{16}}=\frac{\sqrt{170}}{4}$

$g=-\frac{5}{4}, f=\frac{11}{4}, c=\frac{25}{16}+\frac{121}{16}-\frac{170}{16}=\frac{-24}{16}=\frac{-3}{2}$

$x^2+y^2-2.\frac{5}{4}x+2.\frac{11}{4}y-\frac{3}{2}=0$

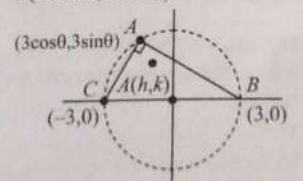
$2x^2+2y^2-5x+11y-3=0$

8. (a) Circle is

$x^2+y^2=9$

$\therefore$ co-ordinate of point

$A(3\cos\theta, 3\sin\theta)$



centroid of ΔABC is $P(h, k)$ whose coordinate is

$\left(\frac{3+3\cos\theta-3}{3}, \frac{0+0+3\sin\theta}{3}\right)\equiv(\cos\theta, \sin\theta)$

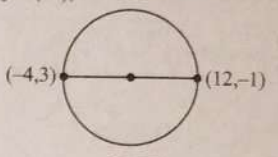
$h=\cos\theta, k=\sin\theta$

$h^2+k^2=1 \quad \Rightarrow \quad x^2+y^2=1$

9. (c) $(x+4)(x-12)+(y-3)(y+1)=0$

$x^2+y^2-8x-2y-51=0$

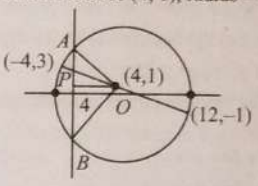
$f=(-1), c=-51$



y intercept $\sqrt{f^2-c}=2\sqrt{1+51}=2\sqrt{52}$

$=2=4\sqrt{13}$

Aliter: Centre (4, 1), radius = $\sqrt{68}$



$AP=\sqrt{68-16}=\sqrt{52}$

$AB=2(AP)=2\sqrt{52}=4\sqrt{3}$

10. (a) $y^2-2y+2xy=0$ represent normals

$\{(y(y-2)-2x(y-2)=0)$

$(y-2)(y-2x)=0\}$

Intersection point is centre

$y=2$ and $y=2x \quad \Rightarrow \quad x=1, y=2$

centre (1, 2), passing through (2, 1)

$r=\sqrt{(-1)^2+1^2}=\sqrt{2}$

$(x-1)^2+(y-2)^2=2$

$x^2+y^2-2x-4y+3=0$

11. (b)

12. (b) $BC=5\sqrt{2}$

$CA=5\sqrt{2}$

$AB=6\sqrt{2}$

In-Centre of The Circle $\left(\frac{9}{2}, \frac{9}{2}\right)$

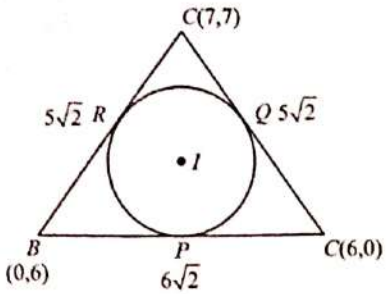
The in radius r is equal to the perpendicular distance from in center to any of the sides.

Equation of the line AB

$x+y-6=0$

So the in radius is $\frac{3}{\sqrt{2}}$

So the equation of circle is $x^2+y^2-9x-9y+36=0$



$\because$ $P(3, 3)$

$$\therefore I\left(\frac{6.(5\sqrt{2})+0+7.6\sqrt{2}}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}}, \frac{0+6.(5\sqrt{2})+7(6\sqrt{2})}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}}\right)$$

$$I\left(\frac{9}{2},\frac{9}{2}\right), r = IP = \sqrt{\left(\frac{9}{2}-3\right)^2+\left(\frac{9}{2}-3\right)^2} = \frac{3}{\sqrt{2}}$$

$$\Rightarrow \left(x-\frac{9}{2}\right)^2+\left(y-\frac{9}{2}\right)^2 = \frac{9}{2}$$

$$\Rightarrow x^2+y^2-9x-9y+\frac{81}{2}-\frac{9}{2}=0$$

$$\Rightarrow x^2+y^2-9x-9y+36=0$$

13. (*b*) Let the equation to the circle be

$x^2+y^2+2gx+2fy+c=0$ ... (*i*)

Since the three points lie on the circle, we have

$2g+4f+c=-5$ ... (*ii*)

$6g-8f+c=-25$ ... (*iii*)

$10g-12f+c=-61$ ... (*iv*)

Subtracting (*ii*) from (*iii*) and (*iii*) from (*iv*), we have

$4g-12f=-20$ and $4g-4f=-36$

Hence $f=-2$ and $g=-11$

Equation (*ii*) then gives $c=25$.

Substituting these values in (*i*), the required equation is

$x^2+y^2-22x-4y+25=0$

Its centre is (11, 2) and radius is 10.

14. (*a*) Point on the line $x+y+13=0$ nearest to the circle $x^2+y^2+4x+6y-5=0$ is foot of $\perp$ from centre

$$\frac{x+2}{1}=\frac{y+3}{1}=-\left(\frac{-2-3+13}{1^2+1^2}\right)=-4$$

$x=6,\quad y=-7$

15. (*c*) Since $S_1 = 10^2+7^2-4\times10-2\times7-20>0$.

So P lies outside the circle. Join P with centre $C(2, 1)$ of the given circle. Suppose PC cuts the circle at A and B. Then PB is greatest distance of P from the circle.

$$PC=\sqrt{(10-2)^2+(7-1)^2}=10$$

CB = radius $=\sqrt{4+1+20}=5$

$\therefore \quad PB = PC+CB = 10+5=15$

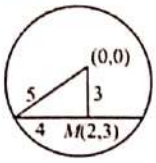
16. (*b*) Let slope of required line is m

$y-3=m(x-2)$

$\Rightarrow \quad mx-y+(3-2m)=0$

length of $\perp$ from origin

$=3$



$$\Rightarrow \quad 9+4m^2-12m=9+9m^2$$

$$\Rightarrow \quad 5m^2+12m=0 \Rightarrow m=0,\ -\frac{12}{5}$$

Hence lines are $y-3=0 \Rightarrow y=3$

$$y-3=-\frac{12}{5}(x-2)$$

$$\Rightarrow \quad 5y-15=-12x+24$$

$$\Rightarrow \quad 12x+5y=39.$$

17. (*b*) From centre (2, –3), length of perpendicular on line $3x+5y+9=0$ is

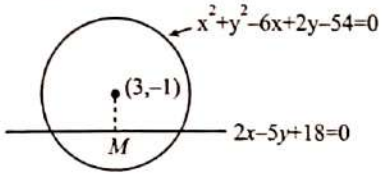
$$p=\frac{6-15+9}{\sqrt{25+9}}=0$$; line is diameter.

18. (*a*) Required point is foot of $\perp$

$$\frac{x-3}{2}=\frac{y+1}{-5}=-\left(\frac{6+5+8}{4+25}\right)=-1$$

$$\Rightarrow \quad x=-2+3=1$$

$x=1, y=4$



19. (*d*) The two diameters intersect at (8, –2) which is the centre of the circle. The circle passes through (6, 2). Therefore its radius $=\sqrt{20}$.

Hence the equation of the circle is

$$(x-8)^2+(y+2)^2=\left(\sqrt{20}\right)^2$$

20. (*b*) Given, $x^2+y^2-2x-6y-\frac{7}{3}=0$

The centre of this circle is (1, 3)

Also, two diameter of this circle are along the lines

$3x+y=c_1$ and $x-3y=c_2$

These two diameters should be passed from (1, 3)

$\therefore \quad c_1=6$ and $c_2=-8$ Hence, $c_1c_2=6\times(-8)=-48$

21. (*c*) Equation of circle is

$(x-4)(x+2)+(y-7)(y+1)=0$

$\Rightarrow \quad x^2-2x-8+y^2+y-7y-7=0$

$\Rightarrow \quad x^2+y^2-2x-6y-15=0$

Here, $g=-1, c=-15$

$\therefore \quad AB=2\sqrt{g^2-c}$

$=2\sqrt{1+15}=8$

22. (*a*) A (10, 0)

$S_A = 10^2-6(10)-11=100-60-11>0$

Point A outside the circle

Similarly point O lying outside the circle

$\therefore$ Line joining of A and B cut the circle.

Centre of the circle

P (3, – 4), A (10, 0), O (0, 0)

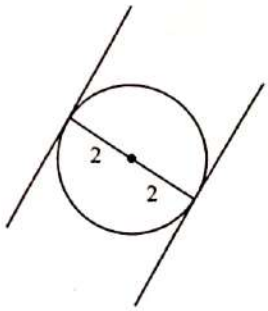
Slope of PA $=\frac{4}{7}$

Slope of AO = 0

$\therefore$ P, A, O

are non-collinear.

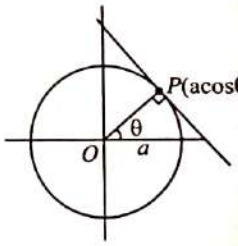
23. (*a*) $4=\left|\frac{c_1-c_2}{\sqrt{1+3}}\right| \Rightarrow |c_1-c_2|=8$



24. (*b*) Point (8,6) lies on circle; $S_1=0$

$\Rightarrow$ one tangent.

25. (*d*) $x^2+y^2=a^2$



$m_N=\tan\theta$

$$m_T=-\frac{1}{m_N}=\frac{1}{\tan\theta}$$

$=-\cot\theta$

26. (*b*) Line parallel to given line $4x+3y+5=0$ is $4x+3y+k=0$

This is tangent to $x^2+y^2-6x+4y-12=0$

$$\left|\frac{12-6+k}{5}\right|=5$$

$6+k=\pm25 \quad \Rightarrow \quad k=19,-31$

Hence required line $4x+3y-31=0, 4x+3y+19=0$

27. (*d*) Equation of tangent $x - 2y = 5$

Let required point be (α, β)

$\alpha x + \beta y - 4(x + \alpha) + 3(y + \beta) + 20 = 0$

$x(\alpha - 4) + y(\beta + 3) - 4\alpha + 3\beta + 20 = 0$

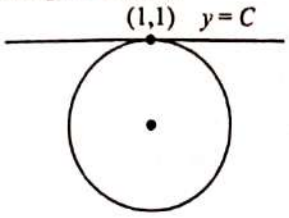
Comparing

$$\frac{\alpha - 4}{1} = \frac{\beta + 3}{-2} = \frac{4\alpha - 3\beta - 20}{5}$$

Similarly $(\alpha, \beta) \equiv (3, -1)$

28. (*a*) $x^2 + y^2 - 2x + 2y - 2 = 0$

Tangent at (1, 1)



$x + y - (x + 1) + (y + 1) - 2 = 0$

$y - 1 + y + 1 - 2 = 0$

$2y - 2 = 0$

$y = 1 \Rightarrow c = 1$

29. (*b*) Tangent at (x_1, y_1) is

$xx_1 + yy_1 = 25$

$3x + 4y = 25 \Rightarrow x_1 = 3, y_1 = 4$

$\Rightarrow (x_1, y_1) = (3, 4)$

30. (*a*) Let tangent from (0, 1) on $x^2 + y^2 - 2x + 4y = 0$

$y - 1 = mx \quad C(1, -2), r = \sqrt{5}$

$\Rightarrow mx - y + 1 = 0$

$$r = \sqrt{5} = \frac{|m + 2 + 1|}{\sqrt{m^2 + 1}}$$

$\Rightarrow 5(m + 1) = (m + 3)^2$

$\Rightarrow 4m^2 - 6m - 4 = 0 \Rightarrow 2m^2 - 3m - 2 = 0$

$\Rightarrow (m - 2)(2m + 1) = 0$

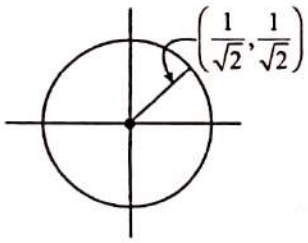
$$\Rightarrow m = 2, -\frac{1}{2},$$

Tangents are $2x - y + 1 = 0$

$x + 2y - 2 = 0$

31. (*c*) Normal is diameter passing through centre (0, 0)

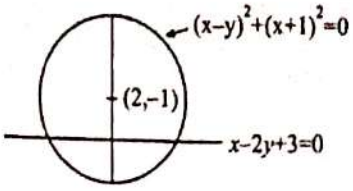
$$\text{and } m = \frac{\frac{1}{\sqrt{2}} - 0}{\frac{1}{\sqrt{2}} - 0} = 1$$



$y = x \Rightarrow x - y = 0$

32. (*b*) Required diameter is $\perp$ to given line.

Hence $y + 1 = -2(x - 2)$



$\Rightarrow 2x + y - 3 = 0$

33. (*b*) $(x - r)^2 + y^2 = r^2$

$\Rightarrow x^2 + y^2 - 2xr = 0$

tangent at (x_1, y_1)

$xx_1 + yy_1 - r(x + x_1) = 0$

$(x_1 - r)x + yy_1 - rx_1 = 0$

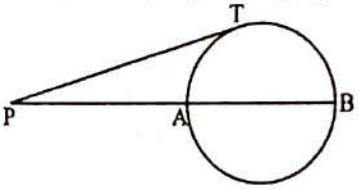
$$\text{slope } m_T \ \frac{r - x_1}{y_1} = \frac{r - x}{y} \qquad (b)$$

$$= \frac{r - x}{y} = \frac{2xr - 2x^2}{2xy} = \frac{x^2 + y^2 - 2x^2}{2xy}$$

$$= \frac{y^2 - x^2}{2xy} \quad (c)$$

34. (*c*) As we know

$PA.PB = PT^2 = (\text{Length of tangent})^2$



Length of tangent $= \sqrt{16 \times 9} = 12$

35. (*a*) Let any point on the circle $x^2 + y^2 + 2gx + 2fy + p = 0$ (α, β)

This point satisfies $\alpha^2 + \beta^2 + 2g\alpha + 2f\beta + p = 0$

Length of tangent from this point to circle $x^2 + y^2 + 2gx + 2fy + q = 0$

length

$$= \sqrt{S_1} = \sqrt{\alpha^2 + \beta^2 + 2g\alpha + 2f\beta + q} = \sqrt{q - p}$$

36. (*c*) $2(x^2 + y^2) - 7x + 9y - 11 = 0, P(2, 3)$

Point lie outside

$$\therefore \quad x^2 + y^2 - \frac{7}{2}x + \frac{9}{2}y - \frac{11}{2} = 0$$

Length of tangent

$$T_1 = \sqrt{s_1} = \sqrt{4 + 9 - 7 + \frac{27}{2} - \frac{11}{2}}$$

$$= \sqrt{6 + 8} = \sqrt{14}$$

37. (*b*) The polar of the point $\left(5, -\frac{1}{2}\right)$ is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$\Rightarrow 5x - \frac{1}{2}y - 2(x + 5) + 0 + 0 = 0$$

$$\Rightarrow 3x - \frac{y}{2} - 10 = 0 \Rightarrow 6x - y - 20 = 0$$

38. (*c*) Chord of contact from (0, 0) and (g, f) are $gx + fy + c = 0$

and $gx + fy + g(x + g) + f(y + f) + c = 0$

$\Rightarrow 2gx + 2fy + g^2 + f^2 + c = 0$

distance between C.O.C.'s

$$= \frac{\left|\frac{g^2 + f^2 + c - c}{2}\right|}{\sqrt{g^2 + f^2}} = \frac{g^2 + f^2 - c}{2\sqrt{g^2 + f^2}}$$

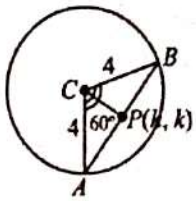
$\{\because g^2 + f^2 - c \geq 0\}$

39. (*c*) Let mid point of cord $P(h, k)$

$x^2 + y^2 - 2x - 4y - 11 = 0$

$C(1, 2), r = 4$

$$CP = 4 \cos 30° = 4\frac{\sqrt{3}}{2} = 2\sqrt{3}$$



We know that locus is circle

whose radius is CP and centre (1, 2)

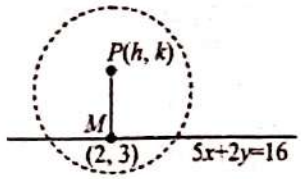
$(x - 1)^2 + (y - 2)^2 = (2\sqrt{3})^2$

$\Rightarrow x^2 + y^2 - 2x - 4y - 7 = 0$

M-II equation of chord $T = S_1$ have a distance from centre is $2\sqrt{3}$ and get the locus.

40. (*a*) Let the centre $P(h, k)$

$$m_{PH} = \frac{-1}{m_2} = \frac{-1}{-\frac{5}{2}} = \frac{2}{5}$$



$$\frac{k - 3}{h - 2} = \frac{2}{5}$$

$2h - 5k + 11 = 0$

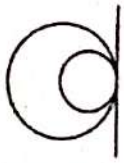
$2x - 5y + 11 = 0 \rightarrow$ Line PM.

41. (*d*) The equation ot the required chord is

$-2x + 4y - 3(x - 2) + 5(y + 4) - 9 = 4 + 16 + 12 + 40 - 9$

or $5x - 9y + 46 = 0$

42. (*b*) $C_1C_2 = 5, r_1 = 7, r_2 = 2$



$C_1C_2 = |r_1 - r_2|$ one common tangent

43. (*b*) Equation of common tangent at point of contact is $S_1 - S_2 = 0$

$\Rightarrow 10x + 24y + 38 = 0$

$\Rightarrow 5x + 12y + 19 = 0$

44. **(d)** Given, $x^2 + y^2 = 6x$ ... (i)

and $x^2 + y^2 + 6x + 2y + 1 = 0$ ... (ii)

From Eq. (i), $x^2 - 6x + y^2 = 0$

$\Rightarrow (x-3)^2 + y^2 = 3^2$

$\therefore$ Centre $(3, 0)$, $r = 3$

From Eq. (ii),

$x^2 + 6x + y^2 + 2y + 1 + 3^2 = 3^2$

$\Rightarrow (x+3)^2 + (y+1)^2 = 3^2$

$\therefore$ Centre $(-3, -1)$ radius $= 3$

Now, distance between centres

$= \sqrt{(3+3)^2 + 1} = \sqrt{37} > r_1 + r_2 = 6$

$\therefore$ Circles do not cut each other

$\Rightarrow$ 4 tangents (two direct and two transversal) are possible

45. **(c)** Equation of pair of tangents is given by $SS_1 = T^2$.

Here

$S = x^2 + y^2 + 20(x + y) + 20,\ S_1 = 20$

$T = 10(x + y) + 20$

$\therefore SS_1 = T^2$

$\Rightarrow 20\{x^2 + y^2 + 20(x + y) + 20\}$

$= 10^2(x + y + 2)^2$

$\Rightarrow 4x^2 + 4y^2 + 10xy = 0$

$\Rightarrow 2x^2 + 2y^2 + 5xy = 0$

46. **(a)** $SS_1 = T^2$

$\Rightarrow (x^2 + y^2 - 2x + 4y + 3)$

$(36 + 25 - 12 - 20 + 3)$

$= (6x - 5y - x - 6 + 2(y - 5) + 3)^2$

$\Rightarrow 7x^2 + 23y^2 + 30xy + 66x + 50y - 73 = 0$

47. **(a)** $S_1 \Rightarrow C_1(1, 0), r_1 = \sqrt{2}$

$S_2 \Rightarrow C_2(0, 1), r_2 = 2\sqrt{2}$

$C_1C_2 = \sqrt{1^2 + 1^2} = \sqrt{2}$

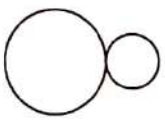
$C_1C_2 = |r_2 - r_1|$

$\sqrt{2} = \sqrt{2}$

Internally touch

$\therefore$ common tangent is one.

48. **(a)** $x^2 + y^2 = 9$



$\Rightarrow C_1(0, 0), r_1 = 3$

$x^2 + y^2 + 6y + c = 0$

$C_2(0, -3), r_2 = \sqrt{9 - c}$

If circle are externally touching

$c_1c_2 = r_1 + r_2$

$B = 3 + \sqrt{9 - c}$

$\Rightarrow c = 9$

If circle are internally touching

$C_1C_2 = |r_1 - r_2|$

$3 = +3 - \sqrt{9 - c}$ or $3 = -3 + \sqrt{9 - c}$

$\Rightarrow c = 9 \Rightarrow 6 = \sqrt{9 - c} \Rightarrow c = -27$

$c = 9, -27$

Aliter: Commo tangent of S_1 and S_2

$6y + c + 9 = 0$

$3 = \left|\frac{c+9}{\sqrt{6^2}}\right| \Rightarrow 18 = |c + 9|$

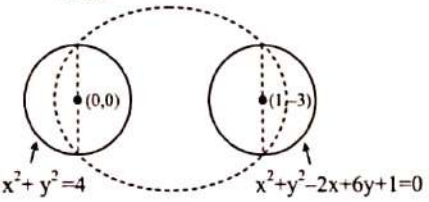
$\Rightarrow c = 9, -27$

49. **(a)** Let required circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Hence common chord with $x^2 + y^2 - 4 = 0$ is $2gx + 2fy + c + y = 0$

This is diameter of circle $x^2 + y^2 = 4$ hence $c = -4$.

Now again common chord with other circle



$2x(g + 1) + 2y(f - 3) + (c - 1) = 0$

This is diameter of $x^2 + y^2 - 2x + 6y + 1 = 0$

$2(g + 1) - 6(f - 3) + 5 = 0$

$2g - 6f + 15 = 0$

locus $2x - 3y - 15 = 0$ which is st. line.

50. **(c)** $S_1 - S_3 = 0$

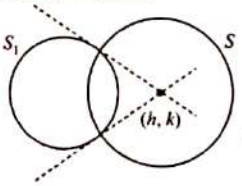
$\Rightarrow 16y + 120 = 0$

$\Rightarrow y = \frac{-120}{16}$

$\Rightarrow y = -\frac{15}{2} \Rightarrow x = 8$

Intersection point of radical axis is $\left(8, \frac{-15}{2}\right)$

51. **(a)** Let point of intersection of tangents is (h, k) family of circle.



$x^2 + y^2 - (\lambda + 6)x + (8 - 2\lambda)y - 3 = 0$

Common chord is $S - S_1 = 0$

$\Rightarrow -(\lambda + 6)x + (8 - 2\lambda)y - 2 = 0$

$\Rightarrow (\lambda + 6)x + (2\lambda - 8)y + 2 = 0$ ... (i)

C.O.C. from (h, k) to S_1: $x^2 + y^2 = 1$ is

$hx + ky = 1$ ... (ii)

(i) & (ii) are same equation

$\frac{\lambda + 6}{h} + \frac{2(\lambda - 4)}{k} = \frac{2}{-1}$

$\Rightarrow \lambda = -2h - 6, \quad \lambda = -k + 4$

$\therefore -2h - 6 = -k + 4$

$\Rightarrow 2h - k + 10 \Rightarrow$ Locus : $2x - y + 10 = 0$

52. **(a)** $S_1 \equiv x^2 + y^2 - 4x - 6y - 12 = 0$

$\Rightarrow C_1(2, 3), r = 5$

$S_2 \equiv x^2 + y^2 + 6x + 4y - 12 = 0$

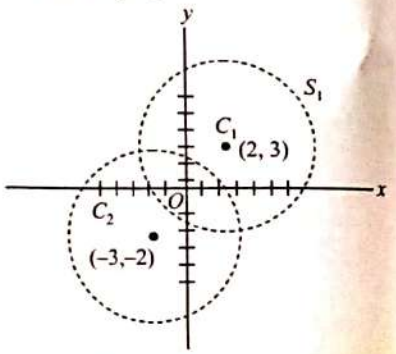
$C_2(-3, -2), r = 5$

$L = x + y = 0$

$S_1 - S_2 = 0$

$-10x - 10y = 0$

$\Rightarrow x + y = 0$



(a) Origin inside both cirlce

(b) L is common chord

(c) L is radical Axis

(d) $m_{C_1C_2} = \frac{5}{5} = 1$ & $m_L = -1$

$C_1C_2 \perp L$

53. **(a)** $S_1 - S_2 = 0$

$\Rightarrow 7x - 8y + 16 = 0$

$S_2 - S_3 = 0$

$\Rightarrow 2x - 4y + 20 = 0$

$S_3 - S_1 = 0$

$\Rightarrow 9x - 12y + 36 = 0$

On solving centre (8, 9)

Length of tangent

$= \sqrt{S_1} = \sqrt{64 + 81 - 16 + 27 - 7} = \sqrt{149}$

$= (x - 8)^2 + (y - 9)^2 = 149$

$= x^2 + y^2 - 16x - 18y - 4 = 0$

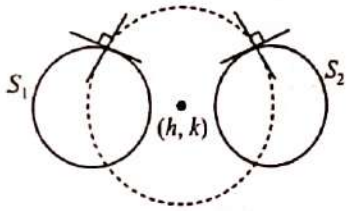
54. **(c)** Let centre (h, k) and circle

$x^2 + y^2 + 2gx + 2fy + c = 0$

$h = -g, k = -f$

For $S_1 : g_1 = 2, f_1 = -3, c_1 = 9$,

For $S_2 : g_2 = -\frac{5}{2}, f_2 = 2, c_2 = -2$



$\therefore \quad 2.g.2 + 2.f(-3) = c + 9$

$\Rightarrow \quad 4g - 6f = c + 9 \qquad \ldots (i)$

and $2g\left(\frac{-5}{2}\right) + 2.f(2) = c - 2$

$\Rightarrow \quad -5g + 4f = c - 2 \qquad \ldots (ii)$

Subtract (ii) from (i)

$-9g + 10f = 11 \quad \Rightarrow \quad 9x - 10y + 11 = 0$

Prabal (JEE Main Level)

1. (b) Let the centre of the required circle be (x_1, y_1) and the centre of given circle is $(1, 2)$. Since radii of both circles are same, therefore, point of contact $(5, 5)$ is the mid point of the line joining the centres of both circles. Hence $x_1 = 9$ and $y_1 = 8$. Hence the required equation is $(x-9)^2 + (y-8)^2 = 25$

$\Rightarrow \quad x^2 + y^2 - 18x - 16y + 120 = 0.$

Trick : The point $(5, 5)$ must satisfy the required circle. Hence the required equation is given by (2).

2. (d) Let the centre be (h, k), then radius $= h$

Also $CC_1 = R_1 + R_2$

or $\sqrt{(h-3)^2 + (k-3)^2} = h + \sqrt{9+9-14}$

$\Rightarrow (h-3)^2 + (k-3)^2 = h^2 + 4 + 4h$

$\Rightarrow k^2 - 10h - 6k + 14 = 0$

or $y^2 - 10x - 6y + 14 = 0$

3. (a) Centre and the radius of the given circle are

$C(5, 7)$ and $r = \sqrt{25+49+51} = 5\sqrt{5}$ respectively

Let $P = (-7, 2)$ be the given point.

Now $CP = \sqrt{12^2 + 5^2} = 13$

Thus $p = CP + r = 13\ 5\sqrt{5}$ and $q = CP - r = 13 - 5\sqrt{5}$

Hence G.M. of p and q is

$= \sqrt{pq} = \sqrt{13^2 - 125} = 2\sqrt{11}$

4. (d) Lines $x = -g$ and $y = -f$ will be diameter for the circle. Equation of circle will be $(x+g)^2 + (y+f)^2 = r^2$. Now this intersect $x^2 + y^2 + 2gx + 2fy + c = 0$ orthogonally.

Hence $2g.g_1 + 2f.f = c + k$

$\Rightarrow \quad k = 2g^2 + 2f^2 - c$

5. (a) $S_1 = x^2 + y^2 + 4x + 1 = 0$

$S_2 = x^2 + y^2 + 6x + 2y + 3 = 0$

Common chord

$\equiv S_1 - S_2 = 0 \Rightarrow 2x + 2y + 2 = 0$

$\Rightarrow x + y + 1 = 0$

6. (c) Equation of common chord is

$ax - by = 0$

Now length of common chord

$= 2\sqrt{r_1^2 - p_1^2} = 2\sqrt{r_2^2 - p_2^2}$

where r_1 and r_2 are radii of given circles and p_1, p_2 are the perpendicular distances from centres of circles to common chords.

Hence required length

$= 2\sqrt{a^2 - \frac{a^4}{a^2+b^2}} = \frac{2ab}{\sqrt{a^2+b^2}}$

7. (d) $\Delta REO \cong \Delta RDO'$

$\frac{1}{r} = \frac{OR}{O'R}$

$\frac{1}{r} = \frac{OR}{O'R} \quad ER = \sqrt{3}$

$OR^2 = 1 + (\sqrt{3})^2 = 4$

$OR = 2$

$O'R = g + 1 + r = g + r$

$\frac{1}{r} = \frac{2}{3+r}$ $\boxed{r = 3}$

8. (c) Here the intersection point of chord and circle can be found by solving the equation of circle with the equation of given line, therefore, the points of intersection are $(-4, -3)$ and $\left(\frac{24}{5}, \frac{7}{5}\right)$. Hence the midpoint is

$\left(\frac{-4+\frac{24}{5}}{2}, \frac{-3+\frac{7}{5}}{2}\right) = \left(\frac{2}{5}, -\frac{4}{5}\right).$

9. (b) Required equation is

$(x^2 + y^2 + 13x - 3y) +$

$\lambda(2x^2 + 2y^2 + 4x - 7y - 25) = 0$

which passes through $(1, 1)$, so $\lambda = \frac{1}{2}$

Hence required equation is

$4x^2 + 4y^2 + 30x - 13y - 25 = 0.$

10. (a) Let equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ be with $x^2 + y^2 = p^2$ cutting orthogonally, we get $0 + 0 = +c - p^2$ or $c = p^2$ and passes through (a, b), we get

$a^2 + b^2 + 2ga + 2fb + p^2 = 0$

or $2ax + 2by - (a^2 + b^2 + p^2) = 0$

Required locus as centre $(-g, -f)$ is changed to (x, y).

11. (b) Given circle is $\left(2, \frac{3}{2}\right), \frac{5}{2} = r_1$ (say)

Required normals of circles are $x + 3 = 0, x + 2y = 0$

which intersect at the centre $\left(-3, \frac{3}{2}\right)$, $r_2 =$ radius (say).

2nd circle just contains the 1st

i.e., $C_2C_1 = r_2 - r_1 \Rightarrow r_2 = \frac{15}{2}$.

12. (d) $(x-1)^2 + (y+2)^2 = 2$

$CQ = \sqrt{2}$

$\angle QSR = \frac{17}{4}$

co-ordiante of point Q & S are

$\left(1 \pm \sqrt{2}\cos 45°, -2 \pm 52 \sin 45°\right)$

$(2, -1)$ & $(0, -3)$

Co-ordiante of P & R are given by

$\left(1 \pm \sqrt{2}\cos 135°, -2 \pm \sqrt{2}\sin 135°\right)$

$= \left(1 \pm \sqrt{2}\left(-\frac{1}{\sqrt{2}}\right), -2 \pm \sqrt{2} \cdot \frac{1}{\sqrt{2}}\right)$

$(0, -1)$ & $(2, -3)$

13. (b) $x^2 + y^2 + 2x - 1 \le 0 \quad x - y + c \ge 0$

To contain only one point in common the line should be a tangent to the circle.

$\Rightarrow x^2 + y^2 + 2x + 1 - 1 - 1 \le 0$

$\Rightarrow (x+1)^2 + y^2 \le 2$

Centre $= (-1, 0)$

Radius $= \sqrt{2}$

$(-1, 0) \quad x - y + c \ge 0$

$d = \left|\frac{-1+c}{\sqrt{2}}\right| = \sqrt{2}$

$\Rightarrow |c - 1| = 2$

$c - 1 = 2 \qquad c - 1 = -2$

$c = 3 \qquad c = -1$

$\therefore c = 3, -1$

14. (b) $C : x^2 + y^2 + 2gx + 2fy + c = 0$

$x^2 + y^2 = 4$

$2(g_1g_2 + f_1f_2 = C_1 + C_2$

$2(0 + 0) = C - 4 \quad \Rightarrow \quad C = 4$

also $2x - 2y + 9 = 0$

$2(-g) - 2(-f) + 9 = 0$

$2f = 2g - 9$

$\therefore \quad x^2 + y^2 + 2gx + (2g - 9)y + 4 = 0$

$\therefore \quad (x^2 + y^2 - 9y + 4) + 2g(x + y) = 0$

$\therefore \quad x^2 + y^2 - 9y + 4 = 0$ and $x + y = 0$

$\therefore \quad x^2 + x^2 + 9x + 4 = 0$

$\Rightarrow \quad 2x^2 + 9x + 4 = 0$

$\Rightarrow \quad (2x + 1)(x + 4) = 0$

$\Rightarrow \quad x = \frac{-1}{2}, -4.$

$\therefore \quad$ Point $\left(\frac{-1}{2}, \frac{1}{2}\right), (-4, 4).$

15. **(d)** $S: x^2 + y^2 + 2gx + 2fy + c = 0$

$S_1: \quad x^2 + y^2 - 1 = 0$

$S_2: \quad x^2 + y^2 - 2x - 2y - 1 = 0$

$(S_1S_1) \quad 2(g)(0) + 2(0)(f) = c - 1$

$\boxed{c = 1}$

$(S_2, S_2) \qquad g + f = 0$

$\Rightarrow \left(x - \frac{1}{\sqrt{2}}\right)^2 + \left(y - \frac{1}{\sqrt{2}}\right)^2 = 0$

16. **(c)** $C_1 \equiv r \Rightarrow 1$

$C_2 \equiv r \Rightarrow 2$

$C_3 \equiv r \Rightarrow 3$

Therefore after we join the centres we'll get a right angle triangle of sides 3,4 and 5

$\therefore d = \frac{abc}{2 \times \text{Area}}$

$\Rightarrow d = \frac{3 \times 4 \times 5}{2 \times 6} = 5$

$r = \frac{d}{2} = 2.5$ units

17. **(d)** Equation of circle is,

$|x - 2| + |y - 3| = 4$

$\Rightarrow |x - 4| + |y - 4| = 1$ will represent square in the xy plane

Centered at (a, b) of side length being $\sqrt{2}$ units

Radius of circle = 1/2 × side length of squre

$= 1/2 \times \sqrt{2} = 1 units$

$= \frac{1}{\sqrt{2}} \times (4)$

$= 2\sqrt{2}$

Hence equation of circle

$(x - a)^2 + (y - b)^2 = r^2$

$(x - 2)^2 + (y - 3)^2 = (2\sqrt{2})^2$

$(x - 2)^2 + (y - 3)^2 = 8$

18. **(c)** Equation of the two circles be $(x - r)^2 + (y - r)^2 = r^2$ i.e.

$x^2 + y^2 - 2rx - 2ry + r^2 = 0$ where $r = r_1$ and r_2. Condition of orthogonality gives

$2r_1r_2 + 2r_1r_2 = r_1^2 + r_2^2$

$\Rightarrow \quad 4r_1r_2 = r_1^2 + r_2^2$

Circle passes through (a, b)

$\Rightarrow a^2 + b^2 - 2ra - 2rb + r^2$

$= 0$ i.e. $r^2 - 2r(a + b) + a^2 + b^2 = 0$

$r_1 + r_2 = 2(a + b)$ and $r_1r_2 = a^2 + b^2$

$\therefore 4(a^2 + b^2)$

$= 4(a + b)^2 - 2(a^2 + b^2)$ i.e. $a^2 - 4ab + b^2 = 0$

19. **(b)** Given, radius of circle = a and centre $(0, 0)$

$CO^2 = AO^2 + AC^2$

$\left(\sqrt{(\alpha - 0)^2 + (\beta - 0)^2}\right)^2 = a^2 + AC^2$

$\alpha^2 + \beta^2 - a^2 = AC^2$

$\Rightarrow AC = \sqrt{\alpha^2 + \beta^2 - a^2} = \sqrt{s_1}$

In $\Delta\, COA$

$\Rightarrow \tan\theta/2 = \frac{A0}{AC} = \frac{a}{\sqrt{s_1}}$

$\theta/2 = \tan^{-1}\left[\frac{a}{\sqrt{s_1}}\right]$

$\theta = 2\tan^{-1}\left[\frac{a}{\sqrt{s_1}}\right].$

20. **(b)** $s = 4xy - 3y^2 + 12h$

$\frac{\partial 5}{\partial x} - 4y = 0$

$\frac{\partial s}{\partial y} = 4x - 6y = 0$

point of intersection $\equiv (0,0)$

lines $y = 0$ and $y = \frac{4}{3}x$

and $h = 12$

$(h, k) \equiv (12, r)$

Now, $\left|\frac{4 \times 12 - 3r}{5}\right| = r$

$\Rightarrow r = 6$

so, $r + k = 2r = 12$

21. **(b)** $C_3: \quad \frac{x+2}{\cos\lambda} = \frac{y-1}{\sin\lambda} = 1$

$(x + 2)^2 + (y - 1)^2 = 1$

equation of distance circle

$(x + 2)^2 + (y - 1)^2 = 2$

$y = 4$

$p(t, 4)$

Circumcentre $\equiv \left(\frac{t+0}{2} \quad \frac{4+0}{2}\right)$

22. **(a)** The correct option is $A\ (x + 1)^2 + y^2 = 1$

Using $T = S_1$ take mid-point (h, k) then equation of PQ, is

$xh + yk - (h^2 + k^2) = 0 \ldots\ldots (i)$

$mx - y + 2m = 0 \ldots\ldots (ii)$

Compare (i) and (ii) eliminate m

23. **(b)** $\left(1 + a^2\right)x^2 + 2a^2x + a^2 + b^2 - 1 = 0$ has

Root of opposite sign

Product of Root = negative

$\frac{a^2 + b^2 - 1}{1 + a^2} < 0$

$\Rightarrow \quad \boxed{a^2 + b^2 < 1}$

24. **(d)** Let $C(h,k)$ be the center of the circle.

Circle touches the x-axis then its radius is $r_1 = k$.

Also circle touches the circle with centre $C_2(0, 3)$ and radius $r_2 = 2$.

$\therefore |C_1C_2| = r_1 + r_2$

$\Rightarrow \sqrt{(h - 0)^2 + (k - 3)^2} = |k + 2|$

squaring

$h^2 - 10k + 5 = 0$

$\Rightarrow$ Locus is $x^2 - 10y + 5 = 0$, which is parabola.

25. **(c)** $x^2 + y^2 = 25$

Let $x, y \geq 0$

$y = \sqrt{25 - x^2}$

$x = 5, y = 0 \quad (5,0)$

$x = 4, y = 3 \quad (4,3)$

$x = 3, y = 4 \quad (3,4)$

$x = 0, y = 5 \quad (0,5)$

on quadrant $4 \times 2 = 8 + 4 = 12$

26. **[45]** $C_1(r, 0)$

$C_2(R, 0)$

Equation of $AB\ y = r$

Equation of circle $C_2(x - R)2 + y^2 = R^2$

$A\left(R - \sqrt{R^2 - r^2}, r\right)$ using $R^2 = 2r^2$

$B\left(R + \sqrt{R^2 - r^2}, r\right) A(R - r, r), B(R + r, r)$

Slope of $OA = \frac{r}{R - r} = m_1$

Slope of $OB = \frac{r}{R + r} = m_2$

$\tan\theta = \frac{m_1 - m_2}{1 + m_1m_2} = 1$

$\theta = 45°$

27 **[5]**

$OC = 1$ radius of circle $x^2 + y^2 = 1$

$OA = AP$

$\therefore \quad \angle AOP = \angle OPA = 45°$

AP PC radius of circle

In ΔPCA,

$\therefore \angle PCA + \angle PAC + \angle CPA = 180°$

$\Rightarrow 2\angle PCA + 45° = 180°$

$\Rightarrow \angle PCA = \frac{135^\circ}{2}$

$\Rightarrow \angle OCA = 180^\circ - \angle PCA$

$\Rightarrow \angle OCA = 180^\circ - \frac{135^\circ}{2} = \pi - \frac{3\pi}{8}$

$\Rightarrow \angle OCA = \frac{5\pi}{8}$

28. **[8]** Length of each tangent

$L^2 = (4)^2 + (5)^2 - (4\times4) - (2\times5) - 11$

$L = 2$

$r = \sqrt{2^2 + 1^2 - (-11)}$

$r = 4$

Area $= L \cdot r = 8$ sq. units.

29. **[49]** $x^2 + y^2 - 5x + 2y - 5 = 0$

$\Rightarrow \left(x - \frac{5}{2}\right)^2 + (y+1)^2 - 5 - \frac{25}{4} - 1 = 0$

$\Rightarrow \left(x - \frac{5}{2}\right)^2 + (y+1)^2 = \frac{49}{4}$

$\Rightarrow$ So the axes are shifted to $\left(\frac{5}{2}, -1\right)$

New equation of circle must be $x^2 + y^2 = \frac{49}{4}$

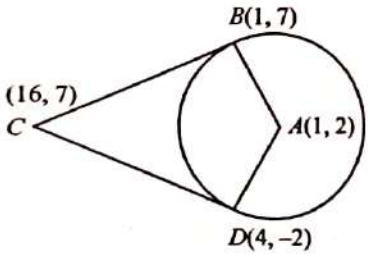
30. **[75]** Given circle $x^2 + y^2 - 2x - 4y - 20 = 0$

Tangent at $B(1, 7)$ is

$x + 7y - (x + 1) - 2(y + 7) - 20 = 0$

$5y - 35 = 0$

$\Rightarrow y = 7$



at $D(4, -2)$

$4x - 2y - (x + 4) - 2(y - 2) - 20 = 0$

$3x - 4y = 20$

Hence $c(16, 7)$

Area of quadrilateral $ABCD = AB \times BC$

$= 5 \times 15 = 75$ square units.

[15] Centre and radius of the given circle are $C(2,1)$

and $\sqrt{4+1+20} = 5$ respectively

Now $CP = \sqrt{8^2 + 6^2} = 10$ Hence greatest distance of point P from the given circle is $= 10 + r = 15$

32. **[1]** $S_1 : x^2 + y^2 + 2ax + 2by + c = 0 \Rightarrow$ represents the circle with centre

$C_1 \equiv (-a, -b)$ & radius $r_1 = \sqrt{a^2 + b^2 - c}$

$S_2 : x^2 + y^2 + 2bx + 2ay + c = 0 \Rightarrow$ represents the circle with centre

$C_2 \equiv (-b, -a)$ & radius $r_2 = \sqrt{a^2 + b^2 - c}$

Distance between centre

$e = \sqrt{(a-b)^2 + (a-b)^2} = \sqrt{2}(a-b)$

Case I

$\sqrt{2}(a-b) = r_1 + r_2 = 2\sqrt{a^2 + b^2 - c}$

(touching externally)

$2a^2 + 2b^2 - 4ab = 4a^2 + 4b^2 - 4c$

$2a^2 + 2b^2 + 4ab = 4c$

$(a+b)^2 = 2c$

Case - 2

$\sqrt{2}(a-b) = r_1 - r_2 = 0$

(touching internally)

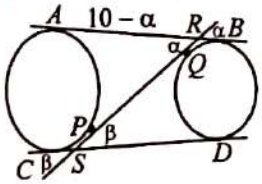
33. **[10]** $\therefore RP = RA = 10 - \alpha$

$\Rightarrow RS = 10 - \alpha + \beta$ .....(*i*)

Also $SQ = SD = 10 - \beta$

$\Rightarrow RS = 10 - \alpha + \beta$ .....(*ii*)

(*i*) and (*ii*) $\Rightarrow \alpha = \beta$, Hence $RS = 10$



34. **[2]** If $\lambda x - y + 1$ and $x - 2y + 3 = 0$ intersect the axes at con cyclic points then $\lambda 1 = (-1)(-2)$ so $\lambda = 2$

35. **[1]**

$(x-\alpha)^2 + (y-\beta)^2 = r^2$

$x^2 + y^2 - 2\alpha x - 2\beta y + \alpha^2 + \beta^2 - r^2 = 0$

$x^2 + y^2 - 2x - 2y - 7 = 0$

$x^2 + y^2 + 4x - 6y - 3 = 0$

$2\alpha + 2\beta = \alpha^2 + \beta^2 - r^2 - 7$ ...(1)

$-4\alpha + 6\beta = \alpha^2 + \beta^2 - x^2 - 3$ ...(2)

from (1) & (2)

$3\alpha - 2\beta + 2 = 0$

$\frac{3}{4}\alpha - \frac{\beta}{2} = -\frac{1}{2}$

$\left|\frac{3\alpha}{4} - \frac{\beta}{2}\right| = \frac{1}{2}$

36. **[4]** $(x+2)(x-1) + (y-1)(y-2) = 0$

x = intercept = 1

y = intercept = 1

sum of intercept = 1 + 3 = 4

Parikshit (JEE Advanced Level)

1. **(c)**

$C_1 : ax^2 + 2bxy + by^2 - 2gx - 2 + y + c = 0$

$C_2 : a'x^2 - 2bxy + (a' + a - b)y^2 - 2g'x - 2f'y + c = 0$

$C_1 + kC_2 = 0$, KE constant

$(a + a'k)\cdot x^2, 2(1-k)hxy + \{b + (a' + a - b)\}y^2$

$+2(g + kg')x - 2(f + f'k)y + c(1 + k) = 0$

Coefficient of x^2 = coefficient of y^2

$a + a'k = b + a'k + ak - bk$

$a + bk = ak + b$

$\Rightarrow k = 1$

Coefficeint of $xy = 0 \Rightarrow k = 1$

$(a + a')x^2 + (a + a')y^2 - 2(g + g')2 - 2(f + f')y + 2c = 0$

$x^2 + y^2 - 2\left(\frac{g+g'}{a+a'}\right)x - 2\left(\frac{f+f'}{a+a'}\right)y + \frac{2c}{a+a'} = 0$

Centre $= \left(\frac{g+g'}{a+a'}, \frac{f+f'}{a+a'}\right)$

$\frac{PA^2}{PD^2} + \frac{PB^2}{PD^2} + \frac{PC^2}{PD^2} = \frac{r^2}{r^2} + \frac{r^2}{r^2} + \frac{r^2}{r^2} = 3$

2. **(b)**

$x^2 + y^2 = 1$

At 't' varies from 1 to $1 + \sqrt{2}$

The equation of line joining (0, 1) and $(1 + \sqrt{2}, 0)$ is,

$y - 1 = \left(\frac{0-1}{1+\sqrt{2}-1}\right)(x - 0)$

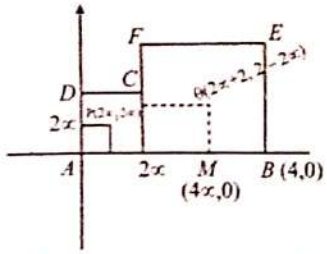
$x + (1 + \sqrt{2})y - (1 + \sqrt{2}) = 0$

It intersects the circle at $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ as shown in the figure.

Here, $\tan\theta = \frac{\left(\frac{1}{\sqrt{2}} - 0\right)}{\left(\frac{1}{\sqrt{2}} - 0\right)} = 1$

$\theta = \frac{\pi}{4}$

3. (a)



$S_1 : (x-2\alpha)^2 + (y-2\alpha)^2 = 4\alpha^2 + 4\alpha^2$

or $x^2 + y^2 - 4\alpha x - 4\alpha y = 0$

$S_2 : (x-(2\alpha+2))^2 + (y-(2-2\alpha))^2$

$= (2-2\alpha)^2 + (2-2\alpha)^2$

or $x^2 + y^2 - 4(\alpha+1)x - 4(1-\alpha)y$

$= (2-2\alpha)^2 - (2\alpha+2)^2 = -16\alpha$

Common chord of S_1 and S_2 is

$4x + 4(1-2\alpha)y = 16\alpha$

or $x + (1-2\alpha)y = 4\alpha$

$x + y - 2\alpha(y-2) = 0$

This line always passes through intersection of $x + y$

$= 0$ and $y = 2$

or $(-2, 2)$

$\therefore$ coordinates of R are $(-2, 2) = (a, b)$

$\Rightarrow a + b = 0$

4. (b)

$r = OA\cos 30° = \dfrac{\sqrt{3}}{2}$

Equation of circle is $x^2 + y^2 = \dfrac{3}{4}$,

Where the centre is (0, 0) and radius is $\dfrac{\sqrt{3}}{2}$

$PA^2 + PB^2 + PC^2 + PD^2$

$(x_1-0)^2 + (y_1-1)^2 (x_1+\sqrt{3}) + (y_1-0)^2$

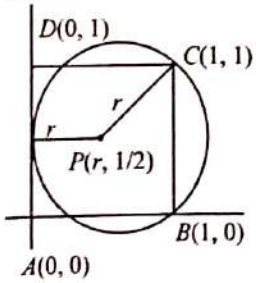
$+ x_1^2 + (y_1+1^2) + (x_1-\sqrt{3})^2 + (y_1-0)^2$

$4x_1^2 + 4y_1^2 + 8 = 4(x_1^2 + y_1^2) + 8$

Since x_1 and y_1 lies on the circle thus putting the value of $x^2 + y^2 = \dfrac{3}{4}$ in the above equation, we have

$PA^2 + PB^2 + PC^2 + PD^2 = 4 \times \dfrac{3}{4} + 8 = 11$

5. (d)



$PC = r$

$PC^2 = r^2$

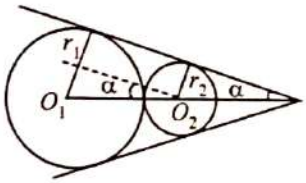
$(r-1)^2 + \left(\dfrac{1}{2}-1\right)^2 = r^2$

$1 - 2r + \dfrac{1}{4} = 0$

$r = \dfrac{5}{8}$

6. (b) $\sin\alpha = \dfrac{r_1 - r_2}{r_1 + r_2}$

$\Rightarrow \theta = 2\sin^{-1}\left(\dfrac{r_1 - r_2}{r_1 + r_2}\right)$



7. (b) Equation of (1) $x^2 + y^2 = 2$

Equation of (2) $(x-2)^2 + y^2 = 1$

$A\left(\dfrac{5}{4}, \dfrac{\sqrt{7}}{4}\right)$

$x_B = \dfrac{x_A + x_C}{2}$

$y_B = \dfrac{y_A + y_C}{2}$

$x_c^2 + y_c^2 = 2$

$\left(\dfrac{x_c + \frac{5}{4}}{2} - 2\right)^2 + \left(\dfrac{y_c + \frac{\sqrt{7}}{4}}{2}\right)^2 = 1$

$AC = \sqrt{\dfrac{7}{2}}$.

8. (d) Radius $= a, a \in R$

Centre $= (a, a)$

MN $\perp$ BC, MN $\perp$ AD

Equation of MN is $y = x$.

Equation of MN in distance form is

$\dfrac{x-a}{\cos 45} = \left(a + \dfrac{a}{\sqrt{2}}, a + \dfrac{a}{\sqrt{2}}\right)$

Similarly, coordinates of N is,

$\left(a + \dfrac{a}{\sqrt{2}}, a + \dfrac{a}{\sqrt{2}}\right)$

x intercept of AD is $2a + \sqrt{2}a$

x intercept of BC is $2a - \sqrt{2}a$

Area of trapezium, $ABCD$

$= \dfrac{1}{2}(AD + BC)$ MN

$\Rightarrow 900\sqrt{2} = \dfrac{1}{2} \times \sqrt{2}(4a) \times 2a$

$\Rightarrow a = 15$

9. (b) Let 'r' and 'R' be radius of required and given circle, respectively and let centre is (h, k).

By given condition,

$\sqrt{(h-1)^2 + (k-2)^2} = R - r$

Now, $\dfrac{r}{AB} = \tan 30°$

$r = AB\tan 30°$

$= (R-r)\dfrac{1}{\sqrt{3}}$ $(AB = R - r)$

$\Rightarrow \sqrt{(h-1)^2 + (k-2)^2}$

$= R - \dfrac{R}{1+\sqrt{3}} = R\left(\dfrac{\sqrt{3}}{1+\sqrt{3}}\right)$

Now, $R = 1 + \sqrt{3}$

$\Rightarrow \sqrt{(h-1)^2 + (k-2)^2} = \sqrt{3}$

Therefore, locus is $(x-1)^2 + (y-2)^2 = 3$.

10. (c)

$x^2 + y^2 = 50$

A (−5, 5)

B (7, − 1)

$c(5\sqrt{2}\cos\theta, 5\sqrt{2}\sin\theta)$

$G = \left(\dfrac{2 + 5\sqrt{2}\cos\theta}{3}, \dfrac{4 + 5\sqrt{2}\sin\theta}{3}\right)$

$\dfrac{2 + 5\sqrt{2}\cos\theta}{3} = \dfrac{h}{3}$ $\dfrac{k}{3} = \dfrac{4 + 5\sqrt{2}\sin\theta}{3}$

$\dfrac{h-2}{5\sqrt{2}} = \cos\theta$ $\dfrac{k-4}{5\sqrt{2}} = \sin\theta$

$(h-2)^2 + (k-4)^2 = 50$

Replace h by x & k by y

$(x-2)^2 + (y-4)^2 = 50$

$x^2 + y^2 + 4x - 8y - 30 = 0$

11. (b)

Equation of MH

$y = 2x - 6$ ...(1)

Centre (3,0)

$(x-3)^2 + y^2 = 25$ ...(2)

From (1) & (2)

$x = 3 \pm \sqrt{5}$

$y = -2\sqrt{5}$

$MH = \sqrt{(2 + 2\sqrt{5})^2 + (-1 - \sqrt{5})^2}$

$MH = (1 + \sqrt{5})\sqrt{5}$

Area of $\Delta = \frac{1}{2}\sqrt{5}(1+\sqrt{5})\ \ 4\sqrt{5}$

$= 10(1+\sqrt{5})$

12. **(c)** Let 'P' be the point (a, b). The equation of the line through 'P' with slope $\tan\theta$ is $\frac{x-\alpha}{\cos\theta} = \frac{y-\beta}{\sin\theta} = r$ any point on this line is $(a+r\cos q, b+r\sin q)$

if then point lies on the curve, then

$(a+r\cos q, b+r\sin q)+b(b+r\sin q)^2 = 1.$

This is a quadratic equation in 'r' where value give the coordinates of Q and R,

Where, $PQ = r_1$ and $PQ = r_2$

$$r_1 r_2 = \frac{a^2+2ha\beta+b\beta^2-1}{a\cos^2\theta+2h\cos\theta\sin\theta+b\sin^2\theta}$$

$$= \frac{a^2+2ha\beta+b\beta^2-1}{a\cos^2\theta+2h\cos\theta\sin\theta+b\sin^2\theta}$$

$$= \frac{a\alpha^2+3hr\beta+b\beta^2-1}{\frac{a+b}{2}+\left[\sqrt{\left(\frac{a-b}{2}\right)^2+h^2}\right]\cos(2\theta+0)}$$

Which is independent of 'q' if

$\left(\frac{a-b}{2}\right)^2 + h^2 = 0$ and

$a = b, h = 0$

Hence given curve is circle.

13. **(a,b)** $PM + 3QN = 4$

$ax + by = 1$

$$P_1 = \left|\frac{1}{\sqrt{a^2+b^2}}\right| \quad P_2 = \left|\frac{4a-1}{\sqrt{a^2+b^2}}\right|$$

$$\left|\frac{1}{\sqrt{a^2+b^2}}\right| + 3\left|\frac{4a\cdot 1}{\sqrt{a^2+b^2}}\right|$$

Now (0,0) & (4,0) must be given the same sign

$$\frac{1}{\sqrt{a^2+b^2}} + \frac{3(1-4a)}{\sqrt{a^2+b^2}} = k$$

$$\left|\frac{4(1-3a)}{\sqrt{a^2+b^2}}\right| = k$$

$$\left|\frac{3a-1}{\sqrt{a^2+b^2}}\right| = \frac{k}{4}$$

Hence the centre of the fixed circle c is (3,0) lies on the line segment AB.

If $k = 4$, then radius $= \frac{4}{4} = 1.$

14. **(a,d)** Two fixed pts. are point of intersection of

$x^2 + y^2 - 2x - 2 = 0$ & $y = 0$

Point $x^2 - 2x - 2 = 0$

$(x-1)^2 - 3 = 0$

$\Rightarrow x - 1 = \sqrt{3}, x - 1 = \sqrt{3}$

$(1+\sqrt{3}, 0)(1+\sqrt{3}, 0)$

15. **(a, c)**

Mid point of C_1C_2 is centre of the circle passing through P, Q, R, S

$(PM)^2 + (OM)^2 = (OP)^2$

$$OP^2 = \frac{96}{4} + 9 = \frac{96+36}{4}$$

$$OP^2 = \frac{132}{4}$$

$\lambda^2 = 33$

16. **(c)**

The correction option is (c) $(3, 0), \sqrt{5}$.

Let the equation of the circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ...(1)

The line $lx + my + 1 = 0$ will touch circle

(i) if the length of perpendicular from the centre $(-g, -f)$ of the circle on the line is equal to its radius i.e.,

$$\frac{|-gl-mf+1|}{\sqrt{l^2+m^2}} = \sqrt{g^2+f^2-c}$$

$(gl+mf-1)^2 = (l^2+m^2)(g^2+f^2-c)$

or,

$(c+f^2)l^2 + (c-g^2)m^2 - 2gl - 2fm + 2gflm + 1 = 0$

But the given condition of tangency is,

$4l^2 - 5m^2 + 6l + 1 = 0$ ...(3)

Comparing (2) and (3) we get,

$c - f^2 = 4, c - g^2 = -5, -2g = 6,$

Solving we get,

$f = 0, g = -3, c = 4$

Substituting these values in (1), the equation of the circle is,

$x^2 + y^2 - 6x + 4 = 0$

$\Rightarrow (x-3)^2 + y^2 = 5$

17. **(b)** Circle $x^2 + y^2 - 6x + 4 = 0$

any point on the line $x + y = 1$ is $(t, 1-t)$. $t \in IR$

Chord of contact generated by this point for circle is $tx + y(1-t) - 3(t+x) + 4 = 0$

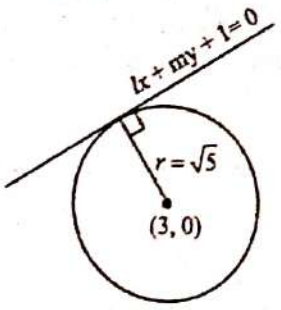
$t(x-y-3) + (-x+y+4) = 0$ which are concurrent at poitn of intersection of the lines $x - y - 3 = 0$ & $-3x + y + 4 = 0$ for all values of t.

$\Rightarrow \left(\frac{1}{2}, -\frac{5}{2}\right)$

18. **(a)** Given $4l^2 - 5m^2 + 6l + 1 = 0 (l, m \in R)$

$\Rightarrow (3l+1)^2 = 5(l^2 + m^2)$

$$\Rightarrow \frac{|3l+1|}{\sqrt{l^2+m^2}} = \sqrt{5},$$



So, clearly the line $lx + my + 1 = 0$ is tangent to a fired circle $S = 0$

i.e., $(x-3)^2 + (y-0)^2 = (\sqrt{5})^2$, whose centre is $(3, 0)$ and $r = \sqrt{5}$

(i) $\Rightarrow$ Circle is $x^2 + y^2 - 6x + 4 = 0$ ...(1)

(ii) Any point on line $x + y - 1 = 0$ is $(t, 1-t), t \in R$.

$\therefore$ The equation of chord of contact for the circle (1) w.r.t. $(t, 1-t)$ is

$tx + (1-t)y - 3(t+x) + 4 = 0$

i.e. $t(x-y-3) + (-3x+y+4) = 0$, which passes through $\left(\frac{1}{2}, \frac{-5}{2}\right)$

(iii) As line $x - 2y + c = 0$ intersects the circle S orthogonally so the line must passes through centre of circle S.

$\Rightarrow 3 - 2(0) + c = 0$

$\Rightarrow c = -3$

Alternative:

Let the required equation of circle S be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ...(1)

As line

$lx + my + 1 = 0$ ...(2)

is tangent to circle (1), so

$$\frac{|-gl-mf+1|}{\sqrt{l^2+m^2}} = \sqrt{g^2+f^2-c}$$

$\Rightarrow (gl+mf-1)^2 = (l^2+m^2)(g^2+f^2-c)$

$\Rightarrow (c-f^2)l^2 + (c-g^2)m^2 - 2g\cdot l - 2f\cdot m + 2gf\cdot lm + 1 = 0$ ...(3)

But, we are given

$4l^2 - 5m^2 + 6l + 1 = 0$ ...(4)

$\therefore$ On comparing (3) and (4), we get

$$\frac{c-f^2}{4} = \frac{c-g^2}{-5} = \frac{-2g}{6} = \frac{-2f}{0} = \frac{2fg}{0} = \frac{1}{1}$$

$\Rightarrow g = -3, f = 0, c = -5 + g^2 = 4$

$\Rightarrow$ The equation of fixed circles $x^2 + y^2 - 6x + 4 = 0$

19. (*c*) $2\tan^{-1}\frac{r}{\sqrt{S_1}} = 2\tan^{-1}\frac{1}{3}$

$= \tan^{-1}\frac{2/3}{1-(1/3)^2} = \tan^{-1}\left(\frac{3}{4}\right)$

20. (*b*) Centre of circle C_2 is $\left(3\frac{1}{2},1\right)$

21. (*b*)

$PA^2 = \frac{85}{4} - 1$

$\boxed{PA = \frac{9}{2}}$

22. (*c*), **23.** (*c*), **24.** (*c*)

$AQ = TQ = 1$ as they are radii of circle

Let $PQ = x$

$PT = \sqrt{x^2-1}$

In ΔTQP & ΔAPO

$\angle TPQ = \angle APO$ ..... (common angle)

$\angle A = \angle T$ .....(right angle)

$\Delta TPO \cong \Delta APO$

$\frac{TQ}{PT} = \frac{AO}{PA}$

$\frac{1}{\sqrt{x^2-1}} = \frac{AO}{AQ+PQ}$

$AO = \frac{1+x}{\sqrt{x^2-1}}$

As AO & OT are tangent on a circle from same point $AO = OT = \frac{x+1}{\sqrt{x^2-1}}$

Perimeter of ΔAOP is 8.

$AO + AP + OP = 8$

$\frac{x+1}{\sqrt{x^2-1}} + 1 + x + OT + TP = 8$

$(x+1)^2 = (7-x)\left(\sqrt{x^2-1}\right)$

$(x+1)^4 = (7-x)^2(x-1)(x+1)$

$(3x-5)^2 = 0$

$x = \frac{5}{3}$

$PQ = \frac{5}{3}$

$AO = \frac{1+x}{\sqrt{x^2-1}} = \frac{8/3}{4/3} = 2$

Centre Q is given by (2,1)

$(x-2)^2 + (y-1)^2 = 1$

$R\left(\frac{3}{2},2\right)$

$M\left(\frac{3}{4},1\right)$

$\left(x-\frac{3}{4}\right)^2 + (y-1)^2 = \left[\left(\frac{3}{4}\right)^2 + 1^2\right]$

$\left(x-\frac{3}{4}\right)^2 + (y-1)^2 = \frac{25}{16}$

25. (*b*) By homogenisation,

$x^2 + y^2 + 2(3x - 5y)$

$\left(\frac{y-ax}{b}\right) + \left(\frac{y-ax}{b}\right)^2 = 0$

As the angle subtended at origin $= \pi/2$

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow 1 + 1 - \frac{6a}{b} - \frac{10}{b} + \frac{1}{b^2} + \frac{a^2}{b_2} = 0$

$\Rightarrow a^2 + 2b^2 - 6ab - 10b + 1 = 0$

$\therefore$ locus of (a, b) will be

$g(x, y) = x^2 + 2y^2 - 6xy - 10y + 1 = 0$

26. (*d*) $y = 1$ ciut the curve $g(x, y)$ at $x^2 - 6x - 7 = 0$

$\Rightarrow x = 7, -1$

$\therefore$ in first quadrant at (7, 1)

$\therefore \frac{dy}{dx} = -\left(\frac{x-3y}{2y-3x-5}\right)_{(7,1)} = \frac{-4}{-24} = \frac{1}{6}$

27. (*c*) $B(0,2), A(0, 8)$

Equation of AB : $S_1 - S_2 = 0$

$\Rightarrow 2x - 2y - 6 = 0 \Rightarrow x + y - 3 = 0$

from $C_1(1,1), C_1D \perp AB$

$\therefore C_1D = \left|\frac{1+1-3}{\sqrt{1^2+1^2}}\right| = \frac{1}{\sqrt{2}}$

$c_1A = \sqrt{g^2+f^2-c} = \sqrt{1^2+1^2+7} = 3$

$\therefore AD = \sqrt{3^2 - \left(\frac{1}{\sqrt{x}}\right)^2} = \sqrt{9-\frac{1}{2}} = \sqrt{\frac{17}{2}}$

$\therefore AB = 2AD = 2\sqrt{\frac{17}{2}} = \sqrt{34}$

28. (*b*) Equation of circle passing through AB is given by

$S_1 + \lambda L = 0$ (or $S_2 + \lambda L = 0$) $\lambda \in R$

i.e., $x^2 + y^2 - 2x - 2y - 7 + \lambda(x + y - 3) = 0$

i.e., $x^2 + y^2 + (\lambda - 2)x + (\lambda - 2)y - (7 + 3\lambda) = 0$

$C \equiv \left(\frac{-(\lambda-2)}{2}, \frac{-(\lambda-2)}{2}\right)$

$\because$ AB is a diameter of this circle

C lies on AB

$\therefore -(\lambda - 2) = 3 \Rightarrow \lambda = -1$

$\therefore$ equation of the circle is $x^2 + y^2 - 3x - 3y - 4 = 0$

29. (*b*) Mid point is (3/2, 3/2).

30. (*a*) $PT_2 = PT = \sqrt{(-4)^2 + 0^2 - 4} = 2\sqrt{3}$

Circumcentre Of Triangle PT_1T_2 is Midpoint of PO As

$\angle PT_1O = \angle PT_2O = 90^\circ$

So, $\left(\frac{-4+0}{2}, \frac{0+0}{2}\right) = (-2,0)$

31. (*d*) P be a point on the circle

$30 = \frac{\pi}{2} \rightarrow \theta = \frac{\pi}{6}$

Area of the rhombus

$= 2(3 \times \sqrt{3}) = 6\sqrt{3}$

Ratio is 1 : 1

32. (*b*), **33.** (*a*), **34.** (*d*)

The correct option is C $2x + y = 3$

$\because PQ = PR$ i.e. parallelogram $PQRS$ is a rhombus

$\therefore$ Mid point of OR = mid point of PS and $QR \perp PS$

$\therefore S$ is the mirror image of P w.r.t. QR

$\because L \equiv 2x + y = 6$

Let $P \equiv (k, 6-2k)$

$\because \angle PQO = \angle PRO = \frac{\pi}{2}$

$\therefore$ OP is diamter of circumcircle PQR, then centre is mid point of OP

$= \left(\frac{k}{2}, 3-k\right)$

$\therefore x = \frac{k}{2}$

$\Rightarrow k = 2x$

$\Rightarrow y = 3 - k$

$\Rightarrow 2x + y = 3$

If $y = 4$ then locus of circumcentre of ΔPQR is $x = 2$.

$P(2,3)$

$Q(2,0)$

Circumcentre of ΔPQR is

$\left(\frac{2+0}{2}, \frac{3+0}{2}\right) \equiv \left(1, \frac{3}{2}\right)$

line QR : $2x + 3y = 4$

Image of $\left(1, \frac{3}{2}\right)$ about $2x + 3y = 4$

$\frac{x-1}{2} = \frac{y-3/2}{3} = \frac{-2\left(2+\frac{9}{2}-4\right)}{4+9}$

$$\frac{x-1}{2}=\frac{2y-3}{6}=\frac{-2}{13}\left(\frac{5}{2}\right)=-\frac{5}{13}$$

$$x=-\frac{10}{13}+1=\frac{3}{13}$$

$$y=\left(-\frac{30}{13}+3\right)\frac{1}{2}=\left(\frac{9}{26}\right)$$

$$\left(\frac{3}{13},\frac{9}{26}\right)$$

35. *(a)*

(A) $S_1 - S_2 = 0$ is the required common chord i.e $2x = a$

Make homogeneous, we get $x^2 + y^2 - 8.4$

$$\frac{x^2}{a^2}=0$$

As pair of lines substending angle of 90° at origin

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

$\therefore$ $a = \pm 4$

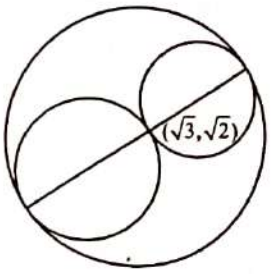
(B) $y = 22\sqrt{3}(x - 1)$ passes through centre (1, 0) of circle

(C) Three lines are parallel

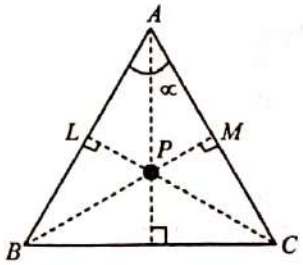
(D) $2(r_1 + r_2) = 4$

$r_1 + r_2 = 2$

$$\frac{r_1+r_2}{2}=1$$



36. *(b)* *(A)* $\arg\dfrac{Z_2-Z}{Z_1-Z}=\propto$



AMPL is

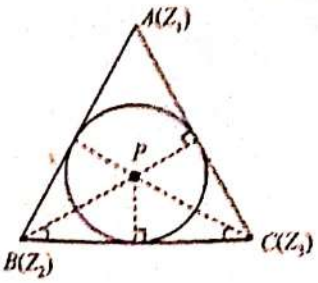
Cyclic quadrilateral

So, $\angle LPM = \pi - d$

So, $\arg\dfrac{Z_2-Z_p}{Z_1-Z_p}=\pi-d$

Hence p.

(B)



$$\angle BPC = x-\frac{B}{2}-\frac{C}{2}$$

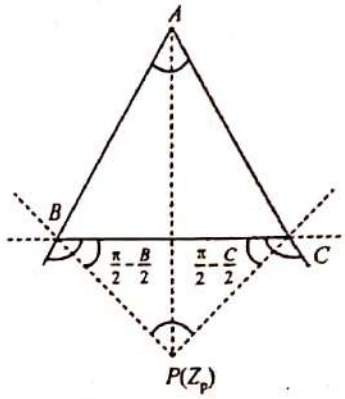
$$=\pi-\left(\frac{x-A}{2}\right)$$

$$=\frac{x}{2}+\frac{A}{2}$$

$$=\frac{\pi}{2}+\frac{\alpha}{2}$$

So $\arg\dfrac{z_2-z_p}{z_1-z_p}=\dfrac{\pi}{2}+\dfrac{\alpha}{2}$

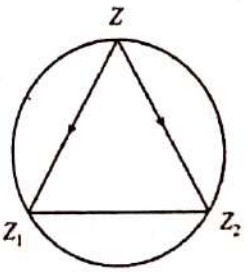
(C)



$$\angle BPC=\pi-\left(\pi-\left(\frac{B+C}{2}\right)\right)$$

$$=\frac{B+C}{2}=\frac{\pi}{2}-\frac{A}{2}=\frac{\pi}{2}=-\frac{\pi}{2}$$

(D) are $\dfrac{Z_2-Z}{Z_1-Z}=\propto$



$$\left(\frac{Z_1+Z_2+Z}{3}\right)=Z_p$$

$$Z=3Z_p-Z_1-Z_2$$

$$\frac{Z_2-Z}{Z_1-Z}=\frac{Z_2(3Z_p-Z_1-Z_2)}{Z_1(3Z_p-Z_1-Z_2)}$$

$$=\frac{2Z_2+Z_1-3Z_p}{2Z_1+Z_2-3Z_p}$$

$$=\frac{\frac{2Z_2+Z_1-Z_p}{3}}{\frac{2Z_1+Z_2-Z_p}{3}}$$

37. *(c)*

(A) $C_1(1, 0)$,

$r_1 = 1$ and $C_2(-3, 3), r_2 = 4$

distance between centres C_1 and $C_2 = d = 5$

$d = r_1 + r_2 = 5$

$\Rightarrow$ 3 common tangents

(B) $C_1(2, 5), r_1 = 5$ and $C_2(3, 6), r_2 = 10$

distance between centres C_1 and $C_2 = d = \sqrt{2}$

$d < |r_1 - r_2|$

$\Rightarrow$ no common tangent

(C) $C_1(1, 2), r_1 = \sqrt{5}$ and $C_2(0, 4), r_2 = 2\sqrt{5}$

distance between centres C_1 and $C_2 = d = \sqrt{5}$

$|r_1 - r_2| = d$

number of common tangents is 1

(D) $C_1(-1, 4), r_1 = 2$ and $C_2(3, 1), r_2 = 2$

distance between centres C_1 and $C_2 = d = 5$

$d > r_1 + r_2$

$\Rightarrow$ number of direct common tangents is 2

38. *(d)* $(2-3)^2+(4-4)^2 \le x^2$

(A) $\max\left\{\dfrac{4x}{7}-\dfrac{5y}{7};(x,y)\{s\}\right\}=6$ at

(B)

$$\min\left\{\frac{\left(x^2+y^2+1\right)}{2}+(x-y);(x,y)\in s\right\}=4$$

(C) $\dfrac{3x}{7}+\dfrac{4y}{7}$ will be maximum at

$x = 4, 2$ & $y = 5, 6$

$$\max\left\{\frac{3x}{7}+\frac{4y}{7};(x,y)\in s\right\}=6$$

(D) $$\min\left\{\frac{\sqrt{3}y+1x-31}{|x-3|};(x,y)\in s\right\}=4$$

39. **[0]**

The given circles are

$(x-1)^2+y^2=4$ *and* $(x-1)^2+y^2=16$

The Points $\left(a+1,\sqrt{3}a\right)$ lie on the line

$y=\sqrt{3}(x-1)$

whose slope = $\sqrt{3}$ hence makes angle $60°$ with x-axis.

$A=(1+2\cos 60°, 2\sin 60°)=(2,\sqrt{3})$,

$B=(1+4\cos 60°, 4\sin 60°)=(3,2\sqrt{3})$

Hence there is no point on the line segment AB

40. **[153.86]**

$px^2+2(p+q-7)x+2q=0$

Less than 2 real rectum $\Rightarrow$ 1 real solution

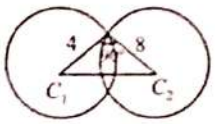
$D=0$

$(p-7)^2+(q-7)^2=7^2$

Area of the region = 49π

$= 153.86$

41. **[16]**



$C_1C_2=\sqrt{80}$

Area $=\frac{1}{2}\times 4\times 8=\frac{1}{2}\times\sqrt{80}\times\frac{\ell}{2}$

$\ell=\frac{64}{\sqrt{80}}=\frac{16}{\sqrt{5}}$

PYQ's (Past Year Questions)

1. *(a)* Only possibility $\alpha=0, \beta=1$

Equation of circle by figure with AB as diameter $(x-0)(x-1)+(y-0)(y-1)=0$

Image of center of circle $\left(\frac{1}{2},\frac{1}{2}\right)$ in $x+y+2=0$ is

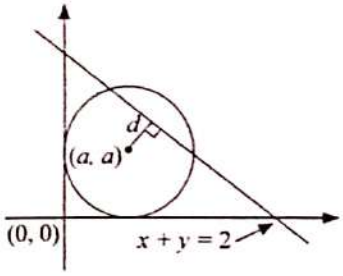
$\left(\frac{-5}{2},\frac{-5}{2}\right)$ and radius will be same.

So, $x^2+y^2+5x+5y+12=0$

2. **[7]** Circle $(x-a)^2+(y-a)^2=a^2$

$x^2+y^2-2ax-2ay+a^2=0$

Intercept = 2



Where d = perpendicular distance of center from line $x+y=2$

$\Rightarrow 2\sqrt{a^2-\left(\frac{a+a-2}{\sqrt{2}}\right)^2}=2$

$\Rightarrow a^2-\frac{(2a-2)^2}{2}=1$

$\Rightarrow 2a^2-4a^2+8a-4=2$

$\Rightarrow 2a^2-8a+6=0 \Rightarrow a^2-4a+3=0$

Here, a = radius.

$\therefore r_1+r_2=4$ and $r_1r_2=3$

$\therefore r_1^2+r_2^2-r_1r_2=(r_1+r_2)^2-3r_1r_2$

$=16-9=7$

3. *(b)* Let $P(h, k)$ be the orthocentre of ΔABC

Then

$h=\frac{\sin t+\cos t+a}{3}$, $k=\frac{-\cos t+\sin t+b}{3}$

(orthocentre coincide with centroid in an equilateral triangle)

$\therefore (3h-a)^2+(3k-b)^2=2$

$\therefore \left(h-\frac{a}{3}\right)^2+\left(k-\frac{b}{3}\right)^2=\frac{2}{9}$

$\because$ orthocentre lies on circle with centre $\left(1,\frac{1}{3}\right)$

$\therefore a=3, b=1$

Hence, $a^2-b^2=8$

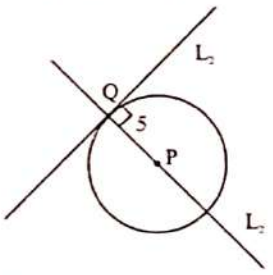
4. **[11]** $L_1: 4x+3y+2=0$

$L_2: 3x-4y-11=0$

Since circle C touches the line L_2 at Q. Also, intersection point of L_1 and L_2 is $Q(1,-2)$

P lies at L_1

$\therefore P\left(x,-\frac{1}{3}(2+4x)\right)$



Now,

$PQ=5 \Rightarrow (x-1)^2+\left(-\frac{4x+2}{3}+2\right)^2=25$

$\Rightarrow (x-1)^2\left[1+\frac{16}{9}\right]=25$

$\Rightarrow (x-1)^2=9$

$\Rightarrow x=4, -2$

At $x=4$

$y=-6$

And at $x=-2, y=2$

$\because$ The circle lies below the x-axis so centre of the circle is $P(4,-6)$

Now distance of P from $5x-12y+51=0$

$=\left|\frac{20+72+51}{13}\right|=\frac{143}{13}=11$

5. *(c,d)* $\sin\left(\frac{2\pi}{2n}\right)=\frac{r}{R+r} \Rightarrow \frac{R+r}{r}=\frac{1}{\sin\left(\frac{\pi}{n}\right)}$

$\frac{R}{r}+1=\operatorname{cosec}\frac{\pi}{n} \Rightarrow \frac{R}{r}=\left(\operatorname{cosec}\frac{\pi}{n}-1\right)$

For $n=4, \frac{R}{r}=(\sqrt{2}-1) \Rightarrow R=r(\sqrt{2}-1)$

For $n=8, \frac{R}{r}>(\sqrt{2}-1) \Rightarrow R>(\sqrt{2}-1)r$

For $n=5, \frac{R}{r}=(\operatorname{cosec}36°-1)<1 \Rightarrow R<r$

For $n=12, \frac{R}{r}=(\operatorname{cosec}15°-1)$

$\Rightarrow \frac{R}{r}=\left(\sqrt{2}\left(\sqrt{3}+1\right)-1\right)<\sqrt{2}\left(\sqrt{3}+1\right)$

$\Rightarrow R<\sqrt{2}\left(\sqrt{3}+1\right)r$

6. **[3]** $x^2+y^2-2x-6y+6=0$ centre $(1,3)$

$r_1=\sqrt{1+9-6}=2$

$CM=\sqrt{1+4}=\sqrt{5}$

$r_2=\sqrt{5+4}=3$

$\therefore$ Radius of another circle = 3cm

7. *(a)* Let circle be $S_1+\lambda S_2=0$

$x^2+y^2-6x+\lambda(x^2+y^2-4y)=0$

$\Rightarrow (1+\lambda)x^2+(1+\lambda)y^2-6x-4\lambda y=0$

Centre $=\left(\frac{3}{1+\lambda},\frac{2\lambda}{\lambda+1}\right)$

Centre lies on $2x-3y+12=0$

$\frac{6}{\lambda+1}-\frac{6\lambda}{\lambda+1}+12=0$

$\Rightarrow \lambda=-3$

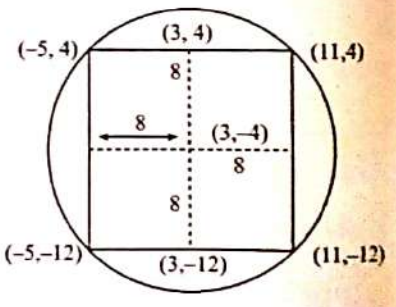
$C=-2x^2-2y^2-6x+12y=0$

$x^2+y^2+3x-6y=0$

It passes through $(-3, 6)$

8. *(c)* Centre $(3,-4)$,

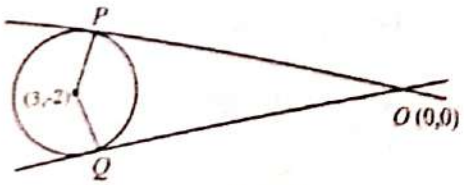
$r=\sqrt{9+16+103}=\sqrt{128}$ and side = $r\sqrt{2}=16$



$\therefore$ $(-5, 4)$ will be nearer to the $(0, 0)$

$\therefore$ $\sqrt{5^2+4^2} = \sqrt{25+16} = \sqrt{41}$

9. *(b)*



Circumcircle of ΔOPQ:

$(x-0)(x-3)+(y-0)(y+2)=0$

$x^2+y^2-3x+2y=0$

It passes through $\left(\alpha, \frac{1}{2}\right)$

$\therefore \alpha^2+\frac{1}{4}-3\alpha+1=0$

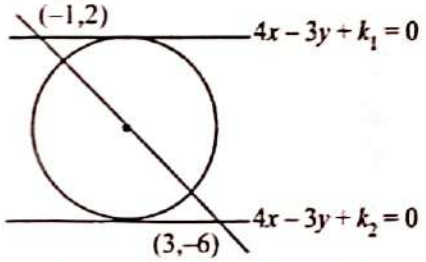
$\Rightarrow \alpha^2-3\alpha+\frac{5}{4}=0$

$\Rightarrow 4\alpha^2-12\alpha+5=0$

$\Rightarrow (2\alpha-1)(2\alpha-5)=0$

$\Rightarrow \alpha=\frac{1}{2}, \frac{5}{2}$

10. *(c)*



L_1: $4x-3y+k_1=0$ passes via $(-1, 2)$

$-4-6+k_1=0$

$\Rightarrow k_1=10$

L_2: $4x-3y+k_2=0$ passes via $(3, -6)$

$12+18+k_2=0$

$\Rightarrow k_2=-30$

Distance between L_1 & L_2 will be dia of circle.

$\left|\frac{10+30}{\sqrt{16+9}}\right| = 2r \Rightarrow r=4$

And center is the mid-point of A and B.

$C \equiv \left(\frac{3-1}{2}, \frac{2-6}{2}\right) \equiv (1,-2)$

Equation of circle: $(x-1)^2+(y+2)^2=16$

11. *(b)* Given that equation of circle $x^2+y^2+ax+2ay+c=0$

$; a<0$

Now, x- intercept $= 2\sqrt{g^2-c} = 2\sqrt{2}$

$\Rightarrow \sqrt{\left(\frac{a}{2}\right)^2-c} = \sqrt{2}$

$\Rightarrow \frac{a^2}{4}-c=2$

$\Rightarrow a^2=8+4c$ ...(i)

Also, since y- intercept $= 2\sqrt{5}$

$\Rightarrow f^2-c=5$

$\Rightarrow a^2-c=5$

$\therefore a^2=5+c$ ...(ii)

From (i) and (ii)

$8+4c=5+c$

$\Rightarrow 3c=-3$

$\Rightarrow c=-1$

$\Rightarrow a^2=5+(-1)$

$\Rightarrow a^2=4 \Rightarrow a=\pm 2$

as $a<0 \Rightarrow a=-2$

Therefore, equation of circle is

$x^2+y^2-2x-4y-1=0$

$\Rightarrow (x-1)^2+(y-2)^2=6$

Equation of tangent to the circle is perpendicular to the line $x+2y=0$ slope of the tangent $=2$

Therefore, equation of tangent:

$(y-2)=2(x-1)\pm\sqrt{6}\sqrt{1+4}$

$\Rightarrow 2x-y\pm\sqrt{30}=0$

Shortest between L_1 and $L_2 = \frac{\sqrt{30}}{\sqrt{5}} = \sqrt{6}$

12. [165] Let $C_1 \circ (x-1)^2+(y-1)^2=1$

$C_2 \circ (x-9)^2+(y-1)^2=4$.

Since centres $C_1(1, 1)$ and $C_2(9, 1)$ lies opposite sides of the line $3x+4y=\alpha$

$((3+4-\alpha)((27+4-\alpha)<0 \Rightarrow \alpha \in (7,31)$ ... (i)

Also length of perpendicular from centre of the circle is greater than radius of the circle.

$\frac{|3+4-\alpha|}{5} \geq 1 \Rightarrow \alpha \leq 2$ *or* $\alpha \geq 12$ ... (ii)

And $\frac{|27+4-\alpha|}{5} \geq 2 \Rightarrow \alpha \leq 21$ *or* $\alpha \geq 41$ ... (iii)

From (i), (ii) and (iii) we get $\alpha \in [12,21]$

$\therefore$ sum of all integers = 165.

13. [36] Two circles touches each other if $C_1C_2 = |r_1 \pm r_2|$

Distance between $C_2(3, 0)$ and $C_1(0, 4)$ is either $\sqrt{k}+1$

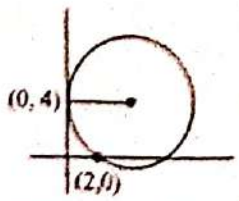
or $\left|\sqrt{k}-1\right| (C_1C_2 = 5)$

$\Rightarrow \sqrt{k}+1=5$ or $\left|\sqrt{k}-1\right|=5$

$\Rightarrow k=16$ or $k=36$

$\Rightarrow$ maximum value of k is 36

14. *(d)*



Equation of family of circle

$(x-0)^2+(y-4)^2+\lambda x=0$

$\Rightarrow$ passes $(2, 0)$

$4+16+2\lambda=0$

$\Rightarrow \lambda=-10$

$x^2+y^2-10x-8y+16=0$

centre $(5, 4)$, $r=\sqrt{25+16-16}=5$

Check the options

Option *(d)*

$\left|\frac{4\times4+3\times4-8}{5}\right| = \frac{24}{5} \neq 5$

15. *(c)* Tangent to $y^2=4x\sqrt{2}$ is

$y=mx+\frac{\sqrt{2}}{m}$ $\left[\because y=mx+\frac{a}{m}\right]$

It is also tangent to $x^2+y^2=1$

$\perp$ distance from origin is equal to radius

$\Rightarrow \left|\frac{\sqrt{2}/m}{\sqrt{1+m^2}}\right| = 1$

$\Rightarrow m=\pm 1$

$\Rightarrow$ Tagent will be $y=x+\sqrt{2}$

or $y=-x-\sqrt{2}$

Compare with $y=-ax+C$

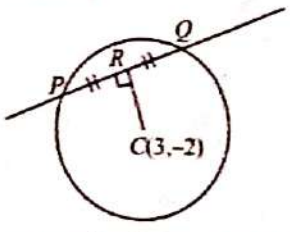
$\Rightarrow a=\pm 1 \& c=\pm\sqrt{2}$

16. *(b)* $R \equiv \left(-\frac{3}{5}, \frac{-3m}{5}+1\right)$

So, m $\left(\frac{-\frac{3m}{5}+3}{-\frac{3}{5}-3}\right) = -1$

$\Rightarrow m^2-5m+6=0$

$\Rightarrow m=2, 3$



17. [2] $C_1: x^2+y^2-4x-2y+(5-\alpha)=0$

Centre $O_1=(2, 1)$, radius$(r_1)=\sqrt{\alpha}$

$C_2: x^2+y^2-2fx-2gy+\frac{36}{5}=0$

Centre $O_2=(f, g)$ radius (r_2)

$=\sqrt{f^2+g^2-\frac{36}{5}}$

$\Rightarrow \frac{f-2}{2} = \frac{g-1}{-1} = -2\left(\frac{2\times 2-1+1}{2^2+1^2}\right) = \frac{-8}{5}$

$\Rightarrow f = \frac{-6}{5}$ and $g = \frac{13}{5}$

$\Rightarrow \alpha = 1$ and $r = 1$

Hence, $\alpha + r = 2$

18. [10] Given,

$x^2 + y^2 - 2\sqrt{2}x - 6\sqrt{2}y + 14 = 0$

Centre $= \left(\sqrt{2}, 3\sqrt{2}\right)$

Radius $= \left((\sqrt{2})^2 + (3\sqrt{2})^2 - 14\right)^{1/2}$

$= (2 + 18 - 14)^{1/2} = \sqrt{6}$

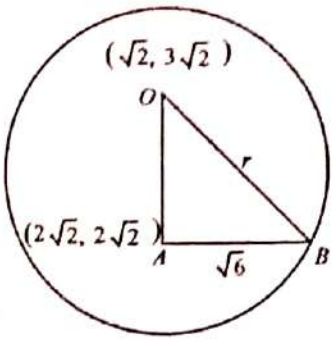
Now, $(x-2\sqrt{2})^2 + (y-2\sqrt{2})^2 = r^2$

Centre $= \left(2\sqrt{2}, 2\sqrt{2}\right)$

$OA = \sqrt{(2\sqrt{2}-\sqrt{2})^2 + (2\sqrt{2}-3\sqrt{2})^2}$

$= \sqrt{2+2} = 2$

$r^2 = (\sqrt{6})^2 + (2)^2 = 6 + 4 = 10$



19. **(b)** $S_1 : x^2 + y^2 - x - y - \frac{1}{2} = 0$

$C_1 : \left(\frac{1}{2}, \frac{1}{2}\right)$

$r_1 = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{2}} = 1$

$S_2 : x^2 + y^2 - 4y + \frac{7}{4} = 0 \quad C_2 : (0,2)$

$r_2 = \sqrt{4 - \frac{7}{4}} = \frac{3}{2}$

$S_3 : x^2 + y^2 - 4x - 2y + 5 - r^2 = 0$

$C_3 : (2,1)$

$r_3 = \sqrt{4 + 1 - 5 + r^2} = |r|$

$C_1C_3 = \sqrt{\frac{5}{2}}, C_2C_3 = \sqrt{5}$

S_1 must be inside $S_3 \therefore \sqrt{\frac{5}{2}} \le |r-1|$

$\Rightarrow r \le 1 + \sqrt{\frac{5}{2}}, \; r \ge \frac{3}{2} + \sqrt{5}$ and

also S_2 must be inside $S_3 \therefore \sqrt{5} \le \left|r - \frac{3}{2}\right|$

$\Rightarrow r - \frac{3}{2} \ge \sqrt{5}, \; r - \frac{3}{2} \le -\sqrt{5}$

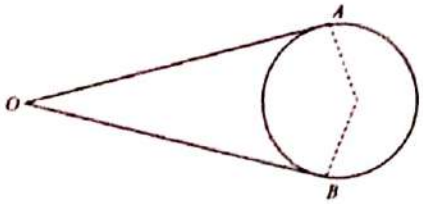
20. **(c)** Length of tangent

$= L = \sqrt{S_1} = \sqrt{16} = 4$

Radius of circle $R = \sqrt{16 + 4 - 16} = 2$

Length of $AB = \frac{2RL}{\sqrt{L^2 + R^2}} = \frac{16}{\sqrt{20}}$

Hence, $AB^2 = \frac{64}{5}$



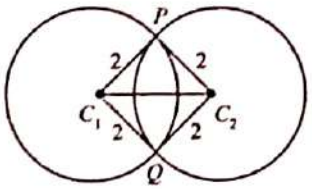
21. **(d)** $C_1(1,1), C_2(3,3), C_1C_2$

$= \sqrt{2^2 + 2^2} = 2\sqrt{2}$

$r_1 = \sqrt{1+1+2} = 2$

$r_2 = \sqrt{9+9-14} = 2$ and we observe

$C_1C_2^2 = PC_1^2 + PC_2^2$



So, circles are orthogonal and have equal radii therefore the quadrilateral PC_1QC_2 is a square.

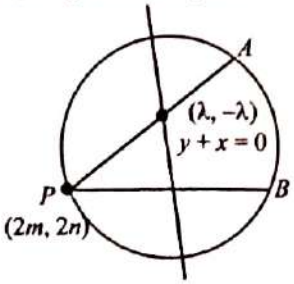
Hence area $= 2 \times 2 = 4$sq. units

22. **(a)** $x^2 + y^2 - \frac{(1+a)x}{2} - \frac{(1-a)y}{2} = 0$

Centre $\left(\frac{1+a}{4}, \frac{1-a}{4}\right) \Rightarrow (m,n)$

$P\left(\frac{1+a}{2}, \frac{1-a}{2}\right) \Rightarrow (2m, 2n)$

So, $x^2 + y^2 - 2mx - 2ny = 0$



Let mid point of chord is $(1 - \lambda)$

Equation of chord $\Rightarrow T = S_1$

$\Rightarrow (x-y)\lambda - m(x+\lambda) - n(y-\lambda) = 2\lambda^2 - 2m\lambda + 2n\lambda$

$(2m-2n)\lambda - m(2n+\lambda) - n(2k-\lambda) = 2\lambda^2 - 2m\lambda + 2k\lambda$

$\Rightarrow 2\lambda^2 - 4m\lambda + 4k\lambda + m\lambda - n\lambda + 2m^2 + 2n^2 = 0$

$\Rightarrow 2\lambda^2 - 3\lambda m + 3n\lambda + 2m^2 + 2n^2 = 0$

$\Rightarrow 2\lambda^2 - 3\lambda(n-m) + 2m^2 + 2n^2 = 0$

$D > 0$

$9(n-m)^2 - 16(m^2 + n^2) > 0$

$\Rightarrow 9n^2 + 9m^2 - 18mn - 16m^2 - 16n^2 > 0$

$\Rightarrow 7n^2 + 7m^2 + 18mn < 0$

$\Rightarrow 7\left(\frac{1-a}{4}\right) + 7\left(\frac{1+a}{4}\right)^2 + 18\left(\frac{1-a^2}{16}\right) < 0$

$\Rightarrow \frac{7}{16}\left(1 + a^2 - 2a\right) + \frac{7}{16}\left(1 + a^2 + 2a\right) + \frac{18}{16}\left(1 - a^2\right) < 0$

$\Rightarrow \frac{7}{16}2\left(1 + a^2\right) + \frac{18}{16}\left(1 - a^2\right) < 0$

$\Rightarrow 14 + 14a^2 + 18 - 18a^2 < 0$

$\Rightarrow 32 - 4a^2 < 0 \Rightarrow a^2 - 8 > 0 \Rightarrow a^2 > 8$

23. **(b)** $x^2 + y^2 - 4x + 3 = 0$

$\Rightarrow (x-2)^2 + y^2 = 1$

$AO = \sqrt{(OC)^2 - (AC)^2} = \sqrt{4-1} = \sqrt{3}$

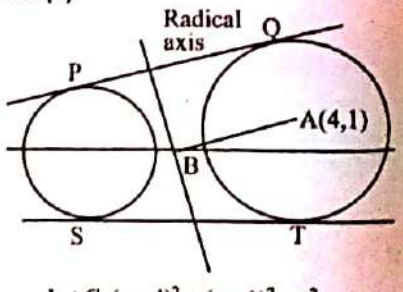
$\sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$

Also, $AO = BO$

$\Delta OAB = \frac{1}{2} \cdot OA \cdot OB \sin 60^\circ$

$= \frac{1}{2} \times \sqrt{3} \cdot \sqrt{3} \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$

24. [2]



Let $C_2 (x-4)^2 + (y-1)^2 = r^2$

Equation of radical axis $8x + 2y - 17 = 1 - r^2$

$8x + 2y = 18 - r^2$

Thue, $B\left(\frac{18-r^2}{8}, 0\right) A(4,1)$

$AB = \sqrt{5}$

$$\sqrt{\left(\frac{18-r^2}{8}-4\right)^2+1}=\sqrt{5}$$

hence, $r^2=2$

25. [1] $x^2+y^2-10x-10y+41=0$

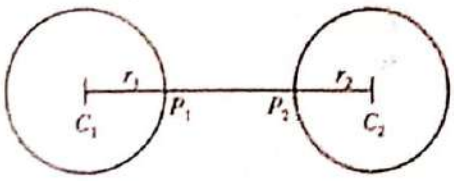
$C_1(5,5), r_1=3$

and $x^2+y^2-24x-10y+160=0$

$C_2(12,5), r_2=3$

$\Rightarrow C_1C_2=\sqrt{49+0}=7$

$\therefore P_1P_2=7-(3+3)=1$



26. (*b*) Circle touches internally

Equation of common tangent is

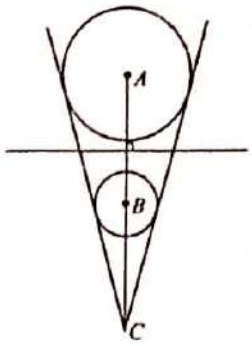
$S_1-S_2=0$

$6x+8y-20=0 \Rightarrow 3x+4y=10$

$(6,-2)$ satisfy it

27. [10] Distance of point A from given line $=\frac{5}{2}$

$$\frac{CA}{CB}=\frac{2}{1} \Rightarrow \frac{AC}{AB}=\frac{2}{1}$$



$\Rightarrow AC=2\times 5=10$

28. (*a*) Slope of tangent to C_1 at $m=-3/4$

Equation of Tangent, $y=mx-2\sqrt{(1+m^2)}$

$y=-3x/4-3\sqrt{1+9/16}=-3x/4-15/4$

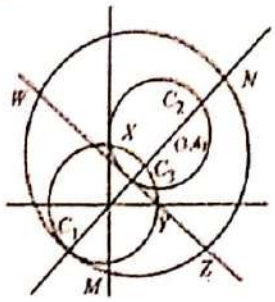
$x=-4y/3-5$ ...(*i*)

Tangent to $x^2=4(2\alpha)y$ is $x=m^2y=2\alpha/m^2$ ...(*ii*)

On comparing (*i*) and (*ii*), we get

$m^2=-4/3$ and $2\alpha/m^2=-5 \Rightarrow \alpha=10/3$

29. (*d*)

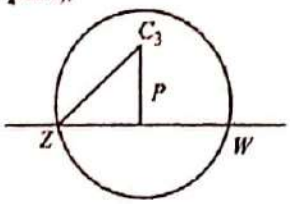


$MC_1+C_1C_2+C_2N=2r$

$\Rightarrow 3+5+4=2r=6$

$\Rightarrow$ Radius of $C_1=6$

Suppose centre of C_3 be $(0+r_4\cos\theta, 0+r_4\sin\theta)$,



$$\left\{\begin{matrix} r_4=C_1C_3=3 \\ \tan\theta=\frac{4}{3} \end{matrix}\right\}$$

$C_3=\left(\frac{9}{5},\frac{12}{5}\right)C_3$

$=(h,k)$

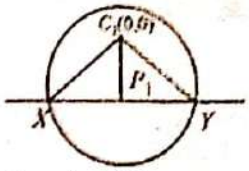
$\Rightarrow 2h+k=6$

Equation of ZW and XY is $3x+4y-9=0$

(Common chord of circle $C_1=0$ and $C_2=0$)

$ZW=2\sqrt{r^2-p^2}=\frac{24\sqrt{6}}{5}$ (where $r=6$ and $p=\frac{6}{5}$)

$XY=2\sqrt{r_1^2-p_1^2}=\frac{24}{5}$ (where $r_1=3$ and $p_1=\frac{9}{5}$)



$$\frac{\text{Length of } ZW}{\text{Length of } XY}=\sqrt{6}$$

Let length of perpendicular from M to ZW be λ,

$\lambda=3+\frac{9}{5}=\frac{24}{5}$

$$\frac{\text{Area of } \Delta MZN}{\text{Area of } \Delta ZMW}=\frac{\frac{1}{2}(MN)\times\frac{1}{2}(ZW)}{\frac{1}{2}\times ZW\times\lambda}=\frac{1}{2}\frac{MN}{\lambda}=\frac{5}{4}$$

$C_3:\left(x-\frac{9}{5}\right)^2+\left(y-\frac{12}{5}\right)^2=6^2$

$C_1:x^2+y^2-9=0$

Common tangent to C_1 and C_3 is common chord of C_1 and C_3 is $3x+4y+15=0$.

Now $3x+4y+15=0$ is tangent to parabola $x^2=8\alpha y$.

$x^2=8\alpha\left(\frac{-3x-15}{4}\right)$

$\Rightarrow 4x^2+24\alpha x+120\alpha=0$

$D=0 \Rightarrow \alpha=\frac{10}{3}$

By Aditya Mohod
username - muje kabhi dekha nahi

By Aditya Mohod
username - maje_kabhi_dekha_nahi

CHAPTER 12

Conic Sections: Parabola

Prarambh (Topicwise)

1. (*d*) $a = 4$, = (0, 0) vertex, focus = (0, –4)

2. (*c*) $S \equiv (5, 0)$. Therefor, latus rectum = $4a = 20$

3. (*b*) Distance between focus and directrix is

$$= \left|\frac{3-4-2}{\sqrt{2}}\right| = \frac{3}{\sqrt{2}}$$

Hence latus rectum $= 3\sqrt{2}$

(Since latus rectum is two times the distance between focus and directrix).

4. (*c*) Vertex

$\Rightarrow$ focus is $(2 + 2, 0) = (4, 0)$.

5. (*c*) The point (–3, 2) will satisfy the equation $y^2 = 4ax$

$$\Rightarrow \quad 4 = -12a, \Rightarrow 4a = \frac{4}{3}$$

(Taking positive sign).

6. (*c*) $x^2 = -8y$

$\Rightarrow \quad a = -2$, So, focus = (0, –2)

Ends of latus rectum = (4, –2), (–4, –2).

Trick: Since the ends of latus rectum lie on parabola, so only points (–4, –2) and (4, –2) satisfy the parabola.

7. (*a*) Given equation is $x^2 = -8ay$.

Here $A = 2a$

Focus of parabola $(0, -A)$ i.e. $(0, -2a)$

Directrix $y = A$ i.e., $y = 2a$.

8. (*d*) Distance between tangent at vertex and $LR = a$

Clearly; $a = \left|\frac{-8+12}{\sqrt{1+1}}\right| = \frac{4}{\sqrt{2}}$

Length of latus rectum $= 4a$

$$= 4 \times \frac{4}{\sqrt{2}} = 8\sqrt{2}.$$

9. (*a*) $(x+1)^2 = 4a(y+2)$

Passes through (3, 6)

$$\Rightarrow \quad 16 = 4a.8 \Rightarrow a = \frac{1}{2}$$

$$\Rightarrow \quad (x+1)^2 = 2(y+2)$$

$$\Rightarrow \quad x^2 + 2x - 2y - 3 = 0$$

10. (*d*) The parabola is $(x - 2)^2 = (3y - 6)$. Hence axis is $x - 2 = 0$

11. (*c*) Given equation $\Rightarrow (y + B)^2 + 2Ax + C - B^2 = 0$

$$\Rightarrow (y+B)^2 = -2A\left(x + \frac{C-B^2}{2A}\right)$$

$\Rightarrow$ So, equation of latus rectum is:

$$x = \frac{B^2 - C}{2A} - \frac{A}{2} = \frac{B^2 - A^2 - C}{2A}$$

12. (*a*) Say the point is (h, k)

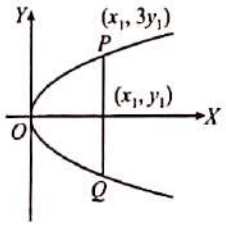
So, $(h - a)^2 + k^2 = h^2$

$$\Rightarrow \quad k^2 - 2ah + a^2 = 0$$

$\therefore$ Locus is $y^2 - 2ax + a^2 = 0$

13. (*a*) Required locus is $(3y)^2 = 4ax$

$$\Rightarrow 9y^2 = 4ax$$



14. (*c*) The equation of parabola can be written as

$$(y+2)^2 = -4\left(x - \frac{1}{2}\right)$$

$\Rightarrow y^2 = -4x$ where $X = x - \frac{1}{2}$, $Y = y + 2$

An equation of its directrix is $X = 1$

$\therefore$ Required directrix is $x = \frac{3}{2}$

15. (*c*) Let $k = 64$

$\therefore y = x^2 - 2 \times 8x + 64$

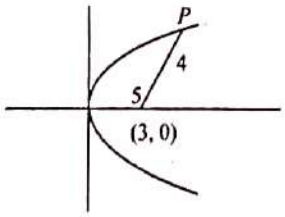
$\Rightarrow y = (x - 8)^2$

$\Rightarrow$ It has vertex on x-axis

16. (*c*) $h^2 = ab$

$\Rightarrow 4 = \lambda.1 \Rightarrow \lambda = 4$

17. (*b*)



Let the point P is $(3t^2, 6t)$

and $PS = 3 + 3t^2 = 4$

$t^2 = 1/3$

$$t = \pm\frac{1}{\sqrt{3}}$$

$\therefore$ Points are

$(1, 2\sqrt{3})$ & $(1, -2\sqrt{3})$

18. (*b*) Parametric equations of $y^2 = 4ax$ are $x = at^2$, $y = 2at$

Hence if equation is $y^2 = 8x$, then parametric equations are $x = 2t^2$, $y = 4t$

19. (*c*) Any point on the parabola $y^2 = 4ax$ is $(at^2, 2at)$

$$\therefore at^2 = \frac{9}{2}$$

and $2at = 6 \quad \Rightarrow \quad t = \frac{3}{a}$ ... (*i*)

$$\therefore a\left(\frac{3}{a}\right)^2 = \frac{9}{2} \Rightarrow a = 2$$

On putting the value of a in Eq. (*i*), we get

$$t = \frac{3}{2}$$

$\therefore$ Parameter of the point P is $\frac{3}{2}$

20. (*c*) Semi latus rectum is harmonic mean between segments of focal chords of a parabola.

$$\therefore b = \frac{2ac}{a+c} \quad \Rightarrow \quad a, b, c \text{ are in H.P.}$$

21. (*d*) Given parabola is $y^2 = 12x$

Here, $a = 3$ For point $P(x, y)$, $y = 6$

This point lie on the parabola

$\therefore (6)^2 = 12x \quad \Rightarrow \quad x = 3$

Thus, focal distance of point P is 6

22. (*b*) $S_1 \equiv x^2 - 108y = 0$

$T \equiv xx_1 - 54(y + y_1) = 0$

$$\Rightarrow \quad xx_1 - 54\left(y + \frac{x_1^2}{108}\right) = 0$$

$S_2 \equiv y^2 - 32x = 0$

$T \equiv yy_2 - 16(x + x_2) = 0$

$$\Rightarrow \quad yy_2 - 16\left(x + \frac{y_2^2}{32}\right) = 0$$

$$\therefore \frac{x_1}{16}=\frac{54}{y_2}=\frac{-x_1^2}{y_2^2}=r$$

$$\Rightarrow x_1=16r \text{ and } y_2=\frac{54}{r}$$

$$\therefore \frac{-(16r)^2}{(54/r)^2}=r \Rightarrow r=-\frac{9}{4}$$

$$x_1=-36,\ y_2=-24,\ y_1=\frac{(36)^2}{108}$$
$$=12,\ x_2=18$$

$\therefore$ Equation of common tangent

$$(y-12)=\frac{-36}{54}(x+36)\Rightarrow 2x+3y+36=0$$

Second Method: Using direct formula of common tangent

$yb^{1/3}+xa^{1/3}+(ab)^{2/3}=0$, where $a = 8$ and $b = 27$. Hence the required tangent is $3y + 2x + 36 = 0$.

23. **(c)** $m = \tan\theta$. The tangent to $y^2 = 4ax$ is $y = x\tan\theta + c$

Hence $c=\dfrac{a}{\tan\theta}=a\cot\theta$

$\therefore$ The equation of tangent is

$y = x\tan\theta + a\cot\theta$

24. **(b)** Equation of parabola is $Y^2 = 4X$, where $X = x+\dfrac{5}{4}$

Tangent parallel to $Y = 2X + 7$ is

$$Y=2X+\frac{a}{m}$$

$$\Rightarrow y=2\left(x+\frac{5}{4}\right)+\frac{1}{2}\Rightarrow y=2x+3$$

i.e., $2x - y + 3 = 0$

25. **(a)** $m = \tan\theta = \tan 60° = \sqrt{3}$

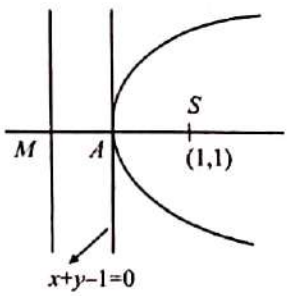
The equation of tangent at (h, k) to is $y^2 = 4ax$ is $yk = 2a(x + h)$

Comparing, we get $m=\sqrt{3}=\dfrac{2a}{k}$ or $k=\dfrac{2a}{\sqrt{3}}$ and $h=\dfrac{a}{3}$

26. **(a)** Any point on $y^2 = 4ax$ is $(at^2, 2at)$, then tangent is

$$2aty = 2a(x+t^2)\Rightarrow yt = x+at^2$$

27. **(a)**



Point A is $\left(\dfrac{1}{2},\dfrac{1}{2}\right)$

$\therefore$ M is $(0, 0)$

$\therefore$ Eq. of Diretrix is $x + y = 0$

$\therefore$ Eq. of parabola is $(x-1)^2+(y-1)^2=\left(\dfrac{x+y}{\sqrt{2}}\right)^2$

Length of latus vectrum = 2($\perp r$ distance from focus to the directrix)

$$=2\cdot\left|\frac{1+1}{\sqrt{2}}\right|=2\sqrt{2}$$

28. **(b)** a = $\perp^r$ distance from (3, 4) to the tangent at vertex

$$=\left|\frac{3+4-7-5\sqrt{2}}{\sqrt{2}}\right|$$

$a = 5$

$LR = 4a = 20$

29. **(a)** Normal at (h, k) to the parabola $y^2 = 8x$ is

$$y-k=-\frac{k}{4}(x-h)$$

Gradient $=\tan 60°=\sqrt{3}=-\dfrac{k}{4}$

$\Rightarrow k=-4\sqrt{3}$ and $h = 6$

Hence required point is $(6,-4\sqrt{3})$

30. **(c)** $$y-\frac{2a}{m}=-\frac{2a/m}{2a}\left(x-\frac{a}{m^2}\right)$$

$$\Rightarrow y-\frac{2a}{m}=\frac{-1}{m}\left(x-\frac{a}{m^2}\right)$$

$$\Rightarrow m^3y+m^2x-2am^2-a=0$$

31. **(d)** Let normal at (h, k) be $y = mx + c$ then, $k = mh + c$ also $k^2 = 4a(h - a)$

slope of tangent at (h, k) is m_1 then on differentiating equation of parabola.

$$2ym_1=4a\ \Rightarrow\ m_1=\frac{2a}{k}\ \text{also}\ mm_1=-1$$

$\Rightarrow m=-\dfrac{k}{2a}$, solving and replacing (h, k) by (x, y)

$$\Rightarrow y=m(x-a)-2am-am^3$$

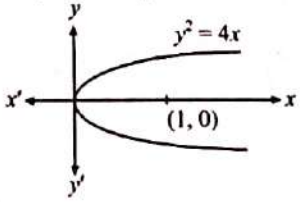
32. **(d)** We have $t_2=-t_1-\dfrac{2}{t_1}$

Since $a = 2$, $t_1 = 1$ $\therefore$ $t_2 = -3$

$\therefore$ The other end will be $(at_2^2, 2at_2)$ *i.e.*, $(18, -12)$.

33. **(d)** Given curve is $y^2 = 4x$

Also, point (1, 0) is the focus of the parabola. It is clear from the graph that only normal is possible



34. **(d)** Point $(am^2, -2am)$, where $m = \pm 1$

$\therefore$ point is (1, 2)

35. **(c)** Line : $y = -2x - \lambda$

Parabola : $y^2 = -8x$

$c = -2am - am^3$ (condition for line to be normal to parabola)

$-\lambda = -2 \times -2 \times -2 - (-2)(-8)$

$-\lambda = -8 - 16$

$\lambda = 24$

36. **(d)** The given point (–1, –60) lies on the directrix $x = -1$ of the parabola $y^2 = 4x$. Thus the tangents are at right angle.

37. **(c)** Equation of tangent at (1, 7) to $y = x^2 + 6$

$$\frac{1}{2}(y+7)=x.1+6\ \Rightarrow\ y=2x+5 \quad ...(i)$$

This tangent also touches the circle

$x^2 + y^2 + 16x + 12y + c = 0$ ... (*ii*)

Now solving (*i*) and (*ii*), we get

$\Rightarrow x^2+(2x+5)^2+16x+12(2x+5)+c=0$

$\Rightarrow 5x^2+60x+85+c=0$

Since, roots are equal so

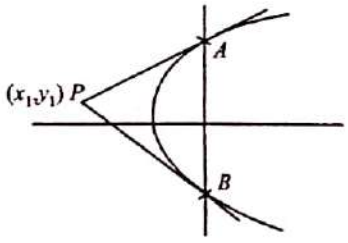
$b^2-4ac=0\ \Rightarrow\ (60)^2-4\times5\times(85+c)=0$

$\Rightarrow 85+c=180\ \Rightarrow\ 5x^2+60x+180=0$

$$\Rightarrow x=-\frac{60}{10}=-6\ \Rightarrow\ y=-7$$

Hence, point of contact is (–6, –7)

38. **(c)**



Eq. of AB is:

$T = 0$

$yy_1 = 2(x + x_1)$

$2x - yy_1 + 2x_1 = 0$ ... (i)

$4x - 7y + 10 = 0$ ... (ii)

equ. (i) and (ii) are identical

$$\therefore \frac{2}{4}=\frac{y_1}{7}=\frac{2x_1}{10}$$

$$y_1=\frac{7}{2}\ \&\ x_1=\frac{5}{2}$$

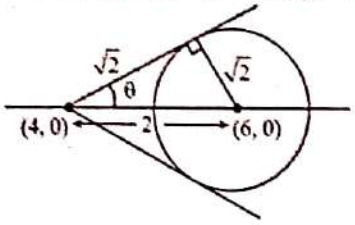
39. **(c)** Equation of chord of contact of tangent drawn from a point (x_1, y_1) to parabola $y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$ so that $5y = 2 \times 2(x + 2) \Rightarrow 5y = 4x + 8$. Point of intersection of chord of contact with parabola $y^2 = 8x$ are $\left(\dfrac{1}{2},2\right),(8,8)$, so that length $=\dfrac{3}{2}\sqrt{41}$.

40. (*a*) The combined equation of the lines joining the vertex to the points of intersection of the line $lx + my + n = 0$ and the parabola $y^2 = 4ax$ is $y^2 = 4ax\left(\frac{lx+my}{-n}\right)$ or $4alx^2 + 4amxy + my^2 = 0$

This represents a pair of perpendicular lines, if $4al + n = 0$.

41. (*a*) Focus of parabola is (4, 0)

From diagram, $\theta = 45° \Rightarrow$ Slope $= \pm 1$.

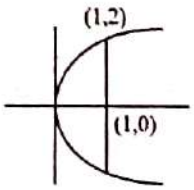


42. (*b*) Any line through origin (0,0) is $y = mx$. It intersects $y^2 = 4ax$ in $\left(\frac{4a}{m^2}, \frac{4a}{m}\right)$.

Mid point of the chord is $\left(\frac{2a}{m^2}, \frac{2a}{m}\right)$

$x = \frac{2a}{m^2}, y = \frac{2a}{m} \Rightarrow \frac{2a}{x} = \frac{4a^2}{y^2}$ or $y^2 = 2ax$, which is a parabola.

43. (*c*)



$y^2 = 4x$, the other end of focal chord will be (1, –2) and this satisfy options (*a*) (*b*) & (*d*)

44. (*d*) $y^2 + 4y - 6x - 2 = 0$

$y^2 + 4y + 4 - 6x - 6 = 0; a = \frac{3}{2}$

$(y + 2)^2 = 6(x + 1)$

$Y^2 = 6X$ vertex (–1, –2)

POI of tangents $t_1 t_2 = -1$

$[at_1t_2, a(t_1 + t_2)]$

$h + 1 = at_1t_2$

$h + 1 = -\frac{3}{2}$

$2h + 2 = -3$

$2h + 5 = 0 \Rightarrow 2x + 5 = 0$

45. (*b*) Eq. of chord is $T = S_1$

$ky - 2(x + h) = k^2 - 4h$ ... (i)

$\because$ above eq. passes through focus (1, 0)

$\therefore 0.k - 2(1 + h) = k^2 - 4h$

$-2 - 2x = y^2 - 4x$

$y^2 = 2(x - 1)$

46. (*c*) By properatry of parabola, harmonic menas of lengths of two parts into which focus divides the focal charod is equal to length of semi latus rectum.

$\therefore \frac{1}{SP} + \frac{1}{SQ} = \frac{2}{\left(\frac{L}{2}\right)}$

$\Rightarrow \frac{1}{3} + \frac{1}{6} = \frac{4}{L}$

$\Rightarrow L = 4 \times \frac{6 \times 3}{(3+6)}$

$\therefore L = 8$

Hence , {C is correct}

47. (*a*) The axis of parabola $y^2 = 4ax$ is x-axis and focus is $(a, 0)$ Equation of line passing through $(a, 0)$ and making and angle θ with x-axis is $y = \tan\theta\,(x - a)$

Let this line intersect the parabola at (x_1, y_1) and (x_2, y_2)

$\therefore$ Length of focal cord

$$= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$= \sqrt{\left(\frac{y_1^2}{4a} - \frac{y_2^2}{4a}\right)^2 + (y_1 - y_2)^2}$$

$$= (y_1 - y_2)\sqrt{\left(\frac{y_1 + y_2}{4a}\right)^2 + 1}$$

Equation of line is

$y = \tan\theta\,(x - a)$

$\Rightarrow x = y\cot\theta + a$

Substituting this in the equation of parabola we get

$y^2 = 4ax = 4a(y\cot\theta + a)$

$\therefore y^2 - 4a\cot\theta y - 4a^2 = 0$

$\Rightarrow y_1 + y_2 = 4a\cot\theta$

$\Rightarrow y_1 - y_2 = \sqrt{(y_1 + y_2)^2 - 4y_1y_2}$

$= \sqrt{16a^2\cot^2\theta + 16a^2}$

$= \sqrt{16a^2\,\mathrm{cosec}^2\theta}$

$= 4a\,\mathrm{cosec}\,\theta$

Lenght of focal chord

$$l = (y_1 - y_2)\sqrt{\left(\frac{y_1 + y_2}{4a}\right)^2 + 1}$$

$$= 4a\,\mathrm{cosec}\,\theta\sqrt{\left(\frac{4a\cot\theta}{4a}\right)^2 + 1}$$

$$= 4a\,\mathrm{cosec}^2\theta$$

Prabal (JEE Main Level)

1. (*d*) Eq. of the parabola is $= \sqrt{(x+3)^2 + y^2} = |x + 5|$

$x^2 + 6x + 9 + y^2 = x^2 + 25 + 10x$

$y^2 = 4(x + 4)$

2. (*d*) $(x - 2)^2 + (y - 3)^2 = \left|\frac{3x - 4y + 7}{5}\right|^2$

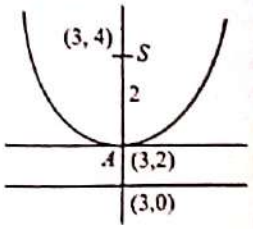
$\therefore$ focus is (2, 3) and directrix is $3x - 4y + 7 = 0$

latus rectum $= 2 \times \perp_r$ distance from focus to directrix $= 2 \times \frac{1}{5} = 2/5$

3. (*a*) $y^2 - 12x - 4y + 4 = 0$

$y^2 - 4y = 12x - 4$

$(y - 2)^2 = 12x$



$Y^2 = 12X$

$x^2 = 4ay$

$(x - 3)^2 = 4 \times 2(y - 2)$

$x^2 - 6x + 9 = 8y - 16$

$x^2 - 6x - 8y + 25 = 0$

4. (*b*) $x = t^2 + 1; y = 2t \Rightarrow t = \frac{y}{2}$

$x = \frac{y^2}{4} + 1$ ... (i)

$x = 2s; y = \frac{2}{s} \Rightarrow s = \frac{2}{y}$

$x = \frac{4}{y} \Rightarrow \frac{4}{y} = \frac{y^2}{4} + 1$

$y^3 + 4y - 16 = 0 \Rightarrow \left.\begin{matrix} y = 2 \\ x = 2 \end{matrix}\right\} POI$

5. (*a*) Length of chord

$$= \frac{4}{m^2}\sqrt{a(a - mc)(1 + m^2)}$$

$m = \tan 60° = \sqrt{3}$

Length of chord $= \frac{4}{3}\sqrt{3(3 - \sqrt{3} \times 0)(1 + 3)}$

$= \frac{4}{3}\sqrt{36} = 8$

6. (*a*) $y^2 = 4x \quad a = 1$

$P(t^2, 2t) \quad t_1t_2 = -1$ For focal chord

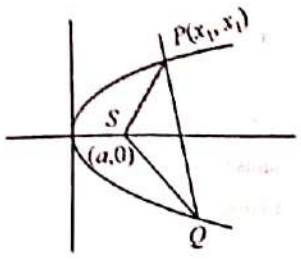
$t_2 = -\frac{1}{t}$

$Q\left(\frac{1}{t^2}, \frac{-2}{t}\right)$

$PQ = \sqrt{\left(t^2 - \frac{1}{t^2}\right)^2 + \left(2t + \frac{2}{t}\right)^2}$

$\left(t + \frac{1}{t}\right)\sqrt{\left(t - \frac{1}{t}\right)^2 + 4} = \left(t + \frac{1}{t}\right)^2$

7. (*a*)



$y^2 = 4ax$

$x_1^2 = 4ax_1$

$x_1 = 0, 4a$

$P(4a, 4a)$

$\therefore$ Q is $(9a, -6a)$ $\left\{\text{using } t_2 = -t_1 - \frac{2}{t_1}\right\}$

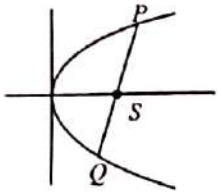
$\Rightarrow x^2 - 4mx - \frac{4}{m} = 0$

$D = 0 \Rightarrow 16m^2 + \frac{16}{m} = 0$

$\Rightarrow m = -1$

slope of $PS \times$ slope of $QS = -1$

8. (*d*)



From the property, $\frac{1}{PS} + \frac{1}{QS} = \frac{1}{a}$

$\frac{1}{3} + \frac{1}{2} = \frac{1}{a}$

$a = \frac{6}{5}$

$\therefore$ Latus rectum $= 4a = \frac{24}{5}$

9. (*d*) Slope of tangent $= \frac{1-0}{4-3} = 1$

also $\frac{dy}{dx} = 2(x-3)$

$\left(\frac{dy}{dx}\right)_{(x_1,y_1)} = 2(x_1 - 3) = 1 \Rightarrow 3 = \frac{1}{2}$

$x_1 = \frac{7}{2}$

$\therefore$ $y_1 = \left(\frac{7}{2} - 3\right)^2 = \frac{1}{4}$

Equation of tangent is

$y - \frac{1}{4} = 1\left(x - \frac{7}{2}\right)$

$4y - 1 = 2(2x - 7)$

$4x - 4y = 13$

10. (*c*) Let the equation of tangent to the parabola $y^2 = 4x$ is

$y = mx + \frac{1}{m}$ ... (i)

solving equation (1) with parabola $x^2 = 4y$

$\Rightarrow x^2 = 4\left(mx + \frac{1}{m}\right)$

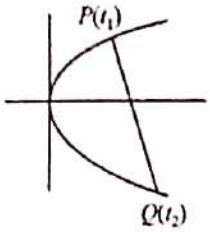
Now put $D = 0$ and find the value of m

11. (*b*) Normal at $P(at^2, 2at_1)$

$a = 1$

$P(t_1^2, 2t_1)$

$y + t_1 x = 2t_1 + t_1^3$ ... (i)



slope $= 1 = -t_1$

$t_1 = -1$

$P(1, -2)$ $t_2 = -t_1 - \frac{2}{t_1}$

$Q(t_2^2, 2t_2)$ $t_2 = 1 + 2 = 3$

$Q(9, 6)$

$PQ = \sqrt{(9-1)^2 + (6+2)^2} = 8\sqrt{2}$

12. (*c*) Use $T^2 = SS_1$

$\Rightarrow [y.0 - 4(x-2)]^2 = (y^2 - 8x)(0 - 8(-2))$

$\Rightarrow 16(x-2)^2 = 16(y^2 - 8x)$

$\Rightarrow y = \pm (x + 2)$

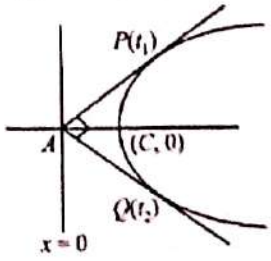
13. (*d*) $y^2 = x - c$; $a = 1/4$

Slope of tangent $= \frac{1}{t}$

so $\frac{1}{t_1 t_2} = -1$

$t_1 t_2 = -1$ ...(i)

$A(at_1 t_2 + C, a(t_1 + t_2))$



$at_1 t_2 + C = 0$

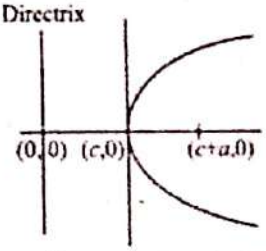
$C = -at_1 t_2$

$C = a$

$C = \frac{1}{4}$

Aliter $\frac{c + a + 0}{2} = c$

Directrix

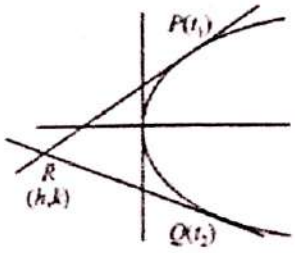


$c + a = 2c \Rightarrow c = a \Rightarrow c = 1/4$

14. (*a*) $y^2 = 4ax$

Slope $= \frac{1}{t}$

$\frac{1}{t_1} = \frac{2}{t_2}$



$\Rightarrow t_2 = 2t_1$ ... (i)

$R[at_1 t_2, a(t_1 + t_2)]$

$h = at_1 t_2, k = a(t_1 + t_2)$

$k = 3at_1 \Rightarrow t_1 = \frac{k}{3a}$

$h = 2at_1^2$

$h = 2a\frac{k^2}{9a^2}$

$\Rightarrow k^2 = \frac{9}{2}ah \Rightarrow y^2 = \frac{9}{2}ax$

15. (*c*) Let point $P(x_1, y_1)$

$x_1 - y_1 + 3 = 0$

C.O.C. w.r.t. (x_1, y_1) of $y^2 = 4ax$

$yy_1 = 4(x + x_1)$

$y(x_1 + 3) = 4x + 4x_1$

$yx_1 + 3y - 4x - 4x_1 = 0$

$(3y - 4x) + x_1(y - 4) = 0$

$L_1 + \lambda L_2 = 0$

$L_1 = 0$ & $L_2 = 0$

$3y = 4x$ $y = 4$

$x = 3$

point (3, 4)

16. (*c*) Equation of PQ

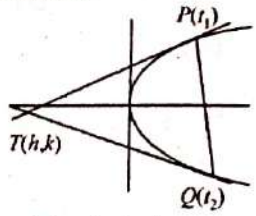
$(t_1 + t_2)y = 2x + 2at_1 t_2$

passes through $(-a, b)$

$b(t_1 + t_2) = -2a + 2at_1 t_2$ ... (i)

$h = at_1 t_2$ & $k = a(t_1 + t_2)$

POI of tangents



$h = at_1t_2$ & $k = a(t_1 + t_2)$

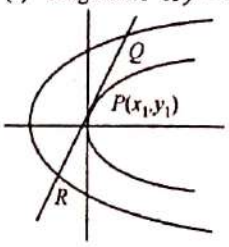
$\frac{bk}{a} = -2a + 2h$

$bk = -2a^2 + 2ah$

$by = -2a^2 + 2ax$

$by = 2a(x - a)$

17. **(c)** Tangent at P of $y^2 = 4ax$



$yy_1 = 2a(x + x_1)$ ... (i)

Let Mid point (h, k)

$T = S_1$

$yk - 2a(x + h) - 4ab = k^2 - 4a(h + b)$

$yk - 2ax - 2ah + 4ah - k^2 = 0$

$yk - 2ax + 2ah - k^2 = 0$ ... (ii)

(i) & (ii) are same

$\frac{k}{y_1} = \frac{-2a}{-2a} = \frac{2ah - k^2}{-2ax_1}$

$k = y_1$; $-2ax_1 = 2ah - k^2$

$-2ax_1 = 2ah - y_1^2$; $y_1^2 = 4ax_1$

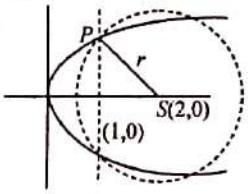
Mid point $-2ax_1 = 2ah - 4ax_1$

(x_1, y_1) $2ah = 2ax_1$

$h = x_1$

18. **(b)** $P(1, 2\sqrt{2})$ Intersection point of $x = 1$ with

$y^2 = 8x$



$r^2 = SP^2$

$= (1 - 2)^2 + (2\sqrt{2})^2$

$= 1 + 8 = 9$

equation of circle as centre (2, 0); $r = 3$

$(x - 2) + y^2 = 9$

19. **(a)** From the property : the feet of the $\perp r$ will lie on the tangent at vertex of the parabola.

$y = (x - 1)^2 - 3 - 1$

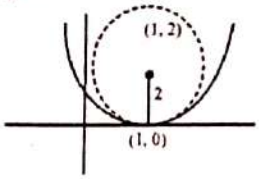
$(x - 1)^2 = (y + 4)$

Tangent at vertex of above parabola is $y + 4 = 0$.

20. **(d)** $(x - 1)^2 = 8y$; $a = 2$

$x^2 = 8y$;

$x = 1, y = 2$



vertex (1, 0)

Focus (1, 2)

Radius of circle = 2

$(x - 1)^2 + (y - 2)^2 = 4$

$x^2 + y^2 - 2x - 4y + 1 = 0$

21. **(c)** $y^2 = 4a(x - \ell_1)$, $x^2 = 4a(y - \ell_2)$

let the POC (h, k)

$2yy' = 4a$, $2x = 4ay$

$y' = \frac{2a}{y}\Big|_{(h,k)} = \frac{2a}{k}$ ...(i) $y' = \frac{x}{2a}\Big|_{(h,k)}$ ...(ii)

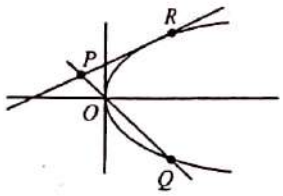
(i) and (ii) are equal = $\frac{h}{2a}$

$\frac{2a}{k} = \frac{h}{2a}$

$hk = 4a^2$

$xy = 4a^2$

22. **(c)**



$y = mx + \frac{a}{m}$ ... (i)

equation of OP is

$y = -\frac{1}{m}x$ ... (ii)

$OP = \frac{a/m}{\sqrt{1 + m^2}}$

equation (ii) meets the parabola at Q

$\frac{1}{m^2}x^2 = 4ax$

$\Rightarrow x = 4am^2, y = -4am$

$\therefore OQ = 4am, \sqrt{1 + m^2}$, $OP.\ OQ = 4a^2$

23. **(c)** Tangent $\equiv 2x - y + 2(x + 2) + 3(y - 1) - 7 = 0$

$\Rightarrow \quad 4x + 2y = 6$

$2x + y = 3$

Image of $2x + y = 3$ about $y - x - 2 = 0$

$= \quad y = \frac{-x}{2} + \frac{5}{2}$

If parabola is $y^2 = 4ax$

then equation of tangent $\equiv y = mx + a/m$

Here $m = -\frac{1}{2} \Rightarrow \frac{a}{m} = \frac{5}{2}$

$\therefore \quad a = -\frac{5}{4}$

parabola is $y^2 = -5x$

24. **(a)** $y^2 = 4(x - 10)$

$\Rightarrow 2y\,dy/dx = 4$

$\Rightarrow dy/dx = 2/y$

Slope of line $x + y = 4$ is -1 [as $y = -x + 4$]

Hence slope of tangent parabola = stope of line $x + y = 4$

(as shortest distance would be perpendicular)

$\Rightarrow dy/dx = 2/y = -1 \Rightarrow y = -2$

$(-2)^2 = 4(x - 10) \Rightarrow 1 = x \to 0 \Rightarrow x = 11$

(11, –2) is the point

(11, –2) is the point on parabola slope of perpendicular line = 1 (line of shortest distance)

$\Rightarrow y - (-2) = 1(x - 11)$

$\Rightarrow y + 2 = x - 11$

$\Rightarrow x - y = 13$

$x - y = 13$

$x + y = 4$

$\Rightarrow 2x = 17 \Rightarrow x = 17/2$

$\Rightarrow y = -9/2$

Point (17/2, –9/2) is nearest to the parabola

$y^2 = 4(x - 10)$ on line $x + y = 4$

25. **(b)**

(a) $x = 3\cos t \Rightarrow \cos t = \frac{x}{3}$

$y = 4\sin t \Rightarrow \sin t = \frac{y}{4}$

We know the identify , $\sin^2 t + \cos^2 t = 1$

$\Rightarrow \left(\frac{y}{4}\right)^2 + \left(\frac{x}{3}\right)^2 = 1$

$\Rightarrow \frac{y^2}{16} + \frac{x^2}{9} = 1$

$\Rightarrow \frac{x^2}{9} + \frac{y^2}{16} = 1$ represents and equation of ellipse

(b) $x^2 - 2 = -2\cos t$

$\Rightarrow x^2 = 2 - 2\cos t$

$\Rightarrow x^2 = 2(1 - \cos t)$

$\Rightarrow x^2 = 2\left(1-\left(1-2\sin^2\frac{t}{2}\right)\right)$

$\Rightarrow x^2 = 4\sin^2\frac{t}{2}$

We have $y = 4\cos^2\frac{t}{2}$

$\Rightarrow \cos^2\frac{t}{2} = \frac{y}{4}$

We know the identity, $\sin^2\frac{t}{2}+\cos^2\frac{t}{2}=1$

$\Rightarrow \frac{x^2}{4}+\frac{y}{4}=1$

$\Rightarrow x^2 = 4 - y$ represents a parabolic profile.

(c) $\sqrt{x} = \tan t, \sqrt{y} = \sec t$

We know that $\sec^2 t - \tan^2 t = 1$

$\Rightarrow (\sqrt{y})^2 - (\sqrt{x})^2 = 1$

$\Rightarrow y - x = 1$ or $x - y + 1 = 0$ is the equation of a straight line

(d) $x = \sqrt{1-\sin t} = \sqrt{\sin^2 t + \cos^2 t - \sin t}$

using the identity $\sin^2 t + \cos^2 t = 1$

$= \sqrt{\sin^2 t + \cos^2 t - 2\sin\frac{t}{2}\cos\frac{t}{2}}$

using multiple angle formula

$\sin t = 2\sin\frac{t}{2}\cos\frac{t}{2}$

$= \sqrt{\left(\sin\frac{t}{2}-\cos\frac{t}{2}\right)^2}$

$= \left|\sin\frac{t}{2}-\cos\frac{t}{2}\right|$

$\therefore x = \sin\frac{t}{2}-\cos\frac{t}{2}$ and $x = \sin\frac{t}{2}+\cos\frac{t}{2}$

we have $y = \sin\frac{t}{2}+\cos\frac{t}{2}$

$\Rightarrow x + y = \sin\frac{t}{2}-\cos\frac{t}{2}+\sin\frac{t}{2}+\cos\frac{t}{2}$

$= 2\sin\frac{t}{2}$

$\Rightarrow \sin\frac{t}{2} = \frac{y+x}{2}$

$\Rightarrow x - y = \sin\frac{t}{2}-\cos\frac{t}{2}-\sin\frac{t}{2}-\cos\frac{t}{2}$

$= -2\cos\frac{t}{2}$

$\Rightarrow \cos\frac{t}{2} = \frac{y-x}{2}$

We have the identity

$\sin^2\frac{t}{2}+\cos^2\frac{t}{2}=1$

$\Rightarrow \left(\frac{y+x}{2}\right)^2+\left(\frac{y-x}{2}\right)^2=1$

$\Rightarrow x^2 + y^2 + 2xy + x^2 + y^2 - 2xy = 4$

$\Rightarrow x^2 + y^2 = 2$ is the equation of the circle.

26. (a) The line $\frac{x}{p}+\frac{y}{q}=1$ be a normal to the parabola

$y^2 = 4ax$

If, for some value of m, it is identical with

$y = mx - 2am - am^3$ i.e. $mx - y = (2am + am^3)$

Comparing coefficeints, we get

$\frac{m}{1/p} = \frac{-1}{1/q} = \frac{2am+am^3}{1} \rightarrow mp = -q$

$\therefore m = -q/p$ and $mp = m(2a + am^2)$

or $p = 2a + \frac{aq^2}{p^2}$ or $p^3 = 2ap^2 + aq^2$

27. (d) The equation of any normal to $y^2 = -8x$ is

$y = mx + 4m + 2m^3$ ..... (1)

[using $y = mx - 2am - am^3$]

Then given line is $2x + y + k = 0$

$\Rightarrow y = -2x - k$ .... (ii)

on comparing equations (i) and (ii), we get

$m = -2$ and $k = -4m - 2m^3$

$\therefore k = 8 + 16 = 24$

28. (a) Given that

line passing throught (21, 30) & normal to curve $y = 2\sqrt{x}$

To find No. of slopes

$y = 2\sqrt{x}$

$\therefore y^2 = 4x$

There fore it is the equation of parabola having vertex (0, 0) & focus (1, 0)

Therefore

equaton of normal of parabola

$y = mx - 2am - am^3$

$\Rightarrow 30 = 2ln - 2m - m^3$

$\Rightarrow m^3 - 19m + 30 = 0$

Since, $y = 2\sqrt{x}$

then $x \neq$ negative

Graph will be

Since, Normals will have only negative value then, m = slope = –5

29. (c) Point of intersection of the tangents drawn at the ends of a chord of $y^2 = 4ax$ is $R = (at_1t_2, a(t_1 + t_2))$

Let the two fixed point of contact be

$P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$

And the point of contact of variable tangent be $A(at^2, 2at)$ and the mid point of portion be (h, k)

Point of intersection of tangent at P and A is $B(att_1, a(t + t_1))$

Point of intersection of tangent at Q and A is $C(att_2, a(t + t_2))$

Mid point of BC is

$\left(\frac{att_1+att_2}{2}, \frac{a(t+t_1)+a(t+t_2)}{2}\right)$

$\left(\frac{at(t_1+t_2)}{2}, \frac{2at+a(t_1+t_2)}{2}\right)$

$\frac{at(t_1+t_2)}{2} = h$

$\Rightarrow at = \frac{2h}{t_1+t_2}$ ...(i)

$\frac{2at+a(t_1+t_2)}{2} = k$

$2at + a(t_1 + t_2) = 2k$

$\frac{4h}{t_1+t_2} + a(t_1 + t_2) = 2k$

$4x - 2y(t_1 + t_2) + a(t_1 + t_2)^2 = 0$

As t_1 and t_1 are fixed therefore the equation represents a straight line

Hence proved

30. (b) Chord joining point(2, 2) and (8, –4) is

$y - 2 = \frac{6}{-6}(x - 2)$

$\Rightarrow y - 2 = -x + 2$

$\Rightarrow y + x - 4 = 0$

For $(a^2, a - 2)$ to lie inside the parabola

$(a-2)^2 - 2(a^2) < 0$

$\Rightarrow a^2 + 4a - 4 > 0$

$\Rightarrow a \in (-\infty, -2-2\sqrt{2}) \cup (-2+2\sqrt{2}, \infty)$ ...(i)

Now for chord, (0,0) and $(a^2, a - 2)$ should lie on same side of origin

$\Rightarrow y + x - 4 = 0$

$0 + 0 - 4 < 0$

and $a^2 + a - 6 < 0$

$\Rightarrow (a + 3)(a - 2) < 0$

$\Rightarrow a \in (-3, 2)$ ...(ii)

31. (a) Intersection points are

$(x_r, y_r) : r = 1, 2, 3, 4$

$y^2 = 4ax$ and the circle is

$x^2 + y^2 + 2gx + 2fy + c = 0$

From parabola equation, we have

$x = \frac{y^2}{4a}$

Now, $\frac{y^4}{16a^2} + y^2 + \frac{2gy^2}{4a} + 2fy + c = 0$

Above equation has four roots.

As coefficient of y^3 is 0,

so, $y_1 + y_2 + y_3 + y_4 = 0$

32. (*b*) We have $\sqrt{px}+\sqrt{qy}=1$

$\Rightarrow (\sqrt{px}+\sqrt{qy})^2=1$

$\Rightarrow px+qy+2\sqrt{(pq)(xy)}=1$

$\Rightarrow (px+qy-1)^2=4(pq)(xy)$

$\Rightarrow p^2x^2-2(pq)(xy)+q^2y^2-2px-2qy+1=0$

On comparing the is equation with the equation

$ax^2+2hxy+by^2+2gx+2fy+c=0$, we get

$a=p^2, b=q^2, c=1, g=-p$

$f=-q$ and $h=-pq$

$\Delta=abc+2fgh-af^2-b^2-c^2$

$p^2q^2-2p^2q^2-p^2q^2-p^2q^2-p^2q^2$

$=-4p^2q^2=0$

and $h^2-ab=p^2q^2-p^2q^2=0$

Thus, we have $\Delta\equiv 0$ and $h^2=ab$

33. (*a*) $m_{OA}\times m_{OB}=-1$

$t_1t_2=-1$

$t_1+t_2=h$

$K=(t_1+t_2)^2-2t_1t_2$

$K=h^2+2\Rightarrow y=x^2+2$

34. (*a*) Given : $y^2=x-1$ and $x^2=y-1$

By observation, if we swap x by y in $y^2=x-1$, then we get $x^2=y-1$

So, they are symmetric about $y=x$

Finding the equation of tangents on both the curve which is parallel to $y=x$

Now for $y^2=x-1$

$2yy'=1$

$\Rightarrow y'=\frac{1}{2y}=1$

$\Rightarrow x=\frac{1}{2}\Rightarrow y=\frac{5}{4}\Rightarrow x-y-\frac{3}{4}=0$

Now, for $x^2=y-1$

$y'=2x$

$\Rightarrow 2x=1$

$\Rightarrow x=\frac{1}{2}\Rightarrow y=\frac{5}{4}$

$\Rightarrow x-y+\frac{3}{4}=0$

Distance between parallel lines

$\left|\frac{-\frac{3}{4}-\frac{3}{4}}{\sqrt{1^2+1^2}}\right|=\frac{3\sqrt{2}}{4}$

35. [8] We know parabola passes through its vertex

$\Rightarrow 0+\lambda+\mu=0\Rightarrow \lambda+\mu=0$ ...(i)

Now parabola is, $x^2+\lambda y+\mu=0$

$\Rightarrow x^2=-\lambda(y-1)$ [Using (i)

This directrix of this parabola is

$y=1+\frac{\lambda}{4}=2$ (given)

$\Rightarrow \lambda=4$

$\therefore \mu=-4$

$(\lambda-\mu)=(4-(-4))=8$

36. [2] Let AB be a normal chord where $A\equiv(at_1^2,2at_1)$ and $B\equiv(at_2^2,2at_2)$

$2h=a(t_1^2+t_2^2)$

$=a[(t_1+t_2)^2-2t_1t_2]$

and $2k=2a(t_1+t_2)$

We also have, $t_2=-t_1-\frac{2}{t_1}$

$\Rightarrow t_1+t_2=\frac{-2}{t_1}$ and $t_1t_2=-t_1^2-2$

$\Rightarrow \frac{k}{a}=\frac{-2}{t_1}$

$\Rightarrow t_1=-\frac{2a}{k}$

So, $2h=a\left[\left(-\frac{2}{t_1}\right)^2+2t_1^2+4\right]$

$\Rightarrow h=a\left(\frac{2}{t_1^2}+t_1^2+2\right)$

Thus, required locus is

$x=a\left(\frac{4a^2}{y^2}+2+\frac{y^2}{2a^2}\right)$

37. [2] Equation of normal at $(am^2, 2am)$ is

$(y-2am)=-m(x-am^2)$

$\Rightarrow y=-mx+am^3+2am$ ...(i)

It meets parabola again at $a(am'^2,2am')$ is satisfy eq. (i)

$\therefore m'=-m-\frac{2}{m}$

Slop of tangetn at $Q=\frac{-1}{m'}=\frac{m}{m^2+2}$

$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=\left|\frac{m}{2}\right|\Rightarrow\theta=\tan^{-1}\left|\frac{m}{2}\right|$

Parikshit (JEE Advanced Level)

1. (*b*) Equation of the curve passing through the points A, B, C, D is $(3x+4y-24)(4x+3y-24)-\lambda xy=0$

If it represents a parabola then the quadratic terms must form a perfect square $\Rightarrow \lambda=1$ or 49.

2. (*b*) Any point on parabola is (t^2, t)

If (x, y) be the image of the point (t^2, t) in the line $x+y+4=0$

$\frac{x-t^2}{1}=\frac{y-t}{1}=\frac{-2(t^2+t+4)}{1^2+1^2}$

$x-t^2=-t^2-t-4$ and $y-t=-t^2-t-4$

$x=-t-4$ and $y=-t^2-4$

$(x+y)=-t$ and $y+4=t^2$

$(x+y)^2=t^2$

$\Rightarrow (y+4)=-(x+4)^2$

3. (*a*) If at $x=1$, $y=2x$ is a tangent to parabola $y=ax^2+bx+c$

The repsecitve value of a, b, c are

At $x=1, y=2$

$2=a+b+c$ ...(1)

Equation of tangent at (1,2) is

$\frac{y}{2}+\frac{2}{2}=ax+\frac{b}{2}x+\frac{b}{2}+c$

$y=(2a+b)x+(b+2c-2)$

Comparing with $y=2x$

$2a+b=2$ ...(2)

$b+2c-2=0$ ...(3)

Solve (1)(2)(3)

We get

$a=\frac{1}{2}$

$b=1$

$c=\frac{1}{2}$

Hence a, b, c are $\frac{1}{2}, 1, \frac{1}{2}$

4. (*c*) As, orthocentrs of ΔABC and ΔDEF are equidrant from the a-xis of parabola

Axis of $y^2=4ax$ is $x=0$

$\therefore d=4$

5. (*c*) $(x+1)^2=-4(y+3)$

Point on axis $\equiv(-1,-H), (H>0)$

for there normal

$H-3>2$

$H>5\Rightarrow -H<-5$

so option (c) $(-1,-k), K\in(-\infty,-5)$

6. (*b*) Let $y_1=5+\sqrt{1-x_1^2}$ and $y_2=\sqrt{4x_2}$

or $x_1^2+(y_1-5)^2=1$ and $y_2^2=4x_2$

Thus (x_1, y_1) lies on the circle $x^2+(y-5)^2=1$

and (x_2, y_2) lies on the parabola $y^2=4x$.

Thus, given expression is the shortest distance between the curve $x^2+(y-5)^2=1$ and $y^2=4x$.

Now the shortest distance always occur along common normal to the curves and normal to circle passes through the centre of the circle. Normal to the parabola $y^2 = 4x$ is $y = mx - 2m - m^3$ passes through (0,5) gives

$m^3 + 2m + 5 = 0$, which has only one root m = –2.

Hence, corresponding point on the parabola is (4,4)

Thus required minimum distance

$= \sqrt{4^2 + 8^2} - 1 = 4\sqrt{5} - 1$

7. **(c)** Given equation of parabola is

$y^2 = 4x$

Let $P(x_1, y_1) \equiv \left(t_1^2, 2t_1\right)$ and

$Q(x_2, y_2) \equiv \left(t_2^2, 2t_2\right)$

Let the normal at point meets the parabola again at $R(t^2, 2t)$

$\Rightarrow t = -t_1 - \frac{2}{t_1}$ ...(*i*)

$\Rightarrow t = -t_2 - \frac{2}{t_2}$ ...(*ii*)

From equations (*i*) and (*ii*)

$-t_1 - \frac{2}{t_1} = -t_2 - \frac{2}{t_2}$

$\Rightarrow t_1 t_2 = 2$

Now given that $x_1 + x_2 = 4$

$\Rightarrow t_1^2 + t_2^2 = 4$

$\Rightarrow (t_1 + t_2)^2 = 4 + 4 = 8$

$\Rightarrow |t_1 + t_2| = 2\sqrt{2}$

$\Rightarrow |y_1 + y_2| = 4\sqrt{2}$

8. **(b)** Let PQ and RS be two normal chords of the parabola Let the concyclic point be $P(t_1)$, $Q(t_2)$, $R(t_3)$ and $S(t_1)$

$\Rightarrow t_1 + t_2 + t_3 + t_4 = 0$

Here, t_2 and t_4 are feet of the normals.

$\Rightarrow t_2 = -t_1 - \frac{2}{t_1}$ and $t_4 = -t_3 - \frac{2}{t_3}$

$\Rightarrow t_1 + t_2 = -\frac{2}{t_1}$ and $t_4 + t_3 = \frac{2}{t_3}$

adding.

$-2\left(\frac{1}{t_1} + \frac{1}{t_3}\right) = 0$

$\Rightarrow t_1 + t_3 = 0$

$\Rightarrow t_2 + t_4 = 0$

Now, equation of tangents at the feet of normals are

$t_2 y = x + at_2^2$

$t_4 y = x + at_4^2$

$\Rightarrow t_2 y - at_2^2 = t_4 y - at_4^2$

$\Rightarrow y = -a(t_4 + t_2) = 0$

$\Rightarrow y = -a(t_4 + t_2) = 0$

So, $x = -at_2^2 = at_2 t_4$ (since $t_2 + t_4 = 0$)

$\Rightarrow$ Point of intersection of tangents at t_2 and t_4 is $(at_2\, t_4, 0)$. Hence, point of intersection lies on the axis of the parabola.

9. **(a)** Equation of the circle is

$\lambda(2x + 3y + 1) + (x - 1)^2 + (y + 1)^2 = 0$

It passes through (1, 0)

$\Rightarrow \lambda = -1/3$

Equation of circle is $3x^2 + 3y^2 - 8x + 3y + 5 = 0$

10. **(c)** Given equation of parabola is

$y^2 = 4ax$

Focus is at (a, 0)

Equation of tangent to the parabola at $P(at^2, 2at)$ is $ty = x + at^2$ ...(*i*)

Since, the tangent meets x-axis i.e. y = O

$\Rightarrow x = -at^2$

So, coordinates of point T is $(-at^2, 0)$ is

$y = -tx + 2at + at^3$ ...(*ii*)

Since, the normal meets x-axis i.e. y = 0

$\Rightarrow x = 2a + at^2$

So, coordinates of G are $(2a + at^2, 0)$

Since, the circle passes through *P, T, G* so, its center is same as the focus.

So centre of the circle is $(a,0)$

Since $\angle TPG = 90°$

$(90° - \theta)$is the angle between PT and OP

Slope $PT = \frac{2at}{2at^2} = \frac{1}{t}$

Slope of $OP = \frac{2at}{a\left(t^2 - 1\right)} = \frac{2t}{t^2 - 1}$

$$\therefore \tan\left(90° - \theta\right) = \left|\frac{\frac{1}{t} - \frac{2t}{t^2 - 1}}{1 + \frac{1}{t}\left(\frac{2t}{t^2 - 1}\right)}\right| = \frac{1}{t}$$

$\therefore \cot\theta = \frac{1}{t} \Rightarrow \tan\theta = t$

$\Rightarrow \theta = \tan^{-1}(t)$

11. **(b,c)** Let $A = (3m_1^2, -6m_1), B = (3m_2^2 - 6m_2), D = (3m_3^2, -6m_3)$

A and B lies on $x + y = 1$

$\therefore \quad 3m_1^2 - 6m_1 = 1$ and $3m_2^2 - 6m_2 = 1$

$3m_1^2 - 3m_2^2 - 6(m_1 - m_2) = 0$

$m_1 + m_2 = 2 \Rightarrow -m_3 = 2$

$\therefore$ Coordinates of D are (12, 12)

12. **(c,d)** Clearly tangents are perpendicular

$\Rightarrow$ their point of intersection i.e. (0, 0) lies at directrix

$\Rightarrow AB$ is focal chord of the parabola

$\Rightarrow$ focus is foot of perpendicular from (0, 0) on focal chord AB whose equation is $2x - y - 3 = 0$

$\Rightarrow \frac{\alpha - 0}{2} = \frac{\beta - 0}{-1} = \frac{-(0 - 0 - 3)}{2^2 + 1^2}$

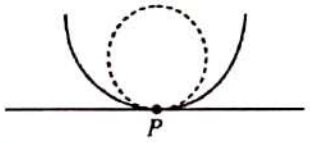
$\Rightarrow$ Focus is $\left(\frac{6}{5}, -\frac{3}{5}\right)$

Slope of axis = Slope of line joining (0, 0) and midpoint of AB.

i.e. $\frac{1 - 0}{2 - 0} = \frac{1}{2}$

Equation of directrix $\Rightarrow (y - 0) = -2(x - 0)$

13. **(a,d)** $at^2 = 2at$



point

$t = 0, \qquad t = 2$

(0, 0), (4, 4)

(I) when $P \equiv (0, 0)$

$x^2 + y^2 + \lambda(x) = 0$

pass the (1, 0) $\lambda = -1$ equation tagent *at* (0, 0)

$y^2 = 4x$

Equ. $x^2 + y^2 - x = 0$

$y.y_1 = 2(x + x_1)$

$x = 0$

(II) when point (4, 4)

$2x - 2y + 8 = 0$

$(x - 4)^2 + (y - 4)^2 + \mu(2x - 2y + 8) = 0$

pass (1, 0)

Equation

$x^2 + y^2 - 13x + 2y + 12 = 0$

14. **(a,b,d)**

Let two perpendicular chords through (x_1, y_1) be PQ and RS

Equation of PQ is $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$

Where tanθ = slope of PQ

Any point on this line may be taken as $\left(x_1 + r\cos\theta, y_1 + r\sin\theta\right)$

As the point lies on $y^2 = 4ax$

$\therefore \quad (y_1 + r\sin\theta)^2 = 4a\left(x_1 + r\cos\theta\right)$

$\Rightarrow r^2\sin^2\theta + 2y_1\sin\theta r - 4a\cos\theta r + y_1^2 - 4ax_1 = 0$

$\therefore \quad r_1 r_2 = \frac{\left(y_1^2 - 4ax_1\right)}{\sin^2\theta}$

Similarly $r_3r_4 = \dfrac{\left(y_1^2 - 4ax_1\right)}{\sin^2\alpha}$ where $\tan\alpha$

= slope of

Since PQ is perpendicular to RS.

$\therefore \quad \alpha = 90 + \theta$ and $\theta - 90$

In either $\sin^2\alpha = \cos^2\theta$

Now

$$\frac{1}{r_1r_2} + \frac{1}{r_3r_4} = \frac{\sin^2\theta + \cos^2\theta}{\left(y_1^2 - 4ax_1\right)} = \frac{1}{\left(y_1^2 - 4ax_1\right)}$$

which constant

15. **(a,b,c)** Let

$$P = \left(t_1^2, 2t_1\right) \text{ and } Q = \left(\frac{1}{t_1^2}, \frac{-2}{t_1}\right)$$

be the extremities of the focal chord. By theorem 4, 19, part (4), the circle touches the directirx $x + 1 = 0$. The equaton of the cirlce is

$$\left(x - t_1^2\right)\left(x - \frac{1}{t_1^2}\right) + \left(y - 2t_1\right)\left(y + \frac{2}{t_1}\right) = 0$$

$$\Rightarrow x^2 - \left(t_1^2 + \frac{1}{t_1^2}\right)x + 1 + y^2 - 2\left(t_1 - \frac{1}{t_1}\right)y - 4 = 0$$

The slope of the focal chord PQ is equal to 1. So we have

$$\frac{2\left(t_1 + 1/t_1\right)}{t_1^2 - 1/t_1^2} = 1$$

$$\Rightarrow \frac{2}{t_1 - 1/t_1} = 1$$

$$\Rightarrow t_1 - \frac{1}{t_1} = 2$$

$$\Rightarrow t_1^2 - 2t_1 + 1 = 0$$

$$\Rightarrow t_1 = 1$$

Therefore, $P = (1, 2)$ and $Q = (1, 2)$. From Eq. (1). the equation of the circle is $x^2 + y^2 - 2x - 3 = 0$. Hence, the radius is 2 and the centre is (1, 0).

16. **(a,b,c,d)**

Let $P = (at^2, 2at)$ and $Q = (at^2, -2at)$. The normal at P is $tx + y = 2at + at^3$ ... (1)

The line passing through Q and parallel to the axis (i.e., x-axis) is $y = -2at$ ...(2)

From Eqs. (1) and (2), we have

$G = (4a + at^2, -2at) = (h,k)$ (say)

Therefore

$h = 4a + at^2$ and $k = -2at$

$$\Rightarrow h - 4a = a\left(\frac{k^2}{4a^2}\right) = \frac{k^2}{4a}$$

$$\Rightarrow k^2 = 4a(h - 4a)$$

$$\Rightarrow y^2 = 4a(x - 4a)$$

Hence, the vertex is $(4a, 0)$, latus rectum is $4a$, the directrix is $x - 4a = -a$ or $x - 3a = 0$ and the focus $(x - 4a = a, 0) = (Sa, 0)$.

17. **(b)**

18. **(a)**

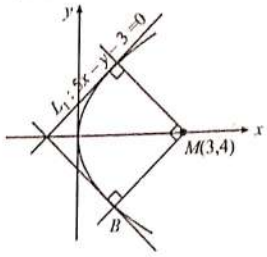
19. **(c)** $L_1 : 5x - y - 3 = 0$

$L_2 : x + 5y - 11 = 0$

since, L_1 and L_2 are perpendicular

$\therefore$ they meet on the directrix

Also point of intersection of L_1 and L_2 is (1, 2).



$M(3, 4)$ lies on the axis of the parabola.

(i) $\therefore$ Equations of directrix is

$y - 2 = -1(x - 1) = x + y = 3$

(ii) Length of latus rectum = Distance between (1, 2) and (3, 4) = $\sqrt{4 + 4} = 2\sqrt{2}$

(iii) Equation of circumcircle of ΔAMB is

$\Rightarrow (x - 1)(x - 3) + (y - 2)(y - 4) = 0$

$x^2 + y^2 - 4x - 6y + 11 = 0$

20. **(a)** $y = px^2 + qx + r$

$c_n = pa_n^2 + qa_n + r$

$d_n = pb_n^2 + qb_n + r$

$$\Rightarrow \quad c_n - d_n = p\left(a_n^2 - bn^2\right) + q\left(a_n - b_n\right)$$

$$\Rightarrow \frac{n}{2}[2a_1 + (n-1)2] - \frac{n}{2}[2b_1 + (n-1)2]$$

$$= \left(a_n - b_n\right)\left(p\left(a_n + b_n\right) + q\right)$$

$$\Rightarrow \frac{n}{2}[2a_1 - 2b_1) = \left(a_1 - b_1\right)$$

$$\left[p\left(a_1 + b_1 + 4(n-1)\right) + q\right)$$

$$\Rightarrow \quad n = p\left(a_1 + b_1 + 4(n-1)\right] + q$$

For $n = 1 \Rightarrow 1 = P(a_1 + b_1) + q$ ... (1)

For $n = 2 \Rightarrow 2 = p\left(a_1 + b_1 + 4\right) + q$ ... (2)

subtract (1) from (2)

$1 = 4p \qquad \boxed{p = \frac{1}{4}}$

21. **(c)** If $r = 0 \Rightarrow y = px^2 + qx$

$c_n = pa_n^2 + qa_n$

For $n = 1$ $c_1 = pa_1^2 + qa_1$

$$a_1 = \frac{1}{4}a_1^2 + qa_1$$

$$1 = \frac{1}{4}a_1 + q \text{ or } a_1 = 0$$

For $n = 2$

$$c_2 = a_1 + a_2 = \frac{1}{4}a_2^2 + qa_2$$

$$2a_1 + 2 = \frac{1}{4}\left(a_1 + 2\right)^2 + q\left(a_1 + 2\right)$$

$$2a_1 + 2 = \frac{a_1^2}{4} + 1 + a_1 + qa_1 + 2q$$

$$0 = \frac{a_1^2}{4} + a_1(q - 1) + 2q - 1$$

Comparing we get $a_1 = 0$, $b_1 = 2$

22. **(a)**

(a) Let end point of the chord is $\left(at_1^2, 2at_1\right)$ and $\left(at_2^2, 2at_2\right)$

Since it is normal to parabola so

$$t_2 = -t_1 - \frac{2}{t_1}$$

equation of normal of parabola

$y = -tx + 2at + at^3$

Slope is $-t_1$ which is given acute so $-t_1 > 0$ or t_1

Now chord make right angel at origin so

$$\frac{2at_1 - 0}{at_1^2 - 0}\,\frac{2at_1 - 0}{at_2^2 - 0} = -1$$

$t_1t_2 = -4$

$$t_1\left(-t_1 - \frac{2}{t_1}\right) = -4$$

$t_1 = -\sqrt{2}$

Hence

$t^2 = 2$

(b) $y^2 = 4x \Rightarrow x = \dfrac{y^2}{4}$

ordinates are 1, 2 & 4

$\Rightarrow$ abscica are $\dfrac{(1)^2}{4}, \dfrac{(2)^2}{4}, \dfrac{(4)^2}{4} = \dfrac{1}{4}, 1, 4$

$\Rightarrow$ vertices are $\left(\dfrac{1}{4}, 1\right), (1,2), (4,4)$

$$\Delta = \frac{1}{2}\begin{vmatrix} \frac{1}{4} & 1 & 1 \\ 1 & 2 & 1 \\ 4 & 4 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left[\left[\frac{1}{4}(2-4) - 1(1-4) + 1(4-8)\right]\right]$$

$$= \frac{1}{2}\left|-\frac{1}{2} + 3 - 4\right| = \frac{1}{2}\left|-\frac{3}{2}\right| = \frac{3}{4}$$

(c) $y^2 = 4x$

$d = 1$

$N : y = mx - 2am - am^3$

$y = mx - 2m - m^3$

at $\left(\frac{11}{4}, \frac{1}{4}\right)$

$\Rightarrow \frac{1}{4} = \frac{11m}{4} - 2m - 3m^2$

$\Rightarrow \frac{1}{4} = \frac{11m - 8m - 4m^3}{4}$

$\Rightarrow 1 = 3m - 4m^3$

$\Rightarrow 4m^3 - 3m + 1 = 0$

$m = -1 \Rightarrow -4 + 3 + 1 = 0$

$\therefore (m + 1)(4m^2 - 4m + 1) = 0$

$(m + 1)(2m - 1)^2 = 0$

$\therefore \quad m = -1, \frac{1}{2}$

2 values of m is possible.

(d) $y^2 = 4ax$

$2y.y' = 4a$

$y' = \frac{2a}{y}$

Hence, $y'_{a,2a} = 1$

Hence slope of the normal will be -1.

Hnece the equation of the normal will be

$\frac{y - 2a}{x - a} = -1$

$y - 2a = -x + a$

$x + y = 3a$

$y = 3a - x$..... (i)

$y^2 = x^2 - 6ax + 9a^2$

$4ax = x^2 - 6ax + 9a^2$

$x^2 - 10ax + 9a^2 = 0$

$(x - 9a)(x - a) = 0$

$x = 9a$ and $x = a$

Therefore $y = 3a - x$

$y = -6a, 2a$

Now, $2at = -6a$

$|t - 1| = |-3 - 1| = 4$

23. **[35]** The given parabola equation is $Y^2 = 8X$ where $Y = y$ and $X = x - 1$

Every point on this parabola is

$\left(x = 2t^2, y = 4t\right) = \left(2t^2 + 1, 4t\right)$

Normal at $(2t^2 + 1, 4t)$ is

$tX + Y = 2(2t) + 2t^3$

$\Rightarrow t(x - 1) + y = 4t + 2t^3$ ...(i)

Suppose the tangents at the ends of normal chord intersect at $P(x_1, y_1)$. Then, the normal chord is the chord of contact of $P(x, y)$ and hence its equation is

$Yy_1 - 4(X + x_1) = 0$

$\Rightarrow Yy_1 - 4(x - 1 + x_1) = 0$

$\Rightarrow Yy_1 - 4x + 4 - 4x_1 = 0$ ...(ii)

Equations (i) and (ii) represent the same straight line.

$\therefore \frac{t}{-4} = \frac{1}{y_1} = \frac{4t + 2t^3}{4x_1 - 4}$

$\Rightarrow t = -\frac{4}{y_1}$ and $-1 = \frac{4 + 2t^2}{x_1 - 1}$

$\Rightarrow -(x_1 - 1) = 4 + 2\left(\frac{16}{y_1^2}\right)$

$\Rightarrow -x_1 - 3 = \frac{32}{y_1^2}$

$\Rightarrow y_1^2(x_1 + 3) + 32 = 0$

Hence the locus at $(x_1, y_1)$is

$y^2(x + 3) + 32 = 0$

24. **[2]** Let the point of contact of tangents be $P\left(at_1^2, 2at_1\right)$ and $\left(at_2^2, 2at_2\right)$

Let centre of the circle through $T\,P\,Q$ be (h, k)

Then point of intersection of tangents is $T(at_1\,12, a(t_1 + t_2))$

Equation of chord P Q is

$(t_1 + t_2)y = 2x + 2t_1t_2$

$2x - (t_1 + t_2)y + 2a_1t_2 = 0$

Equation of circle throuh TPQ is

$\left(x - at_1^2\right)\left(x - at_2^2\right) + (y - 2t_1)(y - 2at_2) + \lambda\left(2x - (t_1 + t_2)y + 2at_1t_2\right) = 0$ ...(i)

It passes through T

$\left(at_1t_2 - at_1^2\right)\left(at_1t_2 - at_2^2\right) + \left(a(t_1 + t_2) - 2at_1\right)\left(a(t_1 + t_2) - 2at_2\right) + \lambda\left(2at_1t_2 - (t_1 + t_2)a(t_1 + t_2) + 2at_1t_2\right) = 0$

$\Rightarrow \lambda = -a\left(t_1t_2 + 1\right)$

Simplifying (i)

$x^2 + y^2 - x\left(at_1^2 + at_2^2 - 2\lambda\right) - y\left(2at_1 + 2at_2 + (t_1 + t_2)\lambda\right)$

$a^2t_1^2t_2^2 + 4a^2t_1t_2 + 2att_1t_2\lambda = 0$

Comparing the centre of the circle (h, k) with general equation of circle

$\Rightarrow 2h = at_1^2 + at_2^2 + 2\lambda$

substituting λ

$2h = at_1^2 + at_2^2 - 2\left(-a\left(t_1t_2 + 1\right)\right)$

$2h = at_1^2 + at_2^2 + 2at_1t_2 + 2a$

$2h = a\left(t_1 + t_2\right)^2 + 2a$

$2k = 2at_1 + 2at_2 + (t_1 + t_2)\lambda$

$2k = 2at_1 + 2at_2 + (t_1 + t_2)\left(-a\left(t_1t_2 + 1\right)\right)$

$2k = a\left\{2(t_1 + t_2) - 3(t_1 + t_2)\right\}$

$2k = -a(t_1 + t_2)$

$t_1 + t_2 = \frac{2k}{a}$

Substituting in (i) $2h = a\left(-\frac{2k}{a}\right)^2 + 2a$

$ah = 2k^2 + a^2$

$2k^2 = a(h - a)$

Generalising the equation

$2y^2 = a(x - a)$

Hence proved

25. **[1]** Let the equation of circle be $x^2 + y^2 = a^2$ with ends of diameter $M(a, 0)$ and $M'(-a, 0)$

Equation of concentric circle is $x^2 + y^2 = b^2$ ...(i)

Equation of tangent to (i) is

$y = mx + b\sqrt{m^2 + 1}$

$mx - y + b\sqrt{m^2 + 1} = 0$

Which is the directrix of parabola

let the foucs be $S(h, k)$

For parabola we have $PS = PM$

$\sqrt{(h - a)^2 + k^2} = \frac{m(a) - 1(0) + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}}$

$\sqrt{(h - a)^2 + k^2} = \frac{am + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}}$ ... (ii)

Also $PS = PM'$

$\sqrt{(h + a)^2 + k^2} = \frac{m(-a) - 1(0) + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}}$

$\sqrt{(h + a)^2 + k^2} = \frac{-am + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}}$ ... (iii)

Adding (ii) and (iii)

$\sqrt{(h - a)^2 + k^2} + \sqrt{(h + a)^2 + k^2} = 2b$

$\sqrt{(h + a)^2 + k^2} = 2b - \sqrt{(h - a)^2 + k^2}$

Squaring both sides

$h^2 + 2ah + a^2 + k^2 = 4b^2 + h^2 - 2ah + a^2 + k^2 - 4b\sqrt{(h - a)^2 + k^2}$

$4b^2 - 4ah = 4b\sqrt{(h - a)^2 + k^2}$

$b^2 - ah = b\sqrt{(h - a)^2 + k^2}$

$b - \frac{ah}{b} = \sqrt{(h - a)^2 + k^2}$

Again squaring both sides

$b^2 + \frac{a^2h^2}{b^2} - 2ah = h^2 - 2ah + a^2 + k^2$

$$b^2 - a^2 = \left(1 - \frac{a^2}{b^2}\right)h^2 + k^2$$

$$\left(\frac{b^2 - a^2}{b^2}\right)h^2 + k^2 = b^2 - a^2$$

$$\frac{h^2}{b^2} + \frac{k^2}{b^2 - a^2} = 1$$

generalising the equation

$$\frac{x^2}{b^2} + \frac{y^2}{b^2 - a^2} = 1$$

26. **[0]** Equation of tangent to parabola $y^2 = 4ax$ at point (x_1, y_1) is given by $yy_1 = 2a(x + x_1)$ ...(1)

Also (x_1, y_1) lies on parabola

$\Rightarrow y_1^2 = 4ax_1$ ... (2)

Equation of given circle is

$x^2 + y^2 = r^2$

Let (h, k) be the mid point of PQ

$\because$ equation of chord whose mid point is (h,k) is given by

$T = x_1$ ie. $xh + yk = h^2 + k^2$ ... (3)

Equation (1) and (3) are identical

Therefore $\frac{2a}{h} = -\frac{y_1}{k} = \frac{2ax}{-\left(h^2 + k^2\right)}$

$\Rightarrow y_1 = \frac{-2ak}{h} \rightarrow (4)$

and $x_1 = -\frac{\left(h^2 + k^2\right)}{h}$

Putting the value of x_1 and y_1 from equation (4) and (5) is (2).

we get $\left(\frac{-2ak}{h}\right)^2 = 4a\left(\frac{-\left(h^2 + k^2\right)}{h}\right)$

$\Rightarrow \frac{4a^2k^2}{h^2} = -\frac{4a}{h}\left(h^2 + k^2\right)$

$4ah\left(h^2 + k^2\right) + 4a^2k^2 = 0$

$\Rightarrow ah\left(h^2 + k^2\right) + a^2k^2 = 0$

$\Rightarrow$ Locus of mid-point of PQ is given by

$x\left(x^2 + y^2\right) + ay^2 = 0$

27. **[4]** Let P be the point of intersection of the tangents to the parabola. The equation of the pair of tangents. PQ and PR is

$$\left[(y_1) - 2a(x + x_1)\right]^2 = \left(y^2 - 4^a x\right)\left(y_1^2 - 4ax_1\right)$$

These lines meet the directix

$x + 1 = 0$ or $x = a$

Substituting this we get,

$$\left[(y_1) - 2a(-a + x_1)\right]^2 = \left(y^2 - 4a^2\right)\left(y_1^2 - 4x_1\right)$$

On expanding we get,

$$(y_1)^2 + \left[2a(x_1 - a)\right]^2 - 4ayy_1(x - a)$$

$$= (y_1)^2 - 4x_1y^2 + 4ax_1y^2 + 4a^2y_1^2 - 16a^3x_1$$

$$\text{(i.e)}\ y^2 + \left[y_1^2 - y_1^2 + 4x_1\right] - y\left[4a(x_1 - a)y_1\right]$$

$$+\left[4a^2(x_1 - a)^2 - 4a^2\left(y_1^2 - 4x_1\right)y\right] = 0$$

$$x_1y^2 - y_1(x_1 - a)y + a\left[(x_1 - a)^2 - \left(y_1^2 - 4ax_1\right)\right] = 0$$

$$\Rightarrow x_1y^2 - y_1(x_1 - a) + a\left[(x_1 - a)^2 - y_1^2\right] = 0$$

Here y_1 and y_2 are the ordiantes of the point of intersection of tangents with directix $x + a = 0$.

Sum of the ordinates is $y_1 + y_2 = \frac{-b}{a}$

$= (x_1 - a)y_1$

Product of the ordinates is

$$y_1y_2 = \frac{a\left[(x_1 + a)^2 - y_1^2\right]}{x_1}$$

$$\therefore d^2 = (y_1 - y_2)^2 = (y_1 + y_2)^2 - 4y_1y_2$$

Now substituting the values we get,

$$d^2 = (x_1 - a)^2 y_1^2 - 4ax_1\left[(x_1 + a)^2 - y_1^2\right]$$

$$x_1^2d^2 = y_1^2\left[(x_1 - a)^2 + 4ax_1\right] - 4ax_1(x_1 + a)^2$$

$$= y_1^2(x_1 + a)^2 - 4ax_1(x_1 + a)^2$$

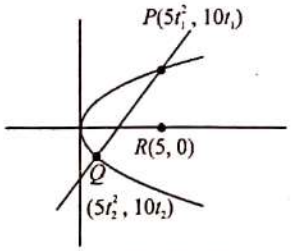
$$= (x_1 + a)^2(y_1 - 4ax_1)$$

Hence the locus of(x_1, y_1) is

$$x^2d^2 = (y - 4ax_1)(x + a)^2$$

PYQ's (Past Year Questions)

1. ***(b)***



Since, centroid is (10, 10)

$$\frac{5\left(t_1^2 + t_2^2 + 1\right)}{3} = 10$$

$\therefore t_1^2 + t_2^2 = 5$ ...(i)

$$\frac{10(t_1 + t_2)}{3} = 10$$

$\therefore t_1 + t_2 = 3$ ...(ii)

$\therefore t_1 = 1, t_2 = 2$

$P = (5, 10)$ $Q = (20, 20)$

Equation of PQ : $y - 10 = \frac{10}{15}(x - 5)$

$3y - 30 = 2x - 10$

$y = \frac{2}{3}x + \frac{20}{3}$

$\therefore PQ^2 = (20 - 5)^2 + (20 - 10)^2 = 225 + 100 = 325$

2. **[16]** $y^2 = 8x + 4y + 4$

$(y - 2)^2 = 8(x + 1)$

$Y^2 = 4aX$

where, $a = 2, X = x + 1, Y = y - 2$

focus $\equiv (2, 0)$ and $x + 1 = 2, y - 2 = 0$

focus $(1, 2)$

$y - 2 = m(x - 1)$

Put $(3, 0)$ in the above line

$\Rightarrow m = -1$

Length of focal chord = 16

3. ***(d)*** We have vertex (5,4) and directrix $3x + y - 29 = 0$

Let foot of perpendicular of (5,4) on directrix be (α, β)

$$\frac{\alpha - 5}{3} = \frac{\beta - 4}{1} = \frac{-(-10)}{10}$$

$\therefore (\alpha, \beta) = (8, 5)$

Since, vertex is midpoint of the foot of perpendicular to the directrix and focus.

So, focus of parabola will be $S = (2, 3)$

For any point $P(x, y)$ on the parabola,

$$\Rightarrow (x - 2)^2 + (y - 3)^2 = \frac{(3x + y - 29)^2}{10}$$

$\Rightarrow 10(x^2 + y^2 - 4x - 6y + 13)$

$= 9x^2 + y^2 + 841 + 6xy - 58y - 174x$

$\Rightarrow x^2 + 9y^2 - 6xy + 134x - 2y - 711 - 0$

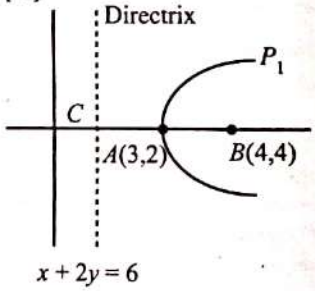
Also equation of the parabola is

$x^2 + ay^2 + bxy + cx + dy + k = 0$

So, $a = 9$, $b = -6$, $c = 134$, $d = -2$, $k = -711$

Hence, $a + b + c + d + k = -576$

4. **[10]**



$x + 2y = 6$

A is mid point of BC

So, $C \equiv (6 - 4, 4 - 4) \equiv (2, 0)$

Equation of directrix

$x + 2y = \lambda$ pass via $C(2, 0)$

So, $2 + 0 = \lambda$

$\Rightarrow x + 2y = 2$

Image of (2,0) about

$x + 2y = 6$ is

$$\frac{x-2}{1} = \frac{y-0}{2} = \frac{-2(2+0-6)}{1+4}$$

$$x = \frac{8}{5} + 2 = \frac{18}{5}; y = \frac{16}{5}$$

So Directrix of P_2 is

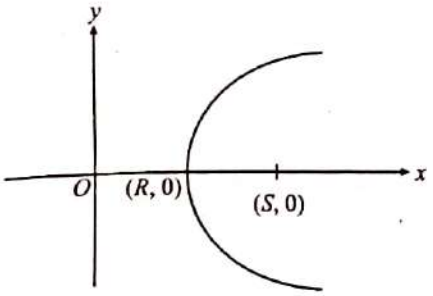
$x + 2y = \lambda_1$ which

pass via $\left(\frac{18}{5}, \frac{16}{5}\right)$

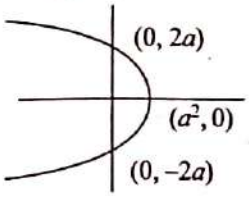
$$\frac{18+32}{5} = \lambda_1 \Rightarrow \lambda_1 = 10$$

So, $x + 2y = 10$

5. **(d)** Length of latus rectum = 4 (distance between vertex and focus) = $4(S - R)$



6. **(d)** $y^2 = -4(x - a^2)$



Vertices of triangle are $(a^2, 0)$ and $(0, 2a)$ and $(0, -2a)$

Area $= \frac{1}{2}(a^2)(4a) = 250$

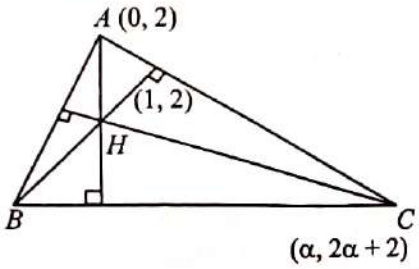
$\Rightarrow a^3 = 125$

7. **(b)** $AB : (\lambda + 1)x + \lambda y = 4$

$AC : \lambda x + (1 - \lambda)y + \lambda = 0$

Vertex A is on y-axis

$\Rightarrow x = 0$



So $y = \frac{4}{\lambda}, y = \frac{\lambda}{\lambda - 1}$

$\Rightarrow \frac{4}{\lambda} = \frac{\lambda}{\lambda - 1} \Rightarrow \lambda^2 - 4\lambda + 4 = 0$

$\Rightarrow \lambda = 2$

Point $A\left(0, \frac{4}{\lambda}\right) \equiv (0, 2)$

AB: $3x + 2y = 4$

AC: $2x + y + 2 = 0$

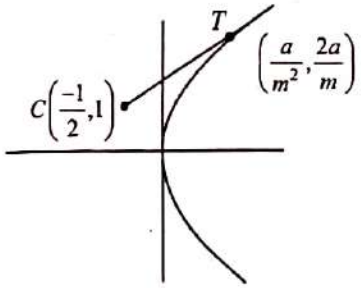
$\Rightarrow A(0,2)$ Let $C\ (\alpha, 2\alpha + 2)$

Now (Slope of Altitude through C) $\left(-\frac{3}{2}\right) = -1$

$(\because m_{CH} \times m_{AB} = -1)$

$$\left(\frac{2\alpha}{\alpha - 1}\right)\left(-\frac{3}{2}\right) = -1 \Rightarrow \alpha = -\frac{1}{2}$$

So $C\left(-\frac{1}{2}, 1\right)$



Let Equation of tangent be $y = mx + \frac{3}{2m}$,

it will pass through $\left(-\frac{1}{2}, 1\right)$

$m^2 + 2m - 3 = 0$

$\Rightarrow m = 1, -3$ (rejected)

So Point of contact of tangent which touches in first quadrant at T is

$$T \equiv \left(\frac{a}{m^2}, \frac{2a}{m}\right)$$

$$\equiv \left(\frac{3}{2}, 3\right)$$

$\Rightarrow CT = \sqrt{4+4} = 2\sqrt{2}$

8. **[10]** $S_1 : y^2 = 2x, S_2 : x^2 + y^2 = 4x$

$P(2, 2)$ is common point on S_1 & S_2

T_1 is tangent to S_1 at P

$\Rightarrow T_1 : y \cdot 2 = x + 2$

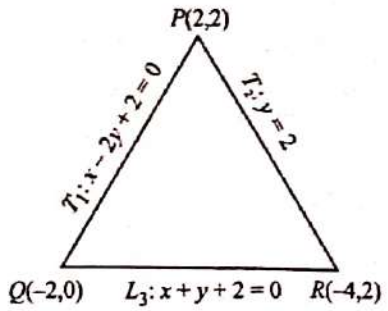
$\Rightarrow T_1 : x - 2y + 2 = 0$

T_2 is tangent to S_2 at P

$\Rightarrow T_2 : x.2 + y \cdot 2 = 2(x + 2)$

$\Rightarrow T_2 : y = 2$

& $L_3 : x + y + 2 = 0$ is third line



$PQ = a = \sqrt{20}$

$QR = b = \sqrt{8}$

$RP = c = 6$

$$\text{Area}(\Delta PQR) = \Delta = \frac{1}{2}\begin{vmatrix} 2 & 2 & 1 \\ -2 & 0 & 1 \\ -4 & 2 & 1 \end{vmatrix}$$

$= \frac{1}{2} \times 6 \times 2 = 6$ sq. units

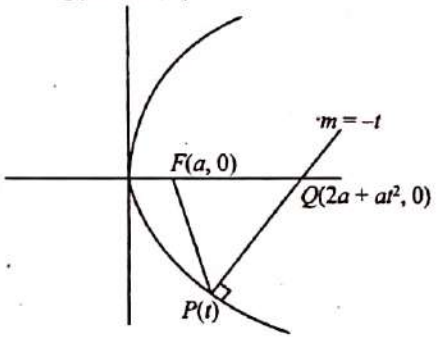
$\therefore r = \frac{abc}{4\Delta} = \frac{\sqrt{160}}{4} = \sqrt{10} \Rightarrow r^2 = 10$

9. **(a)** Let point $P(at^2, 2at)$

normal at P is $y = -tx + 2at + at^3$

$y = 0, x = 2a + at^2$

$Q(2a + at^2, 0)$



Area of triangle $= \frac{1}{2} \times$ base $\times$ height

Area of $\Delta PFQ = \left|\frac{1}{2}\left(a + at^2\right)(2at)\right| = 120$ sq. units

$\because m = -t$

$\because a^2[1 + m^2]m = 120, (a, m) = (2, 3)$ will satisfy

10. **(c)** $y = x - x^2 = -\left(x - \frac{1}{2}\right)^2 + \frac{1}{4}$

$$\therefore \quad y - \frac{1}{4} = -\left(x - \frac{1}{2}\right)^2$$

So, vertex V is $\left(\frac{1}{2}, \frac{1}{4}\right)$.

Now, line $y = 4 + kx$ touches the parabola.

Solving line and parabola, we get

$4 + kx = x - x^2$

$\therefore x^2 + (k - 1)x + 4 = 0$

This equation must have equal roots.

$\therefore (k - 1)^2 - 16 = 0 \Rightarrow k = 5$ (as $k > 0$)

For $k = 5$, $x = -2$ and $y = -6$.
So, point P is $(-2, -6)$.

$\therefore$ Slope of PV is $= \dfrac{-6-\frac{1}{4}}{-2-\frac{1}{2}} = \dfrac{5}{2}$.

11. **[9]** Both, the parabola and the circle, are symmetrical about x-axis.
So, common tangents will also be symmetrical about x-axis.

Parabola is $y^2 = -\dfrac{x}{2}$.

Equation of tangent having slope m is $y = mx - \dfrac{1}{8m}$.

This tangent passes through the point $(2, 0)$.

$\therefore m = \pm\dfrac{1}{4}$

So, one of the tangents is $x - 4y - 2 = 0$.
This line touches circle also.

$\therefore \dfrac{|5-0-2|}{\sqrt{1+16}} = \dfrac{3}{\sqrt{17}} = \sqrt{r}$

$\Rightarrow 17r = 9$

12. **[10]** Equation of tangent to parabola $75x^2 = 64(5y - 3)$ at $\left(\dfrac{8}{5}, \dfrac{6}{5}\right)$ is

$$75x.\frac{8}{5} = 64\left(5.\frac{y+\frac{6}{5}}{2} - 3\right)$$

or $3x - 4y = 0$
Equation of family of circles touching the line $3x - 4y = 0$ at $\left(\dfrac{8}{5}, \dfrac{6}{5}\right)$ is

$$\left(x-\frac{8}{5}\right)^2 + \left(y-\frac{6}{5}\right)^2 + \lambda(3x-4y) = 0$$

$$\text{or } x^2 + y^2 + x\left(3\lambda - \frac{16}{5}\right) + y\left(-4\lambda - \frac{12}{5}\right) + 4 = 0$$

Circle touches y-axis.
Putting $x = 0$, we get

$$y^2 + y\left(-4\lambda - \frac{12}{5}\right) + 4 = 0$$

This equation must have equal roots.

$$\therefore \left(4\lambda + \frac{12}{5}\right)^2 = 16 \qquad [\because D = 0]$$

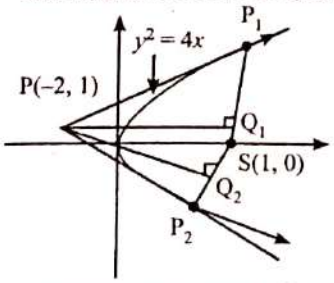
$$\Rightarrow \lambda = \frac{2}{5} \text{ or } -\frac{8}{5}$$

For, $\lambda = \dfrac{2}{5}, r = 1$

For $\lambda = -\dfrac{8}{5}, r = 4$

So, sum of diameters $= 2(1 + 4) = 10$

13. **(b,c,d)**
Let a rough graph to refer the question



Let parametric point at $P_1(t^2, 2t)$ then tangent at P_1 $ty = x + t^2$
Since it passes through $(-2, 1)$
$\therefore t^2 - t - 2 = 0$
$\therefore t = 2, -1$
$\therefore P_1(4, 4)$ and $P_2(1, -2)$
$\therefore SP_1 : 4x - 3y - 4 = 0$
and $SP_2 : x - 1 = 0$
and for Q_1 :

$$\frac{x_1+2}{4} = \frac{y_1-1}{-3} = \frac{-(-8-3-4)}{25} = \frac{3}{5}$$

$$\therefore x_1 = \frac{2}{5}, y_1 = \frac{-4}{5}$$

and $Q_2 = (1, 1)$s

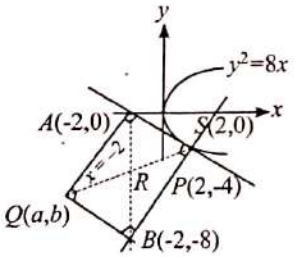
$$\text{So, } SQ_1 = \sqrt{\left(1-\frac{2}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 1$$

$$Q_1Q_2 = \sqrt{\frac{9}{25} + \frac{81}{25}} = \sqrt{\frac{90}{25}} = \frac{3\sqrt{10}}{5}$$

$$PQ_1 = \sqrt{\frac{144}{25} + \frac{81}{25}} = \frac{15}{5} = 3$$

$SQ_2 = 1$

14. **(c)** Equation of tangent at $P\ (2, -4)$:
$y(-4) = 4(x + 2)$
$\Rightarrow x + y + 2 = 0$ ...(i)
So, point of intersection of equ (i) and $x = -2$ is $A\ (-2, 0)$



Equation of normal at P :
$y + 4 = 1\ (x - 2)$
$x - y = 6$ ...(ii)
So, point of intersection of equ (ii) and $x = -2$ is $B\ (-2, -8)$
For square, mid-point of AB = mid-point of PQ

$$\Rightarrow \frac{a+2}{2} = -2 \Rightarrow a = -6$$

$$\frac{b-4}{2} = \frac{-8}{2} \Rightarrow b = -4$$

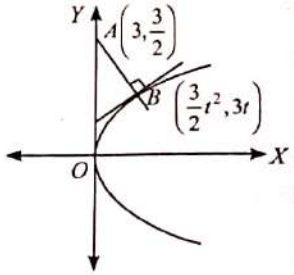
So, $2a + b = -16$

15. **[9]** Let equation of the normal AB is $y = -tx + 3t + \dfrac{3}{2}t^3$ and it is passing through

$$A\left(3, \frac{3}{2}\right) \therefore t = 1$$

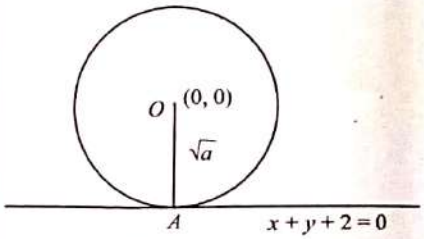
$$\therefore B\left(\frac{3}{2}, 3\right) \Rightarrow \alpha = \frac{3}{2} \text{ and } \beta = 3$$

$2(\alpha + \beta) = 9$



16. **[2]** The tangent to the parabola $y^2 = 8x$ at $(2, 4)$ is $-4y = 4(x + 2)$
$x + y + 2 = 0$

$$OA = \sqrt{a} \Rightarrow \left|\frac{0+0+2}{\sqrt{2}}\right| = \sqrt{a}$$



$\sqrt{2} = \sqrt{a}$
$a = 2$

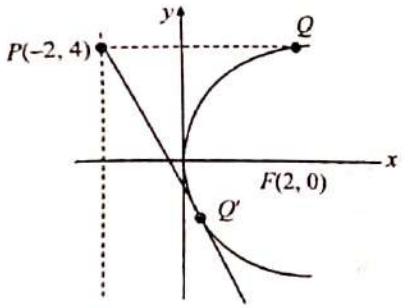
17. **(a, b, d)**
$E = y^2 - 8x = 0$
$a = 2$
Let $Q(2t_1^2, 4t_1), Q'(2t_2^2, 4t_2)$
$t_1t_2 = -1, t_1 + t_2 = 2$
$t_1 = 1 + \sqrt{2}, t_2 = 1 - \sqrt{2}$
(a) (slope of PF) (slope of FQ) $= -1$

$$\Rightarrow \angle PFQ = \frac{\pi}{2}$$

(b) (slope of PQ') (slope of PQ) $= -1$

$$\angle QPQ' = \frac{\pi}{2}$$

(c) $PF = 4\sqrt{2}$
(d) slope of $Q'F$ = slope of FQ
$\Rightarrow Q, F, Q'$ are collinear



18. (*a*) Let $\left(\frac{1}{2},-2\right)$ is $(2t^2, 4t)$

$\Rightarrow \quad t=\frac{-1}{2}$

Parameter of other end of focal chord is 2 $[\because t_1t_2=-1]$

$\Rightarrow$ point is (8, 8)

$\Rightarrow$ equation of tangent is $8y-4(x+8)=0$

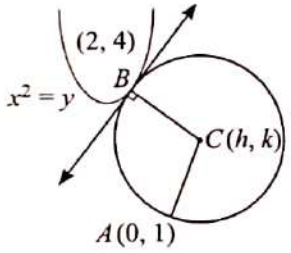
$\Rightarrow 2y-x=8$

19. (*c*) Let $c(h, k)$ be the centre of circle touching $x^2 = y$ at $B(2, 4)$

Equation of common tangent at B is

$2x=\frac{1}{2}(y+4) \Rightarrow 4x-y=4$

Since, radius is perpendicular to this $x^2 = y$ tangent



$\therefore 4\left(\frac{k-4}{h-2}\right)=-1 \Rightarrow 4k=18 \quad ...(i)$

Also, $AC = BC \Rightarrow h^2 + (k-1)^2 = (h-2)^2 + (k-4)^2$

$\Rightarrow 4h + 6k = 19 \quad ...(ii)$

On solving, centre $(h, k) = \left(\frac{-16}{5}, \frac{53}{10}\right)$

20. **[0.5]** The given curve $y = (x-1)(x-2)$, intersects the x-axis at $A(1,0)$ and $B(2,0)$

$\therefore \frac{dy}{dx}=2x-3; \left(\frac{dy}{dx}\right)_{(x=1)}=-1$ and $\left(\frac{dy}{dx}\right)_{(x=2)}=1$

Equation of tangent at A (1,0)

$y = -1(x-1) \Rightarrow x + y = 1$

Equation of tangent at $B(2, 0)$

$y = 1(x-2) \Rightarrow x - y = 2$

So, $a = 1$ and $b = 2$

$\Rightarrow \frac{a}{b}=\frac{1}{2}=0.5$

21. (*c*) Give $y^2 = 4x \quad ...(i)$

and $x^2 + y^2 = 5 \quad ...(ii)$

by (*i*) and (*ii*)

$x^2 + 4x = 5$

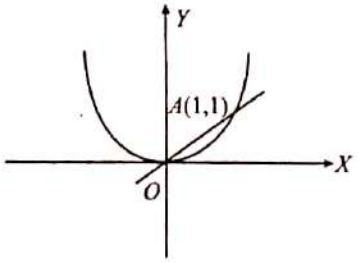
$\Rightarrow x = 1$ or -5 (Neglecting $x = -5$)

$\Rightarrow \quad x = 1$ and $y = 2$

Equation of tangent at (1,2) to $y^2 = 4x$ is $y = x + 1$

22. (*c*) Put $x - 2 = X$ & $y + 1 = Y$

$\therefore$ given curve becomes $Y = X^2$ and $Y = X$



tangent at origin is X-axis

and tangent at $A(1, 1)$ is $Y + 1 = 2X$ $\left[\because \frac{y+1}{2}=x\cdot 1\right]$

$\therefore$ there intersection is $\left(\frac{1}{2},0\right) \equiv (X, 0)$

$\therefore x-2=\frac{1}{2}$ & $y+1=0$

therefore $x=\frac{5}{2}, y=-1$

23. (*c*) $x^2 = 8y \quad \Rightarrow \quad \frac{dy}{dx}=\frac{x}{4}=\tan\theta$

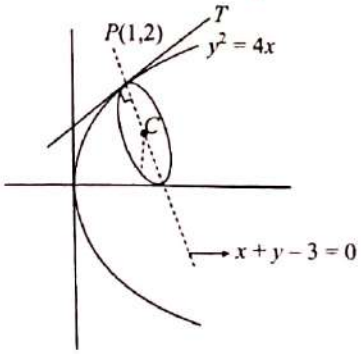
$\therefore x_1 = 4\tan\theta, y_1 = 2\tan^2\theta$

Equation of tangent: $y - 2\tan^2\theta = \tan\theta(x - 4\tan\theta)$

$\Rightarrow x = y\cot\theta + 2\tan\theta$

24. (*b*) Tangent at (1, 2) on $y^2 = 4x$

$\Rightarrow y(2) = 2(x+1) \Rightarrow y = x + 1$



Equation of circle is

$(x-1)^2 + (y-2)^2 + \lambda(x - y + 1) = 0$

$\Rightarrow x^2 + y^2 + x(\lambda - 2) + y(-4 - \lambda) + (5 + \lambda) = 0$

As circle touches x-axis then $g^2 - c = 0$

$\frac{(\lambda-2)^2}{4}=(5+\lambda)$

$\Rightarrow \lambda^2 + 4 - 4\lambda = 20 + 4\lambda \Rightarrow \lambda^2 - 8\lambda - 16 = 0$

$\lambda = 4 \pm 4\sqrt{2}$

Put λ and get least radius which will be given by $\lambda = 4 - 4\sqrt{2}$

$r=\frac{4+\lambda}{2}$

$r = 2 + 2 - 2\sqrt{2}$

$r = 4 - 2\sqrt{2}$

Area $= \pi r^2$

$= 8\pi\left(3-2\sqrt{2}\right)$

25. (*d*) Given parabola $x = 4y^2$

$\Rightarrow y^2=\frac{x}{4} \Rightarrow y^2=4\cdot\frac{1}{16}x$

$\therefore a=\frac{1}{16}$

Equation of normal

$y = -tx + 2at + at^3$

$\Rightarrow y=-tx+\frac{2}{16}t+\frac{1}{16}t^3$

It passes through (0, 33)

$\Rightarrow 33=\frac{t}{8}+\frac{t^3}{16} \Rightarrow t^3 + 2t - 528 = 0$

$\Rightarrow (t-8)(t^2 + 8t + 66) = 0$

$\therefore t = 8$

$P\left(at^2, 2at\right)=\left(\frac{1}{16}\times 64, 2\times\frac{1}{16}\times 8\right)$

$=(4,1)=(h,k)$

Parabola:

$y^2 = 4(x + y)$

$\Rightarrow y^2 - 4y = 4x \Rightarrow (y-2)^2 = 4(x+1)$

Equation of directix

$x + 1 = -1 \Rightarrow x + 2 = 0$

Distance of point (4, 1) from $x + 2 = 0$

$=\left|\frac{4+2}{1}\right|=6$

26. **[45]** $f'(x)=\frac{4x^2-1}{x}$,

So $f(x)$ is decreasing in $\left(0,\frac{1}{2}\right]$ and increasing in $\left(\frac{1}{2},\infty\right) \Rightarrow a=\frac{1}{2}$

Tangent at $y^2 = 2x$ is $y = mx + \frac{1}{2m}$ it is passing through (4, 3) therefore

$3 = 4m + \frac{1}{2m} \Rightarrow 8m^2 - 6m + 1 = 0$

$(4m - 1)(2m - 1) = 0$

$m=\frac{1}{2}$ or $\frac{1}{4}$

So tangent may be

$y=\frac{1}{2}x+1$ or $y=\frac{1}{4}x+2$ but $y=\frac{1}{2}x+1$

passes through (–2, 0) so rejected.

$P\left(\frac{a}{m^2},\frac{2a}{m}\right)\equiv\left(\frac{1\times 16}{2\times 1}, 2\times\frac{1}{2}\times\frac{4}{1}\right)\equiv(8,4)$

Normal

$y = -4x + \lambda$

$4 = -32 + \lambda \Rightarrow \lambda = 36$

Normal

$y = -4x + 36$

$\Rightarrow 4x + y = 36$

$\Rightarrow \frac{x}{9} + \frac{y}{36} = 1$

Equation of normal $\frac{x}{9} + \frac{y}{36} = 1$

$\therefore \quad \alpha + \beta = 9 + 36 = 45$

27. **(c)** The parabola: $\left(x - \frac{1}{2}\right)^2 = y - \frac{3}{4}$

$\Rightarrow y = x^2 - x + 1$ ...(i)

For $x = \frac{-1}{2}, y = \frac{7}{4}$, We get $P\left(-\frac{1}{2}, \frac{7}{4}\right)$

$\frac{dy}{dx} = 2x - 1$

$\left(\frac{dy}{dx}\right)_P = -2$

$\therefore m_T = -2, m_N = \frac{1}{2}$

Equation of Normal: $y - \frac{7}{4} = \frac{1}{2}\left(x + \frac{1}{2}\right)$

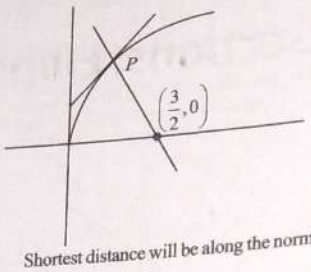
$\Rightarrow y = \frac{x}{2} + 2$ ...(ii)

From equ (i) & (ii) we get, $Q(2, 3)$

$PQ^2 = \frac{125}{16}$

28. **(a)** Let P be the point nearest to $\left(\frac{3}{2}, 0\right)$, then normal at P will pass through $\left(\frac{3}{2}, 0\right)$.

Let Co-ordinates of P be $\left(\frac{t^2}{4}, \frac{t}{2}\right)$



Shortest distance will be along the normal.

Hence equation of normal is $y + tx = \frac{t}{2} + \frac{t^2}{4}$

The line passes through $\left(\frac{3}{2}, 0\right)$

$\frac{3t}{2} = \frac{t}{2} + \frac{t^3}{4}$

$\Rightarrow \quad t = 2$ ($\{-2, 0\}$ are rejected)

Hence nearest point is $(1, 1)$ for $t = 2$.

distance $\sqrt{\left(\frac{3}{2} - 1\right)^2 + (1 - 0)^2} = \frac{\sqrt{5}}{2}$

29. **(a)** $y^2 = 24x$

On comparing with $y^2 = 4ax$, we get

$a = 6$

Given curve:

$xy = 2$

Tangent on Parabola

$AB \equiv ty = x + 6t^2$ ...(i)

$AB \equiv T = S_1$

chard with nud point for $xy = 2$

$kx + hy = 2hk$ ...(ii)

From (i) and (ii)

$\frac{k}{1} = \frac{h}{-t} = \frac{2hk}{-6t^2}$

$\Rightarrow$ then locus is $y^2 = -3x$

Therefore directrix is $4x = 3$

30. **(b)** Let (h, k) be the midpoint then,

$h = \frac{at^2 + a}{2}, k = \frac{2at + 0}{2}$

$\Rightarrow t^2 = \frac{2h - a}{a}$ and $t = \frac{k}{a}$

$\Rightarrow \frac{k^2}{a^2} = \frac{2h - a}{a}$

$\Rightarrow$ Locus of (h, k) is $y^2 = a(2x - a)$

$\Rightarrow y^2 = 2a\left(x - \frac{a}{2}\right)$

Its directrix is $x - \frac{a}{2} = -\frac{a}{2} \Rightarrow x = 0$

By Aditya Mohod
username - maje_kabhi_dekha_nahi

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 13

Conic Sections: Ellipse

Prarambh (Topicwise)

1. (*b*) $ae = 2 \Rightarrow a = \frac{2}{e} = \frac{2}{1/2} = 4$

$b^2 = a^2(1-e^2) = 16(1-1/4)$

Now equation is $\frac{x^2}{16} + \frac{y^2}{16\left(1-\frac{1}{4}\right)} = 1$

i.e. $\frac{x^2}{16} + \frac{y^2}{12} = 1$

2. (*b*) $9x^2 + 5(y^2 - 6y + 9) = 45$

$\Rightarrow \frac{x^2}{5} + \frac{(y-3)^2}{9} = 1$

$a^2(1-e^2) = b^2$

$\Rightarrow 9(1-e^2) = 5$

$\Rightarrow 1 - e^2 = \frac{5}{9} \Rightarrow e^2 = \frac{4}{9} \Rightarrow e = \frac{2}{3}$

3. (*c*) $a = 6, b = 2\sqrt{5}$

$b^2 = a^2(1-e^2)\frac{20}{36} = (1-e^2)$

$\Rightarrow e = \sqrt{\frac{16}{36}} = \frac{2}{3}$

But directrices are $x = \pm\frac{a}{e}$

Hence distance between them is $2 \cdot \frac{6}{2/3} = 18$.

4. (*b*) $\frac{x^2}{(48/3)} + \frac{y^2}{(48/4)} = 1$

$a^2 = 16, b^2 = 12 \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{1}{2}$

Distance is $2ae = 2 \cdot 4 \cdot \frac{1}{2} = 4$.

5. (*b*) Vertex (0,7), directrix $y = 12$, $\therefore b = 7$

Also $\frac{b}{e} = 12 \Rightarrow e = \frac{7}{12}, a = 7\sqrt{\frac{95}{144}}$

Hence equation of ellipse is

$144x^2 + 95y^2 = 4655$

6. (*a*) The equation of the ellipse is $16x^2 + 25y^2 = 400$

or $\frac{x^2}{25} + \frac{y^2}{16} = 1$

Here $a^2 = 25, b^2 = 16 \Rightarrow e = \frac{3}{5}$.

Hence the foci are $(\pm 3, 0)$

7. (*a*) Let point $P(x_1, y_1)$

So, $\sqrt{(x_1+2)^2 + y_1^2} = \frac{2}{3}\left(x_1 + \frac{9}{2}\right)$

$\Rightarrow (x_1+2)^2 + y_1^2 = \frac{4}{9}\left(x_1 + \frac{9}{2}\right)^2$

$\Rightarrow 9[x_1^2 + y_1^2 + 4x_1 + 4] = 4\left(x_1^2 + \frac{81}{4} + 9x_1\right)$

$\Rightarrow 5x_1^2 + 9y_1^2 = 45 \Rightarrow \frac{x_1^2}{9} + \frac{y_1^2}{5} = 1,$

Locus of (x_1, y_1) is $\frac{x^2}{9} + \frac{y^2}{5} = 1$, which is equation of an ellipse.

8. (*c*) In the first case, eccentricity $e = \sqrt{1-(25/169)}$

In the second case, $e' = \sqrt{1-(b^2/a^2)}$

According to the given condition,

$\sqrt{1-b^2/a^2} = \sqrt{1-(25/169)}$

$\Rightarrow b/a = 5/13, \ (\because a > 0, b > 0)$

$\Rightarrow a/b = 13/5.$

9. (*b*) $4(x-2)^2 + 9(y-3)^2 = 36$

Hence the centre is (2, 3).

10. (*a*) The ellipse is

$4(x-1)^2 + 9(y-2)^2 = 36$

Therefore, latus rectum $= \frac{2b^2}{a} = \frac{2.4}{3} = \frac{8}{3}$

11. (*b*) Foci $= (3,-3) \Rightarrow ae = 3 - 2 = 1$

$= (4,-3) \Rightarrow a = 4 - 2 = 2 \Rightarrow e = \frac{1}{2}$

$\Rightarrow b = a\sqrt{\left(1-\frac{1}{4}\right)} = \frac{2}{2}\sqrt{3} = \sqrt{3}$

Therefore, equation of ellipse with centre (2, –3) is

$\frac{(x-2)^2}{4} + \frac{(y+3)^2}{3} = 1$

12. (*b*) Check $\Delta \neq 0$ and $h^2 < ab$.

13. (*a*) $\frac{(x+1)^2}{\frac{225}{25}} + \frac{(y+2)^2}{\frac{225}{9}} = 1$

$a = \sqrt{\frac{225}{25}} = \frac{15}{5}, b = \sqrt{\frac{225}{9}} = \frac{15}{3}$

$\Rightarrow e = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$

Focus $= \left(-1, -2 \pm \frac{15}{3} \cdot \frac{4}{5}\right) = (-1, -2 \pm 4)$

$= (-1, 2); (-1, -6).$

14. (*c*) $3x^2 - 12x + 4y^2 - 8y = -4$

$\Rightarrow 3(x-2)^2 + 4(y-1)^2 = 12$

$\Rightarrow \frac{(x-2)^2}{4} + \frac{(y-1)^2}{3} = 1$

$\Rightarrow \frac{X^2}{4} + \frac{Y^2}{3} = 1$

$\therefore e = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}.$

$\therefore$ Foci are $\left(X = \pm 2 \times \frac{1}{2}, Y = 0\right)$

i.e., $(x - 2 = \pm 1, \ y - 1 = 0) = (3,1)$ and $(1,1)$.

15. (*c*) Given equation of ellipse is,

$25x^2 + 9y^2 - 150x - 90y + 225 = 0$

$\Rightarrow 25(x-3)^2 + 9(y-5)^2 = 225$

$\Rightarrow \frac{(x-3)^2}{9} + \frac{(y-5)^2}{25}$

$= 1$. Here $b > a$

$\therefore$ Eccentricity

$e = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$

16. (*c*) Coordinates of a any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose eccentric angle is θ are $(a\cos\theta, b\sin\theta)$.

The coordinates of the end points of latus rectum are and $\left(ae, \pm\frac{b^2}{a}\right)$.

$\therefore a\cos\theta = ae$ and $b\sin\theta = \pm\frac{b^2}{a}$

$\Rightarrow \tan\theta = \pm\frac{b}{ae} \Rightarrow \theta = \tan^{-1}\left(\pm\frac{b}{ae}\right)$

17. (*b*) $\because ae = \pm\sqrt{5} \Rightarrow a = \pm\sqrt{5}\left(\frac{3}{\sqrt{5}}\right) = \pm 3$

$\Rightarrow a^2 = 9$

$\therefore b^2 = a^2(1-e^2) = 9\left(1 - \frac{5}{9}\right) = 4$

By Aditya Mohoa
username - muje_kabhi

Hence, equation of ellipse

$\frac{x^2}{9}+\frac{y^2}{4}=1 \Rightarrow 4x^2+9y^2=36$

18. (*b*) $(4x-8)^2+16y^2=(x+\sqrt{3}y+10)^2$

$\Rightarrow (x-2)^2+y^2=\left(\frac{1}{2}\right)^2\frac{(x+\sqrt{3}y+10)^2}{4}$

Focus is $(h,k)=(2,0)$, $e=\frac{1}{2}$ and directrix

$x+\sqrt{3}y+10=0$

$\frac{a}{e}-ae$ = perpendicular distance from $(2,0)$ to directrix $x+\sqrt{3}y+10=0$

$\Rightarrow 2a-\frac{a}{2}=\frac{|2+0+10|}{2}$

$\Rightarrow \frac{3a}{2}=6 \Rightarrow a=4$

$\therefore$ Distance between directrix $=\frac{2a}{e}=\frac{2.4}{1/2}=16$

19. (*a*) Centre is $(3,0)$, $a=8$,

$b=\sqrt{64\left(1-\frac{1}{4}\right)}=4\sqrt{3}$

Now $x=3+8\cos\theta$

$y=4\sqrt{3}\sin\theta$

$(3+8\cos\theta,\ 4\sqrt{3}\sin\theta)$

20. (*b*) $(x-2)^2+9\left(y+\frac{1}{3}\right)^2=1$

$(x-2)^2+\frac{\left(y+\frac{1}{3}\right)^2}{\left(\frac{1}{3}\right)^2}=1$

Let, $x=2+1\cos\theta$

$y=-\frac{1}{3}+\frac{1}{3}\sin\theta$

So $(4x-9y)=4(2+\cos\theta)-9\left(-\frac{1}{3}+\frac{1}{3}\sin\theta\right)$

$=8+3+4\cos\theta-3\sin\theta=11+5=16$

21. (*a*) Since $S_1>0$. Hence the point is outside the ellipse.

22. (*b*) $y=3x\pm\sqrt{\frac{3.5}{3.4}\times 9+\frac{5}{3}\times\frac{4}{4}}$

$\Rightarrow y=3x\pm\sqrt{\frac{155}{12}}$

23. (*a*) From the given options it can the easily said

Alternative:

$\frac{x^2}{16}+\frac{y^2}{9}=1$

As pair of lines of $T^2=SS_1$

$\left(\frac{x}{8}+\frac{y}{3}-1\right)^2=\left(\frac{x^2}{16}+\frac{y^2}{9}-1\right)\left(\frac{1}{4}+1-1\right)$

$\Rightarrow \frac{x^2}{64}+\frac{y^2}{9}+1-\frac{x}{4}-\frac{2y}{3}+\frac{xy}{12}$

$=\frac{x^2}{64}+\frac{y^2}{36}-\frac{1}{4}$

$\Rightarrow \frac{y^2}{12}-\frac{2y}{3}-\frac{x}{4}+\frac{xy}{12}+\frac{5}{4}=0$

$\Rightarrow (y-3)(x+y-5)=0$

24. (*d*) By symmetry the quadrilateral is a rhombus. So area is four times the area of the right angled triangle formed by the tangent and axes in the Ist quadrant. Now,

$ae=\sqrt{a^2-b^2} \Rightarrow ae=2$

$\Rightarrow$ Tangent (in first quadrant) at end of latus rectum $\left(2,\frac{5}{3}\right)$ is $\frac{2}{9}x+\frac{5}{3}\frac{y}{5}=1$

i.e., $\frac{x}{9/2}+\frac{y}{3}=1$

Area $=4.\frac{1}{2}.\frac{9}{2}.3=27$ sq. unit.

25. (*a*) $y=\frac{-l}{m}x+\frac{n}{m}$ is tangent to

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$,

if $\frac{n}{m}=\pm\sqrt{b^2+a^2\left(\frac{l}{m}\right)^2}$

or $n^2=m^2b^2+l^2a^2$.

26. (*c*) $SS_1=T^2$

$\tan\theta=2\frac{\sqrt{h^2-ab}}{a+b}$, $a=9$, $b=-4$ and $h=-12$.

27. (*c*) The locus of point of intersection of two perpendicular tangents drawn on the ellipse is $x^2+y^2=a^2+b^2$, which is called 'director-circle'.

Given ellipse is $\frac{x^2}{9}+\frac{y^2}{4}=1$,

$\therefore$ Locus is $x^2+y^2=13$.

28. (*c*) Change the equation

$9x^2+5y^2-30y=0$ in standard form

$9x^2+5(y^2-6y)=0$

$\Rightarrow 9x^2+5(y^2-6y+9)=45$

$\Rightarrow \frac{x^2}{5}+\frac{(y-3)^2}{9}=1$

$\because a^2<b^2$, so axis of ellipse on y-axis.

At y axis, put $x=0$, so we can obtained vertex.

Then $0+5y^2-30y=0 \Rightarrow y=0, y=6$

Therefore, tangents of vertex $y=0, y=6$.

29. (*d*) For $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, equation of normal at point (x_1,y_1),

$\Rightarrow \frac{(x-x_1)a^2}{x_1}=\frac{(y-y_1)b^2}{y_1}$

$\therefore (x_1,y_1)\equiv(0,3)$, $a^2=5$, $b^2=9$

$\Rightarrow \frac{(x-0)}{0}5=\frac{(y-3).9}{3}$ or $x=0$ i.e., y-axis.

30. (*a*) Given, equation of ellipse is $4x^2+9y^2=36$

Tangent at point $(3,-2)$ is $\frac{(3)x}{9}+\frac{(-2)y}{4}=1$

or $\frac{x}{3}-\frac{y}{2}=1$

$\therefore$ Normal is and it passes through point $(3,-2)$

$\therefore \frac{3}{2}-\frac{2}{3}=k \Rightarrow k=\frac{5}{6}$

$\therefore$ Normal is, $\frac{x}{2}+\frac{y}{3}=\frac{5}{6}$

31. (*a*) We know that the equation of the normal at point $(a\sin\theta, b\cos\theta)$ on the curve $x^2+\frac{y^2}{4}=1$ is given by

$-\frac{ax}{\sin\theta}+\frac{by}{\cos\theta}=-a^2+b^2$

$\Rightarrow -\frac{1\cdot x}{\sin\theta}+\frac{2y}{\cos\theta}=3$ ...(*i*)

Comparing equation (*i*) with $2x-\frac{8}{3}\lambda y=-3$. We get,

$-\frac{1}{2\sin\theta}=-\frac{2\cdot 3}{8\lambda\cos\theta}=-\frac{3}{3}$

$\Rightarrow \sin\theta=\frac{1}{2}$ and $\cos\theta=\frac{3}{4\lambda}$

$\Rightarrow \pm\frac{\sqrt{3}}{2}=\frac{3}{4\lambda}$

$\Rightarrow \lambda=\pm\frac{\sqrt{3}}{2}$

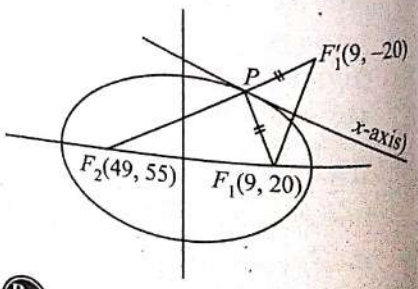
32. (*c*) $\frac{(3x-4y-2)^2}{100}+\frac{(4x-3y+5)^2}{625}=1$

$\Rightarrow \frac{\left(\frac{3x+4y-2}{5}\right)^2}{4}+\frac{\left(\frac{4x-3y+5}{5}\right)^2}{25}=1$

$\Rightarrow a^2=4$ and $b^2=25$

Radius of director circle $=\sqrt{a^2+b^2}=\sqrt{29}$

33. (*a*)



By Aditya Mohod
username - muje_kabhi_dekha_nahi

Image of F_1 in tangent (x-axis) is F_1'. From the property, points F_2, P, F_1' are collinear.
$2a = F_1P + F_2P = F_1'P + F_2P = F_1'F_2 = 85$

34. (*b*) Let the equation of the ellipse be

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Equation of a normal at any point $P(\theta)$ on the ellipse is $(a \sec\theta)\, x - (b \operatorname{cosec}\theta)\, y = a^2e^2$

Let normal is passing through a point $A(h, k)$
$\Rightarrow (ah \sec\theta - a^2e^2)^2 = (bk \operatorname{cosec}\theta)^2$
$\Rightarrow a^2h^2 \sec^2\theta - 2a^3e^2h \sec\theta + a^4e^4$

$\Rightarrow b^2k^2 \operatorname{cosec}^2\theta = b^2k^2\left(\frac{\sec^2\theta}{\sec^2\theta - 1}\right)$

$\Rightarrow a^2h^2 \sec^4\theta - 2a^3e^2h \sec^3\theta + (a^4e^4 - a^2h^2 - b^2k^2)$
$\sec^2\theta + 2a^3e^2h \sec\theta - a^4e^4 = 0$ ...(*i*)

If α, β, γ, and δ be the roots of the above equation, then

$\Sigma \sec\alpha = \frac{2a^3e^2h}{a^2h^2} = \frac{2ae^2}{h}$

Multiplying equation (*i*) by $\cos^4\theta$, it reduces to
$a^4e^4\cos^4\theta - 2a^3e^2h \cos^3\theta - (a^4e^4 - a^2gh^2 - b^2k^2) \cos^2\theta + 2a^3e^2\, h \cos\theta - a^2h^2 = 0$ ...(*ii*)

Then $\Sigma \cos\alpha = \frac{2a^3e^2h}{a^4e^4} = \frac{2h}{ae^2}$

Hence, we have

$(\Sigma \sec\alpha)(\Sigma \cos\alpha) = \frac{2ae^2}{h} \cdot \frac{2h}{ae^2} = 4$

35. (*a*) Ellispe is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Let ellipse is horizontal

$\Rightarrow 2ae = 4\sqrt{5}$ ...(*i*)

and $\frac{2a}{e} = 5\sqrt{5}$ ...(*ii*)

Multiply both (*i*) and (*ii*)
$4a^2 = 100 \Rightarrow a^2 = 25$
(i) divide by (ii)

$\frac{2ae}{\frac{2a}{e}} = \frac{4}{5}$

$e^2 = \frac{4}{5} = 1 - \frac{b^2}{a^2}$

$\Rightarrow \frac{b^2}{a^2} = 1 - \frac{4}{5} = \frac{1}{5}$

$\Rightarrow b^2 = \frac{a^2}{5} = 5$

$\Rightarrow a = 5, \quad b = \sqrt{5}$

36. (*a*) $PS = ePM$

$\sqrt{(x-1)^2 + (y+1)^2} = \frac{1}{2}\left|\frac{x - y - 3}{\sqrt{1^2 + 1^2}}\right|$

Squaring, we have
$7x^2 + 7y^2 + 7 - 10x + 10y + 2xy = 0$

37. (*d*) $4x^2 + 9y^2 + 8x + 36y + 4 = 0$
$4(x^2 + 2x + 1) + 9[y^2 + 4y + 4] = 36$
$4(x+1)^2 + 9(y+2)^2 = 36$
$\frac{(x+1)^2}{9} + \frac{(y+2)^2}{4} = 1 \Rightarrow e = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$

38. (*c*) $2 \times \frac{a}{e} = 3 \times 2ae$

$e^2 = \frac{1}{3} \Rightarrow e = \frac{1}{\sqrt{3}}$

39. (*b*) $\frac{x^2}{r-2} + \frac{y^2}{5-r} = 1$

$r - 2 > 0 \Rightarrow r > 2,$
$5 - r > 0 \Rightarrow r < 5$
Taking intersection of both for ellipse
$2 < r < 5$

40. (*a*) $x = 3(\cos t + \sin t)$ $y = 4(\cos t - \sin t)$

$\Rightarrow \frac{x}{3} = \cos t + \sin t;\ \frac{y}{4} = \cos t - \sin t$

square & add $\frac{x^2}{9} + \frac{y^2}{16} = 2$

Ellipse Equation $\frac{x^2}{18} + \frac{y^2}{32} = 1$

41. (*c*) $4(x^2 - 4x + 4) + 9(y^2 - 6y + 9) = 36$
$4(x-2)^2 + 9(y-3)^2 = 36$

$\frac{(x-2)^2}{9} + \frac{(y-3)^2}{4} = 1.$

Equation of major axis $y = 3$.
Equation of minor axis $x = 2$

42. (*d*) $3x + 5x^2 = 15$

$\frac{x^2}{5} + \frac{y^2}{3} = 1$

Equation of director circle.
$x^2 + y^2 = 5 + 3 = 8$
clearly (2, 2) lies on it

here $\angle\theta = \frac{\pi}{2}$

43. (*b*) $(S_1F_1) \cdot (S_2F_2) = b^2 = 3$

44. (*b*) $2ae = \frac{2b^2}{a}$

$a^2e = b^2$

$e = \frac{b^2}{a^2} = 1 - e^2$

$e^2 + e - 1 = 0$

$e = \frac{-1 \pm \sqrt{5}}{2} \qquad (\because 0 < e < 1)$

$e = \frac{\sqrt{5} - 1}{2}$

45. (*d*) $\frac{x^2}{4} + \frac{y^2}{3} = 1$; $e^2 = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow e = \frac{1}{2}$

$\therefore 3h = 2\cos\theta \quad \Rightarrow \cos\theta = \frac{3h}{2}$

and $3k = \sqrt{3}\sin\theta \Rightarrow \sin\theta = \sqrt{3}k$

Now, on squaring and adding, we get

$1 = \frac{9x^2}{4} + 3y^2$ or $\frac{x^2}{\frac{4}{9}} + \frac{y^2}{\frac{1}{3}} = 1$

Now verify alternatives.

46. [9]

$(x-2)^2 + (y+3)^2 = \left(\frac{1}{2}\right)^2\left(\frac{3x - 4y + 7}{5}\right)^2$

is an ellipse, whose focus is (2, –3), directrix $3x - 4y + 7 = 0$ and eccentricity $\frac{1}{2}$.

Length of the perpendicular from the focus to the directrix is $\frac{3\times2 - 4\times(-3) + 7}{5} = 5$

so that $\frac{a}{e} - ae = 5 \Rightarrow 2a - \frac{a}{2} = 5 \Rightarrow a = \frac{10}{3}$

So length of the major axis is $\frac{20}{3}$

47. (*b*) $4 = 9(1 - e^2) \Rightarrow e = \sqrt{5}/3$

Distance between the directrices

$= \frac{2 \times \text{semi major axis}}{e} = \frac{2\times3\times3}{\sqrt{5}} = \frac{18}{\sqrt{5}}$

Prabal (JEE Main Level)

1. (*a*) $e = \frac{5}{8}$; $2ae = 10 \Rightarrow 2a = \frac{10}{e}$

$\Rightarrow 2a = 16$

Latus rectum $= \frac{2b^2}{a} = \frac{2a^2(1-e^2)}{a}$

$= 2a(1 - e^2) = 16$

$\Rightarrow \left(1 - \frac{25}{64}\right) = \frac{39}{4}$

2. (*c*) $F_1(3, 3)$; $F_2(-4, 4)$
$2ae = F_1F_2$

$2ae = \sqrt{(3+4)^2 + (3-4)^2}$

$2ae = 5\sqrt{2}$ ...(1)

mid point of P_1P_2 will be centre of ellipse

centre $\left(-\frac{1}{2}, \frac{7}{2}\right)$

Ellipse $\frac{\left(x + \frac{1}{2}\right)^2}{a^2} + \frac{\left(y - \frac{7}{2}\right)^2}{b^2} = 1$

Passing through origin $\frac{1}{4a^2} + \frac{49}{4b^2} = 1$ ...(2)

From (1) and (2) $e = \frac{5}{7}$

3. **(c)** $\frac{x^2}{64}+\frac{y^2}{16}=1$

4. **(c)** Equation of major axis: $x - y = \lambda$, passes through $(1, 1) \Rightarrow \lambda = 0$

$\therefore x - y = 0$

Hence, centre is (2, 2)

Now, CS = ae

$\Rightarrow \sqrt{2} = a \cdot \frac{1}{2} \Rightarrow a = 2\sqrt{2}$

Equation of directrix: $x + y = \lambda$

perpendicular from centre to directrix

$= \frac{a}{e}$

$\Rightarrow \left|\frac{\lambda-4}{\sqrt{2}}\right| = \frac{2\sqrt{2}}{1/2} = 4\sqrt{2}$

$\lambda - 4 = \pm 8$

$\lambda = 12$ or -4

$\therefore$ equation of directrix are $x + y - 12 = 0$ and $x + y + 4 = 0$

$\therefore$ combined equation: $(x + y - 12)(x + y + 4) = 0$

$(x + y)^2 - 8(x + y) - 48 = 0$

5. **(b)** Max. area $= \frac{1}{2} \times 2ae \times b$

$= \frac{1}{2} \times 2 \times 3 \times 4 = 12$

6. **(b)** Let $P(a\cos\theta, b\sin\theta)$

$OP = 2$

$\Rightarrow OP^2 = 4$

$\Rightarrow a^2\cos^2\theta + b^2\sin^2\theta = 4$

$\Rightarrow 6\cos^2\theta + 2\sin^2\theta = 4$

$\cos\theta = \pm\frac{1}{\sqrt{2}} \Rightarrow \theta = \pm\frac{\pi}{4}$

7. **(d)** $\frac{de}{dt} = -0.1$

$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{3}{4} = \frac{1}{4}$

$e = -0.1\,t + c \quad \Rightarrow \quad e = 1/2$

when $t = 0$, $e = 1/2$

$\Rightarrow c = 0.5$

$e = -0.1\,t + 0.5$

ellipse become auxiliary circle where $e \to 0$

$0 = -0.1\,t + 0.5 \Rightarrow t = 5$ sec.

8. **(b)** $M_{OP} = \frac{b\sin\theta_1}{a\cos\theta_1} = \frac{b}{a}\tan\theta_1$

$M_{OQ} = \frac{b}{a}\tan\theta_2$

$M_{OP} \times M_{OQ} = \frac{b^2}{a^2}\tan\theta_1\tan\theta_2$

$= \left(\frac{b^2}{a^2}\right)\left(\frac{-a^2}{b^2}\right) = -1$

So right angle at entre.

9. **(b)** Let eccentric angle be θ, then equation of tangent is

$\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$ ...(1)

given equation is

$\frac{x}{a} + \frac{y}{b} = \sqrt{2}$ ...(2)

comparing (1) and (2)

$\cos\theta = \sin\theta = \frac{1}{\sqrt{2}}$

$\Rightarrow \theta = 45°$

10. **(b)** $C = \pm\sqrt{8\times4+4} = \pm 6$

11. **(d)** $3x^2 + 4y^2 = 1$

$3xx_1 + 4yy_1 = 1$

given $3x + 4y = -\sqrt{7}$

comparing

$\because \frac{3x_1}{3} = \frac{4y_1}{4} = \frac{1}{-\sqrt{7}}$

$x_1 = -\frac{1}{\sqrt{7}}$

$y_1 = -\frac{1}{\sqrt{7}}$

12. **(d)** Equation of normal

$ax\sec\phi - by\,\mathrm{cosec}\phi = a^2 - b^2$ ...(1)

$x\cos\alpha + 4\sin\alpha = p$ ...(2)

$\frac{a\sec\phi}{\cos\alpha} = \frac{-by\,\mathrm{cosec}\,\phi}{\sin\alpha} = \frac{a^2-b^2}{p}$

$\Rightarrow \cos\phi = \frac{ap}{(a^2-b^2)} \times \sec\alpha$ ...(3)

$\Rightarrow \sin\phi = \frac{-bp}{(a^2-b^2)} \times \mathrm{cosec}\alpha$ ...(4)

squaring and adding

$1 = \frac{p^2}{(a^2-b^2)^2}[a^2\sec^2\alpha + b^2\mathrm{cosec}^2\alpha]$

13. **(a)** $\frac{x^2}{16}+\frac{y^2}{9}=1$

Let the point $P(4\cos\theta, 3\sin\theta)$

Tangent at P

$\frac{x}{4}\cos\theta + \frac{y}{3}\sin\theta = 1$

$A\left(\frac{4}{\cos\theta}, 0\right); B\left(0, \frac{3}{\sin\theta}\right)$

Let the middle point $M(h, k)$

$2h = \frac{4}{\cos\theta} \Rightarrow \cos\theta = \frac{2}{h}$

$2k = \frac{3}{\sin\theta} \Rightarrow \sin\theta = \frac{3}{2k}$

square & add

$\frac{4}{h^2} + \frac{9}{4k^2} = 1$

$16k^2 + 9h^2 = 4h^2k^2$

$16y^2 + 9x^2 = 4x^2y^2$

14. **(b)** $y = mx \pm \sqrt{(a^2+b^2)m^2+b^2}$ ...(1)

$y = mx \pm \sqrt{a^2m^2 + (a^2+b^2)}$ ...(2)

Eq^n (1) and (2) are same

$(a^2 + b^2)m^2 + b^2 = a^2m^2 + a^2 + b^2$

$m^2 = a^2/b^2 \quad \Rightarrow \quad m = \pm a/b$

$\Rightarrow by = ax \pm \sqrt{a^4 + b^4 + a^2b^2}$

15. **(a)** Tangent to $y^2 = 4x$ is $y = mx + \frac{1}{m}$ ...(i)

and tangent to $\frac{x^2}{4}+\frac{y^2}{3}=1$ will be

$y = mx \pm \sqrt{4m^2+3}$ ...(ii)

$\because$ (i) and (ii) represent same line for common tangent $\frac{1}{m} = \sqrt{4m^2+3}$

$\Rightarrow 1 = 4m^4 + 3m^2 \Rightarrow 4m^4 + 3m^2 - 1 = 0$

$\Rightarrow (4m^2 - 1)(m^2 + 1) = 0$

$\Rightarrow m^2 = \frac{1}{4} \Rightarrow m = \pm\frac{1}{2}$

$\therefore$ Common tangent will be:

$y = \frac{1}{2}x + 2 \Rightarrow x - 2y + 4 = 0$

$y = \frac{-1}{2}x - 2 \Rightarrow x + 2y + 4 = 0$

16. **(b)** Equation of normal

$\frac{a^2x}{ae} - \frac{b^2ya}{b^2} = a^2 - b^2$

$\frac{ax}{e} - ay = a^2 - b^2$

$x - ey = ae^3$

17. **(c)** $\frac{x}{a}\cos\phi + \frac{y}{b}\sin\phi = 1$ ...(1)

$x^2 + y^2 = a^2$

$ax\cos\phi + ay\sin\phi = a^2$

$x\cos\phi + y\sin\phi = a$

$\frac{x}{a}\cos\phi + \frac{y}{b}\sin\phi = 1$ ...(2)

Solving (1) and (2) $y = 0$

18. **(c)** Let the mid-point of chord be (α, β), so its equation is $\alpha x + \frac{\beta y}{7} = \alpha^2 + \frac{\beta^2}{7}$

Now, on comparing with $7x + y = 20$, we get

$\frac{\alpha}{7} = \frac{\beta/7}{1} \Rightarrow \alpha = \beta$

So, the required line be $y = \frac{\beta x}{\alpha} \Rightarrow y = x$

19. (b) $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$

$\text{slope} = \dfrac{a\sec\theta}{b\operatorname{cosec}\theta} = \dfrac{5}{3}$

$\dfrac{\sec\theta}{\operatorname{cosec}\theta} = 1$

$\tan\theta = 1 \Rightarrow \theta = \dfrac{\pi}{4}$

20. (c) $P(a\cos\theta, b\sin\theta)$

Normal at P; $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$

$3x\sec\theta - y\operatorname{cosec}\theta = 8$ ...(i)

Equation of diameter is given as (as it is passing through origin (0, 0))

$(y-0) = \dfrac{-\sin\theta}{3\cos\theta}(x-0)$

$y = \dfrac{-x\sin\theta}{3\cos\theta}$ ...(ii)

On solving equation (i) and (ii),

$x = \dfrac{24}{10}\cos\theta,\ y = \dfrac{-8}{10}\sin\theta$

Now for locus (h, k)

$h = \dfrac{3\cos\theta + \dfrac{24}{10}\cos\theta}{2} \Rightarrow \cos\theta = \dfrac{10}{27}h$

$k = \dfrac{\sin\theta - \dfrac{8}{10}\sin\theta}{2} \Rightarrow \sin\theta = 10k$

As are know, $\sin^2\theta + \cos^2\theta = 1$

$\dfrac{h^2}{\left(\dfrac{27}{10}\right)^2} + \dfrac{k^2}{\left(\dfrac{1}{10}\right)^2} = 1$, it's an equation of ellipse.

21. (b) Ellipse $-2x^2 + 5y^2 = 20$, mid point (2, 1) using $T = S_1$

$2x(2) + 5(y \times 1) - 20 = 2(2)^2 + 5(1)^2 - 20$

$4x + 5y = 13$

22. (a) $P(a\cos\alpha, b\sin\alpha)$

$Q(a\cos\alpha, a\sin\alpha)$

Tangent at Q point

$x\cos\alpha + y\sin\alpha = a$

$SN = \left|\dfrac{ae\cos\alpha - a}{\sqrt{\cos^2\alpha + \sin^2\alpha}}\right| \Rightarrow |ae\cos\alpha - a|$

$SP = \sqrt{(ae - a\cos\alpha)^2 + b^2\sin^2\alpha}$

$= \sqrt{a^2e^2 + a^2\cos^2\alpha - 2a^2e\cos\alpha + b^2 - b^2\cos^2\alpha}$

$= \sqrt{a^2 + \cos^2\alpha(a^2 - b^2) - 2a^2e\cos\alpha}$

$= |ae\cos\alpha - a|$

$\Rightarrow SP = SN$

23. (a) Given equation is $2x^2 + 3y^2 = 5$

$\Rightarrow \dfrac{2x^2}{5} + \dfrac{3y^2}{5} = 1$

So, compairing with the general equation,

$a^2 = \dfrac{5}{2}$ and $b^2 = \dfrac{5}{3}$

ie, The coordinates of the focii are $\left(\pm\sqrt{a^2 - b^2}, 0\right)$

$\Rightarrow \left(\pm\sqrt{\dfrac{5}{2} - \dfrac{5}{3}}, 0\right) \Rightarrow \left(\pm\sqrt{\dfrac{5}{6}}, 0\right)$

So, the rays are coming from one of the focii of the ellipse. Then according to the reflection property of the ellipse, The reflected rays go through the other focii.

ie, $\left(-\sqrt{\dfrac{5}{6}}, 0\right)$

24. (a) Any point on ellipse $x^2 + 2y^2 = 2$ is

$x = \sqrt{2}\cos\theta,\ y = \sin\theta$

$\alpha = x^2 + y^2 + \sqrt{2}xy - 1$

$= 2\cos^2\theta + \sin^2\theta + \sin^2\theta - 1$

$= \dfrac{1 + \cos 2\theta}{2} + \sin 2\theta$

25. (a) It is the square of minimum distance between 2 curves $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ and $y = x - 5$.

Tangent to ellipse parallel to given line $y = x - 5$ is $y = x \pm 5$

Since $y = x - 5$ is tangent to ellipse, required distance is 0

26. (d) Equation of normal to ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ at $P(3\cos\theta, 2\sin\theta)$ is

If it is tangent to circle $x^2 + y^2 = 3$, then

$\Rightarrow \dfrac{5}{\sqrt{9\sec^2\theta + 4\operatorname{cosec}^2\theta}} = \sqrt{3}$

$9\sec^2\theta + 4\operatorname{cosec}^2\theta = 9 + 4 + 9\tan^2\theta + 4\cot^2\theta$

$= 25 + (3\tan\theta - 2\cot\theta)^2$

$\therefore (9\sec^2\theta + 4\operatorname{cosec}^2\theta)_{\text{min.}} = 25$

$\therefore$ no such θ exists.

Hence no tangents to circle which is normal to ellipse.

27. (b) $\left(7 - \dfrac{5}{4}\alpha, \alpha\right)$ lies inside the ellipse $\dfrac{x^2 + y^2}{25} = 1$

$\Rightarrow \dfrac{\left(7 - \dfrac{5}{4}\alpha\right)^2}{25} + \dfrac{\alpha^2}{16} - 1 < 0$

$\Rightarrow \dfrac{(28 - 5\alpha)^2}{400} + \dfrac{\alpha^2}{16} - 1 < 0$

$\Rightarrow (28 - 5\alpha)^2 + 25\alpha^2 - 400 < 0$

$\Rightarrow 50\alpha^2 - 140\alpha - 192 < 0$

$\Rightarrow (5\alpha - 12)(5\alpha - 16) < 0$

$\Rightarrow \dfrac{12}{5} < \alpha < \dfrac{16}{5}$

28. (d) Equation of the ellipse is

$\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$

$\Rightarrow e = \sqrt{1 - \dfrac{9}{16}} = \dfrac{\sqrt{7}}{4}$

Hence, coordinates of foci are $S(\sqrt{7}, 0)$ and $S'(-\sqrt{7}, 0)$.

Normal at P is a bisector of angle between S'P and SP So, applying angular bisector theorem (angular bisector divides opposite side in the ration of its corresponding sides) in ΔS 'p S

i.e., $\dfrac{SQ}{S'Q} = \dfrac{SP}{S'P}$

$\dfrac{SQ}{S'Q} = \dfrac{a - ex_1}{a + ex_1}$

$\Rightarrow \dfrac{SQ}{S'Q} = \dfrac{4 - \dfrac{\sqrt{7}}{2}}{4 + \dfrac{\sqrt{7}}{2}}$

$\Rightarrow \dfrac{SQ}{S'Q} = \dfrac{8 - \sqrt{7}}{8 + \sqrt{7}}$

29. [85]

Center of ellipse = (29, 75/2)

foot of perpendicular from focii lie on auxillary circle

equaton of auxillary circle

$(x - 29)^2 + (y - 75/2)^2 = a^2$

$\downarrow$ (9,0) foot of perpendicular

$2a = 85$.

Parikshit (JEE Advanced Level)

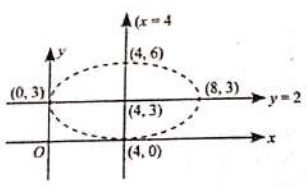
1. (b) Let $(4\cos\theta + 4, 3\sin\theta + 3)$ be any point on the ellipse $\dfrac{(x-4)^2}{16} + \dfrac{(y-3)^2}{9} = 1$

Image of $(4\cos\theta + 4, 3\sin\theta + 3)$ about the line $x - y - 2 = 0$ is (h, k), then

$\dfrac{h - 4\cos\theta - 4}{1} = \dfrac{k - 3\sin\theta - 3}{-1}$

$= \dfrac{2(4\cos\theta - 3\sin\theta - 1)}{2}$

$\Rightarrow h = 3\sin\theta + 5$ and $k = 4\cos\theta + 2$



Locus of (h, k) is $\left(\frac{x-5}{3}\right)^2+\left(\frac{y-2}{4}\right)^2=1$

$\Rightarrow 16x^2+9y^2-160x-36y+292=0$

$\Rightarrow k_1+k_2=25$

2. **(c)** Equation of circle will be

$x^2+y^2=(ae)^2$ ...(1)

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ...(2)

$\frac{(ae)^2-y^2}{a^2}+\frac{y^2}{b^2}=1$

$y=\frac{b}{e}\sqrt{1-e^2}=\frac{a}{e}(1-e^2)\,2a=17$

$PM=\frac{b}{e}(1-e^2)$

Area of $\Delta PF_1F_2=30$

$\frac{1}{2}(F_1F_2)\times PM=30\ F_1F_2=2ae$

$\frac{1}{2}(2ae)\times\frac{a}{e}\sqrt{1-e^2}=30=17\times\frac{13}{17}$

$e=\frac{13}{17}F_1F_2=13$

3. **(a)** *Eq*^{n} of *CF*:

$\frac{x}{6}+\frac{y}{b}=1$

$p=r$

$\left|\frac{\frac{1}{6}+\frac{1}{b}-1}{\sqrt{\frac{1}{36}+\frac{1}{b^2}}}\right|=1$

$\Rightarrow b=5/2\Rightarrow 2b=5$

$ae=6$

$e^2=1-b^2/a^2\ \Rightarrow\ a^2e^2=a^2-b^2\Rightarrow 36=a^2-25/4$

$\Rightarrow a^2=169/4\ \Rightarrow\ a=\frac{13}{2}$

$2a=13\ \Rightarrow\ (AB)\,(CD)=13\times 5=65$

4. **(c)** $\frac{\left(\frac{x-y+1}{\sqrt{2}}\right)^2}{10}+\frac{\left(\frac{x+y-3}{\sqrt{2}}\right)^2}{5/2}=1$

Here $a^2=10$ and $b^2=5/2$ and centre is (1,2)

∵ Locus of feet of perpendicular lie on auxillary circle of ellipse

∵ equation of circle is $(x-1)^2+(y-2)=10$

$x^2+y^2-2x-4y-5=0$

5. **(d)** $A'S=SS'=SA$

$2ae=a-ae$

$3ae=a$

$e=1/3$

$1-b^2/a^2=1/9$

$\Rightarrow b^2/a^2=8/9$

$\Rightarrow 8/a^2=8/9$

$\Rightarrow a=3$

6. **(b)** If the first foci is (3, 7) and the second is (*d*,7), then $C=\left(\frac{d+3}{2},7\right)$. *C* is the centre of the ellipse. The point where the ellipse tocuhes the *x*-axis $\left(\frac{d+3}{2},0\right)$ We know that *P*(any point on the ellipse) = *T* So,

$2\sqrt{\left(\frac{d-3}{2}\right)^2+7^2}=d+3$

$\sqrt{\left(\frac{d-3}{2}\right)^2(4)+7^2(4)}=d+3$

$\sqrt{(d-3)^2+196}=d+3$

Then, by squaring both sides, we get

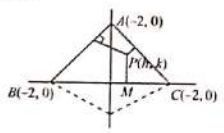
$(a-3)^2+196-a^2+9$

$a^2-6a^2+9+196-a^2+6d+9$

$12d=196$

$d=\frac{49}{3}$

7. **(a, c)**



$PM=k$

Equation of $AB\equiv -x+y=2$

Equation of $AC\equiv x+y=2$

According to question

$\left(\frac{2-h-k}{\sqrt{2}}\right)\left(\frac{2+h+k}{\sqrt{2}}\right)=2k^2$

$\Rightarrow h^2+3k^2+4k=4$

$\Rightarrow h^2+3\left(k^2+\frac{4}{3}k+\frac{4}{9}\right)=4+\frac{4}{3}$

$\Rightarrow h^2+3\left(k+\frac{2}{3}\right)^2=\frac{16}{3}$

$\Rightarrow \frac{h^2}{16/3}+\frac{\left(k+\frac{2}{3}\right)^2}{16/9}=1$

$\Rightarrow$ Ellipse with $e=\sqrt{\frac{2}{3}}$ and $D\equiv\left(0,-\frac{2}{3}\right)$.

8. **(a,c,d)**

$\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$

$\frac{y}{b}\sin\theta=1-\cos\theta\ \Rightarrow\ y=\frac{b(1-\cos\theta)}{\sin\theta}$

$AV.A'V'=\frac{b(1-\cos\theta)}{\sin\theta}\times\frac{b(1+\cos\theta)}{\sin\theta}=b^2$

$\angle V'SV=90°$ so $V'S'SV$ is a cyclic quadrilateral

9. **(a, b)**

Let *P* and *Q* be the points (α,β) and (α_1,β_1)

$\Rightarrow$ Equations of *AB* and *CD* are $\frac{x}{a}\alpha+\frac{y}{b}\beta$ and $\frac{x}{a}\alpha_1+\frac{y}{b}\beta_1=1$ (Chord of contact)

These lines are parallel

$\Rightarrow \frac{\alpha}{\alpha_1}=\frac{\beta}{\beta_1}=k$

Also $\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}=\frac{\alpha_1^2}{a^2}+\frac{\beta_1^2}{b^2}$

$\Rightarrow \frac{\alpha}{\alpha_1}=\frac{\beta}{\beta_1}=-1$

$\Rightarrow PQ$ passes through origin and is bisected at the origin.

10. **(a, b, c, d)**

Orthogonal trajectory of $y^2=kx$ which passes through (2, 0) is $C: x^2+\frac{y^2}{2}=4$

11. **(a,c,d)** $2ae=2$

Coordinate of foci $=(-1\pm ae,0)=(-2,0)$ and (0, 0)

Equation of directrix $x=-1\pm\frac{a}{e}$

$\Rightarrow (x+5)\,(x-3)=0$

12. **(b, d)** Tangent to ellipse

$y=mx+\sqrt{25m^2+9}$ ...(1)

If (1) is also a tangent to the circle x^2+

$y^2=16$ then $p=r\left|\frac{\sqrt{25m^2+9}}{\sqrt{1+m^2}}\right|=4$

$25m^2+9=16(1+m^2)$

$9m^2=7$

$\tan^2\theta=\frac{7}{9}\Rightarrow\tan\theta=\frac{\sqrt{7}}{3}$

$\theta=\tan^{-1}\frac{\sqrt{7}}{3}$ or $\cos^{-1}\frac{3}{4}\Rightarrow$ B

Again let $L=AB$

Also $L^2=(OA)^2-(OB)^2$

$=(a^2\cos^2\alpha+b^2\sin^2\alpha)-16$

(α is the eccentric angle of point *A*)

$=25\cos^2\alpha+9\sin^2\alpha-16$

$L^2=16\cos^2\alpha+9-16=9-16\sin^2\alpha$ ...(2)

To find α equation of a tangent at $A(\alpha)$ is

$\frac{x\cos\alpha}{5}+\frac{y\cos\alpha}{3}=1,$

By Aditya Mohad
username - muje_kabhi_dekha_nahi

For this to be tangent perpendicular distance from (0, 0) should be equal to 4.

$$\therefore \left|\frac{1}{\sqrt{\frac{\cos^2\alpha}{25}+\frac{\sin^2\alpha}{9}}}\right| = 4$$

$$\Rightarrow \left|\frac{15}{\sqrt{9\cos^2\alpha+25\sin^2\alpha}}\right| = 4$$

$\Rightarrow 225 = 16[9\cos^2\alpha + 25\sin^2\alpha]$

$\Rightarrow 225 = 16[9 + 16\sin^2\alpha]$

$\Rightarrow 256\sin^2\alpha = 225 - 144 = 81$

$$\Rightarrow \frac{81}{16.16} = \sin^2\alpha \Rightarrow \sin\alpha = \frac{9}{16}$$

Hence $L^2 = 9 - \frac{81}{16} = \frac{63}{16}$

$$\Rightarrow L = \frac{3\sqrt{7}}{4}$$

13. **(a,b)**

Normal at point $P(a\cos\theta, b\sin\theta)$ is

$$\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2 \quad ...(i)$$

It meets axes at $Q\left(\frac{(a^2-b^2)\cos\theta}{a}, 0\right)$

and $R\left(0, -\frac{(a^2-b^2)\sin\theta}{b}\right)$

Let $T(h,k)$ is a midpoint of QR.

Then $2h = \frac{(a^2-b^2)\cos\theta}{a}$

and $2k = -\frac{(a^2-b^2)\sin\theta}{b}$

$$\Rightarrow \cos^2\theta + \sin^2\theta = \frac{4h^2a^2}{(a^2-b^2)^2} + \frac{4k^2b^2}{(a^2-b^2)^2} = 1$$

$$\Rightarrow \text{Locus is } \frac{x^2}{\frac{(a^2-b^2)^2}{4a^2}} + \frac{y^2}{\frac{(a^2-b^2)^2}{4b^2}} = 1 \quad ...(ii)$$

Which is an ellipse, having eccentricitye' given by

$$e'^2 = 1 - \frac{\frac{(a^2-b^2)^2}{4a^2}}{\frac{(a^2-b^2)^2}{4b^2}} = 1 - \frac{b^2}{a^2} = e^2$$

$e' = e$

14. **(a,c)**

Let Δ be the area of the triangle whose vertices are $A(a\cos\theta, b\sin\theta)$, $B(a\cos(\theta+a), b\sin(\theta+a))$ and $C(a\cos(\theta+2a), b\sin(\theta+2a))$

We have

$$\Delta = \frac{1}{2}\begin{vmatrix} a\cos\theta & b\sin\theta & 1 \\ a\cos(\theta+a) & b\sin(\theta+a) & 1 \\ a\cos(\theta+2a) & b\sin(\theta+2a) & 1 \end{vmatrix}$$

$R_1 \to R_1 - R_1$ and $R_3 \to R_3 - R_1$

$$= \frac{ab}{2}\begin{vmatrix} \cos\theta & \sin\theta & 1 \\ \cos(\theta+\alpha)-\cos\theta & \sin(\theta+\alpha)-\sin\theta & 0 \\ \cos(\theta+2\alpha)-\cos\theta & \sin(\theta+2\alpha)-\sin\theta & 0 \end{vmatrix}$$

$$= \frac{ab}{2}\begin{vmatrix} -2\sin(\theta+a/2)\sin a/2 & 2\sin a/2\cos(\theta+a/2) \\ -2\sin a\sin(\theta+a) & 2\sin a\cos(\theta+a) \end{vmatrix}$$

$$= 2ab\sin\theta\sin\frac{a}{2}\begin{vmatrix} \sin(\theta+a/2) & -\cos(\theta+a/2) \\ \sin(\theta+a) & -\cos(\theta+a) \end{vmatrix}$$

$$= 2ab\sin a\sin\frac{a}{2}\sin\frac{a}{2}$$

$$= 2ab\sin^2\frac{a}{2}\sin a$$

$$\Rightarrow \Delta = 2ab\sin^2\frac{a}{2}\sin a$$

It is independent of θ

Now $\Delta = 2ab\sin^2\frac{a}{2}\sin a$

$= ab(1 - \cos a)\sin a$

$= ab(\sin a - \sin a\cos a)$

$$\frac{d}{da}(\Delta) = ab(\cos a - \cos^2 a + \sin^2 a)$$

$$\Rightarrow \frac{d}{da}(\Delta) = ab(\cos a - \cos 2a)$$

and $\frac{d^2}{da^2}(\Delta) = ab(-\sin a + 2\sin 2a)$

For $\frac{d}{da}(\Delta) = 0$

$\cos a - \cos 2a = 0$

$\Rightarrow \cos 2\alpha - \cos(2\pi - \alpha)$

$\Rightarrow 2a = 2\pi - a$

$$\Rightarrow a = \frac{2\pi}{3}$$

For $a = \frac{2\pi}{3}$

$$\frac{d^2}{da^2}(\Delta) = ab\left(-\sin\frac{2\pi}{3} + 2\sin\frac{4\pi}{3}\right) < 0$$

Therefore Area of ΔABC is maximum when $a = \frac{2\pi}{3}$

15. **(a,b)** $\frac{x^2}{4} + \frac{y^2}{1} = 1$

$x = 2\cos\theta, y = \sin\theta$

$S = 4\cos^2\theta + \sin^2\theta - 2\sin\theta\ln\theta =$

$S = 3\sin^2\theta - \sin 2\theta + 1$

$$= \frac{3}{2}(1 + \cos 2\theta) - \sin 2\theta + 1$$

$$= \frac{5}{2} + \frac{3}{2}\cos 2\theta - \sin 2\theta$$

$$a = \frac{5}{2} + \sqrt{\frac{9}{4}+1} = \frac{5+\sqrt{13}}{2}$$

$$b = \frac{5}{2} - \sqrt{\frac{9}{4}+1} = \frac{5-\sqrt{13}}{2}$$

16. **(a)**

17. **(b)**

Sol. (16 to 17)

$$\left(\frac{3x-4y+10}{5}\right)^2 \times \frac{25}{2} + \left(\frac{4x+3y-15}{5}\right)^2 \times \frac{25}{3} = 1$$

$$a^2 = \frac{2}{25}$$

$$\Rightarrow a = \frac{\sqrt{2}}{5} \text{ minor axis} = 2a = \frac{2\sqrt{2}}{5}$$

$$b^2 = \frac{3}{25}$$

$$\Rightarrow b = \frac{\sqrt{3}}{5} \text{ major axis} = 2b = \frac{2\sqrt{3}}{5}$$

$$e = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{2}{3}} = \frac{1}{\sqrt{3}}$$

centre is point of intersection of

$3x - 4y + 10 = 0, 4x + 3y - 15 = 0$

$\left(\frac{6}{5}, \frac{17}{5}\right)$

18. **(b)** $\because \angle QAP = \angle PBQ = 90°$

hence a circle drawn taking 'PQ' as diamete will pass through B, A, P, Q

$\therefore$ center will be mid point of PQ

19. **(a)** $m_1 + m_2 = \frac{2hk}{h^2 - a^2}$

and $\cot\theta_1 + \cot\theta_2 = \lambda$

$$\Rightarrow \frac{1}{\tan\theta_1} + \frac{1}{\tan\theta_2} = \lambda$$

$$\Rightarrow \frac{\tan\theta_1 + \tan\theta_2}{\tan\theta_1\tan\theta_2} = \lambda$$

$$\Rightarrow \frac{\frac{2hk}{h^2-a^2}}{\frac{k^2-b^2}{h^2-a^2}} = \lambda$$

$\Rightarrow 2hk = \lambda(k^2 - b^2)$

$2xy = \lambda(y^2 - b^2)$

20. **(c)**

21. **(b)**

Sol. (20 to 21)

Equation of parabola is $y^2 = 4\sqrt{3}(x + \sqrt{3})$

Normal to the parabola at B meets the x axis at $N(2\sqrt{3}, 0)$ and normal to the ellipse at B is the y-axis.

$\therefore$ Required area

$$= \frac{1}{2} \times 2\sqrt{3} \times 2\sqrt{3} = 6 \text{ sq.units}$$

$$\frac{A_1}{A_2} = \frac{\frac{1}{2} \cdot \pi \cdot 4 \cdot 2\sqrt{3} - \frac{2}{3} \cdot 12}{\frac{1}{2}\pi \cdot 4 \cdot 2\sqrt{3} + \frac{2}{3} \cdot 12} = \frac{\pi\sqrt{3} - 2}{\pi\sqrt{3} + 2}$$

22. **(a)** The equation of common tangent between C_1 and C_2 having negative gradient in the first quadrant is

$2x + \sqrt{3}y = 4\sqrt{7}$

So, x-intercept

$= 2\sqrt{7}$

23. **(b)** As quadrilateral formed by common tangents between C_1 and C_2 is a rhombus, so area of rhombus

$$= \frac{1}{2} d_1 \times d_2 = \frac{1}{2}(4\sqrt{7})\left(\frac{8\sqrt{7}}{\sqrt{3}}\right)$$

$$= \frac{16 \times 7}{\sqrt{3}} = \frac{112}{3}\sqrt{3}$$

24. **(d)** Director circle of C_1 is $x^2 + y^2 = 32$. So,

any point on it is $(4\sqrt{2}\cos\theta, 4\sqrt{2}\sin\theta)$

Also, auxiliary circle of C_2, is $x^2 + y^2 = 25$.

So, equation of chord of contact, is

$x \cdot (4\sqrt{2}\cos\theta) + y(4\sqrt{2}\sin\theta) = 25$ ...(1)

Let, mid-point of chord of contact be (h, k).

So, also equation of chord of contact is

$hx + ky = h^2 + k^2$ [Using $T = S_1$] ...(2)

As, equation (1) and equation (2) represents same line, so on comparing, we get

$$\frac{4\sqrt{2}\cos\theta}{h} = \frac{4\sqrt{2}\sin\theta}{k} = \frac{25}{h^2 + k^2}$$

$$\therefore \cos\theta = \frac{25h}{4\sqrt{2}\left(h^2 + k^2\right)},$$

$$\sin\theta = \frac{25k}{4\sqrt{2}\left(h^2 + k^2\right)}$$

Now, on squaring and adding, we get

$$\frac{625h^2}{32\left(h^2 + k^2\right)^2} + \frac{625k^2}{32\left(h^2 + k^2\right)^2} = 1$$

$$\Rightarrow 625(h^2 + k^2) = 32(h^2 + k^2)^2$$

$$\Rightarrow h^2 + k^2 = \frac{625}{32}$$

$\Rightarrow$ Locus of (h, k) is $x^2 + y^2 = \frac{625}{32}$

25. **(b)** $2x^2 - 2xy + 4y^2 - (3 + \sqrt{2}) = 0$

compare with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$\Rightarrow a = 2, 2h = -2, b = 4$

We know that if θ is inclination then

$$\tan 2\theta = \frac{2h}{a - b} \Rightarrow \tan 2\theta = \frac{-2}{2 - 4} = 1$$

$$\Rightarrow 2\theta = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{8}$$

26. **(d)** $x = x'\cos\theta - y'\sin\theta$

$2(x'\cos\theta - y'\sin\theta)^2 - 2(x'\ln\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta) + 4(x\sin\theta + y'\sin\theta)^2 = 3 + \sqrt{2}$

$\Rightarrow 2[(x'\cos\theta)^2 + 2y(y'\cos\theta)^2 - 2x'y'\sin\theta \cdot \ln\theta]$

$-2[x'^2\cos\theta\sin\theta + x'y'\cos^2\theta - x'y'\sin^2\theta - y'^2\sin\theta\cos\theta) + 4(x'^2\sin^2\theta + y'\cos\theta + 2xy^2\sin\theta\cos\theta) = 3 + \sqrt{2}$

$2x'\cos^2\theta + 2y^2\sin^2\theta - 4x'y'\sin\theta\cos\theta - 2x^2\ln\theta\sin\theta - 2x'y'\cos^2\theta + 2x'y'\sin^2\theta + 2y'^2\sin\theta\cos\theta + 4x^2\sin\theta + 4y'^2\cos^2\theta + 8x'y'\sin\theta\cos\theta = 3 + \sqrt{2}$

$x'^2\left[2\cos^2\theta - Q\sin 2\theta + 4\sin^2\theta\right] + y'^2\left[2\sin^2\theta + \sin 2\theta + 4\cos^2\theta\right] + x'y'[-4\sin\theta\cos\theta - 2\cos 2\theta + 8\sin\theta\cos\theta] = 3 + 12x$

$x'^2[2 + (1\cos 2\theta) - \sin 2\theta] + y'^2[3 + \cos 2\theta + \sin 2\theta] + x'y'[\theta] = 3 + 12$

$x^2[3 - (\sqrt{2})] + y^2[3 + \sqrt{2}] = 3 + i_2$

$2[x'^2 + y'^2 - 1] = [2[1 - y'^2 + x'^2]$

$3(x^2 + y^2 - 1) + 12[-x'^2 + y'^2 - 1] = 0$

27. **(a)** Ellispe is

$\sqrt{2}(x^2 - y^2 + 1) = 3(y^2 + x^2 - 1)$

$\sqrt{2}(x^2 - y^2 + 1) = 3(y^2 + x^2 - 1)$

$$\frac{x^2}{\frac{3 + \sqrt{2}}{3 - \sqrt{2}}} + \frac{y^2}{1} = 1$$

Length of minor axis is 2

$\therefore$ area of circle $= \pi(1)^2 = \pi$

28. **(c)** $ax^2 + 2hxy + by^2 = 1$

$x = x'\ln\theta - y'\sin\theta$

$y = x'\sin\theta + y'\cos\theta$

$a(x'^2\cos^2\theta + y'\sin^2\theta - 2x^2y\sin\theta\cos\theta) + Lh(x'^2\cos\theta\sin\theta - y'^2\sin\theta\cos\theta + x^2\cos^2\theta - x'y'\sin^2\theta) + b(x'^2\sin^2\theta + y^2\cos^2\theta + 2x'y'\sin\theta - \cos\theta)$

$x'^2[\sin^2\theta + 2h\sin\theta\sin\theta + b\sin^2\theta] + y^2[a\sin^2\theta - 2h\sin\theta\ln\theta + b\cos^2\theta]$

$xy[-a^2\sin\theta\ln\theta + 2h\cos 2\theta + b\sin 2\theta]$

$(b - a)\sin 2\theta + 2h\cos 2\theta$

$\tan^2\theta = \frac{2h}{a - b}$

$x'^2\left[a\cos^2\theta + b\sin^2\theta + h\sin^2\theta\right]$

$+y'^2\left[a\sin^2\theta - h\sin 2\theta + b\cos 2\theta\right] = 1$

$\frac{1}{\alpha^2} = a\cos^2\theta + b\sin^2\theta + h\sin 2\theta$

$\frac{1}{\beta^2} = a^2\sin^2\theta + 5\cos^2\theta - h\sin 2\theta$

$$\boxed{\frac{1}{\alpha^2} + \frac{1}{\beta^2} = a + b}$$

29. **(b)** Ellipse $\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$ $\tan 2\theta = \frac{2h}{a - b}$

$$e^2 = 1 - \frac{\beta^2}{\alpha^2} = 1 - \frac{a\cos^2\theta + b\sin^2\theta + h\sin 2\theta}{a\sin^2\theta + b\cos^2\theta - h\sin 2\theta}$$

$$= \frac{b\cos 2\theta - a\cos 2\theta - 2h\sin 2\theta}{a\sin^2\theta + b\cos^2\theta - h\sin 2\theta}$$

$$= \frac{2\sqrt{(a - b)^2 + 4h^2}}{a + b + \sqrt{(a - b)^2 + 4h^2}}$$

30. **(a)** $(a) \to (r)$, $(b) \to (p)$, $(c) \to (s)$, $(d) \to (q)$

(a) $y = mx \pm \sqrt{a^2m^2 + b^2}$

$$y = -\frac{4}{3}x \pm \sqrt{18 \times \frac{16}{9} + 32} \Rightarrow y = -\frac{4}{3}x \pm 8$$

Distance between tangent

$$= \frac{16}{\sqrt{1 + \frac{16}{9}}} = \frac{16 \times 3}{5} = \frac{48}{5}$$

(b) $y = -\frac{4}{3}x + 8$ $A(6, 0)$ $B(0, 8)$

Area of $\Delta AOB = \frac{1}{2} \times 6 \times 8 = 24$

(c) point of contact

$$\left(-\frac{a^2m}{\sqrt{a^2m^2 + b^2}}, \frac{b^2}{\sqrt{a^2m^2 + b^2}}\right)$$

product of coordinates

$$= -\frac{a^2b^2m}{a^2m^2 + b^2} = -\frac{18 \times 32 \times \left(-\frac{4}{3}\right)}{64} = 12$$

(d) $4x + 3y = 24$ $\ell = \frac{4}{24}$ $m = \frac{3}{24}$

$$\frac{4}{24}x + \frac{3}{24}y = 1\ell + m = \frac{7}{24}$$

31. **(b)** (a) p; (b) q; (c) r; (d) s

Let $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$ be the normal

$\Rightarrow 4x\sec\theta - 2\sqrt{3}\,y\operatorname{cosec}\theta = 4$ ...(1)

$\because$ it passes through $\left(\frac{1}{2\sqrt{2}}, \frac{\sqrt{2}}{2\sqrt{3}}\right)$

$\Rightarrow \sqrt{2}\sec\theta - \sqrt{2}\operatorname{cosec}\theta = 4$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

$\Rightarrow \sec\theta - \operatorname{cosec}\theta = 2\sqrt{2}$

$\Rightarrow \theta = -\frac{\pi}{4}, \frac{5\pi}{12}, \frac{13\pi}{12}$

Now, $m = \frac{4\sec\theta}{2\sqrt{3}\operatorname{cosec}\theta} = \frac{2\tan\theta}{\sqrt{3}}$ (from (1))

$m = -\frac{2}{\sqrt{3}}, \frac{4}{\sqrt{3}}, -2, \frac{4}{\sqrt{3}}, +2$

32. (c) (A) Let $P(h,k)$ be the foot of perpendicular to a tangent

$y = mx + \sqrt{a^2m^2 + b^2}$ from centre

$\therefore \frac{k}{h} \cdot m = -1 \Rightarrow m = -\frac{h}{k} \because P(h,k)$ lies on tangent

$\therefore k = mh + \sqrt{a^2 m^2 + b^2}$

from equation (*ii*) and (*iii*) we get

$\left(k + \frac{h^2}{k}\right)^2 = \frac{a^2 h^2}{k^2} + b^2$

$\Rightarrow$ locus is $\left(x^2 + y^2\right)^2 = a^2x^2 + b^2y^2$

(*B*) Let the centre of the ellipse be *O* and the foot be *R*. Also let point *P* is $(a\cos\theta, b\sin\theta)$ then Q is

$\left(a\cos\left(\theta + \frac{\pi}{2}\right), b\sin\left(\theta + \frac{\pi}{2}\right)\right)$ or

$(-a\sin\theta, b\cos\theta)$ then *PQ* is

$\frac{y - b\cos\theta}{x + a\sin\theta} = \frac{b(\sin\theta - \cos\theta)}{a(\sin\theta + \cos\theta)}$

or $\frac{\frac{y}{b} - \cos\theta}{\frac{x}{a} + \sin\theta} = \frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta}$

Also *OR* is

$\frac{y}{x} = -\frac{a(\sin\theta + \cos\theta)}{b(\sin\theta - \cos\theta)}$ or $\frac{by}{ax} = -\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta}$

From these two equations we have

$x^2 + y^2 = -ax\sin\theta + by\cos\theta$

Divide both sides by

$\sqrt{a^2x^2 + b^2y^2}$ and let $\tan\phi = \frac{by}{ax}$ then we have

$\sin(\theta - \phi) = \frac{x^2 + y^2}{\sqrt{a^2x^2 + b^2y^2}}$

or $\theta = \sin^{-1}\left(\frac{x^2 + y^2}{\sqrt{a^2x^2 + b^2y^2}}\right) + \phi$

Also from the second equation

$\frac{y}{x} = -\frac{a}{b}\frac{\tan\theta + 1}{-1 + \tan\theta}$

Substitute θ in the last equation using tangent compound angles and simplify, we have the following equation of the locus provided x and y not equal to zero.

$a^2x^2 + b^2y^2 = 2(x^2 + y^2)^2$

(*C*) Let $Q = (\cos\theta, \sin\theta)$

$\therefore m = (\cos\theta, 0)$

Let $P \equiv (n, k)$

$h = \frac{a\cos\theta + b\cos\theta}{a+b} \Rightarrow k = \frac{b\sin\theta}{a+b}$

$\Rightarrow \cos\theta = h \quad ; \sin\theta = \frac{(a+b)k}{b}$

$\sin^2\theta + \cos^2\theta = 1$

(*D*) Let $P = (h, k)$ be the point of intersection of tangents at *A* & *B* equation of chord of contact AB

$\frac{xh}{a^2} + \frac{yk}{b^2} = 1$ is which touches the paraboia equation of tangent of parabola

$ay^2 = 4axy = mx + \frac{a}{m} \Rightarrow mx - y = -\frac{a}{m}$

Equation (*i*) and (*iii*) as musrt be same

$\therefore \frac{m}{\left(\frac{h}{a^2}\right)} = \frac{-1}{\left(\frac{k}{b^2}\right)} = \frac{-\frac{a}{m}}{1}$

$\Rightarrow m = -\frac{hb^2}{ka^2}m = \frac{ak}{b^2} \therefore -\frac{hb^2}{ka^2} = \frac{ak}{b^2} \Rightarrow$

Locus of P is $y^2 = -\frac{b^4}{a^3} \cdot x$

33. (*b*) (*A*) Solving given line with ellipse, we get

$\frac{(4y - \lambda)^2}{9\times16} + \frac{y^2}{9} = 1 \frac{2y^2}{9} - \frac{y\lambda}{18} + \frac{\lambda^2}{144} - 1 = 0$

Since, line intersect the parabola at two distinct points,

$\therefore$ roots of above equation are real & distinct

$\therefore D > 0 \Rightarrow \frac{\lambda^2}{(18)^2} - \frac{8}{9}\left(\frac{\lambda^2}{144} - 1\right) >$

$0 \Rightarrow -12\sqrt{2} < \lambda < 12\sqrt{2}$

(*B*) $\frac{x^2}{25} + \frac{y^2}{16} = 1$

Since point $\left(7 - \frac{5}{4}\alpha, \alpha\right)$ lies inside the ellipse

$\therefore S_1 < 0$

$\Rightarrow 16\left(7 - \frac{5}{4}\alpha\right)^2 + 25\cdot\alpha^2 < 400$

$\Rightarrow (28 - 5\alpha)^2 + 25\alpha^2 < 400$

$\Rightarrow 50\alpha^2 - 280\alpha + 384 < 0$

$\Rightarrow 25\alpha^2 - 140\alpha + 192 < 0$

$\Rightarrow (5\alpha - 12)(5\alpha - 16) < 0$

$\Rightarrow \alpha \in \left(\frac{12}{5}, \frac{16}{5}\right)$

(*C*) given equation of ellipse is

$\frac{x^2}{9} + \frac{y^2}{4} = 1$

The equations of the pair of tangents from *P* to the ellipse can be found using $SS_1 = T_2$.

$\Rightarrow \left(\frac{x^2}{9} + \frac{y^2}{4} - 1\right)\left(\frac{9}{9} + \frac{16}{4} - 1\right) = \left(\frac{3x}{9} + \frac{4y}{4} - 1\right)^2$

$\Rightarrow \left(\frac{x^2}{9} + \frac{y^2}{4} - 1\right)(4) = \left(\frac{x}{3} + y - 1\right)^2$

$\Rightarrow \frac{4x^2}{9} + y^2 - 4 = \frac{x^2}{9} + y^2 + 1 - \frac{2x}{3} - 2y + \frac{2xy}{3}$

$\Rightarrow x^2 + 2x + 6y - 2xy - 15 = 0$

$\Rightarrow x(x-3) + 5(x-3) - 2y(x-3) = 0$

$\Rightarrow (x-3)(x - 2y + 5) = 0$

Thus equations of tangents from *P* are $x - 3 = 0$ and $x - 2y + S = 0$.

Also, $x - 3 = 0$ meets ellipse at point $A(3, 0)$. Point of tangency for $x - 2y + S = 0$ is $B\left(\frac{-9}{5}, \frac{8}{5}\right)$

Therefore, the orthocentre of triangle *PAB* can be found by using perpendiculars to *PA* and *PB*.

Slope of $PA = \infty \Rightarrow$ slope of perpendicular to *PA* is *O* and passing through *B*. Let equation of this perpendicular be $L_1 \equiv y - mx + c$

$\Rightarrow L_1 \equiv y = \frac{8}{5}$

Similarly, let $L_2 \equiv y - mx + c$ be the equation of perpendicular to *PB* from *C*. The slope for L_2 is -2

$\Rightarrow L_2 \equiv 2x - y = 6$

The point of intersection of L_1 and L_2 is $\left(\frac{11}{5}, \frac{8}{5}\right)$

(*D*) For the ellipse $4x^2 + 9y^2$ the required slope of the tangent is 8/9 Differentiating the ellipse equation w.r.t x, we get

$8x + 18y\frac{dy}{dx} = 0$

$\therefore 4x + 9y \times \frac{8}{9} = 0$

$4x + 8y = 0$ or $x = -2y$

Substituting this in the ellipse equation, we get $4(-2y)^2 + 9y^2 - 1$

$\therefore 25y^2 = 1$ or $y = \pm\frac{1}{5}$

Thus the point are $\left(\frac{-2}{5}, \frac{1}{5}\right)$ and $\left(\frac{2}{5}, \frac{-1}{5}\right)$

34. **[7]** $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ...(1)

$\frac{x}{7}+\frac{y}{2}=1$ meets x-axis at A (7, 0), line

$\frac{x}{3}-\frac{y}{5}=1$ meets y-axis at B (0, –5).

Ellipse passes through A and B

$\Rightarrow \frac{49}{a^2}+0=1, 0+\frac{25}{b^2}=1$

$a^2=49, b^2=25; b^2=a^2(1-e^2)$

$25=49(1-e^2)$

$\Rightarrow e^2=\frac{24}{49} \Rightarrow e=\frac{2\sqrt{6}}{7}$

35. **[2]** $2a=10 \Rightarrow a=5; 2b=8 \Rightarrow b=4$

$e=\sqrt{1-\frac{b^2}{a^2}}=3/5$

Focus $(\pm ae, 0)$

$\Rightarrow (\pm 3, 0)$

$r=5-3=2$

$\Rightarrow r=2$

36. **[186]**

Equation of parabola,

$(x-3)^2=k(y+11)$

which is passing through

$(7,-4) \Rightarrow k=16/7$

$\therefore 16y=7(x-3)^2-176$

$\Rightarrow a+h+k=186$

37. **[1]** $P_1P_2=b^2$

$\Rightarrow b^2=1$

$b=1$

And are $=4ab=8$

$a=2$

$e=\sqrt{1-\frac{1}{4}}=\frac{\sqrt{3}}{2} \Rightarrow \frac{2e}{\sqrt{3}}=1$

38. **[3]** Let the perpendicular distance of P from the line be h,

$\therefore \frac{1}{2}\times h\times 5=6(\sqrt{2}-1)$

$\therefore h=\frac{12}{5}(\sqrt{2}-1)$

Also tangent parallel to the given line is

$4y+3x=12\sqrt{2}$

Its distance from the line $4y+3x=12$

$h=\frac{12}{5}(\sqrt{2}-1)$ Hence there are three points as show in the figure.

39. **[15]** (α, β) lies on the directoex circle of the ellipse is on

$x^2+y^2=9$

So, we can assume, $\alpha=3\cos\theta, \beta=\sin\theta$

$F=12\cos\theta+a\sin\theta=3(4\cos\theta+3\sin\theta)$

$-15 \le F \le 15$

PYQ's (Past Year Questions)

1. **(b)** Equation of normal is $ax\sec\theta - by\,\text{cosec}\theta = a^2-b^2\ 2x\sec\theta - by\,\text{cosec}\theta = 4-b^2$

Distance from origin

$(d)=\left|\frac{4-b^2}{\sqrt{4\sec^2\theta+b^2\text{cosec}^2\theta}}\right|$

Distance is maximum if $4\sec^2\theta + b^2\,\text{cosec}^2\theta$ is minimum

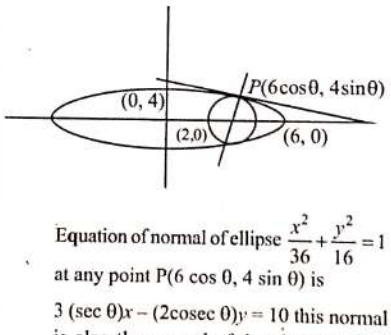
$\Rightarrow \tan^2\theta=\frac{b}{2}$

$\Rightarrow \frac{4-b^2}{\sqrt{4\cdot\frac{b+2}{2}+b^2\cdot\frac{b+2}{b}}}=1$

$\Rightarrow 4-b^2=b+2 \Rightarrow b=1$

$\Rightarrow e=\sqrt{1-\frac{b^2}{a^2}}=\frac{\sqrt{3}}{2}$

2. **[118]**



Equation of normal of ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$ at any point P(6 cos θ, 4 sin θ) is

$3(\sec\theta)x-(2\text{cosec}\,\theta)y=10$ this normal is also the normal of the circle passing through the point (2, 0) So,

$6\sec\theta=10$ or $\sin\theta=0$ (Not possible)

$\cos\theta=\frac{3}{5}$ and $\sin\theta=\frac{4}{5}$ so point

$P=\left(\frac{18}{5},\frac{16}{5}\right)$

So the largest radius of circle

$r=\sqrt{\left(\frac{18}{5}-2\right)^2+\left(\frac{16}{5}\right)^2}$

$r=\frac{\sqrt{320}}{5}$

So the equation of circle

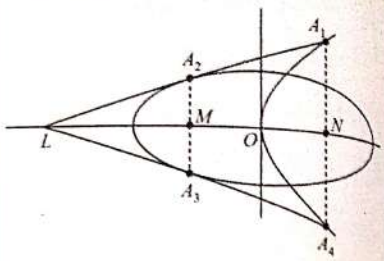
$(x-2)^2+y^2=\frac{64}{5}$

Passing it through $(1, \alpha)$

Then $\alpha^2=\frac{59}{5}$

$10\alpha^2=118$

3. **(a,c)**



Equation of tangent of parabola

$y=mx+\frac{3}{m}$

$C_2=a^2m^2+b^2\ \left(c=\frac{3}{m}, a=\sqrt{6}, b=\sqrt{3}\right)$

$\frac{9}{m^2}=6m^2+3 \Rightarrow m^2=1$

Equation T_1 & T_2

$y=x+3, y=-x-3$

Cuts x-axis at (–3, 0)

$A_1(3, 6)$

$A_4(3, -6)$

$A_2(-2, 1)$

$A_3(-2, -1)$

$A_1A_4=12, A_2A_3=2, MN=5$

Area of quad $=\frac{1}{2}$ (sum of parallel side) × height

Area $=\frac{1}{2}(12+2)\times 5=35$ sq. unit

Ans. (a, c)

4. **(b)** Line L is given by

$(bx+10y-8)+\lambda(2x-3y)=0$

$\Rightarrow (b+2\lambda)x+(10-3\lambda)y-8=0$

As, line L is passing through (1, 1)

$\Rightarrow b+2\lambda+10-3\lambda-8=0$

$\Rightarrow b+2=\lambda$

Now, L becomes

$(b+2(b+2))x+(10-3(b+2))y-8=0$

$\Rightarrow (3b+4)x-(3b-4)y-8=0$

Now, L is also a tangent to circle $17(x^2+y^2)=16$

$\therefore \frac{|0-0-8|}{\sqrt{(3b+4)^2+(3b-4)^2}}=\sqrt{\frac{16}{17}}$

$\Rightarrow \frac{64}{9b^2+16+24b+9b^2+16-24b}=\frac{16}{17}$

$\Rightarrow 18b^2+32=68 \Rightarrow 18b^2=36$

$\Rightarrow b^2=2$

$\therefore \frac{x^2}{5}+\frac{y^2}{2}=1$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

Now, $2 = 5(1 - e^2)$

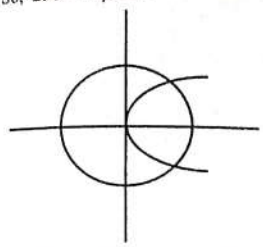
$e^2 = \frac{3}{5}$

$e = \sqrt{\frac{3}{5}}$

5. **[2929]** $\frac{x^2}{\frac{1}{25}} + \frac{y^2}{\frac{1}{4}} = 1$

Point (α, β) lies on ellipse

So, $25\alpha^2 + 4\beta^2 = 1$ ...(*i*)



Equation of tangent to parabola

$y = mx + \frac{1}{m}$ passes through (α, β)

$\alpha m^2 - \beta m + 1 = 0$

$m_1 + m_2 = \frac{\beta}{\alpha}$, $4m_1^2 = \frac{1}{\alpha}$ $[\because m_2 = 4m_1]$

$\Rightarrow 4\beta^2 = 25\alpha$ ...(*ii*)

$\therefore$ from (*i*) and (*ii*)

$25\alpha^2 + 25\alpha = 1$

$\Rightarrow \alpha^2 + \alpha = \frac{1}{25}$

$(10\alpha + 5)^2 = 100(\alpha^2 + \alpha) + 25 = 29$

and $(16\beta^2 + 50)^2 = 2500[4(\alpha^2 + \alpha) + 1]$
$= 2900$

Hence required answer is $2900 + 29$
$= 2929$.

6. (c) Equation of ellipse is $\frac{x^2}{4} + \frac{y^2}{1} = 1$.

The equation of tangent at $P(2\cos\theta, \sin\theta)$ is

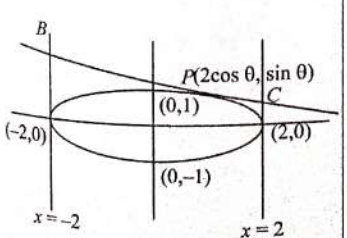
$(\cos\theta)x + (2\sin\theta)y = 2$ ...(*i*)

For point '*B*', solve equation of tangent with equation of

$x = -2$, we get

$-2\cos\theta + 2\sin\theta\, y = 2$

$\Rightarrow y = \frac{1 + \cos\theta}{\sin\theta}$



Therefore point '*B*' is

$B\left(-2, \frac{1 + \cos\theta}{\sin\theta}\right) \equiv B\left(-2, \cot\frac{\theta}{2}\right)$

For point '*C*', solve equation of tangent with equation of $x = 2$

$y = \frac{1 - \cos\theta}{\sin\theta}$

Therefore point '*C*' is

$C\left(2, \frac{1 - \cos\theta}{\sin\theta}\right) \equiv C\left(2, \tan\frac{\theta}{2}\right)$

Now BC is the diameter of circle

$\therefore (x+2)(x-2) + \left(y - \cot\frac{\theta}{2}\right)\left(y - \tan\frac{\theta}{2}\right) = 0$

Only $(\sqrt{3}, 0)$ satisfying equation of circles

7. (*a*) $y^2 = 3x^2$ and $x^2 + y^2 = 4b$

On solving both, we get

$x^2 = b$

Ellipse: $\frac{x^2}{16} + \frac{4^2}{b^2} = 1$

$\Rightarrow \frac{x^2}{16} + \frac{3x^2}{b^2} = 1$

$\Rightarrow \frac{b}{16} + \frac{3}{b} = 1$

$\Rightarrow b^2 - 16b + 48 = 0$

$\Rightarrow (b - 12)(b - 4) = 0$

$\Rightarrow b = 12,\ b = 4,\ b > 4$

8. (*a*) Let, point on ellipse $(2\sin\theta, 3\cos\theta)$ and the mid point of line segment joining $(-3, -5)$ and $(2\sin\theta, 3\cos\theta)$ will be (h, k)

$\frac{2\sin\theta - 3}{2} = h$ and $\frac{3\cos\theta - 5}{2} = k$

$2\sin\theta = 2h + 3$ and $3\cos\theta = 2k + 5$

$\sin\theta = \frac{2h+3}{2}$ and $\cos\theta = \frac{2k+5}{3}$

$\because \sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \left(\frac{2h+3}{2}\right)^2 + \left(\frac{2k+5}{3}\right)^2 = 1$

$\Rightarrow 36x^2 + 16y^2 + 108x + 80y + 145 = 0$

9. **[3]** $E: \frac{x^2}{9} + \frac{y^2}{4} = 1$

$C: x^2 + y^2 = \frac{31}{4}$

Equation of tangent to ellipse is

$y = mx \pm \sqrt{9m^2 + 4}$ ... (*i*)'

Equation of tangent to circle is

$y = mx \pm \sqrt{\frac{31}{4}m^2 + \frac{31}{4}}$ ... (*ii*)

Comparing equation (*i*) & (*ii*) $9m^2 + 4$

$= \frac{31}{4}m^2 + \frac{31}{4}$

$\Rightarrow 36m^2 + 16 = 31m^2 + 31$

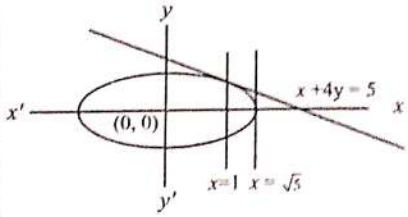
$\Rightarrow 5m^2 = 15$

$\Rightarrow m^2 = 3$

10. **[1.25]** Required Area

$= \frac{1}{2}\left(\sqrt{5} - 1\right)\frac{9 - \sqrt{5}}{4} - \int_1^{\sqrt{5}} \frac{1}{2}\sqrt{5 - x^2}\,dx$

$= \frac{1}{8}\left(-14 + 10\sqrt{5}\right) - \frac{1}{2}\int_{\cos^{-1}\frac{1}{\sqrt{5}}}^{0} \left(-\sin^2\theta\right)d\theta$



$A = \frac{-14}{8} + \frac{5}{4}\sqrt{5} - \left(\frac{5}{4}\cos^{-1}\frac{1}{\sqrt{5}} - \frac{1}{2}\right)$

$= \frac{5}{4}\sqrt{5} - \frac{5}{4} - \frac{5}{4}\cos^{-1}\frac{1}{\sqrt{5}}$

$\alpha = \frac{5}{4}, \beta = \frac{-5}{4}, \gamma = \frac{-5}{4}$

$|\alpha + \beta + \gamma| = \frac{5}{4}$

11. (*b*) $3x + 4y = 12\sqrt{2}$

$\Rightarrow 4y = -3x + 12\sqrt{2} \Rightarrow y = -\frac{3}{4}x + 3\sqrt{2}$

Condition of tangency

$c^2 = a^2 m^2 + b^2$

$\Rightarrow 18 = a^2 \times \frac{9}{16} + 9 \Rightarrow a^2 \times \frac{9}{16} = 9$

$\Rightarrow a^2 = 16 \Rightarrow a = 4$

Now, $e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$

$\therefore ae = \frac{\sqrt{7}}{4}.4 = \sqrt{7}$

$\therefore$ focus are $(\pm\sqrt{7}, 0)$

$\therefore$ distance between foci $2\sqrt{7}$

12. (*c*) $2b = \frac{4}{\sqrt{3}} \Rightarrow b = \frac{2}{\sqrt{3}}$

Equation of tangent $\equiv y = mx \pm \sqrt{a^2m^2 + b^2}$

Comparing with $\equiv y = \frac{-x}{6} + \frac{4}{3}$

$m = \frac{-1}{6}$ and $a^2m^2 + b^2 = \frac{16}{9}$

$\Rightarrow \frac{a^2}{36} + \frac{4}{3} = \frac{16}{9} \Rightarrow \frac{a^2}{36} = \frac{16}{9} - \frac{4}{3} = \frac{4}{9}$

$\Rightarrow a^2 = 16$

$$e=\sqrt{1-\frac{b^2}{a^2}}$$

$$e=\sqrt{1-\frac{4}{3\times16}}=\sqrt{\frac{11}{12}}=\frac{1}{2}\sqrt{\frac{11}{3}}$$

13. **(b)** $2ae=6$ and $\frac{2a}{e}=12$

$\Rightarrow ae=3$ and $\frac{a}{e}=6\Rightarrow a^2=18$

$\Rightarrow b^2=a^2-a^2e^2=18-9=9$

$\Rightarrow$ $L.R.=\frac{2b^2}{a}=\frac{2\times9}{3\sqrt{2}}=3\sqrt{2}.$

14. **(c)** Ellipse $=\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$

Given $\frac{2b^2}{a}=10\Rightarrow b^2=5a$ ...(*i*)

Now $\phi(t)=\frac{5}{12}+t-t^2$

$\phi'(t)=1-2t=0\Rightarrow t=\frac{1}{2}$

$\phi''(t)=-2<0$ $\Rightarrow$ maximum

$\Rightarrow \phi(t)_{max}=\frac{5}{12}+\frac{1}{2}-\frac{1}{4}=\frac{8}{12}=\frac{2}{3}=e$ (given)

where e is the eccentricity of ellipse

Now $b^2=a^2(1-e^2)$

$5a=a^2\left(1-\frac{4}{9}\right)$

$5a=\frac{5a^2}{9}\Rightarrow a^2-9a=0$

$\Rightarrow a=9$ $\Rightarrow$ $a^2=81$ $\Rightarrow$ $b^2=45$

$\Rightarrow a^2+b^2=81+45=126$

15. **(c)** Equation of ellipse is $\frac{x^2}{4}+\frac{y^2}{2}=1$

The equation of tangent is given as

$y=mx+\sqrt{4m^2+2}$

The line passes through (h, k).

$(k-mh)^2=4m^2+2$ ...(*i*)

The slope of normal at $(h, k)=-\frac{1}{m}$

The equation of normal at foci $\left(\sqrt{2},0\right)$ is given as

$(y-0)=-\frac{1}{m}(x-\sqrt{2})$

$\Rightarrow$ $x+my=\sqrt{2}\Rightarrow(x+my)^2=2$

The line passes through (h, k)

$\therefore$ $(h+mk)^2=2$ ...(*ii*)

On adding both the equation (*i*) and (*ii*), to find the locus.

Locus will be $x^2+y^2=4$ (auxiliary circle)

$\therefore$ $\left(-1,\sqrt{3}\right)$ satisfies the given equation

(Property of ellipse is that locus of foot of perpendicular from any foci to tangent lies on auxiliary circle of ellipse)

16. **(d)** Normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at $\left(ae,\frac{b^2}{a}\right)$ is

$x-ey=\frac{e(a^2-b^2)}{a}$

$\because (0,-b)$ lies on it, so

$be=\frac{e(a^2-b^2)}{a}$

$ab=a^2-b^2$

$[b^2=a^2(1-e^2)\Rightarrow a^2-b^2=a^2e^2]$

$\Rightarrow ab=a^2e^2$

$\Rightarrow b=ae^2\Rightarrow\frac{b^2}{a^2}=e^4\Rightarrow1-e^2=e^4$

$\Rightarrow e^4+e^2-1=0$

17. **(d)** $E:\frac{x^2}{16}+\frac{y^2}{9}=1$

$e=\sqrt{1-\frac{9}{16}}=\frac{\sqrt{7}}{4}\Rightarrow4e=\sqrt{7}$

$\because\left(\pm\sqrt{7},0\right)$ are the foci of given ellipse.

So for any point P on it; $PA+PB=2a$

$\Rightarrow PA+PB=2(4)=8$

18. **(b)** Given $x=4$

$\Rightarrow \pm\frac{a}{e}=4$ $[\because$ Directrix $x=\frac{a}{e}]$

$\Rightarrow a=4e$

$\Rightarrow a=4\times\frac{1}{2}=2$

$\Rightarrow a=2$

Now, $b^2=a^2(1-e^2)$

$\Rightarrow b^2=4\left(1-\frac{1}{4}\right)=4\times\frac{3}{4}=3$

So, equation $\frac{x^2}{4}+\frac{y^2}{3}=1$

$\Rightarrow 3x^2+4y^2=12$ ...(*i*)

Now $P(1,\beta)$ lies on it

$\Rightarrow 3+4\beta^2=12$

$\Rightarrow 4\beta^2=9$

$\Rightarrow \beta=\frac{3}{2}$

So, $P\left(1,\frac{3}{2}\right)$

Now differentiating (*i*)

$6x+8y\frac{dy}{dx}=0$

$\Rightarrow\frac{dy}{dx}=\frac{-3x}{4y}\Rightarrow\left.\frac{dy}{dx}\right]_{\left(1,\frac{3}{2}\right)}=-\frac{1}{2}$

So, Slope of normal $=2$

So, equation of normal : $y-\frac{3}{2}=2(x-1)$

$\Rightarrow 2y-3=4x-4\Rightarrow4x-2y=1$

19. **(d)** Let P be (x_1, y_1)

Equation of normal at P is $=\frac{x}{2x_1}-\frac{y}{y_1}=\frac{-1}{2}$

It passes through

$\left(\frac{-1}{3\sqrt{2}},0\right)\Rightarrow\frac{-1}{6\sqrt{2}x_1}=\frac{-1}{2}$

$\Rightarrow x_1=\frac{1}{3\sqrt{2}}$

So, $y_1=\frac{2\sqrt{2}}{3}$ (as P lies in I[st] quadrant)

So, $\beta=\frac{y_1}{2}=\frac{\sqrt{2}}{3}$.

20. **(c)** $2b=\frac{4}{\sqrt{3}}\Rightarrow b=\frac{2}{\sqrt{3}}$

Equation of tangent $\equiv y=mx\pm\sqrt{a^2m^2+b^2}$

Comparing with $\equiv y=\frac{-x}{6}+\frac{4}{3}$

$m=\frac{-1}{6}$ and $a^2m^2+b^2=\frac{16}{9}$

$\Rightarrow\frac{a^2}{36}+\frac{4}{3}=\frac{16}{9}$

$\Rightarrow\frac{a^2}{36}=\frac{16}{9}-\frac{4}{3}=\frac{4}{9}\Rightarrow a^2=16$

$e=\sqrt{1-\frac{b^2}{a^2}}$

$e=\sqrt{1-\frac{4}{3\times16}}=\sqrt{\frac{11}{12}}=\frac{1}{2}\sqrt{\frac{11}{3}}$

21. **(d)** Curve is $C\equiv\frac{x^2}{5}+\frac{y^2}{4}=1$

Let a point on curve be $(\sqrt{5}\cos\theta, 2\sin\theta)$

$PQ^2=\left(\sqrt{5}\cos\theta\right)^2+(-4-2\sin\theta)^2$

$=5\cos^2\theta+4\sin^2\theta+16+16\sin\theta$

$PQ^2=21+16\sin\theta-\sin^2\theta$

$=21+64-(\sin\theta-8)^2$

$=85-(\sin\theta-8)^2$

For PQ^2 to be maximum $\sin\theta=1$

$PQ^2_{max}=85-49=36$

22. **(c)** Let equation of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$2a-2b=10$ ...(*i*)

$be=5\sqrt{3}$

$\frac{2a^2}{b}=?$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

$a^2 = b^2(1-e^2)$, $b^2e^2 = b^2 - a^2$

$25 \times 3 = (b-a)(b+a)$

$b - a = 5$ ...(ii)

$b + a = 15$ ...(iii)

On solving (ii) and (iii), $b = 10$, $a = 5$

$\frac{2a^2}{b} = 5$

23. (d) Given that $be = 2$ and $a = 2$

(here $a < b$)

$\because a^2 = b^2(1-e^2) \Rightarrow a^2 = b^2 - (be)^2$

$\therefore b^2 = 8 =$ equation of ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$

24. (c) $3x^2 + 5y^2 = 32$

$6x + 10y.\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{-6x}{10y}$

$\left.\frac{dy}{dx}\right|_{(2,2)} = -\frac{3}{5}$

Tangent : $y - 2 = -\frac{3}{5}(x-2)$ x-intersection

$\Rightarrow Q\left(\frac{16}{3}, 0\right)$

Normal : $y - 2 = \frac{5}{3}(x-2)$ x-intersection

$\Rightarrow R\left(\frac{4}{5}, 0\right)$

Area is $= \frac{1}{2}(QR) \times 2 = QR = \frac{16}{3} - \frac{4}{5} = \frac{68}{15}$

25. (d) $3x^2 + 4y^2 = 12 \Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$

$x = 2\cos\theta$, $y = \sqrt{3}\sin\theta$

Let $P(2\cos\theta, \sqrt{3}\sin\theta) \equiv (x_1, y_1)$

Equation of normal is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$

$\Rightarrow 2x\sin\theta - \sqrt{3}\cos\theta y = \sin\theta\cos\theta$

Slope: $\frac{2}{\sqrt{3}}\tan\theta = -2$

$\therefore \tan\theta = -\sqrt{3}$ ...(i)

Equation of tangent and it passes through (4, 4)

$3xx_1 + 4yy_1 = 12$

$3x\cos\theta + 2\sqrt{3}\sin\theta\, y = 6$

Tangent passes via (4, 4)

$12\cos\theta + 8\sqrt{3}\sin\theta = 6$ ...(ii)

from (i) and (ii)

$\cos\theta = -\frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$

$\therefore \theta = 120°$

Hence point P is $(2\cos 120°, \sqrt{3}\sin 120°)$

$P\left(-1, \frac{3}{2}\right)$, $Q(4, 4)$

$PQ = \sqrt{25 + \frac{25}{4}} = \frac{5\sqrt{5}}{2}$

26. (a) Tangent at $\left(3, -\frac{9}{2}\right)$

$\frac{3x}{a^2} - \frac{9y}{2b^2} = 1$ $\left[\because \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1\right]$

Comparing this with $x - 2y = 12$

$\frac{3}{a^2} = \frac{9}{4b^2} = \frac{1}{12}$

We get $a = 6$ and $b = 3\sqrt{3}$

Latus Rectum $= \frac{2b^2}{a} = 9$

27. (a) $m_{SB} \cdot m_{S'B} = -1$

$\Rightarrow b^2 = a^2e^2$ ...(i)

Area $= \frac{1}{2}S'B.SB = 8$

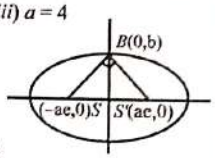
$\Rightarrow a^2e^2 + b^2 = 16$ [$\because SB^2 = ae^2 + b^2 = S'B^2$] ... (ii)

$b^2 = a^2(1-e^2)$ ...(iii)

using (i), (ii), (iii) $a = 4$

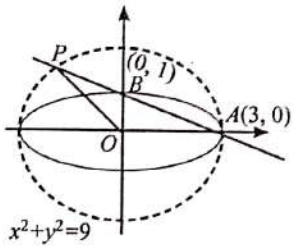
$b = 2\sqrt{2}$

$e = \frac{1}{\sqrt{2}}$



$\therefore \ell(L.R) = \frac{2b^2}{a} = 4$

28. (c)



Equation of line AB or AP =

$\frac{x}{3} + \frac{y}{1} = 1 \Rightarrow x = 3 - 3y$

For intersection point of line AB: $x + 3y = 3$ and circle: $x^2 + y^2 = 9$

$(3-3y)^2 + y^2 = 9$

$\Rightarrow 10y^2 - 18y = 0$

$\Rightarrow 5y^2 - 9y = 0 \Rightarrow y(5y-9) = 0$

$\Rightarrow y = 0, \frac{9}{5}$

Hence, $x = 3(1-y)$

$= 3\left(1 - \frac{9}{5}\right) = -\frac{12}{5}$

$\therefore P = \left(\frac{-12}{5}, \frac{9}{5}\right)$ or (3, 0)

Area of $\Delta AOP = \frac{1}{2} \times OA \times$ height

$= \frac{1}{2} \times 3 \times \frac{9}{5} = \frac{27}{10} = \frac{m}{n}$

$\therefore m - n = 17$

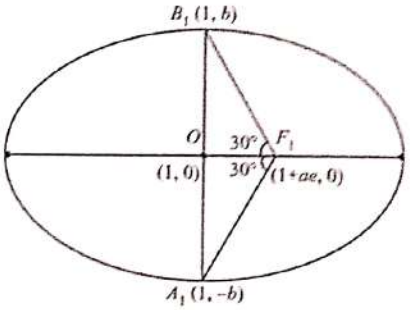
29. [9] Length of Latus Rectum $= \frac{2b^2}{a} = \frac{1}{2}$

$\Rightarrow 4b^2 = a$ ... (i)

Ellipse $\frac{(x-1)^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{b}{ae} = \tan 30° \Rightarrow \frac{b}{ae} = \frac{1}{\sqrt{3}}$

$\Rightarrow 3b^2 = a^2e^2 = a^2 - b^2$



$\Rightarrow 4b^2 = a^2$ ...(ii)

From (i) and (ii)

$a = a^2$

$\therefore a = 1$

$\Rightarrow b^2 = \frac{1}{4} \Rightarrow b = \frac{1}{2}$

Hence, $((2a) + (2b))^2 = 9$

30. [13] Ellipse is

$\frac{x^2}{2} + \frac{y^2}{4} = 1; e = \frac{1}{\sqrt{2}}; S = \left(0, -\sqrt{2}\right)$

Chord of contact is

$\frac{x}{\sqrt{2}} + \frac{\left(2\sqrt{2}-2\right)y}{4} = 1$

$\Rightarrow \frac{x}{\sqrt{2}} = 1 - \frac{\left(\sqrt{2}-1\right)y}{2}$

Solving with ellipse

$\Rightarrow y = 0, \sqrt{2} \therefore x = \sqrt{2}, 1$

$P \equiv \left(1, \sqrt{2}\right), Q \equiv \left(\sqrt{2}, 0\right)$

$\therefore (SP)^2 + (SQ)^2 = 13$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 14

Conic Sections: Hyperbola

Prarambh (Topicwise)

1. **(c)** $\frac{x^2}{10-a}+\frac{y^2}{4-a}=1$ represents on hyperbola if

$10-a>0$ & $4-a<0$

$4<a<10$.

2. **(a)** Directrix of hyperbola, $x=\frac{a}{e}$

where $e=\sqrt{\frac{b^2+a^2}{a^2}}=\frac{\sqrt{b^2+a^2}}{a}$

3. **(a)** Conjugate axis is 5 and distance between foci

$=13 \Rightarrow 2b=5$ and $2ae=13$.

Now, also we know for hyperbola

$b^2=a^2(e^2-1) \Rightarrow \frac{25}{4}=\frac{(13)^2}{4e^2}(e^2-1)$.

$\Rightarrow \frac{25}{4}=\frac{169}{4}-\frac{169}{4e^2}$ or $e^2=\frac{169}{144} \Rightarrow e=\frac{13}{12}$

or $a=6, b=\frac{5}{2}$ or hyperbola is

$\frac{x^2}{36}-\frac{y^2}{25/4}=1$

$\Rightarrow 25x^2-144y^2=900$

4. **(c)** Vertices $(\pm4,0)\equiv(\pm a,0) \Rightarrow a=4$

Foci $(\pm6,0)\equiv(\pm ae,0) \Rightarrow e=\frac{6}{4}=\frac{3}{2}$

5. **(c)** $(4x+8)^2-(y-2)^2=-44+64-4$

$\Rightarrow \frac{16(x+2)^2}{16}-\frac{(y-2)^2}{16}=1$

Transverse and conjugate axes are $y=2$, $x=-2$

6. **(c)** Foci $(0,\pm4)\equiv(0,\pm be) \Rightarrow be=4$

Vertices

$(0,\pm2)\equiv(0,\pm b) \Rightarrow b=2 \Rightarrow a=2\sqrt{3}$

Hence equation is $\frac{-x^2}{(2\sqrt{3})^2}+\frac{y^2}{(2)^2}=1$ or $\frac{y^2}{4}-\frac{x^2}{12}=1$.

7. **(a)** $(x-2)^2+(y-1)^2=4\left[\frac{(x+2y-1)^2}{5}\right]$

$\Rightarrow 5[x^2+y^2-4x-2y+5]$

$=4[x^2+4y^2+1+4xy-2x-4y]$

$\Rightarrow x^2-11y^2-16xy-12x+6y+21=0$

8. **(d)** We have $16(x^2-2x)-3(y^2-4y)=44$

or $16(x-1)^2-3(y-2)^2=48$

or $\frac{(x-1)^2}{3}-\frac{(y-2)^2}{16}=1$

This equation represents a hyperbola with eccentricity

$e=\sqrt{1+\frac{16}{3}}=\sqrt{\frac{19}{3}}$

9. **(d)** Given Hyperbola

$9(x^2+2x+1)-16(y^2-2y+1)$

$=151+9-16 \Rightarrow \frac{(x+1)^2}{16}-\frac{(y-1)^2}{9}=1$

foci (4, 1), (–6, 1)

10. **(d)** For the ellipse, $a=5$ and

$e=\sqrt{\frac{25-9}{25}}=\frac{4}{5}$

$\therefore \quad ae=4$

Hence, the foci are (–4, 0) and (4, 0).

For the hyperbola, $ae=4$, $e=2$

$\therefore \quad a=2$

$b^2=4(4-1)=12$ or $b=\sqrt{12}$

11. **(a)** $(x-\alpha)^2+(y-\beta)^2$

$=k(lx+my+n)^2$

or $\sqrt{(x-\alpha)^2+(y-\beta)^2}$

$=\sqrt{k}\sqrt{l^2+m^2}\,\frac{(lx+my+n)}{\sqrt{l^2+m^2}}$

or $\frac{PS}{PM}=\sqrt{k}\sqrt{l^2+m^2}$ where $P(x, y)$ is any point on the curve.

Fixed point $S(\alpha, \beta)$ is focus and fixed line $lx+my+n=0$ is directrix.

If $k(l^2+m^2)=l$, P lies on a parabola.

If $k(l^2+m^2)<l$, P lies on an ellipse.

If $k(l^2+m^2)>l$, P lies on a hyperbola.

If $k=0$, P lies on a point circle.

12. **(b)** $\frac{9}{a^2}-\frac{4}{b^2}=1$

and $\frac{289}{a^2}-\frac{144}{b^2}=1$

we get $a^2=1$, $b^2=\frac{1}{2} \Rightarrow 2a=2$.

$\Rightarrow$ Length of transverse axis is $2a=2$.

13. **(c)** We must have $ae=a'e'$

$e^2=1-\frac{b^2}{16}$, $e'^2=1+\frac{81/25}{144/25}=\frac{25}{16}$

$\Rightarrow 4e=\frac{12}{5}e'$

Squaring both sides, $16e^2=\frac{144}{25}e'^2$

$\Rightarrow$ On solving we get $b^2=7$

14. **(a)** $e=\sqrt{1-\frac{5}{9}}$, $e'=\sqrt{1+\frac{45/4}{45/5}}$

$e=\frac{2}{3}$, $e'=\frac{3}{2}$

$\therefore \quad ee'=1$

15. **(d)** As, $r>1$, $1-r<0$ and $-(1+r)<0$

So, L.H.S. = negative, but R.H.S. = positive. So that is not possible.

16. **(d)** $x^2-8x-4y^2-16y-4$

$\frac{(x-4)^2}{4}-(y+2)^2=1$

$=4+2\sec\theta=-2+\tan\theta$

17. **(b)** For hyperbola $\frac{x^2}{5}-\frac{y^2}{5\cos^2\alpha}=1$

We have,

$e_1^2=1+\frac{b^2}{a^2}=1+\frac{5\cos^2\alpha}{5}=1+\cos^2\alpha$

For ellispe

$\frac{x^2}{25\cos^2\alpha}+\frac{y^2}{25}=1$

We have $e_2^2=1-\frac{25\cos^2\alpha}{25}=\sin^2\alpha$

$\Rightarrow \quad e_1=\sqrt{3}e_2$

$\Rightarrow \quad e_1^2=3e_2^2$

$\Rightarrow \quad 1+\cos^2\alpha=3\sin^2\alpha$

$\Rightarrow \quad 2a=4\sin^2\alpha$

$\Rightarrow \quad \sin\alpha=\frac{1}{\sqrt{2}}$

$\Rightarrow \quad \alpha=\frac{\pi}{4}$

18. (*b*) $x=\frac{a}{2}\left(t+\frac{1}{t}\right) \quad y=\frac{b}{2}\left(t-\frac{1}{t}\right)$

$\left(\frac{2x}{a}\right)^2-\left(\frac{2y}{b}\right)^2=4$

$\left(\frac{x}{a}\right)^2-\left(\frac{y}{b}\right)^2=1$

$x^2-y^2=4$

$x^2-y^2=6+2\cos t+2-4\cos^2\frac{t}{2}$

$x^2-y^2=6+2(1+\cos x)-4\cos^2\frac{t}{2}$

$x^2-y^2=6+4\cos^2\frac{\pi}{2}-4\cos^2\frac{\pi}{2}$

$x^2-y^2=6$

19. (*a*) The equation is

$(x-0)^2+(y-0)^2=a^2$.

20. (*c*) If $y=2x+\lambda$ is tangent to given hyperabola, then

$\lambda=\pm\sqrt{a^2m^2-b^2}=\pm\sqrt{(100)(4)-144}$.

$=\pm\sqrt{256}=\pm16$

21. (*a*) Suppose point of contact be (h, k), then tangent is

$hx-4ky-5=0\equiv3x-4y-5=0$

or $h=3, k=1$

Hence the point of contact is (3, 1).

22. (*a*) Tangent to $\frac{x^2}{1}-\frac{y^2}{3}=1$ and perpendicular to $x+3y-2=0$ is given by

$y=3x\pm\sqrt{9-3}=3x\pm\sqrt{6}$.

23. (*a*) According to the question,

$\frac{2\sqrt{9m^2-49}}{\sqrt{1+m^2}}=2$

or $9m^2-49=1+m^2$

or $8m^2=50$

or $m=\pm\frac{5}{2}$

24. (*c*) Equation of normal to hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at $(a\sec\theta, b\tan\theta)$ is

$\frac{a^2x}{a\sec\theta}+\frac{b^2y}{b\tan\theta}=a^2+b^2$

25. (*d*) Applying the formula, the required normal is

$\frac{16x}{8}+\frac{9y}{3\sqrt{3}}=16+9$ *i.e.*, $2x+\sqrt{3}y=25$

Trick: This is the only equation among the given options at which the point $(8, 3\sqrt{3})$ is located.

26. (*b*) We know that the equation of the normal of the conic $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at point $(a\sec\theta, b\tan\theta)$

is $ax\cos\theta+by\cot\theta=a^2+b^2$

or $y=\frac{-a}{b}\sin\theta\,x+\frac{a^2+b^2}{b\cot\theta}$ Comparing above equation with equation

$y=mx+\frac{25\sqrt{3}}{3}$ and taking $a=4, b=3$.

we get,

$\frac{a^2+b^2}{b\cot\theta}=\frac{25\sqrt{3}}{3}\Rightarrow\tan\theta=\sqrt{3}\Rightarrow\theta=60^\circ$

and $m=-\frac{a}{b}\sin\theta=\frac{-4}{3}\sin60^\circ$.

$=\frac{-4}{3}\times\frac{\sqrt{3}}{2}=\frac{-2}{\sqrt{3}}$

27. (*a*) Tangent to $y^2=8x\Rightarrow y=mx+\frac{2}{m}$

Tangent to

$\frac{x^2}{1}-\frac{y^2}{3}=1\Rightarrow y=mx\pm\sqrt{m^2-3}$

On comparing, we get

$m=\pm2$ or tangent as $2x\pm y+1=0$.

28. (*d*)

Normal at P $(\theta)\equiv$ P $(2\sec\theta, 2\tan\theta)$

$2x\cos\theta+2y\cot\theta=8$

$\Rightarrow x\cos\theta+y\cot\theta=4$

$\therefore$ $G(4\sec\theta, 0)$, $g(0, 4\tan\theta)$ and $c(0, 0)$

$PG=\sqrt{4\sec^2\theta+4\tan^2\theta}=PC=Pg$

29. (*b*) $\frac{x^2}{18}-\frac{y^2}{9}=1$ given line is $y=x$

$\therefore$ slope of tangent

$\therefore$ equation is

$y=mx\pm\sqrt{a^2m^2-b^2}\quad\Rightarrow\quad y=-x\pm3$

30. (*d*) is a tangent to the hyperobla $^2-3^2=1$

$\frac{x^2}{1}-\frac{y^2}{\frac{1}{3}}=1$

$c^2=a^2m^2-b^2$

$c^2=m^2-\frac{1}{3}$

$m^2-c^2=\frac{1}{3}$

$\boxed{m^2>c^2}$

31. (*b*) According to question,

$S\equiv25x^2-16y^2-400=0$

Equation of required chord is $S_1=T$... (*i*)

Here, $S_1=25(5)^2-16(3)^2-400$

$=625-144-400=81$

and $T\equiv25xx_1-16yy_1-400$, where $x_1=5, y_1=3$

$=25(x)(5)-16(y)(3)-400$

$=125x-48y-400$

So from (*i*), required chord is

$125x-48y-400=81$

or $125x-48y=481$.

32. (*d*) $T=S_1\Rightarrow5.\ 5x-3(-3)y=5.5^2-3(-3)^2\Rightarrow25x+9y=98$

33. (*d*) Equation of chord of contact from (1, 1) and (2, 2) to the hyperbola are given by.

$\frac{x_1}{a^2}-\frac{y_1}{b^2}=1$ and $\frac{x_2}{a^2}-\frac{y_2}{b^2}=1$

Therefore slopes of these lines are

$m_1=\frac{b^2}{a^2}\cdot\frac{x_1}{y_1}$

and $m_2=\frac{b^2}{a^2}\cdot\frac{x_2}{y_2}$

Given both lines are perpendicular

$\Rightarrow m_1\cdot m_2=-1\Rightarrow\frac{x_1x_2}{y_1y_2}=-\frac{a^4}{b^4}$

34. (*a*) Let mid point be (h, k)

Equation of a chord whose mid point is (h, k) would be ,

or $3xh-2yk+2(x+h)-3(y+k)$

$=3h^2-2k^2+4h-6k$

$\Rightarrow x(3k+2)-y(2k+3)-2h+3k$

$-3h^2+2k^2=0$

It's slope is $\frac{3h+2}{2k+3}=2$ (given)

$\Rightarrow\quad3=4+4$

$\Rightarrow$ required locus is $3-4=4$.

35. (*d*) Given, equation of hyperbola $2x^2+5xy+2y^2+4x+5y=0$ and equation of

asymptotes

$2x^2+5xy+2y^2+4x+5y+\lambda=0$ ... (*i*), which is the equation of a pair of straight lines. We know that the standard equation of a pair of straight lines is

$ax^2+2hxy+by^2+2gx+2fy+c=0$.

Comparing equation (*i*) with standard equation, we get

$a=2, b=2,\ h=\frac{5}{2}, g=2, f=\frac{5}{2}$

and $c=\lambda$.

We also know that the condition for a pair of straight lines is

$abc+2fgh-af^2-bg^2-ch^2=0$.

Therefore $4\lambda+25-\frac{25}{2}-8-\frac{25}{4}\lambda=0$

or $-\frac{9\lambda}{4}+\frac{9}{2}=0$ or $\lambda=2$. Substituting value of λ in equation (*i*), we get

$2x^2+5xy+2y^2+4x+5y+2=0$.

36. (*d*) Transverse axis is the equation of the angle bisector containing point (2, 3), which is given by

$$\frac{3x-4y+5}{5}=\frac{12x+5y-40}{13}$$

or $21x+77y=265$

37. (*b*) $xy=c^2$ as $c^2=\frac{a^2}{2}$. Here,co-ordinates of focus are

$(ae\cos 45°, ae\sin 45°)\equiv(c\sqrt{2}, c\sqrt{2})$,

$\{\because e=\sqrt{2}, a=c\sqrt{2}\}$

Similarly other focus is $(-c\sqrt{2}, -c\sqrt{2})$

Note: Students should remember this question as a fact.

38. (*c*) Multiplying both, we get $x^2-y^2=a^2$. This is equation of rectangular hyperbola as $a=b$.

39. (*b*) Transverse axis is along the line $y=x$. Solving $y=x$ and $xy=18$, we have $x^2=18$ or $x=\pm 3\sqrt{2}$. Then the two vertices of the hyperbola are $(\pm 3\sqrt{2}, \pm 3\sqrt{2})$. The distance between them is $\sqrt{72+72}=12$

40. (*b*) Tangent at $(a\sec\theta, b\tan\theta)$ is,

$$\frac{x}{(a/\sec\theta)}-\frac{y}{(b/\tan\theta)}=1$$

or $\frac{a}{\sec\theta}=1, \frac{b}{\tan\theta}=1$

$\Rightarrow a=\sec\theta \quad b=\tan\theta$

or (a, b) lies on $x^2-y^2=1$

41. (*c*) Since the general equation of second degree represents a rectangular hyperbola, if $\Delta\neq 0$, $h^2>ab$ and coefficient of x^2 + coefficient of $y^2=0$. Therefore the given equation represents a rectangular hyperbola, if $\lambda+5=0$ i.e., $\lambda=-5$

42. (*d*) $\because$ Distance between directrices $=\frac{2a}{e}$.

$\because$ Eccentricity of rectangular hyperbola $=\sqrt{2}$

$\therefore$ Distance between directrics $=\frac{2a}{\sqrt{2}}$.

Given that, $\frac{2a}{\sqrt{2}}=10 \Rightarrow 2a=10\sqrt{2}$

Now, distance between foci

$=2ae=(10\sqrt{2})(\sqrt{2})=20$.

43. (*b*) Let equation of circle is $x^2+y^2=a^2$

Parametric form of $xy=c^2$ are

$x=ct, \quad y=\frac{c}{t}$

$\Rightarrow c^2t^2+\frac{c^2}{t^2}=a^2 \Rightarrow c^2t^4-a^2t^2+c^2=0$

Product of roots will be, $t_1t_2t_3t_4=\frac{c^2}{c^2}=1$

44. (*c*) Tangent to the *hyp.* $xy=c^2$

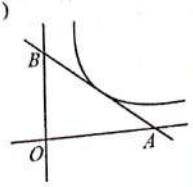
$\frac{x}{x_1}+\frac{y}{y_1}=2$ (16, 1)

$\frac{x}{16}+\frac{y}{1}=2$

$x+16y=32$

$a(32, 0)$

$a(0, 2)$



Area $=\frac{1}{2}\times 2\times 32=32$ Sq. unit

45. (*d*) Equation of hyperbola will be $(x-3)(y-5)+k=0$

This passes through (7, 8)

$\Rightarrow k=-12$

Hence equation is $xy-3y-5x+3=0$

46. (*a*) $T=0 \Rightarrow xy_1+yx_1=18$

$\Rightarrow 3x+2y=18$

47. (*a*) Let it be $xy-4x-3y+\lambda=0$,

Now $(x-3)(y-4)=xy-4x-3y+12$,

So $\lambda=12$ and the asymptotes are $x=3$ and $y=4$.

48. (*d*) y' for hyperbola $=\frac{x_1}{y_1}$ so, y' for ellipse $=-\frac{y_1}{x_1}=-\frac{b^2}{a^2}\frac{x_1}{y_1}$ (orthogonal cut)

$\Rightarrow b^2x_1^2=a^2y_1^2$, also, $b^2x_1^2+a^2y_1^2=a^2b^2$

$\Rightarrow x_1^2=\frac{a^2}{2}$ and $y_1^2=\frac{b^2}{2}$ also $x_1^2-y_1^2=c^2 \Rightarrow a^2-b^2=2c^2$.

Prabal (JEE Main Level)

1. (*b*) Given hyperbola

$(x-2)^2-(y-2)^2=-16$

Rectangular hyperbola

$\therefore \quad e=\sqrt{2}$.

2. (*c*) If e_1 & e_2 are eccentircities of two congugate hyperbolas

then $\frac{1}{e_1^2}+\frac{1}{e_2^2}=1$

$\therefore \quad e_1=\sec\alpha$ & $e_2=\operatorname{cosec}\alpha$

3. (*c*) $\frac{2b^2}{a}=8$ ... (*i*)

and $2b=\frac{2ae}{2}$ ... (*ii*)

and $e^2=1+\frac{b^2}{a^2}$ ... (*iii*)

by (*i*), (*ii*), (*iii*) $e=\frac{2}{\sqrt{3}}$

4. (*d*) $\sqrt{3}x-y-4\sqrt{3}k=0$ ... (*i*)

$\sqrt{3}kx+ky-4\sqrt{3}=0$ ... (*ii*)

Solve (1) and (2)

$x=2\frac{(1+k^2)}{k}$ and $y=\frac{2\sqrt{3}(1-k^2)}{k}$

$\frac{x^2}{4}-\frac{y^2}{12}=4 \Rightarrow \frac{x^2}{16}-\frac{y^2}{48}=1$ Hyperbola

5. (*a*) $\frac{2b^2}{a}=8; e=\frac{3}{\sqrt{5}}$

$\Rightarrow b^2=4a; e^2=\frac{9}{5}$

$1+\frac{b^2}{a^2}=\frac{9}{5} \Rightarrow \frac{b^2}{a^2}=\frac{4}{5} \Rightarrow a=5 \Rightarrow b^2=20$

Hyp. $\frac{x^2}{25}-\frac{y^2}{20}=1 \Rightarrow 4x^2-5y^2=100$

6. (*c*) $C(0, 0)$ $A_1(4, 0)$ $F_1(6, 0)$

$CA_1=4$ $\quad CF_1=6$

$\Rightarrow a=4$ $\quad ae=6$

$a^2e^2=36 \Rightarrow a^2\left(1+\frac{b^2}{a^2}\right)=36$

$\Rightarrow b^2=36-16 \Rightarrow b^2=20$

Hyp. $\frac{x^2}{16}-\frac{y^2}{20}=1$ or $5x^2-4y^2=80$

7. (*b*) Centre of hyp. will be mid point of A_1 & A_2

$=\left(\frac{10+0}{2}, 0\right)=(5, 0)$ & check options

8. (*c*) $(x-3)^2+(y+1)^2=(4x+3y)^2$

or $(x-3)^2+(y+1)^2=25\left(\frac{4x+3y}{5}\right)^2$ or

PS = 5PM

Therefore, the directrix is $4x+3y=0$ and the focus is $(3, -1)$.

So, the equation of transverse axis is

$y+1=\frac{3}{4}(x-3)$

or $3x-4y=13$

9. (*c*) $2a=7 \Rightarrow a=\frac{7}{2}$

Let the Equation of hyp.

$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

passes through $(5, -2)$

$\frac{25}{a^2}-\frac{4}{b^2}=1$

$\frac{25}{a^2}-1=\frac{4}{b^2}$

$b^2=\frac{4a^2}{25-a^2}=\frac{4\times\frac{49}{4}}{25-\frac{49}{4}}=\frac{196}{51}$

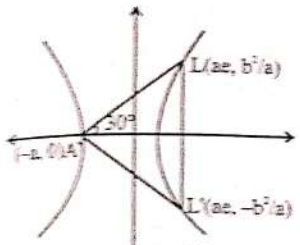
Equation $\frac{4x^2}{49}-\frac{51y^2}{196}=1$

10. (b) The equation of the hyperbola is

$$\frac{\left\{\frac{(2x-y+4)}{\sqrt{5}}\right\}^2}{\frac{1}{2}} = \frac{\left\{\frac{(x+2y-3)}{\sqrt{5}}\right\}^2}{\frac{1}{3}}$$

or $\frac{2}{5}(2x-y+4)^2 - \frac{3}{5}(x+2y-3)^2 = 1$

11. (d) $\tan 30° = \frac{b^2/a}{a+ae}$ or $\frac{1+e}{\sqrt{3}} = e^2 - 1$



or $e-1 = \frac{1}{\sqrt{3}}$ or $e = \frac{\sqrt{3}+1}{\sqrt{3}}$

12. (a) Any tangent to hyperbola forms a triangle with the asymptotes which is constant area ab.

$\Rightarrow ab = a^2 \tan\lambda$.

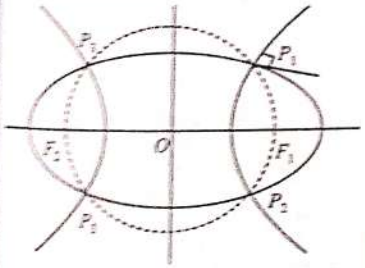
$= \frac{b}{a} = \tan\lambda$.

$e = \sqrt{1+\frac{b^2}{a^2}}$

$\Rightarrow e = \sqrt{1+\tan^2\lambda} = \sec\lambda$.

13. (a) If they intersect at right angles then circle will pass through its focus

Circle will be



$x^2 + y^2 = (OF_1)^2$

$x^2 + y^2 = (\sqrt{5})^2$

$x^2 + y^2 = (\sqrt{5})^2$, $F_1(ae, 0)\ e = \sqrt{5}$

$x^2 + y^2 = 5$; $F_1(\sqrt{5}, 0)$

14. (a) $\sqrt{2}^2 \sec^2\theta + \sqrt{2}^2 \tan^2\theta = 6$

$\Rightarrow 1 + 2\tan^2\theta = 3$

$\therefore \theta = \pi/4$ for first quadrant

15. (b) Since $\frac{e}{2}$ and $\frac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate hyperbola, we have

$\frac{4}{e^2} + \frac{4}{e'^2} = 1$

or $4 = \frac{e^2 e'^2}{e^2 + e'^2}$

The line passing through the points $(e, 0)$ and $(0, e')$ is $e'x + ey - ee' = 0$

It is tangent to the circle $x^2 + y^2 = r^2$.

Hence, $\frac{ee'}{\sqrt{e^2+e'^2}} = r$

or $r^2 = \frac{e^2 e'^2}{e^2 + e'^2} = 4$

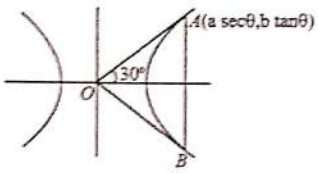
or $r = 2$

16. (d) $\theta = 30°$

$\frac{b \tan\theta}{a \sec\theta} = \tan 30°$

$\frac{b}{a}\sin\theta = \frac{1}{\sqrt{3}}$

$\frac{b}{a} = \frac{1}{\sqrt{3}\sin\theta}$



$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{1}{3\sin^2\theta}$

$e^2 > 1 + \frac{1}{3}$

$e > \frac{2}{\sqrt{3}}$

17. (a) $4x^2 - 9y^2 = 36 \Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1$

$5x + 2y - 10 = 0$

$m = \frac{-5}{2} \Rightarrow m' = \frac{2}{5}$

Equation of tangent $y = m'x \pm \sqrt{a^2(m')^2 - b^2}$

$y = \frac{2}{5}x \pm \sqrt{9 \times \frac{4}{25} - 4}$

$y = \frac{2x}{5} \pm \sqrt{-ve}$ so not possible

18. (c) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Tangent

$y = mx \pm \sqrt{a^2m^2 - b^2}$ ... (i)

$\frac{x^2}{(-b^2)} - \frac{y^2}{(-a^2)} = 1$

$y = mx \pm \sqrt{(-b^2)m^2 + a^2}$ ... (ii)

(i) and (ii) are same

$\frac{1}{1} = \frac{1}{1} = \frac{\sqrt{a^2m^2 - b^2}}{\sqrt{a^2 - b^2m^2}}$

$a^2 - b^2m^2 = a^2m^2 - b^2$

$m^2 = 1 \Rightarrow m = \pm 1$

$y = \pm x \pm \sqrt{a^2 - b^2}$

19. (c) The equation of the hyperbola is

$\frac{x^2}{9} - \frac{y^2}{16} = 1$

The equation of the tangent is

$y = mx + \sqrt{9m^2 - 16}$

or $\sqrt{9m^2 - 16} = 2\sqrt{5}$

or $m = \pm 2$ or $a + b =$ Sum of roots $= 0$

20. (d) Locus of the feet of the $\perp^n$ drawn from any focus of the the hyp. upon any tangent is its auxilary circle

Hyp. $\frac{x^2}{\left(\frac{1}{16}\right)} - \frac{y^2}{\left(\frac{1}{9}\right)} = 1$

Auxiliary circle $x^2 + y^2 = \frac{1}{16}$

21. (b) $\frac{hx}{16} - \frac{ky}{9} = 1$

Let the point R is (h, k)

So the equation of chord of contact.

$\frac{hx}{16} - \frac{ky}{9} = 1$

It passes through (2, 1) so $\frac{2h}{16} - \frac{k}{9} = 1$

$\frac{h}{8} - \frac{k}{9} = 1$

so locus of R is $9x - 8y = 72$

22. (c) Let the point be (h, k).

Then the equation of the chord of contact is $hx + ky = 4$

Since $hx + ky = 4$ is tangent to $xy = 1$,

$x\left(\frac{4 - hx}{k}\right) = 1$ has two equal roots.

Therefore,

$hx^2 - 4x + k = 0$

or $hk = 4$

Thus, the locus of (h, k) is $xy = 4$.

Hence, $c^2 = 4$

23. (d) The equation of the normal to the hyperbola

$\frac{x^2}{4} - \frac{y^2}{1} = 1$

at $(2\sec\theta, \tan\theta)$ is $2x\cos\theta + y\cot\theta = 5$.

The slope of the normal is $-2\sin\theta = -1$

or $\sin\theta = \frac{1}{2}$ or $\theta = \frac{\pi}{6}$

y-intercept of the normal $= \frac{5}{\cot\theta} = \frac{5}{\sqrt{3}}$

As it touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

we have $a^2 + b^2 = \frac{25}{9}$

24. **(a)** Equation of the normal is $ax\cos\theta + by\cot\theta = a^2 + b^2$

The normal at P meets the coordinate axes at $G\left(\frac{a^2+b^2}{a}\sec\theta, 0\right)$ and $g\left(0, \frac{a^2+b^2}{b}\tan\theta\right)$

$$\therefore \quad PG^2 = \left(\frac{a^2+b^2}{a}\sec\theta - a\sec\theta\right)^2 + (b\tan\theta - 0)^2$$

$$PG^2 = \frac{b^2}{a^2}(b^2\sec^2\theta + a^2\tan^2\theta)$$

When $\tan\theta = 0$

$$PG = \frac{b^2}{a}$$

25. **(a)** Director circle of the circle $x^2 + y^2 = a^2$ is $x^2 + y^2 = 2a^2$.

The semi-transverse axis is $\sqrt{3}a$.

The radius of the circle is $\sqrt{2}a$.

Hence, director circle and hyperbola do not intersect.

26. **(d)** The product of perpendiculars drawn from the foci upon any of its tangents is 9.

Therefore, $b^2 = 9$

Also, $\frac{b}{a} = \tan 30^\circ = \frac{1}{\sqrt{3}}$

$\therefore \quad a^2 = 3b^2 = 27$

Therefore, the required locus is the director circle of the hyperbola which is given by

$x^2 + y^2 = 27 - 9$

or $x^2 + y^2 = 18$

If $b/a = \tan 60^\circ$, then

$a^2 = \frac{b^2}{3} = \frac{9}{3} = 3$

Hence, the required locus is $x^2 + y^2 = 3 - 9 = -6$ which is not possible.

27. **(d)** The asymptotes of a rectangular hyperbola are perpendicular to each other.

Given one asymptote $x + y + c = 0$

Let the other asymptote be $x - y + \lambda = 0$

We also know that the asymptotes pass through the center of the hyperbola. Therefore, the line $2x - y = 0$ and the asymptotes must be concurrent.

Thus, we have $\begin{vmatrix} 2 & -1 & 0 \\ 1 & 1 & c \\ 1 & -1 & \lambda \end{vmatrix} = 0$

or $\lambda = -\frac{c}{3}$, $3x - 3y - c = 0$

28. **(c)** The foci of the hyperbola lie on $y = x$.

So, the major axis is $y = x$.

The major axis of hyperbola bisects the asymptote. Therefore, the equation of other asymptote is $x = 2y$.

Thus, the equation of hyperbola is $(y - 2x)(x - 2y) + k = 0$.

Given that it passes through (3, 4), we get $k = 10$. Hence, the required equation is $2x^2 + 2y^2 - 5xy + 10 = 0$

29. **(d)** Let the circle on which P, Q, R, S lie be

$x^2 + y^2 + 2gx + 2fy + C_1 = 0$

How let $\left(ct, \frac{c}{t}\right)$ lie on it

$\Rightarrow \quad c^2t^4 + 2gct^3 + C_1t^2 + 2fct + c^2 = 0$

where t_1, t_2, t_3, t_4 represents the parameters for P, Q, R, S

$\therefore \quad t_1t_2t_3t_4 = 1$

also since orthocentre of ΔPQR be

$\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right) \Rightarrow (-x_4, -y_4)$

30. **(a)** Let $A\left(ct_1, \frac{c}{t_1}\right), B\left(ct_2, \frac{c}{t_2}\right), C\left(ct_3, \frac{c}{t_3}\right)$

then orthocentre be

$H\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right)$ which lies on $xy = c^2$

31. **(a)** The equation of normal at any point $\left(ct, \frac{c}{t}\right)$ is

$ct^4 - xt^3 + ty - c = 0$

$\therefore \quad$ Slope of normal $= t^2$

Let P be (h, k)

Then, $ct^4 - ht^3 + tk - c = 0$

or $\Sigma t_i = \frac{h}{c}$ and $\Sigma t_i t_j = 0$

or $\Sigma t_i^2 = (\Sigma t_i)^2$

or $h^2 = c^2\lambda$.

Therefore, the required locus is $x^2 = \lambda c^2$.

32. **(a)** The midpoint of the chord is $\left(\frac{(x_1+x_2)}{2}, \frac{(y_1+y_2)}{2}\right)$.

The equation of the chord in terms of its midpoint is $T = S_1$

or $x\left(\frac{y_1+y_2}{2}\right) + y\left(\frac{x_1+x_2}{2}\right) - c^2$

$= 2\left(\frac{x_1+x_2}{2}\right)\left(\frac{y_1+y_2}{2}\right) - c^2$

or $x(y_1 + y_2) + y(x_1 + x_2) = (x_1 + x_2)(y_1 + y_2)$

or $\frac{x}{x_1+x_2} + \frac{y}{y_1+y_2} = 1$

33. **(a)** Curve $xy = c^2$

Point $P(ct, \frac{c}{t})$ Point $Q(ct', \frac{c}{t'})$

Equation of normal $xt^3 - yt = c(t^4 - 1)$

Point Q satisfy the equation $ct't^3 - \frac{c}{t'}t = c(t^4 - 1)$

$t't^3 - \frac{t}{t'} = t^4 - 1$

$(t')^2t^3 - t = t'(t^4 - 1)$

$t'^2t^3 + t' - t - t't^4 = 0$

$\Rightarrow \quad t'(t't^3 + 1) - t(1 + t't^3) = 0$

$t' = t \quad$ or $\quad t' = -\frac{1}{t^3}$

so only possibility $t' = -\frac{1}{t^3}$

34. **(a)** by $T = S_1$

$\frac{xk + yh}{2} = hk$

$\Rightarrow \quad \frac{x}{h} + \frac{y}{k} = 2$

$\therefore \quad m = \frac{-1/h}{+1/k}$

$\Rightarrow \quad k + mh = 0$

$\Rightarrow \quad y + mx = 0$

35. **(d)** Equation of tangent at $P(ct, c/t)$ is

$\frac{x}{t} + ty = 2c$

or $x + t^2y = 2ct$ ...(i)

Slope of tangent $= -\frac{1}{t^2}$

$\therefore$ Equation of perpendicular to this tangent from (0, 0) is $y = t^2x$ ...(ii)

Squaring (i), $(x + t^2y)^2 = 4c^2t^2$

Putting the value of t^2 from (ii) in above we get

$\left(x + \frac{y^2}{x}\right) = 4c^2\frac{y}{x}$

$\Rightarrow \quad (x^2 + y^2) = 4c^2xy$

36. **(a)** Since ellipse and hyperbola intersect orthogonally, they are confocal.

Hence, $a = 2$ (equating foci).

Let the point of intersection in the first quadrant be $P(x_1, y_1)$

P lies on both the curves. Therefore, $4x_1^2 + 9y_1^2 = 36$ and $4x_1^2 - y_1^2 = 4$

Adding these two results, we get

$8(x_1^2 + y_1^2) = 40$

or $x_1^2 + y_1^2 = 5$ or $r = \sqrt{5}$

Hence, the equation of the circle is $x^2 + y^2 = 5$

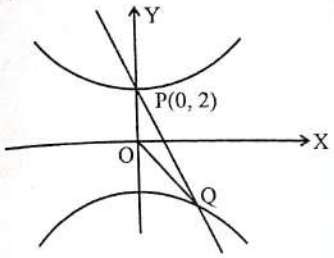
37. **(c)** Homogenizing the hyperbola using the straight line, we get a pair of straight lines OP and $O.Q$, which are given by

$$y^2 - x^2 = 4\left(\frac{\sqrt{3}x + y}{2}\right)^2$$

or $y^2 - x^2 = 3x^2 + y^2 + 2\sqrt{3}xy$

or $4x^2 + 2\sqrt{3}xy = 0$

or $x = 0$ and $2x + \sqrt{3}y = 0$



Angle between the lines is

$$\frac{\pi}{2} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right) = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

38. **(a)** Let the point $P(x_1, y_1)$

tangent at P

$xx_1 - 9yy_1 = 9$

$$x\left(\frac{x_1}{9}\right) - y(y_1) = 1 \qquad ...(i)$$

$$\left(\frac{5}{19}\right)x + \left(\frac{12}{19}\right)y = 1 \qquad ...(ii)$$

By comparing (i) & (ii)

$$x_1 = \frac{45}{19} : y_1 = \frac{-12}{19}$$

39. **(b)** For a triangle formed by joining any three ccrnormal points, orthocentre consides with the fourth point.

40. **(a)** Equation of tangent at the point (a secϕ, b tanϕ)

is $\frac{x\sec\phi}{a} - \frac{y\tan\phi}{b} = 1$

Finding its intersection with transverse axis (y = 0)

$\therefore$ Distance of the point T from the focus (, 0)

$= ae - a\cos\phi$ [$\because a\cos\phi \le a$, $ae > a$]

similarly distance from another focus is cos(α)

41. **(a)** Let middle point of chord is (h, k)

Thus equation of chord with mid point P is, = 2

Given slope of this chord is m

$$\Rightarrow \frac{-k}{h} = m \Rightarrow k + mh = 0$$

Thus locus P (h, k) is, = 0

42. **[1]** Equation of the tangent with slope m to the parabola 2 = 4

$$y = mx + \frac{a}{m} \qquad ...(i)$$

This line is also tangent to the hyperbola 2 ...(ii)

We can find the point of contact on the hyperbola by substituting $y = mx + \frac{a}{m}$ in equation (ii)

$$x\left(mx + \frac{a}{m}\right) = c^2$$

$m^2x^2 + ax - mc^2 = 0$

There is one point of contact, so the above quation will have two equal roots.

Discriminant = 0

$a^2 + 4m^3c^2 = 0$

$$m^3 = -\frac{a^2}{4c^2}$$

$$m^3 = -\frac{a^2}{4c^2}$$

$$m = -\left(\frac{a}{2c}\right)^{\frac{2}{3}} \qquad ...(iii)$$

There is only one value of the slope m, which is negative. So, there will be only one common tangent.

43. **[2]** Product of perpendiuclars drawn from any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to asymptotes is $\frac{a^2b^2}{a^2 + b^2}$. Given hyperbola

$$\frac{x^2}{2} - \frac{y^2}{1} = 1$$

$\therefore$ Product $= \frac{2}{3}$

Hence $k = 2$.

44. **[0]** Equation of tangent $\perp r$ to $5x + 2y - 3 = 0$ is $2x - 5y + k = 0$ By using $c^2 = a^2m^2 - b^2$ then we get k^2 is negative which is not possible. So total no of tangents to $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is zero.

45. **[5]** The equation of tangents to the hyperbola having slope m are

$y = mx \pm \sqrt{9m^2 - 49}$

Distance between the tangents is 2.

Therefore,

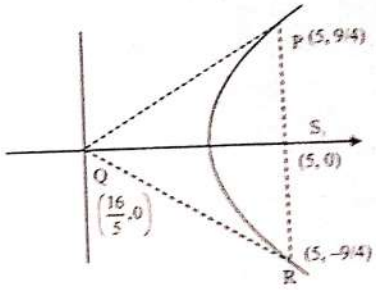
$$\frac{2\sqrt{9m^2 - 49}}{\sqrt{1 + m^2}} = 2$$

or $9m^2 - 49 = 1 + m^2$

or $8m^2 = 50$ or $m = \pm\frac{5}{2}$

46. **[324]** $e = \frac{5}{4}$

$$PS_1 = \frac{b^2}{a} = \frac{9}{4}$$



$$A = \frac{1}{2}\cdot\frac{9}{2}\cdot\frac{9}{5} \Rightarrow 80A = 324$$

47. **[6]** The equation of hyperbola is

$$\frac{x^2}{16} - \frac{y^2}{18} = 1$$

or $9x^2 - 8y^2 - 144 = 0$

Homogenizing this equation using

$$\frac{x\cos\alpha + y\sin\alpha}{p} = 1$$

we have

$$9x^2 - 8y^2 - 144\left(\frac{x\cos\alpha + y\sin\alpha}{p}\right)^2 = 0$$

Since these lines are perpendicular to each other, we have

$9p^2 - 8p^2 - 144(\cos^2\alpha + \sin^2\alpha) = 0$

$p^2 = 144$ or $p = \pm 12$

$\therefore$ Radius of circle = 12

$\therefore$ Diameter of circle = 24

48. **[9]** $H: \frac{x^2}{9} - \frac{y^2}{4} = 1$

Equation of AB : $xx_1 + yy_1 = 9$ ...(i)

(Chord of contact)

$xh + yk = h^2 + k^2$ ...(ii)

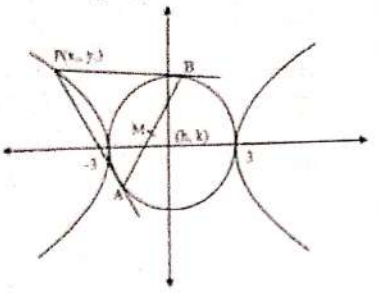
$(T = S_1)$

On comparing equation (i) and (ii)

$$\frac{x_1}{h} = \frac{y_1}{k} = \frac{9}{h^2 + k^2}$$

$$x_1 = \frac{9h}{h^2 + k^2} = 3\sec\theta$$

$$y_3 = \frac{9k}{h^2+k^2} = 2\tan\theta$$



On using $\sec^2\theta - \tan^2\theta = 1$, we get

$$\left(\frac{3h}{h^2+k^2}\right)^2 - \left(\frac{9k}{2(h^2+k^2)}\right)^2 = 1$$

$$\Rightarrow \frac{81x^2}{9} - \frac{81y^2}{4} = \left(x^2+y^2\right)^2$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2+y^2}{9}\right)^2$$

$\therefore k = 9$

49. [4] Normal to hyperbola

$xy = c^2$ at $\left(ct, \frac{c}{t}\right)$

is $y - \frac{c}{t} = t^2(x - ct)$ solving with $xy = -c^2$, we get

$$\Rightarrow x\left\{\frac{c}{t} + t^2(x-ct)\right\} + c^2 = 0$$

$$\Rightarrow t^2x^2 + \left(\frac{c}{t} - ct^3\right)x + c^2 = 0$$

Since line touches the curve,above equation has equal roots

$$\therefore \Delta = 0, \left(\frac{1}{t} - t^3\right)^2 - 4t^2 = 0$$

$$\Rightarrow \left(1 - t^4\right)^2 - 4t^4 = 0$$

$$\Rightarrow t^2 = \frac{2 \pm \sqrt{8}}{2} \text{ or } \frac{-2 \pm \sqrt{8}}{2}$$

$$\Rightarrow t^2 = 1 + \sqrt{2} \text{ or } -1 + \sqrt{2}$$

Thus four such values of t are possible

50. [9] $\frac{x-2}{\frac{1}{\sqrt{2}}} = \frac{y-1}{\frac{1}{\sqrt{2}}} = 4\sqrt{2}$ or $-4\sqrt{2}$

$\therefore A(-2,5)$ & $B(6,-3)$

$\therefore$ area of $\Delta CAB = 20$

$\Rightarrow$ sum $= 2\sqrt{20} = 4\sqrt{5}$

Parikshit (JEE Advanced Level)

1. (*a*)

(*i*) $\left(\frac{dy}{dx}\right)_{(h,k)} = -\frac{k}{h}$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

$$\Rightarrow \ln(xy) = c$$

$\Rightarrow xy = \lambda \Rightarrow$ Hyperbola

(*ii*) $\left(\frac{dy}{dx}\right)_{(h,k)} = -\frac{h}{2k}$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{2y}$$

$\Rightarrow y^2 + \frac{x^2}{2} = \lambda \Rightarrow$ Ellipse

(*iii*) $(TP)^2 = y^2 + \left(\frac{y}{\left(\frac{dy}{dx}\right)}\right)^2$

$$\Rightarrow y^2 + \left(\frac{y}{\left(\frac{dy}{dx}\right)}\right)^2 = x^2 + y^2$$

$$\Rightarrow \frac{dy}{dx} = \pm\frac{y}{x}$$

$\Rightarrow xy = \lambda$ or $y = \lambda x$

$\Rightarrow$ hyperbola or straight line

(*iv*) $x^2 + y^2 = y^2 + \left(y\frac{dy}{dx}\right)^2$

$$\Rightarrow \frac{dy}{dx} = \pm\frac{x}{y}$$

$y^2 - x^2 = \lambda$

or $y^2 + x^2 = \lambda$

$\Rightarrow$ hyperbola or circle

2. (*a*) (-2, 4) & (4,6) $\rightarrow$ focii

centre (1,5)

Equation of transverse is

$x - 3y + 14 = 9$

Equation of transverse axis

$3x + y - 8 = 0$

$2ae = \sqrt{40}$

$$\frac{(3x+y-8)^2}{20} - \frac{(x-3y+14)^2}{80} = 1.$$

3. (*d*) consider to hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

now, equation of tangent is

$$\frac{x'x}{a^2} + \frac{y'y}{b^2} = 1$$

Give that this tangent passes through positive vertex $(a,0)$

$$\therefore \frac{ax}{a^2} + \frac{0y}{b^2} = 1$$

$\Rightarrow x = a$

conjugate hyperbola is given by

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Intersecting point of tangent and the C.H is.

$$-\frac{a^2}{a^2} + \frac{y^2}{b^2} = 1$$

$y = \pm b\sqrt{2}$

$\therefore$ point of intersection is $(a, \pm b\sqrt{2})$

4. (*b*) Equation of asymptotes to hyperbola is

$$\frac{(x-7)^2}{a^2} - \frac{(y+3)^2}{b^2} = 0 \text{ or}$$

$$\frac{b^2(x-7)^2}{a^2} - (y+3)^2 = 0$$

or $\frac{9(x-7)^2}{4} - (y+3)^2 = 0$

Now point $P\left(\alpha+7, \alpha^2-4\right)$ is such that two tangents can be drawn one to each branch of the given hyperbola, then

$$\frac{9(\alpha+7-7)^2}{4} - \left(\alpha^2-1\right)^2 < 0$$

$$\Rightarrow \left(\alpha^2-1\right)^2 > \frac{9\alpha^2}{4}$$

$$\Rightarrow \left(2\alpha^2-3\alpha-2\right)\left(2\alpha^2+3\alpha-2\right) > 0$$

$$\Rightarrow (2\alpha+1)(\alpha-2)(2\alpha-1)(\alpha+2) > 0$$

$$\Rightarrow \alpha \in (-\infty,-2) \cup (-1/2, 1/2) \cup (2,\infty)$$

5. (*a, b, c*)

$\left(px^2+qy^2+r\right)\left(4x^2+4y^2-8x-4\right) = 0$

$$\Rightarrow 4x^2+4y^2-8x-4 = 0$$

$\Rightarrow (x-1)^2 + y^2 = 1$ or $px^2 + qy^2 + r = 0$ will represents

(*i*) two straight lines if $r = 0$ and p, q are of opposite sign.

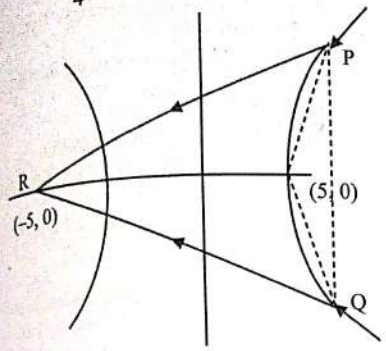
(*ii*) a circle if $p = q$ and r is of opposite sign that of p.

(*iii*) a hyperbola if p and q are of opposite sign and $r \neq 0$.

(*iv*) an ellipse if p and q are unequal but of same sign and r is of sign opposite to that of p.

6. **(b, d)** $\frac{x^2}{16}-\frac{4^2}{9}=1\Rightarrow e^2=1+\frac{9}{16}$

$e=\frac{5}{4}\Rightarrow(\pm ae,0)=(\pm5,0)$



$A=\frac{1}{2}\times6\sqrt{3}\times(5+8)=39\sqrt{3}$

$\theta=2\tan^{-1}\frac{3}{4}=\tan^{-1}\frac{2\cdot\frac{3}{4}}{1-\frac{9}{16}}=\tan^{-1}\frac{24}{7}$

$P(8,3\sqrt{3})$

$A=\frac{1}{2}\times6\sqrt{3}\times(5+8)=39\sqrt{3}$

7. **(a, b, c)** Let mid point be $P(h, k)$.Chord with mid point P for $x^2-y^2=9: hx-ky-9=h^2-k^2-9$

$\Rightarrow\ y=\frac{hx}{k}-\frac{(h^2-k^2)}{k}$

for tangent to $y^2=12x: y=mx+\frac{3}{m}$

$m=\frac{h}{k};\ \frac{3}{m}=-\frac{(h^2-k^2)}{k}$

$\therefore\ \frac{3k}{m}=-\frac{(h^2-k^2)}{k}$

$\Rightarrow\ x^3+3y^2-xy^2=0$

$\therefore\ l_1=-1; l_2=3$

8. **(a,b,c)**

Given $B(-2, -2)$ and $C(6, 2/3)$ on hyperbola

Let $A(2t, 2/t)$

Differentiating curve $xy'+y=0$

or $y'=-y/x=-1/t^2$

Slope of normal at A is t^2

Normal is parallel to BC, which has slope

$\frac{\frac{2}{3}+2}{6+2}=\frac{1}{3}=t^2$

$\therefore\ t=\frac{1}{\sqrt{3}}$

$\therefore\ A\left(\frac{2}{\sqrt{3}},2\sqrt{3}\right)$

Also (slope of B) × (slope of AC) $=-1$

$\therefore\ \Delta ABC$ is a right angled Δ.

Circum-center of ΔABC is mid-point BC, $(2, -2/3)$

Circum-circle of ΔABC is circle on BC as diamter.

Orthocenter of ΔABC is at A.

9. **(b,c,d)** Equation of normal at $\left(ae,\frac{b^2}{a}\right)$

$\frac{a}{e}x-ay=a^2e^2$

$x-ey=ae^3$

Put $y=\frac{9}{x}, x-9\frac{e}{x}=ae^3$

$\Rightarrow x^2-ae^3x-9e=0\le x_2$

If $x_1=6, x_1x_2=-9e$

$x_2=-\frac{3e}{2}$

$y_2=\frac{9}{x_2}=-\frac{6}{e}$

$\Rightarrow$ (slope of) (slope of)

$=\left(\frac{y_1}{x_1}\right)\left(\frac{y_2}{x_2}\right)=\frac{9}{x_1^2}\frac{9}{x_2^2}=\frac{81}{x_1^2x_2^2}>0$

10. **(b, c, d)** For the ellipse, $\frac{x^2}{2^2}+\frac{y^2}{1^2}=1$

We have $1^2=2^2(1-e^2)$

or, $e=\frac{\sqrt{3}}{2}$

Therefore, the eccentricity of the hyperbola is $\frac{2}{\sqrt{3}}$.

So, for hyperbola, $b^2=a^2\left(\frac{4}{3}-1\right)$ or $3b^2=a^2$

One of the foci of the ellipse is $(\sqrt{3},0)$.

The hyperbola passes through $(\sqrt{3},0)$.

$\frac{3}{a^2}=1$ or $a^2=3$ and $b^2=1$

Therefore, the equation of the hyperbola is,

$\frac{x^2}{3}-\frac{y^2}{1}=1$ or $x^2-3y^2=3$

The focus of the hyperbola is,

$(ae,0)\equiv\left(\sqrt{3}\times\frac{2}{\sqrt{3}},0\right)\equiv(2,0)$

11. **(a,d)** $x=Kt, y=\frac{K}{t}, K>0$

$\frac{dx}{dt}=K, \frac{dy}{dt}=-\frac{K}{t^2}$.

Hence slope of tangent at point 't' is,

$m=\frac{\frac{dy}{dt}}{\frac{dx}{dt}}=-\frac{1}{t^2}<0\forall t\in R$

Hence for line $\frac{x}{a}+\frac{y}{b}=1$ to be tangent to the given curve It slope should be negative

$\frac{-b}{a}<0\Rightarrow\frac{b}{a}>0$

$a>0, b>0$ or $a<0, b<0$

12. **(a, d)** Distance between foci

$=\sqrt{19^2+5^2}=\sqrt{386}$

Now by $PS+S'P=2a$(for ellipse)

(take point P at origin) we get a = 19

$\therefore\ 2ae=\sqrt{386}\ \Rightarrow\ e=\frac{\sqrt{386}}{38}$

If conic is hyperbola

$|PS-PS'|=2a\ \Rightarrow\ a=6$

by $2ae'=\sqrt{386}$

$e'=\frac{\sqrt{386}}{12}$

13. **(a, d)**

$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ Asyp. $y=\pm\frac{b}{a}x$

$m_1=\frac{b}{a}$ and $m_2=-\frac{b}{a}$

$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=\left|\frac{\frac{b}{a}+\frac{b}{a}}{1-\frac{b^2}{a^2}}\right|$

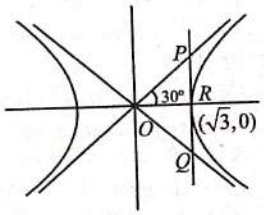
$\tan\theta=\frac{2ab}{a^2-b^2}\ \Rightarrow\ \tan\frac{\theta}{2}=\frac{b}{a}$ and $-\frac{a}{b}$

$\sec\frac{\theta}{2}=\sqrt{1+\frac{b^2}{a^2}}$ and $\sec\frac{\theta}{2}=\sqrt{1+\frac{a^2}{b^2}}=e=\frac{1}{e}$

14. **(b, c)**

$\frac{x^2}{3}-\frac{y^2}{1}=1$

Asyp $y=\pm\frac{1}{\sqrt{3}}x$



ΔOPQ will be equilateral triangle.

$PR=1$

area of $\Delta OPQ=\frac{1}{2}\times\sqrt{3}\times(2)=\sqrt{3}$ sq. units

15. **(a, b, c, d)**

Solving $xy=c^2$ and $x^2+y^2=a^2$, we have

$x^2+\frac{c^4}{x^2}=a^2$

or $x^4 - a^2x^2 + c^4 = 0$

or $x = \frac{21}{5}$ and $x = -\frac{11}{5}$

Length of latus rectum $= \frac{2b^2}{a} = \frac{9}{2}$

The foci are given by $X = \pm ae$, $Y = 0$

or (6, 1), (–4, 1)

16. **(b, d)**

Differentiating $xy = 1$ w.r.t. x, we have

$\frac{dy}{dx} = -\frac{1}{x^2} < 0$

Hence, the slope of normal at any point $P(x_1, y_1)$ is $x_1^2 > 0$.

Therefore, the slope of normal must always be positive.

Hence, the possible lines are (b) and (d).

17. **(a, c)**

For the given ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$

we have $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

Hence, the eccentricity of the hyperbola is $\frac{5}{3}$

Let the hyperbola be $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$

Then

$B^2 = A^2\left(e^2 - 1\right) = A^2\left(\frac{25}{9} - 1\right) = \frac{16}{9}A^2$

Therefore, the equation of the hyperbola is

$\frac{x^2}{A^2} - \frac{9y^2}{16A^2} = 1$

As it passes through (3, 0), we get $A^2 = 9$

and $B^2 = 16$. The equation is $\frac{x^2}{9} - \frac{y^2}{16} = 1$

The foci of the hyperbola are

$(\pm ae, 0) \equiv (\pm 5, 0)$

The vertex of the hyperbola is (3, 0).

18. **(a, c)**

Let tangent give by $y = mx + \sqrt{m^2 - 5}$

$\because$ it passes through (2, 8)

$(8 - 2m)^2 = m^2 - 5$

$3m^2 - 32m + 69 = 0$

$\Rightarrow m = 3$ or $23/3$

$\therefore$ tangent can be

$3x - y + 2 = 0$

or $23x - 3y - 22 = 0$

19. **(a, c)**

For the given ellipse, $\frac{x^2}{25} + \frac{y^2}{16} = 1$

We have, $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

Hence, the eccentricity of the hyperbola is $\frac{5}{3}$.

Let the hyperbola be $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$

Then, $B^2 = A^2(e^2 - 1)$

$= A^2\left(\frac{25}{9} - 1\right) = \frac{16}{9}A^2$

Therefore, the equation of the hyperbola is,

$\frac{x^2}{A^2} - \frac{9y^2}{16A^2} = 1$

As it passes through (3, 0) we get $A^2 = 9$

and $B^2 = 16$. The equation is $\frac{x^2}{9} - \frac{y^2}{16} = 1$

The foci of the hyperbola are

$(\pm ae, 0) \equiv (\pm 5, 0)$.

The vertex of the hyperbola is (3, 0).

20. **(c)** $PQ - PA = PB - PQ$

$\Rightarrow QA = BQ$

$\therefore$ Q is mid point of AB, Let $Q = (h, k)$

Equation of chord AB

$T = S_1$

$\frac{1}{2}(xk + yh) = hk$

It passes through $P(-1, 2)$

$\therefore$ locus of Q is $2x - y = 2xy$

21. **(b)** $\frac{x+1}{\cos\theta} = \frac{y-2}{\sin\theta} = r$

$x = r\cos\theta - 1$, $y = 2 + r\sin\theta$

Putting it in $xy = c^2$

$r^2\sin\theta\cos\theta + r(2\cos\theta - \sin\theta) - 2 - c^2 = 0$

$PA.PB = \frac{-(2+c^2)}{\sin\theta\cos\theta} = PQ^2$

$2 + c^2 + (PQ\sin\theta)(PQ\cos\theta) = 0$

$2 + c^2 + (y - 2)(x + 1)$

$xy + y - 2x + c^2 = 0$

22. **(a)** $\frac{2}{PQ} = \frac{PA + PB}{PAPB}$

Gives $2x - y = 2c^2$

23. **(c)**, **24.** **(b)**, **25.** **(d)**

Let $(2t, 2/t)$ be a point on the hyperbola.

Equation of the tangent at this point $x + yt^2 = 4t$.

$\therefore$ $A = (4t, 0)$, $B = (0, 4/t)$

Locus of circumcentre of triangle is $xy = 16$

Its eccentricity is $\sqrt{2}$

Shortest distance exist along the common normal.

$\therefore$ $t^2 = \frac{1}{2}$ or $t = \frac{1}{\sqrt{2}}$

$\therefore$ Foot of the perpendicular is $C(\sqrt{2}, 2\sqrt{2})$

Shortest distance is distance of C from the given line which is $\frac{4(\sqrt{2} - 1)}{\sqrt{5}}$.

Given line intersect the x-axis at $R(2, 0)$

Any point on this line at distance 'r' from R is $(2 + r\cos 0, r\sin 0)$

If this point lies on hyperbola, then we have

$(2 + r\cos\theta)$

Product of roots of above quadratic in 'r' is $r_1r_2 = 8/|\sin 2\theta|$, which has minimum value 8

$\therefore$ Minimum value of $RS \times RT$ is 8

26. **(c)**, **27.** **(b)**

Portion of tangent to hyperbola intercepted between asymptotes is bisected at its point of contact.

$\Rightarrow$ $PA = PB = 3\sqrt{2}$

$\Rightarrow A \equiv \left(1 + 3\sqrt{2}\left(-\frac{1}{\sqrt{2}}\right), 1 + 3\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)\right)$

$A \equiv (-2, 4)$

$B \equiv \left(1 - 3\sqrt{2}\left(-\frac{1}{\sqrt{2}}\right), 1 - 3\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)\right) \equiv (4, -2)$

$B \equiv (4, -2)$

Equation of asymptotes of H are $y = -\frac{1}{2}x$ and $y = -2$ (The equations are and)

Equation of pair of asymptotes of H is

$(2y + x)(y + 2x) = 0$

$\Rightarrow 2x^2 + 5xy + 2y^2 = 0$

Equation of w.r.t = as -axis and

$y = -x$ as y-axis $\frac{X - Y}{\sqrt{2}} = x, \frac{X + Y}{\sqrt{2}} = y$

$\Rightarrow 2\left(\frac{(x-y)^2 + (x+y)^2}{2}\right) + 5\left(\frac{x^2 - y^2}{2}\right) = 9$

$\frac{x^2}{2} - \frac{y^2}{18} = 1$

$2\left(e^2 - 1\right) = 18 \Rightarrow e = \sqrt{10}$

Foot to directirex are

$\left(0 \pm \frac{\sqrt{2}}{\sqrt{10}}\frac{1}{\sqrt{2}}, 0 \pm \frac{\sqrt{2}}{\sqrt{10}\sqrt{2}}\right)$

$= \left(\frac{1}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right), \left(-\frac{1}{\sqrt{10}}, -\frac{1}{\sqrt{10}}\right)$

Equation of directrices are

$x + y = \frac{2}{\sqrt{10}}, x + y = -\frac{2}{\sqrt{10}}$

Equation of pair or directrices is

$\left(x + y - \frac{2}{\sqrt{10}}\right)\left(x + y + \frac{2}{\sqrt{10}}\right) = 0$

$x^2 + y^2 + 2xy = \frac{4}{10} = \frac{2}{5}$

$5x^2 + 5y^2 + 10xy - 2 = 0$

28. (*b*), **29.** (*c*), **30.** (*d*)

Perpendicular tangents intersect at the centre of rectangular hyperbola. Hence, centre of hyperbola is (I, · I) and equation of asymptotes are x -I = 0 and –1 = 0

Let the equation of hyperbola is $xy - x_1 - y + 1 + \lambda = 0$. It passes through (3,2) hence $\lambda = 2$

so the equation of hyperbola is $= + 1$

From the centre of hyperobla we can draw two real tangents to the rectangular hyperbola

31. (*a*)

(*A*) $y = mx \pm \sqrt{a^2 m^2 - b^2}$

$y = x \pm \sqrt{5 - b^2}$

$\therefore \quad b = 0, \pm 1, \pm 2$

b can not be zero

$\therefore$ four values are possible

(*B*) We have, $a = 3$ and $\frac{b^2}{a} = 4$, $b^2 = 12$

Hence, the equation of the hyperbola is

$\frac{x^2}{9} - \frac{y^2}{12} = 1$

$4x^2 - 3y^2 = 36$

(*C*) The product of the lengths of the perpendiculars from the two focii on any tangent to the hyperbola

$\frac{x^2}{25} - \frac{y^2}{3} = 1$ is 3

$\therefore 3 = \sqrt{k}$, hence $k = 9$

(*D*) Equation of the hyperbola can be written as

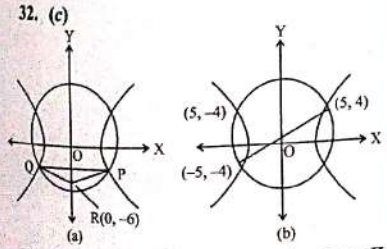
$\frac{X^2}{5^2} - \frac{Y^2}{4^2} = 1$

where $X = x - 3$ and $Y = y - 2$.

$\therefore$ tangent $Y = X \pm \sqrt{25 - 16}$

$\Rightarrow y = x + 2$ or $y = x - 4$

32. (*c*)



(*A*) Obviously, all the points in column II are common to the hyperbola and circle.

(*B*) The chord of contact of hyperbola w.r.t(0, –9/4) is

$\theta(x) - \left(-\frac{9}{4}\right) y = 9$ or $y = 4$

solving this with hyperbola, we have

$x^2 - 16 = 9$ or $x^2 = 25$ or $x = \pm 5$

hence, the points of contact are (± 5, 4)

(*C*) Obviously, the required point is (–5, –4)

(*D*) Let the points on the hyperbola be $P(h, k)$ and $Q(-h, k)$. Then

Area of triangle $= \frac{1}{2}|2h||-6-k| = 10$

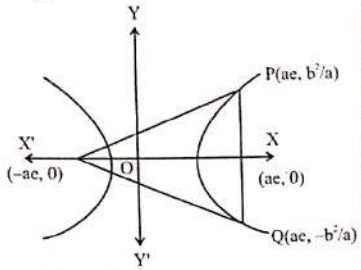
or $h|6 + k| = 10$ ...(i)

Also, points P and Q lie on the hyperbola.

Hence, $h^2 - k^2 = 9$ ...(ii)

Obviously, points (±5, –4) satisfy both (i) and (ii).

33. (*d*)



(*A*) $\text{Im}(z^2) = 3$

or $\text{Im}((x + iy)^2) = 3$

or $2xy = 3$

which is a rectangular hyperbola having eccentricity $\sqrt{2}$.

(*B*) $\tan 30° = \frac{b^2/a}{2ae}$

or $\frac{2}{\sqrt{3}} e = e^2 - 1$

or $\sqrt{3}e^2 - 2e - \sqrt{3} = 0$

or $e = \frac{2 \pm \sqrt{4 + 12}}{2\sqrt{3}} = \frac{2 \pm 4}{2\sqrt{3}}$

or $e = \frac{3}{\sqrt{3}} = \sqrt{3}$

(*C*) Eccentricity of hyperbola

$= \frac{AB}{PA - PB} = \frac{6}{4} = \frac{3}{2}$

If the eccentricity of conjugate hyperbola is e', then

$\frac{1}{(3/2)^2} + \frac{1}{e'^2} = 1$

or $e' = \frac{3}{\sqrt{5}}$

(*D*) Angle between the asymptotes is

$\tan^{-1}\left|\frac{2ab}{a^2 - b^2}\right| = \frac{\pi}{3}$

or $\left|\frac{2 \times a/b}{(a^2/b^2) - 1}\right| = \sqrt{3}$

or $\frac{2\sqrt{e'^2 - 1}}{|e'^2 - 2|} = \sqrt{3}$

where e' is the eccentricity of conjugate hyperbola. Therefore, $e' = 2$.

34. **[2]** $\frac{x^2}{9} - \frac{y^2}{4} = 1$

$y = mx \pm \sqrt{a^2 m^2 - b^2}$

passes throught $(6, 2\sqrt{3})$

$\Rightarrow m = \frac{4\sqrt{3}}{9}$

$y = \frac{4\sqrt{3}}{9} x \pm \frac{2\sqrt{3}}{3}$

for $= 3$

$y = 2\sqrt{3}, \frac{2}{\sqrt{3}}$

For $= 3$

$y = -2\sqrt{3}, -\frac{2}{\sqrt{3}}$

$\boxed{k = 2}$

35. **[1]** The eccentricity e_1 of the given hyperbola is obtained from

$b^2 = a^2 (e_1^2 - 1)$ ... (*i*)

The eccentricity e_2 of the conjugate hyperbol is given by

$a^2 = b^2 (e_2^2 - 1)$ ... (*ii*)

Multiply (*i*) and (*ii*), we get

$1 = (e_1^2 - 1)(e_2^2 - 1) \Rightarrow 0 = e_1^2 e_2^2 - e_1^2 - e_2^2$

$\Rightarrow e_1^{-2} + e_2^{-2} = 1$

36. **[22]** Let (x_1, y_1) be the pt, of contact of tangent

$3x - 4y = 5$ to $x^2 - 4y^2 = 5$ Solving we have

$\Rightarrow \quad (x_1, y_1) = (3, 1)$

Now any tangent to $\frac{x^2}{25} - \frac{y^2}{16} = 1$ is

$y = mx \pm \sqrt{25m^2 - 16}$

$\Rightarrow \quad y^2 + m^2 x^2 - 2mxy = 25m^2 - 16$ ... (*i*)

$\because$ (1) passes through (3, 1)

$\therefore \quad 16m^2 + 6m - 17 = 0$ ... (*ii*)

Let m_1 & m_2 be the roots of (*ii*) and

$m_1 + m_2 = -\frac{3}{8}$ and $m_1 m_2 = \frac{-17}{16}$

$\therefore \quad 32(m_1 + m_2 - m_1 m_2) = 22$

37. **[4]** $3x^2 - 2y^2 = 6$

$\frac{x^2}{2} - \frac{y^2}{3} = 1$

Let the equation of tangent

$y = mx + \sqrt{a^2 m^2 - b^2}$

passes through (α, β)

$(\beta - m\alpha)^2 = a^2 m^2 - b^2$

$m^2\alpha^2 + \beta^2 - 2m\alpha\beta = a^2 m^2 - b^2$

$m^2(\alpha^2 - a^2) - 2m\alpha\beta + \beta^2 + b^2 = 0$

$m_1 m_2 = \frac{\beta^2 + b^2}{\alpha^2 - a^2} = 2$

$2\alpha^2 - 2a^2 = \beta^2 + b^2$

or $2\alpha^2 - 4 = \beta^2 + 3$

$\beta^2 = 2\alpha^2 - 7 \quad (\because \beta = 2)$

PYQ's (Past Year Questions)

1. (*a*) Distance between foci

$2ae = \left|\left(1+\sqrt{2}\right) - \left(1-\sqrt{2}\right)\right| = 2\sqrt{2}$

$\Rightarrow ae = \sqrt{2}$

$\Rightarrow a = 1 \qquad \left[\because e = \sqrt{2}\right]$

$b = a$ when $e = \sqrt{2} \Rightarrow$ Hyperbola is rectangular

$\Rightarrow L \cdot R = \dfrac{2b^2}{a} = 2$

2. [2] Hyperbola (H): $2x^2 - 2y^2 = 1$

$x^2 - y^2 = \dfrac{1}{2}$

$\therefore e_H = \sqrt{2}$

$e_E = \dfrac{1}{\sqrt{2}}$

Since the curves intersect each other orthogonally

The ellipse and the hyperbola are confocal

$H: \dfrac{x^2}{1/2} - \dfrac{y^2}{1/2} = 1$

$\Rightarrow$ foci $= (1, 0)$

For ellipse $ae_E = 1$

$\Rightarrow a = \sqrt{2}$

$\because \left(e_E\right)^2 = \dfrac{1}{2} \Rightarrow 1 - \dfrac{b^2}{a^2} = \dfrac{1}{2} \Rightarrow \dfrac{b^2}{a^2} = \dfrac{1}{2}$

$\Rightarrow b^2 = 1$

Length of L.R. $= \dfrac{2b^2}{a} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$

Hence, square of length of the latus-rectum of the ellipse is 2.

3. (*b*) $H: \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$

Foci : $S(ae, 0), S'(-ae, 0)$

Foot of directrix of parabola is $(-ae, 0)$

Focus of parabola is $(ae, 0)$

Now, semi latus rectum of parabola $= SS' = |2ae|$

Given, $4ae = e\left(\dfrac{2b^2}{a}\right)$

$\Rightarrow b^2 = 2a^2$ ... (*i*)

Given, $\left(2\sqrt{2}, -2\sqrt{2}\right)$ lies on H

$\Rightarrow \dfrac{1}{a^2} - \dfrac{1}{b^2} = \dfrac{1}{8}$ ... (*ii*)

From equ (*i*) and equ (*ii*), we get

$a^2 = 4, b^2 = 8$

$\because b^2 = a^2(e^2 - 1)$

$\therefore e = \sqrt{3}$

Hence, equation of parabola is $y^2 = 8\sqrt{3}x$

4. [3] Now $e_E = \sqrt{1 - \dfrac{b^2}{a^2}}$ and $e_H = \sqrt{2}$

$\Rightarrow e_E = \dfrac{1}{e_H} \Rightarrow \dfrac{a^2 - b^2}{a^2} = \dfrac{1}{2}$

$\Rightarrow 2(a^2 - b^2) = a^2 \Rightarrow 2a^2 - 2b^2 = a^2$

$\Rightarrow a^2 = 2b^2$

and $y = \sqrt{\dfrac{5}{2}}x + k$ is tangent to ellipse then

$K^2 = a^2 \times \dfrac{5}{2} + b^2 = \dfrac{3}{2}$

$\Rightarrow 6b^2 = \dfrac{3}{2} \Rightarrow b^2 = \dfrac{1}{4}$ and $a^2 = \dfrac{1}{2}$

$\therefore 4(a^2 + b^2) = 3$

5. (*c*) Given hyperbola is

$16\left(x^2 + 2x + 1\right) - 9\left(y^2 - 4y + 4\right)$

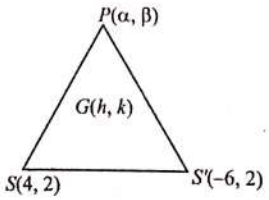
$= 16 - 36 + 164$

$\Rightarrow \dfrac{(x+1)^2}{9} - \dfrac{(y-2)^2}{16} = 1$

$\therefore e = \sqrt{1 + \dfrac{16}{9}} = \dfrac{5}{3}$ and focii: $x + 1 = \pm 5$ & $y - 2 = 0$

$\therefore S(4, 2), S'(-6, 2)$

$\Rightarrow S(4, 2)$ and $S'(-6, 2)$



Let point P on the hyperbola.

$\therefore$ Centroid $\equiv \left(\dfrac{\alpha + 4 - 6}{3} + \dfrac{\beta + 2 + 2}{3}\right)$

$\equiv \left(\dfrac{\alpha - 2}{3} + \dfrac{\beta + 4}{3}\right) = (h, k)$

$\Rightarrow \alpha = 3h + 2$ and $\beta = 3k - 4$

$\because \alpha, \beta$ lies on hyperbola.

$\therefore \dfrac{(3h + 2 + 1)^2}{9} - \dfrac{(3k - 4 - 2)^2}{10} = 1$

$\therefore 16x^2 - 9y^2 + 32x + 36y - 36 = 0$

6. (*a*) $2b = 5$ and $2ae = 13$

$(ae)^2 = a^2 + b^2$

$\Rightarrow a^2 = (ae)^2 - b^2 = \dfrac{169}{4} - \dfrac{25}{4}$

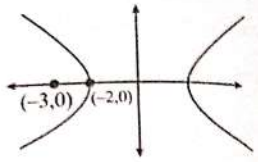
$\Rightarrow a = 6$

$e = \dfrac{ae}{a} = \dfrac{13}{12}$

7. (*d*) Equation of hyperbola is

$\dfrac{x^2}{4} - \dfrac{y^2}{b^2} = 1$ and $ae = 3$

We know that $a^2e^2 = a^2 + b^2$



$9 = 4 + b^2 \Rightarrow b^2 = 5$

Hence, equation of hyperbola is

$\dfrac{x^2}{4} - \dfrac{y^2}{b^2} = 1$

Hence, $\left(6, 5\sqrt{2}\right)$ does not lie on

$\dfrac{x^2}{4} - \dfrac{y^2}{5} = 1$

8. (*c*) $3x^2 - 4y^2 = 36 \Rightarrow \dfrac{x^2}{12} - \dfrac{y^2}{9} = 1$

$x = \sqrt{12}\sec\theta$

or

$y = 3\tan\theta$

$3x + 2y = 1$

Slope of line is $m = -\dfrac{3}{2}$

$\dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \dfrac{3\sec\theta}{\sqrt{12}\cdot\tan\theta} = \dfrac{3}{\sqrt{12}\sin\theta}$

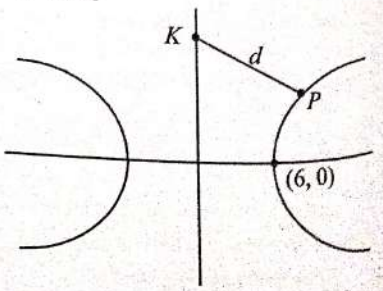
$\Rightarrow \dfrac{3}{\sqrt{12}} \times \dfrac{1}{\sin\theta} = \dfrac{-3}{2}$

$\sin\theta = -\dfrac{1}{\sqrt{3}}$

$\left(\sqrt{12}\cdot\sec\theta, 3\tan\theta\right)$

$\left(\sqrt{12}\cdot\dfrac{\sqrt{3}}{\sqrt{2}}, -3\times\dfrac{1}{\sqrt{2}}\right) \equiv \left(\dfrac{6}{\sqrt{2}}, \dfrac{-3}{\sqrt{2}}\right)$

9. [216]



$H: \frac{x^2}{36} - \frac{y^2}{9} = 1$

Equation of normal is $6x\cos\theta + 3y\cot\theta = 45$

Slope $= -2\sin\theta = -\sqrt{2}$

$\left[\text{Slope of line } \sqrt{2}x + y = 2\sqrt{2} \text{ is } -\sqrt{2}\right]$

$\Rightarrow \theta = \frac{\pi}{4}$

Equation of normal is $\sqrt{2}x + y = 15$

$P:(a\sec\theta, b\tan\theta) \quad [\because a = 6, b = 3]$

$\Rightarrow P\left(6\sqrt{2}, 3\right)$ and $K(0,15)$

$d^2 = 216$

10. *(b)* For hyperbola, $a^2\left(e^2 - 1\right) = b^2$

$b^2 = \frac{3a^2}{2} \Rightarrow \left(e = \sqrt{\frac{5}{2}}\right)$ ...(*i*)

Length of latus rectum $\frac{2b^2}{a} = 6\sqrt{2}$

$\frac{2\left(\frac{3a^2}{2}\right)}{a} = 6\sqrt{2} \Rightarrow a = 2\sqrt{2}$

$\Rightarrow b = 2\sqrt{3}$

$y = 2x + c$ is tangent to hyperbola

On comparing it with

$y = mx + \sqrt{a^2m^2 - b^2}$ we get

$m = 2$ and $c^2 = a^2m^2 - b^2$

$\therefore c^2 = a^2m^2 - b^2 = \left(2\sqrt{2}\right)^2 2^2 - \left(2\sqrt{3}\right)^2$

$= 20$

11. [88] Given $2a + 2b = 4\left(2\sqrt{2} + \sqrt{14}\right)$

$\Rightarrow 2a + 2b = 4\sqrt{2}\left(2 + \sqrt{7}\right)$ ... (*i*)

$b^2 = a^2\left(\frac{11}{4} - 1\right)$

$4b^2 = 7a^2$

$b = \frac{\sqrt{7}}{2}a$ ... (*ii*)

On solving equation (*i*) and equation (*ii*), we get

$a = 4\sqrt{2}, b = 2\sqrt{14}$

Hence, $a^2 + b^2 = 88$

12. [12] Equation of L_1 is

$\frac{x\sec\theta}{4} - \frac{y\tan\theta}{2} = 1$

Equation of line L_2 is

$\frac{x\tan\theta}{2} + \frac{y\sec\theta}{4} = 0$

$\because$ Required point of intersection of L_1 and L_2 is (x_1, y_1) then

$\frac{x_1\sec\theta}{4} - \frac{y_1\tan\theta}{2} - 1 = 0$

and $\frac{y_1\sec\theta}{4} + \frac{x_1\tan\theta}{2} = 0$

From equations (*iii*) and (*iv*)

$\sec\theta = \frac{4x_1}{x_1^2 + y_1^2}$ and $\tan\theta = \frac{-2y_1}{x_1^2 + y_1^2}$

$\therefore$ Required locus of (x_1, y_1) is

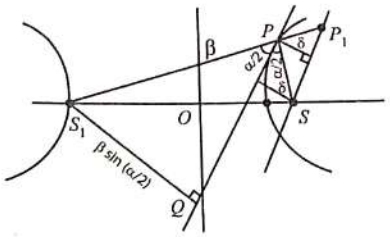
$(x^2 + y^2)^2 = 16x^2 - 4y^2$

$\therefore \alpha = 16, \beta = -4$

$\therefore \alpha + \beta = 12$

13. [7] In ΔS_1QP, sin

$\frac{\alpha}{2} = \frac{S_1Q}{\beta} \Rightarrow S_1Q = \beta\sin\frac{\alpha}{2}$



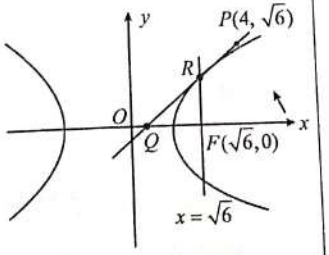
Product of distances of any tangent from two foci $= b^2$

$\delta \cdot S_1Q = \delta \times \beta\sin\frac{\alpha}{2} = b^2$

So, $\frac{\beta\delta\sin\frac{\alpha}{2}}{9} = \frac{b^2}{9} = \frac{64}{9}$

$\therefore \left[\frac{64}{9}\right] = 7$ s

14. *(b)*



$H: x^2 - 2y^2 = 4$

$\Rightarrow \frac{x^2}{4} - \frac{y^2}{2} = 1 \Rightarrow e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{3}{2}}$

$\therefore$ Focus $F(ae, 0) \Rightarrow F(\sqrt{6}, 0)$

Equation of tangent at P to the hyperbola is

$2x - y\sqrt{6} = 2$

$\because$ Tangent meet x-axis at $Q(1, 0)$

Latus rectum at F, $x = \sqrt{6}$ cuts the tangent at $y = \frac{2\left(\sqrt{6} - 1\right)}{\sqrt{6}}$

$R\left(\sqrt{6}, \frac{2}{\sqrt{6}}(\sqrt{6} - 1)\right)$

Area of

$\Delta QFR = \frac{1}{2}(\sqrt{6} - 1)\cdot\frac{2}{\sqrt{6}}(\sqrt{6} - 1) = \frac{7}{\sqrt{6}} - 2$ sq. units

15. [2] $\sqrt{3}kx + ky = 4\sqrt{3}$

$\Rightarrow k = \frac{4\sqrt{3}}{\sqrt{3}x + y}$ ....(*i*)

$\sqrt{3}\cdot x - y = 4\sqrt{3}k = 0$

$\Rightarrow k = \frac{\sqrt{3}x - y}{4\sqrt{3}}$ ...(*ii*)

from (*i*) and (*ii*)

$\therefore \frac{4\sqrt{3}}{\sqrt{3}x + y} = \frac{\sqrt{3}x - y}{4\sqrt{3}}$

$3x^2 - y^2 = 48$

$\Rightarrow \frac{x^2}{16} - \frac{y^2}{48} = 1 \Rightarrow 48 = 16(e^2 - 1)$

$\Rightarrow e = \sqrt{4} = 2$

16. *(b)* Here ellipse: $3x^2 + 4y^2 = 12$

$\Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1 \Rightarrow b^2 = a^2(1 - e^2)$

$\Rightarrow e = \frac{1}{2}$

Foci $= (\pm 1, 0)$ $(\because \pm ae, 0)$

Now for Hyperbola

$2a = \sqrt{2} \Rightarrow a = \frac{1}{\sqrt{2}}$ and

$ae = 1 \Rightarrow e = \sqrt{2} \Rightarrow b = \frac{1}{\sqrt{2}}$

So equation of hyperbola: $\frac{x^2}{\frac{1}{2}} - \frac{y^2}{\frac{1}{2}} = 1$

$\Rightarrow x^2 - y^2 = \frac{1}{2}$

So, option (*b*) does not satisfy it.

17. *(d)* Vertex is at $(\pm 6, 0)$

$\therefore a = 6$

Let the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Putting point $P(10, 16)$ on the hyperbola

$\frac{100}{36} - \frac{256}{b^2} = 1 \Rightarrow b^2 = 144$

$\therefore$ Hyperbola is $\frac{x^2}{36} - \frac{y^2}{144} = 1$

$\therefore$ Equation of normal is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$

$\therefore$ Putting we get $2x + 5y = 100$.

18. **(d)** General tangent to hyperbola in slope form is

$y = mx \pm \sqrt{100m^2 - 64}$

and that of circle is

$y = mx \pm 6\sqrt{1+m^2}$

For common tangent

$36(1 + m^2) = 100m^2 - 64$

$100 = 64m^2$

$\Rightarrow m^2 = \frac{100}{64}$

$\therefore\ c^2 = 36\left(1+\frac{100}{64}\right) = \frac{164\times 36}{64} = \frac{369}{4}$

$\Rightarrow 4c^2 = 369$

19. **(b)** Hyperbola : $\frac{x^2}{4} - \frac{y^2}{2} = 1$

So, let the point on it be $(2\sec\theta, \sqrt{2}\tan\theta) = (x_1, y_1)$

So equation of tangent

$$= \frac{2x\sec\theta}{4} - \frac{\sqrt{2}y\tan\theta}{2} = 1$$

$\Rightarrow$ slope $= \frac{1}{\sqrt{2}\sin\theta}$

Which will be equal to the slope of given line $2x - y = 0$

$\Rightarrow \frac{1}{\sqrt{2}\sin\theta} = 2 \Rightarrow \sin\theta = \frac{1}{2\sqrt{2}}$

$\Rightarrow \cos^2\theta = 1 - \sin^2\theta = 1 - \frac{1}{8} = \frac{7}{8}$

$\Rightarrow \sec^2\theta = \frac{8}{7} \Rightarrow \tan^2\theta = \frac{8}{7} - 1 = \frac{1}{7}$

Now $x_1^2 + 5y_1^2 = 4\sec^2\theta + 10\tan^2\theta$

$= 4\times\frac{8}{7} + 10\times\frac{1}{7} = \frac{42}{7} = 6$

20. **(a, d)**

Since Normal at point P makes equal intercept on co-ordinate axes, therefore slope of Normal $= -1$

Hence slope of tangent $= 1$

Equation of tangent $= 1$

$y - 0 = 1(x - 1)$

$y = x - 1$

Equation of tangent at (x_1, y_1)

$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

$x - y = 1$ (equation of Tangent)

on comparing $x_1 = a^2, y_1 = b^2$

Also $a^2 - b^2 = 1$ ... (i)

Now equation of normal at $(x_1, y_1) = (a^2, b_1^2)$

$y - b^2 = -1(x - a^2)$

$x + y = a^2 + b^2$ ...(Normal)

point of intersection with x-axis is $B(a^2 + b^2, 0)$

Now $e = \sqrt{1+\frac{b^2}{a^2}}$

$e = \sqrt{1+\frac{b^2}{b^2+1}}$ $\left[\text{from}(i)\ \frac{b^2}{b^2+1} < 1\right]$

$1 < e < \sqrt{2}$ Option (a)

$\Delta = \frac{1}{2}.AB.PQ$

and $\Delta = \frac{1}{2}\left(a^2 + b^2 - 1\right).b^2$

$\Delta = \frac{1}{2}\left(2b^2\right)b^2$ (from (1) $a^2 - 1 = b^2$)

$\Delta = b^4$ so option (d)

21. **(a)** Let us Suppose equation of hyperbola is

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$e = 2, e^2 = 1 + \frac{b^2}{a^2} \Rightarrow b^2 = 3a^2$ ...(i)

passing through (4, 6)

$\Rightarrow \frac{16}{a^2} - \frac{36}{b^2} = 1$ ...(ii)

on solving (i) and (ii)

$\Rightarrow a^2 = 4, b^2 = 12 \Rightarrow$ equation of tangent

$x - \frac{y}{2} = 1 \Rightarrow 2x - y - 2 = 0$

22. **(a)** Equation of tangents

$y^2 = 12x \Rightarrow y = mx + \frac{3}{m}$ $\left[\because y = mx + \frac{a}{m}\right]$

$\frac{x^2}{1} - \frac{y^2}{8} = 1 \Rightarrow y = mx \pm \sqrt{m^2 - 8}$

$\left[\because y = mx \pm \sqrt{a^2m^2 - b^2}\right]$

Since they are common tangent

$\therefore \frac{3}{m} = \pm\sqrt{m^2 - 8}$

$\Rightarrow m^4 - 8m^2 - 9 = 0$

$m = \pm 3$

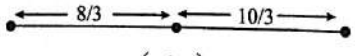
$\therefore y = 3x + 1$

or $y = -3x - 1$

$P\left(-\frac{1}{3}, 0\right)$, $S=(3,0)$, $S'=(-3,0)$ $\Rightarrow \frac{x^2}{1} - \frac{y^2}{8} = 1$

$e = 3$

$ae = 3$



$S'(-3,0)$ $P\left(-\frac{1}{3}, 0\right)$ $S(3,0)$

23. **(a)** $\frac{x^2}{\cos^2\theta} - \frac{y^2}{\sin^2\theta} = 1$

$\because e > 2$ (given)

$\because e^2 > 4 \Rightarrow 1 + \frac{\sin^2\theta}{\cos^2\theta} > 4$

$\Rightarrow 1 + \tan^2\theta > 4 \Rightarrow \tan^2\theta > 3$

$\because \theta \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

Latus rectum $= 2\frac{\sin^2\theta}{\cos\theta} = 2\tan\theta\sin\theta$

$\because$ for $\theta \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$, $2\tan\theta\sin\theta$ is increasing function

Hence latus rectum $\in (3, \infty)$

24. **(c)** $\frac{x^2}{9} - \frac{y^2}{16} = 1$

$a = 3, b = 4$ and

$e = \sqrt{1+\frac{16}{9}} = \frac{5}{3}$ $\left[\because e = \sqrt{1+\frac{b^2}{a^2}}\right]$

Corresponding focus will be $(-ae, 0)$ i.e. $(-5, 0)$

25. **(b)** $\frac{y^2}{1+r} - \frac{x^2}{1-r} = 1$

$r > 1 \Rightarrow$ ellipse

$e = \sqrt{1-\left(\frac{r-1}{r+1}\right)} = \sqrt{\frac{2}{r+1}}$

26. **(c)** $\frac{x^2}{24} - \frac{y^2}{18} = 1 \Rightarrow a = \sqrt{24}; b = \sqrt{18}$

Parametric normal:

$[ax\cos\theta + by\cot\theta = a^2 + b^2]$

$\sqrt{24}\cos\theta.x + \sqrt{18}.y\cot\theta = 42$

At $x = 0 : y = \frac{42}{\sqrt{18}}$ $\tan\theta = 7\sqrt{3}$ (from given equation)

$\Rightarrow \tan\theta = \sqrt{\frac{3}{2}} \Rightarrow \sin\theta = \pm\sqrt{\frac{3}{5}}$

Slope of parametric normal

$$= \frac{-\sqrt{24}\cos\theta}{\sqrt{18}\cot\theta} = m$$

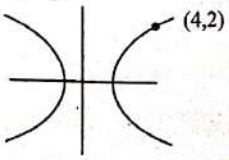
$\Rightarrow m = -\sqrt{\frac{4}{3}}\sin\theta = -\frac{2}{\sqrt{5}}\ or\ \frac{2}{\sqrt{5}}$

27. **(a)** Given hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ ...(i)

Given, $2a = 4 \Rightarrow a = 2$

Equation (i) Satisfying the point (4, 2)

$\frac{16}{4} - \frac{4}{b^2} = 1$



$\Rightarrow b^2 = \frac{4}{3}$ Now, $e = \sqrt{1+\frac{b^2}{a^2}}$

$= \sqrt{1+\frac{4}{3\times 4}} = \frac{2}{\sqrt{3}}$

28. **(c)** Equation of tangent to parabola $y^2 = 4x$ is

$y = mx + \frac{1}{m}$ ... (i)

Now solve if with $xy = 2$

$\left(mx + \frac{1}{m}\right)x = 2$

$\Rightarrow mx^2 + \frac{1}{m}x - 2 = 0$

For common tangent $D = 0$

$\Rightarrow \frac{1}{m^2} + 8m = 0$

$m = -\frac{1}{2}$

Put it in equation (i) $y = -\frac{1}{2}x - 2$

$\Rightarrow x + 2y + 4 = 0$

29. **(a)** Hyperbola is $\frac{x^2}{5} - \frac{y^2}{4} = 1$

Equation of its tangent in slope from is

$y = mx \pm \sqrt{5m^2 - 4}$ $\left[\because y = mx \pm \sqrt{a^2m^2 - b^2}\right]$

Hence tangent with slope 1 is $y = x \pm 1$

30. **(b)** Let $P(h, k)$ be the mid point of the chord $x^2 - y^2 = 4$

$T = S_1$

$\therefore$ its equation is $xh - yk = h^2 - k^2$

$\Rightarrow y = \left(\frac{h}{k}\right)x + \frac{k^2 - h^2}{k}$

We know, if $y = mx + c$ is tangent to $y^2 = 4ax$ then $c = \frac{a}{m}$

If this line is tangent to $y^2 = 8x$

then $\frac{k^2 - h^2}{k} = \frac{2}{h/k} = \frac{2k}{h}$

$\Rightarrow h(k^2 - h^2) = 2k^2$

$\therefore$ Required locus is $2y^2 = x(y^2 - x^2)$

$\Rightarrow x^3 = y^2(x - 2)$

By Aditya Mohod
username - maje_kabhi_dekha_nahi

By Aditya Mohad
username - muje_kabhi_dekha_nahi

CHAPTER 15

Complex Number-II

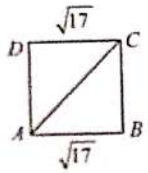
Prarambh (Topicwise)

1. **(a)** New $z = \frac{3}{2}(-4+5i)e^{in} = 6 - \frac{15}{2}i$

2. **(a)** z_1, z_2, z_3 are three complex numbers joined to form an equilateral triangle the coordinates of z_1 are (1, 1) and that of, z_2 are (–1, 1) and that of z^3 are (x, y) (say) There's no harm in considering complex numbers as coordinates in a real plane. After all, there isn't much distinction while representing graphically.Now using the distance formula,we can write
$(x-1)^2 + (y-1)^2 = (x+1)^2 + (y+1)^2$
$= 8$ Solving we get
$x + y = 1$ ...(i)
$x^2 + y^2 = 8 - 2 = 6$ ... (ii)
Solving for x & y we get
$x = \pm\sqrt{3} \quad y = \pm\sqrt{3}$
$\therefore z_3 = (\sqrt{3} + i\sqrt{3})$ or $(\sqrt{3} - i\sqrt{3})$

3. **(c)** Given the vertices of quadrilateral $A(1+2i), B(-3+i), C(-2-3i)$ and $D(2-2i)$

Now, $AB = \sqrt{16+1} = \sqrt{17}, BC = \sqrt{16+1} = \sqrt{17}$



$CD = \sqrt{16+1} = \sqrt{17}, DA = \sqrt{16+1} = \sqrt{17}$

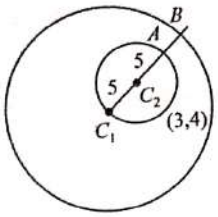
$AC = \sqrt{9+25} = \sqrt{34}, BD = \sqrt{25+9} = \sqrt{34}$

$\frac{1}{2}d^2 = \left(\sqrt{17}\right)^2$

$d = \sqrt{34}$

Hence it is a square.

4. **(b)** The two circles are $C_1(0,0), r_1 = 12$, and $C_2(3,4), r_2 = 5$ and it passes through origin, the centre of C_1.



$C_1C_2 = 5 < r_1 - r_2 = 7$
Hence circle C_2 lies inside circle.
Therefore minimum distance between them is
$AB = C_1B - C_1A = r_1 - 2r_2 = 12 - 10 = 2.$

5. **(a)** Equation of line through z_o and making angle α is
$\arg(z - z_0)$ is
Here $z_0 = 2 + i$, $\alpha = 60° = \pi/3$
$\therefore$ equation of line
$\arg(z - 2 - i) = \pi/3$

6. **(b)** Equation of line joining z_1 and z_2 is
$= z = z_1 + k(z_2 - z_1), k \in \mathbb{Z}$
here $z_1 = -2 - i, z_1 = 3 + i$
$\therefore z = (-2-i) + k((3+i) - (-2-i)), k \in \mathbb{Z}$
$= (-2-i) + k(5+2i), k \in \mathbb{Z}$

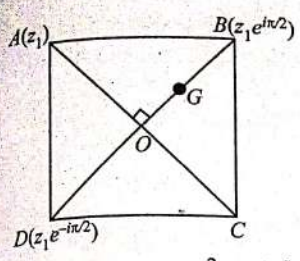
7. **(c)** Equation of circle in diameter form is $\arg\left(\frac{z-z_1}{z-z_2}\right) = \pm\frac{\pi}{2}$
Where z_1 and z_2 are end points of diameter
here $z_1 = 3+i, \quad z_2 = 2-i$
Therefore $\arg\left(\frac{z-(3+i)}{z-(2-i)}\right) = \pm\frac{\pi}{2}$

8. **(c)** Given Z, $\arg(z) = -\pi/3$
Where z is a complex number and can be represented as it where
$\mathrm{Re}(z) = a$
$\mathrm{Im}(z) = b$
And we can represent it in the complex plane
Now
$\arg(z) = \tan^{-1}\left|\frac{\mathrm{Im}(z)}{\mathrm{Re}(z)}\right| = \tan^{-1}\frac{b}{a} = -\pi/3$
$\Rightarrow \tan\tan^{-1}(b/a) = \tan -\pi/3$
$\Rightarrow b/a = -\tan\pi/3 = -\sqrt{3}$
$\Rightarrow b = -\sqrt{3}a$
Replace b with y & a with x to obtain the locus
$\Rightarrow \sqrt{3}x + y = 0; x > 0 y < 0$

9. **(a)** Since it is equilateral triangle so
$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$
Hence $z^2 + (-z)^2 + (1-z)^2$
$= z(-z) + z(1-z) - z(1-z)$
$3z^2 - 2z + 1 = -z^2 + z - z^2 - z + z^2$
$4z^2 - 2z + 1 = 0$
$\therefore \quad z = \frac{-(-2) \pm \sqrt{(-2)^2 - 4\cdot4\cdot1}}{2\cdot4}$
$\Rightarrow \quad \mathrm{Re}(z) = \frac{2}{8} = \frac{1}{4}$

10. **(b)** $z_1z_2 = \frac{\lambda}{2}$ and $z_1 + z_2 = 1$
Now since o, z_1, z_2 form an equilateral triangle
$z_1(z_2) + z_2(0) + z_1(0) = z_1^2 + z_2^2 + 0^2$
$z_1^2 + z_2^2 = z_2z_1$
$(z_1 + z_2)^2 - 2z_1z_2 = z_1z_2$
$1 = 3z_1z_2$
$1 = 3\left(\frac{\lambda}{2}\right)$
$\lambda = \frac{2}{3}$

11. **(b)** ΔABC is isosceles triangle
So centroid divides median BO in ratio 2 : 1
Centroid $G = \frac{z_1e^{i\pi/2}}{3} = \frac{z_1}{3}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$
Also centroid $G = \frac{z_1e^{-i\pi/2}}{3} =$
$\frac{z_1}{3}\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right) = \frac{z_1}{3}(0-i) = \frac{z_1}{3}$



12. (*d*) $z_1 - z_2 = 1 \Rightarrow a = \frac{2}{3}, b = \frac{-5}{3}$

$\text{centroid} = \frac{1}{3}(z_1 + z_2 + z_3) = \frac{1}{9}(1+7i)$

13. (*a*) $|z - z_1| = |z - z_2| = |z - z_3|$

$\therefore z$ must be mid point of z_1 and z_2.

$\Rightarrow z, z_1, z_2$ are collinear.

z is circumcentre of Δ formed by z_1, z_2, z_3.

$\Rightarrow \arg\left(\frac{z_3 - z_2}{z_3 - z_1}\right) = +\frac{\pi}{2}$

$\arg(z-a)$=

14. (*a*) $|z-5i| = |z+5i|$ implies perpendicular bisector of line joining $5i$ & $-5i$ which is real axis.

15. (*b*) Given that, $|z^2 - 1| = |z|^2 + 1$

$\Rightarrow |z^2 + (-1)| = |z^2| + |-1|$

It shows that the origin, -1 and z^2 lies on a line and z^2 and -1 lies on one side of the origin, therefore z^2 is a negative number. Hence, z will be purely imaginary. So, we can say that z lies on y-axis.

16. (*a*) Obvious (A) $|z| \le 1$ is region interior of circle of radius 1 unit and centre (0, 0) and arg. of given region is $-\pi/2 \le \arg z \le \pi/2$

17. (*c*) $|z-1|^2 + |z+1|^2 = 2$

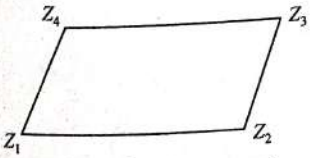
$\Rightarrow z\bar{z} + 1 - (z + \bar{z}) + z\bar{z} + 1 + (z + \bar{z}) = 2$

$\Rightarrow z\bar{z} = 0 \Rightarrow |z| = 0 \Rightarrow z = 0$

$\Rightarrow x = 0, y = 0$

18. (*c*) $A(1, 0)$ and $B(-1, 0)$ are two fixed point at a distance $\sqrt{2}$ & $P(z)$ moves so that $PA + PB \le 4$. Hence locus of P is ellipse.

19. (*b*)



Mid points of diagonals are equal

$\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2} \Rightarrow z_1 + z_3 = z_2 + z_4$

20. (*a*) $|z - 2i| > 1$ depicts all the points outside the circle centered at (0, 2) and having a radius 3.

$|z - 2i| < 5$ depicts all the points outside the circle centered at (0,2) and having a radius 5.

Hence

$1 < |z - 2i| < 5$ points within the concentric region between these two circles.

21. (*c*) Let $z = x + iy$

$x + iy + i = ix - y + 1$

$x = -(y-1) = 1 - y$ ...(*i*)

and

$ix = i(y+1)$

$x = (y+1)$ ...(*ii*)

Mulitplying i and ii, we get

$x^2 = 1 - y^2 \; x^2 = 1 - y^2$

Or,

$x^2 + y^2 = 1$

Now argument is positive, hence it represents the semicircular arc of the circle (1st and 4th quadrant)

22. (*d*) $\arg(x-3) + i(y-4) = \frac{\pi}{3}$

$\tan^{-1}\left(\frac{y-4}{x-3}\right) = \frac{\pi}{3}$

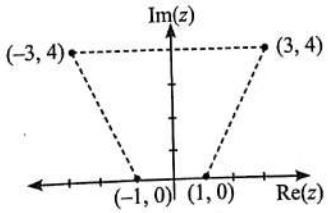
$y - 4 = \sqrt{3}(x-3)$

23. (*c*) Given point

$1 + 0i, -1 + 0i, 3 + 4i, \frac{25}{-3-4i}$

Rationalising $\frac{25}{-3-4i} \times \frac{-3+4i}{-3+4i} =$

$\frac{25(-3+4i)}{9+16} = -3 + 4i$



Clearly, $AB = CD = \sqrt{16+4} = 2\sqrt{5}$

$\Rightarrow$ trapezium is isosceles

Hence, cyclic

24. (*d*) We have $\frac{z_1}{z_2} + \frac{z_2}{z_1} = 1$

Let us assmue the origin be z_3

$\Rightarrow z_1^2 + z_2^2 = z_1 z_2$

$\Rightarrow z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$

Using the fact that three point a_1, a_2 and a_3 are points of eqilateral triangle if and only

If $\Rightarrow a_1^2 + a_2^2 + a_3^2 = a_1 a_2 + a_2 a_3 + a_3 a_1$

$\Rightarrow z_1, z_2$ and the origin form an equilateral triangle.

25. (*c*) $z_1 = e^{i\alpha}, z_1 = e^{i\alpha/2}, z_1 = e^{i\alpha/3}$

Lies on unit circle, thus circumcentre = origin = 0

$\text{centroid} = \frac{z_1 + z_2 + z_3}{3} = G$

Since $\underset{H}{\overset{2:1}{\text{---}}}\underset{G}{\text{---}}\underset{C}{}$

i.e. centroid (G) divides orthorcentre (H) and circumcentre (C) in 2 : 1.

$\therefore \; G = \frac{2\times C + 1\times H}{3} = \frac{2\times 0 + 1\times H}{3} = \frac{H}{3}$

$\Rightarrow \frac{H}{3} = \frac{z_1 + z_2 + z_3}{3} \Rightarrow H = z_1 + z_2 + z_3$

$= e^{i\alpha/1} + e^{i\alpha/2} + e^{i\alpha/3}$

26. (*c*) $z^2 + z|z| + |z|^2 = 0$

$\Rightarrow \left(\frac{z}{|z|}\right)^2 + \frac{z}{|z|} + 1 = 0$

$\Rightarrow \frac{z}{|z|} = \omega, \omega^2$

$\Rightarrow Z = \omega |z|$ or $z = \omega |z|$

$\Rightarrow x + iy = |z|\left(\frac{-1}{2} + \frac{i\sqrt{3}}{2}\right)$ or $x + iy$

$= |z|\left(\frac{-1}{2} - \frac{i\sqrt{3}}{2}\right)$

$\Rightarrow x = -\frac{1}{2}|z|, y = |z|\frac{\sqrt{3}}{2}$

or $x = -\frac{|z|}{2}, y = -\frac{|z|\sqrt{3}}{2}$

$\Rightarrow y + \sqrt{3}x = 0$ or $y - \sqrt{3}x = 0$

$\Rightarrow y^2 - 3x^2 = 0$

27. (*d*) $z = \frac{3}{(\cos\theta + 2) + i(\sin\theta)}$

$= \frac{3[(\cos\theta + 2) - i\sin\theta]}{(\cos\theta + 2)^2 + \sin^2\theta}$

$= 3\frac{(\cos\theta + 2) - i\sin\theta}{5 + 4\cos\theta}$

$= \frac{3}{5 + 4\cos\theta}((\cos\theta + 2) - i\sin\theta)$

$= a(\cos\theta + 2) - i\sin\theta)$

Now

$x = a(\cos\theta + 2)$

$\frac{x}{a} - 2 = \cos\theta$

And $y = -a\sin\theta$

$\frac{-y}{a} = \sin\theta$

Hence $\left(\frac{x}{a}-2\right)^2+\frac{y^2}{a^2}=1$

$(x^2-2a)+y^2=a^2$

Hence it is a circle,centered at $(2a, 0)$

Where $a=\frac{3}{5+4\cos\theta}$

28. (*a*) $a=1-\text{i}$ and $y=\sqrt{(1+t)^2+1}$

Hence $t=(1-x)$

Now

$y^2=(1+t)^2+1=(1+1,-x)^2$

$y^2-(x-2)^2=1$

Hence this resembles as hyperbola

29. (*d*) Given that $|z_1|=1$ and $|z_2|=2$

We know that $||z_1|-|W||\le|z-W|\le|z|+|W|$

Now, $||z_1|-|2z_2||\le|z_1-2z_2|\,|z_1|+|2z_2|$

$|1-2(2)|\le|z_1-2z_2|\le 1+2(2)$

$\Rightarrow 3\le|z_1-2z_2|\le 5$

and $|z_1|-|z_2||\le||z_1+z_2|\le|z_1|+|z_2|$

$\Rightarrow|1-(2)|\le|z_1+z_2|\le 1+(2)$

$\Rightarrow 1|z_1+z_2|\le 3$

and $|z_1-3z_2|\ge||z_1|+|3z_2||$

$\Rightarrow|z_1-3z_2|\,|1-3(2)|$

$\Rightarrow|z_1-3z_2|\ge 5$

and $||z_1-z_2|\ge||z_1|-|3z_2||$

$\Rightarrow|z_1-3z_2|\ge|1-(2)|$

$\Rightarrow|z_1-3z_2|\ge 1$

30. (*d*) z = locus of centre of circle, r = radius of circle

$|z-z_1|=r-a$

$|z-z_2|=r-b$

$||z-z_1|\,|z-z_2|=|a-b|$

31. (*c*) $1\le|z-4|\le 9$

32. (*a*) Given, $\Delta=\begin{vmatrix}1&\omega^n&\omega^{2n}\\ \omega^n&\omega^{2n}&1\\ \omega^{2n}&1&\omega^n\end{vmatrix}$

$=1(\omega^{3n}-1)-\omega^n(\omega^{2n}-\omega^{2n})+\omega^{2n}(\omega^n-\omega^{4n})$

$=1(1-1)-0+\omega^{2n}(\omega^n-\omega^n)=0$

33. (*a*) $\omega=2\cdot i^{1/3}=2\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)^{1/3}$

$=2\left[\cos\frac{2n\pi+\frac{\pi}{2}}{3}+i\sin\frac{2n\pi+\frac{\pi}{2}}{3}\right]$

p u t n = 1

$=2\left[\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right]=-\sqrt{3}+i$ Ans.

34. (*d*) We have, $|z|=|\omega|$ and $\arg z=\pi-\arg\omega$

Let $\omega=re^{i\theta}$. Then $z=re^{i(\pi-\theta)}$

$\Rightarrow z=re^{i\pi}e^{-i\theta}=(re^{-i\theta})(\cos\pi+i\sin\pi)=\bar{\omega}(-1)=-\bar{\omega}$

35. (*b*) $-\frac{y}{x}=1-\sqrt{2}$

$\Rightarrow y=(\sqrt{2}-1)x$

$\tan\theta=\sqrt{2}-1$

$\Rightarrow\tan 2\theta=1$

$2\theta=\frac{\pi}{4}$

Number of side $=\frac{2\pi}{n}=\frac{\pi}{4}$

$\Rightarrow\boxed{n=8}$

36. (*c*) $x_n=\cos\left(\frac{\pi}{2^n}\right)+i\sin\left(\frac{\pi}{2^n}\right)\cdots$

$x_1x_2x_3\ldots\infty\,x_1x_2x_3\ldots\infty$

$=\cos\left(\frac{\pi}{2^1}+\frac{\pi}{2^2}+\frac{\pi}{2^3}.+\ldots\infty\right)$

$+\sin\left(\frac{\pi}{2}+\frac{\pi}{2^2}+\frac{\pi}{2^3}+\ldots\infty\right)$

$=\cos\pi+1\sin\pi-1$

37. (*b*) $\alpha=\cos\left(\frac{8\pi}{11}\right)+i\sin\left(\frac{8\pi}{11}\right)=e^{i\frac{8\pi}{11}}$

α is one of the 11^{th} root of the unity

$$=\frac{(\alpha+\bar{\alpha})+(\alpha^2+\bar{\alpha}^{-2})+(\alpha^3+\bar{\alpha}^3)+(\alpha^4+\bar{\alpha}^{-4})+(\alpha^5+\alpha^{-5})}{2}$$

$$=\frac{-1+(1+\alpha+\alpha^2+\alpha^3+\alpha^4+\alpha^5+\alpha^{10}+\alpha^9+\alpha^2+\alpha^7+\alpha^6)}{2}$$

$=-\frac{1}{2}$

38. (*a*) $x^2-x+1=0$

$\swarrow\quad\searrow$

$-\omega\qquad-\omega^2$

$=\sum_{n=1}^{5}\left(x^n+\frac{1}{x^n}\right)^2=\sum_{n=1}^{5}\left(x^{2n}+\frac{1}{x^{2n}}+2\right)$

$\Rightarrow x^2+x^4+x^0+x^{10}+\frac{1}{x^2}+\frac{1}{x^4}+\frac{1}{x^6}+\frac{1}{x^8}+\frac{1}{x_{10}}+2\times 5$

$=\left(x^2+\frac{1}{x^2}\right)+\left(x^4+\frac{1}{x^4}\right)+\left(x^{10}+\frac{1}{x^{10}}\right)+10$

$=(-1)+(-1)+(2)+(-1)+(-1)+10=8$

39. (*a*) $\sum_{r=0}^{50}z^k=0$

$z^0+z^{1+}z^2+z^{50}=0$

$\frac{z^{51}-1}{z-1}=0$

$\Rightarrow z^{51}-1=0$

$z^{51}-1=(z-t_1)(z-z_1)(z-z_2)\text{–}(z-z_{50})$

$1+z+z^{2}\text{++}z^{50}=(z-z_1)(z-z_2)+(z-z^{50})$

Take log both side adn differentate

$\frac{1}{1+z+z^2+\ldots+z^{50}}$

$\frac{1}{z-z_1}+\frac{1}{z-z_2}+\cdots\frac{1}{z-z_{z_0}}$

$\Rightarrow\sum\frac{1}{zr-1}=-25$

40. (*c*) $\sum_{k=1}^{n}\cos(k\theta)=\text{Re}\sum_{k=1}^{n}e^{ik\theta}$

$=\text{Re}\left\{e^{i\theta}\left(\frac{e^{in\theta}-1}{e^{i\theta}-1}\right)\right\}$

$=\text{Re}\left\{e^{i\theta}\frac{e^{in\theta/2}\left(e^{in\theta/2}-e^{-in\theta/2}\right)}{e^{i\theta/2}\left(e^{i\theta/2}-e^{-i\theta/2}\right)}\right\}$

$=\text{Re}\left\{e^{i\theta}\frac{e^{in\theta/2}}{e^{i\theta/2}}\frac{2i\sin(n\theta/2)}{2i\sin(\theta/2)}\right\}$

$=\text{Re}\left\{e^{i(n+1)\theta/2}\frac{\sin(n/2)}{\sin(\theta/2)}\right\}$

$=\cos\left(\frac{(n+1)\theta}{2}\right)\frac{\sin(n\theta/2)}{\sin(\theta/2)}$

41. (*d*) From the hypothesis we have

$z=\frac{\sqrt{3}}{2}-\frac{i}{2}=i\left(-\frac{1}{2}-\frac{i\sqrt{3}}{2}\right)=i\omega$

where $\omega=\left(-\frac{1}{2}\right)-\left(\frac{i\sqrt{3}}{2}\right)$ which is a cube root of unity. Now $z^{95}=(i\omega)^{95}=-i\omega^2$

(since $\omega^3=1$) and $i^{67}=i^3=-i$.

Therefore, $z^{95}+i^{67}=-i(1+\omega^2)=(-i)(-\omega)=i\omega$

$(z^{95}+i^{67})^{94}=(i\omega)^{94}=i^2\omega=-\omega$

Now $-\omega=z^n=(i\omega)^n\Rightarrow i^n\cdot\omega^{n-1}=-1$

$\Rightarrow n=2,6,10,14,\ldots$ and $n-1=3,6,9,\ldots$

Therefore $n=10$ is the required least positive integer.

Prabal (JEE Main Level)

1. (*a*) $\left(\frac{e^{i\alpha}}{e^{-i\alpha}}\right)^n-\left(\frac{e^{in\alpha}}{e^{-in\alpha}}\right)$

$\Rightarrow\quad e^{i2n\alpha}-e^{i2n\alpha}=0$

2. (*c*) We have

$\alpha^{2n}+2^n\alpha^n+2^{2n}=2^{2n}\left[\left(\frac{\alpha}{2}\right)^{2n}+\left(\frac{\alpha}{2}\right)^n+1\right]$

$=2^{2n}(\omega^{2n}+\omega^n+1)\left[\because\frac{\alpha}{2}=\frac{-1}{2}+\frac{i\sqrt{3}}{2}\right.$

$\left.\text{and}\left(\frac{\alpha}{2}\right)^3=1\right]$

$=2^{2n}(0)=0$ (Since 3 does not divide n)

3. (c) Upon expanding, we get

$$(a+b)+(2a+b)w+(3a+b)w^2+\ldots (na+b)w^{n-1}$$

$$=a\left(1+2w+3w^2+\ldots nw^{n-1}\right)+b\left(1+w+w^2+\ldots w^{n-1}\right)$$

$$=a\left(1+2w+3w^2+\ldots nw^{n-1}\right)+b\left(\frac{1-w^n}{1-w}\right)$$

$$=a\left(1+2w+3w^2+\ldots nw^{n-1}\right)+0$$

Let

$$S=a\left(1+2w+3w^2+\ldots nw^{n-1}\right)$$

$$Sw=a\left(w+2w^2+3w^3+\ldots(n-1)w^{n-1}-nw^n\right)$$

$$S(1-w)=a\left(1+w+w^2\ldots w^{n-1}\right)-anw^n$$

$$S(1-w)=a\left(\frac{1-w^n}{1-w}\right)-anw^n$$

$$S(1-w)=a(0)-an$$

$$S=-\frac{an}{1-w}=\frac{an}{w-1}$$

4. (b) $S=1+2a+3a^2+\ldots na^{n-1}$

$$S(a)=a+2a^2+3a^3+\ldots(n-1)a^{n-1}+na^n$$

$$S(1-\alpha)=1+a+a^2+\ldots a^{n-1}-na^n$$

$$S(1-a)=\frac{1-a^n}{1-a}-na^n$$

$$S(1-\alpha)=0-n$$

$$S=\frac{-n}{1-a}$$

$$S=\frac{n}{a-1}$$

5. (b) Let $z=x+iy$

$$\Rightarrow \mathrm{Re}\left(\frac{x+iy+21}{ix-y+2}\right)\le 4$$

$$\Rightarrow \mathrm{Re}\left(\frac{x+(y+2)i}{(2-y)+ix}\times\frac{2-y-ix}{2-y-ix}\right)$$

$$\Rightarrow \mathrm{Re}\left(\frac{(x+(y+2)i)(2-y-ix)}{(2-y)^2+x^2}\right)$$

$$\Rightarrow \mathrm{Re}\left(\frac{2x-xy-ix^2+2iy-y^2i+xy+4i}{(2-y)^2+x^2}-2iy+2x\right)$$

$$\Rightarrow \frac{yx}{x^2+(2-y)^2}\le 4$$

$$\Rightarrow 4x\le 4x^2+4\left(4+y^2-4y\right)$$

$$\Rightarrow x\le x^2+4+y^2-4y$$

$$\Rightarrow x^2+y^2-x-4y+4\ge 0$$

$\therefore$ Locus is a circle

6. (c) Hence centre is (1,0) is also the mid-poin of diagonal of square.

$$\Rightarrow \frac{z_1+z_2}{2}=z_0$$

Where $z_0=1+0i$

$$\Rightarrow z_2=-\sqrt{3}i \text{ and } \frac{z_3-1}{z_1-1}=e^{\pm\frac{i}{2}}$$

$$\Rightarrow z_3=1+(1+\sqrt{3}i)\cdot\left(\cos\frac{\pi}{2}\pm i\sin\frac{\pi}{2}\right)$$

$$\left(\because z_1=2+\sqrt{3}i\right)$$

$$=1\pm i(1+\sqrt{3}i)=(1\mp\sqrt{3})\pm i$$

$$z_3=(1-\sqrt{3})+i \text{ and } z_4=(1+\sqrt{3})-i$$

7. (a) $$\frac{z-i}{z-3i-z}=\frac{|z-i|}{|z-3i-z|}\pm e^{i\frac{2\pi}{3}}$$

$$\frac{z-i}{3-3i-z}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$$

$$z=\frac{1}{2}\ [-3+4\sqrt{3}+(3\sqrt{3}+6)i]$$

8. (a) Let $A\left(re^{i\alpha}\right)\&\beta\left(re^{i\beta}\right)$ Point of intersection of tangents at 'α', 'β' to circle $x^2+y^2=r^2$ is

$$\left(r\cdot\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}-i\frac{r\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$$

$$=\frac{re^{i\left(\frac{\alpha+\beta}{2}\right)}}{\cos\frac{\alpha-\beta}{2}}=\frac{2z_1z_2}{z_1+z_2}$$

9. (a) $$z_1-z_2=(z_3-z_2)e^{i\frac{\pi}{3}}$$

$$z_1-z_3=(z_2-z_3)e^{i\frac{\pi}{3}}$$

$$\frac{2z_1-z_2-z_3}{z_3-z_1}=\frac{(z_1e-z_2)+(z_1-z_3)}{z_3-z_2}$$

$$=\frac{(z_3-z_2)^{ei\frac{\pi}{3}}+(z_2-z_3)^{e-\frac{i\pi}{3}}}{(z_3-z_2)}$$

$$=\frac{(z_3-z/2)\left[e^{i\frac{\pi}{3}}-e^{-i\frac{\pi}{3}}\right]}{(z_3-z_2)}$$

$$\left(e^{i\theta}=\cos\theta+i\sin\theta\right)$$

$$=\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}-\left[\cos\left(-\frac{\pi}{3}\right)+i\sin\left(-\frac{\pi}{3}\right)\right]$$

$$=\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}-\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}$$

$$=2i\sin\frac{\pi}{3}=2i\frac{\sqrt{3}}{2}$$

$$=i\sqrt{3}$$

$$\arg(i\sqrt{3})=\tan^{-1}\left(\frac{\sqrt{3}}{0}\right)=\tan^{-1}(0)$$

$$=\frac{\pi}{2}$$

10. (a) In the Argand plane, P is represented by $e^{i\theta}$ and Q is represented by $e^{i(\alpha-\theta)}$

Now rotation about a line with angle α is given by $e^{i\theta}\to e^{i(2\alpha-\theta)}$. Therefore, Q is obtained from P by reflection in the line making angle α/2

11. (a) For a complex number z,

$$z(\bar{z})=|z|^2$$

Where $\bar{z}$ is the conjugate of z and |z| is the modulus of z

$$\Rightarrow \frac{1}{z}=\frac{\bar{z}}{|z|^2}$$

For point F, $z=a+bi$ (where a and b are positive)

so 1/z will be of form $x-iy$ (where x and y are positive) and will lie in fourth quadrant.

Now, $|z|>1$

$$\frac{1}{|z|}\frac{|\bar{z}|}{|z|}=\frac{1}{|z|^2}=\frac{1}{|z|}$$

So $\left|\frac{1}{z}\right|<1$

Therefore the reciprocal of F will lie in the fourth quadrant and within the unit circle

12. (a) Let $z=x+iy$

$$|z-2|=3\Rightarrow|x+iy-2|=3$$

$$\Rightarrow|(x-2)+iy|=3$$

$$\Rightarrow\sqrt{(x-2)^2+y^2}=3$$

$$\Rightarrow(x-2)^2+y^2=9$$

$$\Rightarrow x^2+y^2-4x+4-9=0$$

$$\Rightarrow S_1: x^2+y^2-4x-5=0$$

Again $|z-2-3i|=4$

$$\Rightarrow|x+iy-2-3i|=4$$

$$\Rightarrow|(x-2)+i(y-3)|=4$$

$$\Rightarrow\sqrt{(x-2)^2+(y-3)^2}=4$$

$$\Rightarrow(x-2)^2+(y-3)^2=16$$

$\Rightarrow x^2+y^2-4x-6y+4+9-16=0$

$\Rightarrow S_2=x^2+y^2-4x-6y-3=0$

Radical axis $=S_2-S_1=0$

$\Rightarrow x^2+y^2-4x-6y-3-x^2-y^2+4x+5=0$

$\Rightarrow -6y+2=0$

$\Rightarrow 2(3y-1)=0$

$\therefore 3y-1=0$ is the radical axis.

$|z-|z+1||=|z+|z-1||$

13. **(a)** Let $|z+1|=\alpha$ and $|z-1|=\beta$

$\Rightarrow |z-\alpha|=|z+\beta|$

Z is a line

14. **(b)** $x_1=3, y_1=\sqrt{3}$

and $x_2=2\sqrt{3}, y_2=6$

Angle between live is 30°

So Slope of Bisector is 45°

$y=x$

15. **(a)** $z_2=z_1e^{i\alpha}$

$z_1+z_2=-p \Rightarrow z_1\left(1+e^{i\alpha}\right)=-p$

$z_1z_2=q \Rightarrow z^2e^{i\alpha}=q$

So, $|z_1|\left|(1+\cos\alpha+i\sin\alpha)\right|=|p|$

$|z_1|^2=|q|$

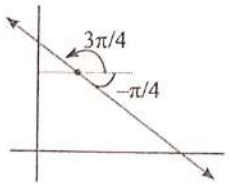
$\Rightarrow \left|2\cos^2\alpha_{12}+2\sin\alpha/2\ln\alpha/2\right|^2|z_1|^2=|p|^2$

$\Rightarrow 4|q|\cos^2\alpha/2=|p|^2$

OR $\arg\left(\dfrac{\alpha_1}{\alpha_2}\right)=\dfrac{\pi}{2}$

16. **(d)** opposite vertex

17. **(d)** Ray originated from



18. **(d)** $Z_D=\omega e^{i\theta}$

$Z_D=\bar{\omega}=z_De^{i\theta}$

$\dfrac{z_D}{\bar{\omega}}=\dfrac{\omega}{z_D}$

$z_D^2=\omega\bar{\omega}=1$

$z_D=-1$

$\omega+\bar{\omega}=-1$

19. **(b)** ω is purely Imaginary

$\omega+\bar{\omega}=0$

$\dfrac{z-i}{2z+1}+\dfrac{\bar{z}+i}{2\bar{z}+1}=0$

$(z-i)(2\bar{z}+1)+(\bar{z}+i)(2z+1)$

$\Rightarrow 2z\bar{z}+z-2\bar{z}i-i+2z\bar{z}+\bar{z}+2iz+i=0$

$\Rightarrow 4z\bar{z}+z+\bar{z}+2iz-2\bar{z}i=0$

$z\bar{z}+\dfrac{z+\bar{z}}{4}+2i\left(\dfrac{z-\bar{z}}{4}\right)=0$

$x^2+y^2+\dfrac{x}{2}-y=0$

20. **(a)** $z^3+(3+i)z^2-3z-(m+i)=0, m\in R$

$z=a=$ Real

$z=\Sigma$

$z^3+(3-i)z^2-3\bar{z}-(m-i)=0$, mex

So

$\Rightarrow z^3+(3-i)z^2-3z-(m-i)=0$

So subtract

$z=\pm1$

Now find m by put $z=\pm1$

$m=1, 5$

21. **(a)** $w_1=e^i \; w_1=|w_1|e^{i\theta}$

$w_2=|w_2|e^{i\alpha}$

$=\dfrac{p}{q}\left|\dfrac{w_1}{w_2}\right|e^{i(\theta-\alpha)}+\dfrac{q}{p}\left|\dfrac{w_2}{w_1}\right|e^{i(\alpha-\theta)}$

$=e^{i(\theta-\alpha)}+e^{i(\alpha-\theta)}$

$=e^{i(\theta-\alpha)}+e^{-i(\theta-\alpha)}$

$=2\cos(\theta-\alpha)$

$=$ Real

22. **(a)** $\alpha_1\bar{z}+\bar{\alpha}_1z+\beta_1=0$

$\alpha_2\bar{z}+\bar{\alpha}_2z+\beta_2=0$

complex slope $=-\dfrac{\alpha_1}{\bar{\alpha}_1}$

$-\dfrac{\alpha_1}{\bar{\alpha}_1}+\left(\dfrac{\alpha_2}{\bar{\alpha}_2}\right)=0$

$\alpha_1\overline{\alpha_2}+\alpha_2\overline{\alpha_1}=0$

$\dfrac{\alpha_1}{\alpha_2}+\left(\dfrac{\alpha_1}{\alpha_2}\right)=0 \Rightarrow \dfrac{\alpha_1}{\alpha_2}=$ purely Imaginary

23. **[6]** $\sum_{r=1}^{6}z_r^2=\beta z_0^2$

$z_1^2+z_2^2+z_3^2+z_4^2+z_5^2+z_6^2$

$|z_1-z_2|^2=|z_2-z_3|^2+|z_3-z_1|^2=a^2$

$z_1^2+z_2^2+z_3^2=3z_0^2$

When z_1, z_2, z_3 are vertices of a equilateral Δ

So, centroid will form a regular Hexagon.

$z_1^2+z_2^2+z_0^2=3z_{01}^2$

$z_2+z_3^2+z_0^2=3z_{02}^2$

$z_6+z_5^2+z_0^2=3z_{06}^2$

$2\left(z_1^2+z_2^2++z_6^2\right)+6z_0^2=3\left(z_{01}^2+z_{02}^2+\right.$

$\left.+z_{02}^2+z_{03}^2+z_{04}^2+z_{05}^2+z_{06}^2\right)$

$\Rightarrow z_1^2+z_2^2+z_3^2+z_4^2+z_5^2+z_6^2=6z_0^2$

$\beta=6$

24. **[3]** $Z_H \xrightarrow{2} Z_G \xrightarrow{1} Z_C$

$\dfrac{z_H+2z_C}{3}=z_4$

$z_H=3z_4=z_1+z_2+z_3$

$a+b+c=3$

25. **[4]** $z_1-z_2=i(z_3-z_2)$

$z_2=\dfrac{-z_1}{-1+i}+\dfrac{i}{-1+i}z_3$

$2z_2=(1+i)z_1-(i-1)z_3$

$2z_2=(1+i)z_1+(1-i)z_3-(1)$

Similarity

$2z_4=(1-i)z_1+(1+i)z_3$

$a=2=b$

$a+b=4$

26. **[7]** $|z_1+z_2|=|z_1-z_2|$

$\Rightarrow \left(\dfrac{z_1}{z_2}\right)$ is purely Imaginary

ΔOAB is Right Angled Tragle

$z_H=\dfrac{z_1+z_2}{2}$

$\alpha=7$

27. **[9]** $r^2+r^2=4$

$r^2=2 \Rightarrow r=\sqrt{2}$

$\text{re}=\dfrac{3}{4}\pi(2)$

$A=\dfrac{3}{2}\pi=\dfrac{2k^2}{\pi^2}$

Perimeter

$=\dfrac{3}{4}(2\pi r)$

$=\dfrac{3}{2}\pi\sqrt{2}=\dfrac{2k^2}{\pi^2}$

$\Rightarrow \dfrac{3}{4}\pi^3\sqrt{2}=k^2$

28. [1] $|z| = \min(|z-1|, |z+1|)$

$|z| = |z-1|$

$\text{Re}(z) = \frac{1}{2}$

and $|z| = |z+1|$

$Re(z) = -\frac{1}{2}$

$|z+\bar{z}| = |2\text{Re}(z)| = 1$

29. [7] $|iz + z_1| = |z - iz_1|$

$= |z - (-3+5i)|$

max value $= 5 + 2 = 7$

30. [2] $|z-2| = 2$ and $z(1-i) + \bar{z}(1+i) = 4$

$z = x + iy$

$y = \pm\sqrt{2}$

Solve both Equation

$z = (2-\sqrt{2}) + i\sqrt{2}$

31. [1] $z^3 - 3\alpha z^2 + 3\beta z + \gamma$

$z_1 + z_2 + z_3 = 3\alpha$

$z_1z_2 + z_2z_3 + z_3z_1 = 3\beta$

$(z_1 + z_2 + z_3)^2 = z_1^2 + z_2^2 + z_3^2 +$

$2(z_1z_2 + z_2z_3 + z_3z_1)z = (2+\sqrt{2}) - i\sqrt{2}$

$9\alpha^2 \neq z_1^2 + z_2^2 + z_3^2 + 6\beta$

$9\alpha^2 - 6\beta = z_1^2 + z_2^2 + z_3^2$

Condition for Equilateral Δ

$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

$9\alpha^2 - 6\beta = 3\beta$

$\alpha^2 = \beta$

Parikshit (JEE Advanced Level)

1. (c) $(z+1)^4 = 16z^4$

$(z+1)^2 = \pm 4z^2$

$z+1 = \pm 2z$ or $z+1 = \pm 2$ iz

$z+1 = 2z, z+1 = -2z, z+1 = 2iz, z+1 = -2iz$

$z = 1, -\frac{1}{3}, -\frac{1}{1-2i}, -\frac{1}{1+2i}$

$\therefore z = 1, -\frac{1}{3}, -\frac{(1+2i)}{5}, -\frac{(1-2i)}{5}$

$\therefore$ Points on the argand plane are

$(1,0), \left(-\frac{1}{3}, 0\right), \left(-\frac{1}{5}, -\frac{2}{5}\right), \left(-\frac{1}{5}, \frac{2}{5}\right)$

$\therefore \left(\frac{1}{3}, 0\right)$ is a point equidistant from

$(1,0), \left(-\frac{1}{3}, 0\right), \left(-\frac{1}{5}, -\frac{2}{5}\right), \left(-\frac{1}{5}, \frac{2}{5}\right)$

2. (a) $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$

$\Rightarrow \left(\frac{z_1}{z_2}\right)^2 - 2\left(\frac{z_1}{z_2}\right) + 2 = 0$

$\Rightarrow \frac{z_1}{z_2} = \frac{2 \pm \sqrt{4-8}}{2} = (1 \pm i)$

$\Rightarrow |z_1| = \sqrt{2}|z_2|$ ..... (i)

Again $z_1 - z_2 \pm iz_2$

$\Rightarrow (z_1 - z_2)$ is $\perp$ to z_2 .....(ii)

and $|z_1 - z_2| = |z_2|$

$\Rightarrow$ O, z_1 and z_2 form an isosceles right angled triangle.

3. (c) $Q = \sqrt{2|z|^2}\,\text{cis}\left(\frac{\pi}{4} + 0\right)$

$= |Z|\sqrt{2}\left[\cos\left(\frac{\pi}{4}+0\right) + i\sin\left(\frac{\pi}{4}+0\right)\right]$

$= |Z|[(\cos 0 - \sin 0) + i(\sin 0 + \cos 0)]$

$= |Z|[(\cos 0 + i\sin 0) + |z|i/i(-\sin 0 + i\cos 0)]$

$= |Z|[\cos 0 + i\sin 0] + |Z|/1(-\cos 0 - i\sin 0]$

$= P + iP$

$\Rightarrow \arg(Q - p/p) = \pi/2$

This is right angle triangle with $|P| = |Q - P|$ and right angle at P.

4. (c) $\text{Re}(z_1\bar{z}_2) = 0$ $\text{Re}((a+ib)(c-id)) = 0$

$\Rightarrow ac + bd = 0$

$\Rightarrow \frac{a}{b} = -\frac{d}{c}$

$\Rightarrow \frac{a^2}{b^2} = \frac{d^2}{c^2}$

$\Rightarrow \frac{a^2+b^2}{b^2} = \frac{d^2+c^2}{c^2}$

$\Rightarrow b^2 = c^2$

$|w_1|^2 = a^2 + c^2 = a^2 + b^2 = 1$

$|w_2|^2 = b^2 + d^2 = c^2 + d^2 = 1$

$b^2 = c^2 \Rightarrow b = \pm c$

$w_1\bar{w}_2 = (a+ic)(b-id)$

$= (ab+cd) + i[bc - ad]$

If $b = c$ then $ab + cd = ac + bd = 0$ (given)

5. (a) $z_k = 1 + a + a^2 + \ldots a^{k-1}$

$z_k = \frac{1-a^k}{1-a}$

$z_k - \frac{1}{1-a} = \frac{-a^k}{1-a}$

$\left|z_k - \frac{1}{1-a}\right| = \left|\frac{-a^k}{1-a}\right| < \left|\frac{1}{1-a}\right|$

Hence z_k lies on a circle

$\left|z_k - \frac{1}{1-a}\right| = \left|\frac{1}{1-a}\right|$

6. (a) So we can apply P in log values.

$\log\frac{1}{3}\frac{p^2+2}{11p-2} > 1$

$\frac{p^2+2}{11p-2} > 0$ here

Numerator is always positive, so

$11p - 2 > 0$

then $p > \frac{2}{11}$

$\frac{p^2+2}{11p-2} < \frac{1}{3}$

$\frac{p^2+2}{11p-2} - \frac{1}{3}$

$\frac{3p^2+6-11p+2}{3(11p-2)} < 0$

$\frac{3p^2-11p+8}{3(11p-2)} < 0$

The denominator is positive for the previous condition, then the numerator must be negative

$1 < |z-3| < \frac{8}{3}$

7. (a) $zz_1 + z_2z_3 = 0$

8. (c) $z = -4(i-1) \pm \sqrt{-63-16i}$

$z = -4(i-1) \pm (1-8i)$

$z = 5 - 12i, 3 + 4i$

Hence area

$(\Delta z_1z_2) = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 5 & -12 & 1 \\ 3 & 4 & 1 \end{vmatrix} = \frac{1}{2} \times 56 = 28$

9. **(d)** The line is

$z(1+i)=\bar{z}(1-i)$ or $z(-1+i)+\bar{z}(-1-i)=0$

If the reflection of a number ω in the line $a\bar{z}+\bar{a}z+b=0, b\in R$ be ω' then

$\omega'\bar{a}+\bar{\omega}a+b=0$

Now, $\omega=\dfrac{2+i}{3-i}=\dfrac{(2+i)(3+i)}{10}=\dfrac{1+i}{2}$

So, $\omega'(-1+i)+\left(\dfrac{1-i}{2}\right)(-1-i)=0$

$\therefore \omega'(-1+i)=\dfrac{(1-i)(1-i)}{2}=1$

$\Rightarrow \omega'=\dfrac{1}{-1+i}=\dfrac{-1-i}{2}=\dfrac{2+i}{-3+i}$

10. **(b,c)** $A(z_1)$ $B(z_2)$ $C(z_3)$

$|z_3-z_2|\le|z_3-z_1|$

$\therefore \dfrac{|z_3|-|z_2|}{|z_3-z_2|}\ge\dfrac{|z_3|-|z_2|}{|z_3-z_1|}$

11. **(a,b)** $C_1: |z-1-i|=1$

Centre of C_1 is (1, 1) and radius is 1

$C_2: |z-1+i|=1$

Centre of C_2 is (1, -1) and radius is 1

Circle C with radius 2

C_1 and C_2 with radius 1 touch this circle C internally.

Circle C_3 and C_4 touch all the three circles.

Using Pythagoras theorem,

$(1)^2+(2-r)^2=(1+r)^2$

$\Rightarrow 1+4-4r+r^2=1+2r+r^2$

$\Rightarrow r=\dfrac{2}{3}$

Centre of C_3 and C_4 are $\left(-1+\dfrac{2}{3},0\right)$ and $\left(3-\dfrac{2}{3},0\right)$ respectively.

i.e centres of C_3 and C_4 are $\left(\dfrac{-1}{3},0\right)$ and $\left(\dfrac{7}{3},0\right)$ respectively.

Required equation of circle is

$\left|z+\dfrac{1}{3}\right|=\dfrac{2}{3}$ or $\left|z-\dfrac{7}{3}\right|=\dfrac{2}{3}$

12. **(a,b)** $\dfrac{z-4}{z-2i}$ is purely imagination

Circle is $(x-2)^2+(y-1)^2=5$

$Oz_1\perp Cz_1$

So $\dfrac{z_1}{z_1-2-i}=ik$

and $\dfrac{0+z}{2}=z_1$

$\Rightarrow z=2z_1$

$\dfrac{z-4}{z-2i}=$ purely .....

$\dfrac{2z_1-4}{2z_1-2i}=ki,\quad k\in R-\dfrac{1}{2}\{0\}$

$\dfrac{z_1-2}{z_1-i}=ki, k\in R-\{0\}$

13. **(b,c,d)** λ can be 1, 2, 3.

14. **(a,b,c,d)**

$Z_H^2 z_G z^1 z_c$

$z_H+2\cdot z_C=3Z_G$

$Z_H=z_1+z_2+z_3$

(d) If Δ ABC is right angle

$Z_H=Z_2$

$\Rightarrow Z_1+Z_3=0$

(b) $|z_1|=|z_2|=|z_3|=|z|$

$\Rightarrow z_1\bar{z}_1=z_2\bar{z}_2=z_3\bar{z}_3=z\bar{z}$

$\because$ AP $\perp^r$ BC

$\therefore \dfrac{z-z_1}{\bar{z}-\bar{z}_1}+\dfrac{z_2-z_3}{\bar{z}_2-\bar{z}_3}=0$

$\Rightarrow \dfrac{z-z_1}{\frac{z_1\bar{z}_1}{z}-\bar{z}_1}+\dfrac{z_2-z_3}{\frac{z_3\bar{z}_3}{z_7}-\bar{z}_3}=0$

$\Rightarrow -\dfrac{z}{\bar{z}_1}-\dfrac{z_2}{\bar{z}_3}=0$

$\Rightarrow z=-\dfrac{\bar{z}_1 z_2}{\bar{z}_3}=-\dfrac{z_2 z_3}{z_1}$

15. **(a,c)** centre $=\dfrac{k^2\beta-\alpha}{k^2-1}$

16. **(a,b)** $|z_1|=|z_2|=1; |z_1+z_2|=1$

$\Rightarrow (z_1+z_2)(\bar{z}_1+\bar{z}_2)=1$

$|z_1+z_2|=|z_1||1+e^{i\theta}|=1$

$|z_2-z_1|=|z_1||e^{i\theta}-1|$

17. **(a,c,d)** Since $P_k=e^{i\theta_k}, \theta_k=\dfrac{(2k)\pi}{n}$

Let $A_k=x_k+iy_k=a\cos\theta_k+ib\sin\theta_k$

$\Rightarrow \dfrac{x_k}{a}=\cos\theta_k, \dfrac{y_k}{b}=\sin\theta_k$

which implies that

$\dfrac{x_k^2}{a^2}+\dfrac{y_k^2}{b^2}=1, A_k$ lies on ellipse

$\Rightarrow \sum_{k=1}^{n}A_kS=\sum_{k=1}^{n}(a-ex_k)=na-e\sum_{k=1}^{n}x_k=na$

$\Rightarrow \sum_{k=1}^{n}(A_kS)^2=\sum_{k=1}^{n}(a-ex_k)^2$

$=na^2-2ae\sum_{k=1}^{n}x_k+e^2\sum_{k=1}^{n}x_k^2$

$=na^2+a^2e^2\sum_{k=1}^{n}\cos^2\theta_k=na^2+\dfrac{na^2e^2}{2}$

$+\dfrac{1}{2}a^2e^2\sum_{k=1}^{n}\cos2\theta_k$

$=na^2+\dfrac{na^2}{2}\left(1-\dfrac{b^2}{a^2}\right)=\dfrac{n}{2}\left(3a^2-b^2\right)$

$\left(\because \sum_{k=1}^{n}\cos2\theta_k=0$ sum of roots of $z^{n/2}-1=1\right)$

18. **(c)** **19.** **(d)** **20.** **(a)** **21.** **(a),** **22.** **(c)**

23. **(a)** **24.** **(a)**

Where $\alpha=e^{i\frac{2\pi}{7}}$

$G_1\equiv\dfrac{1+\alpha+\alpha^4}{3}$

$G_2\equiv\dfrac{\alpha^2+\alpha^5+\alpha^6}{3}$

Centroid of ΔOG_1G_2

$=\dfrac{0+\frac{1+\alpha+\alpha^4}{3}+\frac{\alpha^2+\alpha^5+\alpha^6}{3}}{3}$

$P=-\dfrac{\alpha^3}{9}\quad\therefore\quad \angle POA_1=\dfrac{\pi}{7}\ \&\ OP=\dfrac{1}{9}$

$\dfrac{G_1+G_2+G_3}{3}=0$

$\Rightarrow G_3=-\dfrac{1}{3}\left(1+\alpha+\alpha^2+\alpha^5+\alpha^6\right)=\dfrac{\alpha^3}{3}$

$\Rightarrow OG_3=\dfrac{1}{3}$

$PA_1=\left|1+\dfrac{\alpha^3}{9}\right|=\left|1+\dfrac{1}{9}\left(\cos\dfrac{6\pi}{7}+i\sin\dfrac{6\pi}{7}\right)\right|$

$=\dfrac{1}{9}\sqrt{82+18\cos\dfrac{6\pi}{7}}=\dfrac{1}{9}\sqrt{82-18\cos\dfrac{\pi}{7}}$

25. **(c)** $\angle IAB=\dfrac{\theta}{2}, \angle IAC=\dfrac{\theta}{2}$

$C(z_3)$, $I(z_4)$, $A(z_1)$, $B(z_2)$, $D\left(\dfrac{z_1+z_2}{2}\right)$

$$\frac{z_2-z_1}{|z_2-z_1|}=\frac{z_4-z_1}{|z_4-z_1|}e^{-\frac{i\theta}{2}}$$

$$\frac{z_3-z_1}{|z_3-z_1|}=\frac{z_4-z_1}{|z_2-z_1|}e^{\frac{i\theta}{2}}$$

$$\frac{(z_2-z_1)(z_3-z_1)}{|z_2-z_1||z_3-z_1|}=\frac{(z_4-z_1)^2}{|z_4-z_1|^2}e^{\theta}$$

$$\therefore \frac{(z_2-z_1)(z_3-z_1)}{(z_4-z_1)^2}=\frac{AB\cdot AC}{(IA)^2}$$

$$=\left(\frac{AB}{IA}\right)^2\left(\frac{AC}{AB}\right)$$

26. (*a*) $\dfrac{(z_2-z_1)(z_3-z_1)}{(z_4-z_1)^2}=2\left(\dfrac{AD}{IA}\right)^2\left(\dfrac{AC}{AD}\right)$

(since $AB=2AD$)

$$(z_4-z_1)^2(1+\cos\theta)\sec\theta=(z_2-z_1)(z_3-z_1)$$

27. (*b*) $\omega_1=\omega_2 e^{i\frac{4\pi}{3}}$

$$\Rightarrow \omega_1^3=\omega_2^3$$

$$\Rightarrow \omega_1^3\bar{\omega}_1^2\bar{\omega}_2^2=\omega_2^3\bar{\omega}_1^2\bar{\omega}_2^2$$

$$\Rightarrow \omega_2\bar{\omega}_1^2=\omega_1\bar{\omega}_2^2$$

28. (*b*) Since $i\beta$ is real

$\therefore \beta$ is purely imaginary

29. (*d*) $-\dfrac{\alpha}{\alpha}=e^{\pm i\frac{\pi}{2}}=\pm i$

$$\therefore (1+i)\left(-\frac{2\theta}{\alpha}\right)=\pm 2(-1+i)$$

30. (*c*), **31.** (*c*), **32.** (*d*)

(IV) Let,

$$z_1=\frac{2}{\sqrt{3}}e^{i\frac{\pi}{2}},z_2\frac{2}{\sqrt{3}}e^{-i\frac{\pi}{6}},z_3=\frac{2}{\sqrt{3}}e^{-i\frac{5\pi}{6}}$$

Clearly, the points lie on the circle $|z|=\dfrac{2}{\sqrt{3}}$ and ΔABC is equilateral and its centroid coincides with circumcentre. Hence,

$z_1+z_2+z_3=0$ and $\bar{z}_1+\bar{z}_2+\bar{z}_3=0$

Since, triangle is equilateral, inradius $r=OD=\dfrac{1}{\sqrt{3}}$. The equation of incircle is

$$|z|=\frac{1}{\sqrt{3}}$$

Let P(z) be any pint on the incircle. Now,

$AP^2=|z-z_1|^2=|z_1|^2-(z\bar{z}_1+\bar{z}z_1)$

Similarly,

$BP^2=|z|^2+|z_2|^2-(z\bar{z}_3+\bar{z}z_3)$

$\therefore AP^2+BP^2+CP^2$

$=3|z|^2+|z_1|^2+|z_2|^2+|z_3|^2-z(\bar{z}_1+\bar{z}_2+\bar{z}_3)-\bar{z}(z_1+z_2+z_3)$

$=3\times\dfrac{4}{3}+\dfrac{1}{3}+\dfrac{1}{3}+\dfrac{1}{3}-z(0)-\bar{z}(0)=5$

(*iii*) $an=\sqrt{1+\left(1+\dfrac{1}{n}\right)^2}+\sqrt{1+\left(1-\dfrac{1}{n}\right)^2}$

$$an=\frac{\sqrt{2n^2+2n+1}+\sqrt{2n^2+1-2n}}{n}$$

$$\frac{1}{an}=\frac{n}{\sqrt{2n^2+2n+1}+\sqrt{2n^2-2n+1}}\times\frac{\sqrt{2n^2+2n+1}-\sqrt{2n^2-2n+1}}{\sqrt{2n^2+2n+1}-\sqrt{2n^2-2n+1}}$$

$$=\frac{n(\sqrt{2n^2+2n+1}-\sqrt{2n^2-2n+1})}{2n^2+2n+1-2n^2+2n-1}$$

$$\frac{1}{an}=\frac{1}{4}\left[\sqrt{2n^2+2n+1}-\sqrt{2n^2-2n+1}\right]$$

$$\frac{1}{an}=\frac{1}{4}\left[\sqrt{n^2+(n+1)^2}-\sqrt{(n-1)^2-n^2}\right]$$

$$\left(\sum_{n=1}^{20}\frac{1}{an}\right)-3=\frac{1}{4}\left[\sqrt{(20)^2+(21)^2-1}\right]-3$$

$$=7-3=4$$

(*iv*) **Correction option (a)**

$9x^3+9x^2y-45x^2=4y^3+4xy^2-20y^2$

$9x^2(x+y-5)=4y^2(x+y-5)$

$9x^2(x+y+5)-4y^2(x+y-5)=0$

$(9x^2-4x^2)(x+y-5)=0$

$(3x+2y)(3x-2x)(x+y-5)=0$

So the three lines are

$x=\dfrac{2}{3}y, x=\dfrac{-2}{3}y$ and $x+y-5$

from equation of lines, we get the vertices of triangle formed by the lines

A(0, 0), B(−10, 15) and C(2, 3)

Area of the triangle

$$=\frac{1}{2}(x_a-x_c)(y_b-y_a)-(x_a-x_b)(y_c-y_a)$$

$$=\frac{1}{2}(x_ay_b+x_by_c+x_cy_a-x_ay_c-x_cy_b-x_by_a)$$

$$=\frac{1}{2}(0\times 15+-10\times 3+2\times 0-0\times 3-2\times 15-(-10\times 0))$$

$$=\frac{1}{2}\times -60$$

Since, area is non-negativengle

So, the area of the triangle = 30 unit^2

(III)

$|z|^2+|z-3|^2+|z-6i|^2$

$x^2+y^2+(x-3)^2+y^2+x^2+(y-6)^2$

$f(x,y)=3x^2+2y^2-6x-12y+9+36$

$=3[x^2-2x]+3[y^2-4y]+45$

$=3[(x-1)^2-1]+3[(y-2)^2-4]+45$

$=3(x-1)^2-3+3(y-2)^2-12+45$

$=3(x-1)^2+3(y-2)^2+30$

min value = 30

$\dfrac{x}{10}=3$

(*ii*) $2g_1g_2+2f_1f_2=c_1+c_2$

$2(n_1)\left(\dfrac{n_2}{2}\right)+2(1)\left(\dfrac{n_2}{2}\right)=\dfrac{1}{2}+\dfrac{1}{2}$

$n_1n_2+n_2=1$

$n_2(n_1+1)=1$

$n_2=1, n_1+1=1$ or $n_2=-1$

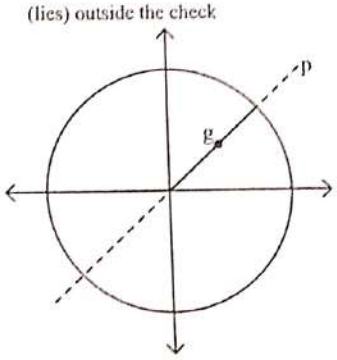
or, $n_1+1=-1$

$n_2=1, n_1=0$ or $n_2=1, n_1=-2$

Two rendered pain

(*i*) $P(\sec\alpha, \cos\alpha)$

(lies) outside the check



$\sqrt{\sec^2 x+\operatorname{cosec}^2 x}-\sqrt{3}=$ Minimum distance

$$D=\sqrt{\frac{\sin^2 x+\cos^2 x}{\sin^2 x.\cos^2 x}}-\sqrt{3}$$

$$=\sqrt{\frac{1}{\sin^2\times\cos^2 x}}-\sqrt{3}$$

$$D=\sqrt{\sec^2 x\operatorname{cosec}^2 x}-\sqrt{3}$$

33. (*c*) → (q), (*b*) → (t), (*c*) → (p), (*d*) → (r)

(a) $\text{Re}\left(\dfrac{iz+1}{iz-1}\right)=2 \Rightarrow \text{Re}\left(\dfrac{z-i}{z+i}\right)=2$

Let $z=x+iy$ then

$$\Rightarrow \text{Re}\left(\frac{x+(y-1)i}{x+(y+1)i}\right)=2$$

$$\Rightarrow \text{Re}\left(\frac{x^2+y^2-1-i2x}{x^2+(y+1)^2}\right)=2$$

$$\Rightarrow x^2+y^2-1=2x^2+2(y+1)^2$$

$$\Rightarrow x^2+y^2+4y+3=0$$

(b) It is a circle with diameter whose end points are $z_1(6, 1)$ and $z_2(4, -3)$ then equation of circle is

$\Rightarrow (x-6)(x-4)+(y-1)(y+3)=0$

$\Rightarrow x^2+y^2-10x+2y+21=0$

$\Rightarrow (x-5)^2+(y+1)^2=5$

(c) Let $z=x+iy$

$$\Rightarrow \left\{\left(\frac{(2x+1)+2iy}{(1-y)^2+x^2}\right)((1-y)-ix)\right\}$$

$\Rightarrow -x(2x+1)+2y(1-y)=2(1-y)^2+2x^2$

$\Rightarrow 4x^2+4y^2+x-6y+2=0$

(d) $\left|\frac{2z-i}{z+1}\right|=1$

Let $z=x+iy$

$\Rightarrow |2z-i|^2=|z+1|^2$

$\Rightarrow |2x+2iy-i|^2=|(x+1)+iy|^2$

$\Rightarrow (2x)^2+(2y-1)^2=(x+1)^2+y^2$

$\Rightarrow 3(x^2+y^2)-2x-4y=0$

$\Rightarrow z_1, z_2, z_3, \ldots\ldots, z_n$ are roots of

$\alpha^n=1$

Which from regular polygon. So distance is zero.

34. ***(b)***

(A) $|(x-2)+i(y-1)|$

$$=|z|\left|\frac{1}{\sqrt{2}}\cos\theta-\frac{1}{\sqrt{2}}\sin\theta\right|$$

Where θ = argz.

$\sqrt{(x-2)^2+(y-1)^2}=\frac{1}{\sqrt{2}}|x-y|$ which is parabola

(B) $(2-3+i)(\overline{z}-3-i)=5$

$|z-(3-i)|^2=5$

$\therefore$ circle

(C) $3|z-2+1|=7$

$|z-(2-i)|=\frac{7}{3}$.

$\therefore$ circle

(D) $|z-3|=2$

circle

35. ***(d)*** (A) $\left|z-i|z|\right|=\left|z+i|z|\right|$

$\left|\frac{z}{|z|}-i\right|=\left|\frac{z}{|z|}+i\right|$

$\left|e^{i\theta}-i\right|=\left|e^{i\theta}+i\right|$

$|z_1-i|=|z_1+i|$

$z_1 \rightarrow$ is line (x-axis)

Im(z) = 0

(B) $|z+4|+|z-4|=10 \quad (B \rightarrow p)$

$2ae=8$

$10(e)=8$

$e=\frac{8}{10}=\frac{4}{5}$

(C) $z=\omega-\frac{1}{\omega}^{(c\rightarrow p),+}$

$z=\omega-\frac{\overline{\omega}}{4}$

$z=x+iy, \quad w=u+iv$

$x+iy=u+iv-\frac{u-iv}{4}$

$x+iy=\frac{3}{4}u+i\frac{5}{4}v$

$x=\frac{3}{4}u, y=\frac{5}{4}v$

$u^2+v^2=4$

$\frac{x^2}{9}+\frac{y^2}{25}=\frac{1}{4}$

$e=\frac{4}{5}$

$|z|=\left|w-\frac{1}{w}\right| \le |w|+\frac{1}{|w|} \le 2+\frac{1}{2}<3$

(D) $|w|=1 \Rightarrow w\overline{w}=1$

$w=u+iv$

$z=x+iy$

$z=w+\overline{w}=2\,\mathrm{Re}(w)$

$\Rightarrow x+iy=z=2u$

$x=2u$

$y=0$

$\frac{x^2}{4}+v^2=1$

$x^2+4v^2=4$

$x^2=4-4v^2$

$x^2=2\sqrt{1-v^2}$

$|\mathrm{Re}(z)| \le 2$

36. ***(a)***

(A) when z = 7

(B) when z = –4

(D) $|z-3|^2+|z-(5-2i)|^2+|z-(1-i)|^2$

$f(x,y)=(x-3)^2+y^2+(x-5)^2+(y+2)^2+(x-1)^2+(y+1)^2$

$=3x^2+9+25+1-6x-10x-2x+3y^2+4+1+4y+2y$

$=3x^2+3y^2-18x+6y+40$

$=3[x^2-6x]+3[y^2+2y]+40$

$=3[(x-3)^2-9]+3[(y+1)^2-1]+40$

$=3(x-3)^2-27+3(y+1)^2-3+40$

$=3(x-3)^2+3(y+1)^2+10$

least value = 10

37. **[0]** $1, \alpha, \alpha^2, \ldots\ldots, \alpha^{n-1}$

Given

$1+\alpha+\alpha^2+\ldots\ldots+\alpha^{n-1}$

$=0 \Rightarrow \alpha^n-1=0$

38. **[4]** $\frac{z_3-z_1}{z_2-z_1}=e^{i\alpha}$

$\Rightarrow \frac{z_3-z_1}{z_2-z_1}=\cos\alpha+i\sin\alpha$

$\Rightarrow \frac{z_3-z_1}{z_2-z_1}-1=\cos\alpha-1+i\sin\alpha$

$\Rightarrow \frac{z_3-z_2}{z_2-z_1}=-2\sin^2\frac{\alpha}{2}+2i\sin\frac{\alpha}{2}\cdot\cos\frac{\alpha}{2}$

$\Rightarrow \frac{z_3-z_2}{z_2-z_1}=2i\sin\frac{\alpha}{2}\left(\cos\frac{\alpha}{2}+i\sin\frac{\alpha}{2}\right)$

Squaring both sides

$\frac{(z_3-z_2)^2}{(z_2-z_1)^2}=-4\sin^2\frac{\alpha}{2}(\cos\alpha+i\sin\alpha)$

$\frac{(z_3-z_2)^2}{(z_2-z_1)^2}=-4\sin^2\frac{\alpha}{2}\left(\frac{z_3-z_1}{z_2-z_1}\right)$

$\Rightarrow (z_3-z_2)^2=4\sin^2\frac{\alpha}{2}(z_3-z_1)(z_1-z_2)$

39. **[2]** In right angled triangle ABC

$\frac{z_2-z_3}{z_1-z_3}=\frac{BC}{AC}e^{i\pi/2}$

$\Rightarrow \frac{(z_2-z_3)}{(z_1-z_3)}=i \quad (\because BC=AC)$

$\Rightarrow (z_2-z_3)=i(z_1-z_3)$ squaring

$(z_2-z_3)^2=-(z_1-z_3)^2$

$\Rightarrow z_2^2+z_3^2-2z_2z_3=-z_1^2-z_3^2+2z_1z_3$

$\Rightarrow z_1^2+z_2^2-2z_1z_2=2z_1z_3+2z_2z_3+2z_2z_3-2z_1z_2-2z_3^2$

$\Rightarrow (z_1-z_2)^2=2(z_1-z_3)(z_3-z_2)$

40. **[30]** $|z|+|z-1|+|2z-3| \ge |z+z-1+3-2z|=2$

$\therefore \lambda=2$

$y=2[x]+3=3[x-2]$

$2[x]+3=3[x]-6$

$[x]=9$

$\Rightarrow y=2[x]+3=21$

$[x+y]=30$

41. **[163]** $\sum_{k=1}^{5}z_k-\sum_{k=1}^{5}w_k=0$

$\sum_{k=1}^{5}z_k=3+504i$

Each $z_k=x_k+y_ki$ lies on the complex line

$y=mx+3$, so we can rewrite this as

$$\sum_{k=1}^{5} z_k = \sum_{k=1}^{5} x_k + \sum_{k=1}^{5} y_k i$$

$$3+504i = \sum_{k=1}^{5} x_k + i\sum_{k=1}^{5}(mx_k+3)$$

Matching the real parts and the imaginary parts, we get that

$$\sum_{k=1}^{5} x_k = 3 \quad \text{and} \quad \sum_{k=1}^{5}(mx_k+3) = 504 \ .$$

Simplifying the second summation, we find that $m\sum_{k=1}^{5} x_k = 504 - 3\cdot 5 = 489$, and substituting, the answer is

$m\cdot 3 = 489 \Rightarrow m = 163.$

42. **[10]** $\tan\alpha = \frac{3}{4}$

$Q(3,9)$

$R = (3, 7)$

$z = 3 + 7i$

PYQ's (Past Year Questions)

1. **(d)** Given, $\left|\frac{z-2i}{z+i}\right| = 2$

$\Rightarrow (z-2i)(\overline{z}+2i) = 4(z+i)(\overline{z}-i)$

$\Rightarrow z\overline{z} + 4 + 2i(z-\overline{z}) = 4(z\overline{z}+1+i(\overline{z}-z))$

$\Rightarrow 3z\overline{z} - 6i(z-\overline{z}) = 0$

Put $z = x + iy$ and $\overline{z} = x - iy$,

$\Rightarrow x^2 + y^2 - 2i(2iy) = 0$

$\Rightarrow x^2 + y^2 + 4y = 0$

This is a equation of circle where centre $(0,-2)$

2. **[1680]** $\left(\frac{z^2+8iz-15}{z^2-3iz-2}\right) \in R$

$\Rightarrow 1 + \frac{(11iz-13)}{(z^2-3iz-2)} \in R$

Put $z = \alpha - \frac{13}{11}i \Rightarrow 11iz - 13 = 11\alpha i$

$\Rightarrow (z^2 - 3iz - 2)$ is imaginary

Put $z = x + iy$

$\Rightarrow (x^2 - y^2 + 2xyi - 3ix + 3y - 2) \in$ Imaginary

$\Rightarrow \text{Re}(x^2 - y^2 + 3y - 2 + (2xy - 3x)\, i) = 0$

$\Rightarrow x^2 - y^2 + 3y - 2 = 0$

$\Rightarrow x^2 = y^2 - 3y + 2$

$\Rightarrow x^2 = (y-1)(y-2) \therefore z = \alpha - \frac{13}{11}i$

Now, Put $x = \alpha, y = \frac{-13}{11}$ in $x^2 = (y-1)(y-2)$

$\Rightarrow \alpha^2 = \left(\frac{-13}{11} - 1\right)\left(\frac{-13}{11} - 2\right)$

$\Rightarrow \alpha^2 = \frac{(24\times 35)}{121}$

$\Rightarrow 242\alpha^2 = 48 \times 35 = 1680$

3. **[2]** For circle:

$|z-z_1|^2 + |z-z_2|^2 = |z_1 - z_2|^2$

Here, $z_1 = \alpha, z_2 = \beta$

$\therefore r = \frac{|z_1 - z_2|}{2} = \frac{|\alpha - \beta|}{2} = \sqrt{\lambda - 1}$

$2\lambda = |\alpha - \beta|^2$

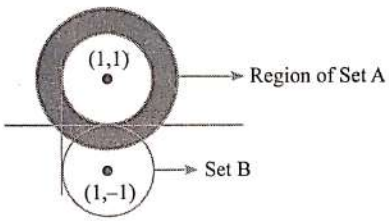
$\because |\alpha - \beta| = 2r = 2\sqrt{\lambda - 1}$

$|\alpha - \beta|^2 = 4\lambda - 4 \Rightarrow 2\lambda = 4\lambda - 4$

$\Rightarrow \lambda = 2$

$\Rightarrow |\alpha - \beta|^2 = 4$ Hence, $|\alpha - \beta| = 2$

4. **(d)** $A = \{z \in C : 1 \le |z - (1+i)| < 2\}$



$B = \{z \in A : |z - (1-i)| = 1\}$

$A \cap B$ has inifiite set.

5. **(c)** $C_1 : |z - (4+3i)| = 2$ and $C_2 : |z| + |z-4| = 6, z \in C$

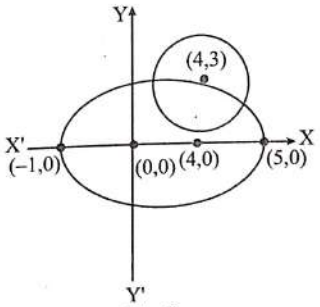
C_1 is a circle with centre (4,3) radius 2.

C_2 is an ellipse with focii (0,0), (4,0), length of major axis = 6,

and length of semi- major axis = $2\sqrt{5}$

Now, (4,1) which lie on the circle is inside the ellipse. Also, centre of the circle (4,3) is outside the ellipse.

$\therefore$ Number of point of intersection = 2



6. **(c)** $|z-i| - |z+5i| = 0$

$\Rightarrow |x + (y-1)i| = |x + (y+5)i|$

$\Rightarrow x^2 + (y-1)^2 = x^2 + (y+5)^2$

$\Rightarrow (y-1)^2 - (y+5)^2 = 0$

$\Rightarrow (2y+4)(-6) = 0$

$y = -2$

$\therefore x^2 + (-2)^2 = 4$ $\quad [\because |z| = 2]$

$x = 0$

$z \equiv (0,-2)$, check option

7. **(c)** $\overline{z_1} = i\overline{z_2}$

$z_1 = -iz_2$

Given, $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$

or $\arg(z_1) + \arg(z_2) = \pi$

$\Rightarrow \arg\left(-i\frac{z_2}{\overline{z}_2}\right) = \pi$

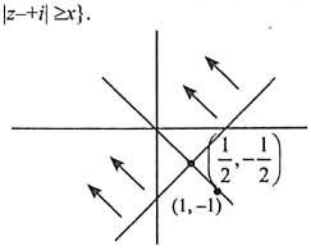
$\Rightarrow -\frac{\pi}{2} + \theta + \theta = \pi$ $\quad [\because$ Let $\arg(z_2) = \theta]$

$\Rightarrow 2\theta = \frac{3\pi}{2}$

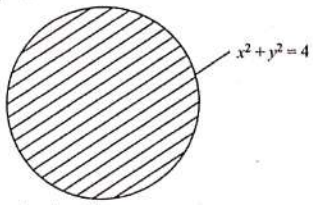
$\Rightarrow \arg(z_2) = \theta = \frac{3\pi}{4}, \arg z_1 = \frac{\pi}{4}$

$[\because \arg(z_1) + \arg(z_2) = \pi]$

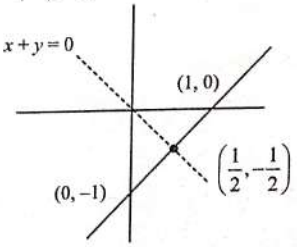
8. **(b)** S={$z=x+iy : |z-1+i| \ge |z|, |z|<2, |z+i| = |z-+i| \ge x$}.

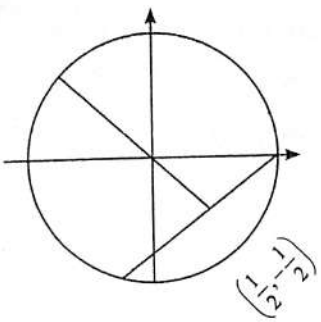


$|z| < 2$



$|z - i| = |z - 1|$





$\because \omega \in S$ and $\omega = 2x + iy$

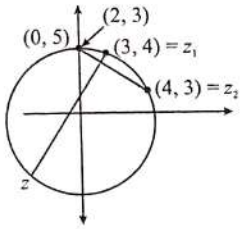
$2x < \frac{1}{2} \ \therefore x < \frac{1}{4} \Rightarrow (2x)^2 + (-2x)^2 < 4$

$\Rightarrow 4x^2 + 4x^2 < 4$

$\Rightarrow x^2 < \frac{1}{2} \Rightarrow x \in \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$\therefore x \in \left(-\frac{1}{2}, \frac{1}{4}\right]$

9. ***(b)***



$\frac{5-3}{0-4} = \frac{2}{-4} = \frac{-1}{2}$

Slop of 1 line is 2

$(y-4) = 2(x-3)$

$\Rightarrow y = 2x - 2$

Eq^n of circle is

$x^2 + y^2 = 25$

$x^2 + 4(x-1)^2 = 25$

$5x^2 - 8x - 21 = 0$

$c = 3, \ \frac{-7}{5}$

$z = \frac{-7}{5} + \left(2 \times \left(\frac{-7}{5}\right) - 2\right)$

$z = \frac{-7}{5}, -\frac{24}{5}$

$\arg(z) = \theta = \pi = \tan^{-1}\left(\frac{24}{7}\right) - \pi$

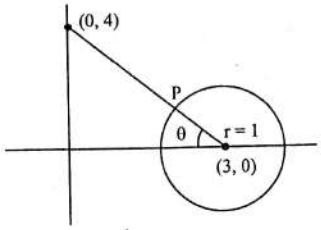
10. **[80]** Here $|z-3| < 1$

$\Rightarrow (x-3)^2 + y^2 < 1$ and

$z(4+3i) + \bar{z}(4-3i) \le 24$

$\Rightarrow 4x - 3y \le 12$

$\tan\theta = \frac{4}{3}$



$\therefore$ coordinate of P = $(3 - \cos\theta, \sin\theta)$

$= \left(3 - \frac{3}{5}, \frac{4}{5}\right)$

$\therefore \alpha + i\beta = \frac{12}{5} + \frac{4}{5}i$

$\therefore 25(\alpha + \beta) = 80$

11. **[2]** Given, $z + \bar{z} = iz^2 + z^2$

Consider $z = x + iy$

$2x = (i+1)(x^2 - y^2 + 2xyi)$

$\Rightarrow 2x = x^2 - y^2 - 2xy$ and $x^2 - y^2 + 2xy = 0$

$\Rightarrow 2x = -4xy$

$\Rightarrow x = 0$ or $y = \frac{-1}{2}$

Case I : $x = 0$

$\Rightarrow y = 0$ here $z = 0$

Case II : $y = \frac{-1}{2}$

$\Rightarrow 4x^2 - 4x - 1 = 0$

$(2x-1)^2 = 2$

$2x - 1 = \pm\sqrt{2}$

$x = \frac{1 \pm \sqrt{2}}{2}$

Here $z = \frac{1+\sqrt{2}}{2} - \frac{i}{2}$ or $z = \frac{1-\sqrt{2}}{2} - \frac{i}{2}$

Sum of squares of modulus of z

$= 0 + \frac{(1+\sqrt{2})^2 + 1}{4} + \frac{(1-\sqrt{2})^2 + 1}{4} = \frac{8}{4} = 2$

12. ***(b)*** $\frac{|z|+11}{(|z|-1)^2} \ge \frac{1}{2}$

$2|z| + 22 \ge (|z|-1)^2$

$2|z| + 22 \ge |z|^2 - 2|z| + 1$

$|z|^2 - 4|z| - 21 \le 0$

$(|z|-7)(|z|+3) \le 0$

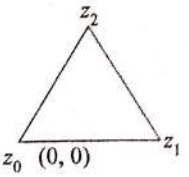
$\Rightarrow |z| \le 7$ ($\because |z|$ cannot be negative)

$\therefore \ |z|_{max} = 7$

13. **[6]** In equilateral triangle,

$z_0^2 + z_1^2 + z_2^2 = z_0z_1 + z_1z_2 + z_2z_0$

$\Rightarrow (z_1 + z_2)^2 = 3z_1 \cdot z_2 \Rightarrow z_1^2 + z_2^2 = z_1z_2$



Since z_1 and z_2 are the roots of $z^2 + az + 12 = 0$

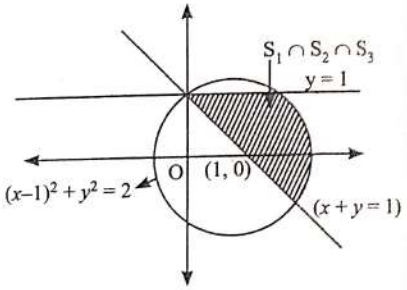
$\therefore \ z_1 + z_2 = (-a)$ and $z_1 \cdot z_2 = 12$

On substituting the values, we get

$a^2 = 3 \times 12 \Rightarrow a^2 = 36$

$\Rightarrow |a| = 6$

14. ***(a)***



Let, $z = x + iy$

$S_1 \equiv (x-1)^2 + y^2 \le 2$ ...(*i*)

$S_2 \equiv x + y \ge 1$ ...(*ii*)

$S_3 \equiv y \le 1$ ...(*iii*)

$\Rightarrow S_1 \cap S_2 \cap S_3$ has infinitely many elements.

15. ***(d)*** $a|z|^2 + \overline{\bar{\alpha}z + \alpha\bar{z}} + d = 0$

$az \cdot \bar{z} + \alpha\bar{z} + \bar{\alpha}z + d = 0$

$z.\bar{z} + \frac{\alpha}{a}\bar{z} + \frac{\bar{\alpha}}{a}z + \frac{d}{a} = 0$

Centre $= \frac{-\alpha}{a}$

Radius $= \sqrt{\left|\frac{-\alpha}{a}\right|^2 - \frac{d}{a}}$

$\therefore \left|\frac{-\alpha}{a}\right|^2 - \frac{d}{a} \ge 0$

Hence, $|\alpha|^2 - ad \ge 0, a \in R - \{0\}$

16. ***(c)*** $S_1 : |z - 3 - 2i|^2 = 8$

$|z - 3 - 2i| = (2\sqrt{2})$

$(x-3)^2 + (y-2)^2 = (2\sqrt{2})^2$

$S_2 : \text{Re}(z) \ge 5$

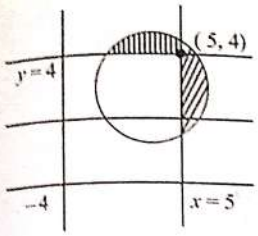
$\Rightarrow x \ge 5$

$S_3 : |z - \bar{z}| \ge 8$

$\Rightarrow |2iy| \ge 8 \Rightarrow 2|y| \ge 8$

$\Rightarrow |y| \ge 4 \Rightarrow y \ge 4$ or $y \le -4$

$\therefore n(S_1 \cap S_2 \cap S_3) = 1$



17. (*b*) We have,

$$\arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{4}$$

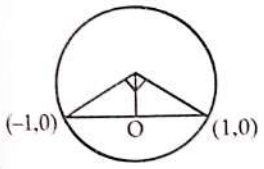
$$\Rightarrow \arg(z-1)-\arg(z+1)=\frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x-1}\right)-\tan^{-1}\left(\frac{y}{x+1}\right)=\frac{\pi}{4}$$

$$\Rightarrow \frac{\frac{y}{x-1}-\frac{y}{x+1}}{1+\frac{y^2}{x^2-1}}=1$$

$$\Rightarrow 2y=x^2+y^2-1 \Rightarrow x^2+(y-1)^2=2$$

$\Rightarrow$ Centre : (0, 1) & radius = $\sqrt{2}$



18. [98] Let $z = x + iy$

$$\arg\left(\frac{x-2+iy}{x+2+iy}\right)=\frac{\pi}{4}$$

$$\arg(x-2+iy)-\arg(x+2+iy)=\frac{\pi}{4}$$

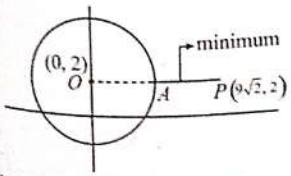
$$\Rightarrow \tan^{-1}\left(\frac{y}{x-2}\right)-\tan^{-1}\left(\frac{y}{x+2}\right)=\frac{\pi}{4}$$

$$\Rightarrow \frac{\frac{y}{x-2}-\frac{y}{x+2}}{1+\left(\frac{y}{x-2}\right)\cdot\left(\frac{y}{x+2}\right)}=\tan\frac{\pi}{4}=1$$

$$\Rightarrow \frac{xy+2y-xy+2y}{x^2-4+y^2}=1$$

$$\Rightarrow 4y=x^2-4+y^2 \Rightarrow x^2+y^2-4y-4=0$$

Locus is a circle with center (0, 2) & radius = $2\sqrt{2}$



min. value = $(AP)^2 = (OP - OA)^2$

$$=\left(9\sqrt{2}-2\sqrt{2}\right)^2$$

$$=\left(7\sqrt{2}\right)^2=98$$

19. [1] Equation of circle

$(x^2 - y^2) + 2y^2 + 2x = 0$

$x^2 + y^2 + 2x = 0$

Centre = (–1, 0)

Equation of parabola

$x^2 - 6x - y + 13 = 0$

$(x - 3)^2 = (y - 4)$

Vertex = (3, 4)

Equation of line passing through centre and vertex

$$\frac{(y-0)}{(x+1)}=\frac{4-0}{3+1}$$

$y = x + 1$

y-intercept = 1

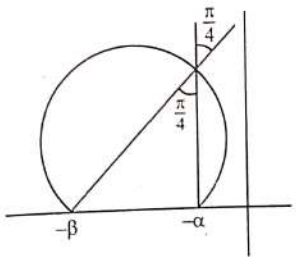
20. (*a*)

21. (*b,d*) $\arg\left(\frac{z+\alpha}{z+\beta}\right)=\frac{\pi}{4}$ implies z is on arc and $(-\alpha, 0), (-\beta, 0)$ subtend $\frac{\pi}{4}$ on z.

And z lies on $x^2 + y^2 + 5x - 3y + 4 = 0$

So put $y = 0$;

$x^2 + 5x + 4 = 0 \Rightarrow x = -1; x = -4$



Now,

$$\arg\left(\frac{z+\alpha}{z+\beta}\right)=\frac{\pi}{4} \Rightarrow z+\alpha=(z+\beta)\cdot r\cdot e^{i\frac{\pi}{4}}$$

So, $z + \beta = z + 4 \Rightarrow \beta = 4$

$z + \alpha = z + 1 \Rightarrow \alpha = 1$

22. (*d*) Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$

$$\Rightarrow x_1=\sqrt{(x_1-1)^2+y_1^2}$$

$$\Rightarrow x_1^2=x_1^2+1-2x_1+y_1^2$$

$$\Rightarrow y_1^2-2x_1+1=0 \quad \ldots(i)$$

Similarly

$$y_2^2-2x_2+1=0 \quad \ldots(ii)$$

$(i) - (ii), (y_1 - y_2)(y_1 + y_2) = 2(x_1 - x_2)$

$$\Rightarrow y_1-y_2=\frac{2(x_1-x_2)}{y_1+y_2} \Rightarrow \frac{y_1-y_2}{x_1-x_2}=\frac{2}{y_1+y_2}$$

Now,

$$\arg(z_1-z_2)=\frac{\pi}{6}$$

$$\Rightarrow \tan^{-1}\left(\frac{y_1-y_2}{x_1-x_2}\right)=\frac{\pi}{6}$$

$$\Rightarrow \frac{y_1-y_2}{x_1-x_2}=\frac{1}{\sqrt{3}} \Rightarrow \frac{2}{y_1+y_2}=\frac{1}{\sqrt{3}}$$

$y_1 + y_2 = 2\sqrt{3}$

So, $Im = (z_1 + z_2) = y_1 + y_2 = 2\sqrt{3}$

23. (*c*) $u=\frac{2(x+iy)+i}{(x+iy)-ki}=\frac{2x+i(2y+1)}{x+i(y-k)}\times\frac{x-i(y-k)}{x-i(y-k)}$

$$\therefore \mathrm{Re}(u)=\frac{2x^2+(y-k)(2y+1)}{x^2+(y-k)^2}$$

and $$\mathrm{Im}(u)=\frac{-2x(y-k)+x(2y+1)}{x^2+(y-k)^2}$$

Also Re(*u*) + Im(*u*) = 1

$\Rightarrow 2x^2 + 2y^2 - 2ky + y - k - 2xy + 2kx + 2xy + x$

$\Rightarrow x^2 + y^2 + k^2 - 2ky$

Let y_1 and y_2 are roots of equations if $x = 0$

$\Rightarrow y^2 + y - k(k + 1) = 0 < \frac{y_1}{y_2}$

Given $PQ = 5 \Rightarrow |y_1 - y_2| = 5$

$\Rightarrow 4k^2 + 4k - 24 = 0 \Rightarrow k = 2$ or -3

as $k > 0 \quad k = 2$

24. (*a*) Let $z = x + iy$

$\because z^2 = i|z|^2$

$x^2 - y^2 + 2ixy = i(x^2 + y^2)$

$\Rightarrow x^2 - y^2 = 0$ and $2xy = x^2 + y^2$

$\Rightarrow (x - y)(x + y) = 0$ and $(x - y)^2 = 0$

$\Rightarrow (x - y) = 0$

$\Rightarrow x = y$

25. (*b*) $z = x + iy$

$\sqrt{x^2 + y^2} + x + iy = 3 + i$

$\Rightarrow y = 1$

and $\sqrt{x^2 + 1} + x = 3$

$\Rightarrow x^2 + 1 = 9 - 6x + x^2$

$\Rightarrow x = \frac{4}{3} \Rightarrow |z| = \sqrt{x^2 + y^2} = \frac{5}{3}$

26. **(a)** $|z_1| = 9, |z_2 - (3+4i)| = 4$

$C_1(0, 0)$, (radius) $r_1 = 9$

$C_2(3,4)$, radius $r_2 = 4$

$C_1C_2 = |r_1 - r_2| = 5$

$\therefore$ Circle touches internally

$\therefore |z_1 - z_2|_{min} = 0$

27. **(a)** $z = \dfrac{\alpha + i}{\alpha - i}$

$|z| = \dfrac{\sqrt{\alpha^2+1}}{\sqrt{\alpha^2-1}} = 1 \Rightarrow |z| = 1$

28. **(a)** $\dfrac{z-\alpha}{z+\alpha} = -\left(\dfrac{\bar{z}-\alpha}{\bar{z}+\alpha}\right)$

($z + \bar{z} = 0$ when z is purely imaginary)

$\Rightarrow z\bar{z} + z\alpha - \alpha\bar{z} - \alpha^2 = -z\bar{z} + \alpha z - \alpha\bar{z} + \alpha^2$

$\Rightarrow \alpha^2 = |z|^2$

$\Rightarrow \alpha = \pm 2$

Rotation Theorem

29. **(d)** $i \rightarrow$ Rotate complex number to 90° angle.

$w_1 = z_1 i = (5 + 4i)i = -4 + 5i$ ...(i)

$w_2 = (z_2)(-i) = (3 + 5i)(-i) = 5 - 3i$...(ii)

$w_1 - w_2 = -9 + 8i$

Principal argument $= \pi - \tan^{-1}\left(\dfrac{8}{9}\right)$

[$\because$ Complex in 2nd quadrant]

30. **(d)** $\dfrac{z_2 - 0}{(1+2i) - 0} = \dfrac{|OB|}{OA} e^{\frac{i\pi}{4}}$

$\Rightarrow \dfrac{z_2}{1+2i} = \sqrt{2} e^{\frac{i\pi}{4}}$

OR $z_2 = 1(1 + 2i)(1 + i) = -1 + 3i$

$\arg z_2 = \pi - \tan^{-1} 3$

$|z_2| = \sqrt{10}$

$z_1 - 2z_2 = (1 + 2i) + 2 - 6i = 3 - 4i$

$\arg(z_1 - 2z_2) = -\tan^{-1}\dfrac{4}{3}$

$|2z_1 - z_2| = |2 + 4i + 1 - 3i| = |3 + i|$

$= \sqrt{10}$

By Aditya Mohod

username - maje_kabhi_dekha_nahi

By Aditya Mohad
username - muje_kabhi_dekha_nahi

CHAPTER 16

Limits and Derivatives

Prarambh (Topicwise)

1. (*d*) To find L.H.L. at $x = 1$. i.e.,

$\lim_{x\to1^-} f(x) = \lim_{h\to0} f(1-h) = \lim_{h\to0}(1-h)^2$

$\lim_{h\to0}(1+h^2-2h) = 1$ i.e.,

$= \lim_{x\to1^-} f(x) = 1$ ... (*i*)

Now find R.H.L. at $x = 1$ i.e.,

$\lim_{x\to1^+} f(x) = \lim_{h\to0} f(1+h) = 1$ i.e.,

$\lim_{x\to1^+} f(x) = 1$ ...(*ii*)

From (*i*) and (*ii*), L.H.L. = R.H.L.

$\Rightarrow \lim_{x\to1} f(x) = 1$.

2. (*c*) L.H.L. $= \lim_{x\to2^-} \frac{|x-2|}{x-2} = \lim_{h\to0} \frac{|2-h-2|}{2-h-2}$

$= \lim_{h\to0} \frac{h}{-h} = -1$ ... (*i*)

and, R.H.L. $= \lim_{x\to2^+} \frac{|x-2|}{x-2} = \lim_{h\to0} \frac{|2+h-2|}{2+h-2}$

$= \lim_{h\to0} \frac{h}{h} = 1$ ... (*ii*)

From (*i*) and (*ii*) L.H.L. $\neq$ R.H.L. *i.e.* $\lim_{x\to2} \frac{|x-2|}{x-2}$ does not exist.

3. (*c*) $\lim_{x\to3+} f(x) = 5-3 = 2$ and

$\lim_{x\to3-} f(x) = \frac{2}{5-3} = 1$

4. (*d*) LHL $= \lim_{x\to1-0} f(x) = \lim_{h\to0} f(1-h) = \lim_{h\to0} 3(1-h)$

$= \lim_{h\to0}(3-3h) = 3-3.0 = 3$

R.H.L. $= \lim_{x\to1+0} f(x) = \lim_{h\to0} f(1+h)$

$= \lim_{h\to0}[5-3(1+h)]$

$= \lim_{h\to0}(2-3h) = 2-3.0 = 2$

Hence $\lim_{x\to1} f(x)$ does not exists.

5. (*d*) $\because \lim_{x\to0-} \frac{|x|}{x} = -1$ and $\lim_{x\to0+} \frac{|x|}{x} = 1$, ,

hence limit does not exists.

6. (*a*) $\lim_{x\to1}(3x^2+4x+5) = 3(1)^2 + 4(1) + 5 = 12.$

7. (*c*) $\lim_{x\to2} \frac{3^{x/2}-3}{(3^{x/2})^2-(3)^2}$

$= \lim_{x\to2} \frac{(3^{x/2}-3)}{(3^{x/2}-3)(3^{x/2}+3)} = \frac{1}{6}.$

8. (*d*) $\lim_{x\to3} \frac{x-3}{\sqrt{x-2}-\sqrt{4-x}}$

$= \lim_{x\to3} \frac{(x-3)(\sqrt{x-2}+\sqrt{4-x})}{(\sqrt{x-2})^2-(\sqrt{4-x})^2}$

$= \lim_{x\to3} \frac{(x-3)(\sqrt{x-2}+\sqrt{4-x})}{(2x-6)}$

$= \lim_{x\to3} \frac{\sqrt{x-2}+\sqrt{4-x}}{2} = \frac{1+1}{2} = 1.$

9. (*d*) Let $y = \lim_{x\to3} \frac{x^3-x^2-19}{x-3}$

$= \lim_{x\to3}(x^2+2x+6) = 9+6+6 = 21$

10. (*a*) $\lim_{x\to2} \frac{x^3-8}{x^2-4} = \lim_{x\to2} \frac{(x^2+2x+4)(x-2)}{(x+2)(x-2)}$

$= \lim_{x\to2} \frac{x^2+2x+4}{x+2} = \frac{4+4+4}{2+2} = 3.$

11. (*c*) $\lim_{x\to0}\left(\frac{x}{\sqrt{1+x}-\sqrt{1-x}}\right)$

$= \lim_{x\to0}\left(\frac{x}{\sqrt{1+x}-\sqrt{1-x}} \times \frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$

$= \lim_{x\to0}\left(\frac{x(\sqrt{1+x}+\sqrt{1-x})}{1+x-1+x}\right)$

$= \lim_{x\to0}\left(\frac{(\sqrt{1+x}+\sqrt{1-x})}{2}\right)$

$= \frac{2}{2} = 1$

12 (*b*) $\lim_{x\to a} \frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$

$= \lim_{x\to a}\left(\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}\right)$

$\times\left(\frac{\sqrt{a+2x}+\sqrt{3x}}{\sqrt{a+2x}+\sqrt{3x}}\right)\times\left(\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{3a+x}+2\sqrt{x}}\right)$

$= \lim_{x\to a}\left\{\frac{\sqrt{3a+x}+2\sqrt{x}}{3(\sqrt{a+2x}+\sqrt{3x})}\right\} = \frac{2}{3\sqrt{3}}$

13. (*c*) We have

$f(x)+g(x)+h(x)$

$= \frac{x^2-4x+17-4x-2}{x^2+x-12}$

$= \frac{x^2-8x+15}{x^2+x-12} = \frac{(x-3)(x-5)}{(x-3)(x+4)}$

$\therefore \lim_{x\to3}[f(x)+g(x)+h(x)]$

$= \lim_{x\to3} \frac{(x-3)(x-5)}{(x-3)(x+4)} = -\frac{2}{7}.$

14. (*a*) Apply the L-Hospital's rule,

$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$.

15. (*a*) $\lim_{x\to0} \frac{x^3}{\sin x^2} = \lim_{x\to0} \frac{x^2}{\sin x^2} \cdot x$

$= \left(\lim_{x\to0} \frac{x^2}{\sin x^2}\right)\left(\lim_{x\to0} x\right) = 1.0 = 0.$

16. (*c*) $\lim_{x\to0} \frac{\sin 3x + \sin x}{x} = \lim_{x\to0} \frac{\sin 3x}{x} + \lim_{x\to0} \frac{\sin x}{x}$

$= \lim_{x\to0} \frac{\sin 3x}{3x} \cdot 3 + \lim_{x\to0} \frac{\sin x}{x} = 1.3+1 = 4.$

17. (*c*) $\lim_{x\to1}(1-x)\tan\left(\frac{\pi x}{2}\right)$, Put $1-x = y$

$\Rightarrow$ as $x \to 1, y \to 0$

Thus $\lim_{y\to0} y\tan\frac{\pi(1-y)}{2} = \lim_{y\to0} \frac{2}{\pi} \cdot \frac{\left(\frac{\pi y}{2}\right)}{\tan\left(\frac{\pi y}{2}\right)}$

$= \frac{2}{\pi} \times 1 = \frac{2}{\pi}.$

18. (*d*) $f(1+) = \lim_{h\to0} f(1+h)$

$= \lim_{h\to0} \frac{\sqrt{1-\cos 2h}}{h}$

$= \lim_{h\to0} \sqrt{2}\frac{\sinh}{h} = \sqrt{2}$

$$f(1-)=\lim_{h\to 0} f(1-h)=\lim_{h\to 0}\frac{\sqrt{1-\cos(-2h)}}{-h}$$

$$=\lim_{h\to 0}\sqrt{2}\frac{\sin h}{-h}=-\sqrt{2}.$$

$\therefore$ limit does not exist because left hand limit is not equal to right hand limit.

19. (*a*) $\lim_{x\to 0}\frac{2\sin^2 x\sin 5x}{x^2}\frac{3x}{\sin 3x}\frac{5x}{3x}\frac{}{5x}$

$$=\lim_{x\to 0}\frac{2\sin^2 x}{x^2}\cdot\frac{3x}{\sin 3x}\cdot\frac{\sin 5x}{5x}\cdot\frac{5x}{3x}=2\cdot\frac{5}{3}=\frac{10}{3}.$$

20. (*a*)

$$\lim_{\theta\to\pi/2}\frac{1-\sin\theta}{\cos\theta}$$

$$=\lim_{\theta\to\pi/2}\frac{\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right)^2}{\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right)\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)}=0$$

21. (*d*) $\lim_{x\to 0}\frac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}=\lim_{x\to 0}\frac{|\sin x|}{x}$

So, $\lim_{x\to 0+}\frac{|\sin x|}{x}=1$ and $\lim_{x\to 0-}\frac{|\sin x|}{x}=-1$

Hence limit does not exist.

22. (*c*) Using L-Hospital's rule,

$$\lim_{\theta\to\frac{\pi}{2}}\frac{-1}{-\text{cosec}^2\theta}=1.$$

23. (*c*) $y=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}$

$$\Rightarrow\frac{dy}{dx}=0+1+x+\frac{x^2}{2!}+\cdots+\frac{x^{n-1}}{(n-1)!}$$

$$\Rightarrow\frac{dy}{dx}+\frac{x^n}{n!}=1+x+x\frac{x^2}{2!}+\cdots+\frac{x^n}{n!}$$

$$\Rightarrow\frac{dy}{dx}=y-\frac{x^n}{n!}$$

24. (*b*) $f(x)=\sqrt{1+\cos^2(x^2)}$

$$\Rightarrow f'(x)=\frac{1}{2\sqrt{1+\cos^2(x)^2}}(2\cos x^2)$$

$$(-\sin x^2)(2x)$$

$$\Rightarrow f'(x)=\frac{-x\sin 2x^2}{\sqrt{1+\cos^2(x)^2}}$$

$$\Rightarrow f'\left(\frac{\sqrt{\pi}}{2}\right)=\frac{-\frac{\sqrt{\pi}}{2}\sin\frac{2\pi}{4}}{\sqrt{1+\cos^2\frac{\pi}{4}}}=\frac{\frac{\sqrt{\pi}}{2}1}{\sqrt{\frac{3}{2}}}$$

$$\therefore f'\left(\frac{\sqrt{\pi}}{2}\right)=-\sqrt{\frac{\pi}{6}}$$

25. (*d*) $y=a\sin x+b\cos x$

Differentating with respect to x, we get

$$\frac{dy}{dx}=a\cos x-b\sin x$$

Now, $\left(\frac{dy}{dx}\right)^2=(a\cos x-b\sin x^2)$

$a^2\cos^2 x+b^2\sin^2 x-2ab\sin x\cos x$, and

$y^2=(a\sin x+b\cos x)^2$

$=a^2\sin^2 x+b^2\cos^2 x+2ab\sin x\cos x$

So, $\left(\frac{dy}{dx}\right)^2+y^2=a^2(\sin^2 x+\cos^2 x)$

$$+b^2(\sin^2 x+\cos^2 x)$$

$$\Rightarrow\left(\frac{dy}{dx}\right)^2+y^2=(a^2+b^2)=\text{constant}.$$

26. (*c*) We have $y=\sqrt{\frac{1-x}{1+x}}$

Differetiating *w.r.t.* x, we get

$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{1-x}{1+x}\right)^{1/2-1}\frac{d}{dx}\left(\frac{1-x}{1+x}\right)$$

$$=\frac{1}{2}\sqrt{\frac{1+x}{1-x}}\times\frac{(1+x)-(1-x)(1)}{(1+x)^2}$$

$$=-\sqrt{\frac{1+x}{1-x}}\frac{1}{(1+x)^2}$$

$$\Rightarrow(1-x^2)\frac{dy}{dx}=-\sqrt{\frac{1+x}{1-x}}\frac{1}{(1+x)^2}(1-x^2)$$

$$\Rightarrow(1-x)^2\frac{dy}{dx}=-\sqrt{\frac{1-x}{1+x}}$$

$$\Rightarrow(1-x)^2\frac{dy}{dx}=-y$$

$$\Rightarrow(1-x^2)\frac{dy}{dx}+y=0$$

27. (*c*) In neighbourhood of

$x=\frac{2\pi}{3}$, $|\cos x|=-\cos x$ and $|\sin x|$

$=\sin x$

$y=-\cos x+\sin x$

$$\Rightarrow\frac{dy}{dx}=\sin x+\cos x$$

$\Rightarrow$ At $x=\frac{2\pi}{3}$, $\frac{dy}{dx}=\sin\frac{2\pi}{3}+\cos\frac{2\pi}{3}=\frac{\sqrt{3}-1}{2}$.

28. (*b*) Given,

$$f(x)=1+x+\frac{x^2}{2}+\ldots+\frac{x^{100}}{100}$$

$$\therefore f'(x)=0+1+2\times\frac{x}{2}+\ldots+100\frac{x^{99}}{100}$$

$f'(x)=1+x+x^2+\ldots+x^{(n)}$

Now, $f(1)=1+1+1+\ldots+1$(100 time)

$=100$

29. (*d*) Given,

$$f(x)=\frac{x^n-a^n}{x-a}$$

$$\therefore f'(x)=\frac{(x-a)nx^{n-1}-(x^n-a^n)(1)}{(x-a)^2}$$

[by quotient rule]

$$\Rightarrow f'(x)=\frac{nx^{n-1}(x-a)-x^n+a^n}{(x-a)^2}$$

Now, $\Rightarrow f'(a)=\frac{na^{n-1}(0)-a^n+a^n}{(x-a)^2}$

$$\Rightarrow f'(a)=\frac{0}{0}$$

So, $f'(a)$ does not exist,

Since, $f(x)$ is not defined at x = a.

Hence, $f'(x)$ at $x=a$ does not exist.

30. (*a*) Given, $f(x)=x^{100}+x^{99}+\ldots+x+1$

$\because f'(x)=100x^{99}+99x^{98}+\ldots+1+0$

$=100x^{99}+99x^{98}+\ldots+1$

Now, $f'(1)=100+99+\ldots+1$

$$=\frac{100}{2}[2\times 100+(100-1)(-1)]$$

$=50[200-99]$

$$\left[\because S_n=\frac{n}{2}\{2a+(n-1)d\}\right]$$

$=50\times 101=5050$

31. Given, $\lim_{x\to 1/2}\frac{4x^2-1}{2x-1}=\lim_{x\to 1/2}\frac{(2x)^2-(1)^2}{2x-1}$

$$=\lim_{x\to 1/2}\frac{(2x+1)(2x-1)}{(2x-1)}=\lim_{x\to 1/2}(2x+1)$$

$$=2\times\frac{1}{2}+1=1+1=2$$

32. Given, $\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}$

$$=\lim_{h\to 0}\frac{(x+h)^{1/2}-(x)^{1/2}}{x+h-x}$$

$$=\lim_{h\to 0}\frac{(x+h)^{1/2}-(x)^{1/2}}{(x+h)-x}$$

$$\left[\because\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\right]$$

$$=\frac{1}{2}x^{\frac{1}{2}-1}=\frac{1}{2}x^{-1/2}\left[\because h\to 0\Rightarrow x+h\to x\right]$$

$$=\frac{1}{2\sqrt{x}}$$

33. Given, $\lim_{x\to a}\frac{(2+x)^{5/2}-(a+2)^{5/2}}{x-a}$

$$=\lim_{x\to a}\frac{(2+x)^{5/2}-(a+2)^{5/2}}{(2+x)-(a+2)}$$

$$=\frac{5}{2}(a+2)^{\frac{5}{2}-1}\left[\because\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\right]$$

$$=\frac{5}{2}(a+2)^{\frac{3}{2}}\quad[\because x\to a\Rightarrow x+2\to a+2]$$

34. Given, $\lim_{x\to1}\frac{x^4-\sqrt{x}}{\sqrt{x}-1}=\lim_{x\to1}\frac{\sqrt{x}\left[(x)^{7/2}-1\right]}{\sqrt{x}-1}$

$=\lim_{x\to1}\frac{(x)^{7/2}-1}{\sqrt{x}-1}\cdot\lim_{x\to1}\sqrt{x}$

$\left[\because \lim_{x\to a}f(x)\cdot g(x)=\lim_{x\to a}f(x)\cdot\lim_{x\to a}g(x)\right]$

$=\lim_{x\to1}\frac{\frac{x^{7/2}-1}{x-1}}{\frac{(x)^{1/2}-1}{x-1}}\cdot1$

$=\frac{\lim_{x\to1}\frac{x^{7/2}-1}{x-1}}{\lim_{x\to1}\frac{(x)^{1/2}-1}{x-1}}\left[\because \lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}\right]$

$=\frac{\frac{7}{2}(1)^{\frac{7}{2}-1}}{\frac{1}{2}(1)^{\frac{1}{2}-1}}=\frac{\frac{7}{2}}{\frac{1}{2}}=7$

35. Given,

$\lim_{x\to0}\frac{\sqrt{1+x^3}-\sqrt{1-x^3}}{x^2}$

$=\lim_{x\to0}\frac{\sqrt{1+x^3}-\sqrt{1-x^3}}{x^2}\cdot\frac{\sqrt{1+x^3}+\sqrt{1-x^3}}{\sqrt{1+x^3}+\sqrt{1-x^3}}$

$=\lim_{x\to0}\frac{(1+x^3)-(1-x^3)}{x^2(\sqrt{1+x^3}+\sqrt{1-x^3})}$

$=\lim_{x\to0}\frac{1+x^3-1+x^3}{x^2(\sqrt{1+x^3}+\sqrt{1-x^3})}$

$=\lim_{x\to0}\frac{2x^3}{x^2(\sqrt{1+x^3}+\sqrt{1-x^3})}$

$=\lim_{x\to0}\frac{2x}{(\sqrt{1+x^3}+\sqrt{1-x^3})}$

$=0$

36. Given,

$=\lim_{x\to-3}\frac{\frac{x^3-(-3)^3}{x-(-3)}}{\frac{x^5-(-3)^5}{x-(-3)}}=\frac{\lim_{x\to-3}\frac{x^3-(-3)^3}{x-(-3)}}{\lim_{x\to-3}\frac{x^5-(-3)^5}{x-(-3)}}$

$\left[\because \lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}\right]$

$=\frac{3(-3)^{3-1}}{5(-3)^{5-1}}=\frac{3}{5}\frac{(-3)^2}{(-3)^4}$

$\left[\because \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\right]$

$=\frac{3}{5(-3)^2}=\frac{3}{45}=\frac{1}{15}$

37. Given,

$\lim_{x\to0}\frac{1-\cos mx}{1-\cos nx}=\lim_{x\to0}\frac{1-1+2\sin^2\frac{mx}{2}}{1-1+2\sin^2\frac{nx}{2}}$

$\left[\because \cos mx=1-2\sin^2\frac{mx}{2}\right.$

$\left.\text{and } \sin nx=1-2\sin^2\frac{nx}{2}\right]$

$=\lim_{x\to0}\frac{\sin^2\frac{mx}{2}}{\sin^2\frac{nx}{2}}$

$=\lim_{x\to0}\frac{\frac{\sin^2\frac{mx}{2}}{\left(\frac{mx}{2}\right)^2}\cdot\left(\frac{mx}{2}\right)^2}{\frac{\sin^2\frac{nx}{2}}{\left(\frac{nx}{2}\right)^2}\cdot\left(\frac{nx}{2}\right)^2}$

$=\frac{m^2}{n^2}\cdot\frac{\lim_{x\to0}\left(\frac{\sin\frac{mx}{2}}{\frac{mx}{2}}\right)^2}{\lim_{x\to0}\left(\frac{\sin\frac{nx}{2}}{\frac{nx}{2}}\right)^2}=\frac{m^2}{n^2}$

$\left[\because \lim_{x\to0}\frac{\sin x}{x}=1\right]$

$[\because x\to0\Rightarrow kx\to0]$

38. Given,

$\lim_{x\to\pi/6}\frac{\sqrt{3}\sin x-\cos x}{x-\frac{\pi}{6}}=\lim_{x\to\pi/6}\frac{2\left(\frac{\sqrt{3}}{2}\sin x-\frac{1}{2}\cos x\right)}{\left(x-\frac{\pi}{6}\right)}$

$[\because \sin A\cos B-\cos A\sin B=\sin(A-B)]$

$=2\left[\because \lim_{x\to0}\frac{\sin x}{x}=1\right]$

$\left[\because x\to\frac{\pi}{6}\Rightarrow\left(x-\frac{\pi}{6}\right)\to0\right]$

39. Given,

$\lim_{x\to1}\frac{x^4-1}{x-1}=\lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$

$\Rightarrow\ 4(1)^{4-1}=\lim_{x\to k}\frac{\frac{x^3-k^3}{x-k}}{\frac{x^2-k^2}{x-k}}$

$\left[\because \lim\frac{x^n-a^n}{x-a}=na^{n-1}\right]\Rightarrow 4=\frac{\lim_{x\to k}\frac{x^3-k^3}{x-k}}{\lim_{x\to k}\frac{x^2-k^2}{x-k}}$ I

$\left[\because \lim_{x\to a}\frac{f(x)}{g(x)}=\frac{\lim_{x\to a}f(x)}{\lim_{x\to a}g(x)}\right]$

$\Rightarrow 4=\frac{3k^2}{2k}\Rightarrow 4=\frac{3}{2}k$

$\therefore\ k=\frac{4\times2}{3}=\frac{8}{3}$

Differentiate each of the functions w.r.t. in following questions

40. Given,

$\lim_{x\to\pi}\frac{1-\sin\frac{x}{2}}{\cos\frac{x}{2}\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)}$

$=\lim_{x\to\pi}\frac{\cos^2\frac{x}{4}+\sin^2\frac{x}{4}-2\cdot\sin\frac{x}{4}\cdot\cos\frac{x}{4}}{\cos\frac{x}{2}\cdot\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)}$

$\left[\because \sin^2\theta+\cos^2\theta=1\sin2\theta=2\sin\theta\cos\theta\right]$

$=\lim_{x\to\pi}\frac{\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)^2}{\left(\cos^2\frac{x}{4}-\sin^2\frac{x}{4}\right)\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)}$

$\left[\because \cos^2 2\theta=\cos^2\theta-\sin^2\theta\right]$

$=\lim_{x\to\pi}\frac{\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)}{\left(\cos\frac{x}{4}+\sin\frac{x}{4}\right)\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)}$

$[\because\ a^2-b^2=(a+b)(a-b]$

$\lim_{x\to\pi}\frac{1}{\cos\frac{x}{4}+\sin\frac{x}{4}}=\frac{1}{\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}}=\frac{\sqrt{2}}{2}=\frac{1}{\sqrt{2}}$

41. Given, $\lim_{x\to x/4}\frac{|x-4|}{x-4}$

LHL $=\lim_{x\to\frac{\pi}{4}}\frac{-(x-4)}{x-4}=-1$

$[\because |x-4|=-(x-4), x<4]$

RHL $=\lim_{x\to\frac{\pi}{4}}\frac{(x-4)}{x-4}=1$

$[\because |x-4|=(x-4), x>4]$

$\therefore$ LHL $\neq$ RHL

So, limit does not exist.

42. Given,

$f(x)=\frac{\tan x}{x-\pi}=\lim_{x\to\pi}\frac{\tan x}{x-\pi}$

$$= \lim_{x\to 0} \frac{-\tan(\pi - x)}{-(\pi - x)}$$

$$\left[\because \lim_{x\to 0} \frac{\tan x}{x} = 1\right] = 1$$

43. Given, $\lim_{x\to 0}\left(\sin mx \cot\frac{x}{\sqrt{3}}\right) = 2$

$$\Rightarrow \lim_{x\to 0} \frac{\sin mx}{mx} \cdot mx \cdot \frac{1}{\tan\frac{x}{\sqrt{3}}} = 2$$

$$\Rightarrow \lim_{x\to 0} \frac{\sin mx}{mx} \cdot mx \cdot \frac{\frac{x}{\sqrt{3}}}{\tan\frac{x}{\sqrt{3}}} \cdot \frac{1}{\frac{x}{\sqrt{3}}} = 2$$

$$\Rightarrow \lim_{x\to 0} \frac{\sin mx}{mx} \cdot \lim_{x\to 0} \frac{\frac{x}{\sqrt{3}}}{\tan\frac{x}{\sqrt{3}}} \cdot \lim_{x\to 0} \frac{mx}{\frac{x}{\sqrt{3}}} = 2$$

$$\sqrt{3}m = 2$$

$$\therefore m = \frac{2\sqrt{3}}{3}$$

44. $$\frac{d}{dx}\left(\frac{x^4 + x^3 + x^2 + 1}{x}\right)$$

$$= \frac{d}{dx}\left(x^3 + x^2 + x + \frac{1}{x}\right)$$

$$= \frac{d}{dx}x^3 + \frac{d}{dx}x^2 + \frac{d}{dx}x + \frac{d}{dx}\left(\frac{1}{x}\right)$$

$$= 3x^2 + 2x + 1 + \left(-\frac{1}{x^2}\right)$$

$$= 3x^2 + 2x + 1 - \frac{1}{x^2} = \frac{3x^4 + 2x^3 + x^2 - 1}{x^2}$$

45. Let $y = \left(x + \frac{1}{x}\right)^3$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}\left(x + \frac{1}{x}\right)^3 = 3\left(x + \frac{1}{x}\right)^{3-1} \frac{d}{dx}\left(x + \frac{1}{x}\right)$$

[by chain rule]

$$= 3\left(x + \frac{1}{x}\right)^2 \left(1 - \frac{1}{x^2}\right)$$

$$= 3x^2 - \frac{3}{x^2} - \frac{3}{x^4} + 3$$

46. Let $y = (3x + 5)(1 + \tan x)$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}[(3x+5)(1+\tan x)]$$

$$= (3x+5)\frac{d}{dx}(1+\tan x) + (1+\tan x)\frac{d}{dx}(3x+5)$$

[by product rule]

$$= (3x + 5)(\sec^2 x) + (1 + \tan x)\cdot 3$$

$$= (3x + 5)\sec^2 x + 3(1 + \tan x)$$

$$= 3x\sec^2 x + 5\sec^2 x + 3\tan x + 3$$

47. Let

$$y = \frac{x^5 - \cos x}{\sin x}$$

$$\therefore \frac{dy}{x} = \frac{\sin x \frac{d}{dx}(x^5 - \cos x) - (x^5 - \cos x)\frac{d}{dx}\sin x}{(\sin x)^2}$$

[by quotient rule]

$$= \frac{\sin x(5x^4 + \sin x) - (x^5 - \cos x)\cos x}{\sin^2 x}$$

$$= \frac{5x^4 \sin x + \sin^2 x - x^5 \cos x + \cos^2 x}{\sin^2 x}$$

$$= \frac{5x^4 \sin x - x^5 \cos x + \sin^2 x + \cos^2 x}{\sin^2 x}$$

$$= \frac{5x^4 \sin x - x^5 \cos x + 1}{\sin^2 x}$$

48. Let $y = \frac{x^2 \cos\frac{\pi}{4}}{\sin x} = \frac{\frac{x^2}{\sqrt{2}}}{\sin x}$

$$y = \frac{1}{\sqrt{2}} \cdot \frac{x^2}{\sin x}$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{2}}\left[\frac{\sin x \cdot \frac{d}{dx}x^2 - x^2 \frac{d}{dx}\sin x}{\sin^2 x}\right]$$

[by quotient rule]

$$= \frac{1}{\sqrt{2}}\left[\frac{\sin x \cdot 2x - x^2 \cdot \cos x}{\sin^2 x}\right]$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{2x\sin x - x^2 \cos x}{\sin^2 x}$$

$$= \frac{x}{\sqrt{2}}[2\operatorname{cosec} x - x\cot x \operatorname{cosec} x]$$

$$= \frac{x}{\sqrt{2}}\operatorname{cosec} x[2 - x\cot x]$$

49. Let $y = (ax^2 + \cot x)(p + q\cos x)$

$$\therefore \frac{dy}{dx} = (ax^2 + \cot x)\frac{d}{dx}(p + q\cos x) + (p + q\cos x)\frac{d}{dx}(ax^2 + \cot x)$$

[by product rule]

$$= (ax^2 + \cot x)(-q\sin x) + (p + q\cos x)(2ax - \operatorname{cosec}^2 x)$$

$$= -q\sin x(ax^2 + \cot x) + (p + q\cos x)(2ax - \operatorname{cosec}^2 x)$$

50. Let $y = (\sin x + \cos x)^2$

$$\therefore \frac{dy}{dx} = 2(\sin x + \cos x)(\cos x - \sin x)$$

[by chain rule]

$$= 2(\cos^2 x - \sin^2 x) = 2\cos 2x \quad [\because \cos 2x = \cos^2 x - \sin^2 x]$$

51. Let $y = (2x - 7)^2 (3x + 5)^3$

$$\frac{dy}{dx} = (2x-7)^2 \frac{d}{dx}(3x+5)^3 + (3x+5)^3 \frac{d}{dx}(2x-7)^2$$

[by product rule]

$$= (2x - 7)^2 (3)(3x + 5)^2 (3) + (3x + 5)^3 2(2x - 7)(2)$$

[by chain rule]

$$= 9(2x - 7)^2 (3x + 5)^2 + 4(3x + 5)^3 (2x - 7)$$

$$= (2x - 7)(3x + 5)^2 [9(2x - 7) + 4(3x + 5)]$$

$$= (2x - 7)(3x + 5)^2 (18x - 63 + 12x + 20)$$

$$= (2x - 7)(3x + 5)^2 (30x - 43)$$

52. Let $y = \sin^3 x \cos^3 x$

$$\therefore \frac{dy}{dx} = \sin^3 x \frac{d}{dx}\cos^3 x + \cos^3 x \frac{d}{dx}\sin^3 x$$

[by product rule]

$$= \sin^3 x \cdot 3\cos^2 x(-\sin x) + \cos^3 x \cdot 3\sin^2 x \cos x$$ [by chain rule]

$$= -3\cos^2 x \sin^4 x + 3\sin^2 x \cos^4 x$$

$$= 3\sin^2 x \cos^2 x(\cos^2 x - \sin^2 x) = 3\sin^2 x \cos^2 x \cos 2x$$

$$= \frac{3}{4}(2\sin x \cos x)^2 \cos 2x$$

$$= \frac{3}{4}\sin^2 2x \cos 2x$$

53. Let $y = \frac{1}{ax^2 + bx + c} = (ax^2 + bx + c)^{-1}$

$$\therefore \frac{dy}{dx} = -(ax^2 + bx + c)^{-2}(2ax + b)$$

[by chain rule]

$$= \frac{-(2ax + b)}{(ax^2 + bx + c)^2}$$

54. Given, $y = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$

$$\therefore \frac{dy}{dx} = 0 + 1 + \frac{2x}{2} + \frac{3x^2}{6} + \frac{4x^3}{4!}$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \ldots$$

$$1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

$$y = 1$$

55. $$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h\to 0} \frac{2\cos\left(\frac{2x+h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} = \cos x$$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER 17

Statistics

By Aditya Mahod
username - muje_kabhi_dekha_nahi

Prarambh (Topicwise)

1. (c) $\bar{x} = \dfrac{3+4+x+7+10}{5} = 6$

$x + 24 = 30$

$\therefore\ x = 6$

2. (b) $\bar{x} = \dfrac{1+2+4.....+2^n}{n+1}$

$\therefore\ \bar{x} = \dfrac{\left(\dfrac{2^{n+1}-1}{2-1}\right)}{n+1} = \dfrac{2^{n+1}-1}{n+1}$

3. (b) $\bar{x} = \dfrac{(a)+(a+d)+(a+2d)...n \text{ terms}}{n}$

$\therefore\ \bar{x} = \dfrac{na + d(1+2+...+(n-1))}{n}$

$\bar{x} = a + \dfrac{n(n-1)}{2} \cdot \dfrac{d}{n}$

$\therefore\ \bar{x} = a + \dfrac{(n-1)}{2}d$

4. (a) $\bar{x} = \dfrac{1^2+2^2+...+n^2}{n}$

$\therefore\ \bar{x} = \dfrac{n(n+1)(2n+1)}{6n}$

$\therefore\ \bar{x} = \dfrac{(n+1)(2n+1)}{6} = \dfrac{46n}{11}$ (given)

$\Rightarrow 22n^2 + 33n + 11 = 276\,n$

$\therefore 22n^2 - 243n + 11 = 0$

$\therefore\ n^2 - \left(\dfrac{243}{22}\right)n + \dfrac{1}{2} = 0$

$\Rightarrow (n-11)\left(n - \dfrac{1}{22}\right) = 0$

but n is integer

$\therefore n = 11$

5. (a) $\bar{x} = \dfrac{\Sigma f_i x_i}{\Sigma f_i}$

$= \dfrac{4\times67 + 3\times70 + 2\times72 + 2\times73 + 1\times75}{4+3+2+2+1}$

$= \dfrac{268+210+144+146+75}{12}$

$= \dfrac{843}{12} = 70.25$ kg

6. (a) 100 workers

$n_1 = 60; \bar{x}_1 = 40 \qquad n_2 = 40;\ \bar{x}_2 = ?$

60 workers + 40 workers

$n = 100;\ \bar{x} = 38$

$\because\ n\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$

$\bar{x}_2 = \dfrac{n\bar{x} - n_1\bar{x}_1}{n_2} = \dfrac{3800-2400}{40} = 35$

7. (c) $\sum(x_i - \bar{x}) = \sum x_i - n\bar{x} = n\bar{x}$

$-\bar{x}\cdot n = 0$

8. (c) Let

Data	Mean
x_i	$\bar{x}$
$\dfrac{x_i}{\alpha}$	$\dfrac{\bar{x}}{\alpha}$
$\dfrac{x_i}{\alpha} + 10$	$\dfrac{\bar{x}}{\alpha} + 10$

$\therefore$ Mean $= \dfrac{\bar{x} + 10\alpha}{\alpha}$

9. (b) $\bar{x} = \dfrac{\Sigma x_i}{n} = \dfrac{\Sigma x}{n} = \dfrac{n(n+1)}{2n}$

$\therefore\ \bar{x} = \dfrac{(n+1)}{2} = \dfrac{(n+7)}{3}$ (Given)

$\therefore\ n = 11$

10. (d) $\because\ n\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$

$\therefore\ \bar{x} = \dfrac{3\times14 + 2\times18}{5} = \dfrac{78}{5} = 15.6$

11. (c)

x_i	w_i	$x_i w_i$
0	0	0
1	1	1^2
2	2	2^2
3	3	3^2
4	4	4^2
⋮	⋮	⋮
n	n	n^2

$$\frac{\sum x_i w_i}{\sum w_i} = \frac{\dfrac{n(n+1)(2n+1)}{6}}{\dfrac{n(n+1)}{2}} = \boxed{\frac{2n+1}{3}}$$

12. (c)

x_i	f_i
1	1
2	1
3	1
⋮	⋮
n	1

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1+2+3+....n}{n} = \frac{n(n+1)}{2n}$$

$$= \left(\frac{n+1}{2}\right)$$

13. (d) $G = (x_1 \cdot x_2 \cdot x_3 ...)^{1/n}$

$G = (2^1 \cdot 2^2 ... 2^n)^{1/n}$

$G = (2^{1+2...+n})^{1/n}$

$G = 2^{\frac{n(n+1)}{2}\frac{1}{n}} = 2^{\frac{n+1}{2}}$

14. (c) $H = \dfrac{3}{(1/4)+(1/8)+(1/16)}$

$\therefore\ H = \dfrac{3\times16}{4+2+1} = \dfrac{48}{7}$

$\therefore = 6.85$

15. (a) Arranged Data in increasing order is

6, 8, 9, $\boxed{10}$, 11, 12, 14

↓ Median

16. (a) Data arranged in ascending order is

$$\alpha - \frac{7}{2}, \alpha - 3, \alpha - \frac{5}{2}, \alpha - 2, \alpha - \frac{1}{2}, \alpha + \frac{1}{2}, \alpha + 4, \alpha + 5$$

$$\therefore \text{Median} = \frac{(\alpha - 2) + \left(\alpha - \frac{1}{2}\right)}{2} = \alpha - \frac{5}{4}$$

17. (c) Data in ascending order is

3, 4, 6, 9, 10, 11, 18, 22

$\therefore$ Median $= \dfrac{9+10}{2} = 9.5$

18. (c)

Marks (x)	Frequency	Cumulative Frequency
0-10	10	10
10-20	20	30
20-30	30	60
30-40	50	110
40-50	40	150
50-60	30	180

$N = 180$

then $\frac{N}{2} = 90^{th}$ item

Median in Class = 30 – 40, $l = 30$, $C = 60$, $f = 50$ and $i = 10$

$$\text{Median} = l + \frac{\left(\frac{N}{2} - C\right)}{f} \times i = 36$$

19. (a) Highest frequency data is 6

$\therefore$ Mode = 6

20. (a) Mode class = 30 – 40

$$\text{Mode} = 30 + \frac{45-30}{90-30-35} \times 10$$

$$= 30 + \frac{150}{25}$$

$$= 30 + 6 = 36$$

21. (b) For skew distribution

$\because$ mode = 3(median) – 2(mean)

$\therefore$ Median = $\frac{1}{3}$ (Mode + 2(mean))

$\therefore M = \frac{1}{3}$

22. (b)

Data	Mean
x	σ
$\frac{ax}{c}$	$\left\|\frac{a}{c}\right\|\sigma$
$\frac{ax}{c} + \frac{b}{c}$	$\left\|\frac{a}{c}\right\|\sigma$

23. (c) Arrange data is in increasing order is 34, 38, 42, 44, 46, 48, 54, 55, 63, 70

$\therefore$ Median = $\frac{46+48}{2} = 47$

x_i	$D_i = x_i - \text{median}$	$\lvert d_i \rvert$
34	–13	13
38	–9	9
42	–5	5
44	–3	3
46	–1	1
48	+1	1
54	+7	7
55	+8	8
63	+16	16
70	+23	23

$$\text{M.D.} = \frac{\sum |d_i|}{n} = \frac{86}{10} = 8.6$$

24. (c) $\bar{x} = \frac{-1+0+4}{3} = 1$

$$\therefore M.D. = \frac{\sum |x_i - \bar{x}|}{n}$$

$$= \frac{|-1-1| + |0-1| + |4-1|}{3} = \frac{2+1+3}{3} = 2$$

25. (b) $S.D. : \sigma = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$

$$= \sqrt{\frac{1^2 + 2^2 \ldots + 7^2}{7} - \left(\frac{1+2+\ldots 7}{7}\right)^2}$$

$$= \sqrt{\frac{7.8.15}{6.7} - \left(\frac{7.8}{2.7}\right)^2} = \sqrt{20-16} = \sqrt{4} = 2$$

26. (b) When each data is multiplied by 'c' then S.D. all gets multiplied $|c|$.

$\therefore$ new S.D. = $|c|.\sigma$

27. (a) $\Sigma x = n\bar{x} = 100 \times 50 = 5000$

S.D. = $\sqrt{\sigma^2}$

$4 = \sqrt{\sigma^2}$

$$= \sqrt{\frac{1}{n}\Sigma x_i^2 - \bar{x}^2} = \sqrt{\frac{\Sigma x_i^2}{100} - (50)^2}$$

$$16 = \frac{\Sigma x_i^2}{100} - 2500$$

$$(16 + 2500) \cdot 100 = \Sigma x_i^2$$

$$\boxed{251600 = \Sigma x_i^2}$$

28. (c) $n_1 = 100, n_2 = 150$

$\bar{x}_1 = 50 \quad \bar{x}_2 = 40$

$\sigma_1^2 = 5 \quad \sigma_2^2 = 6$

$n\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$

$= 100 \times 50 + 150 \times 40$

$= 5000 + 6000$

$\bar{x} = \frac{11000}{250} = 44$

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$d_1 = 50 - 44 = 6$

$d_2 = 40 - 44 = -4$

$$\sigma^2 = \frac{100(25+36) + 150(36+16)}{250}$$

$$= \frac{6100 + 7800}{250} = 55.6$$

$\sigma = \sqrt{55.6} = 7.46$

29. (b) $\sigma^2 = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2$

$$= \frac{1^2 + 2^2 + 3^2 + \ldots + n^2}{n} - \left(\frac{1+2+3+\ldots+n}{n}\right)^2$$

$$= \frac{n(n+1)(2n+1)}{6n} - \left(\frac{n(n+1)}{2n}\right)^2$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{n^2(n+1)^2}{4n^2}$$

$$= \frac{n^2-1}{12} = 10$$

$\Rightarrow n^2 = 12 \times 10 + 1 = 121 \Rightarrow n = 11$

(n is never negative)

30. (c) Mean of series is

$$\bar{x} = \frac{a + (a+d) + (a+2d) + \ldots + (a+2nd)}{(2n+1)}$$

$\bar{x} = a + nd$

$$\therefore \sum_{i=0}^{2n} |x_i - \bar{x}| \Rightarrow \frac{2d(n)(n+1)}{2}$$

$\Rightarrow n(n+1)d$

$\therefore$ Mean deviation = $\frac{n(n+1)d}{(2n+1)}$

Prabal (JEE Main Level)

1. (c) Let $x_1, x_2, x_3 \ldots x_9$ are 9 items then

$$\Rightarrow \frac{x_1 + x_2 + x_3 + \ldots\ldots + x_9}{9} = 15$$

$$\therefore \sum_{i=1}^{9} x_i = 135$$

if one more item x_{10} is added then

$$\sum_{i=1}^{10} x_i = 135 + x_{10}$$

$$\therefore \text{mean} = \bar{x} = \frac{135 + x_{10}}{10} = 16$$

$\therefore x_{10} = 160 - 135 = 25$

2. (*d*) Given : Mean ($\bar{x}$) = 2.5 and discrete data

Variable x :	0	1	2	3	4	5
Frequency f :	k	20	40	40	20	4

We know that

$\Sigma f = k + 20 + 40 + 40 + 20 + 4 = 124 + k$

We also know that

$\Sigma fx = (0 \times k) + (1 \times 20) + (2 \times 40) + (3 \times 40) + (4 \times 20) + (5 \times 4)$

$= 0 + 20 + 80 + 20 = 320$

Therefore, mean ($\bar{x}$)

$$= \frac{\Sigma f x}{\Sigma f} = \frac{320}{124+k} \text{ or } 2.5 = \frac{320}{124+k}$$

or $124 + k = \frac{320}{2.5} = 128$

or $k = 128 - 124 = 4$

3. (*d*) $\frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \bar{x}$

$\therefore$ New mean is $\bar{x}'_B = \frac{\sum_{r=1}^{n}(x_i + 2r)}{n}$

$= \bar{x} + \frac{2n(n+1)}{2n}$

$\bar{x}' \Rightarrow \bar{x} + (n+1)$

4. (*d*) If variable is decreased by 1 then there is no effect on standard deviation.

5. (*a*) $\sum (x_i - \bar{x})^2 = 250$

$\therefore \sigma^2 = \frac{250}{10} = 25$

$\therefore \sigma^2 = 25 \Rightarrow \sigma = 5$

$\Rightarrow$ coefficient of variation $= \frac{\sigma}{\bar{x}} \times 100$

$= \frac{5}{50} \times 100 = 10\%$

6. (*a*) Given $\frac{x_1 + x_2 + x_3 + \ldots\ldots + x_n}{n} = \bar{x}$

then new mean is

$\frac{x_1 + x_2 + x_3 + \ldots + x_n + 1^2 + 2^2 + \ldots + n^2}{n}$

$= \bar{x} + \frac{(n+1)(2n+1)}{6}$

7. (*d*) Odd divisors of 360 are

$360 = 2^3 \times 3^2 \times 5$

$\Rightarrow$ 1, 3, 5, 9, 15, 45

so mean $= \frac{1+3+5+9+15+45}{6} = 13$

median $= \frac{5+9}{2} = 7$

mean of 13 and 7 $= \frac{13+7}{2} = 10$

8. (*c*) $n_1 = 35$, $\bar{x}_1 = 80$, $\sigma_1 = 4$

$n_2 = 65$, $\bar{x}_2 = 60$, $\sigma_2 = 3$

$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_1^2)}{n_1 + n_2}$

Combined mean

$= \frac{35 \times 80 + 65 \times 60}{100} = 67$

$\therefore d_1 = 80 - 67 = 13$, $d_2 = 60 - 67 = -7$

$\therefore \sigma^2 = \frac{35(16+169) + 65(9+49)}{100}$

$\sigma^2 = 102.45$

$\therefore \sigma = \sqrt{102.45} = 10.12$

9. (*d*) Number of candidates = 200

mean = 40

s.d. = 15

Correct data = 40

Incorrect data = 50

(*i*) Incorrect mean $\frac{\sum x_i}{200} = 40$

Incorrect $\sum x_i = 8000$

correct sum $\sum x_i = 8000 - 50 + 40$

$= 7990$

correct mean $= \frac{\sum x_i}{200} = 39.95$

(*ii*) Incorrect s.d. $= \sqrt{\frac{1}{n}\sum x_i^2 - \left(\frac{1}{n}\sum x_i\right)^2}$

$\Rightarrow \left[(\text{s.d.})^2 + \left(\frac{1}{n}\sum x_i\right)^2\right] n = \sum x_i^2$

$\Rightarrow (225 + 1600) \times 200 = \sum x_i^2$

= Incorrected sum

$\Rightarrow \sum x_i^2 = 365000$

$\Rightarrow$ corrected sum

$\Rightarrow \sum x_i^2 - (50)^2 + (40)^2$

correcte.d $\sigma^2 = \left[\frac{364100}{200} - (39.95)^2\right]$

$= 1820.5 - 1596.0025 = 224.4975$

$\sigma = 14.98$

10. (*d*) Given: Numbers $x_1, x_2, x_3, \ldots x_n$ and their average = M. We know that the average of n numbers

$M = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$

or $\frac{nM + x}{n} = \frac{x_1 + x_2 + x_3 + \ldots + x_n + x}{n}$

or $\frac{nM + x}{n} = \frac{x_1}{n} + \frac{x_2 + x_3 + \ldots\ldots + x_n + x}{n}$

or $\frac{nM + x - x_1}{n} = \frac{x_2 + x_3 + \ldots + x_n + x}{n}$

or new average

$= \left(\frac{x_2 + x_3 + \ldots + x_n + x}{n}\right) = \frac{nM - x_1 + x}{n}$

11. (*d*) Data are multipled by 5 then variance sould be 25 times

so new variance = 25 × 9 = 225

12. (*b*) Mean = n items is $\bar{x}$

New mean

$= \frac{x_1 + x_2 + \ldots x_n}{n} + \frac{3 + 3^2 + 3^1 + \ldots 3^n}{n}$

$= \bar{x} + \frac{3(3^n - 1)}{2n}$

13. (*b*) Since the total number students = 100 and the number of boys = 70

$\therefore$ Number of girls = (100 – 70) = 30

Now, the total marks of 100 students = 100 × 72 = 7200

And total marks of 70 boys = 70 × 75 = 5250

$\Rightarrow$ Total marks of 30 girls = 7200 – 5250 = 1950

$\therefore$ Average marks of 30 girls = $\frac{1950}{30}$ = 65

14. (*d*) Since the median is the 5th and the increase is made in the last four terms, hence the median remains unchanged.

15. (*a*) Given : $N = 15$

$\Sigma x^2 = 2830$, $\Sigma x = 170$

One observation 20 was replaced by 30, then

$\Sigma x^2 = 2830 - 400 + 900 = 3330$

$\Sigma x = 170 - 20 + 30 = 180$

$\therefore$ Variance, σ^2

$= \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2 = \frac{3330}{15} - \left(\frac{180}{15}\right)^2$

$= 222 - 144 = 78$

16. (*c*) In the $2n$ observations, half of them are equal to a and the remaining half are equal to $-a$. Then, then mean of the total $2n$ observations is equal to zero.

$\therefore$ S.D. $= \sqrt{\frac{\Sigma(x - \bar{x})^2}{N}}$

$\Rightarrow 2 = \sqrt{\frac{\Sigma x^2}{2n}}$

$\Rightarrow 4 = \frac{\Sigma x^2}{2n} = \frac{2na^2}{2n}$

$\Rightarrow a^2 = 4$

$\Rightarrow |a| = 2$

17. (c) Each variable is increased by 20 so there will be no effect on standard deviation.

18. (a) Total number of terms = $n+1$ = odd

$\therefore$ median = $\dfrac{n+1+1}{2}$

$\Rightarrow \left(\dfrac{n}{2}+1\right)^{\text{th}}$ term

$\Rightarrow {}^{2n}C_{\frac{n}{2}}$

19. [10.1]

$$\bar{x} = \frac{\text{sum of quantities}}{n} = \frac{\frac{n}{2}(a_1+a_n)}{n}$$

$$= \frac{1}{2}[1+1+100d] = 1+50d$$

$$\therefore \text{MD} = \frac{1}{n}\Sigma|x_i - \bar{x}|$$

$$\Rightarrow 255 = \frac{1}{101}[50d+49d+\ldots+d$$
$$+0+d+49d+50d]$$

$$= \frac{2d}{101}\left[\frac{50\times 51}{2}\right]$$

$$\Rightarrow d = \frac{255\times 101}{50\times 51} = 10.1$$

20. [16] $\bar{x} = 5$

$$\text{Variance} = \frac{1}{n}\Sigma x_i^2 - \left(\bar{x}^2\right)$$

$$0 = \frac{1}{n}\cdot 400 - 25$$

$$\Rightarrow n = \frac{400}{25} = 16$$

21. [10] Given that

$n_1 = 10, \bar{x}_1 = 12, n_2 = 20, \bar{x}_2 = 9$

$$\therefore \bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1+n_2} = \frac{10\times 12+20\times 9}{10+20}$$

$$= \frac{120+180}{30} = \frac{300}{30} = 10$$

22. [30.1]

Given that, $\Sigma_{i=1}^{20}(x_i - 30) = 2$

$$\Rightarrow \sum_{i=1}^{20} x_i - \sum_{i=1}^{20}(30) = 2$$

$$\Rightarrow \bar{x} = \frac{20.30}{20} + \frac{2}{20} = 30 + 0.1 = 30.1$$

23. [24] Given series is 148, 146, 144, 142, ... whose first term and common difference is

$a = 148, d = (146 - 148) = -2$

$$S_n = \frac{n}{2}\left[2a+(n-1)d\right] = 125\,n \text{ (given)}$$

$$\Rightarrow 125n = \frac{n}{2}\left[2\times 148+(n-1)\times(-2)\right]$$

$$\Rightarrow n^2 - 24n = 0$$

$$\Rightarrow n(n-24) = 0$$

$$\Rightarrow n = 24 \ (n \neq 0)$$

24. [5] Given that,

$n_1 = 4, \bar{x}_1 = 7.5, n_1 + n_2 = 10, \bar{x} = 6$

$$\therefore 6 = \frac{4\times 7.5 + 6\times \bar{x}_2}{10} \Rightarrow 60 = 30 + 6\bar{x}_2$$

$$\Rightarrow x_2 = \frac{30}{6} = 5$$

25. [44.46] Total of corrected observations = 4500 – (91 + 13) + (19 + 31) = 4446

$$\therefore \text{Mean} = \frac{4446}{100} = 44.46$$

26. [63] Total marks of 10 failed students = $28 \times 10 = 280$

and Total marks of 50 students = 2800

$\therefore$ Total marks of 40 passed students = 2800 – 280 = 2520

$\therefore$ Average marks of 40 passed students

$$= \frac{2520}{40} = 63$$

PYQ's (Past Year Questions)

1. [35] Given, $\bar{X} = \dfrac{\sum_{i=1}^{25} x_i}{25} = 40$

$$\Rightarrow \sum_{i=1}^{25} x_i = 1000$$

After retirement of a teacher

$$\sum_{i=1}^{24} x_i = 1000 - 60 = 940$$

New mean, $\overline{X'} = \dfrac{\sum_{i=1}^{24} x_i + x}{25}$

$$\Rightarrow 39 = \frac{940+x}{25} \Rightarrow x = 35$$

2. (a) $\Sigma f_i = 20 = (x+1)^2 + (2x-5) + (x^2-3x) + x = 2x^2 + 2x - 4$

$$\Rightarrow 2x^2 + 2x - 24 = 0$$

$$\Rightarrow (x-3)(x+4) = 0$$

$\Rightarrow x = 3, -4$ (rejected)

Marks	2	3	5	7
Frequency	16	1	0	3

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i} = 2.8$$

3. (a)

x_i	3	4	5	8	10	11
f_i	5	4	4	2	2	3

(A) Mean

(B) Median

(C) Mean deviation about mean

(D) Mean deviation about median

x_i	f_i	$x_i f_i$	C.F.	$\lvert x_i - \text{mean}\rvert$	$f_i\lvert x_i - \text{mean}\rvert$	$\lvert x_i - \text{Median}\rvert$	$f_i\lvert x_i - \text{median}\rvert$
3	5	15	5	3	15	2	10
4	4	16	9	2	8	1	4
5	4	20	13	1	4	0	0
8	2	16	15	2	4	3	6
10	2	20	17	4	8	5	10
11	3	33	20	5	15	6	18
	$\Sigma f_i = 20$	$\Sigma f_i x_i = 20$			$\Sigma f_i\lvert x_i - \text{Mean}\rvert = 54$		$\Sigma f_i\lvert x_i - \text{Median} = 48$

(A) Mean $= \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{120}{20} = 6$

(B) Median $= \left(\frac{20}{2}\right)^{th}$ observation

$= 10^{th}$ observation $= 5$

(C) Mean deviation about mean

$= \frac{\Sigma f_i |x_i - Mean|}{\Sigma f_i} = \frac{54}{20} = 2.70$

(D) Mean deviation about median

$= \frac{\Sigma f_i |x_i - Median|}{\Sigma f_i} = \frac{48}{20} = 2.40$

4. [142] $\sum_{i=1}^{2x} = \frac{3\left(1-\left(\frac{1}{2}\right)^{20}\right)}{1-\frac{1}{2}} = 6\left(1-\frac{1}{2^{20}}\right)$

$= \sum_{i=1}^{20}(x_i - i)^2$

$= \sum_{i=1}^{20}\left[(x_i)^2 + (i)^2 - 2x_i i\right]$

Now,

$\sum_{i=1}^{20}(x_i)^2 = \frac{9\left(1-\left(\frac{1}{4}\right)^{20}\right)}{\left(1-\frac{1}{4}\right)} = 12\left(1-\frac{1}{2^{40}}\right)$

$\sum_{i=1}^{20} i^2 = \frac{1}{6}\times 20\times 21\times 41 = 2870$

$\sum_{i=1}^{20} x_i i = s = 3 + 2\cdot 3\frac{1}{2} + 3\cdot 3\cdot\frac{1}{2^2} + \ldots + 20\times 3\times\frac{1}{2^{19}} = 6\left(2-\frac{22}{2^{20}}\right)$

Therefore,

$\bar{x} = \frac{12 - \frac{12}{2^{40}} + 2870 - 12\left(2-\frac{22}{2^{20}}\right)}{20}$

$\Rightarrow \bar{x} = \frac{2858}{20} + \left(\frac{-12}{2^{40}} + \frac{22}{2^{20}}\right)\times\frac{1}{20}$

$\Rightarrow [\bar{x}] = 142$

5. [164]

Class	Frequency	Cummulative Frequency
10-20	α	α
20-30	110	$110+\alpha$
30-40	54	$164+\alpha$
40-50	30	$194+\alpha$
50-60	β	$194+\alpha+\beta$
	$\Sigma f = 584$	

Given, $\alpha + 110 + 54 + 30 + \beta = 584$

$\Rightarrow \alpha + \beta = 390$ ... (i)

Median $= L + \left(\frac{\frac{N}{2} - C}{f}\right)\times h = 45$

$L = 40, N = 584, C = \alpha + 164,$

$f = 30, h = 10$

$\Rightarrow \beta = 277$ [From (i)]

$\therefore |\alpha - \beta| = 164$

6. [6] Mean $= \frac{\sum x_i f_i}{\sum f_i}$

$= \frac{0.{}^nC_0 + 2.{}^nC_1 + 2^2.{}^nC_2 + \ldots + 2^n.{}^nC_n}{{}^nC_0 + {}^nC_1 + \ldots + {}^nC_n}$

For finding sum of numerator

Let $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots + {}^nC_n x^n$ ...(i)

Put $x = 2 \Rightarrow 3^n - 1 = 2.{}^nC_1 + 2^2.{}^nC_2 + \ldots + 2^n.{}^nC_n$

For sum of denominator

Put $x = 1$ in equation (i)

$2^n = {}^nC_0 + {}^nC_1 + \ldots + {}^nC_n$

$\therefore \frac{3^n - 1}{2^n} = \frac{728}{2^n} \Rightarrow 3^n = 729 \Rightarrow n = 6$

7. (a) Median = 35

$= \frac{\left(\frac{n}{2}\right)th + \left(\frac{n}{2}+1\right)th}{2}, n = even$

$\frac{34+x}{2} = 35$

$\Rightarrow x = 36$

Mean = 42

$= \frac{10+22+26+29+34+36+42+67+70+y}{10}$

$\Rightarrow 420 - 336 = y \Rightarrow y = 84$

$\Rightarrow \frac{y}{x} = \frac{84}{36} = \frac{7}{3}$

8. (a) $\Sigma f_i = 62 = 3k^2 + 4k - 2 = 62$

$\Rightarrow 3k^2 + 16k - 12k - 64 = 0$

$\Rightarrow k(3k+16) - 4(3k+16) = 0$

$k = 4$ or $-\frac{16}{3}$ (rejected)

$\mu = \frac{\Sigma f_i x_i}{\Sigma f_i}$

$\mu = \frac{8 + 2(15) + 3(15) + 4(17) + 5}{62} = \frac{156}{62}$

Variance $= \sigma^2 = \frac{\Sigma f_i x_i^2}{\Sigma f_i} - \mu$

$= \frac{8\times 1^2 + 15\times 13 + 17\times 16 + 25}{62} - \left(\frac{156}{62}\right)^2$

$\sigma^2 = \frac{500}{62} - \left(\frac{156}{62}\right)^2$

$\Rightarrow \sigma^2 + \mu^2 = \frac{500}{62} = 8.064 \left[\because \mu = \frac{156}{62}\right]$

Hence, $\left[\sigma^2 + \mu^2\right] = 8$

9. [398] Given, $7\times 8, 10\times 10, 13\times 12, 16\times 14, \ldots$

$t_r = (3r+4)(2r+6) = 6r^2 + 26r + 24$

$\therefore \sum_{r=1}^{10} t_r = \frac{6\times 10\times 11\times 21}{6} + \frac{26\times 10\times 11}{2} + 24\times 10 = 10\times 398$

$\therefore$ Mean $= \frac{10\times 398}{10} = 398$

10. (d) Mean = 5

$\Rightarrow 1 + 2 + 4 + 5 + x + y = 6\times 5$

$\Rightarrow x + y = 18$ {$\because$ Mean = 5} ...(i)

Variance =

$10 = \frac{1+4+16+25+x^2+y^2}{6} - (5)^2$

$\Rightarrow x^2 + y^2 = 164$ ...(ii)

By solving (i) and (ii), we get

$x = 8, y = 10$

Mean deviation about mean

$= \frac{\sum |x_i - \bar{x}|}{6}$

$= \frac{4+3+1+3+5}{6} = \frac{8}{3}$

11. [8] Given mean = 5

$5 = \bar{x} = \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{4 + 72 + 140 + 7\alpha + 72}{64 + \alpha}$

$\Rightarrow 320 + 5\alpha = 288 + 7\alpha \Rightarrow 2\alpha = 32 \Rightarrow \alpha = 16$

M.D. $(\bar{x}) = \frac{\Sigma f_i |x_i - \bar{x}|}{\Sigma f_i}$

where $\Sigma f_i = 64 + 16 = 80$

M.D. $(\bar{x}) = \frac{4\times 4 + 24\times 2 + 28\times 0 + 16\times 2 + 8\times 4}{80}$

$$m = \frac{128}{80} = \frac{8}{5}$$

$$\text{Variance} = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}$$

$$= \frac{4\times16 + 24\times4 + 0 + 16\times4 + 8\times16}{80} = \frac{352}{80}$$

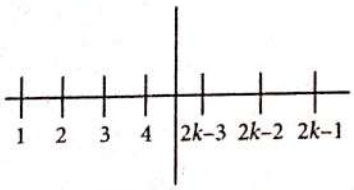
$$\therefore \frac{3\alpha}{m + \sigma^2} = \frac{3\times16}{\frac{128}{80} + \frac{352}{80}} = 8$$

12. **[21]** Mean $= \dfrac{n(n+1)/2}{n}$

$$\left[\text{Sum} = 1 + 2 + 3 + n = \frac{n(n+1)}{2}\right]$$

For $n = 2k-1, k \in N$

Mean $= k$



$k-2, k-1, \quad k \quad k-1, k-2$

Mean deviation = avg. of deviations

$$= \frac{2(0 + 1 + 2 + \ldots + (k-1))}{2k-1} = \frac{(k-1)k}{2k-1}$$

$$= \frac{5(2k)}{2k-1}$$

Put $n = 2k - 1$

$$\left[\because \frac{5(n+1)}{n} = \frac{5(2k)}{2k-1}\right]$$

$\Rightarrow k = 11 \Rightarrow n = 2k - 1 = 21$

13. *(d)* Given $\sum_{i=1}^{50}(x_i - 30) = 50$

$$\Rightarrow \sum x_i = 30(50) + 50 \Rightarrow \frac{\sum x_i}{50} = 31$$

$$\left[\because \sum_{i=1}^{n} 1 = n\right]$$

14. **(c)** Mean $= \dfrac{\sum x}{12} = \dfrac{9}{2} \Rightarrow \sum x = 54$

Variance $= 4$

$$\Rightarrow \frac{\sum x^2}{12} - \left(\frac{9}{2}\right)^2 = 4 \Rightarrow \sum x^2 = 291$$

$\sum x_{new} = 54 - (9 + 10) + (7 + 14) = 56$

$\sum x_{new}^2 = 219 - (81 + 100)(49 + 196) = 355$

$$\sigma_{new}^2 = \frac{355}{12} - \left(\frac{56}{12}\right)^2$$

$$= \frac{281}{36} = \frac{m}{n}\,(\text{Given})$$

$\therefore m = 281$ and $n = 36$

Hence, $m + n = 317$

15. **[25]**

x_i	f_i	$f_i x_i$	$f_i x_i^2$
2	4	8	16
4	4	16	64
6	α	6α	36α
8	15	120	960
10	8	80	800
12	β	12β	144β
14	4	56	784
16	5	80	1280

$N = \sum f_i = 40 + \alpha + \beta$

$\sum f_i x_i = 360 + 6\alpha + 12\beta$

$\sum f_i x_i^2 = 3904 + 36\alpha + 144\beta$

$$\text{Mean}(\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = 9$$

$\Rightarrow 360 + 6\alpha + 12\beta = 9(40 + \alpha + \beta)$

$\Rightarrow 3\alpha = 3\beta \Rightarrow \alpha = \beta$

$$\sigma^2 = \frac{\sum f_i x_i^2}{\sum f_i} - \left(\frac{\sum f_i x_i}{\sum f_i}\right)^2$$

$$\Rightarrow \frac{3904 + 36\alpha + 144\beta}{40 + \alpha + \beta} - (\bar{x})^2 = 15.08$$

$$\Rightarrow \frac{3904 + 180\alpha}{40 + 2\alpha} - (9)^2 = 15.08$$

$\Rightarrow \alpha = 5$

Now, $\alpha^2 + \beta^2 - \alpha\beta = \alpha^2 = 25$

16. *(b)* Mean $=$

$$\frac{4 + 5 + 6 + 6 + 7 + 8 + x + y}{8} = 6$$

$\therefore x + y = 12 \qquad \ldots(i)$

and Variance $=$

$$\frac{2^2 + 1^2 + 0^2 + 0^2 + 1^2 + 2^2 + (x-6)^2 + (y-6)^2}{8}$$

$$= \frac{9}{4}$$

$\therefore (x-6)^2 + (y-6)^2 = 8 \qquad \ldots(ii)$

From equation *(i)* and *(ii)*,

$x = 4$ and $y = 8$

$\therefore x^4 + y^2 = 320$

17. **[17]** Given, Mean $= \dfrac{\sum_{i=1}^{15} x_i}{15} = 8$

$\Rightarrow \sum_{i=1}^{15} x_i = 120 \qquad \ldots(i)$

Also, S. D, $\sigma = 3$

$$\therefore \sigma^2 = 9 = \frac{1}{15}\sum_{i=1}^{15} x_i^2 - (8)^2$$

$\Rightarrow \sum_{i=1}^{15} x_i^2 = 15(9 + 64) = 15 \times 73$

$= 1095$

Given 20 has misread as 5

$\therefore$ in new case

$\sum_{i=1}^{15} x_i^2 = 15 \times 73 - 25 + 400 = 1470$

Also, mean in new case

$$\bar{x} = \frac{120 - 5 + 20}{15} = 9$$

$\therefore$ Variance in new case

$$\sigma_{new}^2 = \frac{1}{15}(1470) - 81 = 17$$

18. *(a)* For a, b, c

Mean, $\bar{x} = \dfrac{a + b + c}{3}$

$\Rightarrow \bar{x} = \dfrac{2b}{3} \qquad [\because b = a + c]$

S.D. $(a + 2, b + 2, c + 2)$

$=$ S.D. $(a, b, c) = d$

$$\Rightarrow d^2 = \frac{a^2 + b^2 + c^2}{3} - (\bar{x})^2$$

$$\Rightarrow d^2 = \frac{a^2 + b^2 + c^2}{3} - \frac{4b^2}{9}$$

$$\Rightarrow 9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$$

$$\Rightarrow b^2 = 3(a^2 + c^2) - 9d^2$$

19. **[12]** Given, mean $= 5$

$$\Rightarrow \frac{3 + 7 + x + y}{4} = 5$$

$\Rightarrow x + y = 10 \qquad \ldots(i)$

Also, variance $= 10$

$$\Rightarrow \frac{1}{4}\left(9 + 49 + x^2 + y^2\right) - 25 = 10$$

$\Rightarrow x^2 + y^2 = 82 \qquad \ldots(ii)$

Required mean is $\bar{x}$

$$\therefore \bar{x} = \frac{(3 + 2x) + (7 + 2y) + (x + y) + (x - y)}{4}$$

$$= \frac{10 + 4x + 2y}{4}$$

On solving equ. *(i)* & *(ii)*, we get $x = 9$, $y = 1$

$$\therefore \bar{x} = \frac{48}{4} = 12$$

20. (*d*) $\because \bar{x} = \dfrac{\sum_{r=1}^{17} r}{17} = \dfrac{17\times18}{17\times2} = 9$

$\because \bar{y} = a\bar{x} + b = 17$

$\Rightarrow 9a + b = 17$ ...(*i*)

$\because$ Var (x) =

$$\frac{\sum_{r=1}^{17} r^2}{17} - (\bar{x})^2 = \frac{17\times18\times35}{17\times6} - 9^2 = 24$$

and Var $(y) = a^2$. Var$(x) \Rightarrow 24a^2 = 216$

$\Rightarrow a = 3$ ...(*ii*)

Clearly $b = -10$.

Hence $a + b = -7$

21. (*b*) If variate varries from a to b then variance

$$\text{var}(x) \le \left(\frac{b-a}{2}\right)^2$$

$$\Rightarrow \text{var}(x) < \left(\frac{10-0}{2}\right)^2$$

$$\Rightarrow \text{var}(x) < 25$$

$\Rightarrow$ Standard deviation < 5

Clearly standard deviation cannot be 6.

22. [4]

x_i	15	25	35
f_i	2	x	2

Let A = 25

d_i	−10	0	10
f_i	2	x	2

$$\bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$$

$$= 25 + \frac{-10\times2 + 0\times x + 10\times2}{4+x} = 25$$

Given $\sigma^2 = 50 = \dfrac{\sum f_i x_i}{\sum f_i} - (\bar{x})^2$

$$\Rightarrow 50 = \frac{450 + 625x + 2450}{4+x} - 625$$

$$\Rightarrow 675 = \frac{2900 + 625x}{4+x} \Rightarrow 50x = 200$$

$\therefore x = 4$

23. (*b*) $\bar{x} = \dfrac{1+3+8+x+y}{5} = 5$

$\Rightarrow 25 = 12 + x + y \Rightarrow x + y = 13$ ...(*i*)

Variance $= \sigma^2 = \dfrac{\Sigma x_i^2}{n} - \left(\dfrac{\Sigma x_i}{n}\right)^2$

$$\Rightarrow 9.2 = \frac{1+9+64+x^2+y^2}{5} - 25$$

$\Rightarrow 34.2 \times 5 = 74 + x^2 + y^2 \Rightarrow 171 = 74 + x^2 + y^2$

$\Rightarrow 97 = x^2 + y^2$ ...(*ii*)

Now, $(x+y)^2 = x^2 + y^2 + 2xy$

$\Rightarrow 169 - 97 = 2xy \Rightarrow xy = 36$

$x = 4, y = 9$ or $y = 4, x = 9$

So ratio is $\dfrac{4}{9}$ or $\dfrac{9}{4}$

24. (*b*)

$\alpha - \beta$	Case	P
5	(6, 1)	1/36
4	(6, 2) (5, 1)	2/36
3	(6, 3) (5, 2) (4, 1)	3/36
2	(6, 4) (5, 3) (4, 3) (3, 1)	4/36
1	(6, 5) (5, 4) (4, 3) (3, 2) (2, 1)	5/36
0	(6, 6) (5, 5) ... (1, 1)	6/36
−1	—	5/36
−2	—	4/36
−3	—	3/36
−4	(2, 6) (1, 5)	2/36
−5	(1, 6)	1/36

Variance (σ^2)

$$= \sum x^2 P(x) = 2\left[\frac{25}{36} + \frac{32}{36} + \frac{27}{36} + \frac{16}{36} + \frac{5}{36}\right]$$

$$= \frac{105}{18} = \frac{35}{6} = \frac{p}{q}$$

$\mu = \sum(x) = 0$ as data is symmetric

$p = 35$

Sum of divisors $= (5^0 + 5^1)(7^0 + 7^1) = 6 \times 8 = 48$

25. [238] $\bar{x} = \dfrac{\Sigma x_i}{40} = 30 \Rightarrow \Sigma x_i = 1200$ ...(*i*)

$$\sigma^2 = \frac{1}{40}\Sigma x_i^2 - (30)^2 = 25$$

$$\left[\because \sigma^2 = \frac{\Sigma x_i^2}{40} - (\bar{x})^2\right]$$

$\Rightarrow \Sigma x_i^2 = 37000$

After omitting two wrong observations

$\Sigma y_i = 1200 - 12 - 10 = 1178$

$\Sigma y_i^2 = 37000 - 144 - 100 = 36756$

Variance after omitting

$$\therefore \sigma^2 = \frac{\Sigma y_i^2}{38} - \left(\frac{\Sigma y_i}{38}\right)^2$$

$$\sigma^2 = \frac{36756}{38} - \left(\frac{1178}{38}\right)^2$$

$\therefore 38\sigma^2 = 36756 - 36518 = 238$

26. [68] Let $X_1, X_2 \ldots\ldots X_{2n}$ be the first 2n observations and

$Y_1, Y_2 \ldots\ldots Y_n$ be the last n observations

$$\therefore \frac{\Sigma x_i}{2n} = 6 \text{ and } \frac{\Sigma y_i}{n} = 3$$

$\therefore \Sigma x_i = 12n$ ...(*i*)

and $\Sigma y_i = 3n$ ...(*ii*)

$$\therefore \frac{\Sigma x_i + \Sigma y_i}{3n} = 5 \Rightarrow \Sigma x_i + \Sigma y_i = 15n$$

$$\frac{\Sigma x_i^2 + \Sigma y_i^2}{3n} - 5^2 = 4$$

$\Rightarrow \Sigma x_i^2 + \Sigma y_i^2 = 87n$

According to question,

$$\frac{\Sigma(x_i + 1) + \Sigma(y_i - 1)}{3n}$$

$$= \frac{15n + 2n - n}{3n} = \frac{16}{3}$$

and $\dfrac{\Sigma(x_i+1)^2 + \Sigma(y_i-1)^2}{3n} - \left(\dfrac{16}{3}\right)^2$

$$= \frac{87n + 2(9n) + 3n}{3n} - \left(\frac{16}{3}\right)^2 = \frac{68}{9}$$

$$\therefore \frac{68}{9} = k \Rightarrow 9k = 68$$

27. (*d*) Combined variance

$$= \frac{n_1\sigma^2 + n_2\sigma^2}{n_1 + n_2} + \frac{n_1 n_2 (m_1 - m_2)^2}{(n_1 + n_2)^2}$$

$$13 = \frac{15.14 + 15.\sigma^2}{30} + \frac{15.15(12-14)^2}{30\times30}$$

$$\Rightarrow 13 = \frac{14 + \sigma^2}{2} + \frac{4}{4}$$

$\sigma^2 = 10$

28. [37] Given $\dfrac{x_1 + x_2 + \ldots + x_7}{7} = 8$

$$\frac{x_1 + x_2 + x_3 \ldots + x_6 + 14}{7} = 8$$

$\Rightarrow x_1 + x_2 + \ldots + x_6 = 42$

$$\therefore \frac{x_1 + x_2 \ldots + x_6}{6} = \frac{42}{6} = 7 = a$$

$$\frac{\sum x_i^2}{7} - 8^2 = 16$$

$$\sum x_i^2 = 560$$

$$\Rightarrow x_1^2 + x_2^2 + \ldots + x_6^2 = 364$$

$$b = \frac{x_1^2 + x_2^2 + \ldots + x_6^2}{6} - 7^2$$

$$= \frac{364}{6} - 49 \Rightarrow b = \frac{70}{6}$$

$$\therefore a + 3b - 5 = 7 + 3 \times \frac{70}{6} - 5 = 37$$

29. **[603]** $\because n(X) = 31$

$n(Y) = 31$

$$\bar{x} = \frac{\sum_{i=11}^{41} i}{31} = \frac{1}{31} \times \frac{31}{2}\left(\frac{11+41}{2}\right) = 26$$

$$\bar{y} = \frac{\sum_{j=61}^{91} j}{31} = \frac{1}{31} \times \frac{31}{2}\left(\frac{61+91}{2}\right) = 76$$

Combined mean, $\mu = \frac{31 \times 26 + 31 \times 76}{31 + 31}$

$$= \frac{26 + 76}{2} = 51$$

$$\sigma^2 = \frac{1}{62} \times \left(\sum_{i=1}^{31} (x_i - \mu)^2 + \sum_{i=1}^{31} (y_i - \mu)^2 \right)$$

Since, $x_i \in X$ are in A.P. with 31 elements & common difference 1 , same is $y_1 \in y$, when written in increasing order.

$$\therefore \sum_{i=1}^{31} (x_i - \mu)^2 = \sum_{i=1}^{31} (y_i - \mu)^2$$

$$= 10^2 + 11^2 + \ldots. + 40^2$$

$$= \frac{40 \times 41 \times 81}{} - \frac{9 \times 10 \times 19}{} = 21855$$

Hence, $\sigma^2 = \frac{21855}{62} + \frac{21855}{62} = 705$

$$\therefore \left| \bar{x} + \bar{y} - \sigma^2 \right| = \left| 26 + 76 - 705 \right| = 603$$

30. **[54]** Mean

$$\bar{x} = \frac{3 + 7 + 9 + 12 + 13 + 20 + x + y}{8} = 10$$

$$\Rightarrow x + y = 16 \quad \ldots(i)$$

Variance $\sigma^2 = \frac{\sum (x_i)^2}{8} - (\bar{x})^2 = 25$

$$\frac{9 + 49 + 81 + 144 + 169 + 400 + x^2 + y^2}{8} - 100 = 25$$

$$\Rightarrow x^2 + y^2 = 148 \quad \ldots(ii)$$

$(x + y)^2 = (16)^2$

$\Rightarrow x^2 + y^2 + 2xy = 256 \Rightarrow x \cdot y = 54$

By Aditya Mohod

username - muje_kabhi_dekha_nahi

By Aditya Mohad
username - muje_kabhi_dekha_nahi

CHAPTER 18 Introduction to Three Dimensional Geometry

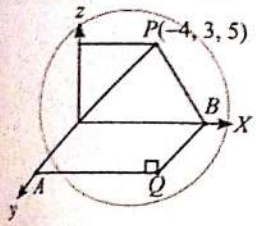
Prarambh (Topicwise)

1. **(a)** If p lies in x-axis then its y and z-coordinates = 0

2. **(b)** In xz-plane y-coordinate = 0
This implies $y=0$

Sol. (3 to 7)

3. **(b)** 4. **(a)** 5. **(b)** 6. **(a)** 7. **(d)**



Let PQ be the perpendicular to the xy-plane and QA be perpendicular from Q to the y-axis. PA will be perpendicular to the x-axis.

Also, $QA = |3|$ and $PQ = |5|$

Now, distance of P from the x-axis:

$$PB = \sqrt{BQ^2 + QP^2}$$
$$= \sqrt{3^2 + 5^2}$$
$$= \sqrt{9+25} = \sqrt{34}$$

Similarly,

From the right-angled ΔPAQ distance of P from the y-axis:

$$PA = \sqrt{AQ^2 + QP^2}$$
$$= \sqrt{(-4)^2 + (5)^2}$$
$$= \sqrt{16+25} = \sqrt{41}$$

Similarly, the length of the perpendicular from P to the z-axis $= \sqrt{(-4)^2 + (3)^2}$
$= \sqrt{16+9} = \sqrt{25} = 5$

8. **(c)** $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
$= \sqrt{(2+2)^2 + (1-3)^2 + (2-1)^2}$
$= \sqrt{(4)^2 + (-2)^2 + (1)^2}$
$= \sqrt{16+4+1}$
$= \sqrt{21}$ units

9. **(b)** **(i)** If R divides PQ internally in the ratio 2 : 3, then

$$x = \frac{2\times 4 + 3\times 2}{2+3}, y = \frac{2\times 3 + 3x - 1}{2+3},$$
$$z = \frac{2\times 2 + 3\times 4}{2+3}$$
$$\Rightarrow x = \frac{14}{5}, y = \frac{3}{5}, z = \frac{16}{5}$$

So, the coordinates of point Rare

$$\left(\frac{14}{5}, \frac{3}{5}, \frac{16}{5}\right)$$

10. **(c)** Suppose the line joining the points $P(1, 2, 3)$ and $Q(-3, 4, -5)$ is divided by the xy-plane at a point R in the ratio $\lambda : 1$. Then, the coordinates of R are

$$\left(\frac{-3\lambda+1}{\lambda+1}, \frac{4\lambda+2}{\lambda+1}, \frac{-5\lambda+3}{\lambda+1}\right)$$

Since R lies on xy-plane i.e. $z = 0$,
Therefore,

$$\frac{-5\lambda+3}{\lambda+1} = 0 \Rightarrow \lambda = \frac{3}{5}$$

So, the required ratio is

$\frac{3}{5}$: 1 or, 3 : 5. Putting $\lambda = \frac{3}{5}$ in (i) we obtain the coordinates of R as $(-1/2, 11/4, 0)$.

11. **(a)**

12. **(b)** We know that the x-coordinate on yz-plane is 0.

The coordinates of the foot of the perpendicular drawn from the point $P(3, 4, 5)$ on the yz-plane are $(0, 4, 5)$. Hence, the correct answer is option (b).

13. **(a)** We know that the y and z coordinates on x-axis are 0

The coordinates of the foot of the perpendicular from a point $P(6,7, 8)$ on x-axis are $(6, 0, 0)$ Hence, the correct answer is option (a).

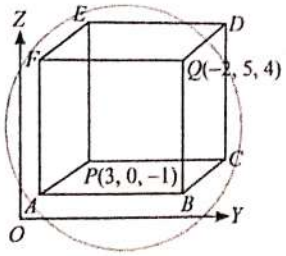
14. **(a)** 15. **(c)** 16. **(b)**

17. **(d)** Let $P(x, y, z)$ be any point, the sum of whose distance from the points $A(4,0,0)$ and $B(-4, 0, 0)$ is equal to 10. Then, $PA + PB = 10$

$$\Rightarrow \sqrt{(x-4)^2 + (y-0)^2 + (z-0)^2} + \sqrt{(x+4)^2 + (y-0)^2} = 10$$
$$\Rightarrow \sqrt{(x+4)^2 + (y-0)^2 + (z-0)^2} = 10 - \sqrt{(x-4^2) + (y-0)^2 + (z-0)^2}$$
$$\Rightarrow \sqrt{(x^2 + 8x + 16 + y^2 + z^2} = 10 - \sqrt{x^2 - 8x + 16 + y^2 + z^2}$$
$$\Rightarrow x^2 + 8x + 16 + y^2 + z^2 = 100 + x^2 - 8x + 16 + y^2 + z^2$$
$$-20\sqrt{x^2 - 8x + 16 + y^2 + z^2}$$
$$\Rightarrow 16x - 100 = -20\sqrt{x^2 - 8x + 16 + y^2 + z^2}$$
$$\Rightarrow 4x - 25 = -5\sqrt{x^2 - 8x + 16 + y^2 + z^2}$$
$$\Rightarrow 16x^2 - 200x + 625 = 25(x^2 - 8x + 16 + y^2 + z^2)$$
$$\Rightarrow 16x^2 - 200x + 625 = 25x^2 - 200x + 400 + 25y^2 + 25z^2$$
$$\Rightarrow 9x^2 + 25y^2 + 25z^2 - 225 = 0$$

$\therefore 9x^2 + 25y^2 + 25z^2 - 225 = 0$ is the required locus.

18. **(a)**



Let $P = (3, 0, -1)$, $Q = (-2, 5, 4)$
PE = Distance between the parallel planes $ABCP$ and $FQDE = |4 + 1| = 5$
(These planes are perpendicular to the z-axis)
PA = Distance between the parallel planes $ABQF$ and $PCDE = |-2 - 3| = 5$
(These planes are perpendicular to the x-axis)
Similarly, $PC = |5 - 0| = 5$
Thus, the length of the edges of the parallelepiped are 5, 5 and 5

19. **(b)**

20. **(a)** Let $P(x, y, z)$ be the required point. Then, $OP = PA = PB = PC$.

Now, $OP = PA$

$\Rightarrow OP^2 = PA^2$

$\Rightarrow x^2 + y^2 + z^2 = (x-a)^2 + (y-0)^2 + (z-0)^2$

$\Rightarrow 0 = -2ax + a^2 \Rightarrow x = a/2$

Similarly, $OP = PB$

$\Rightarrow y = b/2$ and $OP = PC \Rightarrow z = c/2$

Hence, the coordinates of the required point are $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$.

21. **(a)** Suppose the plane $2x + 2y - 2z = 1$ divides the line joining the points $A(2, 1, 5)$ and $B(3, 4, 5)$ at a point C in the ratio $\lambda:1$. Then, the coordinates of C are

$$\left(\frac{3\lambda+2}{\lambda+1}, \frac{4\lambda+1}{\lambda+1}, \frac{3\lambda+5}{\lambda+1}\right)$$

Since point C lies on the plane $2x + 2y - 2z = 1$. Therefore, coordinates of C must satisfy the equation of the plane i.e.

$$2\left(\frac{3\lambda+2}{\lambda+1}\right)+2\left(\frac{4\lambda+1}{\lambda+1}\right)-2\left(\frac{3\lambda+1}{\lambda+1}\right)=1$$

$$\Rightarrow 8\lambda - 4 = \lambda + 1 \Rightarrow \lambda = \frac{5}{7}$$

So, the ratio is $\frac{5}{7}:1$ or $5:7$. Putting $\lambda = \frac{5}{7}$ in (i), the coordinates of the point C of division are $\left(\frac{29}{12}, \frac{9}{4}, \frac{25}{6}\right)$.

22. **(b)** Let M be the point on the z-axis.

Then, the coordinates of M will be $(0, 0, z)$

Let M be equidis from the points $A(1, 5, 7)$ and $B(5, 1, -4)$

$$AM = \sqrt{(0-1)^2 + (0-5)^2 + (z-7)^2}$$

$$= \sqrt{(-1)^2 + (-5)^2 + (z-7)^2}$$

$$= \sqrt{1 + 25 + z^2 - 14z + 49}$$

$$= \sqrt{z^2 - 14z + 75}$$

$$BM = \sqrt{(0-5)^2 + (0-1)^2 + (z+4)^2}$$

$$= \sqrt{(-5)^2 + (-1)^2 + z^2 + 8z + 16}$$

$$= \sqrt{25 + 1 + z^2 + 8z + 16}$$

$$= \sqrt{z^2 + 8z + 42}$$

Then, $AM = Bm$

$$\therefore \sqrt{z^2 - 14z + 75} = \sqrt{z^2 + 8z + 42}$$

$$z^2 - 14z + 75 = z^2 + 8z + 42$$

$$-14z - 8z = 42 - 75 - 22z = -33$$

$$z = \frac{33}{22} \Rightarrow z = \frac{3}{2}$$

Thus, the coordinates of M are $\left(0, 0, \frac{3}{2}\right)$

23. **(a)** Let the point on the y-axis be $Y(0, y, 0)$ which is equidistant from the points $P(3, 1, 2)$ and $Q(5, 5, 2)$ Then, $PY = QY$

Now,

$$\sqrt{(0-3)^2 + (y-1)^2 + (0-2)^2}$$

$$= \sqrt{(0-5)^2 + (y-5)^2 + (0-2)^2}$$

$$\Rightarrow \sqrt{(-3)^2 + y^2 + 2y + 1 + (-2)^2}$$

$$= \sqrt{(-5)^2 + (y-5)^2 + (-2)^2}$$

$$\Rightarrow \sqrt{9 + y^2 + 2y + 1 + 4}$$

$$= \sqrt{25 + y^2 + 10y + 25 + 4}$$

$$\Rightarrow \sqrt{y^2 + 2y + 14} = \sqrt{y^2 + 10y + 54}$$

$$\Rightarrow y^2 + 2y + 14 = y^2 + 10y + 54$$

$$2y - 10y = 54 - 14$$

$$-8y = 40$$

$$-y = \frac{40}{8}$$

$$y = -5$$

Thus, the required point on the y-axis $(0, -5, 0)$

24. **(c, d)**

Let the point be $A(0, 0, z)$. Then,

$$AP = \sqrt{21}$$

$$\Rightarrow \sqrt{(0-1)^2 + (0-2)^2 + (z-3)^2} = \sqrt{21}$$

$$\Rightarrow (-1)^2 + (-2)^2 + (z-3)^2 = 21$$

$$\Rightarrow 1 + 4 + (z-3)^2 = 21$$

$$\Rightarrow (z-3)^2 = 21 - 5$$

$$\Rightarrow (z-3)^2 = 16$$

$$\Rightarrow z - 3 = \pm 4$$

$$\Rightarrow z - 3 = 4 or z - 3 = -4$$

$$\Rightarrow z = 7 or z = -1$$

Hence the coordinates of the required point are $(0, 0, 7)$ and $(0, 0, -1)$

25. **(a)**

26. **(a)** Suppose the plane $x + y + z = 5$ divides the line joining the points $A(2, -1, 3)$ and $B(-1, 2, 1)$ at a point C in the ratio $\lambda : 1$

Then coordinates of C are as follows:

$$\left(\frac{-\lambda+2}{\lambda+1}, \frac{2\lambda-1}{\lambda+1}, \frac{\lambda+3}{\lambda+1}\right)$$

Now, the point C lies on the plane $x + y + z = 5$.

Therefore, the coordinates of C must satisfy the equation of the plane.

$$\frac{-\lambda+2}{\lambda+1} + \frac{2\lambda-1}{\lambda+1} + \frac{\lambda+3}{\lambda+1} = 5$$

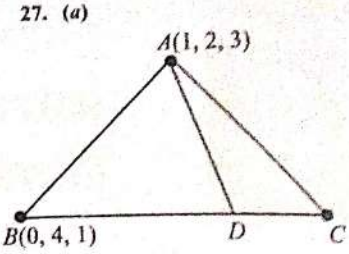
$$\Rightarrow -\lambda + 2 + 2\lambda - 1 + \lambda + 3 = 5\lambda + 5$$

$$\Rightarrow 2\lambda + 4 = 5\lambda + 5$$

$$\Rightarrow -3\lambda = 1 \quad \therefore \lambda = \frac{-1}{3}$$

So, the required ratio is 1:3 (externally).

27. **(a)**



$$AB = \sqrt{(-1)^2 + 2^2 + (-2)^2}$$

$$= \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$AC = \sqrt{(-2)^2 + (-3)^2 + (-6)^2}$$

$$= \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

AD is the internal bisector of $\angle BAC$

$$\therefore \frac{BD}{DC} = \frac{AB}{AC} = \frac{3}{7}$$

Thus, Ddivides BC internally in the ratio 3:7.

$$\therefore D \equiv \left(\frac{3(-1)+3\times 0}{3+7}, \frac{3(-1)+7(4)}{3+7}, \frac{3(-3)+7\times 1}{3+7}\right)$$

$$\Rightarrow D \equiv \left(\frac{-3}{10}, \frac{25}{10}, \frac{-2}{10}\right)$$

$$\Rightarrow D \equiv \left(\frac{-3}{10}, \frac{5}{2}, \frac{-1}{5}\right)$$

28. **(c)** Let $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ be the vertices of the given triangle, and let $D(1, 2, -3)$ $E(3, 0, 1)$ and $F(-1\ 1, -4)$ be the mid points of the sided BC, CA and AB respectively. D is the mid point of BC

$$\therefore \frac{x_2 + x_3}{2} = 1, \frac{y_2 + y_3}{2} = 2 \text{ and } \frac{z_{22} + z_3}{2} = -3$$

$$\Rightarrow x_2 + x_3 = 2, y_2 + y_3 = 4 \text{ and } z_2 + z_3 = -6 \quad ...(i)$$

E is the mid point of CA

$$\therefore \frac{x_1 + x_3}{2} = 3, \frac{y_1 + y_3}{2} = 0 \text{ and } \frac{z_1 + z_3}{2} = 1$$

$$\Rightarrow x_1 + x_3 = 6, y_1 + y_3 = 0 \text{ and } z_1 + z_3 = 2 \quad ...(ii)$$

F is the mid point of AB

$$\therefore \frac{x_1 + x_2}{2} = -1, \frac{y_1 + y_2}{2} = 1 \text{ and } \frac{z_1 + z_2}{2} = -4$$

$$\Rightarrow x_1 + x_2 = -2, y_1 + y_2 = 2 \text{ and } z_1 + z_2 = -8 \quad ...(iii)$$

Adding first three equations we get,

$2(x_1 + x_2 + x_3) = 6$, $2(y_1 + y_2 + y_3) = 6$

and $2(z_1 + z_2 + z_3) = -12$

$\Rightarrow x_1 + x_2 + x_3 = 3, y_1 + y_2 + y_3 = 3$

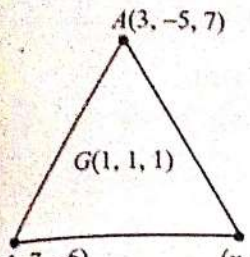
and $z_1 + z_2 + z_3 = -6$

The coordinate of the centroid is given by

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$$

$\Rightarrow (1, 1, -2)$

29. (b)



Let G be the centroid of ΔABC.

Given: $G \equiv (1, 1, 1)$

Let $C \equiv (x, y, z)$

Then, $1 = \frac{3-1+x}{3}$

$\Rightarrow 3 = 3-1+x$

$\Rightarrow 3 = 2+x \Rightarrow x = 1$

and $1 = \frac{-5+7+y}{3}$

$\Rightarrow 3 = 2+y$

$\Rightarrow y = 1$

and $1 = \frac{7-6+z}{3}$

$\Rightarrow 3 = 1+z$

$\Rightarrow z = 2$

$\therefore C \equiv (1, 1, 2)$

30. (b)

31. (a) Suppose C divides AB in the ratio $\lambda : 1$ Then the coordinates of C are $\left(\frac{8\lambda+2}{\lambda+1}, \frac{-3}{\lambda+1}, \frac{10\lambda+4}{\lambda+1}\right)$

The z-coordinate of C is 8.

$\therefore \frac{10\lambda+4}{\lambda+1} = 8$

$\Rightarrow 10\lambda+4 = 8\lambda+8$

$\Rightarrow 2\lambda = 4$

$\therefore \lambda = 2$

Hence, the coordinates of C are as follows:

$\left(\frac{8\lambda+2}{\lambda+1}, \frac{-3}{\lambda+1}, \frac{10\lambda+4}{\lambda+1}\right)$

$\Rightarrow \left(\frac{8\times2+2}{2+1}, \frac{-3}{2+1}, \frac{10\times2+4}{2+1}\right)$

$\Rightarrow \left(\frac{18}{3}, \frac{-3}{3}, \frac{24}{3}\right)$

$\therefore (6, -1, 8)$

Hence, the coordinates of C are $(6, -1, 8)$.

32. (d) $PQ^2 = (4-0)^2 + (-2-1)^2 + (1-2)^2 = 16+9+1 = 26$

$OP^2 = (0-0)^2 + (1-0)^2 + (2-0)^2 = 0+1+4 = 5$

$QO^2 = (0-4)^2 + (0+2)^2 + (0-1)^2 = 16+1+4 = 21$

Since, $PQ^2 = OP^2 + QO^2$

Hence, $\angle POQ = \frac{\pi}{2}$

33. (c) Length of diagonal $= \sqrt{(2-1)^2 + (-3+2)^2 + (5-3)^2}$

$= \sqrt{1+1+4} = \sqrt{6}$

$\therefore$ Length of the side $= \frac{\text{Length of diagonal}}{\sqrt{2}}$

$= \frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}$

34. (d) Suppose:

$A\,(5, -4, 2)$

$B\,(4, -3, 1)$

$C\,(7, 6, 4)$

$D\,(8, -7, 5)$

$AB = \sqrt{(4-5)^2 + (-3+4)^2 + (1-2)^2}$

$= \sqrt{(-1)^2 + (1)^2 + (-1)^2}$

$= \sqrt{1+1+1} = \sqrt{3}$

$BC = \sqrt{(7-4)^2 + (6=3)^2 + (4-1)^2}$

$= \sqrt{(3)^2 + (9)^2 + (3)^2}$

$= \sqrt{9+81+9} = \sqrt{99} = 3\sqrt{11}$

$CD = \sqrt{(8-7)^2 + (-7-6)^2 + (5-4)^2}$

$= \sqrt{(1)^2 + (-13)^2 + (1)^2}$

$= \sqrt{1+169+1} = \sqrt{171}$

$DA = \sqrt{(8-5)^2 + (-7+4)^2 + (5-2)^2}$

$= \sqrt{(3)^2 + (-3)^2 + (3)^2}$

$= \sqrt{9+9+9} = \sqrt{27} = 3\sqrt{3}$

We see that none of the sides are equal.

35. (b)

36. (c)

Suppose d is the diameter of the sphere. Then

$d^2 = (-1-3)^2 + (2-4)^2 + (3+1)^2$

$\Rightarrow d^2 = (-4)^2 + (-2)^2 + (4)^2$

$\Rightarrow d^2 = 16+4+16$

$\Rightarrow d^2 = 36$

$\Rightarrow d = 6$

Hence, radius of the sphere is 3 units.

37. The seven coordinates are as follows:

$(-3, 2, 5)$

$(3, 2, -5)$

$(-3, -2, 5)$

$(3, -2, -5)$

$(-3, 2, -5)$

$(-3, -2, -5)$

$(3, 2, 5)$

38. We know that the z-coordinate of every point on the xy-plane is zero. So, Let $P(x, y, 0)$ be a point on the xy-plane such that $PA = PB = PC$

Now, $PA = PB$

$\Rightarrow PA^2 = PB^2$

$\Rightarrow (x-1)^2 + (y+1)^2 + (0-0)^2$

$= (x-2)^2 + (y-1)^2 + (0-2)^2$

$\Rightarrow x^2 - 2x + 1 + y^2 + 2y$

$= x^2 - 4x + 4 + y^2 - 2y + 1 + 4$

$\Rightarrow -2x + 4x + 2y + 2y + 2 - 9 = 0$

$\Rightarrow 2x + 4y - 7 = 0$

$\Rightarrow 2x + 4y = 7$ ...(i)

$PB = PC \Rightarrow PB^2 = PC^2$

$\Rightarrow (x-2)^2 + (y-1)^2 + (0-2)^2$

$= (x-3)^2 + (y-2)^2 + (0+1)^2$

$\Rightarrow x^2 - 4x + 4 + y^2 - 2y + 1 + 4$

$= x^2 - 6x + 9 + y^2 - 4y + 4 + 1$

$\Rightarrow -4x + 6x - 2y + 4y + 9 - 14 = 0$

$\Rightarrow 2x + 2y - 5 = 0$

$\Rightarrow x + y = \frac{5}{2}$

$\Rightarrow x = \frac{5}{2} - y$ ...(ii)

Putting the value of x in equation (i) :

$2\left(\frac{5}{2} - y\right) + 4y = 7$

$\Rightarrow 5 - 2y + 4y = 7$

$\Rightarrow 5 + 2y = 7 \Rightarrow 2y = 2$

$\Rightarrow y = \frac{2}{2}$

$\therefore y = 1$

Putting the value of y in equation (ii) :

$x = \frac{5}{2} - 1$

$x = \frac{5-2}{2}$

$x = \frac{3}{2}$

Hence, the required point is $\left(\frac{3}{2}, 1, 0\right)$.

(ii) We know that the x-coordinate of every point on the yz-plane is zero. So, let $P(0, y, z)$ be a point on the yz-plane such that $PA = PB = PC$

Now, $PA = PB$

$\Rightarrow (0-1)^2 + (y+1)^2 + (z-0)^2$

$= (0-2)^2 + (y-1)^2 + (z-2)^2$

$\Rightarrow 1 + y^2 + 2y + 1 + z^2 = 4 + y^2 - 2y + 1 + z^2 - 4z + 4$

$\Rightarrow 2y + 2 = -2y - 4z + 9$

$\Rightarrow 2y + 2y - 4z = 9 - 2$

$\Rightarrow 4y - 4z = 7$

$\Rightarrow y - z = 7/4$ ...(*i*)

$PB = PC$

$\Rightarrow PB^2 = PC$

$\Rightarrow (0-2)^2 + (y-1)^2 + (z-2)^2$

$= (0-3)^2 + (y-2)^2 + (z+1)^2$

$\Rightarrow 4 + y^2 - 2y + 1 + z^2 - 4z + 4$

$= 9 + y^2 - 4y + 4 + z^2 + 2z + 1$

$\Rightarrow -2y - 4z + 9 = -4y + 2z + 14$

$\Rightarrow -2y + 4y - 4z - 2z = 14 - 9$

$\Rightarrow 2y - 6z = 5$

$$\Rightarrow y - 3z = \frac{5}{2}$$

$$\therefore y = \frac{5}{2} + 3z \quad \text{...(ii)}$$

Putting the value of y in equation (*i*):

$$y - z = \frac{7}{4} \Rightarrow \frac{5}{2} + 3z - z = \frac{7}{4}$$

$$\Rightarrow 2z = \frac{7}{4} - \frac{5}{2} \Rightarrow 2z = \frac{7-10}{4}$$

$$\Rightarrow 2z = \frac{-3}{4}$$

$$\therefore z = \frac{-3}{8}$$

Putting the value of z in equation (*ii*):

$$y = \frac{5}{2} + 3z \Rightarrow y = \frac{5}{2} + 3\left(\frac{-3}{8}\right)$$

$$\Rightarrow y = \frac{5}{2} - \frac{9}{8} \Rightarrow y = \frac{20-9}{8}$$

$$\therefore y = \frac{11}{8}$$

Hence, the required point is $\left(0, \frac{11}{8}, \frac{-3}{8}\right)$

(*iii*) We know that the y-coordinate of every point on the zx-plane is zero. So, let $P(x, 0, z)$ be a point on the zx-plane such that $PA = PB = PC$

Now, $PA = PB$

$\Rightarrow (x-1)^2 + (0+1)^2 + (z-0)^2$

$= (x-2)^2 + (0-1)^2 + (x-2)^2$

$\Rightarrow x^2 + 1 - 2x + 1 + z^2 = x^2 - 4x + 4 + 1 + z^2 - 4z + 4$

$\Rightarrow -2x + 2 = -4x - 4z + 9$

$\Rightarrow -2x + 4x - 4z = 7$

$\Rightarrow 2x - 4z = 7$

$\Rightarrow x - 2z = 7/2$ ...(*i*)

$PB = PC$

$\Rightarrow PB^2 = PC^2$

$\Rightarrow (x-2)^2 + (0-1)^2 + (z-2)^2$

$= (x-3)^2 + (0-2)^2 + (z+1)^2$

$\Rightarrow x^2 - 4x + 4 + 1 + z^2 - 4z + 4$

$= x^2 - 6x + 9 + 4 + z^2 + 2z + 1$

$\Rightarrow -4x - 4z + 9 = -6x + 2z + 14$

$\Rightarrow -4x + 6x - 4z - 2z = 14 - 9$

$\Rightarrow -4x + 6x - 4z - 2z = 14 - 9$

$\Rightarrow 2x - 6z = 5 \Rightarrow x - 3z = \frac{5}{2}$

$$\therefore x = \frac{5}{2} + 3z \quad \text{...(ii)}$$

Putting the value of x in equation (*i*):

$$x - 2z = \frac{7}{2} \Rightarrow \frac{5}{2} + 3z - 2z = \frac{7}{2}$$

$$\Rightarrow \frac{5}{2} + z = \frac{7}{2} \Rightarrow z = \frac{7}{2} - \frac{5}{2}$$

$$\Rightarrow z = \frac{7-5}{2} \Rightarrow z = \frac{2}{2}$$

$\therefore z = 1$

Putting the value of z in equation (*ii*):

$$x = \frac{5}{2} + 3z \Rightarrow x = \frac{5}{2} + 3(1)$$

$$\Rightarrow x = \frac{5}{2} + 3 \Rightarrow x = \frac{5+6}{2}$$

$$\therefore x = \frac{11}{2}$$

Hence, the required point is $\left(\frac{11}{2}, 0, 1\right)$

39. Suppose C divides AB in the ratio $\lambda : 1$

Then, the coordinates of C are

$$\left(\frac{-\lambda+2}{\lambda+1}, \frac{2\lambda+3}{\lambda+1}, \frac{-3\lambda+4}{\lambda+1}\right)$$

But, the coordinates of C are $(-4, 1, -10)$ ·10

$$\therefore \frac{-\lambda+2}{\lambda+1} = -4, \frac{2\lambda+3}{\lambda+1} = 1,$$

$$\frac{-3\lambda+4}{\lambda+1} = -10$$

$\Rightarrow -\lambda + 2 = -4\lambda - 4, 2\lambda + 3 = \lambda + 1,$

$-3\lambda + 4 = -10\lambda - 10$

$\Rightarrow 3\lambda = -6, \lambda = -2, 7\lambda = -14$

$\therefore \lambda = -2, \lambda = -2, \lambda = -2$

From these three equations, we have:

$\lambda = -2$

So, C divides AB in the ratio 2 : 1 (externally).

By Aditya Mohod
username - muje_kabhi_dekha_nahi

By Aditya Mahad
username - muje_kabhi_dekha_nahi

CHAPTER 19

Probability

Prarambh (Topicwise)

1. (c) Probability of getting 1 in first throw $= \frac{1}{6}$

Probability of not getting 1 in second throws $= \frac{5}{6}$

Both are independent events, so the required probability $= \frac{1}{6} \times \frac{5}{6} = \frac{5}{36}$.

2. (b) Total number of ways = 36

Favourable numbers of cases are (1 4), (2, 3), (3, 2), (4, 1), (1, 5), (2, 4), (3, 3), (4, 2), (5, 1) = 9

Hence the required probability

$$= \frac{9}{36} = \frac{1}{4}.$$

3. (a) In a non-leap year's there are 365 days which have 52 weeks and 1 day. If this day is a Tuesday or Wednesday, then the year will have 53 Wednesday.

∴ Required probability $= \frac{1}{7}$

4. (b) Since, the set of three consecutive numbers from 1 to 20 are 123, 234, 345, ... 18, 19, 20 i.e., 18.

P(numbers are consecutive)

$$= \frac{18}{{}^{20}C_3} = \frac{18}{1140} = \frac{3}{190}$$

P(three numbers are not consecutive)

$$= 1 - \frac{3}{190} = \frac{187}{190}$$

5. (b) Favourable cases for one are three *i.e.* 2, 4 and 6 and for other are two *i.e.* 3, 6.

Hence required probability

$$= \left[\left(\frac{3\times 2}{36}\right) 2 - \frac{1}{36}\right] = \frac{11}{36}$$

{As same way happen when dice changes numbers among themselves}

6. (d) P (at least 1H) = 1 − P (No head)

1 − P (four tail) $= 1 - \frac{1}{16} = \frac{15}{16}$

7. (c) Since in a back of 52 cards 26 are red colour and 26 are black colour.

∴ P(both cards of opposite colour)

$$= \frac{26}{52} \times \frac{26}{51} \times \frac{26}{51} \times \frac{26}{51} = 2 \times \frac{26}{52} \times \frac{26}{51} = \frac{26}{51}$$

8. (a) Total number of ways $= 2^n$

If head comes odd times, then favourable ways $= 2^{n-1}$

∴ Required probability $= \frac{2^{n-1}}{2^n} = \frac{1}{2}$.

9. (c) Required probability

$$= \frac{{}^3C_3 + {}^7C_3 + {}^4C_3}{{}^{14}C_3}$$

$$= \frac{1+35+4}{14.13.2} = \frac{40}{14.26} = \frac{10}{91}.$$

10. (c) If two persons sit next to each other, then consider these two person as 1 group.

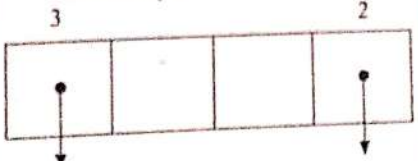
Now, we have to arrange 6 persons.

∴ Number of arrangement = 2! × 6!

Total number of arrangement of 7 person = 7!

Required probability $= \frac{2!6!}{7!} = \frac{2}{7}$

11. (d) We have, to form four-digit number using the digit 0, 2, 3 and 5 which are divisible by 5.



If 0 is fixed at units place = 3 × 2 × 1 = 6

If 5 is fixed at units place = 2 × 2 × 1 = 4

Total four-digit numbers divisible by 5 = 6 + 4 = 10

∴ Required probability $= \frac{10}{18} = \frac{5}{9}$

12. (b) Total ways of arrangements $= \frac{8!}{2!.4!}$

•w•x•y•z•

Now 'S' can have places at dot's and in places of w, x, y, z we have to put 2A's, one I and one N.

Therefore favourable ways $= 5\left(\frac{4!}{2!}\right)$

Hence required probability

$$= \frac{5.4!\,2!\,4!}{2!\,8!} = \frac{1}{14}$$

13. (a) Required probability

$$= \frac{{}^3C_1}{{}^7C_1} \times \frac{{}^2C_1}{{}^5C_1} = \frac{1}{7}$$

14. (a) For mutually exclusive events

$P(A \cap B) = 0$

$P(A \cup B) = P(A) + (B) - P(A \cap B)$

$\Rightarrow P(A \cup B) = P(A) + P(B)$

$\Rightarrow P(A) + P(B) \le 1$

$\Rightarrow P(A) + 1 - P(\overline{B}) \le 1$

$[\because P(B) = 1 - P(\overline{B})]$

$\therefore P(A) \le P(\overline{B})$

15. (c) Required probability

$$= \frac{{}^{37}C_2}{{}^{38}C_3} = \frac{\binom{37}{2}}{\binom{38}{3}}.$$

16. (a) Given, $P(A \cup B) = P(A \cap B)$

$P(A) + P(B) - P(A \cap B) = P(A \cap B)$

17. (c) Since we have $P(A + B) = P(A) + P(B) - P(AB)$

$\Rightarrow 0.7 = 0.4 + P(B) - 0.2 \Rightarrow P(B) = 0.5$

18. (d) Required probability is

$P(\text{Red} + \text{Queen}) - P(\text{Red} \cap \text{Queen})$

$= P(\text{Red}) + P(\text{Queen}) - P(\text{red} \cap \text{Queen})$

$$= \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$$

19. (a) Since we have

$P(A \cup B) + P(A \cap B) = P(A) + P(B) = P(A) + \frac{P(A)}{2}$

$$\Rightarrow \frac{7}{8} = \frac{3P(A)}{2} \Rightarrow P(A) = \frac{7}{12}.$$

20. (c) $P(A \cup B) = 0.9$ and $P(A \cap B) = 0.3$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow 0.9 = 1 - P(A') + 1 - P(B') - 0.3$

$\Rightarrow P(A') + P(B') = 0.8$

21. (b) $P(A') = 0.3$, $\therefore P(A) = 0.7$

$P(B') = 0.6$, $P(B) = 0.4$ and $P(A \cap B') = 0.5$

$P(A \cup B') = P(A) + P(B') - P(A \cap B')$

$= 0.7 + 0.6 - 0.5 = 0.8$.

22. (c) If all girls sit together, then considered it as 1 group.

∴ Arrangement of 6 + 1 = 7 person in a row is 7! And the girls interchanges their seets in 6! Ways.

∴ Required probability = $\frac{6!7!}{12!} = \frac{1}{132}$

23. (b) Total number of alphabet in the word probability = 11

Number of vowels = 4

P(letter is vowel) = $\frac{4}{11}$

24. (c) Given, P(A fail) = 0.2

and P(B fail) = 0.3

∴ P(either A or B fail) ≤ P(A fail) + P(B fail)

≤ 0.2 + 0.3

≤ 0.5

25. (c) Given,

$P(A \cup B) = 0.6$ and $P(A \cap B) = 0.2$

∴ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow 0.6 = P(A) + P(B) - 0.2$

$\Rightarrow P(A) + P(B) = 0.8$

∴ $P(\bar{A}) + P(\bar{B}) = 1 - P(A) + 1 - P(B)$

$= 2 - [P(A) + P(B)] = 2 - 0.8 = 1.2$

26. (b) If M and N are any two events.

∴ $P(M \cup N) = P(M) + P(N) - P(M \cap N)$

27. (c) Since sum of $1 + 2 + 3 + ... 9 = \frac{9 \times 10}{2}$

= 45 is divisible by 9, hance all no. will be divisible by 9.

28. (a) Max sum = 12

$\left.\begin{matrix} 6+6=12 \\ 6+5=11 \\ 6+4=10 \\ 5+5=10 \\ 5+6=11 \\ 4+6=10 \end{matrix}\right\}$ 6 cases

$P = \frac{6}{36} = \frac{1}{6} = \frac{1}{6}$

29. (d) Required Probability

$= \frac{{}^{7}C_3 - 2}{{}^{10}C_3} = \frac{11}{40}$

30. (a) Required probability

$= \frac{{}^{3}C_2 \times 1 \times 1}{2^3 - 1} = \frac{3}{7}$

Prabal (JEE Main Level)

1. (c) Since one die can be thrown in six ways to obtain any one of the six numbers marked on its six faces. Therefore, if three dice are thrown, the total number of elementary events = 6×6×6=216.

Let A be the event of getting a total of at lest 6. Then, $\bar{A}$ denotes the event of getting a total of less than 6 *i.e.*, 3, 4, 5.

∴ $\bar{A}$ = [(1, 1, 1), (1, 1, 2), (2, 1, 1), (1, 1, 3), (3, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1)]

So, favourable number of elementary events = 10

∴ $P(\bar{A}) = \frac{10}{216}$

$P(A) = \frac{10}{216} \Rightarrow P(A) = 1 - \frac{10}{216} = \frac{103}{108}$.

2. (b) Let S be the sample space associated with the given random experiment and A denote the event "getting a ticket bearing a number which is a multiple of 3 or 7. Then,

S = [1, 2, ..., 20] and A = [3, 6, 9, 12, 15, 18, 7, 14]

∴ Total number of elementary vents = 20

Favourable number of elementary events = 8

Hence, required probability = 8/ 20 = 2/5

3. (a) There are 20 balls in the bag out of which 2 balls can be drawn in ${}^{20}C_2$ ways.

So, total number of elementary events = ${}^{20}C_2 = 190$.

(i) There are 9 red balls out of which 2 balls can be drawn in ${}^{9}C_2$ ways.

∴ Favourble number of elementary events = ${}^{9}C_2 = 36$.

So, required probability = $\frac{36}{190} = \frac{18}{95}$.

(ii) There are 7 white balls out of which one white can be drawn in ${}^{7}C_1$ ways. One ball from the remaining 13 balls can be drawn in ${}^{13}C_1$ ways. Therefore, one white and one other colour ball can be drawn in ${}^{7}C_1 \times {}^{13}C_1$ ways.

So, favourable number of elementary events = ${}^{7}C_1 \times {}^{13}C_1$

Here, required probability

$= \frac{{}^{7}C_1 \times {}^{13}C_1}{{}^{20}C_2} = \frac{91}{190}$.

(iii) Two balls drawn are of the same colour means that either both are red or both are white or both are black. Out of 9 red balls two red balls can be drawn in ${}^{9}C_2$ ways. Similarly, two white balls can be drawn from 7 white balls in ${}^{7}C_2$ ways and two black balls from 4 black balls in ${}^{4}C_2$ ways. Therefore

The number of ways of drawing 2 balls of the same colour = ${}^{9}C_2 + {}^{7}C_2 + {}^{4}C_2 = 36 + 21 + 6 = 63$

∴ Favourable number of elementary events = 63.

So, required probability = $\frac{63}{190}$.

(iv) Out of 7 white balls one white ball can be drawn in ${}^{7}C_1$ ways and out of 9 red balls one red ball can be drawn in ${}^{9}C_1$ ways. Therefore,

One white and one red ball can be drawn in ${}^{7}C_1 \times {}^{9}C_1$ ways.

So, favourable number of elementary events = ${}^{7}C_1 \times {}^{9}C_1 = 63$.

So, required probability = $\frac{63}{190}$.

4. (a) There are 9 persons viz. 3 men, 2 women and 4 children. Out of these 9 persons 4 persons can be selected in ${}^{9}C_4 = 126$ ways.

∴ Total number of elementary events = 126

(i) 1 man, 2 woman and 2 children can be selected in ${}^{3}C_1 \times {}^{2}C_1 \times {}^{4}C_2 = 36$ ways.

∴ Favourable number of elementary events = 36

So, required probability = $\frac{36}{126} = \frac{2}{7}$

(ii) Exactly 2 children means : 2 children out of 4 children and 2 persons from 5 persons consisting of 3 men and 2 women. This can be done in ${}^{4}C_2 \times {}^{5}C_2$ ways,

∴ Favourable number of elementary events

= ${}^{4}C_2 \times {}^{5}C_2 = 60$

So, required probability = $\frac{60}{126} = \frac{10}{21}$

(iii) we gave to select 4 persons of which 2 are women and the remaining 2 are chosen from 7 persons consisting of 3 men and 4 children. This can be done in ${}^{2}C_2 \times {}^{7}C_2$ ways.

∴ Favourable number of elementary events

= ${}^{2}C_2 \times {}^{7}C_2 = 21$

So, required probability = $\frac{21}{126} = \frac{1}{6}$

5. (b) Total number of elementary events associated with the random experiment of throwing three dice simultaneously is $6 \times 6 \times 6 = 216$.

(i) A total of 5 can be obtain in one of the following ways:

(1, 1, 3), (3, 1, 1), (1, 3, 1), (2, 2, 1), (1, 2, 2), (2, 1, 2)

∴ Favourable number of elementary events = 6

Hence, required probability

= $\frac{6}{216} = \frac{1}{36}$

obability $= \frac{63}{190}$.

white balls one white ball in ${}^7C_1$ ways and out of 9 d ball can be drawn in ${}^9C_1$ e,

ne red ball can be drawn in

number of elementary $C_1 = 63$.

obability $= \frac{63}{190}$.

9 persons viz. 3 men, 2 children. Out of these rsons can be selected in

of elementary events

n and 2 children can be ${}^2C_1 \times {}^4C_2 = 36$ ways, number of elementary

obability $= \frac{36}{126} = \frac{2}{7}$

dren means : 2 children en and 2 persons from isting of 3 men and 2 an be done in ${}^4C_2 \times {}^5C_2$

umber of elementary

obability $= \frac{60}{126} = \frac{10}{21}$

ect 4 persons of which nd the remaining 2 are persons consisting of 3 ren. This can be done in

number of elementary

bability $= \frac{21}{126} = \frac{1}{6}$

er of elementary events the random experiment e dice simultaneously is

be obtain in one of the

(1, 3, 1), (2, 2, 1), (1, 2,

number of elementary

probability

ons MATHEMATICS

(iii) A total of at most 5 can be obtained in any one of the following ways: (1, 1, 1), (1, 1, 2), (2, 1, 1), (1, 2, 1), (1, 1, 3), (1, 3, 1), (3, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1)

$\therefore$ Favourable number of elementary events = 10

Hence, required probability

$= \frac{10}{216} = \frac{5}{108}$

(iii) Let A be the event "getting a total of at least 5". Then, P(A) = 1 – $P(\bar{A})$ = 1 – P (Getting a total of at most 4)

A total of at most 4 can be obtained in any one of the following ways: (1, 1, 1), (1, 1, 2), (2, 1, 1), (1, 2, 1).

So, Favourable number of elementary events to $\bar{A}$ is 4.

$\therefore P(\bar{A}) = \frac{4}{216} \Rightarrow 1 - P(A) = \frac{4}{216}$

$\Rightarrow P(A) = 1 - \frac{4}{216} = \frac{212}{216} = \frac{53}{54}$

6. (*a*) It given that the odds is favour of an event are 3:5. Therfore,

Favourable number of elementary events = 3x.

Unfavourable number of elementary events = 5x.

So, total number of elementary events = 3x + 5x = 8x.

Hence, probability of the occurrence of the event $= \frac{3x}{8x} = \frac{3}{8}$

7. (*d*) It is given that a PIN is a sequence of four symbols selected from 36 (26 letters and 10 digits) symbols. Therefore,

Total numbers of PINs = 36 × 36 × 36 × 36

$= 36^4 = 1,679,616$

Total number of PINs with distinct symbols

$= 36 \times 35 \times 34 \times 33 = 1,413,720$.

The number of PINs that contain at least one repeated symbol = 1,679,616 – 1, 413,720 = 2,65,896

Hence,

The probability that a randomly chosen PIN contains a respected symbol $= \frac{2,65,896}{1,679,616} = 0.1583$

8. (*a*) Multiple of 2 from 1 to 1000 are 2, 4, 6, 8, ..., 1000

Let n be the number of terms of above series.

$\therefore$ nth term = 1000

$\Rightarrow 2 + (n-1)^2 = 1000$

$\Rightarrow 2 + 2n - 2 = 1000 \Rightarrow 2n = 1000$

$\therefore n = 500$

Since, the number of multiple of 2 are 500.

So, the multiple of 9 are 9, 18, 27, ..., 999

Let m be the number of term in above series.

$\therefore$ m^{th} term = 999

$\Rightarrow 9 + (m-1)9 = 999$

$\Rightarrow 9 + 2m - 9 = 999 \Rightarrow 9m = 999$

$\therefore m = 111$

Since, the number of multiple of 9 are 111. So, the multiple of 2 and 9 both are 18, 36, ..., 990

Let p be the number of terms above series.

$\therefore$ pth term = 990

$\Rightarrow 18 + (p-1)18 = 990 \Rightarrow 18 + 18p - 18 = 990$

$\Rightarrow 18p = 990$

$\therefore p = \frac{990}{18} = 55$

Since, the number of multiple of 2 and 9 are 55.

$\therefore$ Number of multiple of 2 or 9 = 500 + 111 – 55 = 556

$\therefore$ Required probability

$= \frac{n(E)}{n(S)} = \frac{556}{1000} = 0.556$

9. (*a*) In a through of a die there is 6 sample points.

(i) If 2 appears on the kth roll of the die.

So, first $(k-1)$ roll have 5 outcomes each and kth results 2 i.e., 1 outcome.

$\therefore$ Number of element of sample space correspond to the event that 2 appears on the kth roll of the die = 5^{k-1}

(ii) If we consider that 2 appears not later than kth roll of the die, then it is possible that 2 comes I first throw *i.e.*, 1 outcome.

i.e., 1 outcome, possible outcome = 5 × 1 = 5

Similarly, if 2 does not appear in second throw and appears in third throw.

$\therefore$ Possible outcomes = 5 × 5 × 1

Given, series = $|1 + 5 + 5 \times 5 + 5 \times 5 + \ldots + 5^{k-1}$

$= 1 + 5 + 5^2 + 5^3 + \ldots + 5^{k-1}$

$= \frac{1(5^k - 1)}{5 - 1} = \frac{5^k - 1}{4}$

10. (*a*) It is given that, 2 × Probability of even number = Probability of odd number

$\Rightarrow P(O) = 2P(E)$

$\Rightarrow P(O) : P(E) = 2 : 1$

$\therefore$ Probability of occurring odd number, $P(O) = \frac{2}{2+1} = \frac{2}{3}$ and probability of occurring 5each number,

$P(E) = \frac{1}{2+1} = \frac{1}{3}$

Now, G be the even that a number greater than 3 occur in a single roll of die.

So, the possible outcomes are 4, 5 and 6 out of which are even and one odd.

$\therefore$ Required probability = $P(G) = 2 \times P(E) \times P(O)$

$= 2 \times \frac{1}{3} \times \frac{2}{3} = \frac{4}{9}$

11. (*d*) Let E_1 be the event that family own colour television sett and E_2 be the event that family

It is given that, $P(E_1) = 0.87$

$P(E_2) = 0.36$ and $P(E_1 \cap E_2) = 0.30$

We have to find probability that a family owns either anyone or both kind of sets *i.e.*, $P(E_1 \cup E_2)$.

Now, $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

[by addition theorem]

$= 0.87 + 0.36 - 0.30 = 0.93$

12. (*b*) Let E_1 = John promoted

E_2 = Rita promoted

E_3 = Aslam promoted

E_4 = Gurpreet promoted

Given sample space, S = {John promoted, Rita promoted, Aslam promoted, Gurpreet promoted}

i.e.,

It is given that chances of John's promotion is same as that of Gurpreet.

$P(E_1) = P(E_4)$

Rita's chances of promotion are twice as likely as John.

$P(E_2) = 2P(E_1)$

And Aslam's chances of promotion are four times that of John.

$P(E_3) = 4P(E_1)$

Now,

$P(E_1) + P(E_2) + P(E_3) + P(E_4) = 1$

$\Rightarrow P(E_1) + 2P(E_1) + 4P(E_1) + P(E_1) = 1 \Rightarrow 8P(E_1) = 1$

$\therefore P(E_1) = \frac{1}{8}$

(i) P(John promoted) = $P(E_1) = \frac{1}{8}$

P(Rita promoted)

$= P(E_2) = 2P(E_1) = 2 \times \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$

P(Aslam promoted) $P(E_3) = 4P(E_1)$

$= 4 \times \frac{1}{8} = \frac{1}{2}$

P(Gurpreet promoted) = $P(E_4) = P(E_1) = \frac{1}{8}$

(ii) A = John promoted or Gurpreet promoted

$A = E_1 \cup E_4$

$P(A) = P(E_1 \cup E_4) = P(E_1) + P(E_4) - P(E_1 \cap E_4)$

$= \frac{1}{8} + \frac{1}{8} - 0 \quad [\because P(E_1 \cap E_4) = 0]$

$= \frac{2}{8} = \frac{1}{4}$

13. (*a*) It is given that one of the two urn is chosen, then a ball is randomly chosen from the urn, then a second ball is chosen at random from the same urn without replacing the first ball

(i) $\therefore$ Sample space S

= {B_1B_2, B_1W, B_2W_1, B_2W, WB_1, WB_2, BW_1, BW_2, W_1B, W_1W_2, W_2B, W_2W_1}

$\therefore$ Total sample point = 12

(ii) If two black ball are chosen.

So, the favorable events are B_1, B_2, B_3, B_4 i.e., 2

$\therefore$ Required probability $= \frac{2}{12} = \frac{1}{6}$

(iii) If two balls of opposite colour are chosen.

So, the favorable events are B_1W_1, B_2W_1, WB_1, WB_2, BW_1, BW_2, W_1B_1, W_2B i.e., 8.

$\therefore$ Required probability $= \frac{8}{12} = \frac{2}{3}$

14. (*b*) $\because$ Number of red balls = 8

And number of white balls = 5

(i) P(all the three balls are white)

$$= \frac{{}^5C_3}{{}^{13}C_3} = \frac{\frac{5!}{3!2!}}{\frac{13!}{3!10!}} = \frac{5!}{3!2!} \times \frac{3!10!}{13!}$$

$$= \frac{5 \times 4 \times 3 \times 2!}{2!} \times \frac{10!}{13 \times 12 \times 11 \times 10!} = \frac{5 \times 4 \times 3}{13 \times 12 \times 11}$$

$$= \frac{5}{13 \times 11} \times \frac{5}{143}$$

$$= \frac{5 \times 4 \times 3}{13 \times 12 \times 11} = \frac{5}{13 \times 11} = \frac{5}{143}$$

(ii) P(All the three balls are red)

$$= \frac{{}^8C_3}{{}^{13}C_3} = \frac{\frac{8!}{3!5!}}{\frac{13!}{3!10!}} = \frac{8!}{3! \times 5!} \times \frac{3!10!}{13!}$$

$$= \frac{8 \times 7 \times 6 \times 5!}{5!} \times \frac{10!}{13 \times 12 \times 11 \times 10!}$$

$$= \frac{8 \times 7 \times 6}{13 \times 12 \times 11} = \frac{28}{143}$$

(iii) P(One ball is red and two balls are white)

$$= \frac{{}^8C_1 \times {}^5C_2}{{}^{13}C_3} = \frac{8 \times 10}{13 \times 6 \times 11} = \frac{40}{143}$$

15. (*d*) Total number of letters in the word 'ASSASSINATION' are 13.

Out of which 3A's, 4S's, 2I's, 2N's, 1 T's and 1O.

(i) If four S's come consecutively in the word, then we considers these 4 S's as 1 group.

Now, the number of laters is 10.

S	S	S	S	A	A	A	I	I	N	N	T	O
1				9								

Number of words when all S's are together $= \frac{10!}{3!2!2!}$

Total number of word using letters of the word 'ASSASSINATION' $= \frac{13!}{3!4!2!2!}$

$\therefore$ Required probability $= \frac{10!}{3!2!2! \times 13!}$

$$= \frac{10! \times 4!}{13!} = \frac{4!}{13 \times 12 \times 11} = \frac{24}{1716} = \frac{2}{143}$$

(ii) If 2 I's and 2N's come together, then there as 10 alphabets.

Number of would when 2 I's and N's are come together $= \frac{10!}{3!4!} \times \frac{4!}{2!2!}$

$\therefore$ Required probability

$$= \frac{\frac{10!4!}{3!4!2!2!}}{\frac{13!}{3!4!2!2!}} = \frac{4!10!}{2!2!3!4!} \times \frac{3!4!2!2!}{13!}$$

$$= \frac{4!10!}{13!} = \frac{4!}{13 \times 12 \times 11} = \frac{24}{13 \times 12 \times 11} = \frac{2}{143}$$

(iii) If all A's are coming together, then there are 11 alphabets.

Number of words when all A's come together

$$= \frac{11!}{4!2!2!}$$

Probability when all A's come together

$$= \frac{\frac{11!}{4!2!2!}}{\frac{13!}{4!3!2!2!}} = \frac{11!}{4!2!2!} \times \frac{4!3!2!2!}{13!}$$

$$= \frac{11 \times 3!}{13!} = \frac{6}{13 \times 12} = \frac{1}{26}$$

Required probability when all A's does not come together

$$= 1 - \frac{1}{26} = \frac{25}{26}$$

(iv) If no two A's are together, then first we arrange the alphabets except A's.

S	S	S	S	I	N	T	I	O	N

All the alphabets except A's are arranged in $\frac{10!}{4!2!2!}$.

There are 11 vacant place between these alphabets.

So, 3 A's can be place in 11 places in ${}^{11}C_3$ ways $= \frac{11!}{3!8!} \times \frac{10!}{4!2!2!}$

$\therefore$ Total number of words when no two A's together

Required probability $= \frac{11!}{3!8!} \times \frac{10!}{4!2!2!}$

$$= \frac{10 \times 9}{13 \times 12} = \frac{90}{156} = \frac{15}{26}$$

16. (*d*) $S = \{E_1, E_2, E_3, E_4, E_5, E_6, E_7, E_8, E_9\}$

$A = \{E_1, E_5, E_8,\}$, $B = \{E_2, E_5, E_8, E_9\}$

$P(E_1) = P(E_2) = 0.08$

$P(E_3) = P(E_4) = P(E_5) = 0.1$

$P(E_6) = P(E_7) = P(E_8) = P(E_9) = 0.7$

(i) $P(A) = P(E_1) + P(E_5) + P(E_8)$

$= 0.08 + 0.1 + 0.07 = 0.25$

(ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ ... (i)

Now, $P(B) = P(E_2) + P(E_5) + P(E_8) + P(E_9)$

$= 0.08 + 0.1\ 0.07 + 0.07 = 0.32$

$A \cap B = \{E_5, E_8\}$

$P(A \cap B) = P(E_5) + P(E_8) = 0.1 + 0.7 = 0.17$

On substituting these values in Eq. (i), we get

$P(A \cup B) = 0.25 + 0.32 - 0.17 = 0.40$

(iii) $A \cup B = \{E_1, E_2, E_5, E_8, E_9\}$

$P(A \cup B) = P(E_1) + P(E_2) + P(E_5) + P(E_8) + P(E_9)$

$= 0.08 + 0.08 + 0.1 + 0.07 + 0.07 = 0.40$

(iv) $\because P(\overline{B}) = 1 - P(B) = 1 - 0.32 = 0.68$

and $\overline{B} = \{E_1, E_3, E_4, E_6, E_7\}$

$\therefore P(\overline{B}) = P(E_1) + P(E_3) + P(E_4) + P(E_6) + P(E_7)$

$= 0.08 + 0.1 + 0.1 + 0.2 + 0.2 = 0.68$

17. (*a*) Total number of words formed $= \frac{6!}{3! \times 2!} = 60$. The number of words containing the pattern

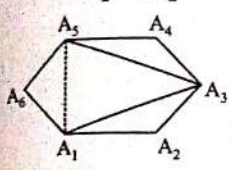
BAN $= \frac{4!}{2!} = 12$.

So, the required probability $= \frac{60-12}{60} = \frac{4}{5}$

18. *(b)* Probability $= \frac{5\times4}{9\times8} + \frac{4\times5}{9\times8} = \frac{5}{9}$ (EE or OE)

19. *(b)* (i) 6 vertical & 5 horizontal lines.

$p = \frac{5\times4+4\times3+3\times2+2\times1}{{}^6C_2 \quad {}^5C_2} = \frac{4}{15}$



(ii) $A_1A_3A_5$ or $A_2A_4A_6 = \frac{2}{{}^6C_3} = \frac{1}{10}$

20. *(a)*

(i) $p(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 0.4 + 0.48 - 0.16 = 0.72$

(ii) $P(B) - P(A \cap B) = 0.32$

(iii) $p(\overline{A} \cap \overline{B}) = p(A \cup B)^c = 1 - p(A \cup B)$
$= 1 - 0.72$
$= 0.28$
insert values and obtain answer

(iv) $P(A \cup B) - P(A \cap B) = 0.56$

21. (a) The sample space associated with the random experiment is given in example 1.
We have,
A = The sum is even
= The sum is either 2 or 4 or 6, or 8 or 10 or 12
= [(1, 1), (2, 2), (1, 3), (1, 5), (2, 4), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (4, 6), (5, 1), (5, 3), (5, 5), (6, 2), (6, 4), (6, 6)]
B. = The sum is a multiple of 3
= The sum is either 3 or 6 or 9 or 12
= ((1, 2), (2, 1), (1, 5), (5, 1), (2, 4), (4, 2), (3, 3), (3, 6), (6, 3), (4, 5), (5, 4), (6, 6))
C. = The sum is less then 4 = (The sum is 2 or 3 = (1, 1), (1, 2), (2, 1))
D. = The sum is greater than 11 = The sum is 12 = [(6, 6)]
We observe that A ∩ B = [(1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (6, 6)] ≠ ϕ. So, A and B are not mutually exclusive events.
Similarly, we observe that A ∩ C ≠ ϕ, A ∩ D ≠ ϕ, B ∩ C ≠ ϕ, B ∩ D ≠ ϕ. and, C ∩ D = ϕ.
Hence, C and D are mutually exclusive events.

22. **[7]** P (Not 8) = 1 − P (8) $\begin{matrix}6+2\\5+3\\4+4\end{matrix}\}$ #5 = 1

$-\frac{5}{36} = \frac{31}{36}$

P (Not 11) = 1 − P (11) 6 + 5 # 2

$= 1 - \frac{2}{36} = \frac{34}{36}$. Total cases of obtaining 8 or 9 are 7.

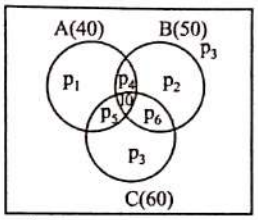
Here for cases = 29.

$P = \frac{29}{36}$.

23. **[1]** We known that

$n(A-B) = n(A) - n(A \cap B) = n(A \cup B) - n(B)$
$= n(A \cap \overline{B}) = n(S) - n(\overline{A} \cap B)$ dividing by n(S) in total we get
$P(A-B) = P(A) - P(A \cap B) = P(A \cup B) - P(B)$
$= P(A \cap \overline{B}) = 1 - P(\overline{A} \cap B)$

24. **[16]**



$p_1 + p_2 + p_3 = 70$ and $p_4 + p_5 + p_6 = 25$
A : Event that he has membership of exactly two clubs $P(A)$

$= \frac{25}{70+25+10} = \frac{25}{115} = \frac{5}{21}$

PYQ's (Past Year Questions)

1. *(b)* $P = x(66-x)$, Quadratic expression's minimum value exists at $x = 33$

$M = 33 \times 33$

$x(66-x) \geq \frac{5}{9} \times 33 \times 33$

$11 \leq x \leq 55$

A : {12,15,18, ... 54}

$P(A) = \frac{15}{45} = \frac{1}{3}$

2. *(a)* Total number of ways = $6^3 = 216$

Favourable outcomes = ${}^6P_3 = 120$

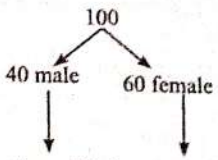
$\therefore$ Probability $= \frac{120}{216} = \frac{5}{9}$

Given, probability $= \frac{p}{q}$

$\therefore p = 5, q = 9$

Hence, $q - p = 4$

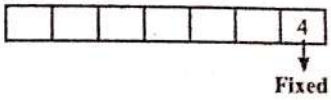
3. *(a)*



20 male qualified 40 female qualified

Probability that chosen candidate is a female $= \frac{40}{60} = \frac{2}{3}$

4. *(a)* Given digits are, 3, 3, 4, 4, 4, 5, 5
To make even seven digits number in remaining six places you have to arrange 3, 3, 4, 4, 5, 5



$\therefore$ Number of ways= $n(E) = \frac{6!}{2!2!2!}$

Total number of seven digits number

$= n(S) = \frac{7!}{2!3!2!} \times 1$

Hence Req. prob, $P(E) = \frac{n(E)}{n(S)}$

$= \frac{\frac{6!}{2!2!2!}}{\frac{7!}{2!3!2!}} = \frac{6!3!}{2!7!} = \frac{3}{7}$

5. *(a)* Number of six digits integers formed with 0, 1, 2, 3, 4, 5, 6 = 6 × 6 × 5 × 4 × 3 × 2 × 1 = 6 × 6!

(I) Total no of integers formed with digit 1, 2, 3, 4, 5, 6 = 6!

(II) Total no of integers formed with digit 0, 1, 2, 3, 4, 5 = 5.5!

(III) Total no of integers formed with digit 0, 1, 2, 4, 5, 6 = 5.5!

Total no. of integers formed which is divisible by 3 = 5! × (6 + 5 + 5) = 16 × 5!

Required probability =

$$\frac{\text{No. of favourable outcome}}{\text{Total no. of outcomes}}$$

$= \frac{16\times5!}{6\times6!} = \frac{16\times5!}{6\times6\times5!} = \frac{4}{9}$

6. *(b)* Total two digit number are 90

Possible outcome = ${}^{90}C_1 = 90$

Now, $(2^n - 2) = (3-1)^n - 2$

${}^nC_0 3^n - {}^nC_1 3^{n-1} + \ldots + (-1)^{n-1} \cdot {}^nC_{n-1} \times 3 + (-1)^n \cdot {}^nC_n - 2$

$3(3^{n-1} - n \cdot 3^{n-2} + \ldots + (-1)^{n-1} \cdot n) + (-1)^n - 2$

$(2^n - 2)$ is multiple of 3 only when n is odd

$\therefore$ Required probability $= \frac{45}{90} = \frac{1}{2}$

7. (*d*) Sample space $= 9 \times 10^4$

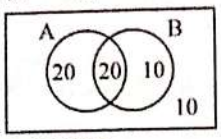
Case-I: Out of exactly two digits selected one is zero then favourable cases $= {}^9C_1(2^4 - 1)$

Case-II: Both selected digits are non-zero then favourable cases $= {}^9C_2(2^3 - 2)$

$$\text{Prob.} = \frac{9(2^4 - 1) + \frac{9.8}{2}(2^3 - 2)}{9 \times 10^4}$$

$$= \frac{15 + 120}{10^4} = \frac{135}{10^4}$$

8. (*a*)



A → opted NCC

B → opted NSS

$\therefore$ P (neither A nor B) $= \frac{10}{60} = \frac{1}{6}$

9. (*b*) Sum of all elements of S = 210

$$\left[sum = \frac{20 \times 21}{2} = 210\right]$$

So X be a "nice" set if

$X = \{S - \{7\}, S - \{1, 6\}, S - \{2, 5\}, S - \{3, 4\}, S - \{1, 2, 4\}\}$

$P(X) = \frac{5}{2^{20}}$ [$\because$ Total subset $= 2^{20}$]

10. (*a*) $2^N < N!$ is satisfied for $N \geq 4$

Required probability

$P(N \geq 4) = 1 - P(N < 4)$

$N = 1$(Rejected)

$N = 2\ (1, 1)$

$\Rightarrow P(N = 2) = \frac{1}{36}$

$N = 3\ (1, 2), (2, 1) \Rightarrow P(N = 3) = \frac{2}{36}$

$\therefore P(N < 4) = \frac{1}{36} + \frac{2}{36} = \frac{3}{36} = \frac{1}{12}$

$\therefore P(N \geq 4) = 1 - \frac{1}{12} = \frac{11}{12} = \frac{m}{n}$

$\therefore m = 11, n = 12$

Hence, $4m - 3n = 8$

11. (*c*) Required probability of obtaining a total sum $= 7 = \frac{13}{96}$ (given)

Probability of obtaining a total sum = probability of getting opposite faces

$$\Rightarrow 2\left(\left(\frac{1}{6} + x\right)\left(\frac{1}{6} = x\right) + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6}\right) = \frac{13}{96}$$

$$\Rightarrow x = \frac{1}{8}$$

12. (*c*) A bag has 6 white and 4 black balls.

Probability that all drawn ball white is

$$\frac{1}{6} \times \left[\frac{{}^6C_1}{{}^{10}C_1} + \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^6C_3}{{}^{10}C_3} + \frac{{}^6C_4}{{}^{10}C_4} + \frac{{}^6C_5}{{}^{10}C_5} + \frac{{}^6C_6}{{}^{10}C_6}\right]$$

$$= \frac{1}{6}\left(\frac{126 + 70 + 35 + 15 + 5 + 1}{210}\right) = \frac{42}{210} = \frac{1}{5}$$

13. [27] $64x^2 + 5Nx + 1 = 0$

$D = 25N^2 - 256 < 0$

$\Rightarrow N^2 < \frac{256}{25} \Rightarrow N < \frac{16}{5}$

$\therefore N = 1, 2, 3$

$\therefore$ Probability

$$= \frac{1}{4} + \frac{3}{4} \times \frac{1}{4} + \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} = \frac{37}{64}$$

14. (*b*) Total number of 5 digit number

10,000, 10,001 ..., 99999

$x = 90{,}000$

Now, find digit numbers which are divisible by 7.

10,003, 10,101, 10,017, ..., 99955

$n_1 = 12{,}857$

Now, find 5-digit numbers which are divisible by 35

100,10, 10045, ..., 99995

$n_2 = 2572$

Now, $n_1 - n_2$ = will give number of which are divisible by 7 but not by 5

10,285

$p = 10{,}285$

90000

$\Rightarrow 9p = 1.0285$

15. (*c*) No. of ways to select and arrange x_1, x_2, x_3, x_4, x_5 from 1, 2, 3.... 18

$n(s) = {}^{18}C - 5$

$x_1 \quad (x_2) \quad x_3 \quad (x_4) \quad x_5$

$7 \qquad 11$

$n(E) = {}^6C_1 \times {}^3C_1 \times {}^7C_1$

$$P(E) = \frac{6 \times 3 \times 7}{{}^{18}C_5}$$

$$\Rightarrow \frac{1}{17 \times 4} = \frac{1}{68}$$

16. (*d*) Let $P(H) = x$ then $P(T) = 1 - x$

$\Rightarrow P(4H\,1T) = P(5H)$

$\Rightarrow {}^5C_1(x)^4(1 - x)^1 = {}^5C_5x^5$

$\Rightarrow 5(1 - x) = x$

$\Rightarrow 6x - 5 = 0 \Rightarrow x = \frac{5}{6}$

P(atmost 2H)

$= P(0H, 5T) + (1H, 4T) + P(2H, 3T)$

$$= {}^5C_0\left(\frac{1}{6}\right)^5 + {}^5C_1\frac{5}{6}\left(\frac{1}{6}\right)^4 + {}^5C_2\left(\frac{5}{6}\right)^2\left(\frac{1}{6}\right)^3$$

$$= \frac{1}{6^5}(1 + 25 + 250) = \frac{276}{6^5} = \frac{46}{6^4}$$

17. (*a*) For $x^2 + \alpha x + \beta > 0\ \forall\ x \in R$ to hold, we should have $\alpha^2 - 4\beta < 0$

If $\alpha = 1$, β can be 1, 2, 3, 4, 5, 6 i.e., 6 choices

If $\alpha = 2$, β can be 2, 3, 4, 5, 6 i.e., 5 choices

If $\alpha = 3$, β can be 3, 4, 5, 6 i.e., 4 choices

If $\alpha = 4$, β can be 5 or 6 i.e., 2 choices

If $\alpha = 6$, No possible value for β i.e., 0 choices

Hence total favourable outcomes

$= 6 + 5 + 4 + 2 + 0 + 0 = 17$

Total possible choices for α and $\beta = 6 \times 6 = 36$

Required probability $= \frac{17}{36}$

18. (*b*) Probabilaity always lies between [0,1]

$$0 \leq \frac{2 + 3p}{6} \leq 1 \Rightarrow p \in \left[-\frac{2}{3}, \frac{4}{3}\right], 0 \leq \frac{2 - p}{8} \leq 1$$

$\Rightarrow p \in [-6, 2]$ and $0 \leq \frac{1 - p}{2} \leq 1 \Rightarrow p \in [-1, 1]$

$0 \leq P(E_1) + P(E_2) + P(E_3) \leq 10 \leq$

$\Rightarrow \frac{13}{12} - \frac{p}{8} \leq 1$

$\Rightarrow p \in \left[\frac{2}{3}, \frac{26}{3}\right]$ Taking intersection to all $p \in \left[\frac{2}{3}, 1\right]$

$\therefore p_1 + p_2 = \frac{5}{3}$

19. (*a*) Total number of 3-digit number $= 900$

Here, atleast 2 digits are odd means exactly 2 digits are odd or exactly 3 digits are odd.

Exactly 2 digit are odd

Case 1: E O O $\rightarrow 4 \times 5 \times 5$

$= 100$ (ignore O in this case)

Case 2: O E O $\rightarrow 5 \times 5 \times 5 = 125$

Case 3: O O E $\rightarrow 5 \times 5 \times 5 = 125$

Total numbers are 350

Exactly 3 digits are odd

Total numbers $= 5 \times 5 \times 5 = 125$

Probability $= \frac{350+125}{900} = \frac{475}{900} = \frac{19}{36}$

20. (*d*) Total number of elements $= 2022$

$2022 = 2 \times 3 \times 337$

HCF $(n, 2022) = 1$ is feasible when the value of 'n' and 2022 has no common factor.

A = Number which are divisible by 2 from {1, 2, 3..., 2022}

$n(A) = 1011$

B = Number which are divisible by 3 by from {1, 2, 3..., 2022}

$n(B) = 674$

$A \cap B$ = Number which are divisible by 6 from {1, 2, 3..., 2022}

6, 12, 18 ..., 2022

$n(A \cap B) = 337$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 1011 + 674 - 337 = 1348$

C = Number which divisible by 337 from {1, ...1022}

$C = \{337, 674, 1011, 1348, 1685, 2022\}$

Already counted in Set (A∪B) | Already counted in Set (A∪B) | Already counted in Set (A∪B)

Total elements which are divisible by 2 or 3 or 337

$= 1348 + 2 = 1350$

Favourable cases = Element which are neither divisible by 2, 3 or 337

$= 2022 - 1350 = 672$

Required probability $= \frac{672}{2022} = \frac{112}{337}$

21. [33] Total number of numbers from given condition $= n(S) = 2^6$

Every required number is of the form

$A = 7 \cdot (10^{a_1} + 10^{a_2} + 10^{a_3} + \ldots) + 111111$

Here 11111 is always divisible by 21 .

$\therefore$ If A is divisible by 21 then

$10^{a_1} + 10^{a_2} + 10^{a_3} + \ldots$. must e divisible by 3 .

For this we have

${}^6C_0 + {}^6C_3 + {}^6C_6$

cases are there

$\therefore n(E) = {}^6C_0 + {}^6C_3 + {}^6C_6 = 12$

$\therefore$ Required probability

$= \frac{22}{2^5} = p$

$\therefore \frac{11}{32} = p$

$\therefore 96p = 33$

22. [19] We have, $P(E_n) = n/36$ for $n = 1, 2, 3..., 8$

Let A is subset of S such that $P(A) = k \geq \frac{4}{5}$

$\Rightarrow k \geq 29$

Now, $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36$

No. of ways of selecting A from S if no element is left in S then no. of ways of selecting $A = 1$

If 1 element is left in S then, no. of ways of selecting $A = {}^7C_1 = 7$. [as 8 cannot be left]

If 2 elements are left in S then we can remove only (6, 1), (5, 1), (5, 2), (4, 1), (4, 2), (4, 3), (3, 1), (3, 2) or (2, 1)

So, no. of ways of selecting A in this case $= 9$

If 3 Elements are left in S then we can remove only (4, 2, 1) or (3, 2, 1)

So, no. of ways of selecting A in this case $= 2$

4 elements cannot be left as minimum possibility is to remove $1 + 2 + 3 + 4 = 10$ which would lead to $k < 29$

Thus, number of ways of selecting $A = 7 + 9 + 2 + 1 = 19$ ways

Hence, total elements in set = 19

23. (*b*) $n(s) = 6^2 = 36$

$E = (1, 1), (1, 2), (1, 3), (1, 5), (1, 7), (2, 1), (2, 2)$

$(2, 3), (2, 5), (3, 1), (3, 2), (3, 3), (3, 5), (5, 1), (5, 2), (5, 3), (7, 1)$

$\therefore n(E) = 17$

Required prob. $P(E) = \frac{n(E)}{n(S)} = \frac{17}{36}$

24. (*b*) Total number formed using the given letters $n(s), = \frac{11!}{2!2!2!}$

Total number formed when M is fixed at 4th place

$n(E) = \frac{10!}{2!2!2!}$

Required probability $P(E)$

$= \frac{n(E)}{n(S)} = \frac{10!/2!2!2!}{11!/2!2!2!} = \frac{1}{11}$

25. (*c*) Total number of possibilities $n(S) = 2^5 \times 2^5$

Farounable case $n(E) = {}^5C_2 \times 3^3 = 10 \times 3^3$

$\therefore$ required probability $P(E)$

$= \frac{10 \times 3^3}{2^5 \times 2^5} = \frac{5 \times 27}{2^9} = \frac{135}{2^9}$

26. (*a*) Let $f(x) = x^2 + 2(a+4)x - 5a + 64$

for $f(x) > 0$, we have,

$D < 0$

$\Rightarrow 4[(a+4)^2 - (64 - 5a)] < 0$

-16 | 3

$\Rightarrow a^2 + 13a - 48 < 0$

$\Rightarrow (a + 16)(a - 3) < 0$

$\Rightarrow -16 < a < 3$

$\because a \in [-5, 30]$

$\Rightarrow -5 \leq a \leq 3$

$\Rightarrow a = -5, -4, \ldots, 1, 2$ (integers)

$\therefore n(E) = 8$ and $n(S) = 36$

Probability $P(E) = \frac{8}{36} = \frac{2}{9}$

27. (*d*) Let n be the number of times.

$p = \frac{1}{2}, q = \frac{1}{2}$

According to question, we have

${}^nC_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{n-7} = {}^nC_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{n-9}$

$\Rightarrow {}^nC_7 = {}^nC_9 \Rightarrow n - 7 = 9 \Rightarrow n = 16$

$P(x = 2) = {}^{16}C_2 \left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^{14} = \frac{15}{2^{13}}$

28. (*d*)

29. (*b*) Total numbers with given condition

Case-I | 7 | | | |

$= 1 \times 9 \times 9 \times 9 = 1 \times 9^3$

Case-II | | | | |

8 ways

$({}^3C_1 \times 8) \times 9 \times 9$

Total $= 9^3 + 24 \times 9^2 = 81 \times 33$

In favourable events, the last digit of the number must be either 2 or 7.

Favourable cases $= 8 \times 9^2 + 9^2 + 8 \times 9 \times 2 = 873$,

Probability $= \frac{873}{81 \times 33} = \frac{97}{297}$

30. (*a*) Given, $P(E) < \frac{1}{2}$

General form: $\sum_{r=n}^{8} {}^8C_r \left(\frac{1}{2}\right)^{8-r} \left(\frac{1}{2}\right)^r < \frac{1}{2}$

$\Rightarrow \sum_{r=n}^{8} {}^8C_r \left(\frac{1}{2}\right)^8 < \frac{1}{2}$

$\Rightarrow {}^8C_n + {}^8C_{n+1} + \ldots . {}^8C_8 < 2^7$

$\Rightarrow 256 - \left({}^8C_0 + {}^8C_1 + \ldots . {}^8C_{n-1}\right) < 128$

$\Rightarrow {}^8C_0 + {}^8C_1 + {}^8C_{n-1} > 128$

$\Rightarrow n - 1 \geq 4$

Hence, $n \geq 5$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

CHAPTER
20
Properties of Triangle

Prarambh (Topicwise)

1. (*b*) A, B, C are in A. P. then angle B = 60°.

$\cos B = \frac{a^2+c^2-b^2}{2ac}$,

{since A + B + C = 180° and A + C = 2B ⇒ B = 60°}

$\Rightarrow \frac{1}{2} = \frac{a^2+c^2-b^2}{2ac} \Rightarrow a^2+c^2-b^2 = ac$

$\Rightarrow b^2 = a^2+c^2-ac$

2. (*a*) $a\sin(B-C)+b\sin(C-A)+c\sin(A-B)$

$= k(\Sigma \sin A \sin(B-C)$

$= k\{\Sigma \sin(B+C)\sin(B-C)\}$

$= k\left\{\Sigma \frac{1}{2}(\cos 2C - \cos 2B)\right\} = 0$

3. (*a*) $(b^2-c^2)\cot A = (b^2-c^2)\frac{\cos A}{\sin A}$

$= \frac{(b^2-c^2)(b^2+c^2-a^2)}{2bc.ka}$

Hence L.H.S. $= \frac{1}{2k\,abc}$

$[(b^4-c^4)+(c^4-a^4)+(a^4-b^4)-\{a^2(b^2-c^2)+b^2(c^2-a^2)+c^2(a^2-b^2)\}] = 0$.

4. (*b*) Largest side is $\sqrt{p^2+pq+q^2}$.

If largest angle is θ, then

$\cos\theta = \frac{p^2+q^2-p^2-pq-q^2}{2pq} = -\frac{1}{2} = \cos\left(\frac{2\pi}{3}\right)$

$\Rightarrow \theta = \frac{2\pi}{3}$

5. (*b*) $\sqrt{\frac{b+c}{4c}} = \sqrt{\frac{\sin 3C+\sin C}{4\sin C}}$

$\Rightarrow \sqrt{\frac{2\sin 2C\cos C}{4\sin C}} = \cos C$

$\frac{b-c}{2c} = \frac{\sin 3C-\sin C}{2\sin C} = \frac{2\cos 2C\sin C}{2\sin C}$

$= \cos 2C = \sin\frac{A}{2}$

6. (*b*) $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13} = \lambda$ (Let)

$\therefore\ b+c = 11\lambda$ ... (*i*)

$c+a = 12\lambda$ ... (*ii*)

and $a+b = 13\lambda$ ... (*iii*)

From (*i*) + (*ii*) + (*iii*), $2(a+b+c) = 36\lambda$.

$\therefore\ a+b+c = 18\lambda$ ... (*iv*)

Now, (*iv*) − (*i*) gives, $a = 7\lambda$.

(*iv*) − (*ii*) gives, $b = 6\lambda$.

(*iv*) − (*iii*) gives, $c = 5\lambda$.

7. (*d*) We have, $b = \sqrt{3}$, $c = 1$, $\angle A = 30°$

$\cos A = \frac{b^2+c^2-a^2}{2bc} \Rightarrow \frac{\sqrt{3}}{2} = \frac{(\sqrt{3})^2+1^2-a^2}{2.\sqrt{3}.1}$

$\therefore\ a = 1, b = \sqrt{3}, c = 1$

b is the largest side. Therefore, the largest angle B is given by

$\cos B = \frac{a^2+c^2-b^2}{2ac} = \frac{1+1-3}{2.1.1} = -\frac{1}{2}$

$\therefore\ B = 120°$

8. (*b*) $2ac\sin\frac{A-B+C}{2} = 2ac\sin\frac{\pi-2B}{2}$

$= 2ac\cos B$

$2ac\frac{c^2+a^2-b^2}{2ca} = c^2+a^2-b^2$

9. (*b*) $\cos A = \frac{\sin B}{2\sin C} \Rightarrow \frac{b^2+c^2-a^2}{2bc} = \frac{b}{2c}$

$\Rightarrow\ b^2+c^2-a^2-b^2 = 0 \Rightarrow c^2 = a^2$

10. (*a*) $a^4+b^4+c^4-2a^2c^2-2b^2c^2+2a^2b^2 = 2a^2b^2$

$\Rightarrow (a^2+b^2-c^2)^2 = (\sqrt{2}ab)^2$

$\Rightarrow a^2+b^2-c^2 = \pm\sqrt{2}ab$

$\Rightarrow \frac{a^2+b^2-c^2}{2ab} = \pm\frac{\sqrt{2}ab}{2ab} = \pm\frac{1}{\sqrt{2}}$

$\Rightarrow \cos C = \cos 45°$ or $\cos 135°$

$\Rightarrow C = 45°$ or $135°$

11. (*c*) $\because (b+c)^2-a^2 = \lambda bc$

$\Rightarrow b^2+c^2-a^2+2bc = \lambda bc$

$\Rightarrow 2bc\cos A + 2bc = \lambda bc$ (from cosine rule)

$\Rightarrow 2(\cos A+1) = \lambda$

$\Rightarrow \cos A = \frac{\lambda}{2}-1$

But $-1 < \cos A < 1$

$\Rightarrow -1 < \frac{\lambda}{2}-1 < 1$

$\Rightarrow 0 < \frac{\lambda}{2} < 2 \Rightarrow 0 < \lambda < 4$

12. (*c*) $\because \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$\Rightarrow \frac{a}{\sin 45°} = \frac{b}{\sin 75°} = \frac{c}{\sin 60°}$

$\Rightarrow a\sqrt{2} = \frac{b\cdot 2\sqrt{2}}{\sqrt{3}+1} = \frac{2c}{\sqrt{3}}$

$\Rightarrow a = \frac{2b}{\sqrt{3}+1}, c = \frac{\sqrt{6}b}{\sqrt{3}+1}$

$\Rightarrow a+c\sqrt{2} = \frac{2b}{\sqrt{3}+1}+\frac{2b\sqrt{3}}{\sqrt{3}+1} = 2b$

13. (*c*) Given that $2B = A + C$ and $\frac{b}{c} = \frac{\sqrt{3}}{\sqrt{2}}$

$\Rightarrow \frac{b}{\sqrt{3}/2} = \frac{c}{1/\sqrt{2}}$

$\Rightarrow \frac{b}{\sin 60°} = \frac{c}{\sin 45°}$

$\Rightarrow \angle B = 60°, \angle C = 45°$

$120° = A + 45°$

$\Rightarrow A = 75° = \frac{5\pi}{12}$

14. (*a*) $\frac{\sin(A-B)}{\sin(A+B)} = \frac{\sin^2 A-\sin^2 B}{\sin^2 A+\sin^2 B}$

$\Rightarrow \frac{\sin(A-B)}{\sin C} = \frac{\sin(A+B)\sin(A-B)}{\sin^2 A+\sin^2 B}$

$\Rightarrow \sin(A-B)\left[\frac{1}{\sin C}-\frac{\sin C}{\sin^2 A+\sin^2 B}\right] = 0$

either $\sin(A-B) = 0 \Rightarrow A = B$

or $\sin^2 A + \sin^2 B = \sin^2 C$

$\Rightarrow a^2+b^2+c^2$

Triangle is right angled or isosceles

15. (*d*) From sine rule we take 1st term

$$\Rightarrow \frac{\lambda^2 \sin^2 A \sin(B-C)}{\sin B + \sin C}$$

$$\Rightarrow \frac{\lambda^2 \sin A \sin(B+C)\sin(B-C)}{\sin B + \sin C}$$

$$\Rightarrow \frac{\lambda^2 \sin A(\sin^2 B - \sin^2 C)}{\sin B + \sin C}$$

$\lambda^2 \sin A(\sin B - \sin C)$

similarly we solve remain term and add we get

$\Rightarrow \lambda^2[\sin A(\sin B - \sin C) + \sin B(\sin C - \sin A) + \sin C(\sin A - \sin B)] = 0$

16. (*b*) $E = \frac{[(b+c)^2 - a^2][a^2 - (b-c)^2]}{4b^2c^2}$

$$= \frac{[b^2+c^2-a^2+2bc][a^2-b^2-c^2+2bc]}{4b^2c^2}$$

$$= \frac{-[(b^2+c^2-a^2)^2 - 4b^2c^2]}{4b^2c^2}$$

$$= \left[\left(\frac{b^2+c^2-a^2}{2bc}\right)^2 - 1\right]$$

$= -[\cos^2 A - 1] = \sin^2 A$

17. (*a*) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$

$$\Rightarrow \tan\left(\frac{90^\circ}{2}\right) = \frac{\sqrt{3}-1}{\sqrt{3}+1}\cot\frac{A}{2}$$

$$\Rightarrow \tan\left(\frac{A}{2}\right) = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{3+1-2\sqrt{3}}{2} = 2-\sqrt{3}$$

$\Rightarrow \frac{A}{2} = 15^\circ \Rightarrow A = 30^\circ$.

18. (*c*) $\sin^2 B - \sin^2 A = \sin^2 C - \sin^2 B$

$\therefore \sin(B+A)\sin(B-A) = \sin(C+B)\sin(C-B)$

or $\sin C(\sin B\cos A - \cos B\sin A)$

$= \sin A(\sin C\cos B - \cos C\sin B)$

Divide by sinA sinB sinC

$\therefore \cot A - \cot B = \cot B - \cot C$. Hence the result.

19. (*a*) From the given relation

$$\sin C = \frac{1-\cos A\cos B}{\sin A\sin B} \le 1 \quad \ldots (i)$$

$\Rightarrow 1 \le \cos A\cos B + \sin A\sin B$

$\Rightarrow \cos(A-B) \ge 1 \quad \because \cos\theta \not> 1$ $\ldots (ii)$

$A - B = 0$ or $A = B$

Hence from (i),

$$\sin C = \frac{1-\cos^2 A}{\sin^2 A} = \frac{\sin^2 A}{\sin^2 A} = 1$$

$\therefore C = 90^\circ \Rightarrow A + B = 90^\circ$ or $A = B = 45^\circ$ {by (*ii*)}

Hence,

$a : b : c = \sin A : \sin B : \sin C = 1 : 1 : \sqrt{2}$.

20. (*d*) We have,

$$\frac{\tan\left(\frac{B}{2}\right)}{\cot\left(\frac{C-A}{2}\right)} = \frac{\sin\frac{B}{2}\sin\left(\frac{C-A}{2}\right)}{\cos\frac{B}{2}\cos\left(\frac{C-A}{2}\right)}$$

$$= \frac{\sin C - \sin A}{\sin C + \sin A} = \frac{kc-ka}{kc+ka} = \frac{c-a}{c+a} = \frac{a}{3a} = \frac{1}{3}, \quad (\because c = 2a)$$

21. (*d*) $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$;

$$\frac{5}{\sin\left(\frac{\pi}{2}+B\right)} = \frac{4}{\sin B}$$

$$\frac{5}{\cos B} = \frac{4}{\sin B}, \quad \therefore \tan B = \frac{4}{5}$$

$$\tan A = \tan\left(\frac{\pi}{2}+B\right) = -\cot B = \frac{-5}{4}$$

$\tan C = \tan(\pi - (A+B))$

$= -\tan(A+B), [A+B+C = \pi]$

$$= -\frac{(\tan A + \tan B)}{1 - \tan A.\tan B} = \frac{-\left(-\frac{5}{4}+\frac{4}{5}\right)}{1+1} = \frac{9}{40}$$

$$C = \tan^{-1}\left(\frac{\left(2.\frac{1}{9}\right)}{1-\left(\frac{1}{9}\right)^2}\right), \quad \therefore C = 2\tan^{-1}\left(\frac{1}{9}\right)$$

22. (*a*) We have, $2s = a+b+c$, $A^2 = s(s-a)(s-b)(s-c)$

$\because$ A.M. $\ge$ G.M.

$$\Rightarrow \frac{s-a+s-b+s-c}{3} \ge \sqrt[3]{(s-a)(s-b)(s-c)}$$

$$\Rightarrow \frac{3s-2s}{3} \ge \frac{(A^2)^{1/3}}{s^{1/3}} \Rightarrow \frac{s^3}{27} \ge \frac{A^2}{s} \Rightarrow A \le \frac{s^2}{3(\sqrt{3})}$$

23. (*b*) $s - a = 3 \Rightarrow b + c - a = 6$ $\ldots (i)$

$s - c = 2 \Rightarrow a + b - c = 4$ $\ldots (ii)$

Adding these two equations, we get $b = 5$

Since B is a right angle

$\therefore b^2 = a^2 + c^2 \Rightarrow a^2 + c^2 = 25$ $\ldots (iii)$

Solving, we get $a = 3$, $c = 4$

24. (*b*) We have $\cos A = \frac{b^2+c^2-a^2}{2bc}$

$\Rightarrow b^2 - 2bc\cos A + (c^2 - a^2) = 0$

It is given that b_1 and b_2 are roots of this equation.

Therefore $b_1 + b_2 = 2c\cos A$ and $b_1 b_2 = c^2 - a^2$

$\Rightarrow 3b_1 = 2c\cos A$, $2b_1^2 = c^2 - a^2$

$(\because b_2 = 2b_1$ (given) $2\left(\frac{2c}{3}\cos A\right)^2 = c^2 - a^2$

$\Rightarrow 8c^2(1 - \sin^2 A) = 9c^2 - 9a^2$

25. (*d*) $a^2\sin 2C + c^2\sin 2A$

$= a^2(2\sin C\cos C) + c^2(2\sin A\cos A)$

$= 2a^2\left(\frac{2\Delta}{ab}\cos C\right) + 2c^2\left(\frac{2\Delta}{bc}\cos A\right)$

$\left(\because \Delta = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A,\right.$

$\left.\therefore \sin C = \frac{2\Delta}{ab}, \sin A = \frac{2\Delta}{bc}\right)$

$= 4\Delta\left\{\frac{a\cos C + c\cos A}{b}\right\} = 4\Delta\left(\frac{b}{b}\right) = 4\Delta$

26. (*b*) A.M. $= \frac{a+b+c}{3}$

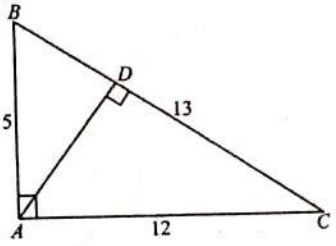
Altitudes are $\frac{2\Delta}{a}, \frac{2\Delta}{b}, \frac{2\Delta}{c}$

$$\text{H.M.} = \frac{3}{\frac{a}{2\Delta}+\frac{b}{2\Delta}+\frac{c}{2\Delta}} = \frac{6\Delta}{(a+b+c)}$$

(A.M.) $\times$ (H.M.)

$$= \frac{(a+b+c)}{3} \times \frac{3\times 2\Delta}{(a+b+c)} = 2\Delta$$

27. (*b*) Given c = 5, b = 12, a = 13



$$\cos A = \frac{12^2+5^2-13^2}{2(12)(5)} = 0$$

i.e. $A = 90^\circ$

Altitude $AD = \frac{2\Delta}{a} = \frac{2\times 30}{13} = \frac{60}{13}$

$\{\Delta = \frac{1}{2}\times 5\times 12 = 30\}$

28. (*b*) $\cos\frac{\pi}{6} = \frac{b^2+c^2-a^2}{2bc} = \frac{\sqrt{3}}{2} \Rightarrow a = 1$

$2R = \frac{a}{\sin A} \Rightarrow R = 1$

29. (*b*) $a^2(2\cos^2 B - 1) + b^2(2\cos^2 A - 1) + 2ab\cos(A-B)$

$= 2(a^2\cos^2 B + b^2\cos^2 A) - a^2 - b^2 + 2ab\cos(A-B)$

$2(a\cos B + b\cos A)^2 - a^2 - b^2 + 4ab\cos A\cos B + 2ab\cos(A-B) = 2c^2 - a^2 - b^2 - 2ab\cos A\cos B + 2ab\sin A\sin B$

$= 2c^2 - a^2 - b^2 - 2ab\cos(A+B)$

$= 2c^2 - a^2 - b^2 + 2ab\cos C = 2c^2 - c^2 = c^2$

30. **(a)** $\because \tan\left(\frac{B-C}{2}\right)=\frac{b-c}{b+c}\cot\left(\frac{A}{2}\right)$

$\Rightarrow \tan 45^\circ=\frac{\sqrt{3}-1}{\sqrt{3}+1}\cot\left(\frac{A}{2}\right)$

$\Rightarrow \cot\frac{A}{2}=\frac{\sqrt{3}+1}{\sqrt{3}-1}=\cot 15^\circ$

$\Rightarrow \frac{A}{2}=15^\circ \Rightarrow A=30^\circ$

31. **(a)** $\frac{(s-a)(s-b)}{s(s-c)}=1$

$\Rightarrow \tan^2\left(\frac{C}{2}\right)=1 \Rightarrow \tan\left(\frac{C}{2}\right)=1$

$\Rightarrow \frac{C}{2}=45^\circ \Rightarrow C=90^\circ$

32. **(c)** $\frac{1}{\sin^2\frac{A}{2}},\frac{1}{\sin^2\frac{B}{2}},\frac{1}{\sin^2\frac{C}{2}}$ are in A. P.

$\Rightarrow \frac{1}{\sin^2\frac{C}{2}}-\frac{1}{\sin^2\frac{B}{2}}=\frac{1}{\sin^2\frac{B}{2}}-\frac{1}{\sin^2\frac{A}{2}}$

$\Rightarrow \frac{ab}{(s-a)(s-b)}-\frac{ac}{(s-a)(s-c)}$

$=\frac{ac}{(s-a)(s-c)}-\frac{bc}{(s-b)(s-c)}$

$\Rightarrow \left(\frac{a}{s-a}\right)\left(\frac{b(s-c)-c(s-b)}{(s-b)(s-c)}\right)$

$=\left(\frac{c}{s-c}\right)\left(\frac{a(s-b)-b(s-a)}{(s-a)(s-b)}\right)$

$\Rightarrow abs-abc-acs+abc=acs-abc$

$-bcs+abc$

$\Rightarrow ab-ac=ac-bc \Rightarrow ab+bc=2ac$

or $\frac{1}{c}+\frac{1}{a}=\frac{2}{b}$, *i.e.*, a, b, c are in H. P.

33. **(b)**

$$\frac{\tan\frac{A}{2}-\tan\frac{B}{2}}{\tan\frac{A}{2}+\tan\frac{B}{2}}=\frac{\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}-\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}}{\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}+\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}}$$

$$=\frac{(s-b)\sqrt{s(s-c)}-(s-a)\sqrt{s(s-c)}}{(s-b)\sqrt{s(s-c)}+(s-a)\sqrt{s(s-c)}}$$

$$=\frac{\sqrt{s(s-c)}(s-b-s+a)}{\sqrt{s(s-c)}\ (s-b+s-a)}=\frac{a-b}{c}$$

34. **(c)** $ab^2\cos A+ba^2\cos B+ac^2\cos A$

$+bc^2\cos B+b^2c\cos C+ca^2\cos C$

$=ab(b\cos A+a\cos B)+ac\,c(\cos A+a\cos C)$

$+bc(c\cos B+b\cos C)$

$=abc+abc+abc=3abc$

35. **(c)** $\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{a^2-(b^2-c^2)}{2ac}$

Now $AD=\frac{abc}{b^2-c^2}$

$\therefore \cos B=\frac{a^2-\frac{abc}{AD}}{2ac}$

Also, $AD=b\sin 23^\circ$

$\therefore \cos B=\frac{a-\frac{c}{\sin 23^\circ}}{2c}$

By sine formula, $\frac{a}{c}=\frac{\sin(B+23^\circ)}{\sin 23^\circ}$

$\therefore \cos B=\left(\frac{\sin(B+23^\circ)}{\sin 23^\circ}-\frac{1}{\sin 23^\circ}\right)\div 2$

$\Rightarrow \sin(23^\circ-B)=-1=\sin(-90^\circ)$

$\Rightarrow 23^\circ-B=-90^\circ$ or $B=113^\circ$

36. **(a)** $a=3,b=5,c=4,\ s=\frac{a+b+c}{2}$

$=\frac{12}{2}=6$

$\sin\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{ca}}=\sqrt{\frac{2.3}{12}}=\sqrt{\frac{1}{2}}$

$\cos\frac{B}{2}=\sqrt{\frac{s(s-b)}{ca}}=\sqrt{\frac{6.1}{12}}=\sqrt{\frac{1}{2}}$

$\therefore \sin\frac{B}{2}+\cos\frac{B}{2}=\frac{2}{\sqrt{2}}=\sqrt{2}$

37. **(c)** $\frac{b-c}{a}=\frac{\sin B-\sin C}{\sin A}$

$=\frac{2\sin\frac{B-C}{2}\cos\frac{B+C}{2}}{2\sin\frac{A}{2}\cos\frac{A}{2}}=\frac{\sin\frac{B-C}{2}}{\cos\frac{A}{2}}$

$\Rightarrow (b-c)\cos\frac{A}{2}=a\sin\frac{B-C}{2}$

38. **(b)** $\cot\frac{B}{2}.\cot\frac{C}{2}$

$=\sqrt{\frac{s(s-b)}{(s-a)(s-c)}\cdot\frac{s(s-c)}{(s-a)(s-b)}}$

$=\frac{s}{s-a}$ {Since $3a=b+c$ or $a+b+c$

$=2s=4a=2a/a=2$

39. **(c)** $\frac{\sin A}{a}=\frac{\sin B}{b}=\frac{\sin C}{c}=\frac{1}{2R}$

$\Rightarrow \sin A=\frac{a}{2R}$ and so on.

Therefore $2R^2\sin A\sin B\sin C$

$=2R^2\frac{a}{2R}.\frac{b}{2R}.\frac{c}{2R}$

$=\frac{abc}{4R}=\Delta$

40. **(b)** $r=4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\Rightarrow r=4R\sin^3 30^\circ$; {$\because A=B=C=60^\circ$}

$\Rightarrow r=\frac{R}{2}$

41. **(a)** $a=5k, b=6k$ and $c=5k$

$s=\frac{a+b+c}{2}=\frac{5k+6k+5k}{2}=8k$

$r=\frac{\Delta}{s}=\sqrt{\frac{s(s-a)(s-b)(s-c)}{s}}$

$r=\sqrt{\frac{8k(8k-5k)(8k-6k)(8k-5k)}{8k}}$

$r=\frac{3k}{2}\Rightarrow k=\frac{2r}{3}=\frac{2\times 6}{3}=4$

42. **(b)** $R=\frac{abc}{4\Delta}$, where

$\Delta=\sqrt{s(s-a)(s-b)(s-c)}$

$a=13,b=12,c=5,s=\frac{30}{2}=15$

$\Delta=\sqrt{15(2)(3)10}=3\times 2\times 5=30$

$\therefore R=\frac{13\times 12\times 5}{4\times 30}=\frac{13}{2}$

43. **(d)** $\because \frac{\cos A}{a}=\frac{\cos B}{b}$

$\Rightarrow \frac{b^2+c^2-a^2}{2abc}=\frac{a^2+c^2-b^2}{2abc}$

$\Rightarrow a^2=b^2\Rightarrow a=b$

similarly $\frac{\cos B}{b}=\frac{\cos C}{c}\Rightarrow b=c$

so $a=b=c=2$ given

triangle is equilateral so

Area $=\frac{\sqrt{3}}{4}a^2=\sqrt{3}$

44. **(a)** We know that

$\frac{b}{\sin B}=\frac{c}{\sin C}$

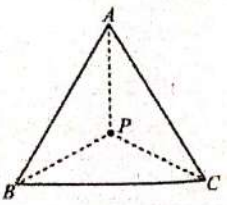
$\Rightarrow \sin C=\frac{c\sin B}{b}$

given that $b<c\sin B$

$\Rightarrow$ $\sin C>1$ which is impossible

$\Rightarrow$ No triangle is possible

45. **(b)**



Given that $AB = BC = CA = 2$

x_a, x_b and x_c are the distances of P from the sides BC, CA and AB

$\Rightarrow$ area of ΔBPC

$= \frac{1}{2} BC \cdot x_a = \frac{1}{2} \cdot 2x_a = x_a$

similarly

Area of $\Delta CPA = x_b$ and Area of $\Delta APB = x_c$ then the area of ΔABC or (ΔABC) = ar (ΔBPC) + ar (ΔCPA) + ar (ΔAPB)

$\because ABC$ is equilateral

ar $(\Delta ABC) = \sqrt{3}$ $\quad (\because AB = 2)$

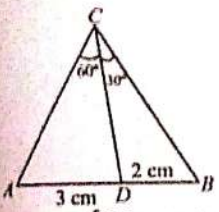
$x_a + x_b + x_c = \sqrt{3}$

46. **(b)** $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$\Rightarrow c^2 - 2bc \cos A + b^2 - a^2 = 0$

Let c_1 and c_2 be two values of c then $c_1 + c_2 = 2b\cos A$.

47. **(a)**



Using $(m+n)\cot\theta = n\cot\beta - m\cot\alpha$ we get

$(3+2)\cot(\angle CDA) = 2\cot 30 - 3\cot 60$

$\Rightarrow \cot(\angle CDA) = \frac{\sqrt{3}}{5}$

$\Rightarrow \sin(\angle CDA) = \frac{5}{2\sqrt{7}}$

Apply sine rule in ΔACD, we get

$\frac{AC}{\sin\left(\sin^{-1}\frac{5}{2\sqrt{7}}\right)} = \frac{AD}{\sin 60}$

$AC = 3 \cdot \frac{2}{\sqrt{3}} \cdot \frac{5}{2\sqrt{7}} = 5\sqrt{\frac{3}{7}}$

Prabal (JEE Main Level)

1. **(c)** $\because$ A : B : C = 3 : 5 : 4

$\therefore$ A = 45°, B = 75°, C = 60°

$\because$ from Sine - rule

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$

$\Rightarrow \frac{a}{\frac{1}{\sqrt{2}}} = \frac{b}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \frac{c}{\frac{\sqrt{3}}{2}} = k$

$(\because \sin 75° = \sin(45° + 30°))$

$\therefore a = \frac{k}{\sqrt{2}}, b = \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)k$ and $c = \frac{k\sqrt{3}}{2}$

$\therefore a + b + c\sqrt{2} = \frac{k}{\sqrt{2}} + \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)k + \left(\frac{k\sqrt{3}}{2}\right)\sqrt{2}$

$= \frac{k}{2\sqrt{2}}\left[2 + (\sqrt{3}+1) + 2\sqrt{3}\right] = \frac{3k(\sqrt{3}+1)}{2\sqrt{2}} = 3b$

2. **(c)** Given $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$ ... (i)

$\because a = k \sin A, b = k \sin B, c = k \sin C$

$\therefore$ (i) becomes

$\frac{\cot A}{k} = \frac{\cot B}{k} = \frac{\cot C}{k}$

$\therefore A = B = C$

$\therefore \Delta ABC$ is an equilateral triangle

3. **(c)** $\because \frac{bc\sin^2 A}{\cos A + \cos B \cos C}$

$= \frac{k^2 \sin B \sin C \sin^2 A}{-\cos(B+C) + \cos B \cos C}$

$= \frac{k^2 \sin B \sin C \sin^2 A}{\sin B \sin C} = k^2 \sin^2 A = a^2.$

4. **(c)** $(a+b+c)(b+c-a) = kbc$

$\Rightarrow (b+c+a)(b+c-a) = kbc$

$\Rightarrow (b+c)^2 - a^2 = kbc$

$\Rightarrow b^2 + c^2 - a^2 + 2bc = kbc$

$\Rightarrow b^2 + c^2 - a^2 = (k-2)bc$

$\Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = \frac{k-2}{2}$

$\because -1 < \frac{k-2}{2} < 1$

$\Rightarrow -2 < k - 2 < 2 \Rightarrow 0 < k < 4$

5. **(b)** $\because A = \frac{2\pi}{3}$, $b - c = 3\sqrt{3}$ and

Area $= \frac{9\sqrt{3}}{2}$ cm^2

$\because \Delta = \frac{1}{2} bc \sin A \Rightarrow \frac{9\sqrt{3}}{2} = \frac{1}{2} bc \sin\frac{2\pi}{3}$

$\Rightarrow bc = 18$

$\because \cos\frac{2\pi}{3} = \frac{b^2 + c^2 - a^2}{2bc} = -\frac{1}{2}$

$\Rightarrow \frac{(b-c)^2 + 2bc - a^2}{2bc} = -\frac{1}{2}$

$\Rightarrow a = 9$

6. **(c)** $AD = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

$4AD^2 = 2b^2 + 2c^2 - a^2$

$\Rightarrow 4BE^2 = 2c^2 + 2a^2 - b^2$

$\Rightarrow 4CF^2 = 2a^2 + 2b^2 - c^2$

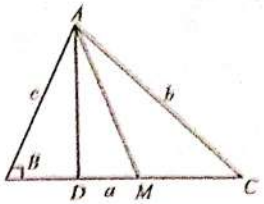
$\therefore 4(AD^2 + BE^2 + CF^2) = 3(a^2 + b^2 + c^2)$

$\Rightarrow \frac{AD^2 + BE^2 + CF^2}{BC^2 + CA^2 + AB^2} = \frac{3}{4} = 3:4$

7. **(b)** $BM = a/2$

$BD = c \cos B$

$DM = \frac{a}{2} - c\cos B$



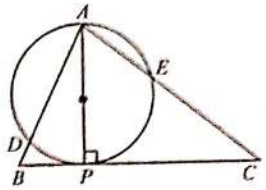
$= \frac{a}{2} - c\frac{(c^2 + a^2 - b^2)}{2ca}$

$= \frac{a^2 - c^2 - a^2 + b^2}{2a} = \frac{b^2 - c^2}{2a}$

8. **(d)** Sine Rule in ΔADE

$\frac{DE}{\sin A} = AP$

$\Rightarrow \frac{2\Delta}{a}\sin A$



$= 2\Delta \times \frac{1}{R} = \frac{\Delta}{R} \quad \left\{\frac{\sin A}{a} = \frac{1}{2R}\right\}$

9. **(b)** Given In a ΔABC, $\sin A = \frac{2\Delta}{bc}$

$\Rightarrow \sin A = \frac{2[a^2 - b^2 - c^2 + 2bc]}{bc}$

$\Rightarrow \sin A = 2[2 - 2\cos A]$

$\Rightarrow \sin A = 4[1 - \cos A]$

$\Rightarrow 2\sin\frac{A}{2}\cos\frac{A}{2} = 4 \cdot 2\sin^2\frac{A}{2}$

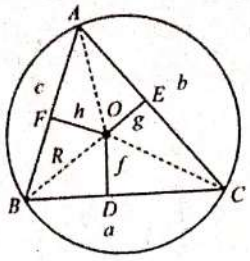
$\Rightarrow \tan\frac{A}{2} = \frac{1}{4}$

$\Rightarrow \tan A = \frac{2\tan\frac{A}{2}}{1 - \tan^2\frac{A}{2}} = \frac{2 \times \frac{1}{4}}{1 - \left(\frac{1}{4}\right)^2}$

$= \frac{\frac{1}{2}}{\frac{15}{16}} = \frac{1}{2} \times \frac{16}{15} \Rightarrow \tan A = \frac{8}{15}$

10. **(a)** In ΔBOD,

$$\tan B = \frac{\frac{a}{2}}{f} = \frac{a}{2f}$$



$\Sigma \tan A = \pi \tan A$

$$\Rightarrow \frac{a}{2f}+\frac{b}{2g}+\frac{c}{2h}=\frac{abc}{8fgh}$$

$$\Rightarrow \frac{a}{f}+\frac{b}{g}+\frac{c}{h}=\frac{abc}{4fgh}$$

$$\Rightarrow \frac{a}{f}+\frac{b}{g}+\frac{c}{h}=\lambda\frac{abc}{fgh} \quad \therefore \lambda=\frac{1}{4}$$

11. **(b)** $\because s-a=3$ ... (i)

and $s-c=2$ ... (ii)

by (1) – (2), we get

$c-a=1$

(1) + (2), we get $2s-a-c=5$

$\Rightarrow b=5$

$\because$ ΔABC is right angled at B

$\therefore a^2+c^2=25$ ... (iii)

$\therefore (c-a)^2+2ac=25$

$ac=12$ ... (iv)

$\therefore a(1+a)=12 \Rightarrow a^2+a-12=0$

$\Rightarrow (a+4)(a-3)=0$

$\Rightarrow a=3$ and $c=4$.

12 **(a)** $\because b\cos^2\frac{A}{2}+a\cos^2\frac{B}{2}=\frac{3}{2}c.$

$$\Rightarrow b\frac{s(s-a)}{bc}+a\frac{s(s-b)}{ac}=\frac{3}{2}c.$$

$$\Rightarrow \frac{s}{c}[s-a+s-b]=\frac{3}{2}c$$

$$\Rightarrow \frac{s}{c}\times c=\frac{3}{2}c$$

$$\Rightarrow \frac{a+b+c}{2}=\frac{3c}{2} \Rightarrow a+b=2c$$

$\Rightarrow$ a, c, b are in A.P.

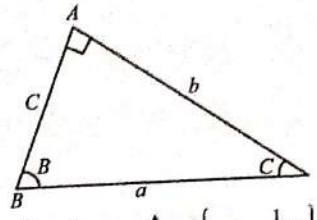
13. **(c)** $\because \cos\frac{B}{2}\cot\frac{C}{2}$

$$=\sqrt{\frac{s(s-b)}{(s-a)(s-c)}}\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

$$=\frac{s}{s-a}=\frac{2s}{2s-2a}$$

$$=\frac{a+b+c}{b+c-a}=\frac{4a}{2a}=2 \quad (\because b+c=3a)$$

14. **(d)** ΔABC, $\angle A=\frac{\pi}{2}$



$$\tan\frac{C}{2}=\frac{r}{s-c}=\frac{\Delta}{s(s-c)} \quad \left\{\because \Delta=\frac{1}{2}bc\right\}$$

$$\tan\frac{C}{2}=\frac{\frac{1}{2}bc}{\frac{(a+b+c)}{2}\frac{(a+b-c)}{2}}=\frac{2bc}{2b^2+2ab}$$

$\{\because a^2=b^2+c^2\}$

$$=\frac{2bc}{2b(b+a)}=\frac{c}{a+b}\times\frac{a-b}{a-b}$$

$$=\frac{c(a-b)}{a^2-b^2}=\frac{c(a-b)}{c^2}=\frac{a-b}{c}$$

15. **(b)** $\because \frac{b^2-c^2}{2aR}=\frac{4R^2(\sin^2 B-\sin^2 C)}{2.2R\sin A.R}$

$$=\frac{\sin(B+C).\sin(B-C)}{\sin A}=\sin(B-C)$$

16. **(a)** $\frac{a\cos A+b\cos B+c\cos C}{a+b+c}$

$$=\frac{R(\sin 2A+\sin 2B+\sin 2C)}{2R(\sin A+\sin B+\sin C)}$$

$$=\frac{4\sin A\sin B\sin C}{8\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}$$

$$=4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=\frac{r}{R}.$$

17. **(b)** Given $a:b:c=3:7:8$

We know $R=\frac{a}{2\sin A}$ & $r=\frac{\Delta}{s}$

$$\Rightarrow \frac{R}{r}=\frac{as}{2\Delta\sin A}=\frac{a(a+b+c)}{4\times\frac{1}{2}bc\sin^2 A}$$

$$=\frac{a(a+b+c)}{2bc\sin^2 A} \quad \left\{\Delta=\frac{1}{2}bc\sin A\right\}$$

$\{\because a=3k, b=7k, c=8k\}$

$\because \cos A=\frac{b^2+c^2-a^2}{2bc}$

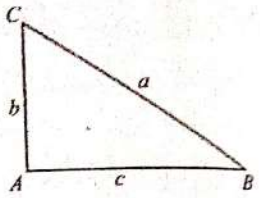
$$=\frac{k^2}{k^2}\frac{49+64-9}{2.7.8}=\frac{104}{2.7.8}=\frac{13}{14}$$

$$\therefore \sin^2 A=1-\frac{13^2}{14^2}=\frac{27}{14^2}$$

$$\Rightarrow \frac{R}{r}=\frac{3(3+7+8)}{2\times7\times8\sin^2 A}\frac{k^2}{k^2}=\frac{3\times18\times14\times14}{2\times7\times8\times27}$$

$$\Rightarrow \frac{R}{r}=\frac{14}{4}=\frac{7}{2} \Rightarrow R:r=7:2$$

18. **(b)** $\because R=\frac{a}{2}$



$\because$ $r=(s-a)\tan\frac{A}{2}$

$$\Rightarrow r=(s-a) \Rightarrow r=s-2R \Rightarrow R=\frac{s-r}{2}$$

19. **(c)** Given $\Delta=100$ sq. cm, $r_1=10$ cm, $r_2=50$ cm

$$r_1=\frac{\Delta}{s-a} \Rightarrow s-a=\frac{100}{10} \Rightarrow s-a=10 \quad \text{... (i)}$$

$$r_2=\frac{\Delta}{s-b} \Rightarrow s-b=\frac{100}{50}$$

$$\Rightarrow s-b=2 \quad \text{... (ii)}$$

(i) – (ii) $\Rightarrow b-a=10-2=8$

20. **(b)** $r=\frac{\Delta}{s}, r_1=\frac{\Delta}{s-a}, r_2=\frac{\Delta}{s-b},$

$$r_3=\frac{\Delta}{s-c}$$

$$\Rightarrow r.r_1.r_2.r_3=\frac{\Delta}{s}\cdot\frac{\Delta}{(s-a)}\cdot\frac{\Delta}{(s-b)}\cdot\frac{\Delta}{(s-c)}$$

$\Rightarrow r\,.\,r_1\,.\,r_2\,.\,r_3=\Delta^2 \because \{\Delta^2=s(s-a)(s-b)(s-c)\}$

21. **(a)** If $r_1>r_2>r_3 \Rightarrow \frac{\Delta}{s-a}>\frac{\Delta}{s-b}>\frac{\Delta}{s-c}$

$\Rightarrow (s-a)<(s-b)<(s-c)$

$\Rightarrow -a<-b<-c$

$\Rightarrow a>b>c$

22. **(a)** Given $\angle B=\frac{\pi}{2}$

$$r=(s-b)\tan\frac{B}{2}=(s-b)\tan\frac{\pi}{4}$$

$$=\frac{a+b+c}{2}-b=\frac{c+a-b}{2}=\frac{AB+BC-CA}{2}$$

23. **(a)** H is orthocentre of ΔABC

Radius of circumcircle of ABC,

BHC, CHA, AHB are same and equal to R

$\angle BHC = B + C$

In ΔBHC, by sin law

$$\Rightarrow \frac{a}{\sin(B+C)}=2R \;\&\; \frac{a}{\sin A}=2R$$

$\Rightarrow \sin(B+C)=\sin A$

$\Rightarrow R_1=R$

similarly $R_2=R$ & $R_3=R$

24. (*a*) $\frac{a-b}{b-c}=\frac{s-a}{s-c}$

$\Rightarrow \frac{(s-b)-(s-a)}{(s-c)-(s-b)}=\frac{(s-a)}{(s-c)}$

$\Rightarrow (s-b)(s-c)-(s-a)(s-c)$

$=(s-a)(s-c)-(s-a)(s-b)$

$\Rightarrow (s-b)(s-c)+(s-a)(s-b)=2(s-a)(s-c)$

divided by $(s-a)(s-b)(s-c)$

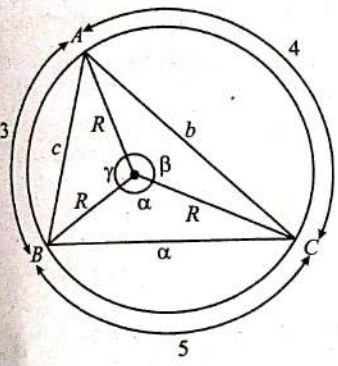
$\Rightarrow \frac{1}{(s-a)}+\frac{1}{(s-c)}=\frac{2}{(s-b)}$

$\Rightarrow \frac{\Delta}{(s-a)}+\frac{\Delta}{(s-c)}=\frac{2\Delta}{(s-b)}$

$\Rightarrow r_1+r_3=2r_2 \Rightarrow r_1, r_2, r_3$ in A.P.

25. (*a*) $2\pi R=3+4+5 \Rightarrow R=\frac{6}{\pi}$

$\alpha=\frac{3\times 360}{3+4+5}=\frac{3\times 360}{12}$



$\alpha=90, \beta=120, \gamma=150$

$\Delta ABC=\Delta OBC+\Delta OCA+\Delta OAB$

$\Delta ABC=\frac{1}{2}R.R\ \sin\alpha+\frac{1}{2}\ R.R\ \sin\beta+\frac{1}{2}$ R.R sin γ

$=\frac{1}{2}R^2[\sin 90^\circ+\sin 120^\circ+\sin 150^\circ]$

$=\frac{R^2}{2}\left[1+\frac{\sqrt{3}}{2}+\frac{1}{2}\right]=\frac{1}{2}\left(\frac{6}{\pi}\right)^2\left[\frac{3+\sqrt{3}}{2}\right]$

$=\frac{36}{4\pi^2}(3+\sqrt{3})=\frac{9\sqrt{3}}{\pi^2}(1+\sqrt{3})$

26. (*b*) $r=\frac{\Delta}{s}, r_1=\frac{\Delta}{s-a}, s=\frac{a+b+c}{2}$

$\frac{r}{r_1}=\frac{s-a}{s}=\frac{1}{2} \Rightarrow s=2a$

$\because \tan\frac{A}{2}=\frac{r}{s-a}$

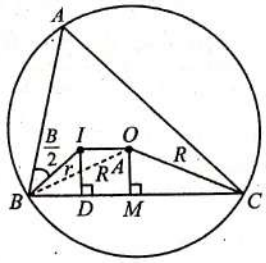
$\tan\frac{A}{2}\left[\tan\frac{B}{2}+\tan\frac{C}{2}\right]=\frac{r}{s-a}\left[\frac{r}{s-b}+\frac{r}{s-c}\right]$

$=\frac{r^2(2s-b-c)}{(s-a)(s-b)(s-c)}=\frac{r^2(a+b+c-b-c)}{\left(\frac{\Delta^2}{s}\right)}$

$=\frac{\Delta^2}{s^2}\frac{s}{\Delta^2}\times a=\frac{a}{s}=\frac{a}{2a}=\frac{1}{2}$

27. (*b*) In ΔOBM, $\cos A=\frac{r}{R}$

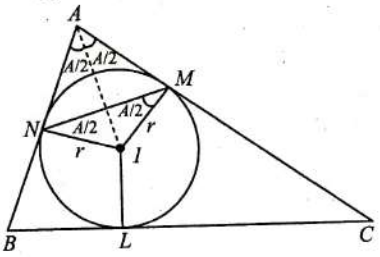
$\Rightarrow \cos A+\cos B+\cos C=1+4\pi\sin\frac{A}{2}$



$\Rightarrow \frac{r}{R}+\cos B+\cos C=1+\frac{r}{R}$

$\Rightarrow \cos B+\cos C=1$

28. (*c*) *r* is in radius



Circumradius is x

$x=\frac{r}{2\sin\frac{A}{2}}, y=\frac{r}{2\sin\frac{B}{2}}\ \&\ z=\frac{r}{2\sin\frac{C}{2}}$

$\Rightarrow xyz=\frac{r^3}{8\left(\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)}=\frac{r^3}{8\frac{r}{4R}}=\frac{1}{2}r^2R$

29. (*a*) We have

$a+b+c=\frac{6(\sin A+\sin B+\sin C)}{3}$

$\Rightarrow k(\sin A+\sin B+\sin C)=2(\sin A+\sin B+\sin C)$

Where $K=\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$

$\Rightarrow K=2$ .....$(\sin A+\sin B+\sin C=0)$

$\Rightarrow \frac{a}{\sin A}=2$

$\Rightarrow \sin A=\frac{1}{2}$

$\Rightarrow A=\frac{\pi}{6}$

30. (*a*) Given: a^2, b^2, c^2 are in A.P

$\Rightarrow b^2-a^2=c^2-b^2$

$\Rightarrow k^2\sin^2B-k^2\sin^2A$

$=k^2\sin^2C-k^2\sin^2B$

using sine rule $a=k\sin A, b=k\sin B$ and $c=k\sin C$

$\Rightarrow \sin(B+A)\sin(B-A)=\sin(C+B)\sin(C-B)$

$\Rightarrow \sin(\pi-C)\sin(B-A)$

$=\sin(\pi-A)\sin(C-B)$ since $A+B+C=\pi$

$\Rightarrow \sin C\sin(B-A)=\sin A\sin(C-B)$

$\Rightarrow \frac{\sin(B-A)}{\sin A}=\frac{\sin(C-B)}{\sin C}$

$\Rightarrow \frac{\sin(B-A)}{\sin A\sin B}=\frac{\sin(C-B)}{\sin B\sin C}$

$\Rightarrow \frac{\sin B\cos A-\cos B\sin A}{\sin A\sin B}=\frac{\sin C\cos B-\cos B\sin C}{\sin B\sin C}$

$\Rightarrow \cot A-\cot B=\cot B-\cot C$

$\therefore$ cot*A*, cot*B*, cot*C* are in A.P

31. (*a*) Let *r* be the radius of the circle.

$\therefore$ its area $A_1=\pi r^2$.

Length of one side of a regular polygon of *n* sides

= (perimeter of the circle)/*n*

$=\frac{2\pi r}{n}=k$(say)

Area of the polygon

$=A_2=\frac{\pi k^2}{4}\cot\left(\frac{\pi}{n}\right)$

$=\frac{r^2\pi^2}{n}\cot\left(\frac{\pi}{n}\right)$

$\therefore A_1:A_2=\pi r^2:\frac{r^2\pi^2}{n}\cot\left(\frac{\pi}{n}\right)$

$=\tan\left(\frac{\pi}{n}\right):\frac{\pi}{n}$

32. (*c*) $=(a+(b+c))((b+c)-a)$

$=\left(\frac{6+c)^2-a^2}{b^2+c^2-a^2+2bc}=\lambda-2\right.$

$\Rightarrow b^2+c^2-a^2+2bc=\lambda bc$

$\Rightarrow \frac{b^2+c^2-a^2}{bc}=\lambda-2$

$\Rightarrow \frac{b^2+c^2-a^2}{bc}=\cos A=\frac{\lambda-2}{2}$

$-1<\cos A<1$

$\Rightarrow -1<\frac{\lambda-2}{2}<1$

$\Rightarrow -2<\lambda-2<2$

$\Rightarrow 0<\lambda<4$

Option (*c*) is correct

33. **(b)** In ΔADC, $AD = b\sin C$

$\Rightarrow \frac{abc}{b^2-c^2} = b\sin C$

$\Rightarrow \frac{\sin C}{c} = \frac{a}{b^2-c^2}$

$\frac{\sin A}{a} = \frac{a}{b^2-c^2}$

$\Rightarrow \sin A = \frac{a^2}{b^2-c^2} \left[\because \frac{\sin C}{c} = \frac{\sin A}{a}\right]$

$\Rightarrow \sin A = \frac{\sin^2 A}{\sin^2 - \sin^2 C}$

$\Rightarrow \sin A = \frac{\sin^2 A}{\sin(B+C)\sin(B-C)}$

$\Rightarrow \text{in}(B-C) = 1 \ [\because \sin(B+C) = \sin A]$

$\Rightarrow B - C = 90°$

$\Rightarrow B = 90° + C = 90° + 23° = 113°$

34. **(c)** $a^3\cos(B-C) = a^2(a\cos(B-C))$

Using sine rule for $a = k\sin A$, we get

$\Rightarrow a^3\cos(B-C) = a^2(k\sin A.\cos(B-C))$

We have $A+B+C = \pi$ and $A = \pi-(B+C)$

$\Rightarrow \sin A = \sin(\pi-(B+C)) = \sin(B+C)$

$\therefore a^3\cos(B-C) = a^2k\sin(B+C)\cos(B-C)$

$= a^2\frac{k}{2}(\sin 2B + \sin 2C)$

$= a^2\frac{k}{2}(2\sin B\cos B + 2\sin C\cos C)$

$= a^2(k\sin B\cos B + k\sin C\cos C)$

$= a^2(b\cos B + c\cos C)$ ....(i)

Now, $b^3\cos(C-A)$

$= b^2k\sin(C+A)\cos(C-A)$

$= b^2\frac{k}{2}(\sin 2C + \sin 2A)]$

$= b^2\frac{k}{2}(2\sin C\cos C + 2\sin A\cos A)$

$= b^2\frac{k}{2}(k\sin C\cos C + k\sin A\cos A)$

$= b^2(c\cos C + a\cos A)$ ....(ii)

Again, $b^3\cos(C-A)$

$= c^2k\sin(A+B)\cos(A-B)$

$= c^2\frac{k}{2}(\sin 2A + \sin 2B)$

$= c^2\frac{k}{2}(2\sin A\cos A + 2\sin B\cos B)$

$= c^2(k\sin A\cos A + k\sin B\cos B)$

$= c^2(a\cos A + b\cos B)$ ....(iii)

Adding equs(i), (ii) and (iii) we get

$a^3\cos(B-C) + b^3\cos(C-A) + c^3\cos(A-B)$

$= a^2(b\cos B + c\cos C) + b^2(c\cos C + a\cos A)$

$+ c^2(a\cos A + b\cos B)$

Taking common, we get

$= ab(a\cos B + b\cos A) + bc(b\cos C + c\cos B) + ca(c\cos A + a\cos C)$

We know that

$a\cos B + b\cos A = c$, $b\cos C + c\cos B = a$

and $c\cos A + a\cos C = b$

$= abc + abc + abc = 3abc$

35. **(a)** $\because \left(\sqrt{a}+\sqrt{b}-\sqrt{c}\right)$

$= \frac{\left(\sqrt{a}+\sqrt{b}-\sqrt{c}\right)\left(\sqrt{a}+\sqrt{b}+c\right)}{\left(\sqrt{a}+\sqrt{b}+\sqrt{c}\right)}$

$= \frac{\left(\sqrt{a}+\sqrt{b}\right)^2 - \left(\sqrt{c}\right)}{\left(\sqrt{a}+\sqrt{b}+\sqrt{c}\right)}$

$= \frac{a+b+2\sqrt{ab}-c}{\left(\sqrt{a}+\sqrt{b}+\sqrt{c}\right)} = \frac{(a+b-c)+2\sqrt{ab}}{\left(\sqrt{a}+\sqrt{b}+\sqrt{c}\right)}$

Since $a+b>c$ we have

$= \frac{(a+b-c)+2\sqrt{ab}}{\left(\sqrt{a}+\sqrt{b}+\sqrt{c}\right)} > 0$

$\therefore \left(\sqrt{a}+\sqrt{b}-\sqrt{c}\right) > 0$

36. **(c)** Area of ΔABC = Area (ΔACD) + Area (ΔBCD)

$\Delta = \frac{1}{2}\times b\times r + \frac{1}{2}\times a\times r = \frac{1}{2}r(a+b)$

$\Rightarrow \frac{2\Delta}{a+b} = r$

$\boxed{\therefore r = \frac{2\Delta}{a+b}}$

$\therefore$ option A is incorrect.

$\therefore$ option C is correct.

37. **(c)** Given $\cos A, \cos B, \cos C$ are in AP.

Then $2\cos B = \cos A + \cos C$

$\Rightarrow 2\left(1-2\sin^2\frac{B}{2}\right) = 2\cos\frac{A-C}{2}\sin\frac{B}{2}$

$\Rightarrow 1-\sin^2\frac{B}{2} = \sin\frac{B}{2}\left[\sin\frac{B}{2}+\cos\left(\frac{A-C}{2}\right)\right]$

$\cos^2\frac{B}{2} = \sin\frac{B}{2}\left[\cos\left(\frac{A+C}{2}\right)+\cos\left(\frac{A-C}{2}\right)\right]$

$\Rightarrow \cot\frac{B}{2} = \frac{2\cos\frac{A}{2}\cos\frac{C}{2}}{\sin\left(\frac{A+C}{2}\right)}$

$\Rightarrow 2\tan\frac{B}{2} = \tan\frac{A}{2}+\tan\frac{C}{2}$

$\Rightarrow \cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in HP

$\Rightarrow 2\left(s\tan\frac{B}{2}\right) = \left(s\tan\frac{A}{2}\right)+\left(s\tan\frac{C}{2}\right)$

$\Rightarrow 2r_2 = r_1 + r_2$

$\Rightarrow r_1, r_2, r_3$ are in AP

38. **(c)** $\therefore \sin A + \sin B = \frac{C(a+b)}{C^2} = \frac{a+b}{c}$

$\therefore \sin A\sin B = \frac{ab}{c^2}$

Also By rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$

$\therefore \sin A\sin B = \frac{ab}{k^2} = \frac{ab}{c^2} \rightarrow$ Given

$C^2x^2 - C(a+b)x + ab = 0(\sin A, \sin B)$

$\therefore k = c$

$\therefore \frac{a+b}{2} = \frac{k(\sin A+\sin B)}{k(\sin c)}$

$= \frac{\sin A+\sin B}{\sin C}$

$k = 2R = C$

Right angled triangle

39. **(c)** $\sqrt{a^2+b^2}$ is the largest side.

Consine of largest angle is

$\cos\theta = \frac{(a-b)^2+(a+b)^2-(\sqrt{a^2+b^2})^2}{2(a-b)(a+b)}$

$\cos\theta = -\frac{1}{2}$

$\theta = \frac{2\pi}{3}$

40. **(c)** Using sin rule

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

Numerator of given expression

$a\text{ Cos }A + b\text{ Cos }B + c\text{ Cos }C$

$\Rightarrow a\text{ Cos }A + R(2\text{Sin}B\text{ Cos}B + 2\text{Sin}C\text{ Cos}C)$

$\Rightarrow a\text{ Cos }A + R(\text{Sin }2B + \text{Sin}2C)$

$\Rightarrow a\text{ Cos }A + 2R\text{ Sin}(B+C)\text{ Cos}(B-C)$

$\Rightarrow a\text{ Cos}(\pi-(B+C)) + 2R\text{ Sin}(\pi-A)\text{ Cos}(B-C)$

$\Rightarrow -a\text{Cos}(B+C) + 2R\text{ Sin }A\text{ Cos}(B-C)$

$\Rightarrow -a\text{Cos}(B+C) + a\text{Cos}(B-C)$

$\Rightarrow a\text{ Cos }(B-C) - a\text{Cos}(B-C)$

$\Rightarrow 2a\text{ Sin }B\text{ Sin }C$

$\Rightarrow 2a\times\frac{2\Delta}{ac}\times\frac{2\Delta}{ab} = \frac{8\Delta^2}{abc} = \frac{8\Delta^2}{4R\Delta} = \frac{2\Delta}{R}$

denominator of the given expression.

$a+b+c = \frac{2\Delta}{r}$

$\Rightarrow \frac{a\text{Cos }A+b\text{Cos }B+c\text{Cos }C}{a+b+c} = \frac{\frac{2\Delta}{R}}{\frac{2\Delta}{r}} = \boxed{\frac{r}{R}}$

41. **(b)** Let AD, BC, CF be the altitudes

$\angle HCD = 90° - B$

$\angle DHC = B$

$\sec B = \frac{HC}{HD}$ ... (i)

in ΔCDH

$\angle HCF = 90° - A$

$\angle EHC = A$

$\sec A = \frac{HC}{HE}$ ... (ii)

from (i) and (ii)

$HE = \sec HD = HE \sec A$

$\frac{HD}{HE} = \frac{\sec A}{\sec B}$

$\frac{HE}{HF} = \frac{\sec B}{\sec C}$

$\therefore \quad HD : HE : HF = \sec A : \sec B : \sec C$

$= \cos A : \cos B : \cos C$

42. **(b)** Given P_1, P_2, P_3 are perpendicular from vertices of triangle

Let a, b, c are length of sides of triangle

area $\Delta = \frac{1}{2} a p_1 \Rightarrow p_1 = \frac{2\Delta}{a}$

$\Delta = \frac{1}{2} b p_2 \Rightarrow p_2 = \frac{2\Delta}{b}$

$\Delta = \frac{1}{2} c p_3 \Rightarrow p_3 = \frac{2\Delta}{c}$

$P_1 P_2 P_3 = \frac{2\Delta}{a} \cdot \frac{2\Delta}{b} \cdot \frac{2\Delta}{c}$

$= \frac{8(\Delta^3)}{abc}$

$\left(\because \Delta = \frac{abc}{4R}\right) = \frac{8(a^2b^2c^2)}{(4R)^3}$

$= \frac{a^2b^2c^2}{8R^3}$

$P_1 P_2 P_3 = \frac{a^2b^2c^2}{8R^3}$

43. **[30°]** $\angle A + \angle B + \angle C = 180°$

$2\angle B = \angle A + \angle C$

$\frac{b}{c} = \frac{\sqrt{3}}{\sqrt{2}}$

$3\angle B = 180°$

$\angle B = 60°$

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$\frac{\sin A}{a} = \frac{\sqrt{3}\sqrt{2}}{2\sqrt{3c}} = \frac{\sin c}{\sqrt{2c}}$

$\sin c = 1$

$c = 90°$

$90 + 60 + x = 180°$

$x = 30°$

44. **(b)** Given that a, b, c are lengths of ΔABC and a^2, b^2, c^2 are the roots of the cubic equation $x^3 - Px^2 + Qx - R = 0$.

In a cubic equation, $ax^3 + bx^2 + cx + d = 0$

Sum of roots can be calculated using the formula, $\frac{-b}{a}$ and

Product of the roots can be calculated using the formula, $\frac{-d}{a}$ So in the given cubic equation,

Sum of roots $= a^2 + b^2 + c^2 = P$.

Product of Roots $= a^2b^2c^2 = R$

In a triangle ABC, using the Cosine Rule we get

$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$\cos B = \frac{a^2 + c^2 - a^2}{2ac}$

$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

On squaring them we get

$\Rightarrow \quad \frac{\cos A}{a} = \frac{b^2 + c^2 - a^2}{2abc}$ ...(i)

$\Rightarrow \quad \frac{\cos B}{b} = \frac{a^2 + c^2 - b^2}{2abc}$ ....(ii)

$\Rightarrow \quad \frac{\cos C}{c} = \frac{a^2 + b^2 - c^2}{2abc}$ .....(iii)

Adding equations (i), (ii) and (iii) we get

$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$

$= \frac{P}{\sqrt{2R}}$

Parikshit (JEE Advanced Level)

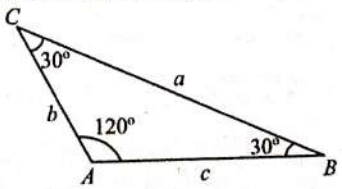
1. **(a)** Here, ratio of angles are 4 : 1 : 1

$\Rightarrow \quad 4x + x + x = 180°$

$\Rightarrow \quad x = 30°$

$\therefore \quad \angle \geq A = 120°, \angle B = \angle C = 30°$

Thus the ratio of longest side to perimeter



$= \frac{a}{a+b+c}$

Let $b = c = x$

$\Rightarrow a^2 = b^2 + c^2 - 2bc\cos A$

$\Rightarrow a^2 = 2x^2 - 2x^2 \cos A = 2x^2 (1 - \cos A)$

$a^2 = 4x^2 \sin^2 A/2$

$\Rightarrow a = 2x \sin A/2$

$a = 2x \sin 60° = \sqrt{3}x$

2. **(a)** $\because \frac{b-c}{a}\cos\frac{A}{2} = \frac{\sin B - \sin C}{\sin A}\cos\frac{A}{2}$

$= \frac{2\cos\frac{B+C}{2}\sin\frac{B-C}{2}}{2\sin\frac{A}{2}\cos\frac{A}{2}}\cos\frac{A}{2}$

$= \sin\frac{B-C}{2}$

L.H.S. option (a) is correct.

(b) Clearly (b) is wrong.

(c) $\because \frac{a}{b+c}\cos\frac{A}{2}$

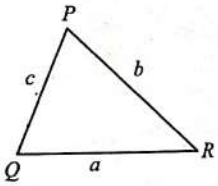
$= \frac{\sin A}{\sin B + \sin C}\cos\frac{A}{2}$

$= \frac{2\cos\frac{A}{2}\sin\frac{A}{2}}{2\sin\frac{B+C}{2}\cos\frac{B-C}{2}}\cos\frac{A}{2}$

$= \frac{\sin A}{2\cos\frac{B-C}{2}}$

Clearly (c) and (d) are wrong.

3. **(c)**



$\left.\begin{matrix} a = 2 \\ b = 7/2 \\ c = 5/2 \end{matrix}\right\} \Rightarrow s = 4$

$\frac{2\sin P - 2\sin P\cos P}{2\sin P + 2\sin P\cos P} = \frac{1-\cos P}{1+\cos P}$

$\Delta = \sqrt{4 \cdot 2 \cdot \frac{1}{2} \cdot \frac{3}{2}} = \sqrt{6}$

$= \tan^2 \frac{P}{2}$

$= \left(\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}\right)^2 = \frac{\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)}{4(2)} = \frac{3}{32}$

$= \left(\frac{9}{16.6}\right) = \left(\frac{3}{4\Delta}\right)^2$

4. **(a)** Given expression is

$\frac{1}{2}\sum\frac{2a}{b+c-a} = \frac{1}{2}\sum\left(\frac{2a}{b+c-a} + 1\right) - \frac{3}{2}$

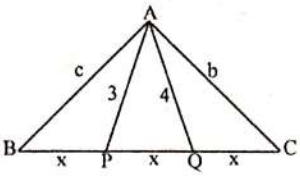
$= \frac{1}{2}(a+b+c)\sum\left(\frac{1}{b+c-a}\right) - \frac{3}{2}$

Now, as $(a+b+c) = \sum(b+c-a)$

Applying A.M. $\geq$ H.M. Minimum value of the expression $= \frac{1}{2}\times 9 - \frac{3}{2} = 3$

5. **(c)** In $\Delta ABP\ 9 = c^2 + x^2 - 2cx\cos B$; but $\cos B = \frac{c}{3x}$

$9 = c^2 + x^2 - 2cx\ \frac{c}{3x}$



$9 = x^2 + \frac{c^2}{3}$ ...(*i*)

$16 = x^2 + \frac{b^2}{3}$ ...(*ii*)

(i) + (ii)

$25 = 2x^2 + \frac{1}{3}(b^2 + c^2) = 2x^2 + 3x^2$

$\Rightarrow 5x^2 = 25$

$\Rightarrow x = \sqrt{5}$;

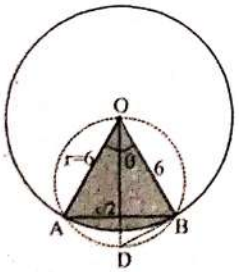
$\therefore BC = 3\sqrt{5} = \sqrt{45}$

6. **(d)** $R = \frac{abc}{4\Delta}; \Delta = \frac{1}{2}\cdot 6\cdot 6\sin\theta = 18\sin\theta$

$a = b = 6$

$\sin\frac{\theta}{2} = \frac{c}{12} \Rightarrow c = 12\sin\frac{\theta}{2}$

$R = \frac{36\cdot 12\sin(\theta/2)}{4\cdot 18\cdot\sin\theta} = 3\sec\frac{\theta}{2}$ *Ans.*



Alternatively: $OD^2 = 6^2 + x^2$

$\frac{x}{6} = \tan$

$OD^2 = 62\cdot\sec^2(\theta/2)$

$4r^2 = 36\sec^2(\theta/2) \Rightarrow r = 3\sec(\theta/2)$

7. **(d)** [Hint put $p = 1$, we get $a^2 = 4$

$\Rightarrow b = 4$

put $p = 2$, we get $a^3 = 4 \Rightarrow c = 4$

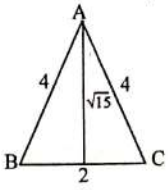
Hence the ΔABC is isosceles

now $\Delta = \sqrt{15}$

$r_1 = \frac{\Delta}{s-a} = \frac{\sqrt{15}}{3}$ and $r^2 = \frac{\Delta}{s-b}$

$= \frac{\sqrt{15}}{1} = r^3$

hence $r^2 = r^3 = 3r_1$]

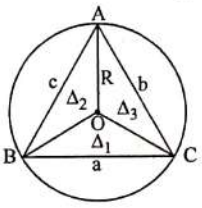


8. **(c)** [Hint : Using $R = \frac{abc}{4\Delta}$

$\Rightarrow \frac{a}{R} = \frac{4\Delta}{bc}$

$\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3} = \frac{4}{R^2}\ (D^1 + D^2 + D^3)$

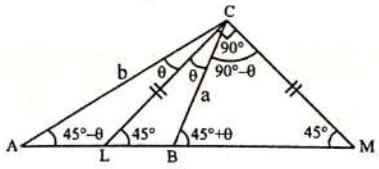
$= \frac{4\Delta}{R^2} \Rightarrow C$]



9. **(c)** $a^2 + b^2 = 4R^2[\sin^2(45° - \theta) + \sin^2(135° - \theta)]$

$= 4R^2[\sin^2(45° - \theta) + \cos^2(45° - \theta)]$

$= 4R^2$ Ans.]



10. **(a, b)**

(a) $\because \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$

$= \frac{b^2+c^2-a^2}{2abc} + \frac{c^2+a^2-b^2}{2abc} + \frac{a^2+b^2-c^2}{2abc}$

$= \frac{a^2+b^2+c^2}{2abc}$

(b) $\because \frac{\sin A}{a} + \frac{\sin B}{b} + \frac{\sin C}{c}$

$= \frac{a}{2R.a} + \frac{b}{2R.b} + \frac{c}{2R.c} = \frac{3}{2R}$

(c) $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$

$\Rightarrow \cot A = \cot B = \cot C$

$\Rightarrow A = B = C$ true for equilateral triangle only

(d) $\frac{\sin 2A}{a^2} = \frac{\sin 2B}{b^2} = \frac{\sin 2C}{c^2}$

$\Rightarrow \frac{2\sin A\cos A}{k^2\sin^2 A} = \frac{2\sin B\cos B}{k^2\sin^2 B} = \frac{2\sin C\cos C}{k^2\sin^2 C}$

$\Rightarrow \cot A = \cot B = \cot C$

$\Rightarrow A = B = C$ true for equilateral triangle only

11. **(a, b, c)**

(a) $\because \tan\left(\frac{A-B}{2}\right) = \left(\frac{a-b}{a+b}\right)\cot\frac{C}{2}$ ...(*i*)

$\because \tan^2\left(\frac{A-B}{2}\right) = \frac{1-\cos(A-B)}{1+\cos(A-B)} = \frac{1-\frac{31}{32}}{1+\frac{31}{32}} = \frac{1}{63}$

$\therefore \tan\left(\frac{A-B}{2}\right) = \frac{1}{3\sqrt{7}}$ $\because a = 5$ and $b = 4$

$\therefore$ from equation (*i*), we get

$\frac{1}{3\sqrt{7}} = \left(\frac{5-4}{5+4}\right)\cot\frac{C}{2} \Rightarrow \frac{1}{3\sqrt{7}} = \frac{1}{9}\cot\frac{C}{2}$

$\Rightarrow \cot\frac{C}{2} = \frac{3}{\sqrt{7}}$

$\because \cos C = \frac{1-\tan^2 C/2}{1+\tan^2 C/2} = \frac{1-7/9}{1+7/9} = \frac{2}{16} = \frac{1}{8}$

$\because \cos C = \frac{b^2+a^2-c^2}{2ab}$

$\Rightarrow c^2 = a^2 + b^2 - 2ab\cos C \Rightarrow c = 6$

(b), (c) $\because$ Area $= \frac{1}{2}$ ab sin C

$\because \cos C = \frac{1}{8}$

$\Rightarrow \sin C = \sqrt{1-\frac{1}{64}} = \frac{3\sqrt{7}}{8}$

Area $= \frac{1}{2}\times 5\times 4\times\frac{3\sqrt{7}}{8}$

Area $= \frac{15\sqrt{7}}{4}$ sq. unit. $\because$ From Sine rule

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$\Rightarrow \sin A = \frac{a\sin C}{c} = \frac{5\times 3\sqrt{7}}{6\times 8}$

$\therefore \sin A = \frac{5\sqrt{7}}{16}$

12. **(a, c, d)**

$\because \beta_a = \frac{2bc}{b+c}\cos\frac{A}{2}$

(a) correct

(b) incorrect

(c) $\frac{abc\cos ec\frac{A}{2}}{2R(b+c)} = \frac{abc\cos ec\frac{A}{2}}{\frac{a}{\sin A}.(b+c)}$

$= \frac{bc\,2\sin\frac{A}{2}\cos\frac{A}{2}}{\sin\frac{A}{2}.(b+c)} = \frac{2bc}{(b+c)}\cos\frac{A}{2}$

(d) $\because \frac{2\Delta}{(b+c)}\ \text{cosec}\ \frac{A}{2} = \frac{bc\sin A}{(b+c)}.\frac{1}{\sin\frac{A}{2}}$

$= \frac{2bc\sin\frac{A}{2}\cos\frac{A}{2}}{(b+c)}.\frac{1}{\sin\frac{A}{2}} = \frac{2bc}{b+c}\cos\frac{A}{2}$

13. **(b, d)** $r_1 = 2r_2 = 3r_3$

$\Rightarrow \frac{\Delta}{s-a} = \frac{2\Delta}{s-b} = \frac{3\Delta}{s-c}$

$\Rightarrow \frac{1}{s-a} = \frac{2}{s-b} = \frac{3}{s-c}$

(i) (ii) (iii)

From (i) and (ii) we get $a - b = c/3$ ... (i)

From (i) and (iii), we get $2a - b = 2c$ ... (ii)

From (ii) and (iii), we get $a - 5b = -5c$ ... (iii)

let $c = k$, then from (i) and (ii), we get

$a = \frac{5k}{3}$ and $b = \frac{4k}{3}$

$\therefore \frac{a}{b} = \frac{5}{4};\ \frac{a}{c} = \frac{5}{3}.$

14. **(a, b, c, d)**

(a) $\frac{\Delta}{s-b} + \frac{\Delta}{s-c} = \frac{\Delta}{s-a} - \frac{\Delta}{s}$

$\Rightarrow \frac{1\cdot}{(s-b)(s-c)} = \frac{1}{s(s-a)}$

$\Rightarrow \tan^2\frac{A}{2} = 1 \Rightarrow A = 90°$

(b) $4R^2(\sin^2A + \sin^2B + \sin^2C) = 8R^2$

$1 - (\cos^2A - \sin^2B) + \sin^2C = 2$

$1 - \cos(A+B)\cos(A-B) + 1 - \cos^2C = 2.$

$\cos C\cos(A-B) - \cos^2C = 0$

$\cos C[\cos(A-B) - \cos C] = 0$

$\cos C[\cos(A-B) + \cos(A+B)] = 0$

$2\cos A\cos B\cos C = 0$

$\Rightarrow$ A = 90° or B = 90° or C = 90°

(c) $r_1 = s.$

$s\tan A/2 = s \Rightarrow \tan A/2 = 1$

$\Rightarrow A = 90°$

(d) $\frac{a}{\sin A} = \frac{a\Delta}{s(s-a)} \Rightarrow \frac{1}{\sin A} = \tan\frac{A}{2}$

$\Rightarrow 2\sin^2A/2 = 1$

$\Rightarrow 1 - \cos A = 1 \Rightarrow \cos A = 0 \Rightarrow A = 90°$

15. **(a, b)**

$\because \sin C = \frac{1-\cos A\cos B}{\sin A\sin B} \le 1$

$\Rightarrow \cos(A-B) \ge 1$

$\Rightarrow \cos(A-B) = 1$

$\Rightarrow A - B = 0 \qquad \Rightarrow A = B$

$\therefore \sin C = \frac{1-\cos^2 A}{\sin^2 A} = 1$

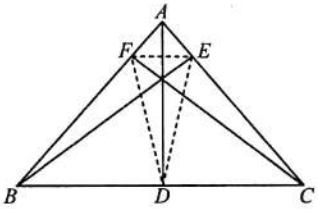
$\Rightarrow C = 90°$

16. **(a, b, c, d)**

$\because$ FE = $a\cos A = R\sin 2A$

DE = $c\cos C = R\sin 2C$

FD = $b\cos B = R\sin 2B$



(a) $= \frac{R\left(\sum\sin 2A\right)}{a+b+c}$

$= \frac{R(4\sin A\sin B\sin C)}{2R\left(4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}\right)}$

$= \frac{8\left(\prod\sin\frac{A}{2}\right)\left(\prod\cos\frac{A}{2}\right)}{2\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}} = \frac{r}{R}$

(b) $\because$ Area of ΔDEF

$= \frac{1}{2}$ FD × DE sin $(\pi - 2A)$

$= \frac{1}{2}\ b\cos B.c\cos C.\sin 2A$

$= \frac{1}{2}\ bc\cos B.\cos C.2\sin A.\cos A$

$= 2\left(\frac{1}{2}bc\sin A\right)\cos A.\cos B.\cos C$

$= 2\Delta\cos A.\cos B.\cos C$

(c) Area of $\Delta AEF = \frac{1}{2}AE \times AF\sin A$

$= \frac{1}{2}$ (c cos A) (b cos A) sin A

$= \left(\frac{1}{2}bc\sin A\right)\cos^2A = \Delta\cos^2A$

(d) $R_{DEF} = \frac{FE \times DE \times FD}{4\Delta_{DEF}}$

$= \frac{abc\cos A\cos B\cos C}{4\times 2\Delta\cos A\cos B\cos C} = \frac{abc}{8\Delta} = \frac{4R\Delta}{8\Delta} = \frac{R}{2}$

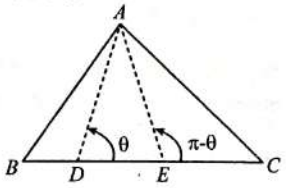
17. **(b, d)**

Product of distances of incenter from angular points

$= \frac{r^3}{\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}} = \frac{r^3}{r/4R} = 4Rr^2$

$= \frac{abc}{\Delta}r^2 = \frac{(abc)(r)}{\frac{\Delta}{r}} = \frac{(abc)(r)}{s}$

18. **(a, c, d)**



if we apply *m-n* Rule in Δ ABE, we get

$(1 + 1)\cot\theta = 1.\cot B - 1.\cot\theta$

$2\cot\theta = \cot B - \cot\theta$

$3\cot\theta = \cot B$

$\tan\theta = 3\tan B$ ... (i)

Similarly, if we apply *m-n* Rule in Δ ACD, we get

$(1 + 1)\cot(\pi - \theta) = 1.\cot\theta - 1.\cot C.$

$\cot C = 3\cot\theta$

$\Rightarrow \tan\theta = 3\tan C$ ... (ii)

form (i) and (ii) we can say that

$\tan B = \tan C \Rightarrow B = C$

$\because A + B + C = \pi$

$\therefore A = \pi - (B + C)$

$= \pi - 2B \because B = C$

$\therefore \tan A = -\tan 2B$

$= -\left(\frac{2\tan B}{1-\tan^2 B}\right) = -\frac{\frac{2\tan\theta}{3}}{1-\frac{\tan^2\theta}{9}}$

$\Rightarrow \tan A = \frac{6\tan\theta}{\tan^2\theta - 9}$

19. **(a, c, d)**

$\because r_1 - r = \frac{\Delta}{s-a} - \frac{\Delta}{s} = \frac{\Delta a}{s(s-a)} = a\tan\frac{A}{2}$

$\therefore \Pi(r_1 - r) = abc\tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}$

$$= abc\,\Pi \tan\frac{A}{2} = abc\,\frac{\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}$$

$$= \frac{(abc)r}{4R.\frac{(\sin A+\sin B+\sin C)}{4}}$$

$$= \frac{(abc)r}{R\left(\frac{a}{2R}+\frac{b}{2R}+\frac{c}{2R}\right)}$$

$$= \frac{2(abc)r}{2s} = \frac{4R\Delta r}{s} = 4r^2$$

20. **(b,c)** $\tan\frac{C-B}{2} = \frac{c-b}{c+b}\cot\frac{A}{2}$

$\Rightarrow \tan\frac{C-B}{2} = \frac{1}{5}\cot 30^\circ$

$= \frac{\sqrt{3}}{5} = \tan\alpha$ given.

$\Rightarrow C-B = 2\alpha$ and $C+B = 180^\circ - 60^\circ = 120^\circ$

Solving we get $C = 60^\circ + \alpha$, $\beta = 60^\circ - \alpha$

21. **(b,c)** $\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}$

$= \frac{1}{3}\cot\frac{C}{2}$

$$\therefore \cos(A-B) = \frac{1-\tan^2\frac{A-B}{2}}{1+\tan^2\frac{A-B}{2}}$$

$$\Rightarrow \frac{4}{5} = \frac{1-\frac{1}{9}\cot^2\frac{C}{2}}{1+\frac{1}{9}\cot^2\frac{C}{2}}$$

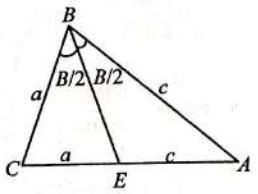
$\Rightarrow \tan^2\frac{C}{2} = 1 \Rightarrow C = \frac{\pi}{2}$

$ar(\triangle ABC) = \frac{1}{2}ab = \frac{1}{2}\cdot 6\cdot 3 = 9$

Also $\sin A = \frac{6}{\sqrt{3^2+6^2}} = \frac{6}{3\sqrt{5}}$

$A = \sin^{-1}\frac{2}{\sqrt{5}}$

22. **(c)**



Let $BE = x$

$\because \Delta = \Delta_1 + \Delta_2$

$\Rightarrow \frac{1}{2}ac\sin B = \frac{1}{2}ax\sin\frac{B}{2} + \frac{1}{2}cx\sin\frac{B}{2}$

$\Rightarrow x = \frac{ac\sin B}{(a+c)\sin\frac{B}{2}} = \frac{2ac\cos\frac{B}{2}}{a+c}$

23. **(a)** we know that

$BE = \frac{2ac}{a+c}\cos\frac{B}{2}, AD = \frac{2bc}{b+c}\cos\frac{A}{2}$

$CF = \frac{2ab}{a+b}\cos\frac{C}{2}$

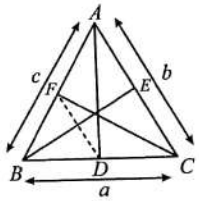
$\because \frac{1}{AD}\cos\frac{A}{2} + \frac{1}{BE}\cos\frac{B}{2} + \frac{1}{CF}\cos\frac{C}{2}$

$= \frac{b+c}{2bc} + \frac{a+c}{2ac} + \frac{a+b}{2ab}$

$= \frac{2(ab+bc+ca)}{2abc} = \frac{1}{a}+\frac{1}{b}+\frac{1}{c}$

24. **(b)**

Area of $\Delta BDF = \frac{1}{2}\times BD\times BF\times\sin B$



$\frac{BD}{CD} = \frac{c}{b} = x$ (let) $\Rightarrow \frac{BD}{c} = \frac{CD}{b} = x$

$\Rightarrow BD = cx$ ...(i)

and $CD = bx$ ...(ii)

$BD + CD = cx + bx = a$ (from (i) & (ii))

$\Rightarrow (x)(b+c) = a$

$\Rightarrow x = \frac{a}{b+c}$

$BD = cx = \frac{ac}{b+c}$ (from (i))

$BD = \frac{ac}{b+c}$ ...(iii)

Similarly,

$\frac{AF}{b} = \frac{BF}{a} = y$(let)

$AF = by$ ...(iv)

and $BF = ay$ ...(v)

$AF + BF = y(a+b) = c$ {From (iv) & (v)}

$\Rightarrow y = \frac{c}{a+b}$

$BF = ay = \frac{ac}{a+b}$ (from (v))

$BF = \frac{ac}{a+b}$ ...(vi)

ar $(\Delta BDF) = \frac{1}{2}\times BD\times BF\sin B$

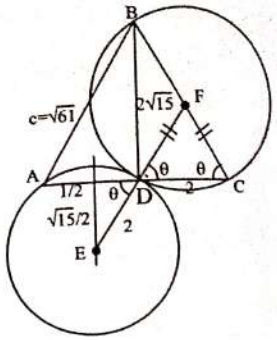
$= \frac{1}{2}\times\frac{ac}{b+c}\times\frac{ac}{a+b}\sin B$

$= \frac{\Delta ac}{(a+b)(b+c)}$, ($\Delta$ = area of ΔABC)

25. **(a)** $b = AC = 3$ and ratio is 1 : 2,

Hence, $AD = 1$ and $DC = 2$

Now $\sin\theta = \frac{\sqrt{15}}{4} \Rightarrow \theta = \sin^{-1}\frac{\sqrt{15}}{4}$



26. **(d)** Obviously $\angle FDC = \theta = \cos^{-1}\frac{1}{4}$

$= \sin^{-1}\frac{\sqrt{15}}{4} = \angle FCD$

hence A, B, C are correct.

now area $\Delta ADE = 1\cdot\frac{\sqrt{15}}{2}\cdot\frac{1}{2} = \frac{\sqrt{15}}{4}$

now $\tan\theta = \frac{BD}{2}$ (in ΔBDC)

$\therefore \sqrt{15} = \frac{BD}{2} \Rightarrow BD = 2\sqrt{15}$

$\tan A = \frac{2\sqrt{15}}{1} = 2\sqrt{15}$

$\sin A = \frac{2\sqrt{15}}{\sqrt{61}}$ ...(i)

Also $\sin A = \frac{2\sqrt{15}}{c} = \frac{2\sqrt{15}}{\sqrt{61}} \Rightarrow c = \sqrt{61}$

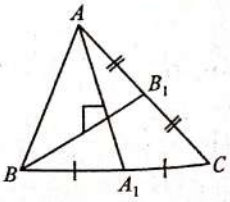
Area of $\Delta DBC = \frac{2\cdot 2\sqrt{15}}{2} = 2\sqrt{15}$

$\therefore \frac{\text{area. }\Delta ADE}{\text{area. }\Delta DBC} = \frac{\sqrt{15}/4}{2\sqrt{15}} = \frac{1}{8} \Rightarrow (D)$

27. **(c)** In ΔABC, $\frac{c}{\sin\theta} = 2R$;

$R = \frac{c}{2\sin\theta} = \frac{\sqrt{61}\cdot 4}{2\sqrt{15}} = \frac{2\sqrt{61}}{\sqrt{15}}$

28. **(a)** (a) → (s), (b) → (p), (c) → (r), (d) → (q)



Match the column

(a) $\because AA_1$ and BB_1 are perpendicular

$\therefore a^2 + b^2 = 5c^2$

$\therefore c^2 = \frac{a^2+b^2}{5} = 5 \Rightarrow c = \sqrt{5}$

$\because \cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{16+9-5}{2\times4\times3} = \frac{5}{6}$

$\because \sin C = \frac{\sqrt{11}}{6}$

$\therefore \Delta = \frac{1}{2}$ ab sin C $= \sqrt{11}$

$\therefore \Delta^2 = 11$

(*b*) $\because$ line joining the circumcentre and orthocentre is parallel to side AC

$\Rightarrow$ R cos B = 2R cos A cos C

$\Rightarrow -\cos(A+C) = 2\cos A\cos C$

$\Rightarrow \sin A \sin C = 3\cos A\cos C$

$\Rightarrow \tan A \tan C = 3$

(*c*) $\because \tan^2\frac{C}{2} = \frac{(s-a)(s-b)}{s(s-c)}$

$\because a = 5, b = 4 \because 2s = 9 + c$

$= \frac{(9+c-10)(9+c-8)}{(9+c)(9-c)} = \frac{c^2-1}{81-c^2}$

$\Rightarrow \frac{7}{9} = \frac{c^2-1}{81-c^2} \Rightarrow c^2 = 36 \Rightarrow c = 6$

(*d*) $\because 2a^2 + 4b^2 + c^2 = 4ab + 2ac.$

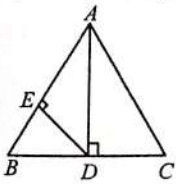
$\Rightarrow (a-2b)^2 + (a-c)^2 = 0$

$\Rightarrow a = 2b = c$

$\because \cos B = \frac{a^2+c^2-b^2}{2ac} = \frac{7}{8}$

$\therefore 8\cos B = 7$

29. (*c*) A $\to$ R, B $\to$ P, C $\to$ S, D $\to$ Q



(A) Area of $\Delta ADB = \frac{1}{2}(AD)\ (BD)$

$\because c\cos B = BD,\ c\sin B = AD$

Area $= \frac{1}{2}c^2 \sin B\cos B = \frac{c^2}{4}\sin 2B$

(B) Area of $\Delta ADC = \frac{1}{2}(AD)\ (DC)$

$\because AD = b\sin C,\ DC = b\cos C$

Area $= \frac{1}{2}b^2 \sin C\cos C = \frac{b^2}{4}\sin 2C$

(C) Area of $\Delta ADE = \frac{1}{2}(AE)(ED)$

$\because ED = BD\sin B = C\sin B\cos B$

$\because (AE)^2 = (AD)^2 - (ED)^2 = c^2\sin^4 B$

$\Rightarrow AE = c\sin^2$ B

Area $= \frac{1}{4}c^2 \sin^2 B \sin^2 B$

(D) Area of (ED) (BE)

$ED = c\sin B\cos B$

$(BE)^2 = (BD)^2 - (ED)^2$

$(BE)^2 = c^2\cos^2 B - c^2\sin^2 B\cos^2 B = c^2\cos^4 B$

$BE = \cos^2 B$

Area $= \frac{1}{4}c^2 \cos^2 B \sin 2B$

30. [7]

$\cos A = \frac{9+16-4}{2\cdot3\cdot4} = \frac{7}{8} \qquad s = \frac{9}{2}$

$D = \sqrt{s(s-a)(s-b)(s-c)} = \frac{3\sqrt{15}}{4}$

HA $= 2R\cos A = 2\cdot\left(\frac{abc}{4\Delta}\right)\cdot\cos A = 2\left(\frac{2\cdot3\cdot4}{4\cdot\frac{3\sqrt{5}}{4}}\right)\frac{7}{8}$

$= \frac{14}{\sqrt{15}} \Rightarrow \frac{\sqrt{15}\,\text{HA}}{2} = 7$ Ans.

31. [2]

$\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{25+36-16}{60} = \frac{3}{4}$

Similarly, $\cos B = \frac{9}{16}$ and $\cos C = \frac{1}{8}$

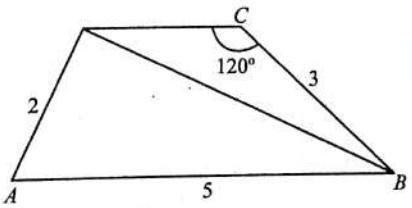
$\therefore \cos A + \cos B + \cos C$

$= \frac{23}{16} = \frac{K}{M}$ then $\left(\frac{K}{M}\right) = 2$

(where (.) denotes least integer)

32. [2] $\cos 60° = \frac{2^2+5^2-BD^2}{2(2)(5)}$, $BD^2 = 19$

Now $\cos 120° = \frac{x^2+3^2-19}{(2)(3)(x)}$



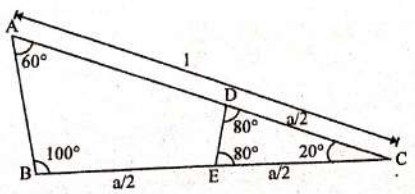
$\Rightarrow x^2 + 3x - 10 = 0$

$\Rightarrow x = -5, 2$

33. [67] Δ_1 = area of ΔABC

Δ_2 = area of ΔCDE

$\Delta_1 = \frac{1}{2}\cdot1\cdot c\sin 60° = \frac{c\sqrt{3}}{4}$



now, $\frac{a}{\sin 60°} = \frac{c}{\sin 20°} = \frac{1}{\sin 100°}$

$\Delta_1 = \frac{1}{2}(\sin 60°)\frac{\sin 20°}{\sin 100°}$

$\Delta_1 = \frac{1}{2}\cdot\frac{\sqrt{3}}{2}\cdot\frac{2\sin 10°\cdot\cos 10°}{\cos 70°}$

$\Delta_1 = \frac{\sqrt{3}}{2}\sin 10°$ ...(*i*)

$\Delta_2 = \frac{1}{2}\cdot\frac{a}{2}\cdot\frac{a}{2}\cdot\sin 2\ \ 0° = \frac{1}{8}\cdot$

$\sin 20°\cdot a^2 = \frac{1}{8}\cdot\sin 20°\cdot$

$\left(\frac{\sin 60°}{\sin 100°}\right) = \frac{1}{8}\cdot\sin 20°\cdot\frac{3}{4\cos^2 10°}$

$= \frac{3}{32}\cdot\frac{2\sin 10°\cdot\cos 10°}{\cos^2 10°} = \frac{3}{16}\cdot\frac{\sin 10°}{\cos 10°}$

$2\Delta_2 = \frac{3}{8}\cdot\frac{\sin 10°}{\cos 10°}$

now $\Delta_1 + 2\Delta_2 = \frac{\sqrt{3}}{2}\sin 10° + \frac{3}{8}\cdot\frac{\sin 10°}{\cos 10°}$

$= \frac{\sqrt{3}}{8}\left(4\sin 10° + \sqrt{3}\frac{\sin 10°}{\cos 10°}\right)$

$= \frac{\sqrt{3}}{8}\left(\frac{2\sin 20° + \sqrt{3}\sin 10°}{\cos 10°}\right)$

$= \frac{\sqrt{3}}{8}\left(\frac{2\sin 20° + 2\sin 60°\cdot\sin 10°}{\cos 10°}\right)$

$= \frac{\sqrt{3}}{8}\left(\frac{\overline{2\sin 20°} + \cos 50° - \overline{\cos 70°}}{\cos 10°}\right)$

$= \frac{\sqrt{3}}{8}\left(\frac{\sin 20° + \sin 40°}{\cos 10°}\right)$

$= \frac{\sqrt{3}}{8}\left(\frac{2\sin 30°\cos 10°}{\cos 10°}\right) = \frac{\sqrt{3}}{8} = \sqrt{\frac{3}{64}}$

$\Rightarrow \quad m = 3, n = 64$

$\Rightarrow \quad (m+n) = 3 + 64 = 67$ Ans.]

34. [12] Given, the equation $ax^2 + bx + c = 0$ has equal roots, so

$b^2 = 4ac$ ...(*i*)

As, $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$= \frac{a^2+c^2-4ac}{2ac}$ (Using equation (*i*))

$\Rightarrow 2\cos^2\frac{B}{2} - 1 = \frac{a^2+c^2}{2ac} - 2$

$\Rightarrow 2\cos^2\frac{B}{2} = \frac{a^2+c^2}{2ac} - 1$

$\Rightarrow \cos^2\frac{B}{2} = \frac{a^2+c^2}{4ac} - \frac{1}{2} = \frac{1}{4}\left(\frac{a}{c}+\frac{c}{a}-2\right)$

As, $0 < \cos^2\frac{B}{2} < 1 \Rightarrow 0 < \frac{1}{4}\left(\frac{a}{c}+\frac{c}{a}-2\right) < 1$

$\Rightarrow 0 < \left(\frac{a}{c}+\frac{c}{a}-2\right) < 4$

$\Rightarrow 2 < \frac{a}{c} + \frac{c}{a} < 6$

Also, $\frac{\sin A}{\sin C} + \frac{\sin C}{\sin A} = \frac{a}{c} + \frac{c}{a}$

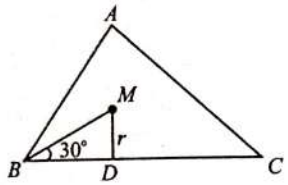
(Using sine law in ΔABC)

Hence, possible integers in the range = 3, 4, 5.

So, sum of integers = 3 + 4 + 5 = 12.

PYQ's (Past Year Questions)

1. (*a*)



If $\cos 2A + \cos 2B + \cos 2C$ is minimum then $A = B = C = 60°$, So $\sin 2A = \sin A$, $\sin 2B = \sin B$, $\sin 2C = \sin C$

So ΔABC is equilateral

Now in-radias $r = 3$

So in ΔMBD we have

$\tan 30° = \frac{MD}{BD} = \frac{r}{a/2} = \frac{6}{a}$

$BM = 6$

$1/\sqrt{3} = \frac{6}{a} \Rightarrow a = 6\sqrt{3}$

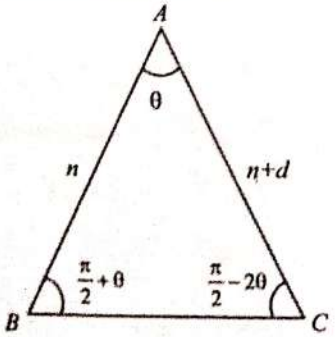
$\overrightarrow{MA}.\overrightarrow{MB} = 36\cos 120° = -18$

Perimeter of $\Delta ABC = 18\sqrt{3}$

Area of $\Delta ABC = \frac{\sqrt{3}}{4}a^2 = 27\sqrt{3}$

2. **[1008.00]**

Let us consider $A < C < B$



$n - d = 2\sin\theta$ ...(*i*)

$n + d = 2\sin\left(\frac{\pi}{2} + \theta\right)$

$\Rightarrow n + d = 2\cos\theta$ ...(*ii*)

$n = 2\sin\left(\frac{\pi}{2} - 2\theta\right)$

$\Rightarrow n = 2\cos 2\theta$ ...(*iii*)

$\Rightarrow 2\cos 2\theta = 2\sin\theta = \cos\theta$

$\Rightarrow 2(\cos\theta - \sin\theta) = 1$

$\Rightarrow \sin 2\alpha = \frac{3}{4}$

Then, $a = \frac{1}{2}\, n.\,(n + d).\,\sin\theta$

$= \frac{1}{2}.\, 2\cos 2\theta.\, 2\cos\theta.\sin\theta$

$= \sin 2\theta.\, \cos 2\theta$

$= \frac{3}{4} \times \frac{\sqrt{7}}{4} = \frac{3\sqrt{7}}{16}$

Hence, value of $64a^2$ will be

$= \left(64 \times \frac{3\sqrt{7}}{16}\right)^2 = 16 \times 9 \times 7 = 1008$

3. **[0.25]** From above equation

We know, $a = \frac{1}{2}n \cdot (n + d) \cdot \sin\theta$

$r = \frac{\Delta}{s} = \frac{1}{2}\frac{n(n+d)\sin\theta}{\left(\frac{3n}{2}\right)}$

$= \frac{(n+d)\cdot\sin\theta}{3}$

$= \frac{2\cos\theta\sin\theta}{3}$

(from previous we know $n+d = 2\cos\theta$)

$r = \frac{\sin 2\theta}{3} = \frac{1}{4}$

4. (*a*) $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13} = \frac{a+b+c}{18} = s$

$\Rightarrow a = 7s,\ b = 6s,\ c = 5s$

Now, $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{1}{5}$

Similarly, $\cos B = \frac{19}{35}$, $\cos C = \frac{5}{7}$

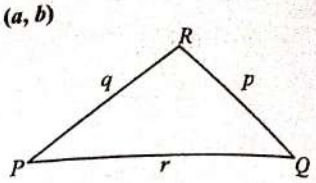
Now, $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos c}{\lambda}$

So, $\frac{1}{5\alpha} = \frac{19}{35\beta} = \frac{5}{7\lambda}$

$\Rightarrow \frac{7}{35\alpha} = \frac{19}{35\beta} = \frac{25}{35\gamma}$

$\alpha : \beta : \gamma = 7 : 19 : 25$

5. (*a*, *b*)



(*a*) $\cos P = \frac{q^2 + r^2 - p^2}{2qr} = \frac{q^2 + r^2}{2qr} - \frac{p^2}{2qr} \geq 1 - \frac{p^2}{2qr}$

(as $p^2 + q^2 \geq$ (AM $\geq$ GM), so (*a*) is correct}

(*b*) $(p+q)\cos R \geq (q-r)\cos P + (p-r)\cos Q$

$\Rightarrow (p\cos R + r\cos P) + (q\cos R + r\cos Q) \geq q\cos P + p\cos Q$

$\Rightarrow q + p \geq r$

So (*b*) is correct

(*c*) $\frac{q+r}{p} = \frac{\sin Q + \sin R}{\sin P} \geq \frac{2\sqrt{\sin Q \times \sin R}}{\sin P}$

So (*c*) is incorrect

(*d*) $\cos Q > \frac{p}{r} \Rightarrow \sin R\cos Q > \sin P$

$\Rightarrow \sin P + \sin(R - Q) > 2\sin P$

$\Rightarrow \sin(R - Q) > \sin P$

need not necessarily hold true if R < Q

Hence (*a*), (*b*)

6. (*c*) 5, 5*r*, 5*r*^{2} sides of triangle, Now triangle - Inequality

$5 + 5r > 5r^2$ ...(*i*)

$5 + 5r^2 > 5r$ ...(*ii*)

$5r + 5r^2 > 5$ ...(*iii*)

From (*i*)

$r^2 - r - 1 < 0$

$r \in \left(\frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2}\right)$ ...(*iv*)

From (*ii*) $r^2 - r + 1 > 0 \Rightarrow r \in R$ ...(*v*) [$\because D < 0$]

From (*iii*) $r^2 + r - 1 > 0$

So, $\left(r + \frac{1-\sqrt{5}}{2}\right)\left(r + \frac{1+\sqrt{5}}{2}\right) > 0$

$r \in \left(-\infty, \frac{-1-\sqrt{5}}{2}\right) \cup \left(\frac{-1+\sqrt{5}}{2}, \infty\right)$ ...(*vi*)

From (*iv*), (*v*), (*vi*) $r \in \left(\frac{-1+\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2}\right)$ $\approx (0.61, 1.61)$

Now check options, option (*c*) will satisfy the condition.

7. **(c)** $a<b<c$ are in A.P.

$\angle C = 2\angle A$(Given)

$\Rightarrow \sin C = \sin 2A \Rightarrow \sin C = 2\sin A.\cos A$

$\Rightarrow \dfrac{\sin C}{\sin A} = 2\cos A$

$\Rightarrow \dfrac{c}{a} = 2\dfrac{b^2+c^2-a^2}{2bc}$

put $a = b-\lambda, c = b+\lambda, \lambda > 0$

$\Rightarrow \lambda = \dfrac{b}{5}$

$\Rightarrow a = b-\dfrac{b}{5} = \dfrac{4}{5}b, c = b+\dfrac{b}{5} = \dfrac{6b}{5}$

$\Rightarrow a = b\dfrac{b}{5} = \dfrac{4}{5}b, c = b+\dfrac{b}{5} = \dfrac{6b}{5}$

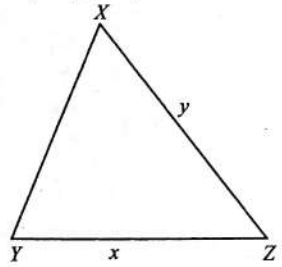
$\Rightarrow$ required ratio = 4 : 5 : 6

8. **(b, c)** $\tan\dfrac{x}{2}+\tan\dfrac{z}{2} = \dfrac{2y}{x+y+z}$

$\dfrac{\Delta}{s(s-x)}+\dfrac{\Delta}{s(s-z)} = \dfrac{2y}{2s}$

$\dfrac{\Delta}{s}\left(\dfrac{2s-(x+z)}{(s-x)(s-z)}\right) = \dfrac{y}{s}$

$\Rightarrow \dfrac{\Delta y}{s(s-x)(s-z)} = \dfrac{y}{s}$



$\Rightarrow \Delta^2 = (s-x)^2(s-z)^2$

$\Rightarrow s(s-y) = (s-x)(s-z)$

$\Rightarrow (x+y+z)(x+z-y)$

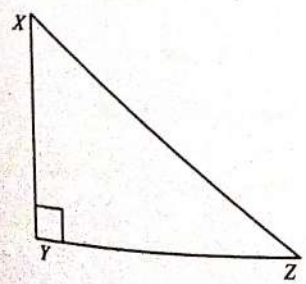
$= (y+z-x)(x+y-z)$

$\Rightarrow (x+z)^2 - y^2 = y^2 - (z-x)^2$

$\Rightarrow (x+z)^2 + (x-z)^2 = 2y^2$

$\Rightarrow x^2+z^2 = y^2 \quad \Rightarrow \angle Y = \dfrac{\pi}{2}$

$\Rightarrow \angle Y = \angle X + \angle Z$



$\tan\dfrac{x}{2} = \dfrac{\Delta}{S(S-x)}$

$\tan\dfrac{x}{2} = \dfrac{\frac{1}{2}xz}{\frac{(y+z)^2-x^2}{4}}$

$\tan\dfrac{x}{2} = \dfrac{2xz}{y^2+z^2+2yz-x^2}$

$\tan\dfrac{x}{2} = \dfrac{2xz}{2z^2+2yz} \quad$ (using $y^2 = x^2+z^2$)

$\tan\dfrac{x}{2} = \dfrac{x}{y+2}$

9. **[5]** $x\sin A + y\sin B + z\sin C = 18$ and $x\sin A + y\sin 2B + z\sin 2C + 9$

Put $x = y = z$

$\sin A + \sin B + \sin C = \dfrac{18}{x}$

$\text{Sin}\,2A + \sin 2B + \sin 2C = \dfrac{9}{x}$

$\therefore \sin A + \sin B + \sin C = 2(\sin 2A + \sin 2B + \sin 2C)$

$\Rightarrow 4\cos A/2\ \cos B/2\ \cos C/2$

$= 2(4\sin A\sin B\sin C)$

$\Rightarrow 16\sin A/2\sin B/2\sin C/2 = 1$

10. **(d)** We have,

$2\sin\left(\dfrac{\pi}{8}\right)\sin\left(\dfrac{2\pi}{8}\right)\sin\left(\dfrac{3\pi}{8}\right)\sin\left(\dfrac{5\pi}{8}\right)$

$\sin\left(\dfrac{6\pi}{8}\right)\sin\left(\dfrac{7\pi}{8}\right)$

$2\sin\dfrac{\pi}{8}\sin\dfrac{2\pi}{8}\sin\dfrac{3\pi}{8}\sin\left(\pi-\dfrac{5\pi}{8}\right)$

$\sin\left(\pi-\dfrac{6\pi}{8}\right)\sin\left(\pi-\dfrac{7\pi}{8}\right)$

$= 2\sin\dfrac{\pi}{8}\sin\dfrac{2\pi}{8}\sin\dfrac{3\pi}{8}\sin\dfrac{3\pi}{8}\sin\dfrac{2\pi}{8}\sin\dfrac{\pi}{8}$

$= 2\sin^2\dfrac{\pi}{8}\sin^2\dfrac{2\pi}{8}\sin^2\dfrac{3\pi}{8}$

$= \left(\dfrac{1}{\sqrt{2}}\right)^2 \cdot 2\sin^2\dfrac{\pi}{8}\sin^2\dfrac{3\pi}{8}$

$= \dfrac{1}{4}\cdot\left(2\sin\dfrac{\pi}{8}\sin\dfrac{3\pi}{8}\right)^2$

$= \dfrac{1}{4}\left[\cos\dfrac{\pi}{4}-\cos\dfrac{\pi}{2}\right]^2 = \dfrac{1}{4}\times\left(\dfrac{1}{\sqrt{2}}\right)^2 = \dfrac{1}{8}$

11. **(a)** $\dfrac{a+b}{7} = \dfrac{b+c}{8} = \dfrac{c+a}{9} = \lambda$

$a+b = 7k, b+c = 8k, a+c = 9k$

$\Rightarrow a+b+c = 12k$

Now $a = 4k, b = 3k, c = 5k,$

It is observed that

$\because c^2 = b^2+a^2$

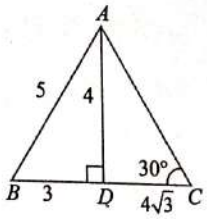
$\angle C = 90°$

$\Delta = \dfrac{1}{2}ab\sin C = \dfrac{1}{2}ab$

$\dfrac{R}{r} = \dfrac{c}{2\sin C}\times\dfrac{s}{\Delta} = \dfrac{c}{2}\times\dfrac{6k}{\frac{1}{2}ab} = \dfrac{c}{ab}\times 6k = \dfrac{5}{2}$

$\left[\because \dfrac{C}{\sin C} = 2R, r = \dfrac{\Delta}{S}\right]$

12. **(a)** Given, $\cos B = \dfrac{3}{5}, \Rightarrow AD = 5$ units



using Pythagoras theorem we have $AD = 4$ units.

$\therefore \dfrac{5}{\sin C} = 10 \Rightarrow C = 30°$

$\Delta = \dfrac{1}{2}(4)\left(3+4\sqrt{3}\right) = 6+8\sqrt{3}$

13. **(c)** Let smallest angle is $R = \theta$

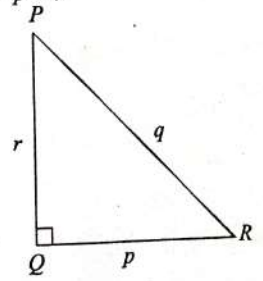
Therefore the angle $P = 90° - \theta$

and angle $Q = 90°$

i.e. $q > p > r$

Here, $p = 2R\cos\theta, q = 2R, r = 2R\sin\theta$

$q^2 = p^2 + r^2$



according to question,

$\dfrac{1}{r^2} = \dfrac{1}{p^2}+\dfrac{1}{q^2} \Rightarrow \dfrac{p^2q^2}{r^2} = p^2+q^2$

$\Rightarrow \dfrac{\cos^2\theta}{\sin^2\theta} = \cos^2\theta + 1$

$\Rightarrow 1-\sin^2\theta = \sin^2\theta\,(2-\sin^2\theta)$

$\Rightarrow \sin^4\theta - 3\sin^2\theta + 1 = 0$

$\Rightarrow \sin^2\theta = \dfrac{3-\sqrt{5}}{2}$

$\Rightarrow \sin\theta = \dfrac{\sqrt{5}-1}{2}$

14. **[15]** $\Delta = \frac{1}{2} \times 5.12.\sin A = 30$

$\sin A = 1$

$\Rightarrow A = 90°$

$BC = \sqrt{5^2 + (12)^2}$

$BC = 13$

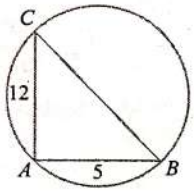
$\Rightarrow 2R = BC = 13$

Now, $r = (s-a)\tan\left(\frac{90}{2}\right)$

$\Rightarrow r = (s-a)$

$\therefore 2R + r = BC + s - BC = s$

$= \frac{a+b+c}{2} = \frac{13+12+5}{2} = 15$



15. **(b)** Let $a, b, c,$ are sides.

$x = a+b,\ y = ab$

$x^2 - c^2 = y$

$\Rightarrow (a+b)^2 - c^2 = ab$

$\Rightarrow a^2 + b^2 - c^2 = -ab$

$\Rightarrow \frac{a^2+b^2-c^2}{2ab} = -\frac{1}{2}$

$\Rightarrow \cos C = -\frac{1}{2} \left[\because \cos c = \frac{a^2+b^2-c^2}{2ab}\right]$

$\Rightarrow C = \frac{2\pi}{3} \quad \Rightarrow \sin C = \frac{\sqrt{3}}{2}$

Now,

$\frac{c}{\sin C} = 2R \Rightarrow R = \frac{c}{2\sin C} = \frac{c}{\sqrt{3}}$

[by sine rule]

16. **(c)** $a = 2 = QR$

$b = \frac{7}{2} = PR$

$c = \frac{5}{2} = PQ \Rightarrow s = \frac{a+b+c}{2} = \frac{8}{4} = 4$

$\frac{2\sin P - 2\sin P\cos P}{2\sin P + 2\sin P\cos P} = \frac{2\sin P(1-\cos P)}{2\sin P(1+\cos P)}$

$= \frac{1-\cos P}{1+\cos P} = \frac{2\sin^2\frac{P}{2}}{2\cos^2\frac{P}{2}} = \tan^2\frac{P}{2}$

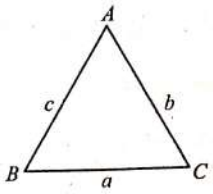
$= \frac{(s-b)(s-c)}{s(s-a)} = \frac{(s-b)^2(s-c)^2}{\Delta^2}$

$= \frac{\left(4-\frac{7}{2}\right)^2\left(4-\frac{5}{2}\right)^2}{\Delta^2} = \left(\frac{3}{4\Delta}\right)^2$

17. **(b, d)**

18. **(b)** $a + b = x$

$ab = y$



$x^2 - c^2 = y$

$(a+b)^2 - c^2 = ab$

$a^2 + b^2 + ab = c^2$

$a^2 + b^2 - c^2 = -ab$

$\frac{a^2+b^2-c^2}{2ab} = \frac{7}{2}$

$\cos C = \frac{-1}{2}$

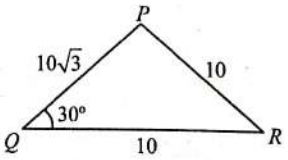
$\Rightarrow C = \frac{2\pi}{3}$

$\frac{r}{R} = \frac{\Delta \times 4\Delta}{s \times abc} = \frac{4 \times \frac{1}{4}a^2b^2\sin^2 C}{(a+b+c)abc} = \frac{3ab}{4c(x+c)}$

$= \frac{3y}{4c(x+c)}$

19. **(b, c, d)**

$\cos 30° = \frac{(10\sqrt{3})^2 + (10)^2 - (PR)^2}{2 \times 10\sqrt{3} \times 10}$



$\Rightarrow PR = 10$

$\because QR = PR$

$\Rightarrow \angle PQR = \angle QPR$

$\angle QPR = 30°$

(b) area of $\Delta PQR = \frac{1}{2} \times 10\sqrt{3} \times 10 \times \sin 30°$

$= \frac{1}{2} \times 10 \times 10\sqrt{3} \times \frac{1}{2} = 25\sqrt{3}$

$\angle QRP = 180° - (30° + 30°) = 120°$

(c) $r = \frac{\Delta}{S} = \frac{25\sqrt{3}}{\left(\frac{10+10+10\sqrt{3}}{2}\right)}$

$= \frac{25\sqrt{3}}{10+5\sqrt{3}}$

$= 5\sqrt{3}.(2-\sqrt{3}) = 10\sqrt{3} - 15$

(d) $R = \frac{a}{2\sin A} = \frac{10}{2\sin 30°} = 10$

$\therefore$ Area $= \pi R^2 = 100\pi$

20. **(b,c,d)**

$\frac{\sin P}{\sqrt{3}} = \frac{\sin Q}{1} = \frac{1}{2R} = \frac{1}{2}$

$\Rightarrow P = \frac{\pi}{3}$ or $\frac{2\pi}{3}$ and $Q = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

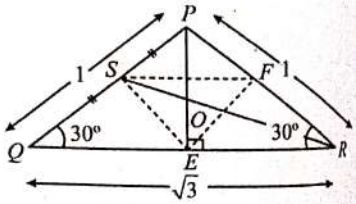
Since $p > q \Rightarrow P > Q$

So, if $P = \frac{\pi}{3}$ and $Q = \frac{\pi}{6}$

$\Rightarrow R = \frac{\pi}{2}$ (not possible)

Hence, $P = \frac{2\pi}{3}$ and $Q = R = \frac{\pi}{6}$

$r = \frac{\Delta}{s} = \frac{\frac{1}{2}(1)(\sqrt{3})\left(\frac{1}{2}\right)}{\left(\frac{\sqrt{3}+2}{2}\right)} = \frac{\sqrt{3}}{2}(2-\sqrt{3})$



Now, area of $\Delta SEF = \frac{1}{4}$ area of ΔPQR

$\Rightarrow$ area of $\Delta SOE = \frac{1}{3}$ area of ΔSEF

$= \frac{1}{12}$ area of ΔPQR

$= \frac{1}{12}.\frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{48}$

$RS = \frac{1}{2}\sqrt{2(3)+2(1)-1} = \frac{\sqrt{7}}{2}$

$OE = \frac{1}{3}PE = \frac{1}{3}.\frac{1}{2}\sqrt{2(1)^2 + 2(1)^2 - 3} = \frac{1}{6}$

By Aditya Mohod
username - muje_kabhi_dekha_nahi

By Aditya Mohad
username - muje_kabhi_dekha_nahi

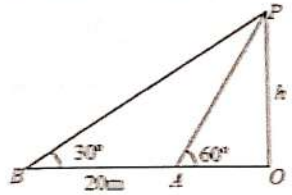
CHAPTER 21

Height & Distances

Prarambh (Topicwise)

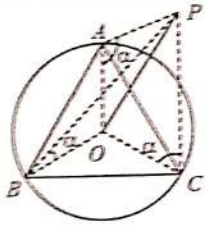
1. (c) $OA = h\cot 60°$, $OB = h\cot 30°$

$OB - OA = 20 = h(\cot 30° - \cot 60°)$



$$h = \frac{20}{\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)} = \frac{20\sqrt{3}}{2} = 10\sqrt{3}$$

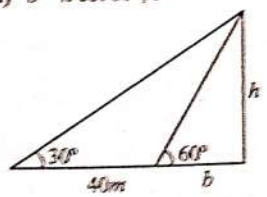
2. (d) Since the tower makes equal angles at the vertices of the triangle, therefore foot of the tower is at the circumcentre.



From ΔOAP, we have $\tan\alpha = \frac{OP}{OA}$

$\Rightarrow OP = OA\tan\alpha$ $OP = R\tan\alpha$

3. (a) $b = h\cot 60°$, $b + 40 = h\cot 30°$

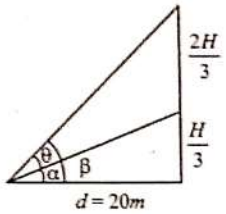


$$\Rightarrow \frac{b}{b+40} = \frac{\cot 60°}{\cot 30°}$$

$\Rightarrow b = 20$ m

4. (b) $\frac{H}{3}\cot\alpha = d$ and $H\cot\beta = d$

or $\frac{H}{3d} = \tan\alpha$ and $\frac{H}{d} = \tan\beta$



$$\tan(\beta - \alpha) = \frac{1}{2} = \frac{\frac{H}{d} - \frac{H}{3d}}{1 + \frac{H^2}{3d^2}}$$

$$\Rightarrow 1 + \frac{H^2}{3d^2} = \frac{4H}{3d} \Rightarrow H^2 - 4dH + 3d^2 = 0$$

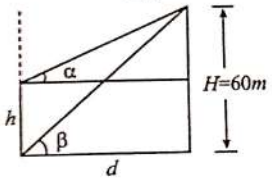
$\Rightarrow H^2 - 80H + 3(400) = 0$

$\Rightarrow H = 20$ or 60m

5. (d) $H = d\tan\beta$ and $H - h = d\tan\alpha$

$$\Rightarrow \frac{60}{60-h} = \frac{\tan\beta}{\tan\alpha}$$

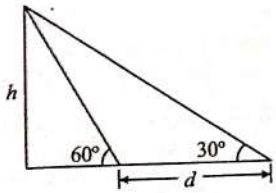
$$\Rightarrow -h = \frac{60\tan\alpha - 60\tan\beta}{\tan\beta}$$



$$\Rightarrow h = \frac{60\sin(\beta-\alpha)}{\cos\alpha\cos\beta\frac{\sin\beta}{\cos\beta}}$$

$\Rightarrow x = \cos\alpha\sin\beta$

6. (c) $d = h\cot 30° - h\cot 60°$ and time = 3 min.



$\therefore$ Speed $= \frac{h(\cot 30° - \cot 60°)}{3}$ per minute

It will travel distance $h\cot 60°$ in

$$\frac{h\cot 60° \times 3}{h(\cot 30° - \cot 60°)} = 1.5 \text{ minute}$$

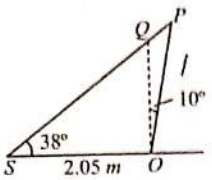
7. (a) $64\cot\theta = d$

Also $(100 - 64)\tan\theta = d$

or $(64)(36) = d^2$

$\therefore d = 8 \times 6 = 48$ m.

8. (a) $\frac{\sin 38°}{l} = \frac{\sin(SPO)}{2.05}$



$$= \frac{\sin(180° - 38° - 90° - 10°)}{2.05}$$

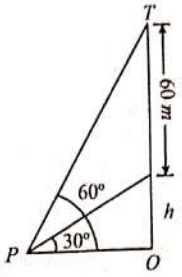
$$\Rightarrow l = \frac{2.05\sin 38°}{\sin 42°}$$

9. (b) Required distance

$$= 60\cot 15° = 60\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$$

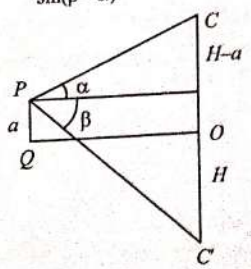
10. (a) $(60 + h)\cot 60° = h\cot 30°$

$\Rightarrow h = 30$ m



11. (b) $(H + a)\cot\beta = (H - a)\cot\alpha$

$$H = \frac{a\sin(\alpha+\beta)}{\sin(\beta-\alpha)}$$

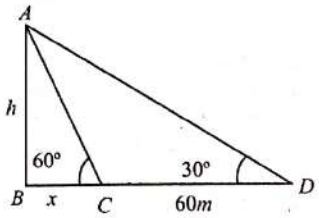


Using $\cot\alpha+\cot\beta=\frac{\sin(\alpha+\beta)}{\sin\alpha\sin\beta}$

and $\cot\alpha-\cot\beta=\frac{\sin(\beta-\alpha)}{\sin\alpha\sin\beta}$

12. **(d)** $\tan 30^o=\frac{h}{x+60}, \frac{1}{\sqrt{3}}=\frac{h}{x+60}$

$x+60=\sqrt{3}\,h, x=\sqrt{3}\,h-60$

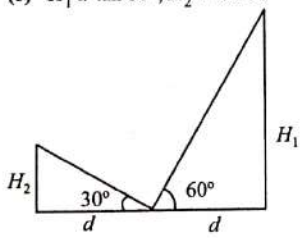


$\tan 60^o=\frac{h}{x}, \; x=\frac{h}{\sqrt{3}}$

$\Rightarrow \sqrt{3}h-60=\frac{h}{\sqrt{3}} \Rightarrow 3h-60\sqrt{3}=h$

$\Rightarrow h=\frac{60\sqrt{3}}{2}=30\sqrt{3}=51.96\approx 52$ ms

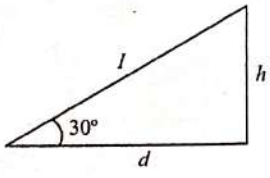
13. **(c)** $H_1\, d\tan 60°, H_2\, d\tan 30°$



$\frac{H_1}{H_2}=\frac{\tan 60^o}{\tan 30^o}=\frac{3}{1}$

14. **(c)** $H=20=l+h,\; l=\frac{d}{\cos 30^o},$

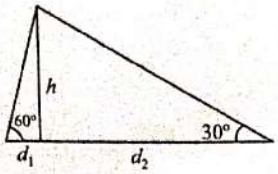
$h=d\tan 30^o$



$\therefore\; d=\frac{20}{(\sec 30^o+\tan 30^o)}=\frac{20}{\sqrt{3}}$

and hence $h=d\tan 30^o=\frac{20}{3}$m

15. **(b)** $d_2=h\cot 30^o=500\sqrt{3},\; d_1=\frac{500}{\sqrt{3}}$

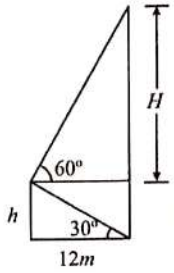


Diameter $D=500\sqrt{3}+\frac{500}{3}\sqrt{3}=\frac{2000}{\sqrt{3}}$m

16. **(b)** $h=12\tan 30^o=\frac{12}{\sqrt{3}}$ and

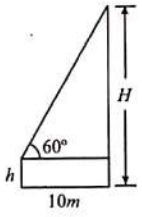
$H=12\tan 60^o+\frac{12}{\sqrt{3}}$

$12\sqrt{3}+\frac{12}{\sqrt{3}}=16\sqrt{3}$m



17. **(a)** $H=(10\tan 60^o+1.5)=$

$(10\sqrt{3}+1.5)$m

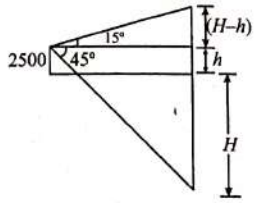


18. **(a)** **Trick:** From $H=1\tan\alpha.\tan\beta$, the height of tower is $h\tan 30^o\cot 60^o$ or $\frac{h}{3}$

19. **(c)** Obviously, the length of the tree is equal to $10+10\sqrt{2}=10\,(1+\sqrt{2})m$

20. **(a)** $(H-h)\cot 15^o=(H+h)\cot 45^o$

or $H=\frac{h(\cot 15^o+1)}{(\cot 15^o-1)}$

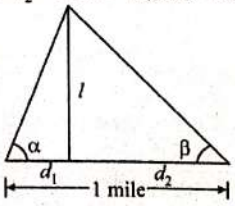


Since $h=2500$ and substitute

$\cot 15^o=2+\sqrt{3}$,we get, $H=2500\sqrt{3}$

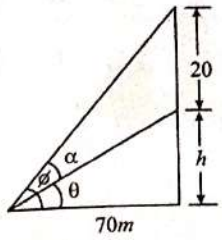
21. **(d)** $d_1=h\cot\alpha$ and $d_2=h\cot\beta$

$d_1+d_2=1$ mile $=h(\cot\alpha+\cot\beta)$



$\Rightarrow h=\frac{1}{(\cot\alpha+\cot\beta)}=\frac{\tan\alpha.\tan\beta}{\tan\alpha+\tan\beta}$

22. **(c)**



$\tan\alpha=\tan(\varphi-\theta)$

$\tan\alpha=\frac{1}{6}=\frac{\frac{20+h}{70}-\frac{h}{70}}{1+\frac{(20+h)h}{(70)^2}}$

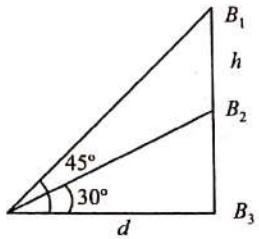
$\Rightarrow (70)^2+20h+h^2=(6)(70)(20)$

$\Rightarrow h^2+20h+70(70-120)=0$

$\Rightarrow h^2+20h-(50)(70)=0$

$\Rightarrow h=\frac{-20\pm\sqrt{400+(4)(50)(70)}}{2}=50$m

23. **(b)**



$B_1B_2=h=(d\tan 45^o-d\tan 30^o)$

Time taken = 10 *min*

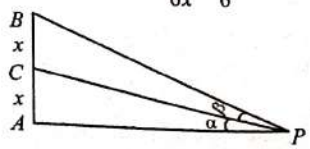
Rate $=4=\frac{d}{10}\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)$

$\Rightarrow d=\frac{40\sqrt{3}}{\sqrt{3}-1}=20(3+\sqrt{3})m.$

24. **(b)** Let $AC=x=CB, AP=3AB=6x.$

Let $\angle CPA=\alpha$

In $\Delta ACP, \tan\alpha=\frac{x}{6x}=\frac{1}{6}$



In $\Delta ABP,\; \tan(\alpha+\beta)=\frac{2x}{6x}=\frac{1}{3}$

Now $\tan\beta=\tan\{(\alpha+\beta)-\alpha\}=$

$\frac{\tan(\alpha+\beta)-\tan\alpha}{1+\tan(\alpha+\beta)\tan\alpha}$

$=\frac{\frac{1}{3}-\frac{1}{6}}{1+\frac{1}{3}.\frac{1}{6}}=\frac{1}{6}\times\frac{18}{19}=\frac{3}{19}$

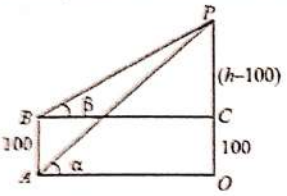
25. **(d)** Let the two roads intersect at A. If the bus and the car are at B and C on the two roads respectively, then $c = AB = 2$km, $b = AC = 3$ km. The distance between the two vehicles $= BC = a$ km

Now $\cos A = \cos 60^\circ = \dfrac{b^2 + c^2 - a^2}{2bc}$

$\Rightarrow \dfrac{1}{2} = \dfrac{3^2 + 2^2 - a^2}{2.3.2} \Rightarrow a = \sqrt{7}$ km.

26. **(c)** If OP = h, then CP = h – 100

Now equate the values of OA and BC.



$h \cot \alpha = (h - 100) \cot \beta$

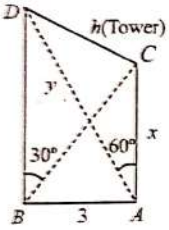
$\therefore h = \dfrac{100 \cot \beta}{\cot \beta - \cot \alpha}$

27. **(d)** From ΔCDA, $x = h \cot 60^\circ = \dfrac{h}{\sqrt{3}}$

From ΔCDB, $y = h \cot 30^\circ = \sqrt{3}h$

From ΔABC, by Pythagoras theorem

$x^2 + 3^2 = y^2$



$\Rightarrow \left(\dfrac{h}{\sqrt{3}}\right) + 3^2 = (\sqrt{3}h)^2 \Rightarrow h = \dfrac{3\sqrt{6}}{4}$ km.

28. **(c)** In ΔABC, $\tan 30^\circ = \dfrac{AC}{BC}$

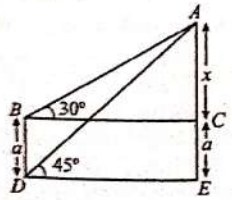
or $\dfrac{1}{\sqrt{3}} = \dfrac{x}{BC}$,

where $AC = x$ or $BC = x\sqrt{3}$ and in

ΔADE, $\tan 45^\circ = \dfrac{a + x}{DE}$

or $1 = \dfrac{a + x}{x\sqrt{3}}$ or $x\sqrt{3} = a + x$, $x(\sqrt{3} - 1) = a$

or $x = \dfrac{a}{\sqrt{3} - 1}$.



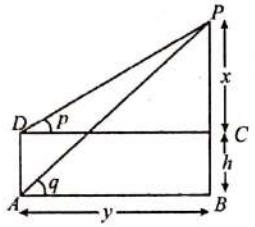
Therefore height of the tower,

$a + x = a + \dfrac{a}{\sqrt{3} - 1}$

$= a\left[\dfrac{\sqrt{3} - 1 + 1}{\sqrt{3} - 1}\right] = \dfrac{a\sqrt{3}}{\sqrt{3} - 1} \times \dfrac{\sqrt{3} + 1}{\sqrt{3} + 1}$

$= \dfrac{a(3 + \sqrt{3})}{2}$.

29. **(b)**



Let AD be the building of height h and BP be the hill then $\tan q = \dfrac{h + x}{y}$ and $\tan p = \dfrac{x}{y}$

$\Rightarrow \tan q = \dfrac{h + x}{x \cot p}$

$\Rightarrow x \cot p = (h + x) \cot q$

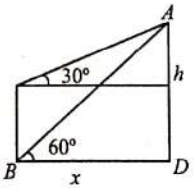
$\Rightarrow x = \dfrac{h \cot q}{\cot p - \cot q}$

$\Rightarrow h + x = \dfrac{h \cot p}{\cot p - \cot q}$

30. **(a)** Total distance from temple

$= \sqrt{x^2 + (240)^2}$ where $x = \dfrac{h}{\tan 60^\circ} = \dfrac{h}{\sqrt{3}}$

So distance $= \sqrt{\dfrac{h^2}{3} + (240)^2}$

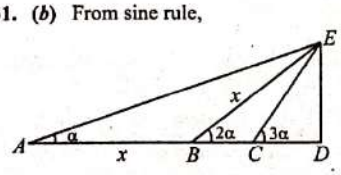


but $\dfrac{h}{\sqrt{\dfrac{h^2}{3} + (240)^2}} = \dfrac{1}{\sqrt{3}}$

$\Rightarrow \dfrac{h^2}{\dfrac{h^2}{3} + (240)^2} = \dfrac{1}{3}$

After solving, $h = 60\sqrt{6}$ m.

31. **(b)** From sine rule,



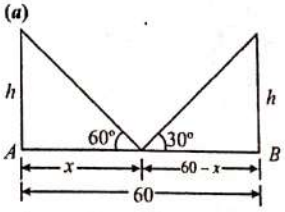
$\Rightarrow \dfrac{BE}{\sin(180^\circ - 3\alpha)} = \dfrac{BC}{\sin \alpha}$

$\Rightarrow \dfrac{AB}{\sin 3\alpha} = \dfrac{BC}{\sin \alpha}$ (Since $BE = AB$)

$\Rightarrow \dfrac{AB}{BC} = \dfrac{\sin 3\alpha}{\sin \alpha} = 3 - 4\sin^2 \alpha$

$= 3 - 2(1 - \cos 2\alpha) = 1 + 2\cos 2\alpha$.

32. **(a)**



$\tan 60^\circ = \dfrac{h}{x} \Rightarrow \dfrac{\sqrt{3}}{1} = \dfrac{h}{x} \Rightarrow h = \sqrt{3}x$ .....(*i*)

$\tan 30^\circ = \dfrac{h}{60 - x} \Rightarrow \dfrac{1}{\sqrt{3}} = \dfrac{h}{60 - x}$

$\Rightarrow 60 - x = \sqrt{3}h$ ...(*ii*)

From equation (*i*) and (*ii*),

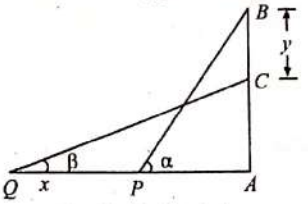
$60 - x = \sqrt{3}(\sqrt{3}x)$

$\dfrac{60}{4} = x \Rightarrow x = 15$

Then $h = \sqrt{3}x \Rightarrow h = 15\sqrt{3}$ metre.

33. **(a)** $PB = QC = 1$ (Length of ladder)

$\Rightarrow PA = 1 \cos \alpha$, $QA = 1 \cos \beta$



$\Rightarrow AC = l \sin \beta$, $AB = l \sin \alpha$

$\Rightarrow CB = AB - AC = l(\sin \alpha - \sin \beta)$

$\Rightarrow y = l(\sin \alpha - \sin \beta)$

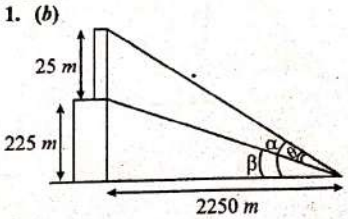
and $QP = x = AQ - AP = 1, (\cos \beta - \cos \alpha)$

$\Rightarrow \dfrac{CB}{QP} = \dfrac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha} = \dfrac{y}{x} = \dfrac{2\sin\left(\dfrac{\alpha - \beta}{2}\right)\cos\left(\dfrac{\alpha + \beta}{2}\right)}{2\sin\left(\dfrac{\alpha + \beta}{2}\right)\sin\left(\dfrac{\alpha - \beta}{2}\right)}$

$\Rightarrow \dfrac{y}{x} = \cot\left(\dfrac{\alpha + \beta}{2}\right) \Rightarrow x = y \tan\left(\dfrac{\alpha + \beta}{2}\right)$

Prabal (JEE Main Level)

1. **(b)**



$$\tan\alpha = \frac{250}{2250} = \frac{1}{9}$$

$$\tan\beta = \frac{225}{2250} = \frac{1}{10}$$

$$\theta = \alpha - \beta$$

$$\tan\theta = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta}$$

$$= \frac{\frac{1}{9} - \frac{1}{10}}{1+\frac{1}{90}}$$

$$\tan\theta = \frac{1}{91}$$

$$\Rightarrow \boxed{\tan\theta = \frac{1}{91}}$$

2. (d)

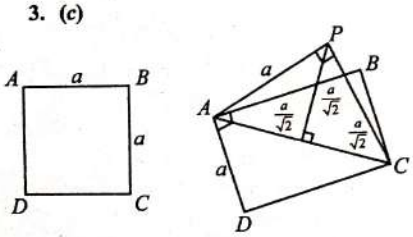
$$\tan\left(\frac{\pi}{3}\right) = \frac{h}{r}$$

Area of hexagen $= \frac{3\sqrt{3}}{2}r^2$

Area of circle $= \pi r^2 = A$

$$\Rightarrow \frac{3\sqrt{3}}{2}\cdot\left(\frac{A}{\pi}\right) = \frac{3\sqrt{3}\,A}{2\pi}\text{m}^2$$

3. (c)



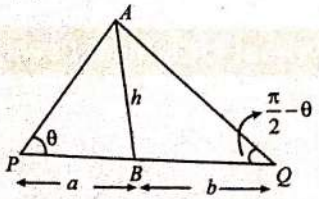
$\Rightarrow$ AP = AD = PD = a

$\Rightarrow$ angle subtended

by aside $= \frac{\pi}{3}$

($\because$ equilateral Δu)

4. (a)



In ΔAPB,

$$\tan\theta = \frac{h}{a} \qquad \ldots(i)$$

In ΔABQ,

$$\tan\left(\frac{\pi}{2}-\theta\right) = \frac{h}{b}$$

$$\cot\theta = \frac{h}{b} \qquad \ldots(ii)$$

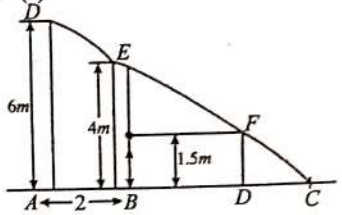
from (i) & (ii),

$$\tan\theta.\cot\theta = \frac{h}{a}\times\frac{h}{b}$$

$$\frac{h^2}{ab} = 1$$

$$h = \sqrt{ab} \;= (\alpha-\beta/\alpha+\beta)$$

5. (a)



$$BD \Rightarrow \frac{BE}{BC} = \frac{FD}{CD}$$

$$\frac{AD}{AC} = \frac{BE}{BC}$$

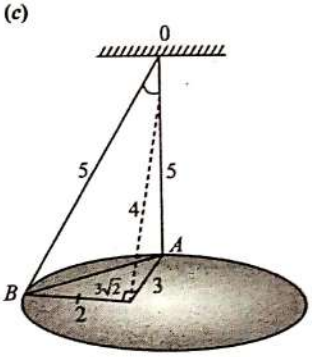
$$\frac{4}{2_1} = \frac{1.5}{4-y} \qquad \frac{6}{2+x} = \frac{y}{x}$$

$4 - y = 1.5 \Rightarrow y = 2.5 = \boxed{\frac{5}{2}}$ $\quad 6x = 8 + 4x$

$2x = 8$

$x = 4$m

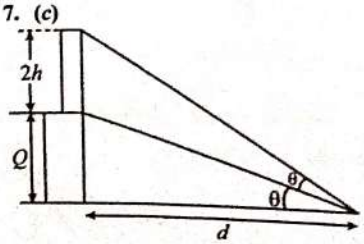
6. (c)



$$\cos\theta = \frac{5^2+5^2-(3\sqrt{2})^2}{2(5)(5)} = \frac{25+25-18}{50}$$

$$\cos\theta = \frac{32}{50} = \frac{16}{25}$$

7. (c)



$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\tan 2\theta = \frac{3b}{d}$$

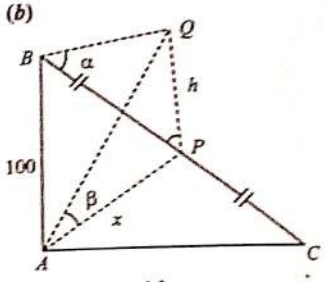
$$\tan\theta = \frac{h}{d}$$

$$\Rightarrow \frac{3h}{d} = \frac{\frac{2h}{d}}{1-\frac{h^2}{d^2}} \qquad \Rightarrow 1-\left(\frac{h}{d}\right)^2 = \frac{2}{3}$$

$$\Rightarrow \boxed{\frac{1}{\sqrt{3}} = \frac{h}{d}}$$

8. (a) $\tan 30° = h \Rightarrow h = \frac{1}{\sqrt{3}}$ km

9. (b)



$\cot\alpha = 3.2 = \frac{16}{5}$; $\operatorname{cosec}\beta = 2.6 = \frac{13}{5}$

Let $AP = x$ (metre)

$$\cot\alpha = \frac{x}{h} = \frac{16}{5} \Rightarrow x = \frac{16}{5}h$$

In ΔAPB,

$BP^2 + x^2 = 100^2 \Rightarrow BP^2 = 100^2 - x^2$

In ΔBPQ, $BP^2 + PQ^2 = BQ^2$

$BQ^2 = 10^4 - x^2 + h^2$

$$\text{Sin}\,\beta = \frac{PQ}{BQ} \Rightarrow \frac{h}{\sqrt{10^4 - x^2 + h^2}} = \frac{5}{13}$$

$169\,h^2 = 25\,(10^4 - x^2 + h^2)$

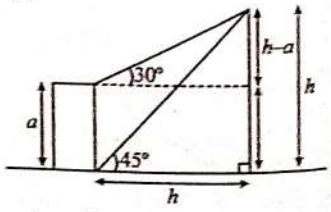
$144h^2 = 25.10^4 - 25x^2$

$$= 25.10^4 - 25.\frac{256}{25}h^2 \qquad \text{(from (i))}$$

$(144 + 256)h^2 = 25.10^4$

$$h^2 = \frac{25.10^4}{400} \Rightarrow h = \frac{5.100}{20} = 25\text{ m}$$

10. (c)



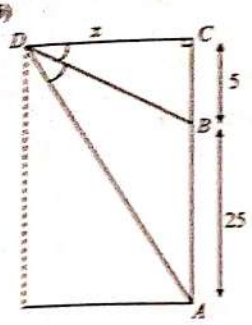
$$\frac{1}{\sqrt{3}} = \frac{h-a}{h}$$

$h = h\sqrt{3} - a\sqrt{3} \Rightarrow a\sqrt{3} - h(\sqrt{3}-1)$

$$\Rightarrow h = \frac{a\sqrt{3}}{(\sqrt{3}-1)} = a(3-\sqrt{3})$$

$$\boxed{h = \frac{a\sqrt{3}}{(\sqrt{3}-1)}}$$

11. **(b)**



In ΔBCD in ΔACD

$\tan\alpha = \frac{5}{x}$; $\tan 2\alpha = \frac{30}{x}$

$\frac{2\tan\alpha}{1-\tan^2\alpha} = \frac{30}{x}$

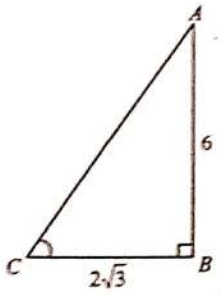
$\frac{2x\frac{5}{x}}{1-\frac{25}{x^2}} = \frac{30}{x}$

$\frac{10}{x^2}\cdot\frac{x^2}{(x^2-25)} = \frac{30}{x} \Rightarrow x^2 = 3x^2 - 75$

$75 = 2x^2$

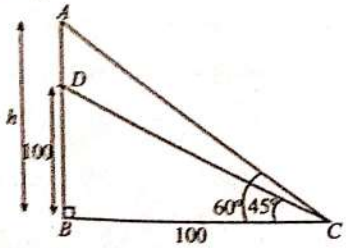
$x = 5\sqrt{\frac{3}{2}}$

12. **(a)**



$\tan\theta = \frac{6}{2\sqrt{3}} \Rightarrow \theta = 60°$

13. **(c)**

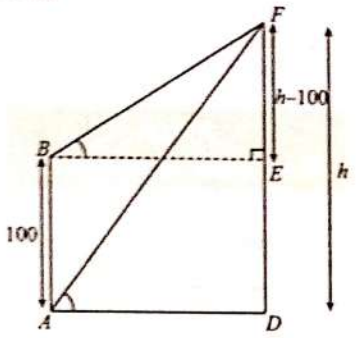


$\frac{h}{100} = \tan 60° \Rightarrow h = 100\sqrt{3}$

Height increased by $(100\sqrt{3} - 100)$

$= 100(\sqrt{3} - 1)$

14. **(c)**

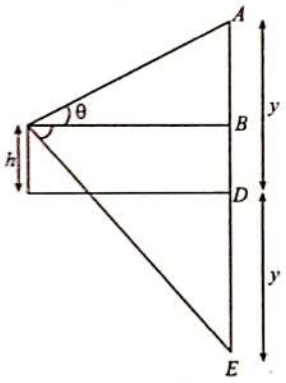


Same as previous question.

15. **(b)** In ΔABC at BCE

$(y - h)\cot\theta = (y + h)\cot\phi$

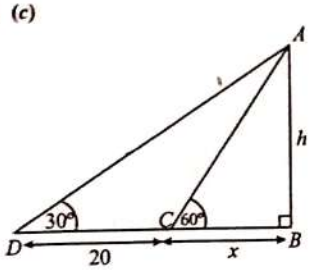
$y(\cot\theta - \cot\phi) = (\cot\phi + \cot\theta)h$



$\Rightarrow y = \frac{(\cot\phi + \cot\theta)h}{\cot\theta - \cot\phi}$

$y = \frac{\sin(\theta+\phi)}{\sin(\phi-\theta)}h$

16. **(c)**



In Δ ABC

$\frac{h}{x} = \tan 60°$

$\boxed{h = \sqrt{3}x}$ ...(i)

$2\ \Delta ABD$

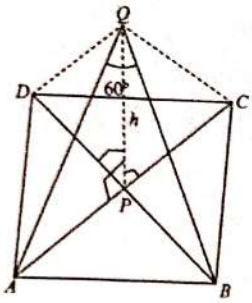
$\frac{h}{x+20} = \tan 30° \Rightarrow \sqrt{3}g = x + 20$ ...(ii)

from (i) & (ii)

$h = \sqrt{3}(\sqrt{3}h - 20) \Rightarrow 2h = 20\sqrt{3}$

$\Rightarrow \boxed{h = 10\sqrt{3}}$

17. **(b)**



$\angle AQB = 60°$ and $AQ = BQ$

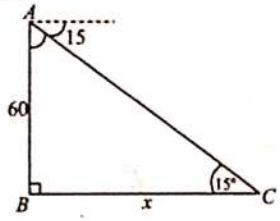
$\Rightarrow ABQ$ is equilateral triangle

In ΔDPQ

$AQ^2 = h^2 + AP^2$

$Q^2 = h^2 + \frac{a^2}{2} \Rightarrow \boxed{2h^2 = a^2}$

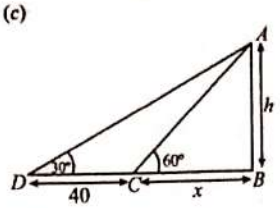
18. **(b)**



If ΔABC $\frac{x}{60} = \cot 15°$

$\Rightarrow x = (2+\sqrt{3})\,60 = \frac{\sqrt{3}+1}{\sqrt{3}-1}.60$ metres

19. **(c)**



Total x

If ΔABC ; If ΔABD

$h = \sqrt{3}x$ ...(i)

$\sqrt{3}h = 40 + x$ ...(ii)

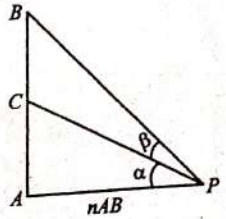
$h = \sqrt{3}x = \frac{40+x}{\sqrt{3}}$

Simplify.

$\sqrt{3}\times\sqrt{3}x = 40 + x$

$2x = 40 \Rightarrow x = 20$ m

20. **(a)**



In ΔACP

$$\tan\alpha = \frac{AC}{AP} = \frac{AB/2}{nAP} = \frac{1}{2n}$$

In ΔABP

$$\tan(\alpha+\beta) = \frac{AB}{AP} = \frac{AB}{nAB} = \frac{1}{n}$$

$$\tan(\alpha+\beta) = \frac{1}{n}$$

$$\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta} = \frac{1}{n}$$

$$\frac{\frac{1}{2n}+\tan\beta}{1-\frac{1}{2n}\tan\beta} = \frac{1}{4}$$

$$\frac{1+2n\tan\beta}{\frac{2n}{2n-\tan\beta}} = \frac{1}{n}$$

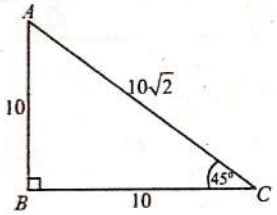
$$n(1+2n\tan\beta) = 2n-\tan\beta$$

$$n+2n^2\tan\beta = 2n-\tan\beta$$

$$\tan\beta(2n^2+1) = 2n-n$$

$$\tan\beta = \frac{n}{2n^2+1}$$

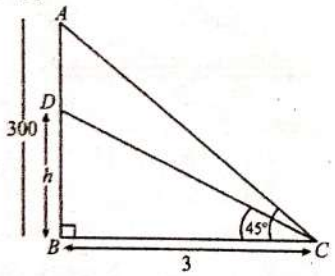
21. *(c)*



Total height of true = AB + AC

$= 10(\sqrt{2}+1)$

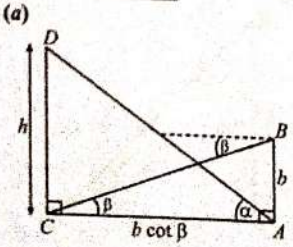
22. *(a)*



$$BC = 300\cot 60° = \frac{100}{\sqrt{3}}$$

If Δ BCD $\boxed{h = \frac{300}{\sqrt{3}}} = 100\sqrt{3}$

23. *(a)*



In ΔABC

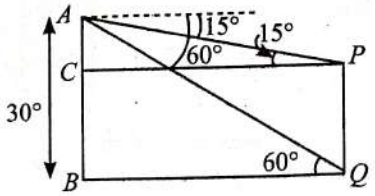
$AC = b\cot\beta$

$h\,\Delta ACD$

$h = b\cot\beta\tan\alpha.$

PYQ's (Past Year Questions)

1. *(a)*



In ΔABQ

$$\tan 60° = \frac{AB}{BQ} \Rightarrow \sqrt{3} = \frac{30}{BQ}$$

$BQ = 10\sqrt{3}\,m = CP$

In ΔACP,

$$\tan 15° = \frac{AC}{CP} \Rightarrow 2-\sqrt{3} = \frac{AC}{CP}$$

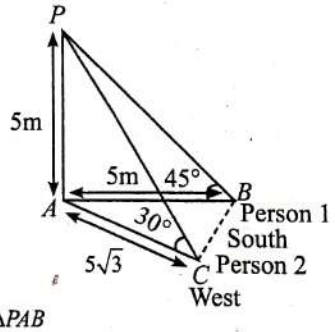
$AC = 10\sqrt{3}(2-\sqrt{3})$

$\therefore BC = 60-20\sqrt{3}$

In ΔACP,

$\therefore$ Area $= 10\sqrt{3}(60-20\sqrt{3}) = 600(\sqrt{3}-1)$

2. *(a)*



In ΔPAB

$$\tan 45° = \frac{5}{AB}$$

$\Rightarrow AB = 5$

In ΔPAC

$$\tan 30° = \frac{AB}{AC} = \frac{5}{AB}$$

$AB = 5\sqrt{3}$

Now, ΔABC

$$BC = \sqrt{5^2+(5\sqrt{3})^2}$$

Distance $BC = 10$

(By Pythagoras theorem)

3. **[6]** Let $AB = CD = x$

In ΔCAB

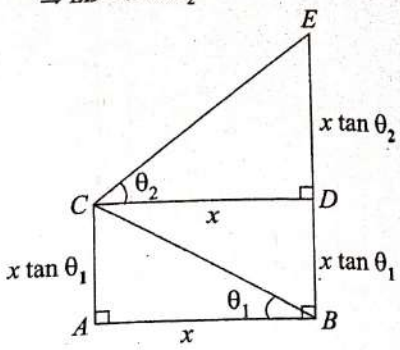
$$\tan\theta_1 = \frac{AC}{x}$$

$\Rightarrow AC = x\tan\theta_1$

In ΔCDE

$$\tan\theta_2 = \frac{ED}{x}$$

$\Rightarrow ED = x\tan\theta_2$



It is given $\sqrt{3}BE = 4AB$

$ar(\Delta CAB) = 2\sqrt{3}-3$

$$\Rightarrow \frac{1}{2}x^2\tan\theta_1 = 2\sqrt{3}-3$$

$BE = BD + DE$

$= x(\tan\theta_1 + \tan\theta_2)$

$BE = AB(\tan\theta_1 + \cot\theta_1)$

$$\frac{4}{\sqrt{3}} = \tan\theta_1+\cot\theta_1 \Rightarrow \tan\theta_1 = \sqrt{3}, \frac{1}{\sqrt{3}}$$

$$\left[\because \theta_1+\theta_2 = \frac{\pi}{2}\right]$$

$$\theta_1 = \frac{\pi}{6} \text{ and } \theta_2 = \frac{\pi}{3}$$

or $\theta_1 = \frac{\pi}{3}$ and $\theta_2 = \frac{\pi}{6}$

As $\frac{\theta_2}{\theta_1}$ is largest $\therefore \theta_1 = \frac{\pi}{6}$; $\theta_2 = \frac{\pi}{3}$

$$\therefore x^2 = \frac{(2\sqrt{3}-3)\times 2}{\tan\theta_1} = \frac{\sqrt{3}(2-\sqrt{3})\times 2}{\tan\frac{\pi}{6}}$$

$x^2 = 12-6\sqrt{3} = (3-\sqrt{3})^2$

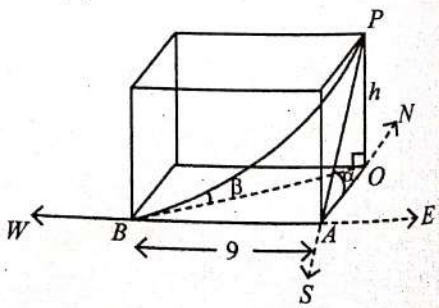
$\Rightarrow x = 3-\sqrt{3}$

Perimeter of ΔCED

$= CD + DE + CE$

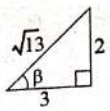
$= 3-\sqrt{3}+(3-\sqrt{3})\sqrt{3}+(3-\sqrt{3})\times 2 = 6$

4. *(a)*

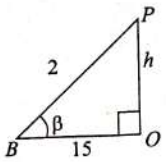


Given, $OB = 15$

$$\cos\beta = \frac{3}{\sqrt{13}}$$

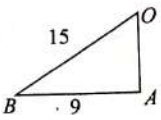


$\tan\beta = \frac{2}{3}$



$\tan\beta = \frac{h}{15} \Rightarrow \frac{2}{3} = \frac{h}{15}$

$\Rightarrow h = 10$



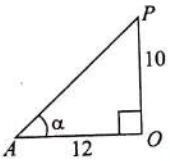
$OA^2 + AB^2 = 225$

$\Rightarrow OA^2 + 81 = 225$

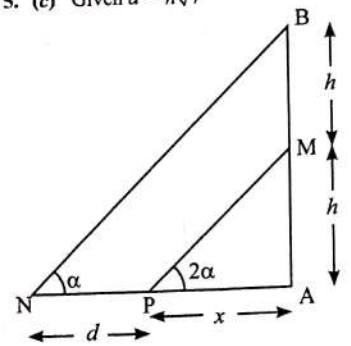
$OA = 12$

$\tan\alpha = \frac{10}{12}$

$\cot\alpha = \frac{12}{10} = \frac{6}{5}$



5. (*c*) Given $d = h\sqrt{7}$



$\tan 2\alpha = \frac{h}{x}$ ...(*i*)

$\tan\alpha = \frac{2h}{x+d}$ ...(*ii*)

Now, from eqn (*ii*)

$\tan\alpha = \frac{2h}{x + h\sqrt{7}}$

$\Rightarrow \tan\alpha = \frac{2\left(\frac{h}{x}\right)}{1+\left(\frac{h}{x}\right)\sqrt{7}}$ [$\because d = h\sqrt{7}$]

$\Rightarrow \tan\alpha + \tan\alpha.\tan 2\alpha\sqrt{7} = 2\tan 2\alpha$

$\Rightarrow \tan\alpha \tan\alpha.$

$\frac{2\tan\alpha}{1-\tan^2\alpha}\sqrt{7} = 2.\frac{2.\tan\alpha}{1-\tan^2\alpha}$

$\Rightarrow 1 + \frac{2\tan\alpha.\sqrt{7}}{1-\tan^2\alpha} = \frac{4}{1-\tan^2\alpha}$

$\Rightarrow 1 - \tan^2\alpha + 2\sqrt{7}\tan\alpha = 4$

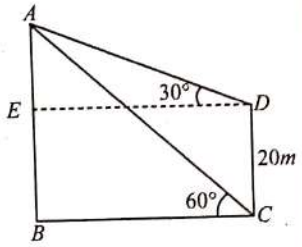
$\Rightarrow \tan^2\alpha - 2\sqrt{7}\tan\alpha + 3 = 0$

$\Rightarrow \tan\alpha = \frac{2\sqrt{7} \pm \sqrt{28-12}}{2}$

$\Rightarrow \tan\alpha = \sqrt{7} \pm 2$

So, $\tan\alpha = \sqrt{7} - 2$

6. (*d*)



Let CD be a pole of height $20m$.

In rt. ΔABC

$\tan 60° = \frac{AB}{BC}$

$BC\sqrt{3} = AB$ ...(*i*)

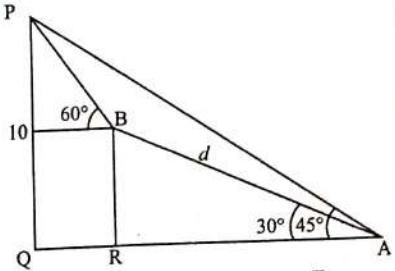
In rt. ΔAED

$\tan 30° = \frac{AE}{ED}$

$\Rightarrow BC = (AB - 20)\sqrt{3}$ ($\because ED = BC$)...(*ii*)

On solving (*i*) and (*ii*), we get $AB = 30m$

7. (*a*)



QA = 10, RA = $d\cos 30° = \frac{\sqrt{3}d}{2}$

QR = $10 - \frac{\sqrt{3}d}{2}$, BR = $d\sin 30° = \frac{d}{2}$

$\therefore \tan 60 = \frac{PQ - BR}{QR} = \frac{10 - \frac{d}{2}}{10 - \frac{\sqrt{3}d}{2}}$

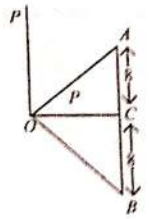
$\Rightarrow \sqrt{3} = \frac{20 - d}{20 - \sqrt{3}d}$

$\Rightarrow d = 10(\sqrt{3} - 1)$

Area of PQRB = $\alpha = \frac{1}{2}(PQ + BR)QR$

$= \frac{1}{2}\left(10 + \frac{d}{2}\right).\left(10 - \frac{\sqrt{3}d}{2}\right) = 25$

8. (*b*) $\frac{OP}{OA} = \tan 15°$



$\Rightarrow OA = OP\cot 15°$

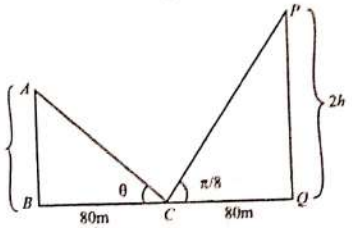
$\frac{OP}{OC} = \tan 45° \Rightarrow OP = OC$

Now, $OP = \sqrt{OA^2 - 64}$

$OP^2 = OP^2\cot^2 15° - 64$

$OP^2 = \frac{32}{\sqrt{3}}\left(2 - \sqrt{3}\right)$

9. (*c*)



In ΔPQC,

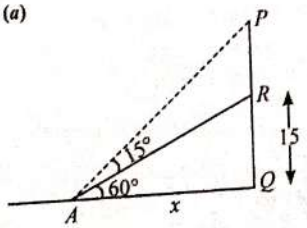
$\tan\frac{\pi}{8} = \frac{2h}{80}$ ... (*i*)

In ΔABC, $\tan\theta = \frac{h}{80}$ ... (*ii*)

$\Rightarrow \tan\frac{\pi}{8} = 2\tan\theta \Rightarrow 2\tan\theta = \left(\sqrt{2} - 1\right)$

$\therefore \tan^2\theta = \frac{3 - 2\sqrt{2}}{4}$

10. (*a*)



Let, $AQ = x$

$$\tan 60° = \frac{15}{x} \Rightarrow x = \frac{15\sqrt{3}}{3} = 5\sqrt{3}$$

Now, $\tan 75° = \frac{PQ}{x}$

$$2A\sqrt{3} = \frac{PQ}{5\sqrt{3}} \Rightarrow PQ = 10\sqrt{3} + 15$$

11. **(c)** Since, angle of elevation of top of pole from A, B, C are equal So, O must be circumcentre of ABC.

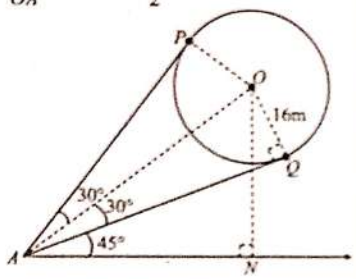
Let height of the pole be h

$$\therefore \frac{h}{2} = \tan\frac{\pi}{3} = \sqrt{3}$$

$$\Rightarrow h = 2\sqrt{3}$$

12. **(b)** In ΔOQA,

$$\frac{OQ}{OA} = \sin 30° = \frac{1}{2} \Rightarrow OA = 32\,\text{m}$$



Again $ON = OA \sin 75°$

$= 32 \sin(45° + 30°)$

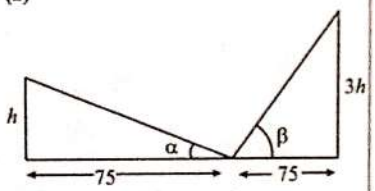
$$= 32\left[\frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}\right] = \frac{32(\sqrt{3}+1)}{2\sqrt{2}}$$

$$= 8\left(\sqrt{6} + \sqrt{2}\right)$$

Height of top most point

$$= 8\left(\sqrt{6} + \sqrt{2}\right) + 16 = 8\left(\sqrt{6} + \sqrt{2} + 2\right)$$

13. **(d)**



we hvae, $\alpha + \beta = 90°$

$\Rightarrow \alpha = 90° - \beta$

$\Rightarrow \tan\alpha = \tan(90° - \beta)$

$\Rightarrow \tan\alpha = \cot\beta$

$\Rightarrow \tan\alpha.\tan\beta = 1$

$$\Rightarrow \frac{h}{75}\cdot\frac{3h}{75} = 1 \Rightarrow h^2 = 75 \times 25$$

$$\therefore h = 25\sqrt{3}\,\text{m}$$

14. **(a)**

15. **(c)** In ΔBCD,

$$\tan\beta = \frac{x}{a+b}$$

In ΔAPC,

$$\tan(\alpha+\beta) = \frac{x}{b}$$

$$\Rightarrow \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{x}{b}$$

$$\Rightarrow \frac{\frac{1}{2} + \frac{x}{a+b}}{1 - \frac{1}{2}\cdot\frac{x}{a+b}} = \frac{x}{b}\left[\because \text{Given } \tan\alpha = \frac{1}{2}\right]$$

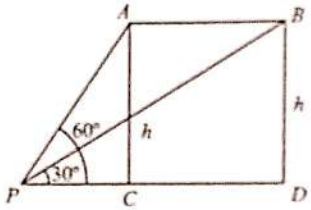
$$\Rightarrow \frac{a+b+2x}{2a+2b-x} = \frac{x}{b}$$

$$\Rightarrow ab + b^2 + 2xb = 2ax + 2xb - x^2$$

$$\Rightarrow x^2 - 2ax + ab + b^2 = 0$$

$$\Rightarrow x^2 - 2ax + b(a+b) = 0$$

16. **(b)**



$$V = 432 \times \frac{1000}{3600}\,\text{m/s}$$

$$= 120\,\text{m/s}$$

$AB = CD = 120\,\text{m/s} \times 20\text{s} = 2400\,\text{m}$

In ΔPAC, $\tan 60° = \frac{h}{PC} \Rightarrow PC = \frac{h}{\sqrt{3}}$

In ΔPBD, $\tan 30° = \frac{h}{PD} \Rightarrow PD = h\sqrt{3}$

$$CD = PD - PC = h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$$

$$\frac{2h}{\sqrt{3}} = 2400 \Rightarrow h = 1200\sqrt{3}\,\text{m}$$

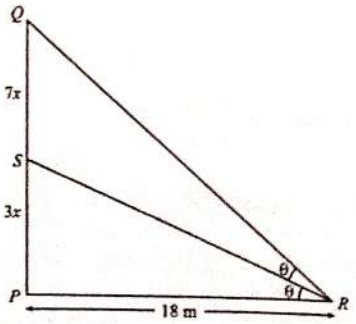
17. **(c)** Let PQ represent a pole.

In ΔPSR,

$$\tan\theta = \frac{3x}{18} = \frac{x}{6} \qquad \ldots(i)$$

In ΔPQR,

$$\tan 2\theta = \frac{10x}{18} = \frac{5x}{9} \qquad \ldots(ii)$$



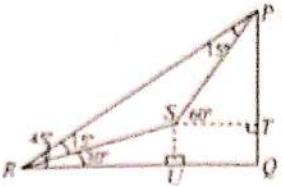
$$\Rightarrow \frac{2\tan\theta}{1-\tan^2\theta} = \frac{5x}{9}$$

$$\Rightarrow \frac{2\left(\frac{x}{6}\right)}{1-\frac{x^2}{36}} = \frac{5x}{9} \qquad \left(\text{Use } \tan\theta = \frac{x}{6}\right)$$

$$\Rightarrow x = \sqrt{\frac{72}{5}}$$

$\therefore$ Height of pole $= 10x = 12\sqrt{10}$

18. **(b)** $\because SP = SR = 1$ km



In ΔSPT,

$$\frac{PT}{SP} = \sin 60°$$

$$\Rightarrow PT = \frac{\sqrt{3}}{2}\,\text{km}$$

In ΔRSU, $\sin 30° = \frac{SU}{RS}$

$$\Rightarrow SU = \frac{1}{2}\,\text{km}$$

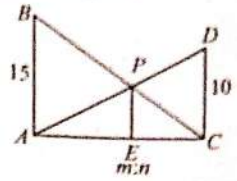
Since, $TQ = SU = \frac{1}{2}$ km

$\therefore$ Height of mountain $PQ = PT + TQ$

$$= \left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right) = \left(\frac{\sqrt{3}+1}{2}\right)\text{km}$$

$$= \frac{1}{\sqrt{3}-1}\,\text{km}$$

19. **(a)** Refer to diagram,



Let $PE \perp AC$

and $\frac{AE}{EC} = \frac{m}{n}$

So; $PE = \frac{10m}{m+n}$ ...(i)

(because ΔACD and ΔAEP are similar)

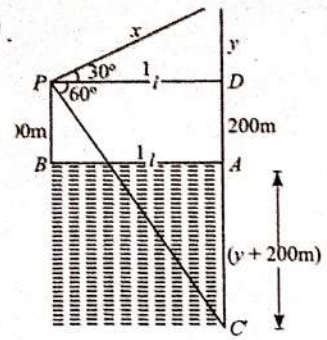
Similarly $PE = \frac{15n}{m+n}$ ...(ii)

From (*i*) and (*ii*)

$10m = 15n \Rightarrow m = \frac{3}{2}n$

So $PE = 6$

20. (*a*)



Here in ΔPCD

$\sin 30° = \frac{y}{x} \Rightarrow y = \frac{x}{2}$

$\cos 30° = \frac{l}{x} \Rightarrow l = \frac{\sqrt{3}x}{2}$

Now, in right $\Delta PC'D$

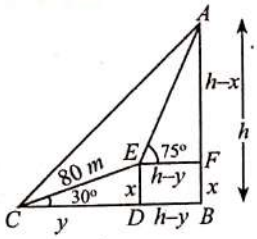
$\tan 60° = \frac{y+400}{l}$

$\Rightarrow y + 400 = \sqrt{3}l = \sqrt{3} \times \frac{\sqrt{3}x}{2}$

$\Rightarrow \frac{x}{2} - \frac{3x}{2} = -400 \Rightarrow x = 400$ m

So, $PC = 400$ m

21. (*b*) In right angled ΔCDE,



Let height = h m

$\sin 30° = \frac{x}{80} \Rightarrow x = 40$

$\cos 30° = \frac{y}{80}$

$\Rightarrow y = 40\sqrt{3}$

Now, In ΔAEF

$\tan 75° = \frac{h-x}{h-y} \Rightarrow (2+\sqrt{3}) = \frac{h-40}{h-40\sqrt{3}}$

$(2+\sqrt{3})(h-40\sqrt{3}) = h-40$

$\Rightarrow 2h - 80\sqrt{3} + \sqrt{3}h - 120 = h - 40$

$\Rightarrow h + \sqrt{3}h = 80 + 80\sqrt{3}$

$\Rightarrow (\sqrt{3}+1)h = 80(\sqrt{3}+1)$

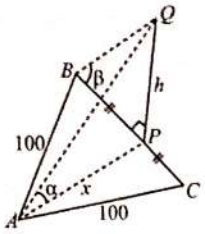
Hence, $h = 80$ m

22. (*c*) $\cot\alpha = 3\sqrt{2}$ & $\operatorname{cosec}\beta$

$= 2\sqrt{2} \Rightarrow \tan\beta = \frac{1}{\sqrt{7}}$

In ΔAPQ

$\cot\alpha = \frac{AP}{PQ}$



So, $\frac{x}{h} = 3\sqrt{2}$ ...(*i*)

and $\tan\beta = \frac{PQ}{PB}$ in ΔPBQ

$\frac{h}{\sqrt{10^4 - x^2}} = \frac{1}{\sqrt{7}} = \tan\beta$

$\left[\because PB = \sqrt{BA^2 - AP^2}\right]$ ...(*ii*)

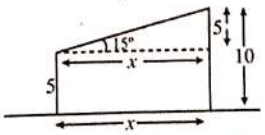
So, from (*i*) & (*ii*)

$\Rightarrow \frac{h}{\sqrt{10^4 - 18h^2}} = \frac{1}{\sqrt{7}}$

$\Rightarrow 25h^2 = 100 \times 100$

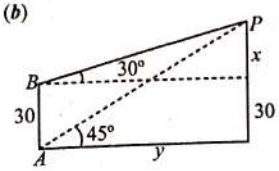
$\Rightarrow h = 20m$

23. (*c*) $\tan 15° = \frac{5}{x}$



$\Rightarrow 2 - \sqrt{3} = \frac{5}{x} \Rightarrow x = 5(2+\sqrt{3})$

24. (*b*)



$\tan 45° = 1 = \frac{x+30}{y} \Rightarrow x + 30 = y$ ...(*i*)

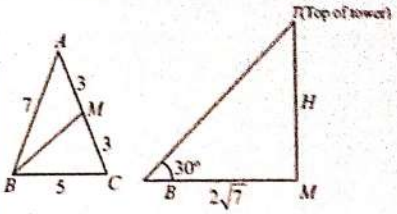
$\tan 30° = \frac{1}{\sqrt{3}} = \frac{x}{y} \Rightarrow x = \frac{y}{\sqrt{3}}$ ...(*ii*)

From (*i*) and (*ii*) $y = 15(3+\sqrt{3})$

25. (*b*) Length of median BM

$= \frac{1}{2}\sqrt{2(BC^2 + BA^2) - (AC)^2}$

$= \frac{1}{2}\sqrt{2(25+49) - 36}$



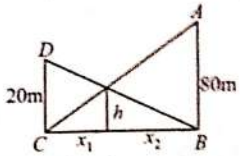
$= \frac{1}{2}\sqrt{112} = \sqrt{\frac{112}{4}} = \sqrt{28} = 2\sqrt{7}$

Let H be height of tower, given

$\tan 30° = \frac{H}{2\sqrt{7}}$

$\Rightarrow h = 2\sqrt{\frac{7}{3}} = \sqrt{\frac{28}{3}} = \frac{2}{3}\sqrt{21}$

26. (*c*)



by similar triangle

$\frac{h}{x_1} = \frac{80}{x_1 + x_2}$ ...(*i*)

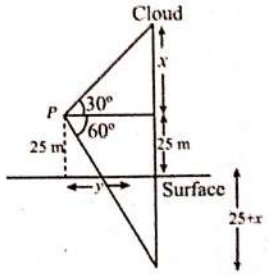
by $\frac{h}{x_2} = \frac{20}{x_1 + x_2}$ ...(*ii*)

by (*i*) and (*ii*)

$\frac{x_2}{x_1} = 4$ or $x_2 = 4x_1$

$\Rightarrow \frac{h}{x_1} = \frac{80}{5x_1}$ or $h = 16$ m

27. (*b*)



$\tan 30° = \frac{x}{y} \Rightarrow y = \sqrt{3}x$ ...(*i*)

$\tan 60° = \frac{25 + x + 25}{y}$

$\Rightarrow \sqrt{3}y = 50 + x$

$\Rightarrow 3x = 50 + x$

$\Rightarrow x = 25$ m

$\therefore$ Height of cloud from surface

$= 25 + 25 = 50$ m

By Aditya Mohod
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